

DEEP PRIMER

FTAI Aviation and the CFM56 Aftermarket — A Deep Primer

A first-principles reference on FTAI Aviation Ltd. (NASDAQ: FTAI) and the narrowbody engine aftermarket it operates in. Descriptive throughout; every figure cited; disagreements between sources reported, never averaged. Takes no view.

Research as of 3 October 2026

Spine first, then twelve Parts. Built from eleven research dossiers, a reconciliation pass and a term registry, all published alongside.

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THE SPINE

The chain in one pass

The whole chain in one reading: the engine, what it costs to keep flying, who pays, why the fleet behind the world's older narrowbodies is still flying in 2026, how that became a shop-visit plateau and a parts shortage, who sells the substitutes for new parts, and how FTAI Aviation built a module business, a capital business and a power business on top of it. Terms a reader cannot do without are defined in a box at first use; every number carries its source and date, and where two sources disagree both are given and nothing is averaged. Ten disagreements bear directly on the economics; each is presented where the chain reaches it and all are listed at the end. Each section ends with a pointer to the Part that carries the depth. Nothing is ranked or recommended.

S1. The machine: a turbofan, its modules, the life limits that force it open, and the margin that erodes

Everything in this primer follows from four physical facts about one engine. It is built in separable sections. Inside them sit parts scrapped on a clock. Its hot section loses temperature margin as it wears. And it is only as usable as its paperwork.

TURBOFAN: FAN, BYPASS AND CORE

A turbofan is a jet engine with a large fan at the front. Most of the air the fan moves passes around the hot middle, the bypass stream; a small share goes through the middle, where fuel is burned and turbines are driven. The turbines spin the fan, and the fan does most of the pushing. [Part I §1]

CFM56 (-7B AND -5B), THE 737NG AND A320CEO IT POWERS, AND THE V2500 (THE A320CEO'S ALTERNATIVE ENGINE)

Counts of CFM56s "in service" are about 23,000 (CFM website, undated), about 24,000 (Aviation Business News, Sept 2025) and more than 22,800 (Safran, end-2025); none says whether spares are included. The -7B is the Boeing 737NG's (Next Generation) only engine; the -5B powers about six in ten Airbus A320ceo (current engine option) aircraft, the rest carrying the V2500 from the IAE consortium. The last -7B was delivered by May 2019 and -5B production for airframes ended in 2022, so every CFM56 that flies tomorrow exists today. [CFM; Aviation Business News; Safran, 13 Feb 2026; GE Aerospace, 2019; ISTAT (the International Society of Transport Aircraft Trading)]

ORIGINAL EQUIPMENT MANUFACTURER (OEM), AND CFM INTERNATIONAL

The original equipment manufacturer (OEM) holds the regulator's approval of the engine's design. It is the sole source of new parts at its published catalogue price, controls the repair manual, and runs its own shops and service contracts. For the CFM56 the OEM is CFM International, the joint venture of GE Aerospace and Safran, described everywhere as 50/50: GE builds the core, Safran the fan, booster, low-pressure turbine and gearbox. The same venture makes the LEAP (CFM's new engine) for the A320neo and the 737 MAX (the re-engined 737). [CFM website; Part I §2; Part IV §25]

MODULE, AND THE CORE AS THE HOT MODULE

A module is a section of the engine built to be removed and replaced as a unit, bolted to its neighbours at flanges, with its own serial number and records. The CFM56 has three gas-path major modules, the fan and booster, the core and the low-pressure turbine (LPT), plus an accessory gearbox. The core is the hot, fast, expensive middle: the high-pressure compressor (HPC), the combustor and the high-pressure turbine (HPT). Sources count three to six modules depending on the level counted; FTAI's releases convert at three per engine ("450 modules (150 engines)", 26 Feb 2025). In FTAI's exchange business "core" also means the unit a customer hands back, called the "returned core" here. [Aircraft Commerce Issue 58; FTAI release, 26 Feb 2025; Part I §3; reconciliation R-035]

LIFE-LIMITED PART (LLP) AND THE ENGINE FLIGHT CYCLE (EFC)

An engine flight cycle (EFC) is one take-off and landing; a mature short-haul narrowbody flies roughly 1,000 to 1,500 cycles a year (Aircraft Value News). A life-limited part (LLP) is a disk, spool or shaft whose failure the engine case cannot be designed to contain. It is certified for a fixed number of cycles and must be scrapped at that limit whatever its condition: 30,000 in the fan and booster, 20,000 in the compressor and high-pressure turbine, 25,000 in the low-pressure turbine on the -7B. These are the innermost parts, so reaching a limit forces the engine open. [Aircraft Commerce Issue 34; records mini-packs; Aircraft Value News]

EXHAUST-GAS TEMPERATURE (EGT) MARGIN

Every engine has a certified exhaust-gas temperature (EGT) red line. EGT margin is the gap between that limit and the temperature the engine reaches at full take-off thrust on a hot day. As blade tips and seals wear the engine runs hotter and the margin shrinks; at zero the engine must come off, and restoring the high-pressure turbine's blades, vanes and seals recovers most of it. [Aircraft Commerce Issue 27; Part I §5]

TIME ON WING

Time on wing is the interval between an engine's installation and its removal, set by whichever clock runs out first: the life-limited part with the fewest cycles remaining or the temperature margin. A mature CFM56 has a performance restoration every 10,000 to 15,000 cycles and a full overhaul with life-limited-part replacement at 20,000 to 25,000 (Leeham, Mar 2024): roughly seven to fifteen years between restorations at 1,000 to 1,500 cycles a year. [Leeham News, 8 Mar 2024; Part I §5–§6]

WORKED EXAMPLE S.1 — THE SHORTER CLOCK GOVERNS

Inputs: a CFM56-7B whose high-pressure turbine disk has used 18,500 of its 20,000 cycles (limit from the mini-packs; cycles used assumed); 1,500 cycles a year (top of Aircraft Value News's range); temperature margin for another 5,000 cycles (assumed).

cycles remaining on the disk = $20,000 - 18,500 = 1,500$
time to the limit = $1,500 \div 1,500$ a year = about one year
margin remaining = 5,000 cycles, worth nothing to this owner

The disk sets the removal date at about a year, and 5,000 cycles of margin go unused. On a fresh disk and a worn hot section the margin governs instead, and the disk's life goes into the shop unused. Two inputs are assumed; the point is the rule. [Part I worked example 1.1]

Three consequences follow. An engine at a shop visit is really three or four assets with different remaining lives. The quickest return to service is often to replace the one module that has run out. And the usable supply is the documented supply: a module with a gap in its records cannot be installed. The fleet behind all of this only shrinks, since every engine retired, torn down or converted leaves the pool and nothing enters it. [Part I §3–§4, §9]

Depth: Part I §1–§5.

S2. What a shop visit is and costs, and why the parts bill is where the money sits

When either clock runs out, the engine comes off and goes to a shop. The bill has five lines, and the mix among them decides most of what follows.

SHOP VISIT

A shop visit is any removal of the engine for work in an engine shop, from a visit to fix one finding to a complete rebuild. The shop disassembles the engine, inspects every piece, replaces or repairs what is out of limits, rebuilds it and runs it in a test cell before release. "Teardown" is reserved here for dismantling an engine to sell its parts. [Part I §6]

MAINTENANCE, REPAIR AND OVERHAUL (MRO), AND THE INDEPENDENT MRO

MRO is the industry's generic term for maintenance, repair and overhaul work and the companies that do it. An independent MRO is an engine shop not owned by the engine's manufacturer: a stand-alone company such as StandardAero, an airline's maintenance arm, or another engine maker's shop such as MTU (a German engine maker). All work to the manufacturer's manual. [Part I §6; Part IV §25]

WORKSCOPE, PERFORMANCE RESTORATION AND OVERHAUL

The workscope specifies which modules are opened, how deep, and which parts are replaced new, repaired or swapped for used material. A performance restoration opens the core to recover temperature margin; the fan and low-pressure turbine are inspected but not fully disassembled. An overhaul disassembles every module and replaces the life-limited parts that would fall due in the next run ("zero-time": a complete new set). A hospital or quick-turn visit fixes one thing and returns the engine; a module exchange is one form of it. [Part I §6; registry C-7]

TURNAROUND TIME (TAT) AND THE INDUCTION SLOT

Turnaround time (TAT) runs from an engine's induction into a shop to its release: 90 to 120 days for a full CFM56 overhaul in 2025 against about 60 before the pandemic (trade press); "sustaining about 90 days" in GE's shops (May 2026). The induction slot is the date a shop commits to start; when shops are full the wait for one, two to six months (Bain, 2024), is additional to the turnaround, and every day away needs a spare engine or grounds an aircraft. [Aviation Business News, 2025; GE Aerospace, 27 May 2026; Safe Fly Aviation, 2026 (consultancy blog); Bain, 2024]

THE SHOP-VISIT COST BUILD

A visit's cost is labour, new OEM parts, repairs, life-limited-part replacement (new or used) and the test-cell run, plus freight, consumables and the shop's margin. Material is 60 to 70% of the total, labour 20 to 30% and repairs 10 to 20% (Air Cargo Week); a consultancy blog adds that life-limited parts are 40 to 60% of parts cost when replaced and high-pressure turbine blades and vanes 15 to 25%. [Air Cargo Week, 2025/26; Safe Fly Aviation, 2026; Part I §7]

CATALOGUE LIST PRICE AND ESCALATION

The catalogue list price is the manufacturer's published price for every new spare part, repriced at least annually; shops pass it through at list plus a handling margin, and used and alternative parts are priced as discounts to it. Escalation is the annual increase: around 10% a year on new spares in 2022 and 2023 (Aviation Week); Safran's chief executive described pricing "above inflation by 3 to 4 points"; and the published price of a full -7B life-limited-part set compounded at about 7% a year from 2008 to 2025, derived in the research. [Aviation Week, late 2023; Part I worked example 1.3; Part IV §28; reconciliation R-018]

EXHIBIT S.2 — WHAT A CFM56 SHOP VISIT COSTS: THE BINDING RANGES WITH THEIR SOURCE LEVEL

Quantity	Figure	Source and level	Note
Performance restoration	\$1.2–1.6M on one page; \$1.8–2.5M on another	Safe Fly Aviation, 2026, two pages (consultancy blog)	Part IV uses the first, Part I the second; never averaged (R-013)
Heavy visit with life-limited parts	\$2.1–2.8M; ">\$3.5M"	Safe Fly Aviation, two pages	Neither fits a full new set at current list
Full overhaul ("zero-time")	\$3.5–4.2M	Safe Fly Aviation	Full new set by the label
Full -7B life-limited-part set at list	\$1.775M (2008); \$3.4M (2018); "\$4m" (Nov 2019); \$5.7M (June 2025, scope not visible)	Aircraft Commerce Issues 58 and 120; Aircraft Value News; MyAirTrade	Part IV computes with \$4M, Part I with \$5.7M (R-016)
Cost shares	Material 60–70%, labour 20–30%, repairs 10–20%; LLPs 40–60% of parts, HPT blades and vanes 15–25%	Air Cargo Week; Safe Fly Aviation	Two secondary sources (R-014)

Source: Safe Fly Aviation, 2026 (consultancy blog); Aircraft Commerce Issues 58 and 120; Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Air Cargo Week, 2025/26. Reconciliation E.2, rules 4 to 6. Accessed 2026-10-03.

Every cost figure in the exhibit is from trade press or one consultancy blog, and that blog disagrees with itself on the same scope; no manufacturer's or shop's price list was read. [reconciliation R-013] The set prices supply a check: a restoration at \$1.4M plus a full new set at \$4M is \$5.4M, above every quoted "with LLPs" range, so those totals must assume partial replacement, used parts or older prices. [Part I §7.1]

WORKED EXAMPLE S.2 — STUB LIFE, AND WHY THE LIFE-LIMITED-PART DECISION OUTWEIGHS THE WHOLE RESTORATION

Inputs: an engine arrives for a restoration at 12,000 cycles (assumed); its core parts have a 20,000-cycle limit (sourced), so 8,000 remain; the next run is planned at 12,000 cycles (assumed, inside Leeham's range). A full new set lists at \$5.7M (MyAirTrade, June 2025, scope not visible) or about \$4.0M (Aircraft Value News, Nov 2019). The restoration costs \$2.15M or \$1.4M, the midpoints of the two consultancy ranges.

keep the parts: the engine comes off again after 8,000 cycles, forced by the limit, with its restoration two-thirds used
 replace them new now: 8,000 of each part's 20,000 cycles (40%) are scrapped unused
 cost per cycle, full life used: $5,700,000 \div 20,000 = \285 (at \$4.0M: \$200)
 cost per cycle, scrapped at 12,000: $5,700,000 \div 12,000 = \475 (at \$4.0M: \$333)
 restoration alone, per cycle of the next run: $2,150,000 \div 12,000 = \179 (at \$1.4M: \$117)

The shorter remaining lives already inside the engine, the "stub life" (Aircraft Commerce), add a third to a half to the cost of every cycle if the parts are scrapped early, a penalty larger than the whole restoration on either set price. The third option, which the used-parts market supplies, is used life-limited parts with 12,000 to 15,000 cycles remaining; their prices were not retrieved. [Aircraft Commerce Issue 34; Part I worked examples 1.2 and 1.6; reconciliation R-016, R-017]

The parts bill is where the money sits for three reasons. Material is 60 to 70% of a visit and is priced off the catalogue, so escalation flows into every visit whoever performs it. Life-limited parts are 40 to 60% of parts cost when replaced and no alternative maker produces one for the CFM56, so that line is single-source. And the high-pressure turbine's blades and vanes, the next largest line, sit where most of the lost margin is recovered (S8). A shop's levers against the manufacturer are therefore all on the material line: used parts, approved alternative parts, repair instead of replacement, and exchange of a module instead of overhaul of the engine. [Part I §7; Part VI §38]

Depth: Part I §6–§7; Part IV §29.

S3. Where the money sits across the chain: a first margin map, bases stated, nothing ranked

Before the chain goes further, its links and the profit each reports give the segment sections a frame. This is a map, not an analysis.

MARGIN POOL

The margin pool is this primer's term for how the profit earned in keeping an engine type flying is distributed across the links of the chain: the manufacturer, the shops, the repairers, the alternative-parts makers, the used-material traders, the lessors and the servicers of third-party capital. No company reports a "margin pool"; the term names a way of reading the filings together. [Part I §11; Part XI §86]

FTAI'S TWO REPORTABLE SEGMENTS: AEROSPACE PRODUCTS AND AVIATION LEASING

Aerospace Products "develops and manufactures, repairs/refurbishes, and sells aircraft engines and aftermarket components primarily for the CFM56-7B, CFM56-5B, and V2500" through its Module Factory (S9); FY2025 segment revenue was \$1,936.2M. Aviation Leasing "owns and manages aircraft and engines and leases and sells them, directly and through an equity-method investment"; it held 290 assets at 31 December 2025 (S10). FTAI Power appears as a third line in the 2027 guidance; whether it is a reportable segment was not confirmed (S11). [FTAI 10-K FY2025; FTAI release, 29 Jul 2026; Part VII §45; Part VIII §58]

ADJUSTED EBITDA (CONSOLIDATED AND BY SEGMENT) AND GUIDANCE

Adjusted EBITDA is earnings before interest, taxes, depreciation and amortisation as FTAI adjusts them: items treated as non-recurring or non-cash are removed and the company's pro-rata share of Adjusted EBITDA from its equity-method investments is added back. FTAI reports it consolidated (\$597.3M in 2023, \$862.1M in 2024, \$1,190.9M in 2025) and by segment (Aerospace Products \$160.0M, \$380.6M, \$671.3M; Aviation Leasing \$500.1M and \$608.9M for 2024 and 2025), and sets guidance in the segment measure. The segments do not sum to the total (Corporate and Other about – \$89.2M in 2025). [FTAI 10-K FY2025; Part X §80, §84]

EXHIBIT S.3 — INDICATIVE MARGIN FIGURES ALONG THE CHAIN, EACH ON ITS OWN STATED BASIS

Link in the chain	Company and segment	Revenue	Margin	Basis	Source
A shop visit, by slice	Any CFM56 visit	n/a	Material 60–70%; labour 20–30%; repairs 10–20%	Share of cost, not a margin	Air Cargo Week, 2025/26
Manufacturer	GE Aerospace, Commercial Engines & Services	\$33,314M; services \$25,010M	26.6%	GE segment profit (GE's own measure)	GE 10-K FY2025
Approved alternative parts, with repair and distribution	HEICO (a maker of approved alternative parts), Flight Support Group	\$3,117.3M	24.1%	Segment operating income, derived; year to 31 Oct 2025	HEICO 10-K FY2025

Link in the chain	Company and segment	Revenue	Margin	Basis	Source
Independent engine shop	StandardAero, Engine Services	\$5,354.0M	13.2%	Segment Adjusted EBITDA, StandardAero's definition	StandardAero 10-K FY2025
Independent shop (context, non-SEC)	MTU Aero Engines, commercial maintenance	€6.0B	8.0%	Adjusted EBIT (earnings before interest and tax)	MTU, 24 Feb 2026
Teardown and used material	AerSale, Asset Management Solutions	\$211.6M	35.0%	Gross margin, derived	AerSale 10-K FY2025
Module exchange, module and engine sales	FTAI, Aerospace Products	\$1,936.2M (FY2025); \$875.0M (Q2 2026)	34.7%; 28.5%	FTAI-defined segment Adjusted EBITDA, derived; includes related-party Maintenance, Repair and Exchange (MRE) Contract revenue (S9, S10)	FTAI 10-K FY2025; Q2 2026 release
Engine and aircraft lessor	FTAI, Aviation Leasing	Lease income \$235.2M	Segment Adjusted EBITDA \$608.9M; no margin computable	Includes gains, maintenance revenue and the equity-method share	FTAI 10-K FY2025

Source: as listed in each row; compiled from Part I Exhibit 1.14 and Part XI Exhibit 11.4. Bases and fiscal years differ. Accessed 2026-10-03.

Three cautions travel with the table. The margins sit on five bases: an EBITDA margin exceeds an EBIT margin by the depreciation share of revenue, about nine points at FTAI in 2025 (\$225.8M on \$2,507.4M). A share of cost is not a margin. And a parts pass-through compresses a percentage: a shop billing \$2M of the manufacturer's parts at a small margin beside \$1M of its own labour shows a thin percentage, a parts maker a wide one, and the dollars may be similar. FTAI's segment sells a module or engine it owns, so its margin is built more like a parts business's than a shop's. [FTAI 10-K FY2025; Part X §80; Part XI §86.9]

One structural fact belongs on the map. Safran describes about 85% of the CFM56's aftermarket revenue as spare parts and time-and-materials work, against about 40% for the LEAP, and about 15% of the CFM56 fleet is under long-term flight-hour agreements against about 50% of the LEAP fleet. On this engine the manufacturer earns mainly by selling parts at list into visits it does not perform or control, which leaves room for every substitute in S8 and S9; on a contracted engine the manufacturer chooses the parts. [Safran via Aviation Week, c. 2023/24; Part IV §28]

Depth: Part I §11; Part XI §86.

S4. Who owns engines and how they pay: leases, reserves, engine values, green time

Almost every dollar an engine earns passes through a lease, and almost every dollar its owner borrows is secured on that lease. The lease is where the life an engine consumes is turned into cash, and where its value is written down part by part.

OPERATING LEASE

An operating lease rents an aircraft or engine from its owner, the lessor, to an operator, the lessee, for a fixed term and periodic rent; the lessor keeps title and the risk that the asset is worth more or less than expected when it comes back. Nearly all of FTAI's leases are operating leases. At the end of the term the lessor extends, re-leases, sells or parts out. [FTAI 10-K FY2025; Part II §12]

MAINTENANCE RESERVES

Maintenance reserves are payments the lessee makes in addition to rent, set per flight hour or cycle and collected per component (each engine's restoration and life-limited parts, the airframe's checks, the landing gear). They pre-fund the eventual shop visit; the lessor reimburses a completed event up to the balance held and keeps any unclaimed balance at lease end. FTAI books reserves it no longer has to return as "maintenance revenue" (\$218.5M in 2025). [AerCap 20-F FY2013; FTAI 10-K FY2025; Part II §13]

RETURN CONDITIONS

Return conditions are the contractual minimum state of a leased engine at redelivery: minimum hours and cycles to the next restoration, minimum cycles remaining on each life-limited part, and records in order. With the reserve regime they decide who pays for the life consumed. They may also exclude parts and repairs that are not the manufacturer's; current practice on that clause was not found. [Part II §13; Part VI §42]

LEASE RATE FACTOR (LRF)

The lease rate factor (LRF) is monthly rent divided by the asset's market value; 1.0% a month is a 12% gross annual yield before depreciation, interest and maintenance. No source states a factor for FTAI, for Willis Lease or for the CFM56; the primer's figures are derived: 0.70% a month (mixed bases), 0.87 to 1.1% (FTAI's 2025 book), about 1.0% (Willis Lease), and 0.9% assumed in one fee example. [Part II §12, §14; reconciliation R-030]

HALF-LIFE VALUE AND MAINTENANCE-ADJUSTED VALUE

Half-life is an appraisal convention, not a physical midpoint: the value of an engine with every module halfway between restorations and every life-limited part at half its life, so that no adjustment has been made for actual maintenance status (ISTAT); IBA (an appraiser) publishes CFM56 values on that basis. Maintenance-adjusted value adds, for each item, the actual remaining fraction less one half, times the cost to restore it. [ISTAT Jetrader, 2010; IBA, Sept 2025; Part I §8]

GREEN TIME

Green time is the usable life left on an engine or module before its next mandatory shop visit, set by whichever runs out first: the shortest-lived part, the temperature margin or a hard inspection. A green-time engine is one the owner will not restore, flown or leased as-is and then torn down. The pool is fed by retirements, so it shrinks exactly when a slow-retiring fleet most wants it. [Part II §16; registry C-2]

EXHIBIT S.4 — CFM56-7B VALUES AND RENTS: EACH FIGURE WITH ITS BASIS, SOURCE LEVEL AND DATE

Quantity	Figure	Basis	Source and level	Date
-7B24 and -7B27 market value	\$5.7M; \$6.4M; "levelled off after approximately 18 months of increases"	Half-life, H2 2025	IBA, Engine Values 2025B and H1 2026 (appraiser)	Sept 2025; 2026
-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", internally inconsistent	Safe Fly Aviation (consultancy blog)	2026
Run-out -7B as feedstock	Bought \$0.8–1.2M; yields \$1.6–2.2M of used material; net \$0.4–0.8M	Exhausted engine	Safe Fly Aviation	2026
-7B spare-engine rent	~\$100,000 a month, from ~\$75,000 in 2019	Spare engine; condition not stated	IBA via AviTrader	Apr 2024

Quantity	Figure	Basis	Source and level	Date
-7B rent, alternative reading	\$42,000–48,000 a month; "about 34% above 2019"; up 15–20% "in recent months"	Beside green-time values; may be green-time	Safe Fly Aviation	2026
12-year-old 737-800 rent	~\$228,000 a month, down 11% year on year	Whole aircraft	IBA via India Seatrade News	Jul 2025

Source: IBA, Sept 2025, H1 2026 and via AviTrader, 25 Apr 2024; Safe Fly Aviation, 2026 (consultancy blog); India Seatrade News, 6 Sep 2026. Compiled from Part I Exhibit 1.11 and Part II Exhibit 2.3. Reconciliation R-028, R-029, R-031. Accessed 2026-10-03.

Two disagreements in the exhibit reach every later Part. On rent, the appraiser's \$100,000 a month and the consultancy's \$42,000 to \$48,000 agree on the rise since 2019, about a third, and disagree on the level; the lower figure sits beside green-time values and may price a part-worn engine. On value, the appraiser's half-life figures have "levelled off" while the blog's green-time value comes with rents still rising; the bases differ and so does the direction. [IBA, Apr 2024, Sept 2025, H1 2026; Safe Fly Aviation, 2026; reconciliation R-028, R-029]

Two -7B spare engines at the appraiser's rent would command \$200,000 a month, 88% of the \$228,000 a 12-year-old 737-800 earned in July 2025 (IBA); at the consultancy's rent, \$84,000 to \$96,000, 37 to 42%. The first mixes 2024 and 2025 rents and the second may price part-worn engines, so both are carried, but neither disputes the direction: the engines are separable, their value is dominated by maintenance condition, no new CFM56 is being built, and a 15-to-20-year-old airframe has few other uses. FTAI states the consequence: it "owns and leases jet aircraft which often facilitates the acquisition of engines at attractive prices". [IBA, Apr 2024 and Jul 2025; Safe Fly Aviation, 2026; FTAI Q2 2026 release; Part II worked example 2.6]

SPECIAL-PURPOSE VEHICLE (SPV), ASSET-BACKED SECURITIZATION (ABS) AND WAREHOUSE FACILITY

A special-purpose vehicle (SPV) is a bankruptcy-remote entity that owns only a defined pool of assets and the debt secured on them. An asset-backed securitization (ABS) sells leased aircraft or engines to such a vehicle, which issues notes in tranches paid in order of seniority; the sponsor usually keeps the equity and services the leases for a fee, and rating agencies size the senior notes by stressing rents, downtime, maintenance and residual values, so an appraiser's note that values have levelled off reaches the debt a pool can carry. A warehouse facility is a revolving bank loan to an SPV that funds assets until there are enough to securitize. FTAI's Strategic Capital Initiative (SCI) vehicles use all three (S10); on 22 May 2026 the first of them priced FTAI MRE 2026-1, \$612M of notes on 48 A320ceo and 737NG aircraft on lease to 23 airlines, with Series A expected to be rated Asf by Fitch and A(sf) by KBRA (a rating agency) and Series B in the BBB (triple-B) category, BBB+sf, by Fitch. [FTAI release, 22 May 2026, via Finviz and AviTrader; Part II §17]

Depth: Part II §12–§17.

S5. Why 2026 looks the way it does: the fleet that only shrinks, retirements that did not come, new engines that did not relieve, airframers that did not deliver

Nothing in S1 to S4 changes between a quiet year and a tight one; what changes is the quantities. Between 2019 and 2026 four things moved them the same way.

INSTALLED BASE, RETIREMENT RATE AND THE SHOP-VISIT FORECAST

The installed base is the population of engines fitted to aircraft in the operating fleet; the three CFM56 counts of S1 may include spare and stored engines, and a consultancy blog's about 14,200 "active" and more than 19,000 "flying" are labelled consultancy figures and never the fleet figure. The retirement rate is the share of a fleet withdrawn in a year: GE's planning assumption for 2026 moved from 3–4% of the base, to 2–3%, to 1.5–2% (GE CFO via Aviation Week, 2026); GE reported about 1.5% retired in 2025 and expected about 2% in 2026 (via Leeham, 22 Jan 2026); first-quarter 2026 retirements ran "below 1%" (GE investor-relations page). The shop-visit forecast counts expected inductions a year: 2,300 to 2,400 CFM56 visits a year in 2026 to 2028 with the apex around 2027 to 2028 and a gradual fade (GE via Aviation Week, 2026), against Safran's earlier peak of about 2,500 in 2025 to 2026 (c. 2023/24). [Part III §18, §19, §23; reconciliation R-001, R-004, R-006]

Because no new -5B or -7B is built, the base falls by exactly the number retired or converted each year, and an engine that does not retire both accumulates cycles toward the visits of S2 and withholds its parts from the used-material market: the same number sits on both sides of the aftermarket.

The fleet count is contested, and every per-engine figure in the chain divides by it: none of the three manufacturer-level figures states its basis, and the consultancy's 14,200 was read in the research both as the whole active fleet and as the -7B fleet alone. [reconciliation R-001, R-002] The retirement record is contested twice over. GE's planning number for 2026 fell three times within a year and the realised rate fell further, with the two venues differing by half a point on the 2026 expectation. At aircraft level an appraiser said fewer than 200 aircraft retired in 2024 and a data provider recorded 587, with the scope visible in neither; arithmetic on other sources gives about 410 and about 1,130 a year, and Cirium's own series ran 502, 465, 333 and 325 for 2019 to 2022. [GE CFO via Aviation Week, 2026; Leeham, 22 Jan 2026; IBA via AviTrader, 24 Sep 2024; Cirium via Informa and via Aerodynamic Advisory, IATA (the airlines' trade body) conference, 2023; reconciliation R-004, R-005]

WORKED EXAMPLE S.5 — THE GAP BETWEEN THE PLANNED AND THE REALISED RETIREMENT RATE, IN ENGINES

Inputs: a base of about 24,000 (Aviation Business News, Sept 2025) or more than 22,800 (Safran, end-2025); rates at the midpoints of GE's ranges, 3.5% for "3–4%" and 1.75% for "1.5–2%" (midpoints are the primer's choice); the 2025 actual of about 1.5% and 2026 expectation of about 2% (GE via Leeham).

on 24,000: 3.5% = 840 engines a year; 1.75% = 420; 1.5% = 360; 2.0% = 480; below 1% = fewer than 240
 on 22,800: 3.5% = 798; 1.75% = 399; 1.5% = 342; 2.0% = 456
 plan less realised: $24,000 \times (0.035 - 0.0175) = 420$ a year; over 2024–26 about 1,200 to 1,260

About 400 engines a year on any manufacturer-level base, roughly 1,200 over three years, are still flying that the manufacturer's first plan had retired: each is a future shop visit and a teardown that did not happen. The three-year figure is an upper bound. [Part III worked example 3.2]

Why did so few retire? An aircraft is retired when its next heavy event costs more than the aircraft is worth kept, and when a replacement is available. From 2023 to 2026 the second condition failed: an airline with no new aircraft pays for the visit rather than shrink, and a lessor whose lessee cannot find a replacement extends. Cirium's count that only about a dozen mid-life 737NGs and fewer than 25 A320ceos were available to rent anywhere measures how closed the gate was. [Cirium Ascend via Bloomberg; Part III §19]

THE CONTEXT SHOCK: THE GEARED TURBOFAN (GTF) POWDER-METAL ISSUE, LEAP DURABILITY AND ITS KIT, AND THE 737 MAX RATE CAP

Three failures of replacement kept the old fleet flying. Pratt & Whitney's geared turbofan (GTF) on the A320neo had turbine discs made from contaminated metal powder, disclosed by RTX (Pratt & Whitney's parent) in 2023; 835 GTF-powered A320neos were grounded or stored in late 2025 on Cirium's count against RTX's planning figure of about 350, and aircraft-on-ground (AOG) levels were down more than 20% from the 2025 peak by January 2026. CFM's LEAP came off wing early, at 2,000 to 6,000 cycles, in hot and harsh environments; CFM certified a LEAP-1B durability kit with the Federal Aviation Administration (FAA) on 18 July 2026, with production cutover expected in early 2027. And the FAA capped 737 production at 38 a month from January 2024, raising it to 42 in October 2025 and completing the review to 47 in May 2026, against a 2018 peak of 52; Airbus delivered 793 aircraft in 2025 against a target of 820 and expects A320neo rate 75 only by the end of 2027. [Cirium via The Flying Engineer, 2026; Leeham, 27 Jan 2026; Visual Approach; CFM release, 18 Jul 2026; TravelPulse, Oct 2025; CAPA (the Centre for Aviation), May 2026; Wikipedia compilation; Argus Media; Air Data News; Part III §20–§21]

The mechanism from each shock to the CFM56 is indirect and specific. A grounded A320neo needs its flying replaced by an A320ceo on the same pilot rating, which works the -5B and the V2500 harder, not the -7B. An early LEAP removal is a LEAP visit in the manufacturer's

network, but an operator whose 737 MAX fleet is less available keeps its 737NGs, which touches the -7B. A cap of 38 a month against 52 holds back on the order of 170 737NG replacements a year at a one-for-one ceiling. Each shock carries its own reversal: the groundings were easing by January 2026, the LEAP-1B kit cuts over in early 2027, and the backlogs are owed rather than cancelled, so when rates rise the deferred retirements arrive with them. [Part III §20.3, worked example 3.4]

The lessors closed the loop. With 75 to 90% of expiries extended for four to six years, the maintenance events that would have fallen at a lease transition were paid by the incumbent airline inside the extended term, and only 10 to 25 aircraft of every 100 came back to the used market as acquisition candidates or teardown feedstock. The resulting forecast is contested between the two CFM partners, a peak of about 2,500 against a plateau of 2,300 to 2,400 through 2028, and either depends on a retirement rate that moved three times in a year. [Cirium Ascend via Bloomberg; IBA, Nov 2023; Part III worked example 3.5; reconciliation R-006, R-007]

Depth: Part III §18–§23.

S6. The translation table in one page: each driver, the requirement it creates, what it does not stress

Part III turns each documented driver into a numbered row stating the requirement it creates and what it leaves untouched or pushes the other way; a driver that raises shop-visit demand but removes teardown feedstock has done two things.

EXHIBIT S.6 — THE TRANSLATION TABLE: FOURTEEN DRIVERS, THE REQUIREMENT EACH CREATES, AND WHAT EACH DOES NOT STRESS

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-1	Low CFM56 retirements: ~1.5% in 2025, ~2% expected 2026, "below 1%" in 1Q 2026, against a 3–4% plan (GE)	More engines accruing cycles: sustains 2,300–2,400 visits a year through 2028; more LLP replacements and full restorations; module demand	Starves teardown feedstock (T-9); does not change shop capacity
T-2	GTF groundings: 835 aircraft (Cirium, late 2025); easing from the 2025 peak (RTX)	A320ceo lift: ceo extensions and short leases; V2500 and -5B visits brought forward; spare and green-time modules	No -7B demand except through general scarcity; adds no used material

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-3	LEAP durability: removals at 2,000–6,000 cycles; LEAP-1B kit certified 18 Jul 2026, cutover early 2027	LEAP visits in the manufacturer's network; indirectly, slower ceo and 737NG retirements	No CFM56 visits, modules or used material; the indirect support fades after cutover
T-4	737 MAX cap: 38 → 42 → 47 a month; 260 and 440 delivered in 2024 and 2025	737NG extensions and utilisation: -7B visits, LLP replacements, spares, green-time modules	Nothing on -5B or V2500; reverses as the cap lifts toward 52
T-5	Airbus shortfall: 793 against 820 in 2025; rate 75 only by end-2027	A320ceo extensions and utilisation: -5B and V2500 visits and spares; low ceo part-out	Does not stress the -7B; reverses as the rate rises
T-6	Lessor extensions: 75–90% of expiries; ~12 737NGs and <25 A320ceos available; terms 4–6 years	Maintenance paid by the incumbent inside the term; higher mid-life lease rates; capital held in aircraft on lease	Only 10–25 of every 100 aircraft return: fewer acquisition candidates and less feedstock; rent direction contested (R-031)
T-7	OEM escalation: ~10% a year 2022–23; inflation plus 3–4 points (Safran); -7B LLP set ~7% a year 2008–25	Demand for substitutes on the material line, 60–70% of every visit: used material, approved alternative (PMA) parts, repair, module exchange	Does not change the removal count; leaves the LLP line unsubstituted, since no PMA exists for a CFM56 LLP
T-8	Capacity shortfall: slot waits 2–6 months; overhaul 90–120 days; demand to exceed capacity by >17% by decade end (Bain)	Spare engines and leases in proportion to the delay; exchanges and quick-turns in place of overhauls	Does not change removals; GE reports turnaround improving (R-019, R-020)
T-9	Used-material scarcity: supply "remains thin" (Aviation Maintenance magazine, 2025); AerSale whole-asset sales 20 engines → 13; values up as much as 50% in two years (ePlaneAI)	Higher used-part and green-time prices; more new parts bought; PMA as the other substitute; capital to acquire feedstock	A consequence of T-1, T-4, T-5 and T-6, reversing when they do

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-10	Data-center power and the turbine shortage: data-center electricity to ~945 TWh by 2030 (International Energy Agency (IEA)); GE Vernova backlog plus reservations 116 gigawatts (GW) at Q2 2026, slots sold for 2031	A new sink for CFM56 cores: 100 Mod-1 units in 2027 would consume ~100 engines a year at one core per unit	No aviation visits and no used material; ~100 engines is 21–28% of a year's retirements at 1.5–2%; engines per unit undisclosed
T-11	Freighter conversion: >260 737-800 freighters in service, >300 in the pipeline; IBA warns of "oversupply" (6 Sep 2026)	Green-time -7B modules and visits for 20-plus-year-old airframes	Removes airframes and engines from the teardown pool; contested (R-034)
T-12	Mature-narrowbody utilisation "rising significantly" (IBA, Sept 2024)	Visits and LLP replacements earlier in calendar time; faster margin erosion	No used material; ends when replacements arrive
T-13	GTF groundings easing: AOG down 25% year to date, output up 40% (RTX, Jan 2026)	Returning neos reduce ceo lift: ceo rates soften; more -5B and V2500 teardown supply later	Does not reduce visits already induced; resolution put at 2027–28
T-14	CFM56 plateau then decline: 2,300–2,400 through 2028, apex 2027–28, gradual fade (GE); ~2,000 visits by 2030 implied (Safran)	Long run: shop capacity moves to the LEAP; teardown supply rises as retirements normalise (~2.7% a year from 2028, McKinsey)	Does not stress 2026–27 demand; timing depends on a retirement rate revised three times in a year

Source: Part III Exhibit 3.14 and the sources cited there; rows T-7, T-9 and T-10 draw on Part I §4, Part V §33 and Part IX §71. Reconciliation R-001 to R-007, R-019, R-020, R-031, R-034. Accessed 2026-10-03.

Read down the right-hand column and the chain's central tension is visible before any segment is described. The drivers that raise demand for visits, modules and spares (T-1, T-4, T-5, T-6) are the drivers that cut the supply of used parts and green-time engines (T-9), because the engine that is not retired is both. T-7 alone acts on price, raising the value of every visit and the demand for every substitute for a new part. And the one driver that removes engines for a non-aviation reason (T-10) competes with the teardown trade and the exchange business for the same run-out cores. The rows are answered by the shop-visit market (S7: T-1, T-7, T-8), the substitutes (S8: T-1, T-6, T-7, T-9, T-11), the Module Factory (S9: T-1, T-2, T-4, T-7, T-8), leasing and capital (S10: T-2, T-4, T-5, T-6, T-8, T-9) and FTAI Power (S11: T-10). [Part III §24, Exhibit 3.15]

Depth: Part III §24.

S7. The shop-visit market and the OEM's pricing: demand plateau, capacity shortage, list-price escalation, the OEM's own offerings

The market the engine creates has four kinds of shop, the manufacturer's network, the large independents, the airlines' maintenance arms and, since 2022, FTAI's Module Factory, and one price-setter, the manufacturer, which earns from nearly every visit without performing it.

Demand is a plateau whose height and length are contested. Safran's earlier framing had CFM56 visits peaking at about 2,500 a year in 2025 and 2026 and then falling; GE's 2026 framing has 2,300 to 2,400 a year, "probably closer to 2,400", held through 2028 with the apex around 2027 to 2028 and a gradual fade. The later view is lower at its top and later in reaching it; the CFM56 is about a quarter of all engine shop work by count (about 2,300 of about 9,000 visits in 2025). The shape moved with the retirement assumption, so the fade's start date is the least certain number in the chain. [Safran and GE via Aviation Week; Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI; reconciliation R-006, R-007]

WORKED EXAMPLE S.7 — RECONCILING 2,300 VISITS A YEAR WITH THE INSTALLED BASE

Inputs: 2,300 visits a year (GE via Aviation Week, 2026, low end); bases of about 24,000, more than 22,800, more than 19,000 "flying" and about 14,200 "active" (the last two consultancy, labelled); a mature restoration every 10,000–15,000 cycles at 1,000–1,500 cycles a year (Leeham; Aircraft Value News), roughly 6.7 to 15 years; spares at about 10% of installed engines (MTU).

on 24,000: $2,300 \div 24,000 = 9.6\%$ a year $\rightarrow$ about 10.4 years between visits
on 22,800: 10.1% $\rightarrow$ 9.9 years; on 19,000: 12.1% $\rightarrow$ 8.3 years; on 14,200:
16.2% $\rightarrow$ 6.2 years
removing 10% spares from 24,000: $2,300 \div 21,600 = 10.6\% \rightarrow 9.4$ years

An implied interval of about ten years matches the mature interval only at the long end of its range; about six years on the consultancy base is shorter than any sourced interval. So the 24,000 base is consistent with the sourced interval only if it includes a few thousand spare, stored or green-time engines, or if long first runs are depressing the visit rate: about 70% of the fleet had seen at most one shop visit in 2024 (Safran). Which base is right decides every per-engine figure in the chain, including FTAI's share of it. [Part III worked example 3.6; reconciliation R-001, R-008]

Capacity has two readings and both are carried. Trade press and consultancies describe a full system: overhauls at 90 to 120 days, slot waits of two to six months, demand set to exceed engine-maintenance capacity by more than 17% by the end of the decade (Bain, 2024), and two-thirds of operators short of technicians (Oliver Wyman, April 2026). GE's filing and conference material read the other way: turnaround for the LEAP, CFM56 and GE90 (a widebody engine) improved by more than 10% year on year in the fourth quarter of 2025, and CFM56 turnaround

was "sustaining about 90 days" in May 2026. The consequence does not depend on which is right: a removal that returns in five to ten months ties up a spare for that long, so the shortage of slots is itself demand for spare engines and for anything that returns the aircraft to service in weeks, which is the exchange of S9. [Aviation Business News, 2025; Bain, 2024; Oliver Wyman, Apr 2026; GE 10-K FY2025; GE Aerospace, 27 May 2026; reconciliation R-019, R-020]

The manufacturer's model on this engine has four elements. Spare parts are sold to any shop at the published catalogue price, which escalated about 10% a year in 2022 and 2023 and, for -7B parts, 20 to 30% after August 2023 on one shop's account (Magnetic via Aviation Week); the 2024 to 2026 steps were not found. About 85% of CFM56 aftermarket revenue is transactional, bought visit by visit with the operator or its shop choosing the parts, against about 40% for the LEAP; the remainder sits under long-term service agreements (LTSAs), multi-year contracts at a rate per flight hour under which the manufacturer performs or pays for the visits and chooses the shop and the parts, covering about 15% of the CFM56 fleet and targeted at 60 to 70% of the LEAP fleet by 2030 (Safran via Aviation Week). GE sells the same services under TrueChoice, with Flight Hour, Overhaul, Material and Transitions lines, the last covering green-time leases, exchanges and material buy-back; GE says "GE and CFM are the largest used serviceable material provider for their products" and calls the CFM56 "an open network ... with multiple providers who compete". And it signs materials agreements with shops: FTAI's of 22 January 2026 "secures OEM replacement part supply, thrust performance upgrades and component repair", terms undisclosed. [Aviation Week, late 2023, 2024 and c. 2023/24; GE Aerospace statements, 2026; FTAI release, 22 Jan 2026; Part IV §28; reconciliation R-009, R-018]

Put the elements together and the manufacturer's capture on this engine is plain. Most visits are bought visit by visit; within each, 60 to 70% of the bill is material; most material is bought new at list. So the manufacturer receives the largest slice of nearly every CFM56 visit without performing it, and its escalation lands on that slice whoever runs the shop. The substitutes of S8 and the exchanges of S9 therefore address the 85% of the aftermarket that is transactional, and would address far less of a fleet contracted the way the LEAP is being contracted; the manufacturer is on the substitutes' side too, through TrueChoice and the January 2026 agreement. [Part IV §28–§29]

An operator whose engine is at its life limit has four options: a full visit with new life-limited parts after the slot wait and the turnaround; a module exchange (S9); a green-time engine leased to fly out the remaining months; or a sale of the engine for its parts and the purchase of another. The last three exist because the engine is modular and a secondary market exists, and low retirements shrink the supply behind all three just when the slot shortage makes the first slow and escalation makes it expensive. [Part IV §30]

Depth: Part IV §25–§30.

S8. The substitutes for new parts: used material from teardowns, approved alternative parts (PMA) and designated-representative (DER) repairs

Three things can replace a new part from the manufacturer: a used part from a dismantled engine, a new part made by someone else under a regulator's approval, and a repair that keeps the old part in service. All three are priced against the manufacturer's list, and all three depend on a supply or an approval the manufacturer does not control.

USED SERVICEABLE MATERIAL (USM)

Used serviceable material (USM) is any part that has been in service, removed, inspected and if necessary repaired to the manual, and released on an airworthiness tag with its full history; a part with a gap in its paperwork is scrap. Used material trades at 60 to 80% of a new part's price on average and above list when the manufacturer cannot deliver (Aviation Week); a specialist trader sells repaired engine material to an airline at 70 to 75% of list (Aircraft Commerce). Because the base is the catalogue, the used price rises with every revision even if the discount does not move.

[Aviation Week, undated; Aircraft Commerce Issue 120; Part V §31, §34; reconciliation R-022]

TEARDOWN, PART-OUT AND FEEDSTOCK

Part-out is the owner's decision to dismantle a retired aircraft or engine and sell its components; teardown is the disassembly. Feedstock is the trade's word for the aircraft and engines bought to be torn down; FTAI's filings extend it to the unserviceable module a customer returns in an exchange and to engine inventory held for module production. A teardown module is valued like an engine in miniature: parts at used-material discounts, plus the life-limited parts' remaining life (life fraction times list, discounted), less the repair to make it serviceable. Supply is a lagged function of retirements. [Part V §32–§33, §36; registry C-14]

A run-out CFM56 bought for \$0.8 to \$1.2M yields \$1.6 to \$2.2M of used material (Safe Fly Aviation, 2026, consultancy blog), 25 to 39% of a half-life -7B's \$5.7M to \$6.4M (IBA, Sept 2025); no owner tears down a \$6M engine for \$2M of parts, so engines with life are sold or leased whole and only run-out engines reach the teardown market. Gross retirements on the manufacturer-level bases are roughly 340 to 480 engines a year (1.5 to 2% of 22,800 to 24,000), of which an unknown share is torn down: some are sold whole, some leased for green time first, and from 2026 some earmarked as power-generation cores (S11). [Part V worked examples 5.1 and 5.4; Part II §16]

The 2023 to 2026 shortage follows in the steps the sources describe: the retirement wave did not arrive, would-be feedstock was leased as green time or overhauled, teardown companies competed for a smaller pool, and prices rose, with CFM56 values up "as much as 50%" in two

years on one trade-press account, "a lessors' market" (IBA, April 2024), "a seller's market for used parts" (Aviation Week) and AerSale's whole-asset sales down from 20 engines in 2024 to 13 in 2025 in "a hypercompetitive feedstock environment". [ePlaneAI; IBA via AviTrader, 25 Apr 2024; Aviation Week; AerSale, 5 Mar 2026; Part V §33] FTAI sits on both sides of the trade: it buys engines and whole aircraft for their material, and it sells the used modules and parts it does not build into modules through an exclusive partnership the annual report does not name; AAR (a parts distributor and maintenance company) announced on 27 March 2025 that the two had extended their exclusive Serviceable Engine Products agreement for CFM56 material through 2030, covering more than 450 engines as trade coverage carried it; the agreement's terms were not read. [FTAI 10-K FY2025; AAR via PRNewswire, 27 Mar 2025, body as carried by ePlaneAI; Part V §33.6, §35]

PARTS MANUFACTURER APPROVAL (PMA)

A Parts Manufacturer Approval (PMA) is the FAA's combined design-and-production approval that lets a company other than the type-certificate holder legally make and sell a replacement part, with no access to the manufacturer's data. Approvals have concentrated in consumables and simple parts. A cooled, coated high-pressure turbine blade is the hard case: its life depends on an alloy, a crystal structure and a coating the applicant must reproduce and prove in an engine test programme it funds itself. Chromalloy's approval of a CFM56-5B/7B blade on 30 October 2025, "the only aftermarket alternative to the OEM manufactured new part", is treated in the trade as that milestone. No PMA exists for any CFM56 life-limited part. [eCFR, 14 CFR (Code of Federal Regulations) Part 21 Subpart K; HEICO 10-K FY2025; Chromalloy release, 30 Oct 2025; Part VI §37–§38, §41; reconciliation R-024]

DESIGNATED ENGINEERING REPRESENTATIVE (DER) REPAIR

A Designated Engineering Representative (DER) is an engineer appointed by the FAA to approve engineering data on its behalf; a DER repair is a major repair done to data a DER approved rather than to the manufacturer's manual, so the part stays the manufacturer's part and what the manual would scrap is restored. The cadence decides which lever applies: at a first visit, around 20,000 cycles, airlines typically repair the turbine blades; at a second, around 30,000, they cannot be repaired again and a new part is needed, which is where an approved alternative blade competes. [FAA Order 8110.37F; In Practise, undated; Part VI §40]

EXHIBIT S.8 — THE SUBSTITUTES: WHAT EACH REPLACES, HOW IT IS PRICED, WHO SELLS IT, AND WHAT THE RESEARCH COULD AND COULD NOT SOURCE

Substitute	Replaces and pricing basis	Who sells it	Binding figures; not retrieved
Used serviceable material	New OEM parts, and LLPs with cycles remaining; 60–80% of new; 70–75% of list from a trader; LLPs at life fraction × list, discounted	Teardown traders (AerSale and others); lessors' part-outs; the manufacturer (TrueChoice); FTAI through its AAR partnership	Run-out CFM56 \$0.8–1.2M → \$1.6–2.2M of material; 2025 market \$5.87B, \$7.7B or \$8.55B by three vendors. Not retrieved: used LLP package prices; engines torn down a year
PMA parts	New OEM blades, vanes, seals, consumables; never a disk, spool or shaft; a discount to list no source discloses	HEICO (300–500 or 400–550 approvals a year); Chromalloy, with FTAI as partner	HPT blades and vanes 15–25% of parts cost; Chromalloy's CFM56 roster is three announced approvals by one reading, four parts by another. Not retrieved: the discount; the saving per visit
DER repair	A new part, by keeping the old one in service; repair price against list	Turbine-part repair houses; airline engineering; independents; the manufacturer (about 400 CFM56 repairs industrialised since early 2023)	Repairs are 10–20% of a visit. Not retrieved: a repair price for any CFM56 hot-section part
FTAI–Chromalloy joint venture	Hot-section parts in FTAI's own modules; "at cost" per investor blogs, not in filings	FTAI "develops and manufactures PMA parts through a joint venture" (10-K)	Blogs claim "up to \$2 million per shop visit"; FTAI's only quantified lever is "up to \$75,000" a visit from its Bristol accessories venture. Not retrieved: partner, ownership and accounting; the saving per module

Source: Aviation Week, undated; Aircraft Commerce Issue 120; Safe Fly Aviation, 2026 (consultancy blog); three market-research vendors as in Part V §35; HEICO 10-K FY2025; Chromalloy, 30 Oct 2025; CFM International; FTAI 10-K FY2025; Insider Monkey, Q3 2025 call; Komodo Capital and Kairos Research (investor blogs). Compiled from Parts V and VI. Reconciliation R-021 to R-027. Accessed 2026-10-03.

The alternative-parts lever touches only the on-condition parts line, since life-limited parts remain single-source and labour, test and margin are untouched, and it applies only at visits

where blades are scrapped rather than repaired, which the interview places at second and later visits, exactly the visits a fleet kept past its planned retirement undergoes. The claimed saving is contested at three source levels: investor blogs say "up to \$2 million per shop visit"; a set of 80 blades priced from an interview is \$1.6M; the consultancy's share puts blades and vanes at \$0.11 to \$0.63M a visit; and FTAI's one quantified saving is \$75,000 a visit from a different venture. Against it stand CFM's TRUEngine designation and lessors' manufacturer-only return conditions; an appraiser records that approved alternatives "continue to undermine full-life maintenance adjustments". [Komodo Capital; Kairos Research; In Practise; Safe Fly Aviation, 2026; Insider Monkey, Oct 2025; Aircraft Value News, undated; Part VI §42, §44; reconciliation R-015, R-026, R-027]

Depth: Part V §31–§36; Part VI §37–§44.

S9. FTAI's answer: the Module Factory and the Aerospace Products segment

FTAI's module business is the fourth lever of S2 organised as a production line. It rebuilds modules before any customer engine arrives, holds them in inventory, and sells them into engines whose one expired module would otherwise send the whole engine through a 90-to-120-day visit after a two-to-six-month wait.

MODULE FACTORY

The Module Factory is FTAI's name for its CFM56-7B and -5B modular repair and refurbishment programme, first described in 2022 at a 500,000-square-foot Montréal facility "with a capacity for up to 300 shop visits per year" and bought for \$170.0M on 9 September 2024. It is a place, a programme, and a process that builds and holds an inventory of serviceable modules for exchange, fed by returned cores, engines retired from FTAI's leasing fleet and the capital vehicles, and engines and aircraft bought for the purpose. [FTAI 10-K FY2025; FTAI release, 9 Sep 2024; Part VII §46]

MODULE EXCHANGE

A module exchange separates a customer's engine at the flanges, removes the module that triggered the removal, installs a serviceable module from inventory, tests the engine and returns it. The customer pays an exchange fee and gives up the unserviceable module, the returned core, which becomes FTAI's feedstock. The industry prices an exchange as the fee plus the difference in life-limited-part life between the modules swapped. Under US revenue rules the returned core is noncash consideration at fair value, so FTAI books revenue equal to the fee plus the core's value and carries the core in inventory until it is sold again. No provider's exchange fee, FTAI's included, was found in any source. [Part VII §46–§47; reconciliation cluster K-01]

MAINTENANCE, REPAIR AND EXCHANGE (MRE), AND MRE CONTRACT REVENUE

MRE is FTAI's name for its offer, a contrast with MRO: instead of restoring a customer's engine while the customer waits, FTAI swaps in a serviceable module or engine and takes the unserviceable unit as feedstock. It is sold as engine, module and parts sales on the "Aerospace products revenue" line (\$455.0M, \$1,079.8M and \$1,600.5M for 2023 to 2025), not split further in any retrieved filing. MRE Contract revenue is a second line, new in 2025, for sales to the vehicles of the 2025 Partnership that FTAI manages and part-owns: \$335.8M for 2025 and \$221.2M, \$182.8M and \$404.0M for the first quarter, second quarter and half of 2026. Two filings define it differently: the Q3 2025 10-Q describes "stand-ready obligations to provide replacement CFM56-5B and CFM56-7B engines to customers as they become unserviceable", and the Q2 2026 10-Q "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership"; one is an obligation and the other a sale, and both are carried. [FTAI 10-K FY2025; FTAI 10-Qs Q3 2025 and Q2 2026; Part VII §45, §48; reconciliation R-044]

EXHIBIT S.9 — AEROSPACE PRODUCTS BY PERIOD: REVENUE, THE RELATED-PARTY LINE, ADJUSTED EBITDA, MARGIN AND MODULES

Period	Aerospace products revenue	MRE Contract revenue	Segment revenue	Segment Adjusted EBITDA	Margin (derived)	Modules produced
FY2023	\$455.0M	\$0	\$455.0M	\$160.0M	35.2%	not retrieved
FY2024	\$1,079.8M	\$0	\$1,079.8M	\$380.6M	35.2%	Q4 ≈136 (derived)
FY2025	\$1,600.5M	\$335.8M	\$1,936.2M	\$671.3M	34.7%; Q4 stated 35%	757 (target 750); Q2 184; Q4 228
Q1 2026	\$522.6M	\$221.2M	\$743.8M	\$222.6M	29.9%	≈270 (derived)
Q2 2026	\$692.2M (derived)	\$182.8M	\$875.0M	\$249.7M	28.5%	296
1H 2026	\$1,214.8M	\$404.0M	≈\$1,618.8M	≈\$472.3M	≈29.2%	566
2026 guidance				\$1,050M (25 Feb 2026, reaffirmed 29 Jul 2026)	"can approach 40%" (call, 26 Feb 2026)	1,050 (Feb release); raised to 1,200 on the July call (call coverage)
2027 guidance				\$1,400M (29 Jul 2026)		

Source: FTAI 10-K FY2025; 10-Qs Q1 and Q2 2026; earnings releases 29 Jul 2025 to 29 Jul 2026; call coverage (Motley Fool, MarketBeat, GuruFocus, Nasdaq.com). Derived figures are arithmetic on filed numbers. Compiled from Part VII Exhibits 7.4 to 7.18. Reconciliation B-1, R-036, R-043. Accessed 2026-10-03.

Four tensions sit in that table. Segment revenue equals the products line plus the MRE line by arithmetic (\$1,936,244 thousand is exactly \$1,600,456 plus \$335,788 thousand); the explicit segment-note mapping was not retrieved. [reconciliation B-1, R-043] The 2026 module target has two figures: 1,050 was set on 25 February 2026, a 39% increase on 757, and coverage of the 30 July 2026 call reported it raised to 1,200; the July release text was not found to carry 1,200. [FTAI release, 25 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat, 30 Jul 2026; reconciliation R-036] Cost of sales for 2025 has two figures in one filing: \$1,240,368 thousand in the segment expenses table and \$1,349,719 thousand on the consolidated line, which also carries leased assets sold; the segment figure gives a derived gross margin of 35.9%. [FTAI 10-K FY2025; reconciliation R-045] And the margin fell from 35% in the fourth quarter of 2025 to about 29 to 30% in the first half of 2026, when the related-party share of revenue was highest (17.3% for 2025; 29.7%, 20.9% and 25.0% for the first quarter, the second and the half); the research recorded the two side by side and did not infer a cause, since the filings disclose neither margin by line nor the elimination's composition (S10). [Part VII §48, §53]

Capacity is stated in four units at four dates for different footprints: "up to 300 shop visits per year" for Montréal (2022 language); 1,350 module overhauls and more than 500 engine tests a year for Montréal plus Miami (release, 9 Sep 2024); 1,800 modules and 600 engines after Rome Fiumicino was added (release, 26 Feb 2025); and 3,000 modules a year across all sites and partners (call coverage, July 2026). FTAI's releases convert at three modules per engine; Part I counts four major modules including the gearbox, so 1,200 modules is 400 engines on one convention and 300 on the other. [FTAI 10-K FY2025; FTAI releases, 9 Sep 2024 and 26 Feb 2025; GuruFocus and Quartr, Jul 2026; reconciliation R-035, R-037, R-041]

WORKED EXAMPLE S.9 — REVENUE AND ADJUSTED EBITDA PER MODULE: A BOUND, NOT A PRICE

Inputs: products revenue of \$1,600.5M (FY2025), \$522.6M (Q1 2026) and \$692.2M (Q2 2026, derived); segment revenue of \$1,936.2M, \$743.8M and \$875.0M; segment Adjusted EBITDA of \$671.3M, \$222.6M and \$249.7M; modules of 757, about 270 (derived) and 296; 2026 guidance of \$1,050M over targets of 1,050 and 1,200 modules.

products revenue per module: FY2025 $1,600.5 \div 757 = \$2.11\text{M}$; Q1 2026 $\$1.94\text{M}$; Q2 2026 $692.2 \div 296 = \$2.34\text{M}$
 segment revenue per module: FY2025 $\$2.56\text{M}$; Q2 2026 $875.0 \div 296 = \$2.96\text{M}$
 Adjusted EBITDA per module: FY2025 $671.3 \div 757 = \$0.89\text{M}$; Q1 2026 $\$0.82\text{M}$; Q2 2026 $\$0.84\text{M}$
 2026 guidance per target module: $1,050 \div 1,050 = \$1.00\text{M}$; $1,050 \div 1,200 = \$0.875\text{M}$
 second half of 2026: $1,050 - 566 = 484$ modules, 242 a quarter against 296 in Q2; for 1,200: 317 a quarter

For every module produced the segment booked about \$2M of product revenue and earned about \$0.8 to \$0.9M of Adjusted EBITDA; with the related-party line, revenue per module is \$2.6 to \$3.0M. Five limits make each an upper bound: the numerator includes whole engines, parts, used material and V2500 work while the denominator counts only CFM56 modules; a module produced is not a module sold; the profit on related-party sales is counted before elimination; exchange revenue includes the fair value of returned cores, which is not cash; and the core-to-fan mix is undisclosed.

[Part VII §56; reconciliation R-046]

The segment's share of its market is framed three ways on three denominators: about 5% of a \$22B addressable market as of 2024 (release, 26 Feb 2025); a goal of 25% of CFM56 shop visits, year unstated (call, July 2026); and, derived, about 17% of a 2,350-visit year at 1,200 modules and three per engine. Management's three levers for moving the margin toward 40% in 2026 are all on the material line of S2: the approved alternative blade, more used material and the CFM materials agreement; the first half ran at about 29%. [FTAI release, 26 Feb 2025; GuruFocus, Jul 2026; MarketBeat and Motley Fool, 26 Feb 2026; Part VII §49, §52, §57; reconciliation R-042]

THE MUDDY WATERS SHORT REPORT AND THE AUDIT COMMITTEE REVIEW

Muddy Waters Research, a short seller, published a report on FTAI on 15 January 2025 alleging, as third-party summaries and the later complaint describe it, that whole-engine sales were presented as module sales, that margins were inflated and that sales were pushed to intermediaries ahead of demand; the report itself was not read. FTAI said it "strongly disagrees"; on 18 January 2025 its Audit Committee "determined to begin a review, which included engaging independent advisors", and in February 2025 announced that "the allegations made against the Company are without merit". A securities class action in the Southern District of New York, No. 25-cv-00541, covers 23 July 2024 to 21 January 2025; the defendants' motion to dismiss, filed 20 November 2025, was pending at the research date. The committee engaged a new auditor, KPMG (an audit firm), replacing Ernst & Young LLP, effective 17 June 2025. [FTAI 8-Ks, Jan and Feb 2025; FTAI 10-Q Q3 2025; FTAI 10-K FY2025; Benzinga, 15 Jan 2025; law-firm case pages; Part VII §54; reconciliation R-063, R-064, R-066]

The definitional point behind the allegations is one the reader can now state. Three modules built into an engine and sold outright is an engine sale; the same three installed one at a time into three customers' engines are three exchanges. The revenue may be similar, and the filings present engines, modules and parts on one line, so a reader cannot tell how much is whole engines. That is the boundary the dispute concerns; nothing in the primer resolves it. [Part VII §47.4]

Depth: Part VII §45–§57.

S10. FTAI's capital: Aviation Leasing, the Strategic Capital vehicles, the money that flows between them, and the 2026 securitization

The other half of FTAI owns things. Aviation Leasing held 290 assets at 31 December 2025, 47 aircraft and 243 engines, with utilization of about 77% on the company's equity-weighted definition. Its 2025 lines were lease income of \$235.2M, maintenance revenue of \$218.5M and asset sales of \$106.9M against segment Adjusted EBITDA of \$608.9M, so rent cannot account for the earnings; maintenance revenue, asset-sale margins, a \$46.4M gain on sales to the first capital vehicle and the equity-method share must. [FTAI 10-K FY2025; Part VIII §58] The aircraft is bought for its two engines (S4), so the book is both rent and a feedstock pipeline for S9; in 2025 the company began selling it into vehicles funded by others while keeping three roles in them.

STRATEGIC CAPITAL INITIATIVE (SCI): THE 2025 PARTNERSHIP, THE 2026 SPV AND THE SEED ASSETS

The Strategic Capital Initiative (SCI), launched 30 December 2024, raises third-party capital into closed-end vehicles that buy on-lease mid-life 737NG and A320ceo aircraft; FTAI manages each as Servicer, co-invests as a limited partner, and supplies every engine shop visit for the fleet through exchanges with its MRE business. The 2025 Partnership closed on 27 October 2025 at a \$2.0B equity hard cap with a \$2.5B asset-level debt commitment, and by 17 August 2026 had "approximately \$6.0 billion of total capital across over 300 aircraft", fully committed and in "harvest phase". The Seed Assets are the 46 on-lease aircraft FTAI sold from its own book to start it, "at an estimated net purchase price of \$549 million". The 2026 SPV closed a \$2.0B warehouse facility with an accordion to \$3.0B on 14 August 2026; its equity and FTAI's stake were undisclosed. [FTAI releases, 30 Dec 2024, 26 Feb 2025, 27 Oct 2025, 17 Aug 2026; FTAI 10-K FY2025; Part VIII §60–§62; reconciliation R-055]

SERVICER AND SERVICING FEE

The Servicer manages a vehicle's assets for a fee: leasing and remarketing the aircraft, collecting rent and reserves, overseeing condition and selling. FTAI "as the Servicer, provides aircraft management services to the 2025 Partnership, and receives customary, market-based compensation": \$10.2M for 2025, \$6.99M for the second quarter of 2026 and \$12.85M for the first half. The fee's rate, its base and the income-statement line it sits on are not disclosed. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; Part VIII §61, §63]

EQUITY METHOD AND PROFIT ELIMINATION

Under the equity method an investor with significant influence but not control carries its stake at cost plus its share of earnings, less distributions and eliminated profit, on one line, with none of the investee's revenue, assets or debt consolidated. FTAI's stake in the 2025 Partnership is 19% in the FY2025 10-K and the 2026 10-Qs and 20% in the Q3 2025 10-Q. Profit elimination removes from current earnings the profit on sales to such an investee and recognises it later: "the profit from the MRE Contract revenue is eliminated through equity method earnings and will be recognized over time as the 2025 Partnership generates income". Eliminations were \$10.0M in the first quarter of 2026 and \$6.6M in the second, \$16.6M for the half; whether that is 19% of the profit or all of it is not stated. [FTAI 10-Qs Q3 2025, Q1 and Q2 2026; FTAI 10-K FY2025; Part VIII §63–§64; reconciliation R-047, R-048]

ASSET-LIGHT MODEL AND HARVEST PHASE

"Asset-light" is management's term for owning fewer leased assets directly and earning servicing fees, maintenance work and equity income from vehicles that own them; the July 2026 cut to Leasing guidance was explained as "reflecting a continued shift to an asset-light business model". Harvest phase is the period after a closed-end vehicle has finished buying, in which it collects rent, pays distributions and sells assets until it winds up; the 2025 Partnership's term and exit route were not located. [FTAI releases, 29 Jul 2026 and 17 Aug 2026; FTAI 10-Q Q2 2026; Part VIII §60–§61; reconciliation R-058]

EXHIBIT S.10 — THE STREAMS BETWEEN FTAI AND THE 2025 PARTNERSHIP, BY PERIOD, AND WHERE EACH LANDS

Stream	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands
Gain on aircraft sold to the Partnership	\$46.4M (the three 2025 quarters sum to \$50.1M on 45 aircraft)	n/l	n/l	n/l	A gain line outside revenue, under topic 610-20 of the US Accounting Standards Codification (ASC), described as "non-recurring"; inside Leasing Adjusted EBITDA

Stream	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands
Servicing fees	\$10.2M	\$5.9M (derived)	\$7.0M	\$12.85M	Line not confirmed
MRE Contract revenue, sales to and from the Partnership	\$335.8M	\$221.2M	\$182.8M	\$404.0M	MRE Contract revenue line, Aerospace Products (S9)
Profit elimination on those sales	n/l	\$10.0M	\$6.6M	\$16.6M	Through equity-method earnings
Carrying value; stake	\$281.7M; 19% (20% in Q3 2025)	\$279.4M; 19%	\$365.5M; 19%		Equity-method investment, balance sheet
Leasing guidance for 2026	\$575M (25 Feb 2026)		\$475M (29 Jul 2026)		Segment Adjusted EBITDA; 2027 guidance \$450M

Source: FTAI 10-K FY2025; 10-Qs Q1 2025 to Q2 2026; releases of 25 Feb 2026 and 29 Jul 2026. n/l = not located. Compiled from Part VIII Exhibits 8.6, 8.10 and 8.11. Reconciliation R-047 to R-049, R-054, R-057. Accessed 2026-10-03.

Three disagreements in the table bear directly on the economics. The stake is 19% in the later filings and 20% in the Q3 2025 10-Q; the research suggests dilution at the final close and marks that as a guess. The 2026 Leasing guidance was \$575M on 25 February 2026 and \$475M on 29 July 2026; a search reading of the July release that returned \$575M was established on follow-up as a mis-read of the February figure, so the path is \$575M cut to \$475M, with \$450M for 2027. And the profit elimination supports two readings of one sentence. If FTAI eliminates only the share of profit corresponding to its stake, the usual treatment, then \$16.6M implies about \$87M of gross profit on \$404.0M of first-half sales to the Partnership, a margin of about 22% (at 20%, \$83.0M); if it defers all of the profit, the gross profit was \$16.6M and the margin about 4%. The filing supports either, so the segment's margin on Partnership volume cannot be compared with its margin on third-party volume; the related-party pricing of aircraft, exchanges and the fee is not disclosed. [FTAI 10-Q Q2 2026; FTAI releases, 25 Feb 2026 and 29 Jul 2026; reconciliation R-047, R-048, R-049, R-057]

WORKED EXAMPLE S.10 — WHAT A \$6.0B VEHICLE PAYS ITS MANAGER: THE DISCLOSED RUN-RATE AGAINST THREE PEER SCALES

Inputs: servicing fees of \$6.988M for Q2 2026 and \$12.849M for 1H 2026 (10-Q); total capital about \$6.0B and equity \$2.0B (releases); a lease rate factor of 0.9% a month (assumed; alternatives 0.70% and 1.0%, S4); peer fee scales: Carlyle's aviation securitizations at about 2% of rents plus incentives

up to 5% of rents and 3% of sale proceeds; Willis Lease's engine trusts at 13.5% of net rents; a private-fund convention of 1.5% of committed equity (assumed, not an SCI term).

disclosed run-rate: $6.988 \times 4 = \$27.95\text{M}$ a year; $12.849 \times 2 = \$25.70\text{M}$
on total capital: $27.95 \div 6,000 = 0.47\%$; on equity: $27.95 \div 2,000 = 1.40\%$

rent at 0.9% a month on \$6.0B: $0.009 \times 12 \times 6,000 = \648M a year (at
0.70%: \$504M; at 1.0%: \$720M)

Carlyle scale: servicing 2% of rents = \$13.0M; incentive at the 5% cap =
\$32.4M; 3% on an assumed \$600M of disposals = \$18.0M

private-fund scale: $1.5\% \times 2,000 = \$30.0\text{M}$; Willis trust scale, for
contrast: $13.5\% \times 648 = \$87.5\text{M}$

The disclosed run-rate of \$26 to \$28M sits between the servicing-only securitization figure of about \$13M and a 1.5% management fee of \$30M, and well below the Willis scale. The placement rests on assumed inputs; FTAI's share of earnings is allocated "based on the contractual profit-sharing arrangement", and whether a promote exists is undisclosed. [FTAI 10-Q Q2 2026; Carlyle 10-K FY2025; Willis Lease 10-Ks FY2019 and FY2020; Part VIII worked example 8.5; reconciliation R-030]

The maintenance stream is the one the releases describe but never size. A 300-aircraft fleet carries 600 engines; on the intervals of S1 that is 40 to 100 removals a year, a few hundred million dollars of exchanges at the per-module bounds of S9. The disclosed sales to the Partnership, \$404.0M in the first half of 2026 alone, sit above every case built that way; the record offers four candidates without choosing: the line covers purchases as well as sales, aircraft may have arrived with engines already due, whole engines may have been sold to equip the fleet, and the per-module figures are bounds. [Part VIII worked example 8.6] The vehicle's borrowing, the February 2025 commitment, the \$612M securitization of May 2026 and whatever bank debt remains, never reaches FTAI's balance sheet; FTAI's own \$3.45B of long-term debt is its senior notes (S12). [Part VIII §61, §64] The WestJet transaction of 28 September 2026 shows the structure in one deal: 17 737-700s on lease went to the 2026 SPV, and 10 off-lease aircraft went to FTAI for their 20 CFM56-7B engines; terms were not disclosed. [FTAI release, 28 Sep 2026; Part VIII §67]

Depth: Part VIII §58–§67; Part X §82.

S11. FTAI Power: a second life for cores, the J&F order, the market it sells into

The newest business removes CFM56 engines from aviation for good. It sits on the same side of the ledger as retirement and teardown, and competes with both for the same run-out cores.

FTAI POWER: MOD-1, HEAT RATE, BEHIND-THE-METER BRIDGING POWER, AND J&F POWER SYSTEMS

FTAI Power, launched 30 December 2025, converts CFM56 engines into aeroderivative generator sets: the fan is removed and replaced by a small compressor, a power turbine and generator are fitted, and the fuel system is changed for gas. Mod-1 is its 25 megawatt (MW) trailer-mounted unit, described in call coverage as 35 to 40% efficient at a heat rate (fuel energy per kilowatt-hour) of about 9,000 and as giving "near-fully depreciated" engines a 10-to-20-year second life; the release-level statements are capacity for "over 100 units annually", first delivery by the fourth quarter of 2026 and 100 units in 2027. Behind-the-meter (BTM) generation is power made on the customer's side of the utility meter; bridging power is temporary on-site generation until a grid connection arrives. J&F Power Systems, the venture with Jereh Group (a Chinese maker of trailer-mounted gas-turbine packages) that packages and sells Mod-1, is "FTAI's joint venture" to FTAI and "its subsidiary" in Jereh's exchange disclosure as reported; the ownership split and consolidating party are undisclosed. On 22 July 2026 J&F received a five-year master supply agreement and an initial \$1.465 billion purchase order from "a leading international cloud service provider" (FTAI's release) or "a U.S. hyperscaler" (call summaries), for delivery in batches through November 2027; the unit count was not disclosed. [FTAI releases, 30 Dec 2025 and 22 Jul 2026; FTAI 10-Q Q2 2026; TheValueist and Aviation Week, call coverage, 26 Feb 2026; FilingReader, 22 Jul 2026; Part IX §68–§71; reconciliation R-061]

The market is defined by a shortage the sources date. The International Energy Agency expects global data-center electricity use to more than double to about 945 TWh by 2030. GE Vernova's gas-turbine backlog plus slot reservations rose from 50 GW in 2025 to 116 GW at the second quarter of 2026, with slots sold for 2031 delivery, and Siemens Energy quotes lead times of three years and more. A buyer that needs power in 2026 or 2027 cannot get a new frame turbine; that is the opening for a 25 MW unit built from a used airline engine, as for piston plants and fuel cells. The 2027 guidance puts FTAI Power's Adjusted EBITDA at \$450M in the release text; one call summary gives "\$450 million to \$750 million", and both are carried. [IEA, Apr 2025; Power Engineering, Jul 2026; Oilprice, 2026; FTAI release, 29 Jul 2026; GuruFocus, 30 Jul 2026; Part IX §71; reconciliation R-062]

WORKED EXAMPLE S.11 — THE IMPLIED PRICE OF A MOD-1, AND THE CFM56 CORES A HUNDRED OF THEM CONSUME

Inputs: order value \$1.465 billion (release, 22 Jul 2026); 2027 target 100 units (release, 25 Feb 2026); unit count in the order undisclosed, so 100, 80 and 60 assumed; 25 MW per unit; one gas generator per unit (assumed, by analogy with the LM2500 (GE's aeroderivative of the same class)); bases and rates as in S5; three modules per engine.

at 100 units: $1,465 \div 100 = \$14.65\text{M a unit} = \586 per kW ; at 80: $\$18.3\text{M}, \$733/\text{kW}$; at 60: $\$24.4\text{M}, \$977/\text{kW}$
 2027 Power Adjusted EBITDA per unit at 100 units: $450 \div 100 = \$4.5\text{M (at } \$750\text{M: } \$7.5\text{M)}$
 cores at 100 units: 100 engines = 0.42% of 24,000; 21% (at 2%) to 29% (at 1.5%) of a year's retirements on OEM-level bases; 35% on $14,200 \times 2\% = 284$
 module-equivalents: $100 \times 3 = 300$ a year = 29% of the 1,050 target, 25% of 1,200, 40% of 2025's 757

The implied price sits below the \$2,000 to \$2,500 per kW of plant-level filings and near Wood Mackenzie's \$600 per kW turbine-only projection for end-2027, on differing scopes. Every line scales with the undisclosed unit count, and the margin line holds only if FTAI consolidates J&F, which is undisclosed. The cores line joins this business to the chain: a hundred engines a year is under half a percent of the base and a fifth to a third of a year's retirements, so the business competes with the teardown trade and the exchange business for the same run-out engines. [Part IX §74; reconciliation R-001, R-004, R-035, R-061, R-062]

Depth: Part IX §68–§74.

S12. The company, its numbers and what remains unknown

INTERNALIZATION, AND THE COMPANY IN ONE PARAGRAPH

FTAI Aviation Ltd. is a Cayman Islands company listed on Nasdaq. It began as Fortress Transportation and Infrastructure Investors LLC, listed on the New York Stock Exchange on 15 May 2015 and externally managed by a Fortress Investment Group subsidiary; it spun off its infrastructure businesses on 1 August 2022 and became a Cayman parent on 10 November 2022. Internalization is the termination of an external management agreement, the hiring of the manager's people, and a payment for the fee stream given up; FTAI's closed on 28 May 2024 for a stated \$300.0M, \$150.0M in cash paid on 17 June 2024 and 1,866,949 shares valued at \$150.0M, so expense lines before and after mid-2024 are not like for like. [FTAI 10-Q Q3 2024; FTAI 8-K, 28 May 2024; FTAI Infrastructure 10-K FY2024; Part X §75–§77]

The 2025 consolidated statement shows the chain's money in one place. Revenues were \$2,507.4M, of which Aerospace products \$1,600.5M, MRE Contract \$335.8M, lease income \$235.2M, maintenance revenue \$218.5M and asset sales \$106.9M; cost of sales was \$1,349.7M, depreciation and amortisation \$225.8M and interest expense \$247.8M; consolidated Adjusted EBITDA was \$1,190.9M. The company reported net income in 2024 and 2025 and consumed cash in operations in both years (+\$129.0M in 2023, −\$188.0M in 2024, −\$310.7M in 2025), because buying assets to sell is an operating outflow while buying assets to lease is an investing

outflow: a factory holding modules ahead of demand shows its purchases as operating cash out until they are sold. Inventory stood at \$1,364.3M at 31 March 2026, about a year of 2025's consolidated cost of sales. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; Part X §79–§81]

The balance sheet carries five series of senior unsecured notes due 2028 to 2033 at coupons of 5.50% to 7.875%, long-term debt net of \$3,451.1M and cash of \$412.2M at 31 March 2026, an undrawn revolver maturing 22 May 2027, and perpetual preferred shares (Series D at 9.50%, \$65.0M, remains; Series C's redemption was announced on 15 May 2026). Credit ratings and covenants were not retrieved; about \$3.0B of net debt is 2.6 to 2.7 times an annualised quarter of Adjusted EBITDA, an illustration only. The dividend rose at four consecutive results, from \$0.30 a quarter to \$0.35 (27 Oct 2025), \$0.40 (25 Feb 2026), \$0.45 (29 Apr 2026) and \$0.50 (29 Jul 2026); a \$500M repurchase program was announced on 15 September 2026. On 6 March 2026 Nicholas McAleese was appointed Chief Financial Officer, succeeding Eun (Angela) Nam. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; GlobeNewswire release titles; FTAI release, 6 Mar 2026; Part X §82–§83]

Guidance is given in segment Adjusted EBITDA, before Corporate and Other. For 2025 it was \$1.1 to \$1.15B on 30 December 2024, and both segments delivered above the range. For 2026 it moved from \$1.4B, to \$1.525B (\$1.0B and \$525M, 27 October 2025), to \$1.625B (\$1.05B and \$575M, 25 February 2026), and back to \$1.525B when Leasing was cut to \$475M on 29 July 2026. For 2027 it is \$2.3B: Aerospace Products \$1.4B, FTAI Power \$450M, Aviation Leasing \$450M. [FTAI releases, 30 Dec 2024, 27 Oct 2025, 25 Feb 2026, 29 Jul 2026; Part X §84]

EXHIBIT S.12 — THE BINDING NUMBERS OF THE CHAIN IN ONE TABLE, EACH AS THE RESEARCH FIXED IT

Quantity	Binding presentation	Source
CFM56 in service	~23,000 (CFM, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); >35,000 delivered; consultancy ~14,200 and >19,000 labelled	CFM; Aviation Business News; Safran; Safe Fly
Life limits and intervals, -7B	Fan 30,000; HPC and HPT 20,000; LPT 25,000 cycles; 1,000–1,500 cycles a year; restoration every 10,000–15,000, overhaul at 20,000–25,000	Mini-packs; Aircraft Value News; Leeham
Full -7B LLP set at list	\$1.775M (2008); \$3.4M (2018); "\$4m" (Nov 2019); \$5.7M (Jun 2025); ~7% a year compounded (derived)	Aircraft Commerce; Aircraft Value News; MyAirTrade

Quantity	Binding presentation	Source
Shop-visit cost and shares	Restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M"; overhaul \$3.5–4.2M; material 60–70%, labour 20–30%, repairs 10–20%; LLPs 40–60% of parts, HPT blades and vanes 15–25%	Safe Fly; Air Cargo Week
Turnaround and escalation	90–120 days overhaul (trade press); ~90 sustaining (GE); slot wait 2–6 months (Bain); spares ~10% a year 2022–23 (Aviation Week)	Aviation Business News; GE; Bain; Aviation Week
-7B value and rent	Half-life \$5.7M / \$6.4M (IBA, Sept 2025); green-time \$2.8–3.4M, run-out \$0.8–1.2M (Safe Fly); rent ~\$100,000 (IBA, 2024) and \$42,000–48,000 (Safe Fly, 2026)	IBA; Safe Fly
Used-material price	60–80% of new; 70–75% of list from a trader; above list when the OEM is constrained	Aviation Week; Aircraft Commerce
Retirement rate and visits	GE planning 3–4% → 2–3% → 1.5–2%; 2025 ~1.5%; 2026 ~2%; 1Q 2026 "below 1%"; visits 2,300–2,400 in 2026–28 (GE) against ~2,500 peak 2025–26 (Safran); just above 2,000 in 2019	Aviation Week; Leeham; GE investor-relations page
OEM model	~85% of CFM56 aftermarket transactional vs ~40% LEAP; ~15% vs ~50% of fleets under long-term agreements	Safran via Aviation Week
Modules and capacity	757 (FY2025, target 750); ~270 (Q1 2026); 296 (Q2 2026); 566 (1H 2026); target 1,050 raised to 1,200 (call coverage); capacity 300 visits / 1,350 / 1,800 modules (600 engines) / 3,000 (call coverage); three modules per engine	FTAI releases and calls
Aerospace Products	Revenue \$455.0M / \$1,079.8M / \$1,936.2M (2023–25, incl. MRE \$335.8M); Adjusted EBITDA \$160.0M / \$380.6M / \$671.3M; Q2 2026 \$875.0M and \$249.7M; cost of sales FY2025 \$1,240.4M (segment) and \$1,349.7M (consolidated); MRE Contract revenue \$404.0M (1H 2026)	FTAI 10-K FY2025; 10-Qs; releases
Aviation Leasing and SCI	290 assets (47 aircraft, 243 engines), ~77% utilization; Adjusted EBITDA \$500.1M / \$608.9M; guidance \$575M → \$475M; 2025 Partnership \$2.0B equity, \$2.5B debt, ~\$6.0B total, 300+ aircraft; seed 46 for ~\$549M; stake 19% / 20%; fees \$10.2M and \$12.85M; eliminations \$16.6M; 2026 SPV \$2.0B warehouse; ABS \$612M on 48 aircraft	FTAI 10-K; 10-Qs; releases
FTAI Power	Launched 30 Dec 2025; Mod-1 25 MW; >100 units a year; first delivery Q4 2026; 100 units 2027; J&F \$1.465B (22 Jul 2026); 2027 Adjusted EBITDA \$450M (" \$450–750M" in one summary)	FTAI releases; call coverage

Quantity	Binding presentation	Source
Consolidated and corporate	2025 revenues \$2,507.4M; Adjusted EBITDA \$1,190.9M; operating cash flow −\$310.7M; inventory \$1,364.3M, long-term debt \$3,451.1M, cash \$412.2M (31 Mar 2026); guidance 2026 \$1.525B, 2027 \$2.3B; internalization 28 May 2024, \$300.0M; dividend \$0.30 → \$0.50	FTAI filings and releases

Source: reconciliation section E.2 and the orchestrator's resolutions, as presented in the sections above; each figure's underlying source is cited where it first appears in this Spine. Accessed 2026-10-03.

Read backwards from this table, the chain is one sentence long. A company that owns aircraft bought for their engines, that rebuilds the modules of an engine no longer in production into exchanges for a fleet that did not retire on schedule, that sells those exchanges in part to vehicles it manages and part-owns, and that has begun turning run-out cores into generators for a power market short of turbines, depends on three quantities: how many CFM56 engines there are, how fast they leave, and what a new part from their only manufacturer costs. Each is contested, and the ten disagreements below are where the contest sits.

Depth: Part X §75–§85; Part XII §94.

Tensions carried in the Spine

Id	The two (or more) figures	Where in the Spine
R-001 / R-002	CFM56 in service: ~23,000 (CFM); ~24,000 (Aviation Business News); >22,800 (Safran); consultancy ~14,200 and >19,000 (Safe Fly)	S1 box; S5 box and text; S7 worked example; S8; S11
R-004 / R-005	GE 3–4% → 2–3% → 1.5–2%; 2025 ~1.5%, 2026 ~2%; 1Q 2026 "below 1%"; 2024 aircraft retirements "fewer than 200" (IBA) against 587 (Cirium)	S5 box, text and worked example S.5; S8; S11
R-013	Restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M" (one consultancy blog, two pages)	S2, Exhibit S.2, worked example S.2
R-016 / R-017	Full -7B LLP set \$5.7M (MyAirTrade, Jun 2025) or ~\$4M (Aircraft Value News, Nov 2019); \$285 or \$200 a cycle	S2, Exhibit S.2, worked example S.2
R-028 / R-029	-7B rent ~\$100,000/month (IBA) against \$42,000–48,000 (Safe Fly, may be green-time); value \$5.7M/\$6.4M "levelled off" (IBA) against green-time \$2.8–3.4M and rising rents (Safe Fly)	S4, Exhibit S.4; S8
R-035 / R-037	Four major modules (anatomy) against three per engine (FTAI releases); capacity 300 visits / 1,350 / 1,800 / 3,000 modules in different units, footprints and dates	S1 box; S9; S11

Id	The two (or more) figures	Where in the Spine
R-036	2026 module target 1,050 (release, 25 Feb 2026) against 1,200 (call coverage, 30 Jul 2026)	S9, Exhibit S.9, worked example S.9
R-043 / R-044	Segment revenue = products + MRE by arithmetic, mapping not retrieved; MRE Contract revenue as "stand-ready obligations" (Q3 2025 10-Q) against "the transaction price for sales" (Q2 2026 10-Q)	S9 box, Exhibit S.9
R-045	Cost of sales FY2025 \$1,240,368K (segment table) against \$1,349,719K (consolidated line); 35.9% derived	S9; S12
R-047 / R-048 / R-049	Eliminations of \$16.6M imply ~\$87M of gross profit if proportional to 19% or \$16.6M if fully deferred; stake 19% against 20%; Leasing guidance \$575M cut to \$475M (a \$575M July reading was a mis-read)	S10, Exhibit S.10

Unknowns carried in the Spine

The register of everything the research could not find is Part XII §94. Nineteen clusters group the entries that ask one question; the five largest are these. K-02, appraiser values for CFM56 engines (five entries): no half-life base, -5B, full-life or mid-life aircraft value was reached, so every valuation example stands on IBA's two -7B values. K-01, module-exchange pricing (three entries): no provider's exchange fee was found, so every per-module figure is a bound. K-17, segment figures by year (three entries): the 2023 Leasing lines, the Adjusted EBITDA reconciliation and net income by year. K-05, a stated PMA discount to list (three entries): neither Chromalloy's nor HEICO's, so the alternative-parts saving cannot be sized. K-18, the capital tied up in module inventory (two entries): the inventory change and the investing lines of the cash-flow statement. Also open, because S10 turns on them: the servicing fee's rate and base (K-09) and the elimination method.

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PART I

First principles: the CFM56 as a machine

This Part explains the engine that the whole FTAI Aviation story is built on: how it works, how it comes apart, what wears out, what is scrapped on a clock, what a repair costs and what the engine is worth at any point in its life. A reader who finishes it will be able to read a shop-visit quote, an engine records summary and an appraiser's value with every term understood.

WHERE THIS PART SITS IN THE BUILD

The primer is built in layers. Parts I and II are first principles: Part I is the engine as a physical and economic object, and Part II is how money moves around that object through leases, reserves and financing. Part III is the context, meaning the fleet, the retirements that did not happen, the new engines that arrived late, and a translation table that turns each of those facts into a requirement on the aftermarket. Parts IV to IX are the segments, three for the market around FTAI and three for FTAI's own businesses, and each of them opens by naming the translation-table rows it answers. Parts X and XI cover the company and the money, and Part XII is reference. Nothing in this Part depends on anything written later; everything written later depends on this Part.

1. What a turbofan does, functionally

A jet engine takes in air, squeezes it, burns fuel in it and lets the hot gas rush out of the back. The rush of gas pushes the engine forward. A turbofan is a jet engine with a large fan bolted on the front. The fan pushes most of the air around the outside of the hot central part of the engine, not through it. Only a small share of the air goes through the middle, where the fuel is burned. That burning drives turbines, and the turbines turn the fan. So the hot middle of the engine exists mainly to spin the fan, and the fan does most of the pushing.

TURBOFAN, BYPASS AND HIGH-BYPASS TURBOFAN

A turbofan is a jet engine in which a large front fan pushes most of the incoming air around the hot core rather than through it. The air that goes around is the bypass stream. A high-bypass turbofan is one in which the bypass stream is several times larger than the core stream; the CFM56 is one. [D1 §2.1]

The split between the two streams has a name. The bypass ratio is the mass of air the fan pushes around the core divided by the mass of air that passes through the core. A ratio of five means five kilograms of air go around for every kilogram that goes through. A higher ratio moves more air more slowly, which is the efficient way to make thrust at airliner speeds. The research for this primer classified the CFM56 as a high-bypass design but did not retrieve its bypass ratio as a figure, so none is quoted here.

CORE (GAS GENERATOR)

The core is the hot, fast, expensive middle of the engine: the high-pressure compressor, the combustor and the high-pressure turbine, treated as one module. The industry also calls it the gas generator. In Part VII the same word has a second meaning, the unserviceable unit a customer hands back in an exchange; that Part flags the double meaning where it arises. [D1 §2.3; registry C-6]

Thrust is the push. It is quoted in pounds-force, written lbf. A CFM56-7B is sold at ratings between 19,500 and 27,300 lbf according to one secondary source, and the CFM website summary returned by search gave a ceiling of 33,000 lbf; the dossier records that the higher figure may be a transcription from the -5B family and did not resolve it. [ePlaneAI, undated; CFM website, undated; D1 T1] The important point about thrust for this Part is that one set of hardware is sold at several ratings.

THRUST AND THRUST RATING

Thrust is the engine's forward push, in pounds-force (lbf). A thrust rating is the thrust level at which a given set of hardware is sold; the engine's electronic control limits how hard it may run, and the model suffix gives the rating in thousands of lbf (a -7B24 is rated at 24,000 lbf, a -7B27 at 27,300). A higher rating is the same metal working hotter, so it has less temperature margin and a shorter life between overhauls, and it is worth more because it can serve heavier aircraft and be turned down to a lower rating; a low-rated engine cannot be turned up without paying the manufacturer. [D1 §2.1; IBA (an appraiser), Sept 2025, via D3 §2.4]

The CFM56 has two rotating assemblies, one inside the other, and they are not geared together. The outer assembly carries the fan at the front and a turbine at the back. The inner assembly carries the main compressor and a smaller, hotter turbine. The two spin at different speeds. The only thing that connects them is the air passing from one to the other.

TWO-SHAFT (TWO-SPOOL) DESIGN; LOW-PRESSURE AND HIGH-PRESSURE SPOOLS

A spool is a rotating assembly of compressor and turbine stages on one shaft. The CFM56 has two, one inside the other. The low-pressure (LP) spool carries the fan, a small booster compressor and the rear turbine on the outer shaft. The high-pressure (HP) spool carries the main compressor and the front turbine on a shorter inner shaft. The two are mechanically independent and connected only by the air flowing through them. [D1 §2.1]

Two units of time govern everything that follows. The first is the flight cycle: one take-off and one landing. The second is the flight hour: one hour with the engine running. They are not interchangeable. A short-haul aircraft flies many cycles per hour of operation; a long-haul aircraft flies few. The parts that wear by heat and rotation count hours. The parts that are certified to a

life limit count cycles, because the stress that matters is the spin-up and heat-up at take-off, not the steady hours in cruise.

ENGINE FLIGHT CYCLE (EFC) AND ENGINE FLIGHT HOUR (EFH)

An engine flight cycle (EFC), or simply a flight cycle (FC) or "cycle", is one take-off and landing. An engine flight hour (EFH), or flight hour (FH), is one hour of engine operation. Life limits on rotating parts are certified in cycles, because the stress peak that consumes their life is the spin-up and heat-up at take-off. Maintenance reserves and cost-per-unit arithmetic (Part II §13) use both units. A mature short-haul narrowbody flies roughly 1,000 to 1,500 cycles a year. [D1 §2.4; Aircraft Value News, undated]

Everything in Part I follows from four physical facts. The engine is built as separable sections. Inside those sections sit parts certified to a fixed number of cycles. The hot section loses temperature margin as it wears. And the engine is only as usable as its paperwork says it is. Sections 3 to 9 take those facts in turn. Section 10 sets two other engines beside the CFM56 for contrast, and section 11 gives a first map of where money is made along the chain, which Part XI §86 treats in full.

2. The CFM56 family and its fleet

The CFM56 is built by a company owned equally by two others. That matters because the owner of an engine's design is the only party allowed to make new parts for it, and it controls the manual that says how the engine may be repaired.

ORIGINAL EQUIPMENT MANUFACTURER (OEM)

The OEM is the company that holds the type certificate for the engine, meaning the regulator's approval of its design, and that built it. It is the sole source of new parts at its published catalogue price, it runs its own network of repair shops and branded service agreements, and GE states that it is also the largest supplier of used material for its own products. Part IV §25 carries the full treatment. [D2 §2.2–2.3; D4 §2.1]

CFM INTERNATIONAL (CFM)

CFM International is the joint venture of GE Aerospace and Safran Aircraft Engines (formerly Snecma) that is the OEM and type-certificate holder of the CFM56 and of its successor, the LEAP (CFM's new-generation engine, treated in §10). CFM describes itself as a joint venture; the 50/50 split is how it has always been described and is carried here as the widely published figure, not one the research re-verified. Within the venture GE builds the core and Safran the fan, booster, low-pressure turbine and gearbox. Part IV §25 carries the OEM network. [CFM website, undated; D1 §2.2; D2 §3.1; D4 §2.1]

The family has five commercial members. Two of them matter for FTAI: the -7B, which is the only engine ever fitted to the Boeing 737 Next Generation, and the -5B, which powers about six in ten of the Airbus A320 family's current-engine-option aircraft. The other three are older or belong to aircraft types that have largely left service.

CFM56 FAMILY (-3, -5A, -5B, -5C, -7B)

The CFM56 is a two-shaft, high-bypass turbofan of which more than 35,000 have been delivered since entry into service in 1982, with roughly 23,000 to 24,000 in service depending on the source (§2.2). The -7B is the sole engine of the 737NG, with more than 15,000 built and the last engine for a commercial 737NG delivered in 2019. The -5B powers about 60% of A320ceo-family aircraft, and production of installed -5B engines ended in 2022. No new CFM56 is being built for an airframe, so every CFM56 that flies tomorrow is one that exists today. [CFM website, undated; GE Aerospace, 2019; ISTAT (the International Society of Transport Aircraft Trading), undated]

737NG AND A320CEO

The 737NG (Next Generation) is the Boeing 737-600, -700, -800 and -900ER, built from about 1997 to 2019 and powered only by the CFM56-7B. The A320ceo (current engine option) is the Airbus A318, A319, A320 and A321 family built from about 1988 to 2020, powered by either the CFM56-5B or the V2500 (the alternative engine from the IAE consortium, §10). These two airframe families are what the whole chain described in this primer serves. [D1 §2.2; D6 §2.4]

2.1 THE VARIANTS

EXHIBIT 1.1 — THE CFM56 FAMILY: AIRFRAMES, THRUST, PRODUCTION STATUS AND DELIVERIES

Variant	Airframe	Thrust class (lbf)	Production status	Delivered	Source
CFM56-3	Boeing 737 Classic (-300/-400/-500); smaller fan for ground clearance	18,500–23,500	Last engine shipped 1999	3,974 engines on 1,987 aircraft	CFM/GE release, 1999
CFM56-5A	Airbus A320 (early build)	22,000–34,000, quoted jointly for -5A/-5B/-5C	Ended; year not retrieved	Not retrieved	CFM website, undated
CFM56-5B	A318/A319/A320/A321 (A320ceo); "nearly 60%" of A320ceo ordered	Within 22,000–34,000	Installed-engine production ended 2022	>4,100 -5B-powered aircraft	CFM website; ISTAT, undated
CFM56-5C	Airbus A340-200/-300 (sole engine)	Upper end of 22,000–34,000	A340 out of production; end year not retrieved	Not retrieved	CFM website; CFM release on the 100th A340
CFM56-7B	Boeing 737NG (-600/-700/-800/-900ER), sole engine	19,500–27,300 per a secondary source; "19,500–33,000" per the CFM website summary (D1 T1)	Last engine for a commercial 737NG delivered by May 2019; 15,000th built 2019	>15,000 engines; >7,000 aircraft	CFM website; GE Aerospace, 2019; ePlaneAI

Source: CFM International, "The CFM56 engine family", undated, accessed 2026-10-03; CFM/GE, "Last CFM56-3 rolls off production line", 1999; GE Aerospace, "1 billion flight hours", Paris Air Show 2019; ISTAT, "CFM International CFM56-5B/-7B", undated; ePlaneAI, "Behind the numbers", undated. Compiled in D1 §2.2 and §4.1.

Two dates in that table do most of the work in this primer. The last -7B for a commercial 737NG was delivered by May 2019, and installed-engine production of the -5B ended in 2022. [GE Aerospace, 2019; ISTAT, undated] From those dates onward the number of CFM56 engines in the world

can only fall. Every engine retired, torn down for parts, or converted to some other use leaves the pool, and nothing enters it. Part III §18 and §19 treat the consequences.

2.2 HOW MANY THERE ARE

Counting the fleet sounds simple and is not. The sources disagree, and the disagreement is partly about what is being counted.

INSTALLED BASE

The installed base is the population of engines fitted to aircraft in the operating fleet. Published figures for the CFM56 "in service" range from about 23,000 to about 24,000 and may include spare and stored engines as well as installed ones, since none of the sources states its basis. Because no new -5B or -7B is being built, the base falls by exactly the number of engines retired or converted each year. Part III §18 carries the full treatment. [D9 §2.1–2.2; registry C-11]

EXHIBIT 1.2 — CFM56 FLEET COUNTS, BY SOURCE, BASIS AND DATE

Figure	What it counts	Source and level	Date
>35,000	CFM56 delivered, all variants	CFM website (OEM web page)	Undated, accessed 2026-10-03
~23,000	CFM56 "in service"	CFM website (OEM web page)	Undated
~24,000	CFM56 "in service", of >35,000 delivered	Aviation Business News (trade press)	September 2025
>22,800	CFM56 in Safran's civil installed base	Safran FY2025 results presentation (OEM investor material)	13 February 2026 (end-2025 figure)
"over 20,000"	CFM56-5B and -7B in service	Chromalloy press release (parts maker)	30 October 2025
>19,000	CFM56 "flying"	Safe Fly Aviation (consultancy blog)	2026
~14,200	"active" CFM56, read by one dossier as all variants (≈6,200 -5B, ≈7,500 -7B) and by another as the -7B fleet alone	Safe Fly Aviation (consultancy blog)	2026
"in excess of 8,000"	CFM56-7B "currently installed in 737NGs"	CFM product page (OEM web page)	Undated
>15,000	CFM56-7B delivered	CFM website; GE Aerospace article on the 15,000th engine	2019 and later

Source: CFM International, "The CFM56 engine family" and "CFM56-5B & CFM56-7B" product page, undated; Aviation Business News, "CFM56 turbofan aircraft engine", Sept 2025; Safran, FY2025 Results & Investor Update, 13 Feb 2026; Chromalloy, release of 30 Oct 2025; Safe Fly Aviation, "CFM56 Engine Market Report 2026"; GE Aerospace, 2019. Reconciliation R-001, R-002, R-003 and R-069.

The three OEM-level figures sit close together: about 23,000 on CFM's own page, about 24,000 in trade press citing September 2025, and more than 22,800 in Safran's end-2025 count. [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026] None of them says whether spare engines in warehouses and engines on parked aircraft are included. The research recorded that question as open. [D9 U3; reconciliation R-001] The two Safe Fly Aviation figures come from a consultancy blog and are labelled as such wherever they appear in this primer; the 14,200 figure was read by one dossier as the whole active CFM56 fleet and by another as the -7B fleet alone, and this Part reports both readings without choosing. [Safe Fly Aviation, 2026; reconciliation R-002]

The -7B count has its own puzzle. CFM's product page says "in excess of 8,000 CFM56-7Bs are currently installed in 737NGs", with no date visible, while the same company reports more than 15,000 -7B engines delivered and more than 7,000 737NG aircraft, each carrying two. [CFM product page, undated; GE Aerospace, 2019; reconciliation R-003] Two 737NGs per 8,000 engines would be 4,000 aircraft, far fewer than were delivered; the statement may be old, or may count only one subset. The research did not resolve it. Total deliveries are given as more than 35,000 by CFM and the trade press, and as "well over 30,000" in one dossier's unsourced general statement; the OEM figure is the one carried. [reconciliation R-069]

2.3 WHY THE FLEET ONLY SHRINKS

An engine leaves the base in one of three ways. It is retired with its aircraft and torn down for parts (Part V §32). It is sold on as a whole engine to another operator, which does not reduce the base. Or it is converted to a non-aviation use, which FTAI's power business has begun to do with CFM56 cores (Part IX). The rate at which engines leave is the retirement rate.

RETIREMENT RATE

The retirement rate is the share of a fleet withdrawn from service in a year. GE's planning assumption for CFM56 retirements in 2026 moved from 3–4% of the installed base, to 2–3%, to 1.5–2%; GE reported about 1.5% retired in 2025 and expected about 2% in 2026, and first-quarter 2026 retirements ran "below 1%". Other sources use other denominators (all commercial aircraft, or the active-plus-stored passenger fleet), so every figure must carry its base. Part III §19 is the full treatment. [GE CFO via Aviation Week, 2026; GE Q4 2025 results via Leeham, 22 Jan 2026; GE Aerospace investor-relations page, 2026; registry C-10]

The arithmetic that connects the base to the rest of this primer is short. On a base of about 24,000 engines, a retirement rate of 1.75% removes about 420 engines a year, and a rate of 3.5%

removes about 840. [D9 §2.2, illustrative arithmetic on sourced figures] Every engine that does not retire keeps flying, keeps accumulating cycles, and will need the work described in sections 4 to 7. Every engine that does not retire is also one that is not torn down, so its parts do not reach the used-material market described in Part V. The same number sits on both sides of the aftermarket at once, which is why Part III makes so much of it.

3. Modules: how the engine comes apart, and why modules are interchangeable

The CFM56 was designed to be taken apart at a few large joints rather than stripped to its last bolt every time something inside it needs attention. The sections between those joints are modules.

MODULE

A module is a section of the engine designed to be removed and replaced as a unit. It is bolted to its neighbours at flanges, and it carries its own part number, its own serial number and its own maintenance record. A module can move from one engine to another when its configuration, its records and its part numbers allow. Section 3.2 sets out the conditions. [D1 §2.3; D3 §2.7]

MAJOR MODULE AND SUB-MODULE

The manufacturer's shop manual arranges the engine as a hierarchy. The major modules (the fan and booster, the core, the low-pressure turbine, and the accessory gearbox) are each built from numbered sub-modules. The full numbered sub-module list for the -7B and -5B was not reached by the research, which recorded the commonly cited count of 17 as unverified; this primer does not quote a count. [D1 §2.3, U1]

3.1 THE MODULES AND WHAT EACH ONE DOES

Air enters at the fan and leaves at the back of the low-pressure turbine. The modules sit in that order along the engine, with the gearbox hung beneath.

BOOSTER, OR LOW-PRESSURE COMPRESSOR (LPC)

The booster is a small three-stage compressor mounted directly behind the fan on the low-pressure shaft. It gives the air bound for the core a first squeeze before the main compressor takes over. The fan and booster together form the first major module. [Aircraft Commerce Issue 58, 2008; D1 §2.1]

FAN FRAME

The fan frame is the structural front assembly of the engine, built from four main pieces: the containment case that would catch a released fan blade, the outlet guide vanes that straighten the fan's airflow, the frame itself, and the housing for the radial shaft that drives the gearbox. It ducts both airstreams, passes the engine's thrust to the aircraft, and carries the bearings of the low-pressure rotor. It is the manual's example of a numbered sub-module. [Aircraft Commerce Issue 58, 2008]

HIGH-PRESSURE COMPRESSOR (HPC)

The high-pressure compressor is the nine-stage compressor on the high-pressure spool, driven by the high-pressure turbine behind it. It raises the core airflow to the pressure at which fuel can be burned efficiently. Its spools and disks are life-limited parts with a 20,000-cycle limit on the -7B (\$4). [Aircraft Commerce Issue 58, 2008; D1 §2.4]

COMBUSTOR

The combustor is the chamber between the high-pressure compressor and the high-pressure turbine in which fuel is sprayed into the compressed air and burned. It is part of the core module. [D1 §2.1]

HIGH-PRESSURE TURBINE (HPT)

The high-pressure turbine is the single-stage turbine immediately behind the combustor. It takes power from the hottest gas in the engine and uses it to drive the high-pressure compressor. It is the hottest, fastest-wearing and most expensive module to restore: its blades and vanes are 15–25% of a shop visit's parts cost, and most of the engine's lost temperature margin is recovered there (§5). Its disk is a 20,000-cycle life-limited part. [Safe Fly Aviation, 2026; D1 §2.5]

LOW-PRESSURE TURBINE (LPT)

The low-pressure turbine is the four-stage turbine at the rear of the engine. It takes the remaining energy from the gas after the high-pressure turbine and drives the fan and booster through the long low-pressure shaft. With the turbine rear frame it forms the third major module. Its disks are 25,000-cycle life-limited parts. [Aircraft Commerce Issue 58, 2008; D1 §2.4]

ACCESSORY GEARBOX (AGB)

The accessory gearbox is a train of gears driven off the high-pressure spool by a radial shaft running out through the fan frame. It turns the fuel pump, the oil pumps, the hydraulic pump and the aircraft's electrical generator. It is the fourth module in the manual's hierarchy and sits outside the gas path. [D1 §2.1, §2.3]

EXHIBIT 1.3 — THE CFM56-7B MODULES: CONTENTS, FUNCTION AND LIFE-LIMIT CLOCK

Major module	Contents on the -7B	What it does	Life-limited parts and limit (EFC)	Source
Fan and booster	61-inch fan with 22 solid titanium wide-chord blades; three-stage booster; fan frame sub-module	Moves the bypass air that produces most of the thrust; booster pre-compresses the core air; fan frame carries thrust and the LP bearings	Fan disk and booster spool, 30,000	Aircraft Commerce Issue 58, 2008; AJW (AJW Group, an engine trader) and StandardAero mini-packs
Core (HPC, combustor, HPT)	Nine-stage HPC; combustor; single-stage HPT	Compresses, burns, and extracts the power that drives the HPC; the hottest and most expensive section	HPC stage 1–2 spool, stage 3 disk, stage 4–9 spool, HPT disk, all 20,000	Same
Low-pressure turbine	Four-stage LPT and turbine rear frame	Drives the fan and booster through the LP shaft	LPT stage 1 disk, 25,000 (other LPT disks and shafts listed in the packs without limits in the returned text)	Same
Accessory gearbox	Gear train driven by a radial shaft from the HP spool	Powers fuel, oil and hydraulic pumps and the generator	None appears in the records packs the research reached	D1 §2.1, §2.3

Source: Aircraft Commerce, CFM56-7B Owner's & Operator's Guide, Issue 58, 2008; CFM56 training material (four-module description), Scribd, undated; AJW Group mini-pack for ESN (engine serial number) 889979, 2022; StandardAero mini-pack for ESN 892820, Mar 2025. Compiled in D1 §2.1, §2.3 and §2.4.

The -5B is described in CFM-derived training material as the same four modules: the fan and low-pressure compressor; the high-pressure compressor, combustion chamber and high-pressure turbine; the low-pressure turbine and turbine rear frame; and the accessory gearbox. [CFM56 training material, Scribd, undated] The -7B has the same architecture. [Aircraft Commerce Issue 58, 2008]

3.2 HOW MANY MODULES AN ENGINE HAS, AND WHY THE SOURCES DISAGREE

The dossiers behind this primer describe the CFM56 as having three, four, four to five, or six modules. The disagreement is not about the hardware. It is about which level of the hierarchy is being counted and whether the gearbox is included.

EXHIBIT 1.4 — "HOW MANY MODULES?": EACH SOURCE'S LIST AND THE LEVEL IT COUNTS AT

Count	List as given	Where it comes from	What the count includes
4	Fan and booster; core (HPC + combustor + HPT); LPT; accessory gearbox	D1 (CFM-derived training material and Aircraft Commerce); D3 §2.7 uses the same four	Three gas-path major modules plus the gearbox
3 (+1)	Fan/booster, core and LPT "as the three major modules, with the accessory gearbox as a fourth"	D2 §2.1	The gas-path modules, gearbox set aside
3 per engine	"450 modules (150 engines)"; "1,800 modules (600 engines)"	FTAI release on QuickTurn Europe, 26 Feb 2025; stated as FTAI's convention in D8	FTAI's counting convention: fan, core, LPT
4 to 5	Fan and booster; HPC; combustor plus HPT ("the core"); LPT; plus the gearbox	D5 §2.1, §2.4; D9 §2.1 ("four to five major sub-assemblies")	Splits the HPC out of the core; gearbox optional
5	Fan; HPC; HPT; LPT; plus accessory gearbox	D7 §2.4	Counts compressor and turbine of the core separately
4	Fan; HPC; HPT; LPT	D10 §2.4	Same as D7 without the gearbox
6	Fan; booster; HPC; combustor; HPT; LPT	D11 §2.3	Every gas-path component counted separately, no gearbox

Source: dossiers D1, D2, D3, D5, D7, D8, D9, D10 and D11 as cited in each row; FTAI Aviation / GlobeNewswire release of 26 Feb 2025 on QuickTurn Europe (via orchestrator notes). Reconciliation R-035 and B-3; registry C-5.

Read down the fourth column and the pattern is plain. Every list contains the same hardware. The six-module list names each gas-path component; the four-module list groups three of them into the core; the three-module list leaves out the gearbox because it carries no life-limited parts and is not part of the gas path. FTAI's own releases convert modules to engines at three per engine, which matches the gas-path count: 450 modules for 150 engines and 1,800 modules for 600 engines. [FTAI release, 26 Feb 2025, via orchestrator notes; D8 §2.10; reconciliation B-3] This Part uses four major modules, the three gas-path modules plus the accessory gearbox, each built from sub-modules. Whenever a later Part converts a module count into engines it names the convention it is using, because 1,200 modules is 400 engines at three per engine and 300 at four. [reconciliation R-035; E.1 rule 7]

3.3 WHAT HAS TO BE TRUE FOR A MODULE TO MOVE

Three things must hold before a shop can take the fan module off engine A and bolt it onto engine B.

The first is mechanical and configuration compatibility. The two modules must be the same model and at compatible build standards.

BUILD STANDARD (CONFIGURATION)

The build standard is the set of service-bulletin states a module has been built to. The manufacturer issues service bulletins that change hardware over the years, and two modules at different standards may not mate, or may not be approved to run together. [D1 §2.3]

AIRWORTHINESS DIRECTIVE (AD) AND SERVICE BULLETIN (SB)

A Service Bulletin is a manufacturer's instruction to modify or inspect; an Airworthiness Directive is a regulator's order making a fix or inspection mandatory. Compliance records for both must travel with life-limited and hard-time parts. Part V §31 carries the full treatment. [D3 §2.1; D1 §2.3]

CFM's TRUEngine programme exists to certify exactly this kind of conformity, on the stated ground that "the engine's configuration, material content, maintenance history, and supportability impact overall value as it changes ownership". [CFM International, TRUEngine launch release]

TRUEENGINE

TRUEngine is CFM's designation for an engine whose configuration and material content conform to CFM's own definition: OEM parts and OEM-approved repairs. CFM states that the designation "is being embraced by industry's leading asset valuation providers", so appraisers can treat a non-conforming configuration as a discount (§8). A lease clause requiring TRUEngine status is an OEM-only clause by another name (Part VI §42). [CFM International, TRUEngine launch release; D1 §2.3, §2.9]

The second condition is serialisation and records. Each module, and each life-limited part inside it, carries a serial number and a record of the cycles it has consumed. A module can only move if the receiving engine's records can absorb it, with every life-limited part's remaining life known and documented. This is the reason records travel with the module rather than with the engine. An engine is an assembly whose parts have different histories; the history belongs to the part. Section 9 sets out the documents. The practical consequence is that a module with a gap in its records is physically present but commercially absent: it cannot be installed, so it does not count as supply. [D1 §2.3, §2.9, §5]

The third condition is common part numbers. The -5B and -7B share core hardware to the point that used-parts dealers advertise a single "CFM56-5B & -7B Core LLP Package". [Aviation Fleet

Support, undated listing] That commonality is what lets a shop building -7B cores draw on -5B material and vice versa. The exact list of parts common to both variants was not retrieved. [D1 U2]

FEEDSTOCK

Feedstock is the trade's word for the aircraft and engines bought, or taken on consignment, to be torn down for parts. Later Parts extend it to the unserviceable module a customer returns in an exchange and to engine inventory held for module production, and say so when they do. Part V §33 carries the full treatment. [D3 §2.2; registry C-14]

3.4 MODULE EXCHANGE AGAINST FULL DISASSEMBLY

A module exchange separates the engine at the major-module flanges, removes one module and installs a serviceable one. A full disassembly strips every module to piece parts, inspects and repairs or replaces each one, and rebuilds the engine. The difference is time and exposure. An exchange takes days to a few weeks and touches only the module exchanged. A full disassembly takes months and opens every module to inspection findings that must then be repaired. [D1 §2.3]

MODULE EXCHANGE

A module exchange separates the engine at the major-module flanges, removes a time-expired or damaged module and installs a serviceable one from a pool, so the aircraft returns to service in days to weeks instead of months. The incoming unserviceable module becomes the provider's feedstock. The industry prices an exchange as a fee plus the difference in life-limited-part life between the module handed over and the module received, adjusted for the condition of the other hardware. Part VII §46 and §47 carry the full treatment; GE sells the same thing under its TrueChoice Transitions offering (§9). [D1 §2.3; D2 §2.1, §2.5; D5 §2.3]

StandardAero, an independent engine shop, lists "engine module changes" among its quick-turn shop-visit services, beside borescope inspection, blending out small blade damage, removal and installation of accessories, and fan, top-case, bottom-case, hot-section and low-pressure-turbine repairs. [StandardAero CFM56-7B brochure, Oct 2022] Whether a given module change requires a run in a test cell depends on the shop manual's procedure for that module, and the research did not verify it. [D1 U14]

FTAI's business is built on this distinction. Its annual report describes a "Module Factory", "a dedicated commercial maintenance program designed to focus on modular and parts repair and refurbishment of CFM56-7B and CFM56-5B engines", and says the company "targets assets which require maintenance repairs that can be performed through [its] proprietary Module Factory process". [FTAI 10-K FY2025]

MODULE FACTORY

The Module Factory is FTAI's name for the CFM56-7B and -5B modular repair and refurbishment programme quoted above from its FY2025 annual report, based at its Montréal shop. Functionally it is a production-line approach that builds and holds an inventory of serviceable modules for exchange. Part VII §46 carries the full treatment. [FTAI 10-K FY2025; D5 §2.2]

WHY THE MODULE IS THE UNIT

Four facts make the module, not the engine, the natural unit of this market. The engine is designed to split at the module flanges. Each module carries its own serial number and records, so it can be valued and traded on its own. The life-limit clocks inside the modules were deliberately set at different lengths (30,000 cycles in the fan, 20,000 in the core, 25,000 in the turbine, §4), so a fan and a core from the same engine are never at the same point in their lives. And a shop can install a serviceable module without stripping the rest of the engine. Put together, these mean that an engine at a shop visit is really three or four assets with three or four different remaining lives, and that the quickest way to return it to service is often to replace the one that has run out rather than to restore all of them. [D1 §2.3, §2.4, §2.8]

4. Life-limited parts: limits, prices, escalation, stub life

Most parts in an engine are replaced when inspection finds them worn. A small number are replaced on a clock, whatever their condition, because the consequence of their failing is a disk fragment the engine casing cannot contain. These are the life-limited parts, and much of the economics of the CFM56 aftermarket turns on them.

LIFE-LIMITED PART (LLP)

A life-limited part is a rotating or pressure-bearing part (a disk, a spool, a shaft, certain seals) whose failure the engine case cannot be designed to contain. It is certified for a fixed number of flight cycles and must be removed and scrapped at that limit regardless of its condition. On the -7B the limits are 20,000 cycles in the high-pressure compressor and turbine, 25,000 in the low-pressure turbine and 30,000 in the fan and booster. Reaching a limit forces the engine open. [Aircraft Commerce Issue 34, 2004; AJW mini-pack, 2022; StandardAero mini-pack, Mar 2025]

DISK AND SPOOL

A disk is the heavy wheel to which a row of blades attaches. A spool is several disks machined as a single piece. Together with the shafts and certain seals, these are the life-limited parts. [D1 §2.4]

4.1 THE LIMITS, MODULE BY MODULE

CFM's stated design policy is "target lives of 30,000 EFC in the fan/booster module, 20,000 EFC in the HPC and HPT modules and 25,000 EFC in the LPT module". [Aircraft Commerce Issue 34, 2004] Engine-specific documents confirm those limits in service on the -7B. The documents in question are records summaries that engine traders publish for engines they are selling.

MINI-PACK (RECORDS MINI-PACK)

A mini-pack is an engine records summary that lists each life-limited part with its certified limit and the cycles it has consumed. Traders publish them so that a buyer can see the engine's remaining life part by part before asking for the full records. [AJW Group mini-pack, 2022; StandardAero mini-pack, Mar 2025]

EXHIBIT 1.5 — CFM56-7B LIFE-LIMITED PARTS AND CERTIFIED LIMITS, BY MODULE

Life-limited part	Module	Certified limit (EFC)	Source
Fan disk	Fan and booster	30,000	AJW mini-pack ESN 889979 (-7B26), 2022; StandardAero mini-pack ESN 892820, Mar 2025
Booster spool	Fan and booster	30,000	Same
HPC stage 1–2 spool	Core (HPC)	20,000	Same
HPC stage 3 disk	Core (HPC)	20,000	Same
HPC stage 4–9 spool	Core (HPC)	20,000	Same
HPT disk	Core (HPT)	20,000	Same
LPT stage 1 disk	LPT	25,000	Same
Shafts, remaining LPT disks, seals	Various	Listed in the packs; individual limits not in the returned text	Same
CFM56-5B, whole engine	All	"18 different sets of LLPs with life limits varying between 20,000 and 30,000 flight cycles"; per-part table not reached (D1 U3)	Leeham News, 8 Mar 2024

Source: AJW Group, ESN 889979 CFM56-7B26 mini-pack, 2022; StandardAero, ESN 892820 mini-pack, Mar 2025; LeeHam News, Bjorn's Corner Part 49, 8 Mar 2024; CFM design policy per Aircraft Commerce Issue 34, 2004. Compiled in D1 §2.4.

Why does a limit force the engine open? Because the life-limited parts are the innermost parts. The high-pressure turbine disk sits behind the combustor with the turbine blades mounted on it; the compressor spools are buried in the core. To change the turbine disk the core must come out of the engine, the turbine module must be split from the compressor and combustor, the blades must be pulled, and the disk replaced; the rotor is then rebalanced and rebuilt. There is no procedure for doing any of this on the wing. [D1 §2.4] So when the lowest remaining life among all installed life-limited parts reaches zero, the engine comes off and goes to a shop whether or not it is still performing well. In practice the operator removes it earlier, at a planned shop visit, and replaces the parts that would otherwise run out before the next planned visit.

WORKED EXAMPLE 1.1 — THE LIFE-LIMIT CLOCK SETS THE REMOVAL DATE

Inputs: a CFM56-7B26 whose high-pressure turbine disk has consumed 18,500 cycles against a 20,000-cycle limit (limit sourced: mini-packs; cycles consumed assumed for illustration). The aircraft flies 1,500 cycles a year (sourced as the upper end of Aircraft Value News's 1,000–1,500 cycles a year for a mature narrowbody). The engine's temperature margin (§5) would otherwise allow another 5,000 cycles (assumed).

$$\begin{aligned}\text{cycles remaining} &= 20,000 - 18,500 = 1,500 \\ \text{time to limit} &= 1,500 \div 1,500 \text{ per year} = \text{about one year}\end{aligned}$$

So the disk sets the removal date at about one year, and the 5,000 cycles of temperature margin the engine still has are worth nothing to this owner: the engine must come off when the disk runs out. Limits of the example: the cycles consumed and the margin are assumed; the point is the mechanism, which is that the shorter of the two clocks always governs. [D1 §2.4, example A]

4.2 STUB LIFE, AND WHY IT CREATES A MARKET FOR USED PARTS

An engine rarely arrives at a shop with all of its life-limited parts at exactly zero. It arrives with some parts near their limits and others with thousands of cycles left. The shortest of those remaining lives has a name.

STUB LIFE

Stub life is "the shortest life remaining of all LLPs installed in an engine". In its pejorative sense it is the "several thousand EFCs of remaining LLP life that will not be used and the parts will be scrapped" when life-limited parts are replaced early so that they do not cut short the next run. Scrapping unused cycles raises the amortised cost per cycle of the part. [Aircraft Commerce Issue 34, 2004]

The mechanism, in Aircraft Commerce's words: "shop visit input may occur at a time when LLPs still have a few thousand EFCs remaining, and LLP replacement at this stage will be necessary if remaining lives are not to limit the subsequent removal interval"; and "if scrapped with excessive stub life the amortised cost per EFC and EFH is increased". [Aircraft Commerce Issue 34, 2004] The remaining cycles on an engine's life-limited parts "influence mid-life lease pricing, redelivery conditions, and asset value protection for lessors". [AviTrader, 23 Apr 2026]

MID-LIFE NARROWBODY

A mid-life narrowbody is a 737NG or A320ceo roughly 10 to 18 years old, or more loosely one in the middle of a 25-to-30-year economic life, when an increasing share of the aircraft's value sits in its engines and those engines are approaching or past a shop visit. Part II §15 carries the full treatment. [D6 §2.4; D7 §2.4; registry C-13]

CYCLES REMAINING (CR) AND CYCLES SINCE NEW (CSN)

Cycles remaining is a life-limited part's certified limit minus the cycles it has consumed; used-part packages are advertised on it ("6235 CR"). Cycles since new is the number of cycles an engine or part has flown since manufacture. Part IV §30 uses the second term in a table. [D1 §2.4; D2 §4.6]

USED LLPS (PART-LIFE LLPS; LLP PACKAGE)

Used LLPs are life-limited parts removed from another engine with full records and remaining life, sold as packages with a stated number of cycles remaining and chosen to match the buyer's planned next run. Dealers advertise them as such: "CFM56-7B27E/B1F Engine LLP Package ... 7,445 Cycles Remaining" and "CFM56-5B & -7B Core LLP Package 6235 CR". Prices of these packages were not shown in the listings the research reached. [Salvex listing, undated; Aviation Fleet Support listing, undated; D1 U4]

WORKED EXAMPLE 1.2 — STUB LIFE AND AMORTISED COST PER CYCLE

Inputs: an engine arrives for a performance restoration (§6) at 12,000 cycles since new (assumed). Its core life-limited parts have a 20,000-cycle limit (sourced: mini-packs), so 8,000 cycles remain. The operator plans a next run of 12,000 cycles (assumed; inside Leeham's 10,000–15,000-cycle mature interval, §6). A part lists at price P.

Option one, keep the parts. The engine comes off again after 8,000 cycles, forced by the life limit, with its restored performance only two-thirds used.

Option two, replace them now with new parts at list. Then 8,000 of each part's 20,000 cycles (40%) are scrapped unused.

$$\begin{aligned}\text{cost per cycle, full life used} &= P \div 20,000 \\ \text{cost per cycle, scrapped at 12,000} &= P \div 12,000 \\ \text{increase} &= (20,000 \div 12,000) - 1 = 67\%\end{aligned}$$

Option three, which is the option the used-parts market exists to supply: install used life-limited parts with 12,000 to 15,000 cycles remaining, matched to the planned run, from a package of the kind the dealers advertise above. The arithmetic for how such a part is priced is in §4.4 and Part V §34. Limits of the example: the cycles consumed and the planned run are assumed; the limit and the listings are sourced; the package price is unknown (D1 U4). [D1 §2.4, example B]

The stub problem also explains why the limits are what they are. Engines "whose inherent design includes LLPs with high stub-lives will tend to experience lower rates of engine removals compared to engines with lower LLP stub-lives". [Aircraft Commerce Issue 34, 2004] CFM's choice of 20,000 cycles for the core and 30,000 for the fan means the fan's parts span a core overhaul cycle and a half. At the first core replacement, near 20,000 cycles, the fan disk still has about 10,000 cycles left. That mismatch is one of the physical reasons a fan module and a core module from the same engine have different values at the same moment, which §8 takes up. [D1 §2.4]

4.3 WHAT A FULL SET COSTS, YEAR BY YEAR

New life-limited parts come from one source, at a published price that is revised upward on a schedule.

CATALOGUE LIST PRICE (CLP), OR LIST PRICE

The catalogue list price is the manufacturer's published price for every new spare part, repriced at least annually. A shop passes it through at list plus a handling margin, and used and alternative parts are priced as discounts to it. Part IV §28 carries the full treatment. [D2 §2.2–2.3; D3 §2.1]

ESCALATION (CATALOGUE ESCALATION; LLP ESCALATION)

Escalation is the manufacturer's annual increase in catalogue list prices. Trade press put new spare-parts price increases at around 10% a year in 2022–23 (a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023), and Safran's chief executive described a rule of pricing "above inflation by 3 to 4 points". On the figures in Exhibit 1.6, a full -7B life-limited-part set compounded at about 7% a year from 2008 to 2025. Because 60–70% of a shop visit is OEM-priced material, escalation flows into every visit whoever performs it. Part IV §28 carries the full treatment. [Aviation Week, late 2023; Air Cargo Week, 2025/26; D1 example C]

EXHIBIT 1.6 — FULL LIFE-LIMITED-PART SET FOR A CFM56-7B AT LIST PRICE, BY YEAR, WITH SOURCES AND CAVEATS

Year	Figure	Scope as the source gives it	Source and level	Caveat
2008	\$1.775M	Full -7B set, "shipset" (the source's word for one engine's complete set)	Aircraft Commerce Issue 58 (trade press)	None
2013	\$1.7M	"A full set of LLPs for a CFM56-7B has a list price of \$1.7 million"	Aircraft Commerce Issue 87 (trade press)	Below the same publisher's 2008 figure; D3 T4 says the excerpt may cover a subset; not used for arithmetic
2017	"north of \$3m"	Whole-engine CFM56 LLP stack; HPT exchange "easily \$0.5m"; fan/booster "\$0.5m–\$0.7m"	Leeham News, 3 Mar 2017 (trade press)	Rounded; variant not stated
Oct/Nov 2018	\$3.4M	Full -7B set	Aircraft Commerce Issue 120 (trade press)	None
Nov 2019	"\$4m for CFM56-7"	Headline; article paywalled	Aircraft Value News (trade press)	Implies +18% in one year against the 2018 figure (D1 T2); Part IV uses this figure in its option arithmetic
Aug 2023	-7B parts +20–30% in 2023	Parts generally, after CFM's catalogue revision; not LLPs specifically	Magnetic Group via Aviation Week, 2024 (trade press)	Not a set price
Jun 2025	\$5.7M	"LLPC \$m: 5.700" (LLPC is the table's label for LLP cost) in an engine-status table	MyAirTrade (market-data site)	Methodology, rating and exact scope not visible (D1 T2); this Part uses it as its current case
Apr 2023	\$9,878,733 full life; \$4,939,366 half life	CFM56-5B engine LLPs for an A320, 2023 dollars	Mark Calver, Cargo Facts (conference presentation)	Per engine or per aircraft (two engines) unstated (D3 U3)

Source: Aircraft Commerce Issues 58 (2008), 87 (2013) and 120 (Oct/Nov 2018); Leeham News, Bjorn's Corner, 3 Mar 2017; Aircraft Value News, Nov 2019; Aviation Week, "Magnetic expects CFM56 market challenges", 2024; MyAirTrade engine-status resource, June 2025; Mark Calver, Cargo Facts Session 3, Apr 2023. Reconciliation R-016; E.2 "LLP full set".

The sources disagree, and the primer carries the disagreement rather than resolving it. [reconciliation R-016] The 2013 figure of \$1.7M sits below the 2008 figure from the same publisher, which cannot both be full-set prices at list unless the scope differed; it is quoted here only with that caveat. The current figure is either about \$4M (Aircraft Value News, November 2019, as used by Part IV) or \$5.7M (MyAirTrade, June 2025, as used by this Part), and the two are never mixed in one calculation. Where this Part computes with \$5.7M it shows \$4M as the alternative case and says what changes. [E.1 rule 5] The Calver figure for the -5B is carried only with the statement that the presentation does not say whether it covers one engine or an aircraft's two.

WORKED EXAMPLE 1.3 — DERIVING THE ESCALATION RATE FROM THE PRICE SERIES

Inputs: \$1.775M in 2008 (sourced: Aircraft Commerce Issue 58); \$3.4M in late 2018 (sourced: Issue 120); \$5.7M in mid-2025 (sourced: MyAirTrade, scope not visible). Periods taken as ten years, seven years and seventeen years (assumed, from the publication dates).

2008 to 2018: $3.4 \div 1.775 = 1.916$; $1.916^{(1/10)} - 1 = 6.7\%$ a year
 2018 to 2025: $5.7 \div 3.4 = 1.676$; $1.676^{(1/7)} - 1 = 7.7\%$ a year
 2008 to 2025: $5.7 \div 1.775 = 3.21$; $3.21^{(1/17)} - 1 = 7.1\%$ a year

The 2019 point of \$4.0M would imply an 18% rise in one year, which is out of line with the trend and may reflect a rounded headline or a different scope (D1 T2). Ishka reports that one recent escalation step on -5B/-7B life-limited parts was about 12%, without showing the year, and that "engine OEMs calculate LLP escalation based on a formula which factors in labour costs, the price of raw materials and energy costs". Because a life-limited-part set is 40–60% of the parts bill in a heavy shop visit (Safe Fly Aviation, consultancy blog, 2026), a 7% annual escalation on the set alone lifts the cost of a heavy visit by roughly 3–4% a year before any labour or repair inflation; that last figure is derived, not sourced. Limits of the example: the dates are publication dates, the 2025 scope is not visible, and the years of each catalogue step between 2019 and 2025 were not retrieved (D1 U10; D2 U1). [D1 example C; Ishka, undated; Safe Fly Aviation, 2026]

Two readings of escalation sit side by side in the research. One is life-limited-part specific: about 7% a year compounded on the set prices above, a 12% single step reported by Ishka without a year, and a 20–30% rise in -7B parts prices generally after CFM's August 2023 catalogue revision. [D1 example C; Ishka, undated; Magnetic Group via Aviation Week, 2024] The other covers all spares: around 10% a year, a double-digit increase on 1 November 2022, a high-single-digit increase in August 2023, and Safran's stated rule of inflation plus three to four points "at least for some years". [Aviation Week, late 2023] They cover different parts, years and bases, and the primer reports both. [reconciliation R-018]

4.4 WHAT THE ESCALATION DOES TO THE COST OF A CYCLE

Dividing a set price by the cycles it buys gives the life-limited-part cost of each take-off and landing. The research produced two such figures from two set prices, and both are shown.

at \$5.7M: $5,700,000 \div 20,000 = \285 per cycle (D1)
 at \$4.0M: $4,000,000 \div 20,000 = \200 per cycle (D2)
 at \$5.7M, scrapped after 12,000 cycles: $5,700,000 \div 12,000 = \475 per cycle
 at \$4.0M, scrapped after 12,000 cycles: $4,000,000 \div 12,000 = \333 per cycle

The first two lines inherit the disagreement over the set price; the second two show the stub-life penalty of worked example 1.2 in dollars. [D1 example D; D2 §2.5; reconciliation R-017] At 1,500 cycles a year the \$200-per-cycle figure is \$300,000 per engine per year, which is D2's statement of it. [D2 §2.5]

This arithmetic is what creates a market for used life-limited parts and for engine cores whose remaining lives match a buyer's planned run. A part with 60% of its life left is worth roughly 60% of its list price, less a discount for being used; that is how an engine shop prices a used part, and how an appraiser adjusts an engine's value for its life-limited parts (§8). [D3 §2.4] Used material in general trades at 60–80% of a new part's price on average, or at 70–75% of catalogue list price when a specialist trader sells repaired engine material to an airline; Part V §34 sets out those figures and their tension. [Aviation Week, undated; Aircraft Commerce Issue 120; reconciliation R-022] No alternative manufacturer makes a life-limited part for the CFM56: the research found approvals for turbine blades and vanes but none for a disk or spool, so new life-limited parts remain a single-source purchase at the escalating list price (Part VI §38). [D4 U7; D1 §5]

5. Exhaust-gas temperature margin and time on wing

The life-limit clock of §4 is one of two clocks that take an engine off the wing. The other is a thermometer. An engine that has worn cannot make its rated thrust without running hotter, and there is a temperature it may not exceed. The gap between the two is the margin, and the margin is spent a little on every flight.

EXHAUST-GAS TEMPERATURE (EGT)

Exhaust-gas temperature is the temperature of the gas leaving the turbine, measured by probes behind the low-pressure turbine. Every engine has a certified EGT red line that it may not exceed.

[D1 §2.5]

EGT MARGIN (EGTM)

EGT margin is "the difference between the maximum temperature limit on the engine and the actual temperature read at the time of take-off", measured at full take-off thrust on a hot day so that engines can be compared. It shrinks as the engine wears, fastest in the first 1,000 to 2,000 cycles after a shop visit. At zero margin the engine can no longer make rated thrust within its limit and must come off the wing. A performance restoration (§6) recovers most of it. [Jurnal Teknik Mesin, June 2023; Aircraft Commerce Issue 27, 2003]

HOT SECTION

The hot section is the combustor, the high-pressure turbine blades and vanes (the vanes are also called nozzles; the stage-1 vane is the HPT nozzle), and the adjacent parts that combustion gas touches directly. It is where margin is lost and where most of it is recovered. [D1 §2.5; D4 §8]

5.1 WHY MARGIN ERODES AND HOW IT IS RESTORED

As the engine wears, its compressor and turbine move less air for each unit of fuel. The control system burns more fuel to make the same thrust, and the gas runs hotter, so the margin shrinks. The wear is concentrated at the blade tips and the seals they run against. "Rates of deterioration [are] highest in the first 1,000 EFC of operation as blade tips and seals are worn and clearances increase. This leads to leakage around blade tips and causes EGT margin to erode. EGT margin deterioration rates slow down after the first 1,000–2,000 EFC on-wing." [Aircraft Commerce Issue 27, 2003] A measured case on an older CFM56-3C1 found 26.2°C of margin lost since the last repair, of which the gap between the rotor blade tips and the casing accounted for 58.1%. [Jurnal Teknik Mesin, June 2023] Sand and salt in the operating environment erode blades faster, and a higher thrust rating starts with less margin to begin with. [D1 §2.5]

Restoration reverses the same wear. "Shop visit maintenance will restore hardware and clearances between blade tips and engine casings, and most of the original EGT margin will therefore be restored." [Aircraft Commerce Issue 27, 2003] In practice that means new or repaired high-pressure turbine blades and vanes, new honeycomb seals (the soft rub-in rings that blade tips run against), new shroud segments (the ring pieces that surround the turbine blade tips) and restored compressor blade tips. The high-pressure turbine is where most of the margin comes back, which is why its blades and vanes are the second-largest cost line in a shop visit at 15–25% of parts cost, and why an approved alternative to the manufacturer's turbine blade matters as much as it does. [Safe Fly Aviation, 2026; D1 §2.5]

PARTS MANUFACTURER APPROVAL (PMA)

A Parts Manufacturer Approval is the US Federal Aviation Administration's (FAA) combined design-and-production approval that lets a company other than the type-certificate holder legally make and sell a replacement part for a certificated engine. Historically PMA covered consumables and simple parts; an approved alternative to the CFM56 high-pressure turbine blade, announced on 30 October 2025, is treated in the trade as a milestone because that blade sits where most of a shop visit's margin and cost are. No PMA exists for any CFM56 life-limited part. Part VI §37 carries the full treatment.

[D4 §2.1–2.3; Chromalloy release, 30 Oct 2025]

The research did not retrieve the as-new margin for each -7B and -5B thrust rating, or the typical loss per thousand cycles, so this Part quotes no figures in degrees beyond the single measured case above. [D1 U5]

5.2 WHICH CLOCK SETS THE REMOVAL

Time on wing is the shorter of the two clocks.

TIME ON WING

Time on wing is the number of cycles or hours between an engine's installation on an aircraft and its removal. It is the smaller of the life-limit clock (§4) and the EGT-margin clock (§5), or, less often, a hard inspection falling due. Shortfalls in time on wing for the new-generation engines are Part III's subject (§10 gives the figures). [D1 §2.5; D9 §2.7]

The sources disagree about which clock usually governs, and the disagreement is partly about era and variant. Aircraft Commerce's maintenance analyses of 2003 and 2008 state that "EGT margin erosion is generally the most influential factor in maintenance removal intervals for short-haul engines, while the effect of hardware deterioration is greater on engines used on medium-haul operations". [Aircraft Commerce Issue 27, 2003, and Issue 58, 2008] The same publisher's later analysis of the -5B finds that for the latest variants at most ratings "the exhaust gas temperature (EGT) margin is high enough ... for it to remain on-wing up to the life-limited part (LLP) engine flight cycle (EFC) life limits", so that "it is possible for certain engines to only come due their third planned removal and shop visit after more than 25 years of operation". [Aircraft Commerce Issue 50] The shops report the same thing: "shop visits for the latest CFM56-5B/7B variants tended to be driven by life-limited parts (LLPs) reaching their ceiling rather than a need to restore engine performance", and "especially first-run engines have flown almost to LLP limit". [FlightGlobal, undated] The primer carries both statements; they describe different variants, ratings and eras, and the reconciliation records them as an unresolved tension. [reconciliation R-011, B-7] Part III's translation table reaches the same place from the fleet side: a fleet kept flying past its planned retirement date is a fleet that hits its life limits next.

WORKED EXAMPLE 1.4 — TURNING CYCLES ON WING INTO YEARS ON WING

Inputs: a mature CFM56's performance-restoration interval of 10,000 to 15,000 cycles and a life-limit-driven overhaul at 20,000 to 25,000 cycles (sourced: Leeham News, 8 Mar 2024); utilisation of 1,000 to 1,500 cycles a year for a mature short-haul narrowbody (sourced: Aircraft Value News); a core life limit of 20,000 cycles (sourced: mini-packs).

restoration interval in years = $10,000 \div 1,500$ to $15,000 \div 1,000$ = 6.7 to 15 years
 first core life limit in years = $20,000 \div 1,500$ to $20,000 \div 1,000$ = 13.3 to 20 years

So a mature engine flying 1,200 cycles a year reaches a restoration roughly every 8 to 12 years and its first core life limit at about 17 years. Two other readings exist. Aviation Week's rule of thumb is that a CFM56 needs a shop visit "roughly every eight years", which makes 24 years a de facto retirement age. And 2,300 shop visits a year on a base of about 24,000 engines is 9.6% of the base, which, if every engine were on a regular cycle, would imply an average interval of about ten years; the dossier that made that calculation records that the base figure may include spare and stored engines and does not resolve the gap. A third dossier assumed one restoration every six years and labelled it an assumption to be replaced. Limits of the example: utilisation varies by operator; first-run intervals run longer than mature ones (§6); the three readings are reported, not reconciled.

[Leeham News, 8 Mar 2024; Aircraft Value News, undated; Aviation Week via D3; D9 §2.2; D7 §5.2; reconciliation R-008]

6. Shop visits: types, workscopes, intervals, turnaround

When either clock runs out, or something breaks, the engine comes off the aircraft and goes to a shop. Everything that happens there has a name, and the names are not interchangeable.

SHOP VISIT (SV)

A shop visit is any removal of the engine from the aircraft for work in an engine shop, from a visit to fix one finding to a complete rebuild. Part VIII uses one dossier's phrase "teardown and restoration" for the same event; this primer reserves "teardown" for dismantling an engine to sell its parts (Part V §32). [D1 §2.6; D2 §2.1; registry C-8]

ENGINE SHOP

An engine shop is a facility holding a regulatory repair-station approval (§9) and, in practice, the manufacturer's technical data and tooling. It disassembles engines on stands, cleans and inspects every piece, replaces or repairs what is out of limits, reassembles the engine and runs it in a test cell before release. Part IV §25 carries the full treatment. [D2 §2.1]

MRO AND INDEPENDENT MRO

MRO is the industry's generic term for maintenance, repair and overhaul work and for the companies that do it. An independent MRO is an engine shop not owned by the engine's manufacturer: a stand-alone company such as StandardAero, an airline's maintenance subsidiary that sells to third parties, or an engine maker's shop that is independent on the CFM56, such as MTU (Germany's MTU Aero Engines) through its MTU Maintenance arm (§10). Part IV §25 carries the full treatment. [D2 §3.2]

WORKSCOPE

The workscope is the specification of what a shop visit will do: which modules are opened, how deep the work goes in each, and which parts are replaced new, repaired, or swapped for used material. It runs from a light quick-turn to a full overhaul. [D1 §2.6; D2 §2.1]

6.1 THE TYPES, FROM LIGHTEST TO HEAVIEST**HOSPITAL VISIT (QUICK-TURN VISIT)**

A hospital or quick-turn visit fixes one thing and returns the engine: a borescope finding, a damaged fan blade, a module exchange, an accessory change. Trade press reports that such visits "are being used as an alternative to avoid substantial financial and time investments into full-performance shop visits", at about 45 days' turnaround. The lower-case term is a workscope; the capitalised "QuickTurn" is the brand of FTAI's Miami and Rome businesses (Part VII §50). Part VII §46 carries the module exchange as the quick-turn. [Aviation Week, 2024; Aviation Business News, 2025; registry C-20]

BORESCOPE AND BOROBLEND

A borescope inspection looks inside the engine with a camera passed through ports in the casing, on wing or in a shop. A boroblend repair grinds out small blade damage found that way, through the same ports, without disassembly. [D1 §2.3, §2.6]

LINE-REPLACEABLE UNIT (LRU) AND QUICK-ENGINE-CHANGE KIT (QEC)

A line-replaceable unit is an accessory that can be changed on the flight line without opening the engine. The quick-engine-change kit is the set of mounts, plumbing and accessories that dresses a bare engine for installation on a particular aircraft. [D1 §2.3, §8]

TOP-CASE REPAIR

In a top-case repair the high-pressure compressor case is split horizontally and the upper half lifted off, giving access to the compressor blades and vanes for repair or replacement without removing the rotor. StandardAero lists top-case and bottom-case repairs as separate services. [D1 §2.6; StandardAero brochure, Oct 2022]

PERFORMANCE RESTORATION (PR) AND CORE PR

A performance restoration is a shop visit in which the core is opened and the high-pressure compressor, combustor and high-pressure turbine are restored to recover EGT margin; the fan and low-pressure turbine may be inspected but are not fully disassembled. A core PR is this scope confined to the core module. Replacing life-limited parts is a separate decision, which a life-limit-driven removal combines with the restoration; one dossier describes a PR as "typically with LLP replacement", and this primer keeps the two decisions distinct. [D1 §2.6; D2 §2.1; registry C-7]

OVERHAUL (HEAVY OR FULL SHOP VISIT; "ZERO-TIME")

An overhaul disassembles every module to piece parts, inspects, repairs and rebuilds, and replaces the life-limited parts that would fall due within the next run. It is done when the life limits are due, at roughly 20,000 to 25,000 cycles. "Zero-time" is one source's term for an overhaul with a complete new set of life-limited parts. [D1 §2.6; D2 §2.1; Leeham News, 8 Mar 2024]

EXHIBIT 1.7 — SHOP-VISIT TYPES: WHAT IS OPENED, WHAT IS DONE, AND TYPICAL TURNAROUND

Type	What is opened	What is done	Life-limited parts	Turnaround	Source
Hospital / quick-turn	Limited; one module or an access port	One finding fixed: borescope finding, fan blade, module exchange, accessory	None	~45 days (trade press, 2025)	D1 §2.6; Aviation Business News, 2025; StandardAero, 2022
Top-case repair	Upper half of the HPC case	HPC blades and vanes repaired or replaced, rotor in place	None	Not retrieved	D1 §2.6; StandardAero, 2022
Performance restoration	Core (HPC, combustor, HPT); fan and LPT inspected	Hot-section rebuild to recover EGT margin	Separate decision; combined when a limit is near	75–90 days (Safe Fly, consultancy blog, 2026)	D1 §2.6; Safe Fly, 2026
Heavy visit with LLPs	Core plus whichever modules hold parts at or near limit	PR plus replacement of the parts at or near limit	Those at or near limit	120–150 days (Safe Fly, 2026)	D2 §2.1; Safe Fly, 2026

Type	What is opened	What is done	Life-limited parts	Turnaround	Source
Full overhaul ("zero-time")	Every module, to piece parts	All modules to overhaul standard	Full new set	90–120 days for a full overhaul (trade press, 2025); ~60 pre-pandemic	D1 §2.6; D2 §2.1; Aviation Business News, 2025

Source: D1 §2.6 and D2 §2.1; StandardAero, CFM56-7B engine services brochure, Oct 2022; Aviation Business News, "Inside the engine MRO supply chain", 2025; Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026" and "CFM56-7B engine availability report 2026" (consultancy blog). The cost of each type is in Exhibit 1.9.

6.2 INTERVALS: FIRST RUN AND MATURE RUNS

FIRST RUN AND MATURE RUN

The first run is the interval from new to the first shop visit. Mature runs are the later intervals, which are shorter because repaired hardware does not match new hardware and because operators plan them around the life-limit clock. On the latest -5B and -7B variants the first run reaches the life limit before the margin runs out. [D1 §2.6; FlightGlobal, undated]

Leeham's description of the cadence is the cleanest: "a mature engine like the CFM56 has a performance shop visit at half the LLP flight cycle limit, and then a full engine overhaul is done when the LLPs are due for replacement". [Leeham News, 8 Mar 2024] Intervals by thrust rating and by region were not retrieved. [D1 U6]

6.3 TURNAROUND, AND THE WAIT BEFORE IT

TURNAROUND TIME (TAT)

Turnaround time is the calendar time from an engine's induction into a shop to its release. For a full CFM56 overhaul it was about 60 days before the pandemic and 90 to 120 days in 2025; light work takes about 45 days. Every extra day needs a spare engine or grounds an aircraft, which is the economic hook for a module exchange. Part IV §27 carries the full treatment, including the wait for an induction slot, which is additional to the turnaround. [Aviation Business News, 2025; D2 §2.1]

EXHIBIT 1.8 — INTERVALS AND TURNAROUND: THE FIGURES AND THEIR SOURCES

Metric	Value	Source and level	Date
Mature CFM56, performance-restoration interval	10,000–15,000 cycles	Leeham News (trade press)	8 Mar 2024

Metric	Value	Source and level	Date
Mature CFM56, overhaul with LLP replacement	20,000–25,000 cycles	Leeham News	8 Mar 2024
First-run -5B/-7B, latest variants	"almost to LLP limit"	FlightGlobal (trade press)	Undated
-5B third planned removal	Possible after more than 25 years	Aircraft Commerce Issue 50 (trade press)	Undated
Rule of thumb	A shop visit "roughly every eight years"; 24 years a de facto retirement age	Aviation Week (trade press), via D3	Undated
Implied average interval	~10 years (2,300 visits a year on ~24,000 engines, 9.6%)	D9 derived; base may include spares and stored engines	2026
Reserve-model assumption	One restoration every six years, labelled an assumption	D7 §5.2	2026
Full overhaul TAT, pre-pandemic	~60 days	Aviation Business News (trade press)	2025
Full overhaul TAT, current	90–120 days	Aviation Business News	2025
Light work TAT	~45 days	Aviation Business News	2025
CFM56 TAT in GE shops	"sustaining ~90 days"	GE Aerospace, Bernstein conference deck (OEM)	27 May 2026
TAT for LEAP, CFM56 and GE90 (a widebody engine)	Improved by more than 10% year on year in 4Q 2025	GE 10-K FY2025 (OEM filing)	Feb 2026
PR visit TAT	75–90 days	Safe Fly Aviation (consultancy blog)	2026
Heavy visit with LLPs TAT	120–150 days	Safe Fly Aviation	2026
Wait for an induction slot	Up by two to three months, up to six	Bain (consultancy)	2024

Source: Leeham News, 8 Mar 2024; FlightGlobal, undated; Aircraft Commerce Issue 50; Aviation Week via D3 §2.6; D9 §2.2; D7 §5.2; Aviation Business News, 2025; GE Aerospace, 27 May 2026; GE 10-K FY2025; Safe Fly Aviation, 2026; Bain, 2024. Reconciliation R-008 and R-019.

The turnaround figures come from different shops and worksopes and the primer records all of them without choosing. [reconciliation R-019] The 2025 trade press attributes the lengthening since the pandemic to "long turnaround times for material repairs ... primarily due to parts scarcity issues in the MRO supply chain". [Aviation Business News, 2025] Bain adds that the wait for a slot,

before the turnaround even begins, had grown by two to three months and in some cases six. [Bain, 2024]

6.4 WHAT THE DOWNTIME COSTS: SPARE ENGINES

While an engine is in a shop, the aircraft needs another engine or it does not fly. Airlines hold or lease spares for this.

SPARE ENGINE AND SPARE RATIO

A spare engine is one held or leased to replace an installed engine while it is in the shop. The spare ratio is spares as a percentage of installed engines: airlines "used to have about 15% of spare engines of their installed engines, but this figure is now down to 10% and is expected to further decrease to about 7–8% for newer engine types" (MTU). Longer turnarounds tie up more spares. Part II §14 carries the full treatment. [MTU AEROREPORT, undated; D6 §2.3]

ENGINE LEASE, SHORT-TERM AND LONG-TERM

A short-term engine lease runs from a few months to about three years and covers a shop visit or an aircraft grounded for want of an engine. A long-term engine lease runs for years and provides a permanent spare. Part II §14 carries the full treatment. [D6 §2.1, §2.3]

WORKED EXAMPLE 1.5 — THE RENT PAID WHILE AN ENGINE IS AWAY

Inputs: turnaround of 90 days (sourced: GE Aerospace, May 2026, "sustaining ~90 days") or 120 days (sourced: trade press upper figure, 2025), taken as three or four months (assumed rounding). A CFM56-7B spare engine rent of about \$100,000 a month in 2024 (sourced: IBA, an appraiser, April 2024, up from about \$75,000 in 2019) or \$42,000 to \$48,000 a month in 2026 (sourced: Safe Fly Aviation, consultancy blog; the figure sits beside green-time values and may describe a green-time engine, which the source does not say). A slot wait of two to three months (sourced: Bain, 2024).

at \$100,000 a month: 3 months = \$300,000; 4 months = \$400,000
 at \$42,000–48,000 a month: 3 months = \$126,000–144,000; 4 months =
 \$168,000–192,000
 slot wait of 2–3 months adds: \$200,000–300,000 (IBA rate) or \$84,000–
 144,000 (Safe Fly rate)

So the rent for the downtime of one restoration is somewhere between about \$126,000 and \$700,000 depending on which rent, which turnaround and whether a slot wait is included. The two rents are not averaged: they may price different engines (a full-life spare against a green-time engine) and come from different source levels, and Part II §14 carries them as a pair. A quick-turn of about 45 days, which is what a module exchange from inventory aims to achieve, would cut the three-to-four-month figures by roughly half to two-thirds at either rent. Limits of the example: the turnaround and rent inputs are each contested; an owned spare carries a capital cost rather than a rent; and the aircraft's

own lost flying is not counted. [GE Aerospace, 27 May 2026; Aviation Business News, 2025; IBA, Apr 2024; Safe Fly Aviation, 2026; Bain, 2024; reconciliation R-019, R-028]

TEST CELL (TEST-CELL RUN)

A test cell is an instrumented enclosure in which a rebuilt engine is run at power to confirm its thrust, EGT margin and vibration before release. Test-cell hours are a physical capacity limit for a shop: FTAI's releases quote capacity in engine tests a year alongside modules (Part VII §49). [D1 §2.7; D2 §2.1]

7. The cost of a shop visit: the build and a worked example

A shop visit's bill has five lines, and the mix among them decides most of what later Parts discuss.

COST BUILD (SHOP VISIT)

The cost of a shop visit is the sum of labour (hours times the shop's rate) to disassemble, inspect, assemble and test; new OEM parts that are scrapped and replaced (blades, vanes, seals, bearings); repairs, meaning work on parts that can be restored rather than replaced, done in-house or by specialist vendors under the manufacturer's manual or under separately approved repair schemes (Part VI §40); life-limited-part replacement, new or used; and the test-cell run; plus freight, consumables and the shop's margin. Material is 60–70% of the total. [D1 §2.7; Air Cargo Week, 2025/26]

7.1 THE RANGES, AND WHY THEY DISAGREE

Every cost figure the research reached is from trade press or a consultancy blog. No manufacturer or shop price list was read; one shop's fixed-price catalogue was located and not read. [D1 U7; reconciliation R-013] Safe Fly Aviation, a consultancy blog, publishes two sets of ranges on different pages, and the research carries both.

EXHIBIT 1.9 — CFM56 SHOP-VISIT COST RANGES, BY LABELLED SCOPE, WITH BOTH CONSULTANCY SETS AND THE OTHER FIGURES REACHED

Scope as labelled by the source	Range	Source and level	Date	Note
Light / quick-turn: limited opening, targeted repair, no LLPs	\$650k–900k	Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026" (consultancy blog)	2026	Carried by D2 only

Scope as labelled by the source	Range	Source and level	Date	Note
Performance restoration: hot-section rebuild; "may replace some LLPs if convenient"	\$1.2–1.6M	Safe Fly Aviation, same page ("2026 industry data")	2026	Part IV's option arithmetic uses this range
Performance restoration, excluding LLPs	\$1.8–2.5M	Safe Fly Aviation, "What determines aircraft engine overhaul costs?" and "CFM56-7B availability report 2026"	2026	This Part's worked example uses the midpoint, \$2.15M
Heavy shop visit with LLPs: PR plus replacement of parts at or near limit	\$2.1–2.8M	Safe Fly Aviation ("2026 industry data")	2026	None
Full heavy overhaul with complete LLP replacement	">\$3.5M"	Safe Fly Aviation (other page)	2026	None
Full overhaul ("zero-time"): all modules to overhaul standard, full LLP stack	\$3.5–4.2M	Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026"	2026	Carried by D2 only
CFM56-5 light performance restoration	~\$300k, "up to the double for a more thorough visit"	Leeham News (trade press)	17 Mar 2017	Different workscope and year; not comparable
V2500 heavy shop visit (contrast)	\$2–3M "depending on the LLP profile"	Aircraft Commerce Issue 70 (trade press)	2010	Fifteen years old; \$10
Performance restoration, reserve-model input	\$4.5M	D6 §2.2, labelled "ASSUMED (illustrative)"	2026	Sits above every sourced range
A shop visit	"costed in the single-digit millions"	D7 §2.4, description	2026	Not a figure

Source: Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026", "What determines aircraft engine overhaul costs?" and "CFM56-7B engine availability report 2026" (2026, consultancy blog); Leeham News, Bjorn's Corner Part 3, 17 Mar 2017; Aircraft Commerce Issue 70, 2010; D6 §2.2; D7 §2.4. Reconciliation R-013; E.1 rule 4.

Two things stand out in that table. The first is that the two consultancy sets disagree with each other on the same scope: a restoration is \$1.2–1.6M on one page and \$1.8–2.5M on another, and a heavy visit with life-limited parts is \$2.1–2.8M on one and more than \$3.5M on the other. [Safe Fly Aviation, 2026; D1 T3] The second is that none of the "with LLPs" totals is consistent with a full new set of life-limited parts at 2025 list. A full set is \$5.7M on the figure this Part uses (or about \$4M on Part IV's); a restoration at \$1.4M plus a full set at \$4M is already \$5.4M, above every quoted "with LLPs" range. [D2 §2.1; D1 T3] So the quoted totals must assume partial replacement, used parts, or older price levels. The primer does not pick a range. Part I's worked example uses the \$1.8–2.5M midpoint and Part IV's option arithmetic uses \$1.2–1.6M, and each names the other. [E.1 rule 4]

7.2 THE SHARES: MATERIAL, LABOUR, REPAIRS

EXHIBIT 1.10 — WHERE A SHOP VISIT'S COST SITS: TWO SHARE BREAKDOWNS AND THE TURBINE-BLADE TENSION

Item	Figure	Source and level	Date
Material share of a visit	60–70%	Air Cargo Week (trade press)	2025/26
Labour share of a visit	20–30%	Air Cargo Week	2025/26
Repairs and other shop activities	10–20%	Air Cargo Week	2025/26
LLPs as share of parts cost (CFM56)	40–60%	Safe Fly Aviation (consultancy blog)	2026
HPT blades and vanes as share of parts cost	15–25%	Safe Fly Aviation	2026
Labour share (CFM56)	15–20%	Safe Fly Aviation	2026
HPT blades and vanes in dollars, at the Safe Fly share applied to the visit total as an approximation	\$0.18–0.40M on a \$1.2–1.6M visit; \$0.27–0.63M on a \$1.8–2.5M visit	Derived from the two rows above	2026
New OEM HPT blade, price each	About \$20,000	In Practise expert interview (interview digest), via D4	Undated
New OEM HPT blade, alternative listing	About €35,000	Web page of weak provenance, condition unclear, via D4	Undated
HPT blades per set	80, an unverified recollection (D4 U16)	D4	Not dated

Item	Figure	Source and level	Date
Implied HPT blade set	\$1.6M at \$20,000; \$3.08M at about \$38,500	Derived in D4 §2.8	Not dated
HPT blade and vane set	"up to a few million dollars"	In Practise, via D4	Undated
Labour-rate inflation, 2025	5.5–6.0%	Oliver Wyman MRO survey (consultancy)	Apr 2026
Labour hours per workscope; shop rate per hour	Not found	D1 U7	Not dated

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26; Safe Fly Aviation, 2026 (consultancy blog); In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA", undated; web listing of weak provenance (D4 [72]); Oliver Wyman, "The new MRO supply paradigm", Apr 2026. Reconciliation R-014 and R-015.

The two share breakdowns come from different secondary sources and neither names a shop-level data set. [reconciliation R-014] One dossier reads them as consistent, since life-limited parts and turbine airfoils together are the bulk of material and material is the bulk of the visit. [D2 §2.2] The turbine-blade line is a direct contradiction at the level of one cost item. Priced per blade from an interview and a listing, a set of 80 blades is \$1.6M to \$3.1M, which is close to or above the whole restoration; priced as 15–25% of parts cost from the consultancy shares, the blades and vanes together are \$0.2M to \$0.6M. [D4 §2.8; Safe Fly Aviation, 2026; reconciliation R-015] The blade count is a recollection and the list price of a full blade-and-vane set was not retrieved, so the primer reports both sides and stops. [D4 U16]

7.3 A WORKED EXAMPLE, INPUT BY INPUT

WORKED EXAMPLE 1.6 — A PERFORMANCE RESTORATION WITH LIFE-LIMITED-PART REPLACEMENT, CFM56-7B26, SECOND SHOP VISIT, 2025 PRICES

Every input is labelled. The engine is at 12,000 cycles since new (assumed); its core life-limited parts have a 20,000-cycle limit (sourced: mini-packs) and so 8,000 cycles left; the planned next run is 12,000 cycles (assumed, inside LeeHam's 10,000–15,000-cycle range).

Step 1, the restoration workscope excluding life-limited parts. Take the midpoint of the \$1.8–2.5M range, \$2.15M (sourced: Safe Fly Aviation, consultancy blog, 2026; midpoint is this Part's choice under E.1 rule 4). The alternative range is \$1.2–1.6M, midpoint \$1.4M (sourced: Safe Fly Aviation, other page; used by Part IV). Within the \$2.15M: labour at 15–20% is \$320,000–430,000 (sourced share; hours and rate unknown, D1 U7); turbine blades and vanes at 15–25% are \$320,000–540,000 (sourced share); the remainder of \$1.2–1.5M is other new parts, repairs, test and consumables (derived).

Step 2, the life-limited parts. Three options. (a) A full new set at list, \$5.7M (sourced: MyAirTrade, June 2025, scope not visible). This would replace fan and booster parts with 18,000 cycles still on them, so a shop would not do it; it is the ceiling case. (b) New core parts only (compressor spools and disk, turbine disk, shafts). The core group's share of the set was not retrieved (D1 U8), so this line cannot be priced. (c) A used core package with about 12,000 cycles remaining, matched to the next run. Such packages are advertised; their prices were not shown (D1 U4).

case (a) at \$5.7M: $2,150,000 + 5,700,000 = \$7,850,000$
 case (a) at the \$4.0M alternative set price: $2,150,000 + 4,000,000 = \$6,150,000$
 case (a) at the 2018 list of \$3.4M: $2,150,000 + 3,400,000 = \$5,550,000$
 restoration alone at the alternative midpoint: $\$1,400,000$

Step 3, what the totals say about the quoted ranges. The "heavy shop visit with LLPs" figures of \$2.1–2.8M and more than \$3.5M (sourced: Safe Fly Aviation) are reachable only with partial replacement, used parts or older prices; a full new set at either current figure puts the visit above \$6M.

Step 4, cost per cycle over the next 12,000-cycle run.

restoration only: $2,150,000 \div 12,000 = \179 per cycle (alternative:
 $1,400,000 \div 12,000 = \117)
 full set over its 20,000-cycle life: $5,700,000 \div 20,000 = \285
 (alternative set: \$200)
 full set scrapped after 12,000 cycles: $5,700,000 \div 12,000 = \475
 (alternative set: \$333)

So the life-limited-part decision moves the engine's maintenance cost per cycle by more than the whole restoration workscope does, on either set price. That is the arithmetic that creates a market for used life-limited parts and for cores whose remaining lives match the buyer's planned run (Part V §34 and §36).

Step 5, time. The restoration takes 75–90 days (sourced: Safe Fly) or 90–120 days (sourced: trade press); against that, a core module from inventory is a quick-turn (sourced: StandardAero's menu). The owner's alternative cost is the spare-engine rent of worked example 1.5. Limits of the example: both consultancy ranges are contested; the set price has two readings; the core-only and used-package prices are unknown; labour hours and rates are unknown. [D1 example D; Safe Fly Aviation, 2026; MyAirTrade, June 2025; Aircraft Value News, Nov 2019; Aircraft Commerce Issue 120; reconciliation R-013, R-016, R-017]

Who collects each of those lines is the subject of Part IV §29, and §11 of this Part gives a first map. For now the structure is enough: 60–70% of the bill is material, most material is priced off the manufacturer's catalogue, and the catalogue is repriced upward every year, so escalation flows into every shop visit regardless of who performs it. [Air Cargo Week, 2025/26; D2 §2.2] The

levers a shop other than the manufacturer has are all on the material line: used parts instead of new, approved alternative parts instead of the manufacturer's, repair instead of replacement, and exchange instead of overhaul. Parts V, VI and VII take those four levers in turn.

8. What an engine is worth: half-life, full-life, maintenance-adjusted value

An engine's value depends on how much life is left in it, and the trade has a standard way of saying so. Independent valuers publish figures for a reference condition and then adjust them for the engine in front of them.

APPRAISER

An appraiser is an independent valuer that publishes base and market values and lease rates for aircraft and engines, at half-life unless stated; the four this primer cites are IBA, mba Aviation, Ascend by Cirium and AVITAS (a US appraisal firm). The market trades on their figures, and securitisation rating agencies test deals against them. Part II §12 carries the full treatment. [D1 §2.8; D3 §2.4]

ISTAT

ISTAT, the International Society of Transport Aircraft Trading, is the trade body whose definitions standardise appraisal terms such as half-life and full-life. [D1 §2.8]

8.1 THE THREE VALUE CONCEPTS

HALF-LIFE (BASE) VALUE

Half-life is an appraisal convention, not a physical midpoint. In ISTAT's words it is "a standard appraisal industry term to indicate that no value adjustment has been made for the actual maintenance status ... the assumption being that the airframe, engines (modules & LLPs), landing gear, and other major maintenance events are in half-life status. It does not indicate that the aircraft is half-way through its useful life." For an engine it assumes every module halfway between restorations and every life-limited part at half its certified life. One source defines it as every scheduled event at its mid-point and every life-limited part at the mid-point of its ultimate life; another as each component midway between overhauls with half its cycles remaining. Two caveats recur: full life is "theoretical, even on a new aircraft", and the concept becomes "fluid" as aircraft pass mid-life because the market stops paying for maintenance status it will never use. [ISTAT Jetrader, Jul/Aug 2010; Calver, Apr 2023; D6 §2.1; Aircraft Value News, undated; registry C-1]

FULL-LIFE VALUE

Full-life value is the value with every maintenance item fresh and every life-limited part at zero cycles: "full-life engines are brand new or in recently restored condition". It equals the half-life value plus half the cost of a full restoration and a full set of life-limited parts. It is theoretical even for a new engine, which has already consumed test cycles. [VREF (an aircraft valuation service), undated; Calver, Apr 2023; D1 §2.8]

MAINTENANCE-ADJUSTED VALUE

Maintenance-adjusted value is "the fair market value of an Aircraft or Engine based on half-life values as adjusted by the relevant Aircraft Appraiser for the actual maintenance conditions and specifications". For each item the adjustment is (actual remaining fraction – 0.5) multiplied by the cost to restore that item, summed module by module. On this Part's inputs the swing between a fresh and a run-out -7B of the same age is about \$2.8M (worked example 1.7). [Law Insider, "Maintenance Adjusted CMV (current market value)"; D1 §2.8]

BASE VALUE AND MARKET VALUE

Base value is an appraiser's opinion of an asset's underlying economic value in an open, unrestricted, stable market with balanced supply and demand. Market value is the appraiser's opinion of the most likely trading price in the actual current market between willing, informed parties; it sits above base value in a tight market and below it in a soft one. Both are quoted at half-life unless stated. The research reached market values for the -7B (below) and no base values (D1 U9). Part II §12 carries the full treatment. [D3 §2.4]

Because the modules are separable (§3) and carry clocks that deliberately do not align (§4), the adjustment is naturally computed module by module. Engine value equals the reference value plus the sum over modules of each module's restoration and life-limited-part adjustments. [D1 §2.8]

SUM OF MODULES (SUM OF PARTS)

Sum of modules is the engine's value computed module by module, or as parts. It can exceed the whole-engine value when the modules' remaining lives are mismatched: a fan with 18,000 cycles of life attached to a core with 1,500 is worth more as a fan module sold to someone who needs a fan plus a core sold for its stub life than as one engine that must go into a shop. [D1 §2.8]

MODULE AS A UNIT OF TRADE

A serviceable module from a dismantled engine is valued like a whole engine in miniature: its serviceable parts at a discount to list, plus the remaining life of the life-limited parts inside it, less the repair needed to make it serviceable. It is tradeable because the design lets a shop fit it without disassembling the rest, and because its records travel with it. Part V §36 carries the full treatment. [D3 §2.7]

PART-OUT AND TEARDOWN

Part-out is the owner's decision to dismantle a retired aircraft or engine and sell its components individually; teardown is the physical disassembly. A run-out CFM56 bought for \$0.8–1.2M is reported to yield \$1.6–2.2M of used material, net \$0.4–0.8M before repair cost (consultancy blog, 2026). Part V §32 carries the full treatment. [D3 §2.2; Safe Fly Aviation, 2026]

8.2 THE PUBLISHED VALUES, AND THE BASES THEY SIT ON**GREEN TIME**

Green time is the remaining usable life on an engine or module before its next mandatory shop visit, set by whichever runs out first: the life-limited part with the fewest cycles remaining, the EGT margin, or a hard inspection or directive. A green-time engine is one the owner will not restore, flown or leased "as-is" to use up that life and then torn down. Part II §16 carries the full treatment. [D3 §2.3; D2 §2.3; registry C-2]

EXHIBIT 1.11 — CFM56-7B VALUES AND LEASE RATES: EACH FIGURE WITH ITS BASIS, SOURCE LEVEL AND DATE

Quantity	Figure	Basis	Source and level	Date
CFM56-7B24 market value	\$5.7M	Half-life, H2 2025	IBA, Engine Values Release 2025B (appraiser)	Sept 2025
CFM56-7B27 market value	\$6.4M	Half-life, H2 2025	IBA, 2025B (appraiser)	Sept 2025
CFM56-7B value change	"Around 20% from 2023 to 2024", the largest gain among engines covered	Market value	IBA (appraiser)	Apr 2024
CFM56-5B/-7B value trend	"Have levelled off after approximately 18 months of increases"	Market value	IBA H1 2026 update (appraiser)	H1 2026

Quantity	Figure	Basis	Source and level	Date
CFM56-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", which the research notes is internally inconsistent with "green-time"	Safe Fly Aviation (consultancy blog)	2026
CFM56-5B "green-time value"	\$2.6–3.2M	Same label	Safe Fly Aviation	2026
Low-life engine as teardown feedstock	\$0.8–1.2M acquisition; \$1.6–2.2M of used serviceable material (USM) recovered; net \$0.4–0.8M	Run-out engine	Safe Fly Aviation	2026
CFM56 values	Up "as much as 50% over the past two years"	Not stated	ePlaneAI (trade press)	Date not captured
LLP life as share of a used engine's value	60–80%; two -7Bs at 10,000 cycles since new differ by \$1.6M on LLP life alone	Used engine	Safe Fly Aviation	2026
CFM56-7B spare-engine lease rate	~\$100,000 a month in 2024, from ~\$75,000 in 2019	Spare engine	IBA (appraiser)	Apr 2024
CFM56-7B lease rate	\$42,000–48,000 a month; "about 34% above 2019"; up 15–20% "in recent months"	Quoted beside green-time values; basis not stated	Safe Fly Aviation	2026
CFM56-5B lease rate	\$38,000–44,000 a month	Same	Safe Fly Aviation	2026
V2500-A5 lease rate	\$70,000–80,000 a month	Typical	IBA, 2025B	Sept 2025
CFM56-5B LLP value	\$9,878,733 full life; \$4,939,366 half life (2023 dollars)	Per engine or per aircraft unstated (D3 U3)	Calver, Cargo Facts (conference presentation)	Apr 2023

Source: IBA, "Engine Values Release September 2025 (2025B)"; IBA, "Engine and Lease Rate Update H1 2026"; IBA, "It's a lessors' market", Apr 2024 (and AviTrader, 25 Apr 2024); Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); ePlaneAI, "CFM56 engine values highlight investor opportunities and risks", date not captured; Mark Calver, Cargo Facts Session 3, Apr 2023. Reconciliation R-028 and R-029; E.1 rule 6.

The exhibit holds a conflict the primer does not resolve. On one side, the appraiser: IBA gives half-life market values of \$5.7M for the -7B24 and \$6.4M for the -7B27 for the second half of 2025, describes the -7B market as having "a distinct lack of availability" with values and lease rates above long-term trend, recorded a rise of about 20% from 2023 to 2024, and in its first-half 2026 update says -5B and -7B values "have levelled off after approximately 18 months of increases". [IBA, Sept 2025; IBA, Apr 2024; IBA, H1 2026] On the other, a consultancy blog gives a -7B "green-time value (new condition)" of \$2.8–3.4M and a teardown value of \$0.8–1.2M, and reports rents still rising by 15–20% "in recent months", while a trade-press item has values up by as much as 50% over two years. [Safe Fly Aviation, 2026; ePlaneAI, date not captured] The figures sit on different bases: a half-life market value is not a green-time value, and neither is a teardown value. The label "(new condition)" on a green-time figure is internally inconsistent, as the research noted. And the sources disagree on direction, levelled off against still rising. [D3 T3; reconciliation R-029] Both sides are carried; the primer takes no view on which describes the market better. The same is true of the two lease rates, \$100,000 a month from the appraiser in 2024 and \$42,000–48,000 from the blog in 2026, which may price different engines. [reconciliation R-028] One dossier divided the \$45,000 midpoint by the \$6.4M value to get 0.70% a month and noted that the calculation mixes a green-time rent with a half-life value; no lease rate factor is stated as a single number anywhere in this primer, and Part II §14 carries the derived figures with their bases. [D3 §2.3; E.1 rule 6]

8.3 A VALUATION EXAMPLE

The dossier that built the valuation arithmetic wrote it against a half-life base value, which the research never reached; what it reached later was IBA's half-life market value, a different quantity. The arithmetic is identical and the reference point differs, so the example below uses the market value and says so. [D1 U9; reconciliation C.1 item 2]

WORKED EXAMPLE 1.7 — MAINTENANCE-ADJUSTED VALUE OF A MID-LIFE CFM56-7B27

Inputs: half-life market value \$6.4M for a -7B27, H2 2025 (sourced: IBA, Sept 2025; used here as the half-life reference). Full set of life-limited parts \$5.7M (sourced: MyAirTrade, June 2025, scope not visible; alternative \$4.0M, Aircraft Value News, Nov 2019). Restoration cost \$2.15M (sourced midpoint of Safe Fly Aviation's \$1.8–2.5M; alternative \$1.4M from the other range). Restoration interval 12,000 cycles (assumed, within Leeham's range). The engine's life-limited parts average 65% of life remaining (assumed), and it has flown 3,000 cycles since its last restoration, so 9,000 of 12,000 remain, 75% (assumed).

$$\begin{aligned}
 \text{LLP adjustment} &= (0.65 - 0.50) \times 5.7\text{M} = +\$0.855\text{M} && (\text{alternative set:} \\
 &\quad (0.15) \times 4.0\text{M} = +\$0.60\text{M}) \\
 \text{PR adjustment} &= (0.75 - 0.50) \times 2.15\text{M} = +\$0.54\text{M} && (\text{alternative PR:} \\
 &\quad (0.25) \times 1.4\text{M} = +\$0.35\text{M}) \\
 \text{maintenance-adjusted value} &= 6.4\text{M} + 0.855\text{M} + 0.54\text{M} = \$7.8\text{M} && (\text{alternative} \\
 &\quad \text{inputs: } 6.4\text{M} + 0.95\text{M} = \$7.35\text{M})
 \end{aligned}$$

The same engine at 9,000 cycles since restoration (25% remaining) and 35% life-limited-part life:

$$\begin{aligned}
 (0.35 - 0.50) \times 5.7\text{M} &= -\$0.855\text{M}; && (0.25 - 0.50) \times 2.15\text{M} = -\$0.54\text{M}; && \text{value} \\
 &\quad = 6.4\text{M} - 1.4\text{M} = \$5.0\text{M} \\
 \text{swing between the two states} &= \$2.8\text{M} && (\text{alternative inputs: } \$1.9\text{M}) \\
 \text{full-life value} &= 6.4\text{M} + 0.5 \times (5.7\text{M} + 2.15\text{M}) = \$10.3\text{M} && (\text{alternative} \\
 &\quad \text{inputs: } 6.4\text{M} + 2.7\text{M} = \$9.1\text{M})
 \end{aligned}$$

So the maintenance adjustment is driven mostly by the life-limited-part line, and the swing between a fresh and a run-out engine of the same model and age is about \$2.8M on this Part's inputs, or about \$1.9M on Part IV's. For comparison, the consultancy blog's statement that two -7Bs at 10,000 cycles since new can differ by \$1.6M on life-limited-part life alone falls between the two LLP swings here (\$1.71M and \$1.20M); the figures are reported side by side, not reconciled. A -5B version of the same arithmetic exists in the research: on the Calver figure read as a pair of engines, \$4.94M of life-limited parts per engine at full life, an engine at 75% life is worth $0.25 \times \$4.94\text{M} = \1.23M more than half-life and one at 25% is worth \$1.23M less, with the per-engine-or-per-aircraft caveat attached. Limits of the example: the reference is a market value standing in for a base value; the set price and restoration cost each have two readings; the remaining-life fractions are assumed; and the -7B24 at \$5.7M would shift every total down by \$0.7M. [D1 example E; IBA, Sept 2025; Calver, Apr 2023, via D3 §2.4; Safe Fly Aviation, 2026]

Two further caveats come from the appraisers themselves. For engines whose life limits outlast the aircraft, the "half" reference is fluid: on the CFM56-3, 20,000 cycles at 1,000 to 1,500 cycles a year "equates to more than 15 years, considerably longer than the envisaged economic life", so appraisers may not treat 10,000 cycles as half-life. [Aircraft Value News, undated] And the full-life adjustment assumes the manufacturer's list price for every part, which is why the same publisher records that approved alternative parts "continue to undermine full-life maintenance adjustments" (Part VI §42), and why CFM's TRUEngine designation, "embraced by industry's leading asset valuation providers", is the manufacturer's lever on this calculation. [Aircraft Value News, undated; CFM International, TRUEngine release]

9. Records, release tags and the shop manual

An engine is only worth its paperwork. A module with a gap in its records cannot be installed, however good its metal. This section is brief because Part V owns what makes a used part

serviceable and Part VI owns alternative parts and repairs; what belongs here is the set of documents a module carries and why each exists.

BACK-TO-BIRTH TRACEABILITY (BTB)

Back-to-birth traceability is a continuous record, for each life-limited part, of every cycle it has flown on every engine since manufacture, with no gaps. A gap means the part's remaining life cannot be proven and it is treated as unusable. The research reached industry descriptions of the term and no formal regulatory definition. [D1 §2.9, U11; AviTrader, 23 Apr 2026]

SERVICEABLE AND UNSERVICEABLE

A part or module is serviceable when an approved organisation has inspected and released it and it is eligible for installation. It is unserviceable when it has been removed pending repair, or scrapped. The release is evidenced by a tag. [D1 §2.9; D3 §2.1]

RELEASE TAG (SERVICEABLE TAG)

The release tag is the signed airworthiness release, on FAA Form 8130-3 or on the European Union Aviation Safety Agency (EASA) Form 1, by which an approved organisation makes a part legally serviceable. It is necessary and not sufficient: the buyer also needs the part's documented history, and because the forms "can be easily generated" buyers vet the supplier too. Part V §31 carries the full treatment. [D3 §2.1; D1 §2.9]

9.1 THE TWO RELEASE FORMS

PRODUCTION CERTIFICATE (PC) AND TYPE CERTIFICATE (TC)

A type certificate is the regulator's approval of an engine's design; its holder is the manufacturer (CFM International for the CFM56). A production certificate, or production approval, is the regulator's approval under which the type-certificate holder, or a PMA holder, manufactures parts. A replacement part may be installed only if it was produced under one, or under narrow exceptions. Part VI §37 carries the full treatment. [D4 §2.1]

PART 145 REPAIR STATION

A Part 145 repair station is a maintenance organisation certificated by the FAA, or its European equivalent under the European Union Aviation Safety Agency (EASA), to perform maintenance and to issue release tags. It is the approval an engine shop or a component repair shop holds. Part IV §25 carries the full treatment. [D2 §2.1; D3 §2.1]

FAA FORM 8130-3 (AIRWORTHINESS APPROVAL TAG; AUTHORIZED RELEASE CERTIFICATE)

Form 8130-3 is the US tag. For a new part it states that the part, produced under an FAA production approval, conforms to its design and is in a condition for safe operation; for a used part it is the maintenance release by an FAA-approved repair station. "Products and articles not produced under an FAA production approval are not eligible to receive an FAA Form 8130-3." [Aviation Suppliers Association, 2016]

EASA FORM 1

EASA Form 1 is the European agency's authorised release certificate, the equivalent of Form 8130-3, issued by an EASA-approved maintenance or production organisation. New parts made under a European production organisation approval (POA) "are typically ineligible for an 8130-3 tag and are more properly released on an EASA Form 1". [Aviation Suppliers Association, 2016; D1 §2.9]

DUAL RELEASE

A dual release is one tag signed under both FAA and EASA authority, so that a part can move between US- and EU-registered aircraft without re-certification. The research reached industry descriptions only. [D1 §2.9, U11]

The trade press lists the documents a buyer most commonly requires: the release form, "Certificates of Conformance, manufacturer certifications (PMA, PC, TC)", ATA (Air Transport Association) Specification 106 sheets, "and return-to-service records from operators or repair stations". [Aviation Business News, undated] Incorrectly issued 8130-3 tags are a recurring compliance problem in the parts trade, which is why buyers check the issuing organisation and not only the form. [Aviation Maintenance magazine, undated]

9.2 THE MANUALS AND WHO MAY GO BEYOND THEM**ENGINE SHOP MANUAL (ESM)**

The Engine Shop Manual is the manufacturer's manual defining how the engine is disassembled, inspected, repaired and rebuilt, including the limits a part must meet to stay in service. Regulators treat it as the reference: an Australian airworthiness directive notes that the "CFM56-7B ESM contains instructions for calculating remaining life of each engine stationary part within certain time frames and thresholds". A repair the manual does not cover needs separate approval. [Civil Aviation Safety Authority (CASA) of Australia, AD 2014-0130; D1 §2.9]

COMPONENT MAINTENANCE MANUAL (CMM)

The Component Maintenance Manual is the manufacturer's document that sets inspection and repair limits for an individual component; the Engine Shop Manual is the engine-level equivalent. Part V §31 carries the full treatment. [D3 §2.1]

DESIGNATED ENGINEERING REPRESENTATIVE (DER) AND DER REPAIR

A Designated Engineering Representative is an engineer appointed by the FAA to examine engineering data and make compliance findings on the FAA's behalf. A DER repair is a major repair whose data a DER approved rather than being taken from the manufacturer's manual; the part stays the manufacturer's part, and the repair saves money by restoring what the manual would scrap or restoring it more cheaply. Part VI §40 carries the full treatment. [D4 §2.6; D2 §2.2]

Who performed a repair, and under which approval, is recorded on the module and matters when the module changes hands. An engine maintained only with the manufacturer's parts and manual repairs can carry the TRUEngine designation (§3), and GE sells service agreements to the same end. [CFM International, TRUEngine release; MRO Global, Safair TrueChoice agreement]

TRUECHOICE

TrueChoice is GE Aerospace's branded menu of CFM56 and LEAP services: Flight Hour (a rate-per-flight-hour agreement), Overhaul (time-and-materials shop visits in GE's network), Material (new and used OEM parts and repairs sold to third-party shops and airlines) and Transitions (green-time leases, exchanges, material buy-back and shorter builds using more used material). Part IV §28 carries the full treatment. [D2 §2.3; GE Aerospace releases]

RETURN CONDITIONS (REDELIVERY CONDITIONS)

Return conditions are the contractual minimum state of a leased engine at redelivery: minimum hours and cycles to the next restoration, minimum cycles remaining on each life-limited part, a borescope and a test-cell run, and records in order. They may exclude parts and repairs that are not the manufacturer's. Part II §13 carries the full treatment and Part VI §42 the clause on alternative parts. [D6 §2.1; D4 §2.9]

EXHIBIT 1.12 — THE DOCUMENTS A MODULE CARRIES: WHAT EACH PROVES AND WHO ISSUES IT

Document	Who issues it	What it proves	Why a module exchange needs it
Back-to-birth trace for each life-limited part	Every operator and shop in the part's history	Every cycle flown since manufacture, without gaps	Remaining life cannot be proven without it; the part is treated as unusable
FAA Form 8130-3	Production-approval holder (new part) or FAA-approved repair station (used part)	Conformity to design, or maintenance release	Makes the part legally serviceable in the US system
EASA Form 1	EASA-approved production or maintenance organisation	The European equivalent	Needed for EU-registered aircraft; a dual release serves both
Service-bulletin and airworthiness-directive compliance records	Operator or shop	Build standard and mandatory inspections complied with	Two modules at different standards may not mate or be approved together
Repair records with their approval basis	Shop	Whether a repair followed the manual or a separately approved scheme	A lessor's return conditions may exclude non-manufacturer repairs
Certificates of conformance; manufacturer certifications; ATA 106 sheets; return-to-service records	Seller, manufacturer, operator, repair station	Provenance and release history	Standard elements of a used-part documentation package

Source: D1 §2.9; Aviation Suppliers Association, 8130-3 workshop, 28 Jun 2016, and member bulletin, May 2016; Aviation Business News, "The evolution of airworthiness documentation for aircraft parts", undated; D3 §2.1. Part V §31 carries the full treatment.

The practical conclusion for the rest of the primer is the one §3 reached from the other direction. The usable supply of modules is the documented supply, not the physical one. [D1 §5]

10. The V2500 and the LEAP, for contrast

Two other engines frame the CFM56. One is its contemporary and competitor on the A320ceo. The other is its replacement. Part III §20 owns what each did to demand for CFM56 work; this section sets out the machines and the figures the research reached.

10.1 THE V2500

V2500 AND IAE

The IAE V2500-A5 is the A320ceo family's alternative engine, a two-shaft modular turbofan built by the IAE consortium, IAE International Aero Engines. Aircraft Commerce in 2010 gave the shareholders as Pratt & Whitney 32.5%, Rolls-Royce 32.5%, the Japanese Aero Engines Corporation (JAEC) 23% and MTU (Germany's MTU Aero Engines) 12%; a 2024 RTX (Pratt & Whitney's parent) release names the current members as Pratt & Whitney, Pratt & Whitney Aero Engines International, JAEC and MTU Aero Engines. Rolls-Royce's exit was widely reported and not retrieved by the research; the 2010 table is stale. [Aircraft Commerce Issue 70, 2010; RTX release, 6 Jun 2024; reconciliation R-070]

The V2500 pool is smaller than the CFM56 pool. About 5,286 V2500s were in service in 2023, of which 5,260 were -A5s, against roughly 23,000 to 24,000 CFM56s of all variants, so the V2500 pool is about a quarter the size and a smaller fraction still of the A320ceo-only pool. [Aviation Week 2023 Fleet & MRO Forecast; D1 §2.10] A market-research page gives ">13,000 units" in service in mid-2026; the research flagged that page as low grade and the primer cites it only as such. [MarketIntel, 2026; reconciliation R-010] The research did not retrieve the V2500's module list or a current shop-visit cost. The one cost source reached is fifteen years old: a heavy visit at \$2–3M "depending on the life limited part (LLP) profile required" and a full set of life-limited parts at \$1.7M list, both as of 2010, which should not be set beside the 2025 CFM56 figures in §7. [Aircraft Commerce Issue 70, 2010; D1 U13]

Shop-visit counts for the V2500 disagree. Aviation Week's data tool projected a spike to about 1,400 V2500 shop visits in 2025 and more than 7,800 overhaul visits plus 3,142 life-limited-part visits over the ten years from 2023; RTX's chief financial officer, reported by the same publication in 2026, put 2025 at "slightly more than 800" with 2026 expected within about 20 of that. [Aviation Week data tool, 2023; Aviation Week, 2026; reconciliation R-009] A forecast and a reported outturn, both via one publication; the primer carries both. MTU Maintenance, an engine maker's shop that is independent on this engine, performed 38% of all V2500 shop visits in 2024 and marked its 7,000th V2500-A5 visit in 2025. [MTU Aero Engines, 2025]

ENGINEWISE

EngineWise is Pratt & Whitney's aftermarket services brand, under which IAE's network performs V2500 shop visits. Part IV §28 carries the full treatment. [RTX release, 6 Jun 2024; D2 §2.3]

FTAI'S V2500 PROGRAM (THE IAE ENGINEWISE AGREEMENT)

On 6 June 2024 IAE and FTAI signed a five-year EngineWise agreement covering more than 100 full performance-restoration shop visits on V2500 engines in IAE's network, described by RTX as "one of IAE's largest engine maintenance agreements by number of engines with a non-airline customer". It is a different model from the Module Factory: a large engine owner buying manufacturer-network restorations at a committed volume. Part VII §51 carries the full treatment. [RTX release, 6 Jun 2024; D5 §2.5]

10.2 THE LEAP**LEAP (LEAP-1A AND LEAP-1B)**

The LEAP is CFM's new-generation engine: the LEAP-1A is one of two options on the Airbus A320neo family, and the LEAP-1B is the sole engine of the Boeing 737 MAX (the re-engined 737). About 5,500 were in service at the date of one trade-press piece, with about 50% under long-term service agreements against about 15% of the CFM56 fleet. Part III §20 carries the full treatment. [Aviation Week, undated; D9 §2.5]

The LEAP's durability sets the CFM56's remaining working life, because an airline that cannot keep a new aircraft flying keeps its old one in service instead. Operators have reported "a wave of early LEAP-1A/1B shop visits", with removals between 2,000 and 6,000 cycles and "a planning base of 4,000", against a mature CFM56's restoration interval of 10,000 to 15,000 cycles. [Visual Approach, undated; Leeham News, 8 Mar 2024] The problems concentrate in hot and harsh environments and in the high-pressure turbine blade and the fuel nozzles. [D9 §2.5]

LEAP DURABILITY KIT (DURABILITY IMPROVEMENT PACKAGE)

The durability kit is CFM's package of hardware changes to lengthen the LEAP's time on wing: a new high-pressure turbine stage-1 blade and nozzle and a forward inner nozzle support, "designed to more than double time on wing, especially in hot and harsh environments". The LEAP-1A kit is certified and "now incorporated into all deliveries and shop visits". For the LEAP-1B, one source reached by the research gave certification as targeted for the first half of 2026; CFM then announced FAA and EASA certification on 18 July 2026, with full production cutover expected in early 2027. Part III §20 carries the full treatment. [InsideFlyer / ePlaneAI, undated; CFM International release, 18 Jul 2026; D1 §2.10; D9 §2.5]

By May 2026 GE Aerospace was stating that LEAPs "being shipped now will match the durability of the venerable CFM56". [Leeham News, 27 May 2026] Both the problem and the claimed fix are recorded here; whether delivered LEAPs reach CFM56-level time on wing is Part III's question, not this Part's. [D1 §2.10]

Scale is the other contrast. By December 2017 CFM had orders and commitments for more than 14,270 LEAPs and had delivered 459 in 2017; GE's annual report gives 1,802 LEAP deliveries in 2025 against 1,407 in 2024; Safran reported a LEAP backlog of more than 12,900 units. The cumulative total delivered was not retrieved. [Safran release, 6 Feb 2018; GE 10-K FY2025; Safran, 13 Feb 2026; D1 U12] The other engine on the A320neo, Pratt & Whitney's geared turbofan, had its own durability problem; Part III §20 carries it.

EXHIBIT 1.13 — THREE ENGINES SIDE BY SIDE: THE FIGURES THE RESEARCH REACHED

Item	CFM56 (-5B, -7B)	V2500-A5	LEAP-1A / -1B
Airframes	A320ceo (-5B, ~60%); 737NG (-7B, sole)	A320ceo (alternative)	A320neo (one of two); 737 MAX (sole)
Maker	CFM International (GE Aerospace and Safran)	IAE (Pratt & Whitney, P&W Aero Engines International, JAEC, MTU)	CFM International
In service	~23,000 (CFM, undated); ~24,000 (trade press, Sept 2025); >22,800 (Safran, end-2025)	5,286 in 2023 (Aviation Week); ">13,000" mid-2026 (low-grade vendor page)	~5,500 (Aviation Week, undated)
Production for airframes	Ended 2019 (-7B) and 2022 (-5B)	A320ceo out of production	1,802 delivered in 2025 (GE 10-K); backlog >12,900 (Safran)
Mature restoration interval	10,000–15,000 cycles (Leeham, 2024)	Not retrieved	Early removals at 2,000– 6,000 cycles, planning base 4,000 (Visual Approach)
Shop visits a year	2,300–2,400 in 2026–28 (GE via Aviation Week, 2026); ~2,500 peak 2025–26 (Safran, c. 2023/24)	>800 in 2025 (RTX CFO via Aviation Week, 2026); ~1,400 forecast (Aviation Week data tool)	Not retrieved
Share under long-term agreements	~15%	Not retrieved	~50%

Item	CFM56 (-5B, -7B)	V2500-A5	LEAP-1A / -1B
Heavy shop visit cost	Exhibit 1.9 (2026 ranges)	\$2–3M (2010)	Not retrieved
Full LLP set at list	\$5.7M (2025) or ~\$4M (2019); see Exhibit 1.6	\$1.7M (2010)	Not retrieved

Source: D1 §2.2, §2.10 and §4.1; Aviation Week 2023 Fleet & MRO Forecast and V2500 data tool; Aviation Week, "Pratt & Whitney sees high-single-digit MRO growth", 2026; Aviation Week, "CFM56 overhaul demand remains strong", 2026; Aviation Week, "Safran sees Leap aftermarket emerging by mid-decade", c. 2023/24; Aviation Week, "Leap aftermarket poised for growth", undated; MarketIntel, 2026; Aircraft Commerce Issue 70, 2010; GE 10-K FY2025; Safran, 13 Feb 2026; Visual Approach, undated. Reconciliation R-006, R-007, R-009, R-010.

11. Orientation: where the money sits in this chain

This section is a first map, not an analysis. Part XI §86 carries the full treatment of the margin pool with every figure on its own basis; here the reader needs only to see the links of the chain and a few indicative numbers, so that the segment Parts have a frame. Nothing is ranked.

MARGIN POOL

The margin pool is this primer's term for how the profit earned in keeping an engine type flying is distributed across the links of the chain: the manufacturer's new parts and services, the shop's labour, the component repairers, the makers of approved alternative parts, the traders in used material and teardown, the lessors of engines and aircraft, and the servicers of third-party capital. No dossier uses the phrase; its inputs are the slice-by-slice capture of a shop visit (D2 §2.2) and the peers' reported margins on their own bases (D11). Part XI §86 is the full treatment. [registry 1c]

Start from the shop visit of §7. Material is 60–70% of it, labour 20–30%, repairs and other shop work 10–20%. [Air Cargo Week, 2025/26] Each slice has a different collector. New parts go to the manufacturer at catalogue list price, whoever runs the shop; life-limited parts, 40–60% of parts cost when replaced, go to the manufacturer alone because no alternative exists. [D2 §2.2; Safe Fly Aviation, 2026] Used serviceable material (USM, parts recovered from dismantled engines and re-released) substitutes for new parts and is sold by teardown traders and by the manufacturer itself, which states that "GE and CFM are the largest used serviceable material provider for their products". [GE Aerospace, 2026, via D2 §2.2] Approved alternative parts (PMA, §5) substitute for the manufacturer's blades and vanes and are sold by their makers. Component repairs go to repair vendors, some under the manufacturer's licence and some under separately approved schemes. Labour and test go to the shop. And the CFM56 aftermarket is mostly transactional: Safran describes about 85% of its revenue as spare parts and time-and-materials work, against about

40% for the LEAP, so on this engine the manufacturer's main capture is parts rather than contracts. [Safran via Aviation Week, c. 2023/24; D2 §2.3]

Around the shop visit sit the owners. Lessors own engines and aircraft and collect rent and maintenance reserves (Part II). FTAI's two reportable segments sit on both sides of this picture.

AEROSPACE PRODUCTS SEGMENT

FTAI's reportable segment that develops, repairs and sells CFM56 and V2500 engines, modules and parts through the Module Factory; FY2025 segment revenue \$1,936.2M and segment Adjusted EBITDA (FTAI's adjusted earnings measure) \$671.3M; Adjusted EBITDA is earnings before interest, taxes, depreciation and amortisation as the company adjusts them, defined in Part X §80. Part VII §45 carries the full treatment. [FTAI 10-K FY2025, via orchestrator notes]

AVIATION LEASING SEGMENT

FTAI's reportable segment that owns, leases and sells aircraft and engines, directly and through an equity-method investment; 290 assets at 31 Dec 2025 and segment Adjusted EBITDA \$608.9M for FY2025. Part VIII §58 carries the full treatment. [FTAI 10-K FY2025, via D6 and orchestrator notes]

EXHIBIT 1.14 — INDICATIVE MARGIN FIGURES ALONG THE CHAIN, EACH ON ITS OWN STATED BASIS

Link in the chain	Company and segment	Margin figure	Basis	Source and date
Shop visit, by slice	Any CFM56 shop visit	Material 60–70%; labour 20–30%; repairs 10–20% of the bill	Share of cost, not a margin	Air Cargo Week, 2025/26
Manufacturer, new engines, parts and services	GE Aerospace, Commercial Engines & Services (CES)	26.6% on \$33,314M; services \$25,010M of revenue	GE segment profit (GE's own measure)	GE 10-K FY2025
Approved alternative parts	HEICO (a PMA manufacturer), Flight Support Group (FSG)	24.1% on \$3,117.3M	Segment operating income, derived	HEICO 10-K FY2025 (year to 31 Oct 2025)
Independent engine shop	StandardAero, Engine Services	13.2% on \$5,354.0M	Segment Adjusted EBITDA, StandardAero's definition	StandardAero 10-K FY2025

Link in the chain	Company and segment	Margin figure	Basis	Source and date
Independent engine shop (context, non-SEC)	MTU Aero Engines, commercial MRO	8.0% on €6.0B	Adjusted EBIT (after depreciation; not comparable with an EBITDA margin)	MTU, 24 Feb 2026
Independent engine shop (context, non-SEC)	Lufthansa Technik	7.5% on €8.049B	Adjusted EBIT	Lufthansa Technik, FY2025 results
Teardown and used material	AerSale, Asset Management Solutions (AMS)	35.0% on \$211.6M	Gross margin, derived	AerSale 10-K FY2025
Module exchange and engine sales	FTAI, Aerospace Products	34.7% on \$1,936.2M (FY2025, derived); 28.5% on \$875.0M (Q2 2026, derived)	Segment Adjusted EBITDA (FTAI's definition; includes MRE (Maintenance, Repair and Exchange) Contract revenue to a related party, Part VII §48)	FTAI 10-K FY2025; FTAI release, 29 Jul 2026

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26 (D2 §2.2); GE 10-K FY2025; HEICO 10-K FY2025; StandardAero 10-K FY2025; MTU Aero Engines, FY2025 results, 24 Feb 2026; Lufthansa Technik, FY2025 results release; AerSale 10-K FY2025; FTAI 10-K FY2025 (segment table via orchestrator notes) and Q2 2026 release (D11 §1 and §4.2). Adjusted EBITDA is earnings before interest, taxes, depreciation and amortisation as each company adjusts it; the five margin bases in this table are not interchangeable, since an EBITDA margin exceeds an EBIT margin by the depreciation share of revenue, and the fiscal years differ (December, October). Part XI §88.

Three cautions travel with the table into Part XI. The margins are on five different bases and are not comparable with each other without adjustment. [D11 §2.3, §2.5; E.1 rule 35] FTAI's segment figure is a derived ratio of two filed numbers and includes revenue from sales to a partnership in which FTAI holds a 19% stake, which Part VII §48 and Part VIII §63 treat. [reconciliation R-043, R-047] And a share of cost is not a margin: the first row says where the bill goes, not what anyone earns on it. Part XI §86 builds the pool link by link from these inputs and from the teardown and alternative-parts economics of Parts V and VI.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-001	CFM56 in service: ~23,000 (CFM, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); "over 20,000" -5B/-7B (Chromalloy); >19,000 and ~14,200 (Safe Fly, consultancy blog)	§2.2, Exhibit 1.2
R-002	Safe Fly's ~14,200 read as the -7B fleet (D3) or all CFM56 (D8)	§2.2, Exhibit 1.2
R-003	-7B: "in excess of 8,000" installed (CFM product page) against >15,000 delivered and >7,000 aircraft	§2.2
R-069	Deliveries >35,000 (CFM) against "well over 30,000" (unsourced)	§2.2
D1 T1	-7B thrust ceiling 27,300 lbf (secondary) against "33,000" (CFM website summary)	§1, Exhibit 1.1
R-035	Module count: 4 (D1, D3); 3 plus gearbox (D2); 3 per engine (FTAI release, D8); 4–5 (D5, D9); 5 (D7); 4 (D10); 6 (D11)	§3.2, Exhibit 1.4
R-016	Full -7B LLP set: \$1.775M (2008), \$1.7M (2013, caveat), \$3.4M (2018), ~\$4M (2019), \$5.7M (2025); Calver -5B \$9.88M (basis unstated)	§4.3, Exhibit 1.6
R-017	LLP cost per cycle: \$200 (D2, \$4M set) and \$285 (D1, \$5.7M set); \$333 and \$475 with stub life	§4.4, worked example 1.6
R-018	Escalation: LLP set ~7%/yr compounded, Ishka ~12% step, -7B parts +20–30% (2023) against spares ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023, inflation +3–4 points	§4.3
R-011	Removal driver: EGT margin "most influential" (Aircraft Commerce 2003/2008) against LLP-driven for latest -5B/-7B (Aircraft Commerce Issue 50; FlightGlobal)	§5.2
R-008	Interval: 10,000–15,000 cycles PR, 20,000–25,000 overhaul (Leeham) against "every eight years" (Aviation Week), ~10 years implied (D9), six years assumed (D7)	§5.2, worked example 1.4; Exhibit 1.8
R-019	Turnaround: 90–120 days (trade press, 2025); ~90 sustaining (GE, May 2026); PR 75–90, heavy 120–150 (Safe Fly); slot wait +2–3 months (Bain)	§6.3, Exhibit 1.8
R-028	-7B lease rate: ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time)	§6.4, worked example 1.5; Exhibit 1.11
R-013	Shop-visit cost: PR \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M"; full overhaul \$3.5–4.2M; light \$650–900k (all Safe Fly); D6's \$4.5M assumed	§7.1, Exhibit 1.9; worked example 1.6

Id	The two (or more) figures	Where in this Part
R-014	Cost shares: material 60–70% / labour 20–30% / repairs 10–20% (Air Cargo Week) against LLPs 40–60% of parts / HPT blades and vanes 15–25% / labour 15–20% (Safe Fly)	§7.2, Exhibit 1.10
R-015	HPT blade set \$1.6M or \$3.08M (per-blade prices × 80, unverified) against \$0.2–0.6M implied by the 15–25% share	§7.2, Exhibit 1.10
R-029	-7B value: half-life market value \$5.7M/\$6.4M, levelled off (IBA) against green-time \$2.8–3.4M, teardown \$0.8–1.2M, rents up 15–20% (Safe Fly) and values +50% in two years (ePlaneAI)	§8.2, Exhibit 1.11
R-009	V2500 shop visits 2025: >800 (RTX CFO via Aviation Week) against ~1,400 forecast (Aviation Week data tool)	§10.1, Exhibit 1.13
R-010	V2500 in service: 5,286 (Aviation Week, 2023) against ">13,000" (low-grade vendor page)	§10.1, Exhibit 1.13
R-070	IAE shareholders: 2010 table with Rolls-Royce against the 2024 RTX release without it	§10.1

Unknowns carried in this Part

- **U-D1-01 (D1): U1** Full numbered sub-module list for the -7B and -5B (the ESM breakdown). Resolve with the CFM56-7B/-5B Engine Shop Manual or a CFM training manual.
- **U-D1-02 (D1): U2** Exact list of part numbers common to the -5B and -7B core (basis for cross-variant module and LLP interchange). Resolve with the Illustrated Parts Catalogues or a CFM commonality bulletin.
- **U-D1-03 (D1): U3** Per-part LLP table for the -5B (only the 18-set, 20,000–30,000 range was reached). Resolve with a -5B records mini-pack (AJW, StandardAero publish them).
- **U-D1-04 (D1): U4** Prices of used LLP packages with stated cycles remaining. Resolve with dealer quotes or D3's sources.
- **U-D1-05 (D1): U5** As-new EGT margin by -7B/-5B thrust rating and typical loss per 1,000 cycles. Resolve with Aircraft Commerce Issue 58 (-7B) and Issue 50 (-5B) maintenance analyses in full.
- **U-D1-06 (D1): U6** First-run and mature-run intervals by thrust rating and region. Same sources as U5.

- **U-D1-07 (D1): U7** Labour hours per workscope and shop labour rates (US\$/hour) for CFM56 visits. Resolve with an MRO price catalogue or Aircraft Commerce's maintenance budgets.
- **U-D1-08 (D1): U8** Core-only LLP group price as a share of the full set. Resolve with a CFM catalogue extract.
- **U-D1-09 (D1): U9** Current half-life base values for -7B and -5B by rating (IBA, mba, Cirium). Resolve with an appraiser publication; this session's searches returned definitions, not values.
- **U-D1-10 (D1): U10** Years and amounts of each CFM LLP escalation step 2019–2025 (only "Aug 2023, +20–30% on -7B parts" and "~12% LLP escalation" without a year were reached) [33][34].
- **U-D1-11 (D1): U11** Formal regulatory definition of back-to-birth traceability and of dual release; this session reached industry descriptions only [14][40][41].
- **U-D1-12 (D1): U12** Current LEAP delivery total and the dates/years of -5A and -5C production end.
- **U-D1-13 (D1): U13** V2500 module list and current per-visit cost (2010 figures only) [38].
- **U-D1-14 (D1): U14** Whether a module exchange requires a test-cell run under the CFM56 ESM.
- **U-D2-01 (D2): U1. Catalog escalation 2024, 2025, 2026.** Only the Nov 2022 (double-digit) and Aug 2023 (high-single-digit) increases and Safran's "inflation + 3–4 points" rule were found. Resolve with Aviation Week/Aircraft Commerce coverage of each year's CFM price revision, or GE/Safran earnings call Q&A on "price".
- **U-D3-03 (D3): U3** — Whether Calver's \$9,878,733 / \$4,939,366 CFM56-5B LLP figure is per engine or per aircraft (two engines). The original Cargo Facts presentation would settle it.
- **U-D3-05 (D3): U5** — A current appraiser half-life value for the CFM56-5B (only trend commentary and a trade-press lease-rate range were captured).
- **U-D3-06 (D3): U6** — Current CLP of representative CFM56-7B LLPs (e.g. HPT disk, fan disk) and of a full -7B LLP set. Needed for a precise module-value example.
- **U-D4-16 (D4): U16** — **CFM56-7B HPT blade count per set** (80 is a recollection, unverified here) and the OEM list price for a full HPT blade and vane set.
- **U-D9-03 (D9): U3** — **Definition of the ~24,000 CFM56 "in service" figure [S78]** (installed only, or installed plus spares plus stored) and its split between -5B, -7B and the older -3/-5A/-5C variants. Resolve: GE Aerospace or CFM investor materials with a fleet breakdown; the GE Bernstein presentation (27 May 2026) [S6] may contain it but its content did not surface.

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PART II

First principles: how money moves around an engine

This Part explains how an aircraft or an engine turns into cash flows: the lease that rents it out, the reserves that pay for the life a lessee uses up, the settlement at the end of the lease, the rent an engine earns on its own, the trade in engines with a little life left, and the debt that funds the owner. A reader who finishes it will be able to read a lease summary, a lessor's income statement and a securitization term sheet with every term understood.

WHERE THIS PART SITS IN THE BUILD

Part I described the CFM56 (the engine family Part I defined) as a machine: how it comes apart, which parts are scrapped on a clock, what a shop visit costs, and what the engine is worth at any point in its life. This Part takes that object and follows the money around it. Almost every dollar an engine earns for its owner passes through a contract, usually an operating lease, and almost every dollar its owner borrows is secured on that contract. So the lease is the unit of analysis here, the way the module was the unit in Part I. Part III, which follows, supplies the context: the fleet, the retirements that did not happen, the new engines that arrived late, and a translation table that turns each fact into a requirement on the aftermarket. Nothing in this Part depends on anything written after it. Parts VIII and XI, which cover FTAI's leasing segment, its third-party capital vehicles and its peers, depend on every definition here.

12. The operating lease, term by term

An aircraft or engine earns money for its owner in three ways. It earns rent while it is leased. It earns payments for the maintenance life a lessee consumes while flying it. And it earns a price when it is sold, either whole or in parts. The first two flows are written into a lease, and the third is shaped by the condition the lease requires the asset to come back in. This section takes the lease apart term by term. Section 13 works the maintenance terms through a full example, section 14 applies the whole to an engine on its own, and sections 15 to 17 follow the asset to the end of its life and to the lenders who fund it.

The two parties have plain names. The lessor is the owner who rents the asset out. The lessee is the operator, almost always an airline, who rents it. The research found no reason to make either word more technical than that. [definition assembled in the research, D6 §2.1]

OPERATING LEASE

An operating lease is a contract under which the owner (the lessor) rents an aircraft or engine to an operator (the lessee) for a fixed term in exchange for periodic rent, while the lessor keeps title to the asset and keeps the risk that it is worth more or less than expected when it comes back. Nearly all of FTAI's leases, and nearly all of Willis Lease Finance's, are operating leases. The alternative is the finance lease, defined next. [FTAI 10-K FY2025; Willis Lease Finance Q3 2025 release; definition assembled in the research, D6 §2.1]

FINANCE LEASE (SALES-TYPE LEASE)

A finance lease transfers substantially all the economic risks and rewards of owning the asset to the lessee, so that in substance it is a loan secured on the asset rather than a rental. Willis Lease Finance Corporation (WLFC) reports a small book of "investments in sales-type leases", \$16.3M of its \$2,888.5M lease portfolio at 30 September 2025, which are finance leases by another name; FTAI reports none. [WLFC Q3 2025 release; definition assembled in the research, D6 §2.1]

The distinction matters for everything that follows. Under an operating lease the lessor still owns the engine when it comes back, so the lessor cares what condition it comes back in, which is why the maintenance terms in §13 exist. Under a finance lease the lessee effectively owns it and the lessor is a lender. This Part is about operating leases unless it says otherwise.

12.1 TERM, RENT AND THE LEASE RATE FACTOR

The term is the fixed period of the lease. For a mid-life narrowbody it typically runs several years; FTAI reported a weighted-average remaining term of 44 months on its aircraft on lease and 38 months on its engines on lease at 31 December 2025. [FTAI 10-K FY2025] Those are remaining terms, not the terms the leases were signed at; the research did not retrieve new-lease terms at signing or the mix of short against long engine leases in FTAI's book (D6 U15). Engine leases divide into short-term and long-term, which Part I §6.4 defined; §14 carries them.

LEASE RATE (RENT)

The lease rate is the periodic rent, quoted per month. The sourced points in this primer are about \$228,000 a month for a 12-year-old Boeing 737-800 and about \$220,000 for a comparable Airbus A320-200 as of July 2025, and about \$100,000 a month for a CFM56-7B spare engine in 2024, all from IBA (an appraiser), against \$42,000 to \$48,000 a month for a CFM56-7B in 2026 from a consultancy blog that may be describing a green-time engine. The two engine figures are carried as a pair throughout this primer and never averaged. [IBA, Jul 2025, via India Seatrade News, 6 Sep 2026; IBA, Apr 2024, via AviTrader; Safe Fly Aviation, 2026 (consultancy blog); reconciliation R-028]

Rent on its own says little about whether a lease is a good one for the owner, because a \$228,000 rent on a \$15M aircraft is a very different thing from the same rent on a \$30M aircraft. The industry therefore quotes rent as a percentage of what the asset is worth.

LEASE RATE FACTOR (LRF)

The lease rate factor is the monthly rent divided by the asset's current market value, expressed as a percentage. An LRF of 1.0% means annual rent is 12% of value. It is the leasing industry's shorthand for gross yield, before depreciation, interest and maintenance. It rises with asset age, because older assets carry more residual and maintenance risk and lose value faster in percentage terms, and it rises with interest rates. No source the research reached states a lease rate factor for FTAI, for Willis Lease or for the CFM56 as a type; every figure in this primer is derived, and §14 shows each with its basis. [definition assembled in the research, D6 §2.1, D11 §2.2; reconciliation R-030]

WORKED EXAMPLE 2.1 — READING A LEASE RATE FACTOR

Inputs: rent of \$228,000 a month for a 12-year-old 737-800 (sourced: IBA, July 2025). The aircraft's half-life market value was not retrieved by the research (D6 U9), so the value is the unknown in this example and the lease rate factor is the variable.

$$\begin{aligned} \text{LRF} &= \text{rent} \div \text{value} \\ \text{at LRF } 0.8\% \text{ a month: value} &= 228,000 \div 0.008 = \$28.5\text{M} \\ \text{at LRF } 1.0\% \text{ a month: value} &= 228,000 \div 0.010 = \$22.8\text{M} \\ \text{at LRF } 1.2\% \text{ a month: value} &= 228,000 \div 0.012 = \$19.0\text{M} \\ \text{annual gross yield} &= \text{LRF} \times 12: 9.6\%, 12.0\%, 14.4\% \end{aligned}$$

So the same rent is a 9.6% gross yield on a \$28.5M aircraft and a 14.4% gross yield on a \$19.0M one. The identity cuts both ways: a lessor who knows the value quotes the lease rate factor, and a buyer who knows the rent and the factor other lessors accept can back out what the aircraft is worth to them. Limits of the example: the three factors are chosen to show the arithmetic and are not market data; the aircraft value was not sourced; and the factor for a specific lease depends on the lessee's credit, the term and the return conditions below. [IBA, Jul 2025; D6 U9]

12.2 THE SECURITY DEPOSIT**SECURITY DEPOSIT**

A security deposit is cash, or a bank letter of credit, that the lessee posts at signing. The lessor holds it against default and returns it at redelivery if the lessee has performed. It is a credit instrument, not a maintenance instrument: it protects the lessor against unpaid rent, not against life consumed from the engines. FTAI's balance sheet carries "security deposits" as a liability; the amounts by year were not returned by the research (D6 U7). [FTAI 10-K FY2025; definition assembled in the research, D6 §2.1]

The deposit is the first of three pots of the lessee's money that a lessor may hold. The other two, maintenance reserves and end-of-lease compensation, are maintenance instruments and are the subject of §13. They are defined here because the lease contains them.

12.3 MAINTENANCE RESERVES AND END-OF-LEASE COMPENSATION

Every hour an engine flies uses up some of the life that was in it when the lease began: the exhaust-gas temperature margin that a performance restoration will have to recover, and the cycles on each life-limited part (Part I §4 and §5). A lease must say who pays for that. There are two regimes.

MAINTENANCE RESERVES (SUPPLEMENTAL RENT; MAINTENANCE RENTS)

Maintenance reserves are periodic payments the lessee makes to the lessor in addition to basic rent, "calculated with reference to the utilization of airframes, engines and other major life-limited components during the lease". They are set per flight hour or per flight cycle and collected per component: the airframe's heavy checks, the landing gear, the auxiliary power unit, each engine's performance restoration and each engine's life-limited parts. They pre-fund the eventual shop visit so that if the lessee defaults or hands the asset back, the lessor already holds cash for the life consumed. The lessor reimburses the lessee for a completed maintenance event up to the lesser of the reserve balance held for that event and the qualifying cost; any balance not claimed against a qualifying event stays with the lessor at lease end. AerCap calls them maintenance rents; Willis Lease books them as maintenance reserve revenue when they are no longer refundable. [AerCap 20-F FY2013; Spirit Airlines 10-K FY2013; WLFC Q3 2025 release]

END-OF-LEASE (EOL) COMPENSATION

End-of-lease compensation is the alternative regime, used for lessees whose credit is strong enough that the lessor does not demand monthly reserves. The lessee must "re-deliver the aircraft in a similar maintenance condition (normal wear and tear excepted) as when accepted under the lease, with reference to major life-limited components", and "to the extent that such components are redelivered in a different condition than at acceptance, there is an end-of-lease compensation adjustment for the difference at redelivery". The money can flow either way: the lessee pays if the asset comes back with less life than it had at delivery, and the lessor pays or credits if it comes back with more. Many leases combine the two regimes, reserves for some components and an end-of-lease adjustment for others. [AerCap 20-F FY2013]

The difference between the regimes is timing and credit, not amount. Under reserves the lessor collects monthly and holds the cash. Under end-of-lease compensation the lessor collects one lump at the end and bears the lessee's credit in the meantime. Either way the lessee pays for the life it consumes, measured against the condition at delivery, and §13 works both through the same lease.

12.4 RETURN CONDITIONS AND REDELIVERY

Part I §9 defined return conditions, also called redelivery conditions: the contractual minimum state of the asset at redelivery. For an engine that is typically a minimum number of flight hours

and cycles remaining to the next performance restoration, a minimum number of cycles remaining on the life-limited parts, a borescope inspection and a test-cell run. For an airframe it is time remaining to the next heavy check, fresh paint or a paint credit, and records in order.

[definition assembled in the research, D6 §2.1] Redelivery is the act of handing the asset back at the end of the term, and the return conditions are what the lessor checks at that moment.

Return conditions do a specific job in the money flow. Together with the reserve or end-of-lease regime they decide who bears the cost of the maintenance life consumed during the lease. If the engine comes back below the required condition, the lessee must either restore it before redelivery or pay. Return conditions may also exclude parts and repairs that are not the manufacturer's; Part VI §42 carries that clause, and the research recorded that current lessor practice on it, for 2024 to 2026, was not found (D4 U1). [D4 §2.9; reconciliation C.4 item 2]

12.5 EXTENSIONS

A lease does not have to end at its term. The incumbent airline can renew it, and in the years this primer covers most did.

LEASE EXTENSION

A lease extension is the renewal of an expiring operating lease with the incumbent airline instead of taking the aircraft back. In 2023 to 2024, 75–90% of scheduled lease expiries on mid-life narrowbodies were being extended (Cirium Ascend, via Bloomberg); extension terms stretched to four to six years, and extension rents in some cases ran more than 40% above the age-specific rates agreed just before (IBA, November 2023). Part III §22 carries the full treatment. [Cirium Ascend via Bloomberg and AJOT (American Journal of Transportation), date not visible, context 2023–24; IBA Weekly, 24 Nov 2023; Key.Aero, "Lease trends"]

An extension matters here because it changes the return-condition arithmetic. The redelivery that would have triggered a settlement is postponed, the engines fly further toward their next shop visit, and the reserves or compensation keep accruing against a later date. IBA recorded cases in which aircraft due back at year twelve were not expected to return until 2032, at eighteen years of age. [IBA Weekly, 24 Nov 2023] Section 15 carries what that did to rents.

12.6 TRANSFERRING A LEASE: LEASE-ATTACHED TRADING AND NOVATION

A leased aircraft is often sold while the lease is running. The buyer wants the lease as much as the aircraft, because the lease is the cash flow.

LEASE-ATTACHED TRADING

Lease-attached trading is buying an aircraft from another lessor with the existing lease transferred to the buyer, so that the buyer acquires a rent stream and a return-condition obligation rather than an empty aircraft. The lease is what makes the asset financeable: lenders and securitization investors underwrite contracted rent from a known airline. Part VIII §60 and §67 cover FTAI's own transactions of this kind. [definition assembled in the research, D6 §2.4; D7 §2.4]

The legal step by which the lease moves is a novation: the old lessor, the new lessor and the airline agree that the new lessor takes the old lessor's place in the contract, with the airline's consent. No source in the research describes novation specifically; the word is used here in its ordinary legal sense to name the step the dossier describes as the lease being "transferred", and nothing in this primer turns on it. What does turn on it is that an on-lease aircraft changes hands with its deposit, its reserve balances and its return conditions attached, so the buyer inherits the whole money flow described in this section.

12.7 RESIDUAL VALUE

The last term is not in the contract but sits behind every number in it.

RESIDUAL VALUE AND RESIDUAL-VALUE RISK

Residual value is the lessor's forecast of what the asset will be worth, or fetch in a part-out, at the end of its depreciation life. Residual-value risk is the risk that the realised figure differs. Under an operating lease that risk stays with the lessor, which is what distinguishes it from a finance lease. FTAI's depreciation policy for leasing equipment, meaning the useful lives and residual assumptions it uses, was not returned by the research (D6 U8). [definition assembled in the research, D6 §2.1]

The parties who put numbers on residual value, and on current value, are the appraisers Part I §8 defined: IBA, mba Aviation, Ascend by Cirium and AVITAS (a US appraisal firm) publish base values and market values at half-life, and the market trades and the rating agencies test on their figures. Part I §8 also defined base value (the appraiser's opinion of the asset's underlying economic value in a balanced market) and market value (the most likely trading price in the actual current market, above base value when supply is tight and below it when soft). This Part uses those definitions without repeating them. What it adds is the observation that the research reached market values for the CFM56-7B (Exhibit 2.3) and no base values for any engine, and no appraised values of any kind for a 12-to-15-year-old 737-800 or A320ceo (D1 U9; D6 U9).

[IBA, Sept 2025; reconciliation cluster K-02]

EXHIBIT 2.1 — THE OPERATING LEASE, TERM BY TERM: WHAT EACH TERM DOES AND WHICH WAY THE MONEY FLOWS

Term	What it is	Who pays whom	Why it is in the lease	Source
Term	Fixed rental period; FTAI's remaining terms 44 months (aircraft), 38 months (engines) at 31 Dec 2025	n/a	Sets the period of contracted rent lenders underwrite	FTAI 10-K FY2025
Rent (lease rate)	Monthly payment; ~\$228,000 for a 12-yr 737-800 (Jul 2025)	Lessee to lessor	The lessor's primary income	IBA, Jul 2025
Lease rate factor	Rent ÷ market value, % per month	n/a	Gross-yield shorthand; no stated figure retrieved	Derived in the research
Security deposit	Cash or letter of credit at signing, returned on performance	Lessee to lessor, then back	Credit protection against default	D6 §2.1; FTAI 10-K (balance not returned)
Maintenance reserves	Per-hour and per-cycle payments by component; reimbursed up to the lesser of balance and qualifying cost	Lessee to lessor; lessor reimburses events	Pre-funds the life consumed	AerCap 20-F FY2013; Spirit 10-K FY2013
End-of-lease compensation	Cash adjustment at redelivery for the difference in maintenance condition	Either direction	Same purpose, settled at the end	AerCap 20-F FY2013
Return conditions	Minimum hours and cycles to next restoration, minimum LLP cycles, borescope, test run; airframe time to check, paint, records	Lessee restores or pays	Decides who bears the life consumed	D6 §2.1 (Part I §9)
Extension	Renewal with the incumbent; 75–90% of expiries extended, 2023–24	n/a	Postpones redelivery; rents >40% above prior rates in cases	Cirium Ascend; IBA, Nov 2023
Novation / lease-attached sale	Transfer of the lease to a new owner with the airline's consent	n/a	Makes an on-lease aircraft a tradeable cash flow	D6 §2.4 (novation not sourced in the research)

Term	What it is	Who pays whom	Why it is in the lease	Source
Residual value	Lessor's forecast end-of-life value	n/a	The risk the lessor keeps	D6 §2.1; FTAI policy not returned

Source: FTAI 10-K FY2025; IBA via India Seatrade News, 6 Sep 2026; AerCap 20-F FY2013; Spirit Airlines 10-K FY2013; Cirium Ascend via Bloomberg/AJOT; IBA Weekly, 24 Nov 2023; definitions assembled in D6 §2.1 and §2.4. Accessed 2026-10-03.

13. Maintenance reserves, return conditions and the lease-end settlement (worked example)

The maintenance terms are where most of the money in a mid-life lease actually sits. The reason is arithmetic. A performance restoration on a CFM56 costs between \$1.2M and \$2.5M on the two consultancy ranges Part I carried, and more with life-limited parts; an aircraft carries two such engines; and a six-year lease consumes a large fraction of a restoration interval on each. So the payments for life consumed can approach half of the basic rent, and the settlement at the end can be a multi-million-dollar negotiation. This section explains the mechanism, then works it through the one example the research built.

13.1 HOW A RESERVE IS SET AND SPENT

A reserve is collected by component because each component has its own clock. The airframe has heavy checks on a calendar and flight-hour schedule. The landing gear has an overhaul interval. The auxiliary power unit has its own hours. Each engine has two clocks of its own: the performance-restoration clock, measured in hours or cycles since the last restoration, and the life-limited-part clock, measured in cycles against each part's certified limit (Part I §4 and §5).

[mba Aviation, "Maintenance Matters in Aircraft ABS Deals", undated; ABS is the asset-backed securitization §17 defines; Maheshwari & Co., undated]

AUXILIARY POWER UNIT (APU)

The auxiliary power unit is the small gas turbine in the aircraft's tail that supplies electrical power and compressed air on the ground and in emergencies. It is one of the components a lease reserves for separately, and one of the higher-value items when an airframe is parted out. Part V §32 carries the part-out treatment. [D3 §2.2; D6 §2.1]

The rate for each component is set so that the balance accumulated by the next event roughly covers it. For an engine restoration the rate is per flight hour; for life-limited parts it is per flight cycle, because the parts are certified in cycles (Part I §4). No source the research reached gives

the rates a lessor actually charges on a CFM56 lease; the example below uses rates labelled as assumed, and the research recorded the gap (D6 U7; reconciliation C.2 item 6). [D6 §2.2]

Spending the reserve follows one rule, quoted from a lessee's own filing: the lessor reimburses "the lesser of (1) the amount of the maintenance reserve held by the lessor associated with the specific maintenance event or (2) the qualifying costs related to the specific maintenance event". [Spirit Airlines 10-K FY2013] Two consequences follow. If the event costs more than the balance, the lessee funds the difference. If the event costs less, the lessor keeps the rest. And because the reserve covers only life consumed since the lease began, not life consumed by earlier operators, a reserve rarely covers a whole shop visit unless the engine was delivered fresh from one. [D6 §2.2]

For the lessor, reserve receipts are not income when they arrive. They are a liability, carried as maintenance deposits, until the event is reimbursed or the lease ends with the balance unclaimed, at which point the unclaimed balance becomes revenue. [WLFC Q3 2025 release; D11 §2.2] The size of that line at an engine lessor shows how much of the money is maintenance rather than rent. Willis Lease reported maintenance reserve revenue of \$76.1M in the third quarter of 2025 alone, up 52.8% on a year earlier, against lease rent of \$76.6M in the same quarter; for the full year 2025 its combined "core lease rent and maintenance reserve revenues" were \$523.6M, of which lease rent was \$291.6M, leaving \$232.0M of maintenance reserve revenue, a figure derived in the research by subtraction. [WLFC Q3 2025 release; WLFC 10-K FY2025; derived in D11 §2.2] At AerCap, whose fleet is young, "maintenance rents and other receipts" were \$690M in 2025 against basic lease rents of \$6,679M, about 9%. [AerCap Q4 2025 release] The gap between 9% and 44% is the gap between a young fleet far from its shop visits and a mid-life engine book close to them. [D11 §2.2]

13.2 THE TWO REGIMES, SIDE BY SIDE

EXHIBIT 2.2 — MAINTENANCE RESERVES AGAINST END-OF-LEASE COMPENSATION: HOW EACH REGIME HANDLES THE SAME EVENTS

Event	Reserve regime	End-of-lease compensation regime	Source
Lessee flies 250 hours in a month	Pays rent plus a per-hour restoration reserve and a per-cycle LLP reserve for each engine	Pays rent only; the life consumed is tallied for settlement at redelivery	AerCap 20-F FY2013; D6 §2.2
Lessee performs a shop visit mid-lease	Claims reimbursement up to the lesser of the balance and the qualifying cost; funds any excess	Funds the visit; the engine's improved condition is credited at redelivery	Spirit 10-K FY2013; AerCap 20-F FY2013

Event	Reserve regime	End-of-lease compensation regime	Source
Lessee defaults mid-lease	Lessor holds the accrued balances and the security deposit	Lessor holds only the security deposit; the maintenance claim is unsecured	D6 §2.1
Redelivery below the return condition	Lessor keeps unclaimed balances and pursues any shortfall; lessee may restore before return	Lessee pays the agreed rate per hour and per cycle for life consumed below delivery condition	D6 §2.2; AerCap 20-F FY2013
Redelivery above the delivery condition	Balances are normally non-refundable; the lessee has pre-paid for life it did not consume	Lessor pays or credits the lessee for the excess life	D6 §2.1; AerCap 20-F FY2013
Who the regime suits	Lessees whose credit the lessor does not want to carry	Stronger-credit lessees; many leases mix the two by component	AerCap 20-F FY2013

Source: AerCap Holdings N.V., Form 20-F FY2013 (maintenance rents and end-of-lease compensation definitions); Spirit Airlines, Form 10-K FY2013 (lessee-side reserve accounting); mechanics assembled in D6 §2.1–2.2. Accessed 2026-10-03.

13.3 THE WORKED EXAMPLE

The research built one example of a full lease with reserves and a lease-end settlement. Its only sourced input is the rent. Every other input is an assumption chosen to show the arithmetic, and the shop-visit cost it assumed, \$4.5M, sits above every sourced range Part I carried, so the example below shows the sourced alternatives beside it. [D6 §2.2; reconciliation R-013, E.1 rule 4]

WORKED EXAMPLE 2.2 — A 12-YEAR-OLD 737-800 ON A SIX-YEAR OPERATING LEASE WITH MAINTENANCE RESERVES, AND THE SETTLEMENT AT THE END

Inputs. Rent \$228,000 a month (sourced: IBA, July 2025). Term 72 months (assumed). Utilization 250 flight hours (FH) and 140 flight cycles (FC) a month per engine (assumed; that is 1,680 cycles a year, a little above the 1,000–1,500 cycles a year Part I uses for a mature narrowbody from Aircraft Value News, and kept as the research set it). Performance-restoration reserve \$150 per flight hour per engine and life-limited-part reserve \$100 per flight cycle per engine (assumed; no sourced rates, D6 U7). One performance restoration on engine 1 in month 48 (assumed timing). Its cost: \$4.5M as the research assumed, shown beside the two consultancy ranges Part I carries for a restoration, \$1.8–2.5M (midpoint \$2.15M) and \$1.2–1.6M (midpoint \$1.4M), all from Safe Fly Aviation, a consultancy blog, 2026. At delivery both engines had flown 8,000 hours since their last restoration, and the return condition caps redelivery at 12,000 hours since restoration (both assumed).

Step 1, basic rent over the term.

$$228,000 \times 72 = \$16,416,000$$

Step 2, reserves accrued per engine over 72 months, then for two engines.

$$\begin{aligned} \text{restoration reserve: } & 150 \times 250 \times 72 = \$2,700,000 \text{ per engine} \\ \text{LLP reserve: } & 100 \times 140 \times 72 = \$1,008,000 \text{ per engine} \\ \text{two engines: } & 5,400,000 + 2,016,000 = \$7,416,000 \\ \text{reserves as a share of rent: } & 7,416,000 \div 16,416,000 = 45\% \end{aligned}$$

So on these assumptions the maintenance cash flow is not a side-show; it is almost half the size of rent.

Step 3, the month-48 shop visit on engine 1. The restoration balance held for that engine at month 48 is $150 \times 250 \times 48 = \$1,800,000$. The lessor reimburses the lesser of the balance and the qualifying cost.

$$\begin{aligned} & \text{at the research's assumed cost of } \$4,500,000: \text{ lessor reimburses} \\ & \quad 1,800,000; \text{ lessee funds } 2,700,000 \\ & \text{at the } \$2,150,000 \text{ midpoint of the } \$1.8\text{--}2.5\text{M range: lessor reimburses} \\ & \quad 1,800,000; \text{ lessee funds } 350,000 \\ & \text{at the } \$1,400,000 \text{ midpoint of the } \$1.2\text{--}1.6\text{M range: lessor reimburses} \\ & \quad 1,400,000; \text{ lessor keeps } 400,000 \end{aligned}$$

The three lines show the "lesser of" rule doing its work. At either of the higher costs the lessee pays the excess, because the reserve covers only the 48 months of life consumed under this lease, not the 8,000 hours consumed before delivery. At the lowest cost the reserve more than covers the visit and the lessor keeps the surplus. After month 48, engine 1 accrues restoration reserves again from zero.

Step 4, lease end on engine 2, which had no shop visit. It was delivered at 8,000 hours since restoration and flew $250 \times 72 = 18,000$ hours, so it would be at 26,000 hours since restoration. The return condition caps redelivery at 12,000. The lessee must therefore either perform a restoration before redelivery, claiming up to its \$2,700,000 balance against the cost, or negotiate a cash settlement. Under a pure reserve regime, if the lessee redelivers out of condition the lessor keeps the \$2,700,000 restoration balance and the \$1,008,000 LLP balance and pursues the shortfall. Under an end-of-lease compensation regime with no reserves paid, the lessee would owe compensation for 18,000 hours of restoration life consumed at an agreed rate per hour, plus compensation for $140 \times 72 = 10,080$ cycles of life-limited-part life at an agreed rate per cycle, in one lump at redelivery instead of monthly.

Step 5, what the lessor saw in cash over the lease, at the research's assumed visit cost.

$$\begin{aligned} & \text{rent received: } \$16,416,000 \\ & \text{reserves received: } \$7,416,000 \\ & \text{reimbursed: } -\$1,800,000 \\ & \text{net reserve retention: } 7,416,000 - 1,800,000 = \$5,616,000 \end{aligned}$$

That \$5.6M is not profit. It is the lessor's compensation for the maintenance life that walked out of the engines during the lease, to be spent, or forgone, when the lessor or the next lessee opens them. At the \$1.4M visit cost the retention would be \$6.0M, because less was reimbursed, and the lessee would have paid for more life than it used on engine 1.

Limits of the example: only the rent is sourced; the reserve rates, utilization, visit timing and return condition are assumed; the \$4.5M visit cost sits above every sourced range and the two consultancy ranges disagree with each other; the engine's life-limited-part reserve is never spent here because no life-limited part is replaced, which is the common case within a single six-year lease on parts with 20,000-to-30,000-cycle limits. Which set of life-limited-part prices a lessor uses to size that reserve, \$4M or \$5.7M for a full -7B set, is the tension Part I §4 carried (R-016). [D6 §2.2; IBA, Jul 2025; Safe Fly Aviation, 2026; Spirit Airlines 10-K FY2013; reconciliation R-013, R-016]

The example also shows why the maintenance condition at delivery is negotiated as hard as the rent. The lessee's exposure on engine 2 arose because the engine was delivered with 8,000 hours already consumed and the return condition was tighter than the delivery condition. A lessee who takes an engine fresh from a restoration and returns it at the same condition owes nothing; one who takes it part-worn and must return it fresh is buying the previous operator's consumption. In practice the delivery and return conditions are written to match, and the reserve or compensation covers the difference.

13.4 WHERE A LESSOR'S RETURN COMES FROM

A lease generates rent and reserves. The lessor's return is what is left after the asset's loss of value and the cost of the money that funded it. The research set out the decomposition in four lines.

THE FOUR LINES OF A LESSOR'S RETURN

Over the holding period a lessor earns (a) rental yield, meaning rent received, which is the lease rate factor times value, plus any net maintenance-reserve retention; minus (b) depreciation, the fall in the asset's value from purchase to disposal, or the book charge toward a residual; minus (c) financing cost, the interest on the debt that funds the asset and the cost of the equity; plus or minus (d) end-of-life proceeds, the sale price or part-out proceeds against the carrying residual, plus any end-of-lease compensation collected. An older asset has a higher lease rate factor and most of its depreciation behind it, but more maintenance exposure and a shorter remaining lease life, so line (d) matters more for it. In FTAI's structure line (d) is routed into the Aerospace Products segment: the 10 off-lease 737-700s bought from WestJet "will support FTAI's Aerospace Products business by expanding the Company's supply of CFM56-7B engines and modules" (Part VIII §67). [definition assembled in the research, D6 §2.1; FTAI release, 28 Sep 2026]

WORKED EXAMPLE 2.3 — DECOMPOSING A LESSOR'S RETURN INTO YIELD, DEPRECIATION, FINANCING COST AND RESIDUAL

Inputs, every one assumed for illustration and none of them market data: the aircraft is bought at value V ; the lease rate factor is 1.0% a month, so rent is 12% of V a year; straight-line depreciation is 3.5% of V a year toward a residual; debt funds 70% of V at 6.5%, so interest is $0.70 \times 6.5\% = 4.55\%$ of V a year; equity funds the other 30%. For a sense of scale only, the \$228,000 rent of worked example 2.2 at a 1.0% factor implies V of \$22.8M, which is not an appraised value (D6 U9).

rent yield: 12.00% of V
 less depreciation: -3.50% of V
 less interest: -4.55% of V
 net cash return on total capital: $12.00 - 3.50 - 4.55 = 3.95\%$ of V
 on the 30% equity slice: $3.95 \div 0.30 = 13.2\%$ of equity

Now change one input. At a lease rate factor of 0.8% a month, rent is 9.6% of V , and the net return falls to $9.6 - 3.5 - 4.55 = 1.55\%$ of V , or 5.2% on equity. At 1.2% a month it rises to $14.4 - 8.05 = 6.35\%$ of V , or 21.2% on equity. The leverage multiplies both directions: a 0.2-point move in the monthly factor moves the equity return by about eight points on these inputs.

Two further flows sit outside those lines. Net maintenance-reserve retention adds cash, \$5.6M over six years in worked example 2.2, which on V of \$22.8M is about 4.1% of V a year, but it is matched by the restoration cost the lessor will face before re-leasing; it is timing, not margin, unless the asset is parted out instead. End-of-life proceeds against the carrying residual add or subtract, and for an aircraft whose value sits mostly in its engines (§15) that line is largely the engines' value. Downtime between leases, remarketing costs and end-of-lease shortfalls subtract. Limits of the example: every percentage is assumed; FTAI's own depreciation policy and cost of debt by asset were not retrieved (D6 U8); Part VIII §58 works FTAI's implied gross yield from its filed balances and Part XI §88 sets it beside Willis Lease and Air Lease. [D6 §2.2 step 6; D6 §2.7]

The structure of that return explains the behaviour of the specialists this primer describes. A new-aircraft lessor earns a low factor on a slowly depreciating asset with long leases and little maintenance exposure. A mid-life engine lessor earns a higher factor on an asset whose value is close to the sum of its modules, with reserves nearly as large as rent and a part-out decision at every lease end. Section 14 puts filed numbers on both, and Part XI §88 sets FTAI's beside them.

14. Engine leasing: spare ratios, short-term leases, lease-rate factors

Everything in §12 and §13 applies to an engine leased on its own, with two differences. An engine is a smaller, more mobile asset than an aircraft, so its leases can be as short as a few months. And an engine's value is dominated by its maintenance condition, so the lease-end decision is sharper: restore and re-lease, lease out the remaining life, or take it apart. This section

explains why airlines lease engines at all, how the leases are priced, and what the one listed pure-play engine lessor's numbers look like.

14.1 WHY AIRLINES LEASE SPARE ENGINES

An airline needs more engines than it has engine positions, because engines come off the wing for scheduled shop visits and for unscheduled removals. Part I §6.4 defined the spare engine and the spare ratio, spares as a percentage of installed engines, and quoted the statement of the German engine maker MTU Aero Engines (MTU) that airlines "used to have about 15% of spare engines of their installed engines, but this figure is now down to 10% and is expected to further decrease to about 7–8% for newer engine types". [MTU AEROREPORT, undated] A consultancy argues the 10% rule "is no longer sufficient because engines are coming off wing more often and taking longer to repair". [Acumen Aviation, undated (consultancy blog)]

The economic reason to lease rather than own is that a spare earns nothing while it waits. Airlines "are less inclined to keep as many owned spare engines as they used to, mainly because of the significant investment involved in assets that generate very little return". [MTU AEROREPORT, undated] Leasing converts a lumpy multi-million-dollar purchase into a monthly cost, and in a lessor's pool one spare can serve several operators in turn. The sharpest form of the need has its own name.

AIRCRAFT ON GROUND (AOG)

An aircraft on ground is one that cannot fly for want of a serviceable part or engine. Short-term engine leases exist to cover AOG events and planned shop visits; a lessor that can deliver an engine within days is selling the airline the difference between a grounded aircraft and a flying one. [MTU AEROREPORT, undated; FlightGlobal, "Freedom to be flexible", undated]

14.2 SHORT-TERM AND LONG-TERM LEASES, AND SHOP-VISIT COVER

Part I §6.4 defined the two durations. A short-term engine lease runs "a period of a few months to a maximum of three years" and is "ideal if a replacement engine is required for the duration of a shop visit"; a long-term lease runs for years and provides a permanent spare. [MTU AEROREPORT, undated; FlightGlobal, undated] The short-term lease is the one that ties engine leasing to the shop-visit market Part IV describes. A CFM56 restoration takes an engine away for 75 to 120 days on the figures Part I §6.3 carried, and when shops are full the wait for a place in the queue adds two to three months, up to six. [Safe Fly Aviation, 2026 (consultancy blog); Aviation Business News, 2025; GE Aerospace, 27 May 2026; Bain, 2024; reconciliation R-019]

INDUCTION SLOT

An induction slot is the scheduled date on which an engine shop commits to start work on an engine; induction is the acceptance of the engine into work. When shops are full the wait for a slot is additional to the turnaround time and, in 2024, became the binding constraint, adding two to three months and up to six. Part IV §27 carries the full treatment. [Bain, 2024; D2 §2.1, §2.4]

When slots are delayed, "operators often turn to short-term or green-time engine leasing". [MTU AEROREPORT, undated] A green-time engine, which Part I §8.2 defined and §16 of this Part carries, is one with limited life left before its next shop visit, leased out to burn that life rather than be restored; its rent is lower. The lessor's product is therefore time: the months between removal and return, priced per month. FTAI's offer sits beside it as a different answer to the same problem.

MAINTENANCE, REPAIR AND EXCHANGE (MRE)

Maintenance, Repair and Exchange is FTAI's name for its offer, a deliberate contrast with the industry's maintenance, repair and overhaul (MRO). Instead of restoring a customer's engine part by part while the customer waits, or leasing the customer a spare for the duration, FTAI removes the module or engine that triggered the visit, installs a serviceable one from its inventory, tests and returns the engine, and takes the unserviceable unit as feedstock. It is a variant of shop-visit cover in which the exchange replaces the spare. Part VII §46 carries the full treatment. [FTAI 10-K FY2025; D5 §2.2; D6 §2.3]

WORKED EXAMPLE 2.4 — HOW MANY ENGINES A FLEET HAS IN THE SHOP, AND WHAT THE SPARE RATIO MUST COVER

Inputs: a fleet of 100 Boeing 737-800s, each with two CFM56-7B engines, so 200 installed engines (assumed fleet; two engines per aircraft sourced in Part I). Spare ratios of 15%, 10% and 7–8% (sourced: MTU). A mature performance-restoration interval of 10,000–15,000 cycles (sourced: Leeham News, 8 Mar 2024) at 1,000–1,500 cycles a year (sourced: Aircraft Value News), which is between 6.7 and 15 years between restorations; an average of ten years (assumed, close to the roughly ten years implied by 2,300 visits a year on about 24,000 engines, derived in the research) with eight years (Aviation Week's rule of thumb) and six years (the assumption one dossier used and asked to have replaced) as alternatives. Turnaround of 90 days (sourced: GE Aerospace, May 2026) or 120 days (sourced: trade press, 2025), with a slot wait of 60–90 days (sourced: Bain, 2024).

spares at 15%: $0.15 \times 200 = 30$; at 10%: 20; at 7–8%: 14–16
 scheduled removals a year at a 10-year interval: $200 \div 10 = 20$ (8 years: 25; 6 years: 33)
 engines away at any time = removals $\times$ days away $\div$ 365
 10-year interval, 90 days: $20 \times 90 \div 365 = 4.9$
 10-year interval, 120 days: $20 \times 120 \div 365 = 6.6$
 10-year interval, 120 days plus a 90-day slot wait: $20 \times 210 \div 365 = 11.5$
 8-year interval, 90 to 210 days: 6.2 to 14.4
 6-year interval, 90 to 210 days: 8.1 to 19.0

So on the ten-year interval, scheduled restorations alone keep between five and twelve of the fleet's 200 engines away at any moment, and the rest of a 20-engine spare pool (10%) is what covers unscheduled removals and grounded aircraft. Lengthen the time away from 90 to 210 days and the scheduled requirement alone more than doubles. Shorten the interval to the six years one dossier assumed and the requirement at 210 days nearly consumes the whole 10% pool before any unscheduled event. That is the arithmetic behind the consultancy's claim that 10% "is no longer sufficient", and behind the demand for short-term leases when slots are delayed. Limits of the example: the interval, the fleet and the averaging are assumed; unscheduled removal rates were not sourced; an engine exchange from inventory, which aims at a quick-turn of about 45 days, would cut the days-away figures by half or more and is the mechanism Part VII §55 works through. [MTU AEROREPORT, undated; Leeham News, 8 Mar 2024; Aircraft Value News, undated; GE Aerospace, 27 May 2026; Aviation Business News, 2025; Bain, 2024; Aviation Week, "Retirement Uptick", date not captured; reconciliation R-008, R-019; derived in D9 T1]

14.3 LEASE RATES, VALUES AND THE LEASE-RATE-FACTOR ARITHMETIC

The research reached two lease rates for a CFM56-7B that do not agree, and two kinds of value that sit on different bases. Part I §8.2 set them out in Exhibit 1.11 as values; this Part sets them out as rents and factors. The rule the primer follows is that both figures appear together with their sources, that no lease rate factor is stated as a single number, and that every derived factor carries its basis. [reconciliation R-028, R-029, R-030; E.1 rule 6]

EXHIBIT 2.3 — CFM56 AND V2500 (THE A320CEO'S ALTERNATIVE ENGINE, PART I §10) LEASE RATES AND VALUES, WITH THE LEASE RATE FACTOR EACH PAIR IMPLIES AND THE BASIS ON WHICH IT SITS

Quantity	Figure	Basis	Source and level	Date
CFM56-7B spare-engine rent	~\$100,000 a month, from ~\$75,000 in 2019 (+33%)	Spare engine; condition not stated	IBA (appraiser), via AviTrader and AJOT	Apr 2024

Quantity	Figure	Basis	Source and level	Date
CFM56-7B rent	\$42,000–48,000 a month; "about 34% above 2019"; up 15–20% "in recent months"	Quoted beside green-time values; may be green-time; basis not stated	Safe Fly Aviation (consultancy blog)	2026
CFM56-5B rent	\$38,000–44,000 a month	Same	Safe Fly Aviation	2026
V2500-A5 rent	\$70,000–80,000 a month; "slightly higher" than the -5B in 2024	Typical	IBA, Engine Values 2025B; IBA Apr 2024	Sept 2025; Apr 2024
CFM56-7B24 / -7B27 market value	\$5.7M / \$6.4M	Half-life, H2 2025	IBA, 2025B (appraiser)	Sept 2025
CFM56-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", internally inconsistent	Safe Fly Aviation	2026
Run-out CFM56 as teardown feedstock	\$0.8–1.2M	Exhausted engine	Safe Fly Aviation	2026
Value direction	"Levelled off after approximately 18 months of increases"	-5B and -7B market values	IBA H1 2026 update	H1 2026
Implied factor: IBA rent on IBA value	1.56% a month (-7B27); 1.75% (-7B24)	Rent 2024, value H2 2025; dates mixed	Derived in this Part	n/a
Implied factor: Safe Fly rent on IBA value	0.66–0.75% (-7B27); 0.74–0.84% (-7B24); 0.70% at the \$45,000 midpoint	Possibly green-time rent on half-life value; bases mixed	Derived in D3 §2.3 and this Part	n/a
Implied factor: Safe Fly rent on Safe Fly value	1.24–1.71% a month	Both possibly green-time; label inconsistent	Derived in this Part	n/a
Willis Lease portfolio, derived	~1.0% a month on the on-lease portion	\$291.6M rent ÷ \$2,801.7M equipment ÷ 84.9% utilisation	Derived in D11 §2.2 from WLFC 10-K FY2025	FY2025

Quantity	Figure	Basis	Source and level	Date
FTAI leasing book, derived	0.87% a month on cost, 1.1% on net book value (1H 2025 annualised); 0.55% and 0.73% on 1H 2026 income	Lease income ÷ leasing equipment; mixed aircraft and engines	Derived in D6 §2.7; Part VIII §58	FY2025 / 1H 2026
Assumption used elsewhere	0.9% a month	Assumed for a fee example	D7 §5.1; Part VIII §66	n/a

Source: IBA, "It's a lessors' market", Apr 2024, via AviTrader, 25 Apr 2024, and AJOT; IBA, Engine Values Release 2025B, Sept 2025; IBA, Engine and Lease Rate Update H1 2026; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); WLFC 10-K FY2025; FTAI 10-K FY2025 and 10-Q Q1 2026. Derivations in D3 §2.3, D6 §2.7, D11 §2.2 and this Part. Reconciliation R-028, R-029, R-030. Accessed 2026-10-03.

The two rents are a direct contradiction at face value and possibly a difference of definition. An appraiser quoted about \$100,000 a month in April 2024 for a spare engine; a consultancy blog quoted \$42,000 to \$48,000 in 2026 on a page that also quotes "green-time values", and the research noted that the lower rent may describe green-time engines although the source does not say so. [IBA, Apr 2024; Safe Fly Aviation, 2026; D6 T4] The two sources agree on one thing, the change since 2019: IBA's \$75,000 to \$100,000 is a rise of a third, and the blog's rate is "about 34% above 2019". [reconciliation B-5] They disagree on the level, and the primer carries both; they also disagree on direction, the appraiser's "levelled off" against the blog's rents up 15–20% "in recent months", which §16.4 sets out. [IBA, H1 2026; Safe Fly Aviation, 2026; reconciliation R-029]

WORKED EXAMPLE 2.5 — THE LEASE-RATE-FACTOR ARITHMETIC FOR A CFM56-7B SPARE ENGINE, ON BOTH RENTS

Inputs: rent of about \$100,000 a month (sourced: IBA, April 2024) or \$42,000–48,000 a month (sourced: Safe Fly Aviation, consultancy blog, 2026; may be green-time). Half-life market value \$6.4M for a -7B27 and \$5.7M for a -7B24 (sourced: IBA, September 2025). "Green-time value" \$2.8–3.4M (sourced: Safe Fly Aviation, 2026; label internally inconsistent). The research's own derivation uses the \$45,000 midpoint (D3 §2.3).

$$\begin{aligned}
 &\text{IBA rent on IBA value:} \\
 &100,000 \div 6,400,000 = 1.56\% \text{ a month} = 18.8\% \text{ a year} \quad (-7B27) \\
 &100,000 \div 5,700,000 = 1.75\% \text{ a month} = 21.1\% \text{ a year} \quad (-7B24) \\
 &\text{Safe Fly rent on IBA value:} \\
 &42,000 \div 6,400,000 = 0.66\%; \quad 48,000 \div 6,400,000 = 0.75\% \quad (-7B27) \\
 &42,000 \div 5,700,000 = 0.74\%; \quad 48,000 \div 5,700,000 = 0.84\% \quad (-7B24) \\
 &45,000 \div 6,400,000 = 0.70\% \text{ a month} = 8.4\% \text{ a year} \quad (\text{the research's figure}) \\
 &\text{Safe Fly rent on Safe Fly "green-time value":} \\
 &42,000 \div 3,400,000 = 1.24\%; \quad 48,000 \div 2,800,000 = 1.71\% \text{ a month}
 \end{aligned}$$

Read line by line. The first pair mixes a 2024 rent with a late-2025 value, and a spare engine's maintenance condition is not stated, so 1.56–1.75% is indicative, not measured; it is the highest factor in this primer. The second group mixes what may be a green-time rent with a half-life value, which the research itself flagged as a mixed basis, and it gives the lowest factors, 0.66–0.84%. The third pair puts the blog's rent on the blog's own value and gives 1.24–1.71%, close to the first pair, which is what one would expect if both the rent and the value describe a part-worn engine: a cheaper engine renting for less at a similar factor. None of this resolves the contradiction between the two rents; it shows that the factor depends entirely on which rent is set against which value. For comparison, Willis Lease's derived portfolio factor is about 1.0% a month on the on-lease portion of its book, and FTAI's derived book factor is 0.87–1.1% a month on 2025 income, both on mixed books of engines and aircraft at varying condition. Limits of the example: every input is contested or mixed in date or basis; no stated lease rate factor exists in the research for any party (D6 U5; D11 U1); and the V2500-A5 rent of \$70,000–80,000 cannot be turned into a factor because no V2500 value was retrieved. [IBA, Apr 2024; IBA, Sept 2025; Safe Fly Aviation, 2026; derived in D3 §2.3, D6 §2.7, D11 §2.2; reconciliation R-028, R-029, R-030]

GROSS LEASING YIELD

Gross leasing yield is lease income for a period divided by the average leasing equipment at cost over the period, or at net book value (cost less accumulated depreciation). It is the filed-accounts counterpart of the lease rate factor: a company does not state its factor, but its income statement and balance sheet give the yield, and dividing by twelve gives a factor on book rather than on market value. Willis Lease's FY2025 figure is about 10.4% on year-end equipment, or 12.3% on the on-lease portion; Air Lease's, a new-aircraft lessor's, is about 9.0% (\$2,615.4M of rental revenue on \$29.1B of flight equipment at 31 December 2025); FTAI's FY2025 figure is about 10.4% on average cost and 13.4% on average net book value using first-half 2025 income annualised. All three are derived in the research on year-end or average book rather than market value, so they approximate the companies' own measures. Part VIII §58 and Part XI §88 reuse the measure. [derived in D11 §2.2 and D6 §2.7; WLFC 10-K FY2025; Air Lease 10-K FY2025; FTAI 10-K FY2025]

14.4 WILLIS LEASE AS THE REFERENCE

Willis Lease Finance Corporation is the only US-listed company whose primary business is engine leasing, which makes its filings the closest public analogue to an engine book like FTAI's. [D11 §3.1] It owns engines and a small number of aircraft, leases them to airlines and maintenance shops, collects maintenance reserves, sells engines and parts out of the portfolio, and manages engines for third parties. Its numbers appear in this primer at two dates, a quarter-end and a year-end, and the two are not contradictions but different measures; both are quoted because both will be used. [reconciliation R-032]

EXHIBIT 2.4 — WILLIS LEASE FINANCE: THE ENGINE-LESSOR METRICS AT TWO DATES

Metric	At or for 30 Sep 2025 (quarter)	At or for 31 Dec 2025 (year)	Note	Source
Utilisation	86.0% (82.9% a year earlier)	84.9% average for 2025 (82.9% for 2024)	Point-in-time against period average; by book value on lease	WLFC Q3 2025 release; WLFC 10-K FY2025
Lease portfolio	\$2,888.5M, of which \$2,700.4M equipment in operating leases, \$144.8M notes receivable, \$27.0M maintenance rights, \$16.3M sales-type leases	\$2,801.7M equipment held for lease; \$139.9M notes receivable; \$30.6M maintenance rights; \$16.6M sales-type leases	"Maintenance rights" is a balance-sheet line the research did not define (Part XI §87)	Same
Including joint ventures	\$3,302.6M	Not returned		WLFC Q3 2025 release
Owned assets	Not returned	363 engines, 20 aircraft, one marine vessel; 69 lessees in 37 countries		WLFC 10-K FY2025
Third-party managed	Not returned	116 engines and related equipment	The analogue of FTAI's servicing of its vehicles (Part VIII)	WLFC 10-K FY2025
Lease rent revenue	\$76.6M in Q3 (+17.9%, a record)	\$291.6M (+22.4% from \$238.2M)		Same
Maintenance reserve revenue	\$76.1M in Q3 (+52.8%, a record)	\$232.0M, derived as \$523.6M combined line less \$291.6M rent	Reserves recognised when no longer refundable	Same; derived in D11 §2.2
Core lease rent plus maintenance reserve revenue	Not returned	\$523.6M (+15.8% from \$452.1M)	Reserves are 44% of the combined line	WLFC 10-K FY2025
Gain on sale of leased equipment	Not returned	\$54.0M (+19.9%)	Where the sum-of-modules value is realised	WLFC 10-K FY2025

Metric	At or for 30 Sep 2025 (quarter)	At or for 31 Dec 2025 (year)	Note	Source
Total revenue	Not returned	\$730.2M (+28.3%)		WLFC 10-K FY2025
Lease rate factor	Not disclosed	Not disclosed; ~1.0% a month derived	D6 U5; D11 U1	Derived in D11 §2.2

Source: Willis Lease Finance Corp., Q3 2025 earnings release (8-K exhibit 99.1, Nov 2025) and Form 10-Q for the quarter ended 30 Sep 2025; Willis Lease Finance Corp., Form 10-K FY2025. Derivations in D11 §2.2. Reconciliation R-032. Accessed 2026-10-03.

Three things in that table carry into the rest of the primer. First, utilisation is a defined term and the definitions differ by company: Willis measures the share of the lease portfolio, by book value, that is on lease over the period; FTAI measures the percentage of days on lease in the quarter, weighted by the monthly average equity value of its leasing equipment and excluding airframes. The two are not comparable without saying so, and Part VIII §58 and Part XI §88 say so where they set 84.9% beside FTAI's roughly 77%. [WLFC 10-K FY2025; FTAI 10-K FY2025; registry C-3] Second, maintenance reserve revenue at a mid-life engine lessor is of the same order as rent, 44% of the combined line for the year and above rent in one quarter, because the assets are near their shop visits; this is the §13 mechanism showing up in a filing. Third, the gain on sale, \$54.0M in 2025, is the line on which a lessor realises the sum-of-modules value Part I §8 described, and Part V §32 carries what a part-out yields. [WLFC 10-K FY2025; D11 §2.2]

Willis funds itself through a series of rated engine securitizations, the Willis Engine Structured Trust (WEST) deals, which §17 uses as the sourced template for how an engine portfolio is financed. Its lease rate factor, its split between CFM56 and newer engine types, and its spare-parts revenue line were searched for and not returned (D11 U1, U3; D6 U5). [D11 §3.1]

14.5 THE OTHER ENGINE LESSORS

The research named the other participants and retrieved scale for few of them. The list is carried as the research left it, with what was and was not found.

EXHIBIT 2.5 — ENGINE LESSORS NAMED IN THE RESEARCH, AND WHAT WAS RETRIEVED FOR EACH

Lessor	Ownership or affiliation	What the research retrieved	Source
Willis Lease Finance	Listed pure-play	363 engines, 20 aircraft, 116 managed, 84.9% utilisation, \$291.6M lease rent (FY2025); Exhibit 2.4	WLFC 10-K FY2025

Lessor	Ownership or affiliation	What the research retrieved	Source
GE Engine Leasing	GE Aerospace, co-owner of CFM International, the CFM56's original equipment manufacturer (OEM)	A 2019 trade report: "has seen an explosion of financing based on spare engines"; engine leasing "a key part of full-service package deals" from the engine makers; no current scale	AviTrader MRO e-magazine, Jul 2019
AerCap Engines; Shannon Engine Support (SES)	AerCap; SES is AerCap's 50% joint venture with Safran Aircraft Engines	470 owned engines, 9% of AerCap's fleet, at 31 Dec 2025 (20-F); "over 1,200 engines" including engines owned and managed by SES at 30 Sep 2025 (6-K); \$155M of lease rental income from SES in nine months of 2025	AerCap 20-F FY2025; AerCap 6-K Q3 2025
SMBC Aero Engine Lease (SMBC)	Sumitomo Mitsui group, affiliated with SMBC Aviation Capital	Scale not retrieved	D6 §3
Engine Lease Finance (ELF)	Mitsubishi HC Capital; Shannon-based	Scale not retrieved	D6 §3
Rolls-Royce & Partners Finance (RRPF)	Joint venture of Rolls-Royce and GATX Corporation (GATX)	2019 report: market expansion from "airlines becoming more open to spare engine leasing, driven by lower lease rentals in an increasingly competitive market"; no current scale	AviTrader, Jul 2019
MTU Maintenance Lease Services	The leasing arm of MTU's maintenance business	"Short-term leasing, stand-by arrangements, engine pooling, as well as asset management"; no scale	MTU AEROREPORT, undated
FTAI Aviation	Listed; Aviation Leasing segment	243 engines at 31 Dec 2025, 143 on lease; Part VIII §58	FTAI 10-K FY2025
Top five engine lessors together	n/a	"Over 1,000 powerplants with a book value in excess of \$5 billion"	AviTrader, Jul 2019 (dated)

Source: WLFC 10-K FY2025; AviTrader MRO e-magazine, July 2019 (via AJW Group); AerCap Holdings N.V., Form 20-F FY2025 and Form 6-K for the quarter ended 30 Sep 2025; MTU AEROREPORT, "How airlines benefit from engine leasing", undated; FTAI 10-K FY2025. Scale for the unlisted lessors was not retrieved (D6 U14). Accessed 2026-10-03.

Two cautions attach to the table. AerCap's engine count is 470 or more than 1,200 depending on whether the figure is owned engines only or owned plus managed plus the Shannon joint

venture's, and the two dates are a quarter apart; the primer writes "470 owned" unless it states the wider basis. [AerCap 20-F FY2025; AerCap 6-K Q3 2025; reconciliation R-033] And the only figure for the group as a whole, more than 1,000 engines and more than \$5 billion of book value among the top five, is from 2019 and is carried as dated. [AviTrader, Jul 2019; D6 U14] The point that survives the gaps is structural: the engine makers themselves, the largest aircraft lessor, two bank-affiliated specialists, an independent repair shop and one listed pure-play all lease spare engines, and the pure-play's filings are the only place the economics are laid out in full.

15. Mid-life narrowbody economics: why the engines carry the value

A mid-life narrowbody is the aircraft at the centre of this primer: the Boeing 737NG and the Airbus A320ceo, powered by the CFM56-7B and CFM56-5B that Part I described, in the second half of their service lives. Part I §4 boxed the term with both of the research's framings, and they are consistent: roughly 10 to 18 years old (one dossier), or roughly the middle of a 25-to-30-year economic life (another), when an increasing share of the aircraft's value sits in its engines and those engines are approaching or past a shop visit. [D6 §2.4; D7 §2.4; registry C-13] This section explains what has happened to those aircraft's values and rents since 2023, why they are bought with leases attached, why the engines carry the value, and what happens to the airframe at the end.

15.1 VALUES AND LEASE RATES, 2023 TO 2026

The research reached commentary and lease rates from one appraiser at several dates, a lessor's filing, and a data provider's remarks via the financial press. It did not reach an appraised half-life value for a 12-to-15-year-old 737-800 or A320ceo at any date (D6 U9). The sourced points are set out with their dates because the direction changed within the period.

EXHIBIT 2.6 — MID-LIFE NARROWBODY VALUES AND LEASE RATES, 2023 TO 2026: EVERY SOURCED POINT WITH ITS DATE AND LEVEL

Date	Statement or figure	Asset	Source and level
Context 2023–24	75–90% of scheduled lease expiries being extended; only about a dozen mid-life 737NGs and fewer than 25 A320ceos available to rent worldwide	Mid-life narrowbodies	Cirium Ascend (Rob Morris) via Bloomberg/AJOT (data provider via financial press; date not visible)
Context 2023–24	Lease rates on 10-year-old 737-800s 44% above January 2022 levels	10-yr 737-800	Cirium Ascend via Bloomberg/AJOT

Date	Statement or figure	Asset	Source and level
Nov 2023	Extension terms stretched to 4–6 years; extension rents in some cases >40% above the prior age-specific rate; aircraft due back at year 12 not expected until 2032, at age 18	Legacy single-aisles	IBA Weekly, 24 Nov 2023; Key.Aero (appraiser)
Feb 2024	"Values and leases would continue to soar throughout the year, attributed to tight supply driven by lease extensions, engine reliability issues, and a lack of new aircraft deliveries"	Narrowbodies	IBA, Feb 2024 update (appraiser), as summarised in search
Apr 2024	CFM56-7B saw the largest market-value gain among engines covered, "around 20% from 2023 to 2024"; "a lessor's market"	-7B engine	IBA via AviTrader, 25 Apr 2024 (appraiser)
Jul 2025	12-year-old 737-800 about \$228,000 a month (–11% year on year); comparable A320-200 about \$220,000 (–13%); 21-year-old 737-800s "less than \$160,000"; older end "comfortably sub-\$200k"	Mid-life and old narrowbodies	IBA, Jul 2025, via India Seatrade News and AeroIn (appraiser)
Jul 2025	"Greater availability in the secondary market and softening operator demand" pressing lease rates down; "strong demand for engines and component scarcity should support base aircraft values despite the weaker leasing environment"	Narrowbodies	IBA, Jul 2025 (appraiser)
FY2025 (filed early 2026)	Strong demand "helped increase lease rates on new lease agreements and lease extensions during 2025. Lease rates are expected to remain strong as the supply and demand environment for commercial aircraft remains tight"	Air Lease's fleet (new-technology, average age 4.9 years)	Air Lease 10-K FY2025 (filing)
FY2025 (filed early 2026)	"Both Airbus and Boeing have delivery delays which have been compounded by engine manufacturer delays, shorter on-wing engine time of most new technology engines, and ongoing supply chain challenges"	Engines	WLFC 10-K FY2025 (filing)

Date	Statement or figure	Asset	Source and level
H1 2026	-5B and -7B values "have levelled off after approximately 18 months of increases"; "early signs of stabilisation in parts of the narrowbody segment"	Engines	IBA H1 2026 update (appraiser)
6 Sep 2026	IBA "warns of oversupply of narrowbody conversions and lease rate fall"	Narrowbodies; freighter conversions	IBA via India Seatrade News (headline)
2025 (date not captured)	Freighter values rising while narrowbody passenger lease rates receded	Freighters against passenger narrowbodies	IBA via Aeroin (appraiser via trade press)

Source: Bloomberg via AJOT, "Used jets rents surge 44% as airlines pay up to buoy fleets"; IBA Weekly, 24 Nov 2023; Key.Aero, "Lease trends"; IBA, "Aircraft values and lease rates update, February 2024"; AviTrader, "IBA says it's a lessor's market", 25 Apr 2024; India Seatrade News, "IBA warns of oversupply of narrowbody conversions and lease rate fall", 6 Sep 2026; Aeroin, 2025; Air Lease Corp., Form 10-K FY2025; Willis Lease Finance Corp., Form 10-K FY2025; IBA, Engine and Lease Rate Update H1 2026. Reconciliation R-031. Accessed 2026-10-03.

The table holds a divergence the primer carries rather than resolves. A new-aircraft lessor's filing for 2025 says lease rates rose during the year and are expected to remain strong; an appraiser's mid-life commentary from July 2025 has 12-year-old 737-800 rents down 11% year on year and expects more softening; and the same appraiser warned in September 2026 of conversion oversupply and falling rates. [Air Lease 10-K FY2025; IBA, Jul 2025; IBA via India Seatrade News, 6 Sep 2026; reconciliation R-031] The sources describe different asset ages, Air Lease's fleet averaging 4.9 years against the appraiser's 12-to-21-year-old aircraft, and different dates, so the divergence may be a difference of segment and timing rather than a contradiction. [D11 §3.3; D9 §2.8] The appraiser's own July 2025 sentence contains the mechanism this section is about: rents on the whole aircraft were softening while "strong demand for engines and component scarcity" supported the aircraft's base value. The two forces pull a mid-life lessor's rent and its engine-exit value in opposite directions. [IBA, Jul 2025; D6 §5]

Behind the rent movements sits the extension behaviour §12.5 defined. If 100 leases on A320ceos expire in a year and 75 to 90 are extended, only 10 to 25 aircraft come back to lessors for transition, sale or part-out, and that is the whole flow into the used market from that cohort; with fewer than 25 A320ceos available to rent worldwide at the time of Cirium's count, the stock of unplaced aircraft was of the same order as one year's returns from a 100-aircraft cohort, which is the arithmetic behind a 40%-plus rent move. [derived in D9 §2.8; Cirium Ascend via Bloomberg/AJOT] Part III §22 carries the extension story in full and Part III §24 turns it into a row of the translation table.

15.2 WHY MID-LIFE AIRCRAFT ARE BOUGHT WITH A LEASE ATTACHED

A buyer of a 12-year-old aircraft almost always buys it with a lease already on it. There are two routes. In the first, the operating airline sells aircraft it owns to a lessor and rents them straight back, a sale-leaseback, which §17.7 defines; FTAI's September 2026 WestJet transaction, in which a vehicle bought 17 Boeing 737-700s "on lease to WestJet in a sale-leaseback", is the example in the research and belongs to Part VIII §67. [FTAI release, 28 Sep 2026] In the second, lease-attached trading (§12.6), the buyer takes the aircraft from another lessor with its lease transferred. [D6 §2.4]

The lease is what makes the asset financeable. Lenders and securitization investors underwrite contracted rent from a known airline for a known term, and the buyer's return over that term is the decomposition of worked example 2.3. [D6 §2.4] At the end of the term the owner has three choices: re-lease the aircraft, convert it to a freighter, or part it out. The value of that choice is mostly the engines, and the next subsection shows why with the one sourced illustration the research built.

15.3 WHY THE ENGINES CARRY THE VALUE

WORKED EXAMPLE 2.6 — TWO CFM56-7B RENTS AGAINST A 737-800 RENT, ON BOTH ENGINE RENTS

Inputs: a CFM56-7B spare engine rented for about \$100,000 a month in 2024 (sourced: IBA, April 2024), or for \$42,000–48,000 a month in 2026 (sourced: Safe Fly Aviation, consultancy blog; may be green-time); a complete 12-year-old 737-800 rented for about \$228,000 a month in July 2025 (sourced: IBA). A 737-800 carries two engines (sourced in Part I).

on the IBA engine rent:
 two engines: $2 \times 100,000 = \$200,000$ a month
 share of the aircraft's rent: $200,000 \div 228,000 = 88\%$
 left for airframe, landing gear, APU and everything else: $228,000 - 200,000 = \$28,000$ (12%)
 on the Safe Fly engine rent:
 two engines: $2 \times 42,000$ to $2 \times 48,000 = \$84,000$ to $\$96,000$ a month
 share of the aircraft's rent: 37% to 42%
 left for everything else: $\$132,000$ to $\$144,000$ (58% to 63%)

On the appraiser's engine rent, two engines on standalone leases would command 88% of what the whole aircraft commands, and everything else about 12%. On the consultancy blog's rent the engines are 37–42% and the rest 58–63%. The primer carries both. The first calculation mixes an April 2024 engine rent with a July 2025 aircraft rent, and the appraiser's own H1 2026 update says engine values "levelled off", so the 88% is indicative rather than measured; the research flagged exactly this (D6 T5). The second calculation uses a rent that may price a part-worn engine, which would understate what a full-life spare earns. What neither calculation disputes is the direction: a pair of engines earns

a large share of a mid-life narrowbody's rent on either figure, and a pair of fresh ones earns most of it on the appraiser's.

The same comparison in values cannot be completed from the research. Part I's worked example 1.7 put a full-life -7B27 at about \$10.3M on one set of inputs or \$9.1M on the alternative, so a pair of fresh engines is \$18.2–20.6M on those figures; no appraised value for a 12-to-15-year-old 737-800 was retrieved to set beside it (D6 U9). The only per-aircraft dollar figures in the research are the capital per aircraft in FTAI's 2025 vehicle, which range from \$11.9M (the seed portfolio, \$549M over 46 aircraft) to \$23.6M (\$2.1B over 89 aircraft under contract), each derived in the research from a different disclosure and none an appraisal; whether "total capital" includes reserves is unstated. The juxtaposition shows the order of magnitude and stops there. Limits of the example: dates and bases are mixed on both sides; the engine rents are a contested pair; the aircraft value is unknown. [IBA, Apr 2024; IBA, Jul 2025; Safe Fly Aviation, 2026; Part I worked example 1.7; derived in D7 §2.3; reconciliation R-028, R-056; D6 T5]

WHY THE ENGINES CARRY THE VALUE

Four facts from Part I combine. The engines are separable from the airframe and from each other at their module flanges, so they can be sold or leased on their own. Their value is dominated by maintenance condition, so a pair fresh from restoration with full life-limited parts is worth several million dollars more than the same pair run-out. No new CFM56 is being built for an airframe, so every serviceable -7B and -5B that will ever exist already exists, while the airframes they power are being replaced. And the airframe at 15 to 20 years has few other uses, which is why the 21-year-old 737-800 rents for "less than \$160,000" against \$228,000 at 12 years while a -7B rents for the same whether it hangs on a 12-year-old wing or a 21-year-old one. The consequence FTAI states in its own words is that it "owns and leases jet aircraft which often facilitates the acquisition of engines at attractive prices", and the consequence for its third-party vehicles is that the vehicles buy the aircraft and FTAI does the engine work (Part VIII §61). [Part I §3, §8; IBA, Jul 2025; FTAI Q2 2026 release, 29 Jul 2026; D6 §2.4]

The floor under an engine's value is what it fetches taken apart. A consultancy blog puts the price of a run-out CFM56 bought as teardown feedstock at \$0.8–1.2M and the used material it yields at \$1.6–2.2M, and the same blog's "green-time value" of \$2.8–3.4M for a -7B is a floor for engines with little life left; a full-life engine is worth substantially more. [Safe Fly Aviation, 2026 (consultancy blog)] So even at the end of an airframe's life, a pair of run-out -7Bs fetches \$1.6–2.4M as feedstock and yields \$3.2–4.4M of used material on those figures, before counting any life they still have. Section 16 takes up the trade in engines between those two states.

15.4 FREIGHTER CONVERSION AS AN END-OF-LIFE PATH

A passenger 737-800 at 18 to 25 years has one more use before part-out.

FREIGHTER CONVERSION, OR PASSENGER-TO-FREIGHTER (P2F)

Freighter conversion is the structural modification of a used passenger airframe into a cargo aircraft: a main-deck cargo door, a strengthened floor, a rigid barrier between the cockpit and the hold, and a cargo loading system. It extends the airframe's revenue life by 10 to 15 years at low cycles a year. Narrowbodies are converted at 21–22 years old on average; more than 260 Boeing 737-800 freighters were in service in mid-2025 with a pipeline exceeding 300. The conversion cost and the availability of conversion slots were not retrieved (D6 U10). [Cargo Facts, "Passenger-to-freighter conversions are taking place on increasingly older aircraft"; Acumen Aviation, 18 Dec 2025; Cargo Facts, "In-service 737NG freighter fleet surpasses 100"; D6 §2.4]

SUPPLEMENTAL TYPE CERTIFICATE (STC)

A Supplemental Type Certificate is a regulator's approval of a change to a certified aircraft design. A freighter conversion is performed under an STC owned by the conversion house, which is why conversion capacity belongs to a few STC holders. In Part VI §37 the same instrument is one of the approval bases for alternative parts. [D6 §2.4; D4 §2.2]

Conversion changes what the engines are asked to do. A freighter flies few cycles a year, so cargo operators want engines "customized for cargo": lower-cycle, lower-cost worksopes that burn remaining life rather than restore it. On 7 July 2026 FTAI and Aeronautical Engineers Inc. (AEI), a conversion house with "625+ aircraft modified using AEI's STCs", announced a collaboration on a 737-800 freighter solution built on such engines; Part VIII §67 carries it. [FTAI and AEI release, 7 Jul 2026] Two signals about the conversion market sit side by side in the research and are carried as a pair. The type-specialist data has more than 260 737-800 freighters in service and a pipeline above 300, and an earlier appraiser note had freighter values rising while passenger narrowbody rents receded. The same appraiser warned in September 2026 of an "oversupply of narrowbody conversions". [Acumen Aviation, Dec 2025; Cargo Facts; IBA via Aeroin, 2025; IBA via India Seatrade News, 6 Sep 2026; reconciliation R-034] The primer records both and takes no view on which describes the market. What the conversion path does for this primer's subject is mechanical: it keeps -7B engines on wing past passenger retirement age, in a low-cycle duty that suits green-time modules, and it removes those airframes and engines from the teardown pool for a decade or more. Part III §24 carries that as a row of the translation table.

16. Green time and the trade in part-worn engines

Between a fresh engine worth its half-life value plus an adjustment and a run-out engine worth its parts lies a market in time. This section explains what green time is, how it is leased and traded, and how its value differs from the appraiser's half-life figure.

16.1 GREEN TIME, THE THREE CLOCKS, AND THE ENGINES THAT HAVE IT

Part I §8.2 boxed green time: the remaining usable life on an engine or module before its next mandatory shop visit, set by whichever runs out first. The research describes three clocks. The first is the life-limited part with the fewest cycles remaining, because the engine must come off the wing before any such part reaches its limit (Part I §4). The second is exhaust-gas temperature (EGT) margin, the gap between the temperature the engine runs at and its red line, which erodes as the engine wears and which a performance restoration recovers (Part I §5). The third is any hard-time inspection or airworthiness directive that requires the engine to be opened. [D3 §2.3; D2 §2.3; registry C-2] Green time is the quantity; a green-time engine, or a green-time lease, adds an intention. A green-time engine is "one that still has usable time but whose owner does not intend to pay for the next shop visit, either because the engine is near the end of its life or because the owner's fleet is leaving", sold or leased as-is to run down the remaining time and then torn down. [D3 §2.3]

USED SERVICEABLE MATERIAL (USM)

Used serviceable material is a part taken from a retired or dismantled aircraft or engine, inspected and if necessary repaired to the manufacturer's manual, and released on an airworthiness tag with its full history so that it can be installed again. A part with a gap in its paperwork is scrap whatever its condition. Used material is made by the original manufacturer, so it sits outside the alternative-parts debate, but it competes for the same slot in a shop visit's parts bill; its supply is a lagged function of retirements. Part V §31 carries the full treatment. [D3 §2.1; D2 §8]

RUN-OUT (EXHAUSTED) ENGINE

A run-out or exhausted engine is one with no usable time left before a shop visit: one of its clocks has reached zero. It is the only kind of engine that becomes teardown feedstock while engines with life are leased or sold whole, because a half-life -7B at \$5.7–6.4M (IBA, September 2025) is worth several times the \$1.6–2.2M of used material a teardown yields (consultancy blog, 2026). Part V §33 carries the full treatment. [D3 §2.2, §2.3; IBA, Sept 2025; Safe Fly Aviation, 2026]

16.2 THE GREEN-TIME LEASE

A green-time lease is a short fixed-term operating lease, from months to a few years, at a monthly rent, with the engine returned at or near run-out rather than restored. Because the lessee will not restore the engine, the rent is effectively the price of the time consumed, and the return conditions of §12.4 are inverted: the lessor expects the engine back with less life, not more. [D3 §2.3] The sequence has a sourced example. The engine trader APOC Aviation (APOC) leased a CFM56-5A to Condor for twelve months on a green-time lease, with the engine "expected to return to APOC's portfolio for teardown and part-out next year after the lease concludes".

[AviTrader, 28 Jul 2025; Aviation Business News, 2025]

The rents the research reached for green-time engines are the consultancy blog's, labelled as such: CFM56-7B \$42,000–48,000 a month, "about 34% above 2019" and up 15–20% "in recent months"; CFM56-5B \$38,000–44,000. [Safe Fly Aviation, 2026 (consultancy blog)] Whether those are green-time rents is the research's inference from the page's layout; the appraiser's \$100,000 spare-engine rent is the other side of the pair (§14.3, R-028). The appraiser's September 2025 release describes the -7B market as having "a distinct lack of availability", with market values and lease rates above long-term trend, which is the condition in which green-time engines are in demand: when shops are full, "operators often turn to short-term or green-time engine leasing". [IBA, Sept 2025; MTU AEROREPORT, undated] The engine maker sells the same thing: green-time leases are one element of GE's TrueChoice Transitions menu, which Part IV §28 carries. [D2 §2.3]

16.3 THE OWNER'S CHOICE, AND THE TRADE IN PART-WORN ENGINES

An engine lessor faces the same decision at every lease end, and a mid-life aircraft owner faces it for each engine when the airframe retires: restore the engine and lease it again at a full-life rent, lease out its remaining green time, or part it out now. [D6 §2.3] The trade in part-worn engines is the market in which those three options are priced against each other, and it is where the module as a unit of trade (Part I §8; Part V §36) comes from: a core with 80% of its life-limited-part life is worth far more than one with 20%, because the next owner's shop visit will or will not have to open it. [D3 §2.7]

WORKED EXAMPLE 2.7 — TWELVE MONTHS OF GREEN TIME: LEASE IT OUT OR TEAR IT DOWN NOW

Inputs: a CFM56-7B with twelve months of green time (assumed). Green-time rent \$45,000 a month, the midpoint of \$42,000–48,000 (sourced: Safe Fly Aviation, consultancy blog, 2026; the midpoint is the research's choice for illustration), with the appraiser's spare-engine rent of about \$100,000 a month (sourced: IBA, April 2024; a spare engine, not stated to be green-time) as the other side of the pair. Teardown yield \$1.6–2.2M of used material (sourced: Safe Fly Aviation, 2026).

lease twelve months, then tear down:
 rent at \$45,000: $12 \times 45,000 = \$540,000$ (at \$42,000–48,000: \$504,000–576,000)
 total: $540,000 + 1,600,000$ to $2,200,000 = \$2.14\text{M}$ to $\$2.74\text{M}$, over a year
 plus the sell-through period
 the same at the IBA rent: $12 \times 100,000 = \$1,200,000$; total $\$2.8\text{M}$ to $\$3.4\text{M}$
 tear down now: $\$1.6\text{M}$ to $\$2.2\text{M}$, sooner, with no rent

Leasing adds the rent but delays the teardown proceeds by a year and more, and the life-limited parts with twelve months of cycles left are worth slightly more in a teardown now than after another year of flying, since their remaining life is what a buyer pays for; the airfoils and cases are worth the same either way. Which option is chosen depends on the owner's cost of capital, the rent on offer, and whether an engine shop will pay more for the life-limited-part life now than a lessee will pay in rent.

On the blog's rent the lease adds a quarter to a third to the proceeds; on the appraiser's it adds more than half; the primer carries both and takes no view on which rent a green-time engine commands. Limits of the example: the twelve months are assumed; the two rents are a contested pair that may price different engines; the sell-through period for teardown inventory was not sourced (D3 U2); and the teardown yield is a consultancy figure with no appraiser counterpart. [D3 §2.3; Safe Fly Aviation, 2026; IBA, Apr 2024; reconciliation R-028]

16.4 HOW GREEN-TIME VALUE DIFFERS FROM HALF-LIFE VALUE

The appraiser's half-life market value and the consultancy blog's green-time value are not two estimates of one quantity. They are two quantities. Part I §8 gave the arithmetic that connects them: an engine's maintenance-adjusted value is its half-life value plus, for each clock, the actual remaining fraction less one half, times the cost to restore that clock. An engine with full-life life-limited parts is worth the half-life value plus half the full-life set; a run-out engine is worth the half-life value less half the set less half a restoration, which for an old engine "is approximately the teardown value". [D3 §2.4; Part I §8.1] A green-time engine sits between those points, and its value falls toward the run-out figure as its clocks run down.

EXHIBIT 2.7 — GREEN-TIME VALUE AGAINST HALF-LIFE VALUE: THE FIGURES THE RESEARCH REACHED, THE BASES THEY SIT ON, AND THE RUN-OUT VALUE PART I'S INPUTS IMPLY

Quantity	Figure	Basis	Source and level	Date
CFM56-7B27 / -7B24 market value	\$6.4M / \$5.7M	Half-life: every module midway between restorations, every LLP at half its life	IBA (appraiser)	Sept 2025
Full-life -7B27 on Part I's inputs	\$10.3M (\$9.1M on the alternative set and restoration prices)	Half-life plus half of (\$5.7M set + \$2.15M restoration)	Derived in Part I worked example 1.7	2025 prices
Run-out -7B27 on Part I's inputs	\$2.5M (\$3.7M on the alternative inputs)	Half-life less half of (\$5.7M + \$2.15M)	Derived in this Part	2025 prices
Run-out -7B24 on Part I's inputs	\$1.8M (\$3.0M on the alternative inputs)	Same, on the \$5.7M half-life value	Derived in this Part	2025 prices
CFM56-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", which the research notes is internally inconsistent with green time	Safe Fly Aviation (consultancy blog)	2026

Quantity	Figure	Basis	Source and level	Date
CFM56-5B "green-time value"	\$2.6–3.2M	Same label	Safe Fly Aviation	2026
Run-out CFM56 purchase price	\$0.8–1.2M	Exhausted engine bought for teardown	Safe Fly Aviation	2026
Used material from a run-out CFM56	\$1.6–2.2M; net \$0.4– 0.8M after the purchase	Teardown yield	Safe Fly Aviation	2026
LLP life as share of a used engine's value	60–80%; two -7Bs at 10,000 cycles since new differ by \$1.6M on LLP life alone	Used engine	Safe Fly Aviation	2026
Value direction	"Levelled off after approximately 18 months of increases"	-5B and -7B market values	IBA H1 2026	H1 2026
Value direction	Up "as much as 50% over the past two years"; rents up 15–20% "in recent months"	Not stated; green-time rents	ePlaneAI (trade press), date not captured; Safe Fly Aviation, 2026	2026

Source: IBA, Engine Values Release 2025B, Sept 2025; IBA, Engine and Lease Rate Update H1 2026; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); ePlaneAI, "CFM56 engine values highlight investor opportunities and risks", date not captured; Part I worked example 1.7 on IBA, MyAirTrade (June 2025), Aircraft Value News (Nov 2019) and Safe Fly Aviation inputs. Reconciliation R-029. Accessed 2026-10-03.

The derived rows use Part I's inputs and carry Part I's caveats: the life-limited-part set is \$5.7M (a market-data table, June 2025, scope not visible) or \$4.0M (a trade-press headline, November 2019), the restoration is \$2.15M or \$1.4M (two consultancy ranges), and the half-life reference is a market value standing in for a base value the research never reached. [Part I §8.3; reconciliation R-013, R-016, cluster K-02]

run-out -7B27 = $6.4M - 0.5 \times 5.7M - 0.5 \times 2.15M = 6.4 - 2.85 - 1.075 = \$2.5M$
 alternative inputs: $6.4M - 0.5 \times 4.0M - 0.5 \times 1.4M = 6.4 - 2.0 - 0.7 = \$3.7M$
 run-out -7B24: $5.7 - 2.85 - 1.075 = \$1.8M$; alternative: $5.7 - 2.7 = \$3.0M$

Read against the consultancy blog's figures, the derived run-out value of \$1.8–3.7M sits above the blog's \$0.8–1.2M purchase price for an exhausted engine and overlaps its \$2.8–3.4M "green-time value". The research does not reconcile these, and the primer does not. [reconciliation R-029]
 The bases differ in three ways. The appraiser's figure is a half-life market value and the blog's two figures are a green-time value and a feedstock price, so none of the three measures the same engine. The blog's label "(new condition)" on a green-time figure is internally inconsistent, as the

research noted. And the sources disagree on direction, the appraiser's "levelled off" against the blog's rents up 15–20% "in recent months" and a trade-press item's values up by half in two years. [IBA, H1 2026; Safe Fly Aviation, 2026; ePlaneAI, date not captured; D3 T3] What the arithmetic does establish is why the two sides can both be describing something real: the gap between a half-life -7B27 at \$6.4M and a run-out one at \$2.5M is about \$3.9M on Part I's inputs, and that gap is the restoration and the life-limited parts the appraiser's convention assumes are half consumed. A green-time engine is one whose owner has decided not to buy them back.

For FTAI the two ends of this trade are inputs to the same business. Engines it judges "unserviceable for leasing purposes" are "transferred to inventory ... with the intent to tear down the asset into salvageable components that would be manufactured into new modules and parts and sold", and serviceable used modules and parts are "sold through an exclusive partnership responsible for the teardown, repair, marketing and sales of parts from the CFM56 engine pool". [FTAI 10-K FY2025] Part VII §46 carries the module side and Part V §33 and §35 the partnership, which the research identifies from a partner's release as an exclusive Serviceable Engine Products agreement with the parts distributor AAR Corp (AAR), extended through 2030. [AAR release via PRNewswire, 2025 (title located; body not read); reconciliation B-9] The research also records FTAI on the other side of the trade, selling a 15-year-old A320-200 to APOC for disassembly in May 2026. [ePlaneAI, 2026]

17. Financing: securitizations, warehouse facilities, sale-leasebacks (worked tranching example)

A lessor's return in worked example 2.3 had a financing line of 4.55% of value, and the equity return of 13% depended on borrowing 70% of the price at 6.5%. Where that debt comes from, how it is layered, and what it costs the equity when rents fall is the subject of this section. The instruments are general to the industry, and the sourced template is the engine-lessor's: Willis Lease's rated securitizations. FTAI's own vehicles use the same instruments, and the facts specific to them belong to Part VIII; this section defines the instruments and points there.

17.1 THE VEHICLE THAT HOLDS THE ASSETS

Almost every form of asset-level debt in this industry starts with a company that exists only to own a defined set of aircraft or engines.

SPECIAL-PURPOSE VEHICLE (SPV)

A special-purpose vehicle is a bankruptcy-remote legal entity that exists only to own a defined pool of assets and the debt secured on them, insulated from the sponsor's other liabilities. Bankruptcy-remote means that if the sponsor fails, the SPV's assets are not caught up in the sponsor's insolvency, so a lender to the SPV looks only to those assets and their leases. It is the issuer in a securitization, the borrower under a warehouse facility, and the asset-owning layer beneath a partnership such as FTAI's Strategic Capital vehicles, which FTAI's filings describe as acquiring aircraft through "special purpose entities (SPVs) of the 2025 Partnership". Aircraft lessors are usually organised this way, with each aircraft or sub-pool in its own entity so that lenders and lessees deal with a clean owner. Part VIII §61 and §62 carry FTAI's vehicles. [FTAI 10-Q Q1 2026; definition assembled in the research, D6 §2.5, D7 §2.1]

The SPV is the reason a lessor can borrow against a pool of leased engines at a cost that reflects the engines and the airlines renting them rather than the lessor's own credit. Everything in this section is a way of lending to such a vehicle.

17.2 THE SECURITIZATION AND ITS PARTS**ASSET-BACKED SECURITIZATION (ABS)**

An asset-backed securitization is a transaction in which a lessor sells a portfolio of leased aircraft or engines to a bankruptcy-remote SPV, which issues notes to investors secured on the assets and on their lease cash flows. The notes are divided into tranches paid in order of seniority. The lessor usually keeps the equity, the residual claim after the notes, and acts as servicer of the leases for a fee. Rating agencies, KBRA (Kroll Bond Rating Agency) in the two deals this section uses, assign ratings to the notes after testing the structure against stressed assumptions for lease-rate declines, downtime, maintenance costs and residual values. [KBRA, 3 Jun 2025 and 10 Dec 2025; definition assembled in the research, D6 §2.5]

TRANCHE: SENIOR, MEZZANINE AND EQUITY

A tranche is one layer of the claims on a securitization, paid in order of seniority. The senior notes (Class or Series A) are paid first, carry the lowest coupon and the highest rating. The mezzanine notes (Class B, and sometimes C) are paid after the A notes and carry higher coupons and lower ratings. The equity, sometimes issued as E-certificates, is paid last and absorbs the first losses. Because every dollar of shortfall hits the equity first and every dollar of surplus reaches it last, tranching concentrates both the upside and the downside in the layer the lessor keeps. [definition assembled in the research, D6 §2.5; KBRA, 10 Dec 2025]

LOAN-TO-VALUE (LTV)

Loan-to-value is the note balance divided by the appraised value of the assets in the pool. The senior notes might be sized to 60–70% of appraised value and the senior and mezzanine notes together to 75–80%, leaving equity of 20–25%; those ranges are illustrative, because the two deals the research retrieved did not return their loan-to-value ratios (D6 U6). It is distinct from loan-to-cost, which divides debt by the total capital of a vehicle and which Part VIII §61 uses for FTAI's 2025 vehicle. [definition assembled in the research, D6 §2.5; D7 §2.3]

PAYMENT WATERFALL (ABS)

The payment waterfall is the ordered allocation of the pool's collections on each payment date: fees and expenses first, then interest on the A notes, then interest on the B notes, then scheduled principal on the A and B notes, then any C notes, with any surplus to the equity. Part VIII §63 uses the same word for a partnership's distribution order, which allocates cash among partners rather than among note classes; the primer says "payment waterfall" for the first and "distribution waterfall" for the second. [definition assembled in the research, D6 §2.5; registry C-17]

ANTICIPATED REPAYMENT DATE (ARD) AND LEGAL FINAL MATURITY

The anticipated repayment date is the date after which a securitization sweeps all excess cash to pay down principal faster and steps up the coupon on the notes, so that the deal is designed to be refinanced or called at that date. The legal final maturity is the much later last date by which the notes must be repaid. In Willis Lease's ninth engine securitization the two dates are December 2031 and December 2050, a gap of about nineteen years: the structure expects to be refinanced at the first date and treats the second as a backstop. [KBRA, 10 Dec 2025; Asset Securitization Report, Dec 2025]

The rating agencies are the gate. The notes carry ratings because an agency, KBRA in the two deals below, has modelled the pool's rents, the lessees' credit, the time engines spend off lease between lessees, the maintenance they will need and the value they will have at the end, under stresses, and found the senior notes able to pay through those stresses. [KBRA, 3 Jun 2025; KBRA, 10 Dec 2025] That is why the lease-rate and value movements of §14 and §15 matter beyond the lessor's own income statement: an appraiser's report that engine values have "levelled off" or that narrowbody rents fell 11–13% feeds straight into the stress tests that decide how much senior debt a pool can carry. [IBA, H1 2026; IBA, Jul 2025; D6 §5]

17.3 THE SOURCED TEMPLATE: WILLIS ENGINE STRUCTURED TRUST VIII AND IX

WILLIS ENGINE STRUCTURED TRUST (WEST)

The Willis Engine Structured Trust deals are Willis Lease's series of rated engine securitizations: WEST VIII, rated in June 2025, raised \$596M against 64 assets; WEST IX, rated in December 2025, raised \$392.9M against 49 assets. They are the sourced template in this primer for a senior-subordinate engine securitization, and Willis is their servicer and administrative agent, receiving 11.5% of aggregate net rents as servicer plus 2.0% as administrative agent each month and 3.0% of net proceeds on engine sales, according to its 10-K filings for 2019 and 2020. Part XI §87 reuses the deals in the peer comparison. [KBRA, 3 Jun 2025 and 10 Dec 2025; Asset Securitization Report, 2025 and Dec 2025; WLFC 10-K FY2019 and FY2020]

EXHIBIT 2.8 — WEST VIII AND WEST IX: THE STRUCTURE AS THE RATING AGENCY AND TRADE PRESS DESCRIBED IT, AND WHAT WAS NOT RETRIEVED

Item	WEST VIII	WEST IX	Source
Preliminary ratings announced	3 Jun 2025	10 Dec 2025	KBRA
Size	\$596M	\$392.9M	Asset Securitization Report
Collateral	64 assets; proceeds "to refinance the existing WEST IV transaction and acquire a portfolio"	"Financing leases on 49 aircraft assets" (the trade report's wording)	KBRA; Asset Securitization Report
Senior notes	Series A, rated A	Class A, rated A	KBRA
Subordinated notes	Series B, rated BBB (a grade below A)	Class B; "if a series of C notes is issued, it will receive payments subordinate to payments to the series A and B"	KBRA
Repayment structure	Senior-subordinate	"A senior-subordinate repayment structure"	KBRA
Anticipated repayment date	Not returned	December 2031	KBRA
Legal final maturity	Not returned	December 2050	KBRA
Sponsor and servicer	Willis Lease; "the ninth aviation ABS sponsored and serviced by WLFC"	Willis Lease; the tenth	KBRA

Item	WEST VIII	WEST IX	Source
Servicer and administrative-agent fees	11.5% + 2.0% of aggregate net rents, monthly; 3.0% of net engine-sale proceeds	Same source; whether unchanged for WEST IX not retrieved	WLFC 10-K FY2019, FY2020
Note balance per asset (derived)	$\$596\text{M} \div 64 = \9.3M	$\$392.9\text{M} \div 49 = \8.0M	Derived in this Part; a note balance, not a value
Coupons, loan-to-value, equity size	Not returned (D6 U6)	Not returned (D6 U6)	KBRA pre-sale reports not read

Source: KBRA, "Assigns Preliminary Ratings to Willis Engine Structured Trust VIII", BusinessWire, 3 Jun 2025; KBRA, "Assigns Preliminary Ratings to Willis Engine Structured Trust IX", BusinessWire, 10 Dec 2025; Asset Securitization Report (American Banker), "Willis Engine Structured Trust raises \$596 million in aircraft ABS", 2025, and "Willis Engine floats \$392.9 million in aircraft-related ABS", Dec 2025; Willis Lease Finance Corp., Form 10-K FY2020 and FY2019 (WEST fees). Accessed 2026-10-03.

17.4 THE TRANCHING WORKED THROUGH

The research retrieved the sizes, the ratings and the dates of the two deals but not their coupons, their loan-to-value ratios or the size of their equity (D6 U6). The example below therefore takes WEST IX's note total as its one sourced size and applies the illustrative structure the research set out, with every other input labelled assumed. It is arithmetic on a real deal's size, not a description of the deal.

WORKED EXAMPLE 2.8 — TRANCHING A \$392.9M ENGINE SECURITIZATION ON WEST IX'S SIZE, AND WHAT A FALL IN RENTS DOES TO EACH LAYER

Inputs. Total notes \$392.9M on 49 assets (sourced: KBRA and Asset Securitization Report, December 2025). Anticipated repayment date December 2031, legal final December 2050 (sourced: KBRA). Servicer and administrative-agent fees 13.5% of net rents (sourced rate: WLFC 10-K FY2019 and FY2020; applied here to gross rent as an approximation, since the definition of "net rents" was not retrieved). Everything below is assumed for illustration: the senior and subordinated notes together are 75% of appraised value; the Class A notes alone are 65%; the A coupon is 5.5% and the B coupon 7.5%; the pool earns a lease rate factor of 1.0% a month; other expenses (insurance, remarketing, downtime) are \$3.0M a year; scheduled principal is 4% of the notes a year.

implied appraised value: $392.9 \div 0.75 = \$523.9\text{M}$ (per asset: $523.9 \div 49 = \$10.7\text{M}$)

Class A: $0.65 \times 523.9 = \$340.5\text{M}$ (65% LTV)

Class B: $392.9 - 340.5 = \$52.4\text{M}$ (A+B: 75% LTV)

equity: $523.9 - 392.9 = \$131.0\text{M}$ (25%)

annual interest: A $0.055 \times 340.5 = 18.7$; B $0.075 \times 52.4 = 3.9$; total $\$22.6\text{M}$

annual rent at 1.0% a month: $0.01 \times 523.9 \times 12 = \62.9M

servicer and agent fees: $0.135 \times 62.9 = \$8.5\text{M}$

The waterfall in a normal year:

rent 62.9 – fees 8.5 – other expenses 3.0 = 51.4 available for the notes

– A interest 18.7 – B interest 3.9 = 28.8 after interest

– scheduled principal 15.7 = 13.1 to the equity

equity cash yield: $13.1 \div 131.0 = 10.0\%$

coverage of interest plus scheduled principal: $51.4 \div (22.6 + 15.7) = 1.34\times$

Now stress the rent as a rating agency would. If lease rates fall 15%, rent is \$53.4M and the fees fall with it to \$7.2M; the notes' interest and scheduled principal do not move.

at -15%: $53.4 - 7.2 - 3.0 - 22.6 - 15.7 = \4.9M to the equity (3.7%);
coverage $43.2 \div 38.3 = 1.13\times$

at -25%: $47.2 - 6.4 - 3.0 - 22.6 - 15.7 = -\0.5M (nothing to the equity);
coverage $37.8 \div 38.3 = 0.99\times$

A 15% fall in rent cuts the equity's cash by nearly two-thirds, from \$13.1M to \$4.9M, while the A and B notes are paid in full. A 25% fall leaves nothing for the equity and a small shortfall against scheduled principal, which is the point at which a real structure's reserve accounts and cash-trap triggers would operate; they are not modelled here. That asymmetry is the tranching doing its job: the equity concentrates the downside so that the senior notes can carry an A rating, and it concentrates the upside, which is why the lessor keeps it. After the anticipated repayment date in December 2031 the surplus that reached the equity in a normal year would instead be swept to principal and the coupons would step up, so the deal is built to be refinanced then. For scale, WEST VIII's \$596M over 64 assets is \$9.3M of notes per asset against WEST IX's \$8.0M, both derived; neither is a value, and the pools include engine types and aircraft the research did not itemise, so neither should be set against the IBA half-life market values of \$5.7–6.4M for a -7B. Limits of the example: the 75% and 65% loan-to-value ratios, both coupons, the lease rate factor, other expenses and the principal schedule are all assumed; the fee rate is from filings five and six years old; the actual coupons, ratios and equity of WEST IX were not retrieved (D6 U6). [KBRA, 10 Dec 2025; Asset Securitization Report, Dec 2025 and 2025; WLFC 10-K FY2019 and FY2020; illustrative structure from D6 §2.5]

17.5 THE WAREHOUSE FACILITY

A securitization needs a pool large and seasoned enough to rate. The instrument that assembles the pool is a bank line.

WAREHOUSE FACILITY (WAREHOUSE LINE)

A warehouse facility is a revolving bank loan to an SPV, secured on whatever assets the SPV buys during an availability period, with an advance rate and a borrowing base, and with a term after which the balance must be repaid or refinanced, commonly by issuing a securitization. Assets are accumulated in it until there are enough to securitize, which is why it is called a warehouse. A lessor uses it to fund acquisitions one aircraft at a time without going to the bond market for each. FTAI's second Strategic Capital vehicle closed a \$2.0B warehouse facility from 13 lenders on 14 August 2026; Part VIII §62 carries the vehicle. [FTAI release, 17 Aug 2026; definition assembled in the research, D6 §2.5, D7 §2.2]

ADVANCE RATE AND BORROWING BASE

The advance rate is the percentage of each asset's value that the banks will lend against it under a warehouse facility. The borrowing base is the total credit available at any moment, recomputed as assets are added, sold or revalued: the sum over the assets of value times advance rate. If values fall, the borrowing base falls, and the borrower must repay or add collateral. [definition assembled in the research, D6 §2.5]

ACCORDION

An accordion is a pre-agreed option to enlarge a credit facility if lenders agree to fund more. The 2026 vehicle's \$2.0B warehouse carries a \$1.0B accordion, so up to \$3.0B. FTAI's statement that its Strategic Capital vehicles "have now raised \$5.5 billion of warehouse financing in less than two years" matches the disclosed commitments of \$2.5B (February 2025) and \$2.0B (August 2026), which sum to \$4.5B, only if the undrawn accordion is counted; the release does not say how the figure is built. Part VIII §62 carries the full treatment. [FTAI release, 17 Aug 2026; FTAI release, 26 Feb 2025; reconciliation R-055]

WORKED EXAMPLE 2.9 — WHAT A WAREHOUSE LINE BUYS: ADVANCE RATES, THE ACCORDION, AND A VEHICLE'S LOAN-TO-COST

Inputs: a \$2.0B warehouse facility with a \$1.0B accordion (sourced: FTAI release, 17 Aug 2026). Advance rates of 70% and 60% (assumed; the facility's rate, tenor and covenants were not retrieved, D7 U11). For the first vehicle, about \$6.0B of total capital on \$2.0B of equity commitments (sourced: FTAI releases, 27 Oct 2025 and 17 Aug 2026) and a \$2.5B asset-level debt commitment (sourced: FTAI release, 26 Feb 2025).

assets a \$2.0B line funds at a 70% advance rate: $2.0 \div 0.70 = \$2.86\text{B}$,
with \$0.86B of equity
with the \$1.0B accordion drawn: $3.0 \div 0.70 = \$4.29\text{B}$, with \$1.29B of
equity
at a 60% advance rate: $2.0 \div 0.60 = \$3.33\text{B}$ (\$1.33B equity); $3.0 \div 0.60 =$
 $\$5.00\text{B}$ (\$2.00B equity)
first vehicle, loan-to-cost: debt = $6.0 - 2.0 = \$4.0\text{B}$; $4.0 \div 6.0 = 67\%$,
or \$2 of debt per \$1 of equity
if only the \$2.5B commitment were drawn: total 4.5; $2.5 \div 4.5 = 56\%$

The advance rate sets how far a dollar of equity stretches: at 70% each equity dollar buys \$3.33 of aircraft, at 60% \$2.50. The accordion adds buying power only if the lenders agree to fund it, which is why the primer carries "\$5.5 billion" and "\$4.5 billion" side by side rather than choosing. The first vehicle's 67% is a loan-to-cost figure derived in the research from two disclosed totals; it is not a loan-to-value, because no appraised value of the vehicle's aircraft was disclosed, and "total capital" may include reserves for engine exchanges. Limits of the example: the advance rates are assumed; the debt actually drawn by the first vehicle was not disclosed, so 56% and 67% bracket it; Part VIII §61 carries the vehicles' leverage in full. [FTAI releases, 26 Feb 2025, 27 Oct 2025 and 17 Aug 2026; derived in D7 §2.3; reconciliation R-055, U-D7-11]

EXHIBIT 2.9 — THE WAREHOUSE FACILITY AND THE SECURITIZATION, COMPARED AS INSTRUMENTS

Feature	Warehouse facility	Asset-backed securitization	Source
Lender	A bank or syndicate of banks (13 in the 2026 vehicle's facility)	Note investors, in rated tranches	FTAI release, 17 Aug 2026; KBRA
Borrower or issuer	An SPV	An SPV	D6 §2.5
Security	Whatever assets the SPV buys during the availability period	A fixed, identified pool of leased assets and their lease cash flows	D6 §2.5
Sizing	Advance rate on each asset; borrowing base recomputed as the pool changes	Loan-to-value on the appraised pool, by tranche	D6 §2.5
Pricing	Floating bank rate, not retrieved for FTAI's vehicles (D7 U11)	Fixed coupons by tranche, lowest on the senior notes; not retrieved for WEST (D6 U6)	D6 U6; D7 U11

Feature	Warehouse facility	Asset-backed securitization	Source
Term	Availability period, then repay or refinance	Anticipated repayment date (WEST IX: Dec 2031), legal final (Dec 2050)	KBRA, 10 Dec 2025
Flexibility	Revolving: draw as assets are bought, repay as they are sold; accordion to enlarge	Fixed at issue; refinanced at the anticipated repayment date	FTAI release, 17 Aug 2026; D6 §2.5
Role in the sequence	Accumulate and season the pool	Take out the warehouse; release capacity for the next pool	D6 §2.5
Who keeps the equity	The vehicle's equity holders	The sponsor, typically, as E-certificates	D6 §2.5
Example in the research	FTAI 2026 vehicle: \$2.0B, 13 lenders, \$1.0B accordion, closed 14 Aug 2026; first vehicle: \$2.5B asset-level debt commitment, 26 Feb 2025	WEST VIII (\$596M, 64 assets, 2025); WEST IX (\$392.9M, 49 assets, Dec 2025)	FTAI releases; KBRA; Asset Securitization Report

Source: FTAI releases of 26 Feb 2025 and 17 Aug 2026; KBRA, 3 Jun 2025 and 10 Dec 2025; Asset Securitization Report, 2025 and Dec 2025; instrument descriptions assembled in D6 §2.5 and D7 §2.2. Accessed 2026-10-03.

17.6 HOW A LESSOR USES THEM TOGETHER, AND WHAT THAT HAS TO DO WITH FTAI

The sequence a lessor follows is the same whether the lessor is Willis Lease or a vehicle managed by FTAI. Buy assets on the warehouse. Season the leases, meaning let them run long enough to show the collections a rating agency can model. Securitize the pool into a rated deal, which repays the warehouse and releases its capacity for the next pool. Keep the equity and the servicing fee. [D6 §2.5] Willis has done this ten times; the WEST VIII proceeds were partly used to refinance WEST IV, which is the same sequence one generation on. [KBRA, 3 Jun 2025]

SERVICER AND SERVICING FEE

The servicer is the party contracted to manage the assets of a vehicle it does not wholly own: leasing and remarketing them, collecting rent and reserves, overseeing their maintenance condition, and selling them, for a fee. Willis is the servicer of the WEST deals at 11.5% of net rents plus 2.0% as administrative agent and 3.0% of sale proceeds; FTAI is the Servicer of its Strategic Capital vehicles, "receiving customary, market-based compensation", which amounted to \$10.2M in 2025 and \$12.85M in the first half of 2026; the rate and base are undisclosed. Part VIII §61 and §63 carry the full treatment. [WLFC 10-K FY2019 and FY2020; FTAI 10-K FY2025; FTAI 10-Q Q2 2026; D7 §2.5]

FTAI's third-party capital programme is built from the instruments this section has defined, with a partnership layered on top. Three terms are needed to read it, and Part VIII owns all three.

STRATEGIC CAPITAL INITIATIVE (SCI)

The Strategic Capital Initiative is FTAI's asset-management arm, launched on 30 December 2024, which raises third-party capital into closed-end vehicles that buy on-lease, mid-life 737NG and A320ceo aircraft; FTAI manages the vehicles as Servicer, co-invests as a limited partner, and supplies the engine work on their fleets through its Maintenance, Repair and Exchange business. It is not a reportable segment. Part VIII §61 carries the full treatment. [FTAI release, 30 Dec 2024; FTAI 10-K FY2025; D7 §2.1]

THE 2025 PARTNERSHIP AND THE 2026 SPV

The 2025 Partnership is the first Strategic Capital vehicle: a closed-end partnership over asset-owning SPVs that closed at a \$2.0B equity hard cap in October 2025 (against a \$1.5B target), carries a \$2.5B asset-level debt commitment made in February 2025, and had committed about \$6.0B of total capital across more than 300 aircraft by August 2026, when FTAI described it as in its harvest phase; FTAI holds 19% (the FY2025 10-K and 2026 10-Qs) or 20% (the Q3 2025 10-Q) as a limited partner. The 2026 SPV is the second vehicle, which closed the \$2.0B warehouse facility above on 14 August 2026 and made its first disclosed purchase in the WestJet transaction of 28 September 2026; its equity size and FTAI's stake were undisclosed at the research date. Both followed the warehouse-then-securitize shape described above. Part VIII §62 carries the full treatment. [FTAI releases, 26 Feb 2025, 27 Oct 2025, 17 Aug 2026 and 28 Sep 2026; FTAI 10-K FY2025; FTAI 10-Q Q3 2025; reconciliation R-048]

SEED ASSETS

Seed Assets is FTAI's term for the on-lease aircraft it sold from its own book to start the 2025 Partnership: 46 aircraft at "an estimated net purchase price of \$549 million" at launch. The sales are why FTAI's leasing equipment at cost fell from \$2,963.5M to \$2,057.6M during 2025 and why its lease income roughly halved year on year in the first half of 2026. Part VIII §60 carries the full treatment. [FTAI release, 30 Dec 2024; FTAI 10-K FY2025; FTAI 10-Q Q1 2026]

The vehicles carry two layers of debt in the sense this section has defined them: asset-level debt borrowed by the SPVs and secured on the aircraft and their leases, with no recourse to FTAI, and a warehouse facility within that layer for the second vehicle. [FTAI release, 26 Feb 2025; FTAI release, 17 Aug 2026; D7 §2.2] The lessor's incentive to sell assets into such vehicles is to recycle equity: it books a gain on sale, keeps a servicing stream and, in FTAI's case, keeps the engine maintenance work on the fleet it sold. [D6 §2.5; D7 §2.5] The equity it keeps is a 19% or 20% limited-partner stake rather than the whole E-certificate layer a securitization sponsor usually holds, which is one of the differences Part VIII §65 sets beside other platforms. The first vehicle has since used the securitization market itself. On 22 May 2026 FTAI priced FTAI MRE 2026-1, which it called

the Strategic Capital vehicle's inaugural asset-backed securitization: \$612M of notes backed by 48 A320ceo and 737NG aircraft on lease to 23 airlines, in two classes, Series A expected to be rated Asf by Fitch and A(sf) by KBRA and Series B expected BBB+sf by Fitch, with closing expected on 4 June 2026 and both classes described as significantly oversubscribed. The same release put the vehicle's fleet at 292 aircraft. So the deal is a take-out within the vehicle's asset-level layer, of the kind §17.6 describes, not a borrowing by FTAI itself. Its coupons and loan-to-value were not in the research (D7 U12, now partly resolved). [FTAI release, "FTAI Aviation Prices Inaugural Asset-Backed Securitization", 22 May 2026, via Finviz and AviTrader 26 May 2026]

17.7 SALE-LEASEBACKS

The third instrument is not a loan at all from the lessor's side, but it is the airline's way of financing an aircraft and the lessor's way of acquiring one.

SALE-LEASEBACK

A sale-leaseback is a transaction in which an airline sells aircraft it owns to a lessor and leases them straight back, converting owned assets into rent obligations and releasing cash to the airline. For the lessor it is the purest form of lease-attached acquisition: the aircraft arrives with a lease to a known operator already on it, which is what lenders and securitization investors underwrite. FTAI's WestJet transaction of 28 September 2026, in which the 2026 SPV acquired 17 Boeing 737-700s "on lease to WestJet in a sale-leaseback" while FTAI itself acquired 10 off-lease aircraft for their engines and modules, is the example in the research; the price and lease terms were not disclosed. Part VIII §67 carries it. [FTAI release, 28 Sep 2026; definition assembled in the research, D6 §2.4]

The sale-leaseback closes the loop this Part opened. The airline that sells its aircraft and rents it back is now a lessee under an operating lease (§12), paying rent and either reserves or end-of-lease compensation (§13), against return conditions that decide who pays for the engine life it consumes. The lessor that bought it funds the purchase on a warehouse line and later a securitization (§17), whose ratings rest on the appraisers' values (§12.7) and whose equity bears the swings in rent (§14, §15). And at the end of the lease the lessor faces the choice of §16: restore and re-lease, lease the green time, or take the engines out and sell them as modules, which is where Part VII begins.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-028	CFM56-7B rent: ~\$100,000 a month (IBA, Apr 2024, appraiser) against \$42,000–48,000 (Safe Fly Aviation, 2026, consultancy blog; may be green-time)	§12.1 box; §14.3, Exhibit 2.3, worked example 2.5; §15.3, worked example 2.6; §16.2, worked example 2.7
R-029	-7B value: half-life market value \$5.7M/\$6.4M and "levelled off" (IBA) against green-time \$2.8–3.4M "(new condition)", run-out \$0.8–1.2M, rents up 15–20% (Safe Fly) and values +50% in two years (ePlaneAI)	§14.3, Exhibit 2.3; §16.4, Exhibit 2.7
R-030	Lease rate factor: 0.70% (D3, green-time rent on half-life value); 0.87–1.1% (FTAI, derived on book); ~1.0% (Willis, derived); 0.9% (D7 assumed); 1.56–1.75% (IBA rent on IBA value, derived here); no stated figure	§12.1; §14.3, Exhibit 2.3, worked example 2.5; gross leasing yield box
R-031	Mid-life rents: 12-yr 737-800 ~\$228,000 (–11%), A320-200 ~\$220,000 (–13%) and a Sept 2026 oversupply warning (IBA) against rates up in 2025 and "expected to remain strong" (Air Lease 10-K FY2025), 10-yr 737-800 rents +44% vs Jan 2022 (Cirium Ascend) and extension rents >40% above prior (IBA, Nov 2023)	§15.1, Exhibit 2.6
R-032	Willis Lease: utilisation 86.0% at 30 Sep 2025 against 84.9% average for 2025; portfolio \$2,888.5M (Sep) against \$2,801.7M equipment (Dec); quarterly against annual rent and reserve revenue	§14.4, Exhibit 2.4
R-033	AerCap engines: 470 owned (20-F, 31 Dec 2025) against >1,200 including Shannon Engine Support (6-K, 30 Sep 2025)	§14.5, Exhibit 2.5
R-034	Freighter conversion: >260 737-800Fs in service, >300 pipeline (Acumen, Cargo Facts) and freighter values rising (IBA, 2025) against "oversupply of narrowbody conversions" (IBA, 6 Sep 2026)	§15.4
R-013	Restoration cost: D6's \$4.5M assumption against the sourced ranges \$1.2–1.6M and \$1.8–2.5M (Safe Fly, two pages)	§13.3, worked example 2.2
R-016 / R-017	LLP set behind a reserve or a maintenance adjustment: \$5.7M (MyAirTrade, Jun 2025) against ~\$4M (Aircraft Value News, Nov 2019)	§13.3; §16.4, Exhibit 2.7

Id	The two (or more) figures	Where in this Part
R-019	Turnaround: 90–120 days (trade press); ~90 sustaining (GE); 75–90 for a restoration, 120–150 heavy (Safe Fly); slot wait +2–3 months (Bain)	§14.2, worked example 2.4
R-008	Restoration interval: 10,000–15,000 cycles (Leeham) at 1,000–1,500 cycles a year, ~10 years implied (D9), ~8 years (Aviation Week), 6 years (D7 assumption)	§14.2, worked example 2.4
R-055	"\$5.5 billion of warehouse financing" (FTAI release, 17 Aug 2026) against \$2.5B + \$2.0B = \$4.5B disclosed; matches only with the \$1.0B accordion	§17.5 accordion box, worked example 2.9
R-056	Capital per aircraft in the 2025 vehicle: \$11.9M (seed), \$13.9M, \$14.7M, \$20M, \$23.6M on different bases; no appraised aircraft value retrieved	§15.3, worked example 2.6
R-048	FTAI's stake in the 2025 Partnership: 19% (FY2025 10-K, 2026 10-Qs) against 20% (Q3 2025 10-Q)	§17.6 box
D6 T5	Engines ~88% of a 737-800's rent mixes a 2024 engine rent with a 2025 aircraft rent; on the Safe Fly rent the share is 37–42%	§15.3, worked example 2.6
D3 T3	Safe Fly's "green-time value (new condition)" label is internally inconsistent	§14.3, Exhibit 2.3; §16.4, Exhibit 2.7
Registry C-3	Utilisation: Willis by book value on lease (84.9%) against FTAI by equity-value-weighted days on lease excluding airframes (~77%); not comparable without the definitions	§14.4

Unknowns carried in this Part

- **U-D6-05 (D6) — U5 — Lease-rate factors.** No stated LRF for FTAI or WLFC; WLFC engine count, average lease term and FY2025 totals. Resolve: WLFC FY2025 10-K ([S13]) Item 1 and Item 7.
- **U-D6-06 (D6) — U6 — ABS coupons and LTVs.** WEST VIII and IX note coupons, initial LTVs, and E-certificate sizes. Resolve: KBRA pre-sale reports ([S16],[S17]) and WLFC 8-Ks.
- **U-D6-07 (D6) — U7 — Security deposit and maintenance deposit balances.** FTAI's balance-sheet "security deposits" and "maintenance deposits" liabilities by year, and the reserve rates in its leases. Resolve: FY2025 10-K balance sheet and revenue-recognition note.

- **U-D6-08 (D6) — U8 — Depreciation policy.** Useful lives and residual assumptions for FTAI's aircraft and engines. Resolve: FY2025 10-K summary of significant accounting policies.
- **U-D6-09 (D6) — U9 — Half-life market values** for 12–15-year-old 737-800 and A320ceo aircraft, and full-life CFM56-7B/-5B/V2500 values, 2023–2026. Resolve: IBA/Cirium/mba published updates (D3 may hold engine values).
- **U-D6-10 (D6) — U10 — Conversion cost and slot availability** for 737-800 P2F (AEI, Boeing BCF, IAI). Resolve: conversion-house releases, trade press.
- **U-D6-14 (D6) — U14 — Current engine-lessor scale.** Portfolio size and value for GE Engine Leasing, SMBC Aero Engine Lease, ELF, AerCap Engines, RRPf, MTU Maintenance Lease Services. Resolve: company websites and parent filings (D11 for the listed ones).
- **U-D6-15 (D6) — U15 — Average lease terms over time** (new-lease term at signing versus remaining term) and the mix of short- versus long-term engine leases in FTAI's book. Resolve: 10-K MD&A by year.
- **U-D1-09 (D1) — U9** Current half-life base values for -7B and -5B by rating (IBA, mba, Cirium). Resolve with an appraiser publication; this session's searches returned definitions, not values.
- **U-D3-02 (D3) — U2 — Total sell-through period** for teardown inventory (months to recover, say, 80% of proceeds). AerSale's 10-K inventory-turnover disclosure or a trade-press teardown case study would answer it.
- **U-D3-04 (D3) — U4 — Any mba Aviation (or Ascend by Cirium) CFM56-5B/-7B value** or lease-rate figure. None returned.
- **U-D3-05 (D3) — U5 — A current appraiser half-life value** for the CFM56-5B (only trend commentary and a trade-press lease-rate range were captured).
- **U-D4-01 (D4) — U1 — Current lessor return-condition practice (2024–2026).** All lessor statements found are 2009–2016. Resolve with recent lessor 10-K/20-F lease descriptions (AerCap, Air Lease, Willis Lease), ISTAT or IBA surveys, or lessor conference remarks.
- **U-D7-11 (D7) — U11 — Debt terms.** Rate, advance rate, tenor and covenants of the \$2.5B facility and the \$2.0B warehouse; how \$5.5B is built; whether the 2025 facility was upsized.
- **U-D7-12 (D7) — U12 — The May 2026 ABS.** Whether "FTAI Aviation Prices Inaugural Asset-Backed Securitization" [S26] is a 2025-SPV take-out or FTAI's own balance-sheet deal; size, tranches, pricing.

- **U-D11-01 (D11) — U1 — Willis Lease lease rate factor and CFM56/LEAP portfolio split.** Searched the FY2025 10-K [S1]. Not returned. Resolve: read the 10-K's "Lease Portfolio" and "Market" sections directly.
- **U-D11-03 (D11) — U3 — Willis Lease maintenance services and spare-parts revenue lines for FY2025.** The 10-K's revenue table was not returned beyond lease rent, the combined core line and gain on sale. Resolve: read the consolidated statement of income in [S1].

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PART III

The context: fleet, retirements, new engines, and the translation table

This Part explains why the CFM56 aftermarket is where it is in 2026: how many engines there are, why far fewer of them retired than their makers planned, what went wrong with the two engines meant to replace them, Pratt & Whitney's geared turbofan (GTF) and CFM's LEAP (its new-generation engine), how few new aircraft the airframers delivered, and how lessors and airlines behaved as a result. It ends with a translation table that turns each of those facts into a specific requirement on the aftermarket. A reader who finishes it will be able to name the driver behind any shop-visit, used-part, lease or capital figure in Parts IV to XI.

WHERE THIS PART SITS IN THE BUILD

The primer is built in layers. Parts I and II are first principles: the engine as a machine, and the money that moves around it through leases, reserves and financing. Part III is the context shock. It takes the fleet those first principles describe and asks what happened to it between 2019 and 2026: which engines retired and which did not, why the replacement engines and airframes arrived late, and how owners behaved when they could not get new aircraft. None of that changes how a turbofan works or how a lease is priced. It changes the quantities: how many engines need a shop visit, how many are torn down, how many spares are needed and for how long, and what an extension costs. Section 24 turns each documented driver into a numbered row stating the requirement it creates and what it leaves untouched. Parts IV to XI, the segments, open by naming the rows they answer. This Part depends on Parts I and II for its terms and on nothing written later.

18. The fleet and the installed base

Every shop visit, every used part and every lease described in this primer is sold against an engine that already exists. So the first question is how many engines exist, which state each is in, and how hard it flies. This section defines demand and the states a fleet can be in, then sets out every count of the CFM56 fleet the research reached, with what each count covers. It records the count of the V2500 (the A320ceo's alternative engine, Part I §10.1) as an unknown, and closes with how mature the CFM56 fleet is, meaning how many of its engines have never been to a shop.

AFTERMARKET DEMAND

Aftermarket demand for an engine type is the sum of the work and assets needed to keep that type flying: shop visits, modules, spare engines, used serviceable material and leases of engines and aircraft. It is generated by the installed base multiplied by how hard that base flies, net of the engines that leave. Every row of the translation table in §24 names which of these requirements a driver raises. [D9 §2.1, definition assembled in the research]

Part I §2.2 boxed the installed base as the population of engines fitted to aircraft in the operating fleet. It noted that the published "in service" counts may include spare and stored engines, since no source states its basis. That reminder matters here. A spare engine in a warehouse and an engine on a parked aircraft generate no flight cycles and so no maintenance need. The firms that count fleets, Cirium (a data provider) and IBA (an appraiser that also publishes fleet data), sort aircraft into three states.

FLEET STATES: IN SERVICE, PARKED (STORED), RETIRED (WITHDRAWN)

An aircraft in service is assigned to an operator and flying, or in short-term maintenance. A parked or stored aircraft is not flying but has not been formally withdrawn; an aircraft parked for more than about 90 days usually counts as stored, and a stored aircraft can return to service or go on to retirement. A retired or withdrawn aircraft has formally left the operating fleet: it is scrapped whole, parted out so its engines and components can be sold separately (Part V §32), or converted to another use, most commonly a passenger-to-freighter (P2F) conversion (Part II §15). [D9 §2.1]

The states move in both directions except the last. Stored aircraft returned to service in volume after 2021. A retired aircraft does two opposite things to the aftermarket at once. It removes a future customer for shop visits, and it adds supply of used parts and of engines with life remaining. That double effect is why the retirement rate in §19 appears on both sides of the translation table.

AVERAGE AGE

The average age of a fleet is the mean number of years since delivery of its aircraft. Maintenance need rises with age in steps rather than smoothly: the first shop visit, the first life-limited-part replacement, the first heavy airframe check. The retirement decision is largely age-related cost set against what the aircraft is worth kept (§19.1). The average age of the world's commercial aircraft was about 15 years at December 2025, per the airline trade body IATA (the International Air Transport Association) as cited in trade press. [D9 §2.1; IATA Global Outlook, Dec 2025, as cited by Aviation Business News]

18.1 EVERY COUNT OF THE CFM56 FLEET

Counting the CFM56 fleet sounds simple and is not. Part I §2.2 gave the headline counts; this section adds what each count covers and why the basis matters for everything that follows. The research reached thirteen figures from eight sources at three levels: the pages and investor materials of the original equipment manufacturer (OEM), the term Part I §2 boxed for the holder of the engine's design; trade press; and a consultancy blog. None of the "in service" figures states whether spare engines and engines on stored aircraft are included. The research recorded that as an open question. [D9 U3; reconciliation R-001]

EXHIBIT 3.1 — EVERY CFM56 FLEET FIGURE IN THE RESEARCH: WHAT EACH COUNTS, WHETHER SPARES AND STORED ENGINES ARE INCLUDED, SOURCE, LEVEL AND DATE

Figure	What it counts, as the source states it	Spares and stored engines	Source and level	Date
>35,000	CFM56 delivered, all variants, across the 737, A320 and military platforms	Not applicable	CFM website (OEM web page); Aviation Business News (trade press)	Undated; Sept 2025
"well over 30,000"	CFM56 delivered	Not applicable	One dossier's general statement, unsourced (reconciliation R-069)	Not dated
~24,000	CFM56 "in service"	Not stated	Aviation Business News (trade press)	Sept 2025
~23,000	CFM56 "in service"	Not stated	CFM website (OEM web page)	Undated, accessed 2026-10-03
~23,000	CFM56 in service	Not stated	GE Aviation & GECAS (GE's then aircraft-leasing arm) Investor Day (18 Jun 2019) and Safran media (2024); which document gave which reading was not captured	2019; 2024
>22,800	CFM56 in Safran's "civil installed base"	Not stated; "civil" excludes military engines	Safran FY2025 results presentation (OEM investor material)	13 Feb 2026 (end-2025 figure)
"over 20,000"	CFM56-5B and -7B in service, the two variants together	Not stated	Chromalloy press release (a parts maker)	30 Oct 2025
>19,000	CFM56 "flying"	"Flying" suggests installed and active; not stated	Safe Fly Aviation (consultancy blog)	2026

Figure	What it counts, as the source states it	Spares and stored engines	Source and level	Date
~14,200	"Active" CFM56, read by one dossier as all variants ($\approx 6,200$ -5B and $\approx 7,500$ -7B) and by another as the -7B fleet alone	"Active" suggests installed and flying; not stated	Safe Fly Aviation (consultancy blog)	2026
"in excess of 8,000"	CFM56-7B "currently installed in 737NGs"	Installed only, by its wording	CFM product page (OEM web page)	Undated
>15,000	CFM56-7B delivered	Not applicable	CFM website; GE Aerospace article on the 15,000th engine	2019 and later
>7,000 aircraft	737NG aircraft delivered with the -7B, two engines each	Not applicable	CFM product page	Undated
>4,100 aircraft	A320ceo-family aircraft delivered with the -5B, two engines each	Not applicable	CFM product page	Undated

Source: CFM International, "The CFM56 engine family" and "CFM56-5B & CFM56-7B" product page, undated, accessed 2026-10-03; Aviation Business News, "CFM56 turbofan aircraft engine", Sept 2025; GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran, FY2025 Results & Investor Update, 13 Feb 2026; Chromalloy, release of 30 Oct 2025; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); GE Aerospace, 2019. Reconciliation R-001, R-002, R-003, R-069; E.1 rules 3 and 14.

Three figures sit at manufacturer level and close together: about 23,000 on CFM's own page, about 24,000 in trade press citing September 2025, and more than 22,800 in Safran's end-2025 count of its civil installed base. [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026] A fourth figure of about 23,000 was recorded by the shop-visit research from GE's 2019 investor day and from Safran media in 2024, without capturing which document gave which reading. [GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; D2 §4.2] If both the 2019 and the 2024 readings are about 23,000, the count barely moved in five years. That is consistent with the low retirement rates in §19, but it is not evidence of them, because neither figure states its basis.

The two Safe Fly Aviation figures come from a consultancy blog and are labelled as such wherever they appear in this primer. [E.1 rule 3] The research read the 14,200 figure two ways. One dossier labelled it the CFM56-7B active fleet. Another labelled it the whole active CFM56 fleet, split about 6,200 -5B and 7,500 -7B. [Safe Fly Aviation, 2026; reconciliation R-002] The same publisher's ">19,000 flying" cannot be reconciled with 14,200 from the material reached. The gap of nearly

5,000 engines may be stored engines, spares, or the older -3, -5A and -5C variants, and the primer records both figures without choosing. [reconciliation R-001]

Why the basis matters is arithmetic. The forecast curve in §23 counts about 2,300 CFM56 shop visits a year. Divided into 24,000 engines, that is one visit per engine roughly every ten years. Divided into 14,200, it is one every six. Which base is right decides whether the fleet visits shops more often or less often than Part I's mature interval of 10,000 to 15,000 cycles implies. Worked example 3.6 does that arithmetic on every base. Here the point is only that the question is live and the sources do not settle it.

18.2 THE SPLIT BETWEEN -5B AND -7B, AND THE V2500 COUNT

The split of the fleet between the two variants that matter for FTAI is unknown. Only the consultancy blog gives one, about 6,200 -5B and 7,500 -7B, and the research read even that figure two ways. [Safe Fly Aviation, 2026; reconciliation R-002] The manufacturer's product page gives delivery counts by airframe instead: more than 7,000 737NGs delivered with the -7B and more than 4,100 A320ceo-family aircraft delivered with the -5B, each aircraft carrying two engines. [CFM product page, undated] Two engines per aircraft gives more than 14,000 -7Bs and more than 8,200 -5Bs installed at delivery, about 22,200 together before any retirement or any spare engine is counted. That is arithmetic on the manufacturer's own figures, done for this Part. It is a check, not a count. It sits close to the "in service" totals above only if few engines have retired and if those totals include spares, and the research cannot say whether either condition holds.

The same page says "in excess of 8,000 CFM56-7Bs are currently installed in 737NGs", with no date visible. [CFM product page, undated; reconciliation R-003] At two engines per aircraft that is 4,000 737NGs, well under the 7,000 delivered. The research recorded that the statement may be old, from early in the programme, or may count a subset, and did not resolve it. Part I §2.2 carries the same tension. [D9 T5]

The V2500 count is the second unknown. Part I §10.1 boxed the V2500, the A320ceo's alternative engine from the IAE consortium. Aviation Week's 2023 fleet and MRO (maintenance, repair and overhaul) forecast put 5,286 V2500s in service in 2023, of which 5,260 were the -A5 and 26 the older -A1. [Aviation Week, V2500 data tool / 2023 Fleet & MRO Forecast] A market-research page gives ">13,000 units" in service in mid-2026; the research flagged the page as low grade. [MarketIntel, 2026; reconciliation R-010] The research's own check runs as follows. The -5B powered more than 4,100 of the roughly 8,000 or more A320ceos delivered, leaving on the order of 4,000 V2500-powered A320ceos plus the MD-90s, so roughly 8,000 installed V2500s at most before retirements. [D9 T8, arithmetic on the sources' figures] The primer carries 5,286 as the sourced count and the ">13,000" only as a low-grade page. A current count from IAE or from RTX (Pratt & Whitney's parent) was not reached. [D9 U4]

EXHIBIT 3.2 — THE AIRFRAMES AND THE OTHER ENGINES: FLEET-LEVEL FIGURES REACHED, WITH SOURCE AND DATE

Item	Figure	Source and level	Date
A320 family (ceo and neo together): orders / deliveries / in service	20,378 / 12,798 / 11,308	Wikipedia order-and-delivery list (secondary compilation)	July 2026
A320ceo-family aircraft stored 90+ days, and their average age	~570; 17.1 years	Naveo, AFRA (an aircraft-recycling trade association) panel (consultancy presentation)	6 Jun 2023
Share of 737NGs / A320ceos aged ten years or less	47% / 45%	ISTAT (the International Society of Transport Aircraft Trading) via ePlaneAI	Undated; probably 2021–22 (D9 U2)
Average age of the commercial fleet	~15 years	IATA Global Outlook via Aviation Business News	Dec 2025
LEAP (CFM's new-generation engine) in service	~5,500; ~50% under long-term service agreements against ~15% of the CFM56 fleet	Aviation Week (trade press)	Undated
V2500 in service	5,286 (5,260 -A5; 26 -A1)	Aviation Week 2023 Fleet & MRO Forecast	2023
V2500 "in-service fleet"	">13,000 units"	MarketIntelto (market-research page; low grade)	Mid-2026
737-800 freighters in service / conversion pipeline	>260 / >300	Acumen Aviation; Cargo Facts (trade press)	Mid-2025
737-700 freighter conversions to date	<50	Acumen Aviation	18 Dec 2025
Age at narrowbody freighter conversion (737, 757)	21–22 years on average	Cargo Facts	Date not captured
737 MAX (the re-engined 737) cumulative deliveries	2,399	Wikipedia order-and-delivery list	2026

Source: Wikipedia, "List of Airbus A320 family orders and deliveries", July 2026, and "List of Boeing 737 MAX orders and deliveries", 2026; Naveo Consultancy, AFRA Aircraft & Engine Panel presentation, 6 Jun 2023; ISTAT, "CFM International CFM56-5B/-7B", undated; Aviation Business News, "CFM56 turbofan aircraft engine"; Aviation Week, "Leap Aftermarket Poised For Growth", undated, and "Data Tool: The Future Of IAE V2500 Engine"; MarketIntelto, "V2500 MRO Market Research

Report 2034"; Acumen Aviation, 737-700 market insight, 18 Dec 2025; Cargo Facts, two articles, dates not captured. D9 §4.1; reconciliation R-010.

Two things in that table are themselves unknowns. The combined A320 figure of 11,308 in service is not split between ceo and neo, so the number of A320ceos flying in 2026 was not reached. [D9 U1] The "47% / 45% aged ten years or less" statistic is undated; since -7B production ended in 2019 and -5B production in 2022, the share must be lower in 2026, and the research judged the figure to be from about 2021 or 2022. [D9 U2]

18.3 HOW MATURE THE FLEET IS

A second way to describe a fleet is by how many shop visits its engines have had. The research used the phrase "fleet maturity" for this without defining it; here it means the share of the installed base by the number of shop visits endured, none, one, or two or more. It matters because an engine that has not yet had a first visit is a visit still to come, and because the first visit on a mature engine is usually the heaviest event of its life so far. Part I §6.2 explained that first runs are long, with the latest -5B and -7B variants often flown almost to their life-limited-part limit before the first removal.

EXHIBIT 3.3 — CFM56 FLEET MATURITY: SHARE OF ENGINES BY SHOP VISITS ENDURED, THREE READINGS

Reading	No shop visit yet	One shop visit	Two or more	Source and level	Date
GE investor day	57%	21%	22% (by subtraction)	GE Aviation & GECAS Investor Day (OEM investor material)	18 Jun 2019
Safran media	About 70% with zero or one, not split		About 30% (by subtraction)	Safran media (OEM)	2024
Consultancy blog	"over half"	"another quarter"	The remainder	Safe Fly Aviation (consultancy blog)	2026

Source: GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024, via Aviation Week; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog). Compiled in D2 §4.2; reconciliation R-012.

The three readings are not on one base and are not all figures. The two manufacturer readings put the zero-or-one share at 78% in 2019 and about 70% in 2024. The consultancy blog's "over half" plus "another quarter" is about 75% in 2026, which would be higher than Safran's 2024 share. [reconciliation R-012] No engine loses a shop visit, so a rising share can only come from retirements removing multi-visit engines from the denominator, or from the readings being on different bases. The primer records the three readings and stops.

WORKED EXAMPLE 3.1 — HOW MANY CFM56 ENGINES HAVE NEVER BEEN TO A SHOP, ON EACH BASE AND EACH READING

Inputs. Shares (sourced, Exhibit 3.3): 57% with no first visit and 21% with one (GE, June 2019); about 70% with zero or one (Safran, 2024); "over half" with none and "another quarter" with one (Safe Fly Aviation, consultancy blog, 2026). Bases (sourced, Exhibit 3.1): about 23,000 (CFM, undated; also the 2019 and 2024 readings via D2); about 24,000 (Aviation Business News, Sept 2025); more than 19,000 and about 14,200 (Safe Fly Aviation, consultancy blog, 2026). Each share is applied to the base from its own source or date; no share is applied to another publisher's base without saying so.

Step 1, the 2019 reading on the 2019 base.

no first visit: $0.57 \times 23,000 = 13,110$ engines
 one visit: $0.21 \times 23,000 = 4,830$ engines
 two or more: $23,000 - 13,110 - 4,830 = 5,060$ engines

Step 2, the 2024 reading. Safran did not split zero from one.

zero or one visit, on 23,000: $0.70 \times 23,000 = 16,100$ engines
 zero or one visit, on 24,000: $0.70 \times 24,000 = 16,800$ engines

Step 3, the 2026 consultancy reading on the consultancy's own bases. "Over half" is taken as at least 50% and "another quarter" as 25%, which turns words into floor figures.

on >19,000: no visit $\geq 9,500$; one visit $\approx 4,750$
 on ~14,200: no visit $\geq 7,100$; one visit $\approx 3,550$

Step 4, what the readings say together. The zero-or-one share moved from 78% to about 70% between the two manufacturer readings. On a 23,000 base that is about 1,840 engines. It is not a count of visits performed, because retirements of older, multi-visit engines also move the share, in the other direction. The consultancy's 2026 share of about 75% sits above the 2024 share and cannot be reconciled with it on one base (reconciliation R-012).

So on the manufacturer readings, somewhere between 13,000 and 17,000 CFM56 engines had seen at most one shop visit in 2019 to 2024, which is the pool from which the first-shop-visit peak in 2027 (§23.1) is drawn. Limits of the example: every share is on an unstated base; the dates differ by up to seven years; two of the shares are words rather than figures; and the -5B/-7B split of the pool is unknown (§18.2). [GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; Safe Fly Aviation, 2026; D2 §4.2; reconciliation R-001, R-012]

19. Retirements: how the decision is made and what happened 2019–2026

Part I §2.3 boxed the retirement rate as the share of a fleet withdrawn in a year, and gave GE's falling planning assumption in one sentence. This section is the full treatment. It explains how an owner decides to retire an aircraft, sets out what actually retired between 2019 and 2025 with the two data providers' conflicting counts, follows GE's planning number down from 3–4% to "below 1%", and works the arithmetic of what one and three-quarter points of retirement rate mean in engines a year.

19.1 HOW THE DECISION IS MADE

An aircraft is retired by its owner, an airline or a lessor, when keeping it in service costs more than it is worth kept, and when the alternatives of selling it on, converting it to a freighter or parting it out are better. Five inputs enter that decision. [D9 §2.3]

The first input is age, and the way maintenance cost arrives in lumps rather than smoothly. An engine performance-restoration shop visit opens the core to restore exhaust-gas-temperature margin (Part I §5 and §6). A life-limited-part (LLP) replacement forces the engine open when a rotating part reaches its certified cycle limit (Part I §4). A heavy airframe check does the same to the airframe. Each is a multi-million-dollar event. Acumen Aviation's December 2025 note on the 737-700 states that "as the 737-700 fleet ages, heavy checks and major engine shop visits increasingly exceed the residual value of the aircraft, driving earlier retirements or conversions".

[Acumen Aviation, 18 Dec 2025] A 2023 analysis found that aircraft become vulnerable to retirement at 16 to 18 or more years, particularly if the owner does not want to pay for the next heavy check or shop visit, and that the roughly 570 A320ceo-family aircraft then stored for 90 or more days had an average age of 17.1 years. [Naveo, AFRA panel, 6 Jun 2023]

MAINTENANCE CLIFF

The maintenance cliff is the point at which the next heavy maintenance event, an engine shop visit, a life-limited-part replacement or a heavy airframe check, costs more than the aircraft is worth kept. An owner at the cliff retires, converts or sells the aircraft rather than pay for the event. Aircraft become vulnerable to it at about 16 to 18 years and older. Aviation Week's rule of thumb that a CFM56 needs a shop visit roughly every eight years makes 24 years a de facto retirement date, because a third full overhaul is "deemed too expensive". [D9 §2.3; Naveo, 6 Jun 2023; Aviation Week, "Retirement Uptick Will Restock Used Engine Parts", date not captured]

The second input is lease economics. At the end of an operating lease (Part II §12) the lessor chooses between extending with the current airline, moving the aircraft to a new lessee, or selling or parting it out. The airline, for its part, chooses between extending and returning. Both

choices turn on whether a replacement aircraft is available, and between 2023 and 2026 it usually was not (§22).

LEASE TRANSITION

A lease transition is the move of an aircraft from one lessee to the next. Between the two leases the lessor pays for reconfiguration of the cabin, for the maintenance needed to meet the next lessee's delivery condition, and for the downtime while the aircraft earns no rent. Transition is the alternative to extension (§22.3), and its cost is one reason a lessor extends when it can. [D9 §2.3]

The third input is freighter conversion, which Part II §15 owns. A passenger-to-freighter conversion gives a 20-plus-year-old airframe a second life. Cargo Facts reports narrowbody freighters, the 737 and 757, being converted at 21 to 22 years of age on average. [Cargo Facts, date not captured] More than 260 737-800 freighters were in service as of mid-2025 with a conversion pipeline exceeding 300, against fewer than 50 737-700 conversions. [Acumen Aviation, 18 Dec 2025; Cargo Facts] A conversion candidate still needs engines with life on them. So conversion is a source of demand for engines with green time (the remaining usable life on an engine before its next mandatory shop visit, Part II §16) or for shop visits, and not a source of teardown supply.

The fourth input is part-out value (Part V §32). When the value of an aircraft as a sum of parts exceeds its value flying, the owner parts it out. For a mid-life CFM56-powered narrowbody the engines are the dominant share of that sum, and Part V §32 works the numbers. The consultancy figures the research reached are that a run-out CFM56 is bought as feedstock for \$0.8–1.2M, yields \$1.6–2.2M of used material, and nets \$0.4–0.8M before repair cost. [Safe Fly Aviation, 2026, consultancy blog] Rising used-part prices raise part-out value and so, at the margin, encourage retirement. That is a stabilising feedback. The research records that it was not strong enough to lift retirements in 2024 to 2026. [D9 §2.3]

The fifth input is replacement availability, and it governs the other four.

REPLACEMENT AVAILABILITY IS THE GATE

None of the first four inputs triggers a retirement if the airline cannot get a replacement aircraft. An airline at the maintenance cliff with no new aircraft to fly pays for the shop visit anyway, because the alternative is to shrink. A lessor whose lessee cannot find a replacement extends the lease rather than take the aircraft back into a market with no buyer-operators. This is the single mechanism through which the new-engine and airframer problems of §20 and §21 act on the old fleet: they close the gate. Cirium Ascend's count that only about a dozen mid-life 737NGs and fewer than 25 A320neos were available to rent anywhere in the world is the measure of how closed it was. [D9 §2.3; Cirium Ascend via Bloomberg, 2023–24 context]

EXHIBIT 3.4 — INPUTS TO THE RETIREMENT DECISION AND THE FIGURE THE RESEARCH REACHED FOR EACH

Input	Figure reached	Source and level	Date
Age at which aircraft become vulnerable to retirement	16–18+ years	Naveo, AFRA panel (consultancy presentation)	6 Jun 2023
Stored A320ceo-family aircraft (90+ days) and their average age	~570; 17.1 years	Same	6 Jun 2023
Shop-visit rule of thumb and the de facto retirement age it implies	About every 8 years; 24 years	Aviation Week (trade press)	Date not captured
737NG delivery rate in 1998–2002, the cohort now at the cliff	~230 a year	Aviation Week	Date not captured
Age at freighter conversion, 737 and 757	21–22 years on average	Cargo Facts (trade press)	Date not captured
737-800 freighters in service / pipeline	>260 / >300	Acumen Aviation; Cargo Facts	Mid-2025
737-700 freighter conversions to date	<50	Acumen Aviation	18 Dec 2025
Run-out CFM56 as feedstock: purchase / used-material yield / net	\$0.8–1.2M / \$1.6–2.2M / \$0.4–0.8M	Safe Fly Aviation (consultancy blog)	2026
CFM56-7B half-life market value, the "kept flying" side of the comparison	\$5.7M (-7B24); \$6.4M (-7B27)	IBA (appraiser)	Sept 2025
Mid-life aircraft available to rent worldwide (replacement availability)	~12 737NGs; <25 A320ceos	Cirium Ascend via Bloomberg (data provider via press)	2023–24 context

Source: Naveo Consultancy, AFRA panel, 6 Jun 2023; Aviation Week, "Retirement Uptick Will Restock Used Engine Parts", date not captured; Cargo Facts, "Passenger-to-freighter conversions are taking place on increasingly older aircraft" and "In-service 737NG freighter fleet surpasses 100"; Acumen Aviation, 18 Dec 2025; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); IBA, Engine Values Release 2025B, Sept 2025; Bloomberg via AJOT (the American Journal of Transportation), "Used jets rents surge 44%". D9 §2.3; D3 §2.6 and §2.8.

19.2 AIRCRAFT RETIREMENTS, 2019 TO 2025

The research reached aircraft-level counts from two data providers, one appraiser and one consultancy, and an engine-level rate from GE. They agree on direction and disagree on level. [reconciliation B-6, R-005] Exhibit 3.5 sets out every figure with the denominator its source states, because the figures are not on one basis and the reconciliation instructs that they never be mixed.

[registry C-10]

EXHIBIT 3.5 — REALISED RETIREMENTS 2019–2025, EVERY FIGURE WITH ITS DENOMINATOR, SOURCE AND DATE

Period	Figure	Denominator, as stated	Source and level	Date
2010–2019 average	Just under 540 a year; peak 694 in 2013	Aircraft part-outs, all types	Cirium Fleets Analyzer via Aerodynamic Advisory at an IATA maintenance conference (data provider via consultancy)	2023
2019	502	Aircraft retirements	Cirium via Aerodynamic Advisory	2023
2020	465	Same	Same	2023
2021	333	Same	Same	2023
2022	325	Same	Same	2023
2023	"Similar" to 2024's fewer than 200	Aircraft; scope not visible	IBA via AviTrader (appraiser via trade press)	24 Sep 2024
2024	Fewer than 200; 70% below the 2020 peak	Aircraft; scope not visible	IBA via AviTrader	24 Sep 2024
2024	587	Aircraft; scope not visible	Cirium Fleets Analyzer via an Informa forecast page	Undated forecast page
2019 → 2024	2.2% → 0.4%	Passenger aircraft retired as a share of the active-plus-stored fleet	Aviation Week / Bain material; exact document not visible	Undated
2024–2026	About 24% below the 2010–2019 norm	Aircraft, all types	McKinsey (consultancy)	Date not captured (2024–25)
Next five years	1,190 A320ceo-family and 795 737NG aircraft, 35% of all retirements	Aircraft	Cirium / Ascend by Cirium	Date not captured
By 2040	About 72% of all 737NGs retired	737NG aircraft	Ascend by Cirium	Date not captured

Period	Figure	Denominator, as stated	Source and level	Date
2024 and 2025	About 1.5% of the installed base in 2025, "broadly consistent with 2024"	CFM56 engines	GE Q4 2025 results via Leeham News	22 Jan 2026
1Q 2026	"Below 1%"	CFM56 engines	GE Aerospace investor-relations page	2026
2025	46.1%	Global Replacement Ratio, 2015–2024 basis	Cirium via Informa	Undated

Source: IATA Maintenance Cost Conference, Aerodynamic Advisory (Murby), "Retirements, part-outs, dismantling", 2023; AviTrader, "Global airline industry set for record growth in 2024" (IBA data), 24 Sep 2024; Informa Markets, Commercial Forecast story, page 5 (Cirium figures); Aviation Week, "Pockets Of Demand Disrupt Expected Narrowbody Retirement Surge"; McKinsey, "What does the future hold for commercial-aviation maintenance?"; Leeham News, 22 Jan 2026; GE Aerospace, "Recent events: your questions answered", 2026. D9 §2.4 and §4.2; D3 §2.6 and §4.4; reconciliation R-004, R-005; E.1 rule 13.

The 2024 count is the tension. IBA, an appraiser, said in September 2024 that aircraft retirements would stay at fewer than 200 for the year, similar to 2023 and 70% below the 2020 peak. [IBA via AviTrader, 24 Sep 2024] Cirium's Fleets Analyzer, as cited on a forecast page, recorded 587 retirements in 2024. [Cirium via Informa] The two cannot both count the same thing. The scope is visible in neither: one may count passenger jets only, one may be a year-to-date figure at the time of writing, one may apply a size threshold. [D9 T2; reconciliation R-005] The series also disagree on where the peak sits. IBA calls 2020 the peak; in Cirium's series 2019, at 502, stands above 2020, at 465. The reconciliation added two further figures from arithmetic on the sources' own numbers, derived in the research and not counts. McKinsey's "24% below the 2010–2019 norm" applied to the just-under-540 average would be about 410 a year. Cirium's projection that 1,190 A320ceo-family and 795 737NG aircraft will be 35% of retirements over five years implies about 5,670 retirements in total, about 1,130 a year. [reconciliation R-005] So the research holds four numbers for roughly the same quantity, fewer than 200, about 410, 587 and about 1,130, with the scope visible in none. The primer presents them and stops. Year-by-year counts for 2023, 2024 and 2025 from one provider on one basis were not reached. [D3 U16]

The passenger-fleet share tells the same story in a different unit. Retirements as a share of the active-plus-stored passenger fleet fell from 2.2% in 2019 to 0.4% in 2024. [Aviation Week / Bain material, undated] Cirium's Global Replacement Ratio expresses the shortfall from the delivery side.

GLOBAL REPLACEMENT RATIO

The Global Replacement Ratio is the number of deliveries needed to replace retirements, expressed as a share of total deliveries over a period. Cirium put it at 46.1% for 2025 on 2015–2024 data: fewer than half of new deliveries were replacing a retired aircraft, and the rest were growth. A low ratio means the fleet is growing by addition rather than substitution, which is the delivery-side face of low retirements. [Cirium via Informa, undated; D9 §2.4]

Named programmes show the decision being made both ways at once. United, WestJet and Copa have started retiring 737NGs in favour of the 737 MAX 8. [Acumen Aviation, 18 Dec 2025] Wizz Air announced a gradual phase-out of its A321ceo fleet on 24 March 2026. [Wizz Air release, 24 Mar 2026] Against those, Lauda Europe, part of the Ryanair group, is extending A320 leases through the early 2030s because the transition to new aircraft proved too costly, and Lufthansa and Ryanair were reported holding onto older aircraft to fill the gap. [Travel Extra, 2026; Bloomberg via AJOT] IBA noted at the same time that delivery delays had led operators to maximise daily utilisation across all age groups, with mature passenger-narrowbody utilisation "rising significantly". [IBA via AviTrader, 24 Sep 2024] That utilisation effect has its own row in the translation table, because it adds cycles without adding engines.

19.3 THE CFM56 ENGINE RATE: GE'S PATH FROM 3–4% TO BELOW 1%

The figure that the rest of the market plans against is GE's. As one of the two partners in CFM, GE sets a retirement assumption for the CFM56 fleet each year and reports against it. The research captured that assumption at four dates in two venues, and the readings do not quite agree with each other.

EXHIBIT 3.6 — GE'S CFM56 RETIREMENT STATEMENTS BY DATE AND VENUE, WITH THE OTHER DENOMINATORS IN USE

Statement	Figure	Denominator	Venue and date
Original planning assumption for 2026 ("last year")	3–4%	CFM56 installed base	GE CFO Rahul Ghai via Aviation Week, 2026 (exact date not visible)
Revised at the start of 2026	2–3%	Same	Same
"Currently seeing something between"	1.5–2%	Same	Same
2025 actual	About 1.5%, broadly consistent with 2024	CFM56 installed base	GE Q4 2025 results via Leeham News, 22 Jan 2026

Statement	Figure	Denominator	Venue and date
2026 expected	About 2%, "below the 2–3% range previously anticipated"	Same	Same
2026 outlook assumption	About 2% of total fleet	CFM56 fleet	GE Aerospace investor-relations page, 2026
1Q 2026 actual	"Below 1%"	Same	Same
Consultancy reading	"Near 2% per year"	About 14,200 "active" CFM56 (consultancy base)	Safe Fly Aviation, 2026 (consultancy blog)
"Normal" rate expected from 2028	About 2.7% a year	All commercial aircraft	McKinsey, 2024–25
Passenger fleet, 2024	0.4%	Passenger aircraft, active plus stored	Aviation Week / Bain material, undated
V2500	"No retirement surge expected over next five years"	V2500 engines	IAE via Aviation Week, 2024/25

Source: Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Leeham News, "GE Aerospace FY and Q4 2025 Earnings", 22 Jan 2026; GE Aerospace Investor Relations, "Recent events: your questions answered", 2026; Safe Fly Aviation, 2026 (consultancy blog); McKinsey, date not captured; Aviation Week, "Pockets Of Demand Disrupt Expected Narrowbody Retirement Surge"; Aviation Week V2500 coverage via D2 §4.2. Reconciliation R-004; D9 T3; registry C-10; E.1 rule 13.

The path is the same in both venues: the assumption fell, and the outturn fell further. The readings differ in detail. Aviation Week's quotation of the chief financial officer has the 2026 planning number starting at 3–4% "last year", cut to 2–3% at the beginning of the year, and running at 1.5–2% as "currently seeing". [GE CFO via Aviation Week, 2026] Leeham's reading of the fourth-quarter 2025 call has 2025 at about 1.5%, broadly the same as 2024, and 2026 expected at about 2%, "below the 2–3% range previously anticipated". [GE Q4 2025 results via Leeham, 22 Jan 2026] The two differ by half a point on the 2026 expectation, 2% against 1.5–2%, and on whether 3–4% was ever the 2026 number or an earlier year's. [D9 T3] GE's own investor-relations page then reported first-quarter 2026 retirements "below 1%", against the about 2% it assumed for the year. [GE Aerospace investor-relations page, 2026; reconciliation R-004] A single quarter is not a year, and the page does not say whether the quarterly figure is annualised. The primer records all three readings.

Every other rate in the exhibit is on a different denominator and must not be set beside GE's as if it were the same quantity. [registry C-10] GE counts CFM56 engines. McKinsey's 2.7% counts all commercial aircraft and is a forecast of a return to normal from 2028, not an outturn. [McKinsey, 2024–25] The 0.4% counts passenger aircraft against an active-plus-stored base, which is a larger

denominator than the in-service fleet. The consultancy blog's "near 2%" is stated against its own 14,200 "active" base, which §18.1 showed is itself contested. [Safe Fly Aviation, 2026; reconciliation R-002] For the V2500, IAE expected "no retirement surge" over five years, which is a statement of direction without a rate. [IAE via Aviation Week, 2024/25]

WORKED EXAMPLE 3.2 — WHAT A 1.75-POINT DIFFERENCE IN THE RETIREMENT RATE MEANS IN ENGINES A YEAR

Inputs. Installed base (sourced, Exhibit 3.1): about 24,000 (Aviation Business News, Sept 2025); about 23,000 (CFM website, undated); more than 22,800 (Safran, end-2025). Consultancy base (sourced, labelled): about 14,200 "active" (Safe Fly Aviation, 2026). Rates (sourced, Exhibit 3.6): 3.5%, the midpoint of GE's original 3–4%; 2.5%, the midpoint of 2–3%; 1.75%, the midpoint of 1.5–2%; 1.5%, the 2025 outturn; 2%, the 2026 expectation; below 1%, the first quarter of 2026. The midpoints are this Part's choice for arithmetic and are not GE figures. Every base may include spare and stored engines (D9 U3).

Step 1, engines a year on the 24,000 base.

$$\begin{array}{ll} 3.5\%: 24,000 \times 0.035 = 840 & 2.5\%: 24,000 \times 0.025 = 600 \\ 1.75\%: 24,000 \times 0.0175 = 420 & 1.5\%: 24,000 \times 0.015 = 360 \\ 2.0\%: 24,000 \times 0.020 = 480 & \text{below } 1\%: \text{fewer than } 24,000 \times 0.010 = 240 \end{array}$$

Step 2, the same on the other manufacturer-level bases.

$$\begin{array}{l} \text{on } 23,000: 3.5\% = 805; 1.75\% = 403; 1.5\% = 345; 2.0\% = 460 \\ \text{on } 22,800: 3.5\% = 798; 1.75\% = 399; 1.5\% = 342; 2.0\% = 456 \end{array}$$

Step 3, the 1.75-point gap between the original plan and the realised range.

$$\begin{array}{l} \text{on } 24,000: 24,000 \times (0.035 - 0.0175) = 420 \text{ engines a year} \\ \text{on } 22,800: 22,800 \times 0.0175 = 399 \text{ engines a year} \\ \text{over three years 2024–26: about 1,260 (on 24,000) or about 1,200 (on } 22,800) \end{array}$$

The three-year figures are upper bounds, because the base shrinks a little each year as engines leave (D9 §2.2). Step 4, the consultancy base, shown because Part IX's feedstock arithmetic was first done on it.

$$\text{on } 14,200: 3.5\% = 497; 1.75\% = 249; 2.0\% = 284; \text{ the 1.75-point gap} = 249 \text{ engines a year}$$

Step 5, what 420 engines a year are in the aftermarket's units. In aircraft, 420 engines are 210 narrowbody-equivalents at two engines each. In shop visits, each retained engine eventually needs the visit it would otherwise have avoided; if every one needed one restoration during its retained period, 420 is at most about 18% of a 2,300-visit year ($420 \div 2,300 = 0.183$), spread over several years rather than falling in one, so the annual effect is smaller than 18% and the figure is a ceiling. In

used material, 420 fewer teardowns a year at the consultancy's yield of \$1.6–2.2M of used parts per run-out engine is \$0.67–0.92B a year of used parts not produced (illustrative; consultancy input). In FTAI Power's unit, 100 Mod-1 cores a year (Part IX §74) would be 24% of 420 retirements, 28% of 360, 21% of 480, and 35% of the 284 implied on the consultancy base at 2%, which is the pair of readings the reconciliation requires Part IX to show (reconciliation C.8).

So the gap between what the manufacturer planned and what happened is about 400 engines a year on any manufacturer-level base, or about 1,200 engines over 2024 to 2026, that are still flying and will need work, and that were not torn down for parts. Limits of the example: the base is undefined as to spares and stored engines; the rates come from different dates and venues (D9 T3); the midpoints are the Part's, not GE's; the consultancy base is not a manufacturer figure; and the first-quarter 2026 rate may not hold for the year. [GE CFO via Aviation Week, 2026; GE Q4 2025 results via Leeham, 22 Jan 2026; GE Aerospace investor-relations page, 2026; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; Safe Fly Aviation, 2026; D9 §2.2; reconciliation R-001, R-004, C.8]

19.4 BOTH SIDES OF THE LEDGER

The retirement rate is one number with two effects, and the translation table carries them as separate rows. Lower retirements mean more engines accruing cycles, so more shop visits, more life-limited-part replacements as retained engines reach their limits, and more full restorations in place of green-time run-outs; that is row T-1 in §24. Lower retirements also mean fewer teardowns, so less used material and fewer green-time engines, and higher prices for both; that is row T-9. The shop-visit research describes the second effect from the shop's side: "the supply of used serviceable material remains thin, leaving shops more reliant on newly manufactured core parts". [AVM (Aviation Maintenance magazine), 2025, via D2 §2.4] The used-material research describes it from the trader's side, with AerSale reporting a "hypercompetitive feedstock environment" and its whole-asset sales falling from 20 engines and one aircraft in 2024 to 13 engines in 2025. [AerSale, 5 Mar 2026, as relayed in search results; D3 §2.6] Part V §33 carries the supply side in full.

20. New-generation engines: GTF and LEAP durability, and what they did to the old fleet

The A320neo and the 737 MAX (the re-engined 737) were designed to replace the A320ceo and the 737NG. Two things slowed the replacement. The new engines spend less time on wing than planned, time on wing being the interval from an engine's installation to its first removal (Part I §5). And the airframers delivered fewer aircraft than planned (§21). Both keep older aircraft flying longer and harder. This section takes the engines. Part I §10.2 boxed the LEAP (CFM's new-generation engine, the LEAP-1A on the A320neo family and the LEAP-1B as the sole

engine of the 737 MAX) and its durability kit; this section carries what each new engine's problem did to the old fleet, which Part I left to it.

Two new engines are involved, and they are not symmetric. The LEAP comes from the same manufacturer as the CFM56 and sits on both airframe families. The other comes from Pratt & Whitney and sits on one.

GEARED TURBOFAN (GTF)

The geared turbofan is Pratt & Whitney's new-generation engine. A reduction gearbox between the fan and the low-pressure turbine lets each turn at its own best speed, which the CFM56's direct-drive design does not allow. The PW1100G is one of two engine options on the Airbus A320neo family, the other being the LEAP-1A; the PW1500G and PW1900G are the sole engines of the Airbus A220 and the Embraer E2. Pratt & Whitney's parent is RTX. [D9 §2.5]

20.1 THE PRATT & WHITNEY GTF POWDER-METAL ISSUE

GTF POWDER-METAL ISSUE

Certain turbine discs in the GTF are made by a powder-metallurgy route, in which metal powder is consolidated into a disc. Powder produced over a period of years was found to be contaminated. The affected discs must be inspected or replaced, which can be done only with the engine off the wing and opened. RTX disclosed the defect in July to September 2023, estimated that 600 to 700 engines would be pulled for inspection, took a charge of about \$3 billion on 11 September 2023, and later estimates put the total programme cost at about \$7 billion. [D9 §2.5; BusinessDay, 11 Sep 2023; Mobility Outlook, date not captured]

The planning figure and the outturn diverged. RTX's planning figure at the time of disclosure was that around 350 aircraft on average would be grounded across 2024 to 2026. [The Flying Engineer, citing RTX, 2026] Cirium data cited in trade press put 835 PW1100G-powered aircraft grounded or in storage globally as of late 2025. [Cirium via The Flying Engineer, 2026] The same source gives "720 out of 1,912 A320neo family aircraft, nearly four in ten, currently out of service due to engine issues"; the denominator is not defined in the material reached, and the research judged it most plausibly to be the Pratt & Whitney-powered A320neo sub-fleet rather than all A320neos. [D9 T7, U5] An aircraft unable to fly for lack of a serviceable part or engine is said to be on the ground, abbreviated AOG (aircraft on ground), a term Part II §14 carries because short-term engine leases exist to cover such events.

RTX's own framing at its full-year 2025 results on 27 January 2026 was that 2026 is "the third year of elevated AOG levels"; that AOG levels were down more than 20% from their 2025 peak; that PW1100 AOG was down 25% year to date, enabled by a 40% increase in maintenance output and a 23% reduction in turnaround time; that fourth-quarter 2025 GTF maintenance

output was up 39% year on year; and that customer-compensation cash outflows were about \$1 billion in 2025, about \$700 million expected in 2026, and about \$2.8 billion cumulative by the end of 2026. [RTX FY2025 results via Leeham, 27 Jan 2026] Trade commentary expects the issue to be fully worked through only in 2027 to 2028. [The Flying Engineer, 2026] Named effects include Turkish Airlines grounding more than 40 aircraft and Wizz Air signing a compensation agreement with Pratt & Whitney. [Aviacionline; Shares Magazine; dates not captured]

EXHIBIT 3.7 — GTF GROUNDED-AIRCRAFT COUNTS AND RECOVERY INDICATORS, BY SOURCE AND DATE

Item	Figure	Source and level	Date
Engines to be pulled for inspection, initial estimate	600–700	RTX via BusinessDay (OEM via press)	11 Sep 2023
Charge taken / total programme cost	About \$3bn / about \$7bn	RTX via BusinessDay; Mobility Outlook	11 Sep 2023; later
RTX planning figure, aircraft grounded on average 2024–26	About 350	RTX, as cited by The Flying Engineer	2023 figure
PW1100G-powered aircraft grounded or in storage, late 2025	835	Cirium via The Flying Engineer (data provider via blog)	Late 2025
"Out of service due to engine issues"	720 of 1,912, about 38%; denominator undefined	The Flying Engineer	2026
AOG level against the 2025 peak	Down more than 20%	RTX FY2025 results via Leeham News	27 Jan 2026
PW1100 AOG, year to date	Down 25%	Same	27 Jan 2026
GTF maintenance output / turnaround time	Output up 40%; turnaround down 23%; Q4 output up 39% year on year	Same	27 Jan 2026
Customer-compensation cash	About \$1bn (2025); about \$700m (2026); about \$2.8bn cumulative by end-2026	Same	27 Jan 2026
Turkish Airlines aircraft grounded	More than 40	Aviacionline (trade press)	Date not captured

Item	Figure	Source and level	Date
GTF Advantage certification	EASA (the European Union Aviation Safety Agency) certified 2026; entry into service later in 2026	Aviation Today	2026
Expected full resolution	2027–28	The Flying Engineer (trade commentary)	2026

Source: BusinessDay, "RTX takes \$3bn charge over Pratt & Whitney jet engine recall", 11 Sep 2023; Mobility Outlook, "Troubled GTF engines to cost about \$7bn"; The Flying Engineer, "A320neo grounded 2026: Pratt & Whitney crisis" (citing Cirium); Leeham News, "RTX 2025 Earnings", 27 Jan 2026; Aviacionline; Aviation Today, GTF Advantage item. D9 §2.5 and §4.3; D9 T6 and T7.

The counts in that exhibit are in tension and the research did not resolve them. RTX's 2023 planning figure of about 350 aircraft grounded on average is less than half of Cirium's late-2025 count of 835 grounded or in storage. [D9 T6] The two are not the same measure: an average over three years is not a point count, and "in storage" may include aircraft parked for reasons other than the inspection. RTX's January 2026 statement gives a change, down more than 20% from the peak, without the level, so the peak itself and a year-by-year series from RTX's own disclosures were not reached. [D9 U5] The "720 of 1,912" figure cannot be reconciled with 835 without the article that gave it. [D9 T7] The primer carries all three.

GTF ADVANTAGE

The GTF Advantage is Pratt & Whitney's upgraded PW1100G, certified by EASA in 2026 for entry into service later in 2026. It is the fix for the fleet going forward; the discs already in service are dealt with by the inspection programme. [Aviation Today, 2026; D9 §2.5]

The mechanism that connects a grounded A320neo to the CFM56 aftermarket is replacement flying. Every PW1100G-powered A320neo on the ground needs its schedule flown by something else. The practical replacement is an A320ceo. The two share a pilot type rating, the licence endorsement that qualifies a pilot on one aircraft type, the same cabin-crew qualification and the same maintenance-organisation approvals, so an airline can substitute one for the other without retraining or re-approval. [D9 §2.5] That is why the A320ceo lease-extension and used-rent figures in §22 are what they are. The engines affected are the A320ceo's, the V2500 and the CFM56-5B, and not the 737NG's -7B.

WORKED EXAMPLE 3.3 — WHAT THE GTF GROUNDINGS MEAN IN A320CEO FLYING AND IN CEO ENGINE SHOP VISITS

Inputs. Grounded aircraft: 835 PW1100G-powered A320neos grounded or stored in late 2025 (sourced: Cirium via trade press); 350 aircraft on average in 2024–26 (sourced: RTX's 2023 planning

figure). Engines per aircraft: two. Utilisation of a mature narrowbody: 1,000–1,500 cycles a year (sourced: Aircraft Value News, via Part I §1). Mature CFM56 restoration interval: 10,000–15,000 cycles (sourced: Leeham, 8 Mar 2024, Part I §6.2); the V2500's interval was not retrieved, and this example applies the CFM56 interval to the whole ceo fleet as an assumption. Assumption: each grounded neo's flying is replaced one-for-one by a ceo that would otherwise have been parked or retired, at the same utilisation. That is the upper-bound case; in practice some flying was cancelled, some was absorbed by existing ceos flying harder (row T-12 in §24), and some by other types.

Step 1, engines whose flying must be replaced.

on 835 aircraft: $835 \times 2 = 1,670$ engine positions
on 350 aircraft: $350 \times 2 = 700$ engine positions

Step 2, cycles a year flown by the replacing ceo engines that would not otherwise have been flown.

on 1,670 positions: $1,670 \times 1,000$ to $1,670 \times 1,500 = 1.67$ to 2.5 million cycles a year
on 700 positions: $700 \times 1,000$ to $700 \times 1,500 = 0.7$ to 1.05 million cycles a year

Step 3, shop visits those cycles bring forward, at one restoration per 10,000–15,000 cycles.

on 1.67–2.5 million cycles: $1,670,000 \div 15,000 = 111$ to $2,500,000 \div 10,000 = 250$ visits a year
on 0.7–1.05 million cycles: $700,000 \div 15,000 = 47$ to $1,050,000 \div 10,000 = 105$ visits a year

Step 4, scale. The ceo-engine visits brought forward are 47 to 250 a year against a CFM56 forecast of 2,300–2,400 a year and a V2500 outturn of slightly more than 800 in 2025 (§23). They fall on the -5B and the V2500 in proportion to the ceo fleet mix, which is itself unknown (§18.2). Nothing in this example touches the -7B.

So the groundings add, at most, a few hundred ceo engine shop visits a year, and they do so by bringing visits forward in calendar time rather than by creating engines. The same engines would have reached those visits later, or been retired before them. Limits of the example: one-for-one replacement is the ceiling; the CFM56 interval is applied to the V2500 without a source; the 835 count includes stored aircraft that may be parked for other reasons (D9 T6); and the groundings were already falling by January 2026 (Exhibit 3.7), so the figures describe 2024–25 better than 2026.

[Cirium via The Flying Engineer, 2026; RTX via The Flying Engineer; Aircraft Value News, undated; Leeham News, 8 Mar 2024; D9 §2.5; reconciliation R-008]

20.2 LEAP DURABILITY AND THE JULY 2026 KIT

The LEAP-1A and -1B have had shorter time on wing than the CFM56 they replaced, particularly in hot and harsh environments such as the Middle East and India, driven by high-

pressure turbine (HPT) blade and fuel-nozzle issues. [ePlaneAI, two articles, dates not captured; D9 §2.5] Part I §10.2 recorded operators reporting removals at 2,000 to 6,000 cycles with "a planning base of 4,000", against a mature CFM56's restoration interval of 10,000 to 15,000 cycles. [Visual Approach, undated; Leeham News, 8 Mar 2024] The HPT is the hottest rotating part of the core and the usual limiter of time on wing (Part I §5).

The fix is the Durability Improvement Package (DIP), which Part I §10.2 boxed as the LEAP durability kit: a new HPT stage-1 blade, a new HPT stage-1 nozzle and a forward inner nozzle support. On 18 July 2026 CFM announced certification of the LEAP-1B HPT durability kit by the US Federal Aviation Administration (FAA) and by EASA, stating that it "will double time on wing in hot and harsh environments", that industrialisation was beginning, and that full production cutover is expected in early 2027. [CFM International / Safran / GE Aerospace release, 18 Jul 2026] The same release notes initial engine-level certification of a second change for the LEAP-1B.

REVERSE BLEED SYSTEM (RBS)

The reverse bleed system is a LEAP cooling arrangement that reduces the need for on-wing fuel-nozzle replacements, which CFM describes as "bringing on-wing maintenance needs in line with CFM56 standards". It received initial engine-level certification for the LEAP-1B alongside the HPT durability kit on 18 July 2026, and was already being deployed on the LEAP-1A. [CFM International release, 18 Jul 2026; D9 §2.5]

The LEAP-1A was ahead of the -1B. The new HPT blades and the reverse bleed fix were already being deployed on A320neo-family aircraft, with approximately one-third of the LEAP-1A fleet updated at the time of the report. [ePlaneAI, date not captured] By May 2026 GE Aerospace was stating that LEAPs "being shipped now will match the durability of the venerable CFM56". [Leeham News, 27 May 2026] Both the problem and the claimed fix are recorded; whether delivered LEAPs reach CFM56-level time on wing in service is not something the research could observe.

Scale and contract structure differ from the CFM56. About 5,500 LEAPs were in service at the date of one trade-press piece, with about 50% covered by long-term service agreements against about 15% of the CFM56 fleet. [Aviation Week, undated]

LONG-TERM SERVICE AGREEMENT (LTSA)

A long-term service agreement is a multi-year contract under which the airline pays a fixed amount per engine flight hour and the manufacturer or shop commits to perform or pay for the shop visits, taking the volume and cost risk and controlling the workscope, the parts sourcing and the shop. The research's dossiers call it an LTSA, an LTA (long-term agreement), a rate-per-flight-hour agreement or a flight-hour agreement; one concept, several names. It matters here because about half the LEAP fleet is under one and about 15% of the CFM56 fleet is, so a LEAP shop visit mostly flows to the manufacturer's network while a CFM56 visit is mostly bought transactionally. Part IV §28 carries the full treatment. [D2 §2.3; D9 §2.5; registry C-12]

CFM has outlined investments to support growth in LEAP shop visits, and Safran opened a new maintenance shop in Querétaro, Mexico, on 30 June 2026. [CAPA (Centre for Aviation); Safran release, 30 Jun 2026]

EXHIBIT 3.8 — LEAP DURABILITY: THE SHORTFALL, THE FIXES AND THEIR DATES

Item	Figure or date	Source and level	Date
Early LEAP-1A/-1B removals	2,000–6,000 cycles; planning base 4,000	Visual Approach (research house)	Undated
Mature CFM56 restoration interval, for contrast	10,000–15,000 cycles	Leeham News (trade press)	8 Mar 2024
Where the problem concentrates	Hot and harsh environments; HPT blade and fuel nozzle	ePlaneAI (trade press)	Dates not captured
LEAP-1B HPT durability kit certified (FAA and EASA)	18 Jul 2026; "will double time on wing in hot and harsh environments"	CFM / Safran / GE release	18 Jul 2026
LEAP-1B kit full production cutover	Early 2027	Same	18 Jul 2026
LEAP-1B reverse bleed system	Initial engine-level certification	Same	18 Jul 2026
LEAP-1A fleet updated with new HPT blades and RBS	About one-third	ePlaneAI	Date not captured
Manufacturer statement on delivered engines	LEAPs "being shipped now will match the durability of the venerable CFM56"	GE Aerospace via Leeham News	27 May 2026
LEAP in service / under long-term agreements	About 5,500 / about 50% (CFM56: about 15%)	Aviation Week	Undated
LEAP deliveries	1,802 (2025) against 1,407 (2024); Safran backlog >12,900	GE 10-K FY2025; Safran, 13 Feb 2026	Feb 2026

Item	Figure or date	Source and level	Date
New LEAP-capable shop	Safran Querétaro, Mexico, opened	Safran release	30 Jun 2026

Source: Visual Approach, "Wave of early LEAP-1A/1B shop visits frustrating operators", undated; Leeham News, Bjorn's Corner Part 49, 8 Mar 2024, and 27 May 2026; ePlaneAI, two articles; CFM International / Safran / GE Aerospace, "CFM secures certification for LEAP-1B durability upgrades", 18 Jul 2026; Aviation Week, "Leap Aftermarket Poised For Growth", undated; GE 10-K FY2025; Safran, 13 Feb 2026; Safran, Querétaro release, 30 Jun 2026. D9 §2.5 and §4.3; D1 §2.10.

The mechanism for the CFM56 aftermarket is indirect, and that is the point of recording it. A LEAP that comes off wing early needs a shop visit in a LEAP-capable shop, mostly within the manufacturer's network under a long-term agreement, and a spare LEAP while it is away. It does not generate a CFM56 shop visit, a CFM56 module or any CFM56 used material. Its effect on the old fleet runs through the gate described in §19.1: an airline whose A320neo or 737 MAX fleet is less available than planned slows its A320ceo or 737NG retirements. Willis Lease's FY2025 10-K lists "shorter on-wing engine time of most new technology engines" alongside airframer and engine-maker delays as compounding factors in the tight market. [Willis Lease Finance 10-K FY2025]

The timing of the fix bears on the shape of the CFM56 forecast, and here the research holds two readings. Safran's earlier framing, from about 2023 or 2024, had CFM56 shop visits peaking at about 2,500 a year in 2025 to 2026. GE's 2026 framing has a plateau of 2,300 to 2,400 a year through 2028 with the apex around 2027 to 2028 and a gradual fade after. [Safran via Aviation Week, c. 2023/24; GE via Aviation Week, 2026; reconciliation R-006, R-007] The LEAP-1B kit's full cutover in early 2027 is the date from which, on CFM's own statement, new 737 MAX engines should stop coming off wing early, and so the date from which this indirect support for the 737NG fleet begins to fade. That date sits inside GE's plateau and after Safran's earlier peak. The primer records the two curves and the cutover date beside each other and does not say which curve the cutover favours; §23.1 presents both curves in full.

20.3 WHICH OLD ENGINE EACH NEW-ENGINE PROBLEM TOUCHES

The two problems do not land on the same engines, and the table below is the map the translation table in §24 relies on.

EXHIBIT 3.9 — WHICH OLD ENGINE EACH NEW-ENGINE PROBLEM TOUCHES, AND THROUGH WHAT MECHANISM

New-engine problem	New airframe affected	Old airframe that substitutes	Old engines whose demand changes	Mechanism	Status at research date
GTF powder metal (PW1100G)	A320neo family, Pratt & Whitney-powered sub-fleet	A320ceo family	V2500 and CFM56-5B	Replacement flying on a shared type rating; ceo extensions, utilisation and shop visits brought forward	Groundings falling from the 2025 peak; resolution expected 2027–28
LEAP-1A durability	A320neo family, CFM-powered sub-fleet	A320ceo family	V2500 and CFM56-5B	Slower ceo retirement by operators with under-available neos	About one-third of the -1A fleet upgraded
LEAP-1B durability	737 MAX (sole engine)	737NG	CFM56-7B	Slower 737NG retirement by operators with under-available MAX fleets	Kit certified 18 Jul 2026; production cutover early 2027

Source: D9 §2.5 and §5; CFM International release, 18 Jul 2026; The Flying Engineer, 2026 (citing Cirium); ePlaneAI; Willis Lease Finance 10-K FY2025. Compiled for this Part.

So the GTF problem is an A320ceo story and, in engine terms, mostly a V2500 story with a CFM56-5B share. The LEAP-1B problem is a 737NG story and so a CFM56-7B story. Any sentence that attributes 737NG demand to the GTF, or A320ceo demand to the 737 MAX production cap, has the mechanism wrong. The translation table keeps them apart.

21. Airframer output: Boeing and Airbus

The second half of the replacement shortfall is airframes. Even where the new engines performed, there were fewer new aircraft to put them on than the airlines had ordered and planned for. This section gives the production and delivery figures for the two airframers with dates, and the arithmetic that connects a monthly rate to a year of undelivered replacements.

PRODUCTION RATE CAP (737 MAX)

The production rate cap is the monthly ceiling on 737 output imposed by the FAA after the Alaska Airlines door-plug event of January 2024: 38 aircraft a month, raised to 42 in October 2025, with the review to move to 47 completed in May 2026. Boeing's 2018 peak rate was 52 a month. The cap sets the maximum number of 737NG replacements that can be built in a year, and so the minimum number of 737NGs that cannot yet be released. [D9 §2.6; TravelPulse, Oct 2025; CAPA, May 2026]

BACKLOG

Backlog is the count of firm orders not yet delivered. Airbus's stood at a record 8,754 aircraft at the end of December 2025, more than ten years of production at 2026 delivery rates; the 737 MAX backlog exceeds 4,800 aircraft. A large backlog means that each aircraft not delivered this year is still owed, so the replacement is deferred rather than cancelled. [D9 §2.6; All About Industries; CAPA, May 2026; AirInsight]

GLIDER

A glider is a completed airframe parked at the factory awaiting engines. Airbus had about 60 at the end of the first half of 2025 and 32 at the end of the third quarter, and its chief executive said the number would reach zero by year-end. Gliders are the visible form of an engine-limited production line: the airframe exists, the delivery does not. [Leeham News, 30 Oct 2025; Air Data News; D9 §2.6]

21.1 BOEING AND THE 737 MAX

After the January 2024 door-plug event the FAA capped 737 production at 38 aircraft a month. [TravelPulse, Oct 2025] The FAA raised the cap to 42 a month in October 2025. [TravelPulse, Oct 2025] In May 2026 Boeing's chief executive, Kelly Ortberg, said the company had completed what he called the "capstone" review with the FAA, the final review in the agency's sequence, to move from 42 to 47 a month, and that the stabilisation at 42 had not taken long. [CAPA, May 2026] Boeing's stated plan was a further step in September 2026 and another in late 2026, potentially reaching 52 to 53 a month by year-end; Ortberg said he remained cautious about the step to 52 because of inventory concerns. [Travel Extra; AirInsight; CAPA, May 2026]

Deliveries followed the cap with a lag. 737 MAX deliveries were 387 in 2023, 260 in 2024 and 440 in 2025, and the cumulative total reached 2,399 by the date of the compilation the research used. [Wikipedia, 737 MAX orders and deliveries, 2026] Boeing's total deliveries of all types were 348 in 2024; by August 2025 it had delivered 385 aircraft, already more than in all of 2024. [Air Data News, Sep 2025] The compilation notes that Boeing exceeded 38 deliveries in some months by delivering from stored inventory, which is why 2025 deliveries exceed a full year at the capped build rate. [Wikipedia, 2026]

WORKED EXAMPLE 3.4 — WHAT A MONTHLY RATE CAP MEANS IN 737NG REPLACEMENTS A YEAR

Inputs (all sourced from §21.1): caps of 38, 42 and 47 a month and a 2018 peak of 52; MAX deliveries of 260 in 2024 and 440 in 2025. Assumption: every MAX delivered replaces one 737NG. That is the ceiling, since part of the backlog is growth rather than replacement (Cirium's Global Replacement Ratio of 46.1% in §19.2 says fewer than half of deliveries replace a retirement).

Step 1, a full year at each rate.

$$38 \times 12 = 456 \quad 42 \times 12 = 504 \quad 47 \times 12 = 564 \quad 52 \times 12 = 624$$

Step 2, the delivery shortfall against the capped rate.

$$\begin{aligned} 2024: & 456 - 260 = 196 \text{ aircraft short of a full year at 38 a month} \\ 2025: & 440, \text{ roughly a full year at 38 plus deliveries from storage} \end{aligned}$$

Step 3, the gap between the 2018 rate and the cap, as a rough measure of the replacement that did not happen for production-cap reasons alone.

$$(52 - 38) \times 12 = 168 \text{ aircraft a year, or 336 CFM56-7B engine positions not released}$$

So the cap alone held back on the order of 170 737NG replacements a year in 2024 and 2025, before counting the 2019–20 grounding and the 2021–23 ramp. At the replacement ratio of 46.1%, the figure would be about 77 aircraft and 155 engines. Limits of the example: the one-for-one assumption is the ceiling; the replacement ratio is a global all-type figure, not a 737 figure; and the cap is lifting, so the figure shrinks each year. [D9 §2.6, arithmetic on sourced figures; Wikipedia, 2026; CAPA, May 2026; Cirium via Informa]

21.2 AIRBUS AND THE A320NEO FAMILY

Airbus originally targeted 820 deliveries in 2025, later guided to about 790, and delivered 793. [Argus Media; Air Data News; D9 T9] Both the original and the revised targets are recorded, since the figure was revised during the year. It targets around 870 deliveries in 2026; in August 2026 it delivered 57 aircraft, described as leaving "a steep climb" to that target. [Air Data News; Aerocorner, Sep 2026] The engine shortfall behind the gliders came mainly from Pratt & Whitney, whose powder-metal production and inspection load competed with new-engine output, and earlier in 2026 from CFM, which fell behind before catching up. [Air Data News] Airbus was still negotiating Pratt & Whitney engine volumes for 2026 production in late 2025. [Air Data News]

The rate ambition moved. Airbus now expects A320neo-family output to reach 70 to 75 a month by the end of 2027 and to stabilise at 75 thereafter, instead of reaching that rate sooner as earlier plans had it. [Air Data News; Mexico Business News] The backlog was a record 8,754 aircraft at the end

of December 2025. [All About Industries] The A320 family, ceo and neo combined, had 20,378 orders, 12,798 deliveries and 11,308 aircraft in service as of July 2026; the ceo/neo split of the in-service figure was not reached. [Wikipedia, July 2026; D9 U1]

EXHIBIT 3.10 — AIRFRAMER OUTPUT 2023–2027: CAPS, DELIVERIES, GLIDERS, RATES AND BACKLOGS, WITH DATES

Item	Figure	Source and level	Date
737 production cap imposed	38 a month	FAA, via TravelPulse (trade press)	Jan 2024
737 cap raised	42 a month	FAA, via TravelPulse	Oct 2025
Review completed to move to 47	47 a month	Boeing CEO via CAPA	May 2026
Boeing plan for late 2026	52–53 a month; cautious about 52	Travel Extra; AirInsight; CAPA	2026
2018 peak rate	52 a month	CAPA; AirInsight	2018 figure
737 MAX deliveries	387 (2023); 260 (2024); 440 (2025)	Wikipedia compilation	2026
737 MAX cumulative deliveries / backlog	2,399 / more than 4,800	Wikipedia; CAPA	2026
Boeing total deliveries, all types	348 (2024); 385 by Aug 2025	Air Data News	Sep 2025
Airbus 2025 target, original / revised / actual	820 / about 790 / 793	Argus Media; Air Data News	2025
Airbus 2026 target	About 870; 57 delivered in Aug 2026	Air Data News; Aerocorner	Sep 2026
Airbus gliders	About 60 (end 1H 2025); 32 (end Q3 2025); zero targeted (end 2025)	Leeham News; Air Data News	30 Oct 2025
A320neo-family rate ambition	70–75 a month by end-2027; 75 thereafter	Air Data News; Mexico Business News	2026
Airbus backlog	8,754 aircraft	All About Industries	End-Dec 2025
A320 family (ceo and neo) orders / deliveries / in service	20,378 / 12,798 / 11,308	Wikipedia compilation	July 2026

Source: TravelPulse, "FAA Raises Boeing 737 MAX Production Cap", Oct 2025; CAPA, "Boeing testing production rate of 47 aircraft per month", May 2026; Travel Extra; AirInsight; Wikipedia, "List of Boeing 737 MAX orders and deliveries" and "List of Airbus A320 family orders and deliveries"; Air Data News, three articles; Argus Media; Aerocorner, Sep 2026; LeeHam News, "Airbus 3Q2025", 30 Oct 2025; Mexico Business News; All About Industries. D9 §2.6 and §4.3; D9 T9.

21.3 WHAT AIRFRAMER OUTPUT DOES TO THE OLD FLEET

The mechanism is the gate of §19.1 again, seen from the factory. Each A320neo or 737 MAX not delivered in a year is an A320ceo or 737NG that its airline cannot release that year. The airline that waits for a late new aircraft still pays for the old one's maintenance in the meantime. So a delivery shortfall converts the airline's planned capital spending on a new aircraft into aftermarket operating spending on an old one, for as long as the delay lasts. Because the backlogs are owed rather than cancelled, the conversion is a deferral: when the rate rises, from 42 toward 52 at Boeing and toward 75 at Airbus, the deferred replacements arrive and the deferred retirements with them. That is why the translation table's rows for the two airframers (T-4 and T-5 in §24) carry a reversal in their third column. Air Lease's FY2025 10-K puts the lessor's view of the same mechanism in one sentence: "As global air traffic expands and aircraft production volumes remain constrained, lessors are experiencing strong demand for aircraft through new lease requests and lease extension requests." [Air Lease Corp 10-K FY2025]

22. Lessors and airlines: extensions, availability, lease rates

The owners of the old fleet responded to the closed gate in ways that show up directly in aftermarket demand. This section takes the airline's maintenance decision at fleet level, then the decision to extend or return a leased aircraft, then what the lessors did between 2023 and 2026 and what it did to lease rates. Part II owns the operating lease (§12), maintenance reserves and return conditions (§13), engine leasing (§14) and green time (§16); this section uses those terms and adds the context.

22.1 THE AIRLINE'S MAINTENANCE DECISION AT FLEET LEVEL

An airline's engineering and fleet-planning functions do not wait for engines to fail. They run a rolling forecast of when each engine will have to come off.

ENGINE REMOVAL FORECAST

The engine removal forecast is an airline's rolling schedule of each installed engine's projected date of removal for a shop visit. Each engine's date is set by whichever of its limits is reached first: remaining exhaust-gas-temperature margin (Part I §5), remaining life-limited-part cycles (Part I §4), or a hard inspection. From the forecast the airline books shop slots, in 2025 and 2026 some 12 to 18 months ahead, because the wait for a slot runs two to six months. Part IV §27 owns the slot and induction. [D9 §2.7; Air Cargo Week, 2026; Locatory, Mar 2026]

Which of the three clocks runs out first is itself contested in the research, and Part I §5.2 carries the tension. Older trade-press analyses found exhaust-gas-temperature margin erosion "generally the most influential factor" in short-haul removal intervals; later analyses of the latest -5B and -7B variants found margin lasting to the life-limited-part limit, so that removals are LLP-driven, with "especially first-run engines" flown almost to the limit. [Aircraft Commerce, 2003 and 2008; Aircraft Commerce Issue 50; FlightGlobal; reconciliation R-011] The demand research lists the three clocks without ranking them, and so does this Part.

For each removal the airline chooses a workscope (Part I §6), meaning how much of the engine to open and restore. The choice depends on how long the airline expects to keep the aircraft. If the aircraft will be retired or returned to its lessor within a few years, the airline avoids a full performance restoration and buys green time instead: it installs an engine or module with enough margin and LLP life to reach the retirement date, then returns or sells it at or near the end of its life. The green-time engine comes from the used-material market or from an engine lessor. Trade press describes airlines "replacing LLP-expired modules with green-time units to maintain engines on wing while awaiting new aircraft or engine deliveries", through quick-turn or hospital shop visits that do only the essential work (Part I §6; Part VII §46). [Air Cargo Week, 2026; Aviation Business News] If the aircraft will stay for a long time, the airline does the full restoration and replaces the LLPs so the engine runs another full interval.

Green time is a finite pool, and the research records its limit. Bain's 2024 brief on the engine-maintenance capacity crunch notes that at 80% fleet utilisation green-time capacity becomes tight, and that "there is no huge buffer if airlines rely on green time and are getting to 80%". [Bain, 2024] The pool is fed by retirements. When retirements are low the pool shrinks exactly when demand for it is highest, which is the mechanism behind row T-9 of the translation table.

Spare engines are the other quantity the airline manages (Part II §14). An airline carries spares to cover removals; when shop turnaround lengthens, the same removal rate ties up more spares, and the airline buys or leases more. Two filings give the scale at airline level. Spirit Aviation Holdings had 16 spare engines financed under operating leases and owned 18 at 31 December 2025. [Spirit Aviation Holdings 10-K FY2025] American Airlines had agreements for 47 spare engines to be delivered in 2026 and beyond. [American Airlines 10-K FY2025]

22.2 THE EXTEND-OR-RETURN DECISION

When a lease expires the airline weighs three things. The first is the extension rent the lessor asks. The second is the cost of meeting return conditions (Part II §13), the contractual minimum remaining life on engines, LLPs and airframe checks at redelivery, which can force a shop visit just before return. The third is the availability and price of a replacement. [D9 §2.7] Between 2023 and 2026 the third has usually been decisive. With new aircraft late and used aircraft scarce,

extending has been the only way to keep capacity. So airlines extend and perform the maintenance the extended term needs, rather than returning aircraft at the end of their useful life and leaving the shop visit to the next owner. The lessor, facing the same scarcity from the other side, prefers extension to a transition it would have to pay for (§19.1).

22.3 HOW LESSOR BEHAVIOUR CHANGED, 2023–2026

LEASE EXTENSION AND THE EXTENSION SHARE

A lease extension renews an expiring operating lease with the incumbent airline instead of taking the aircraft back for transition, sale or part-out. The extension share is the proportion of scheduled lease expiries that are extended rather than returned. Cirium Ascend's head of consultancy, Rob Morris, said 75 to 90% of scheduled expirations were being extended; IBA found extension terms on legacy single-aisle aircraft stretching to four to six years, with extension rents in some cases more than 40% above the age-specific rates agreed just before the spike. Each extension keeps an aircraft and its two engines out of the used market and inside the maintenance cycle for the term. [Cirium Ascend via Bloomberg/AJOT, 2023–24 context; IBA via Key.Aero and IBA Weekly, Nov 2023; D9 §2.8]

The availability figures measure how far the market tightened. Cirium Ascend estimated that only about a dozen mid-life 737NGs and fewer than 25 A320neos were available to rent anywhere in the world at the time of its count, and that lease rates on ten-year-old 737-800s were 44% above January 2022 levels. [Cirium Ascend via Bloomberg/AJOT; date not visible, context 2023–24] IBA's research found lease lengths being extended, terms growing to four to six years, extension deals running out to 2026 and beyond, and in extreme cases aircraft due back at year 12 not expected to return until 2032, at 18 years of age. [IBA via Key.Aero; IBA Weekly, 24 Nov 2023]

The lessors' own filings describe the same market. Air Lease's FY2025 10-K: "The strong demand for aircraft, combined with elevated interest rates and inflation, helped increase lease rates on new lease agreements and lease extensions during 2025. Lease rates are expected to remain strong as the supply and demand environment for commercial aircraft remains tight." [Air Lease Corp 10-K FY2025] Willis Lease's FY2025 10-K: "Both Airbus and Boeing have delivery delays which have been compounded by engine manufacturer delays, shorter on-wing engine time of most new technology engines, and ongoing supply chain challenges." [Willis Lease Finance 10-K FY2025]

EXHIBIT 3.11 — LEASE-MARKET INDICATORS 2022–2026, BY SOURCE, LEVEL AND DATE

Indicator	Figure	Source and level	Date
Share of scheduled lease expiries extended	75–90%	Cirium Ascend (Rob Morris) via Bloomberg/AJOT (data provider's consultancy via press)	Undated; context 2023–24
Mid-life aircraft available to rent worldwide	About 12 737NGs; fewer than 25 A320neos	Same	Same
Lease rate, ten-year-old 737-800	+44% against January 2022	Same	Same
Extension terms on legacy single-aisles	4–6 years; extension deals to 2026 and beyond; some aircraft due back at year 12 not returning until 2032, at age 18	IBA (appraiser) via Key.Aero; IBA Weekly	Nov 2023
Extension rents against prior age-specific rates	More than 40% above, in some cases	IBA	Nov 2023
Twelve-year-old 737-800 rent	About \$228,000 a month, down 11% year on year	IBA	Jul 2025
Twelve-year-old A320-200 rent	About \$220,000 a month, down 13% year on year	IBA	Jul 2025
Twenty-one-year-old 737-800 rent	Less than \$160,000 a month	IBA	Jul 2025
Narrowbody conversion and lease-rate outlook	Warning of "oversupply of narrowbody conversions and lease rate fall"	IBA via India Seatrade News	6 Sep 2026
New-aircraft lessor on 2025	Rates up on new leases and extensions; "expected to remain strong"	Air Lease Corp 10-K FY2025 (filing)	Feb 2026
Engine lessor on causes	Airframer delays compounded by engine-maker delays, shorter on-wing time, supply chain	Willis Lease Finance 10-K FY2025 (filing)	2026
CFM56-7B spare-engine rent	About \$100,000 a month in 2024, from about \$75,000 in 2019	IBA	Apr 2024

Indicator	Figure	Source and level	Date
CFM56-7B rent, alternative reading	\$42,000–48,000 a month, "about 34% above 2019", up 15–20% "in recent months"; may describe green-time engines	Safe Fly Aviation (consultancy blog)	2026
CFM56-7B half-life market value and its direction	\$5.7M (-7B24), \$6.4M (-7B27); values "levelled off after approximately 18 months of increases"	IBA	Sept 2025; H1 2026
Lauda Europe (Ryanair group) A320 leases	Extended through the early 2030s	Travel Extra (trade press)	2026
Spirit spare engines	16 under operating leases plus 18 owned	Spirit Aviation Holdings 10-K FY2025	31 Dec 2025
American Airlines spare engines on order	47 for 2026 and beyond	American Airlines 10-K FY2025	2026

Source: Bloomberg via AJOT, "Used jets rents surge 44% as airlines pay up to buoy fleets"; Key.Aero, "Lease trends" (IBA research); IBA Weekly, 24 Nov 2023; India Seatrade News, "IBA warns of oversupply of narrowbody conversions and lease rate fall", 6 Sep 2026 (carrying IBA's July 2025 commentary); Air Lease Corp, Form 10-K FY2025; Willis Lease Finance, Form 10-K FY2025; AviTrader, "IBA says it's a lessor's market", 25 Apr 2024; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); IBA, Engine Values Release 2025B, Sept 2025, and H1 2026 update via Air Cargo Update; Travel Extra, 2026; Spirit Aviation Holdings and American Airlines, Forms 10-K FY2025, D9 §2.8 and §4.4; D6 §2.1 and §4; reconciliation R-028, R-029, R-031, R-034.

Three tensions sit in that exhibit, and the primer carries each with both sides. The first is the direction of mid-life narrowbody lease rates. Air Lease, a lessor of new aircraft, reported in its FY2025 filing that rates rose on new leases and extensions in 2025 and were expected to remain strong, and Cirium's 44% rise on ten-year-old 737-800s points the same way. [Air Lease Corp 10-K FY2025; Cirium Ascend via Bloomberg] IBA, an appraiser, put a twelve-year-old 737-800 at about \$228,000 a month in July 2025, down 11% year on year, and a comparable A320-200 at about \$220,000, down 13%, and in September 2026 warned of a lease-rate fall. [IBA, Jul 2025 and 6 Sep 2026, via India Seatrade News; reconciliation R-031] The sources describe different asset ages at different dates, a new-aircraft lessor's whole book against an appraiser's mid-life commentary, and the primer does not reconcile them.

The second is freighter conversion. FTAI and AEI (a freighter-conversion house) announced a collaboration on 7 July 2026 to deliver "a more cost-effective Boeing 737-800 freighter solution", and the type-specialist data show more than 260 737-800 freighters in service with a pipeline above 300. [FTAI and AEI release, 7 Jul 2026; Acumen Aviation; Cargo Facts] IBA's September 2026

note warns of an "oversupply of narrowbody conversions". [IBA via India Seatrade News, 6 Sep 2026; reconciliation R-034] Part VIII §67 carries the collaboration; this Part records that the conversion demand the translation table relies on (row T-11) is contested by an appraiser at the research date.

The third is the engine lease rate, which Part II §14 owns. IBA put a CFM56-7B spare engine at about \$100,000 a month in 2024, up from about \$75,000 in 2019. The consultancy blog gives \$42,000 to \$48,000 a month in 2026, "about 34% above 2019", which the leasing research notes sits beside green-time values and may describe green-time engines, although the source does not say so. [IBA, Apr 2024; Safe Fly Aviation, 2026; reconciliation R-028] The percentage change from 2019 agrees across the two, about a third, while the levels do not. [reconciliation B-5]

WORKED EXAMPLE 3.5 — A 100-AIRCRAFT LEASE COHORT, AND WHAT AN EXTENSION DOES TO THE MAINTENANCE CALENDAR

Inputs. Cohort: 100 A320ceo leases expiring in one year (assumed, for scale). Extension share: 75–90% (sourced: Cirium Ascend). Extension term: 4–6 years (sourced: IBA, Nov 2023). Utilisation: 1,000–1,500 cycles a year per engine (sourced: Aircraft Value News). Mature restoration interval: 10,000–15,000 cycles (sourced: Leeham, Mar 2024). Stock available to rent: fewer than 25 A320ceos worldwide (sourced: Cirium Ascend).

Step 1, the aircraft that come back.

returns: $100 - 75 = 25$ aircraft at a 75% share; $100 - 90 = 10$ aircraft at 90%
engines returned with them: 20 to 50

Those 10 to 25 aircraft are the entire flow into the used market from that cohort for transition, sale or part-out. Step 2, the stock against the flow. Cirium's count of fewer than 25 A320ceos available to rent worldwide is of the same order as one year's returns from a single 100-aircraft cohort. That is the arithmetic behind a 40%-plus move in used rents: the stock was one cohort's returns deep (D9 §2.8).

Step 3, cycles added inside the extension.

$4 \text{ years} \times 1,000 = 4,000$ cycles, to $6 \text{ years} \times 1,500 = 9,000$ cycles per engine

Step 4, whether the next shop visit falls inside the term. Take an engine at expiry with 3,000 cycles of life remaining to its next restoration (assumed). Any extension in the range forces that restoration inside the term, and the incumbent airline pays for it rather than leaving it to redelivery. Take instead an engine with 10,000 cycles remaining (assumed). A four-year extension at 1,000 cycles a year, 4,000 cycles, avoids the visit; a six-year extension at 1,500, 9,000 cycles, nearly reaches it.

engines whose next event now sits inside an extended term: 75 to 90
aircraft $\times 2 = 150$ to 180 engines

So in a 100-aircraft cohort, extension moves the next maintenance event of 150 to 180 engines from a future owner's calendar into the incumbent's, and holds 20 to 50 engines back from the used and teardown market. Limits of the example: the cohort size is assumed; the remaining life at expiry is assumed and in practice is spread across the cohort; return conditions (Part II §13) may already require the visit at redelivery, in which case the extension changes who pays and when, not whether; and the V2500's interval is applied from the CFM56's. [Cirium Ascend via Bloomberg/AJOT; IBA via Key.Aero, Nov 2023; Aircraft Value News, undated; Leeham News, 8 Mar 2024; D9 §2.8]

22.4 FTAI'S POSITION IN THIS CHAIN

FTAI's FY2025 10-K describes the 2025 Partnership, the first vehicle of its Strategic Capital Initiative (Part VIII §60 to §62), as focused on acquiring 737NG and A320ceo aircraft and as the primary buyer of all future on-lease narrowbody aircraft. [FTAI Aviation 10-K FY2025] The lessor behaviour described above cuts two ways for a buyer of mid-life narrowbodies, and the research records both without weighing them. High extension rates and rents support the value of aircraft already owned. The same scarcity makes new acquisitions harder to source, because the 10 to 25 aircraft a year that come back from a 100-aircraft cohort are all there is to buy. [D9 §2.8] Part VIII carries the vehicle; Part II §15 carries why the engines carry the value of a mid-life narrowbody.

23. Forecasts: shop-visit curves and the engine MRO market

The quantities in §18 to §22 feed forecasts. This section sets out every forecast curve the research reached, the measure each uses, and its date, and then reconciles the headline shop-visit count against the installed base. Part IV §26 carries the market's reading of the same curve as "shop-visit demand"; the two sections cross-reference and do not duplicate.

SHOP-VISIT FORECAST AND FIRST SHOP VISIT

A shop-visit forecast counts the expected engine inductions per year for a type. A first shop visit is the first removal of an engine for shop maintenance since new; first runs are usually the longest, so a fleet's first-visit peak comes years after its delivery peak. The two measures can peak in different years: GE and Safran forecast total CFM56 visits of about 2,300 to 2,400 a year in 2026 to 2028, while an independent research house dates the peak in first visits for the CFM56 and V2500 to 2027. [D9 §2.9; GE via Aviation Week, 2026; Visual Approach Analytics]

MRO MARKET AND ENGINE MRO

MRO is maintenance, repair and overhaul, the generic term for aircraft and engine upkeep (Part IV §25). The MRO market is the dollar value of that spending in a year, and engine MRO is the engine share of it. Oliver Wyman put engine MRO at \$70 billion in 2026, rising to \$102.6 billion by 2036. Aviation Week counts the same market in events: about 9,000 engine shop visits of all types in 2025, about 2,300 of them CFM56. [Oliver Wyman via AviationPros and Air Cargo Week, 2026; Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI; D9 §2.9]

UNCONDITIONAL MRO DEMAND

Aviation Week's 2026 forecast gives "unconditional MRO demand of \$853bn over ten years". The forecast's own definition was not retrieved. Read from first principles, unconditional demand is the maintenance the fleet would require if every scheduled event were performed when due, before any constraint of shop capacity, parts or labour is applied; it is the quantity Bain's capacity model is compared against. The primer uses the phrase only in that sense and flags that the publisher's definition is unverified. [Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI; registry section 2]

23.1 THE CFM56 CURVE

The research reached the CFM56 curve from the two CFM partners at different dates, from a trade-press forecast, from an independent research house and from a consultancy blog. They do not describe one shape.

EXHIBIT 3.12 — CFM56 SHOP-VISIT FORECAST STATEMENTS BY SOURCE, MEASURE AND DATE

Period or measure	Figure	Source and level	Date
2019, the pre-downturn level	"Just above 2,000" CFM shop visits, "almost all of them CFM56-5B and -7B"	Safran via Aviation Week (OEM via trade press)	c. late 2023 / early 2024
2020–2023	Not found year by year	D2 U2	Not dated
2024	"Topping 2,000 again in 2024, with some help from the Leap family"	Safran via Aviation Week	c. 2023/24
2024 → 2025	"40 per cent increase in shop visits"; scope (all engines or CFM56) not captured	IBA via Aviation Week headline	2024
2025	CFM56 shop visits grew "mid-single-digit", with "a higher proportion of full work scope shop visits"	Safran FY2025 results release; Forecast International	13 Feb 2026
Peak, earlier framing	About 2,500 a year in 2025–26	Safran via Aviation Week	c. 2023/24

Period or measure	Figure	Source and level	Date
2026 and 2027	"Entered the year projecting about 2,300"; "expanded the range to 2,300–2,400"; "probably closer to 2,400"	GE Aerospace via Aviation Week	2026
Through 2028	2,300–2,400	Same	2026
Apex	"Around the 2027–2028 time period, followed by a gradual fade"	Same	2026
Removals, 1H 2027 against 2026	Up about 10%	GE Aerospace via Aviation Week	2026
First-shop-visit peak, CFM56 and V2500	2027	Visual Approach Analytics (research house headline)	2025/26
2030, all CFM engines	"More than 4,000 shop visits annually, roughly split between Leaps and previous- generation models", so about 2,000 CFM56	Safran via Aviation Week	c. 2023/24
LEAP curve	Shop visits more than double by 2030; peak above 5,000 a year by 2040; LEAP aftermarket revenue parity with CFM56 in 2028	Safran Capital Markets Day via Quartr	2024
737-700 passenger MRO demand	"To decline sharply after 2027"	Acumen Aviation	18 Dec 2025
Engine MRO demand, all types	"Near-term peak in 2026", constrained through the end of the decade	AviationPros (trade press)	2026
Consultancy reading	2,300–2,400 a year through 2028, labelled "CFM56-7B shop visits" by one dossier and all CFM56 by another	Safe Fly Aviation (consultancy blog)	2026

Source: Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24, and "Leap Aftermarket Poised For Growth", undated; Aviation Week, "IBA predicts 40 per cent increase in shop visits from 2024 to 2025", 2024; Safran, FY2025 results release, 13 Feb 2026; Forecast International, 13 Feb 2026; Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Visual Approach Analytics; Quartr, Safran Capital Markets Day 2024 summary; Acumen Aviation, 18 Dec 2025; AviationPros, Engine Stand Utilization Report; Safe Fly Aviation, 2026 (consultancy blog). D2 §4.1; D9 §2.9 and §4.5; reconciliation R-006, R-007; E.1 rule 15.

The level and the shape are both contested, and the reconciliation requires both readings wherever the plateau is discussed. [E.1 rule 15] On level, Safran's earlier framing put the peak at about 2,500 a year in 2025 to 2026; GE's 2026 framing entered the year at about 2,300 for each of 2026 and 2027, widened to 2,300 to 2,400, "probably closer to 2,400", and held that range

through 2028. [Safran via Aviation Week, c. 2023/24; GE via Aviation Week, 2026; reconciliation R-006] On shape, the earlier view is a peak and the later one a plateau with an apex around 2027 to 2028 and a gradual fade. [reconciliation R-007] Visual Approach's 2027 peak is in first visits only, a different measure, so it is not strictly contradictory, but it describes a different shape again. [Visual Approach Analytics; D9 T4] Two further figures are unreconciled with either curve. IBA's headline of a 40% increase in shop visits from 2024 to 2025 does not say whether it counts all engines or the CFM56, and Safran's own full-year 2025 statement has CFM56 visits up only "mid-single-digit", with more full worksopes. [IBA via Aviation Week, 2024; Safran, 13 Feb 2026; reconciliation R-006] The year-by-year counts for 2020 to 2023 that would let a reader see the recovery were not found. [D2 U2]

The long end of the curve depends on the retirement rate, which is why §19 came first. Safran's 2030 figure of more than 4,000 visits across all CFM engines, roughly half LEAP, implies about 2,000 CFM56 visits in 2030, and Safran's 2024 Capital Markets Day put LEAP aftermarket revenue at parity with the CFM56 by 2028. [Safran via Aviation Week, c. 2023/24; Safran Capital Markets Day 2024 via Quartr] Those are forecasts made when the retirement assumption was higher than the outturn, and the research does not say how they would move at 1.5 to 2%. [D9 T3]

23.2 THE V2500 CURVE

The V2500 figures are in direct contradiction and the primer carries both. Aviation Week's V2500 data tool projected a spike to about 1,400 V2500 shop visits in 2025 and more than 7,800 overhaul visits plus 3,142 LLP-related visits over the following ten years. [Aviation Week V2500 data tool, date not visible] RTX's chief financial officer, reported by the same publication in 2026, put 2025 at "slightly more than 800" and expected 2026 to be about the same, "probably within 20". [RTX CFO via Aviation Week, 2026; reconciliation R-009] A published forecast and a reported outturn, both via one publication, one quoting the manufacturer's finance chief. IAE expected "no retirement surge" for the V2500 over the next five years. [IAE via Aviation Week, 2024/25] MTU (MTU Aero Engines, a German engine maker), whose maintenance arm is independent on this engine, performed 38% of all V2500 shop visits in 2024. [MTU Aero Engines, 2025] Part VII §51 carries FTAI's V2500 programme against this market.

23.3 THE MARKET IN DOLLARS AND IN VISITS

EXHIBIT 3.13 — ENGINE MRO MARKET SIZE, SHOP-VISIT COUNTS AND THE CAPACITY SHORTFALL, BY SOURCE AND DATE

Item	Figure	Source and level	Date
Engine shop visits, all types	About 8,800 (2024); about 9,000 (2025), of which about 2,300 CFM56	Aviation Week 2026 Fleet & MRO Forecast via ePlaneAI (trade press)	2026

Item	Figure	Source and level	Date
Ten-year MRO demand	\$853bn "unconditional"; aftermarket surpassing \$1.6 trillion cumulative at a 3.2% compound annual growth rate, as read	Same	2026
Engine MRO market	\$70bn (2026) rising to \$102.6bn (2036)	Oliver Wyman (consultancy) via AviationPros and Air Cargo Week	2026
Global MRO market, all segments	More than \$136bn (2025); about \$140bn (2026)	Oliver Wyman, "The new MRO supply paradigm"	Apr 2026
Narrowbody share of commercial MRO	58.3% (2025)	GM Insights (a market-research vendor; low grade)	2025
Engine MRO capacity shortfall	Demand to exceed available supply by more than 17% by the end of the decade if capacity grows on its current trajectory	Bain (consultancy)	2024
Engine shop visits to 2040	"Set to rise sevenfold"; base not stated	IATA study via AFM (a trade publication)	Undated
V2500 shop visits	Slightly more than 800 (2025); 2026 within about 20 of that	RTX CFO via Aviation Week	2026
V2500 shop visits, forecast	About 1,400 (2025); more than 7,800 overhaul plus 3,142 LLP visits over ten years	Aviation Week data tool	Date not visible

Source: ePlaneAI, "Forecast for US commercial fleet and MRO growth through 2026" (reporting Aviation Week 2026 Fleet & MRO Forecast figures); AviationPros, "6-Month Analysis: Top Engine Maintenance Trends for MROs to Watch in 2026" (Oliver Wyman figures); Air Cargo Week, "From PW1100G to CFM56", 2026; Oliver Wyman, "The new MRO supply paradigm", Apr 2026; GM Insights; Bain & Company, "Commercial aviation forecast" and "Get a step ahead of the engine maintenance capacity crunch", 2024; AFM; Aviation Week, "Pratt & Whitney Sees High-Single-Digit MRO Growth This Year", 2026, and V2500 data tool. D9 §2.9 and §4.5; D2 §4.7; reconciliation R-009; D9 T10.

Two cautions attach to that exhibit. The IATA figure of a sevenfold rise in shop visits by 2040 has no stated base; a sevenfold rise from about 9,000 all-engine visits would be about 63,000, which no manufacturer's curve supports, so the research judged the base to be new-generation engines only and treats the figure with caution until the study is read. [AFM; D9 T10] Bain's shortfall of more than 17% is conditional: it holds if engine MRO capacity keeps growing only on its current trajectory, and Bain's own brief describes the capacity responses under way. [Bain, 2024] The type-level curve for the CFM56 through 2030 inside Aviation Week's forecast, and

Oliver Wyman's narrowbody-engine subtotal by year, were not reached; only the headline totals surfaced. [D9 U7]

23.4 CAPACITY, TURNAROUND AND STANDS

Part IV §27 owns capacity. Four of its figures are needed here because the translation table's row T-8 depends on them, and because the research holds two readings of whether the constraint is easing. Trade press puts a full CFM56 overhaul at 90 to 120 days against about 60 before the pandemic, light work at about 45 days, legacy-engine turnaround times up 35% and new-generation times up more than 150% against five years earlier, and slot waits at two to six months. [Air Cargo Week, 2026; Aviation Business News / AVM, 2025] GE's own filing and conference material read the other way: turnaround for the LEAP, CFM56 and GE90 improved by more than 10% year on year in the fourth quarter of 2025, material input from priority suppliers rose more than 40%, and CFM56 turnaround was "sustaining about 90 days" in May 2026. [GE 10-K FY2025; GE Aerospace, Bernstein conference, 27 May 2026; reconciliation R-019, R-020] Oliver Wyman's April 2026 survey has two-thirds of MRO operators unable to find enough technicians, labour-rate inflation at 5.5 to 6.0%, and more than half expecting attrition to worsen. [Oliver Wyman, Apr 2026] A filing and a conference deck against trade press and a consultancy survey; the primer carries both, and Part IV §27 works them through.

ENGINE STAND

An engine stand is the wheeled fixture that holds an engine or a module during work, storage and transport. Because an engine off the wing must sit on one, stands are a physical limit on work in progress, and their utilisation is a measure of how full the system is: legacy-engine stand use rose by a third in 2025 over 2024, and CFM56-5A/-5B stand use rose 60%. Part IV §27 carries the full treatment. [AviationPros, Engine Stand Utilization Report; D2 §2.4; D9 §6]

Third-party test and repair waits of four to six months, and advice to buyers to plan procurement early for LLPs and critical rotables and to use exchange pools, used material and diversified suppliers, complete the picture the trade press gives. [Locatory, Mar 2026; AVM, 2025] Aero Norway, an independent CFM56 shop, said the long lead times for induction slots were driving interest in quick-turn services, "especially in the case of older-generation engines, such as the CFM56 and V2500". [Aero Norway in AVM, May 2024]

WORKED EXAMPLE 3.6 — RECONCILING 2,300 SHOP VISITS A YEAR WITH ABOUT 24,000 ENGINES

Inputs. Visits: 2,300 a year, the low end of GE's 2,300–2,400 (sourced: GE via Aviation Week, 2026). Bases (sourced, Exhibit 3.1): about 24,000 (trade press, Sept 2025); more than 22,800 (Safran, end-2025); more than 19,000 "flying" and about 14,200 "active" (consultancy blog, labelled). Interval: a mature performance restoration every 10,000–15,000 cycles (sourced: Leeham, Mar 2024)

at 1,000–1,500 cycles a year (sourced: Aircraft Value News), which is 6.7 to 15 years with the central cases, 10,000 at 1,000 to 15,000 at 1,500, at 10 years; an alternative rule of thumb of roughly every eight years (sourced: Aviation Week). Spare ratio: about 10% of installed engines now, from about 15% historically (sourced: MTU AEROREPORT, via Part II's research).

Step 1, the visit rate and the implied interval on each base.

$$\begin{aligned} \text{on } 24,000: 2,300 \div 24,000 &= 9.6\% \text{ a year} \rightarrow 1 \div 0.096 = 10.4 \text{ years between visits} \\ \text{on } 22,800: 2,300 \div 22,800 &= 10.1\% \rightarrow 9.9 \text{ years} \\ \text{on } 19,000: 2,300 \div 19,000 &= 12.1\% \rightarrow 8.3 \text{ years} \\ \text{on } 14,200: 2,300 \div 14,200 &= 16.2\% \rightarrow 6.2 \text{ years} \end{aligned}$$

Step 2, the same at the top of GE's range.

$$2,400 \div 24,000 = 10.0\% \rightarrow 10.0 \text{ years}; \quad 2,400 \div 14,200 = 16.9\% \rightarrow 5.9 \text{ years}$$

Step 3, removing spares from the manufacturer-level base. If about 10% of the 24,000 are spare engines that fly only when installed, the flying base is about 21,600.

$$2,300 \div 21,600 = 10.6\% \rightarrow 9.4 \text{ years}$$

Step 4, comparison with the sourced intervals. An implied interval of about ten years on the 24,000 base matches the mature restoration interval only at the long end of its range, 10,000 cycles at 1,000 a year or 15,000 at 1,500. An interval of 8.3 years on the 19,000 base matches Aviation Week's eight-year rule of thumb. An interval of about six years on the 14,200 base is shorter than any sourced CFM56 interval unless engines fly more than about 1,600 cycles a year ($10,000 \div 6.2 = 1,613$), which is above the sourced utilisation range.

Step 5, the maturity effect. Worked example 3.1 showed about 70% of the fleet with at most one visit in 2024. First runs are longer than mature runs, with the latest variants often flown almost to the LLP limit of 20,000 cycles or more before the first removal (Part I §6.2). A fleet in which most engines are still on a long first run generates fewer visits a year than the mature interval alone would predict, which pushes the implied interval on any base upward. First-run intervals by thrust rating were not retrieved (Part I's unknown U-D1-06).

So the 24,000 base is consistent with the sourced mature interval only if it includes a few thousand spare, stored or green-time engines, or if the long first runs of a still-maturing fleet are depressing the visit rate, or both. The 14,200 base is consistent only with an interval shorter than any the research sourced. The research recorded this and did not resolve it; which base is right is the single most consequential unknown in this Part, because every per-engine figure in Parts IV to IX divides by it. Limits of the example: the visit count and the base come from different publishers and may define "CFM56" differently (all variants or -5B and -7B only); the 10% spare ratio is an industry generality; and the arithmetic assumes every engine is on a regular cycle, which green-time engines with no further visit planned are not. [GE via Aviation Week, 2026; Aviation Business News, Sept 2025; Safran, 13 Feb

2026; Safe Fly Aviation, 2026; Leeham News, 8 Mar 2024; Aircraft Value News, undated; Aviation Week, "Retirement Uptick Will Restock Used Engine Parts"; MTU AEROREPORT; D9 T1; reconciliation R-001, R-002, R-008]

24. The translation table: each driver, the requirement it creates, what it does not stress

Sections 18 to 23 documented the drivers. This section is the device that connects them to the rest of the primer.

TRANSLATION TABLE

The translation table is Part III's organising device: one row per documented demand driver, stating the aftermarket requirement the driver creates and what it does not stress. "Stress" means that the driver raises the quantity or the price of that requirement; the last column records what the driver leaves untouched or pushes the other way. The requirements are the five of aftermarket demand (§18), shop visits, modules, spare engines, used material and leases, with approved alternative parts made under the Parts Manufacturer Approval (PMA) route (Part VI §37) and capital added because several drivers act on them. Rows are numbered T-1 to T-14 and the numbers are stable; Parts IV to XI open by naming the rows they answer. [D9 §5; outline §24]

MOD-1

Mod-1 is FTAI Power's 25 megawatt (MW) trailer-mounted natural-gas generator set built around a CFM56 core, launched as a platform on 30 December 2025, with first delivery targeted for the fourth quarter of 2026 and 100 units planned for 2027. It is named here because it is a new sink for CFM56 engines outside aviation, which makes it a demand driver in this table. Part IX §69 carries the full treatment. [FTAI release, 30 Dec 2025; D8 §2.3]

How to read the table. Each row names the driver with the section of this Part that documents it and its principal source. The middle column states the requirement in the aftermarket's own units: visits, modules, spare engines, used material, approved alternative parts, leases, capital. The last column is as important as the middle one. A driver that raises shop-visit demand but removes teardown feedstock has done two things, and a reader who carries only the first will misread Part V. Nothing is ranked; the rows are in the order the drivers appear in this Part, with the three drivers documented outside this Part at the end.

EXHIBIT 3.14 — THE TRANSLATION TABLE: FOURTEEN DOCUMENTED DRIVERS, THE AFTERMARKET REQUIREMENT EACH CREATES, AND WHAT EACH DOES NOT STRESS

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-1	Low CFM56 retirements: about 1.5% of the installed base in 2025, about 2% expected for 2026, "below 1%" in 1Q 2026, against a 3–4% plan (§19.3; GE via Aviation Week and Leeham, 2026; GE investor-relations page)	More engines accruing cycles → sustains 2,300–2,400 shop visits a year through 2028; more LLP replacements as retained engines reach cycle limits; more full performance restorations in place of green-time run-outs; module demand for LLP-expired modules	Starves teardown feedstock: fewer used parts and green-time engines, higher used-part prices and spare-engine rents (see T-9); does not stress LEAP or GTF spares; does not change shop capacity
T-2	GTF powder-metal groundings: 835 PW1100G-powered aircraft grounded or stored in late 2025; AOG down more than 20% from the 2025 peak by January 2026 (§20.1; Cirium via trade press; RTX via Leeham)	A320ceo lift to replace grounded neos → ceo lease extensions and short-term leases; higher ceo utilisation → V2500 and CFM56-5B shop visits brought forward; spare V2500 and -5B engines; green-time -5B and V2500 modules for airlines that had planned to retire ceos	Does not create CFM56-7B demand except through general narrowbody scarcity; does not add used-material supply, since it suppresses ceo retirements; creates no CFM56 work on LEAP-type engines
T-3	LEAP durability shortfall and the July 2026 kit: early removals at 2,000–6,000 cycles; LEAP-1B HPT kit certified 18 Jul 2026 with production cutover early 2027; about one-third of the -1A fleet upgraded (§20.2; CFM release; Visual Approach; ePlaneAI)	LEAP shop visits earlier than plan, mostly in the manufacturer's network under long-term agreements; LEAP spare engines; indirectly, slower ceo and 737NG retirements by operators whose neo and MAX fleets are under-available	Creates no CFM56 shop visits, modules or used material; once the kit cuts over, LEAP time on wing lengthens and the indirect support for the old fleet fades

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-4	737 MAX production rate cap: 38 a month from January 2024, 42 from October 2025, 47 reviewed May 2026; 260 MAX delivered in 2024 and 440 in 2025 (§21.1; FAA via trade press; CAPA; Wikipedia compilation)	737NG lease extensions; 737NG utilisation held high → CFM56-7B shop visits and LLP replacements; -7B spare engines; -7B green-time modules for carriers deferring 737NG retirement	Does not touch A320ceo, -5B or V2500 demand directly; as the cap lifts toward 52, the effect reverses, with more 737NG retirements, more -7B teardown supply and fewer -7B visits later in the decade; United, WestJet and Copa 737NG retirements are the counter-case
T-5	Airbus delivery shortfall and the rate-75 deferral: 793 delivered against 820 originally targeted in 2025; gliders 60 → 32 → 0; rate 75 only by end-2027 (§21.2; Air Data News; Leeham)	A320ceo extensions and higher ceo utilisation → -5B and V2500 shop visits; ceo spare engines; keeps ceo part-out low	Does not stress -7B; reduces used-material supply; converts planned airline capital spending on new aircraft into aftermarket operating spending on old ones; reverses as the rate rises
T-6	Lessor extension behaviour: 75–90% of expiries extended; about 12 737NGs and fewer than 25 A320ceos available; ten-year-old 737-800 rents up 44% against January 2022; terms 4–6 years (§22.3; Cirium Ascend; IBA; Air Lease 10-K)	Maintenance events paid by the incumbent airline inside the extended term rather than deferred to redelivery → shop visits and LLP replacements; higher lease rates on mid-life aircraft and engines; capital held in aircraft that stay on lease	Reduces the flow of aircraft into used-sale and part-out markets: fewer acquisition candidates for mid-life buyers and less teardown feedstock; fewer end-of-lease return-condition visits; does not change the total cycles the fleet accrues

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-7	OEM parts-price escalation: new spares about 10% a year in 2022–23 (double-digit on 1 Nov 2022, high-single-digit in Aug 2023); Safran's rule of "above inflation by 3 to 4 points"; the -7B LLP set about 7% a year compounded 2008–2025 (Part I §4; Part IV §28; Aviation Week, late 2023)	Demand for substitutes on the material line, which is 60–70% of every visit: used serviceable material, PMA parts, repair instead of replacement, module exchange instead of overhaul; raises the dollar value of every shop visit whoever performs it	Does not change the number of removals; does not stress labour or slots; does not touch LLPs, for which no PMA exists, so the LLP line escalates unsubstituted
T-8	MRO capacity shortfall and turnaround: slot waits 2–6 months; CFM56 overhaul 90–120 days; demand to exceed capacity by more than 17% by the end of the decade (Bain); two-thirds of operators short of technicians (Oliver Wyman) (§23.4; Part IV §27)	Spare engines and spare-engine leases, since each removal ties up a spare for longer; module exchanges and quick-turn or hospital visits as substitutes for full overhauls; exchange pools and used material to shorten shop time; capital tied up in inventory held for exchange	Does not change the number of removals the fleet generates; does not add teardown supply; shifts where and how the work is done, not whether; GE reports turnaround improving (R-019, R-020)
T-9	Used-material and feedstock scarcity: used-material supply "remains thin" (AVM, 2025); a "hypercompetitive feedstock environment" and whole-asset sales down from 20 engines to 13 (AerSale, 5 Mar 2026); CFM56 values up as much as 50% in two years (ePlaneAI); -7B rents about 34% above 2019 (Safe Fly, consultancy) (§19.4; Part V §33)	Higher prices for used parts and green-time engines; more new OEM parts bought because used ones are unavailable; PMA as the other substitute; capital to acquire the feedstock that does exist, including whole aircraft bought for their engines	Does not add shop capacity; does not change the removal count; is a consequence of T-1, T-4, T-5 and T-6 rather than an independent cause, and reverses when they do

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-10	Data-center power demand and the gas-turbine shortage: global data-center electricity more than doubles to about 945 TWh by 2030 (International Energy Agency (IEA), Apr 2025); GE Vernova gas-turbine backlog plus reservations 116 gigawatts (GW) at Q2 2026 with slots selling for 2031; Siemens Energy lead times of three-plus years (Part IX §71; D8)	A new sink for CFM56 cores outside aviation: 100 Mod-1 units in 2027 would consume about 100 engines a year on a one-core-per-unit reading, competing with teardown and exchange for run-out engines; capital for engine inventory held for conversion	Creates no aviation shop visits; adds no used material, since it removes cores from the pool; on a manufacturer-level base at 1.5–2%, 100 engines are about 21–28% of a year's retirements, or about 35% on the consultancy base of 14,200 at 2%; engines per unit undisclosed
T-11	Freighter conversion: more than 260 737-800 freighters in service and more than 300 in the pipeline (mid-2025); conversion at 21–22 years; IBA warns of "oversupply of narrowbody conversions" (6 Sep 2026) (§19.1; §22.3; Acumen; Cargo Facts; IBA)	Engines with green time or a fresh shop visit for 20-plus-year-old airframes: -7B green-time modules and shop visits; keeps -7B engines on wing past passenger retirement age	Removes airframes and their engines from the teardown pool; does not stress -5B or V2500, since A320ceo conversion is smaller and not quantified here; cargo-market dependent, and the appraiser's oversupply warning cuts the other way
T-12	Higher utilisation of mature narrowbodies: mature passenger-narrowbody utilisation "rising significantly" (§19.2; IBA via AviTrader, Sept 2024)	Faster accumulation of cycles and hours → shop visits and LLP replacements earlier in calendar time; faster exhaust-gas-temperature-margin erosion	Creates no used material; does not change the installed base; ends when replacement aircraft arrive and utilisation normalises
T-13	GTF groundings easing: PW1100 AOG down 25% year to date, maintenance output up 40%, turnaround down 23% (RTX, 27 Jan 2026); GTF Advantage certified 2026 (§20.1; LeeHam; Aviation Today)	Over 2026–28, returning neos reduce the need for ceo lift → ceo extension rates and used ceo lease rates soften; more ceo retirements → more -5B and V2500 teardown supply later	Does not reduce the near-term ceo shop visits already induced; the pace is uncertain, with trade commentary putting full resolution at 2027–28

Row	Driver (section; principal source)	Requirement it creates in the aftermarket	What it does not stress, or pushes the other way
T-14	CFM56 plateau-then-decline and LEAP aftermarket parity: 2,300–2,400 visits a year through 2028 with the apex at 2027–28 and a gradual fade (GE, 2026); about 2,000 CFM56 visits by 2030 implied (Safran via Aviation Week); LEAP revenue parity in 2028 (Safran Capital Markets Day 2024) (\$23.1)	Long run: shop capacity reallocates to the LEAP; teardown supply rises as retirements normalise (McKinsey: about 2.7% of the aircraft fleet a year from 2028)	Does not stress 2026–27 CFM56 demand; the timing depends on the retirement rate, which GE revised three times in one year (R-004)

Source: every figure is cited in the section of this Part named in the row, with the sources listed at the end; rows T-7, T-9 and T-10 draw on Part I §4, Part V §33 and Part IX §71 and their dossiers (D2 §4.4, D3 §2.6, D8 §2.10 and §4.3). Structure from D9 §5, extended with three rows for drivers documented by sibling dossiers. Reconciliation R-001, R-004, R-006, R-007, R-019, R-020, R-028, R-031, R-034.

The paragraphs below explain the mechanism in each row, in the same order.

T-1, low CFM56 retirements. The retirement rate drives both sides of the market at once (§19.4). On the demand side, an engine not retired keeps flying and keeps accruing cycles, so it reaches its next performance restoration and, eventually, its LLP limits. Worked example 3.2 put the gap between the planned 3–4% and the realised 1.5–2% at about 400 engines a year on any manufacturer-level base, and about 1,200 engines over 2024 to 2026. Those engines are why GE and Safran's 2,300–2,400 visits a year hold through 2028 rather than fading from 2026. An engine kept past its planned retirement also changes workscope: an owner who had planned a green-time run-out to retirement now needs a full restoration and a new or used LLP set to run another interval, which is Part IV §30's four-option problem. On the supply side the same engine is one not torn down, so the row's third column hands over to T-9. The row does not touch the new engines' spare pools and does not add a single shop slot.

T-2, GTF powder-metal groundings. Every Pratt & Whitney-powered A320neo on the ground needs its flying replaced, and the practical replacement is an A320ceo on the same type rating (§20.1). The ceo that replaces it flies cycles it would not otherwise have flown. Worked example 3.3 put the ceo-engine visits brought forward at 47 to 250 a year depending on whether the grounded count is RTX's planning average of 350 or Cirium's late-2025 point count of 835. Those visits fall on the V2500 and the CFM56-5B in proportion to the ceo fleet's engine mix, which is unknown (§18.2). The row also raises ceo lease extensions, short-term ceo leases and spare -5B and V2500 engines, and it creates demand for green-time -5B and V2500 modules from airlines that had planned to retire ceos and now must keep them through one more interval.

It does not touch the -7B except through general narrowbody scarcity, and because it suppresses ceo retirements it removes ceo teardown feedstock rather than adding it.

T-3, LEAP durability and the July 2026 kit. A LEAP that comes off wing at 4,000 cycles instead of 10,000 generates a LEAP shop visit and needs a LEAP spare, and about half the LEAP fleet is under a long-term agreement, so most of that work flows to the manufacturer's network (§20.2). None of it is CFM56 work. The row's effect on the CFM56 aftermarket is entirely indirect: an operator whose A320neo or 737 MAX fleet is less available than planned keeps its ceo or 737NG flying, which Willis Lease's filing names as a compounding factor. The LEAP-1B kit's certification on 18 July 2026 and its production cutover in early 2027 are the dates at which, on CFM's statement, new MAX engines stop coming off early. From then the indirect support for the 737NG fades, which is why the row's third column carries its own reversal. The LEAP-1A was about a third upgraded at the research date, so the A320ceo side of this row was already fading.

T-4, the 737 MAX production rate cap. Worked example 3.4 showed that the cap alone held back on the order of 170 737NG replacements a year in 2024 and 2025 at a one-for-one replacement ceiling, or about 77 at Cirium's replacement ratio. Each 737NG not replaced is extended, flown hard and maintained: CFM56-7B shop visits, LLP replacements, spare -7B engines, and green-time -7B modules for carriers deferring retirement. This row is the -7B's counterpart to T-2 and T-5, which are -5B and V2500 rows, and the three must not be fused. The cap is lifting, from 42 toward 47 and 52, and the backlog of more than 4,800 is owed rather than cancelled, so the row reverses: more 737NG retirements, more -7B teardown supply, and fewer -7B visits later in the decade. United, WestJet and Copa have already begun the retirements that the reversal implies.

T-5, the Airbus delivery shortfall. Airbus delivered 793 aircraft in 2025 against an original target of 820 and now expects A320neo-family rate 75 only by the end of 2027 (§21.2). Each undelivered neo is a ceo its airline cannot release, so the row raises A320ceo extensions, ceo utilisation, -5B and V2500 shop visits and ceo spare engines, and it keeps ceo part-out low. The airline that waits for a late neo still pays for ceo maintenance in the meantime, which is the conversion of planned capital spending into aftermarket operating spending described in §21.3. The row does not stress the -7B. Like T-4 it reverses as the rate rises, because the 8,754-aircraft backlog is a deferral of replacement, not a cancellation.

T-6, lessor extension behaviour. When 75 to 90% of expiries are extended, the maintenance events that would have fallen at a lease transition fall inside the extended term instead, and the incumbent airline pays for them (worked example 3.5). The row therefore raises shop visits and LLP replacements inside extended terms, raises lease rates on mid-life aircraft and engines, and holds lessors' capital in aircraft that stay on lease rather than being sold. Its third column is the

one that matters for the mid-life buyer and the teardown trader: only 10 to 25 aircraft of every 100 come back, so the flow of acquisition candidates and teardown feedstock shrinks to that trickle. The row does not change the total number of cycles the fleet flies; it changes who pays for the maintenance those cycles cause and when. The direction of mid-life rents is contested between a new-aircraft lessor's filing and an appraiser's commentary (R-031), and the row carries that tension.

T-7, OEM parts-price escalation. Part I §4 and Part IV §28 own this driver; it is in the table because it acts on every row above it. New spare parts rose about 10% a year in 2022 and 2023, with a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023, and Safran's stated rule is pricing "above inflation by 3 to 4 points ... at least for some years"; the -7B LLP set compounded about 7% a year from 2008 to 2025 on the figures Part I assembled. [Aviation Week, late 2023; D1 example C] Because 60 to 70% of a shop visit's bill is material priced off the manufacturer's catalogue, escalation flows into every visit whoever performs it (Part I §7). The requirement it creates is for substitutes on the material line: used serviceable material (Part V), approved alternative parts (Part VI), repair instead of replacement, and module exchange instead of overhaul (Part VII). It does not change how many engines come off wing, it does not stress labour or slots, and it leaves the LLP line unsubstituted, since no PMA exists for a CFM56 life-limited part. [D4 U7]

T-8, MRO capacity shortfall and turnaround. A slot wait of two to six months plus a turnaround of 90 to 120 days means an engine removed today returns in five to ten months, and for that whole period a spare must fly in its place (§23.4). So the first requirement this row creates is spare engines and spare-engine leases, in proportion to the delay rather than to the removal count. The second is substitution of form: a module exchange or a quick-turn visit returns the aircraft to service in weeks rather than months, so the shortage of full-overhaul slots is itself demand for exchanges, exchange pools and the inventory of serviceable modules that an exchange requires, which is capital. The row does not change the number of removals the fleet generates and does not add a single teardown. GE's filing reports turnaround improving and material input up more than 40%, against trade press and a consultancy survey that report the constraint persisting (R-019, R-020); the row carries both.

T-9, used-material and feedstock scarcity. This row is a consequence of T-1, T-4, T-5 and T-6, recorded as its own row because it creates its own requirements and because Part V answers it directly. Fewer retirements mean fewer teardowns; fewer teardowns mean less used material and fewer green-time engines; and the shops that would have used them buy new parts instead, at escalating catalogue prices (T-7). The requirements are higher prices for used parts and green-time engines, more new-parts purchases, approved alternative parts as the other substitute, and capital to acquire what feedstock exists, including whole aircraft bought for their engines. The

observable figures are consultancy and trade-press ones and are labelled so: run-out -7Bs bought at \$0.8–1.2M, -7B rents about 34% above 2019, CFM56 values up as much as 50% in two years, and AerSale's whole-asset sales falling from 20 engines to 13. [Safe Fly Aviation, 2026; ePlaneAI; AerSale, 5 Mar 2026] The row reverses when its causes do, which the third columns of T-4, T-5 and T-13 describe.

T-10, data-center power demand and the gas-turbine shortage. Part IX owns this driver. It is in the table because it is the one driver that removes CFM56 engines from aviation for a reason unrelated to aviation. The International Energy Agency (IEA) expects global data-center electricity use to more than double to about 945 terawatt-hours by 2030; GE Vernova's gas-turbine backlog plus reservations reached 116 GW at the second quarter of 2026 with slots being sold for 2031; Siemens Energy reports lead times of three years and more. [IEA, Apr 2025; Power Engineering, Jul 2026; Oilprice, 2026] Into that shortage FTAI Power sells Mod-1, and 100 units in 2027 would consume about 100 CFM56 engines a year if each unit uses one core, which FTAI has not stated. [D8 §2.10, U3] The requirement created is competition for run-out engines: a core that becomes a generator set is a core that is not torn down for parts and not rebuilt for exchange. Worked example 3.2 gave the scale on both bases the reconciliation requires: 100 engines are about 21 to 28% of a year's retirements on a manufacturer-level base at 1.5 to 2%, and about 35% on the consultancy base of 14,200 at 2%. [reconciliation C.8] The row creates no aviation shop visit and adds no used material.

T-11, freighter conversion. A conversion candidate is a 20-plus-year-old airframe that needs engines with life on them (§19.1). So conversion creates demand for green-time -7B modules and for shop visits on engines that would otherwise have been retired with their passenger airframes, and it keeps those engines on wing past passenger retirement age. Its third column is the teardown trader's: each converted 737-800 is an airframe and two engines removed from the part-out pool. The row is specific to the -7B, because A320ceo conversion is a smaller programme that the research did not quantify, and it is contested at the research date, with more than 260 737-800 freighters in service and more than 300 in the pipeline on one side and IBA's September 2026 warning of oversupply on the other (R-034).

T-12, higher utilisation of mature narrowbodies. IBA observed in September 2024 that delivery delays had led operators to maximise daily utilisation across all age groups, with mature passenger-narrowbody utilisation "rising significantly". [IBA via AviTrader, 24 Sep 2024] Utilisation here means flying intensity, cycles and hours per aircraft per period, not the lessor's on-lease share that Part VIII §58 uses; the registry records three meanings of the word and this row uses only the first. [registry C-3] An engine flown 1,500 cycles a year reaches a 10,000-cycle interval in under seven years; at 1,000 cycles a year it takes ten. The row therefore brings shop visits and LLP replacements forward in calendar time and erodes exhaust-gas-temperature margin faster,

without adding a single engine to the base or a single teardown to supply. It ends when replacement aircraft arrive and utilisation normalises.

T-13, GTF groundings easing. This is the reversal of T-2, recorded separately because it was already under way at the research date. RTX reported PW1100 AOG down 25% year to date, maintenance output up 40% and turnaround down 23% in January 2026, with the GTF Advantage certified for entry into service later in 2026 (§20.1). Each neo that returns to service releases the ceo that was flying in its place. Over 2026 to 2028 that softens ceo extension rates and used ceo lease rates and allows the ceo retirements that T-2 and T-5 suppressed, which in turn adds -5B and V2500 teardown supply later in the decade. The row does not reduce the ceo shop visits already induced, since an engine restored in 2025 has been restored, and its pace is uncertain, with trade commentary putting full resolution at 2027 to 2028.

T-14, the CFM56 plateau-then-decline and LEAP parity. GE's 2026 curve holds 2,300 to 2,400 CFM56 visits a year through 2028 with the apex around 2027 to 2028 and a gradual fade after; Safran's earlier framing implied about 2,000 CFM56 visits in 2030 and LEAP aftermarket revenue at parity with the CFM56 by 2028 (§23.1). The long-run requirement is a reallocation: shop capacity moves to the LEAP, and teardown supply rises as retirements return toward McKinsey's about 2.7% of the aircraft fleet a year from 2028. The row does not stress 2026 to 2027 CFM56 demand. Its timing is the least certain figure in the table, because it depends on the retirement rate, which GE revised three times in one year, and because the two CFM partners' curves disagree on level and shape (R-004, R-006, R-007).

EXHIBIT 3.15 — WHERE EACH AFTERMARKET REQUIREMENT IN THE TABLE IS TREATED IN THE PRIMER

Requirement named in the middle column	Owner Part and sections	Rows in which it appears
Shop visits, worksopes and the shop-visit market	Part IV §25–§30	T-1, T-2, T-4, T-5, T-6, T-8, T-11, T-12, T-14
Used serviceable material, teardown, feedstock and green-time engines	Part V §31–§36	T-1, T-2, T-4, T-5, T-6, T-7, T-9, T-10, T-11, T-13, T-14
Approved alternative parts (PMA) and repairs	Part VI §37–§44	T-7, T-9
Modules, module exchange and the Module Factory	Part VII §45–§57	T-1, T-2, T-4, T-7, T-8
Leases of aircraft and engines, extensions, spare engines and the capital behind them	Part II §12–§17; Part VIII §58–§67	T-2, T-4, T-5, T-6, T-8, T-9, T-13
CFM56 cores as power-generation feedstock	Part IX §68–§74	T-10
Capital: inventory, acquisition, third-party vehicles	Part VIII §61–§66; Part X §81–§82; Part XI §86	T-6, T-8, T-9, T-10

Source: outline.md section numbers; the rows of Exhibit 3.14. Compiled for this Part. The mapping says where each requirement is treated; it does not say which rows each Part will name as answered, which each Part states for itself.

The table closes the first-principles build. Everything a reader needs to interpret a quantity in Parts IV to XI is now in place: how the engine works and what it costs (Part I), how money moves around it (Part II), and which of fourteen documented drivers is pushing which requirement in which direction (this Part). Where two sources disagree on a driver's size, the table carries the tension in its row and the section behind it carries both figures. Where the research could not find a figure, the unknowns list below says so.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-001	CFM56 in service: ~23,000 (CFM, undated; GE 2019 / Safran 2024 via D2); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); "over 20,000" -5B/-7B (Chromalloy, Oct 2025); >19,000 and ~14,200 (Safe Fly, consultancy blog)	§18.1, Exhibit 3.1; worked examples 3.1, 3.2, 3.6
R-002	Safe Fly's ~14,200 read as the -7B fleet (D3) or all CFM56, ≈6,200 -5B and ≈7,500 -7B (D8)	§18.1, §18.2, Exhibit 3.1
R-003	-7B: "in excess of 8,000" installed (CFM product page, undated) against >15,000 delivered and >7,000 aircraft at two each	§18.2
R-069	Deliveries >35,000 (CFM; trade press) against "well over 30,000" (unsourced)	Exhibit 3.1
R-010	V2500 in service: 5,286 (Aviation Week, 2023) against ">13,000" (low-grade vendor page, mid-2026)	§18.2, Exhibit 3.2
R-012	Fleet maturity: 57% / 21% (GE, Jun 2019); ~70% zero or one (Safran, 2024); "over half" / "another quarter" (Safe Fly, 2026)	§18.3, Exhibit 3.3; worked example 3.1
R-004 (D9 T3)	GE retirement rate: 3–4% → 2–3% → 1.5–2% (CFO via Aviation Week, 2026); 2025 ~1.5%, 2026 ~2% (Leeham, 22 Jan 2026); 1Q 2026 "below 1%" (GE investor-relations page)	§19.3, Exhibit 3.6; worked example 3.2
R-005 (D9 T2)	Aircraft retirements 2024: fewer than 200 (IBA, Sept 2024) against 587 (Cirium via Informa); derived ~410 (McKinsey) and ~1,130 (Cirium five-year projection); IBA's 2020 peak against Cirium's 2019 > 2020	§19.2, Exhibit 3.5
D9 T6, T7	GTF grounded aircraft: ~350 average 2024–26 (RTX planning, 2023) against 835 grounded or stored (Cirium, late 2025); "720 of 1,912" with undefined denominator; RTX's "down >20%" without a level	§20.1, Exhibit 3.7; worked example 3.3

Id	The two (or more) figures	Where in this Part
R-006	CFM56 shop visits: 2,300–2,400 a year 2026–28 (GE, 2026) against ~2,500 peak 2025–26 (Safran, c. 2023/24); IBA "+40%" 2024 → 25 and Safran "mid-single-digit" 2025 unreconciled	§20.2, §23.1, Exhibit 3.12
R-007 (D9 T4)	Shape: peak 2025–26 (Safran) against plateau to 2028 with apex 2027–28 (GE) against first-visit peak 2027 (Visual Approach) and 737-700 decline after 2027 (Acumen)	§20.2, §23.1, Exhibit 3.12
R-008 (D9 T1)	Interval: 10,000–15,000 cycles (Leeham) against ~8 years (Aviation Week) against ~10 years implied by 2,300 visits on 24,000 engines	§23.4, worked example 3.6
R-009	V2500 shop visits 2025: >800 (RTX CFO via Aviation Week, 2026) against ~1,400 forecast (Aviation Week data tool)	§23.2, Exhibit 3.13
R-011	Removal driver: exhaust-gas-temperature (EGT) margin "most influential" (Aircraft Commerce, 2003/2008) against LLP-driven for the latest -5B/-7B (Aircraft Commerce Issue 50; FlightGlobal); D9 lists three clocks unranked	§22.1
R-019, R-020	Turnaround and parts: 90–120 days and slot waits 2–6 months (trade press; Bain) against ~90 days sustaining and material input +40% (GE 10-K FY2025; GE, 27 May 2026); Oliver Wyman labour survey	§23.4
R-028 (B-5)	-7B engine rent: ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time); both about a third above 2019	§22.3, Exhibit 3.11
R-031	Mid-life narrowbody rents: rising (Air Lease 10-K FY2025; Cirium +44%) against 12-year 737-800 ~\$228,000 (–11%) and A320-200 ~\$220,000 (–13%) (IBA, Jul 2025) and IBA's Sept 2026 warning	§22.3, Exhibit 3.11
R-034	Conversion demand: FTAI–AEI collaboration (7 Jul 2026), >260 737-800Fs and >300 pipeline against IBA's "oversupply of narrowbody conversions" (6 Sep 2026)	§22.3; row T-11
D9 T5	CFM's "in excess of 8,000" -7B installed against >14,000 implied by >7,000 aircraft	§18.2
D9 T9	Airbus 2025 target: 820 (Argus) against ~790 revised and 793 delivered	§21.2, Exhibit 3.10
D9 T10	"Sevenfold" shop-visit rise by 2040 (IATA via AFM) against Safran's LEAP curve and the >4,000 CFM-stable figure	§23.3

Unknowns carried in this Part

- **U-D9-01 (D9) — U1 — 2026 in-service counts and average ages for the A320ceo family and the 737NG, separately, with parked share.** Only a combined A320 family in-service figure (11,308, July 2026 [S13]) and a 2023 stored-A320ceo figure (~570, avg 17.1 years [S14]) surfaced. Resolve: Cirium Fleets Analyzer or IBA Insight extracts, or the fleet tables in Aviation Week's 2026 Fleet & MRO Forecast.
- **U-D9-02 (D9) — U2 — Date of the "47% of 737NGs / 45% of A320ceos aged ≤10 years" statistic [S77/S78].** Given production ended in 2019/2022, the proportion in 2026 must be lower; the figure is probably from 2021-22. Resolve: read the ISTAT article.
- **U-D9-03 (D9) — U3 — Definition of the ~24,000 CFM56 "in service" figure [S78]** (installed only, or installed plus spares plus stored) and its split between -5B, -7B and the older -3/-5A/-5C variants. Resolve: GE Aerospace or CFM investor materials with a fleet breakdown; the GE Bernstein presentation (27 May 2026) [S6] may contain it but its content did not surface.
- **U-D9-04 (D9) — U4 — V2500 engines in service, 2025-26, from IAE/RTX or Cirium.** Only a low-grade figure surfaced (T8). Resolve: IAE/Pratt & Whitney V2500 programme page or RTX investor day fleet slide.
- **U-D9-05 (D9) — U5 — A year-by-year series of GTF AOG counts (average and peak) for 2024, 2025 and 2026 from RTX's own disclosures.** RTX's statements surfaced as percentages (-20%, -25%) rather than levels [S31]. Resolve: RTX Q4 2024, Q4 2025 and Q2 2026 earnings slides/transcripts [S32].
- **U-D9-06 (D9) — U6 — Historical retirement rates for the A320ceo and 737NG specifically (annual counts 2015-2025).** Only aggregate all-type retirements and the CFM56 percentage surfaced. The sentence "retirements in relation to the share of the active fleet as of 2023 was 31% for the 737 and 26% for the A320" appeared in a result [S59/S60 context] but its meaning (share of total retirements by type?) is not clear. Resolve: Cirium or IBA type-level retirement tables.
- **U-D9-07 (D9) — U7 — The 2026 Aviation Week Fleet & MRO Forecast's type-level shop-visit curve for CFM56 through 2030, and Oliver Wyman's narrowbody-engine subtotal by year.** Only headline totals surfaced [S47, S51]. Resolve: the forecast documents themselves.
- **U-D9-08 (D9) — U8 — Labour figures (engine MRO technician headcount, shortfall, wage inflation).** Oliver Wyman cites skilled-worker retirements qualitatively [S47]. D2 may hold these. (Partly supplied: Oliver Wyman, Apr 2026, two-thirds of operators short of technicians, labour-rate inflation 5.5–6.0%; reconciliation C.9.)

- **U-D9-09 (D9) — U9 — Airline-level fleet plans with CFM56 retirement dates.** Spirit's 2023 fleet plan exhibit surfaced as a URL [SEC fleet plan, Aug 2023] but was not read; the FY2025 10-Ks of Southwest, JetBlue and Frontier surfaced but their fleet-plan tables were not extracted [S73-S75]. Resolve: read the fleet-plan sections.
- **U-D9-10 (D9) — U10 — Boeing/Airbus 20-year market outlook replacement-demand figures for single-aisles (the share of deliveries that replace retirements).** Cirium's 2024 fleet forecast (44,500 deliveries, \$2.7tn, 20 years) surfaced [S83] but without the replacement share. Resolve: Boeing CMO and Airbus GMF 2025/2026 editions.
- **U-D9-11 (D9) — U11 — Average AOG duration (wing-to-wing days) for a GTF powder-metal shop visit, by year.** RTX gave a 23% reduction in turnaround [S31] without the base. Resolve: RTX transcripts.
- **U-D2-02 (D2) — U2. Shop-visit counts 2020–2023.** Only the 2019 level (~2,000+) and the 2024 forecast ("topping 2,000") were found. Resolve from Safran capital-markets-day decks (the safran-group.com media downloads 397180 / 448184 returned in search but not read) or the GE Aerospace Bernstein presentation of 27 May 2026.
- **U-D2-10 (D2) — U10. Dates.** Several Aviation Week articles were returned without publication dates (noted "c. 2023/24" or "2025"). The writer should confirm before quoting.
- **U-D3-16 (D3) — U16 — Actual retirement counts for 2023, 2024 and 2025 from Cirium** (only 2019–2022 counts and the 2024–26 shortfall percentage were captured).
- **U-D1-12 (D1) — U12 Current LEAP delivery total and the dates/years of -5A and -5C production end.**
- **U-D11-16 (D11) — U16 — GE CFM56 shop-visit count, spare-parts price increases, and any "material solutions" or used-serviceable-material programme wording.** Not returned from [S14]. Resolve: read the CES segment discussion in [S14] and the Q4 2025 and Q2 2026 earnings materials.

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PART IV

The shop-visit market and the OEM aftermarket

This Part describes the market that Part I's engine creates: who takes CFM56 engines apart for a living, how many engines come through the doors each year, what stops the shops doing more, and how the engine's manufacturer earns money from every visit whoever performs it. A reader who finishes it will be able to read a capacity announcement, a shop-visit forecast and a parts-price statement with every unit and caveat understood, and will be able to lay out the four choices an operator has when an engine runs out of life.

TRANSLATION-TABLE ROWS ANSWERED

Part III §24 turns each fact about the fleet into a requirement on the aftermarket, one row per driver. This Part answers five of those rows. The row on low retirements (Part III §24, row T-1: low retirements) asks how many shop visits a year the retained fleet generates and for how long; §26 gives the forecasts, the base year and the retirement assumption that moves them. The row on the MRO (maintenance, repair and overhaul) capacity shortage (Part III §24, row T-8: MRO capacity shortfall and turnaround) asks what limits the shops' output and how long an engine waits; §25 names the shops and their announced additions, and §27 gives the slot, labour, parts, stand and turnaround figures. The row on OEM parts escalation (Part III §24, row T-7: OEM parts escalation) asks how the manufacturer's pricing reaches every visit; §28 gives the price steps and the service menu, and §29 shows which slice of a visit's bill they land on. The rows on LEAP (CFM's new-generation engine) durability (Part III §24, row T-3: LEAP durability shortfall) and on the GTF (Pratt & Whitney's geared turbofan) groundings (Part III §24, row T-2: GTF powder-metal groundings) create no CFM56 shop visits directly; they matter here because they keep old aircraft flying and because the new engines are sold under a different commercial model, which §28 sets beside the CFM56's. Part III is being written in parallel with this Part, so the row identifiers above are placeholders to be filled when its table is final. §30 closes with the four options an operator has for an engine at its life limit, which is where the rows meet.

25. Who performs shop visits: the OEM network, the independents, airline shops, FTAI

Part I §6 defined a shop visit as any removal of an engine for work in an engine shop, and an engine shop as a facility holding a Part 145 repair-station approval and, in practice, the manufacturer's data and tooling (Part I §9 has the approval). This section asks who owns those shops. The research sorted them into four groups: the manufacturer's own network, the independents, the airlines' in-house shops, and FTAI, which fits none of the first three neatly. The groups are described in turn, and the section ends with the one number the research most wanted and did not find, the manufacturer's share of CFM56 visits.

EXHIBIT 4.1 — WHO PERFORMS CFM56 SHOP VISITS: THE FOUR GROUPS, THEIR NAMED MEMBERS AND WHAT THE RESEARCH FOUND ABOUT EACH

Group	Named members reached by the research	What the group sells	What was found, and what was not
OEM network	GE Aerospace and Safran Aircraft Engines shops; licensed and joint-venture shops (often with an airline)	Shop visits under time-and-materials or flight-hour agreements; new parts to everyone	GE segment revenue and growth rates (\$28.7); no CFM56 shop-visit count for the network (unknown U3)
Large independents	StandardAero, MTU Maintenance, Lufthansa Technik, ST Engineering, AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir Maintenance & Engineering, GA Telesis; also Aero Norway, FL Technics, Magnetic MRO, SR Technics	Shop visits priced to the customer, with material sourced from the OEM, used material, approved alternative parts and repairs	Capacity statements for StandardAero, MTU and FL Technics (Exhibit 4.2); none for the eight others (unknown U5)
Airline shops	The airline-subsidary MROs above when working on their parent's fleet; airlines maintaining their own engines	Visits for the parent fleet; some third-party work	No count of in-house CFM56 shops or of their share was retrieved
FTAI	Montréal (ex-Lockheed Martin Commercial Engine Solutions), Miami and Rome Fiumicino (QuickTurn), Lisbon, Orange; capacity partnerships with GMF AeroAsia and EgyptAir	Serviceable modules and engines by exchange or sale, rather than a visit priced to the customer (Part VII)	Capacity in shop visits (2022 language), module overhauls (2024, 2025) and modules (2026 call coverage); units differ (Exhibit 4.2)

Source: GE Aerospace, Form 10-K FY2025; Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025; StandardAero, Form 10-K FY2025; FTAI Aviation, Form 10-K FY2025 and releases of 9 Sep 2024 and 26 Feb 2025; compiled in D2 §3.1–3.4 and D5 §2.6. Reconciliation R-037, R-042; unknowns U-D2-03, U-D2-05.

25.1 THE OEM NETWORK

Part I §2 introduced CFM International as the joint venture of GE Aerospace and Safran Aircraft Engines, the manufacturer and type-certificate holder of the CFM56, and boxed the term "original equipment manufacturer", or OEM. Within the venture GE builds the core and Safran builds the fan, booster, low-pressure turbine and gearbox, and each partner sells spare parts for its own content. [D2 §3.1; CFM website, undated] Both partners also run shop networks of their own, and both place their names on the services menus described in §28.

GE reports the aftermarket inside one segment.

COMMERCIAL ENGINES & SERVICES (CES)

Commercial Engines & Services is the GE Aerospace segment that sells new commercial engines and spare parts and performs shop visits, in GE's own shops and under long-term service agreements. In 2025 it reported revenue of \$33,314 million, of which \$25,010 million (75.1%) was services, and segment profit of \$8,861 million, a 26.6% margin. The share of its services revenue earned on the CFM56 is not disclosed. Part XI §87 reuses these figures beside the other peers. [GE Aerospace, Form 10-K FY2025; D11 §3.8]

The segment's growth tells the reader how fast the shop-visit market has been moving at the manufacturer. GE's 10-K attributes 2025 CES revenue growth of \$6.4 billion (24%) to "increased spare parts volume, internal shop visit volume and worksopes, increased engine deliveries and pricing". [GE Aerospace, Form 10-K FY2025] Internal shop-visit revenue grew 24% in 2025, 19% in 2024 and 27% in 2023, and services revenue grew 26% in 2025. [GE Aerospace, Form 10-K FY2025] GE's first-quarter 2026 release headlined services revenue growth of 39%. [GE Aerospace, 1Q 2026 earnings release, Apr 2026] The 10-K also records a long-term materials agreement with Ryanair covering its entire fleet of about 2,000 CFM56 and LEAP engines (LEAP is CFM's new-generation engine, Part I §10), which §28.3 places on GE's services menu. [GE Aerospace, Form 10-K FY2025]

Safran reports the other half. Its 2025 group revenue was €31.3 billion, up 14.7%, with recurring operating income of €5.2 billion at a 16.6% margin. [Safran, 13 Feb 2026] Civil engine spare-parts revenue, which Safran tracks in US dollars, grew 17.6% in 2025, "driven primarily by the CFM56", on "a higher number of shop visits" and "a higher workscope content". [Safran, 13 Feb 2026; Forecast International, 13 Feb 2026] Safran had raised its civil aftermarket guidance twice in 2023 to growth in the "low-30% range", and raised full-year guidance again in October 2025 on what Leeham News called a "booming engine aftermarket". [Aviation Week, late 2023; Leeham News, 24 Oct 2025] Safran's installed base at end-2025 included "more than 22,800 CFM56s", from which it says it "can expect sustained aftermarket revenue from this fleet until 2030". [Safran, FY2025 results presentation, 13 Feb 2026]

The network is wider than the two partners' own buildings.

LICENSED SHOP; JOINT-VENTURE SHOP

A licensed shop is an engine shop that performs CFM56 work under a licence from the manufacturer; a joint-venture shop is one in which the manufacturer holds equity, often alongside an airline. Both operate with guaranteed access to the manufacturer's parts and technical data, and both are counted as part of the OEM network rather than as independents. The research did not retrieve a list of CFM56 licensed and joint-venture shops or their throughput. [D2 §3.1, §8]

GE's own description of the CFM56 market is that "an open network exists with multiple providers who compete and invest to win shop visits", and that GE will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity". [GE Aerospace investor-relations page, 2026] Specific CFM56 shop-visit counts for the OEM network, and therefore the network's share of the market, were not found. [D2 U3] §25.6 returns to that gap.

25.2 THE LARGE INDEPENDENTS

Part I §6 boxed "MRO" as the industry's generic term for maintenance, repair and overhaul and "independent MRO" as an engine shop not owned by the engine's manufacturer. Three kinds of company fit that definition. Stand-alone companies such as StandardAero, Aero Norway and Magnetic MRO do nothing else. Airline subsidiaries such as Lufthansa Technik, AFI KLM E&M, Delta TechOps, Turkish Technik, GMF AeroAsia and EgyptAir Maintenance & Engineering maintain their parent's fleet and sell to third parties. And one engine maker's shop is independent on this engine: MTU Maintenance belongs to MTU Aero Engines, which is a partner in the V2500 (the engine that competes with the CFM56-5B on the A320ceo, Part I §10) but not in CFM, so on the CFM56 it is a customer of the manufacturer like any other shop. [D2 §3.2]

StandardAero is the independent the research knows most about, because it files with the SEC. Its Engine Services segment reported 2025 revenue of \$5,354.0 million, up 15.3%, and segment Adjusted EBITDA (earnings before interest, tax, depreciation and amortisation) of \$706.9 million, on StandardAero's own definition of the measure, a 13.2% margin; the segment covers commercial, business-aviation and military engines, and the filing does not give a CFM56 share. [StandardAero, Form 10-K FY2025; D11 §3.6] Its first-half 2026 report gave revenue of about \$3.2 billion, up 8.8%, and Adjusted EBITDA of \$433.0 million, up 12.3%. [StandardAero, Form 10-Q for the quarter ended 30 Jun 2026]

CENTER OF EXCELLENCE

A Center of Excellence is a shop or line dedicated to one engine type, so that tooling, training, test-cell correlation and parts flow are specialised to it. StandardAero's CFM56-7B Center of Excellence in Dallas, Texas, opened in the second half of 2024 "to service the growing demand on that platform". Part XI §87 uses the term when it profiles StandardAero. [StandardAero, Form 10-K FY2025]

StandardAero states that it has spent "more than \$100 million since 2022 to expand" and to "more than double our shop visit capacity on the CFM56-7B" through the Dallas Center of Excellence, and more than \$100 million on a LEAP line at its 810,000-square-foot San Antonio facility. [StandardAero, Form 10-K FY2025] Its Winnipeg expansion added 70,000 square feet, about 40% more footprint, and came online on 3 September 2025 with capacity for the CF34 (a regional-jet engine) and the CFM56. [GlobalAir, Sept 2025] The LEAP and CFM56 Dallas

programmes "reached profitability" in a 2025 quarter, according to the company's third-quarter 2025 filing. [StandardAero, Form 10-Q for the quarter ended 30 Sep 2025] Its chief commercial officer named labour as the leading constraint. [Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025]

MTU Maintenance overhauls the CFM56 and the V2500 at Hannover and at Zhuhai in China, and has a Berlin-Brandenburg shop; Aviation Week reports it "invested heavily in ramping up capacity, with expansions" at all three. [Aviation Week, 2025] MTU's commercial MRO business reported 2025 adjusted revenue of €6.0 billion, up 18%, and adjusted EBIT (earnings before interest and tax) of €478 million, an 8.0% margin, with geared-turbofan work about 40% of commercial maintenance; no CFM56 share was returned. [MTU Aero Engines, 24 Feb 2026; D11 §3.9] EME Aero, the MTU and Lufthansa Technik joint venture, is a geared-turbofan shop only and targets 500 interventions a year from 2028. [Aviation Week, 2025]

Lufthansa Technik runs a CFM56 engine shop at Hamburg. Its 2025 revenue was €8.049 billion, up 12%, and adjusted EBIT was €603 million, a 7.5% margin; Engine Services "generates almost half" of revenue. No CFM56 capacity or shop-visit figure was returned. [Lufthansa Technik, FY2025 results release, early 2026; D11 §3.9] ST Engineering, the Singapore-based group, named skills shortages and supply chain as its biggest challenges and gave no capacity figure. [Aviation Week, 2025] The three margins just quoted sit on three different bases, segment Adjusted EBITDA for StandardAero and adjusted EBIT for MTU and Lufthansa Technik, and Part XI §88 explains why they cannot be set side by side without saying so. [D11 §2.3]

The smaller specialists fill in the picture. Aero Norway, a CFM56 specialist at Stavanger, wrote in May 2024 that long lead times for induction slots were driving interest in quick-turn services, "especially in the case of older-generation engines, such as the CFM56 and V2500", and it has awarded a repair-management contract to GE (§28.5). [Aero Norway in AVM (a trade magazine), May 2024; GE Aerospace release, undated] FL Technics expanded its Kaunas engine shop for CFM56 demand and reports that about 70% of its work is on the -7B, the remainder on the -5B and the older -3. [FL Technics release, 2025] Magnetic MRO, based in Tallinn, discussed CFM56 "market challenges, opportunities" for 2024 in the trade press. [Aviation Week, 2024] SR Technics in Zurich publishes a CFM56-7B capabilities and price catalogue; a version dated 3 June 2026 was located and not read, which is why no shop's price list appears anywhere in this primer. [SR Technics, 3 Jun 2026; D2 §3.2]

For AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir Maintenance & Engineering and GA Telesis the research established only that each operates a CFM56 shop or a related business. GMF AeroAsia and EgyptAir are FTAI's capacity partners (Part VII §50). EgyptAir holds a GE TrueChoice overhaul-consulting agreement (§28.5). GA Telesis is primarily

a trader in used material with an engine shop attached (Part V §35). No capacity or shop-visit figure was returned for any of the six. [D2 §3.2, U5; Amwal Al Ghad, undated]

25.3 AIRLINE SHOPS

The third group is the hardest to see. Many airlines maintain their own engines in their own shops and never sell a visit to anyone. Where an airline's maintenance arm has become a business in its own right it appears in the previous subsection, since Lufthansa Technik, AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia and EgyptAir Maintenance & Engineering are all airline subsidiaries. The in-house-only shops behind them were not profiled. The research retrieved no count of airline CFM56 engine shops, no estimate of the share of visits performed in-house, and no figure for how much of that work has moved to third parties as fleets have aged. [D2 §3.2, §3.4] The reader should treat the airline group as present in every forecast of §26 and absent from every capacity table of this Part.

25.4 FTAI

FTAI's business on the CFM56 is the Module Factory, which Part I §3 quoted from the FY2025 annual report and which Part VII §46 treats in full. It is listed here as a performer of shop work because its shops take engines and modules apart, repair them and test them, and because its capacity statements sit in the trade press beside the independents'. It differs from the other three groups in what it sells: a serviceable module or engine by exchange or sale, rather than a visit priced to a customer's own engine. [FTAI 10-K FY2025; D5 §2.2]

LMCES; MONTRÉAL

Lockheed Martin Commercial Engine Solutions (LMCES) was the Montréal engine shop that FTAI acquired from Lockheed Martin Canada, closing on 9 September 2024 for total consideration of \$170.0 million; the shop is about 500,000 square feet in the annual report and 526,000 in the closing release. It is the base of the Module Factory. Part VII §50 carries the facility in full. [FTAI release, 9 Sep 2024; FTAI 10-K FY2025; reconciliation R-038, R-039]

QUICKTURN; QUICKTURN EUROPE

QuickTurn is the brand of FTAI's Miami engine business, formed from the assets of iAero Thrust acquired on 4 January 2023 with Unical Aviation as partner; FTAI bought Unical's remaining interest on 1 December 2023 for \$30.3 million in cash. QuickTurn Europe is the 50% joint venture at Rome Fiumicino Airport, Italy, in the former IAG (International Airlines Group) Engine Center, announced 26 February 2025 and closed 5 June 2025 for \$10.5 million, with 200,000 square feet. Lower-case "quick-turn" is a workscope (Part I §6); the capitalised names are these businesses. Part VII §50 carries both. [FTAI releases, 1 Dec 2023, 26 Feb 2025 and 5 Jun 2025; FTAI 10-K FY2025; registry C-20]

FTAI's annual report lists 100%-owned maintenance sites at Montréal, Miami, Lisbon and Orange, the Rome joint venture, and a total of "over one million square feet"; its capacity partnerships with GMF AeroAsia in Indonesia and with EgyptAir were described on calls and their terms were not retrieved. [FTAI 10-K FY2025; D5 U8] Part VII §50 carries what is known about Lisbon and Orange; their deal terms, sizes and opening dates were not returned from any filing or release.

FTAI's capacity has been stated four times in four units, and each statement covers a different footprint. The annual report still carries language first used in 2022 describing Montréal as a facility "with a capacity for up to 300 shop visits per year". [FTAI 10-K FY2025; FTAI 10-K FY2022] At the Montréal closing FTAI said the Montréal and Miami shops together gave "capacity for 1,350 CFM56 module overhauls and >500 engine tests per year". [FTAI release, 9 Sep 2024] With the Rome agreement it said QuickTurn Europe "adds capacity for 450 modules (150 engines) per year, taking FTAI's total to 1,800 modules (600 engines)" and more than 600 engine tests a year. [FTAI release, 26 Feb 2025] On the second-quarter 2026 call management said it had increased CFM56 module production capacity to 3,000 a year, "supporting its 25% market share goal and 100 Mod-1 units annually"; that figure comes from call coverage and was not returned from the release text. [GuruFocus and Quartr coverage of the FTAI call of 30 Jul 2026; orchestrator notes] Lisbon's "113,000 square feet, 300+ modules a year" is also call coverage and unverified against any primary source; the annual report lists Lisbon as owned and the July 2026 call described it as "still ramping". [reconciliation R-037, R-041]

Two cautions govern how these figures are read. First, a shop visit, a module overhaul and a module are different units; FTAI's own releases convert at three modules per engine (450 modules for 150 engines, 1,800 for 600), which matches the three gas-path modules of Part I §3, and no Part of this primer compares the 300-visit figure with the module figures directly. [reconciliation R-035, B-3; registry C-4] Second, the figures are dated statements about growing footprints, not a single capacity revised over time. Exhibit 4.2 carries them beside the independents' statements with the unit each one counts.

Output is reported in modules. FTAI refurbished 757 CFM56 modules in 2025 against a target of 750, 184 in the second quarter of 2025, 228 in the fourth, 296 in the second quarter of 2026 (up 61% year on year) and 566 in the first half of 2026. [FTAI releases, 29 Jul 2025, 25 Feb 2026 and 29 Jul 2026; call coverage, Feb 2026] The 2026 target was set at 1,050 modules with the full-year 2025 results in February 2026 and was raised to 1,200 on the July 2026 call according to call coverage. [FTAI release, 25 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat coverage, 30 Jul 2026; reconciliation R-036]

The research did one piece of arithmetic to size this against the market, and it carries three caveats. At three modules per engine, 1,200 modules is about 400 engine-equivalents, which is about 17% of a 2,350-visit year; 3,000 modules would be about 1,000 engine-equivalents, about

43%. [derived in the research, D2 §3.3] These are not shop-visit shares: modules are sold individually, many go into engines that also visit other shops, and the denominator FTAI uses for its "25% market share goal" was not retrieved. [D5 U5; reconciliation R-042] A third framing, from FTAI's own release of 26 February 2025, is a \$22 billion total addressable market for maintenance, repair and exchange in which FTAI held about 5% as of 2024. [FTAI release, 26 Feb 2025] Dollars, visits and module-equivalents are three different denominators, and Part VII §49 carries all three.

25.5 CAPACITY STATEMENTS AND ADDITIONS, 2022 TO 2026

Before the exhibit, the unit needs a definition, because the independents and FTAI count different things.

CAPACITY (SHOP)

A shop's capacity is the number of engine inductions a year it can take. Four things set it: floor space and engine stands, which bound how many engines can be open at once; licensed technicians; test-cell hours, since every rebuilt engine must be run before release; and parts flow, because a shop that cannot get turbine blades or repaired cases on time fills up with half-built engines that consume stands and floor without producing output. Additions are usually announced in square feet or as a multiple of previous throughput. FTAI's "modules per year" is a different unit, and Part VII §49 states the conversion FTAI's releases use. [D2 §2.4, §5; registry C-4]

EXHIBIT 4.2 — NAMED CAPACITY STATEMENTS AND ADDITIONS, 2022 TO 2026, WITH THE UNIT EACH COUNTS

Shop or site	Statement or addition	Unit counted	Date	Source and level
StandardAero, Dallas (CFM56-7B Center of Excellence)	">\$100 million since 2022"; "more than double our shop visit capacity on the CFM56-7B"; opened second half of 2024	Shop visits (as a multiple) and dollars	Feb 2026 (filing)	StandardAero 10-K FY2025 (filing)
StandardAero, San Antonio (LEAP line)	">\$100 million"; 810,000 sq ft site; LEAP-1A and -1B test-cell correlations completed 2024	Dollars and square feet	Feb 2026	StandardAero 10-K FY2025 (filing)
StandardAero, Winnipeg	+70,000 sq ft (about +40% footprint); CF34 and CFM56; online 3 Sept 2025	Square feet	Sept 2025	GlobalAir (trade press)

Shop or site	Statement or addition	Unit counted	Date	Source and level
MTU Maintenance, Hannover, Zhuhai, Berlin	"Invested heavily in ramping up capacity, with expansions"; CFM56 and V2500 at Zhuhai	None given	2025	Aviation Week (trade press)
EME Aero (MTU and Lufthansa Technik JV)	500 interventions a year from 2028; geared turbofan only	Interventions (visits)	2025	Aviation Week
FL Technics, Kaunas	Expanded for CFM56 demand; about 70% of work CFM56-7B	None given	2025	FL Technics release (company)
Safran, Querétaro, Mexico	New maintenance shop opened 30 June 2026	None given	Jun 2026	Safran release (company)
FTAI, Montréal	"Capacity for up to 300 shop visits per year"	Shop visits	2022 language, still in FY2025 10-K	FTAI 10-K (filing)
FTAI, Montréal plus Miami	"Capacity for 1,350 CFM56 module overhauls and >500 engine tests per year"	Module overhauls; engine tests	9 Sep 2024	FTAI release (company)
FTAI, adding Rome Fiumicino	" +450 modules (150 engines)"; total "1,800 modules (600 engines)"; ">600 engine tests"	Modules; engines at three per engine; tests	26 Feb 2025	FTAI release (company)
FTAI, all sites and partners	"3,000" CFM56 modules a year	Modules	Jul 2026	Call coverage (GuruFocus, Quartr); not in release text
FTAI, Lisbon	"113,000 sq ft, 300+ modules/yr"	Square feet; modules	2026	Call coverage; unverified
FTAI with GMF AeroAsia and EgyptAir	Capacity partnerships	None given	2026	Calls; terms not retrieved (D5 U8)

Shop or site	Statement or addition	Unit counted	Date	Source and level
Lufthansa Technik, AFI KLM E&M, Delta TechOps, Turkish Technic, ST Engineering, GA Telesis	No CFM56 capacity figure returned	None	n/a	D2 U5

Source: StandardAero, Form 10-K FY2025; GlobalAir, "StandardAero's Winnipeg expansion adds CF34 and CFM56 engine capacity", Sept 2025; Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025; FL Technics release, 2025; Safran release, 30 Jun 2026; FTAI Aviation, Form 10-K FY2025 and FY2022; FTAI releases of 9 Sep 2024 and 26 Feb 2025; GuruFocus and Quartr coverage of the 30 Jul 2026 call. Reconciliation R-037, R-041; E.1 rules 2 and 17; E.2 "Capacity".

Two patterns are visible in the table and both are about units rather than facts. The independents announce money and floor space and, once, a multiple of previous throughput; none of them states inductions a year. FTAI states modules, engines and tests, and once, in old language, shop visits. The only way to compare the two sets is to convert FTAI's modules to engines at the convention its own releases use, and even then an engine-equivalent of modules is not an induction. [reconciliation R-037; registry C-4] The table is therefore a list of dated announcements, not a capacity model, and the research built no such model. [D2 §4.8]

25.6 THE OEM SHARE: NOT KNOWN

No source the research reached gives the share of CFM56 shop visits performed in the manufacturer's network against the independents and the airlines. [D2 U3] Three facts bound the question without answering it. Safran describes the CFM56 aftermarket's revenue as 85% spare parts and time-and-materials work, so most visits are not performed under the manufacturer's flight-hour contracts (§28.2). [Safran via Aviation Week, c. 2023/24] Trade press puts about 15% of the CFM56 fleet under long-term agreements against about 50% of the LEAP fleet; the two figures are complements of the same view and are both Aviation Week's reporting of Safran. [Aviation Week, undated; reconciliation B-4] And Safran attributes part of its LEAP spares growth to "the increasing number of shop visits performed by third-party MROs", which implies the manufacturer tracks third-party share internally without disclosing it. [Safran, 13 Feb 2026; D2 §3.4] What would resolve the question is shop-visit-by-provider data from Aviation Week or Cirium, or a provider split in the Oliver Wyman fleet forecast. [D2 U3]

26. Demand: shop visits per year, the peak, and the retirement assumption behind it

Part III §23 carries the forecast curve as the manufacturers and forecasters publish it, and this section reads the same figures as a market: how many engines a year the shops of §25 can expect, when the count stops rising, and which single assumption moves it. The two sections cross-reference each other and neither re-boxes the other's terms. [registry, Part IV rows]

26.1 THE BASE YEAR

The reference point is 2019. Safran, as reported by Aviation Week, put annual CFM shop visits in 2019 at "just above 2,000", "almost all of them CFM56-5B and -7B". [Safran via Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24] Shop-visit counts for 2020 to 2023 were not found year by year. [D2 U2] For 2024 Safran expected total CFM shop visits "topping 2,000 again", a figure that includes some LEAP visits. [Safran via Aviation Week, c. 2023/24] Those are the only pre-2025 points the research holds, and the gap between 2019 and 2024 is carried as an unknown rather than filled.

26.2 THE TWO FORECASTS AND THE PEAK

Two views of the next few years come from the two halves of the same manufacturer at two dates, and the primer carries both without choosing. [reconciliation R-006, R-007; E.1 rule 15]

EXHIBIT 4.3 — CFM56 SHOP VISITS PER YEAR: EVERY FIGURE THE RESEARCH REACHED, WITH POPULATION, SOURCE AND DATE

Period	Figure	Population as the source states it	Source and level	Date
2019	"Just above 2,000"	Annual CFM shop visits, "almost all" CFM56-5B and -7B	Safran via Aviation Week (OEM via trade press)	c. 2023/24
2020 to 2023	Not found	n/a	D2 U2	n/a
2024	"Topping 2,000 again ... with some help from the Leap family"	All CFM, including LEAP	Safran via Aviation Week	c. 2023/24
2024 to 2025	"40 per cent increase in shop visits"	Headline; scope (all engines or CFM56) not captured	IBA (an appraiser) via Aviation Week	2024

Period	Figure	Population as the source states it	Source and level	Date
2025	CFM56 visits grew "mid-single-digit", with "a higher proportion of full work scope shop visits"	CFM56	Safran FY2025 release (OEM)	13 Feb 2026
2025	About 2,300 CFM56 visits within about 9,000 engine shop visits of all types (about 8,800 in 2024)	CFM56 within all engines	Aviation Week 2026 Fleet & MRO Forecast, via ePlaneAI (trade press forecast)	2026
2025 to 2026 (earlier view)	Peak "at about 2,500 per year"	"CFM56-variant shop visits"	Safran via Aviation Week	c. 2023/24
2026 and 2027	"Entered the year projecting about 2,300"; "expanded the range to 2,300–2,400"; "probably closer to 2,400"	CFM56	GE Aerospace via Aviation Week, "CFM56 Overhaul Demand Remains Strong" (OEM via trade press)	2026
Through 2028	"2,300–2,400 range through 2028"	CFM56	Same	2026
Apex	"Around the 2027-2028 time period, followed by a gradual fade"	CFM56	Same	2026
2027	Peak in first shop visits	CFM56 and V2500 first visits only	Visual Approach Analytics (independent research, headline)	2025/26
After 2027	737-700 passenger MRO demand "to decline sharply"	One sub-fleet	Acumen Aviation (appraiser note)	18 Dec 2025
2030	"More than 4,000 shop visits annually, roughly split between Leaps and previous-generation models"	All CFM; implies about 2,000 CFM56 as one dossier reads it	Safran via Aviation Week	c. 2023/24
2028	LEAP aftermarket revenue reaches parity with CFM56	Revenue, not visits	Safran Capital Markets Day 2024, via Quartr	2024

Source: Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade" and "Leap Aftermarket Poised For Growth" (dates not visible, c. 2023/24); Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Aviation Week, "IBA predicts 40 per cent increase in shop visits from 2024 to 2025", 2024; Safran, 13 Feb 2026; ePlaneAI reporting the Aviation Week 2026 Fleet & MRO Forecast; Visual Approach Analytics, 2025/26; Acumen Aviation, 18 Dec 2025; Quartr, Safran Capital Markets Day (CMD) 2024. Compiled in D2 §4.1 and D9 §4.5. Reconciliation R-006, R-007; D2 T6.

Read the two manufacturer views together. Safran's earlier framing, from around 2023 or 2024, has CFM56 visits peaking at about 2,500 a year in 2025 and 2026 and then falling, so that by 2030 the CFM56 would be about half of a 4,000-visit CFM total. [Safran via Aviation Week, c. 2023/24] GE's 2026 framing has a lower and later shape: about 2,300 visits in each of 2026 and 2027 at the start of the year, a range of 2,300 to 2,400 and "probably closer to 2,400" later in the year, that range held through 2028, and an apex "around the 2027-2028 time period, followed by a gradual fade". [GE Aerospace via Aviation Week, 2026] The later view is both lower at its top and later in reaching it. The two are from the two CFM partners at different dates, and the research did not retrieve a statement from either partner reconciling them. [D2 T1; reconciliation R-007]

Two other figures in the exhibit do not fit either curve and are carried as unreconciled. IBA headlined a 40% increase in shop visits from 2024 to 2025, without the scope of the count being visible in what the research captured; if that were CFM56 visits on a 2024 base above 2,000, the 2025 figure would be about 2,800 or more, which is above both manufacturer curves. [IBA via Aviation Week, 2024; D2 T6] Safran's own 2025 report says CFM56 visits grew only mid-single-digit, with a higher share of full worksopes. [Safran, 13 Feb 2026] The most likely explanation recorded in the research is that the two figures describe different populations, but that is a conjecture, and the writer's instruction is to carry both. [D2 T6; E.1 rule 15]

The third shape in the exhibit is a different measure. Visual Approach Analytics dates the peak in first shop visits of the CFM56 and V2500 to 2027. [Visual Approach Analytics, 2025/26] A first visit is one engine's first removal, and the share of the fleet that has not yet had one is treated in §26.4. The Aviation Week forecast gives the context: about 2,300 CFM56 visits inside about 9,000 engine shop visits of all types in 2025, so the CFM56 is roughly a quarter of all engine shop work by count. [Aviation Week 2026 Fleet & MRO Forecast, via ePlaneAI]

A PLATEAU IS NOT A PEAK

The earlier manufacturer view describes a peak: a top of about 2,500 visits in 2025 or 2026 and a fall after it. The later view describes a plateau: 2,300 to 2,400 visits for three years and a "gradual fade" after 2028. The difference matters to a shop deciding whether to add a stand or a test cell, because capacity added into a peak is idle sooner than capacity added into a plateau, and the research records that a shop or module factory adding capacity now "is adding it into a plateau" on the manufacturer's own forecast. What moved the shape between the two dates was not the fleet, which only shrinks, but the assumption about how fast it shrinks, which §26.3 takes up. Where the fade begins, and how steep it is, depends on that same assumption. [D2 §5.7; GE Aerospace via Aviation Week, 2026]

What population each count describes is not stated in the sources. None of them says whether a module exchange performed without a full induction counts as a shop visit, whether visits to the older -3, -5A and -5C variants are included, or whether a quick-turn counts the same as an overhaul. [D2 §4.1] The reader should hold the counts as the sources' own and not as a single series.

26.3 THE RETIREMENT ASSUMPTION BEHIND THE CURVE

Part I §2 boxed the retirement rate and gave GE's sequence; Part III §19 is the full treatment with every denominator. Here the point is mechanical. No new -5B or -7B is being built, so the installed base falls by exactly the number of engines retired or converted each year. [D9 §2.2] Every engine that is not retired keeps flying and keeps accumulating cycles, and eventually it needs the work this Part describes. So the retirement assumption is the one input that sets both the height and the length of the shop-visit curve. [D9 §2.2; D2 §5.7]

EXHIBIT 4.4 — THE RETIREMENT ASSUMPTION AND THE FLEET'S MATURITY: EACH FIGURE WITH ITS BASE, VENUE AND DATE

Item	Figure	Base	Source and level	Date and venue
GE planning assumption for 2026, first	3–4% of the fleet	CFM56 engines	GE CFO via Aviation Week (OEM via trade press)	"Last year", per the 2026 article
GE planning assumption, revised	2–3%	CFM56 engines	Same	Start of 2026
GE planning assumption, current	"Something between 1.5% to 2%"	CFM56 engines	Same	2026
2025 actual	About 1.5%, broadly the same as 2024	CFM56 installed base	GE Q4 2025 results via Leeham News	22 Jan 2026
2026 expected	About 2%, "below the 2–3% range previously anticipated"	CFM56 installed base	Same	22 Jan 2026
1Q 2026 actual	"Below 1%"	CFM56, GE's 2026 outlook assumes about 2%	GE Aerospace investor-relations page (OEM)	2026

Item	Figure	Base	Source and level	Date and venue
Aircraft retirements 2024	"Fewer than 200", similar to 2023, 70% below the 2020 peak	All aircraft, scope not visible	IBA via AviTrader (appraiser)	24 Sep 2024
Aircraft retirements 2024	587	Cirium Fleets Analyzer, scope not visible	Via Informa forecast page (data provider)	2025
Fleet maturity, 2019	57% of CFM56s had no first shop visit; 21% had one	CFM56 engines	GE Aviation & GECAS (GE's former leasing arm) Investor Day (OEM)	18 Jun 2019
Fleet maturity, 2024	About 70% had seen zero or one shop visit	CFM56 engines	Safran media (OEM)	2024
Fleet maturity, 2026	"Over half" with no visit; "another quarter" with one	CFM56 engines	Safe Fly Aviation (consultancy blog)	2026

Source: Aviation Week, "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; Leeham News, 22 Jan 2026; GE Aerospace, "Recent events: your questions answered", 2026; AviTrader (IBA data), 24 Sep 2024; Informa Markets commercial forecast page (Cirium data); GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; Safe Fly Aviation, "CFM56 Engine Market Report 2026". Reconciliation R-004, R-005, R-012; E.1 rule 13.

The sequence in the first three rows is one source disagreeing with itself over time, and the research records it as the mechanism behind the raised shop-visit range rather than as an error. [D2 T3; reconciliation R-004] The two 2026 readings differ by half a point, about 2% in Leeham's account of the January call against 1.5% to 2% in Aviation Week's quotation of the chief financial officer, and the primer carries both with their venues. [E.1 rule 13] The research describes the forecast error as retirements "over-forecast by 1.5–2.5 points in 2025–26", which is the gap between the first assumption and the realised rate. [D2 §5.7] The two aircraft-count rows cannot be reconciled from what was captured, because neither source's scope is visible; Part III §19 carries that tension. [reconciliation R-005]

WORKED EXAMPLE 4.1 — HOW THE RETIREMENT ASSUMPTION MOVES THE NUMBER OF ENGINES THAT STILL NEED SHOP VISITS

Inputs. An installed base of about 24,000 CFM56 engines (sourced: Aviation Business News, Sept 2025) or about 23,000 (sourced: CFM website, undated); the primer carries both and never averages them (E.1 rule 14). Retirement rates taken as the midpoints of GE's stated ranges, 3.5% for "3–4%" and 1.75% for "1.5–2%" (midpoints assumed for illustration; ranges sourced: GE CFO via Aviation

Week, 2026), the 2025 actual of about 1.5% (sourced: Leeham News, 22 Jan 2026) and the first-quarter 2026 rate of "below 1%" (sourced: GE investor-relations page, 2026).

at 3.5% on 24,000: 840 engines a year retired (on 23,000: 805)
 at 1.75% on 24,000: 420 a year (on 23,000: 403)
 at 1.5% on 24,000: 360 a year (on 23,000: 345)
 at under 1% on 24,000: fewer than 240 a year (on 23,000: fewer than 230)
 engines retained each year against the first plan: $840 - 420 = 420$ (at 1.75%); $840 - 360 = 480$ (at 1.5%)
 over three years, 2024 to 2026: roughly 1,260 to 1,440 engines still flying that the first plan had retired

So the fall in the assumed rate keeps something like 1,300 to 1,400 engines in service over three years that the earlier plan had counted out, on the 24,000 base; the 23,000 base gives about 5% less. Against that, the manufacturer's shop-visit range widened by about 100 visits, from "about 2,300" to "2,300–2,400", roughly 4% (derived). The research did not retrieve the model the manufacturer uses to convert retained engines into visits, and the two figures are not expected to match year by year, because a retained engine visits a shop only when its own clocks run out (Part I §5). One consistency check is available: 2,300 visits a year on 24,000 engines is 9.6% of the base, which would imply an average interval of about ten years between visits if every engine were on a regular cycle; CFM56 intervals are usually shorter than that, so either the 24,000 includes spare and stored engines or a large share of engines are flying out their remaining life with no further visit planned (Part I Exhibit 1.8 carries this as a tension). Limits of the example: the midpoints are assumed; the two base figures are contested and may include spares and stored engines; retirements and visits are linked through engine-by-engine clocks that this arithmetic does not model. [Aviation Business News, Sept 2025; CFM website, undated; GE CFO via Aviation Week, 2026; Leeham News, 22 Jan 2026; GE Aerospace investor-relations page, 2026; D9 §2.2 and T1; reconciliation R-001, R-004, R-008]

The same number sits on both sides of the aftermarket. A retained engine is a future shop visit, and it is also an engine that is not torn down, so its parts do not reach the used-material market of Part V and its remaining life does not reach the green-time market of Part II §16. [D9 §2.2; D2 §5.4] Part III §24's row on low retirements is built on that double effect, and §27.4 and §30 of this Part show where it lands in a shop.

26.4 HOW MATURE THE FLEET IS

The second input to the curve is how many engines have never been to a shop.

FLEET MATURITY

Fleet maturity is the distribution of an installed base by the number of shop visits each engine has had: none, one, or more. GE reported in June 2019 that 57% of CFM56 engines had not had a first shop visit and 21% had had one; Safran reported in 2024 that about 70% had seen zero or one. A first visit on the latest -5B and -7B variants comes near the life limit and so combines a performance restoration with life-limited-part replacement (Part I §6), which makes it a heavier visit than a mature-run restoration. [GE Aviation & GECAS Investor Day, 18 Jun 2019; Safran media, 2024; D2 §4.2]

The three maturity readings in Exhibit 4.4 point the same way and come from sources of different weight. GE's 2019 figure and Safran's 2024 figure are manufacturer materials; the 2026 figure, "over half" with no visit and "another quarter" with one, about 75%, is from a consultancy blog and is labelled so. [reconciliation R-012; E.1 rule 3] The research did not resolve how 70% in 2024 and about 75% in 2026 can both hold for a fleet that is only getting older, and records the pair as directionally the same with the later figure from the weaker source. [D2 T7]

Two facts sit beside the maturity figures without the research joining them. Trade sources say first-run -5B and -7B engines of the latest variants fly "almost to LLP limit", so their first visit is a life-limit-driven one (Part I §6). [FlightGlobal, undated; reconciliation B-7] And Safran attributes part of its 2025 spares growth to "a higher proportion of full work scope shop visits". [Safran, 13 Feb 2026] Whether the second is a consequence of the first was not stated by either source, and the primer places them side by side and stops.

27. Capacity: slots, labor, parts, stands, turnaround

§25.5 defined a shop's capacity as the number of inductions a year it can take, set by floor and stands, technicians, test-cell hours and parts flow. Between 2023 and 2026 each of those four limits was reported as binding somewhere in the CFM56 shop network, and this section gives the figure and date for each. It closes with turnaround time, which Part I §6 boxed and which is the quantity an operator actually feels. Every figure below is from a filing, a consultancy survey or the trade press, and the exhibits say which. [D2 §2.4, §5]

27.1 THE FOUR LIMITS, AND THE UNIT THEY SHARE

A shop's output is engines released per year. Each limit caps that output in a different way. Floor space and engine stands cap how many engines can be open at once; a shop that is full of engines waiting for parts is full, whatever its technicians could do. Technicians cap how many engine-hours of work can be done in a month, and the work is licensed, so a vacancy cannot be filled quickly. Test-cell hours cap how many rebuilt engines can be proven and released. Parts flow caps all three, because an engine waiting for a turbine blade or a repaired case occupies a stand,

has consumed labour, and cannot go to test. [D2 §2.4] The industry expresses an addition to any of these as square feet, as §25.5's exhibit showed, because the building is the limit that can be seen and announced. [D2 §5.5]

27.2 INDUCTION SLOTS

INDUCTION; INDUCTION SLOT

Induction is the acceptance of an engine into work at a shop, the point from which turnaround time is counted. An induction slot is the date a shop commits to start a given engine. When shops are full, the wait for a slot is distinct from, and additional to, the turnaround, and in 2023 to 2026 that wait became the binding constraint on an operator's engine availability. In other Parts "slot" also means a production slot for alternative parts (Part VI) or a reserved manufacturing slot for turbines (Part IX); this primer says "induction slot" for the shop meaning. [D2 §2.1, §2.4, §8; registry C-18]

Bain reported in 2024 that "the wait times just to secure an MRO slot rose by two to three months and, in some cases, six months", and projected that engine maintenance demand would exceed available capacity by more than 17% before the end of the decade if capacity kept its current trajectory. [Bain & Company, 2024] Trade press reported shops "at full capacity over the past year" in 2025. [Aviation Week, 2025] Aero Norway's account from the shop floor is that long lead times for induction slots were pushing operators toward quick-turn work, "especially in the case of older-generation engines, such as the CFM56 and V2500". [Aero Norway in AVM, May 2024] An engine-stand utilisation report carried by the trade press expects engine maintenance demand to peak in the near term in 2026 and to remain constrained through the end of the decade. [AviationPros, undated]

WHY THE SLOT, NOT THE TURNAROUND, BECAME THE BINDING CONSTRAINT

An operator planning a removal faces two waits in series. First it waits for a shop to accept the engine; then it waits for the shop to finish. The second wait is the turnaround time of §27.6, which lengthened from about 60 days to 90 to 120 days for a full overhaul. The first wait, which was close to zero when shops had spare stands, grew to two to six months. An aircraft without an engine is grounded or flies on a leased spare for both waits together, so the slot wait added more calendar time to a removal than the longer turnaround did. That is what raised the value of anything that avoids a full induction: a quick-turn visit, a module exchange from inventory, or a green-time engine flown in place of the one that ran out. Each of those substitutes shifts where and how the work is done; none of them changes how many engines reach their limits. [D2 §5.1; Bain, 2024; D9 §5]

27.3 LABOUR

Oliver Wyman's 2026 maintenance survey, of more than 150 respondents, found that "two-thirds of MRO operators struggle to find qualified technicians", that labour-rate inflation had "settled in

at 5.5–6.0%", and that "more than half of survey respondents expect labor attrition to be worse in the coming year". [Oliver Wyman, Apr 2026; ePlaneAI, 2026] StandardAero's chief commercial officer and ST Engineering both named labour as the leading constraint on adding capacity. [Aviation Week, 2025] Oliver Wyman also cites the retirement of skilled workers as a driver of rising maintenance cost, and no headcount or shortfall figure for engine technicians surfaced in the research. [Oliver Wyman via AviationPros, 2026; D9 U8]

Labour is 20% to 30% of a shop visit on one share breakdown and 15% to 20% on another (\$29.1), so a 5.5% to 6.0% rise in labour rates adds roughly 1.1 to 1.8 percentage points to the cost of a visit each year on its own. [derived in the research, D2 §5.2; Air Cargo Week, 2025/26; Safe Fly Aviation, 2026] Worked example 4.2 puts that beside the parts line. One labour data point from FTAI belongs here for scale: the company reported 985 full-time employees and independent contractors at 31 December 2025, of whom 494 were full-time employees in Canada, about 71% of them covered by collective bargaining agreements; the figure mixes employees and contractors, as the filing's wording does. [FTAI 10-K FY2025; E.1 rule 24] Part X §85 carries it.

27.4 PARTS

The parts constraint is the one the sources disagree about, and the disagreement is carried whole.

On one side is the manufacturer's own filing. GE's 10-K for 2025 reports that its operating system raised material input from priority suppliers by more than 40% year on year and that turnaround times for the LEAP, the CFM56 and the GE90 (a widebody engine) improved by more than 10% year on year in the fourth quarter of 2025. [GE Aerospace, Form 10-K FY2025]

FLIGHT DECK (GE'S OPERATING SYSTEM)

FLIGHT DECK is the name GE Aerospace gives its proprietary operating system for running its factories and shops. Its FY2025 annual report credits it with increasing material input from priority suppliers by more than 40% year on year, enabling a 26% rise in services revenue, and with the turnaround improvement quoted above. The research holds no description of what the system consists of beyond the name and those two claims. [GE Aerospace, Form 10-K FY2025; D2 §2.4, §3.1]

On the other side are the surveys and the trade press. Oliver Wyman's April 2026 survey found labour and material shortages remained the primary disruptors; two-thirds of respondents reported that next-generation narrowbody shop-visit costs were running 21% or more above expectation, and a quarter reported 50% or more. [Oliver Wyman, Apr 2026] Aviation Week reported in 2025 that "parts supply remains a capacity constraint with every OEM, due to the recovery in aircraft utilization and the associated growth in MRO demand coming at the same time that engine OEMs are ramping up production". [Aviation Week, 2025] AVM reported repair lead times "surpassing more than 200 days in severe cases" on some new-generation programmes, and the

trade press describes four-to-six-month waits for repairs at third-party facilities, with buyers advised to order life-limited parts and critical rotables early and to use exchange pools and used material. [AVM Magazine, 2025; Air Cargo Week and Locatory, Mar 2026] The same trade press says "the supply of used serviceable material remains thin, leaving shops more reliant on newly manufactured core parts", and Aviation Week ran a column headed "A Seller's Market for Used Parts". [AVM Magazine, 2025; Aviation Week, undated]

The two sides are not necessarily about the same thing. The manufacturer's statement is about its own suppliers and its own shops in one quarter; the survey is about operators' experience across all shops and programmes, and the cost-overrun figure is about new-generation engines. But the research records it as a tension because a reader will meet both: the manufacturer reports easing in late 2025, and the operator survey of April 2026 does not. [D2 T8; reconciliation R-020] What would narrow it is a dated lead-time series for turbine blades, castings and forgings, which the research sought and did not find. [D2 U6]

The used-material side of the parts constraint has a cause this Part has already named. Retirements of 1.5% to 2% rather than 3% to 4% mean fewer teardowns, fewer used parts and fewer green-time engines, so shops are pushed back to new parts at the escalated list price, and the manufacturer's pricing power rises. [D2 §2.4, §5.4] Part V §33 carries the supply side of that and Part VII §52 the agreement FTAI signed with CFM for parts (§28.8).

27.5 ENGINE STANDS AND FLOOR

ENGINE STAND

An engine stand is the wheeled fixture that holds a roughly 2.5-tonne engine, or a module, while it is worked on, stored or transported. Every engine in a shop occupies a stand from induction to release, so the stock of stands is a physical limit on work in progress, and a shortage of stands shows up as engines that cannot be inducted even when technicians are free. Utilisation of stands for legacy engines rose by about a third in 2025 over 2024, with CFM56-5A/B stand usage up 60%, according to one trade-press report. Part III §23 glosses the term at first mention. [D2 §2.1, §8; AviationPros, undated]

The stand figures are the only hard numbers the research holds on this limit. A report carried by AviationPros says engine-stand utilisation for legacy engines rose by a third in 2025 compared with 2024, and that CFM56-5A/B stand usage rose 60%. [AviationPros, "Engine Stand Utilization Report", undated] No figure for a stand shortage, a stand lead time or a stand price was found. [D2 U6] Floor space is reported as the proxy: StandardAero's Winnipeg addition of 70,000 square feet and FTAI's Lisbon site (per call coverage, unverified) are announced in square feet because stands and floor are the limit that an announcement can count. [D2 §5.5; GlobalAir, Sept 2025]

EXHIBIT 4.5 — CAPACITY CONSTRAINTS IN THE ENGINE SHOPS: THE FIGURES, WITH SOURCE, LEVEL AND DATE, AND BOTH SIDES OF THE PARTS QUESTION

Constraint	Figure	Source and level	Date
Slot wait	Up by two to three months, in some cases six	Bain & Company (consultancy)	2024
Demand against capacity	Demand to exceed capacity by more than 17% before 2030, if capacity grows on its current trajectory	Bain	2024
Shops' loading	"At full capacity over the past year"	Aviation Week (trade press)	2025
Technicians	Two-thirds of MRO operators "struggle to find qualified technicians"	Oliver Wyman survey, >150 respondents (consultancy)	Apr 2026
Labour-rate inflation	"Settled in at 5.5–6.0%"	Oliver Wyman	Apr 2026
Attrition	More than half expect labour attrition to be worse in the coming year	Oliver Wyman	Apr 2026
Next-generation shop costs	Two-thirds report 21% or more over expectation; a quarter report 50% or more	Oliver Wyman	Apr 2026
Parts, all OEMs	"Parts supply remains a capacity constraint with every OEM"	Aviation Week	2025
Parts, GE's own supply	Priority-supplier material input up more than 40% year on year	GE Aerospace 10-K FY2025 (OEM filing)	Feb 2026
Turnaround, GE's shops	LEAP, CFM56 and GE90 turnaround improved more than 10% year on year in 4Q 2025	GE Aerospace 10-K FY2025	Feb 2026
Repair lead times, severe cases	More than 200 days, some new-generation programmes	AVM Magazine (trade press)	2025
Third-party repair waits	Four to six months	Air Cargo Week and Locatory (trade press)	Mar 2026
Used material	Supply "remains thin"; "a seller's market for used parts"	AVM Magazine; Aviation Week	2025; undated
Engine stands	Legacy-engine stand utilisation up about a third in 2025; CFM56-5A/B up 60%	AviationPros stand report (trade press)	Undated

Constraint	Figure	Source and level	Date
Engine-stand shortage; blade, casting and forging lead times	Not found as numbers	D2 U6	n/a
Global MRO market	More than \$136 billion (2025); about \$140 billion (2026)	Oliver Wyman	Apr 2026
Engine MRO market	\$70 billion (2026) rising to \$102.6 billion (2036)	Oliver Wyman via AviationPros and Air Cargo Week	2026

Source: Bain & Company, "Get a step ahead of the engine maintenance capacity crunch", 2024; Aviation Week, "Narrowbody Engine Demand Drives MRO Capacity Additions", 2025; Oliver Wyman, "The new MRO supply paradigm", Apr 2026; GE Aerospace, Form 10-K FY2025; AVM Magazine, "The Engines Capacity Crunch" and "Inside the Engine MRO Supply Chain", 2025; Air Cargo Week / Locatory, Mar 2026; Aviation Week, "MRO Memo: A Seller's Market For Used Parts"; AviationPros, "Engine Stand Utilization Report". Compiled in D2 §4.7 and D9 §4.5, §6. Reconciliation R-020.

27.6 TURNAROUND, AND HOW IT DIFFERS FROM THE WAIT

Part I §6 boxed turnaround time (TAT) as the calendar time from induction to release and gave the headline figures. This subsection presents every turnaround figure the research reached, each with its source, because the figures come from different shops, workscopes and dates and the primer does not pick among them. [reconciliation R-019; E.2 "Turnaround"]

EXHIBIT 4.6 — TURNAROUND TIME FOR CFM56 WORK: EVERY FIGURE, WITH WORKSCOPE, SOURCE, LEVEL AND DATE

Workscope or scope	Turnaround	Source and level	Date
Full CFM56 overhaul, pre-pandemic	About 60 days	Aviation Business News / AVM (trade press)	2025
Full CFM56 overhaul, current	90 to 120 days	Same	2025
Light work	About 45 days	Same	2025
CFM56, GE Aerospace's shops	"Sustaining ~90 days"	GE Aerospace, Bernstein conference presentation (OEM)	27 May 2026
LEAP, CFM56 and GE90, GE's shops	Improved by more than 10% year on year in 4Q 2025	GE Aerospace 10-K FY2025 (OEM filing)	Feb 2026
Performance restoration	75 to 90 days	Safe Fly Aviation (consultancy blog)	2026
Heavy visit with LLPs	120 to 150 days	Safe Fly Aviation	2026

Workscope or scope	Turnaround	Source and level	Date
Legacy-engine turnaround, change over five years	Up 35%	Air Cargo Week (trade press)	2026
New-generation-engine turnaround, change over five years	Up more than 150%	Air Cargo Week	2026
Wait for an induction slot, additional to turnaround	Up by two to three months, up to six	Bain (consultancy)	2024
Module exchange from inventory	A quick-turn; FTAI's own exchange turnaround not retrieved	StandardAero menu (Part I §6); D5 U3	2022; n/a

Source: Aviation Business News / AVM, "Inside the engine MRO supply chain", 2025; GE Aerospace, Bernstein Strategic Decisions Conference presentation, 27 May 2026; GE Aerospace, Form 10-K FY2025; Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026" and "CFM56-7B engine availability report 2026" (consultancy blog); Air Cargo Week, "From PW1100G to CFM56: the engine maintenance trends shaping 2026"; Bain & Company, 2024; StandardAero CFM56-7B brochure, Oct 2022. Reconciliation R-019; D5 U3.

The trade press attributes the lengthening since the pandemic to "long turnaround times for material repairs ... primarily due to parts scarcity issues in the MRO supply chain". [Aviation Business News, 2025] Air Cargo Week's figures say legacy-engine turnaround is up 35% against five years earlier and new-generation turnaround up more than 150%, which places the CFM56's problem as the smaller of the two. [Air Cargo Week, 2026] GE's own figures run the other way for its shops: about 90 days sustained for the CFM56 in May 2026 and an improvement of more than 10% in the fourth quarter of 2025. [GE Aerospace, 27 May 2026; GE Aerospace, Form 10-K FY2025] The consultancy figures sit between: 75 to 90 days for a restoration and 120 to 150 days for a heavy visit with life-limited parts. [Safe Fly Aviation, 2026] The figures are not reconciled because they describe different shops and worksopes, and the reader should carry the range.

The economic weight of turnaround is the spare engine. Every day an engine is away is a day the aircraft needs another engine or does not fly, and Part I §6 worked the rent for one restoration at two spare-engine rates, giving a range from about \$126,000 to about \$700,000 depending on the rate, the turnaround and whether a slot wait is included. [Part I, worked example 1.5; IBA, Apr 2024; Safe Fly Aviation, 2026] A module exchange from inventory aims to turn a 90-to-120-day event into a quick-turn of about 45 days, which is the hook Part VII §46 hangs FTAI's product on. [D1 §2.6; D2 §2.1] FTAI's own exchange turnaround was not retrieved. [D5 U3]

WORKED EXAMPLE 4.2 — WHAT A YEAR OF LABOUR AND PARTS INFLATION DOES TO THE PRICE OF ONE SHOP VISIT

Inputs. A performance-restoration visit at the midpoint of the \$1.2–1.6 million range, \$1.4 million (sourced: Safe Fly Aviation, consultancy blog, "2026 industry data"; this Part's range under E.1 rule 4), with the alternative midpoint of \$2.15 million from the same publisher's \$1.8–2.5 million range (sourced: Safe Fly Aviation, other page; Part I's range). Cost shares of material 60–70% and labour 20–30% (sourced: Air Cargo Week, 2025/26), with the alternative labour share of 15–20% (sourced: Safe Fly Aviation). Labour-rate inflation of 5.5–6.0% (sourced: Oliver Wyman, Apr 2026). Parts escalation of about 10% a year, the rate reported for new spares in 2022–23 (sourced: Aviation Week, late 2023), with the alternative of about 7% a year, the compounded rate on full life-limited-part sets 2008–2025 (derived in Part I, worked example 1.3). The 2024, 2025 and 2026 catalogue steps are not known (D2 U1), so both parts rates are illustrative.

labour line: $20\text{--}30\% \times 5.5\text{--}6.0\% = 1.1 \text{ to } 1.8$ points of visit cost a year
 (at the 15–20% share: 0.8 to 1.2 points)
 material line at ~10%: $60\text{--}70\% \times 10\% = 6.0 \text{ to } 7.0$ points a year
 material line at ~7%: $60\text{--}70\% \times 7\% = 4.2 \text{ to } 4.9$ points a year
 total, parts at 10%: 7.1 to 8.8 points a year
 total, parts at 7%: 5.3 to 6.7 points a year
 on a \$1.4M visit: \$74,000 to \$123,000 more each year
 on a \$2.15M visit: \$114,000 to \$189,000 more each year

So on these inputs a visit's price rises by five to nine percent a year before the shop changes anything about how it works, and about four-fifths of that rise comes from the material line that is priced by the manufacturer, not by the shop. That is the arithmetic behind the statement, carried through this primer, that catalogue escalation reaches every shop visit whoever performs it (\$28.1, \$29.5). Limits of the example: the shares are secondary figures that disagree with each other (reconciliation R-014); the parts escalation for 2024–26 is unknown and the two rates shown are from different years and part scopes (R-018); the repairs line (10–20%) is left at zero inflation for want of a figure; and the visit price ranges are themselves contested (R-013). [Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; Oliver Wyman, Apr 2026; Aviation Week, late 2023; Part I worked example 1.3; D2 §5.2, §5.6]

28. The OEM aftermarket model: list prices, escalation, TrueChoice, EngineWise, long-term agreements

The manufacturer earns from a shop visit in two ways. It sells the new parts the visit consumes, at a published price it revises each year, to any shop that performs the work. And it sells services under its own brand: visits in its own shops, parts and repairs to other shops, and multi-year contracts under which it takes over the maintenance of an engine for a price per flight hour. The balance between those two is what distinguishes the CFM56's aftermarket from the LEAP's, and this section takes them in order.

OEM AFTERMARKET MODEL

The OEM aftermarket model is the way the engine's manufacturer earns from its installed base after delivery: spare parts sold at an escalating catalogue list price to every shop, plus branded services (shop visits, parts and repair packages, used material, repair management) and long-term agreements priced per flight hour. On the CFM56 about 85% of aftermarket revenue is spare parts and time-and-materials work, against about 40% for the LEAP, so the CFM56 is an open, transactional market in which the manufacturer's main capture is parts rather than contracts, and the LEAP is being set up as a largely contracted one. [Safran via Aviation Week, c. 2023/24; D2 §2.3, §3.4]

28.1 SPARE PARTS AT LIST PRICE, ESCALATING

Part I §4 boxed the catalogue list price as the manufacturer's published price for every new spare part, repriced at least annually, and escalation as the annual increase in it. The mechanics are simple and the consequence is large. The shop buys the part at list and passes it through at list plus a handling margin, or the airline buys the part directly under a materials agreement; either way the manufacturer receives the list price for every new part a visit consumes. [D2 §2.2] Used and alternative parts are priced as discounts to the same list, so they inherit its escalation (Part V §34, Part VI §44). [D3 §2.5]

EXHIBIT 4.7 — SPARE-PARTS ESCALATION STATEMENTS, BY DATE, SCOPE AND SOURCE

Date or period	Statement	Scope	Source and level
2008 to 2025	Full -7B life-limited-part set compounded at about 7% a year (\$1.775M in 2008; \$3.4M in 2018; \$5.7M in June 2025)	Life-limited parts only	Derived in Part I from Aircraft Commerce Issues 58 and 120 and MyAirTrade (trade press and market-data site)
Year not shown	One escalation step of about 12% on -5B/-7B life-limited parts; escalation set by "a formula which factors in labour costs, the price of raw materials and energy costs"	Life-limited parts	Ishka (trade publication), undated
1 Nov 2022	"Double-digit" increase	New spare parts generally	Aviation Week, late 2023 (trade press)
Aug 2023	"High-single-digit" increase by GE and Safran	New spare parts generally	Aviation Week, late 2023
2023	-7B parts up 20–30% after CFM's catalogue revision	-7B parts generally	Magnetic Group via Aviation Week, 2024

Date or period	Statement	Scope	Source and level
Trend to late 2023	New spare-parts prices rising "around 10% per year"	New spare parts generally	Aviation Week, late 2023
Stated rule	"Always been above inflation by 3 to 4 points ... we'll continue on that path, I would say, at least for some years"	Safran spare-parts pricing	Safran chief executive via Aviation Week, late 2023
Quarter reported in late 2023	GE external spares sales up 35% year on year; record \$42.2 million of spares shipped per day	GE spares	Aviation Week, late 2023
2024, 2025, 2026	Catalogue revision percentages not found	n/a	D2 U1; D1 U10 (cluster K-03)

Source: Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023; Aviation Week, "Magnetic expects CFM56 market challenges, opportunities in 2024"; Ishka, "Pratt & Whitney mulls further LLP escalation hike", undated; Part I, Exhibit 1.6 and worked example 1.3. Reconciliation R-018; E.2 "OEM price escalation".

Two series run through the exhibit and they are not the same thing. One is specific to life-limited parts: a set price that compounded at about 7% a year over seventeen years, a single step of about 12% reported without its year, and a 20% to 30% rise in -7B parts after the August 2023 revision. [Part I worked example 1.3; Ishka, undated; Magnetic Group via Aviation Week, 2024] The other covers all spares: about 10% a year to late 2023, a double-digit step on 1 November 2022, a high-single-digit step in August 2023, and a stated policy of inflation plus three to four points. [Aviation Week, late 2023] They cover different parts, years and bases, and the primer reports both without combining them. [reconciliation R-018]

The gap in the record is recent. No percentage was found for the 2024, 2025 or 2026 catalogue revisions, so the only dated steps the primer can quote are the two of 2022 and 2023, and the only forward statement is Safran's rule. [D2 U1; cluster K-03] Coverage of each year's CFM price revision in Aviation Week or Aircraft Commerce, or a question on price in a GE or Safran earnings call, would close the gap. [D2 U1] In the same late-2023 reporting, GE's external spares sales were up 35% year on year and its spares shipment rate reached a record \$42.2 million a day. [Aviation Week, late 2023]

WORKED EXAMPLE 4.3 — COMPOUNDING THE TWO ESCALATION READINGS ON THE MATERIAL LINE OF ONE VISIT

Inputs. The material line of a performance restoration at this Part's midpoint, \$1.4 million × 60–70% = \$840,000 to \$980,000, taken at its midpoint of \$910,000 (sourced shares: Air Cargo Week, 2025/26; visit price sourced: Safe Fly Aviation, consultancy blog; midpoint assumed). Three forward

escalation readings over three years: about 10% a year (sourced: Aviation Week, late 2023, the 2022–23 observed rate for spares); about 7% a year (derived in Part I on life-limited-part sets); and Safran's rule of inflation plus three to four points, with inflation assumed at 3% for illustration, giving 6% to 7%.

at 10% a year: $910,000 \times 1.10^3 = 910,000 \times 1.331 = \$1,211,000$ (+\$301,000)
 at 7% a year: $910,000 \times 1.07^3 = 910,000 \times 1.225 = \$1,115,000$ (+\$205,000)
 at 6% to 7% a year: $910,000 \times 1.191$ to $1.225 = \$1,084,000$ to $\$1,115,000$
 (+\$174,000 to +\$205,000)

So over three years the material line alone rises by about \$170,000 to \$300,000 on a visit that today costs \$1.4 million, depending on which reading is used, and the increase lands on the same line whether the shop is the manufacturer's, an independent's or an airline's. On the alternative visit price of \$2.15 million (Safe Fly's other range) every dollar figure above scales by about 1.54. Limits of the example: the actual 2024–26 steps are unknown (D2 U1); the 10% rate was observed on all spares in 2022–23 and the 7% rate on life-limited parts over seventeen years, so neither is a forecast; the inflation figure in Safran's rule is assumed; and the material share is a secondary figure (R-014). [Air Cargo Week, 2025/26; Safe Fly Aviation, 2026; Aviation Week, late 2023; Part I worked example 1.3; reconciliation R-013, R-014, R-018]

28.2 TIME-AND-MATERIALS AND LONG-TERM AGREEMENTS

The second distinction in the manufacturer's model is who pays for a visit and when.

TIME-AND-MATERIALS (T&M)

Under a time-and-materials arrangement the operator pays for each shop visit as it occurs: parts at the catalogue list price and labour by the hour, with the workscope agreed visit by visit. The operator carries the cost risk (a heavier workscope costs more) and the volume risk (more removals cost more), and it chooses the shop and, within the shop manual, the source of the parts. It is the pay-per-visit model and the opposite of a long-term agreement. [D2 §2.3, §8]

LONG-TERM SERVICE AGREEMENT (LTSA); ALSO LONG-TERM AGREEMENT (LTA), RATE-PER-FLIGHT-HOUR (RPFH) OR FLIGHT-HOUR AGREEMENT

A long-term service agreement is a multi-year contract under which the operator pays a fixed amount per engine flight hour and the manufacturer, or a shop, commits to perform or pay for the engine's shop visits. The provider takes the volume and cost risk and in return controls the workscope, the parts sourcing and the choice of shop. Revenue under such a contract is recognised against an estimate of the contract's total profitability, so a change in that estimate flows through current-period profit: GE's 2025 segment profit was reduced by "an unfavorable change in estimated profitability of long-term service agreements, primarily from the estimated impact from tariffs". About 15% of the CFM56 fleet is under such agreements against about 50% of the LEAP fleet, with the LEAP targeted at 60–70% by 2030. The dossiers use three names for one concept; this primer uses LTSA and gives LTA and RPFH as aliases once, here. Part III §20 glosses the term at first mention. [D2 §2.3; D9 §2.5; D11 §2.3; GE Aerospace, Form 10-K FY2025; registry C-12]

Safran's description of the CFM56 aftermarket is that 85% of its revenue is spare parts and time-and-materials work, against about 40% for the LEAP, and that the LEAP fleet will reach 60% to 70% coverage by long-term agreements by 2030. [Safran via Aviation Week, "CFM Aftermarket Shifting To Long-term Agreements", c. 2023/24] The fleet-share figures are complements of the same picture: about 15% of CFM56 engines under long-term agreements against about 50% of the roughly 5,500 LEAPs in service at the time of the report. [Aviation Week, "Leap Aftermarket Poised For Growth", undated; reconciliation B-4] Both figures are Aviation Week's reporting of Safran's view, so their agreement is weak as corroboration. [reconciliation B-4]

The distinction matters to every later Part of this primer for one reason. Under a time-and-materials visit the operator, or the shop it chooses, decides whether a part is new from the manufacturer, used, repaired or an approved alternative; under a long-term agreement the manufacturer decides, and it has no reason to choose a competitor's part. [D2 §2.3] So the used-material market of Part V, the alternative parts of Part VI and the module exchanges of Part VII all address the 85% of the CFM56 aftermarket that is transactional, and would address far less of a fleet that was contracted the way the LEAP is being contracted. The research found no source that quantifies how much CFM56 parts demand sits inside long-term agreements and so outside the open market. [D2 §3.4]

28.3 TRUECHOICE: THE FOUR OFFERINGS

Part I §9 boxed TrueChoice as GE Aerospace's branded menu of CFM56 and LEAP services. This subsection gives each of its four offerings in GE's own words and says what each covers, because one of them is the manufacturer's direct equivalent of the product FTAI sells.

EXHIBIT 4.8 — GE AEROSPACE'S TRUECHOICE MENU: THE FOUR OFFERINGS, WHAT EACH COVERS, AND THE REFERENCE CUSTOMERS THE RESEARCH REACHED

Offering	GE's description	What it covers, functionally	Reference customers reached	Where it sits in this primer
TrueChoice Flight Hour	"Customized offerings that help optimize cost of ownership over the entire engine lifecycle with flexible risk transfer and payment options"	A rate-per-flight-hour agreement: the manufacturer takes cost and volume risk and controls workscope and parts	Southwest Airlines, CFM56-7B fleet (agreement expanded)	The long-term agreement of §28.2
TrueChoice Overhaul	"Time and material overhauls with tailored worksopes specific to shop visit objectives, economic priorities and ownership horizon — whether for one engine or an entire fleet"	Pay-per-visit shop visits performed in GE's own network	Safair and TAAG (Angola's flag carrier), CFM56	A time-and-materials visit in an OEM shop
TrueChoice Material	"High-quality new and used OEM parts, advanced repairs and technology upgrades for airlines and MROs that enhance engine performance over the lifecycle and support higher engine residual value"	New and used manufacturer parts, repairs and upgrades sold to third-party shops and to airlines using them; where the manufacturer's used-material business sits	Ryanair, long-term materials agreement, about 2,000 CFM56 and LEAP engines (GE 10-K FY2025)	The parts slice of §29; the used-material competitor of Part V
TrueChoice Transitions	"A broad range of options for changing ownership horizons, including green time leases, exchanges and material buy-back, plus custom worksopes with shorter builds and maximum used material"	Green-time leases, module and engine exchanges, buy-back of material, and shorter, used-material-heavy builds for operators that will not keep the engine long	None named in the research	The direct analogue of module exchange (Part VII §46) and of the green-time trade (Part II §16)

Source: GE Aerospace press releases on Safair, TAAG and Southwest Airlines TrueChoice agreements (undated); GE Aerospace, Form 10-K FY2025 (Ryanair); MRO Global on the Safair agreement. Compiled in D2 §2.3. The quotation marks carry GE's wording, including its own punctuation.

Read down the third column and the menu maps onto the rest of this Part. Flight Hour is the long-term agreement. Overhaul is a time-and-materials visit that happens to be performed in the manufacturer's shop. Material is the parts slice sold to everyone else, and it is where GE's statement that "GE and CFM are the largest used serviceable material provider for their products" belongs. [GE Aerospace investor-relations page, 2026] Transitions is the one that matters most to Part VII: green-time leases, exchanges, material buy-back and "custom worksopes with shorter builds and maximum used material" are, in the manufacturer's own words, the products an operator buys when it does not intend to keep the engine, and they are the same products FTAI's Module Factory and the independents' end-of-life offerings sell. [GE Aerospace press releases; D2 §2.3] The research found no price, volume or customer for Transitions, so the analogy is structural and not quantified. [D2 §2.3, U7]

28.4 MATERIAL SOLUTIONS, AND THE MANUFACTURER AS A SELLER OF USED PARTS

MATERIAL SOLUTIONS

Material solutions is the generic trade phrase for a shop's or a manufacturer's package of used serviceable material, repaired parts and approved alternative parts assembled to reduce the new-parts bill of a shop visit. The whole commercial logic of a shop that is not the manufacturer sits on this line, since material is 60–70% of a visit. The research could not match the phrase to a current CFM or GE programme name; GE's used-material offering sits inside TrueChoice Material, and a historical Pratt & Whitney "Global Material Solutions" venture on the CFM56-3 was found only as a headline. This primer uses the generic sense. [D2 §2.2, §8; D4 U10; registry C-19]

The manufacturer competes on this line too. GE states that "GE and CFM are the largest used serviceable material provider for their products", and that it will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity". [GE Aerospace investor-relations page, 2026] The name, scale and CFM56 role of GE's used-material arm were not captured, and the research records the gap as a cluster shared with Parts V and VI. [D3 U15; D4 U10; cluster K-16] What the pricing of used material against new looks like, 60% to 80% of a new part's price on average or 70% to 75% of list when a trader sells to an airline, is Part V §34's subject, and the two figures are carried there as a pair. [Aviation Week, undated; Aircraft Commerce Issue 120; reconciliation R-022]

28.5 REPAIR MANAGEMENT AND OVERHAUL CONSULTING

REPAIR MANAGEMENT

Repair management is a service the manufacturer sells to a shop that is not its own: it manages the routing of the shop's piece-part repairs through the manufacturer's repair network and data, so the independent shop buys repair capability rather than building it. GE sold such a contract to Aero Norway, a CFM56 independent. A related product is overhaul consulting, under which the manufacturer advises an airline's own shop on its overhauls; EgyptAir signed a TrueChoice overhaul-consulting agreement with GE. The research holds the names of the two contracts and no terms. [GE Aerospace release, "Aero Norway awards repair management contract to GE Aviation", undated; Amwal Al Ghad, undated; D2 §2.3]

These two products show the manufacturer earning from visits it does not perform. A repair-management contract puts the manufacturer's repairs into an independent's visit; an overhaul-consulting agreement puts its workscope judgement into an airline's. Both leave the shop independent in name and bring part of the visit's repair slice (§29.2) back to the manufacturer. [D2 §2.3]

28.6 IAE ENGINEWISE AND FLEETCARE

Part I §10 introduced the V2500 and boxed EngineWise as Pratt & Whitney's aftermarket services brand, under which the IAE network performs V2500 shop visits. IAE is the consortium of Pratt & Whitney, part of RTX (formerly Raytheon Technologies), Pratt & Whitney Aero Engines International, Japanese Aero Engines Corporation and MTU Aero Engines. [D2 §2.3; Part I §10] RTX describes EngineWise as "a variety of aftermarket services to maximize engine performance and fleet availability". [RTX / Pratt & Whitney release, 6 Jun 2024]

The agreement that brings EngineWise into this primer is FTAI's. On 6 June 2024 IAE and FTAI announced a five-year EngineWise agreement covering more than 100 full performance-restoration shop visits on V2500 engines, which RTX called "one of IAE's largest engine maintenance agreements by number of engines with a non-airline customer"; FTAI then owned or was committed to more than 140 V2500 engines and aimed for 200 by 2025. [RTX / Pratt & Whitney release, 6 Jun 2024] Functionally this is the opposite of the Module Factory: a large engine owner buying manufacturer-network restorations at a negotiated price for a committed volume. Part VII §51 carries it. [D5 §2.5]

The V2500 market the agreement sits in has two counts for 2025 and the primer carries both. Aviation Week's V2500 data tool projected a spike to about 1,400 V2500 shop visits in 2025, with more than 7,800 overhaul visits and 3,142 life-limited-part-driven visits over the following ten years. [Aviation Week, "Data Tool: The Future Of IAE V2500 Engine", 2024/25] RTX's chief financial

officer, as reported by Aviation Week in 2026, put 2025 V2500 shop visits at "slightly more than 800", with 2026 expected within about 20 of that. [RTX CFO via Aviation Week, 2026; reconciliation R-009] One is a published forecast and the other a reported outturn, both via the same publication, and the research does not resolve them. IAE separately said it expects no V2500 retirement surge over the next five years. [IAE via Aviation Week, 2024/25]

"FleetCare" was not found as a current IAE or Pratt & Whitney product name; only EngineWise was. [D2 U9] The brief for this primer named it, and the primer records that the name did not surface rather than describing a product it did not find.

28.7 THE MANUFACTURERS' REPORTED NUMBERS

EXHIBIT 4.9 — THE OEM AFTERMARKET IN THE MANUFACTURERS' OWN FIGURES, 2023 TO 1Q 2026

Item	Figure	Source	Date
GE CES revenue, 2025	\$33,314M; equipment \$8,304M; services \$25,010M (75.1%)	GE Aerospace 10-K FY2025	Feb 2026
GE CES segment profit, 2025	\$8,861M; margin 26.6%; up \$1.8B (26%)	Same	Feb 2026
GE CES revenue growth, 2025	Up \$6.4B (24%): "increased spare parts volume, internal shop visit volume and worksopes, increased engine deliveries and pricing"	Same	Feb 2026
GE internal shop-visit revenue growth	+24% (2025); +19% (2024); +27% (2023)	Same	Feb 2026
GE CES services revenue growth, 2025	+26%	Same	Feb 2026
GE commercial engine deliveries	2,386 (2025) vs 1,911 (2024); LEAP 1,802 vs 1,407	Same	Feb 2026
GE services revenue, 1Q 2026	"Services revenue grew 39%"	GE Aerospace 1Q 2026 release	Apr 2026
GE external spares, late 2023	+35% year on year; \$42.2M a day shipped	Aviation Week	Late 2023
GE statement on LEAP	"Expected in the coming years to overtake the mature CFM56 as the industry's largest fleet", driving "a significant increase in shop visits and need for MRO capacity"	GE Aerospace 10-K FY2025	Feb 2026

Item	Figure	Source	Date
CFM56 share of GE CES services	Not disclosed	D2 U4; D11 U16	n/a
Safran group revenue, 2025	€31.3B (+14.7%); recurring operating income €5.2B (16.6%)	Safran release	13 Feb 2026
Safran civil spares (USD), 2025	+17.6%, "driven primarily by the CFM56"	Same	13 Feb 2026
Safran Propulsion services	€10,122M, 64.6% of Propulsion revenue (2024: €8,430M, 61.7%)	Safran FY2025 results presentation	13 Feb 2026
Safran 2026 guidance	Revenue growth low-to-mid teens; recurring operating income €6.1–6.2B	Safran release	13 Feb 2026
Safran civil aftermarket guidance, 2023	Raised twice to "low-30% range" growth	Aviation Week	Late 2023
CFM56 aftermarket mix	85% spares and time-and-materials (LEAP about 40%)	Safran via Aviation Week	c. 2023/24
Long-term agreement coverage	About 15% of CFM56 fleet; about 50% of LEAP fleet; LEAP target 60–70% by 2030	Aviation Week; Safran via Aviation Week	Undated; c. 2023/24

Source: GE Aerospace, Form 10-K FY2025; GE Aerospace, 1Q 2026 earnings release; Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023; Safran, "Safran reports excellent financial performance in 2025", 13 Feb 2026, and FY2025 Results & Investor Update presentation; Forecast International and Leeham News, 13 Feb 2026; Aviation Week, "CFM Aftermarket Shifting To Long-term Agreements" and "Leap Aftermarket Poised For Growth". Compiled in D2 §4.3 and D11 §3.8–3.9. Unknowns U-D2-04, U-D11-16.

The exhibit is the manufacturers' view of the market the shops of §25 work in. GE's services revenue of \$25 billion in 2025 is the size of the pool that FTAI's Aerospace Products segment, at \$875.0 million of revenue in the second quarter of 2026, draws from, and the two are not comparable beyond that sentence (Part XI §89). [GE Aerospace, Form 10-K FY2025; FTAI release, 29 Jul 2026; D11 §3.8] Three things the exhibit does not contain are worth naming. No CFM56 share of GE's services revenue is disclosed, although GE has historically described the CFM56 as its largest services contributor. [D2 U4] No CFM56 shop-visit count, spare-parts price percentage or used-material programme wording was returned from GE's filing. [D11 U16] And Safran's US-dollar spare-parts growth figure for the CFM56 alone was not returned. [D11 U17]

28.8 THE CFM–FTAI MATERIALS AGREEMENT OF JANUARY 2026

On 22 January 2026 FTAI announced a multi-year materials agreement with CFM International under which FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair", described by both companies as strengthening "the open maintenance, repair, and overhaul (MRO) ecosystem". [FTAI release, 22 Jan 2026] The agreement places FTAI, a buyer of used material and a partner in an alternative-parts venture, on the manufacturer's TrueChoice Material side of the ledger at the same time; on FTAI's fourth-quarter 2025 call its chief executive named it, with the approved turbine blade and more used material, as a lever for moving segment margins from the mid-30s toward 40%. [MarketBeat coverage of the FTAI call, 26 Feb 2026] Its duration, volumes, pricing and any effect on FTAI's use of alternative parts were not retrieved, and the release body itself was located and not read. [D4 U14; D5 U18; cluster K-19] Part VII §52 carries the agreement within the segment.

29. Where the money goes in a shop visit, and who captures it

Part I §7 built the cost of a shop visit from five lines: labour, new parts, repairs, life-limited parts and the test-cell run. This section asks who receives each line. The answer is what makes the manufacturer's model of §28 bite on every visit, and it is the frame for Parts V, VI and VII, each of which describes one way of moving money off the manufacturer's line.

29.1 THE SHARES

Two share breakdowns exist and they come from different secondary sources. Air Cargo Week puts material at 60% to 70% of a visit, labour at 20% to 30%, and repairs and other shop activities at 10% to 20%. [Air Cargo Week, "The true cost of engine maintenance", 2025/26] Safe Fly Aviation, a consultancy blog, gives a finer cut for the CFM56: life-limited parts are 40% to 60% of the parts cost when they are replaced, high-pressure turbine blades and vanes are 15% to 25%, and labour is 15% to 20%. [Safe Fly Aviation, 2026] Neither source names a shop-level data set, and the labour shares differ. [reconciliation R-014] One dossier reads the two as consistent, since life-limited parts and turbine airfoils together are the bulk of material and material is the bulk of the visit; the primer reports both and does not combine them. [D2 §2.2]

29.2 SLICE BY SLICE: WHO RECEIVES IT

The exhibit below is the research's own assembly of who captures each slice, from the sources named in each row; it is the input to the margin pool that Part I §11 previewed and Part XI §86 treats in full. [D2 §2.2; registry 1c]

REPAIR VENDORS; COMPONENT REPAIRS

Component repairs are the restoration of parts that can be repaired rather than scrapped, such as turbine airfoils, cases and seals, by welding, re-coating, machining or re-plating. A repair vendor is the specialist shop to which an engine shop subcontracts that work: either the manufacturer's own repair shops, working to the shop manual, or independent specialists holding repairs approved through a Designated Engineering Representative (DER), treated in Part VI §40. Repairs are 10–20% of a visit, and a repaired part costs a fraction of a new one. [D2 §2.2; Air Cargo Week, 2025/26]

EXHIBIT 4.10 — WHERE A SHOP VISIT'S MONEY GOES AND WHO CAPTURES EACH SLICE

Slice	Share of the visit	Who captures it	Mechanism	Where the primer treats it
New OEM parts other than LLPs: blades, vanes, seals, hardware	The largest single slice within material's 60–70%	CFM International, that is GE Aerospace and Safran, at the catalogue list price	Sold through CFM's spares organisation to any shop; the shop passes it through at list plus a handling margin, or the airline buys direct under a TrueChoice Material agreement	§28.1, §28.3
Life-limited parts	40–60% of parts cost when replaced	The manufacturer only, in practice: no approved alternative life-limited part exists for the CFM56	Sold at list; the most visible escalator (the "\$4M stack" headline)	Part I §4; §30
Used serviceable material	Substitutes for new parts	Teardown and used-parts traders (AerSale, GA Telesis, others; lessors' part-outs) and the manufacturer, which calls itself the largest used-material provider for its products	Priced as a discount to list: 60–80% of new on average (Aviation Week); 70–75% of list from a trader to an airline (Aircraft Commerce); above list when the manufacturer cannot deliver	Part V §34
Approved alternative (PMA) parts	Alternative to new OEM parts	PMA makers: HEICO (a maker of approved alternative parts); Chromalloy, with FTAI as its partner	Priced at a discount to list that no retrieved source discloses	Part VI

Slice	Share of the visit	Who captures it	Mechanism	Where the primer treats it
Component repairs	10–20%	Repair vendors: the manufacturer's repair shops (its repair-management offering) and independents holding DER repairs	The shop subcontracts airfoil, case and seal repairs; a repaired part costs a fraction of a new one	§28.5; Part VI §40
Labour and test	20–30% (Air Cargo Week) or 15–20% (Safe Fly)	The engine shop, whoever owns it	Shop rate × hours; labour-rate inflation 5.5–6.0% in 2025 (Oliver Wyman)	§27.3
Shop margin and risk	Embedded in the lines above	The shop under time-and-materials; the manufacturer under a flight-hour agreement	Under T&M the operator bears cost and volume risk; under an LTSA the manufacturer bears it and prices it in	§28.2

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26; Safe Fly Aviation, 2026 (consultancy blog); GE Aerospace investor-relations page, 2026; Aviation Week, "MRO Memo: A Seller's Market For Used Parts"; Aircraft Commerce Issue 120; Oliver Wyman, Apr 2026; D4 on the absence of PMA life-limited parts. Assembled in D2 §2.2. Reconciliation R-014, R-022.

Three of the rows need a name the reader has not yet met in this Part.

CHROMALLOY

Chromalloy is a Florida-based maker of turbine-engine part repairs, coatings and approved alternative parts, with revenue described as "\$1 billion-plus" and about 4,000 employees, owned by Sequa Corporation (a Veritas Capital affiliate since December 2022). In October 2025 it received approval from the FAA (the US Federal Aviation Administration) for a CFM56-5B/7B high-pressure turbine blade, which it calls "the only aftermarket alternative to the OEM manufactured new part". Part VI §41 carries the company. [Chromalloy release, 30 Oct 2025; D4 §3]

FTAI-CHROMALLOY JOINT VENTURE

FTAI's annual report says the company "develops and manufactures PMA parts through a joint venture"; the partner, Chromalloy, is named in Chromalloy's own releases and in trade coverage, and investor commentary describes the arrangement as an exclusive partnership formed in 2018 covering several CFM56 hot-section parts. Whether the "Advanced Engine Repair JV" in FTAI's equity-method note (25%, carrying value \$22.4 million at 31 December 2025) is this venture is unknown. Part VI §43 carries what is filed and what is reported. [FTAI 10-K FY2025; Chromalloy release, 30 Oct 2025; orchestrator notes; reconciliation R-026]

HEICO FLIGHT SUPPORT GROUP

The Flight Support Group is the segment of HEICO (a Florida-based maker of approved alternative parts) Corporation that designs and sells approved alternative parts and repairs for commercial aircraft and engines; it reported fiscal 2025 net sales of \$3,117.3 million and operating income of \$750.4 million (24.1%), and HEICO adds several hundred new part approvals a year, read as 300–500 by one dossier and 400–550 by another from the same filing. Part VI §41 carries it. [HEICO, Form 10-K FY2025; reconciliation R-024]

Two rows in the exhibit are the whole point of the manufacturer's model. The life-limited-part row has one supplier, because the research found approved alternative turbine blades and vanes but no alternative disk or spool for the CFM56, so every new life-limited part is a single-source purchase at the escalating list price (Part I §4, Part VI §38). [D4 U7; D1 §5] The dossier that assembled the exhibit calls the full set of life-limited parts the "LLP stack"; this primer uses Part I's term, the full LLP set, and gives the alias once here. [registry, Part IV rows] The used-material row has the manufacturer on both sides: it sells the new part at list and, by its own statement, is the largest seller of the used part priced as a discount to that list. [GE Aerospace investor-relations page, 2026]

29.3 THE TURBINE-BLADE LINE: A TENSION AT THE LEVEL OF ONE PART

The high-pressure turbine's blades and vanes are the most expensive non-life-limited parts in a visit, and the research holds two readings of what they cost that do not agree. [reconciliation R-015]

On one side is the share. Blades and vanes are 15% to 25% of a visit's parts cost, which on this Part's restoration range of \$1.2–1.6 million is roughly \$0.18–0.40 million and on Part I's range of \$1.8–2.5 million roughly \$0.27–0.63 million. [Safe Fly Aviation, 2026; arithmetic in the research, D1 §2.7 and D2 §2.2] On the other side is the price per blade. An expert interview puts a new manufacturer's CFM56 high-pressure turbine blade at about \$20,000; a web listing of weak provenance, whose condition is unclear, shows about €35,000. [In Practise, undated; web listing via D4] At a commonly cited but unverified 80 blades per rotor, a set is \$1.6 million at the interview price or about \$3.08 million at the listing price, and the same interview says a new set of blades and vanes "can cost up to a few million dollars". [D4 §2.8; In Practise, undated; D4 U16]

The two readings cannot both describe the same visit. A blade set alone at \$1.6 million is more than the whole restoration on this Part's range. The research did not retrieve the manufacturer's list price for a full blade-and-vane set or confirm the blade count, so the primer carries both sides and stops. [reconciliation R-015; D4 U16] The interview adds a cadence that bears on which visits the line applies to: at a first shop visit, around 20,000 cycles, airlines typically repair the turbine blades; at a second, around 30,000, hot-section parts typically cannot be repaired again and new parts are needed, which is where an approved alternative blade competes with a new manufacturer's blade. [In Practise, undated, via D4 §2.8] Part VI §44 works the parts bill with and without the alternative part on both readings.

29.4 WORKED EXAMPLE: ONE RESTORATION, SPLIT BY SLICE AND BY PARTY

WORKED EXAMPLE 4.4 — A CFM56-7B PERFORMANCE RESTORATION SPLIT BY SLICE AND BY WHO RECEIVES EACH SLICE, ON BOTH COST SETS

Inputs. Two visit prices: \$1.4 million, the midpoint of this Part's \$1.2–1.6 million range (sourced: Safe Fly Aviation, consultancy blog, "2026 industry data"), and \$2.15 million, the midpoint of Part I's \$1.8–2.5 million range (sourced: Safe Fly Aviation, other page). Shares of material 60–70%, labour 20–30% and repairs 10–20% (sourced: Air Cargo Week, 2025/26); turbine blades and vanes at 15–25% of parts cost and labour at 15–20% (sourced: Safe Fly Aviation). No life-limited parts are replaced in this visit, so the 40–60% LLP share does not apply. The three Air Cargo Week ranges do not sum to 100% at their ends, so the lines below are each range applied to the total and are not additive.

on \$1.4M: material 840,000–980,000; labour 280,000–420,000 (Safe Fly share: 210,000–280,000); repairs 140,000–280,000
 turbine blades and vanes within material: $15\text{--}25\% \times 840,000\text{--}980,000 = 126,000\text{--}245,000$
 on \$2.15M: material 1,290,000–1,505,000; labour 430,000–645,000 (Safe Fly share: 323,000–430,000); repairs 215,000–430,000
 turbine blades and vanes within material: 194,000–376,000

Who receives each line. The material line goes to the manufacturer at list for every part bought new, so on an all-new-parts visit the manufacturer receives \$840,000 to \$980,000 of a \$1.4 million visit, or \$1.29 to \$1.51 million of a \$2.15 million one, whichever shop performs it. The labour line stays with the shop. The repairs line goes to repair vendors, which may be the manufacturer's repair shops or independents holding their own approved repairs. Now suppose, for illustration only, that the shop sources 20% of the material line's list value as used parts bought at 70% of list (assumed share; discount sourced as the lower end of Aircraft Commerce's 70–75% trader price):

material at midpoint of the \$1.4M visit: 910,000
 list value sourced used: $20\% \times 910,000 = 182,000$; paid to the trader at
 70%: 127,400
 manufacturer receives: $910,000 - 182,000 = 728,000$; shop saves: $182,000 - 127,400 = 54,600$ (3.9% of the visit)

So each fifth of the material line moved from new to used at a 30% discount saves about 4% of the visit's price and moves about \$180,000 of list value from the manufacturer to a trader; an approved alternative part or a repair in place of a replacement moves money the same way, at discounts no retrieved source discloses. Limits of the example: the two visit prices are a contested pair (R-013); the shares are secondary and disagree with each other (R-014); the turbine-blade line is contested at the level of one part (R-015); the 20% used-material share is assumed; and the shop's handling margin on parts passed through at list, which is not zero, is not separated out because no figure was found. [Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; Aircraft Commerce Issue 120; D2 §2.2; reconciliation R-013, R-014, R-015, R-022]

29.5 WHAT FOLLOWS FROM THE SPLIT

Two consequences follow from the exhibit and the example, and the research states both. First, because 60% to 70% of a visit is material and most material is priced by the manufacturer, the manufacturer's annual catalogue escalation flows almost directly into the cost of every shop visit regardless of who performs it; worked example 4.2 put the figure at roughly four to seven points of visit cost a year on the two escalation readings. [D2 §2.2; Air Cargo Week, 2025/26] Second, the only levers a shop other than the manufacturer has to cut the bill are on the material line: used parts instead of new, approved alternative parts instead of the manufacturer's, repair instead of replacement, and module exchange instead of overhaul. [D2 §2.2] Parts V, VI and VII take those four levers in turn, and §30 sets the fourth against the other three in one example.

WHY THE MANUFACTURER'S CAPTURE ON THE CFM56 IS PARTS, NOT CONTRACTS

Put §28.2 and Exhibit 4.10 together. About 85% of the CFM56's aftermarket revenue is spare parts and time-and-materials work, so most visits are chosen and paid for visit by visit by the operator or its shop. Within each of those visits, 60% to 70% of the bill is material, and most material is bought new from the manufacturer at list. So the manufacturer receives the largest slice of nearly every CFM56 visit without performing it, and without a contract that obliges the operator to come back. On the LEAP the structure is reversed: about half the fleet is already under long-term agreements and the target is 60% to 70%, so the manufacturer's capture there moves from the parts line to the contract, under which it also chooses the parts. The CFM56's open structure is what leaves room for the material-line substitutes of Parts V to VII; the LEAP's contracted structure is what Part III §24's rows on LEAP durability and the GTF groundings say does not create demand for those substitutes. [Safran via Aviation Week, c. 2023/24; Aviation Week, undated; Air Cargo Week, 2025/26; D2 §2.3, §3.4; D9 §5]

30. Worked example: four options for an engine at its LLP limit

Everything in this Part meets in one decision. An engine on a mid-life narrowbody is approaching the limit of a life-limited part in its core. The owner can send it to a shop for a full visit, swap the module that has run out, lease a part-worn engine to fly in its place, or sell it and buy another. Each option draws on a different part of the market described above: the first on the shops and slots of §25 and §27 and the parts prices of §28, the second on the module exchange that Part VII sells and GE packages as TrueChoice Transitions, the third and fourth on the green-time and used-engine trade of Part II §16 and Part V. The example below prices all four on the research's figures, with every input labelled sourced or assumed and every contested figure shown with its alternative. It is arithmetic, not advice.

Two aliases from the research need a word before the arithmetic. The dossier behind this section calls the full set of life-limited parts the "LLP stack" and calls the mismatch of remaining lives across those parts "LLP stagger"; Part I §4 treats the same two things as the full LLP set and as stub life, and this section uses Part I's terms with the aliases given once here. [D2 §2.5, §8; registry, Part IV first-use rows] Part I §4 also boxed cycles since new (CSN), which the exhibit below uses, as the cycles an engine or part has flown since manufacture.

WORKED EXAMPLE 4.5 — FOUR OPTIONS FOR A CFM56-7B WHOSE HIGH-PRESSURE TURBINE DISK IS 1,200 CYCLES FROM ITS LIMIT

Setup. An airline operates a 737-800 with two CFM56-7Bs. On one engine the high-pressure turbine disk, a life-limited part with a certified limit of 20,000 cycles (sourced: records mini-packs, Part I Exhibit 1.5) and a new price of \$420,000 (sourced: Safe Fly Aviation, consultancy blog, 2026), has 1,200 cycles remaining (assumed). The engine's temperature margin would otherwise allow about 3,000 more cycles (assumed). The aircraft flies about 1,500 cycles a year (sourced as the upper end of Aircraft Value News's 1,000–1,500 cycles a year for a mature narrowbody). The airline plans to retire the aircraft in 24 months (assumed), so it needs about 3,000 cycles from this engine position. Because the limiting part is in the turbine, any action on it forces the core open (Part I §4).

The per-cycle frame. A full set of -7B life-limited parts is about \$4 million at list (sourced: Aircraft Value News headline, Nov 2019; this Part's figure under E.1 rule 5) or \$5.7 million (sourced: MyAirTrade engine-status table, June 2025, scope not visible; Part I's figure). Over a 20,000-cycle life:

at \$4.0M: $4,000,000 \div 20,000 = \200 per cycle; at 1,500 cycles a year,
\$300,000 per engine per year
at \$5.7M: $5,700,000 \div 20,000 = \285 per cycle; \$427,500 per engine per
year

That cost accrues whether or not the engine ever goes to a shop, which is why lessors collect reserves against it (Part II §13) and why an engine's value is so sensitive to its remaining life (Part I §8).

Option A: a full shop visit with life-limited-part replacement. The restoration workscope is \$1.2–1.6 million, midpoint \$1.4 million (sourced: Safe Fly Aviation, "2026 industry data"; this Part's range under E.1 rule 4); the alternative range from the same publisher is \$1.8–2.5 million, midpoint \$2.15 million (Part I's range). A full new set of life-limited parts is \$4.0 million, or \$5.7 million on the alternative.

$$\begin{aligned} \text{outlay, full set: } & 1,400,000 + 4,000,000 = \$5,400,000 \\ \text{alternatives: } & 1,400,000 + 5,700,000 = \$7,100,000; 2,150,000 + \\ & 4,000,000 = \$6,150,000; 2,150,000 + 5,700,000 = \$7,850,000 \end{aligned}$$

Variant A' replaces only the turbine's life-limited parts, "easily \$0.5m" at 2017 list (sourced: Leeham News, 3 Mar 2017; a 2017 figure, so understated at 2026 prices): \$1.4 million + \$0.5 million = \$1.9 million (or \$2.65 million on the alternative restoration). The other life-limited parts then keep their own, lower, remaining lives, so the next life-limit-forced removal comes sooner; this is the stub-life problem of Part I §4, which the dossier calls LLP stagger. Downtime is the slot wait of two to six months (sourced: Bain, 2024) plus the turnaround, 75–90 days (sourced: Safe Fly) or 90–120 days (sourced: trade press, 2025) or about 90 days (sourced: GE, May 2026). Covering the position with a leased spare for three months of turnaround alone costs \$126,000–144,000 at \$42,000–48,000 a month (sourced: Safe Fly, 2026, a rate that sits beside green-time values and may describe a green-time engine) or about \$300,000 at about \$100,000 a month (sourced: IBA, Apr 2024); the dossier's figure of about \$135,000 is the first of these. Adding a slot wait of two to six months takes the cover to \$210,000–432,000 at the lower rate or \$500,000–900,000 at the higher. The value effect: a low-life -7B trades as teardown feedstock at \$0.8–1.2 million and a green-time -7B at \$2.8–3.4 million (both sourced: Safe Fly, 2026, consultancy blog); IBA's half-life market values of \$5.7 million for a -7B24 and \$6.4 million for a -7B27 (sourced: IBA, Sept 2025) are a reference for an engine with life, on a different basis. What a freshly restored engine with a full new set is worth was not found (D2 U8), so the dossier takes the green-time midpoint as a floor:

$$\begin{aligned} \text{value uplift, floor: } & 3,100,000 - 1,000,000 = \$2,100,000 \\ \text{net economic cost over the horizon, engine sold at exit: } & 5,400,000 + \\ & 135,000 - 2,100,000 \approx \$3,400,000 \\ \text{on the alternative restoration ($2.15M): } & \approx \$4,200,000; \text{ on the} \\ \text{alternative set ($5.7M): } & \approx \$5,100,000; \text{ on both: } \approx \$5,900,000 \end{aligned}$$

The true figure is lower if the fresh engine fetches more than a green-time price, and the airline has bought about 20,000 cycles of life to use 3,000 of them, recovering the other 17,000 only through the resale price.

Option B: a module exchange. The airline swaps the turbine module, or the whole core, for a serviceable one from a provider's inventory carrying, say, 10,000 cycles of remaining life (assumed); the fan and low-pressure turbine stay. The engine is opened only to the core split, so turnaround is

shorter and labour is lower than in Option A. The price structure is an exchange fee plus the value of the life transferred, adjusted for the condition of the other hardware (Part I §3; Part VII §47). The life transferred, if priced at the stack-average new-list rate: $10,000 \times \$200 = \2.0 million (or $10,000 \times \$285 = \2.85 million on the alternative set). For the turbine's own life-limited parts alone, \$0.5 million over 20,000 cycles is \$25 a cycle, so 10,000 cycles is about \$250,000. Safe Fly quotes "removal costs of \$150,000–\$400,000 per module" for life-limited-part modules, whose meaning is unclear (D2 U7). The exchange fee itself, what FTAI, GE under TrueChoice Transitions or MTU charge above the life transferred, was not found in any source, and the research records it as a cluster shared with Parts V and VII (D2 U7; cluster K-01). The arithmetic therefore brackets Option B:

turbine-set exchange: [fee, unknown] + ~250,000 $\approx$ \$500,000 to \$1,000,000
 (the dossier's bracket)
 full-core exchange: higher, up to [fee] + ~2,000,000 if a whole stack's
 life is priced at new list

Downtime for a quick-turn of about 45 days (sourced: trade press, 2025) costs about \$63,000–72,000 of cover at the lower rent or about \$150,000 at the higher. The only FTAI figures the research holds are bounds, not prices: products revenue per module produced of \$2.11 million in 2025 and \$2.34 million in the second quarter of 2026, with the limits Part VII §56 lists (E.1 rule 11). Part VII §55 carries FTAI's own exchange-against-visit example.

Option C: a green-time engine lease. The airline removes the engine and installs a leased green-time -7B, typically carrying 4,000–8,000 cycles (sourced: Safe Fly), enough for the 3,000 it needs. Rent for 24 months:

at \$42,000–48,000 a month: $24 \times 45,000 = \$1,080,000$ (range 1,008,000–1,152,000)
 at about \$100,000 a month: $24 \times 100,000 = \$2,400,000$ (IBA's rate may describe a full-life spare, not a green-time engine)
 sell the removed engine as feedstock: +\$800,000 to +\$1,200,000
 net at the lower rent: $-1,080,000 + 1,000,000 \approx -\$80,000$ (range -\$352,000 to +\$192,000)
 net at the higher rent: $\approx -\$1,200,000$ to $-\$1,600,000$

Both nets are before any per-hour or per-cycle reserves the lessor charges and before redelivery conditions; no sourced reserve rate was found, and the only figures in the research are one dossier's illustrative assumptions (D6 U7). IBA's April 2024 commentary that "it's a lessor's market" and that engine lease rates and values were escalating is the context for both rents.

Option D: sell the engine. The airline sells the life-limited engine at its teardown value of \$0.8–1.2 million and buys a green-time unit at \$2.8–3.4 million (both sourced: Safe Fly): a net outlay of \$1.6–2.6 million, after which it owns an engine with 4,000–8,000 cycles, resaleable after 3,000 at a lower price the research did not find. Or it sells the whole aircraft as it stands and lets the buyer price the deficit: Safe Fly says two otherwise identical -7Bs at 10,000 cycles since new "differ by \$1.6 million

based solely on LLP remaining life", and that 60–80% of a used engine's value is its remaining life-limited-part life (consultancy blog).

Limits of the example. Every cost figure is from a consultancy blog or the trade press and no shop's price was read (R-013); the two restoration ranges and the two set prices are never combined within one line, and each line names its alternative (R-013, R-016, R-017); the two spare-engine rents may price different engines and are carried as a pair (R-028); the value figures sit on three bases (R-029); the exchange fee is a bracket with no source (U7); the reserve rates in Option C are unknown; and the value of a freshly restored engine is unknown (U8). The 2017 figure for the turbine's life-limited parts is understated at 2026 prices by an escalation whose steps for 2024–26 are themselves unknown (U1). [D2 §2.5; Safe Fly Aviation, 2026; Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Leeham News, 3 Mar 2017; Bain, 2024; Aviation Business News, 2025; GE Aerospace, 27 May 2026; IBA, Apr 2024 and Sept 2025; Aircraft Value News, undated (cycles a year); reconciliation R-013, R-016, R-017, R-028, R-029; D2 U7, U8; D6 U7]

EXHIBIT 4.11 — THE FOUR OPTIONS SUMMARISED, ON BOTH SHOP-VISIT COST SETS, WITH THE SOURCES AND THE UNKNOWN IN EACH LINE

Option	Cash out over 24 months (this Part's cost set)	Cash out (Part I's cost set)	Cash or asset back	Indicative net	Downtime	Open items
A. Full visit, full new set	\$5.4M (restoration \$1.4M + set \$4.0M) + \$0.14M cover at the lower rent	\$6.15M (restoration \$2.15M + set \$4.0M); \$7.85M with the \$5.7M set	Engine worth at least the green-time \$3.1M midpoint; full-stack value unknown	About – \$3.4M, or better; –\$4.2M to –\$5.9M on the alternatives	Slot wait 2–6 months plus TAT 75–120 days	Fresh-engine value (U8); which rent applies (R-028)
A'. Restoration plus turbine LLPs only	\$1.9M + \$0.14M cover	\$2.65M + cover	Engine worth more than today, less than A	Not computable: next removal brought forward (stub life)	Same as A	2017 price for the turbine set; stub-life cost not priced

Option	Cash out over 24 months (this Part's cost set)	Cash out (Part I's cost set)	Cash or asset back	Indicative net	Downtime	Open items
B. Turbine or core module exchange	Fee (unknown) + about \$0.25M of life for the turbine set; up to fee + \$2.0M for a whole stack's life at new list	Same, since the restoration ranges do not enter	Engine with the life it needs	About –\$0.5M to –\$1.0M for a turbine-set exchange (dossier bracket)	Quick-turn, about 45 days; FTAI's own TAT not retrieved	Exchange fee (U7, cluster K-01)
C. Green-time lease, 24 months	\$1.0–1.15M rent at \$42–48k a month; \$2.4M at about \$100k	Same	+\$0.8–1.2M from selling own engine	About –\$0.1M (–\$0.35M to +\$0.19M) at the lower rent; about –\$1.2M to –\$1.6M at the higher; before reserves and redelivery	Engine swap only	Reserve rates (D6 U7); which rent (R-028)
D. Sell, buy a green-time engine	\$2.8–3.4M purchase	Same	+\$0.8–1.2M now; resale after 3,000 cycles later	–\$1.6M to –\$2.6M, less an unfound resale value	Engine swap only	Resale value after 3,000 cycles

Source: D2 §2.5, re-worked on both cost sets per reconciliation E.1 rules 4, 5 and 6; Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026", "CFM56 Engine Market Report 2026", "CFM56-7B Engine Availability Report 2026" and LLP pages (consultancy blog); Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Leeham News, 3 Mar 2017; Bain & Company, 2024; IBA via AviTrader and AJOT (a trade publication), 25 Apr 2024; IBA Engine Values 2025B, Sept 2025. CSN is cycles since new (Part I §4).

The arithmetic is not a recommendation, and the point is structural. Options B, C and D exist only because the engine is modular and because a secondary market in modules, green time and used material exists; without those, Option A would be the only option, and its price would be the manufacturer's price. [D2 §2.5] The three rows of Part III §24 that this Part answers meet in the table. Low retirements shrink the supply behind B, C and D, since fewer teardowns mean fewer modules, fewer green-time engines and fewer used parts, at exactly the moment the slot shortage

of §27 makes A slow and the escalation of §28 makes A expensive.[D2 §2.5, §5.4] Which option an operator takes depends on inputs this Part has shown to be contested or unknown: the exchange fee, the lessor's reserve rates, the spare-engine rent, and the value of a restored engine against a part-worn one. Part V prices the used parts and the teardown, Part VI the alternative parts, and Part VII the exchange as FTAI sells it; Part II §16 carries the green-time trade.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-006	CFM56 shop visits a year: 2,300–2,400 in 2026–2028 (GE via Aviation Week, 2026) against about 2,500 peak in 2025–26 (Safran via Aviation Week, c. 2023/24); IBA "+40%" 2024 → 2025 headline (scope not captured) against Safran's "mid-single-digit" 2025 growth	§26.2, Exhibit 4.3
R-007	Peak shape and timing: 2,500 peak in 2025–26 (Safran) against a 2,300–2,400 plateau through 2028 with apex 2027–28 (GE); first-visit peak 2027 (Visual Approach)	§26.2, callout
D2 T6	Safran "topping 2,000 again in 2024" (including LEAP) against IBA's 40% increase 2024 → 2025; if one population, 2025 would be about 2,800	§26.2
R-004	Retirement rate: GE 3–4% → 2–3% → 1.5–2% (CFO via Aviation Week, 2026); 2025 about 1.5%, 2026 about 2% (via Leeham, 22 Jan 2026); 1Q 2026 "below 1%" (GE investor-relations page)	§26.3, Exhibit 4.4, worked example 4.1
R-005	Aircraft retirements 2024: "fewer than 200" (IBA, Sept 2024) against 587 (Cirium via Informa); scope not visible	§26.3, Exhibit 4.4
R-001	Installed base: about 24,000 (Aviation Business News, Sept 2025) against about 23,000 (CFM website, undated); both used in worked example 4.1	§26.3, worked example 4.1
R-008	Implied interval of about ten years (2,300 visits on 24,000 engines) against typical CFM56 intervals (Part I)	§26.3, worked example 4.1
R-012	Fleet maturity: 57% no visit and 21% one (GE, Jun 2019); about 70% zero or one (Safran, 2024); about 75% (Safe Fly, consultancy blog, 2026)	§26.4, Exhibit 4.4
R-037	FTAI capacity: "up to 300 shop visits per year" (10-K, 2022 language); 1,350 module overhauls (release, 9 Sep 2024); 1,800 modules / 600 engines (release, 26 Feb 2025); 3,000 modules (call coverage, Jul 2026)	§25.4, Exhibit 4.2
R-036	2026 module target: 1,050 (release, 25 Feb 2026) against 1,200 (call coverage, 30 Jul 2026)	§25.4
R-041	Lisbon: "planned 113,000 sq ft, 300+ modules" (call coverage, unverified) against "100% owned" and "still ramping" (10-K; call)	§25.4, Exhibit 4.2

Id	The two (or more) figures	Where in this Part
R-042	FTAI share: about 5% of a \$22B market (release, 26 Feb 2025); 25% goal of CFM56 shop visits, denominator not retrieved (call, Jul 2026); about 17% / 43% of a 2,350-visit year at 1,200 / 3,000 modules (derived)	\$25.4
R-019	Turnaround: 90–120 days full overhaul and about 45 light (trade press, 2025); "sustaining ~90 days" (GE, 27 May 2026); more than 10% better in 4Q 2025 (GE 10-K); 75–90 for a performance restoration (PR) and 120–150 heavy (Safe Fly); slot wait +2–3 months, up to 6 (Bain, 2024)	\$27.6, Exhibit 4.6
R-020 (D2 T8)	Parts constraint: priority-supplier input +40% and TAT +10% (GE 10-K FY2025) against labour and material shortages as "primary disruptors", two-thirds reporting next-generation shop costs 21%+ over expectation (Oliver Wyman, Apr 2026) and thin used-material supply (AVM, 2025)	\$27.4, Exhibit 4.5
R-018	Escalation: LLP set about 7% a year 2008–25, Ishka about 12% step (year not shown), -7B parts +20–30% (2023) against spares about 10% a year, double-digit 1 Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Safran)	\$28.1, Exhibit 4.7, worked example 4.3
R-009	V2500 shop visits 2025: "slightly more than 800" (RTX CFO via Aviation Week, 2026) against about 1,400 forecast (Aviation Week data tool, 2024/25)	\$28.6
R-014	Cost shares: material 60–70% / labour 20–30% / repairs 10–20% (Air Cargo Week) against LLPs 40–60% of parts / high-pressure turbine (HPT) blades and vanes 15–25% / labour 15–20% (Safe Fly)	\$29.1, Exhibit 4.10, worked example 4.4
R-015	HPT blade set: \$1.6M or \$3.08M (per-blade price × 80, unverified) against \$0.18–0.63M implied by the 15–25% share on the two visit ranges	\$29.3
R-022	Used-material price: 60–80% of new on average (Aviation Week) against 70–75% of list from a trader (Aircraft Commerce)	\$28.4, Exhibit 4.10, worked example 4.4
R-013	Shop-visit cost: PR \$1.2–1.6M (this Part) and \$1.8–2.5M (Part I); heavy with LLPs \$2.1–2.8M and ">\$3.5M"; full overhaul \$3.5–4.2M; light \$650–900k (all Safe Fly, consultancy blog)	\$27 worked example 4.2, \$29.4, §30, Exhibit 4.11
D2 T4	Full LLP set "north of \$3m" (Leeham, 2017) and "\$4M" (Aircraft Value News) against a "heavy visit with LLPs" of \$2.1–2.8M all-in (Safe Fly), which cannot include a full set at either price	\$30
R-016	Full -7B LLP set: \$4M (Aircraft Value News, Nov 2019; this Part's figure) against \$5.7M (MyAirTrade, June 2025; Part I's figure)	\$30, worked example 4.5

Id	The two (or more) figures	Where in this Part
R-017	LLP cost per cycle: \$200 (on \$4M) against \$285 (on \$5.7M)	§30, worked example 4.5
R-028	CFM56-7B spare rent: about \$100,000 a month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time)	§27.6, §30, worked example 4.5, Exhibit 4.11
R-029	CFM56-7B value: half-life market value \$5.7M / \$6.4M (IBA, Sept 2025) against green-time \$2.8–3.4M and teardown \$0.8–1.2M (Safe Fly, 2026); value of a freshly restored engine unknown	§30, worked example 4.5
R-024	HEICO PMA approvals a year: 300–500 (one reading of the FY2025 10-K) against 400–550 (another)	§29.2, definition box
R-026	FTAI–Chromalloy: "a joint venture" (10-K) against "an exclusive perpetual partnership" (investor commentary); Advanced Engine Repair JV identity unknown	§29.2, definition box

Unknowns carried in this Part

- **U-D2-01 (D2) — U1. Catalog escalation 2024, 2025, 2026.** Only the Nov 2022 (double-digit) and Aug 2023 (high-single-digit) increases and Safran's "inflation + 3–4 points" rule were found. Resolve with Aviation Week/Aircraft Commerce coverage of each year's CFM price revision, or GE/Safran earnings call Q&A on "price".
- **U-D2-02 (D2) — U2. Shop-visit counts 2020–2023.** Only the 2019 level (~2,000+) and the 2024 forecast ("topping 2,000") were found. Resolve from Safran capital-markets-day decks (the safran-group.com media downloads 397180 / 448184 returned in search but not read) or the GE Aerospace Bernstein presentation of 27 May 2026.
- **U-D2-03 (D2) — U3. OEM versus independent share of CFM56 shop visits.** No estimate found. Resolve with Aviation Week/Cirium shop-visit-by-provider data or the Oliver Wyman fleet forecast's provider split.
- **U-D2-04 (D2) — U4. CFM56 share of GE CES services revenue.** Not disclosed in the 10-K. GE has historically described the CFM56 as the largest services contributor; a number would need an investor presentation or call transcript (the 22 Jan 2026 webcast transcript URL was returned, not read).
- **U-D2-05 (D2) — U5. Capacity of the named airline MROs.** AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir M&E, GA Telesis, Lufthansa Technik and ST Engineering: no CFM56 inductions- per-year or expansion figures returned. Resolve via each company's annual report or trade-press shop profiles.

- **U-D2-06 (D2) — U6. Engine-stand shortages and HPT blade / casting / forging lead times.** No numbers found. Resolve via Aviation Week supply-chain coverage, PCC/Howmet commentary, or GE call Q&A.
- **U-D2-07 (D2) — U7. Module-exchange pricing.** No source gives the exchange fee or a worked price for a CFM56 HPT or core exchange; Safe Fly's "\$150,000–\$400,000 removal cost per module" is ambiguous. D5 may find it in FTAI's investor materials or the In Practise interview ("FTAI Aviation: CFM56 repair and module swap", URL returned, not read). Cluster K-01 with U-D3-20 ("U20 — Pricing of a module exchange (exchange fee, LLP-life true-up) from any provider, including FTAI.") and U-D5-03 ("U3 — Exchange turnaround time and exchange pricing.").
- **U-D2-08 (D2) — U8. Value of a freshly overhauled full-stack CFM56-7B** versus a green-time unit. Needed to close the worked example. Resolve via IBA/Ascend/mba engine value tables. Cluster K-02.
- **U-D2-09 (D2) — U9. IAE "FleetCare".** Not found as a current product; only EngineWise. Resolve on the P&W services site.
- **U-D2-10 (D2) — U10. Dates.** Several Aviation Week articles were returned without publication dates (noted "c. 2023/24" or "2025"). The writer should confirm before quoting.
- **U-D1-10 (D1) — U10** Years and amounts of each CFM LLP escalation step 2019–2025 (only "Aug 2023, +20–30% on -7B parts" and "~12% LLP escalation" without a year were reached). Cluster K-03 with U-D2-01.
- **U-D1-07 (D1) — U7** Labour hours per workscope and shop labour rates (US\$/hour) for CFM56 visits. Resolve with an MRO price catalogue or Aircraft Commerce's maintenance budgets.
- **U-D1-14 (D1) — U14** Whether a module exchange requires a test-cell run under the CFM56 ESM.
- **U-D3-15 (D3) — U15** — GE Aerospace's used-material arm: name, scale and CFM56 role not captured (the only hit was a TrueEngine presentation URL whose content was not read). Cluster K-16 with U-D4-10.
- **U-D4-10 (D4) — U10 — Any CFM/GE statement specifically on the Chromalloy HPT blade PMA,** and any GE "material" programme (e.g. TrueChoice Material) positioning against PMA. The brief's phrase "material solutions" was not matched to a current CFM/GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines.
- **U-D4-14 (D4) — U14 — Terms of the FTAI–CFM Jan 2026 materials agreement** (duration, volumes, pricing, whether it restricts FTAI's PMA use). Cluster K-19 with U-D5-18 ("U18 — Terms of the CFM materials agreement mentioned on the Q4 2025 call.").

- **U-D4-16 (D4) — U16 — CFM56-7B HPT blade count per set** (80 is a recollection, unverified here) and the OEM list price for a full HPT blade and vane set.
- **U-D5-05 (D5) — U5 — Denominator of the 25% share goal** and the year by which it is targeted.
- **U-D5-08 (D5) — U8 — GMF AeroAsia and EgyptAir terms** (capacity committed, economics, duration).
- **U-D6-07 (D6) — U7 — Security deposit and maintenance deposit balances.** FTAI's balance-sheet "security deposits" and "maintenance deposits" liabilities by year, and the reserve rates in its leases. Resolve: FY2025 10-K balance sheet and revenue-recognition note. (Carried here for the green-time reserve rates Option C needs.)
- **U-D9-08 (D9) — U8 — Labour figures (engine MRO technician headcount, shortfall, wage inflation).** Oliver Wyman cites skilled-worker retirements qualitatively. D2 may hold these.
- **U-D11-16 (D11) — U16 — GE CFM56 shop-visit count, spare-parts price increases, and any "material solutions" or used-serviceable-material programme wording.** Not returned from the 10-K. Resolve: read the CES segment discussion in the 10-K and the Q4 2025 and Q2 2026 earnings materials.

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PART V

Used serviceable material and teardown

This Part explains where used engine parts come from, what has to be true before one may be installed again, what a retired aircraft or engine is worth when it is taken apart, why the supply of such parts depends on retirements that did not happen, how a used part is priced against a new one, who the traders are, and why the module is the unit in which this material increasingly trades. A reader who finishes it will be able to read a teardown quote, a used-part listing and a used-material market estimate with every term understood.

TRANSLATION-TABLE ROWS ANSWERED

This Part answers the rows of the Part III translation table that run through the supply of used serviceable material (USM), meaning parts made by the original equipment manufacturer (OEM), taken from retired aircraft and engines, and released for reinstallation. (Part III §24, row T-1: low retirements): fewer retirements mean fewer aircraft and engines torn down, so the supply of used parts and of part-worn modules is smaller than the shop-visit demand of Part IV implies; §33 carries the feedstock arithmetic. (Part III §24, row T-6: lease extensions): an aircraft whose lease is extended is not returned, not parted out and not a source of material; §32 and §33. Two rows name the new-generation engines, the geared turbofan (GTF) from Pratt & Whitney and LEAP (CFM's new-generation engine), and one names the 737 MAX (Boeing's successor to the 737NG). (Part III §24, row T-2: GTF groundings) and (Part III §24, row T-3: LEAP durability): the old fleet kept flying consumes used parts at shop visits while releasing none; §33. (Part III §24, row T-4: 737 MAX production cap) and (Part III §24, row T-5: Airbus output): without a replacement aircraft the retirement decision of §33 is not taken, whatever the part-out value; §32 and §33. (Part III §24, row T-7: OEM parts escalation): used parts are priced as a share of the catalogue list price of the original equipment manufacturer (OEM), so every list increase lifts the used price too; §34. (Part III §24, row T-11: freighter conversions): a conversion candidate needs engines with life, so conversion competes with teardown for the same airframes and engines; §32.

31. What makes a used part serviceable

A used part is not a lesser kind of part. It is the manufacturer's own part, taken from an engine or aircraft that has been in service, and put through a process that ends with a signed release. The process is what this section describes. At the end of it the part is "serviceable", a word with a legal meaning, and it can be bolted into another engine. Without that process the same piece of metal is scrap.

USED SERVICEABLE MATERIAL (USM)

Used serviceable material is any aircraft or engine part that has been in service, has been removed, and has been inspected (and repaired if needed) to the manufacturer's manual and released on an airworthiness tag with its full history, so that it is legally fit to be installed again. It is OEM-made, so it sits outside the debate over alternative-manufacturer parts (Part VI), but it competes for the same line in a shop visit's parts bill. A part with a gap in its paperwork is scrap, whatever its physical condition. Its supply is a lagged function of retirements (§33). [AJW Group (AJW), an engine-parts company, via AviTrader, Aug and Oct 2025; D3 §2.1, the USM dossier behind this Part (D3)]

The alternatives to a used part, when a shop needs to replace one, are three. A new OEM part is bought from the manufacturer at its published catalogue list price (CLP), the price Part I §4 glossed and Part IV §28 owns. An alternative-manufacturer part is one made by a third party under a Parts Manufacturer Approval (PMA) from the US Federal Aviation Administration (FAA); Part VI owns it. A repair is the third alternative, either one the manufacturer's manual describes or one approved separately by an FAA Designated Engineering Representative (DER), which Part VI §40 owns. This Part is about the first substitute only, the used part. [D3 §2.1; D4 §2.1, the alternative-parts dossier (D4)]

31.1 HOW A PART LEAVES SERVICE, AND THE THREE PLACES IT CAN GO

A part leaves service in one of three ways. It is removed at a shop visit, when the engine is open and inspection finds it out of limits or finds a better part to install. It is removed from an aircraft at retirement. Or the whole aircraft or engine is torn down, which the trade calls part-out. Part I §8 boxed part-out and teardown: part-out is the owner's decision to dismantle and sell the components; teardown is the physical disassembly. §32 carries the economics. [D3 §2.1; Part I §8]

From there each part goes one of three ways, and the three outcomes are the vocabulary of every teardown report.

SERVICEABLE AS REMOVED

A removed part is serviceable as removed when it has been inspected to the applicable manual and found within limits. It is tagged and sold without repair. It is the cheapest outcome for the trader, because there is no repair bill, and the fastest, because there is no repair-shop queue. [D3 §2.1]

The second outcome is the repairable part. It is outside the manual's limits but can be brought back inside them. For an airframe component the limits and the permitted repairs are in the Component Maintenance Manual (CMM), the manufacturer's document for that component, which Part I §9 boxed. For an engine part they are in the Engine Shop Manual (ESM), also boxed in Part I §9. A repair the manual does not cover may be done under a separately approved DER scheme (Part VI §40). The part goes to a repair shop, is repaired, tested, tagged and sold. The repair cost is the main cost of goods for a trader in used material. [D3 §2.1]

BEYOND ECONOMIC REPAIR (BER)

A removed part is beyond economic repair when it is outside limits and not worth repairing: the repair would cost more than the part would fetch, or no approved repair exists. It is scrapped. [D3 §2.1]

LIFE-EXPIRED

A life-limited part is life-expired when it has reached the cycle limit the manufacturer certified for it. It has no used-material value, however good its condition, and it must be mutilated so that it cannot re-enter the supply chain. Part I §4 carries the limits: 20,000 cycles for the -7B's compressor and turbine disks, 25,000 for the low-pressure turbine, 30,000 for the fan and booster. [D3 §2.1; Part I §4]

EXHIBIT 5.1 — THE THREE OUTCOMES FOR A REMOVED PART, AND WHAT EACH MEANS FOR THE TRADER

Outcome	Test	What happens next	Cost to the trader	Time to sale
Serviceable as removed	Inspected to the CMM or ESM; within limits	Tagged by an approved organisation; sold	Inspection and tagging only	Immediate once tagged
Repairable	Outside limits; a repair exists in the CMM or ESM, or under an approved DER scheme	Sent to a repair shop; repaired; tested; tagged; sold	The repair bill, the main cost of goods in used material	After repair-shop turnaround
Beyond economic repair, or life-expired	Outside limits and not worth repairing; or a life-limited part at its cycle limit	Scrapped; a life-expired life-limited part is mutilated	Disposal	None

Source: AJW Group / AviTrader, "Ensuring the airworthiness of USM", Aug 2025 (PDF) and AviTrader, 16 Oct 2025; D3 §2.1. "DER" is a Designated Engineering Representative (Part VI §40).

The exhibit explains a fact that recurs throughout this Part. Most used material is not serviceable as removed. It passes through a repair shop and is re-certified before it can be sold, so the capacity and price of repair shops (Part IV §27) set both the cost and the pace of the used-parts trade. Rising repair costs are named alongside falling supply as a driver of CFM56 (the engine of Part I) and V2500 (the alternative A320ceo engine of Part I §10) part prices (§34). [Aviation Week, "Magnetic expects CFM56 market challenges", 2024; D3 §5]

31.2 THE RELEASE TAG

A part becomes serviceable in the legal sense when an approved organisation signs it off. Part I §9 boxed the two forms and the release tag itself, so this is a reminder. In the FAA system the form is FAA Form 8130-3, the Authorized Release Certificate, also called the Airworthiness Approval Tag. For a repaired or inspected used part it is issued by a repair station certificated under Part 145 of the US regulations; for a new part it is issued by the holder of a production approval. In the European system the equivalent is Form 1 of the European Union Aviation Safety Agency (EASA), issued by an EASA Part-145 maintenance organisation or a Part-21

production organisation. Both certify that the part complies with the applicable regulations and is safe for use. [Rotabull, 8130-3 instructions, undated; AJW Group via AviTrader, 2025]

Many repair stations hold both approvals and issue a dual release, one tag valid in both systems. That matters commercially. A part tagged for one system only cannot be installed on an aircraft registered in the other without re-certification, so cross-border trade in used parts runs through dual-release shops. [D3 §2.1, §5]

The tag is necessary and not sufficient. Two further things decide whether a buyer will accept the part: its documented history (§31.3) and the identity of the organisation that signed the tag. The trade press is blunt about the second: the 8130-3 and Form 1 documents "can be easily generated", so buyers vet the supplier as well as the paperwork. Delta's published supplier matrix for aftermarket parts is one example of an airline codifying which supplier categories and which tag types it accepts. [AJW Group via AviTrader, 2025; Delta TechOps supplier matrix, Mar 2024]

31.3 TRACEABILITY: WHY A PAPERWORK GAP SCRAPS A PART

No regulator allows "untraceable" used parts. Everything must be documented, certified and compliant. [AJW Group via AviTrader, 2025] The documentation package a buyer requires differs by part class, and the class that matters most for engines is the one with the strictest rule.

HARD-TIME (HT) PART

A hard-time part is one with a fixed overhaul or replacement interval, as distinct from an on-condition part that is replaced when inspection finds it worn. It needs complete operating-history records and evidence of compliance with every applicable Airworthiness Directive and Service Bulletin. [AJW Group via AviTrader, 2025; D3 §2.1]

Part I §3 boxed the two instructions just named. An Airworthiness Directive (AD) is a regulator's order making a fix or inspection mandatory; a Service Bulletin (SB) is the manufacturer's instruction to modify or inspect. Compliance records for both must travel with hard-time parts and with life-limited parts, and an AD inspection can itself be the clock that ends an engine's remaining usable time (Part II §16). [Part I §3; D3 §2.1]

For life-limited parts the standard is the one Part I §9 boxed as back-to-birth (BTB) traceability: an unbroken record of every cycle the part has accumulated since new, on which engine serial numbers and under which operators. The arithmetic is why the rule is absolute. Remaining life is the manufacturer's limit minus the documented cycles consumed. An undocumented period cannot be assumed to contain zero cycles. So a gap means the part cannot be shown to have any life, and a part that cannot be shown to have life has no value. [D3 §2.1; Part I §9]

WHY A PAPERWORK GAP SCRAPS A PART

Consider a high-pressure turbine disk with a 20,000-cycle limit whose records show 11,000 cycles on three engines, then an eighteen-month period with no record, then 2,000 more cycles. The physical disk may have flown 13,000 cycles and have 7,000 left. But nobody can prove the eighteen months contained no flying. A buyer's engineer cannot write a remaining life into the receiving engine's records, so the part cannot be installed; the shop manual's remaining-life calculation has no input. The part is physically present and commercially absent, exactly as Part I §3 said of a module with a gap in its records. The usable supply of used life-limited parts is the documented supply, not the physical one. [D3 §2.1; Part I §3, §9]

Other documents are standard in a used-part package whatever the class: the removal tag from the last operator, the teardown report, a certificate of conformance from the seller, and a declaration that the part has never been through an accident.

NON-INCIDENT STATEMENT

A non-incident statement is the seller's declaration that a part was not installed on an aircraft involved in an accident, a fire or an immersion. It is a standard element of a used-material documentation package, beside the removal tag, the teardown report and the certificate of conformance. [D3 §2.1]

What makes a part unsaleable follows from the above. Missing documents mean the part will not be used. An unclear maintenance history or gaps in the paperwork are grounds for rejection. Mismatched part or serial numbers, unusual formatting on a tag, or an unfamiliar issuing organisation all trigger quarantine. [AJW Group via AviTrader, 2025]

QUARANTINE

Quarantine is the segregation of a part whose documentation is suspect, pending investigation. The triggers named in the trade press are mismatched part or serial numbers, unusual formatting on a release tag, and an issuing organisation the buyer does not recognise. [AJW Group via AviTrader, 2025; D3 §2.1]

EXHIBIT 5.2 — THE USED-PART DOCUMENTATION PACKAGE, BY PART CLASS: WHAT IS REQUIRED, WHO ISSUES IT, AND WHAT A GAP DOES

Part class	Documents required	Who issues them	Effect of a gap
Any used part	Release tag (FAA 8130-3 or EASA Form 1, dual release for both markets); removal tag from the last operator; teardown report; certificate of conformance; non-incident statement	Approved repair station or production approval holder; last operator; teardown company; seller	Rejection; quarantine if the tag's numbers, format or issuer look wrong
Hard-time part	All of the above plus complete operating-history records and AD and SB compliance records	Every operator and shop in the part's history	Rejection: the next interval cannot be set
Life-limited part	All of the above plus back-to-birth traceability: every cycle since new, by engine serial number and operator	Every operator and shop in the part's history	The part has no provable remaining life and no value; at its limit it is mutilated
Module (Part I §3; §36 below)	The above for every life-limited part inside it, plus the module's own serial number, build standard and repair records with their approval basis	Shops and operators in the module's history	The module cannot be installed; its parts may still be sold individually if their own records are complete

Source: AJW Group / AviTrader, Aug and Oct 2025; Rotabull, 8130-3 instructions, undated; Delta TechOps supplier matrix F-840-018-A, Mar 2024; D3 §2.1; Part I Exhibit 1.12. "AD" and "SB" are Airworthiness Directive and Service Bulletin.

31.4 LIFE-LIMITED-PART RECORDS IN PRACTICE

Everything in §31.3 converges on the life-limited part, because it is the part whose used value is highest and whose paperwork rule is strictest. A used disk with life is the direct substitute for a new one at list price, and the next shop visit must replace any disk without enough life for the following run (Part I §4.2). So the trade advertises used life-limited parts on their documented remaining cycles, as Part I §4 showed with listings such as a core package advertised with 6,235 cycles remaining (CR). The records summary that supports such a listing is the mini-pack Part I §4.1 boxed: each life-limited part with its certified limit and its cycles consumed. [Part I §4; D3 §2.2]

Two consequences for a teardown follow. First, the documented fraction of an engine's life-limited parts is a hard cap on what the teardown can yield; §32 returns to this. Second, when a module changes hands as a unit, its documentation must show the history of every life-limited part inside it, which is why §36 can treat the module as an engine in miniature. [D3 §2.7, §5]

32. Teardown and part-out economics

Part-out is a decision; teardown is the work that follows. This section explains how a retired aircraft or engine is bought, where its value sits, how long it takes to turn into cash, and what the one set of published engine figures says. It ends with the full worked example the outline asks for: a 737-800 with two CFM56-7B engines, with every input labelled sourced or unknown.

32.1 HOW A RETIRED ASSET IS BOUGHT: PURCHASE AND CONSIGNMENT

Part I §3 boxed feedstock as the trade's word for the aircraft and engines bought, or taken on consignment, to be torn down for parts. Those are the two commercial routes, and they allocate capital and risk differently. [D3 §2.2; Part I §3]

Under purchase the teardown company buys the asset outright and owns the parts. It bears the capital cost and the inventory risk and keeps the margin. Independent traders such as AerSale and GA Telesis (GA), and the trading arm of AAR (AAR Corp., a parts distributor and maintenance company), do this. Lessors such as Willis Lease and AerCap feed their own retired lease assets into their own material subsidiaries, which is purchase by another name. [D3 §2.2]

CONSIGNMENT

Consignment is teardown and sale of parts on the owner's behalf for a fee or a commission, with title to the parts retained by the owner, who may be an airline, a lessor, a bank or an insurer. It is the alternative to buying feedstock outright and bearing the inventory risk. Willis Aero describes its business as "the acquisition or consignment of aircraft and engines"; the APOC Aviation (APOC) teardown of a 737-800 with Willis Lease offered "packages of parts available for sale and consignment". [Willis Lease Finance, 10-K FY2025; Asian Aviation, 2025; D3 §2.2]

32.2 WHERE THE VALUE SITS

The engines dominate the part-out value of a mature narrowbody. The engine figures in §32.4 put two run-out engines' used-material yield at \$3.2–4.4M; the single largest airframe item, the landing gear, fetches "over \$1 million" as a fully overhauled set, according to the magazine of ISTAT (the International Society of Transport Aircraft Trading). [Safe Fly Aviation, consultancy blog, 2026; ISTAT Jetrader, Autumn 2020] A sourced percentage split between engines and airframe was not captured by the research and is recorded as unknown U-D3-01. [D3, unknown U-D3-01]

Within the airframe the value sits in the rotables.

ROTABLE

A rotatable is a part that is repaired and reused rather than consumed: landing gear, the auxiliary power unit, thrust reversers, avionics boxes, flight-control actuators, wheels and brakes. Rotables are the airframe items a teardown sells; consumables and expendables are not worth the paperwork. [D3 §2.2]

AUXILIARY POWER UNIT (APU)

The auxiliary power unit is the small gas turbine in the aircraft's tail that supplies electrical power and compressed air on the ground and in some flight phases. It is one of the highest-value airframe rotatables in a part-out and one of the components for which lessors collect maintenance reserves (Part II §13). Its value in a 737-800 part-out was not captured by the research (U-D3-10). [D3 §2.2, unknown U-D3-10; Part II §13]

Within an engine the value sits in five places. First, life-limited parts with remaining cycles: fan, booster and compressor disks, shafts and spools, because the next shop visit must replace any that lack life for the following interval and a used disk with life substitutes directly for a new one at list price. Second, the high-pressure turbine (HPT) blades and nozzles and the high-pressure compressor (HPC) airfoils, the most expensive repairable parts; Part I §3 and §7 set out why. Third, fan blades. Fourth, cases and structural parts. Fifth, accessories, of which the fuel-metering unit is the example the trade names.

HYDRO-MECHANICAL UNIT (HMU)

The hydro-mechanical unit is the engine's fuel-metering unit, the accessory that sets fuel flow under the engine control's command. It is a high-value engine accessory and, with CFM56 fan disks and structural engine parts, one of the most searched-for teardown items in one 2026 consultancy report. [Safe Fly Aviation, teardown report, 2026 (consultancy blog); D3 §2.2]

32.3 TIME TO MONETISE, AND TEARDOWN AS THE END OF A LEASE

A 737-800 teardown takes about three months. During that time every part is assessed by technical teams before being sent to specialist repair shops for overhaul, testing and re-certification. Landing gear are among the first serviceable parts to become available. The auxiliary power unit is sold, and parts are offered as packages for sale and consignment. [Asian Aviation and AirlinerGS on the APOC–Willis 737-800 teardown, 2025]

Sale proceeds therefore arrive in layers, and the order matters for a working-capital business. Whole modules and life-limited parts with life can be sold to an engine shop or an exchange pool almost at once, because those buyers are ready-made (§36). Repaired rotatables arrive after repair-shop turnaround. The long tail of slow-moving airframe parts sells over years. A sourced figure for the total sell-through period, the months to recover most of the proceeds, was not captured

and is unknown U-D3-02; AerSale's inventory-turnover disclosure or a trade-press case study would answer it. [D3 §2.2, unknown U-D3-02]

The sequence by which an engine reaches a teardown is visible in one lessor's transactions. APOC leased a CFM56-5A to Condor for twelve months on a green-time lease, with the engine "expected to return to APOC's portfolio for teardown and part-out next year after the lease concludes". Green time, boxed in Part I §8 and owned by Part II §16, is the usable life left before the next mandatory shop visit; a green-time lease rents that life to an operator who will not restore the engine. [AviTrader, 28 Jul 2025; Aviation Business News, 2025; Part II §16] The same lessor acquired its first CFM56-5B for "up-cycling", its own term for buying an engine to tear down for material, in October 2021; has disassembled two CFM56-7Bs; bought four "young" 737 airframes from a large US legacy carrier for teardown; and in May 2026 bought a 15-year-old A320-200 from FTAI for disassembly at Tarmac Aerosave's Toulouse-Francal facility. [AviTrader, 7 Oct 2021; Aviation Business News, undated; ePlaneAI, 2026] That last item is a direct data point on FTAI as a seller of feedstock as well as a buyer (§35).

32.4 THE ENGINE ARITHMETIC THE TRADE PRESS PUBLISHES

One published source gives figures for a CFM56 teardown, and it is a consultancy blog, so it is labelled as such here and wherever it appears in this primer. [reconciliation E.1 rule 3] Safe Fly Aviation's 2026 report says an exhausted engine bought for \$0.8–1.2M can yield \$1.6–2.2M of used material, a net \$0.4–0.8M per engine. [Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog)]

RUN-OUT (EXHAUSTED) ENGINE

A run-out or exhausted engine is one with no usable time left before a shop visit: a life-limited part is at its limit, or the temperature margin is gone, or a hard inspection is due. It is the only kind of engine that becomes teardown feedstock, because an engine with life is worth several times its teardown yield as a whole (§32.5). Part II §16 carries what happens when green time runs out. [D3 §2.2–2.3; Part II §16]

WORKED EXAMPLE 5.1 — THE CONSULTANCY-BLOG ENGINE ARITHMETIC, AND WHAT IT LEAVES OUT

Inputs: purchase price of a run-out CFM56, \$0.8–1.2M; used-material yield, \$1.6–2.2M; stated net margin, \$0.4–0.8M (all sourced: Safe Fly Aviation, consultancy blog, 2026). The implied cost of teardown, repair, certification and selling is not stated by the source; the line below is arithmetic on the source's endpoints and is labelled derived.

low case: yield – purchase = $1.6 - 0.8 = \$0.8\text{M}$; stated net = $\$0.4\text{M}$;
 implied cost = $0.8 - 0.4 = \$0.4\text{M}$
 high case: yield – purchase = $2.2 - 1.2 = \$1.0\text{M}$; stated net = $\$0.8\text{M}$;
 implied cost = $1.0 - 0.8 = \$0.2\text{M}$
 implied cost as a share of yield: $0.4 \div 1.6 = 25\%$; $0.2 \div 2.2 = 9\%$

So in this source's framing, somewhere between a tenth and a quarter of gross used-material proceeds is consumed by the repair, test, tagging and sales effort, if the endpoints pair as written. Limits of the example: the source does not say its low and high cases pair that way; it does not give teardown labour or facility cost (unknown U-D3-09); and the yield is the sum of §32.2's five value pools without a split. Against the appraiser's half-life market values of \$5.7M for a -7B24 and \$6.4M for a -7B27, the whole teardown yield is a quarter to a third of a mid-life engine's worth, which is the point §32.5 makes. [Safe Fly Aviation, 2026 (consultancy blog); IBA (an appraiser), Engine Values 2025B, Sept 2025; D3 §2.2]

32.5 WHY ONLY RUN-OUT ENGINES BECOME FEEDSTOCK, AND THE OWNER'S CHOICE

The teardown yield has to be read against what the same engine is worth whole, and here the research holds a conflict it does not resolve. The appraiser IBA gives half-life market values for the second half of 2025 of \$5.7M for a CFM56-7B24 and \$6.4M for a -7B27, describes the -7B market as having "a distinct lack of availability" with values and lease rates above long-term trend, and in its first-half 2026 update says -5B and -7B values "have levelled off after approximately 18 months of increases". [IBA, Engine Values 2025B, Sept 2025; IBA, H1 2026 update] The consultancy blog gives a -7B "green-time value (new condition)" of \$2.8–3.4M, a label the research notes is internally inconsistent since a green-time engine is by definition not in new condition, and a teardown value of \$0.8–1.2M; a trade-press item has CFM56 values up "as much as 50% over the past two years". [Safe Fly Aviation, 2026 (consultancy blog); ePlaneAI, date not captured; reconciliation R-029] The figures sit on different bases, half-life against green-time against run-out, and they disagree on direction. Part I Exhibit 1.11 carries them side by side and this Part does the same.

WHY ONLY RUN-OUT ENGINES BECOME FEEDSTOCK

An engine near half-life, with a fresh restoration and life-limited parts around half their limits, is worth \$5.7–6.4M on the appraiser's figures. Torn down, the same engine would yield \$1.6–2.2M on the consultancy figures. No rational owner tears down a \$6M engine for \$2M of parts, so engines with life are sold or leased whole, and only engines at or near run-out reach the teardown market. The corollary runs the other way too: when half-life engines are scarce and expensive, as the appraiser describes for 2024–2026, owners of part-worn engines lease them for their remaining months rather than tear them down, and the feedstock pool shrinks further (§33). [IBA, Sept 2025 and H1 2026; Safe Fly Aviation, 2026 (consultancy blog); D3 §2.8 step 2, §5]

That choice, between renting out a part-worn engine for the months it has left and tearing it down now, has no name in the sources. This primer gives it one, because §36 needs it.

HALF-LIFE TRADE

The half-life trade is this primer's term, not the sources', for the trade in engines, modules and life-limited parts that are part-way through their lives and priced on the life they have left: a part-worn engine leased for its remaining months or torn down now; a module valued on the remaining cycles of the life-limited parts inside it; a used disk priced at its life fraction times list price less a discount (§34). The term collects what the dossier describes in three places without naming it. [D3 §2.3, §2.4–2.5, §2.7; registry section 1c]

WORKED EXAMPLE 5.2 — THE OWNER'S CHOICE AT THE END OF A LEASE: RENT THE GREEN TIME OR TEAR DOWN NOW

Inputs: a CFM56-7B with twelve months of green time (assumed term, matching the APOC–Condor twelve-month precedent on a -5A, sourced: AviTrader, 28 Jul 2025). Monthly rent \$45,000, the midpoint of the consultancy blog's \$42,000–48,000 green-time range (sourced: Safe Fly Aviation, 2026; midpoint is the dossier's illustrative choice). The alternative rent is the appraiser's about \$100,000 a month for a -7B spare engine in 2024, which may price a different kind of engine (sourced: IBA, Apr 2024; reconciliation R-028). Teardown yield \$1.6–2.2M (sourced: Safe Fly Aviation, 2026).

lease then tear down, at \$45,000: $12 \times 45,000 = \$540,000$ rent, then
 $\$1.6\text{--}2.2\text{M}$ of material = $\$2.14\text{--}2.74\text{M}$ over about a year plus sell-through
 lease then tear down, at \$100,000: $12 \times 100,000 = \$1,200,000$ rent, then
 $\$1.6\text{--}2.2\text{M} = \$2.8\text{--}3.4\text{M}$
 tear down now: $\$1.6\text{--}2.2\text{M}$, sooner, with no rent

The life-limited parts with twelve months of cycles left are worth slightly more in a teardown today than after another year of flying, since their remaining life is what a buyer pays for; the airfoils and cases are worth the same either way. Which path is chosen depends on the owner's cost of capital, the rent on offer, and whether an engine shop will pay more for the life now than a lessee will pay to burn it. One dossier divided the \$45,000 rent by the \$6.4M half-life value to get a lease rate factor of 0.70% a month, 8.4% a year, and noted that the calculation mixes a green-time rent with a half-life value; Part II §14 carries the lease rate factor and this primer states none as a single number. Limits of the example: the term is assumed; the two rents may describe different engines; and the yield is a consultancy range. [D3 §2.3; Safe Fly Aviation, 2026 (consultancy blog); IBA, Apr 2024; reconciliation R-028, R-030; E.1 rule 6]

32.6 THE FULL WORKED EXAMPLE: A 737-800 WITH TWO CFM56-7B ENGINES

The dossier assembled every input it could source for this example and left the rest open. The exhibit lists them, and the example that follows uses only the sourced ones, with each unsourced input labelled unknown rather than guessed. [D3 §2.8]

EXHIBIT 5.3 — INPUTS FOR A 737-800 PART-OUT: SOURCED FIGURES AND UNKNOWN

Input	Figure	Source, level and date	Status
Airframe purchase price, mid-life or older 737-800 bought for part-out	Not captured	None	Unknown U-D3-08
Engine purchase price, run-out CFM56, each	\$0.8–1.2M	Safe Fly Aviation, CFM56 Engine Market Report 2026 (consultancy blog)	Sourced
Engine used-material yield, run-out CFM56, each	\$1.6–2.2M	Safe Fly Aviation, 2026 (consultancy blog)	Sourced
Engine net margin after teardown, each	\$0.4–0.8M	Safe Fly Aviation, 2026 (consultancy blog)	Sourced
Half-life market value, CFM56-7B24 / -7B27 (ceiling if the engine has life)	\$5.7M / \$6.4M	IBA, Engine Values 2025B, Sept 2025 (appraiser)	Sourced
Green-time rent, CFM56-7B	\$42,000–48,000 a month; alternative about \$100,000 a month	Safe Fly Aviation, 2026 (consultancy blog); IBA, Apr 2024 (appraiser), basis may differ	Sourced, two readings (R-028)
Precedent green-time lease term	12 months (CFM56-5A, APOC to Condor)	AviTrader, 28 Jul 2025 (trade press)	Sourced
CFM56-5B life-limited parts, full life / half life, A320ceo	\$9,878,733 / \$4,939,366 (2023 dollars); per engine or per pair unstated	Mark Calver, Cargo Facts, Apr 2023 (conference presentation)	Sourced, basis unknown (U-D3-03)
CFM56-7B full life-limited-part set at list	\$5.7M (Jun 2025, scope not visible) or about \$4M (Nov 2019)	MyAirTrade (market-data site); Aircraft Value News (trade press)	Sourced, two readings (R-016)
Landing gear, fully overhauled set	"over \$1 million"	ISTAT Jetrader, Autumn 2020 (trade body magazine)	Sourced, 2020 figure

Input	Figure	Source, level and date	Status
Landing-gear overhaul cost	Up to \$400,000	ISTAT Jetrader, Autumn 2020	Sourced, 2020 figure
Airframe teardown duration	About 3 months	Asian Aviation / AirlinerGS, APOC–Willis 737-800 teardown, 2025 (trade press)	Sourced
Teardown labour and facility cost	Not captured	None	Unknown U-D3-09
APU, avionics, thrust-reverser and other rotatable values	Not captured	None	Unknown U-D3-10
Total sell-through period	Not captured	None	Unknown U-D3-02
Split of part-out value between engines and airframe	Not captured	None	Unknown U-D3-01

Source: D3 §2.8 inputs table; Safe Fly Aviation, 2026 (consultancy blog); IBA, Sept 2025 and Apr 2024; AviTrader, 28 Jul 2025; Mark Calver, Cargo Facts Session 3, Apr 2023; MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; ISTAT (the International Society of Transport Aircraft Trading) Jetrader, Autumn 2020, p.45; Asian Aviation and AirlinerGS, 2025. Reconciliation R-016, R-028, R-029.

WORKED EXAMPLE 5.3 — PART-OUT OF A 737-800 WITH TWO CFM56-7B ENGINES, INPUT BY INPUT

Step 1, the engines, both run-out. Purchase $2 \times \$0.8\text{--}1.2\text{M}$; yield $2 \times \$1.6\text{--}2.2\text{M}$; net margin per the source $2 \times \$0.4\text{--}0.8\text{M}$ (all sourced: Safe Fly Aviation, consultancy blog, 2026). These are one source's ranges applied to a pair, not two independent observations.

engine purchase:	$2 \times (0.8 \text{ to } 1.2) = \$1.6\text{--}2.4\text{M}$
engine yield:	$2 \times (1.6 \text{ to } 2.2) = \$3.2\text{--}4.4\text{M}$
engine net:	$2 \times (0.4 \text{ to } 0.8) = \$0.8\text{--}1.6\text{M}$

Step 2, engines with green time. If one engine has twelve months of green time (assumed), the owner can rent it at \$42,000–48,000 a month for \$504,000–576,000 before tearing it down (sourced rent: Safe Fly Aviation, 2026; at the appraiser's about \$100,000 a month, \$1.2M, sourced: IBA, Apr 2024). The purchase price of such an engine would be higher than a run-out engine's by roughly that rent stream, since that is what the seller gives up. If an engine is near half-life, its \$5.7–6.4M market value (sourced: IBA, Sept 2025) is several times its teardown yield and the owner sells or leases it whole; it never enters this example.

Step 3, the life-limited-part life inside the modules. For each engine the life-limited-part value at teardown is the sum over parts of $(\text{cycles remaining} \div \text{life limit}) \times \text{list price}$, then discounted to the

used price (\$34). The only current engine-level figure the research reached is the -5B figure: \$4.94M per engine at full life if Calver's \$9,878,733 is for the aircraft's two engines (sourced: Calver, Apr 2023; basis unknown, U-D3-03). An engine torn down with 30% average life remaining (assumed) carries:

life-proportional value: $0.30 \times 4.94\text{M} = \1.48M
 at the trader's 70–75% of list (sourced: Aircraft Commerce Issue 120):
 $1.48\text{M} \times 0.70 \text{ to } 0.75 = \$1.04\text{--}1.11\text{M}$
 on the -7B set of \$5.7M (sourced: MyAirTrade, Jun 2025, scope not
 visible): $0.30 \times 5.7\text{M} = \$1.71\text{M} \rightarrow \$1.20\text{--}1.28\text{M}$
 on the -7B set of \$4.0M (sourced: Aircraft Value News, Nov
 2019): $0.30 \times 4.0\text{M} = \$1.20\text{M} \rightarrow \$0.84\text{--}0.90\text{M}$

On any of the three set prices the life-limited parts at 30% life are most of the lower bound of the consultancy's \$1.6–2.2M yield, and the remainder is airfoils, cases and accessories. The three set prices are never mixed in one line (E.1 rule 5); the 2013 figure of \$1.7M for a -7B set is below the same publisher's 2008 figure and is not used (R-016).

Step 4, the airframe. The only sourced item is the landing gear: over \$1M for a fully overhauled set, or sold below half-life to a buyer who wants to avoid an overhaul costing up to \$400,000 (sourced: ISTAT Jetrader, Autumn 2020, a 2020 figure). The auxiliary power unit, avionics and the remaining rotatables are unknown (U-D3-10). The airframe purchase price is unknown (U-D3-08), so no airframe margin can be stated.

Step 5, timing. About three months to dismantle (sourced: APOC–Willis teardown, 2025); then repair-shop turnaround for repairable items; whole modules and life-limited parts with life placed with an engine shop or an exchange provider at once; the airframe tail sold over an unsourced period (U-D3-02).

What the example shows even with the gaps: on a run-out airframe the two engines alone return \$3.2–4.4M of used material on \$1.6–2.4M of engine cost, on one consultancy's figures; the single largest airframe item is about a quarter of one engine's yield; and the engine proceeds arrive fastest because the buyers are ready-made. What it cannot show: the airframe's purchase price and margin, the labour cost, the rotatable values other than the gear, and the months to full sell-through. [D3 §2.8; Safe Fly Aviation, 2026 (consultancy blog); IBA, Sept 2025 and Apr 2024; Calver, Apr 2023; MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; Aircraft Commerce Issue 120; ISTAT Jetrader, Autumn 2020; Asian Aviation, 2025; reconciliation R-016, R-028]

One more mechanism belongs here because it competes with teardown for the same airframes. A passenger-to-freighter (P2F) conversion, owned by Part II §15, gives a 20-plus-year-old airframe a second life; narrowbodies are converted at 21–22 years on average, and more than 260 737-800 freighters were in service as of mid-2025 with a pipeline exceeding 300. A conversion candidate still needs engines with life on them. So conversion is a source of demand for green-time engines and shop visits, not a source of teardown supply, and the IBA warning of September 2026 about

an "oversupply of narrowbody conversions" sits beside the pipeline figure without being reconciled in the research. [Cargo Facts and Acumen Aviation via the demand-context dossier (D9) §2.3; IBA, 6 Sep 2026, via the leasing dossier (D6); reconciliation R-034; Part II §15]

33. Supply: retirements, feedstock and the 2023–2026 shortage

Used material comes from aircraft and engines that leave service. Its supply is therefore a function of the retirement rate, lagged by the three months of teardown and the repair and certification period that follow. Part III §19 owns the retirement decision and the realised counts; this section carries what they mean for feedstock, and it presents every contested figure with both sources, as the reconciliation requires. [D3 §2.6; reconciliation E.1 rules 13 and 14]

33.1 THE RETIREMENT DECISION, AS IT BEARS ON FEEDSTOCK

The decision to retire a CFM56-powered aircraft is an economic one, and the trade press states a rule of thumb for it. A CFM56 needs a shop visit roughly every eight years, so the 24-year mark becomes a de facto retirement date because a third full overhaul is "deemed too expensive". Deliveries of the 737NG began in 1997 and ran at about 230 a year in 1998–2002, with A320ceo deliveries at a similar rate, so from 2022 the aircraft and engines of that period began entering their prime retirement window. [Aviation Week, "Retirement Uptick Will Restock Used Engine Parts", date not captured] That rule sits beside a different reading of the interval in the research: a mature CFM56 has a performance restoration at 10,000–15,000 cycles, which at 1,000–1,500 cycles a year is roughly seven to fifteen years, and a full overhaul with life-limited-part replacement at 20,000–25,000 cycles. The two are carried together and Part I §5 and §6 set out why they differ. [Leeham News, 8 Mar 2024; Aircraft Value News, undated; reconciliation R-008]

The rule of thumb has a feedstock consequence. If the third overhaul is the retirement trigger, then the engines reaching teardown are the ones whose life-limited parts are at or near their limits for the second time, and whose yield is therefore mostly airfoils, cases and accessories rather than life. Engines retired early, with life-limited parts part-way through a run, are the ones whose teardown carries life worth selling (§32.6 step 3). Which kind retires depends on the owner's alternatives, which is where the 2023–2026 shortage begins.

Part III §19 lists five inputs to the owner's decision: age and the maintenance cliff, lease economics, freighter conversion, part-out value, and the availability of a replacement. The fourth is this Part's subject. When an aircraft's value as a sum of parts exceeds its value flying, the owner parts it out; rising used-material prices raise part-out value and so, at the margin, encourage retirement. That is a stabilising feedback. The research records that it was not strong enough to lift retirements in 2024–2026, because the fifth input overrode it: none of the others

triggers a retirement if the airline cannot get a replacement aircraft. [D9 §2.3, the demand-context dossier (D9); Part III §19]

33.2 THE COUNTS AND THE RATES, WITH BOTH SOURCES

EXHIBIT 5.4 — RETIREMENTS AND FEEDSTOCK: EVERY FIGURE THE RESEARCH REACHED, WITH ITS BASIS, SOURCE LEVEL AND DATE

Quantity	Figure	Basis	Source and level	Date
Aircraft retirements, 2019 to 2022	502; 465; 333; 325	All commercial aircraft, Cirium Fleets Analyzer	Aerodynamic Advisory presentation to the IATA (the airlines' trade body) Maintenance Cost Conference (data provider via consultancy)	2023
Average annual part-outs, 2010–2019	Just under 540; peak 694 in 2013	Same	Same	2023
Retirement rate, 2024–2026 against 2010–2019	About 24% lower	Aircraft fleet	McKinsey (consultancy)	Date not captured (2024–25)
Normal retirement rate from 2028	About 2.7% of the fleet a year	Aircraft fleet	McKinsey	Same
A320ceo-family and 737NG retirements, next five years	1,190 and 795, together 35% of all retirements	Aircraft	Cirium / Ascend by Cirium (data provider)	Date not captured
737NG retired by 2040	About 72%	Aircraft	Ascend by Cirium	Date not captured
Types about half of retirements through 2025	A320ceo, 737NG, 777, A330 (an Airbus widebody)	Aircraft	Cirium via Informa Markets forecast page	Date not captured
Aircraft retirements, 2024	"Fewer than 200", similar to 2023, 70% below the 2020 peak	Scope not visible	IBA via AviTrader (appraiser)	24 Sep 2024

Quantity	Figure	Basis	Source and level	Date
Aircraft retirements, 2024	587	Scope not visible	Cirium Fleets Analyzer via Informa forecast page (data provider)	Date not captured
Passenger aircraft retired, share of active-plus-stored fleet	2.2% (2019); 0.4% (2024)	Passenger fleet	Aviation Week / Bain material (trade press, consultancy)	2024–25
CFM56 retirement planning assumption for 2026	3–4%, then 2–3%, then 1.5–2% of the installed base	CFM56 engines	GE Aerospace chief financial officer via Aviation Week (OEM via trade press)	2026
CFM56 retirements, 2025 actual; 2026 expected	About 1.5%, "broadly consistent with 2024"; about 2%	CFM56 engines	GE Q4 2025 results via LeeHam News (OEM via trade press)	22 Jan 2026
CFM56 retirements, first quarter 2026	"Below 1%"	CFM56 engines	GE Aerospace investor-relations page (OEM)	2026
CFM56 retirements	"Near 2% per year"	"Active" engines, about 14,200	Safe Fly Aviation (consultancy blog)	2026
CFM56-5B and -7B engines torn down per year, 2023–2025	Not captured	Engines	None	Unknown U-D3-17
Actual aircraft retirement counts, 2023–2025, from Cirium	Not captured (only 2019–2022 and the shortfall percentage)	Aircraft	None	Unknown U-D3-16

Source: IATA Maintenance Cost Conference, "Retirements, part-outs, dismantling", Aerodynamic Advisory (Murby), 2023; McKinsey, "What does the future hold for commercial-aviation maintenance?", date not captured; Cirium retirement-projection page and Informa Markets forecast page, dates not captured; AviTrader, 24 Sep 2024 (IBA data); Aviation Week, "Pockets of demand disrupt expected narrowbody retirement surge", and Bain, "Commercial aviation forecast"; Aviation Week, "CFM56 overhaul demand remains strong, GE Aerospace says", 2026; LeeHam News, 22 Jan 2026; GE Aerospace, "Recent events: your questions answered", 2026; Safe Fly Aviation, 2026. Reconciliation R-004, R-005; E.2 "Aircraft retirements" and "CFM56 retirement rate".

Three things in the exhibit need saying in words, because the figures do not reconcile and the primer does not reconcile them.

First, the two 2024 aircraft counts cannot both describe the same population. IBA's "fewer than 200" and Cirium's 587 differ by a factor of three, and neither snippet states its scope: passenger aircraft only, a size threshold, or year-to-date at the time of writing. They are carried side by side with "scope not visible". [AviTrader, 24 Sep 2024; Informa Markets forecast page; reconciliation R-005; E.1 rule 13]

Second, the aircraft-basis rates imply very different annual numbers. The reconciliation did the arithmetic on the dossier's figures. McKinsey's "24% below the 2010–2019 norm" applied to the Cirium average of just under 540 a year is about 410 aircraft a year. Cirium's projection that 1,190 A320ceo-family and 795 737NG aircraft will be 35% of retirements over five years implies about 5,670 retirements in total, about 1,130 a year. The two sit far apart, and the publication dates and scopes that might explain the gap were not captured. [reconciliation R-005, arithmetic on D3 §2.6]

McKinsey shortfall on the Cirium norm: $540 \times (1 - 0.24) \approx 410$ aircraft a year (derived)

Cirium five-year projection: $(1,190 + 795) \div 0.35 \approx 5,670$ aircraft over five years $\approx 1,130$ a year (derived)

Third, the engine-basis rates are GE's and they moved within one year. GE's planning assumption for 2026 went from 3–4% of the installed base to 2–3% to 1.5–2%; it reported about 1.5% retired in 2025, broadly the same as 2024; it expected about 2% in 2026; and first-quarter 2026 retirements ran "below 1%". [GE chief financial officer via Aviation Week, 2026; Leeham News, 22 Jan 2026; GE Aerospace investor-relations page, 2026; reconciliation R-004] The consultancy blog's "near 2% per year" is a fourth figure on a fourth basis, its own "active" count of about 14,200 engines, which one dossier read as the -7B fleet alone and another as all CFM56; both readings are recorded and neither is chosen. [Safe Fly Aviation, 2026 (consultancy blog); reconciliation R-002]

33.3 FEEDSTOCK ARITHMETIC ON THE INSTALLED BASE

The pool from which feedstock is drawn is the installed base Part I §2 boxed and Part III §18 owns: about 23,000 CFM56 engines in service on CFM's undated web page, about 24,000 in trade press citing September 2025, and more than 22,800 in Safran's end-2025 count; the consultancy blog's about 14,200 "active" and more than 19,000 "flying" are labelled consultancy figures and never the fleet figure. [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; Safe Fly Aviation, 2026; reconciliation R-001; E.1 rule 14] Because no new -5B or -7B is built for an airframe, the base falls by exactly the number retired or converted each year, and that number is the gross supply of engine feedstock.

WORKED EXAMPLE 5.4 — ENGINES LEAVING THE BASE EACH YEAR, ON EACH PUBLISHED BASE AND RATE

Inputs: installed base about 24,000 (sourced: Aviation Business News, Sept 2025), about 23,000 (sourced: CFM website, undated) and more than 22,800 (sourced: Safran, 13 Feb 2026); CFM56 retirement rate about 1.5% (2025 actual) and about 2% (2026 expected) (sourced: GE via Leeham, 22 Jan 2026); consultancy base about 14,200 at "near 2%" (sourced: Safe Fly Aviation, 2026, consultancy blog). The alternative use of a retired core by FTAI's power business, 100 engines a year if one core powers one unit (assumed by the power dossier (D8), Part IX §74), is shown as a share.

on 24,000: $\times 1.5\% = 360$ engines a year; $\times 2\% = 480$
on 23,000: $\times 1.5\% = 345$; $\times 2\% = 460$
on 22,800: $\times 1.5\% = 342$; $\times 2\% = 456$
on the consultancy base, $14,200 \times 2\% = 284$
first quarter 2026 "below 1%" annualised on 24,000: fewer than 240
100 cores a year for power as a share: $100 \div 480 = 21\%$; $100 \div 360 = 28\%$; $100 \div 284 = 35\%$

So the OEM-level bases and GE's realised rate put gross engine retirements at roughly 340 to 480 a year, and the consultancy base and rate at about 284. Not every retired engine is torn down: some are sold whole to another operator, some are leased for their green time first (§32.5), and from 2026 some are earmarked as power-generation cores (Part IX). The number actually torn down per year was not captured (U-D3-17). Limits of the example: the bases may or may not include spare and stored engines (U-D9-03); the rates are a planning assumption, a realised figure and a consultancy figure on different denominators; and one core per power unit is an assumption. The two reconciliation rules that govern the presentation are that every base and every rate carries its source and that the consultancy figures are labelled. [reconciliation R-001, R-002, R-004; C.8 item 1; E.1 rules 3, 13, 14; D8 §2.10]

33.4 THE 2023–2026 SHORTAGE: WHAT THE SOURCES DESCRIBE

The mechanism, as the sources describe it, has six steps. Airlines kept CFM56-powered aircraft flying longer because new narrowbody deliveries slipped and new-generation engines spent time off wing. The expected post-2022 retirement wave therefore did not arrive. Retirements stayed well below the 2010–2019 norm. The engines that would have been feedstock were instead leased as green-time engines or overhauled. Teardown companies competed for a smaller pool. And the price of everything in the pool rose. [D3 §2.6; D9 §2.5–2.8]

Part III owns the first step and this Part takes its headline figures from it. Cirium data cited in trade press put 835 aircraft powered by Pratt & Whitney's geared turbofan (GTF) grounded or stored in late 2025. CFM's LEAP (the CFM56's successor) was removed early at 2,000–6,000 cycles against a mature CFM56's 10,000–15,000-cycle interval, and its 737 durability kit was certified only in July 2026. Boeing's 737 MAX (the 737NG's successor) was capped by the FAA

at 38 a month after January 2024, raised to 42 in October 2025 and reviewed for 47 in May 2026. Airbus delivered 793 aircraft in 2025 against an 820 target. And 75–90% of scheduled lease expiries were being extended, with only about a dozen mid-life 737NGs and fewer than 25 A320neos available to rent worldwide at the time of that survey. [The Flying Engineer citing Cirium, 2026; Leeham News, 8 Mar 2024; CFM release, 18 Jul 2026; TravelPulse, Oct 2025; CAPA (the Centre for Aviation), May 2026; Air Data News, 2026; Bloomberg via AJOT (American Journal of Transportation), date not visible; Part III §20–22]

The consequences in the figures the research captured are these. CFM56 values rose "as much as 50%" in two years on one trade-press account. [ePlaneAI, date not captured] CFM56-7B green-time rents in 2026 stood about 34% above 2019 on the consultancy blog, and the appraiser's spare-engine rent rose from about \$75,000 to about \$100,000 a month from 2019 to 2024, also about a third; the levels differ and the percentage change agrees, which the reconciliation records as one of the few independent corroborations in this area. [Safe Fly Aviation, 2026 (consultancy blog); IBA, Apr 2024; reconciliation B-5, R-028] IBA called it "a lessors' market" in April 2024 as engine lease rates and market values escalated. [IBA via AviTrader, 25 Apr 2024] Aviation Week headlined "a seller's market for used parts" in a column on MRO (maintenance, repair and overhaul). [Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured] AerSale's management described "a hypercompetitive feedstock environment, marked by increased scrutiny and pressure on pricing as more companies enter the teardown sector", and AerSale's whole-asset sales fell from 20 engines and one aircraft in 2024 to 13 engines in 2025. [AerSale, as relayed in search results; AerSale release, 5 Mar 2026] The three descriptions agree that asset owners had the upper hand and teardown buyers did not; they differ only in vantage point, and the dossier recorded that as its sixth tension rather than a contradiction. [D3 tension 6]

Whether the parts constraint was easing by 2026 is contested, and both sides are carried. GE's FY2025 annual report says material input from priority suppliers rose more than 40% and turnaround improved more than 10% in the fourth quarter of 2025. Oliver Wyman's April 2026 survey has two-thirds of maintenance operators unable to find enough technicians and labour-rate inflation at 5.5–6.0%, and a 2025 trade-press report says used-material supply "remains thin". [GE Aerospace 10-K FY2025; Oliver Wyman, Apr 2026; AVM (Aviation Maintenance magazine), 2025; reconciliation R-020]

THE SAME ENGINE ON BOTH SIDES OF THE AFTERMARKET

Every CFM56 that is not retired keeps flying, keeps accumulating cycles, and will need a shop visit (Part IV). Every CFM56 that is not retired is also one that is not torn down, so its life-limited parts, airfoils and modules do not reach the used-material market. The driver that raises demand for used parts therefore lowers their supply at the same moment, with the same engine. Part I §2.3 made the point from the fleet side and Part III §24 turns it into translation-table rows; this Part is where the supply side of it is counted. [D9 §5; Part I §2.3; D3 §2.6]

33.5 THE FORECAST THAT POINTED THE OTHER WAY

In July 2020 Oliver Wyman published a piece titled "A tsunami of used aircraft parts", forecasting a glut of used material from pandemic retirements. The research captured only the title, so none of its figures is reported, and whether Oliver Wyman has since updated the forecast is unknown U-D3-19. [Oliver Wyman, Jul 2020, title only; D3, unknown U-D3-19] The outturn pointed the other way on every count the research reached: Cirium's counts fell every year from 502 in 2019 to 325 in 2022; McKinsey reports 2024–2026 retirements about 24% below the 2010–2019 norm with a return to about 2.7% of the fleet a year only from 2028; GE's realised CFM56 rate was about 1.5% in 2025 against a plan that had started at 3–4%; and AerSale, IBA and Aviation Week all describe scarcity. [IATA Maintenance Cost Conference, 2023; McKinsey, date not captured; LeeHam News, 22 Jan 2026; D3 tension 2; reconciliation B-6] The forecast and the outturn are both carried; the forecast's own numbers are not, because they were not read.

Two Aviation Week headlines in the research point in different directions within the shortage period: "Used Parts Market Shows Signs Of Stabilizing" and "North America Sees Upturn In USM Market". Their dates and bodies were not captured, so the sequence cannot be fixed; IBA's "levelled off" of the first half of 2026 and the consultancy blog's rents "up 15–20% in recent months" sit in the same unresolved tension. [Aviation Week, two titles, dates not captured; IBA, H1 2026; Safe Fly Aviation, 2026; D3 tension 7]

33.6 WHAT GE'S USED-MATERIAL PROGRAMMES MEAN FOR INDEPENDENTS' FEEDSTOCK

The manufacturer is also a used-material trader, and by its own statement the largest. GE Aerospace said in 2026 that "GE and CFM are the largest used serviceable material provider for their products". [GE Aerospace, 2026, via Aviation Week and GE investor materials; D2 §2.2, the shop-visit market dossier (D2)] Part IV §28 owns the programmes; what belongs here is where they touch feedstock.

Two of GE's four branded TrueChoice offerings carry used material. TrueChoice Material sells "high-quality new and used OEM parts, advanced repairs and technology upgrades" to airlines and to third-party shops; Ryanair's long-term materials agreement covering "their entire fleet of ~2,000 CFM56 and LEAP engines" is a contract of that kind. TrueChoice Transitions offers "green time leases, exchanges and material buy-back, plus custom worksopes with shorter builds and maximum used material" to operators changing their ownership horizon. [GE Aerospace press releases, 2019–2025; GE Aerospace 10-K FY2025; D2 §2.3]

For an independent teardown company the programmes mean three things, each a description rather than a judgment. First, the manufacturer competes for retired engines: material buy-back under Transitions takes engines at the end of an operator's horizon into the OEM's own pool before a trader can bid. Second, the manufacturer sells the material those engines yield through Material, into the same shops the independents supply, and it can bundle it with new parts and

repairs, which an independent cannot. Third, a used part sold by the manufacturer carries the manufacturer's release, which matters to a buyer who vets suppliers as §31.2 described. A GA Telesis purchase of \$25M of "OEM-aligned" used engine inventory in October 2025 shows OEM-sanctioned used-material channels running beside independent ones. [D2 §2.3; GA Telesis release, Oct 2025; D3 §3] The name and scale of GE's used-material arm, and its CFM56 role, were not captured; the only hit was a TrueEngine presentation whose content was not read, and the gap is clustered with Part VI's open question on GE's positioning against alternative parts (K-16). [D3, unknown U-D3-15; reconciliation K-16]

34. Pricing versus new OEM parts

A used part has no price list of its own. It is priced as a percentage of the manufacturer's catalogue list price for the same part number, and that one fact explains most of what follows: the used price inherits the manufacturer's annual escalation, the discount narrows when the manufacturer cannot deliver, and it widens when retirements flood the market. [D3 §2.5]

USM PRICING VERSUS NEW (THE USM DISCOUNT TO CLP)

Used serviceable material is priced as a percentage of the OEM's catalogue list price (CLP) for the same part: on average 60–80% of a new part's price; 70–75% of list when a specialist engine trader sells repaired material to an airline; up to 85% of list for high-demand parts early in a platform's life; and above list when the manufacturer cannot deliver. Because the base is the catalogue, the used price rises with every catalogue revision even if the discount does not move. [Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured; Aircraft Commerce Issue 120, date not captured; D3 §2.5]

34.1 THE PUBLISHED FIGURES AND THEIR TENSION

EXHIBIT 5.5 — USED-PART PRICES RELATIVE TO NEW, AS PUBLISHED, WITH BASIS, SOURCE LEVEL AND DATE

Figure	What it describes	Source and level	Date
60–80% of a new part's price	The average used part, all classes	Aviation Week, "MRO Memo: A Seller's Market For Used Parts" (trade press)	Date not captured
Up to 85% of catalogue list price	High-demand used parts early in a platform's life cycle	Aviation Week, same	Date not captured
"More than the catalog list price"	Used parts when the OEM has supply-chain problems	Aviation Week, same	Date not captured

Figure	What it describes	Source and level	Date
70–75% of catalogue list price	Repaired engine used material sold by a specialist trader to an airline	Aircraft Commerce Issue 120 (trade press)	Date not captured
Upward trend "most evident on CFM56 and V2500 platforms"	Driven by rising demand, falling supply of new parts and used parts, and rising repair costs	Aviation Week, "Magnetic expects CFM56 market challenges", 2024 (trade press)	2024
CFM raised its catalogue list price	"A notable shift in pricing strategy for this platform"	Same	Aug 2023
Residual value assigned to "alternative materials"	60–70% of the industry assigns 25% or less	Safe Fly Aviation teardown report (consultancy blog)	2026
Prices of used life-limited-part packages with stated cycles remaining	Not shown in the listings reached	None	Unknown (U-D1-04)

Source: Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured; Aircraft Commerce, Issue 120 maintenance article, date not captured; Aviation Week, "Magnetic expects CFM56 market challenges, opportunities in 2024", 2024; Safe Fly Aviation, "Aircraft Teardown Market Growth", 2026 (consultancy blog); the engine dossier (D1), unknown U-D1-04. Reconciliation R-022.

The figures are consistent as a range and not as a single number. Aviation Week's 60–80% is a general average across every class of used part; Aircraft Commerce's 70–75% is a specific price, a trader selling repaired engine material to an airline; the 85% and above-list cases are the ends of the same spectrum. Which applies depends on the part, the platform and the manufacturer's supply at the time, and the primer carries all four rather than one. [reconciliation R-022; D3 tension 5] The V2500 named in the exhibit is the alternative A320ceo engine of Part I §10.

34.2 LIFE-LIMITED PARTS: LIFE FRACTION FIRST, THEN DISCOUNT

For most used parts the percentage is applied to the list price directly. For a life-limited part there is an extra step, because what the buyer is purchasing is the remaining cycles. The life-proportional value is the fraction of life remaining times the list price of the new part; the used price is that figure less the trader's discount. Part I §4.4 stated the rule from the engine side; here it is from the trader's. [D3 §2.4–2.5; Part I §4.4]

The one current list price the research reached for a CFM56-7B life-limited part is a consultancy figure: a high-pressure turbine disk at \$420,000 new, with its 20,000-cycle limit. The current list price of representative -7B life-limited parts and of a full set was otherwise not captured and is unknown U-D3-06. [Safe Fly Aviation, 2026 (consultancy blog), via D2 §4.4; D3, unknown U-D3-06]

disk with 70% of life left: life-proportional value = $0.70 \times 420,000 = \$294,000$
 at the trader's 70–75% of list: $294,000 \times 0.70$ to $0.75 = \$206,000$ – $221,000$,
 or 49–53% of the new price
 buyer facing a one-year OEM lead time: may pay $0.70 \times 420,000 = \$294,000$ or
 more, the "above list" case in life-adjusted terms

That last line is how the trade-press statement that used parts can sell "for more than the catalog list price" should be read for a life-limited part. The buyer is not paying more than list for a new disk. It is paying the full life-proportional value, or above it, for a used disk it can have now, because the alternative is a grounded engine for the manufacturer's lead time. The mechanism matters more than the specific disk: the discount to list narrows when the manufacturer's own supply is constrained, and widens when retirements flood the market with used parts. [Aviation Week, date not captured; D3 §2.5]

34.3 HOW A USED PART INHERITS THE MANUFACTURER'S ESCALATION

Part I §4.3 boxed escalation and Part IV §28 owns it. Two readings sit in the research and both are carried. The first covers all spare parts: new spare-parts prices rising "around 10% per year", a double-digit increase on 1 November 2022, a high-single-digit increase in August 2023, and Safran's chief executive describing a rule of pricing "above inflation by 3 to 4 points ... at least for some years". [Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023] The second is specific to life-limited parts: a full -7B set compounded at about 7% a year from 2008 to 2025 on the published set prices, a single step of about 12% reported by Ishka without a year, and -7B parts generally up 20–30% in 2023 after CFM's August 2023 catalogue revision. [Part I worked example 1.3; Ishka, undated; Magnetic Group via Aviation Week, 2024; reconciliation R-018] The years and amounts of each catalogue step between 2019 and 2026 were not retrieved (cluster K-03).

Because the used price is a percentage of the catalogue, every one of those steps moves it. The worked example shows the arithmetic over three years on both escalation readings and on both discount readings.

WORKED EXAMPLE 5.5 — ONE PART AT LIST AND AS USED MATERIAL OVER THREE ESCALATION YEARS

Inputs: a CFM56-7B high-pressure turbine disk at \$420,000 new in year 0 (sourced: Safe Fly Aviation, 2026, consultancy blog; the only current -7B life-limited-part price reached). Escalation case A, 7% a year (derived in Part I from the 2008–2025 set prices). Escalation case B, 10% a year (sourced: Aviation Week, late 2023, all spares). Used discount reading one, 70–75% of list (sourced: Aircraft Commerce Issue 120); reading two, 60–80% (sourced: Aviation Week). The example prices the disk as an ordinary used part, that is, before any life-fraction adjustment, to isolate the escalation effect; §34.2 adds the life step.

case A, 7% a year: list 420,000 → 449,400 → 480,858 → 514,518
 used at 70–75%: 294,000–315,000 → 314,580–337,050 → 336,601–
 360,644 → 360,163–385,889
 case B, 10% a year: list 420,000 → 462,000 → 508,200 → 559,020
 used at 70–75%: 294,000–315,000 → 323,400–346,500 → 355,740–
 381,150 → 391,314–419,265
 used at 60–80% of the year-3 list: case A 308,711–411,614; case B
 335,412–447,216
 dollar saving from buying used at 70–75%, year 0 → year 3: case A
 \$105,000–126,000 → \$128,630–154,355; case B → \$139,755–167,706

Three things follow. The used price in dollars rises every year even though the discount is unchanged, by exactly the manufacturer's escalation rate: a trader holding a disk in stock sees its selling price rise with the catalogue. The dollar saving from buying used also grows with escalation, because it is a fixed share of a rising list. And a 20–30% single-year step of the kind reported for -7B parts in 2023 lifts the used price by the same 20–30% in the year it lands. The one thing escalation does not do is close the gap between used and new; only the manufacturer's delivery constraint does that, by narrowing the discount toward 85% of list and above. Limits of the example: the disk price is a consultancy figure; the 7% rate is derived from set prices whose scope differs; the catalogue steps for 2024–2026 are unknown; and the used readings describe different transaction types (R-022). [Safe Fly Aviation, 2026 (consultancy blog); Aviation Week, late 2023; Aircraft Commerce Issue 120; Part I worked example 1.3; reconciliation R-018, R-022; K-03]

HOW THE CATALOGUE SETS THE USED PRICE

The manufacturer never prices a used part, and it sets the price of every used part. It does so because the trade quotes used material as a share of list, and the manufacturer alone sets list and revises it upward on a schedule. A used-material trader therefore carries two exposures it does not control: the manufacturer's escalation, which lifts the value of its stock, and the manufacturer's delivery performance, which sets how close to list it can sell. A third party to the same line in the parts bill is the alternative-manufacturer part of Part VI, which is priced against the same catalogue; the two compete for the same slot, and an engine with alternative parts carries less of the list-price-based maintenance adjustment that appraisers apply, which Aircraft Value News records as undermining full-life adjustments (Part I §8). [D3 §2.5, §5; D4 §2.1; Aircraft Value News, undated]

35. The players

The used-material trade has four kinds of participant: independent traders that buy feedstock and sell parts, lessors that route their own retired assets into captive material arms, the manufacturer's own used-material business, and FTAI, which does something different with the same feedstock. This section gives one paragraph to each named company, with the scale its filings or releases disclose and nothing more, and ends with the market-size estimates. Two listed companies,

AerSale and AAR, are also peers in Part XI §87, which carries their full filing treatment. [D3 §3; D11 §3.5, §3.7, the peers dossier (D11)]

35.1 ONE PARAGRAPH EACH

AerSale. AerSale is the cleanest listed example of a teardown-and-used-material business and is listed on Nasdaq. Its Asset Management Solutions (AMS) segment "extract[s] value from strategic asset acquisitions either as whole assets or by disassembling for used serviceable material"; its TechOps segment is maintenance and engineered products. For 2025 it reported revenue of \$335.3M, down 2.8% from \$345.1M; AMS revenue of \$211.6M against \$215.5M; TechOps revenue of \$123.7M against \$129.6M; used-material revenue of \$137.6M, made up of \$120.1M in AMS and \$17.6M in TechOps (the components sum to \$137.7M, a rounding difference); and flight-equipment sales of \$56.4M from 13 engines, against \$110.1M from 20 engines and one aircraft in 2024. Revenue excluding flight-equipment sales rose 18.7%. [AerSale release, 5 Mar 2026; AerSale 10-K FY2025] Its AMS gross margin was 35.0% in 2025 against 38.3% in 2024, both derived from the filing's gross-profit and sales figures, with the decline attributed to "lower margin on USM sales resulting from changes in the product mix"; engine revenue rose on higher used-material sales in the PW4000 (a Pratt & Whitney widebody engine) and CF6-80 lines, which are not CFM56. Fourth-quarter 2025 feedstock acquisitions were \$15.4M; a full-year purchase total was not returned. [AerSale 10-K FY2025, via D11 §3.5 and §2.4; reconciliation B-8]

WHOLE-ASSET SALE (FLIGHT-EQUIPMENT SALE)

A whole-asset sale is the sale of a complete aircraft or engine from a trader's inventory rather than its disassembly into parts; AerSale's filings call the line "flight equipment sales". AerSale sold 13 engines this way in 2025 against 20 engines and one aircraft in 2024. The choice between selling whole and tearing down is the one §32.5 described from the owner's side. [AerSale release, 5 Mar 2026; D3 §3]

ADJUSTED EBITDA (EBITDA, ADJUSTED)

EBITDA is earnings before interest, taxes, depreciation and amortization. Adjusted EBITDA is that figure further adjusted by the company for items it treats as non-recurring or non-cash, and it is a company-defined profit measure; no two companies' definitions are the same. AerSale reported Adjusted EBITDA of \$46.1M for 2025, 13.8% of revenue, against \$33.4M and 9.7% in 2024. Part X §80 carries FTAI's definition and reconciliation. [AerSale release, 5 Mar 2026; Part X §80]

GA Telesis. GA Telesis (GA) is a privately held trader, lessor and maintenance provider whose Flight Solutions Group buys engines for disassembly and sells the material. In January 2025 it bought eight PW4000 engines from a major US airline for disassembly; in November 2025 it disassembled eight more, one CFM56-5B, two V2500-A5, two PW4000 and three CF6-80C2;

and in October 2025 it bought an additional \$25M of "OEM-aligned" used engine inventory. It signed a CFM56-5B and -7B material agreement with Aero Engine Solutions in January 2018. Its revenue is not disclosed. [GA Telesis releases, Jan and Nov 2025; Aviation Business News, undated; AviTrader, 31 Jan 2018]

VAS Aero Services. VAS Aero Services (VAS) is a private teardown-management and used-material redistribution company that also handles large aircraft: Airbus appointed it to oversee the dismantling and material redistribution of three retiring A380 (Airbus's double-deck widebody) aircraft. Its revenue is not disclosed; a trade-press item quoting it refers to a "\$29 billion MRO market", MRO (maintenance, repair and overhaul) being the industry's generic term, without defining the scope. [ePlaneAI, undated; D3 §3, §4.5]

AAR. AAR is listed on the NYSE and is the scale reference for a parts business that mixes distribution of new manufacturer parts with used material. Its Parts Supply segment, "distribution of new OEM parts and sales and leasing of USM", "accounted for approximately 45% of AAR's sales in fiscal 2026", the year to 31 May 2026. Consolidated sales were \$3,308.0M, derived from \$2,384.1M of commercial and \$923.9M of government sales, which puts Parts Supply at about \$1,489M, also derived. Segment sales rose \$388.1M year on year; new-parts distribution grew 23.9% and used-material activities 32.1%; whole-asset sales in parts trading rose \$27.8M in the first quarter of fiscal 2026; and the segment absorbed the September 2025 acquisition of American Distributors Holding Co. (ADI) for \$137.1M. The split of Parts Supply between distribution and used material, and the segment's operating income, were not returned. [AAR 10-K FY2026; AAR 10-Q for the quarter ended 31 Aug 2025; D11 §3.7, unknown U-D11-14; D3, unknown U-D3-11]

DISTRIBUTION (NEW PARTS)

Distribution is the resale of new manufacturer parts under a distribution agreement: the distributor holds inventory and earns the margin between the manufacturer's transfer price and its selling price. Its margins are structurally thinner than those on used or alternative parts, because the manufacturer sets the cost of goods. AAR's Parts Supply combines it with used-material sales and leasing; FTAI has no distribution business. Part XI §87 carries AAR as a peer. [D11 §2.4, §3.7]

AJW Group. AJW Group (AJW) is a private component-support and used-material company and the author, with AviTrader, of the August 2025 article on used-material airworthiness on which §31 rests. Its scale was not captured and is unknown U-D3-13. [AJW Group via AviTrader, Aug 2025; D3, unknown U-D3-13]

Willis Aero. Willis Aero is the parts subsidiary of Willis Lease Finance, the only US-listed company whose primary business is engine leasing (Nasdaq-listed; Part XI §87). It sells engine parts and materials "through the acquisition or consignment of aircraft and engines". Engines removed from the lease portfolio for part-out are carried in "equipment held for sale", and the

investment is "recovered through the sale of spare parts". It partnered with APOC on a 737-800 teardown. The parent's 2025 figures give its feedstock scale: 363 engines and 20 aircraft in a lease portfolio of \$2,801.7M of equipment held for lease, 116 further engines managed for third parties, average utilisation of 84.9%, revenue of \$730.2M and lease rent of \$291.6M. Willis Aero's own spare-parts revenue was not returned and is clustered as an open question with Part XI (K-15). [Willis Lease Finance 10-K FY2025; Asian Aviation, 2025; D11 §3.1; reconciliation K-15]

Setna iO. Setna iO is a private used-material trader whose reported teardown activity reaches the new generation: it acquired a 2019-vintage A320neo airframe and two sets of quick-engine-change (QEC) components, the mounts, plumbing and accessories that dress a bare engine for installation, for the geared-turbofan engine. Its revenue is not disclosed. [AviTrader, undated item; D3 §3]

GE's used-material arm. The manufacturer's own used-material business sits inside GE Aerospace's TrueChoice Material offering, described in §33.6, and GE states that "GE and CFM are the largest used serviceable material provider for their products". Its name as a business unit, its scale and its CFM56 role were not captured (U-D3-15; K-16). [GE Aerospace, 2026; D2 §2.3; D3, unknown U-D3-15]

AerCap Materials. AerCap Materials is the part-out and used-material arm of AerCap, the largest aircraft lessor, which files a Form 20-F as a foreign private issuer. It was formed when AerCap acquired GECAS (GE Capital Aviation Services) in late 2021, which had a parts business active in teardowns. It dismantles at Greenwood-Leflore County Airport in Mississippi and distributes from Memphis, Tennessee. Its 2023 factsheet records more than 300 aircraft dismantled and more than 15,000 unique parts; its first-quarter 2026 factsheet records more than 250,000 items stocked and more than 850 customers, with dismantling offered "with tailored workscopes, competitive pricing and accelerated turnaround time". Its revenue is not disclosed in the filing text returned. The parent's feedstock scale: 470 owned engines at 31 December 2025, stated as 9% of its fleet, or more than 1,200 engines including those owned and managed by its Shannon Engine Support (SES) joint venture with Safran at 30 September 2025; the two counts are on different bases and both are carried. [AerCap Materials factsheets, 2023 and 1Q 2026; Aviation Week, "Teardown providers expand business", date not captured; AerCap 20-F FY2025; AerCap 6-K, Q3 2025; D11 §3.2; reconciliation R-033]

APOC Aviation. APOC is a private engine and airframe trader and lessor that leases green time and tears down. Its transactions in the research: the first CFM56-5B bought for up-cycling in October 2021; two CFM56-7Bs disassembled; four young 737 airframes bought from a US legacy carrier for teardown; the 737-800 teardown with Willis Lease; the twelve-month CFM56-5A green-time lease to Condor with teardown to follow; and the 15-year-old A320-200 bought from FTAI for disassembly at Tarmac Aerosave's Toulouse-Francal site in May 2026.

Tarmac Aerosave is a dismantling and storage company owned by Airbus, Safran and Suez. [AviTrader, 7 Oct 2021 and 28 Jul 2025; Aviation Business News, undated; Asian Aviation, 2025; ePlaneAI, 2026]

Unical Aviation. Unical is a private used-material trader on which the dossier's own searches returned nothing; the orchestrator's later searches established one fact, which is that Unical was FTAI's joint-venture partner when the assets of iAero Thrust were acquired on 4 January 2023 to form QuickTurn in Miami, and that FTAI bought Unical's remaining interest on 1 December 2023 for \$30.3M in cash. Unical's own scale remains unknown (U-D3-14, partly resolved). [GlobeNewswire, 1 Dec 2023; FTAI 10-K FY2025; orchestrator notes; reconciliation D.3]

FTAI's own sourcing. FTAI is both a buyer and a seller of feedstock, and it does something with it that the traders above do not: it rebuilds the material into modules. Its FY2025 annual report says that engines are "transferred to inventory when determined to be unserviceable for leasing purposes with the intent to tear down the asset into salvageable components that would be manufactured into new modules and parts and sold"; that the Aerospace Products business "focuses on the manufacturing and sale of new assets through the use of inventory purchased from third parties and salvaged modules and parts from generally unserviceable engines"; and that "serviceable used modules and parts are sold through an exclusive partnership responsible for the teardown, repair, marketing and sales of parts from the CFM56 engine pool". [FTAI 10-K FY2025] The filing excerpt does not name the partner. A 2025 PRNewswire release is titled "AAR and FTAI Aviation extend their exclusive Serviceable Engine Products agreement, providing CFM56 engine material to the global aviation aftermarket through 2030"; the title was located by two dossiers and the body was read by neither, so the primer records that the filing's description and the partner's release title are consistent and that the agreement's terms are unknown (U-D3-07). [PRNewswire, 2025, title only; reconciliation B-9, C.3 item 1] FTAI's captive feedstock flows are described in the Parts that own them: a leasing fleet of 243 engines at 31 December 2025 (Part VIII §58); the Strategic Capital Initiative (SCI) vehicles whose engines are powered through exchanges (Part VIII §61); the ten off-lease 737-700s bought from WestJet in September 2026 for their CFM56-7B engines and modules (Part VIII §67); and, from 2026, the power business that consumes CFM56 cores (Part IX §74). On the selling side, the A320-200 sold to APOC for disassembly shows FTAI releasing an airframe it did not need. Part VII §46 owns the Module Factory into which the engine feedstock goes. [FTAI 10-K FY2025, via D6; FTAI release, 28 Sep 2026, via D6; ePlaneAI, 2026; D5 §2.2, the Aerospace Products dossier (D5)]

EXHIBIT 5.6 — THE PLAYERS: WHAT EACH DOES IN USED MATERIAL AND THE SCALE EACH DISCLOSES

Company	Status	Role in used material	Scale disclosed (source, date)
AerSale	Listed (Nasdaq)	Buys mid- and end-of-life aircraft and engines; leases, sells whole or tears down; maintenance (TechOps)	Revenue \$335.3M; AMS \$211.6M at 35.0% gross margin (derived); used-material revenue \$137.6M; 13 engines sold whole; Adjusted EBITDA \$46.1M (release, 5 Mar 2026; 10-K FY2025)
AAR	Listed (NYSE)	Distribution of new parts plus used-material sales and leasing; whole-asset trading	Parts Supply about 45% of sales, about \$1,489M (derived) of \$3,308.0M (derived); used-material activity +32.1% (10-K FY2026)
Willis Lease / Willis Aero	Listed (Nasdaq)	Engine lessor; captive parts arm by acquisition or consignment	363 engines, 20 aircraft, \$2,801.7M equipment held for lease, 84.9% utilisation; parts revenue not returned (10-K FY2025)
AerCap / AerCap Materials	Listed (NYSE; 20-F filer)	Largest aircraft lessor; captive teardown arm since the GECAS acquisition, 2021	300+ aircraft dismantled, 15,000+ unique parts (2023); 250,000+ items, 850+ customers (1Q 2026); 470 owned engines, >1,200 including SES (20-F; 6-K)
GE Aerospace	Listed (NYSE)	OEM; "largest used serviceable material provider for their products"; TrueChoice Material and Transitions	Name and scale of the used-material arm not captured (GE Aerospace, 2026; U-D3-15)
FTAI Aviation	Listed (Nasdaq)	Tears down engines unserviceable for leasing; rebuilds into modules; sells used modules and parts through an exclusive partnership; also sells airframes for teardown	Segment figures in Part VII; partner unnamed in the filing; AAR's release (27 Mar 2025) names AAR as exclusive provider for FTAI's pool of over 450 CFM56 engines, handling teardown, repair, marketing and sales, through 2030 (10-K FY2025; PRNewswire, 2025)
GA Telesis	Private	Buys engines for disassembly; "OEM-aligned" used inventory; leasing; maintenance	16 engines disassembled in two 2025 batches; \$25M inventory purchase, Oct 2025; revenue not disclosed (releases)
VAS Aero Services	Private	Teardown management and redistribution, including widebodies	Three A380 teardowns for Airbus; revenue not disclosed (ePlaneAI)

Company	Status	Role in used material	Scale disclosed (source, date)
AJW Group	Private	Component support; used material	Not captured (U-D3-13)
Setna iO	Private	Used-material trader; new-generation teardown	One A320neo airframe and two GTF QEC sets; revenue not disclosed (AviTrader)
APOC Aviation	Private	Trader and lessor; green-time leases; teardown ("up-cycling")	Transactions listed in §32.3 and §35.1; revenue not disclosed
Unical Aviation	Private	Used-material trader; FTAI's former QuickTurn partner	\$30.3M paid by FTAI for its remaining interest, 1 Dec 2023; own scale unknown (U-D3-14)
Tarmac Aerosave	Private (Airbus, Safran, Suez)	Dismantling and storage facilities	Site of the FTAI-sourced A320-200 teardown (ePlaneAI)

Source: AerSale release, 5 Mar 2026, and 10-K FY2025; AAR 10-K FY2026; Willis Lease Finance 10-K FY2025; AerCap 20-F FY2025 and 6-K Q3 2025; AerCap Materials factsheets 2023 and 1Q 2026; GE Aerospace, 2026, via D2; FTAI 10-K FY2025; PRNewswire, 2025 (title); GA Telesis releases, 2025; ePlaneAI; AviTrader; Asian Aviation; GlobeNewswire, 1 Dec 2023. Compiled from D3 §3 and D11 §3. "GTF" is Pratt & Whitney's geared turbofan.

Three structural observations come from the sources themselves rather than from this primer. AerSale describes more companies entering the teardown sector and a hypercompetitive feedstock market. Lessors have internalised the teardown step, AerCap through AerCap Materials from 2021 and Willis through Willis Aero, so their retired assets no longer reach third-party traders; FTAI's exclusive partnership is a third version of the same move. And GA Telesis's "OEM-aligned" purchase shows the manufacturer's sanctioned channel running beside the independents' (§33.6). [AerSale, as relayed; AerCap Materials factsheets; Willis Lease 10-K FY2025; FTAI 10-K FY2025; GA Telesis, Oct 2025; D3 §3]

35.2 HOW BIG THE MARKET IS: THREE VENDORS AND A CAVEAT

EXHIBIT 5.7 — USED-SERVICEABLE-MATERIAL MARKET SIZE, 2025, AS EACH VENDOR STATES IT

Source and level	2025	Later figure	Stated growth rate	Scope as retrieved
The Business Research Company, via a research-aggregator listing (market-research vendor)	\$5.87B	\$6.22B (2026)	6.1%	"Air Transport USM Global Market Report"; scope definition not retrieved

Source and level	2025	Later figure	Stated growth rate	Scope as retrieved
Market.us (market-research vendor)	\$7.7B	\$12.3B (2035)	4.8% (2026–2035)	"Air Transport USM Market"; scope not retrieved
Fortune Business Insights, as relayed in search results (market-research vendor)	\$8.55B	\$8.95B (2026); \$12.93B (2034)	4.7% (2026–2034)	Report page not isolated; scope not retrieved
Safe Fly Aviation (consultancy blog)	">\$4B/yr, CFM56 largest segment"	None	None	Scope not stated

Source: The Business Research Company via a research-aggregator listing (GII), 2026; Market.us, "Air Transport USM Market", 2025–26; Fortune Business Insights, figures as relayed, URL not isolated; Safe Fly Aviation, 2026, via D2 §4.6. Reconciliation R-021; D3 tension 1 and unknown U-D3-18.

The three vendor figures for the same year differ by 46% from lowest to highest, and their growth rates differ too, 6.1% against 4.7–4.8%. None of the scope definitions was retrieved: what counts as used serviceable material, whether engines, airframes and components are all included, and whether the figures are sales or gross proceeds. The primer carries all three with the consultancy figure labelled as such and does not average or choose; the scope question is unknown U-D3-18. [reconciliation R-021; D3 tension 1, unknown U-D3-18] For a sense of proportion only, and not as a reconciliation, AerSale's used-material revenue of \$137.6M is between 1.6% and 2.3% of the three market figures, and AAR's derived Parts Supply segment of about \$1,489M, most of which is distribution of new parts, is larger than any one of the three estimates is of used material alone.

36. The module as a unit of trade

Part I §3 boxed the module: a section of the engine designed to be removed and replaced as a unit, bolted at flanges, with its own part number, serial number and maintenance record. Part I §8 boxed the idea this section carries in full, the module as a unit of trade, and its companion, the sum of modules. This section explains how a serviceable module from a teardown is valued, how it is sold, and how it feeds the exchange business that Part VII §46 owns.

36.1 WHICH LEVEL IS BEING COUNTED

The CFM56-7B's major modules are the fan and booster, the core (the high-pressure compressor, combustor and high-pressure turbine), the low-pressure turbine (LPT), and the accessory gearbox, each built from numbered sub-modules whose count the research did not reach. That is the anatomical level Part I §3 fixed. FTAI's own releases count three modules per engine, the three gas-path modules without the gearbox: "450 modules (150 engines)" and "1,800 modules (600 engines)". The two counts describe the same hardware at different levels, and this Part names the level each time it converts: a torn-down engine yields four major modules anatomically and three in FTAI's counting convention. [Part I §3.2, Exhibit 1.4; FTAI release, 26 Feb 2025, via orchestrator notes; reconciliation R-035, B-3; E.1 rule 7]

36.2 HOW A MODULE FROM A TEARDOWN IS VALUED

A serviceable module from a torn-down engine is valued the same way a whole engine is, in miniature. Its value is the sum of its serviceable parts (airfoils, cases, bearings, seals) at the used-material discounts to list of §34, plus the remaining life of the life-limited parts inside it (life fraction times list price, discounted), less the cost of any repair needed to make it serviceable as a unit. A core module with turbine and compressor disks at 80% life is worth far more than one at 20% life, because the next owner's shop visit will or will not have to open it to replace them. And its documentation must show the history of every life-limited part inside it, so §31 applies to the module as a whole. [D3 §2.7]

EXHIBIT 5.8 — VALUING A SERVICEABLE MODULE FROM A TEARDOWN: THE COMPONENTS, THE PRICING BASIS AND WHAT THE RESEARCH COULD NOT SOURCE

Component of value	Pricing basis	Figures reached	Source and level	Status
Life-limited parts inside the module	$\Sigma (\text{cycles remaining} \div \text{limit}) \times \text{list price}$, then the used discount	One -7B part: HPT disk \$420,000 new, 20,000-cycle limit; full -7B set \$5.7M (Jun 2025) or about \$4M (Nov 2019); -5B set \$9.88M full life, basis unstated	Safe Fly Aviation, 2026 (consultancy blog); MyAirTrade (market-data site); Aircraft Value News (trade press); Calver, Apr 2023 (conference)	Per-part list prices unknown (U-D3-06); core group's share of the set unknown (U-D1-08)
Serviceable non-life-limited parts	Used discount to list: 60–80%, or 70–75% for repaired engine material sold to an airline	Figures in §34	Aviation Week; Aircraft Commerce Issue 120 (trade press)	Sourced, two readings (R-022)

Component of value	Pricing basis	Figures reached	Source and level	Status
Repair needed to make the module serviceable as a unit	Repair-shop quote	Not captured	None	Unknown (U-D3-09 covers teardown cost; module repair cost not separately logged)
Whether a module change requires a test-cell run under the shop manual	Shop manual procedure	Not verified	None	Unknown (U-D1-14)
Premium for avoided downtime when sold through an exchange	Exchange fee plus the difference in life-limited-part life	No provider's exchange price list was reached, including FTAI's	None	Unknown (K-01: U-D2-07, U-D3-20, U-D5-03)
Whole-module reference	A serviceable module sold to an engine shop or exchange pool "almost immediately" after teardown	No module price was captured	D3 §2.2	Timing sourced; price unknown

Source: D3 §2.7 and §2.8 (the valuation structure, assembled in the research); Safe Fly Aviation, 2026, via D2 §4.4; MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; Mark Calver, Cargo Facts, Apr 2023; Aviation Week and Aircraft Commerce as in Exhibit 5.5; reconciliation K-01, R-016, R-022; D1 unknowns U-D1-08 and U-D1-14. "HPT" is the high-pressure turbine.

The exhibit shows why §36 cannot price a module to the dollar. The two unknowns that would let it, the list price of each life-limited part in the core and any provider's exchange fee, were not reached. What can be shown is the structure, and the one engine-level cross-check the research supports.

WORKED EXAMPLE 5.6 — THE LIFE-LIMITED-PART LEG OF A CORE MODULE, AND THE ENGINE-LEVEL CROSS-CHECK

Inputs: a -7B core module's life-limited parts are the HPC stage 1–2 spool, the HPC stage 3 disk, the HPC stage 4–9 spool and the HPT disk, all with a 20,000-cycle limit (sourced: records mini-packs, Part I Exhibit 1.5). The HPT disk lists at \$420,000 new (sourced: Safe Fly Aviation, 2026,

consultancy blog). The other three list prices are unknown (U-D3-06), and the core group's share of a full set is unknown (U-D1-08). Average life remaining in the core 40% (assumed). Used discount 70–75% of list (sourced: Aircraft Commerce Issue 120).

HPT disk alone: $0.40 \times 420,000 = \$168,000$ life-proportional; $\times 0.70$ to $0.75 = \$117,600\text{--}126,000$
 the other three core parts: $0.40 \times (\text{unknown list}) \times 0.70$ to $0.75 =$
 cannot be stated
 core module life-limited-part leg = $117,600\text{--}126,000 + (\text{three unpriced parts}) =$ a floor, not a value

Because the core's share of the set is unknown, the example steps up to the engine. An engine whose life-limited parts average 40% remaining (assumed) carries, on each of the three set prices never mixed in one line:

on the -7B set of \$5.7M (sourced: MyAirTrade, Jun 2025, scope not visible): $0.40 \times 5.7\text{M} = \$2.28\text{M} \rightarrow$ at 70–75%: $\$1.60\text{--}1.71\text{M}$
 on the -7B set of \$4.0M (sourced: Aircraft Value News, Nov 2019): $0.40 \times 4.0\text{M} = \$1.60\text{M} \rightarrow \$1.12\text{--}1.20\text{M}$
 on the -5B figure read as per engine, \$4.94M (sourced: Calver, Apr 2023; basis unknown, U-D3-03): $0.40 \times 4.94\text{M} = \$1.98\text{M} \rightarrow \$1.38\text{--}1.48\text{M}$

Set against the consultancy's \$1.6–2.2M used-material yield for a run-out engine, the life-limited-part leg at 40% is between half and all of it, before any airfoil, case or accessory is counted. That is why the trade advertises used life-limited parts on cycles remaining and why the module that holds the shorter-lived parts, the core, is the one whose remaining life moves the most money. The mismatch Part I §8 described is the other side of the same arithmetic: a fan module at 18,000 of 30,000 cycles (60%) attached to a core at 1,500 of 20,000 (7.5%) is worth more as a fan sold to someone who needs a fan plus a core sold for its stub life than as one engine that must go into a shop, because the buyer of the fan pays for 60% of the fan's list and the buyer of the core pays only for 7.5% of the core's. Limits of the example: one list price out of four; the life fraction is assumed; the three set prices disagree (R-016); and the non-life-limited parts, the repair cost and any exchange premium are not priced. [D3 §2.7–2.8; D1 §2.8; Safe Fly Aviation, 2026 (consultancy blog); MyAirTrade, Jun 2025; Aircraft Value News, Nov 2019; Calver, Apr 2023; Aircraft Commerce Issue 120; reconciliation R-016, R-022; D1 unknown U-D1-08]

36.3 HOW A MODULE IS SOLD, AND HOW IT FEEDS AN EXCHANGE BUSINESS

A serviceable module from a teardown has three kinds of buyer. An engine shop buys it to install in a customer's engine in place of a module that has run out, which turns a full disassembly into a module change (Part I §3.4). A trader buys it to hold or to break into parts. And an exchange provider buys it, or produces it from its own feedstock, to put into a pool. The first and third can be placed almost at once after teardown, which is why §32.3 said the engine proceeds arrive fastest. [D3 §2.2, §2.7]

The exchange is the transaction that turns a module into a product rather than a part. An engine shop or operator hands over a time-expired or damaged module and receives a serviceable one, paying an exchange fee plus the difference in life-limited-part life between the two. The incoming module becomes the provider's feedstock: the provider tears it down, salvages the serviceable parts and the life, rebuilds, and returns it to the pool. For the shop it converts a full shop visit (open the whole engine, replace parts, re-assemble, test) into a shorter module swap. For the provider it is a way to sell used material and life-limited-part life at module prices, which carry the premium of avoided downtime. Part I §3 boxed module exchange; Part VII §46 and §47 own FTAI's version and its pricing; GE sells the same thing under TrueChoice Transitions (§33.6). [D3 §2.7; D2 §2.3; Part I §3.4]

FTAI's filings describe this chain directly, in the language §35.1 quoted: engines unserviceable for leasing are torn down "into salvageable components that would be manufactured into new modules and parts and sold", and the business sells "engines, modules and parts" under the name Maintenance, Repair and Exchange (MRE). [FTAI 10-K FY2025; D5 §2.2] What no provider publishes is the exchange price. The fee above the life transferred was not found for any provider, including FTAI, which is the cluster K-01 that three dossiers raised; the only bound the research holds is FTAI's products revenue divided by its modules produced, \$2.11M for 2025, which Part VII §56 presents as an upper bound and not a price because the numerator includes whole engines, parts and the fair value of the units handed back. [reconciliation K-01; E.1 rule 11; Part VII §47, §56]

THE TWO MEANINGS OF "CORE"

The word carries two meanings in this primer and the registry flags the conflict. In the anatomical sense, used throughout Parts I to VI, the core is the gas generator: the high-pressure compressor, combustor and high-pressure turbine, the hot module whose disks have the shortest life limits and which therefore dominates the life-limited-part arithmetic above. In the exchange sense, used by FTAI's segment and in Part VII, the core is the unserviceable unit a customer hands back in an exchange, which has value to the exchanger as feedstock and is measured at fair value as noncash consideration. A sentence such as "the core is worth more at 80% life" is about the first; "the exchange fee plus the core" is about the second. This Part writes "core module" or "gas generator" for the first and "returned core" or "the unit handed back" for the second. [registry C-6; D1 §2.3; D5 §2.3; Part VII §47]

Two constraints limit which torn-down modules fit which pools, and both come from the manual rather than the market. A -7B24 and a -7B27 differ in value, \$5.7M against \$6.4M on the appraiser's half-life market values, and the interchange of modules and life-limited parts across thrust ratings and across the -5B and -7B is governed by the shop manual; the -5B and -7B share enough core hardware that dealers advertise a single "CFM56-5B & -7B Core LLP Package", but

the exact list of common parts was not retrieved. [IBA, Sept 2025; Part I §3.3; D3 §5; D1, unknown U-D1-02] Build standard and records, the conditions Part I §3.3 set for a module to move, apply to a module from a teardown exactly as to one from a shop.

The half-life trade of §32.5 closes here. A part-worn engine is leased for its months or torn down; a module from the teardown is valued on the life inside it; the life-limited parts are priced at their fraction of list less a discount; and the module goes into a pool from which an exchange sells that life back to an operator at a price that also buys time. Each step prices the same thing, remaining life, in a different wrapper. Part VII §46 takes the last step in full.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-001	CFM56 in service: ~23,000 (CFM, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); consultancy ~14,200 and >19,000 (Safe Fly)	§33.3, worked example 5.4
R-002	Safe Fly's ~14,200 read as the -7B fleet (D3) or all CFM56 (D8)	§33.2, §33.3
R-004	CFM56 retirement rate: GE 3–4% → 2–3% → 1.5–2% (2026 planning); ~1.5% 2025 actual, ~2% 2026 expected; <1% in 1Q 2026; Safe Fly "near 2%"; McKinsey 24% below norm, 2.7% from 2028	§33.2, Exhibit 5.4; worked example 5.4
R-005	Aircraft retirements 2024: "fewer than 200" (IBA) against 587 (Cirium); Cirium 502/465/333/325 (2019–22); McKinsey-implied ~410 a year against Cirium-projection-implied ~1,130 a year	§33.2, Exhibit 5.4
R-008	Interval: "roughly every eight years", 24-year de facto retirement (Aviation Week) against 10,000–15,000 cycles at 1,000–1,500 a year (Leeham; Aircraft Value News)	§33.1
R-016	Full -7B LLP set: \$5.7M (MyAirTrade, Jun 2025) against ~\$4M (Aircraft Value News, Nov 2019); 2013 \$1.7M (caveat); Calver -5B \$9.88M full life, basis unstated	§32.6 worked example 5.3; §36.2 worked example 5.6; Exhibits 5.3, 5.8
R-018	Escalation: spares ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Aviation Week; Safran) against LLP set ~7%/yr derived, Ishka ~12% step, -7B parts +20–30% in 2023 (Magnetic)	§34.3, worked example 5.5

Id	The two (or more) figures	Where in this Part
R-020	Parts constraint: GE 10-K material input +40%, turnaround +10% against Oliver Wyman labour shortage and AVM "USM supply remains thin"	§33.4
R-021	USM market 2025: \$5.87B (The Business Research Company); \$7.7B (Market.us); \$8.55B (Fortune Business Insights); ">\$4B" (Safe Fly)	§35.2, Exhibit 5.7
R-022	USM price: 60–80% of new, up to 85%, above list (Aviation Week) against 70–75% of list, trader to airline (Aircraft Commerce Issue 120)	§34.1, Exhibit 5.5; worked example 5.5
R-028	-7B rent: ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time)	§32.5 worked example 5.2; §32.6; §33.4
R-029	-7B value: half-life market value \$5.7M/\$6.4M, "levelled off" (IBA) against green-time \$2.8–3.4M "(new condition)", teardown \$0.8–1.2M, rents +15–20% (Safe Fly) and values +50% in two years (ePlaneAI)	§32.5 and its callout; Exhibit 5.3
R-030	Lease rate factor 0.70%/month derived on mixed bases (D3); no stated figure	§32.5 worked example 5.2
R-033	AerCap engines: 470 owned (20-F) against >1,200 including SES (6-K)	§35.1
R-034	Freighter conversion: >260 737-800Fs in service, pipeline >300 (Cargo Facts; Acumen) against IBA's "oversupply of narrowbody conversions" (Sep 2026)	§32.6, closing paragraph
R-035	Module count: four major modules (Part I anatomy) against three per engine (FTAI releases)	§36.1
D3 tension 1	The three vendor market sizes differ by 46%; scopes not retrieved	§35.2
D3 tension 2 (with R-004, R-005, B-6)	Oliver Wyman's 2020 "tsunami of used aircraft parts" (title only) against the 2022–2026 scarcity described by Cirium, McKinsey, GE, AerSale, IBA and Aviation Week	§33.5
D3 tension 5	Same content as R-022	§34.1
D3 tension 6	AerSale's "hypercompetitive feedstock" (buyer's view), IBA's "lessors' market" and Aviation Week's "seller's market" (owner's view): same direction, different vantage	§33.4
D3 tension 7	Direction of USM prices 2024–26: Aviation Week "stabilizing" and "upturn" (undated); IBA "levelled off" (H1 2026); Safe Fly rents +15–20% "in recent months"	§33.5

Unknowns carried in this Part

Entries from the USM dossier (D3), verbatim from the reconciliation register, followed by the entries from other dossiers that the register clusters with them (K-01, K-02, K-15, K-16). Where the orchestrator notes partly resolved an entry, the resolving fact is attached after it.

- **U-D3-01 (D3) — U1** — Sourced split of a 737-800 part-out value between engines and airframe. Would be resolved by an appraiser part-out report or an AerSale/AAR investor presentation with a teardown waterfall.
- **U-D3-02 (D3) — U2** — Total sell-through period for teardown inventory (months to recover, say, 80% of proceeds). AerSale's 10-K inventory-turnover disclosure or a trade-press teardown case study would answer it.
- **U-D3-03 (D3) — U3** — Whether Calver's \$9,878,733 / \$4,939,366 CFM56-5B LLP figure is per engine or per aircraft (two engines). The original Cargo Facts presentation would settle it.
- **U-D3-04 (D3) — U4** — Any mba Aviation (or Ascend by Cirium) CFM56-5B/-7B value or lease-rate figure. None returned.
- **U-D3-05 (D3) — U5** — A current appraiser half-life value for the CFM56-5B (only trend commentary and a trade-press lease-rate range were captured).
- **U-D3-06 (D3) — U6** — Current CLP of representative CFM56-7B LLPs (e.g. HPT disk, fan disk) and of a full -7B LLP set. Needed for a precise module-value example.
- **U-D3-07 (D3) — U7** — The identity and terms of FTAI's "exclusive partnership" for teardown, repair, marketing and sale of CFM56 parts. The 10-K excerpt does not name it. The FTAI 10-K text or an FTAI press release would answer it. *Partly closed (reconciliation C.3 item 1, B-9): a 2025 PRNewswire release is titled "AAR and FTAI Aviation extend their exclusive Serviceable Engine Products agreement, providing CFM56 engine material to the global aviation aftermarket through 2030"; title located, body not read; terms still open.*
- **U-D3-08 (D3) — U8** — Purchase price of a 737-800 bought for part-out in 2025–2026 (and of the airframe alone).
- **U-D3-09 (D3) — U9** — Teardown labour and facility cost per aircraft and per engine.
- **U-D3-10 (D3) — U10** — Values of a 737-800 APU, avionics suite, thrust reversers and other rotables.
- **U-D3-11 (D3) — U11** — AAR FY2026 total revenue and Parts Supply segment revenue in dollars (only the share and growth were captured). *Partly closed (reconciliation C.3 item 6): D11 derives consolidated sales of \$3,308.0M and Parts Supply of about \$1,489M from the FY2026 10-K; the filing's own segment figure was not returned.*

- **U-D3-12** (D3) — **U12** — Willis Lease portfolio value, number of engines, and Willis Aero spare-parts revenue for 2025. *Partly closed (reconciliation C.3 item 5): 363 engines, 20 aircraft, \$2,801.7M of equipment held for lease, 84.9% utilisation (Willis Lease 10-K FY2025, via D11); Willis Aero revenue still open (K-15).*
- **U-D3-13** (D3) — **U13** — AJW Group scale.
- **U-D3-14** (D3) — **U14** — Unical Aviation: nothing returned. *Partly closed (reconciliation D.3): Unical Aviation was FTAI's JV partner in QuickTurn Miami (GlobeNewswire 1 Dec 2023; 10-K). Unical's own USM scale: still open.*
- **U-D3-15** (D3) — **U15** — GE Aerospace's used-material arm: name, scale and CFM56 role not captured (the only hit was a TrueEngine presentation URL whose content was not read).
- **U-D3-16** (D3) — **U16** — Actual retirement counts for 2023, 2024 and 2025 from Cirium (only 2019–2022 counts and the 2024–26 shortfall percentage were captured). *Partly closed (reconciliation C.3 item 4): IBA "fewer than 200" (2024) and Cirium 587 (2024) via D9, unreconciled (R-005); GE ~1.5% of the CFM56 base in 2025.*
- **U-D3-17** (D3) — **U17** — Number of CFM56-5B and -7B engines torn down per year, 2023–2025.
- **U-D3-18** (D3) — **U18** — Scope definitions behind the three USM market-size estimates (what counts as USM; whether engines, airframes and components are all included).
- **U-D3-19** (D3) — **U19** — Oliver Wyman's 2020 forecast figures and any updated Oliver Wyman USM forecast (title only captured).
- **U-D3-20** (D3) — **U20** — Pricing of a module exchange (exchange fee, LLP-life true-up) from any provider, including FTAI. D5 may hold FTAI's own disclosures.
- **U-D2-07** (D2, cluster K-01) — **U7. Module-exchange pricing.** No source gives the exchange fee or a worked price for a CFM56 HPT or core exchange; Safe Fly's "\$150,000–\$400,000 removal cost per module" is ambiguous. D5 may find it in FTAI's investor materials or the In Practise interview ("FTAI Aviation: CFM56 repair and module swap", URL returned, not read).
- **U-D5-03** (D5, cluster K-01) — **U3 — Exchange turnaround time and exchange pricing.** FTAI's quoted turnaround and any published exchange-fee ranges. Trade press (e.g., the In Practise interview [S38]) likely has estimates; not read.
- **U-D1-09** (D1, cluster K-02) — **U9** Current half-life base values for -7B and -5B by rating (IBA, mba, Cirium). Resolve with an appraiser publication; this session's searches returned definitions, not values.
- **U-D2-08** (D2, cluster K-02) — **U8. Value of a freshly overhauled full-stack CFM56-7B** versus a green-time unit. Needed to close the worked example. Resolve via IBA/Ascend/mba engine value tables.

- **U-D6-09** (D6, cluster K-02) — **U9** — **Half-life market values** for 12–15-year-old 737-800 and A320ceo aircraft, and full-life CFM56-7B/-5B/V2500 values, 2023–2026. Resolve: IBA/Cirium/mba published updates (D3 may hold engine values).
- **U-D11-03** (D11, cluster K-15) — **U3** — **Willis Lease maintenance services and spare-parts revenue lines for FY2025**. The 10-K's revenue table was not returned beyond lease rent, the combined core line and gain on sale. Resolve: read the consolidated statement of income in [S1].
- **U-D4-10** (D4, cluster K-16) — **U10** — **Any CFM/GE statement specifically on the Chromalloy HPT blade PMA**, and any GE "material" programme (e.g. TrueChoice Material) positioning against PMA. The brief's phrase "material solutions" was not matched to a current CFM/GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines [51][52].

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PART VI

PMA parts and DER repairs

This Part explains the two legal routes by which a company other than the engine's maker may put a new part into a CFM56 (the engine family Part I describes), or repair one in a way the maker's manual does not describe. It covers the regulation, the hard case of hot-section parts, the European acceptance rules, the market that has grown up around approved alternative parts, how airlines, lessors and the manufacturer have responded, and what is filed and what is only reported about FTAI's own joint venture. A reader who finishes it will be able to read a parts-approval announcement, a lease's parts clause and a claimed per-visit saving with every term understood and every number placed against a total.

TRANSLATION-TABLE ROWS ANSWERED

This Part answers the rows of the Part III translation table that concern the price and supply of new material from the original equipment manufacturer (OEM) for an engine that is kept flying longer than planned. It answers (Part III §24, row T-7: OEM parts escalation), because an approved alternative part is the only legal way to buy a new part for the CFM56 at a price the manufacturer does not set. It answers (Part III §24, row T-9: hot-section material supply), because the high-pressure turbine blade, the largest non-life-limited cost line in a restoration (Part I §7), received its first approved alternative for the -5B and -7B on 30 October 2025. It answers (Part III §24, row T-1: engines kept past planned retirement), because an engine on its third or fourth shop visit needs new hot-section parts that an engine retired on schedule would never have bought. And it touches (Part III §24, row T-6: lease extensions and return conditions), because whether a leased engine may carry an alternative part at all is decided in the lease. What it does not stress is the life-limited-part line: no approved alternative exists for any CFM56 life-limited part, so that line stays single-source at list price (Part I §4). The orchestrator fills the row identifiers.

37. Why PMA exists, and the four approval bases

Every new part that goes into a CFM56 must be made under an approval that a regulator granted to a named company. For most of the engine's life only one company held such an approval for most of its parts: the company that designed it. This section explains why that is so, what it does to prices, and the four ways a second company can obtain an approval of its own.

37.1 THE GATE: WHO MAY MAKE A NEW PART

An aircraft engine is certified as a design, and the company that owns the design is the only one that can make parts to it without a further approval. Part I §9 gave the short form of the two certificates involved; this Part owns them and gives the full form.

TYPE CERTIFICATE (TC) AND TYPE-CERTIFICATE HOLDER (TCH)

A type certificate is the Federal Aviation Administration's (FAA) approval, issued under Title 14 of the Code of Federal Regulations (CFR), of the design of an aircraft, an engine or a propeller, issued after the design has been shown to meet the applicable airworthiness standards. Its holder, the type-certificate holder, is the design's owner and the party responsible for keeping the design airworthy in service. For the CFM56 the holder is CFM International, the 50/50 joint venture of GE Aerospace and Safran Aircraft Engines (Part I §2). The holder's own new spare parts are produced under its production certificate, so for the holder the type certificate is also the gate to the spare-parts business.

[eCFR, 14 CFR Part 21; FAA PMA design-approval page]

PRODUCTION CERTIFICATE (PC) AND PRODUCTION APPROVAL

A production certificate is the FAA's approval of a manufacturer's quality system, under which that manufacturer may produce parts to an approved design and release them as airworthy. The regulations treat the type-certificate holder's production certificate, a Parts Manufacturer Approval and a Technical Standard Order authorization (an approval for standardised articles such as instruments and seat belts) as the three forms of production approval. A replacement or modification part for a certificated product may be installed only if it was produced under one of them, or falls into a narrow exception: standard parts made to a published industry or military specification (the AN/MS hardware of the rules, meaning fasteners and fittings made to published US military AN/MS specifications), parts an owner or operator produces for its own aircraft, and parts a repair station makes in the course of a repair. [eCFR, 14 CFR Part 21 Subpart K; FAA PMA design-approval page]

The practical result is a single source. The manufacturer's new spares come off its own production certificate. Anyone else who wants to sell a new part for the engine must first obtain a Parts Manufacturer Approval, which §37.2 defines. Until that approval exists, the manufacturer is the only legal seller of that part number new. [eCFR, 14 CFR Part 21 Subpart K]

Used material is the other legal source of a part, and it sits outside this debate. Used serviceable material (USM), the parts recovered from retired or dismantled engines and re-released, was made by the manufacturer under its own production certificate, so no further design approval is needed to install it (Part V §31). But it competes for the same slot in a shop visit's parts bill as a new alternative part, and Part V §34 and this Part's §44 should be read together for that reason. [D4 §2.1; D3 §2.1]

Why does the single source matter enough to have produced a regulation, an industry and an antitrust complaint? Because of what it does to prices. The manufacturer publishes a catalogue list price for every spare part and revises it upward on a schedule; Part IV §28 carries the full treatment and Part I §4.3 the life-limited-part series. The figures the research reached are these. New engine spare-parts prices had been rising around 10% a year in 2022 and 2023, with a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023, and Safran's chief executive described the rule as pricing "above inflation by 3 to 4 points" for "at least for some years". [Aviation Week, late 2023] A full CFM56-7B life-limited-part set compounded at about 7% a year from 2008 to 2025 on the trade-press series Part I assembled (derived in Part I §4.3). [Part I, worked example 1.3] One independent shop reported that -7B parts rose 20–30% in 2023 after the August catalogue revision. [Magnetic Group via Aviation Week, 2024] Those readings cover different parts and years and the primer carries all of them. [reconciliation R-018]

The escalation lands on a bill that is mostly material. Material is 60–70% of a shop visit, labour 20–30%, repairs and other shop work 10–20%, according to one trade-press breakdown. [Air Cargo Week, 2025/26] And the CFM56 aftermarket is mostly transactional: Safran describes about

85% of its CFM56 aftermarket revenue as spare parts and time-and-materials work rather than long-term contracts. [Safran via Aviation Week, c. 2023/24] So on this engine the manufacturer's main capture is the parts line, whoever runs the shop, and a shop that is not the manufacturer has only three ways to reduce that line: used parts instead of new (Part V), repair instead of replacement (§40), and an approved alternative part instead of the manufacturer's (§37.2 onward). [D2 §2.2]

WORKED EXAMPLE 6.1 — WHAT CATALOGUE ESCALATION DOES TO ONE BLADE'S PRICE, AND TO THE SAVING FROM AN ALTERNATIVE

Inputs: a new manufacturer's CFM56 high-pressure turbine (HPT) blade at about \$20,000 (sourced: In Practise expert interview, an interview digest, undated; the alternative listing of about €35,000 is in §44). Escalation of about 10% a year (sourced: Aviation Week, late 2023, "around 10% per year" for new spares) or 7% a year (derived: Part I §4.3 on the life-limited-part series). Five years (assumed). A set of 80 blades (a recollection, unverified in the research, D4 U16).

at 10% a year: $20,000 \times 1.10^5 = 32,210$ per blade; set of 80 = 2,576,800
 at 7% a year: $20,000 \times 1.07^5 = 28,051$ per blade; set of 80 = 2,244,080
 starting set of 80: $20,000 \times 80 = 1,600,000$

So a part priced at \$20,000 today is priced at roughly \$28,000 to \$32,000 in five years on either escalation reading, and a set of them moves from \$1.6M to \$2.2–2.6M. If an alternative part sells at a fixed discount d to the manufacturer's price, the dollar saving per set grows with the manufacturer's price: at $d = 30\%$ it is \$480,000 today and \$670,000 to \$770,000 in five years. Limits of the example: the per-blade price comes from an interview, not a catalogue; the blade count is unverified; whether any alternative maker prices as a fixed discount is not disclosed (D4 U2); and §44 shows that the same blade line, priced from a consultancy's cost shares instead of per blade, comes out several times smaller (R-015). The point here is the mechanism: escalation on a single-source part is what gives a second source its market. [In Practise, undated; Aviation Week, late 2023; D4 §2.8]

37.2 WHAT A PARTS MANUFACTURER APPROVAL IS

Part I §5 gave a short definition because the high-pressure turbine blade needed one there. This is the full form.

PARTS MANUFACTURER APPROVAL (PMA)

A Parts Manufacturer Approval is the FAA's combined design-and-production approval, issued under 14 CFR Part 21 Subpart K, that lets a company other than the type-certificate holder legally make and sell a replacement or modification part for a type-certificated aircraft, engine or propeller. It is a design approval, in that the FAA agrees the part's design meets the airworthiness standards that apply to the product it goes on, and a production approval, in that the FAA agrees the holder's quality system can make every unit to that design. It is issued as a letter with a supplement listing each approved part number and the models it is eligible for; the parts are marked "FAA-PMA" with the maker's name and part number, and each shipment is released on FAA Form 8130-3. There are four approval bases (§37.3). PMA has historically concentrated in consumables and expendables; a cooled, coated turbine blade approved in October 2025 is treated in the trade as the hot-section milestone (§38). No PMA exists for any CFM56 life-limited part. [eCFR, 14 CFR Part 21 Subpart K and 14 CFR 45.15; FAA Order 8110.42D; FAA PMA design-approval page; D4 U7]

14 CFR PART 21 SUBPART K

Part 21 Subpart K is the section of the FAA's certification regulations under which Parts Manufacturer Approvals are issued. Section 21.303 sets the design showing: the applicant must provide "test reports and computations necessary to show that the design of the article meets airworthiness requirements, unless the applicant shows that the design is identical to an article covered under a type certificate", and where the design came by a licensing agreement, evidence of that agreement. Section 21.307 requires a quality system meeting the standard of section 21.137 (the same standard a production-certificate holder meets), and section 21.316 lists the holder's continuing obligations once approved. [eCFR, 14 CFR Part 21 Subpart K]

Two FAA offices divide the work. The design side is handled by an Aircraft Certification Office (ACO), or by engineers the FAA has designated to act for it (§40 describes the designee system). The production side is handled by a Manufacturing Inspection District Office (MIDO), which inspects the applicant's quality system and the conformity of the test articles. Both must be satisfied before the supplement is issued. [FAA Order 8110.42D; FAA PMA design-approval page] The release tag that accompanies each shipment is FAA Form 8130-3, which Part I §9 defined: for a new part it certifies that the part was produced under an FAA production approval and conforms to its approved design. A part not produced under a production approval cannot carry one. [Aviation Suppliers Association, 2016, via Part I §9]

37.3 THE FOUR APPROVAL BASES

Section 21.303 draws one line: either the applicant proves the design by its own tests and calculations, or it shows the design is identical to the manufacturer's. FAA guidance, in Advisory Circulars (AC) 21.303-3 and 21.303-4 and Order 8110.42D, turns that line into four approval

bases, which differ in what the applicant has to possess and what it has to prove. [FAA AC 21.303-3; FAA AC 21.303-4; FAA Order 8110.42D]

TEST AND COMPUTATION (T&C)

Test and computation is the PMA basis an applicant uses when it has none of the manufacturer's design data. It reverse-engineers the part: it measures examples of the manufacturer's part, determines the material and any coatings, and writes its own drawings and process specifications. It then proves by its own analysis and testing that the part meets the airworthiness standards that apply to the product (for an engine part, 14 CFR Part 33) and does not degrade the engine's own compliance. It is the only basis on which a company can compete with the manufacturer on a part the manufacturer has not licensed, and Chromalloy's CFM56 high-pressure turbine blade is a test-and-computation part. [FAA AC 21.303-3; FAA AC 33-8; D4 §2.2]

IDENTICALITY (WITHOUT A LICENSING AGREEMENT; BY LICENSING AGREEMENT)

Identity is the PMA basis in which the applicant shows that its design is identical to the type-certificate holder's in every respect that matters. Without a licence, the applicant usually holds the manufacturer's drawings legitimately, for example as a former supplier; FAA Notice 8110.55 of 1996 set the policy. With a licence, the type- or supplemental-type-certificate holder has licensed its design data and the licence itself is the evidence; this is how a manufacturer's subcontractor, or a company the manufacturer has chosen to work with, obtains a PMA. For a critical or life-limited part approved by identity the FAA additionally requires evidence that the manufacturing, inspection and test processes are identical, not only the geometry (§38). [FAA Notice N 8110.55, 1996; FAA Order 8110.42D; FAA AC 21.303-4]

SUPPLEMENTAL TYPE CERTIFICATE (STC)

A Supplemental Type Certificate is the FAA's approval of a change to a type-certificated design. As a PMA basis, the part is the subject of its own design approval, the STC, and the PMA becomes the production approval for that STC's parts. The STC route matters for international acceptance: under the bilateral agreement between the FAA and the European Union Aviation Safety Agency (EASA), a critical part approved by test and computation is accepted in Europe only through an STC that EASA has validated (§39). The same instrument is the approved modification design a conversion house owns for a passenger-to-freighter conversion, which Part II §15 glossed. [FAA AC 21.303-4; FAA-EASA Technical Implementation Procedures (TIP) Rev 7; D4 §2.2, §2.5]

EXHIBIT 6.1 — THE FOUR PMA APPROVAL BASES: WHAT THE APPLICANT HOLDS, WHAT IT MUST SHOW, AND WHO USES EACH

Basis	What the applicant holds	What it must show the FAA	Typical user	CFM56 example in the research
Test and computation	No manufacturer data; its own reverse-engineered drawings and process specifications	Test reports and computations proving the design meets the applicable standards (14 CFR Part 33 for an engine part) without degrading the engine's compliance; for turbine parts, the showings of AC 33-8	An independent competing on an unlicensed part	Chromalloy CFM56-5B/7B HPT blade, approved 30 Oct 2025
Identity without a licence	The manufacturer's drawings held legitimately (e.g. a former supplier)	That the design is identical; for a critical or life-limited part, that processes and inspections are identical too	A former or current supplier acting on its own account	None found
Identity by licence	A licence from the TC or STC holder to its design data	Evidence of the licence, which establishes that the data are FAA-approved and identical	A manufacturer's subcontractor or chosen partner	None found
Supplemental Type Certificate	Its own approved design change	The STC's own showing; the PMA is then the production approval for the STC's parts	Any applicant needing a separate design approval, including for EASA acceptance of a critical part	None found

Source: eCFR, 14 CFR 21.303; FAA Advisory Circulars 21.303-3 and 21.303-4; FAA Order 8110.42D; FAA AC 33-8; FAA Notice N 8110.55 (19 July 1996); Chromalloy release, 30 Oct 2025. Compiled in D4 §2.2. Accessed 2026-10-03.

Read across the rows and the pattern is that the harder the data are to obtain, the more the applicant must prove. A licensee proves almost nothing beyond the licence. A former supplier proves identity. An independent with no data proves airworthiness from scratch. That is why the independent PMA industry grew up in parts whose airworthiness is easy to prove, and why §38 is a separate section. [D4 §2.2–2.3]

WHY THE FOURTH BASIS MATTERS ABROAD

Inside the United States a PMA part approved by any of the four bases is installable on any eligible engine. The bases matter abroad. The European acceptance rules (§39) accept an FAA PMA directly only in defined cases, and for a part the manufacturer designated as critical the only direct route is a Supplemental Type Certificate that the European agency has itself validated. So an applicant's choice of basis is partly a choice of addressable fleet: a test-and-computation approval alone may serve US-registered engines, while serving EU-registered engines may require the fourth route as well. Whether that applies to the CFM56 turbine blade and vane is one of this Part's open questions. [FAA–EASA TIP Rev 7; D4 §2.5, U5]

37.4 THE PRODUCTION SIDE

An approved design is half of a PMA. The other half is an approved factory. The applicant's quality system must meet 14 CFR 21.137, the same standard applied to the type-certificate holder's own production certificate. Before issuing the supplement the FAA checks the conformity of the test articles to the drawings, reviews the test and analysis reports, and inspects the quality system. After issue, section 21.316 obliges the holder to keep the system in place, to report failures and defects, and to allow FAA inspection. [eCFR, 14 CFR 21.307 and 21.316; FAA Order 8110.42D] Every unit then leaves on an 8130-3 that references the PMA. The approval is therefore to a company and a part number, not to a design in the abstract: a second company wanting to make the same part must obtain its own PMA, and a PMA holder that changes its process must show the change does not alter conformity. [FAA Order 8110.42D; D4 §2.2]

38. Critical parts and the hot-section problem

Alternative parts were for decades a business in seals, bushings, fasteners, filters, brackets and non-rotating structure. The parts that cost the most in a CFM56 shop visit are not those. This section explains the two classifications that govern what a maker must prove, why a turbine blade sits at the hard end of both, and what the research found about how long the hard case takes.

38.1 TWO DEFINITIONS THAT ARE EASY TO CONFUSE

Two different categories are in play, and they come from two different documents.

The first is the life-limited part, which Part I §4 defined: a rotating or pressure-bearing part whose failure the engine case cannot contain, certified for a fixed number of flight cycles and scrapped at that limit regardless of condition. On the -7B the limits are 20,000 cycles in the high-pressure compressor and turbine, 25,000 in the low-pressure turbine and 30,000 in the fan and

booster. [Part I §4.1; AJW (an engine trader) and StandardAero records mini-packs] The list of such parts, and each one's limit, sits in a specific section of the engine's certification data.

AIRWORTHINESS LIMITATIONS SECTION (ALS) AND INSTRUCTIONS FOR CONTINUED AIRWORTHINESS (ICA)

The Instructions for Continued Airworthiness are the manufacturer's approved maintenance instructions for a type-certificated product, attached to the type certificate: the Engine Shop Manual and the component maintenance manuals (Part I §9) are part of them. The Airworthiness Limitations Section is the part of those instructions that lists the mandatory life limits and inspection intervals, which no operator may exceed; every life-limited part's cycle limit appears there. The two matter in this Part because a part's appearance in the Airworthiness Limitations Section is the usual trigger for the "critical" designation below, and because a repair that departs from the Instructions for Continued Airworthiness needs separate approval (§40). These definitions are drawn from first principles and from the regulatory structure the dossier cites; the research did not quote the regulation's own text.

[D4 §2.3, §2.6; registry, Part VI row]

The second category comes from the agreement between the FAA and its European counterpart, and it is the one that decides how an approval travels.

CRITICAL COMPONENT (CRITICAL PART)

Under the FAA–EASA bilateral's Technical Implementation Procedures, a critical component is "a part identified as critical by the design approval holder during the product certification process or otherwise by the Authority for the State of Design. Typically, such components include parts for which a replacement time, inspection interval, or related procedure is specified in the Airworthiness Limitations section or certification maintenance requirements." Every life-limited part is therefore a critical component. Whether a given turbine blade or vane is one depends on how the type-certificate holder designated it during certification, and that designation is not public in anything the research retrieved. The classification matters because it selects which European acceptance clause applies (§39) and because, for an identity approval, it adds the requirement that processes and inspections be identical, not only geometry. [FAA–EASA TIP Rev 7, as returned by search; FAA Order 8110.42D; D4 §2.3, U5]

EXHIBIT 6.2 — THREE CLASSES OF CFM56 PART AND WHAT EACH MEANS FOR AN ALTERNATIVE MAKER

Class	Defining document	CFM56 examples	Replaced when	What a PMA applicant must additionally show	PMA found in the research
Life-limited part	Airworthiness Limitations Section of the engine's certification data	Fan disk, booster spool, high-pressure compressor (HPC) spools and disk, HPT disk, low-pressure turbine (LPT) disks, shafts, certain seals	At the certified cycle limit, whatever its condition	Critical by definition under the bilateral; identity requires identical processes; T&C requires a full life substantiation	None for any CFM56 life-limited part (D4 U7)
Critical component that is not life-limited	TCH designation during certification; typically carries an inspection interval in the Airworthiness Limitations Section	Not public for the CFM56 in the sources retrieved	On condition or at an interval	Same process-identity requirement; EASA acceptance only via a validated STC if approved by T&C	Unknown whether the HPT blade and vane are so designated (U5)
On-condition hot-section part	Engine Shop Manual limits	HPT blades, HPT stage-1 vanes (the HPT nozzle), combustor liners, LPT blades and vanes, shroud segments	When inspection finds it beyond repair limits	For T&C: material, cooling, durability and safety-analysis substantiation per AC 33-8	HPT blade (30 Oct 2025); HPT vane 1; LPT vane; HPT shroud per one reading (§41)

Source: FAA–EASA TIP Rev 7 (definition of critical component, as returned by search); FAA Order 8110.42D; FAA AC 33-8; Part I §4 (life-limited parts and limits); Chromalloy releases, Oct–Nov 2025; D4 §2.3 and U5, U7. Accessed 2026-10-03.

The third row is where the money in a restoration sits. The high-pressure turbine blades and vanes are not life-limited; they are replaced when inspection finds them beyond the manual's repair limits, and that happens at the hot end of the engine faster than anywhere else. Part I §5

explained that most of the lost temperature margin is recovered in that module, and one consultancy puts the blades and vanes at 15–25% of a shop visit's parts cost. [Safe Fly Aviation, 2026, consultancy blog; Part I §5] So the on-condition hot-section row, not the life-limited row, is where an approved alternative changes a bill.

38.2 WHY A TURBINE BLADE IS THE HARD CASE

Two regulatory facts make the hot section hard for an applicant with no manufacturer data.

The first is the showing itself. For a test-and-computation approval of a turbine engine part the FAA's product-specific guidance is Advisory Circular (AC) 33-8. Under it the applicant must substantiate the part's material properties, its cooling and thermal behaviour, its durability, and its effect on the engine's safety analysis by its own testing, because it has no access to the manufacturer's data on any of them. [FAA AC 33-8; D4 §2.3] A high-pressure turbine blade is a cooled casting, grown as a single crystal and coated, sitting at the hottest point in the engine. This primer names those three properties and stops, since the depth that matters here is what they do to the approval: each is a property the applicant must measure, reproduce and prove, with no reference data from the designer.

The second is the identity rule for critical and life-limited parts. If the basis is identity, the FAA's procedures require the applicant to show that the manufacturing processes (casting, forging, heat treatment, coating) and the inspection and test procedures produce a part identical to the original, not merely that the drawing matches. [FAA Order 8110.42D, as cited in D4 §2.3] For a part whose life depends on its microstructure, that is a showing few companies outside the manufacturer's own supply chain can make.

WHY A BLADE IS HARDER THAN A BRACKET

A bracket's airworthiness is a matter of geometry, material and static strength, all of which an applicant can measure from an example and prove by a calculation and a pull test. A cooled turbine blade's airworthiness is a matter of how a particular alloy, grown as a single crystal and coated, behaves over thousands of hours at the hottest point in the engine, under centrifugal load, with cooling air flowing through passages whose shape the applicant must infer. The regulation does not forbid an independent from proving that; AC 33-8 tells it how. But the proof is an engine-level test and durability programme the applicant funds itself, and the trade treats the first such approval for the CFM56-5B/7B, announced on 30 October 2025, as a milestone for exactly that reason: the applicant called its blade "the only aftermarket alternative to the OEM manufactured new part" for a platform with "over 20,000 engines" in service. [FAA AC 33-8; Chromalloy release, 30 Oct 2025; D4 §2.3]

Those two facts explain the historical shape of the industry. Approved alternatives concentrated in consumables and expendables, meaning the seals, bushings, fasteners, filters, brackets and non-rotating structural pieces whose airworthiness can be proven with modest test programmes.

[D4 §2.3] They did not reach life-limited rotating parts at all: the research found no PMA for any CFM56 disk, spool or shaft, and that line of the parts bill remains single-source. [D4 U7; Part I §4.4] And they reached the cooled hot-section airfoils only recently, and only after a programme whose length §38.3 sets out as far as the sources allow.

38.3 HOW LONG A PMA TAKES

Neither the FAA nor the companies disclose the interval from application to approval in anything the research retrieved, so this is an open question (D4 U4). Two datapoints bracket it. At the simple end, the annual report of HEICO (the largest independent PMA maker, §41.1) says the company has been adding new approvals at a rate of "approximately 300 to 500" a year by one dossier's reading of the filing, or "approximately 400 to 550" by another's, across a portfolio dominated by less complex parts; simple approvals are evidently a matter of months, not years. [HEICO 10-K FY2025, two readings; reconciliation R-024] At the hard end, third-party commentary dates the FTAI–Chromalloy partnership to 2018 and the FAA approval of the CFM56 turbine blade came on 30 October 2025, about seven years later; the application date is not public, and the 2018 date is itself an investor-blog figure that §43 sets beside what is filed. [Komodo Capital and Kairos Research, investor blogs, undated; Chromalloy release, 30 Oct 2025; D4 §2.4] The FAA's own registry of approvals would carry the dates, and the research did not read it. [D4 U4]

39. EASA and the bilateral

An FAA approval is a US approval. For engines registered in the European Union, acceptance of US approvals is governed by a treaty and the procedures written under it. This section sets out those procedures as the research found them and names the question they leave open for the CFM56 hot-section parts.

TECHNICAL IMPLEMENTATION PROCEDURES (TIP)

The Technical Implementation Procedures are the document by which the FAA and the European Union Aviation Safety Agency (EASA) implement the US–EU Bilateral Aviation Safety Agreement for airworthiness and environmental certification; the current edition is Revision 7 with Amendment 1. Among much else, they set the rules under which EASA accepts FAA Parts Manufacturer Approvals for installation on EASA-certificated or EASA-validated products. As returned by search, EASA accepts an FAA PMA directly, without further showing, when the part is not a critical component and was approved by test and computation or by identity without a licence; or when it conforms to design data licensed from a type- or supplemental-type-certificate holder whose certificate EASA has validated; or when it is a critical component approved through an FAA supplemental type certificate that EASA has validated. [FAA–EASA TIP Rev 7 with Amendment 1; TIP Rev 6 (PDF); D4 §2.5]

A caution on the wording. The dossier recorded that a search summary rendered the first clause as "identity without a licensing agreement" and advised that the exact text of Revision 7 be checked before it is quoted. The research did not read the document itself, so this Part reports the three clauses as the dossier recorded them and does not quote them. [D4 §2.5]

EXHIBIT 6.3 — THE THREE TIP ACCEPTANCE CLAUSES FOR AN FAA PMA PART ON AN EU-REGISTERED ENGINE, AS THE RESEARCH RECORDED THEM

Clause	Condition on the part	Condition on the approval basis	What accompanies the part	Where the CFM56 HPT blade and vane fall
1	Not a critical component	Approved by test and computation, or by identity without a licensing agreement	FAA Form 8130-3 with the EASA-acceptance statement; no further showing	Here, if the TCH did not designate them critical (unknown, D4 U5)
2	Any	Conforms to design data obtained under a licence from a TC or STC holder, where EASA has validated that TC or STC	8130-3; no further showing	Not applicable: the blade is a T&C part, not a licensed copy
3	A critical component	Approved through an FAA STC that EASA has validated	8130-3 referencing the validated STC	Here, if the TCH designated them critical and no validated STC exists (unknown, U5)

Source: FAA–EASA Technical Implementation Procedures Rev 7 with Amendment 1 (web) and Rev 6 with Amendments 1 and 2 (PDF), as returned by search and recorded in D4 §2.5; the exact clause wording was not read. FAA EASA FAQ (PDF) for the Form 1 equivalence. Accessed 2026-10-03.

The practical consequence is that a non-critical test-and-computation part travels to Europe on its FAA release tag, while a critical test-and-computation part needs the supplemental-type-certificate route plus EASA's validation of that certificate. [D4 §2.5] Part I §9 defined EASA Form 1 as the European release certificate, the equivalent of the 8130-3, issued by a holder of a European production organisation approval; a US-made PMA part moving into the EU system under the clauses above does so on its 8130-3, not on a Form 1, since its maker holds an FAA approval rather than a European one. [FAA EASA FAQ; Part I §9]

The question the research could not close is the one that decides the addressable fleet. Whether CFM International designated the CFM56-5B/7B high-pressure turbine blade and stage-1 vane as critical components during certification is not public in the sources retrieved; whether Chromalloy has obtained EASA acceptance for them by any clause was likewise not found. [D4

U5] If the parts are not critical, clause 1 applies and the FAA approval of October 2025 serves EU-registered engines directly. If they are critical, clause 3 applies and a validated supplemental type certificate is required, which the research did not find. This Part presents both branches and does not choose between them. Acceptance by other authorities, including the UK Civil Aviation Authority (CAA), Transport Canada Civil Aviation (TCCA), Brazil's ANAC (its national civil aviation agency), the Civil Aviation Administration of China (CAAC) and India's Directorate General of Civil Aviation (DGCA), runs under separate bilaterals and national rules, and none was retrieved. [D4 U6]

40. DER repairs

A new part, whoever makes it, is one way to restore an engine. Repairing the part already in it is the other, and the manufacturer's manual does not describe every repair that is possible. This section explains who may approve a repair the manual does not cover, on what document, how such a repair differs from one taken from the manual, who develops and uses them, and how they save money. Part I §9 gave the short form of the designee and the repair; this Part owns both.

40.1 THE DESIGNEE SYSTEM

The FAA does not have enough engineers to examine every set of engineering data itself. It appoints private engineers to do so on its behalf within defined limits.

DESIGNATED ENGINEERING REPRESENTATIVE (DER)

A Designated Engineering Representative is an individual engineer appointed by the FAA under 14 CFR Part 183 to examine engineering data and make findings of compliance on the FAA's behalf within a delegated technical area, such as structures, powerplant, engines or systems. A company DER is employed by a manufacturer or repair station and may act only on that company's data; a consultant DER may act for any client. To approve the data for a major repair the designee needs a specific authorisation, "Special – Major Repairs", from the managing Aircraft Certification Office, granted one-time or as part of the designee's standing delegation. The governing document is FAA Order 8110.37, the DER Handbook, currently revision F. [FAA Order 8110.37F; D4 §2.6]

40.2 MANUAL REPAIRS, MAJOR REPAIRS AND THE FORM THAT APPROVES THEM

The manufacturer publishes its approved repairs in the Engine Shop Manual and the component maintenance manuals, which Part I §9 defined and which form part of the Instructions for Continued Airworthiness attached to the type certificate (§38.1). A shop that follows the manual needs no further approval: the manual is already FAA-approved data. [D4 §2.6] The question arises when a shop wants to do something the manual does not describe.

MAJOR REPAIR

A major repair is a repair the manufacturer's manual does not cover, or a repair of a part the manual says must be scrapped, that does not qualify as minor. Major repairs "must be accomplished in accordance with technical data approved by the Administrator", meaning the FAA or someone acting under its delegation. The approval covers the data: the repair scheme, the substantiating analysis, the process specifications and any test. The repair itself is then performed by a certificated repair station under its own certificate. [FAA Order 8110.37F; FAA Order 8300.16; D4 §2.6]

FAA FORM 8110-3 (STATEMENT OF COMPLIANCE WITH AIRWORTHINESS STANDARDS)

Form 8110-3 is the document on which a Designated Engineering Representative records an approval, or a recommendation for approval, of engineering data, including the data for a repair or alteration. The designee keeps the original and sends copies to the FAA's managing office and to the owner, operator or repair station that requested the approval. The form approves data, not a part: the repair station that performs the repair releases the repaired part on FAA Form 8130-3 (Part I §9), and that release references the 8110-3 under which the repair data were approved. The FAA notes that designee-approved data "may not be adequate to cover every aspect of the repair", which is why the repair station's own certificate and release remain necessary. [FAA Order 8110.37F; FAA, Documenting Compliance Findings (Form 8110-3 guidance); D4 §2.6]

DER REPAIR

A DER repair is a major repair whose data a Designated Engineering Representative approved on FAA Form 8110-3, rather than being taken from the manufacturer's shop manual. The approved data may be one-time, covering a specific serial-numbered part, or multiple-use, repeatable on any eligible part; a multiple-use approval is what turns a repair into a product a shop can sell. The part remains the manufacturer's part; nothing is remanufactured. A DER repair saves money by restoring a part the manual would scrap, so that no new part of any origin is bought, or by restoring it more cheaply than the manual's repair. [FAA Order 8110.37F; D4 §2.6–2.7; D2 §2.2]

EXHIBIT 6.4 — MANUAL REPAIR, DER REPAIR AND PMA PART: WHO APPROVES WHAT, ON WHICH DOCUMENT, AND HOW EACH SAVES

Route	What it is	Source of the engineering data	Who approves the data	Document	What leaves the shop	How it saves	Acceptance questions
Manual repair	A repair described in the Engine Shop Manual or component manual	The type-certificate holder	The FAA, through its approval of the manual as part of the Instructions for Continued Airworthiness	The manual itself; part released on 8130-3	The manufacturer's part, repaired to the manufacturer's scheme	Repair instead of a new part	None: accepted everywhere the manual is
DER repair	A major repair outside the manual, or of a part the manual scraps	An independent engineering organisation, repair house, airline or maintenance, repair and overhaul (MRO) shop	An FAA designee with "Special – Major Repairs" authorisation	FAA Form 8110-3 (data); 8130-3 (release) referencing it	The manufacturer's part, repaired to an independently approved scheme	Repairs what the manual would scrap, or repairs more cheaply	Lease clauses may exclude it (§42); EU acceptance of the data not verified (D4 U8)
PMA part	A new part made by someone other than the type-certificate holder	The PMA holder (T&C or identity) or a licensor	The FAA (Aircraft Certification Office and Manufacturing Inspection District Office)	PMA letter and supplement; 8130-3 per shipment	A new, non-manufacturer part marked "FAA-PMA"	New part at a price the manufacturer does not set	Lease clauses may exclude it (§42); TIP clauses decide EU acceptance (§39)

Source: FAA Order 8110.37F (DER Handbook); FAA Order 8300.16; FAA Form 8110-3 guidance; eCFR 14 CFR Part 21 Subpart K; FAA–EASA TIP Rev 7; D4 §2.2, §2.5, §2.6 and §2.9. Accessed 2026-10-03.

How a DER repair differs from a manual repair comes down to three things, and the table shows them: the source of the data (independent engineering against the type-certificate holder), the approving authority (an FAA designee against the FAA's approval of the manufacturer's manual), and acceptance by third parties, meaning lessors and foreign regulators, which §42 and the open question on EU acceptance address. [D4 §2.6] The manufacturer's own 2018 commitments to the

International Air Transport Association (IATA), described in §42.3, refer throughout to "non-CFM parts and/or repairs", treating PMA parts and DER repairs as one category for warranty and service purposes. [IATA–CFM agreement, 2018; MARPA (the Modification and Replacement Parts Association, a PMA trade body), 1 Aug 2018]

40.3 WHO DEVELOPS AND USES THEM, AND HOW THEY SAVE

Three kinds of organisation develop DER repairs. Independent turbine-part repair houses are the first: Chromalloy's historic core business is turbine-part repair and coating, and the trade press describes it as positioned in the "PMA and designated engineering repair sector". [FlightGlobal, Chromalloy profile, undated; D4 §2.6] Airlines with in-house engineering are the second: Southwest Airlines was reported as evaluating PMA and DER options for its CFM56-7B fleet as it phases out its 737-700s. [Aviation Week, c. 2023] Independent MRO shops are the third. [D4 §2.6]

The saving is on the repair line, and the size of that line is the first thing to establish. Repairs and other shop activities are 10–20% of a shop visit in the trade-press breakdown Part I carried; material is 60–70%. [Air Cargo Week, 2025/26] A repair that keeps a part in service avoids a purchase on the material line, so its saving is the new part's price less the repair's cost, and the research did not reach a repair price for any CFM56 hot-section part. What it did reach is the cadence that decides whether a repair is available at all. According to an expert interview, at a first shop visit, around 20,000 cycles, airlines typically repair the high-pressure turbine blades rather than replace them; at a second, around 30,000 cycles, the hot-section parts typically cannot be repaired again and a new part is needed. [In Practise, undated, interview digest] So a repair, manual or DER, is the lever at the first visit, and a new part, manufacturer's or PMA, is the lever at the second. The manufacturer has been working the repair lever too: CFM says it has industrialised about 400 CFM56 repairs since early 2023, meaning it has developed them into repeatable production processes, including automated laser welding of high-pressure turbine blades that cuts repair turnaround by about a third. [CFM International, "CFM boosts HPT blade output and repair", 2025/26, date not captured]

WORKED EXAMPLE 6.2 — SIZING THE REPAIR LINE AND THE NEW-PARTS LINE IN A RESTORATION FROM THE PUBLISHED SHARES

Inputs: a CFM56-7B performance restoration at \$1.2–1.6M (sourced: Safe Fly Aviation, consultancy blog, 2026, the range Part IV uses) or \$1.8–2.5M (sourced: Safe Fly Aviation, other page, 2026, the range Part I uses); both are carried under reconciliation rule E.1.4. Shares: material 60–70%, repairs and other shop work 10–20% (sourced: Air Cargo Week, trade press, 2025/26). HPT blades and vanes 15–25% of parts cost (sourced: Safe Fly Aviation, consultancy blog, 2026).

repair line (10–20% of the visit):

$$0.10 \times 1,200,000 = 120,000; \quad 0.20 \times 1,600,000 = 320,000$$

$$0.10 \times 1,800,000 = 180,000; \quad 0.20 \times 2,500,000 = 500,000$$

material line (60–70% of the visit):

$$0.60 \times 1,200,000 = 720,000; \quad 0.70 \times 1,600,000 = 1,120,000$$

$$0.60 \times 1,800,000 = 1,080,000; \quad 0.70 \times 2,500,000 = 1,750,000$$

HPT blades and vanes (15–25% of the material line):

$$0.15 \times 720,000 = 108,000; \quad 0.25 \times 1,120,000 = 280,000$$

$$0.15 \times 1,080,000 = 162,000; \quad 0.25 \times 1,750,000 = 437,500$$

So on the published shares the repair line of a restoration is \$0.12–0.50M and the material line \$0.72–1.75M, depending on which consultancy range is used, and the turbine blades and vanes within material are \$0.11–0.44M. A DER repair that keeps a blade set in service at a first visit saves against the material line; a PMA blade at a second visit reduces it. Limits of the example: both consultancy ranges are contested (R-013); the shares come from two secondary sources, neither of which names a shop-level data set (R-014); and §44 shows that the same blades and vanes, priced per blade from an interview, cost several times these figures (R-015). The example sizes the lines on one set of sources; it does not reconcile them. [Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; reconciliation R-013, R-014, R-015]

One question the research could not answer belongs here. EASA's equivalent of a DER repair is a repair design approved by a holder of a European Design Organisation Approval, and how FAA designee-approved repair data are accepted on EU-registered engines is governed by the Technical Implementation Procedures; the research did not verify the relevant clause. [D4 U8]

41. The PMA market: HEICO, Chromalloy and others

The approved-alternative-parts business has one listed company large enough to be read from its filings, one private company that holds the CFM56 hot-section approvals, a handful of others, and a set of market-size estimates that do not agree. This section takes them in that order and ends with what the research found about how far alternative parts have penetrated each category of part.

41.1 HEICO FLIGHT SUPPORT GROUP

Part IV §29 glossed the company; this Part carries the full treatment, and Part XI §87 the peer comparison.

HEICO FLIGHT SUPPORT GROUP (FSG)

The Flight Support Group is the larger of HEICO Corporation's two operating groups and the largest independent PMA business. HEICO, listed on the NYSE, describes the group as one that "designs, manufactures, repairs, overhauls and distributes jet engine and aircraft component replacement parts, which are approved by the FAA", using "proprietary technology to design and manufacture jet engine and aircraft component replacement parts for sale at lower prices than those manufactured by OEMs", and describes itself as "the largest independent supplier of non-OEM jet engine and aircraft component replacement parts". For the fiscal year to 31 October 2025 the group reported net sales of \$3,117.3M and operating income of \$750.4M, an operating margin of 24.1% derived in the research. The company says it has been adding new approvals at "approximately 300 to 500" a year by one dossier's reading of the annual report, or "approximately 400 to 550" by another's. It acquired Wencor Group, the other large independent PMA and distribution business, in 2023. Its discount to the manufacturer's price is not disclosed in anything retrieved. [HEICO 10-K FY2025; HEICO Q4 FY2025 release, 18 Dec 2025; HEICO release on Wencor, 2023; D11 §3.4; reconciliation R-024]

EXHIBIT 6.5 — HEICO FLIGHT SUPPORT GROUP: THE FIGURES THE RESEARCH REACHED, WITH THE TWO READINGS OF THE APPROVAL RATE

Metric	FY2024 (to 31 Oct 2024)	FY2025 (to 31 Oct 2025)	Change	Source
Net sales	\$2,639.4M	\$3,117.3M	+18%	HEICO 10-K FY2025; Q4 FY2025 release, 18 Dec 2025
Operating income	\$593.1M	\$750.4M	+27%	Same
Operating margin (derived)	22.5%	24.1%	+1.6 points	Derived in the research from the two lines above
Aftermarket replacement parts, organic growth	n/a	+15%, + \$263.9M	n/a	Q4 FY2025 release
Q1 FY2025 operating margin (quarter to 31 Jan 2025)	n/a	23.3%	n/a	Q1 FY2025 release, Feb 2025
New PMAs added per year	"approximately 300 to 500" (D4's reading) / "approximately 400 to 550" (D11's reading)	Same filing, two readings	n/a	HEICO 10-K FY2025 (R-024)

Metric	FY2024 (to 31 Oct 2024)	FY2025 (to 31 Oct 2025)	Change	Source
Pricing statement	"lower prices than those manufactured by OEMs"; "the functional equivalent" of OEM parts	n/a	n/a	HEICO 10-K FY2025
Stated discount to OEM price; portfolio split by part category; cumulative PMA count	Not disclosed in the text retrieved	n/a	n/a	D4 U9; D11 U5

Source: HEICO Corporation, Form 10-K for the fiscal year ended 31 Oct 2025; HEICO Q4/FY2025 earnings release (8-K exhibit 99.1), 18 Dec 2025; HEICO Q1 FY2025 earnings release, Feb 2025; HEICO release on the acquisition of Wencor Group, 2023. Compiled in D4 §4.1 and D11 §3.4. Reconciliation R-024. Accessed 2026-10-03.

The approval-rate figures need a sentence. Two dossiers read the same annual report and returned two ranges, "approximately 300 to 500" and "approximately 400 to 550" new approvals a year. Both are readings of the FY2025 10-K; the research did not establish whether the filing contains both phrases in different places or whether one is a search-extraction error, and the primer carries both. [HEICO 10-K FY2025, via D4 and D11; reconciliation R-024] Either way the order of magnitude is hundreds of approvals a year across a portfolio dominated by less complex parts, which is the simple end of the interval §38.3 discussed.

WORKED EXAMPLE 6.3 — DERIVING THE MARGIN A MATURE ALTERNATIVE-PARTS BUSINESS REPORTS, AND ITS INCREMENT

Inputs: Flight Support Group net sales \$2,639.4M (FY2024) and \$3,117.3M (FY2025); operating income \$593.1M and \$750.4M (all sourced: HEICO 10-K FY2025 and the Q4 FY2025 release). The group also includes repair and distribution, so these are not pure PMA margins (sourced caveat: HEICO 10-K, via D4 §3).

$$\begin{aligned}
 \text{operating margin FY2024} &= 593.1 \div 2,639.4 = 22.5\% \\
 \text{operating margin FY2025} &= 750.4 \div 3,117.3 = 24.1\% \\
 \text{incremental margin FY2024 to FY2025} &= (750.4 - 593.1) \div (3,117.3 - 2,639.4) = 157.3 \div 477.9 = 32.9\%
 \end{aligned}$$

So the group earned 24.1% of sales as operating income in FY2025, after depreciation and amortisation, and 32.9% on the additional sales it added in the year; both figures are derived in the research. They are on an operating-income basis and are not comparable without adjustment with FTAI's segment Adjusted EBITDA (Part X §80) margin, meaning earnings before interest, taxes, depreciation and amortisation as FTAI adjusts them, which adds back depreciation; Part XI §88 sets the bases side by side. Limits of the example: the group mixes PMA with repair and distribution, whose margins the filing does not separate; HEICO's fiscal year ends in October; and the discount at

which its parts sell is undisclosed, so the margin cannot be decomposed into a price and a cost. [HEICO 10-K FY2025; HEICO Q4 FY2025 release; D11 §2.5, §3.4]

41.2 CHROMALLOY

Part IV §29 glossed the company. This Part carries it in full because it holds the approvals that matter most to the CFM56 and because of its relationship with FTAI (§43).

CHROMALLOY

Chromalloy is a Palm Beach Gardens, Florida, company that repairs and coats turbine-engine parts, makes investment castings (a casting method in which a ceramic mould is formed around a wax pattern of the part), and designs and manufactures PMA parts. Trade and deal coverage put its revenue at "\$1 billion-plus" and its headcount at about 4,000; Chris Celtruda was appointed chief executive in 2024. It is owned by Veritas Capital through its parent Sequa Corporation: Carlyle bought Sequa in 2007 and sold it to a Veritas affiliate in a transaction announced in September 2022 and closed in December 2022, after Bloomberg reported in July 2022 that Carlyle sought about \$2B. It was an early hot-section PMA maker through BELAC (its joint venture with Lufthansa Technik and United Airlines), which made a stage-1 high-pressure turbine blade for the CFM56-3 and for the F108 (the CFM56-3's military designation), and it has held since 30 October 2025 what it calls "the only aftermarket alternative" to the manufacturer's new high-pressure turbine blade for the CFM56-5B/7B, with a stage-1 vane, a low-pressure turbine vane and, by one reading, a shroud approved alongside. Its CFM56 blade production was sold out for the rest of 2025 with slots being reserved into 2026. [GovConWire, Dec 2022; Bloomberg, 19 Jul 2022; FlightGlobal, Chromalloy profile, undated; PRNewswire, BELAC F108 contract, 2000s; Chromalloy release, 30 Oct 2025; D4 §3, §4.2; D5 §2.7]

Two distribution agreements show how its parts reach shops. AAR (a parts distributor and maintenance company listed on the NYSE) signed an engine-parts supply agreement with Chromalloy in November 2024 and an exclusive agreement covering the PW4000 (a Pratt & Whitney widebody engine) in March 2025. [PRNewswire, Nov 2024 and Mar 2025] Chromalloy has also announced an FAA-approved high-pressure turbine blade PMA for the V2500 (the A320ceo's alternative engine, Part I §10), unveiled at an MRO Europe conference whose date the research did not capture. [Aviation Week, MRO Europe, date not captured; D4 U17]

The CFM56-5B/7B roster is the part of the record that matters most to this primer, and the sources disagree on its length.

EXHIBIT 6.6 — CHROMALLOY'S PMA ROSTER FOR THE CFM56 FAMILY, WITH DATES AS THE SOURCES GIVE THEM AND THE THREE-VERSUS-FOUR TENSION

Part	Engine	Approval date as given	What the source says	Source and level
High-pressure turbine blade	CFM56-5B/7B	Announced 30 Oct 2025	"the only aftermarket alternative to the OEM manufactured new part"; orders cover "all planned production for the rest of 2025"; "many customers now reserving manufacturing slots well into 2026"	Chromalloy release (company release); Newswire copy; AviTrader 31 Oct 2025 and Aviation Maintenance Magazine (trade press)
High-pressure turbine stage-1 vane (the HPT nozzle)	CFM56-5B/7B	Chromalloy release undated in the research; third-party commentary dates it to November 2025	FAA approval of "HPT Turbine Vane 1 PMAs"	Chromalloy release (company release); Komodo Capital (investor blog) for the date
Low-pressure turbine vane, described in the headline as an "up-change" PMA (what the term covers was not retrieved)	CFM56	Date not captured	FAA approval	Aeromorning (trade press, undated); D4 U17
High-pressure turbine shroud (the ring segments surrounding the blade tips, Part I §5)	CFM56-5B/7B	Not separately dated	Listed as one of "four PMA parts" Chromalloy "now offers" in one dossier's reading of the Newswire copy of the 30 Oct 2025 release	D5 §2.7 reading of the Newswire copy (company release as read by D5)
Stage-1 high-pressure turbine blade	CFM56-3 / F108	2000s	Made by BELAC LLC; US Air Force one-year contract of \$2.6M for F108 blades	PRNewswire (company release)

Part	Engine	Approval date as given	What the source says	Source and level
High-pressure turbine blade	V2500 (contrast)	Date not captured	FAA-approved, unveiled at MRO Europe	Aviation Week (trade press); D4 U17

Source: Chromalloy, "Chromalloy Secures FAA Approval of CFM56 High Pressure Turbine Blade PMA", 30 Oct 2025, and the Newswire.com copy; Chromalloy, "Chromalloy Receives FAA Approval of CFM56-5B/7B HPT Turbine Vane 1 PMAs", 2025; Aeromorning, LPT vane up-change PMA, undated; AviTrader, 31 Oct 2025; Aviation Maintenance Magazine, 2025; Aviation Week, MRO Europe; PRNewswire, BELAC F108 contract; D4 §4.2; D5 §2.7. Reconciliation R-025. Accessed 2026-10-03.

The count is contested. One dossier, reading Chromalloy's own announcements, found three CFM56-5B/7B approvals: the blade (30 October 2025), the stage-1 vane (November 2025 per commentary) and a low-pressure turbine vane (undated). Another dossier, reading the Newswire copy of the 30 October release, recorded that Chromalloy "now offers four PMA parts on the CFM56-5B/7B: the HPT blade, the HPT nozzle guide vane, the LPT stage-1 nozzle guide vane and the HPT shroud". [D4 §4.2; D5 §2.7; reconciliation R-025] Investor blogs add a third and fourth count: "two approved (HPT, LPT), three pending", and the HPT vane as "the second of five". [Kairos Research and Komodo Capital, investor blogs, undated; D4 T2] These are readings of one release family and of commentary on it, and the primer presents the three-versus-four count as unresolved. Whether the low-pressure turbine vane is one of the "five" the blogs describe is not stated by any source. [D4 T2]

41.3 OTHER MAKERS, AND THE MANUFACTURER THAT ONCE MADE PMA PARTS

Two other names appear in the research. Wencor Group was the other large independent PMA and distribution business until HEICO acquired it in 2023, so the independent industry's two largest portfolios are now under one owner. [HEICO release on Wencor, 2023] And in the late 2000s an engine manufacturer itself obtained alternative-part approvals for a competitor's engine: Pratt & Whitney ran a unit called Global Material Solutions that obtained FAA PMA for CFM56-3 parts, producing a public dispute with CFM at the Farnborough air show that FlightGlobal covered, along with P&W's parts tests. The dates and the unit's eventual wind-down were not re-verified, and the research found the episode only through headlines. [FlightGlobal, "Farnborough: P&W parts display fuels CFM spat" and "P&W to conclude PMA parts tests in six weeks", undated; D4 §3]

A vocabulary note belongs here because two dossiers used one phrase in two senses. "Material solutions", in this primer, means what Part IV §29 means by it: a shop's or manufacturer's package of used material, repaired parts and PMA parts assembled to reduce the new-parts bill. The research did not match the phrase to any current CFM or GE programme by that name; the only programme it found with those words was the historical Pratt & Whitney unit above. [D2 §8; D4 U10; registry C-19]

41.4 MARKET SIZE AND PENETRATION

Every market-size figure the research reached is from a market-research vendor or a trade article, and they do not agree with each other or, in one case, with themselves.

EXHIBIT 6.7 — PMA MARKET-SIZE AND PENETRATION ESTIMATES: THE CONFLICT EXHIBIT

Estimate	Figure	Scope as the source states it	Source and level	Date
Commercial-aircraft PMA market	\$11.8B in 2025, rising to \$16.1B in 2034; compound annual growth rate (CAGR) 3.37%	"Commercial aircraft PMA"; engines stated as the largest segment	IMARC (a market-research vendor)	2025 base
Commercial-aircraft PMA market	\$14B by 2026	"Commercial aircraft PMA"	Global Industry Analysts (GIA), via PRNewswire (market-research vendor)	2022 release
Engine PMA parts plus DER repair market	\$2,952.2M in 2022, rising to \$6,174.7M by 2032	Engine PMA and DER repair together	A vendor report listed by GII (a Korean listing site for market-research reports)	2022 base
PMA share of the engine-parts market	"capped at around 21%"	Engine parts	FlightGlobal (trade press), older article, undated	Undated
Global aircraft aftermarket parts	\$56.5B in 2026, rising to \$91.3B in 2033; engine parts 34.8%	All aftermarket parts, new and used, OEM and PMA	Persistence Market Research (market-research vendor)	2026 base
One airline's PMA saving	\$54M from 341 PMA part numbers in 2022	Southwest Airlines, fleet-wide	Aviation Week (trade press)	c. 2023

Source: IMARC Group, "Commercial Aircraft Parts Manufacturer Approval Market" (2025 base); PRNewswire / Global Industry Analysts, 2022; GII Korea listing, "Global Engine PMA Parts & DER Repair Market Research" (2022 base); FlightGlobal, "Aftermarket shocks", undated; Persistence Market Research, "Aircraft Aftermarket Parts Market" (2026–2033); Aviation Week, "Southwest's 737-700 Engine Strategy As It Phases Out Fleet", c. 2023. Compiled in D4 §4.4. Reconciliation R-023; D4 T4 and T7. Accessed 2026-10-03.

Three disagreements sit in that table and the primer carries all of them. The first is between the two commercial-aircraft figures: \$11.8B in 2025 from one vendor against \$14B by 2026 from another, from a release three years older. [IMARC, 2025; GIA via PRNewswire, 2022; reconciliation R-023] The second is between scopes: a vendor's engine PMA-plus-DER market of about \$2.95B in 2022 sits beside another vendor's statement that engines are the largest segment of an \$11.8B

total, which the two cannot both be unless the scopes differ in ways neither states. [GII listing, 2022 base; IMARC, 2025; D4 T4] The third is between a ceiling and a forecast: a trade article put PMA's share of engine parts at "capped at around 21%", while a vendor forecasts the engine PMA-and-DER market doubling between 2022 and 2032; both are forecasts and neither is sourced to disclosed data. [FlightGlobal, undated; GII listing; D4 T7] The scope definitions behind each figure were not retrieved, and the primer does not average or choose. [D4 §4.4]

Penetration by category of part is better described in words than in numbers, because no source gave a split. Consumables and expendables, the seals, bushings, fasteners, filters and brackets of §38.2, are where PMA has long been established and where HEICO's hundreds of approvals a year mostly fall; HEICO's own split of its portfolio by category was not disclosed. [D4 §2.3, U9] Rotating life-limited parts have zero penetration: no PMA exists for any CFM56 disk, spool or shaft. [D4 U7] The cooled hot-section airfoils sit between: the CFM56-3 blade of the 2000s through BELAC, and the CFM56-5B/7B blade, vane and possibly shroud of late 2025, are the approvals the research found, and the customer uptake behind them is known only as "sold out for 2025". [D4 §4.2, U15] The one operator-level datapoint, Southwest's 341 part numbers and \$54M in 2022, is fleet-wide and mostly non-hot-section, and §42.1 works it. [Aviation Week, c. 2023]

42. Acceptance: airlines, lessors, and the OEM's responses

A PMA part is legal to install the day it is approved. Whether it is installed depends on three other parties: the airline that owns or operates the engine, the lessor whose lease may forbid it, and the manufacturer whose warranty, service and residual-value programmes may discourage it. This section presents each party's position as the sources state it, with dates, and does not adjudicate between them.

42.1 AIRLINES

The operator-level evidence in the research is thin and is set out in full.

EXHIBIT 6.8 — AIRLINE DATAPOINTS ON PMA AND DER USE, AS THE RESEARCH FOUND THEM

Airline	Datapoint	Date	Source and level
Southwest Airlines	341 PMA part numbers in use; \$54M saved in the year; evaluating PMA and DER options for the CFM56-7B fleet as the 737-700s are phased out	2022 figures; article c. 2023	Aviation Week (trade press)

Airline	Datapoint	Date	Source and level
United Airlines	Launch customer for BELAC's CFM56-3 stage-1 HPT blade PMA; a partner in the BELAC venture	2000s	PRNewswire (company release) and search summary
Delta, American, Ryanair and non-US CFM56 operators	No PMA-policy source retrieved for hot-section parts	n/a	D4 U11
Unnamed customers of Chromalloy	Orders covering "all planned production for the rest of 2025"; "many customers now reserving manufacturing slots well into 2026"	30 Oct 2025	Chromalloy release (company release); customers unnamed (D4 U15)

Source: Aviation Week, "Southwest's 737-700 Engine Strategy As It Phases Out Fleet", c. 2023; PRNewswire, BELAC F108 contract release, 2000s; Chromalloy release, 30 Oct 2025; D4 §3 and U11, U15. Accessed 2026-10-03.

WORKED EXAMPLE 6.4 — SOUTHWEST'S SAVING PER PART NUMBER, AND WHY IT CANNOT BE CONVERTED TO A PER-VISIT FIGURE

Inputs: \$54M saved in 2022 from 341 PMA part numbers (sourced: Aviation Week, c. 2023). A fleet of roughly 800 737s (as the dossier states it; unsourced in the research).

$$\begin{aligned}\text{saving per part number per year} &= 54,000,000 \div 341 = 158,358 \\ \text{saving per aircraft per year, at 800 aircraft} &= 54,000,000 \div 800 = 67,500\end{aligned}$$

So the airline saved about \$158,000 per approved part number per year, or about \$67,500 per aircraft per year on the fleet figure the dossier used. Neither figure converts to a saving per shop visit, because the 341 part numbers span airframe and engine, consumables and components, and the article does not give the part mix or the number of shop visits in the year. The figure is a fleet-wide, mostly non-hot-section datapoint from before any CFM56-5B/7B hot-section PMA existed, and §44 does not use it except as a cross-check of scale. Limits of the example: the fleet size is unsourced; the saving's basis (against list price, or against what the airline would otherwise have paid) is not stated.

[Aviation Week, c. 2023; D4 §2.8]

One episode in 2023 should be named here so that it is not confused with the subject of this Part. In 2023 several US airlines, including Delta, United, Southwest and American, found parts with forged documentation in CFM56 engines, supplied through a company called AOG Technics (the abbreviation AOG means aircraft on ground, the trade's term for an urgent parts need), which then faced a lawsuit. Those were unapproved parts passed off with false release paperwork. PMA parts are FAA-approved parts with genuine release paperwork, and the two are different categories. [Fox Business, 2023; D4 §3]

42.2 LESSORS

An operating lease specifies the state in which the aircraft or engine must be returned, and those return conditions, which Part II §13 treats in full and Part I §9 glossed, are where a lessor's view of alternative parts is written down. One lease-management primer calls return conditions "the most important part of any lease", since they, not the rent, protect the lessor's next placement.

[Sofema Online, undated]

OEM-ONLY CLAUSE (NO-PMA / NO-DER CLAUSE; TCH-CONFIGURATION CLAUSE)

An OEM-only clause is a lease return condition requiring that at redelivery the engine contain only parts produced by the type-certificate holder and only repairs from the type-certificate holder's manuals. A lessee that used PMA parts or DER repairs during the term must then either remove them at the last heavy check before return or pay compensation. The sources describe both the clause and the workaround: "when permitted by the lease, lessees can ... use PMA throughout the term of the lease and ... remove PMA parts during the heavy check prior to the return", while for engines "returning a 'PMA-heavy' engine to strict TCH configuration carries significant financial burdens". A lease that requires TRUEngine status (Part I §3) is an OEM-only clause by another name. Engine lessors were reported to forbid PMA and DER "in the vast majority of cases" in 2016; aircraft-lease clauses were reported to be loosening in 2015; practice in 2024–2026 is unknown. [Aviation Suppliers Association blog, undated; Engine Lease Finance Corporation (ELFC), mid-2016; MARPA, 10 Nov 2015; D4 §2.9, U1]

EXHIBIT 6.9 — LESSOR POSITIONS ON PMA AND DER, BY DATE, ASSET CLASS AND SOURCE

Date	Statement as the source gives it	Asset class	Source and level
2009	Operating lessors "most often stipulated the use of OEM parts in the return conditions"	Aircraft leases	Aircraft Commerce, Issue 67 (trade press)
10 Nov 2015	"most lessors permit PMA to be used on leased aircraft (with a few exceptions); the carrier just has to demand the right"; lessors "grow more willing to waive the 'no PMA' clauses"	Aircraft leases	MARPA, the Modification and Replacement Parts Association (a PMA trade association)
Mid-2016	For engines, "lessors, in the vast majority of cases support the OEMs by forbidding the installation of PMA and DER"; "returning a 'PMA-heavy' engine to strict TCH configuration carries significant financial burdens"	Engine leases	Engine Lease Finance Corporation (ELFC), an engine lessor (company article)
Undated	"when permitted by the lease, lessees can ... use PMA throughout the term of the lease and ... remove PMA parts during the heavy check prior to the return"	Aircraft leases	Aviation Suppliers Association blog (a parts-trade association)

Date	Statement as the source gives it	Asset class	Source and level
Undated	PMA parts "continue to undermine full-life maintenance adjustments"	Engine appraisal	Aircraft Value News (trade press), via Parts I and V
2024–2026	No statement retrieved from AerCap, Air Lease, Avolon, Willis Lease or any other aircraft or engine lessor	n/a	D4 U1
FY2025	FTAI lists lessors among the airline, lessor and maintenance customers to whom its PMA portfolio offers "cost savings and flexibility"	n/a	FTAI 10-K FY2025 (filing)

Source: Aircraft Commerce, Issue 67, 2009 (maintenance article on PMA and lessors); MARPA, "Are leases getting looser?", 10 Nov 2015; Engine Lease Finance Corporation, "An overview of the aircraft engine aftermarket in mid 2016"; Aviation Suppliers Association, "Airlines: now is the time to choose lessors who accommodate PMA and DER repairs", undated; Aircraft Value News, "PMA continues to undermine full-life maintenance adjustments", undated; FTAI 10-K FY2025; D4 §2.9, §3 and U1. Reconciliation D4 T6. Accessed 2026-10-03.

Two of those statements pull in opposite directions, and the difference is partly one of asset class and partly one of date. The 2015 statement that "most lessors permit PMA ... with a few exceptions" concerns aircraft leases and comes from a PMA trade association. The 2016 statement that engine lessors forbid PMA and DER "in the vast majority of cases" concerns engine leases and comes from an engine lessor. [MARPA, 10 Nov 2015; ELFC, mid-2016; D4 T6] Both can be true at once, and the primer presents them as such without inferring either onto the other asset class.

What the research could not find is any statement of current practice. Every lessor position retrieved dates from 2009 to 2016. No lessor's 2024–2026 annual report, lease description, survey response (including surveys by ISTAT (the International Society of Transport Aircraft Trading) or IBA (an appraiser)) or conference remark on PMA or DER clauses was reached, so whether the loosening reported in 2015 continued, reversed or reached engine leases is unknown, and the primer says so rather than extrapolating. [D4 U1] The leasing research behind Part II describes return conditions in detail (minimum hours and cycles, life-limited-part cycles, borescope, test-cell run) without any statement on PMA or DER clauses, which is consistent with the gap rather than evidence either way. [D6 §2.1, via reconciliation C.4] The mechanism by which the clause bears on value is documented without being quantified: an appraiser's full-life maintenance adjustment assumes manufacturer-configuration parts, and Aircraft Value News records that PMA content "continue[s] to undermine" that adjustment; no dollar or percentage effect on an engine or aircraft appraisal was found. [Aircraft Value News, undated; D4 U13; Part I §8]

42.3 THE MANUFACTURER'S RESPONSES

CFM International's recorded responses fall into three kinds: commitments it made in 2018 about warranty and service, a configuration designation it markets to lessors and appraisers, and commercial responses in supply, repair and agreements. Each is presented as CFM states it.

CONDUCT POLICIES (CFM–IATA, 2018)

The Conduct Policies are the commitments CFM International published in 2018 to settle a complaint the International Air Transport Association (IATA) had brought to the European Commission about CFM's aftermarket conduct. Under them, installing non-CFM parts or repairs "does not in itself render the warranty void"; CFM honours its warranty on its own parts and repairs in an engine that contains non-CFM parts; and CFM "does not refuse to service engines because they contain non-OEM parts or repairs". The policies also cover CFM's licensing of its shop manual to third-party shops and its sale of CFM parts to them; the research did not re-read those clauses. Trade associations summarised the agreement as CFM agreeing "to permit the use of PMA parts and DER repairs". [IATA, "IATA–CFM Agreement on Engine Maintenance", 2018; IATA, "CFM Conduct Policies and Implementation Measures"; CFM, "External Guidance", Mar 2019; MARPA, 1 Aug 2018; Aviation Suppliers Association bulletin, Oct 2018]

The second response is the designation Part I §3 defined. TRUEngine designates an engine as maintained to CFM-approved configuration, meaning CFM parts and CFM-approved repairs; CFM markets it as supporting residual value and resale, extended it in 2012 to cover subsequent buyers, and lessors can require it in return conditions. [GE Aerospace, TrueEngine presentation; CFM release on the TRUEngine expansion; Aviation International News, 12 Sept 2012] The 2018 policies and the designation are both CFM positions and address different levers: one is a contractual commitment about warranty and service, the other a marketing statement about resale value. The primer records both without treating either as the "real" position. [D4 T8]

The third kind is commercial, and the dated items are in the exhibit.

EXHIBIT 6.10 — CFM INTERNATIONAL'S COMMERCIAL RESPONSES IN THE CFM56 HOT SECTION AND TO INDEPENDENTS, DATED

Date	Response	Figures as CFM gives them	Source and level
2018	Conduct Policies settling IATA's complaint to the European Commission	No warranty voiding for non-CFM parts or repairs; no refusal of service; shop-manual licensing and parts sales to third-party shops covered	IATA and CFM documents (association and company)

Date	Response	Figures as CFM gives them	Source and level
2012 (expansion)	TRUEngine designation extended to subsequent buyers	Configuration-assurance designation marketed for residual value	CFM / GE release; Aviation International News
16 Oct 2023	Upgraded CFM56-5B/7B HPT blade introduced	Increased wall thickness, optimised dovetail loading, tightened tolerances; "up to 25 percent longer life in certain operating environments"	Safran / CFM release, 16 Oct 2023 (company release)
2025/26, date not captured	HPT blade output and repair capacity raised	1,000th and later 1,700th set of new blades shipped; output "almost two and a half times" year over year; about 400 CFM56 repairs industrialised since early 2023, including automated laser welding of HPT blades that cuts repair turnaround by about a third	CFM stories, "Investing in Durability" and "CFM boosts HPT blade output and repair" (company articles)
Same	Positioning statement	The upgraded blades are "designed to keep customers flying with OEM parts they know and trust" and "can help extend time on-wing to optimize cost of ownership and enhance residual value" (Jacey Welsh, CFM executive vice president for CFM56)	CFM story (company article)
FY2025	Material supply to independents	GE's priority-supplier material input up more than 40% year over year under its operating system; "an open network exists with multiple providers who compete and invest to win shop visits"; GE will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity"	GE 10-K FY2025; GE Aerospace, 2026 (filing and company statement)
22 Jan 2026	Multi-year materials agreement with FTAI Aviation	FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair"; both parties describe it as strengthening "the open maintenance, repair, and overhaul (MRO) ecosystem"; terms undisclosed	FTAI release, GlobeNewswire, 22 Jan 2026 (company release)

Date	Response	Figures as CFM gives them	Source and level
n/a	Statement on Chromalloy's CFM56 HPT blade PMA	None found	D4 U10

Source: IATA, 2018 agreement and Conduct Policies; CFM External Guidance, Mar 2019; GE Aerospace TrueEngine presentation; CFM TRUEngine expansion release and Aviation International News, 12 Sept 2012; Safran / CFM, "CFM introduces upgraded HPT blade for CFM56 engines", 16 Oct 2023; CFM International, "Investing in Durability: CFM ships 1,000th set of new CFM56 HPT blades" and "CFM boosts HPT blade output and repair to meet extended demand for CFM56 engines", 2025/26; GE Form 10-K FY2025; GE Aerospace, 2026, via D2; FTAI Aviation, "FTAI Aviation Announces Multi-Year Materials Agreement with CFM International", 22 Jan 2026. Compiled in D4 §3 and D2 §2.3. Accessed 2026-10-03.

The redesigned blade of October 2023 and the alternative blade of October 2025 are two answers to the same shortage, and the manufacturer's figures for its own answer are these. CFM introduced an upgraded CFM56-5B/7B high-pressure turbine blade on 16 October 2023 with increased wall thickness, optimised dovetail loading and tightened tolerances, claiming "up to 25 percent longer life in certain operating environments". [Safran / CFM release, 16 Oct 2023] It later reported shipping the 1,000th and then the 1,700th set of new blades, with output "almost two and a half times" the prior year's, and about 400 CFM56 repairs industrialised since early 2023. [CFM International stories, 2025/26, dates not captured] A longer-lived blade and a faster blade repair both extend the window in which a blade is repaired rather than replaced, which under the cadence in §40.3 is the window in which no new blade, manufacturer's or alternative, is bought. [D4 §5] The research found no CFM statement about the Chromalloy approval itself. [D4 U10]

The last row needs its own definition, because the agreement is one of the levers FTAI's management named (§43) and Part VII §52 owns its treatment within the segment.

CFM MATERIALS AGREEMENT (FTAI-CFM, 22 JANUARY 2026)

The CFM materials agreement is the multi-year agreement FTAI Aviation announced on 22 January 2026 under which, in the release's words, FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair" from CFM International, described by both parties as strengthening "the open maintenance, repair, and overhaul (MRO) ecosystem". Its duration, volumes, pricing, and whether it bears on FTAI's use of PMA parts, were not disclosed in anything retrieved. Part VII §52 carries the full treatment within the Aerospace Products segment. [FTAI release, GlobeNewswire, 22 Jan 2026; D4 U14; reconciliation K-19]

On the Q4 2025 call FTAI's chief executive described it as "a materials agreement with CFM that includes parts and repairs". [MarketBeat call coverage, 26 Feb 2026] That the same company holds, at once, an agreement to buy the manufacturer's new parts and repairs and a joint venture to make

alternative parts is a fact the sources record; what the agreement says about the alternative parts is not. [D4 U14]

43. The FTAI–Chromalloy joint venture: what is filed and what is reported

FTAI's parts venture is described at three levels of source that say different things, and this section keeps them apart. Part IV §29 glossed the venture; this Part carries what is filed and what is reported, and Part VII §52 carries its effect within the segment.

FTAI–CHROMALLOY JOINT VENTURE (THE PMA JOINT VENTURE)

FTAI's FY2025 annual report says only that the company "develops and manufactures Parts Manufacturer Approval ('PMA') parts through a joint venture", as part of a "proprietary portfolio of products that enables it to provide cost savings and flexibility to airline, lessor, and maintenance, repair, and operations customers"; the partner, the ownership split, the accounting treatment, the revenue and the savings are not in the filing text retrieved. The orchestrator's reading of the same annual report's investments note lists an "Advanced Engine Repair JV" held at 25%, with \$15.0M invested in December 2016 and \$13.5M in August 2019 and a carrying value of \$22,429K at 31 December 2025, and records that whether this is the PMA joint venture is unknown. Third-party commentary describes an exclusive, perpetual partnership with Chromalloy formed in 2018 covering five CFM56 hot-section parts supplied to FTAI "at cost", with savings "up to \$2 million per shop visit"; none of that is in FTAI's filings as retrieved. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; orchestrator notes on the FY2025 10-K investments note and the Q3 2025 10-Q note; In Practise, Komodo Capital, Kairos Research, Yahoo Finance bull-case digest (third-party commentary); reconciliation R-026]

43.1 THREE LEVELS OF SOURCE

EXHIBIT 6.11 — THE FTAI–CHROMALLOY RELATIONSHIP AT THREE SOURCE LEVELS

Item	FTAI's filings (10-K FY2025; 10-Q Q2 2026)	Orchestrator's reading of the FY2025 10-K investments note	Third-party commentary (In Practise; Komodo Capital; Kairos Research; Yahoo Finance digest)
Existence	"develops and manufactures Parts Manufacturer Approval ('PMA') parts through a joint venture"	An "Advanced Engine Repair JV" among the equity-method investments; identity with the PMA venture unknown	An "exclusive perpetual partnership" with Chromalloy

Item	FTAI's filings (10-K FY2025; 10-Q Q2 2026)	Orchestrator's reading of the FY2025 10-K investments note	Third-party commentary (In Practise; Komodo Capital; Kairos Research; Yahoo Finance digest)
Partner	Not named in the text retrieved	Not named	Chromalloy
Formation	Not retrieved	\$15.0M invested Dec 2016; \$13.5M more Aug 2019	2018
Ownership and structure	Not retrieved	25%; equity method	A supply partnership; parts supplied "at cost"
Carrying value	Not retrieved	\$22,429K at 31 Dec 2025	n/a
Parts covered	Not retrieved	n/a	Five CFM56 hot-section parts
Approved to date	Not retrieved	n/a	"two approved (HPT, LPT), three pending" (one blog); the HPT vane "the second of five" (another)
Savings	"cost savings and flexibility" (qualitative)	n/a	"up to \$2 million per shop visit"; "5–10 percentage points of additional margin"
Date mismatch	n/a	2016 first investment	2018 formation: inconsistent with the note's date if the two describe one entity

Source: FTAI Aviation Ltd., Form 10-K for FY2025 and Form 10-Q for the quarter ended 30 Jun 2026 (sec.gov); orchestrator notes, "Equity-method investments (10-K FY2025; 10-Q Q3 2025 note R13)"; In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA" and "FTAI & PMA Parts"; Komodo Capital (Substack); Kairos Research (Substack); Yahoo Finance, "FTAI Aviation Ltd. (FTAI): A Bull Case Theory". Compiled in D4 §4.3 and the orchestrator notes. Reconciliation R-026 and R-027; registry 1c. Accessed 2026-10-03.

The three columns describe different things. A filing describes "a joint venture" through which FTAI "develops and manufactures" parts, which is the language of an equity interest in a manufacturing entity. Commentary describes a "partnership" under which parts are supplied "at cost", which is the language of a supply agreement. The orchestrator's reading of the investments

note describes a 25% equity-method stake first funded in December 2016, two years before the commentary's formation date. If the three describe one arrangement, the structure (equity venture against supply agreement) and the date (2016 against 2018) are both unreconciled; if they describe two, the filing's "joint venture" and the note's "Advanced Engine Repair JV" may be different entities. The research did not resolve either, and the primer does not. [reconciliation R-026; registry 1c] The chain map and one dossier call the venture "a joint venture with Chromalloy"; that is an orchestrator-level description that no filing retrieved supports by name. Another dossier's summary says FTAI "teamed up with Chromalloy to develop several CFM56 hot-section parts" and attributes the statement to the chain map and to the Newswire copy of Chromalloy's 30 October 2025 release; the dossier that read that release in full did not record FTAI's name in it. [D10 §3; D5 §2.7; D4 §4.2; orchestrator notes]

The investments note the orchestrator read lists other equity-method stakes, and seeing the list is the only way to see what the "Advanced Engine Repair JV" sits beside.

EXHIBIT 6.12 — FTAI'S EQUITY-METHOD INVESTMENTS AS THE ORCHESTRATOR READ THEM FROM THE FY2025 10-K AND THE Q3 2025 10-Q

Investment	Stake	Invested	Carrying value, 31 Dec 2025	What it is	Where treated
Advanced Engine Repair JV	25%	\$15.0M (Dec 2016) + \$13.5M (Aug 2019)	\$22,429K	Unknown whether this is the Chromalloy PMA joint venture or a separate repair-development venture	This Part §43; Part VII §52; Part X §79
2025 Partnership (Strategic Capital)	19% (FY2025 10-K) / 20% (Q3 2025 10-Q)	\$151.6M in 9M 2025	\$281,740K	The first Strategic Capital vehicle; FTAI is Servicer	Part VIII §61–64
QuickTurn Europe (Rome Fiumicino)	50%	\$10.5M (closed 5 Jun 2025)	\$9,987K	Engine shop joint venture, ex-IAG Engine Center	Part VII §50
Prime Engine Accessories (Bristol, Connecticut, with Bauer)	50%	Not returned	Not returned	Accessory-repair venture; "up to \$75,000 in average savings per shop visit" (Q3 2025 call)	Part VII §50

Source: orchestrator notes, "Equity-method investments (10-K FY2025; 10-Q Q3 2025 note R13)", citing FTAI Form 10-K FY2025 and the Q3 2025 10-Q; Insider Monkey, FTAI Q3 2025 call transcript, Oct 2025 (for the \$75,000 figure, via D5); GlobeNewswire, 5 Jun 2025 (QuickTurn Europe). Reconciliation K-04, K-07, R-048. Accessed 2026-10-03.

The register's instruction is followed here: the Advanced Engine Repair JV is reported as an equity-method investment, its identity with the PMA venture is stated as unresolved, and the two are not equated. [registry 1c; reconciliation K-04] The investments note of the FY2018 to FY2025 annual reports, and any 8-K on the venture, would resolve the question; the research did not read them. [D4 U3]

43.2 THE APPROVALS TIMELINE

EXHIBIT 6.13 — DATED EVENTS BEARING ON FTAI'S PMA PARTS, WITH THE SOURCE LEVEL OF EACH DATE

Date	Event	Source and level
Dec 2016	\$15.0M invested in the "Advanced Engine Repair JV" (if the same entity)	FY2025 10-K investments note, via orchestrator notes (filing)
2018	FTAI–Chromalloy "exclusive perpetual partnership" formed, covering five CFM56 hot-section parts	Komodo Capital; Kairos Research (investor blogs)
By 1 Aug 2018	CFM's agreement with IATA and its Conduct Policies reported, settling IATA's complaint	MARPA, 1 Aug 2018; IATA (association)
Aug 2019	\$13.5M more invested in the "Advanced Engine Repair JV" (if the same entity)	FY2025 10-K, via orchestrator notes (filing)
16 Oct 2023	CFM introduces its upgraded CFM56-5B/7B HPT blade	Safran / CFM release (company release)
30 Oct 2025	Chromalloy announces FAA PMA for the CFM56-5B/7B HPT blade; production sold out for 2025, slots into 2026	Chromalloy release; AviTrader; Aviation Maintenance Magazine (company release; trade press)
Nov 2025	HPT stage-1 vane PMA (date per commentary; Chromalloy's release undated in the research)	Chromalloy release; Komodo Capital (company release; investor blog)
Undated	LPT vane "up-change" PMA; HPT shroud (per one reading of the 30 Oct release)	Aeromorning (trade press); D5 reading of the Newswire copy
22 Jan 2026	FTAI announces the multi-year materials agreement with CFM	FTAI release, GlobeNewswire (company release)

Date	Event	Source and level
26 Feb 2026	On the Q4 2025 call, the chief executive names the PMA HPT blade, more used material and the CFM materials agreement as the levers to "approach 40%" segment margin	MarketBeat call highlights; Motley Fool transcript (call coverage)

Source: as listed in each row; orchestrator notes (equity-method investments); Chromalloy releases, Oct–Nov 2025; Safran / CFM, 16 Oct 2023; FTAI release, 22 Jan 2026; MarketBeat, "FTAI Aviation Q4 earnings call highlights", 26 Feb 2026; Motley Fool, FTAI Q4 2025 call transcript, 26 Feb 2026. Compiled from D4 §4.2–4.3, D5 §2.7 and the orchestrator notes. Accessed 2026-10-03.

43.3 "SOLD OUT THROUGH 2025" AND THE SLOT STATEMENTS

Chromalloy's announcement of 30 October 2025 said that orders covered "all planned production for the rest of 2025" and that "many customers [were] now reserving manufacturing slots well into 2026", without naming any customer. [Chromalloy release, 30 Oct 2025; Newswire copy; AviTrader, 31 Oct 2025] In this primer's vocabulary that is a production slot, a reserved place in a parts maker's output, which is distinct from the induction slot of an engine shop (Part IV §27) and the manufacturing-slot reservation of a turbine maker (Part IX). [registry C-18] Two things the statement does not give are the ones that would size it: Chromalloy's production capacity in blade sets a year, and FTAI's share of that capacity. [D4 U12] The limiting process is single-crystal casting and coating, which §38.2 named; no capacity figure was disclosed. [D4 §5] Customer uptake beyond "sold out" was not found: no named airline or shop, and no count of sets delivered. [D4 U15] If the arrangement is exclusive for the five parts, as commentary alone says, then other shops' access to those parts, and at what price, is a constraint on market-wide penetration that the sources do not describe. [D4 §5, U3]

43.4 WHAT FTAI'S MANAGEMENT HAS SAID

The company's own quantified statements about the venture are few, and they are set beside the commentary's figures so that the gap is visible.

On the Q4 2025 earnings call of 26 February 2026, asked how the Aerospace Products segment's margin would move from the mid-30s toward 40%, chief executive Joe Adams "cited approval of a PMA high-pressure turbine blade as a key factor, along with expanded access to lower-cost parts supply, including used serviceable material and a materials agreement with CFM that includes parts and repairs"; the company said it "can approach 40% aerospace margins in 2026 via parts approvals, lower-cost supply and expanded repair capabilities". [MarketBeat call highlights, 26 Feb 2026; D5 §2.7] The segment's Q4 2025 Adjusted EBITDA was \$195M at a 35% margin, as call coverage reported it, and FY2025 segment Adjusted EBITDA was \$671.3M on segment revenue of \$1,936.2M, a derived 34.7%. [MarketBeat call highlights, 26 Feb 2026; FTAI 10-K FY2025 via

orchestrator notes; D5 §4.1] No dollar figure for the PMA saving per shop visit or per module, no PMA share of parts consumed, and no equity-method earnings line for the venture were retrieved. [D5 U11] The one FTAI-quantified savings lever found anywhere in the research is the Bristol accessories venture, "expected to deliver up to \$75,000 in average savings per shop visit", said on the Q3 2025 call. [Insider Monkey, FTAI Q3 2025 call transcript, Oct 2025; D5 §2.6]

Against that, investor blogs state "up to \$2 million per shop visit" from five parts supplied at cost, and "5–10 percentage points of additional margin". [Komodo Capital; Kairos Research; Yahoo Finance digest; reconciliation R-027] The reconciliation records this as a confidence mismatch: a blog dollar figure against a qualitative management statement. [reconciliation R-027] §44 tests the dollar figure against the parts arithmetic; the worked example here only places the two claims against the segment's reported scale.

WORKED EXAMPLE 6.5 — PLACING THE CLAIMED FIGURES AGAINST THE SEGMENT'S REPORTED SCALE

Inputs: FY2025 Aerospace Products segment revenue \$1,936.2M, of which products revenue \$1,600.5M and MRE (Maintenance, Repair and Exchange) Contract revenue \$335.8M; segment Adjusted EBITDA \$671.3M (all sourced: FTAI 10-K FY2025 via orchestrator notes; Part VII §53). Products revenue per module produced \$2.11M (FY2025) and \$2.34M (Q2 2026), derived in the research on the products-revenue numerator, carried as a bound and not a price (reconciliation R-046). Claimed figures: "5–10 percentage points of additional margin" and "up to \$2 million per shop visit" (sourced: investor blogs). FTAI's one quantified lever: "up to \$75,000 in average savings per shop visit" from the Bristol venture (sourced: Q3 2025 call, via Insider Monkey).

5 to 10 margin points on FY2025 segment revenue (\$M):
 $0.05 \times 1,936.2 = 96.8$; $0.10 \times 1,936.2 = 193.6$
 5 to 10 margin points on FY2025 products revenue (\$M):
 $0.05 \times 1,600.5 = 80.0$; $0.10 \times 1,600.5 = 160.1$
 claimed per-visit saving against products revenue per module:
 $2.0 \div 2.11 = 95\%$; $2.0 \div 2.34 = 85\%$
 claimed per-visit saving against FTAI's quantified lever:
 $2,000,000 \div 75,000 = 27$ times

So the blogs' margin claim, applied to FY2025 revenue, would be about \$80–194M a year depending on the base; and the blogs' per-visit saving is of the same order as the revenue FTAI books per module produced, and 27 times the only per-visit saving FTAI itself has quantified. None of this tests whether the claims are right; it shows their scale against filed figures. Limits of the example: the margin claim's base and period are not stated by the blogs; revenue per module is a bound with the five limits Part VII §56 lists; the blade approval came on 30 October 2025, so no FY2025 figure can contain its effect; and a per-visit saving applies only to visits where blades are replaced rather than repaired (§40.3). [FTAI 10-K FY2025 via orchestrator notes; D5 §2.11, §4.1; Insider Monkey, Oct 2025; Komodo Capital; Kairos Research; reconciliation R-027, R-046]

44. Worked example: the parts bill with and without PMA

This section does what the Part has been building toward: it puts a CFM56-7B performance restoration's hot-section parts bill on the page, prices it with the manufacturer's new parts and then with alternative parts, shows the saving as a function of a discount no source discloses, and tests the claimed "\$2 million per shop visit" against the arithmetic. Every tension that touches the numbers is presented in full, and nothing is resolved.

44.1 THE SETTING AND THE INPUTS

The visit is a performance restoration, in which the core is opened and the high-pressure compressor, combustor and high-pressure turbine are restored to recover exhaust-gas temperature (EGT) margin (Part I §6). The cadence from §40.3 decides whether new blades are bought at all: at a first visit, around 20,000 cycles, blades are typically repaired; at a second, around 30,000, they typically cannot be and new ones are needed. [In Practise, undated, interview digest] The example is therefore a second or later visit, which under Part III's context is exactly the visit an engine kept past its planned retirement undergoes. No shop-level price for a blade or a set is used, because the one fixed-price CFM56-7B catalogue the research located, that of the Zurich engine shop SR Technics (SR), dated 3 June 2026, was not read. [SR Technics catalogue, 3 Jun 2026, located not read; D4 U2] Alternative parts touch only the on-condition hot-section parts of §38.1; they touch no life-limited part, so the life-limited-part line, 40–60% of parts cost when replaced, is unchanged in every case below. [Safe Fly Aviation, 2026, consultancy blog; D4 U7]

EXHIBIT 6.14 — INPUTS TO THE PARTS-BILL EXAMPLE, WITH THE PROVENANCE AND STATUS OF EACH

Input	Value	Source and level	Status
New manufacturer's CFM56 HPT blade, price per blade	About \$20,000	In Practise expert interview (interview digest), undated	Sourced; third-party commentary
Alternative per-blade listing	About €35,000 (about \$38,000–39,000 at the conversion the dossier applied; rate not stated)	Web page on a university host (ATU), undated	Weak provenance; condition (new or serviceable) unclear
HPT blades per CFM56-7B rotor	80	Commonly cited; not confirmed by any source retrieved (D4 U16)	Parameter, unverified

Input	Value	Source and level	Status
New HPT blade and vane set	"up to a few million dollars"	In Practise, undated	Sourced; taken as \$2.5–3.0M (assumed within the phrase)
HPT blades and vanes as a share of a visit's parts cost	15–25%	Safe Fly Aviation (consultancy blog), 2026	Sourced; contradicts the per-blade route (R-015)
Material as a share of a visit	60–70%	Air Cargo Week (trade press), 2025/26	Sourced
Performance-restoration cost, two ranges	\$1.2–1.6M; \$1.8–2.5M	Safe Fly Aviation (consultancy blog), 2026, two pages	Sourced; both carried (R-013)
PMA discount to the manufacturer's price	Not disclosed	No source; neither Chromalloy nor FTAI published a price (D4 U2)	Shown as a function at 20%, 30%, 40% (assumed)
Claimed FTAI saving from five hot-section parts	"Up to \$2 million per shop visit"	Komodo Capital; Kairos Research (investor blogs)	Not in FTAI filings retrieved (R-027)
Life-limited parts as a share of parts cost	40–60%	Safe Fly Aviation, 2026	Sourced; untouched by PMA (D4 U7)

Source: In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA", undated; web listing of weak provenance (D4 source 72); Safe Fly Aviation, 2026, consultancy blog; Air Cargo Week, "The true cost of engine maintenance", 2025/26; Komodo Capital and Kairos Research, investor blogs, undated. Compiled in D4 §2.8 and §4.5. Reconciliation R-013, R-015, R-027. Accessed 2026-10-03.

44.2 THE BILL WITH THE MANUFACTURER'S NEW PARTS, AND THE BLADE-PRICE TENSION

WORKED EXAMPLE 6.6 — A CFM56-7B PERFORMANCE RESTORATION'S HPT PARTS BILL: MANUFACTURER'S NEW PARTS, THEN PMA PARTS AT AN UNDISCLOSED DISCOUNT

Step 1, the manufacturer's blade set, priced per blade. Inputs: 80 blades (unverified parameter) at \$20,000 (sourced: interview digest) or at about \$38,500 (sourced: weak web listing, condition unclear, dossier's conversion of €35,000).

at the interview price:	$80 \times 20,000 = 1,600,000$
at the listing price:	$80 \times 38,500 = 3,080,000$

Step 2, the same blades and vanes priced from the consultancy shares instead. Inputs: HPT blades and vanes 15–25% of parts cost (sourced: Safe Fly Aviation); material 60–70% of a visit (sourced:

Air Cargo Week); visit \$1.2–1.6M or \$1.8–2.5M (sourced: Safe Fly Aviation, two pages). Part I §7.2 applied the share to the visit total as an approximation; the stricter reading applies it to the material line.

$$\begin{aligned}
 &\text{share of the visit total (Part I's approximation):} \\
 &0.15 \times 1,200,000 = 180,000; \quad 0.25 \times 1,600,000 = 400,000 \\
 &0.15 \times 1,800,000 = 270,000; \quad 0.25 \times 2,500,000 = 625,000 \\
 &\text{share of the material line (material 60–70% of the visit):} \\
 &0.15 \times 0.60 \times 1,200,000 = 108,000; \quad 0.25 \times 0.70 \times 1,600,000 = 280,000 \\
 &0.15 \times 0.60 \times 1,800,000 = 162,000; \quad 0.25 \times 0.70 \times 2,500,000 = 437,500
 \end{aligned}$$

This is the tension in full. Priced per blade, the blade set alone is \$1.6M to \$3.1M, which is 100% to 260% of a \$1.2–1.6M restoration and 64% to 171% of a \$1.8–2.5M one. Priced as a share, the blades and vanes together are \$0.11M to \$0.63M. The two routes cannot both describe the same visit at the same prices. The research recorded the possible reasons without establishing any: the 80-blade count is a recollection; the per-blade prices come from an interview and a listing of unclear condition rather than a catalogue; and a share figure drawn from visits at which blades are repaired rather than replaced would be far smaller than a visit at which the whole set is bought new. The primer carries both routes and stops (R-015). [D4 §2.8, T5, U16; Safe Fly Aviation, 2026; Air Cargo Week, 2025/26; Part I §7.2; reconciliation R-015]

Step 3, the PMA blade set at a discount d to the manufacturer's price, on the \$1.6M set. The discount is undisclosed (D4 U2); the three rates are illustrative assumptions.

Assumed PMA discount d	PMA set cost (\$1.6M base)	Saving on the blade set	PMA set cost (\$3.08M base)	Saving on the blade set
20%	\$1,280,000	\$320,000	\$2,464,000	\$616,000
30%	\$1,120,000	\$480,000	\$2,156,000	\$924,000
40%	\$960,000	\$640,000	\$1,848,000	\$1,232,000

Step 4, adding the stage-1 vane set. If blades and vanes together are "up to a few million dollars" (sourced phrase), taken as \$2.5–3.0M (assumed within the phrase), the same discounts give savings on the two part families together of:

$$\begin{aligned}
 &\text{at 20\%: } 0.20 \times 2,500,000 = 500,000 \quad \text{to} \quad 0.20 \times 3,000,000 = 600,000 \\
 &\text{at 30\%: } 0.30 \times 2,500,000 = 750,000 \quad \text{to} \quad 0.30 \times 3,000,000 = 900,000 \\
 &\text{at 40\%: } 0.40 \times 2,500,000 = 1,000,000 \quad \text{to} \quad 0.40 \times 3,000,000 = 1,200,000
 \end{aligned}$$

Step 5, the saving as a share of the visit, divided by both performance-restoration ranges as reconciliation rule E.1.4 requires.

blade set only, \$1.6M base, d = 20–40%: saving 320,000–640,000
 $\div$ visit of 1,200,000–1,600,000 = 20% to 53%
 $\div$ visit of 1,800,000–2,500,000 = 13% to 36%
 blades and vanes, \$2.5–3.0M, d = 20–40%: saving 500,000–1,200,000
 $\div$ visit of 1,200,000–1,600,000 = 31% to 100%
 $\div$ visit of 1,800,000–2,500,000 = 20% to 67%

So on the per-blade inputs, alternative hot-section parts at a 20–40% discount would save 13% to 53% of a restoration's cost on the blades alone and 20% to 100% on blades and vanes together, depending on which visit range is used. A saving approaching or exceeding the whole visit is itself a sign that the per-blade inputs and the visit totals are not on one basis, which is the tension of step 2 seen from the other side. The savings apply only at visits where the blades and vanes are scrapped rather than repaired, which the interview places at second and later visits; at a first visit the lever is a repair (§40). Limits of the example: the blade count, the per-blade price, the vane-set price and the discount are respectively unverified, interview-sourced, assumed within a phrase and assumed; the visit totals are two contested consultancy ranges; and no manufacturer's catalogue price for a blade or a set was read (D4 U16). [D4 §2.8; Safe Fly Aviation, 2026; In Practise, undated; reconciliation R-013, R-015; E. 1 rule 4]

44.3 TESTING THE CLAIMED \$2 MILLION AGAINST THE ARITHMETIC

WORKED EXAMPLE 6.7 — WHAT "UP TO \$2 MILLION PER SHOP VISIT" REQUIRES OF THE FIVE PARTS

Inputs: the claim, "up to \$2 million per shop visit" from five CFM56 hot-section parts supplied to FTAI "at cost" (sourced: investor blogs; not in FTAI filings retrieved). The manufacturer's prices from worked example 6.6: blade set \$1.6M or \$3.08M; blades and vanes \$2.5–3.0M (assumed within the sourced phrase). The other three parts (a low-pressure turbine vane, a shroud by one reading, and a fifth part the commentary does not name): no price retrieved.

Step 1, the effective discount the claim implies if the saving came from the named parts alone.

blade set at \$1.6M: $2,000,000 \div 1,600,000 = 125\%$ (not reachable from the blade set alone)
 blade set at \$3.08M: $2,000,000 \div 3,080,000 = 65\%$
 blades and vanes at \$2.5M: $2,000,000 \div 2,500,000 = 80\%$
 blades and vanes at \$3.0M: $2,000,000 \div 3,000,000 = 67\%$

Step 2, what is left for the other parts at a market-style discount. At d = 40% the blade set saves \$640,000 (\$1.6M base), leaving \$1,360,000 to come from the other four parts; at the same discount blades and vanes together save \$1.0–1.2M, leaving \$0.8–1.0M for the low-pressure turbine vane, the shroud and the fifth part, whose prices the research did not reach.

Step 3, the "at cost" reading. If "at cost" means FTAI pays roughly the manufacturing cost rather than a market PMA price, FTAI's saving per part is the manufacturer's list price less the maker's cost,

which is larger than the discount a third-party buyer of the same part would receive. Whether that difference is large enough to reach the effective discounts in step 1 depends on the maker's cost, which no source states; the primer shows the required effective discount and does not assume a cost.

Step 4, the claim against the visit totals and the only filed-level comparators.

$2,000,000 \div 1,200,000 - 1,600,000 = 125\% \text{ to } 167\% \text{ of a restoration (lower range)}$

$2,000,000 \div 1,800,000 - 2,500,000 = 80\% \text{ to } 111\% \text{ of a restoration (upper range)}$

$2,000,000 \div 75,000 = 27 \text{ times FTAI's quantified Bristol saving per visit Southwest, fleet-wide: } 54,000,000 \div 341 = 158,358 \text{ per part number per year}$

(not convertible to a per-visit figure)

So the claim is reachable from the blade set alone only at the higher per-blade price and a 65% effective discount, is not reachable from the blade set alone at the interview price, and on either visit range is of the same size as, or larger than, the whole restoration as the consultancy prices it. The arithmetic shows that the claim is sensitive to which per-blade price is right (R-015), to what "at cost" means (R-026), to what the other three parts cost (unknown), and to what "shop visit" the blogs mean, since a heavy visit is larger than a restoration (R-013). The example neither confirms nor refutes the claim; it sets out what would have to be true. Limits of the example: every price input is third-party, unverified or assumed; the claim's own basis (against list price, at which visit, "up to") is not stated by its sources; and FTAI has published no per-visit PMA saving. [Komodo Capital; Kairos Research; D4 §2.8 step 4, T1, T5; Insider Monkey, Oct 2025; Aviation Week, c. 2023; reconciliation R-013, R-015, R-026, R-027]

WHAT THE PMA LEVER DOES NOT TOUCH

Whatever the right per-blade price, an approved alternative part changes only the on-condition parts line of a shop visit. Life-limited parts, 40–60% of parts cost when they are replaced, remain single-source at the manufacturer's escalating list price, because no PMA exists for any CFM56 disk, spool or shaft. Labour, test and the shop's margin are untouched. And at a visit where the blades can still be repaired, no new blade is bought from anyone, so the lever is a repair, not a part. The reach of the PMA lever is therefore the new hot-section parts bought at second and later visits, which is a real and growing share of visits on a fleet kept flying past its planned retirement (Part III §19), and no more than that. [Safe Fly Aviation, 2026; D4 U7, §2.7; In Practise, undated]

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-013	Performance restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M"; full overhaul \$3.5–4.2M (all Safe Fly Aviation, consultancy blog, 2026); PMA savings divided by both performance-restoration ranges	§40.3 worked example 6.2; §44 worked examples 6.6 and 6.7
R-014	Material 60–70% / labour 20–30% / repairs 10–20% (Air Cargo Week) against LLPs 40–60% of parts / HPT blades and vanes 15–25% / labour 15–20% (Safe Fly)	§37.1; §40.3; §44.1
R-015	HPT blade set \$1.6M (80 × \$20,000, interview) or \$3.08M (80 × ~\$38,500, weak listing) against \$0.11–0.63M implied by the 15–25% share on either visit range	§37.1 worked example 6.1; §44.2 worked example 6.6 step 2; §44.3
R-018	Escalation: spares ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Aviation Week) against LLP set ~7%/yr compounded (derived) and -7B parts +20–30% in 2023 (Magnetic via Aviation Week)	§37.1
R-023	PMA market: \$11.8B in 2025 (IMARC) against \$14B by 2026 (GIA, 2022); engine PMA+DER \$2.95B in 2022 (GII listing) against "engines the largest segment" of \$11.8B; FlightGlobal ceiling ~21% against a doubling forecast	§41.4, Exhibit 6.7
R-024	HEICO new PMAs per year: "approximately 300 to 500" (D4's reading) against "approximately 400 to 550" (D11's reading), both HEICO 10-K FY2025	§38.3; §41.1, Exhibit 6.5
R-025	Chromalloy CFM56-5B/7B roster: three announced approvals (blade, HPT vane 1, LPT vane; D4) against four parts offered including the HPT shroud (D5's reading of the Newswire copy); blogs' "two approved" and "second of five"	§41.2, Exhibit 6.6; §43.2
R-026	The relationship: "a joint venture" (FTAI 10-K FY2025) against "exclusive perpetual partnership", parts "at cost", formed 2018 (investor blogs) against a 25% "Advanced Engine Repair JV" first funded Dec 2016, identity unknown (orchestrator reading of the 10-K)	§43.1, Exhibits 6.11 and 6.12
R-027	Saving: "up to \$2 million per shop visit" and "5–10 percentage points of margin" (investor blogs) against FTAI's qualitative "cost savings and flexibility" and its one quantified lever, \$75,000 per shop visit (Bristol JV, Q3 2025 call)	§43.4 worked example 6.5; §44.3 worked example 6.7

Id	The two (or more) figures	Where in this Part
D4 T6	"Most lessors permit PMA ... with a few exceptions" (MARPA, Nov 2015, aircraft leases) against engine lessors forbidding PMA and DER "in the vast majority of cases" (ELFC, mid-2016)	§42.2, Exhibit 6.9
D4 T7	PMA share of engine parts "capped at around 21%" (FlightGlobal) against the engine PMA+DER market doubling 2022–2032 (vendor forecast)	§41.4
D4 T8	CFM's 2018 Conduct Policies (no warranty voiding, no service refusal) alongside TRUEngine and residual-value messaging for OEM-only configuration	§42.3

Unknowns carried in this Part

- **U-D4-01 (D4): U1 — Current lessor return-condition practice (2024–2026).** All lessor statements found are 2009–2016. Resolve with recent lessor 10-K/20-F lease descriptions (AerCap, Air Lease, Willis Lease), ISTAT or IBA surveys, or lessor conference remarks.
- **U-D4-02 (D4): U2 — PMA price discount for the Chromalloy HPT blade and vane** (and HEICO's typical discount). Resolve with Chromalloy/FTAI pricing disclosure or MRO price catalogues (the SR Technics 2026 CFM56-7B catalogue may list PMA options [71]).
- **U-D4-03 (D4): U3 — FTAI–Chromalloy JV terms:** formation date, ownership split, accounting (consolidated vs equity method), exclusivity, "at cost" pricing, revenue recognized. Resolve by reading FTAI's FY2018–FY2025 10-Ks for the JV footnote and any 8-K. (Partly and conditionally closed by the orchestrator notes: Advanced Engine Repair JV 25%, \$15.0M Dec 2016 + \$13.5M Aug 2019, carrying value \$22,429K at 31 Dec 2025, if the same entity; reconciliation D.3, K-04.)
- **U-D4-04 (D4): U4 — Time from PMA application to approval** for hot-section parts. Resolve via FAA PMA database (DRS) entries with dates, or company statements.
- **U-D4-05 (D4): U5 — Whether the CFM56 HPT blade/vane PMAs are "critical components"** under the TIP and therefore which EASA acceptance clause applies; whether EASA acceptance has been obtained.
- **U-D4-06 (D4): U6 — Acceptance by other regulators** (UK CAA, TCCA, ANAC, CAAC, DGCA India) of these PMAs.
- **U-D4-07 (D4): U7 — Any PMA for CFM56 life-limited rotating parts.** None found.
- **U-D4-08 (D4): U8 — How DER-approved repair data is accepted on EU-registered engines** under the TIP.

- **U-D4-09 (D4): U9 — HEICO portfolio split** by part category (consumables vs rotating vs hot-section) and HEICO's PMA discount.
- **U-D4-10 (D4): U10 — Any CFM/GE statement specifically on the Chromalloy HPT blade PMA**, and any GE "material" programme (e.g. TrueChoice Material) positioning against PMA. The brief's phrase "material solutions" was not matched to a current CFM/GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines [51][52].
- **U-D4-11 (D4): U11 — PMA policies of Delta, American, United, Ryanair, and non-US CFM56 operators** for hot-section parts.
- **U-D4-12 (D4): U12 — Chromalloy production capacity** (blade sets per year) and FTAI's share of it.
- **U-D4-13 (D4): U13 — Quantified residual-value effect** of PMA/DER content on engine or aircraft appraisals.
- **U-D4-14 (D4): U14 — Terms of the FTAI–CFM Jan 2026 materials agreement** (duration, volumes, pricing, whether it restricts FTAI's PMA use).
- **U-D4-15 (D4): U15 — Customer uptake figures for the Chromalloy blade** (named airlines/MROs, sets delivered). Only "sold out for 2025" was found.
- **U-D4-16 (D4): U16 — CFM56-7B HPT blade count per set** (80 is a recollection, unverified here) and the OEM list price for a full HPT blade and vane set.
- **U-D4-17 (D4): U17 — Dates for the LPT vane PMA and the V2500 HPT blade PMA.**
- **U-D5-11 (D5): U11 — Chromalloy JV contribution in FTAI's numbers:** equity-method earnings line, PMA share of parts, FTAI-stated savings per module. (Cluster K-04 with U-D4-03.)
- **U-D11-05 (D11): U5 — HEICO consolidated FY2025 net sales and operating income; PMA cumulative count; stated discount to OEM; acceptance statements.** Searched [S7, S7a, S8]. Only FSG figures and the 400–550 PMA rate were returned. Resolve: read Item 1 and Item 7 of [S7]. (Cluster K-05 with U-D4-02 and U-D4-09.)
- **U-D10-11 (D10): U-11 — Early asset mix and the shift to aviation.** Figures for Jefferson Terminal, Repauno, Transtar and Long Ridge; dates and prices of QuickTurn, the Chromalloy JV and the Lockheed Martin Commercial Engine Solutions acquisition. Resolve with the FY2021 10-K [S15], the FY2022 annual report and the relevant 8-Ks. (Carried for the Chromalloy JV deal terms only; cluster K-12; the facility items are partly closed and are Part VII's and Part X's.)

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PART VII

FTAI Aerospace Products: the Module Factory

This Part is the fullest treatment in the primer of one business: the segment through which FTAI Aviation sells CFM56 and V2500 engines, modules and parts, and the production process it calls the Module Factory. It sets out what the segment sells in the company's own words, how a module exchange works for a customer, how exchanges and sales are priced and booked, what the related-party line called MRE (Maintenance, Repair and Exchange) Contract revenue is, how many modules were produced and what capacity was claimed at each date, where the shops are, what the segment earned, what a short seller alleged and what the company's board found, and what management has guided for 2026 and 2027. A reader who finishes it will be able to read FTAI's segment table, its module counts and its capacity statements with every term understood and every contested figure labelled.

TRANSLATION-TABLE ROWS ANSWERED

Part III §24 turned each documented demand driver into a requirement on the aftermarket. This Part answers the rows whose requirement is a shorter shop event or a serviceable module from a pool: (Part III §24, row T-8: MRO capacity shortfall and turnaround), MRO being the industry's term for maintenance, repair and overhaul, which creates the requirement for module exchanges and quick-turn visits as substitutes for full overhauls; (Part III §24, row T-1: Low CFM56 retirements), which sustains the shop-visit count the segment addresses while starving the used-material supply the Module Factory draws on; (Part III §24, row T-4: 737 MAX production cap) and (Part III §24, row T-5: Airbus delivery shortfall), which keep 737NG and A320ceo engines flying and hence needing -7B and -5B modules; (Part III §24, row T-2: GTF powder-metal groundings), the GTF being Pratt & Whitney's geared turboprop, which brings forward V2500 and -5B shop visits; (Part III §24, row T-6: Lessor extension behaviour), under which maintenance events fall inside extended terms and are paid by the incumbent airline; (Part III §24, row T-12: Higher utilisation of mature narrowbodies), which accelerates the cycle count; and (Part III §24, row T-11: Freightler conversion), which asks for green-time -7B modules for converted airframes. It also bears on (Part III §24, rows T-3, T-14: LEAP aftermarket parity / CFM56 plateau-then-decline), because the segment's stated capacity and share goal are sized against a shop-visit count that the forecasts have peaking in 2027 and 2028, and on (Part III §24, row T-13: GTF groundings easing), which would soften ceo lift demand later in the decade. It does not answer the rows on new-engine spares or LEAP shop work, which the segment does not serve.

45. What the segment sells, in FTAI's words

FTAI Aviation reports two segments. One owns and leases aircraft and engines (Part VIII §58). The other, Aerospace Products, repairs and sells engines, modules and parts. Part I §11 gave the segment a one-line definition for orientation. This section gives the company's own description, says what the company means by the words in it, and lists the products.

FORM 10-K, FORM 10-Q AND FORM 8-K

The 10-K is the annual report a US-listed company files with the Securities and Exchange Commission (SEC); the 10-Q is the quarterly report; the 8-K is the current report used for press releases and material events. FTAI's earnings releases are filed as exhibits to 8-Ks and also published on GlobeNewswire. Part X §79 carries the full treatment. [D5 §8; D10 §3]

45.1 THE FILING LANGUAGE

The annual report for 2025 describes the segment in one sentence. Aerospace Products "develops and manufactures, repairs/refurbishes, and sells aircraft engines and aftermarket components primarily for the CFM56-7B, CFM56-5B, and V2500 commercial aircraft engines. Engine, module, and parts sales are facilitated through a dedicated commercial maintenance program designed to focus on modular and parts repair and refurbishment of CFM56-7B and CFM56-5B engines". [FTAI 10-K FY2025] The programme named in that sentence is the Module Factory, which the same filing calls "a dedicated commercial maintenance program designed to focus on modular and parts repair and refurbishment" and says the company "targets assets which require maintenance repairs that can be performed through [its] proprietary Module Factory process". [FTAI 10-K FY2025]

Three words in the first sentence carry more weight than they appear to. The first is "manufactures". FTAI's filings describe the business as one that "focuses on the manufacturing and sale of new assets through the use of inventory purchased from third parties and salvaged modules and parts from generally unserviceable engines". [FTAI 10-K FY2025, via D3 §2.7] A module that FTAI has rebuilt from salvaged and purchased parts is, in the filing's framing, a manufactured product, and its sale is a product sale. The second word is "sells". The segment does not describe itself as performing maintenance for a fee on a customer's engine. It describes itself as selling engines, modules and parts. The third is "modular". The unit of the business is the module of Part I §3, not the engine and not the piece part.

The same filing explains where the material comes from. Engines in FTAI's own leasing fleet are "transferred to inventory when determined to be unserviceable for leasing purposes with the intent to tear down the asset into salvageable components that would be manufactured into new modules and parts and sold". [FTAI 10-K FY2025, via D3 §2.7] Material that FTAI does not build into modules has its own outlet: "serviceable used modules and parts are sold through an exclusive partnership responsible for the teardown, repair, marketing and sales of parts from the CFM56 engine pool". [FTAI 10-K FY2025, via D3 §2.7] The filing excerpt the research reached does not name the partner. A PR Newswire release located by title, and not read, says "AAR and FTAI Aviation extend their exclusive Serviceable Engine Products agreement, providing CFM56 engine material to the global aviation aftermarket through 2030". [PR Newswire, 2025, title only; reconciliation B-9]

The 10-K adds that the company "develops and manufactures PMA parts through a joint venture", PMA being the Parts Manufacturer Approval issued by the Federal Aviation Administration (FAA) for a replacement part made by someone other than the engine's manufacturer (Part VI §37). [FTAI 10-K FY2025; D4 §4.3] Section 52 takes that up.

45.2 WHAT MRE MEANS, AND HOW IT DIFFERS FROM MRO

The industry's generic term for maintenance work is MRO, Maintenance, Repair and Overhaul (Part I §6; Part IV §25). FTAI's name for its own offer replaces the last word.

MAINTENANCE, REPAIR AND EXCHANGE (MRE)

MRE is FTAI's name for its offer to engine owners, chosen as a contrast with the industry's MRO. Instead of restoring a customer's engine part by part while the customer waits, FTAI removes the module or engine that triggered the removal, installs a serviceable one from its inventory, tests and returns the engine, and takes the unserviceable unit as feedstock for its own production. The offer is sold as engine, module and parts sales to airlines, lessors and other shops, and to the Strategic Capital vehicles under MRE Contracts that carry stand-ready obligations (§48). The FY2025 10-K uses the phrase when it describes the Rome joint venture as created "to meet increasing demand for MRE services". [FTAI 10-K FY2025; D5 §2.2; D7 §2.4; D10 §2.4; registry C-15]

The distinction is operational, not regulatory. Both an MRO shop and FTAI hold repair-station approvals and work to the manufacturer's manual (Part I §9). The difference is in who waits and for what. In a conventional shop visit the engine's owner waits for its own engine while the shop disassembles, inspects, repairs and rebuilds it; the owner pays for parts at list and labour by the hour, or under a flight-hour agreement (Part IV §28). In an exchange the owner receives a different, already serviceable module or engine, and the wait is the time to swap and test. The owner gives up the unserviceable unit it handed over. Section 46 sets out the sequence step by step.

Two cautions about the vocabulary travel through the rest of this Part. First, the press summaries of the January 2025 short report use "MRO revenue" to describe what FTAI books, while FTAI's own filings say "engine, module, and parts sales" and "MRE Contract revenue"; the two vocabularies name the same transactions and the dispute in §54 is about how they should be classified, not about what they are called. Second, the word "core" has two meanings in this Part: the hot middle module of the engine (Part I §1) and the unserviceable unit a customer hands back in an exchange. Section 47 states the double meaning where it first bears on a price.

45.3 THE PRODUCT SET

Reading the filing language, the earnings releases and the 10-Qs together, the segment's outputs fall into six kinds. The research found no filing table that splits revenue among them. [D5 U2]

EXHIBIT 7.1 — WHAT AEROSPACE PRODUCTS SELLS: THE PRODUCT SET AS THE FILINGS AND RELEASES DESCRIBE IT

Product	What the customer receives	What FTAI receives	Where it appears in the filings	Source
Module exchange	A serviceable CFM56 module (fan and booster, core, or low-pressure turbine, in FTAI's three-module convention) installed in the customer's engine, which is tested and returned	An exchange fee plus the unserviceable module handed over	"Engine, module, and parts sales"; "Aerospace products revenue" line	FTAI 10-K FY2025; D5 §2.2–2.3
Module sale	A serviceable module outright, with no unit coming back	The cash price	Same line	D5 §2.3
Engine exchange or engine sale	A serviceable whole engine of the same type, in exchange for the customer's engine or outright	A fee plus the engine handed over, or the whole-engine price	Same line; and "MRE Contract revenue" when the buyer is a 2025 Partnership SPV	FTAI 10-Q Q2 2026; D7 §2.4
Parts	New, used serviceable or PMA parts; used modules and parts also flow through the exclusive serviceable-parts partnership	Cash	Same line	FTAI 10-K FY2025 via D3 §2.7; PR Newswire (AAR), title only
Repairs	Component repair, including accessory repair at the Bristol joint venture from 2026	Cash, or a lower cost of producing FTAI's own modules	Not separately disclosed	FTAI Q3 2025 call via Insider Monkey; FTAI 10-K FY2025
V2500 shop visits	Performance restorations of FTAI-owned V2500 engines performed in IAE's network under the five-year EngineWise agreement; the restored engines are then leased, sold or sold as material	Restored engines for FTAI's own portfolio	V2500 engines and modules are named in the MRE Contract definition	RTX (Pratt & Whitney's parent) release, 6 Jun 2024; FTAI 10-Q Q2 2026

Source: FTAI Aviation, Form 10-K for FY2025, filed 27 Feb 2026, <https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai-20251231.htm> ; Form 10-Q for the quarter ended 30 Jun 2026; RTX, "IAE AG and FTAI Aviation Sign Strategic V2500 Engine Maintenance Services Agreement", 6 Jun 2024; Insider Monkey, FTAI Q3 2025 call

transcript, Oct 2025; PR Newswire, AAR–FTAI release, 2025 (title only). Compiled from D5 §2.1–2.3, D3 §2.7 and D7 §2.4. Accessed 2026-10-03.

The engine types are the two CFM56 variants of Part I §2 and the V2500 of Part I §10. The -7B is the sole engine of the Boeing 737NG; the -5B powers about six in ten A320ceo-family aircraft; the V2500 powers most of the rest. Chromalloy's October 2025 release put the -5B and -7B in-service fleet at "over 20,000 engines", one of several counts Part III §18 sets side by side. [Chromalloy release, 30 Oct 2025; reconciliation R-001] The Module Factory language in the filings names only the two CFM56 variants; the V2500 work is a different model, which §51 describes.

Who buys is less well documented than what is sold. The research identified four classes of customer: airlines operating 737NGs and A320ceos; aircraft and engine lessors, who must redeliver engines in contracted condition (Part II §13); other shops buying modules or material; and the vehicles of the Strategic Capital Initiative (SCI), which are the counterparties to the MRE Contract revenue line. [D5 §3] FTAI's own Aviation Leasing segment is an internal customer. Two customers of the segment accounted for 16% and 11% of revenue in the third quarter of 2025; their identities, and whether either is the 2025 Partnership, were not retrieved. [FTAI 10-Q Q3 2025, via D10 §4.9; D5 U17]

46. The Module Factory: how an exchange works for a customer

Part I §3.4 boxed the module exchange and the Module Factory in short form. This section is the full treatment: the sequence an exchange follows, what happens to the unit the customer hands back, what makes the flow a factory rather than a shop, and how it differs from the conventional visit of Part IV and from the engine manufacturer's own exchange offer.

46.1 THE FILING'S OWN WORDS, THEN THE FUNCTION

The Module Factory is a place, a programme and a process, and FTAI's filings use the name in all three senses. [registry C-16] The language first used in 2022, when the concept was launched with a third-party operator, and still returned by search against the FY2025 10-K, is: "The Module Factory is a dedicated commercial engine maintenance center focused on modular repair and refurbishment of CFM56-7B and CFM56-5B engines, based at Lockheed Martin Commercial Engine Solutions' 500,000 square foot facility in Montreal, Canada with a capacity for up to 300 shop visits per year". [FTAI 10-K FY2025; FTAI 10-K FY2022] That is the place. The programme is the "dedicated commercial maintenance program" of §45.1. The process is what the rest of this section describes: a production line that builds and holds serviceable modules so that an engine's maintenance becomes a swap from inventory rather than a repair to order. FTAI bought the Montréal facility outright in September 2024 (§50.1).

A conventional performance restoration, as Part I §6 and Part IV describe it, follows one sequence. The engine comes off the aircraft when its temperature margin or a life limit runs out. It waits for an induction slot. The shop disassembles it to the module or piece-part level, inspects every part, repairs or replaces what is out of limits, replaces the life-limited parts that would fall due in the next run, rebuilds the engine and runs it in a test cell. The owner waits for its own engine through all of this. Turnaround for a full CFM56 overhaul was 90 to 120 days in 2025 against about 60 before the pandemic, GE's shops report "sustaining ~90 days", a consultancy blog gives 75 to 90 days for a restoration and 120 to 150 for a heavy visit with life-limited parts, and the wait for a slot before any of that begins had grown by two to three months and in some cases six. [Aviation Business News, 2025; GE Aerospace, 27 May 2026; Safe Fly Aviation, consultancy blog, 2026; Bain, 2024; reconciliation R-019]

The Module Factory inverts the order of the work. FTAI does the disassembly, inspection, repair and rebuilding before any customer engine arrives, on its own schedule, and holds the finished modules in inventory. When a customer engine arrives, the module that triggered the removal is taken off and a serviceable module from inventory is bolted on. The engine is tested and returned. The removed module then joins FTAI's production flow and becomes, in time, a serviceable module for a later customer. [D5 §2.2] The company's own description of the consequence is that "modules produced" is the quantity it reports each quarter, the way a factory reports units (§49). [FTAI Q2 2025 release, 29 Jul 2025; FTAI Q4 2025 call via Motley Fool, 26 Feb 2026]

46.2 THE SEQUENCE, STEP BY STEP

EXHIBIT 7.2 — ONE MODULE EXCHANGE, FROM REMOVAL TO RETURN: WHAT MOVES, WHO HOLDS IT, AND WHAT THE RESEARCH COULD AND COULD NOT SOURCE

Step	What happens	Who holds the engine or module	Sourced	Not retrieved
1. Removal	The engine comes off wing because its exhaust-gas temperature (EGT) margin is exhausted, a life-limited part is at its limit, or a hardware event forces it (Part I §5–§6)	The operator	The three removal drivers (Part I)	Which driver FTAI's exchanges most often answer

Step	What happens	Who holds the engine or module	Sourced	Not retrieved
2. Dispatch	The operator sends the engine, or in some arrangements the unserviceable module alone, to an FTAI shop	In transit to FTAI	"the engine (or, in some arrangements, the unserviceable module alone)" (D5 §2.2)	Whether FTAI ships modules to the customer's own shop instead
3. Split	The engine is separated at the major-module flanges and the expired module is removed	FTAI	The flange-split design (Part I §3)	FTAI's labour hours per exchange
4. Fit	A serviceable module of compatible build standard and documented life-limited-part status is installed from inventory	FTAI	Build-standard and records conditions (Part I §3.3)	FTAI's configuration-management practice; how remaining life is matched to the customer's planned run
5. Test	The rebuilt engine is run in a test cell; capacity quoted as >500 then >600 engine tests a year	FTAI	Test capacity (FTAI releases, 9 Sep 2024 and 26 Feb 2025)	Whether every module change requires a test-cell run under the shop manual (D1 U14)
6. Return	The engine goes back to the operator with one or more different serviceable modules, or a different engine of the same type is supplied	The operator	"the same engine with one or more different, serviceable modules installed, or a different engine of the same type" (D5 §2.2)	FTAI's quoted turnaround (D5 U3)
7. Settlement	The operator pays an exchange fee and gives up the unserviceable module; FTAI books revenue as the fee plus the fair value of the unit received (§47)	Title to the old module passes to FTAI	The industry price structure (D2 §2.5; D3 §2.7)	FTAI's exchange fee or price list (cluster K-01)

Step	What happens	Who holds the engine or module	Sourced	Not retrieved
8. Feedstock	The returned module is torn down, its repairable parts repaired, its expired life-limited parts replaced, its serviceable parts retained or sold as used material; the rebuilt module returns to inventory	FTAI	10-K language on teardown into "salvageable components ... manufactured into new modules"	Yield per returned module; share of parts scrapped

Source: FTAI 10-K FY2025 (via D3 §2.7 and D5 §2.2); FTAI releases of 9 Sep 2024 and 26 Feb 2025 (GlobeNewswire); D2 §2.5; D3 §2.7; D1 §2.3 and U14; D5 §2.2 and U3. Reconciliation cluster K-01.

Step 4 is where Part I's conditions for a module to move bear directly on the business. A module from inventory can go onto a customer's engine only if the two are at compatible build standards, if the module's records are complete for every life-limited part inside it, and if the part numbers match the receiving engine (Part I §3.3). The -5B and -7B share enough core hardware that used-parts dealers advertise one "CFM56-5B & -7B Core LLP Package", which is what lets a pool built for one variant draw on the other. [Aviation Fleet Support, undated listing, via Part I §3.3; D1 U2] How FTAI manages configuration across a pool of hundreds of modules, and how it matches the remaining life in an inventory module to the run a customer plans, were not retrieved. The industry's pricing convention, in which the fee is adjusted for the difference in life-limited-part life between the module handed over and the module received, implies that a mismatch is settled in price rather than avoided (§47). [D2 §2.5; D3 §2.7]

Step 5 produces one of the few numerical cross-checks the research allows. FTAI quotes test capacity in engine tests a year beside module capacity: 1,350 module overhauls and more than 500 engine tests at the September 2024 closing, and 1,800 modules and more than 600 tests after Rome was added in February 2025. [FTAI releases, 9 Sep 2024 and 26 Feb 2025] The ratio of modules to tests is 2.7 at the first date and 3.0 at the second, which is consistent with the three-modules-per-engine convention FTAI's releases use (Part I §3.2; reconciliation B-3) and with one test per engine returned. That is an observation about the company's own figures, not a disclosed rule; whether every exchange is tested was not verified. [D1 U14]

46.3 WHAT HAPPENS TO THE UNIT THE CUSTOMER HANDS BACK

The second meaning of "core" enters at step 7. In the trade an exchange is priced as a fee plus the core, the unserviceable module the customer hands over, which has value to the exchanger as

feedstock. [D5 §2.3; registry C-6] Part I §3.3 glossed feedstock in its narrow sense, the aircraft and engines bought to be torn down, and said later Parts would extend it. This Part extends it to the returned core and to engine inventory held for module production, which is how FTAI's filings use the material. [registry C-14]

A returned module is an engine section in miniature, and FTAI treats it the way Part V §32 describes a teardown. Its repairable parts are repaired, in FTAI's own shops, at specialist repair vendors (Part IV §29), from 2026 at the Bristol accessories joint venture (§50.6), and under the component-repair element of the January 2026 agreement with CFM (§52). Its expired life-limited parts are replaced, either with new parts from CFM at catalogue price or with used packages carrying the cycles a buyer needs (Part I §4; Part V §34). Its parts with remaining life are retained as used serviceable material for FTAI's own builds or sold through the exclusive serviceable-parts partnership of §45.1. The rebuilt module returns to inventory. [FTAI 10-K FY2025, via D3 §2.7; D5 §2.2] Material comes into this flow from four directions at once: customer cores from exchanges, engines retired from FTAI's own leasing fleet, engines on the aircraft the Strategic Capital vehicles own, and engines or aircraft bought for the purpose, such as the ten off-lease 737-700s acquired from WestJet in September 2026 "to expand the Company's supply of CFM56-7B engines and modules available to its Maintenance, Repair and Exchange customers". [FTAI release, 28 Sep 2026, via D6 §4.7; D5 §5]

WHY THE EXCHANGE TURNS MAINTENANCE INTO PRODUCTION

Three things change when the module, not the engine, is the unit of work, and FTAI does the work before the customer arrives rather than after. First, the customer's downtime becomes the swap-and-test interval, not the repair interval, because the repair was done earlier on another unit. Second, FTAI's own work becomes repetitive and schedulable: the same two engine variants, split at the same flanges, rebuilt to a stock of standard configurations, so the shop can be loaded like a production line and its output counted in units. Third, every exchange feeds the next one, because the unserviceable module that comes back is the raw material for a future serviceable module. The cost of this arrangement is the inventory: FTAI must own a stock of serviceable modules large enough to meet exchanges as they arise, and a stock of unserviceable ones in work. Section 53 gives the inventory figure. None of this is unique to FTAI in principle; StandardAero lists "engine module changes" among its quick-turn services and GE sells exchanges under TrueChoice Transitions (§46.5). What FTAI's filings describe as distinctive is the scale and the organisation of the work as a factory. [D5 §2.2; StandardAero brochure, Oct 2022; D2 §2.3]

46.4 DOWNTIME, AND WHAT IS KNOWN ABOUT IT

The whole commercial case for an exchange rests on the time saved, and the research could not source FTAI's own turnaround. [D5 U3] What it could source is the market's figures for the two alternatives. A hospital or quick-turn visit, which Part I §6.1 defined as a visit that fixes one

finding and returns the engine, runs about 45 days in 2025 trade reporting; a module exchange is one of the things such a visit does. [Aviation Business News, 2025; StandardAero brochure, Oct 2022] A performance restoration runs 75 to 90 days on a consultancy blog's figures and a full overhaul 90 to 120 days in trade reporting, with GE at about 90; the slot wait of two to six months comes before either. [Safe Fly Aviation, 2026; Aviation Business News, 2025; GE Aerospace, 27 May 2026; Bain, 2024] FTAI's Miami business is branded "QuickTurn" and the Rome joint venture "QuickTurn Europe", which names the intent without quantifying it; the lower-case workscope and the capitalised brand are kept distinct in this primer. [FTAI release, 5 Jun 2025; registry C-20] Section 55 works the arithmetic of the downtime saving with every input labelled.

46.5 CONTRAST WITH A CONVENTIONAL SHOP VISIT AND WITH TRUECHOICE TRANSITIONS

EXHIBIT 7.3 — THREE WAYS TO DEAL WITH AN EXPIRED CFM56 MODULE: THE CONVENTIONAL VISIT, THE OEM'S TRANSITION MENU, AND FTAI'S EXCHANGE, AS THE SOURCES DESCRIBE EACH

Dimension	Conventional performance restoration (Part IV)	GE TrueChoice Transitions (Part IV §28)	FTAI module exchange (this Part)
Who performs the work	Any Part 145 engine shop: the OEM network, an independent, an airline shop	GE Aerospace's network	FTAI's shops in Montréal, Miami, Lisbon and Orange; QuickTurn Europe in Rome (50%); partner shops in Indonesia and Egypt
What the owner gets back	Its own engine, restored	Per GE: "green time leases, exchanges and material buy-back, plus custom workscope with shorter builds and maximum used material"	Its own engine with a serviceable module from inventory, or a different engine of the same type
What the owner gives up	Nothing; it keeps every part, including remaining life in untouched modules	Under an exchange or buy-back, the unit or material handed over	The unserviceable module (the returned core)
Pricing form	Parts at catalogue list price plus labour by the hour (time-and-materials), or a flight-hour agreement	Not retrieved	Exchange fee plus the fair value of the core; no price list published (K-01)

Dimension	Conventional performance restoration (Part IV)	GE TrueChoice Transitions (Part IV §28)	FTAI module exchange (this Part)
Downtime	75–90 days (restoration, consultancy blog); 90–120 days (overhaul, trade press); plus a 2–6 month slot wait	Not retrieved	FTAI's turnaround not retrieved; quick-turn visits generally ~45 days (trade press)
Material content	New parts from the original equipment manufacturer (OEM) and OEM-approved repairs by default; used serviceable material (USM) and PMA where the owner accepts them	GE describes "maximum used material"; GE states it is "the largest used serviceable material provider" for its products	Used material, PMA parts from the Chromalloy venture, and OEM parts and repairs under the CFM materials agreement
Who bears inventory	The owner, through spare engines held for the turnaround	GE, through its material pool	FTAI, through module inventory (\$1,364.3M at 31 Mar 2026)
Capacity measure	Inductions per year (Part IV §27)	Not retrieved	Modules per year (§49)

Source: D2 §2.1, §2.3 and §2.4 (GE press releases on TrueChoice, GE Aerospace statements, 2026); Safe Fly Aviation, 2026 (consultancy blog); Aviation Business News, 2025; Bain, 2024; FTAI 10-K FY2025; FTAI 10-Q Q1 2026; D5 §2.2–2.3. Reconciliation R-019; registry C-4.

The second column matters because it shows that the engine manufacturer sells a version of the same thing. GE describes TrueChoice Transitions as "a broad range of options for changing ownership horizons, including green time leases, exchanges and material buy-back, plus custom worksopes with shorter builds and maximum used material", and separately states that "GE and CFM are the largest used serviceable material provider for their products" and that on the CFM56 "an open network exists with multiple providers who compete and invest to win shop visits". [GE Aerospace press releases, via D2 §2.3; GE Aerospace, 2026, via D2 §2.3 and §3.4] One dossier calls this offering "the OEM's direct competitor to FTAI's Module Factory". [D2 §2.3] The research did not retrieve TrueChoice Transitions pricing, volumes or turnaround. Aviation Week ran a piece titled "Daily Memo: Will Engine Module Swaps Endure In Leaner Times?", whose date and content were not retrieved; the title records that the trade press treats the module swap as a model whose durability is a live question. [Aviation Week, date not returned, via orchestrator notes] This Part describes the model and stops.

47. Exchanges, module sales and engine sales: pricing and revenue recognition

The segment books three kinds of transaction, and the boundary between them is where the dispute in §54 sits. This section defines the three, sets out how the industry prices an exchange and how US accounting rules book one, records what FTAI has and has not disclosed about its own fees, and states the definitional question the short report raised without resolving it.

47.1 THE THREE TRANSACTIONS

ENGINE SALE, OUTRIGHT MODULE SALE AND EXCHANGE

The segment's three product transactions. An engine sale books the whole-engine price as revenue. An outright module sale books a module's cash price as revenue with no unit coming back. An exchange books an exchange fee plus the fair value of the unserviceable unit received. In all three, revenue is recognised at the point in time when control of the delivered engine or module passes to the customer, and cost of sales is the carrying value of what was delivered. The January 2025 short report's central allegation concerns the boundary between the first and the other two (§54). [D5 §2.3]

EXCHANGE FEE AND THE RETURNED CORE

The exchange fee is the cash price a customer pays to receive a serviceable module in return for an unserviceable one. The returned core is the unserviceable module the customer hands over, which has value to the exchanger as feedstock. The industry prices an exchange as the fee plus the value of life-limited-part life transferred, adjusted for the condition of the other hardware. This is the second meaning of "core" in the primer; the first is the engine's hot middle module (Part I §1), and this Part writes "returned core" or "the unit handed back" whenever the exchange sense is meant. FTAI publishes no exchange price list, so the revenue-per-module figures of §56 are upper bounds on the fee, not prices. [D5 §2.3, §2.10; D2 §2.5, U7; D3 §2.7, U20; registry C-6]

The price structure of an exchange follows from Part I §4 and §8. A module's value is the sum of its serviceable parts at used-material discounts to list plus the remaining life of the life-limited parts inside it, less the repair needed to make it serviceable (Part V §36). [D3 §2.7] When a customer hands over a module whose turbine disk has 1,200 cycles left and receives one whose disk has 10,000, the customer has bought 8,800 cycles of disk life and the price reflects it. The fee above that life transfer, which is what FTAI, GE or MTU Maintenance (the German engine maker's shop) charge for the module itself, the avoided downtime and the margin, was not found in any source the research reached. [D2 U7; D3 U20; D5 U3; reconciliation cluster K-01] One consultancy blog quotes "removal costs of \$150,000–\$400,000 per module" for life-limited modules; the research recorded the phrase as ambiguous and did not use it. [Safe Fly Aviation, 2026, via D2 §2.5] Part IV §30 brackets the life-transfer element at about \$250,000 for a high-pressure-turbine set

carrying 10,000 cycles, or about \$2.0M if a whole stack's life is priced at \$200 a cycle, both derived from contested list prices and with the fee itself unknown. [D2 §2.5; reconciliation R-016, R-017]

47.2 HOW AN EXCHANGE IS BOOKED

ASC 606 AND NONCASH CONSIDERATION

ASC 606, Topic 606 of the US Accounting Standards Codification (ASC), is the US revenue-recognition standard for contracts with customers. Under it, noncash consideration received from a customer is measured at fair value and included in the transaction price. For an exchange, that means revenue equals the cash fee plus the fair value of the returned core, and cost of sales equals the carrying value of the module delivered. FTAI's filings state that MRE Contract sales to the Strategic Capital vehicles are booked under ASC 606 and that the profit on them is eliminated through equity-method earnings (§48). The exact wording of FTAI's revenue-recognition policy for exchanges, including how it values a returned core and whether it quantifies fees separately, was not retrieved.

[D5 §2.3, U1; D7 §2.5]

The consequence for reading FTAI's revenue is that an exchange generates more revenue than cash. If a customer pays a fee of F and hands over a core worth C , FTAI books F plus C as revenue and receives F in cash and a module in kind. The module in kind then becomes inventory at C and, after repair, cost of sales on a later exchange. [D5 §2.3] This is standard accounting, not a peculiarity of FTAI's. It matters here because the segment's revenue per module (§56) includes the fair values of cores received, and because an inventory build (§53) is partly the accumulation of cores.

WORKED EXAMPLE 7.1 — HOW ONE EXCHANGE BOOKS, WITH EVERY NUMBER ASSUMED

Inputs, all assumed for illustration because FTAI publishes none of them (D5 U1, U3; cluster K-01): a customer pays an exchange fee F of \$1.0M for a serviceable core module; the unserviceable core it hands over is valued by FTAI at a fair value C of \$0.6M; the module FTAI delivers is carried in inventory at \$1.1M, being its acquisition cost plus the parts, labour and overhead spent making it serviceable.

$$\begin{aligned} \text{revenue} &= F + C = 1,000,000 + 600,000 = \$1,600,000 \\ \text{cost of sales} &= \text{carrying value delivered} = \$1,100,000 \\ \text{gross profit} &= 1,600,000 - 1,100,000 = \$500,000, \text{ or } 31\% \text{ of revenue} \\ \text{cash received} &= F = \$1,000,000 \\ \text{inventory added} &= C = \$600,000 \text{ (the returned core, before repair)} \end{aligned}$$

So the transaction shows \$1.6M of revenue on \$1.0M of cash, and the \$0.6M difference is a module FTAI now owns and must spend money repairing before it can be sold again. If the same module

were sold outright for \$1.6M with nothing coming back, revenue and gross profit would be identical and cash would be \$0.6M higher. Limits of the example: every input is assumed; FTAI's valuation method for returned cores is not disclosed; the carrying value of a delivered module depends on how much of its content was used material, PMA or new OEM parts (§52); and the sourced revenue per module of \$56 is a blend of exchanges, outright sales, whole engines and parts. [D5 §2.3, §2.10]

47.3 WHAT IS KNOWN AND UNKNOWN ABOUT FTAI'S FEES

Three facts are known. First, the filings describe the outputs as "engine, module, and parts sales" and present them, for all customers other than the Strategic Capital vehicles, on a single income-statement line. [FTAI 10-K FY2025]

AEROSPACE PRODUCTS REVENUE (INCOME-STATEMENT LINE)

The consolidated income-statement line for the segment's sales of modules, engines and parts to customers other than the Strategic Capital vehicles: \$454,970 thousand in 2023, \$1,079,821 thousand in 2024 and \$1,600,456 thousand in 2025. It is distinct from MRE Contract revenue (§48), which the filings show on a separate line from 2025, and from segment revenue, which is the sum of the two. No retrieved filing disaggregates it into engines, modules, parts and repairs. [FTAI 10-K FY2025, segment table via orchestrator notes; D5 §2.3, U2; D10 §2.4]

Second, the only observable bound on the average fee is the ratio of that line to modules produced: \$1,600.5M divided by 757 modules is \$2.11M for 2025, and \$692.2M divided by 296 is \$2.34M for the second quarter of 2026. Those are upper bounds on the average cash fee for a module, because the numerator contains whole-engine sales, parts sales and the fair values of cores received while the denominator counts only CFM56 modules; §56 works the figures and their limits. [D5 §2.10–2.11; reconciliation R-046] Third, the one cost lever FTAI has quantified is on the production side rather than the price side: the Bristol accessories venture is "expected to deliver up to \$75,000 in average savings per shop visit". [FTAI Q3 2025 call, via Insider Monkey]

What is not known is longer. The exchange fee itself, the method for valuing a returned core, whether fees and core values are quantified separately, the split of the revenue line among engines, modules, parts and repairs, the margin by line, and whether disclosure changed after the February 2025 review are all recorded as open. [D5 U1, U2, U3; D2 U7; D3 U20] Trade interviews that may contain estimates were located and not read. [In Practise, "FTAI Aviation: CFM56 repair and module swap", located by D2 and D5, not read]

47.4 THE DEFINITIONAL QUESTION THE SHORT REPORT RAISED

A whole CFM56 is, in FTAI's counting convention, three modules (Part I §3.2). A serviceable engine built from three modules and sold outright is an engine sale. The same three modules installed one at a time into three different customers' engines are three module exchanges. The

revenue booked may be similar; the character of the transaction, and what a reader infers about how repeatable it is, differ. The January 2025 report by Muddy Waters Research alleged, in the words of the later securities complaint, that FTAI had "presented whole engine sales as module sales" and had reported "one-time engine sales as MRO revenue"; FTAI "strongly disagrees", and its Audit Committee found the allegations "without merit". [law-firm case pages via orchestrator notes; Seeking Alpha, Jan 2025; FTAI 8-K exhibit, Feb 2025] This section states only the definitional point: because the filings do not disaggregate the revenue line, a reader cannot tell from the filings alone how much of it is whole engines and how much is modules and parts. [D5 U2] Section 54 gives each party's position with its source.

48. MRE Contract revenue: the Strategic Capital vehicles as customer

From 2025 the segment has a second revenue line whose customer is a partnership that FTAI manages and part-owns. This section defines the line by quoting both filings that describe it, gives every figure the research retrieved, explains how the profit on it is treated, and derives the share of segment revenue it represented by period.

48.1 THE CUSTOMER

Part VIII §60 to §62 describe the Strategic Capital Initiative (SCI) in full. For this section the reader needs four facts. FTAI launched the initiative on 30 December 2024. Its first vehicle, the 2025 Partnership, raises third-party capital into a closed-end partnership whose special-purpose vehicles buy on-lease mid-life 737NG and A320ceo aircraft; it reached a \$2.0B equity hard cap in October 2025 and about \$6.0B of total capital across more than 300 aircraft by August 2026. FTAI is the Servicer, the party contracted to manage the assets for a fee (Part VIII §61), and holds a limited-partner stake described as 19% in the FY2025 10-K and the 2026 10-Qs and as 20% in the Q3 2025 10-Q. [FTAI releases, 30 Dec 2024, 27 Oct 2025 and 17 Aug 2026; FTAI 10-K FY2025; FTAI 10-Qs; reconciliation R-048] And each release on the vehicle repeats one sentence about its engines: they "will be powered exclusively via engine and module exchanges with FTAI's Maintenance, Repair and Exchange (MRE) business". [FTAI releases, 26 Feb 2025, 25 Mar 2025 and 17 Aug 2026, via D7 §2.4]

48.2 THE LINE, DEFINED BY TWO FILINGS

MRE CONTRACT REVENUE

The income-statement line, new in FY2025, for FTAI's sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the special-purpose vehicles of the 2025 Partnership under the MRE Contract: \$335.8M for 2025, \$221.2M in the first quarter of 2026, \$182.8M in the second (against \$69.6M a year earlier) and \$404.0M for the first half of 2026 (against \$170.2M). The Q2 2026 10-Q defines it as "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership" and describes it as covering "the sale and purchase of engines to and from the 2025 Partnership". The Q3 2025 10-Q describes "stand-ready obligations to provide replacement CFM56-5B and CFM56-7B engines to customers as they become unserviceable". Profit on these related-party sales is eliminated through equity-method earnings. [FTAI 10-Q Q2 2026, via orchestrator notes; FTAI 10-Q Q3 2025, via D10 §2.4; FTAI 10-K FY2025; D5 §2.3; registry C-15]

STAND-READY OBLIGATION

A contractual duty to supply on demand. Under the MRE Contracts FTAI must provide replacement CFM56-5B and -7B engines to the customer as the customer's engines become unserviceable, so it must hold a pool of serviceable engines against the obligation whether or not any particular engine fails in a given period. The FY2025 10-K lists these obligations among its risk factors. [FTAI 10-Q Q3 2025, via D10 §2.4 and §5; D10 U-15]

The two filings describe the same line in different registers, and the research recorded the difference as a tension rather than resolving it. [reconciliation R-044] The Q3 2025 10-Q's language is that of an obligation: FTAI stands ready to supply engines as they become unserviceable, which is the language of a service a provider commits to perform over time. The Q2 2026 10-Q's language is that of sales: a transaction price for engines and modules sold to specified buyers, which is the language of a product delivered at a point in time. One dossier reads the line as "a customer pays FTAI to keep its CFM56 fleet powered"; another reads it as "a sale by FTAI of the replacement module or engine ... usually with the run-out unit taken back as feedstock". [D10 §2.4; D7 §2.4] Both quotations are from FTAI's own filings of different dates, both are carried here, and the phrase "sale and purchase of engines to and from" in the later filing indicates that units also flow back from the Partnership to FTAI, which is the returned-core mechanism of §46.3 applied at engine level. Whether the line is recognised at points in time, over time, or in some combination was not retrieved. [D5 U1]

48.3 THE FIGURES BY PERIOD

EXHIBIT 7.4 — MRE CONTRACT REVENUE BY PERIOD, BESIDE SEGMENT REVENUE, WITH THE RELATED-PARTY SHARE DERIVED

Period	MRE Contract revenue	Aerospace products revenue	Segment revenue (sum)	MRE share of segment revenue (derived)	Source
FY2023	\$0	\$455.0M	\$455.0M	0%	FY2025 10-K segment table
FY2024	\$0	\$1,079.8M	\$1,079.8M	0%	FY2025 10-K segment table
Q1 2025	\$100.6M (derived: 170.2 – 69.6)	not retrieved	not retrieved	not computable	Q2 2026 10-Q comparatives
Q2 2025	\$69.6M	≈\$422.0M (derived: 491.6 – 69.6, on a derived base)	≈\$491.6M (derived: 875.0 ÷ 1.78)	≈14.2%	Q2 2026 10-Q; Q2 2026 release (+78%)
1H 2025	\$170.2M	not retrieved	not retrieved	not computable	Q2 2026 10-Q comparatives
2H 2025	\$165.6M (derived: 335.8 – 170.2)	not retrieved	not retrieved	not computable	10-K and 10-Q
FY2025	\$335.8M	\$1,600.5M	\$1,936.2M	17.3%	FY2025 10-K
Q1 2026	\$221.2M	\$522.6M	\$743.8M	29.7%	Q1 2026 10-Q; Q1 2026 release
Q2 2026	\$182.8M	\$692.2M (derived: 1,214.8 – 522.6)	\$875.0M	20.9%	Q2 2026 10-Q; Q2 2026 release
1H 2026	\$404.0M	\$1,214.8M	\$1,618.8M (derived)	25.0%	Q2 2026 10-Q

Source: FTAI Aviation, Form 10-K for FY2025 (segment table, via orchestrator notes; segment geographic table total \$1,936,244 thousand, via D5 §4.5); Form 10-Q for the quarter ended 31 Mar 2026; Form 10-Q for the quarter ended 30 Jun 2026, <https://www.sec.gov/Archives/edgar/data/0001590364/000162828026051412/ftai-20260630.htm> ; Q1 2026 and Q2 2026 earnings releases (GlobeNewswire, 29 Apr 2026 and 29 Jul 2026). Shares and the figures marked derived are arithmetic on the filed numbers, done in the research (D5 §2.8, §4.1) and here. Reconciliation B-1, R-043. Accessed 2026-10-03.

Three documents agree, without quoting each other, that segment revenue is the sum of the two lines: the 10-K's segment geographic table totals \$1,936,244 thousand, which is exactly \$1,600,456 thousand plus \$335,788 thousand; the Q1 2026 release gives segment revenue of \$743.8M, which is the 10-Q's \$522,585 thousand plus \$221,230 thousand; and the Q2 2026 10-Q's \$182.8M of MRE Contract revenue equals the difference between the release's \$875.0M of segment revenue and the derived \$692.2M of products revenue. [reconciliation B-1] The explicit segment-note mapping that assigns the MRE line to the segment was not retrieved, and one dossier treated the placement as open; this Part and Part X each carry that sentence. [D10 T-4; reconciliation R-043; E.1 rule 8]

WORKED EXAMPLE 7.2 — THE RELATED-PARTY SHARE OF SEGMENT REVENUE, PERIOD BY PERIOD

Inputs: MRE Contract revenue of \$335.8M (FY2025), \$221.2M (Q1 2026), \$182.8M (Q2 2026) and \$69.6M (Q2 2025) (all sourced: FY2025 10-K and 2026 10-Qs); segment revenue of \$1,936.2M (FY2025, sourced: 10-K geographic table), \$743.8M (Q1 2026, sourced: release) and \$875.0M (Q2 2026, sourced: release); Q2 2025 segment revenue derived as $\$875.0\text{M} \div 1.78 = \491.6M from the release's "+78%" (derived).

$$\begin{aligned}
 &\text{FY2025: } 335.8 \div 1,936.2 = 17.3\% \\
 &\text{Q2 2025: } 69.6 \div 491.6 = 14.2\% \text{ (on a derived base)} \\
 &\text{Q1 2026: } 221.2 \div 743.8 = 29.7\% \\
 &\text{Q2 2026: } 182.8 \div 875.0 = 20.9\% \\
 &\text{1H 2026: } 404.0 \div 1,618.8 = 25.0\% \\
 &\text{growth in the line, Q2 2026 on Q2 2025: } 182.8 \div 69.6 = 2.63\times; \text{ 1H: } 404.0 \div 170.2 = 2.37\times
 \end{aligned}$$

So between one seventh and three tenths of the segment's revenue in each period from the second quarter of 2025 came from sales to a partnership FTAI manages and part-owns. The segment's Adjusted EBITDA margin (Adjusted EBITDA being FTAI's adjusted earnings measure, boxed in §53.1 and defined in Part X §80) was 35% in the fourth quarter of 2025 and about 29 to 30% in the first half of 2026, the period in which the related-party share was highest. The research recorded that movement and declined to infer a cause, because the filings do not disclose margin by revenue line and because the elimination of profit on these sales (\$48.4) runs through a different line from segment revenue. Limits of the example: the Q2 2025 base is derived from a growth rate; the Q1 2025 and second-half 2025 segment revenues were not retrieved; and the 2025 Partnership was buying aircraft through 2025 and into 2026, so the first exchanges on newly acquired aircraft may not represent a steady state. [D5 §2.8; D7 §5.2; reconciliation R-043, R-047]

48.4 THE PROFIT-ELIMINATION MECHANICS, BOTH READINGS

PROFIT ELIMINATION (INTRA-ENTITY PROFIT ELIMINATION)

When a company sells to an entity it accounts for under the equity method, the share of profit on the sale that corresponds to its own ownership is removed from current earnings and recognised later, as the investee realises the asset's value. FTAI's Q2 2026 10-Q: the elimination is "of profit on sales of engines and modules to the 2025 Partnership under ASC 606. The profit from the MRE Contract revenue is eliminated through equity method earnings and will be recognized over time as the 2025 Partnership generates income from leasing and sales activities". Eliminations were \$10.0M in Q1 2026, \$6.6M in Q2 2026 and \$16.6M for the half. Whether the elimination is proportional to FTAI's stake or covers 100% of the profit is not stated. Part VIII §63 carries the full treatment. [FTAI 10-Q Q2 2026, via D7 §2.5; reconciliation R-047]

The disclosed sentence supports two readings, and the research presents both without choosing. [reconciliation R-047; E.1 rule 12] Under the first, the usual treatment of profit on sales to an equity-method investee, FTAI eliminates only the share of profit that corresponds to its own stake, because the other 81% (or 80%) of the Partnership's owners are third parties who have paid an arm's-length price. Under the second, FTAI defers the whole profit until the Partnership realises it. The arithmetic differs by a factor of about five.

WORKED EXAMPLE 7.3 — WHAT \$16.6M OF ELIMINATIONS IMPLIES ABOUT THE GROSS PROFIT ON MRE SALES TO THE PARTNERSHIP, UNDER EACH READING

Inputs: eliminations of \$16.6M for 1H 2026 (sourced: Q2 2026 10-Q, \$10.0M plus \$6.597M); FTAI's stake of 19% (sourced: FY2025 10-K and 2026 10-Qs) or 20% (sourced: Q3 2025 10-Q); gross MRE Contract revenue to the Partnership of \$404.0M for 1H 2026 (sourced: Q2 2026 10-Q).

Reading A, proportional to the stake:

at 19%: implied gross profit = $16.6 \div 0.19 = \$87.4\text{M}$ for the half
(≈\$175M annualised)

at 20%: implied gross profit = $16.6 \div 0.20 = \$83.0\text{M}$ for the half
(≈\$166M annualised)

implied gross margin on \$404.0M = 21.6% (19%) or 20.5% (20%)

Reading B, 100% deferred:

implied gross profit = \$16.6M for the half (≈\$33M annualised)

implied gross margin on \$404.0M = 4.1%

So the same disclosed number is consistent with the Partnership sales carrying a gross margin in the low twenties, or one in the low single digits, depending on an accounting method the filing does not state. Under reading A about \$87M of half-year gross profit on related-party sales would have been recognised in the segment's revenue and cost of sales and about \$17M of it removed one line lower; under reading B the whole profit would have been removed. A reader comparing the segment's reported margin on Partnership volume with its margin on third-party volume cannot do so from the

filings. Limits of the example: the elimination method is undisclosed; the 19% and 20% figures come from filings of different dates and may reflect dilution at the final close; the half-year may include catch-up exchanges on newly acquired aircraft; and "gross profit" here is the profit the elimination acts on, whose definition the filing does not give. [FTAI 10-Q Q2 2026, via D7 §2.5 and orchestrator notes; reconciliation R-047, R-048; D7 T7]

Two further facts bear on how the line is read. First, the Partnership also pays FTAI servicing fees, \$10.2M in 2025 and \$12.85M in the first half of 2026, and FTAI records equity-method income on its stake; neither sits in Aerospace Products, and Part VIII §63 sets out all the streams. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026, via D7 §2.5] Second, the 10-K accounts for FTAI's sales of aircraft into the Partnership as "non-recurring in nature and not considered part of the Company's ordinary activities", while the engine and module sales under the MRE Contract are ordinary revenue under ASC 606; the two kinds of sale to the same buyer are treated differently, as Part VIII §64 explains. [FTAI 10-K FY2025, via D7 §2.5; reconciliation R-057]

49. Volumes, capacity and the share target, year by year

FTAI reports its output in modules. This section gives every module count the research found, with how each is known; the 2026 target as filed and as reported from a later call; every capacity statement with the unit it is counted in and the footprint it covers; and the share goal against three different denominators. Nothing here is compared across units without saying so.

49.1 MODULES PRODUCED

MODULES PRODUCED (MODULES REFURBISHED)

FTAI's quarterly count of CFM56 modules its Module Factory has refurbished and made serviceable: 184 in the second quarter of 2025, 228 in the fourth, 757 for 2025 against a target of 750, 296 in the second quarter of 2026 and 566 for the first half. A module produced is not a module sold: it may enter inventory, go into an engine in FTAI's own leasing fleet, or be built into a Mod-1 generator set (Part IX). FTAI's releases convert modules to engines at three per engine. Whether modules produced at the 50%-owned Rome shop are included was not retrieved. [FTAI Q2 2025 release; FTAI Q4 2025 call via Motley Fool and MarketBeat; FTAI Q2 2026 release and call; D5 §2.4, U10; registry C-4]

EXHIBIT 7.5 — CFM56 MODULES PRODUCED, BY PERIOD, WITH HOW EACH FIGURE IS KNOWN

Period	Modules	How known	Source
FY2025 plan (set Dec 2024)	≈400 from Montréal	"an average of 100 modules per quarter produced at the Company's Montreal facility in fiscal year 2025"; ×4 derived	FTAI release, 30 Dec 2024
Q4 2024	≈136	Derived: Q4 2025's 228 was "up 68% year-over-year"; $228 \div 1.68 = 135.7$	FTAI Q4 2025 call via Motley Fool, 26 Feb 2026
Q2 2025	184	Stated	FTAI Q2 2025 release, 29 Jul 2025
Q1 + Q3 2025	345	Derived: $757 - 184 - 228$; the split was not retrieved (D5 U4)	FTAI Q4 2025 call summaries
Q4 2025	228	Stated (+68% y/y)	FTAI Q4 2025 call via Motley Fool and MarketBeat
FY2025	757	Stated; target was 750	FTAI Q4 2025 call via Motley Fool and MarketBeat
Q1 2026	≈270	Derived: $566 - 296$	FTAI Q2 2026 release (chain map)
Q2 2026	296	Stated (+61% y/y); $296 \div 1.61 = 183.9$, consistent with Q2 2025's 184	FTAI Q2 2026 release and call via GuruFocus
1H 2026	566	Stated	FTAI Q2 2026 release (chain map)
FY2026 target	1,050	Set with the Q4 2025 results, 25 Feb 2026 (+39% on 757)	FTAI Q4/FY2025 release and call
FY2026 target, later	1,200	"2026 forecast raised to 1,200 modules" on the 30 Jul 2026 call, per call coverage; not returned from the 29 Jul 2026 release text	GuruFocus, Nasdaq.com, MarketBeat, 30 Jul 2026
2022, 2023, 2024 full years; Q1 and Q3 2025	not retrieved	D5 U4	n/a

Source: FTAI Aviation / GlobeNewswire, "FTAI Aviation Launches Strategic Capital Initiative and Announces 2025 Financial Guidance", 30 Dec 2024; FTAI Q2 2025 results release, 29 Jul 2025; Motley Fool, FTAI Q4 2025 earnings call transcript, 26 Feb 2026

2026; MarketBeat, Q4 2025 call highlights, 26 Feb 2026; FTAI Q2 2026 results release, 29 Jul 2026; GuruFocus, Q2 2026 call transcript, 30 Jul 2026. Derivations in D5 §2.4. Reconciliation R-036; E.2 "FTAI modules produced". Accessed 2026-10-03.

WORKED EXAMPLE 7.4 — DERIVING THE TWO MISSING QUARTERS, AND WHAT THE 2026 TARGETS IMPLY FOR THE SECOND HALF

Inputs: Q4 2025 output of 228 modules, "up 68% year-over-year" (sourced: Q4 2025 call); 1H 2026 output of 566 and Q2 2026 of 296 (sourced: Q2 2026 release); FY2025 output of 757 (sourced); the 2026 targets of 1,050 (sourced: Feb 2026 release) and 1,200 (sourced as call coverage of the July 2026 call).

$$\begin{aligned} \text{Q4 2024} &= 228 \div 1.68 = 135.7, \text{ about } 136 \\ \text{Q1 2026} &= 566 - 296 = 270 \\ \text{Q1} + \text{Q3 2025} &= 757 - 184 - 228 = 345 \text{ (split unknown)} \\ \text{second half of 2026 needed for 1,050: } &1,050 - 566 = 484, \text{ or } 242 \text{ a quarter} \\ \text{second half of 2026 needed for 1,200: } &1,200 - 566 = 634, \text{ or } 317 \text{ a quarter} \\ \text{run-rate check: Q2 2026 was 296; Q1 2026 was 270} \end{aligned}$$

So the filed target of 1,050 for 2026 would be met at a second-half pace below the second quarter's, and the 1,200 figure reported from the July call would need a pace about 7% above it. The 2024 full year is not retrievable from these figures: only its fourth quarter is derivable, and the Dec 2024 plan of about 400 modules from Montréal for 2025 was exceeded by the 757 actual, though the actual may include sites the plan did not. Limits of the example: the Q4 2024 figure depends on a rounded growth rate; produced is not sold; and whether Rome's output is counted is unknown (D5 U10). [D5 §2.4; reconciliation R-036]

The 2026 target has two figures and the research keeps both. The company set 1,050 with its fourth-quarter 2025 results on 25 February 2026, a 39% increase on the 757 produced in 2025. [FTAI Q4/FY2025 release, 25 Feb 2026; FTAI Q4 2025 call via Motley Fool] Coverage of the second-quarter 2026 call on 30 July 2026 reported that management had raised the forecast to 1,200; the orchestrator's search of the 29 July 2026 release text did not return that figure, and one summary of the same call still read "2026 goal is 1,050 modules". [GuruFocus, Nasdaq.com and MarketBeat, 30 Jul 2026; Quartr, Q2 2026 summary; orchestrator notes; reconciliation R-036] This primer writes the pair as "1,050 (Feb 2026 release), raised to 1,200 on the July 2026 call (call coverage)" wherever it appears. [E.1 rule 2]

Engine sales counts, as distinct from module counts, were not disclosed in anything the research retrieved, and neither were counts of V2500 shop visits completed under the IAE agreement (§51). [D5 U6]

49.2 CAPACITY: WHAT EACH STATEMENT COUNTS

Part IV §27 defines a shop's capacity as the number of inductions a year it can take, set by floor space and stands, licensed technicians, test-cell hours and parts flow. FTAI's unit is different. Its

releases quote capacity in modules a year, and the two units are never compared directly in this primer. [D2 §2.4; registry C-4] Four capacity statements exist, at four dates, in three units, for different footprints.

EXHIBIT 7.6 — FTAI'S CAPACITY STATEMENTS: UNIT, FOOTPRINT, DATE AND SOURCE LEVEL

Statement	Unit	Footprint covered	Date and source level	Source
"capacity for up to 300 shop visits per year"	Shop visits (whole engines)	Montréal alone, as a conventional shop	Language first used in the FY2022 10-K; still returned from the FY2025 10-K (filing)	FTAI 10-K FY2022 and FY2025
"capacity for 1,350 CFM56 module overhauls and >500 engine tests per year"	Modules; engine tests	Montréal plus Miami, at the closing of the Lockheed Martin Commercial Engine Solutions (LMCES) purchase	9 Sep 2024 (company release)	GlobeNewswire, 9 Sep 2024
QuickTurn Europe "adds capacity for 450 modules (150 engines) per year, taking FTAI's total to 1,800 modules (600 engines) and >600 engine tests per year"	Modules; engines at three per engine; engine tests	Montréal, Miami and Rome	26 Feb 2025 (company release)	GlobeNewswire, 26 Feb 2025
Lisbon "113,000 sq ft, 300+ modules/yr"	Square feet; modules	Lisbon alone	Call coverage via the chain map; unverified against a primary source	Chain map; D2 §3.3
"increased CFM56 module production capacity to 3,000 per year, supporting its 25% market share goal and 100 Mod-1 units annually"	Modules	All sites and partners, including Indonesia and Egypt	30 Jul 2026 call (call coverage)	GuruFocus and Quartr, Q2 2026 call

Source: FTAI Aviation, Form 10-K for FY2022 and FY2025; GlobeNewswire, "FTAI Aviation Closes the Acquisition of LMCES", 9 Sep 2024, <https://www.globenewswire.com/news-release/2024/09/09/2943296/0/en/FTAI-Aviation-Closes-the->

Acquisition-of-LMCES.html ; GlobeNewswire, FTAI Q4/FY2024 results and QuickTurn Europe agreement, 26 Feb 2025; GuruFocus and Quartr coverage of the Q2 2026 call, 30 Jul 2026; chain map (orchestrator). Reconciliation R-037, R-041; E.1 rules 2 and 17. Accessed 2026-10-03.

The statements do not contradict one another; they measure different things. [D5 T2; reconciliation R-037] The 300-visit figure describes one building run as a conventional shop in 2022. The 1,350 and 1,800 figures are module counts for two and then three sites, with the engine count given at three modules per engine. The 3,000 figure is a module count for every site and partner as of mid-2026, reported from a call rather than a filing or release. And the Lisbon figure comes from the chain map's reading of call coverage and was not verified. [E.1 rule 2]

WORKED EXAMPLE 7.5 — PUTTING THE CAPACITY STATEMENTS ON ONE SCALE, UNDER EACH MODULE-PER-ENGINE CONVENTION

Inputs: the four statements above (sourced as labelled); FTAI's conversion of three modules per engine (sourced: the 26 Feb 2025 release's "450 modules (150 engines)"); Part I's anatomical count of four major modules including the gearbox (sourced: D1 §2.3); 1H 2026 output of 566 modules (sourced).

```
300 shop visits at 3 modules per engine = 900 module-equivalents; at 4 =
1,200
1,350 modules at 3 per engine = 450 engines; at 4 = 338
1,800 modules at 3 per engine = 600 engines (the release's own figure);
at 4 = 450
3,000 modules at 3 per engine = 1,000 engines; at 4 = 750
utilisation check: 566 modules in 1H 2026, annualised = 1,132
1,132 ÷ 1,800 (release, Feb 2025) = 63%
1,132 ÷ 3,000 (call coverage) = 38%
```

So, under FTAI's own convention, the 2022 statement about one building would be about 900 module-equivalents, below the 1,350 the company claimed for two buildings in 2024; the figures are consistent with growth, not in conflict. The utilisation check shows that output in the first half of 2026 ran well below either stated capacity, which is what a company adding sites that are "still ramping" would report; it is not a measure of how full any one shop was. Limits of the example: a shop visit is not three module overhauls (an exchange touches one module; an overhaul opens all of them), so the first line is a scale check only; the 3,000 figure is call coverage; and whether partner shops' output counts toward FTAI's reported modules is unknown. [D5 §2.4, T2, U10; reconciliation R-035, R-037; Quartr, Q2 2026 summary]

Two further dimensions of capacity are quoted or implied. Test capacity, as §46.2 noted, is quoted as more than 500 and then more than 600 engine tests a year; the number of test cells and the staffing by site were not retrieved. [GlobeNewswire, 9 Sep 2024 and 26 Feb 2025; D5 U9] And the Mod-1 generator set of FTAI Power draws on the same module production: the 3,000-module statement is described as supporting both the share goal and 100 Mod-1 units a year, and one

dossier computes that 100 units would consume about 100 engines, roughly 300 module-equivalents, a year under the three-per-engine convention. [GuruFocus, Q2 2026 call; D8 §2.10] The allocation rule between aftermarket customers and FTAI Power was not disclosed. [D5 §5]

49.3 THE SHARE GOAL, AGAINST THREE DENOMINATORS

25% SHARE GOAL

FTAI's stated goal of 25% of CFM56 shop visits, which the 3,000-module capacity is said to support alongside 100 Mod-1 units a year. The denominator (the industry's annual shop-visit count, or a module equivalent of it) and the target year were not disclosed in anything the research retrieved. An earlier framing, from a February 2025 release, put the segment's addressable market at \$22B with FTAI holding about 5% as of 2024. [GuruFocus and Quartr, Q2 2026 call; FTAI release, 26 Feb 2025, via D7 §2.4; D5 U5; reconciliation R-042]

EXHIBIT 7.7 — FTAI'S SHARE, AS FRAMED THREE WAYS, WITH THE DENOMINATOR EACH NEEDS

Framing	FTAI's figure	Denominator	Denominator's source and level	What it does not say
Dollar share of an addressable market	~5% as of 2024	\$22B "total addressable market" for MRE	FTAI release, 26 Feb 2025 (company release); basis of the \$22B not stated	Whether the \$22B is CFM56 only, all three engine types, parts and labour or parts alone
Share of CFM56 shop visits	25% goal	2,300–2,400 visits a year in 2026–2028 (GE via Aviation Week, 2026); ~2,500 peak in 2025–26 (Safran via Aviation Week, c. 2023/24)	OEM statements via trade press	The year by which 25% is targeted; whether an exchange of one module counts as a visit
Module-equivalents against visits	1,200 modules ≈ 400 engines ≈ 17% of a 2,350-visit year; 3,000 modules ≈ 1,000 engines ≈ 43%	2,350 visits (midpoint of GE's range) at three modules per engine	Derived in the research (D2 §3.3)	That modules are sold singly into engines that may also visit other shops, so these are material-throughput shares, not shop-visit shares

Source: FTAI / GlobeNewswire, "FTAI's Strategic Capital Initiative Secures Commitment for \$2.5 Billion of Asset-Level Debt Financing", 26 Feb 2025 (via D7 §2.4); GuruFocus and Quartr coverage of the Q2 2026 call; Aviation Week, "CFM56 Overhaul

Demand Remains Strong, GE Aerospace Says", 2026; Aviation Week, "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24; D2 §3.3. Reconciliation R-006, R-042; E.2 "FTAI share framing". Accessed 2026-10-03.

One arithmetical observation about the first row is worth recording because it is derived from two filed or released numbers and nothing more. Five percent of \$22B is \$1.1B, and the segment's 2024 products revenue was \$1,079.8M. [FTAI release, 26 Feb 2025; FTAI 10-K FY2025 segment table] The release does not say how the 5% was computed, so the closeness of the two figures is noted and not interpreted.

WORKED EXAMPLE 7.6 — WHAT 25% OF CFM56 SHOP VISITS MEANS IN MODULES, UNDER EACH DEFINITION OF A MODULE AND OF AN EXCHANGE

Inputs: 2,300–2,400 CFM56 shop visits a year in 2026–2028 (sourced: GE via Aviation Week, 2026; the earlier Safran peak of about 2,500 is the alternative, sourced c. 2023/24); three modules per engine (sourced: FTAI release, 26 Feb 2025); four major modules including the gearbox (sourced: Part I §3.2); 1.5 modules exchanged per shop-visit event (one dossier's labelled assumption, D7 §5.2; FTAI discloses no modules-per-event ratio); capacity of 3,000 modules (call coverage) and the 2026 targets of 1,050 and 1,200.

25% of visits = 575–600 visits a year (on GE's range); 625 on Safran's 2,500

if every one of those visits consumed a full engine's modules:

at 3 per engine: 1,725–1,800 modules a year

at 4 per engine: 2,300–2,400 modules a year

if each visit consumed 1.5 modules (assumed): 863–900 modules a year
against capacity of 3,000 (call coverage):

58–60% utilised at 3 per engine; 77–80% at 4; 29–30% at 1.5

against the 3-per-engine requirement of 1,725:

$1,050 \div 1,725 = 61\%$; $1,200 \div 1,725 = 70\%$

So "25% of shop visits" could require anywhere from about 860 to about 2,400 modules a year depending on how many modules a visit consumes, a figure FTAI does not disclose; at FTAI's own three-per-engine convention and a full set per visit it is 1,725 to 1,800, which the stated 3,000-module capacity exceeds and the 2026 targets fall short of. The target year for the 25% is unknown, and the visit count itself is a plateau that forecasters have peaking in 2027 and 2028 and fading afterwards (Part III §23; Part IV §26). Limits of the example: the 1.5-modules-per-event figure is an assumption from another dossier; an exchange of one module may or may not be counted by FTAI as a share of a "visit"; the 3,000 and 1,200 figures are call coverage; and the visit count carries its own tension (R-006). [D2 §3.3; D5 U5; D7 §5.2; reconciliation R-006, R-035, R-042]

50. Facilities and partners: Montréal, Miami, Rome, Lisbon, Orange, Bristol, Indonesia, Egypt

The 10-K's summary sentence is the frame for this section: "The Company conducts engine maintenance at its 100% owned facilities in Montréal, Miami, Lisbon, and Orange, as well as through its 50% equity ownership in QuickTurn Europe, located in Rome, and 50% equity ownership in Prime Engine Accessories, located in Bristol. Collectively, these facilities span over one million square feet and are equipped with advanced tooling, engine test cells, and engineering capabilities to support a wide range of component repairs and service requirements".

[FTAI 10-K FY2025] Each site gets a subsection with every date, price and size the research found and a statement of what it did not.

EXHIBIT 7.8 — FTAI'S MAINTENANCE SITES AND CAPACITY PARTNERS: OWNERSHIP, DATES, PRICES, SIZES AND STATED CAPACITY

Site	What it is	Ownership and accounting	Date	Price	Size	Stated capacity	Source
Montréal, Canada	Former Lockheed Martin Commercial Engine Solutions (LMCES); home of the Module Factory	100%, consolidated	Closed 9 Sep 2024	\$170.0M total consideration	526,000 sq ft (release); "500,000" (10-K)	"up to 300 shop visits per year" (2022 language); with Miami, 1,350 modules and >500 tests (Sep 2024)	GlobeNewswire 9 Sep 2024; Aviation Week; 10-K

Site	What it is	Ownership and accounting	Date	Price	Size	Stated capacity	Source
Miami, Florida	"QuickTurn" engine maintenance business	100%, consolidated	Assets of iAero Thrust acquired 4 Jan 2023 with Unical Aviation as JV partner; Unical's remaining interest bought 1 Dec 2023	\$30.3M cash for Unical's remaining interest; initial price not retrieved	Not retrieved	Included in the 1,350 (Sep 2024)	GlobeNewswire 1 Dec 2023; 10-K
Rome Fiumicino, Italy	QuickTurn Europe, former IAG Engine Center	50%, equity method; carrying value \$9,987K at 31 Dec 2025	Announced 26 Feb 2025; closed 5 Jun 2025	\$10.5M for 50%	200,000 sq ft	450 modules (150 engines) a year, taking FTAI to 1,800 modules (600 engines) and >600 tests	GlobeNewswire 26 Feb 2025 and 5 Jun 2025; 10-K
Lisbon, Portugal	Maintenance facility	100% (10-K)	Not retrieved; "still ramping" on the Jul 2026 call	Not retrieved	"113,000 sq ft" (chain map from call coverage, unverified)	"300+ modules/yr" (same, unverified)	10-K; Quarterly Q2 2026 summary; chain map
Orange	Maintenance facility; the notes add "(California)" without a source	100% (10-K)	Not retrieved	Not retrieved	Not retrieved	Not retrieved	10-K; orchestrator notes

Site	What it is	Ownership and accounting	Date	Price	Size	Stated capacity	Source
Bristol, Connecticut	Prime Engine Accessories, "an industry-leading MRE repair facility for accessory parts", with Bauer, Inc.	50%, equity method; carrying value not returned	Expected operational "by the end of this year" (Q3 2025 call, i.e. end-2025)	Not retrieved	Not retrieved	"up to \$75,000 in average savings per shop visit"	Insider Monkey Q3 2025 transcript; 10-K
Indonesia	Partnership with GMF AeroAsia (Garuda's maintenance arm)	Partnership; form not retrieved	Referenced on the Jul 2026 call as a 2026 capacity partnership	Not retrieved	Not retrieved	Included in the 3,000 (call coverage)	GuruFocus and Quartr, Q2 2026 call
Egypt	Partnership with EgyptAir Maintenance & Engineering	Partnership; form not retrieved	Same	Not retrieved	Not retrieved	Same	Same
All sites	n/a	n/a	n/a	n/a	"over one million square feet"	n/a	10-K

Source: FTAI Aviation, Form 10-K for FY2025 (facilities sentence; equity-method note via orchestrator notes); GlobeNewswire, "FTAI Aviation Closes the Acquisition of LMCES", 9 Sep 2024; Aviation Week, "FTAI Aviation Snaps Up LMCES For \$170 Million" (title via orchestrator notes); GlobeNewswire, 1 Dec 2023 (Unical interest; via orchestrator notes); GlobeNewswire, FTAI Q4/FY2024 results, 26 Feb 2025; GlobeNewswire, "FTAI Aviation Ltd. Announces Closing of QuickTurn Europe Joint Venture", 5 Jun 2025; Insider Monkey, FTAI Q3 2025 call transcript; GuruFocus and Quartr, Q2 2026 call coverage; chain map. Reconciliation R-038 to R-041; cluster K-12. Accessed 2026-10-03.

50.1 MONTRÉAL: LOCKHEED MARTIN COMMERCIAL ENGINE SOLUTIONS

Montréal is the place the Module Factory language names. From 2022 FTAI ran the programme at Lockheed Martin Commercial Engine Solutions' facility under a third-party arrangement; it then bought the business. The closing release of 9 September 2024 gives total consideration of \$170.0M, a 526,000-square-foot facility, and, with Miami, "capacity for 1,350 CFM56 module overhauls and >500 engine tests per year". [GlobeNewswire, 9 Sep 2024; Aviation Week, title via orchestrator notes] The 10-K rounds the size to about 500,000 square feet; the primer carries both figures. [FTAI

10-K FY2025; reconciliation R-038] An earlier orchestrator entry and one dossier's timeline dated the acquisition to 2023; the release dates the closing to 9 September 2024, the chain map was corrected, and Part X's timeline does not say 2023. [reconciliation R-039; E.1 rule 18] Production at Montréal was planned at "an average of 100 modules per quarter" for 2025. [FTAI release, 30 Dec 2024] The site is also where most of FTAI's unionised workforce sits (§50.8).

50.2 MIAMI: QUICKTURN

Miami is the brand FTAI uses for quick-turn work. The research located the transaction in two steps: on 4 January 2023 FTAI acquired the assets of iAero Thrust together with Unical Aviation as a joint-venture partner, and on 1 December 2023 FTAI bought Unical's remaining interest for \$30.3M in cash. [GlobeNewswire, 1 Dec 2023, via orchestrator notes; FTAI 10-K FY2025] The initial purchase price, the site's size and its own capacity were not retrieved; it is included in the 1,350-module figure of September 2024. [D5 U7] Unical Aviation's own scale as a used-material trader was not retrieved. [D3 U14]

50.3 ROME FIUMICINO: QUICKTURN EUROPE

QuickTurn Europe is "a 200,000 sq ft CFM56 engine MRO facility at Rome Fiumicino Airport", the former IAG (International Airlines Group) Engine Center, held 50% by FTAI. The agreement was announced with the fourth-quarter 2024 results on 26 February 2025 and closed on 5 June 2025 for \$10.5M; the 10-K describes the venture as "established to expand the Company's global engine maintenance capabilities and meet increasing demand for MRE services", and carries the investment at \$9,987 thousand at 31 December 2025. [GlobeNewswire, 26 Feb 2025 and 5 Jun 2025; FTAI 10-K FY2025; orchestrator notes] The release credits it with 450 modules and 150 engines a year. The chain map, and two dossiers that copied it, placed this site in "Rome NY"; the filing and the release place it in Italy, no Part repeats the chain-map entry as fact, and Part XII records the error. [reconciliation R-040; E.1 rule 19] Rome was "still ramping" on the July 2026 call. [Quartr, Q2 2026 summary]

EQUITY METHOD

Accounting for a stake in another company by carrying the investment at cost plus the investor's share of the investee's earnings, less distributions and eliminations, with that share on one income line and none of the investee's revenue consolidated. The conventional range is roughly 20% to 50%, but the test is significant influence: FTAI's 19% stake in the 2025 Partnership, with its Servicer role, uses it. For this Part the consequence is that Rome's and Bristol's revenue is not inside Aerospace Products' segment revenue; only FTAI's share of their earnings is, and FTAI's Adjusted EBITDA adds back its pro-rata share of theirs. Whether modules produced at Rome are counted in FTAI's module totals was not retrieved. Part VIII §64 carries the full treatment. [D5 §2.6, U10; D7 §2.5; D10 §2.4; registry C-9]

50.4 LISBON: WHAT IS KNOWN AND WHAT IS NOT

The FY2025 10-K lists Lisbon among the company's 100%-owned maintenance facilities, and the July 2026 call described it as "still ramping". [FTAI 10-K FY2025; Quartr, Q2 2026 summary] That is the extent of the primary-source record. The chain map, drawing on call coverage, recorded a "planned 113,000 sq ft" site at "300+ modules/yr", and one dossier carried it as planned while the filing lists it as owned; the research marked the size and capacity "unverified against a primary source" and this primer labels them so wherever they appear. [chain map; D2 §3.3; reconciliation R-041; E.1 rule 2] The opening date, the deal terms and the site's nature (engine shop, module shop or component repair) were not retrieved. [D5 U7]

50.5 ORANGE: WHAT IS KNOWN AND WHAT IS NOT

Orange appears in the 10-K's list of 100%-owned maintenance facilities and nowhere else in the research. [FTAI 10-K FY2025] The orchestrator's notes add "(California)" as a location without citing a source; the nature of the site, its size, its acquisition date and its price were not retrieved. [orchestrator notes; D5 U7; cluster K-12] The primer does not describe what it does.

50.6 BRISTOL, CONNECTICUT: PRIME ENGINE ACCESSORIES

Prime Engine Accessories is a 50/50 joint venture with Bauer, Inc., described on the third-quarter 2025 call as "an industry-leading MRE repair facility for accessory parts" and "expected to deliver up to \$75,000 in average savings per shop visit", with operations expected "by the end of this year", meaning the end of 2025. [Insider Monkey, FTAI Q3 2025 call transcript] The 10-K lists the 50% equity ownership; the carrying value was not returned. [FTAI 10-K FY2025; orchestrator notes] The accessories in question are the externally mounted units of Part I §3.1, the gearbox-driven pumps and controls, which are repaired as components rather than overhauled with the gas path. The \$75,000 figure is the only savings lever FTAI has quantified; §52 sets it against the cost of a shop visit.

50.7 INDONESIA AND EGYPT: GMF AEROASIA AND EGYPTAIR

On the second-quarter 2026 call FTAI referred to capacity partnerships with GMF AeroAsia, the maintenance arm of Garuda Indonesia, and with EgyptAir Maintenance & Engineering, both counted within the 3,000-module capacity statement. [GuruFocus and Quartr, Q2 2026 call] Both are airline-owned shops that already overhaul CFM56 engines (Part IV §25), and EgyptAir separately holds a GE TrueChoice overhaul-consulting agreement. [D2 §3.2] The capacity committed, the economics, the duration and the legal form of each partnership were not retrieved, and neither shop's own CFM56 induction capacity was found. [D5 U8; D2 U5]

50.8 THE FOOTPRINT AND THE WORKFORCE

The 10-K says the facilities "span over one million square feet". [FTAI 10-K FY2025] The sizes the research sourced sum to less: 526,000 square feet at Montréal plus 200,000 at Rome is 726,000, and adding Lisbon's unverified 113,000 gives 839,000, so the sites whose sizes were not retrieved (Miami, Orange and Bristol) account for at least 160,000 to 275,000 square feet if the total is taken at one million. That arithmetic is derived here and says nothing about which site holds the balance.

At 31 December 2025 FTAI had 985 full-time employees and independent contractors, of whom 494 were full-time employees in Canada, about 71% of them covered by collective bargaining agreements. [FTAI 10-K FY2025, via orchestrator notes] The first figure mixes employees and contractors and the second counts employees only, so the two are not a split of one total. [reconciliation R-065; E. 1 rule 24] Canada is where Montréal is, so roughly half of FTAI's people and most of its unionised workforce are at the Module Factory's main site; employees by site, and the number of test cells, were not retrieved. [D5 U9; cluster K-13] A search summary attributed "approximately 7,800 people across over 50 facilities" to FTAI; that is StandardAero's profile and never appears in this primer as FTAI's. [D5 T7; reconciliation R-065]

51. The V2500 program

The third engine type in the segment's description is handled on a different model from the two CFM56 variants. Part I §10.1 boxed the engine and FTAI's agreement in short form. This section gives the agreement's terms, what FTAI does with V2500 engines and modules, and the market the programme sits in.

51.1 THE IAE AGREEMENT

On 6 June 2024 IAE AG, the Swiss corporate entity of the IAE consortium, and FTAI announced a five-year EngineWise maintenance services agreement covering "over 100 full-performance restoration shop visits" on V2500 engines. EngineWise is Pratt & Whitney's brand for its aftermarket service agreements (Part IV §28); under it the manufacturer's own network performs the visits. The release states that FTAI "currently owns and is committed to over 140 V2500 engines and looks to grow its portfolio to 200 by 2025", and calls the agreement "one of IAE's largest engine maintenance agreements by number of engines with a non-airline customer".

[RTX / Pratt & Whitney release, 6 Jun 2024; AviTrader, 7 Jun 2024]

EXHIBIT 7.9 — FTAI'S V2500 PROGRAMME: THE AGREEMENT'S TERMS AND THE MARKET AROUND IT

Item	Figure	Source and level	Date
Agreement term	Five years	RTX release (OEM release)	6 Jun 2024
Scope	"over 100 full-performance restoration shop visits" in IAE's EngineWise network	RTX release	6 Jun 2024
FTAI's V2500 fleet at signing	"owns and is committed to over 140 V2500 engines"	RTX release	6 Jun 2024
Stated fleet target	200 by 2025	RTX release	6 Jun 2024
Visits implied per year	≈20 (100 ÷ 5), derived	n/a	n/a
Visits completed to date; current fleet	Not retrieved	D5 U6	n/a
In-house V2500 module work	Not retrieved	D5 U6	n/a
V2500 in service	5,286 in 2023 (5,260 -A5, 26 -A1)	Aviation Week 2023 Fleet & MRO Forecast (trade press)	2023
V2500 in service, alternative	">13,000 units" mid-2026	MarketIntelto (market-research vendor page, flagged low grade)	2026
V2500 shop visits, 2025	"slightly more than 800"; 2026 within about 20 of that	RTX CFO via Aviation Week (OEM via trade press)	2026
V2500 shop visits, 2025, forecast	Spike to ≈1,400; >7,800 overhaul and 3,142 LLP-related visits over ten years	Aviation Week V2500 data tool (trade press)	2023
Largest independent on the engine	MTU Maintenance performed 38% of all V2500 shop visits in 2024; 7,000th -A5 visit in 2025	MTU Aero Engines release	2025
Heavy shop visit cost	\$2–3M "depending on the LLP profile"	Aircraft Commerce Issue 70 (trade press)	2010
Full LLP set at list	\$1.7M	Aircraft Commerce Issue 70	2010
PMA on the engine	Chromalloy FAA-approved V2500 HPT blade, unveiled at MRO Europe	Aviation Week (trade press)	Date not captured

Source: RTX / Pratt & Whitney, "IAE AG and FTAI Aviation Sign Strategic V2500 Engine Maintenance Services Agreement", 6 Jun 2024, <https://www.rtx.com/en/prattwhitney/newsroom/news/2024/06/06/iae-ag-and-ftai-aviation-sign-strategic-v2500->

engine-maintenance-services-agreem ; AviTrader, 7 Jun 2024; Aviation Week, "Data Tool: The Future Of IAE V2500 Engine", 2023; Aviation Week, "Pratt & Whitney Sees High-Single-Digit MRO Growth This Year", 2026; MTU Aero Engines release, 2025; Aircraft Commerce Issue 70, 2010; Aviation Week, "Chromalloy Unveils FAA-Approved V2500 HPT Blade"; MarketIntel, 2026. Reconciliation R-009, R-010; D5 §2.5, U6. Accessed 2026-10-03.

51.2 WHAT FTAI DOES WITH V2500 ENGINES AND MODULES

Functionally the V2500 programme is the reverse of the Module Factory. On the CFM56, FTAI is the shop: it holds the inventory, does the work and sells the module. On the V2500, FTAI is the customer: a large engine owner buying manufacturer-network restorations at a negotiated price for a committed volume, then leasing or selling the restored engines and selling V2500 material. [D5 §2.5] Two things connect the programme to the rest of the segment. First, the MRE Contract definition names "CFM56-5B, CFM56-7B and V2500 engines and related modules" as what FTAI sells to the 2025 Partnership's vehicles, so V2500 engines restored under the IAE agreement are among the units the Partnership's A320ceo aircraft can draw on. [FTAI 10-Q Q2 2026, via orchestrator notes] Second, the Module Factory language names only the two CFM56 variants; whether FTAI performs any V2500 module work in-house, how many of the 100-plus visits have been completed, and the current V2500 fleet against the 200 target were not retrieved. [D5 U6] Chromalloy's FAA-approved V2500 high-pressure-turbine blade is a parts input that could bear on the cost of a V2500 restoration; FTAI's statements on it were not retrieved. [Aviation Week, MRO Europe; D5 §2.5]

51.3 THE MARKET CONTEXT

The V2500 pool is about a quarter the size of the CFM56 pool: 5,286 engines in service in 2023 against roughly 23,000 to 24,000 CFM56s (Part I §10.1; Part III §18). [Aviation Week 2023 Fleet & MRO Forecast; reconciliation R-010] Its shop-visit count carries a direct contradiction that the primer carries in both Parts IV and VII: Aviation Week's data tool projected a spike to about 1,400 visits in 2025, and RTX's chief financial officer, reported by the same publication in 2026, put 2025 at "slightly more than 800" with 2026 expected within about 20 of that. [Aviation Week data tool, 2023; Aviation Week, 2026; reconciliation R-009; E.1 rule 16] Against either figure, FTAI's roughly 20 committed visits a year is a small share: about 2.5% of 800 or 1.4% of 1,400, derived. The largest independent on the engine, MTU Maintenance, performed 38% of all V2500 visits in 2024. [MTU Aero Engines, 2025] Part III §24's translation table assigns the V2500 its demand drivers: the grounding of geared-turbofan-powered A320neos pushed A320ceo utilisation up and brought V2500 and -5B visits forward, and the easing of those groundings from 2026 is expected to soften that lift later in the decade. [D9 §5] The current cost of a V2500 visit was not retrieved; the only figures are fifteen years old and are not set beside the 2026 CFM56 figures of Part I §7. [Aircraft Commerce Issue 70, 2010; D1 U13]

52. PMA and the CFM materials agreement within the segment

Part VI owns the mechanics of approved alternative parts: the four approval bases (§37), the hot-section problem (§38), European acceptance (§39), the makers (§41), acceptance by airlines and lessors (§42) and the FTAI–Chromalloy venture as filed and as reported (§43). This section carries only what bears on the segment: FTAI's own statements about margin levers, the approvals as they affect FTAI's parts bill, and the agreement with CFM.

52.1 FTAI'S STATEMENTS ON THE LEVERS

On the fourth-quarter 2025 call on 26 February 2026, asked how the segment's margin would move from the mid-30s toward 40%, the chief executive, Joe Adams, "cited approval of a PMA high-pressure turbine blade as a key factor, along with expanded access to lower-cost parts supply, including used serviceable material and a materials agreement with CFM that includes parts and repairs". The company said it "can approach 40% aerospace margins in 2026 via parts approvals, lower-cost supply and expanded repair capabilities". [MarketBeat, Q4 2025 call highlights, 26 Feb 2026; Motley Fool transcript, 26 Feb 2026] Those are three levers on one line of the cost build of Part I §7: material, which is 60 to 70% of a shop visit and most of which is priced off the manufacturer's catalogue. [Air Cargo Week, 2025/26]

EXHIBIT 7.10 — THE THREE MATERIAL LEVERS MANAGEMENT NAMED, WHAT EACH SUBSTITUTES FOR, AND WHAT IS DISCLOSED ABOUT EACH

Lever	Substitutes for	What FTAI has disclosed	What third parties say	Not retrieved
PMA high-pressure-turbine blade (Chromalloy venture)	New OEM high-pressure-turbine (HPT) blades, which with vanes are 15–25% of a shop visit's parts cost (consultancy blog)	"develops and manufactures PMA parts through a joint venture" (10-K); the blade's approval as "a key factor" (Q4 2025 call)	Chromalloy: FAA PMA 30 Oct 2025, "the only aftermarket alternative to the OEM manufactured new part"; production sold out for 2025, slots reserved into 2026. Investor blogs: a 2018 "exclusive perpetual partnership", five hot-section parts "at cost", "up to \$2 million per shop visit", "5–10 percentage points of additional margin"	Partner, ownership and accounting of the venture in the filings; the PMA share of parts FTAI consumes; the saving per module; the price discount to the OEM blade (D4 U2; D5 U11)

Lever	Substitutes for	What FTAI has disclosed	What third parties say	Not retrieved
Used serviceable material	New OEM parts generally	"expanded access to lower-cost parts supply, including used serviceable material" (Q4 2025 call); the exclusive serviceable-parts partnership (10-K)	USM trades at 60–80% of a new part's price on average, up to 85% early in a life cycle, above list when the OEM cannot deliver (Aviation Week); traders sell repaired engine USM to airlines at 70–75% of list (Aircraft Commerce). Supply "remains thin" (AVM, Aviation Maintenance magazine, 2025) because retirements are low	FTAI's USM share of content; the AAR agreement's terms
CFM materials agreement (22 Jan 2026)	Third-party or spot sourcing of OEM parts and repairs	Multi-year; FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair" (release); "includes parts and repairs" (call)	None located	Duration, volumes, pricing, and whether it restricts FTAI's use of PMA (D4 U14; D5 U18)
Accessory repair (Bristol venture)	Outsourced accessory repair	"up to \$75,000 in average savings per shop visit" (Q3 2025 call)	None located	Volumes; carrying value

Source: MarketBeat and Motley Fool, FTAI Q4 2025 call coverage, 26 Feb 2026; Insider Monkey, Q3 2025 call transcript; FTAI 10-K FY2025; Chromalloy, "Chromalloy Secures FAA Approval of CFM56 High Pressure Turbine Blade PMA", 30 Oct 2025; Komodo Capital and Kairos Research (investor blogs) and In Practise (interview digest), via D4 §4.3; FTAI / GlobeNewswire, "FTAI Aviation Announces Multi-Year Materials Agreement with CFM International to Further Support CFM56 Engines", 22 Jan 2026; Aviation Week and Aircraft Commerce Issue 120 via D3 §2.5; AVM, 2025, via D2 §2.4; Safe Fly Aviation, 2026 (consultancy blog). Reconciliation R-022, R-025 to R-027; clusters K-04, K-19. Accessed 2026-10-03.

52.2 THE CHROMALLOY APPROVALS AS THEY BEAR ON FTAI

The high-pressure turbine is the module that most often drives a shop visit and the most expensive to restore; its blades and vanes are 15 to 25% of a visit's parts cost on a consultancy

blog's figures, and most of an engine's lost temperature margin is recovered there (Part I §3.1 and §5). [Safe Fly Aviation, 2026; D5 §2.7] An approved non-OEM blade therefore changes the parts bill of exactly the module FTAI's exchanges most often replace. Chromalloy's FAA approval of a CFM56-5B/7B high-pressure-turbine blade on 30 October 2025 is described by Chromalloy as "the only aftermarket alternative to the OEM manufactured new part"; Chromalloy said its order backlog would consume all production for the rest of 2025 with customers "seeking to reserve production slots into 2026". [Chromalloy release, 30 Oct 2025, via Newswire; AviTrader, 31 Oct 2025]

How many CFM56 parts Chromalloy has approved depends on which reading of its releases is taken. One dossier counts three announced approvals (the HPT blade, the HPT stage-1 vane and an LPT vane up-change); another reads the 30 October 2025 release as saying Chromalloy "now offers four PMA parts on the CFM56-5B/7B: the HPT blade, the HPT nozzle guide vane, the LPT stage-1 nozzle guide vane and the HPT shroud", the LPT being the low-pressure turbine. [D4 §4.2; D5 §2.7; reconciliation R-025; E.1 rule 33] Both readings are carried. Part VI §41 and §43 give the roster, the dates and the investor-blog claims that the venture was formed in 2018 as an "exclusive perpetual partnership" covering five hot-section parts supplied to FTAI "at cost". [Komodo Capital; Kairos Research; In Practise, via D4 §4.3; reconciliation R-026]

What FTAI's filings say about the venture is one clause: FTAI "develops and manufactures PMA parts through a joint venture". [FTAI 10-K FY2025] The filings' equity-method note lists an "Advanced Engine Repair JV", 25% owned, with \$15.0M invested in December 2016 and \$13.5M more in August 2019 and a carrying value of \$22,429 thousand at 31 December 2025. Whether that venture is the Chromalloy PMA joint venture is unknown; the dates (2016 against the blogs' 2018) do not match, and this primer reports the figures as an equity-method investment without equating the two. [FTAI 10-K FY2025, via orchestrator notes; reconciliation R-026; registry 1c] No PMA revenue, no equity-method earnings line for the venture and no FTAI-stated saving per module were retrieved. [D5 U11; D4 U3; cluster K-04]

52.3 THE CFM MATERIALS AGREEMENT

CFM MATERIALS AGREEMENT

A multi-year agreement between FTAI and CFM International announced on 22 January 2026 under which, in the release's words, FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair" for CFM56 engines. The chief executive described it on the 26 February 2026 call as "a materials agreement with CFM that includes parts and repairs" and named it, with the PMA blade and more used material, as a lever toward segment margins approaching 40%. Its duration, volumes, pricing and any restriction on FTAI's use of PMA parts were not disclosed. Part VI §42 places it among the OEM's responses to alternative parts. [FTAI / GlobeNewswire, 22 Jan 2026; MarketBeat, 26 Feb 2026; D4 §4.3, U14; D5 §2.7, U18; cluster K-19]

The agreement sits oddly beside the PMA lever only at first sight. The two address different parts of the bill. No alternative manufacturer makes a life-limited part for the CFM56, so every new disk, spool and shaft is a single-source purchase from CFM at catalogue price (Part I §4.4; Part VI §38). [D4 U7] Thrust-rating upgrades, which turn a low-rated engine into a higher-rated one worth more, are sold only by the manufacturer (Part I §1). [D3 §2.4] And the manufacturer's own repairs, of which CFM says it has "industrialized about 400 CFM56 repairs since early 2023", are an input any shop buys. [CFM International, via D4 §2.7] GE's stated posture is that it will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity" within "an open network ... with multiple providers who compete and invest to win shop visits". [GE Aerospace, 2026, via D2 §2.3] Ryanair's "long-term materials agreement" covering about 2,000 CFM56 and LEAP engines is the other named example of such an agreement, under GE's TrueChoice Material brand (Part IV §28). [GE 10-K FY2025, via D2 §2.3] The terms of FTAI's agreement were located as a release title and not read. [D4 [23]; D8 §2.7]

WORKED EXAMPLE 7.7 — THE ONE QUANTIFIED SAVING, AND THE ONE CLAIMED, AGAINST BOTH SHOP-VISIT COST RANGES

Inputs: "up to \$75,000 in average savings per shop visit" from the Bristol accessories venture (sourced: FTAI Q3 2025 call via Insider Monkey); "up to \$2 million per shop visit" from the Chromalloy venture's five parts (sourced as an investor-blog figure, not an FTAI statement: Komodo Capital, Kairos Research); a performance restoration at \$1.2–1.6M on one consultancy page and \$1.8–2.5M on another (sourced: Safe Fly Aviation, consultancy blog, 2026; both ranges carried under E.1 rule 4); HPT blades and vanes at 15–25% of parts cost (sourced: Safe Fly Aviation).

Bristol, \$75,000 against \$1.2–1.6M: 4.7% to 6.3% of the visit
 Bristol, \$75,000 against \$1.8–2.5M: 3.0% to 4.2% of the visit
 blog figure, \$2.0M against \$1.2–1.6M: 125% to 167% of the visit
 blog figure, \$2.0M against \$1.8–2.5M: 80% to 111% of the visit
 HPT blades and vanes at 15–25% of the visit:
 \$0.18–0.40M on the lower range; \$0.27–0.63M on the upper

So the one saving FTAI has quantified is a few percent of a restoration's cost on either range, and the figure third parties attribute to the PMA venture is the size of the whole restoration or larger on either range, and several times the consultancy's estimate of what high-pressure-turbine blades and vanes cost in a visit. The primer records that the two figures sit at different source levels (a management statement against investor blogs) and that neither the price of a Chromalloy blade nor its discount to the OEM's was retrieved, and stops. Limits of the example: both cost ranges are contested consultancy figures; the HPT share is from the same consultancy; the blog saving may count parts beyond the blade and may describe a full engine rather than one module; and FTAI's own saving per module is unknown (D5 U11). [reconciliation R-013, R-015, R-027; E.1 rules 4 and 33]

53. Segment financials: revenue, cost of sales, Adjusted EBITDA, inventory

This section assembles every segment figure the research retrieved: the annual table from the 10-K for 2022 to 2025, the quarters of 2025 and 2026, the two cost-of-sales figures with their scopes, a derived gross margin, the inventory and its consequence for cash flow, capital expenditure where disclosed, the geographic split and customer concentration. It closes with the two listed companies the peer research chose as comparison points, each on its own basis.

53.1 THE MEASURES

ADJUSTED EBITDA (SHORT FORM)

Earnings before interest, taxes, depreciation and amortisation, further adjusted by the company to remove items it treats as non-recurring or non-cash and to add back its pro-rata share of Adjusted EBITDA from equity-method investments. FTAI reports it consolidated (\$597.3M in 2023, \$862.1M in 2024, \$1,190.9M in 2025) and by segment, and sets its guidance in it. The exact FY2025 definition and reconciliation were not retrieved. Part X §80 carries the full treatment. [FTAI 10-K FY2025 segment table, via orchestrator notes; D10 §2.5, U-1]

SEGMENT ADJUSTED EBITDA

FTAI's non-GAAP profit measure reported for each segment: Aerospace Products \$160.0M for 2023, \$380.6M for 2024 and \$671.3M for 2025, and \$249.7M in the second quarter of 2026. It includes the profit on MRE Contract revenue (before the equity-method elimination, which runs through a different line) and FTAI's share of the Rome and Bristol ventures. StandardAero reports a "Segment Adjusted EBITDA" on its own definition, MTU and Lufthansa Technik report adjusted EBIT (earnings before interest and taxes, after depreciation), GE reports segment profit; none is interchangeable with FTAI's. Part X §80 owns the definition; Part XI §88 the comparison bases. [FTAI 10-K FY2025 segment table; FTAI release, 29 Jul 2026; D5 §2.8; D11 §2.3]

53.2 THE ANNUAL TABLE

EXHIBIT 7.11 — AEROSPACE PRODUCTS SEGMENT REVENUE AND ADJUSTED EBITDA, 2021 TO 2025, WITH THE CONSOLIDATED TOTALS (\$ THOUSANDS UNLESS STATED)

Line	2021	2022	2023	2024	2025	Source
Aerospace products revenue	"70,800" (unconfirmed; see note)	85,113	454,970	1,079,821	1,600,456	FY2022 10-K (2021–22); FY2025 10-K segment table (2023–25)
MRE Contract revenue	n/a	n/a	0	0	335,788	FY2025 10-K
Segment revenue (sum)	n/a	85,113	454,970	1,079,821	1,936,244	Derived; 2025 total matches the 10-K geographic table
Segment Adjusted EBITDA	not retrieved	27,377	160,009	380,636	671,252	FY2022 10-K; FY2025 10-K
Segment Adjusted EBITDA margin on segment revenue (derived)	n/a	32.2%	35.2%	35.2%	34.7%	Derived
Segment revenue growth (derived)	n/a	n/a	+435%	+137%	+79% (segment); +48% (products line alone)	Derived

Line	2021	2022	2023	2024	2025	Source
Segment Adjusted EBITDA growth (derived)	n/a	n/a	+484%	+138%	+76% (the release's figure)	Derived; FTAI release 25 Feb 2026
Consolidated total revenues	n/a	n/a	1,170,896	1,734,901	2,507,409	FY2025 10-K
Consolidated Adjusted EBITDA	n/a	n/a	597,282	862,050	1,190,922	FY2025 10-K
Segment share of consolidated Adjusted EBITDA (derived)	n/a	n/a	26.8%	44.2%	56.4%	Derived
Aviation Leasing Adjusted EBITDA, for scale	n/a	107,556	not retrieved	500,062	608,912	FY2022 and FY2025 10-Ks

Source: FTAI Aviation, Form 10-K for FY2025, filed 27 Feb 2026, segment table (via orchestrator notes), <https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai-20251231.htm> ; Form 10-K for FY2022, <https://www.sec.gov/Archives/edgar/data/1590364/000159036423000007/ftai-20221231.htm> ; FTAI Q4/FY2025 results release, 25 Feb 2026. Note on 2021: the search of the FY2022 10-K returned "70,800" three times (Aerospace Products revenue 2021, and Adjusted EBITDA 2021 for both segments); the orchestrator flagged the repetition as a probable extraction artefact and the 2021 segment Adjusted EBITDA as not retrieved, so the 2021 revenue figure is shown as unconfirmed. 2021 and 2022 include the infrastructure businesses spun off in August 2022 in discontinued operations and are not comparable with 2023 onward at the consolidated level. Derivations here and in D5 §2.8. Reconciliation cluster K-17; B-2. Accessed 2026-10-03.

Two independent readings agree on the 2024 figures. One dossier derived 2024 products revenue of \$1,079.8M from the 10-K's statement that 2025 revenue rose by \$520.6M, and derived Adjusted EBITDA of about \$381M from the release's "+76%"; the orchestrator's search of the segment table returned \$1,079,821 thousand and \$380,636 thousand. [reconciliation B-2] The segment grew from \$85.1M of revenue in 2022 to \$1,936.2M in 2025, and its Adjusted EBITDA from \$27.4M to \$671.3M; its share of the company's Adjusted EBITDA rose from about a quarter in 2023 to more than half in 2025. The margin on segment revenue has sat between 32% and 35% in each of the four years for which both figures exist. [FTAI 10-K FY2022 and FY2025]

53.3 THE QUARTERS

EXHIBIT 7.12 — AEROSPACE PRODUCTS BY QUARTER, 2025 TO 1H 2026: REVENUE, MRE CONTRACT REVENUE, ADJUSTED EBITDA AND MARGIN, WITH DERIVED FIGURES LABELLED

Period	Products revenue	MRE Contract revenue	Segment revenue	Segment Adjusted EBITDA	Margin (derived unless stated)	Modules produced	Source
Q2 2025	≈\$422.0M (derived)	\$69.6M	≈\$491.6M (derived: 875.0 ÷ 1.78)	≈\$165.4M (derived: 249.7 ÷ 1.51)	≈33.6%	184	Q2 2026 release (+78%, +51%); Q2 2026 10-Q
Q4 2025	not retrieved	not retrieved	≈\$557M (derived: 195 ÷ 0.35)	\$195M	35% (stated)	228	Q4 2025 call via MarketBeat
FY2025	\$1,600.5M	\$335.8M	\$1,936.2M	\$671.3M	34.7%	757	FY2025 10-K; 25 Feb 2026 release
Q1 2026	\$522.6M	\$221.2M	\$743.8M	\$222.6M	29.9%	≈270 (derived)	Q1 2026 10-Q and release
Q2 2026	\$692.2M (derived)	\$182.8M	\$875.0M (+78% y/y)	\$249.7M (+51% y/y)	28.5%	296	Q2 2026 10-Q and release
1H 2026	\$1,214.8M	\$404.0M	≈\$1,618.8M (derived)	≈\$472.3M (derived)	≈29.2%	566	Q2 2026 10-Q and releases

Source: FTAI Aviation, Q2 2026 results release, 29 Jul 2026 (GlobeNewswire), <https://www.globenewswire.com/news-release/2026/07/29/3335598/35538/en/> ; Form 10-Q for the quarter ended 30 Jun 2026; Form 10-Q for the quarter ended 31 Mar 2026; Q1 2026 results release, 29 Apr 2026; MarketBeat, Q4 2025 call highlights, 26 Feb 2026; Form 10-K for FY2025. Derivations in D5 §2.8 and §4.1. Accessed 2026-10-03.

The margin moved from 35% in the fourth quarter of 2025 to about 29 to 30% in the first half of 2026, over the period in which the related-party share of revenue was highest (\$48.3). The research recorded the two movements side by side and did not infer a link, because the filings disclose neither margin by revenue line nor the composition of the elimination. [D5 §2.8, U2; reconciliation R-047] The reader should also hold in mind that the consolidated figures carry the

segment's related-party profit in two places: in segment revenue and cost of sales, and, with the opposite sign, in the equity-method line (\$48.4).

53.4 COST OF SALES: TWO FIGURES, TWO SCOPES

COST OF SALES (SEGMENT VERSUS CONSOLIDATED)

Two cost-of-sales figures for 2025 exist in the FY2025 10-K. One is \$1,240,368 thousand, which the orchestrator's later reading identified as the Aerospace Products segment's cost of sales in the segment expenses table; one dossier had returned it against a segment query but could not confirm its scope. The other is \$1,349,719 thousand, the consolidated cost-of-sales line, which also carries the carrying value of aircraft and engines the Leasing segment sells. Each is presented with its scope and the two are not interchangeable. [FTAI 10-K FY2025; D5 §2.8, T5; D10 §4.2; orchestrator notes, R-045 resolution; registry C-23]

EXHIBIT 7.13 — COST OF SALES, 2023 TO 1H 2026: THE SEGMENT AND CONSOLIDATED FIGURES, THEIR SCOPES, AND THE MARGINS EACH IMPLIES

Figure (\$ thousands)	Period	Scope	Revenue base	Derived gross margin	Source and caveat
1,240,368	FY2025	Aerospace Products segment cost of sales (segment expenses table, per the orchestrator's reading)	1,936,244 (segment revenue)	35.9%	FY2025 10-K; D5 T5 originally could not confirm the scope
1,160,100 (as \$1,160.1M)	1H 2026	Returned against the same segment query; scope not confirmed	1,618,800 (segment revenue, derived)	28.3%	Q2 2026 10-Q via D5 §2.8; same caveat, not resolved by the notes
1,349,719	FY2025	Consolidated cost of sales, including Leasing asset-sale carrying values	2,507,409 (total revenues)	46.2% (cost of sales 53.8% of revenue)	FY2025 10-K via D10 §4.2
825,884	FY2024	Consolidated	1,734,901	52.4% (47.6%)	FY2025 10-K via D10
502,132	FY2023	Consolidated	1,170,896	57.1% (42.9%)	FY2025 10-K via D10

Figure (\$ thousands)	Period	Scope	Revenue base	Derived gross margin	Source and caveat
1,349,719 – 1,240,368 = 109,351	FY2025	Arithmetic difference between the two scopes	n/a	n/a	Derived here; asset sales revenue for 2025 was 106,945, presented gross with carrying value in cost of sales (D10 §2.4); the 10-K's mapping of the difference was not retrieved

Source: FTAI Aviation, Form 10-K for FY2025 (segment expenses table and consolidated statement of operations, via orchestrator notes and D10 §4.2); Form 10-Q for the quarter ended 30 Jun 2026 (via D5 §2.8). Reconciliation R-045; E.1 rule 10 as modified by the orchestrator's resolution. Accessed 2026-10-03.

WORKED EXAMPLE 7.8 — DERIVED GROSS MARGIN ON EACH COST-OF-SALES FIGURE, AND WHAT EACH MEASURES

Inputs: segment revenue of \$1,936.2M and segment cost of sales of \$1,240.4M for 2025 (sourced: FY2025 10-K, segment scope per the orchestrator's reading); consolidated revenues of \$2,507.4M and consolidated cost of sales of \$1,349.7M (sourced: FY2025 10-K); 1H 2026 segment revenue of \$1,618.8M (derived) and cost of sales of \$1,160.1M (sourced: Q2 2026 10-Q, scope unconfirmed); segment Adjusted EBITDA of \$671.3M for 2025 (sourced).

$$\begin{aligned} \text{segment gross margin, 2025} &= (1,936.2 - 1,240.4) \div 1,936.2 = 35.9\% \\ &\quad \text{(derived)} \\ \text{segment gross margin, 1H 2026} &= (1,618.8 - 1,160.1) \div 1,618.8 = 28.3\% \\ &\quad \text{(derived; scope unconfirmed)} \\ \text{consolidated gross margin, 2025} &= (2,507.4 - 1,349.7) \div 2,507.4 = 46.2\% \\ &\quad \text{(derived; mixes leasing rent, which has no cost of sales, with product sales)} \\ \text{segment Adjusted EBITDA margin, 2025} &= 671.3 \div 1,936.2 = 34.7\% \\ \text{gap between gross margin and Adjusted EBITDA margin, 2025} &= 35.9\% - 34.7\% \\ &= 1.2 \text{ points} \end{aligned}$$

So on the segment figures the gross margin and the Adjusted EBITDA margin for 2025 are within about one point of each other, which is consistent with a segment whose costs below cost of sales are small and whose Adjusted EBITDA adds back items (the equity-method share, any depreciation inside the segment) that are not in cost of sales; the filings do not give the segment's operating expenses, so the 1.2 points cannot be decomposed here. The consolidated margin is not a product margin: lease income of \$235.2M and maintenance revenue of \$218.5M have no cost-of-sales entry, which lifts the consolidated figure above the segment's. Limits of the example: the 1H 2026 cost-of-sales scope is unconfirmed; the 2025 segment scope rests on the orchestrator's reading of the segment

expenses table; and gross margin here includes the fair value of returned cores in revenue (\$47.2).
[D5 §2.8, T5; D10 §2.4, §4.2; orchestrator notes; reconciliation R-045]

53.5 INVENTORY, CASH FLOW AND CAPITAL EXPENDITURE

INVENTORY (THE MODULE FACTORY'S WORKING CAPITAL)

Serviceable modules and parts awaiting exchange or sale, plus unserviceable engines, modules and parts in work: \$1,364.3M net at 31 March 2026. Under US accounting, cash spent buying assets for sale is an operating outflow and cash spent buying assets for lease is an investing outflow, so the inventory build is why FTAI's operating cash flow was negative in 2024 and 2025 while the company reported net income. The balances at 31 December 2025 and 30 June 2026, the segment's capital expenditure and the cost of building a module were not retrieved. Part X §81 carries the consolidated cash-flow treatment. [FTAI 10-Q Q1 2026, via D5 §2.8; D10 §2.6; D5 U12; cluster K-18]

Net cash from operating activities was \$128,982 thousand in 2023, negative \$187,956 thousand in 2024 and negative \$310,745 thousand in 2025. [FTAI 10-K FY2025, via D10 §4.4] The working-capital detail that would show how much of the 2025 outflow was inventory, and the investing lines that would show capital spending on the shops, were not retrieved; the research recorded both as open. [D10 U-2; D5 U12] What can be said is structural. Every exchange requires a serviceable module in stock, every serviceable module began as an unserviceable one that cost money to acquire and repair, and the stand-ready obligations of §48.2 require a pool of serviceable whole engines as well. The inventory of \$1.36B at March 2026 against segment revenue of \$1.94B for 2025 is the visible size of that pipeline. [D5 §5] Segment capital expenditure was not disclosed in anything retrieved; the facility purchases of \$50 (\$170.0M for Montréal, \$30.3M for Unical's Miami interest, \$10.5M for half of Rome) are acquisitions rather than capital spending on existing sites and are the only investment figures the research holds. [D5 U12; orchestrator notes]

53.6 GEOGRAPHY AND CUSTOMER CONCENTRATION

EXHIBIT 7.14 — AEROSPACE PRODUCTS REVENUE BY REGION, FY2025

Region	\$ thousands	Share (derived)
North America	1,132,995	58.5%
Europe	415,145	21.4%
Asia	274,618	14.2%
Africa	71,452	3.7%
South America	42,034	2.2%

Region	\$ thousands	Share (derived)
Total	1,936,244	100.0%

Source: FTAI Aviation, Form 10-K for FY2025, segment geographic table, via D5 §4.5. The total equals Aerospace products revenue plus MRE Contract revenue (reconciliation B-1). Accessed 2026-10-03.

The geographic table's total is the arithmetic that fixes segment revenue at \$1,936.2M, and it also shows that the segment's sales, including the related-party line, were almost three-fifths North American in 2025. [FTAI 10-K FY2025; reconciliation B-1] Whether the 2025 Partnership's vehicles are counted as North American was not retrieved. Two customers of the segment accounted for 16% and 11% of revenue in the third quarter of 2025; annual concentration, the customers' identities and whether either is the Partnership were not retrieved. [FTAI 10-Q Q3 2025, via D10 §4.9; D5 U17] The MRE Contract revenue for the second half of 2025 was \$165.6M, derived, against which the two quarterly percentages cannot be tested because third-quarter segment revenue was not retrieved.

53.7 TWO COMPARISON POINTS, EACH ON ITS OWN BASIS

The peer research chose StandardAero and GE Aerospace as the listed companies whose filings isolate the activities a module business performs. [D11 §2.1] Neither sells what FTAI sells, and Part XI §87 to §89 carry the full comparison with every caveat; two facts from each belong here.

StandardAero, an independent engine shop, reported Engine Services revenue of \$5,354.0M for 2025 with Segment Adjusted EBITDA of \$706.9M, a 13.2% margin on its own definition, and described a CFM56-dedicated "Center of Excellence" in Dallas, opened in the second half of 2024, on which it had spent more than \$100M since 2022 to "more than double our shop visit capacity on the CFM56-7B". [StandardAero 10-K FY2025, via D11 §3.6] Its unit is inductions; FTAI's is modules; and StandardAero sells labour and material on a customer's engine while FTAI sells a module it owns, so the 13.2% and FTAI's 34.7% are on the same EBITDA basis but not for the same activity. [D11 §2.3, §3.6; E.1 rule 35]

GE Aerospace's Commercial Engines & Services segment reported 2025 revenue of \$33,314M, of which services were \$25,010M, and segment profit of \$8,861M, a 26.6% margin on GE's own measure; it attributed growth to "increased spare parts volume, internal shop visit volume and worksopes, increased engine deliveries and pricing". [GE 10-K FY2025, via D11 §3.8] GE is at once the supplier of the new parts FTAI's modules consume, the seller of the TrueChoice Transitions exchanges of \$46.5 and the largest provider of used material for its own products; its services revenue is the size of the pool from which the segment's \$1.9B is drawn. [D11 §3.8; D2 §2.3]

54. The Muddy Waters report, the Audit Committee review and the litigation

On 15 January 2025 a short-selling research firm published a report on FTAI whose allegations concern the segment this Part describes. This section states each allegation as the sources that summarised it put it, FTAI's response, the board review and its conclusion in each document's own words, the timing of the annual report, a change of auditor, and the status of the securities class action. Nothing here has been adjudicated, and every party's position is given with its source. The report itself was not read in the research. [D5 U13; reconciliation R-064; E.1 rule 23]

SHORT REPORT, SHORT SELLER AND THE MUDDY WATERS REPORT

A short seller is an investor positioned to profit if a share price falls; a short report is research such an investor publishes after taking that position. Muddy Waters Research published a report on FTAI on 15 January 2025. As third-party summaries and the later securities complaint describe it, the report alleged that one-time engine sales were reported as maintenance revenue, that whole-engine sales were presented as module sales, that margins were inflated, that sales were pushed to intermediaries ahead of demand, and that depreciation on engines not on lease lowered reported cost of sales. FTAI said it "strongly disagrees"; its Audit Committee found the allegations without merit; a securities class action followed. [Benzinga, 15 Jan 2025; Hedge Fund Alpha; law-firm case pages via orchestrator notes; Seeking Alpha, Jan 2025; FTAI 8-K exhibit, Feb 2025; D5 §2.9]

54.1 THE ALLEGATIONS, AS EACH SOURCE STATES THEM

EXHIBIT 7.15 — THE JANUARY 2025 ALLEGATIONS, SOURCE BY SOURCE, AND FTAI'S RESPONSES

Allegation as stated	Source that states it	Source level	FTAI's position and its source
FTAI "has been recording one-time engine sales as Maintenance Repair & Overhaul (MRO) revenue in its Aerospace Products (AP) segment", making the growth "lower quality than investors give the company credit for"	Benzinga, 15 Jan 2025, summarising the report	Trade press summary	FTAI "strongly disagrees with the assertions" (Seeking Alpha, Jan 2025); Audit Committee: allegations "without merit" (8-K exhibit, Feb 2025)

Allegation as stated	Source that states it	Source level	FTAI's position and its source
Muddy Waters "estimates that the majority of FTAI Aviation's adjusted EBITDA in the Aerospace Products segment comes from gains on sales, which is less recurring in nature"	Benzinga, 15 Jan 2025	Trade press summary	Same
"misrepresentation of sales, inflation of Maintenance, Repair, and Overhaul (MRO) margins (and margins in general), channel stuffing, and a few other accounting red flags"	Crossroads Capital letter via Hedge Fund Alpha	Fund letter summarising the report	Same
The company's financials are "highly misleading"	The Fly via TipRanks, Jan 2025	Trade press summary	Same
"one-time engine sales reported as MRO revenue; whole engine sales presented as module sales; depreciation of engines not on lease lowering reported cost of goods sold"	The amended complaint in S.D.N.Y. No. 25-cv-00541, as summarised on law-firm case pages (ktmc.com, rgrdlaw.com)	Pleading, via secondary pages	Defendants' motion to dismiss filed 20 Nov 2025, pending; Q3 2025 10-Q: "the internal investigation concluded that the allegations of misconduct were all without merit"
"whole-engine sales booked as module sales, expense reclassification" between segments	Chain map (orchestrator summary)	Orchestrator	Same
The report "does not appear to consider" the Strategic Capital Initiative	Wolfe Research via The Fly, Jan 2025	Sell-side comment on the report	Not an allegation; recorded because it bounds the report's scope

Source: Benzinga, "What's Going On With FTAI Aviation Stock Today", 15 Jan 2025; Hedge Fund Alpha, Crossroads Capital letter on FTAI; TipRanks / The Fly, "Muddy Waters goes short FTAI, says financials highly misleading" and "FTAI selloff brings very attractive entry point, says Wolfe Research", Jan 2025; law-firm case pages (ktmc.com, rgrdlaw.com) via orchestrator notes; Seeking Alpha, "FTAI Aviation says it strongly disagrees with assertions in Muddy Waters short report", Jan 2025; FTAI Form 8-K exhibit 99.1, Feb 2025; FTAI Form 10-Q for the quarter ended 30 Sep 2025. The report itself was not read (D5 U13). Reconciliation R-064; D5 T6; D10 T-6. Accessed 2026-10-03.

The summaries differ in content, and the primer reports them side by side rather than merging them. [reconciliation R-064; D5 T6] One source stresses engine sales booked as maintenance revenue

and Adjusted EBITDA from gains on sales; another lists misrepresentation of sales, margin inflation and channel stuffing; the complaint adds the depreciation point; the orchestrator's summary adds reclassification of expenses between segments. The report's own wording on expense classification, on which counterparty was alleged to have been stuffed, and on the "other red flags" was not retrieved. [D5 U13]

CHANNEL STUFFING

Recognising sales to intermediaries or related parties ahead of end demand, so that reported revenue runs ahead of what final customers have bought. It appears among the allegations as third parties summarised the report; the counterparty the report named was not retrieved. [Hedge Fund Alpha, Crossroads Capital letter; D5 §2.9, U13]

The allegations map onto the definitions of §47 and the figures of §53. "Engine sales reported as MRO revenue" and "whole engine sales presented as module sales" are the boundary question of §47.4: the filings present engines, modules and parts on one revenue line and do not disaggregate it. [D5 U2] "Depreciation of engines not on lease lowering reported cost of goods sold" concerns where the cost of an engine sits when it moves from the leasing fleet into module inventory, which the 10-K describes as a transfer "when determined to be unserviceable for leasing purposes" (§45.1); the research did not retrieve the accounting for that transfer. [FTAI 10-K FY2025, via D3 §2.7] "Gains on sales" is a term the segment's own disclosures do not use for its product revenue; FTAI books gains on sale in Aviation Leasing (Part VIII §63). [D7 §2.5] None of this resolves anything; it locates each allegation in the vocabulary of this Part.

54.2 FTAI'S RESPONSE AND THE AUDIT COMMITTEE REVIEW

AUDIT COMMITTEE AND THE AUDIT COMMITTEE REVIEW

The board committee of independent directors that oversees financial reporting and the external auditor. On 18 January 2025 FTAI's Audit Committee "determined to begin a review, which included engaging independent advisors, in response to the assertions alleged in the January 15, 2025 report issued by Muddy Waters Research". In February 2025 it announced that the allegations were "without merit". The 10-Q and 10-K later describe the same exercise as an "internal investigation" by "outside counsel and forensic accountants". Part X §85 cross-refers. [FTAI 8-K, Jan 2025; FTAI 8-K exhibit 99.1, Feb 2025; FTAI 10-Q Q3 2025; FTAI 10-K FY2025, via orchestrator notes; reconciliation R-063]

FORENSIC ACCOUNTING

Investigative accounting that tests transactions and records for misstatement, as distinct from the recurring audit of financial statements. The Audit Committee's advisors were described as "independent legal and forensic accounting advisors" in the February 2025 announcement and as "outside counsel and forensic accountants" in the FY2025 10-K. [FTAI 8-K exhibit, Feb 2025; FTAI 10-K FY2025, via orchestrator notes; D10 §2.11]

The documents describe the review in slightly different words, and the primer quotes each with its date rather than fusing them. [reconciliation R-063; E.1 rule 22] The January 2025 8-K: the committee "determined to begin a review, which included engaging independent advisors". [FTAI 8-K, Jan 2025] The February 2025 8-K exhibit: "After a thorough and comprehensive review with the support of independent legal and forensic accounting advisors, the Audit Committee determined that the allegations made against the Company are without merit. The Company expects to file its Form 10-K timely". [FTAI 8-K exhibit 99.1, Feb 2025] One dossier's reading of the same exhibit has the committee "completed its review, conducted by independent legal and forensic accounting advisors ... and determined that the allegations made against the Company are without merit". [D10 §2.11] The Q3 2025 10-Q: "the internal investigation concluded that the allegations of misconduct were all without merit". [FTAI 10-Q Q3 2025, via D10 §2.11] The FY2025 10-K, as the orchestrator's search returned it: an Audit Committee internal investigation, with outside counsel and forensic accountants, into the January 2025 short-seller reports concluded the allegations of misconduct were "all without merit". [FTAI 10-K FY2025, via orchestrator notes] The announcement's date is fixed only by trade coverage: a piece dated 20 February 2025 reported the shares rising on it, which places the announcement on or about 19 February 2025. [StockStory via Wedbush, 20 Feb 2025; D5 §2.9] The committee's detailed findings were not read. [D10 T-6]

54.3 THE FY2024 10-K, A RESTATEMENT QUESTION, AND THE SEC FILE

The company said it expected to file its 2024 annual report on time, and the FY2024 10-K exists on EDGAR (the SEC's public filing database) under accession number 0001590364-25-000006; its exact filing date was not captured. [FTAI 8-K exhibit, Feb 2025; FTAI 10-K FY2024; D5 U14] No restatement was found or referenced in anything the research retrieved. [D5 §2.9] One disclosure change coincides with the period: the FY2025 10-K presents MRE Contract revenue as a separate line from Aerospace products revenue. That coincides with the Strategic Capital Initiative's first full year, and the research found no statement linking the new line to the review; whether FTAI began disclosing engine sales separately from module sales after the review was not determined. [D5 §2.9, U2] An SEC correspondence file dated 27 November 2024, before the report, exists on EDGAR; its subject was not retrieved. [SEC correspondence, 27 Nov 2024; D5 U15]

54.4 THE AUDITOR

Ernst & Young LLP was FTAI's independent registered public accounting firm from 2016 and was ratified for 2024 in the 2025 proxy statement. [FTAI DEF 14A 2025, via D5 §2.9] The FY2025 10-K, as the orchestrator's search returned it, states that Ernst & Young served from 2016 to 2025 and that the Audit Committee engaged KPMG LLP for the 2025 fiscal year effective 17 June 2025, with the auditor's report dated 27 February 2026. [FTAI 10-K FY2025, via orchestrator notes] The notes' own line assigns 2025 to both firms, and which firm signed the FY2025 report was not confirmed; the primer presents the line as written. [reconciliation R-066; E.1 rule 25] The 10-K's internal-control conclusion was not returned. [D10 U-16; cluster K-11] One other change in the period is recorded without any link asserted: on 6 March 2026 FTAI announced a chief financial officer transition, appointing Nicholas McAleese as Chief Financial Officer and Michael Hazan as Chief Accounting Officer in succession to Eun (Angela) Nam, whose departure the release attributes to a role outside aviation. [FTAI release, 6 Mar 2026, via orchestrator notes; reconciliation R-067] Part X §78 carries the officers.

54.5 THE SECURITIES CLASS ACTION

SECURITIES CLASS ACTION AND LEAD PLAINTIFF

A securities class action is a lawsuit brought on behalf of all investors who bought a company's shares in a defined class period, alleging that false or misleading statements inflated the price. The lead plaintiff is the investor the court appoints to direct the action; the deadline to seek appointment falls 60 days after notice of the first complaint. The action against FTAI is S.D.N.Y. No. 25-cv-00541. Part X §85 carries the status. [BFA Law, 12 Feb 2025; orchestrator notes; D5 §2.9]

Law-firm releases announced an investigation on the day of the report and, on 12 February 2025, a securities-fraud class action with a lead-plaintiff deadline of 18 March 2025. [BFA Law via GlobeNewswire, 15 Jan 2025 and 12 Feb 2025] The case is in the United States District Court for the Southern District of New York, No. 25-cv-00541, before Judge Jeannette A. Vargas; the class period runs from 23 July 2024 to 21 January 2025; the defendants filed a motion to dismiss the amended complaint on 20 November 2025, and the motion was fully briefed and pending at the law-firm pages' last update, whose date was not visible. [law-firm case pages, ktmc.com and rgrdlaw.com, via orchestrator notes] The 10-K's own description of the proceeding was not retrieved, and the primer says so. [orchestrator notes; cluster K-11] No derivative suit, settlement or dismissal was found.

EXHIBIT 7.16 — THE REPORT, THE REVIEW, THE FILINGS, THE AUDITOR AND THE LITIGATION: A DATED SEQUENCE

Date	Event	Source
23 Jul 2024	Start of the class period alleged in the securities complaint	Law-firm case pages via orchestrator notes
27 Nov 2024	SEC correspondence file on EDGAR; subject not retrieved	EDGAR; D5 U15
15 Jan 2025	Muddy Waters Research publishes its report; FTAI says it "strongly disagrees"; law firms announce investigations	Benzinga; Seeking Alpha; BFA Law
18 Jan 2025	Audit Committee "determined to begin a review, which included engaging independent advisors"	FTAI 8-K, Jan 2025
21 Jan 2025	End of the alleged class period	Law-firm case pages
12 Feb 2025	Class action notice; lead-plaintiff deadline 18 Mar 2025	BFA Law via GlobeNewswire
On or about 19 Feb 2025	Audit Committee announces the allegations are "without merit"; company expects to file its 10-K timely	FTAI 8-K exhibit 99.1; StockStory via Wedbush, 20 Feb 2025
Feb/Mar 2025	FY2024 10-K filed (accession 0001590364-25-000006); exact date not captured	EDGAR; D5 U14
17 Jun 2025	Audit Committee engages KPMG LLP for FY2025; Ernst & Young had served from 2016	FTAI 10-K FY2025 via orchestrator notes
Q3 2025 10-Q	"the internal investigation concluded that the allegations of misconduct were all without merit"	FTAI 10-Q Q3 2025
20 Nov 2025	Defendants file motion to dismiss the amended complaint in S.D.N.Y. No. 25-cv-00541	Law-firm case pages
27 Feb 2026	FY2025 10-K filed; auditor's report dated the same day; separate MRE Contract revenue line presented	FTAI 10-K FY2025
6 Mar 2026	CFO transition announced (McAleese succeeds Nam)	FTAI release, 6 Mar 2026
As of research date	Motion to dismiss fully briefed and pending; no adjudication	Law-firm case pages (last-update date not visible)

Source: FTAI Aviation Forms 8-K of January and February 2025 (https://www.sec.gov/Archives/edgar/data/1590364/000114036125001452/ef20041894_8k.htm ; https://www.sec.gov/Archives/edgar/data/1590364/000114036125005179/ef20043996_ex99-1.htm); Form 10-Q for the quarter ended 30 Sep 2025; Form 10-K for FY2024 and FY2025; BFA Law releases via GlobeNewswire, 15 Jan and 12 Feb 2025; Benzinga, 15 Jan 2025; Seeking Alpha, Jan 2025; StockStory via Wedbush, 20 Feb 2025; law-firm case pages (ktmc.com, rgrdlaw.com) via orchestrator notes; FTAI release, 6 Mar 2026. Reconciliation R-063, R-064, R-066, R-067; cluster K-11. Accessed 2026-10-03.

55. Worked example: one module exchange versus a full shop visit

This section works the decision an operator faces when one module of a CFM56-7B has run out of life while the others have life left, from the operator's side and then from FTAI's. Every input is labelled sourced or assumed, both consultancy cost sets are shown, and the two spare-engine rents are carried as a pair. The exchange fee, which decides the answer, is unknown, so the example computes the break-even fee rather than a verdict.

55.1 THE SITUATION AND THE TWO OPTIONS

A 737-800 operator has a CFM56-7B whose high-pressure turbine has consumed its temperature margin, so the engine must come off wing (Part I §5), while the fan and booster and the low-pressure turbine have substantial life-limited-part life left (Part I §4). [D5 §2.10] Option A is a conventional performance restoration: the core is opened and restored, the operator pays the shop, waits the turnaround plus any slot wait, and keeps every part of its own engine including the remaining life in the untouched modules. Option B is a module exchange: FTAI removes the expired core module and installs a serviceable one from inventory, the operator pays an exchange fee and gives up its unserviceable core, and the downtime is the swap-and-test interval. [D5 §2.10]

The operator is indifferent when the two all-in costs are equal:

$$F + C_{\text{core}} + T_B \times c = S + T_A \times c$$

so the exchange is cheaper when $(F + C_{\text{core}} - S) < (T_A - T_B) \times c$

where F is the exchange fee, C_{core} the value the operator gives up in the unserviceable core, S the cost of the restoration, T_A and T_B the two downtimes in days, and c the daily cost of covering the engine position. [D5 §2.10] The left side is the exchange premium, which is unknown because F is unknown. The right side is the downtime saving, which the research can bracket. The example therefore computes the right side under every sourced input and states it as the maximum premium at which the exchange is cheaper all-in.

EXHIBIT 7.17 — INPUTS TO THE EXCHANGE-VERSUS-RESTORATION EXAMPLE, EACH LABELLED SOURCED OR ASSUMED, WITH ITS LEVEL AND THE ALTERNATIVE FIGURE

Symbol	Input	Value used	Alternative value	Status	Source and level
S	Performance restoration, excluding life-limited parts	\$1.2–1.6M, midpoint \$1.4M	\$1.8–2.5M, midpoint \$2.15M (Part I's choice)	Sourced, contested	Safe Fly Aviation, 2026, two pages (consultancy blog); reconciliation R-013

Symbol	Input	Value used	Alternative value	Status	Source and level
T_A	Turnaround of the restoration	90 days	75–90 days (consultancy); 90–120 days (trade press, full overhaul)	Sourced, contested	GE Aerospace, 27 May 2026 ("sustaining ~90 days"); Safe Fly Aviation; Aviation Business News, 2025; reconciliation R-019
W	Wait for an induction slot before T_A begins	60–90 days	Up to 180 days	Sourced	Bain, 2024
T_B	Downtime of a module exchange	45 days	FTAI's own figure not retrieved	Sourced for quick-turn visits generally; assumed to apply to an FTAI exchange	Aviation Business News, 2025; D5 U3
c	Daily cost of a spare CFM56-7B	\$3,300 a day ($\approx \$100,000$ a month $\div 30$)	\$1,400–1,600 a day ($\$42,000$ – $48,000$ a month $\div 30$; may be a green-time rent)	Sourced, contested	IBA, Apr 2024 (appraiser); Safe Fly Aviation, 2026 (consultancy blog); reconciliation R-028
F	Exchange fee	Unknown	Upper bound: revenue per module \$2.11M (FY2025), \$2.34M (Q2 2026), which includes engines, parts and core values	Not sourced	D5 §2.10–2.11; cluster K-01

Symbol	Input	Value used	Alternative value	Status	Source and level
C_core	Value given up in the unserviceable core	Unknown	Part IV §30 brackets the life element of an HPT exchange at about \$0.25M for 10,000 cycles of HPT disk life, or about \$2.0M at a whole-stack \$200 a cycle	Not sourced	D2 §2.5; reconciliation R-016, R-017
L	Life-limited-part life transferred	Settled in F by convention	n/a	Sourced as a convention	D2 §2.5; D3 §2.7

Source: Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026", "What determines aircraft engine overhaul costs?" and "CFM56-7B engine availability report 2026" (consultancy blog); GE Aerospace, Bernstein conference presentation, 27 May 2026; Aviation Business News, "Inside the engine MRO supply chain", 2025; Bain, "Get a step ahead of the engine maintenance capacity crunch", 2024; IBA, "It's a 'Lessors' Market", Apr 2024; D2 §2.5; D5 §2.10. Reconciliation R-013, R-019, R-028; E.1 rules 4 and 6. Accessed 2026-10-03.

55.2 THE ARITHMETIC

WORKED EXAMPLE 7.9 — ONE CORE-MODULE EXCHANGE AGAINST A PERFORMANCE RESTORATION: THE DOWNTIME SAVING, THE BREAK-EVEN PREMIUM, AND FTAI'S SIDE OF THE SAME TRANSACTION

Step 1, the downtime saved. Without a slot wait, $T_A - T_B = 90 - 45 = 45$ days (sourced inputs); with the trade-press upper turnaround of 120 days, 75 days. With Bain's slot wait of 60 to 90 days added to the restoration but not to the exchange (assumed: an exchange from inventory does not queue for a slot in the same way; FTAI's own lead time was not retrieved), the saving is 105 to 135 days on a 90-day turnaround and 135 to 165 days on a 120-day one.

saving in days, no slot wait: 45 (90-day turnaround) to 75 (120-day turnaround)

saving in days, with a 60–90 day slot wait:
105–135 (90-day turnaround) to 135–165 (120-day turnaround)

Step 2, the downtime saving in dollars, at each rent. At IBA's rent of about \$100,000 a month, taken as \$3,300 a day (sourced, appraiser, 2024); at the consultancy's \$42,000–48,000 a month, taken as \$1,400–1,600 a day (sourced, consultancy blog, 2026, possibly a green-time rent). The two are not averaged.

no slot wait, IBA rent: $45 \times 3,300 = \$149,000$; $75 \times 3,300 = \$248,000$
no slot wait, Safe Fly rent:
 $45 \times 1,400-1,600 = \$63,000-72,000$; $75 \times 1,400-1,600 = \$105,000-120,000$
with slot wait, IBA rent: $105 \times 3,300 = \$347,000$; $165 \times 3,300 = \$545,000$
with slot wait, Safe Fly rent: $105 \times 1,400 = \$147,000$; $165 \times 1,600 =$
 $\$264,000$

So the downtime saving of one exchange, measured as spare-engine rent avoided, is between about \$63,000 and about \$545,000 depending on which rent, which turnaround and whether a slot wait is counted. That range is the maximum exchange premium ($F + C_{\text{core}} - S$) at which the exchange is cheaper all-in on the rent measure alone.

Step 3, the restoration cost S the premium is measured against. On the lower consultancy range the restoration is \$1.2–1.6M, midpoint \$1.4M; on the upper range \$1.8–2.5M, midpoint \$2.15M (both sourced, contested; Part I §7 uses the upper midpoint and Part IV §30 the lower). The exchange premium is defined relative to S , so the break-even fee is:

break-even $F = S + \text{downtime saving} - C_{\text{core}}$
on $S = \$1.4\text{M}$: $F^* = 1,400,000 + (63,000 \text{ to } 545,000) - C_{\text{core}} = \1.46M to
 $\$1.95\text{M} - C_{\text{core}}$
on $S = \$2.15\text{M}$: $F^* = 2,150,000 + (63,000 \text{ to } 545,000) - C_{\text{core}} = \2.21M to
 $\$2.70\text{M} - C_{\text{core}}$

Step 4, what is known about F and C_{core} . FTAI publishes no exchange price list (cluster K-01). The only observable is the segment's products revenue per module produced, \$2.11M for 2025 and \$2.34M for the second quarter of 2026 (derived, §56), which is an upper bound on the average cash fee because the numerator includes whole-engine sales, parts and the fair value of cores received while the denominator counts only CFM56 modules. Those bounds sit inside or above the break-even range on the lower S and below most of it on the upper S , which says only that the figures are of the same order; it does not say which way the comparison goes for any real transaction. C_{core} is the value of the operator's unserviceable core as feedstock, which depends on its remaining life-limited-part life and the condition of its airfoils; Part IV §30 brackets the life element of a high-pressure-turbine exchange at about \$0.25M for 10,000 cycles of disk life, or about \$2.0M if a whole stack's life is priced at \$200 a cycle, on contested list prices.

Step 5, what the operator keeps and gives up besides money. Under Option A the operator keeps every part of its own engine, including the remaining life in the untouched modules, and receives back the same serial-numbered engine with its records intact. Under Option B it receives a core of known remaining life, which may be more or less than its planned run, and the mismatch is settled in F ; and it gives up the core it handed over, whose records and remaining life pass to FTAI. Lessors' return conditions (Part II §13) and any OEM-only clauses (Part VI §42) decide whether an exchanged module with used, PMA or repaired content is acceptable at redelivery; the research did not retrieve current practice (D4 U1).

Step 6, FTAI's side of the same transaction. FTAI books revenue of F plus the fair value of the core it receives, and cost of sales equal to the carrying value of the module it delivered (its acquisition cost plus the parts, labour and overhead spent making it serviceable). The segment's fourth-quarter 2025 Adjusted EBITDA margin of 35% applied to \$2.11M of revenue per module implies about \$1.37M of cost per module (derived in the research, with the same mixing caveat). FTAI's cost of producing the module is where the three material levers of §52 act: used material in place of new, the PMA blade in place of the OEM's, and OEM parts and repairs at whatever terms the CFM materials agreement provides.

Limits of the example: the exchange fee and the core value are unknown, so no verdict is possible; both cost ranges and both rents are contested and never averaged; the 45-day exchange downtime is a trade-press figure for quick-turn visits generally, not FTAI's; an owned spare carries a capital cost rather than a rent, and the aircraft's own lost flying is not counted; whether a slot wait applies to an exchange is assumed; and whether the exchange requires a test-cell run is unverified (D1 U14). [D5 §2.10; D2 §2.5; D1 §2.7; Safe Fly Aviation, 2026; GE Aerospace, 27 May 2026; Aviation Business News, 2025; Bain, 2024; IBA, Apr 2024; reconciliation R-013, R-019, R-028, R-046; cluster K-01]

56. Worked example: revenue and EBITDA per module, and its limits

The one unit figure FTAI's disclosures permit is a ratio of a revenue line to a module count. This section computes it on every period and numerator the research used, states the five limits that make it a bound rather than a price, and shows what the two 2026 targets would imply if the 2026 guidance were spread over them.

REVENUE AND ADJUSTED EBITDA PER MODULE (DERIVED)

Aerospace products revenue divided by modules produced (\$2.11M for 2025; \$1.94M for Q1 2026; \$2.34M for Q2 2026) and segment Adjusted EBITDA divided by modules (\$0.89M; \$0.82M; \$0.84M). A second version divides segment revenue including MRE Contract revenue by modules (\$2.96M for Q2 2026). All are upper bounds, because the numerator includes whole engines, parts, used material, V2500 activity and the fair value of returned cores while the denominator counts only CFM56 modules, and because modules produced are not modules sold. [D5 §2.10–2.11; D7 §5.2; reconciliation R-046; E.1 rule 11]

WORKED EXAMPLE 7.10 — REVENUE AND ADJUSTED EBITDA PER MODULE ON BOTH NUMERATORS, AND THE GUIDANCE SPREAD OVER THE 2026 TARGETS

Inputs: Aerospace products revenue of \$1,600.5M (FY2025), \$522.6M (Q1 2026) and \$692.2M (Q2 2026, derived as 1,214.8 – 522.6) (sourced: 10-K and 10-Qs); segment revenue including MRE Contract revenue of \$1,936.2M, \$743.8M and \$875.0M (sourced); segment Adjusted EBITDA of \$671.3M, \$222.6M and \$249.7M (sourced); modules produced of 757, 270 (derived) and 296

(sourced); 2026 segment Adjusted EBITDA guidance of \$1,050M (sourced: Feb 2026 release, reaffirmed Jul 2026); 2026 module targets of 1,050 (sourced) and 1,200 (call coverage).

products revenue per module:
 FY2025: $1,600.5 \div 757 = \$2.11\text{M}$
 Q1 2026: $522.6 \div 270 = \$1.94\text{M}$
 Q2 2026: $692.2 \div 296 = \$2.34\text{M}$
 segment revenue (incl. MRE) per module:
 FY2025: $1,936.2 \div 757 = \$2.56\text{M}$
 Q1 2026: $743.8 \div 270 = \$2.75\text{M}$
 Q2 2026: $875.0 \div 296 = \$2.96\text{M}$ (D7's ceiling figure)
 segment Adjusted EBITDA per module:
 FY2025: $671.3 \div 757 = \$0.887\text{M}$
 Q1 2026: $222.6 \div 270 = \$0.824\text{M}$
 Q2 2026: $249.7 \div 296 = \$0.844\text{M}$
 2026 guidance spread over the targets:
 $1,050 \div 1,050 = \$1.00\text{M}$ of Adjusted EBITDA per module
 $1,050 \div 1,200 = \$0.875\text{M}$ per module

Reading the lines. The first block says that for every CFM56 module FTAI produced, the segment booked about \$2M of product revenue; the second, that including the related-party line the figure is \$2.6M to \$3.0M; the third, that the segment earned about \$0.8M to \$0.9M of Adjusted EBITDA per module produced. The fourth block says that if the 2026 guidance were earned only on modules, it would be about \$1.0M per module on the filed target or \$0.875M on the figure reported from the July call, against \$0.84M to \$0.89M realised; it is a scale check, not a forecast, because the guidance covers engines, parts and the related-party line as well. One dossier's maintenance-stream example assumes \$1.0–2.0M of revenue per module and treats \$2.96M as the ceiling; this Part replaces the assumption with the derived figures above and their limits, as that dossier asked. [D7 §5.2; E.1 rule 36]

The five limits, each of which can move the result by a large fraction. First, the numerator includes whole-engine sales, spare-parts sales, used-material sales and V2500 activity, while the denominator counts only CFM56 modules; revenue per module is overstated by an unknown amount. Second, modules produced in a quarter are not modules sold in it: a produced module may sit in inventory, go into an FTAI-owned leasing engine, or be built into a Mod-1 generator set (Part IX). Third, segment Adjusted EBITDA includes the profit on MRE Contract revenue before the equity-method elimination and FTAI's share of the Rome and Bristol ventures, none of which is a module sale; EBITDA per module is overstated too. Fourth, exchange revenue includes the fair value of cores received, which is not cash (§47.2). Fifth, module mix is not disclosed, and a core module is worth multiples of a fan module. A unit economics of the module would need the disaggregation the research recorded as its first open question (D5 U2). [D5 §2.11; reconciliation R-046]

The per-module figures also bear on the maintenance stream the Strategic Capital vehicles generate, which Part VIII §66 works: 300 aircraft carry 600 engines, and the number of engine events a year, the modules per event and the revenue per module decide how much of the

segment's revenue those vehicles can supply. The first of those inputs comes from Part I §6 (a mature restoration interval of 10,000 to 15,000 cycles at 1,000 to 1,500 cycles a year, or a rule of thumb of one visit roughly every eight years), the second is undisclosed, and the third is bounded above by the figures here. [D7 §5.2; reconciliation R-008, R-046]

57. Guidance: 2026 and 2027

FTAI guides in Adjusted EBITDA by segment. This section gives the path of the segment's guidance with the date and source of each step, the 2027 figure, and management's statements on margin, and works what the guidance implies for the second half of 2026. Part X §84 carries the company-wide guidance history and Part VIII the Leasing segment's.

EXHIBIT 7.18 — AEROSPACE PRODUCTS ADJUSTED EBITDA GUIDANCE AND OUTCOMES, BY DATE SET

Date set	For	Aerospace Products figure	Within total segment guidance of	Outcome or later change	Source
30 Dec 2024	2025	\$600–650M	\$1.1–1.15B (Leasing ~\$500M)	Actual \$671.3M, above the range	FTAI release, 30 Dec 2024 (via D7 §4.3); 10-K FY2025
30 Dec 2024	2025 modules	"an average of 100 modules per quarter produced at the Company's Montreal facility" (≈400)	n/a	Actual 757, all sites	FTAI release, 30 Dec 2024
Dec 2024 / early 2025	2026	not split	\$1.4B	Raised	Orchestrator notes (R-049 resolution)
27 Oct 2025 (per the notes; undated in D5 and D6)	2026	\$1.0B	\$1.525B (Leasing \$525M)	Raised	FTAI Q3 2025 release via Q1 2026 10-Q comparison; D5 T4; D6 [S2]; orchestrator notes
25 Feb 2026	2026	\$1.05B	\$1.625B (Leasing \$575M)	Aerospace reaffirmed Jul 2026	FTAI Q4/FY2025 release, 25 Feb 2026

Date set	For	Aerospace Products figure	Within total segment guidance of	Outcome or later change	Source
25 Feb 2026	2026 modules	1,050 (+39% on 757)	n/a	Raised to 1,200 on the Jul 2026 call, per call coverage	FTAI release and call, 26 Feb 2026; GuruFocus, MarketBeat, Nasdaq.com, 30 Jul 2026
26 Feb 2026 (call)	2026 margin	"can approach 40%" via the PMA blade, USM and the CFM materials agreement	n/a	1H 2026 realised $\approx$ 29.2% (derived)	MarketBeat, Motley Fool
29 Jul 2026	2026	\$1.05B reaffirmed	\$1.525B (Leasing cut from \$575M to \$475M "reflecting its continued shift to an asset-light business model")	n/a	FTAI Q2 2026 release, 29 Jul 2026
29 Jul 2026	2027	\$1.4B	\$2.3B (FTAI Power \$450M; Leasing \$450M)	n/a	FTAI Q2 2026 release, 29 Jul 2026

Source: FTAI Aviation / GlobeNewswire, 30 Dec 2024, 27 Oct 2025, 25 Feb 2026 and 29 Jul 2026 releases; Q2 2026 earnings release 8-K exhibit, <https://www.sec.gov/Archives/edgar/data/1590364/000162828026050622/ftai6302026earningsrelease.htm> ; MarketBeat and Motley Fool, Q4 2025 call coverage, 26 Feb 2026; GuruFocus, MarketBeat and Nasdaq.com, Q2 2026 call coverage, 30 Jul 2026; orchestrator notes (R-049 resolution). The Leasing cut is as read by D5 and D10; D6 records a second search reading of the 29 Jul 2026 8-K exhibit that returned \$575M (reconciliation R-049). The Power 2027 figure is \$450M in the release text; one call summary gives "\$450–750M" (R-062). Reconciliation R-036, R-049, R-050. Accessed 2026-10-03.

The 2026 guidance for the segment has stood at \$1,050M since February 2026 and was reaffirmed in July, while the company-wide total moved because the Leasing figure was cut. [FTAI releases, 25 Feb 2026 and 29 Jul 2026] The step before February, to \$1.0B for Aerospace Products within \$1.525B, appears undated in two dossiers and is dated 27 October 2025 in the orchestrator's later resolution; the primer carries the date with that provenance. [D5 T4; D6 [S2]; orchestrator notes; reconciliation R-050] The 2027 figure of \$1.4B is a third again above the 2026 figure and is given in the release text; the Power and Leasing figures beside it are each \$450M in the release, with one call summary giving a range for Power. [FTAI release, 29 Jul 2026; reconciliation R-062; E.1 rule 30] No guidance item specific to the Strategic Capital vehicles' share of the segment's revenue was located. [D7 U9]

Management's statements on margin are three. The segment ran "\$195 million at a 35% margin" in the fourth quarter of 2025. [MarketBeat, 26 Feb 2026] The company said on the same call that it "can approach 40% aerospace margins in 2026 via parts approvals, lower-cost supply and expanded repair capabilities", naming the PMA high-pressure-turbine blade, used material and the CFM materials agreement (§52). [MarketBeat, 26 Feb 2026; Motley Fool, 26 Feb 2026] And the first half of 2026 was reported at \$472.3M of Adjusted EBITDA on \$1,618.8M of segment revenue, about 29.2%, both derived from the two quarters' filed figures (§53.3). [FTAI releases, 29 Apr 2026 and 29 Jul 2026; D5 §4.1] The primer records the statement and the outturn to date and does not forecast.

WORKED EXAMPLE 7.11 — WHAT THE 2026 GUIDANCE IMPLIES FOR THE SECOND HALF, AND THE 2027 STEP

Inputs: 2026 Aerospace Products Adjusted EBITDA guidance of \$1,050M (sourced: 25 Feb 2026 release, reaffirmed 29 Jul 2026); 1H 2026 segment Adjusted EBITDA of \$472.3M, being \$222.6M plus \$249.7M (sourced); Q2 2026 segment revenue of \$875.0M and margin 28.5% (sourced and derived); 2027 guidance of \$1.4B (sourced); FY2025 actual \$671.3M (sourced).

second half of 2026 implied = $1,050 - 472.3 = \$577.7\text{M}$, or $\$288.9\text{M}$ a quarter

against Q2 2026 actual of \$249.7M: $+15.7\%$ a quarter

2026 guidance on 2025 actual: $1,050 \div 671.3 = +56\%$

2027 guidance on 2026 guidance: $1,400 \div 1,050 = +33\%$

revenue needed for \$288.9M a quarter at the Q2 2026 margin of 28.5%:

$288.9 \div 0.285 = \$1,014\text{M}$ a quarter

revenue needed at 35% (the Q4 2025 margin): $288.9 \div 0.35 = \$825\text{M}$ a quarter

revenue needed at 40% (the aspiration): $288.9 \div 0.40 = \$722\text{M}$ a quarter

So the reaffirmed guidance requires second-half quarters about 16% above the second quarter's Adjusted EBITDA; at the second quarter's margin that is about \$1.0B of quarterly segment revenue, and at the margins management has reported or aspired to it is \$0.72B to \$0.83B against \$0.875B realised in the second quarter. The lines show the arithmetic relationship between revenue and margin that the guidance implies; they do not say which combination the company expects, and the company has not said. Limits of the example: the 2027 figure has no quarterly path; the margin lines mix a derived half-year figure, a stated quarter and an aspiration; and the guidance covers engines, parts and the related-party line as well as modules. [FTAI releases, 25 Feb 2026 and 29 Jul 2026; D5 §2.8, §4.1, §4.2; MarketBeat, 26 Feb 2026]

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-001	CFM56 in service: ~23,000 (CFM); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran); "over 20,000" -5B/-7B (Chromalloy, Oct 2025)	§45.3
R-006	CFM56 shop visits a year: 2,300–2,400 in 2026–28 (GE via Aviation Week, 2026) against ~2,500 peak 2025–26 (Safran via Aviation Week, c. 2023/24)	§49.3, Exhibit 7.7, worked example 7.6
R-008	Restoration interval: 10,000–15,000 cycles (Leeham) against "every eight years" (Aviation Week) and one in six years assumed (D7)	§56
R-009	V2500 shop visits 2025: "slightly more than 800" (RTX CFO via Aviation Week, 2026) against ~1,400 forecast (Aviation Week data tool, 2023)	§51, Exhibit 7.9
R-010	V2500 in service: 5,286 (Aviation Week, 2023) against ">13,000" (low-grade vendor page, 2026)	§51, Exhibit 7.9
R-013	Restoration cost: \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M" (all Safe Fly, consultancy blog)	§52 worked example 7.7; §55, Exhibit 7.17, worked example 7.9
R-015	HPT blade set \$1.6M or \$3.08M (per-blade × 80, unverified) against \$0.2–0.6M implied by a 15–25% share	§52 worked example 7.7
R-016, R-017	Full -7B LLP set \$4M (2019) or \$5.7M (2025); \$200 or \$285 a cycle	§47.1; §55 step 4
R-019	Turnaround: 90–120 days (trade press); ~90 (GE, May 2026); PR 75–90 (Safe Fly); slot wait +2–3 months, up to 6 (Bain)	§46.1, §46.4, Exhibit 7.3; §55
R-022	USM at 60–80% of new, up to 85% (Aviation Week) against 70–75% of list, trader to airline (Aircraft Commerce)	§52, Exhibit 7.10
R-025	Chromalloy CFM56-5B/7B PMA roster: three announced approvals (D4) against four parts offered (D5)	§52.2
R-026	FTAI–Chromalloy: "a joint venture" (10-K) against "exclusive perpetual partnership, parts at cost" (blogs); Advanced Engine Repair JV (25%, \$22.4M) identity unknown	§52.2
R-027	PMA saving: "up to \$2M per shop visit" (blogs) against FTAI's only quantified lever, \$75,000 (Bristol)	§52 worked example 7.7
R-028	-7B spare rent ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time)	§55, Exhibit 7.17, worked example 7.9

Id	The two (or more) figures	Where in this Part
R-035	Module count: four major modules (Part I) against FTAI's three per engine ("450 modules (150 engines)") and D5's "four to five"	§49.2–49.3, worked examples 7.5 and 7.6
R-036	2026 module target: 1,050 (Feb 2026 release) against 1,200 (call coverage of the Jul 2026 call)	§49.1, Exhibit 7.5, worked example 7.4; §57
R-037	Capacity: 300 shop visits (10-K, 2022 language); 1,350 modules (Sep 2024); 1,800 modules / 600 engines (Feb 2025); 3,000 modules (call coverage, Jul 2026)	§49.2, Exhibit 7.6, worked example 7.5
R-038	Montréal: ~500,000 sq ft (10-K) against 526,000 (release)	§50.1, Exhibit 7.8
R-039	Montréal acquisition: "2023" (stale chain-map entry) against closing 9 Sep 2024, \$170.0M (release)	§50.1
R-040	"Rome NY" (chain map) against Rome Fiumicino, Italy (10-K; 5 Jun 2025 release)	§50.3
R-041	Lisbon: "planned 113,000 sq ft, 300+ modules" (chain map, call coverage) against 100%-owned and "still ramping" (10-K; Jul 2026 call)	§50.4, Exhibit 7.8
R-042	Share: ~5% of a \$22B market (release, Feb 2025); 25% of CFM56 shop visits (call, Jul 2026); ~17% of a 2,350-visit year at 1,200 modules (D2 derived)	§49.3, Exhibit 7.7
R-043	Segment revenue = products + MRE (\$1,936.2M = \$1,600.5M + \$335.8M), agreed by three documents; explicit segment-note mapping not retrieved	§48.3, Exhibit 7.4
R-044	MRE Contract revenue: "stand-ready obligations" (Q3 2025 10-Q) against "transaction price for sales of ... engines and related modules" (Q2 2026 10-Q)	§48.2
R-045	Cost of sales FY2025: \$1,240,368K (segment, per the orchestrator's reading; scope unconfirmed in D5) against \$1,349,719K (consolidated)	§53.4, Exhibit 7.13, worked example 7.8
R-046	Revenue per module: \$2.11M / \$1.94M / \$2.34M on products revenue (D5) against \$2.96M on segment revenue incl. MRE (D7)	§56, worked example 7.10
R-047	Profit elimination: \$16.6M in 1H 2026 implies ~\$87M gross profit if proportional to 19%, or \$16.6M if fully deferred	§48.4, worked example 7.3
R-048	Stake in the 2025 Partnership: 19% (10-K; 2026 10-Qs) against 20% (Q3 2025 10-Q)	§48.1, worked example 7.3

Id	The two (or more) figures	Where in this Part
R-049	2026 Leasing guidance: \$575M (25 Feb 2026) → \$475M (29 Jul 2026, as read by D5 and D10) against a second reading of the same release returning \$575M (D6)	§57, Exhibit 7.18
R-050	2026 guidance step to \$1.525B (\$1.0B Aerospace Products, \$525M Aviation Leasing): undated (D5, D6) against 27 Oct 2025 (orchestrator notes)	§57, Exhibit 7.18
R-057	Aircraft sales to the Partnership "non-recurring" (ASC 610-20) while MRE sales to the same buyer are ordinary revenue (ASC 606)	§48.4
R-062	2027 FTAI Power Adjusted EBITDA: \$450M (release) against "\$450–750M" (one call summary)	§57, Exhibit 7.18
R-063	The review: "review ... independent advisors" (8-K) against "internal investigation ... outside counsel and forensic accountants ... all without merit" (10-Q, 10-K); conclusion dated only by trade coverage	§54.2, Exhibit 7.16
R-064	The allegations as Benzinga, a fund letter, the complaint and the chain map each summarise them; the report not read	§54.1, Exhibit 7.15
R-065	Headcount: 985 employees and contractors; 494 full-time in Canada (mixed bases); "7,800" is StandardAero's	§50.8
R-066	Auditor: Ernst & Young "2016–2025" and KPMG engaged for FY2025 from 17 Jun 2025; which firm signed the FY2025 report not confirmed	§54.4
R-067	CFO: Nicholas McAleese (call coverage; 6 Mar 2026 release) against Eun (Angela) Nam as a named executive officer in the proxy	§54.4

Unknowns carried in this Part

- **U-D5-01 (D5): U1 — Revenue-recognition text for exchanges.** The 10-K's exact policy for exchange transactions (how the core is valued, whether exchange fees are separately quantified). Resolves with the 10-K revenue note [S1].
- **U-D5-02 (D5): U2 — Disaggregation of "Aerospace products revenue."** Engines vs modules vs parts vs repairs for 2023, 2024, 2025 and each 2025–2026 quarter; margin by line; whether disclosure changed after the February 2025 review. Resolves with the 10-K and 10-Q revenue notes [S1–S3] or with the Q2 2026 release [S5].

- **U-D5-03 (D5): U3 — Exchange turnaround time and exchange pricing.** FTAI's quoted turnaround and any published exchange-fee ranges. Trade press (e.g., the In Practise interview [S38]) likely has estimates; not read. (Cluster K-01 with U-D2-07 and U-D3-20.)
- **U-D5-04 (D5): U4 — Module counts for 2022, 2023, 2024 and Q1/Q3 2025.** Only Q4 2024 (≈ 136) is derivable. Resolves with the Q4 2024 release [S43], the Q1 2025 release [S42], the Q3 2025 release [S11] and the FY2023 release.
- **U-D5-05 (D5): U5 — Denominator of the 25% share goal** and the year by which it is targeted.
- **U-D5-06 (D5): U6 — Engine sales counts; V2500 shop visits completed; current V2500 fleet vs 200 target; whether any V2500 module work is done in-house.**
- **U-D5-07 (D5): U7 — Facility deal terms.** LMCES purchase price; QuickTurn Miami date and price; what and where "Orange" is and when acquired; Lisbon's size, capacity and opening date (chain map: 113,000 sq ft, 300+ modules/yr); verification of Rome at \$10.5M for 50%. Resolved in part by the orchestrator notes (D.3): LMCES 9 Sep 2024, \$170.0M, 526,000 sq ft; QuickTurn Miami 4 Jan 2023 with Unical, remaining interest \$30.3M on 1 Dec 2023; Rome \$10.5M for 50%, closed 5 Jun 2025, 200,000 sq ft. Orange and Lisbon still open.
- **U-D5-08 (D5): U8 — GMF AeroAsia and EgyptAir terms** (capacity committed, economics, duration).
- **U-D5-09 (D5): U9 — Test-cell count; employees by site and in total.** Resolved in part (D.3): 985 full-time employees and independent contractors at 31 Dec 2025; 494 full-time employees in Canada, $\sim 71\%$ under collective bargaining agreements; test capacity >500 (Sep 2024) and >600 (Feb 2025) engine tests a year. By-site counts and test-cell count still open.
- **U-D5-10 (D5): U10 — Whether Rome's modules are counted in FTAI's module totals,** given equity-method accounting.
- **U-D5-11 (D5): U11 — Chromalloy JV contribution in FTAI's numbers:** equity-method earnings line, PMA share of parts, FTAI-stated savings per module. (Cluster K-04 with U-D4-03; the Advanced Engine Repair JV figures apply only if it is the same entity.)
- **U-D5-12 (D5): U12 — Inventory at 31 Dec 2025 and 30 Jun 2026; segment capex; the cost of building module inventory.** (Cluster K-18 with U-D10-02.)
- **U-D5-13 (D5): U13 — Muddy Waters' own wording** on expense classification, channel stuffing and the "other red flags". Resolved in part (D.3): the complaint's allegations as summarised on law-firm pages; the report's own wording still open.
- **U-D5-14 (D5): U14 — Exact FY2024 10-K filing date** (accession 0001590364-25-000006; company says timely).
- **U-D5-15 (D5): U15 — Subject of the 27 Nov 2024 SEC correspondence** [S20].

- **U-D5-16 (D5): U16 — Securities litigation status** (court, docket, motion to dismiss outcome, any settlement). Resolved in part (D.3): S.D.N.Y. No. 25-cv-00541, Judge Jeannette A. Vargas; class period 23 Jul 2024 to 21 Jan 2025; motion to dismiss the amended complaint filed 20 Nov 2025, fully briefed and pending (law-firm pages; last-update date not visible). Outcome still open. (Cluster K-11 with U-D10-16.)
- **U-D5-17 (D5): U17 — Named customers, long-term agreements, customer concentration** (any customer $\geq 10\%$ of revenue), and whether SCI-vehicle revenue is disclosed as related-party. Resolved in part (D.3): MRE Contract revenue to the 2025 Partnership \$182.8M (Q2 2026) vs \$69.6M; \$404.0M (1H 2026) vs \$170.2M; FY2025 \$335,788K; two Aerospace Products customers at 16% and 11% of revenue in Q3 2025. Names and annual concentration still open.
- **U-D5-18 (D5): U18 — Terms of the CFM materials agreement** mentioned on the Q4 2025 call. (Cluster K-19 with U-D4-14.)
- **U-D5-19 (D5): U19 — FY2023 segment revenue and Adjusted EBITDA**, and the FY2022 baseline (the segment began ramping in 2022–2023 per the 10-Q language [S10 summary]). Resolved in part (D.3): FY2025 10-K segment table, Aerospace products revenue 454,970 (2023), Adjusted EBITDA 160,009 (2023); FY2022 10-K, 85,113 and 27,377 (2022); 2021 figures flagged as a probable extraction artefact. (Cluster K-17.)
- **U-D1-14 (D1): U14** Whether a module exchange requires a test-cell run under the CFM56 ESM.
- **U-D1-13 (D1): U13** V2500 module list and current per-visit cost (2010 figures only) [38].
- **U-D2-05 (D2): U5. Capacity of the named airline MROs.** AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir M&E, GA Telesis, Lufthansa Technik and ST Engineering: no CFM56 inductions- per-year or expansion figures returned. Resolve via each company's annual report or trade-press shop profiles.
- **U-D2-07 (D2): U7. Module-exchange pricing.** No source gives the exchange fee or a worked price for a CFM56 HPT or core exchange; Safe Fly's "\$150,000–\$400,000 removal cost per module" is ambiguous. D5 may find it in FTAI's investor materials or the In Practise interview ("FTAI Aviation: CFM56 repair and module swap", URL returned, not read).
- **U-D3-14 (D3): U14 — Unical Aviation:** nothing returned. Resolved in part (D.3): Unical was FTAI's JV partner in QuickTurn Miami; Unical's own USM scale still open.
- **U-D3-20 (D3): U20 — Pricing of a module exchange** (exchange fee, LLP-life true-up) from any provider, including FTAI. D5 may hold FTAI's own disclosures.
- **U-D4-01 (D4): U1 — Current lessor return-condition practice (2024–2026).** All lessor statements found are 2009–2016. Resolve with recent lessor 10-K/20-F lease descriptions (AerCap, Air Lease, Willis Lease), ISTAT or IBA surveys, or lessor conference remarks.

- **U-D4-02 (D4): U2 — PMA price discount for the Chromalloy HPT blade and vane** (and HEICO's typical discount). Resolve with Chromalloy/FTAI pricing disclosure or MRO price catalogues (the SR Technics 2026 CFM56-7B catalogue may list PMA options [71]).
- **U-D4-03 (D4): U3 — FTAI–Chromalloy JV terms:** formation date, ownership split, accounting (consolidated vs equity method), exclusivity, "at cost" pricing, revenue recognized. Resolve by reading FTAI's FY2018–FY2025 10-Ks for the JV footnote and any 8-K. Resolved in part and conditionally (D.3): Advanced Engine Repair JV 25%, \$15.0M Dec 2016 and \$13.5M Aug 2019, carrying value \$22,429K at 31 Dec 2025, if the same entity; the notes say this is unknown.
- **U-D4-07 (D4): U7 — Any PMA for CFM56 life-limited rotating parts.** None found.
- **U-D4-14 (D4): U14 — Terms of the FTAI–CFM Jan 2026 materials agreement** (duration, volumes, pricing, whether it restricts FTAI's PMA use).
- **U-D7-06 (D7): U6 — MRE revenue from the vehicles.** Dollar revenue and counts of engine and module exchanges to the 2025 Partnership; expected annual volume for 300+ aircraft. Only eliminations located [S12, S13]. Resolved in part (D.3): Q2 2026 \$182.8M (vs \$69.6M); 1H 2026 \$404.0M (vs \$170.2M); definition "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership". Counts of exchanges still open.
- **U-D7-09 (D7): U9 — Guidance.** Any 2025 or 2026 guidance item specific to SCI (fees, gains, MRE volume) and the corresponding actuals. Only total and Aerospace Products figures located [S1, S6].
- **U-D10-01 (D10): U-1 — Full income statement 2021–2025 and 1H 2025 / 1H 2026.** Operating expense lines, depreciation and amortization, interest expense, tax, net income attributable to shareholders by year; annual Adjusted EBITDA, its definition and reconciliation items; segment revenue and Adjusted EBITDA by year; 1H totals. Resolve with the FY2025 10-K Item 8 and Item 7 segment tables [S1], the Q4/FY2025 release [S3], the Q2 2026 release [S4] and the Q2 2025 release [S25]. Also the FY2021 10-K [S15] and FY2022 annual report (EDGAR index) for 2021–2022. Resolved in part (D.3): the FY2025 10-K segment table for 2023–2025 and FY2025 expense lines; the FY2022 10-K segment figures for 2022; see Exhibit 7.11.
- **U-D10-02 (D10): U-2 — Cash flow detail.** Acquisition of leasing equipment, acquisition of property, plant and equipment, proceeds from asset sales, working-capital changes (inventory), and whether the company defines free cash flow. Resolve with the cash flow statement in [S1] and [S2].
- **U-D10-11 (D10): U-11 — Early asset mix and the shift to aviation.** Figures for Jefferson Terminal, Repauno, Transtar and Long Ridge; dates and prices of QuickTurn, the

Chromalloy JV and the Lockheed Martin Commercial Engine Solutions acquisition. Resolve with the FY2021 10-K [S15], the FY2022 annual report and the relevant 8-Ks. Resolved in part (D.3) as for U-D5-07.

- **U-D10-16 (D10): U-16 — Legal proceedings, auditor and internal control.** Status of the securities class actions filed after the Jan 2025 report and any derivative suits; the auditor's name (recollection, unverified: Ernst & Young LLP); whether any material weakness was disclosed; the content of the Jan 2025 EDGAR correspondence [S12 folder] and the later 2025 8-K [S25]. Resolve with Item 3, Item 9A and the auditor's report in [S1]. Resolved in part (D.3): litigation status as for U-D5-16; auditor line "Ernst & Young LLP 2016-2025; Audit Committee engaged KPMG LLP for FY2025 effective 17 Jun 2025; auditor's report dated 27 Feb 2026" with the ambiguity in R-066; internal-control conclusion still open.
- **U-D10-17 (D10): U-17 — Employees and locations.** Headcount at 31 Dec 2025, by geography; properties list (New York headquarters, Montreal, Miami, Rome NY, Lisbon). Resolve with Item 1 "Human Capital" and Item 2 of [S1]. Resolved in part (D.3) as for U-D5-09; properties list still open; "Rome NY" is the chain-map error recorded in R-040.

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PART VIII

FTAI Aviation Leasing and the Strategic Capital Initiative

This Part covers the half of FTAI that owns things: the Aviation Leasing segment, with its aircraft, engines, rent, reserves and the Russian assets it lost, and the Strategic Capital Initiative, through which FTAI has begun to move aircraft ownership into vehicles funded by other people's money while keeping the servicing fee, a minority stake and the engine work. A reader who finishes it will be able to read FTAI's leasing disclosures line by line, trace every dollar that moves between FTAI and the vehicles it manages, and see where each lands in the accounts.

TRANSLATION-TABLE ROWS ANSWERED

Part III §24 turns each demand driver into a requirement on the aftermarket. This Part answers the rows that bear on who owns mid-life aircraft and their engines, and on what terms. Lessors have been extending leases rather than taking aircraft back, which cuts the flow of aircraft into the used-sale market and leaves fewer acquisition candidates for mid-life buyers; the vehicles described in §61 and §62 buy aircraft with their leases attached and so compete for that reduced flow (Part III §24, row T-6: lease extensions). The cap on 737 MAX output and the shortfall in Airbus deliveries keep 737NGs and A320neos flying longer, which holds up the values and rents of exactly the aircraft the vehicles buy; the carriers that are retiring 737NGs regardless, WestJet among them, are the source of the off-lease aircraft FTAI buys for their engines in §67 (Part III §24, row T-4: 737 MAX production cap; row T-5: Airbus delivery shortfall; row T-2: groundings of the Pratt & Whitney geared turbofan (GTF)). Shop capacity is short and turnaround times are long, so the vehicles' engines are powered by module exchanges rather than conventional shop visits, which is the maintenance stream worked in §66 (Part III §24, row T-8: maintenance, repair and overhaul (MRO) capacity shortfall). Low retirements starve the teardown pool and keep engines flying toward their life limits inside the vehicles' holding period (Part III §24, row T-1: low retirements). And the freighter conversion row is the collaboration with the conversion house AEI (Aeronautical Engineers, Inc.) in §67 (Part III §24, row T-11: freighter conversion).

58. The Aviation Leasing segment: fleet, utilization, income

FTAI has two reportable segments. Part VII covered Aerospace Products, the segment that sells engines, modules and parts. This section covers the other one. Aviation Leasing is the segment that owns aircraft and engines and rents them out, and it is the part of FTAI that looks most like a conventional lessor. Part II §12 to §17 explain how an operating lease, a maintenance reserve, a securitization and a warehouse facility work; this Part does not repeat those mechanics beyond a sentence where one is needed.

AVIATION LEASING (AL) SEGMENT

FTAI's reportable segment that, in the FY2025 10-K's words, "owns and manages aircraft and engines and leases and sells them, directly and through an equity-method investment". At 31 December 2025 it held 290 assets, 47 aircraft and 243 engines, including 8 aircraft and 17 engines in Russia; 37 aircraft and 143 engines were on lease and utilization was about 77% on the company's definition. Its revenue lines are lease income, maintenance revenue and asset sales revenue; it also records gains on sale of leasing equipment and its share of an equity-method investment's earnings. Segment earnings before interest, taxes, depreciation and amortisation, as the company adjusts them (Adjusted EBITDA, defined in Part X §80), were \$500.1M in 2024 and \$608.9M in 2025. [FTAI 10-K FY2025, segment description and segment table]

The company's own one-line description of what the segment does is worth quoting, because it explains why a maintenance company owns aircraft at all. FTAI "owns and maintains commercial jet engines with a focus on CFM56 and V2500 engines" and "also owns and leases jet aircraft which often facilitates the acquisition of engines at attractive prices". [FTAI Q2 2026 release, 29 Jul 2026] The aircraft is bought for the two engines hanging under its wings. Part II §15 explains why that is arithmetically sensible for a mid-life narrowbody, where the engines carry most of the value. The leasing book is therefore both a source of rent and a feedstock pipeline for the Module Factory of Part VII.

58.1 THE FLEET AT 31 DECEMBER 2025**EXHIBIT 8.1 — THE AVIATION LEASING FLEET AND ITS OPERATING METRICS, AS FILED**

Metric	Figure	As of	Source
Assets owned or managed	290 (47 aircraft, 243 engines)	31 Dec 2025	FTAI 10-K FY2025
Of which located in Russia	8 aircraft, 17 engines	31 Dec 2025	FTAI 10-K FY2025
On lease	37 aircraft, 143 engines	31 Dec 2025	FTAI 10-K FY2025
Aircraft on lease, by count (derived in the research)	37 of 47 = 79%; 37 of 39 outside Russia = 95%	31 Dec 2025	Arithmetic on the 10-K counts
Engines on lease, by count (derived in the research)	143 of 243 = 59%; 143 of 226 outside Russia = 63%	31 Dec 2025	Arithmetic on the 10-K counts
Utilization, company definition	~77%	Q4 2025	FTAI 10-K FY2025
Utilization, company definition	~79%	nine months to 30 Sep 2024	FTAI 10-Q Q3 2024
Utilization, company definition	~77%	Q4 2023	FTAI 10-K FY2023 (as returned by search)

Metric	Figure	As of	Source
Weighted-average remaining lease term, aircraft	44 months	31 Dec 2025	FTAI 10-K FY2025
Average remaining lease term, engines on lease	38 months	31 Dec 2025	FTAI 10-K FY2025
Assets in Ukraine and Russia	4 aircraft and 1 engine in Ukraine; 8 aircraft and 17 engines in Russia	31 Dec 2022	FTAI 10-K FY2022

Source: FTAI Aviation Ltd., Form 10-K for FY2025, filed 27 Feb 2026; Form 10-Q for the quarter ended 30 Sep 2024; Form 10-K for FY2023; Form 10-K for FY2022. Fleet counts at 31 Dec 2021 to 2024 and at 30 Jun 2026 were not retrieved (D6 U1). Compiled from D6 §2.6 and §4.1.

Two things in that table need explaining. The first is that the count of assets on lease and the utilization figure measure different things. By count, only 59% of the engines were on lease at the year end. By the company's measure, utilization was about 77%. The gap is not an error; it is a definition.

UTILIZATION (FTAI DEFINITION)

FTAI's utilization is "the percent of days on-lease in the quarter weighted by the monthly average equity value of aviation leasing equipment, excluding airframes". Each engine counts in proportion to its book equity value, not one-for-one, and airframes are left out. A run-out engine awaiting a shop visit, or one written to zero in Russia, drags the count ratio down but barely moves the value-weighted figure. Willis Lease measures utilization as the share of its lease portfolio by book value that is on lease, 84.9% for 2025; the two companies' figures are not directly comparable, and Part XI §88 says so when it sets them side by side. [FTAI 10-K FY2025; Willis Lease 10-K FY2025; registry C-3]

So 77% on FTAI's definition is consistent with 59% of engines on lease by count if the off-lease engines are low in equity value. The research recorded this as a definitional gap, not a contradiction, and this Part quotes the 10-K definition rather than the count ratio whenever it uses the word. [D6 T6] The ratio also hints at where the yield of §58.4 comes from: if about a quarter of the value-weighted book earns nothing in a quarter, the engines that are on lease must earn roughly $1 \div 0.77$, or about 1.3 times, the portfolio-wide rate.

The second point is the remaining term. Aircraft had 44 months of lease left on average and engines 38 months, at 31 December 2025. [FTAI 10-K FY2025] Those are the periods over which the rent in §58.2 is contracted. New-lease terms at signing, and the split between short-term and long-term engine leases in FTAI's book, were not retrieved. [D6 U15]

58.2 LEASE INCOME BY PERIOD, AND TWO TENSIONS IN THE RECORD

Lease income is the rent line. It is the first of three revenue lines the segment reports.

LEASE INCOME, MAINTENANCE REVENUE AND ASSET SALES REVENUE

The Aviation Leasing segment's three income-statement lines. Lease income is rent from operating leases of aircraft and engines (\$235.2M for 2025). Maintenance revenue is maintenance reserves and end-of-lease compensation recognised as revenue when FTAI is no longer obliged to reimburse them, typically at lease end or on sale (\$218.5M for 2025); Part II §13 explains the reserve mechanism. Asset sales revenue is the gross proceeds of aircraft and engine sales, with the asset's carrying value in cost of sales (\$106.9M for 2025). Gains on sale of leasing equipment and the equity-method pick-up sit outside these three lines. [FTAI 10-K FY2025, consolidated statement of operations, via the orchestrator's search; D10 §2.4]

EXHIBIT 8.2 — LEASE INCOME BY PERIOD, WITH BOTH READINGS WHERE THE RECORD HOLDS TWO

Period	Lease income (\$ thousands)	Note	Source
FY2021	173,864	Russian lessees contributed "approximately \$39.8 million" of basic lease revenue in 2021	FTAI 10-K FY2022
FY2021	128,328 (Aviation Leasing segment revenue, not lease income)	Returned by a different search of the same 10-K; cannot sit above a \$173.9M rent line (R-053)	FTAI 10-K FY2022, segment table
FY2022	179,314	+\$5,450K despite the Russian terminations	FTAI 10-K FY2022
FY2022	199,040 (Aviation Leasing segment revenue)	Segment revenue, which includes more than rent	FTAI 10-K FY2022, segment table
FY2023	not returned		
FY2024	234,411 or 255,338	Two readings of one filing (R-052)	FTAI 10-K FY2025, two searches
FY2025	235,210	Same figure in both readings	FTAI 10-K FY2025
Q1 2025	68,440		FTAI 10-Q Q1 2026 (comparative)

Period	Lease income (\$ thousands)	Note	Source
Q2 2025	62,439		FTAI Q2 2026 release (comparative)
1H 2025	130,879	Arithmetic	Derived
Q1 2026	39,892	–\$28.5M y/y, "primarily due to decreases in aircraft lease revenue of \$24.9 million, driven by the sale of Seed Assets to the 2025 Partnership"	FTAI 10-Q Q1 2026
Q2 2026	27,765	–56% y/y	FTAI Q2 2026 release, 29 Jul 2026
1H 2026	67,657	–48% y/y, arithmetic	Derived

Source: FTAI Aviation Ltd., Form 10-K for FY2022; Form 10-K for FY2025 (segment table and consolidated statement, two search readings recorded in the orchestrator notes); Form 10-Q for the quarter ended 31 Mar 2026; Q2 2026 earnings release, 29 Jul 2026. Reconciliation R-052 and R-053; E.1 rule 28.

The table carries two tensions, and this Part presents both without choosing. The first concerns 2021. One search of the FY2022 10-K returned lease income of \$173,864K for 2021. Another search of the same filing returned Aviation Leasing segment revenue of \$128,328K for 2021. [FTAI 10-K FY2022, two readings; reconciliation R-053] A rent line cannot be larger than the segment revenue that contains it, so the two figures cannot both be right as read. The reconciliation pass noted that the same search also returned a figure of "70,800" three times for different 2021 lines, which it judged a probable extraction artefact, and it recorded the 2021 segment figures as not reliably retrieved. [orchestrator notes; reconciliation D.3] The 2022 figures from the same filing sit in the expected order: \$179,314K of lease income inside \$199,040K of segment revenue. [FTAI 10-K FY2022] What would resolve 2021 is a direct read of the FY2022 10-K's segment note.

The second tension concerns 2024. The FY2025 10-K's segment table, as returned by one search, gives 2024 lease income of \$234,411K. An earlier search of the same filing returned \$255,338K. [FTAI 10-K FY2025, two readings; reconciliation R-052] The orchestrator's note on the discrepancy offers two possibilities: the two searches may have hit different lines, a total lease-income line against a segment lease-income line, or one of them may be an extraction error. Both figures come from the same filing and the 2025 figure of \$235,210K is the same in both readings. [orchestrator notes] This Part shows the pair wherever 2024 lease income appears and does not average them.

Beyond the tensions, the direction of the series is not in doubt. Rent was flat to rising from 2021 through 2025, roughly \$174M to \$235M, and then fell by almost half in the first half of 2026. The 10-Q attributes the fall to one cause: the sale of the Seed Assets to the 2025 Partnership, which §60 describes. [FTAI 10-Q Q1 2026] Quarterly rent went from \$68.4M in the first quarter of 2025 to \$39.9M a year later, and from \$62.4M in the second quarter of 2025 to \$27.8M a year later. [FTAI 10-Q Q1 2026; FTAI Q2 2026 release] The 2025 annual figure of \$235.2M averaged \$58.8M a quarter; the 2026 run-rate in the second quarter was less than half of that.

58.3 THE OTHER LINES, AND SEGMENT ADJUSTED EBITDA

Rent is not where most of the segment's earnings come from. Two other revenue lines are each of the same order as rent, and the segment's profit measure also includes gains and an equity pick-up that are not revenue at all.

EXHIBIT 8.3 — AVIATION LEASING SEGMENT LINES AND ADJUSTED EBITDA BY YEAR (\$ THOUSANDS)

Line	2022	2023	2024	2025	Q1 2026	Source
Aviation Leasing segment revenue	199,040	n/r	n/r	n/r	n/r	FTAI 10-K FY2022
Lease income	179,314	n/r	234,411 or 255,338	235,210	39,892	10-K FY2022; 10-K FY2025 (two readings); 10-Q Q1 2026
Maintenance revenue	n/r	n/r	n/r	218,499	30,599	10-K FY2025; 10-Q Q1 2026
Asset sales revenue	n/r	n/r	n/r	106,945	fell \$8.8M y/y, "primarily due to an overall decrease in the number of sales transactions"	10-K FY2025; 10-Q Q1 2026
Gain on sale to the 2025 Partnership	—	—	—	46,380	n/r	10-K FY2025
Transactional impairment	n/r	n/r	1,000 (net of redelivery compensation)	0	n/r	10-K FY2025

Line	2022	2023	2024	2025	Q1 2026	Source
Aviation Leasing Adjusted EBITDA	107,556	n/r	500,062	608,912	n/r	10-K FY2022; 10-K FY2025 segment table
Aviation Leasing Adjusted EBITDA, Q2 2026					88,200 (Q2 2026; per call coverage of the 29 Jul 2026 call, GuruFocus and Quartr; not returned from the release text)	Chain map; reconciliation R-051

Source: FTAI Aviation Ltd., Form 10-K for FY2022; Form 10-K for FY2025 (segment table and consolidated statement of operations, via the orchestrator's searches); Form 10-Q for the quarter ended 31 Mar 2026; call coverage of the Q2 2026 call (GuruFocus, Quartr). n/r = not returned by any search in the research (D6 U2; reconciliation K-17). 2022 and 2021 figures include years in which the infrastructure businesses were in discontinued operations and are not comparable with 2023 onward (Part X §79).

Adjusted EBITDA is FTAI's adjusted earnings measure; Part X §80 carries the definition and its reconciliation. For this segment the research recorded that the measure adds back depreciation and includes gains on asset sales and the equity-method pick-up. [D6 §2.6; Part X §80] That composition matters for reading the table. Lease income of \$235.2M cannot produce \$608.9M of Adjusted EBITDA on its own. Maintenance revenue of \$218.5M, asset-sale margins on \$106.9M of proceeds, the \$46.4M gain on sales to the 2025 Partnership, and whatever the equity-method investment contributed must together account for most of the figure. The segment note that would show the composition line by line was not retrieved. [D6 U2; reconciliation K-17]

The same point applies with more force to 2026. Guidance for the segment's 2026 Adjusted EBITDA was \$475M after the July 2026 cut (\$60.4), while first-half 2026 lease income of \$67.7M annualises to about \$135M. [FTAI Q2 2026 release, 29 Jul 2026; derived in the research] On those two figures, rent would be well under a third of the segment's guided earnings, with the balance from gains, asset sales, maintenance revenue and the equity-method share. The research flagged the composition as the single most important unretrieved item for this segment.

One figure in the exhibit is labelled differently from the rest. The Aviation Leasing Adjusted EBITDA of \$88.2M for the second quarter of 2026 came from call-coverage summaries of the 29 July 2026 call, not from the release text, which the research searched and which returned consolidated Adjusted EBITDA of \$291,444K and Aerospace Products Adjusted EBITDA of \$249.7M but no Leasing line. [GuruFocus and Quartr call coverage; FTAI Q2 2026 release; reconciliation R-051] If \$88.2M were right, the two segments would sum to \$337.9M against a consolidated \$291.4M,

implying about −\$46.5M for Corporate and Other. The research described that sign and size as plausible for corporate costs plus the start-up of FTAI Power (Part IX) but did not confirm it. [D6 T3; E.1 rule 2]

58.4 WHAT THE BOOK YIELDS: A WORKED EXAMPLE WITH ITS BOUNDS

A lessor's rent divided by the cost of what it rents out is its gross yield. Part II §14 defines the lease rate factor, the monthly version of the same ratio, and Part II §12 defines the gross leasing yield. Neither FTAI nor Willis Lease states a portfolio lease rate factor in any filing the research reached, so every figure below is derived. [D6 U5; D11 U1; reconciliation K-14]

NET BOOK VALUE (NBV)

Cost less accumulated depreciation and impairments. FTAI's "leasing equipment, net" was \$1,545.8M at 31 December 2025 against \$2,057.6M at cost. A yield on cost strips out depreciation policy; a yield on net book value shows the return on the carrying amount. Part XI §88 uses the same denominator for Air Lease's \$29.1B of flight equipment. [FTAI 10-K FY2025; D11 §2.2]

WORKED EXAMPLE 8.1 — FTAI'S IMPLIED GROSS LEASING YIELD FOR 2025, THREE WAYS

Inputs. Leasing equipment at cost \$2,963.452M at 31 December 2024 and \$2,057.624M at 31 December 2025; net \$2,373.730M and \$1,545.804M (sourced: FTAI 10-K FY2025). FY2025 lease income \$235.210M (sourced: FTAI 10-K FY2025, the same in both readings). 1H 2025 lease income \$130.879M and 1H 2026 lease income \$67.657M (sourced: 10-Q Q1 2026 and Q2 2026 release, summed). Utilization about 77% in Q4 2025 (sourced: 10-K).

average 2025 cost base = $(2,963.452 + 2,057.624) \div 2 = 2,510.538$
 average 2025 net base = $(2,373.730 + 1,545.804) \div 2 = 1,959.767$
 (i) full-year rent on average bases: $235.210 \div 2,510.538 = 9.4\%$ on cost;
 $235.210 \div 1,959.767 = 12.0\%$ on net
 (ii) 1H 2025 rent annualised (as derived in the research): $261.758 \div 2,510.538 = 10.4\%$ on cost; $\div 1,959.767 = 13.4\%$ on net
 (iii) 1H 2026 rent annualised on the 31 Dec 2025 bases: $135.314 \div 2,057.624 = 6.6\%$ on cost; $\div 1,545.804 = 8.8\%$ on net
 monthly equivalents of (i): $9.4 \div 12 = 0.78\%$ a month on cost; $12.0 \div 12 = 1.0\%$ a month on net
 on-lease equivalents of (i), dividing by 0.77: about 12.2% on cost and 15.6% on net

Reading (i) is new to this Part: the research derived (ii) and (iii) before the FY2025 lease-income figure was available and asked that the 10-K figure be substituted, which (i) does. Reading (ii) overstates 2025 because rent fell in the second half after the Seed Asset sales; reading (iii) understates the yield if the cost base kept falling through the first half of 2026, which the guidance language implies. Three things depress the on-cost figure: the Russian assets, which sit at zero net

book value but may still sit in gross cost (not retrieved, D6 U4); off-lease engines; and aircraft bought for their engines and parked or in work rather than leased. Limits of the example: average balances are two-point averages; no 30 June 2026 balance sheet was retrieved; and no external lease rate factor was retrieved for FTAI to check the result against. The derived figures that exist elsewhere in the research are 0.70% a month (a green-time rent on a half-life value, Part II §14), about 1.0% a month for Willis Lease's on-lease book (Part XI §88) and 0.9% a month assumed in the fee example of §66; none is a stated figure. [D6 §2.7; D11 §2.2; reconciliation R-030]

The yield is the bridge between this section and the next three. A book earning 9 to 12 cents a year on each dollar of cost, with about a quarter of its value idle at any time and some of it unrecoverable in Russia, is the asset FTAI chose in 2025 to start selling into vehicles funded by others. The rent fell by half as a result. What replaced it is the subject of §60 to §63.

59. Russia

In 2022, after Russia's invasion of Ukraine and the sanctions that followed, lessors worldwide lost access to aircraft and engines leased to Russian airlines. FTAI's exposure was small in asset count and material in money, and the filings record it in enough detail to follow from loss to recovery.

RUSSIA ASSETS, IMPAIRMENT AND INSURANCE RECOVERIES (CONTINGENT COVER)

In 2022 FTAI "terminated all lease agreements with Russian airlines" and recorded a \$120.0M impairment, "net of maintenance deposits, to write-off the entire carrying value of leasing equipment assets that the company did not expect to recover from Ukraine and Russia". At 31 December 2025, 8 aircraft and 17 engines were "still located in Russia" with an insured value of \$210.7M. Lessees must insure leased assets with FTAI named as insured for total loss, and FTAI buys contingent cover for off-lease periods and for cases where a lessee's policy fails to pay. Recoveries of \$54.3M (nine months to 30 September 2025) and \$44.6M (first quarter of 2026) have been disclosed. [FTAI 10-K FY2022; 10-K FY2025; 10-Q Q3 2025; 10-Q Q1 2026]

59.1 THE TIMELINE AS THE FILINGS GIVE IT

EXHIBIT 8.4 — FTAI'S RUSSIA AND UKRAINE EXPOSURE, 2021 TO 2026

Date or period	Event or figure	Source
FY2021	Russian lessees generated "approximately \$39.8 million" of basic lease revenue	FTAI 10-K FY2022
2022	All leases with Russian airlines terminated; \$120.0M impairment, net of maintenance deposits, writing the assets to zero	FTAI 10-K FY2022

Date or period	Event or figure	Source
31 Dec 2022	4 aircraft and 1 engine in Ukraine; 8 aircraft and 17 engines in Russia	FTAI 10-K FY2022
2023 and 2024	Insurance recoveries, if any: not retrieved	D6 U4
9M 2025	\$54.3M of insurance recoveries	FTAI 10-Q Q3 2025
31 Dec 2025	8 aircraft and 17 engines "still located in Russia"; insured value \$210.7M	FTAI 10-K FY2025
Q1 2026	\$44.6M "in insurance recoveries in connection with the settlement of claims related to the aircraft and engines located in Russia"	FTAI 10-Q Q1 2026
Cumulative disclosed recoveries	\$98.9M (arithmetic)	Derived

Source: FTAI Aviation Ltd., Form 10-K for FY2022; Form 10-Q for the quarter ended 30 Sep 2025; Form 10-K for FY2025; Form 10-Q for the quarter ended 31 Mar 2026. Compiled from D6 §2.6 and §4.2. Reconciliation R-060.

The phrase "net of maintenance deposits" does real work. When the leases were terminated, FTAI held maintenance reserves and security deposits that the Russian lessees had paid in. Part II §13 explains what those are. The write-off of \$120.0M is the carrying value of the lost assets less those balances, which FTAI kept. [FTAI 10-K FY2022; D6 §2.6] The gross carrying value before that netting, and the amount of deposits retained, were not returned by any search. [D6 U4]

The 2022 lease income line shows how small the revenue effect was at the time. Rent rose from \$173.9M in 2021 to \$179.3M in 2022 even though lessees that had paid about \$39.8M in 2021 paid nothing after the terminations. [FTAI 10-K FY2022] Other leases more than filled the gap in the same year.

59.2 WHAT HAS BEEN RECOVERED, AND AGAINST WHAT

WORKED EXAMPLE 8.2 — RECOVERIES AGAINST THE WRITE-OFF AND THE INSURED VALUE

Inputs: write-off \$120.0M net of deposits, 2022 (sourced: 10-K FY2022); insured value of assets remaining in Russia \$210.7M (sourced: 10-K FY2025); recoveries \$54.3M for the nine months to 30 September 2025 (sourced: 10-Q Q3 2025) and \$44.6M in the first quarter of 2026 (sourced: 10-Q Q1 2026).

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cumulative recoveries = 54.3 + 44.6 = 98.9
recoveries ÷ net write-off = 98.9 ÷ 120.0 = 82%
recoveries ÷ insured value = 98.9 ÷ 210.7 = 47%
if the $210.7M is stated before the recoveries: 210.7 - 98.9 = 111.8 of
insured value still to be claimed
if the $210.7M is stated after the 2025 recoveries only: 210.7 - 44.6 =
166.1

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The two conditional lines exist because the filing does not say whether the \$210.7M insured value is before or after the recoveries already received. Nor does it say how many claims have been settled and how many are outstanding, or whether any recoveries were received in 2022, 2023 or 2024. The reconciliation cannot be completed from what was retrieved: the write-off is net of deposits and the insured value is a gross policy figure, so the two are on different bases. Limits of the example: every ratio compares a cash recovery with a figure on a different basis; the income-statement line on which recoveries appear was not retrieved. [D6 §2.6, T8, U4; reconciliation R-060]

The tension the reconciliation recorded is simply that \$120.0M and \$210.7M describe the same assets and differ by \$90.7M. [reconciliation R-060] The difference has two known components, the deposits netted in 2022 and insured values set above carrying values, and the filings do not quantify either. Both figures are carried in this Part with their bases stated.

59.3 WHAT REMAINS, AND WHAT IS UNKNOWN

Eight aircraft and seventeen engines were still in Russia at the end of 2025. [FTAI 10-K FY2025] They are the same counts the FY2022 10-K gave for Russia three years earlier, so none of those assets has been recovered physically. They sit in the fleet count of 290 assets in Exhibit 8.1 at a carrying value of zero. Whether they still sit in the \$2,057.6M of leasing equipment at cost, which would depress the on-cost yield in worked example 8.1, was not retrieved. [D6 U4]

The four aircraft and one engine in Ukraine at the end of 2022 do not reappear in anything the research retrieved from later filings. Whether they were recovered, written off within the same \$120.0M, or remain in Ukraine is unknown; the FY2023 and FY2024 10-Ks' Russia and insurance paragraphs would resolve it. [D6 U4] Part X §85 carries the Russia item among the risk factors the 10-K lists.

60. The shift to asset-light: Seed Assets and the 2025 Partnership

On 30 December 2024 FTAI announced that it would stop being the main owner of the aircraft it manages. The announcement launched the Strategic Capital Initiative (SCI), which §61 describes as a structure, and it named the first transaction: FTAI would sell 46 of its own on-lease aircraft

into the first vehicle. This section follows what that did to the leasing book, to the gains line and to guidance.

60.1 THE SEED ASSETS

SEED ASSETS

FTAI's term for the on-lease aircraft sold from its own balance sheet to start the 2025 Partnership: 46 aircraft "at an estimated net purchase price of \$549 million" in the launch release, about \$11.9M each. The 10-Qs attribute the fall in aircraft lease revenue in 2026 to "the sale of Seed Assets to the 2025 Partnership". The sales cut leasing equipment at cost by \$906M during 2025 and produced a gain of \$46.4M for the year. [FTAI release, 30 Dec 2024; FTAI 10-Q Q1 2026; FTAI 10-K FY2025]

A seed portfolio does for a new fund what it says. A vehicle that has raised commitments but owns nothing has no rent to show investors and no track record. Selling it a portfolio of aircraft already on lease to airlines gives it, on day one, contracted rent, a diversified set of lessees and an asset base against which to borrow. The sponsor, here FTAI, gets cash for assets it was carrying at historical cost, books the difference as a gain, and keeps a minority stake in the buyer. Part II §17 describes the same move in a securitization, where a lessor sells a portfolio to a special-purpose vehicle and keeps the equity and the servicing.

EXHIBIT 8.5 — AIRCRAFT SOLD TO THE 2025 PARTNERSHIP, BY QUARTER, AND WHAT THE LEASING BOOK DID

Period	Aircraft sold	Gain on sale	Proceeds	Source
Q1 2025	4	\$10.9M	\$58.9M	FTAI 10-Q Q1 2025; 10-Q Q1 2026
Q2 2025	33	\$34.6M	n/r	FTAI 10-Q Q2 2025
1H 2025 (as stated)	37	\$45.5M	n/r	FTAI 10-Q Q2 2025
Q3 2025	8	\$4.6M	n/r	FTAI 10-Q Q3 2025
9M 2025 (sum of quarters)	45	\$50.1M	n/r	Arithmetic
FY2025 (as stated)	n/r	\$46.4M (\$46,380K)	n/r	FTAI 10-K FY2025
Q1 2026	n/r	n/r	\$117.3M	FTAI 10-Q Q1 2026
Leasing equipment at cost	\$2,963.5M → \$2,057.6M (−\$905.8M, −31%)			FTAI 10-K FY2025
Accumulated depreciation	\$589.7M → \$511.8M			FTAI 10-K FY2025

Period	Aircraft sold	Gain on sale	Proceeds	Source
Leasing equipment, net	\$2,373.7M → \$1,545.8M (−\$827.9M)			FTAI 10-K FY2025

Source: FTAI Aviation Ltd., Forms 10-Q for the quarters ended 31 Mar 2025, 30 Jun 2025, 30 Sep 2025 and 31 Mar 2026; Form 10-K for FY2025. Balance-sheet figures are 31 Dec 2024 to 31 Dec 2025. Compiled from D7 §2.5 and §4.2 and D6 §4.3. Reconciliation R-054.

The balance sheet shows the footprint. Leasing equipment at cost fell by \$905.8M in 2025, and the net figure by \$827.9M. Accumulated depreciation fell too, by \$77.9M, because the assets sold took their accumulated depreciation off the books with them. [FTAI 10-K FY2025; D6 §2.6] The fall in cost is larger than the \$549M seed price because it also reflects ordinary sales net of purchases during the year; the research did not retrieve the purchase and sale lines of the cash-flow statement that would separate them. [D10 U-2]

60.2 THE GAIN, AND A TENSION IN THE RECORD

GAIN ON SALE OF LEASED EQUIPMENT, AND THE GAIN ON SALE TO THE 2025 PARTNERSHIP

The sale price of an asset sold from the leasing portfolio minus its net book value. FTAI recorded \$46.4M of such gains on sales to the 2025 Partnership in FY2025; the three quarterly gains disclosed for 2025 sum to \$50.1M for nine months, which is a tension (\$60.2). Willis Lease recorded \$54.0M of gains on sale of leased equipment in 2025, which Part XI §88 sets beside FTAI's figure. [FTAI 10-K FY2025; FTAI 10-Qs 2025; Willis Lease 10-K FY2025]

The gain line carries the Part's first arithmetic tension. The 10-Qs for the first three quarters of 2025 disclose gains on sales to the 2025 Partnership of \$10.9M, \$34.6M and \$4.6M, on 4, 33 and 8 aircraft. Those sum to \$50.1M on 45 aircraft for nine months. The FY2025 10-K states a full-year gain of \$46.4M. [FTAI 10-Qs Q1 to Q3 2025; FTAI 10-K FY2025; reconciliation R-054] A nine-month sum \$3.7M above the full-year figure means either that the fourth quarter recorded a net loss on sales to the Partnership of about \$3.7M, or that the quarterly and annual figures count different things, for instance gross gains in one and gains net of some adjustment in the other. The filings as retrieved do not say which. Worked example 8.8 in §66 reconciles the seed sale with this tension in view.

What the quarterly figures do give is one data point on the cost basis, which the research otherwise could not find. [D6 U12] In the first quarter of 2025, proceeds of \$58.9M on four aircraft produced a gain of \$10.9M. The net book value of those four aircraft was therefore \$48.0M, or \$12.0M each, and the sale price was \$14.7M each. The gain was 18.5% of proceeds. [FTAI 10-Q Q1

2026, comparative; derived in this Part] Whether those four were typical of the 46 is not known; the seed average of \$11.9M per aircraft sits below the \$14.7M those four fetched.

60.3 WHERE THE GAIN SITS IN THE ACCOUNTS

The 10-K says the 2025 Partnership "acquires on-lease narrowbody aircraft from the Company ... with aircraft being sold to the 2025 Partnership for a gain", and that FTAI accounts for those sales under a standard for disposals outside ordinary revenue, describing them as "non-recurring in nature and not considered part of the Company's ordinary activities". [FTAI 10-K FY2025] Under that treatment the gain appears on a gain line rather than inside asset sales revenue, and the Aviation Leasing Adjusted EBITDA measure includes it. [D7 §2.5; D6 §2.6] §64 defines the standard and sets the "non-recurring" description beside the company's other statements about how often it expects to do this.

60.4 THE GUIDANCE CUT AND ITS STATED REASON

ASSET-LIGHT MODEL

Management's term for owning fewer leased assets directly and earning instead servicing fees, maintenance work and equity income from vehicles that own them. The Strategic Capital releases say the initiative lets FTAI "maintain an asset-light business model", and the July 2026 cut to Aviation Leasing guidance was explained as "reflecting a continued shift to an asset-light business model". The term is management's; the filings also record a 19% limited-partner interest in the first vehicle carried at \$365.5M at 30 June 2026 (\$63.5). [FTAI releases, 30 Dec 2024, 26 Feb 2025, 25 Mar 2025; FTAI Q2 2026 release, 29 Jul 2026; FTAI 10-Q Q2 2026; reconciliation R-058]

EXHIBIT 8.6 — GUIDANCE FOR AVIATION LEASING ADJUSTED EBITDA, AS DATED

Date set	For	Aviation Leasing	Total business-segment guidance	Wording or reason	Source
30 Dec 2024	2025	~\$500M	~\$1.1–1.15B (~\$600–650M Aerospace Products)	Set with the SCI launch	FTAI release, 30 Dec 2024
Dec 2024 to early 2025	2026	n/r	\$1.4B	Starting point of the 2026 sequence	FTAI 10-Q Q1 2026 (search return); orchestrator notes

Date set	For	Aviation Leasing	Total business-segment guidance	Wording or reason	Source
27 Oct 2025	2026	\$525M	\$1.525B (~\$1.0B Aerospace Products)	Step quoted in the Q1 2026 10-Q without a date; dated to the Q3 2025 release by the orchestrator notes	FTAI 10-Q Q1 2026; Q3 2025 release, 27 Oct 2025
25 Feb 2026	2026	\$575M	\$1.625B (\$1.05B Aerospace Products)	Raised with the FY2025 results	FTAI release, 25 Feb 2026
29 Jul 2026	2026	\$475M	\$1.525B (\$1.05B Aerospace Products reaffirmed)	"updated 2026 Aviation Leasing Adjusted EBITDA guidance from \$575 million to \$475 million, reflecting a continued shift to an asset-light business model"	FTAI Q2 2026 release, 29 Jul 2026
29 Jul 2026	2027	\$450M	\$2.3B (\$1.4B Aerospace Products, \$450M FTAI Power)	First 2027 guidance	FTAI Q2 2026 release, 29 Jul 2026

Source: FTAI Aviation Ltd., releases of 30 Dec 2024, 27 Oct 2025, 25 Feb 2026 and 29 Jul 2026; Form 10-Q for the quarter ended 31 Mar 2026; orchestrator notes, "Resolutions after reconciliation". Reconciliation R-049 and R-050; Part X §84 carries the company-wide guidance history.

Two notes on the table. First, the research initially held the \$525M step without a date, because the Q1 2026 10-Q quoted it as a past change; the orchestrator's later search placed it in the Q3 2025 release of 27 October 2025, the same day the first vehicle hit its hard cap (§62). [reconciliation R-050; orchestrator notes] Second, one search reading of the 29 July 2026 release returned the February figures (\$1.625B and \$575M) rather than the cut, and the reconciliation recorded that as a tension between two readings of one release. The orchestrator's follow-up established that the July release does state the cut to \$475M, and that the \$575M reading was a mis-read of the February figure carried forward. [D6 T1; reconciliation R-049; orchestrator notes] This Part records both the original tension and its resolution, as the notes ask.

The reason given for the cut is the one in the box above: a "continued shift to an asset-light business model". Read with §58.2, the mechanism is plain. Aircraft that produced rent for FTAI in 2025 produce rent for the 2025 Partnership in 2026, and FTAI's share of that rent arrives as 19% of the Partnership's earnings and as a servicing fee, both smaller than the rent itself. The 2027 guidance of \$450M for the segment is lower again. [FTAI Q2 2026 release, 29 Jul 2026]

The reconciliation also recorded a definitional tension in the phrase itself. Management describes the model as asset-light; the balance sheet shows a limited-partner interest of \$281.7M at 31 December 2025 and \$365.5M at 30 June 2026, and the full commitment at 19% of a \$2.0B equity raise would be about \$380M. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; derived in the research; reconciliation R-058] Both statements are as made: fewer aircraft on the balance sheet, and several hundred million dollars of equity in the vehicle that holds them.

61. SCI structure: vehicles, two layers of debt, the Servicer role

STRATEGIC CAPITAL INITIATIVE (SCI)

FTAI's asset-management arm, launched 30 December 2024. It raises third-party capital into closed-end vehicles that buy on-lease, mid-life 737NG and A320ceo aircraft; FTAI manages each vehicle as Servicer, co-invests as a limited partner, and supplies every engine shop visit for the fleet through module and engine exchanges with its Maintenance, Repair and Exchange business (Part VII §46). The Q2 2026 10-Q calls it "an asset management business that manages third-party capital to invest in on-lease aircraft" and says FTAI expects its primary investment activities going forward to run through it. It is not a reportable segment. [FTAI release, 30 Dec 2024; FTAI 10-Q Q2 2026]

The initiative has three parts that are easy to confuse: the vehicles, the debt under them, and FTAI's roles around them. This section takes them in that order and ends with the arithmetic that connects them.

61.1 THE VEHICLE: A PARTNERSHIP ON TOP OF ASSET-OWNING SPVs

2025 PARTNERSHIP (2025 SPV; FTAI SCI I; THE INAUGURAL STRATEGIC CAPITAL VEHICLE)

The first SCI vehicle, launched 30 December 2024: a closed-end partnership whose asset-owning special-purpose vehicles buy on-lease mid-life 737NG and A320ceo aircraft. It closed on 27 October 2025 at a \$2.0B equity hard cap against a \$1.5B target; a \$2.5B asset-level debt commitment was arranged in February 2025; by August 2026 it had "approximately \$6.0 billion of total capital across over 300 aircraft", was fully committed, had paid its first quarterly distribution on 30 June 2026 and was in "harvest phase". FTAI is its Servicer and holds a 19% limited-partner interest (20% in the Q3 2025 10-Q). Its engines are powered "exclusively via engine and module exchanges" with FTAI. [FTAI releases, 30 Dec 2024, 26 Feb 2025, 27 Oct 2025, 17 Aug 2026; FTAI 10-K FY2025; FTAI 10-Qs Q3 2025, Q1 and Q2 2026]

The filings call the first vehicle the "2025 Partnership" and say it acquires aircraft through "special purpose entities (SPVs) of the 2025 Partnership". [FTAI 10-Q Q1 2026] So there are two levels. At the top is a partnership, in which the investors hold interests. Below it are special-purpose vehicles, each a company that exists only to own one aircraft or a small pool of them

and the debt secured on them. Part II §17 defines the special-purpose vehicle; aircraft lessors are usually organised this way so that each lender and each airline deals with a clean owner.

LIMITED PARTNER (LP) AND GENERAL PARTNER (GP)

A limited partner is an investor who commits capital to a partnership, has limited liability and takes no part in managing it. A general partner manages the partnership and bears liability. The 2025 Partnership's limited partners are described as "asset managers, insurance companies, public pensions, foundations, endowments and family offices", plus FTAI at 19%. Which entity acts as general partner, and FTAI's relationship to it, were not located in anything the research reached.

[FTAI release, 27 Oct 2025; FTAI 10-K FY2025; D7 §2.1, §3.1]

CLOSED-END VEHICLE

A fund that raises a fixed amount of equity commitments up to a hard cap, invests them during a deployment period, then stops buying and enters a harvest phase in which it collects rent, distributes cash to its investors and sells assets until it winds up. The 2025 Partnership is one. Its term, the length of its harvest phase, any extension rights and its exit route (a portfolio sale, a securitization, or piecemeal sales) were not located. [D7 §2.1, U15]

EQUITY COMMITMENT AND HARD CAP

An equity commitment is an investor's promise to fund capital when the vehicle calls it; capital is drawn as assets are bought, not on signing. A hard cap is the maximum amount of commitments the sponsor will accept. The 2025 Partnership's target was \$1.5B and it closed at an upsized hard cap of \$2.0B; 19% of \$2.0B is about \$380M, FTAI's full commitment if drawn in proportion (derived in the research). [FTAI release, 27 Oct 2025; D7 §2.5]

DEPLOYMENT, FULLY COMMITTED, AND HARVEST PHASE

Deployment is the buying phase, during which commitments are called and aircraft acquired. "Fully committed" means every dollar of equity has been allocated to signed acquisitions, even if some closings are still pending; it is a different state from "fully deployed", which means closed and funded, and the releases use both words at different dates (§62.1). Harvest phase is what follows: the vehicle collects rent, pays distributions and sells assets until it winds up. The 2025 Partnership was described as "fully committed" in the 29 July 2026 release and as in "harvest phase" in the 17 August 2026 release. [FTAI Q2 2026 release, 29 Jul 2026; FTAI release, 17 Aug 2026; registry C-21]

ON-LEASE

Already under an operating lease to an airline. A buyer of an on-lease aircraft acquires a contracted rent stream and a return-condition obligation rather than an empty aircraft, and lenders can underwrite the contracted rent. Everything the SCI vehicles buy is on-lease; Part II §15 treats the same idea under lease-attached trading. [FTAI releases, 2025–2026; D7 §2.4]

61.2 THE TWO LAYERS OF DEBT

A vehicle of this kind borrows at the asset level, and it borrows in two forms at different stages of its life. Part II §17 explains both instruments; here the point is which one each vehicle used and what it tells the reader.

ASSET-LEVEL DEBT (NON-RECOURSE DEBT)

Borrowing by the asset-owning SPVs, secured on the aircraft and their leases, with no recourse to the sponsor. If the aircraft and their rent do not cover the loan, the lenders' claim stops at the SPV; FTAI's own balance sheet is not exposed. The 2025 Partnership's \$2.5B commitment of February 2025 was asset-level debt, with ATLAS SP Partners committing \$2.0B directly and Deutsche Bank AG, New York Branch co-leading. [FTAI release, 26 Feb 2025]

SYNDICATED LOAN, STRUCTURING AGENT AND LEAD ARRANGER

A syndicated loan is one shared across many lenders under a single agreement, so no bank carries the whole exposure; the 2026 warehouse is syndicated among 13 banks. The structuring agent designs the deal; the lead arranger assembles the lender group. ATLAS SP Partners (a structured-products business majority owned by Apollo funds) was structuring agent and lead arranger of the 2025 facility, and ATLAS SP and Deutsche Bank were co-structuring agents of the 2026 warehouse. [FTAI releases, 26 Feb 2025 and 17 Aug 2026]

The first vehicle's debt was arranged as a \$2.5B commitment in February 2025, eight months before its equity closed. [FTAI release, 26 Feb 2025] The second vehicle's debt is a warehouse facility: a revolving line that funds aircraft one at a time as the vehicle buys them, until the pool is large enough to refinance in a term deal, typically a securitization (Part II §17). It closed on 14 August 2026 at \$2.0B with 13 lenders and an option to grow to \$3.0B. [FTAI release, 17 Aug 2026] The rates, advance rates, tenors and covenants of both facilities were not disclosed in anything the research reached, nor was whether the 2025 facility was later upsized. [D7 U11]

The first vehicle has also taken the step that a warehouse normally leads to. On 22 May 2026 FTAI priced the 2025 Partnership's inaugural asset-backed securitization (ABS), named in the research as FTAI MRE 2026-1, after FTAI's Maintenance, Repair and Exchange (MRE) business (Part VII §46): \$612M of notes backed by 48 A320ceo and 737NG aircraft on lease to 23

airlines, in two classes, a Series A with expected ratings of Asf from Fitch and A(sf) from KBRA (Kroll Bond Rating Agency) and a Series B expected at BBB+sf from Fitch, with closing expected on 4 June 2026 and both classes described as "significantly oversubscribed". [FTAI release, 22 May 2026, via Finviz and AviTrader, 26 May 2026; orchestrator notes] Part II §17 explains what such notes are and sets this deal beside Willis Lease's Willis Engine Structured Trust (WEST) vehicles. For this Part the point is structural: the Partnership's aircraft are now financed in three layers, the asset-level commitment of February 2025, a term securitization on 48 of them, and whatever bank debt remains on the rest. The coupons and the loan-to-value of the notes were not retrieved. [D7 U12, partly resolved by the orchestrator notes]

WHICH LAYER THE SPONSOR SEES

FTAI's consolidated balance sheet shows none of this debt. The \$3.45B of long-term debt it reported at 31 March 2026 (Part X §82) is FTAI's own senior notes; the vehicles' asset-level debt and warehouse borrowings sit inside unconsolidated entities and reach FTAI only through its 19% share of the Partnership's results. That is what "non-recourse" and "unconsolidated" together mean for a reader of FTAI's statements: the leverage described in worked example 8.3 is real, but it is the Partnership's, and whether FTAI was required to consolidate it is the question §64.3 records as open. [FTAI 10-Q Q1 2026; D7 §2.6]

61.3 FTAI'S THREE ROLES

FTAI stands in three relationships to each vehicle. It is a limited partner, with 19% of the 2025 Partnership. It is the Servicer. And it is the exclusive supplier of engine maintenance to the vehicle's fleet.

SERVICER

The party contracted to manage a vehicle's assets: leasing and remarketing the aircraft, collecting rent and reserves, and overseeing their technical condition, for a fee. In a securitization the lessor that sold the assets is usually the servicer (Willis Lease for its WEST trusts, Part II §17). FTAI "as the Servicer, provides aircraft management services to the 2025 Partnership, and receives customary, market-based compensation for providing such services". The compensation's rate and base are not disclosed (§63.2). [FTAI 10-K FY2025; D6 §2.5]

The third role is stated in the same sentence in every SCI release: engines owned by the vehicle "will be powered exclusively via engine and module exchanges with FTAI's Maintenance, Repair and Exchange (MRE) business". [FTAI releases, 26 Feb 2025, 25 Mar 2025, 17 Aug 2026] Part VII §46 explains what an exchange is: FTAI removes the module that triggered the removal, fits a serviceable one from its inventory, and takes the run-out unit as feedstock. For the vehicle this replaces a 90 to 120 day shop visit with an exchange; for FTAI it makes every engine in a 300-aircraft fleet a customer of the Module Factory for as long as the vehicle owns it. In February

2025 FTAI described the market it addresses this way as \$22B, with its own share about 5% as of 2024. [FTAI release, 26 Feb 2025] The accounting consequences of selling maintenance to an entity one partly owns are in §63.3 and §63.4.

WHY THE BUYER OF THE AIRCRAFT AND THE SUPPLIER OF THE ENGINES' MAINTENANCE ARE TIED TOGETHER

A mid-life narrowbody's value is mostly in its engines (Part II §15), and the largest cost its owner faces is the engines' next shop visits (Part I §7). A vehicle that buys 300 such aircraft is buying 600 engines approaching or past those visits. Contracting all of that work to one supplier in advance fixes the vehicle's largest cost line with a party that also holds 19% of its equity and manages its assets. For the supplier, it converts a one-off aircraft sale into a maintenance relationship lasting the vehicle's life. The releases describe the arrangement; the contract's pricing terms are not disclosed (§63.3). [FTAI releases, 2025–2026; D7 §2.4]

61.4 THE LEVERAGE ARITHMETIC, WORKED

LOAN-TO-COST

A vehicle's debt divided by its total capital (equity plus debt). It is distinct from loan-to-value, which divides debt by the appraised value of the assets (Part II §17). For the 2025 Partnership it is derived rather than stated: about 67% if \$6.0B of total capital sits on \$2.0B of equity, or 56% if only the \$2.5B debt commitment is drawn. [D7 §2.3]

WORKED EXAMPLE 8.3 — LEVERAGE OF THE 2025 PARTNERSHIP FROM THE DISCLOSED FIGURES

Inputs: equity commitments \$2.0B (sourced: release, 27 Oct 2025); asset-level debt commitment \$2.5B (sourced: release, 26 Feb 2025); "approximately \$6.0 billion of total capital across over 300 aircraft" (sourced: release, 17 Aug 2026). Total capital is taken as equity plus debt (assumed: the release does not define it).

case A, on the \$6.0B total: $\text{debt} = 6.0 - 2.0 = 4.0$; $\text{loan-to-cost} = 4.0 \div 6.0 = 67\%$; $\text{debt per dollar of equity} = 2.0$
 case B, on the \$2.5B commitment only: $\text{total} = 2.0 + 2.5 = 4.5$; $\text{loan-to-cost} = 2.5 \div 4.5 = 56\%$; $\text{debt per dollar of equity} = 1.25$
 $\text{capital per aircraft, case A: } 6,000 \div 300 = \20M

Case A requires about \$1.5B more debt than the February 2025 commitment, which was not disclosed as a separate event; it may be an upsizing of that facility or a second facility. The \$612M securitization of 22 May 2026 (§61.2) refinances 48 of the Partnership's aircraft and so may replace rather than add to the February borrowing; the release does not say what it repaid. Case B assumes no debt beyond the commitment. Which applies was not disclosed. "Total capital" may also include reserves set aside for engine exchanges rather than purchase prices alone, which would change the

per-aircraft figure. Limits of the example: two inputs are rounded releases figures eighteen months apart; the debt actually drawn at any date was not disclosed. [FTAI releases, 26 Feb 2025, 27 Oct 2025, 17 Aug 2026; D7 §2.3, U11, U12; reconciliation R-056]

The per-aircraft figure in that example is one of five the research produced, and they disagree by a factor of two. §62.3 sets them side by side.

62. Capital raised: the 2025 SPV, the 2026 SPV, OneIM

The capital behind the two vehicles was disclosed in a series of releases over twenty months, each stating a figure on its own basis. This section lays them out in date order, then turns to the second vehicle and to two tensions in the totals.

62.1 THE 2025 VEHICLE, RELEASE BY RELEASE

EXHIBIT 8.7 — CAPITAL OF THE 2025 PARTNERSHIP AS EACH RELEASE STATED IT

Date	Equity	Debt	Total capital as stated	Deployment language	Source
30 Dec 2024	not stated	not stated	"\$3.0+ billion of capital annually" (the market opportunity); 46 seed aircraft at ~\$549M	Launch	FTAI release, 30 Dec 2024
26 Feb 2025	not stated	\$2.5B asset-level commitment (ATLAS SP \$2.0B direct; Deutsche Bank co-lead)	"\$4.0+ billion" of deployment	Engines "exclusively via engine and module exchanges"	FTAI release, 26 Feb 2025
25 Mar 2025	OneIM partnership, size not stated	\$2.5B	"more than \$4 billion"	Partnership "on the inaugural vehicle"	FTAI release, 25 Mar 2025

Date	Equity	Debt	Total capital as stated	Deployment language	Source
27 Oct 2025	\$2.0B hard cap (target \$1.5B)	"current and future debt financing"	"purchasing power of over \$6 billion"	\$1.4B invested in 101 aircraft; \$2.1B under contract; 190 aircraft closed or under letter of intent; "full deployment expected by the end of the first half of 2026"	FTAI release, 27 Oct 2025
25 Feb 2026	\$2.0B	not restated	not restated	Deployment "largely completed"	FTAI release, 25 Feb 2026
22 May 2026	\$2.0B	\$612M ABS priced on 48 aircraft (Series A expected Asf/ A(sf); Series B BBB+sf)	not restated	The vehicle "currently owns 292 aircraft"	FTAI release, 22 May 2026, via Finviz and AviTrader
29 Jul 2026	\$2.0B	not restated	not restated	"completed deployment ... fully committed"; first quarterly distribution 30 June 2026	FTAI Q2 2026 release, 29 Jul 2026
17 Aug 2026	\$2.0B	not restated	"approximately \$6.0 billion of total capital across over 300 aircraft"	"harvest phase"	FTAI release, 17 Aug 2026

Source: FTAI Aviation Ltd. releases via GlobeNewswire on the dates shown. Compiled from D7 §2.3 and §2.7. The "harvest phase" sentence is quoted from the 17 Aug 2026 release; a search summary had attached it to the 29 Jul 2026 release as well, and the research asked that it be sourced to August (D7 T6).

The deployment column needs a word, because the releases use two different states as if they were one. In October 2025 the vehicle expected "full deployment" by mid-2026. In February 2026 deployment was "largely completed". In July 2026 it had "completed deployment" and was "fully committed". [FTAI releases, 27 Oct 2025, 25 Feb 2026, 29 Jul 2026] The sequence is consistent, but "fully committed" describes signed acquisitions, some of which may not yet have closed, while "fully deployed" would mean all capital drawn and all aircraft owned. The results releases do not say how many of the 300-plus aircraft had closed at each date. [D7 T8; registry C-21] One other release does give a count: the securitization release of 22 May 2026 says the vehicle "currently

owns 292 aircraft", against the "over 300 aircraft" of committed capital stated in August 2026. [FTAI release, 22 May 2026, via Finviz and AviTrader; FTAI release, 17 Aug 2026] The two figures are three months apart and describe different states, owned against committed, so they are not a contradiction; they are the gap between the two words. This Part uses each release's own word for its own date.

LETTER OF INTENT (LOI)

A signed agreement to transact that is non-binding, or binding only on conditions, ahead of a definitive purchase contract. At the hard-cap date the 2025 Partnership had "190 aircraft closed or under LOI", of which 101 had closed for \$1.4B and the rest were under contract for \$2.1B. [FTAI release, 27 Oct 2025]

62.2 ONEIM AND THE OTHER INVESTORS

ONEIM (ONE INVESTMENT MANAGEMENT)

An investment manager announced on 25 March 2025 as FTAI's partner "on the inaugural vehicle", in a release that described more than \$4 billion of deployment for the vehicle. OneIM's commitment size, and whether its role is that of an anchor limited partner or a co-general partner, are not disclosed. The ">\$4 billion" is the vehicle's deployment capacity, not OneIM's commitment. [FTAI release, 25 Mar 2025; reconciliation R-059]

The research's chain map and the company dossier's timeline both carried the March 2025 event as "OneIM partnership, \$4B", which reads as if OneIM committed \$4B. The release does not say that. It describes a partnership on the vehicle and a deployment figure for the vehicle as a whole, which the February release had already given as "\$4.0+ billion". [FTAI releases, 26 Feb 2025 and 25 Mar 2025; reconciliation R-059; E.1 rule 31] This Part therefore writes it as the box does: more than \$4 billion of deployment for the inaugural vehicle, with OneIM's commitment and role undisclosed.

The other limited partners are described only by category: "asset managers, insurance companies, public pensions, foundations, endowments and family offices". [FTAI release, 27 Oct 2025] None is named. The advisers are: Kirkland & Ellis, McGuireWoods and Gibson Dunn as legal counsel, and Lincoln International as financial adviser on both the debt and the equity raise. [FTAI releases, 26 Feb 2025 and 27 Oct 2025]

ANCHOR INVESTOR

An early, large limited partner whose commitment seeds a fund-raise and signals to later investors that the terms have been negotiated by someone with scale. The February 2026 results release said fundraising for the second vehicle had launched "with anchor investor commitments"; the anchors' identities and amounts were not disclosed. [FTAI release, 25 Feb 2026; D7 U4]

62.3 TWO TENSIONS IN THE TOTALS

The first tension is a per-aircraft figure that depends on which two disclosed numbers are divided.

EXHIBIT 8.8 — CAPITAL PER AIRCRAFT IN THE 2025 PARTNERSHIP, FIVE BASES

Numerator	Denominator	Per aircraft	What it measures	Source
\$549M estimated net purchase price	46 seed aircraft	\$11.9M	Seed portfolio, FTAI's own book	FTAI release, 30 Dec 2024
\$58.9M proceeds	4 aircraft sold Q1 2025	\$14.7M	First four sales	FTAI 10-Q Q1 2026
\$1.4B invested	101 aircraft closed	\$13.9M	First 101 acquisitions	FTAI release, 27 Oct 2025
\$2.1B under contract	89 aircraft (190 – 101)	\$23.6M	Pipeline at the hard-cap date	FTAI release, 27 Oct 2025; arithmetic
\$3.5B (1.4 + 2.1)	190 aircraft	\$18.4M	Blend of the two above (derived in this Part)	Arithmetic
~\$6.0B total capital	300+ aircraft	~\$20M	Whole vehicle at completion	FTAI release, 17 Aug 2026

Source: FTAI Aviation Ltd. releases of 30 Dec 2024, 27 Oct 2025 and 17 Aug 2026; Form 10-Q for the quarter ended 31 Mar 2026. Derived in D7 §2.3 and T5; reconciliation R-056.

The range runs from \$11.9M to \$23.6M, and the bases are unstated. "Total capital" may include maintenance reserves or capital set aside for exchanges; later purchases may have skewed toward younger aircraft or toward A320ceos; and a purchase price is not the same quantity as capital committed. [D7 T5; reconciliation R-056] Part II §15 notes that no half-life market value for a 12 to 15 year old 737-800 or A320ceo was retrieved, so the research cannot say where in this range a representative aircraft sits. [D6 U9; reconciliation K-02] The exhibit carries all five with their bases, as the reconciliation instructs.

The second tension is in §62.5, because it concerns both vehicles.

62.4 THE 2026 VEHICLE

2026 SPV (SCI II; THE SECOND INVESTMENT VEHICLE)

The second SCI vehicle. Its \$2.0B warehouse facility from 13 lenders, with a \$1.0B accordion to \$3.0B, closed on 14 August 2026; it buys on-lease mid-life 737NG and A320ceo aircraft from August 2026, and its first disclosed deal is 17 737-700s bought on a sale-leaseback with WestJet (\$67). Its equity target, hard cap, anchor investors, FTAI's stake and the timing of its final close are undisclosed. [FTAI releases, 17 Aug 2026 and 28 Sep 2026; D7 U4]

ACCORDION

A pre-agreed option in a credit agreement to enlarge the facility, exercisable if lenders agree to fund the increase; the borrower is not entitled to the extra amount until they do. The 2026 warehouse is \$2.0B of committed capacity with a \$1.0B accordion, so up to \$3.0B. [FTAI release, 17 Aug 2026]

EXHIBIT 8.9 — THE 2026 SPV WAREHOUSE FACILITY AND ITS LENDERS

Item	Detail	Source
Facility	\$2.0B warehouse, closed 14 Aug 2026 (announced 17 Aug 2026)	FTAI release, 17 Aug 2026
Accordion	\$1.0B, to \$3.0B	FTAI release, 17 Aug 2026
Co-structuring agents	ATLAS SP Partners; Deutsche Bank	FTAI release, 17 Aug 2026
Lenders (13)	ATLAS SP Partners, Deutsche Bank, Apple Bank, BNP Paribas, Citibank, Citizens Bank, Goldman Sachs, MUFG Bank, PNC Bank, Royal Bank of Canada, Standard Chartered, Truist Bank, U.S. Bank	FTAI release, 17 Aug 2026
Mandate	On-lease, mid-life 737NG and A320ceo aircraft; engines maintained exclusively through MRE exchanges	FTAI release, 17 Aug 2026
Equity	Fundraising launched by 25 Feb 2026 "with anchor investor commitments"; target, hard cap and close date not disclosed	FTAI release, 25 Feb 2026; D7 U4
First disclosed acquisition	17 Boeing 737-700s on lease to WestJet, sale-leaseback, 28 Sep 2026	FTAI release, 28 Sep 2026
Rate, advance rate, tenor, covenants	Not disclosed	D7 U11

Source: FTAI Aviation Ltd., "FTAI Closes \$2.0 Billion Warehouse Financing Facility for Strategic Capital's Second Investment Vehicle", 17 Aug 2026; FTAI release, 25 Feb 2026; FTAI release, 28 Sep 2026. Compiled from D7 §2.2, §3.1 and §4.1.

The order of events differs from the first vehicle's. The 2025 Partnership arranged its debt commitment in February 2025 and closed its equity in October 2025. The 2026 vehicle's equity raise was launched by February 2026, its warehouse closed in August 2026, and its equity size is still undisclosed at the research date. [FTAI releases, 26 Feb 2025, 27 Oct 2025, 25 Feb 2026, 17 Aug 2026] Part II §17 describes the path a warehouse normally leads to: accumulate aircraft on the line, season the leases, then refinance into a securitization and keep the equity and the servicing. The first vehicle took that step on 22 May 2026 with the \$612M securitization of 48 aircraft described in §61.2; the research dossier had located only the release's title and recorded the issuer as unknown, and the orchestrator's later search established that the deal is the 2025 Partnership's. [FTAI release, 22 May 2026, via Finviz and AviTrader, 26 May 2026; orchestrator notes; D7 U12] The second vehicle, on the pattern of the first, would be expected to follow the same route once its warehouse holds enough seasoned aircraft; nothing in the record says when.

62.5 "\$5.5 BILLION OF WAREHOUSE FINANCING"

The 17 August 2026 release says the Strategic Capital vehicles "have now raised \$5.5 billion of warehouse financing in less than two years". [FTAI release, 17 Aug 2026] The two debt commitments disclosed in the record are the \$2.5B of February 2025 and the \$2.0B of August 2026, which sum to \$4.5B. The \$5.5B figure is reached only if the 2026 facility is counted at its \$3.0B accordion ceiling (\$2.5B + \$3.0B), or if the 2025 facility was upsized by \$1.0B without a separate announcement. The release does not say how its figure is built. [FTAI releases, 26 Feb 2025 and 17 Aug 2026; D7 T2, U11; reconciliation R-055] Both figures are carried here with their derivations, as the reconciliation instructs: \$5.5B as the company's statement, \$4.5B as the sum of disclosed commitments, and the accordion as the only disclosed bridge between them. The \$612M securitization of May 2026 is term debt, not warehouse financing, and so does not bridge the gap on the release's own wording. Note also that the release calls both facilities "warehouse financing" while the February 2025 release described the first as an "asset-level debt financing" commitment; the research treats the first as asset-level debt and the second as a warehouse, following each release's own wording.

63. Money flows between FTAI and the vehicles: gains, fees, MRE sales, equity income, eliminations

Five streams run between FTAI and the 2025 Partnership, and each lands on a different line of FTAI's accounts. This section takes them one at a time, gives the disclosed figures, says where each lands, and marks what is not disclosed. Every one of them is a transaction between FTAI and an entity it manages and part-owns; §64 treats the disclosure and accounting consequences of that.

EXHIBIT 8.10 — DISCLOSED STREAMS BETWEEN FTAI AND THE 2025 PARTNERSHIP, BY PERIOD (\$ THOUSANDS UNLESS NOTED)

Stream	Q1 2025	Q2 2025	Q3 2025	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands	Source
Aircraft sold to the Partnership (count)	4	33	8	n/l	n/l	n/l	n/l	Leasing equipment derecognised	10-Qs 2025
Gain on sale to the Partnership	10,900	34,600	4,600	46,400	n/l	n/l	n/l	Gain line outside revenue (under Accounting Standards Codification (ASC) topic 610-20, §64.2); in Aviation Leasing Adjusted EBITDA	10-Qs 2025; 10-K FY2025
Proceeds from sales to the Partnership	58,900	n/l	n/l	n/l	117,300	n/l	n/l	Cash	10-Q Q1 2026
Servicing fees from the Partnership	n/l	n/l	n/l	10,200	5,861 (derived)	6,988	12,849	Line not confirmed (presumed other revenue, Aviation Leasing)	10-K FY2025; 10-Q Q2 2026
MRE Contract revenue to and from the Partnership	100,600 (derived)	69,600	n/l	335,788	221,230	182,800	404,000	MRE Contract revenue line; Aerospace Products segment (Part VII §48)	10-K FY2025; 10-Qs Q1 and Q2 2026

Stream	Q1 2025	Q2 2025	Q3 2025	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands	Source
Profit elimination on MRE sales	n/l	n/l	n/l	n/l	10,000	6,597	16,597	Through equity-method earnings	10-Qs Q1 and Q2 2026
Equity-method income from the Partnership	n/l	n/l	n/l	n/l	n/l	n/l	n/l	Equity in earnings of unconsolidated entities	D7 U7
Carrying value of the interest, period end	—	—	—	281,740	279,417	365,485	365,485	Equity-method investment, balance sheet	10-K FY2025; 10-Qs 2026
Stake	—	—	20% (Q3 2025 10-Q)	19%	19%	19%	19%		10-Q Q3 2025; 10-K FY2025; 10-Qs 2026

Source: FTAI Aviation Ltd., Forms 10-Q for the quarters ended 31 Mar 2025, 30 Jun 2025, 30 Sep 2025, 31 Mar 2026 and 30 Jun 2026; Form 10-K for FY2025; the Q2 2026 10-Q's MRE figures via the orchestrator notes. n/l = not located in anything the research reached. Q1 2025 MRE revenue is 1H 2025 \$170.2M less Q2 2025 \$69.6M (derived); Q1 2026 servicing fees are 1H less Q2 (derived). Compiled from D7 §4.2. Reconciliation R-047, R-048, R-054.

63.1 GAINS ON AIRCRAFT SALES INTO THE VEHICLE

The first stream is the one §60 described. FTAI sells aircraft from its leasing book into the Partnership and records the difference between price and net book value as a gain. The disclosed figures are the quarterly gains of 2025 and the full-year \$46.4M, with the \$3.7M tension between them (§60.2). [FTAI 10-Qs 2025; FTAI 10-K FY2025; reconciliation R-054] Proceeds were \$58.9M in the first quarter of 2025 and \$117.3M in the first quarter of 2026; the counts and gains for the fourth quarter of 2025 and for 2026 were not located. [FTAI 10-Q Q1 2026; D7 U10]

How the price of an aircraft sold from FTAI to a vehicle it manages is set is the open question in this stream. An appraisal policy, a third-party valuation requirement, or an approval right for the other limited partners or independent directors would each be a normal protection in a related-party sale. None was located in anything the research reached. [D7 U5]

63.2 SERVICING FEES

SERVICING FEE

FTAI's compensation as Servicer of the 2025 Partnership: \$10.2M for FY2025, \$6.988M for the second quarter of 2026 and \$12.849M for the first half of 2026 (so \$5.861M for the first quarter, derived). The 10-K describes it only as "customary, market-based compensation"; its rate, its base (committed equity, invested cost, or rents) and the income-statement line it sits on are not disclosed.

[FTAI 10-K FY2025; FTAI 10-Q Q2 2026; D7 U1, U8]

The fee grew with the vehicle. For 2025, a year in which the Partnership was still buying, it was \$10.2M. For the second quarter of 2026, the first quarter after the vehicle was fully committed, it was \$6.99M, which annualises to \$27.95M; the first half of 2026 annualises to \$25.70M. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; derived in the research] Against the two disclosed capital figures, the second-quarter run-rate is 0.47% of about \$6.0B of total capital, or 1.40% of \$2.0B of equity; on the first-half basis, 0.43% and 1.28%. These ratios say nothing about the fee's actual base, which is undisclosed; they only locate the disclosed dollars against the disclosed capital. §65 sets them beside the fee scales other platforms disclose, and worked example 8.6 in §66 works them through.

Where the fee lands was not confirmed. The research presumed "other revenue" within Aviation Leasing. [D7 U8] One arithmetic observation can be recorded without resolving it: the FY2025 consolidated statement's five named revenue lines sum to \$2,496,898K against total revenues of \$2,507,409K, leaving a residual of \$10,511K that is presumably an "other revenue" line, and FTAI's FY2025 servicing fees were \$10,200K. [FTAI 10-K FY2025, via D10 §2.4; derived in the research] The two figures are close. The filing as retrieved does not say they are the same line, and this Part does not say so either.

63.3 ENGINE AND MODULE SALES UNDER THE MRE CONTRACT

The third stream is the largest in dollars and the one that joins this Part to Part VII. Part VII §48 owns the MRE Contract revenue line; this Part carries the related-party side of it.

The 10-Q for the second quarter of 2026 reports \$182.8M of "MRE Contract revenue for the sale and purchase of engines to and from the 2025 Partnership" for the quarter, against \$69.6M in the same quarter of 2025, and \$404.0M for the first half of 2026 against \$170.2M. [FTAI 10-Q Q2 2026, via the orchestrator notes] The FY2025 10-K's MRE Contract revenue line was \$335,788K and the first quarter of 2026 was \$221,230K. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026] The quarterly figures tie: \$221.2M plus \$182.8M is \$404.0M.

The line is defined in two ways in two filings, and the reconciliation asks both Parts to quote both. The Q3 2025 10-Q describes "stand-ready obligations to provide replacement CFM56-5B

and CFM56-7B engines to customers as they become unserviceable". The Q2 2026 10-Q defines MRE Contract revenue as "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership". [FTAI 10-Q Q3 2025; FTAI 10-Q Q2 2026; reconciliation R-044; E.1 rule 9] The first is a description of an obligation, the second a description of a sale. Both are quoted; the primer does not pick one. Revenue is booked under the revenue-recognition standard for contracts with customers (Part VII §47), with the fair value of any engine or module received back counted as part of the price. [D7 §2.5; D5 §2.3]

The size of this stream relative to the segment is the figure Part VII's margins depend on. MRE Contract revenue was 17.3% of Aerospace Products segment revenue in FY2025 (\$335.8M of \$1,936.2M), 29.7% in the first quarter of 2026 (\$221.2M of \$743.8M), 20.9% in the second (\$182.8M of \$875.0M) and 25.0% for the first half (\$404.0M of \$1,618.8M). [FTAI 10-K FY2025; FTAI Q1 2026 release; FTAI Q2 2026 10-Q; derived in the research] The explicit segment-note sentence mapping the line to the segment was not retrieved; three documents agree arithmetically that segment revenue equals the products line plus the MRE line, which Part VII §48 and Part X §79 record. [reconciliation B-1, R-043] The count of engines and modules behind these dollars, and the expected annual volume for a 300-aircraft fleet, were not disclosed. [D7 U6]

63.4 PROFIT ELIMINATIONS: TWO READINGS OF ONE SENTENCE

PROFIT ELIMINATION (INTRA-ENTITY PROFIT ELIMINATION)

When a company sells to an entity it accounts for under the equity method, the profit on the sale is not all earned yet from the group's point of view, because the company still owns a share of the buyer and so a share of the asset. The profit is removed from the equity-method earnings and recognised later as the investee uses or sells the asset. FTAI: "the profit from the MRE Contract revenue is eliminated through equity method earnings and will be recognized over time as the 2025 Partnership generates income from leasing and sales activities". Eliminations were \$10.0M in the first quarter of 2026 and \$6.6M in the second, \$16.6M for the half. Whether the eliminated amount is 19% of the profit or all of it is not stated. [FTAI 10-Qs Q1 and Q2 2026; reconciliation R-047]

The method matters because the same \$16.6M implies very different margins on the \$404.0M of related-party sales, and the filing sentence supports either reading. [reconciliation R-047; E.1 rule 12]

WORKED EXAMPLE 8.4 — WHAT THE ELIMINATIONS IMPLY UNDER EACH METHOD

Inputs: eliminations \$10.0M (Q1 2026), \$6.597M (Q2 2026), \$16.597M (1H 2026) (sourced: 10-Qs); MRE Contract revenue to and from the Partnership \$221.2M (Q1), \$182.8M (Q2), \$404.0M (1H) (sourced: 10-Qs via the orchestrator notes); stake 19% (sourced: 10-K and 2026 10-Qs) or 20% (sourced: Q3 2025 10-Q).

reading A, proportional to the stake: gross profit = $16.597 \div 0.19 = \$87.4\text{M}$ for 1H 2026 (about \$174M annualised); at 20%, $16.597 \div 0.20 = \$83.0\text{M}$

implied margin on MRE sales to the Partnership: $87.4 \div 404.0 = 21.6\%$
(Q1: $52.6 \div 221.2 = 23.8\%$; Q2: $34.7 \div 182.8 = 19.0\%$)

reading B, all profit deferred: gross profit = \$16.6M for 1H 2026 (about \$33M annualised)

implied margin: $16.6 \div 404.0 = 4.1\%$ (Q1: $10.0 \div 221.2 = 4.5\%$; Q2: $6.6 \div 182.8 = 3.6\%$)

Reading A is the usual treatment of profit on sales to an equity-method investee: the investor eliminates its own share and leaves the rest, because the other owners have paid for it at arm's length. Reading B would mean FTAI defers all of it until the Partnership earns it back. The 10-Q sentence, "recognized over time as the 2025 Partnership generates income", is consistent with either. Under reading A, the margin on related-party sales (about 22%) would sit below the segment's 28.5% Adjusted EBITDA margin for the second quarter (Part VII §53); under reading B it would be far below it. Neither is a reported figure. Limits of the example: the margin lines compare a profit measure of unknown definition with a revenue figure; the second-quarter MRE revenue includes "purchase" as well as "sale" transactions; and the deferred profit will be recognised in later periods on a schedule the filing does not give. [FTAI 10-Qs Q1 and Q2 2026; D7 §2.5, T7; reconciliation R-047, R-048]

The reported effect, whichever reading is right, is the one the research recorded as a constraint: profit on MRE sales to the Partnership is deferred, so the Aerospace Products margin reported on this volume in the period of sale is lower than on the same volume sold to a third party. [D7 §6]

63.5 EQUITY-METHOD INCOME AND THE CARRYING VALUE

The fourth stream is FTAI's share of what the Partnership itself earns. It is the one stream whose amount was not located for any period. [D7 U7; reconciliation K-08] What the filings do give is the carrying value of the interest at three dates and a sentence about how the share is computed.

EXHIBIT 8.11 — FTAI'S INTEREST IN THE 2025 PARTNERSHIP: STAKE AND CARRYING VALUE

Date	Stake	Carrying value (\$ thousands)	Context	Source
9M 2025	20%	\$151.6M invested in the nine months	Deployment under way	FTAI 10-Q Q3 2025
31 Dec 2025	19%	281,740	Hard cap reached 27 Oct 2025	FTAI 10-K FY2025
31 Mar 2026	19%	279,417	Deployment "largely completed"	FTAI 10-Q Q1 2026

Date	Stake	Carrying value (\$ thousands)	Context	Source
30 Jun 2026	19%	365,485	Deployment completed; first distribution 30 Jun 2026	FTAI 10-Q Q2 2026
Derived	19% × \$2.0B	≈380,000	Full commitment if drawn in proportion	Arithmetic

Source: FTAI Aviation Ltd., Form 10-Q for the quarter ended 30 Sep 2025 (note R13, via the orchestrator notes); Form 10-K for FY2025; Forms 10-Q for the quarters ended 31 Mar 2026 and 30 Jun 2026. Reconciliation R-048.

The stake is stated as 20% in the Q3 2025 10-Q and 19% in the FY2025 10-K and both 2026 10-Qs. [reconciliation R-048] The orchestrator's note suggests dilution at the final close as a likely cause, and marks that as a guess; this Part carries both percentages wherever the stake is used and shows the elimination arithmetic at 19% with the 20% alternative. [orchestrator notes; E.1 rule 20]

The carrying value rose by \$86.1M between March and June 2026, from \$279.4M to \$365.5M, in the quarter in which deployment completed. [FTAI 10-Qs Q1 and Q2 2026; derived in the research] Under the equity method a carrying value rises with capital contributed and with the investor's share of earnings, and falls with distributions received and with eliminations. The filings as retrieved do not decompose the movement, so the share of it that was new capital called and the share that was earnings cannot be separated. At 19% of \$2.0B the full commitment would be about \$380M, and \$365.5M is close to it.

FTAI says its share of the Partnership's earnings "is based on the contractual profit-sharing arrangement" rather than being described as a flat 19%. [FTAI 10-Q Q2 2026] That wording is what one would expect if the partnership agreement allocates profit through an ordered set of tiers rather than pro rata.

PAYMENT WATERFALL AND DISTRIBUTION WATERFALL ARE DIFFERENT THINGS

Part II §17 uses "waterfall" for the order in which a securitization pays out collections on each payment date: fees and expenses, senior interest, mezzanine interest, scheduled principal, then any surplus to the equity. A partnership uses the same word for the order in which it allocates distributions among its partners: typically return of capital, then a preferred return to the limited partners, then a disproportionate share to the sponsor. This Part calls the first a payment waterfall and the second a distribution waterfall, and uses only the second from here on. Whether the 2025 Partnership's agreement contains a distribution waterfall, and with what tiers, is not disclosed.

[registry C-17; D7 §2.5]

HURDLE (PREFERRED RETURN)

The return limited partners receive on their capital before the sponsor shares in profit beyond its own pro-rata interest; an 8% annual preferred return is a common private-fund convention, used here only as an illustration in §66. FTAI's filings do not disclose whether the 2025 Partnership has one. [D7 §2.5, U2]

PROMOTE (CARRIED INTEREST; CARRY)

The sponsor's share of a fund's profit above its pro-rata interest, paid once the hurdle is cleared through the distribution waterfall. Carlyle describes its carried interest as "a special residual allocation of profit on third-party capital", generally calculated on realised gains for closed-end funds. Whether SCI has a promote, and its terms, are unknown; FTAI's only words are "contractual profit-sharing arrangement". [Carlyle 10-K FY2025; FTAI 10-Q Q2 2026; D7 U2]

MANAGEMENT FEE, INCENTIVE FEE AND SALES FEE

Three fee types in third-party-capital vehicles: a management or servicing fee on capital committed, assets managed or rents collected; an incentive fee for performance; and a sales fee on disposal proceeds. Peer scales from filings: Carlyle's aviation securitizations pay servicing of about 2% of rents, incentive fees of up to 5% of rents in aggregate and 3% of sales proceeds; Willis Lease's WEST trusts pay 11.5% plus 2.0% of net rents and 3% of sale proceeds. SCI's rates and bases are undisclosed. [Carlyle 10-K FY2025; Willis Lease 10-Ks FY2019 and FY2020; D7 §4.4]

The first distribution to the Partnership's investors was paid on 30 June 2026. Its amount, whether it came from rent or from asset sales, and FTAI's share of it were not located. [FTAI Q2 2026 release, 29 Jul 2026; D7 U13; reconciliation K-08]

63.6 THE FIFTH STREAM: WHAT THE VEHICLE EARNS IN HARVEST

In harvest the Partnership sells aircraft and realises gains or losses of its own. Those reach FTAI only indirectly, through its 19% share and through any promote, and possibly through a sales fee if the servicing agreement has one. A sales fee of 3% of proceeds is disclosed by both Carlyle and Willis for their vehicles; nothing is disclosed for SCI. [Carlyle 10-K FY2025; Willis Lease 10-K FY2020; D7 §2.5] The same harvest also returns engines to the market, and the vehicle's engines are contracted to FTAI for exchanges for as long as it owns them (§61.3).

64. Accounting and segment placement

Three accounting choices shape how everything in §63 appears in FTAI's statements: the Partnership is not consolidated; the aircraft sales into it are booked as disposals outside ordinary revenue; and the related-party streams are disclosed in the 10-Q notes rather than on the face of

the statements. This section states each as the filings do and records the questions the research could not close.

64.1 THE EQUITY METHOD, AND WHY 19% QUALIFIES

EQUITY METHOD (EQUITY-METHOD INVESTMENT; EQUITY PICK-UP; EQUITY IN EARNINGS OF UNCONSOLIDATED ENTITIES)

An accounting method for an investment over which the investor has significant influence but not control. The investment is carried on the balance sheet at cost, increased by the investor's share of the investee's earnings and decreased by distributions received and by eliminated profit; the share of earnings appears on one income-statement line, and none of the investee's revenue, assets or debt is consolidated. A 20% to 50% holding is the conventional range, but the test is influence, not a percentage floor: FTAI's 19% limited-partner interest, held alongside its role as Servicer, is accounted for this way. FTAI's other equity-method investments at 31 December 2025 were the Advanced Engine Repair JV (25%, \$22.4M), QuickTurn Europe (50%, \$10.0M) and Prime Engine Accessories (50%, carrying value not returned), treated in Part VII §50. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; registry C-9; orchestrator notes]

Two of the research dossiers defined the method by the 20% to 50% range and a third observed that FTAI's 19% interest uses it. The registry resolves the conflict in the box above: significant influence is the test, and a Servicer with a 19% stake has it. [registry C-9] The consequence for a reader is the one the box states. The Partnership's roughly \$6.0B of total capital, its 300-plus aircraft, its asset-level debt and its rent appear nowhere in FTAI's consolidated revenue, assets or liabilities. What appears is a single investment line of \$365.5M and a single earnings line whose amount was not located.

One further equity-method investment is referred to without being named. The segment description says Aviation Leasing leases and sells assets "directly and through an equity-method investment". [FTAI 10-K FY2025]

EQUITY-METHOD INVESTMENT IN AVIATION LEASING

The unnamed investment through which the Aviation Leasing segment "leases and sells" assets, in the FY2025 10-K's segment description. Its counterparty, FTAI's ownership percentage, the assets inside it and the income it has contributed by year were not retrieved. Whether it is the 19% interest in the 2025 Partnership, which would be the obvious candidate since the Partnership leases aircraft FTAI formerly owned, is not established in anything the research reached and is carried as an unknown. [FTAI 10-K FY2025; D6 U11; reconciliation K-07]

If the Leasing segment's equity-method investment is the Partnership interest, then the segment's Adjusted EBITDA includes FTAI's share of the Partnership's results, and the Leasing guidance of

§60.4 includes that share. If it is not, the Partnership's contribution sits in Corporate and Other or elsewhere. The FY2025 10-K's investments note would settle it; the research did not retrieve the sentence that does. [reconciliation K-07]

64.2 "NON-RECURRING" SALES UNDER A RECURRING PROGRAMME

ASC 610-20

Topic 610-20 of the US Accounting Standards Codification (ASC), the standard for gains and losses on the derecognition of nonfinancial assets that are not part of a company's ordinary revenue-generating activities. FTAI books aircraft sales to the 2025 Partnership under it, describing them as "non-recurring in nature and not considered part of the Company's ordinary activities". The gain (\$46.4M for FY2025) therefore appears on a gain line rather than inside asset sales revenue, while the segment's Adjusted EBITDA includes it. [FTAI 10-K FY2025; D7 §2.5]

The reconciliation recorded a tension between that accounting description and the company's description of its strategy, and asks that both be set out. On one side, the launch release frames the market as supporting "\$3.0+ billion of capital annually", the Q2 2026 10-Q says FTAI expects its primary investment activities going forward to run through SCI, and a second vehicle began buying in August 2026. On the other, the 10-K describes the aircraft sales into the first vehicle as "non-recurring in nature and not considered part of the Company's ordinary activities". [FTAI release, 30 Dec 2024; FTAI 10-Q Q2 2026; FTAI 10-K FY2025; reconciliation R-057] Both statements are made. The accounting description determines which line the gain sits on; the strategy description determines how often a reader should expect such gains. Part X §80 treats what the distinction does to the composition of Adjusted EBITDA.

64.3 CONSOLIDATION, AND THE QUESTION NOT ANSWERED

VARIABLE INTEREST ENTITY (VIE) AND PRIMARY BENEFICIARY

Under US GAAP, an entity whose equity holders lack the power to direct its activities, or whose equity is too thin to fund it without support, is a variable interest entity, and the party with the power to direct its most significant activities and exposure to its economics (the primary beneficiary) must consolidate it even without a majority of the votes. A sponsor that manages a partnership's assets, supplies its largest service contract and holds 19% of its equity would be expected to have performed this analysis. FTAI does not consolidate the 2025 Partnership; the analysis and its reasoning were not located. [D7 §2.6, U18]

The result is known: the Partnership is unconsolidated and FTAI records an equity-method investment. The basis for it is not. The accounting policies note that would state whether the

Partnership was assessed as a variable interest entity and why FTAI is not its primary beneficiary was not reached. [D7 U18]

64.4 RELATED-PARTY DISCLOSURE, AND WHAT IS NOT DISCLOSED

RELATED-PARTY TRANSACTION

A transaction between a company and an entity it controls, manages or significantly influences, which accounting rules require to be disclosed because it is not necessarily on arm's-length terms. Every SCI stream in §63 is one: FTAI sells aircraft to, services, sells engines and modules to, and holds equity in the same partnership. The 10-Qs disclose the servicing fees received, the aircraft sales and proceeds, the engine and module sales and their profit eliminations, and the carrying value. The pricing policy for aircraft and for exchanges is not disclosed, and no published critique specific to SCI's related-party pricing was located. [FTAI 10-Qs Q1 and Q2 2026; D7 §2.6, U5, U6, U19]

The disclosure that exists is in the 10-Q notes and is the source of nearly every figure in Exhibit 8.10. The items it does not contain are the ones a reader would need to judge the terms: how aircraft are priced when they move from FTAI's book to the Partnership's, how exchanges under the MRE Contract are priced, the servicing fee's rate and base, and whether a promote exists. [D7 U1, U2, U5, U6] The agreements that would answer these were listed as exhibits 10.8 and 10.9 to the Q2 2026 10-Q in the filing index; they were not read. [D7 U1]

The research also looked for outside commentary on these terms and found none. The Muddy Waters short report of January 2025 concerned the reporting of Aerospace Products, treated in Part VII §54 and Part X §85; a broker's comment reported by The Fly at the time said the report "does not appear to consider" SCI. [TipRanks/The Fly, Jan 2025; D7 T9, U19] Sell-side notes, later short-seller updates and the financial press were not reached within the research budget, so the absence of a located critique is a fact about the search, not about the record.

One last placement question is shared with Part VII. The MRE Contract revenue line sits, by arithmetic, inside the Aerospace Products segment: segment revenue equals the products line plus the MRE line in three documents that do not quote each other. The segment note's explicit mapping sentence was not retrieved, and the research treats it as open. [reconciliation B-1, R-043; E.1 rule 8] For this Part the consequence is that the related-party share of Aerospace Products revenue computed in §63.3 rests on that arithmetic rather than on a quoted sentence.

65. Comparables: other third-party-capital platforms

FTAI is not the first lessor to manage aircraft or engines for other people's money and keep a fee. Four listed companies disclose something about how they do it, and their filings are the only

sourced scale against which SCI's undisclosed terms can be read. This section sets out what each filing says and nothing more. Part XI §87 treats the same companies as peers across their whole businesses.

65.1 CARLYLE AVIATION PARTNERS

CARLYLE AVIATION PARTNERS (FEE COMPARABLE)

The aviation arm of The Carlyle Group, which runs closed-end funds that earn carried interest and also sponsors aircraft securitizations that it services. Carlyle's FY2025 10-K states that servicing fees on its aviation securitizations are "generally in the range of 2% of rents, incentive fees up to 5% of rents in the aggregate, and 3% of sales proceeds". It is the one sourced comparable that states a fee rate and base for an aviation vehicle managed for third parties. [Carlyle 10-K FY2025]

Carlyle's filing describes two kinds of vehicle. Its closed-end carry funds pay management fees that "vary by fund strategy" and are offset entirely by transaction and monitoring fees, and the general partner earns carried interest, "a special residual allocation of profit on third-party capital", generally calculated on realised gains for closed-end funds. Its securitization vehicles pay the servicing, incentive and sales fees quoted in the box. [Carlyle 10-K FY2025; D7 §3.2] SCI's first vehicle is a closed-end partnership, the first type; the fee language FTAI uses ("customary, market-based compensation" as Servicer) is the vocabulary of the second. Which scale applies to SCI is not disclosed, and §66 works both.

65.2 WILLIS LEASE AND THE WEST TRUSTS

Willis Lease Finance is the listed pure-play engine lessor and the closest filing analogue to FTAI's leasing segment (Part XI §87). Its securitization vehicles, the Willis Engine Structured Trusts, are described in Part II §17 as the sourced template for engine securitization. For this section the relevant disclosure is the fee: Willis, as servicer and administrative agent of the WEST trusts, receives 11.5% of aggregate net rents as servicer plus 2.0% as administrative agent, monthly, and 3.0% of net proceeds on engine sales. [Willis Lease 10-Ks FY2019 and FY2020] Two features make this a different animal from SCI. Willis consolidates the WEST trusts, so the fee is internal to its own statements rather than income from a third party; and Willis holds all of the equity, not a minority. [D7 §3.2, §5.1(d)] Willis also manages engines for third parties outside the trusts: 116 engines and related equipment at 31 December 2025, against 363 owned. [Willis Lease 10-K FY2025] The fee on those managed engines was not returned.

65.3 AIR LEASE: BLACKBIRD AND THUNDERBOLT

BLACKBIRD CAPITAL I AND II, AND THUNDERBOLT (AIR LEASE)

Air Lease Corporation's two routes to third-party capital. Blackbird Capital I and II are investment funds whose aircraft Air Lease manages and in which it holds non-controlling interests of 9.5% of the equity of each. Thunderbolt is a platform through which Air Lease sells portfolios of mid-life aircraft to investors while it "continue[s] managing these aircraft for a fee", holding about 5.0% non-controlling interests in two Thunderbolt entities. The fee rates were not returned by the research. [Air Lease 10-K FY2024; D7 §3.2]

Air Lease is a new-aircraft lessor with a fleet averaging 4.9 years, so its owned book sits at the opposite end of the age spectrum from FTAI's (Part XI §87). [Air Lease 10-K FY2025] Thunderbolt is nonetheless the closest structural match to SCI in one respect: it moves mid-life aircraft off the sponsor's balance sheet into investor-owned entities that the sponsor keeps managing, with a small retained stake. The stakes differ, 5% to 9.5% at Air Lease against 19% at FTAI. [Air Lease 10-K FY2024; FTAI 10-K FY2025] Air Lease's managed fleet was 45 aircraft at 31 December 2025, and the company was taken private on 8 April 2026, becoming Sumisho Air Lease Corporation, so its FY2025 10-K is likely its last as a public registrant. [Air Lease 10-K FY2025]

65.4 AERCAP, AND THE PLATFORMS NOT REACHED

AerCap, the largest aircraft lessor, offers aircraft asset-management services described in its filings as remarketing, collecting rent and maintenance payments, monitoring maintenance and reviewing lessees; no fee percentage was located. [AerCap 20-F FY2025, search summary; D7 §3.2] Its headline portfolio of 3,500 aircraft, engines and helicopters counts owned, managed and on-order assets together, and its engine business runs partly through Shannon Engine Support, a 50% joint venture with Safran, which Part XI §87 treats. [AerCap 20-F FY2025; AerCap 6-K Q3 2025] Castlelake and Apollo's Merx Aviation, both managers of third-party aviation capital, were not reached within the research budget. [D7 U16]

65.5 READING ACROSS, DESCRIPTIVELY**EXHIBIT 8.12 — FEE AND STAKE TERMS OF THIRD-PARTY-CAPITAL PLATFORMS, AS THEIR FILINGS STATE THEM**

Platform	Structure	Servicing or management fee	Incentive	Sales fee	Sponsor stake	Consolidated by sponsor?	Source
Carlyle Aviation Partners, securitizations	Rated ABS vehicles serviced by Carlyle	~2% of rents	up to 5% of rents in aggregate	3% of sales proceeds	not stated	not stated	Carlyle 10-K FY2025
Carlyle Aviation Partners, carry funds	Closed-end funds	varies by strategy; offset 100% by transaction and monitoring fees	carried interest on realised gains	—	GP interest	no	Carlyle 10-K FY2025
Willis Lease, WEST trusts	Rated engine ABS	11.5% servicer + 2.0% administrative agent, of net rents	—	3.0% of net sale proceeds	100% of equity	yes	Willis Lease 10-Ks FY2019, FY2020
Air Lease, Blackbird I and II	Investment funds	rate not returned	not returned	not returned	9.5% of each	no	Air Lease 10-K FY2024
Air Lease, Thunderbolt	Mid-life portfolio sales, sponsor keeps managing	rate not returned	not returned	not returned	~5.0% in two entities	no	Air Lease 10-K FY2024
AerCap	Asset-management services	not located	not located	not located	—	—	AerCap 20-F FY2025

Platform	Structure	Servicing or management fee	Incentive	Sales fee	Sponsor stake	Consolidated by sponsor?	Source
FTAI, 2025 Partnership	Closed-end partnership over SPVs	"customary, market-based"; \$6.99M in Q2 2026	"contractual profit-sharing arrangement"	not disclosed	19% (20% in Q3 2025 10-Q)	no	FTAI 10-K FY2025; 10-Q Q2 2026

Source: The Carlyle Group Inc., Form 10-K for FY2025; Willis Lease Finance Corporation, Forms 10-K for FY2019 and FY2020; Air Lease Corporation, Form 10-K for FY2024; AerCap Holdings N.V., Form 20-F for FY2025 (search summary only); FTAI Aviation Ltd., Form 10-K for FY2025 and Form 10-Q for the quarter ended 30 Jun 2026. Compiled from D7 §4.4.

EXHIBIT 8.13 — SCALE OF THIRD-PARTY AND OWNED ASSETS AT THE COMPARABLES, LATEST FILED

Company	Owned	Managed for third parties	Date	Source
Willis Lease	363 engines, 20 aircraft; \$2,801.7M of equipment held for lease; utilisation 84.9% (book value basis)	116 engines and related equipment	31 Dec 2025	Willis Lease 10-K FY2025
AerCap	470 engines (9% of fleet); current-technology passenger fleet averaging 15.2 years	Owned, managed and on-order assets counted together: 3,500; >1,200 engines including those owned and managed by Shannon Engine Support (SES) (30 Sep 2025)	31 Dec 2025	AerCap 20-F FY2025; 6-K Q3 2025
Air Lease	490 aircraft averaging 4.9 years; \$29.1B net book value	45 aircraft	31 Dec 2025	Air Lease 10-K FY2025
FTAI	290 assets (47 aircraft, 243 engines); leasing equipment \$2,057.6M at cost, \$1,545.8M net; utilization ~77% (equity-value basis)	2025 Partnership: 300+ aircraft, ~\$6.0B total capital; 2026 SPV: beginning with 17 aircraft	31 Dec 2025; Aug–Sep 2026	FTAI 10-K FY2025; releases 17 Aug and 28 Sep 2026

Source: Willis Lease Finance Corporation, Form 10-K for FY2025; AerCap Holdings N.V., Form 20-F for FY2025 and Form 6-K for the quarter ended 30 Sep 2025; Air Lease Corporation, Form 10-K for FY2025; FTAI Aviation Ltd., Form 10-K for FY2025 and releases of 17 Aug 2026 and 28 Sep 2026. Compiled from D11 §4.1 and D7 §4.1. The utilisation figures are on different bases (registry C-3) and the AerCap engine counts on different bases (reconciliation R-033).

Three descriptive observations follow from the two tables, and none is a judgement. The sponsor's stake runs from 5% to 9.5% at Air Lease, through 19% at FTAI, to 100% at Willis. The fee base, where stated, is rents at both Carlyle and Willis, at 2% in one case and 13.5% in the other, with the difference explained at least in part by Willis's fee being internal to a consolidated vehicle; FTAI's base is undisclosed. And a 3% sales fee appears in both filings that state one, and nowhere in FTAI's. [D7 §3.2] By managed scale, FTAI's 300-plus aircraft in one vehicle after eighteen months compares with 116 engines managed at Willis and 45 aircraft at Air Lease, on the dates shown; the comparison is of counts of different asset types and is given only for orientation.

66. Worked examples: fee income on a \$6B vehicle; the maintenance stream from 300 aircraft

This section works the two streams that the filings leave unquantified, using the disclosed figures where they exist and labelled assumptions where they do not, and ends by reconciling the seed sale. Each example names its binding inputs and the alternative figure where the research holds two. [E.1 rule 36]

66.1 FEE INCOME ON A \$6B VEHICLE

WORKED EXAMPLE 8.5 — WHAT A \$6.0B VEHICLE PAYS ITS MANAGER, AT THE DISCLOSED RUN-RATE AND AT THREE PEER SCALES

(a) The disclosed run-rate. Inputs: servicing fees \$6.988M for Q2 2026 and \$12.849M for 1H 2026 (sourced: 10-Q Q2 2026); total capital about \$6.0B and equity \$2.0B (sourced: releases, 17 Aug 2026 and 27 Oct 2025).

Q2 annualised: $6.988 \times 4 = \$27.95\text{M a year}$; 1H annualised: $12.849 \times 2 = \$25.70\text{M}$

on total capital: $27.95 \div 6,000 = 0.47\%$; $25.70 \div 6,000 = 0.43\%$

on equity: $27.95 \div 2,000 = 1.40\%$; $25.70 \div 2,000 = 1.28\%$

Caveats: Q2 2026 is the first quarter after full commitment, so the run-rate may still be rising as closings complete; and the fee's actual base is undisclosed, so these ratios locate the dollars, not the rate.

(b) Carlyle's securitization scale. This needs an annual rent figure, which the Partnership does not disclose. The research assumed a lease rate factor of 0.9% a month (assumed: D7 §5.1); Part II §14 holds no stated figure for any lessor, and the derived alternatives are 0.70% (a green-time rent on a half-life value, D3), 0.78% on cost and 1.0% on net book value for FTAI's own 2025 book (worked

example 8.1), and about 1.0% for Willis Lease's on-lease book (D11). Monthly rent at each factor, times twelve, on \$6.0B:

at 0.70%: $0.0070 \times 12 \times 6,000 = \504M a year; at 0.78%: \$562M; at 0.9% (assumed): \$648M; at 1.0%: \$720M
 servicing at 2% of rents: 10.1 / 11.2 / 13.0 / 14.4 (\$M)
 incentive at the 5% cap: 25.2 / 28.1 / 32.4 / 36.0
 sales fee at 3% on an assumed \$600M of disposals a year in harvest: 18.0 (assumed volume)
 at the 0.9% base, all three at the top of the range: $13.0 + 32.4 + 18.0 = \$63.4\text{M}$; servicing alone \$13.0M

Two further cautions on the rent line. The lease rates behind it are contested in direction: IBA (an appraiser) reported 12-year-old 737-800 rents of about \$228,000 a month in July 2025, down 11% on the year, and warned in September 2026 of a "lease rate fall", while Air Lease's FY2025 10-K said strong demand "helped increase lease rates on new lease agreements and lease extensions during 2025", and Cirium Ascend via Bloomberg put 10-year-old 737-800 rents 44% above their January 2022 level. Different ages, dates and lessors; both directions are recorded (reconciliation R-031). And a \$6.0B figure is total capital, not necessarily the rent-earning cost of the aircraft.

(c) Private-fund scale. Inputs: a 1.5% management fee on committed equity (assumed: a common convention, not an SCI term) and carried interest of 20% of profit above an 8% hurdle (assumed).

management fee: $0.015 \times 2,000 = \$30.0\text{M}$ a year (at 1.0%: \$20.0M; at 2.0%: \$40.0M)
 carry, illustrative: if the vehicle returns \$2.0B of capital plus \$800M of profit over its life and the hurdle consumes \$500M of that, $20\% \times 300 = \$60\text{M}$, realised at exit rather than annually

(d) Willis WEST scale, for contrast only, since it is a fee inside a consolidated vehicle: $13.5\% \times \$648\text{M} = \87.5M a year. Nothing in the record places SCI near it.

Reading: the disclosed run-rate of \$26–28M sits between the servicing-only securitization figure (about \$10–14M depending on the rent factor) and the private-fund management fee at 1.5% (\$30M). That placement depends entirely on assumed rents and assumed fee conventions, and the filings disclose neither SCI's rate nor its base. Limits: every peer figure is from a different structure; the rent factor is derived or assumed, never stated; the disposals volume is assumed. [FTAI 10-Q Q2 2026; FTAI releases 27 Oct 2025 and 17 Aug 2026; Carlyle 10-K FY2025; Willis Lease 10-K FY2020; IBA via India Seatrade News, Jul 2025 and 6 Sep 2026; Air Lease 10-K FY2025; Cirium Ascend via Bloomberg/AJOT; D7 §5.1; reconciliation R-030, R-031]

66.2 THE MAINTENANCE STREAM FROM 300 AIRCRAFT

The second stream is the one the releases describe but never size: every engine in the vehicle's fleet, powered "exclusively via engine and module exchanges" with FTAI. The research built the

example from three inputs it did not have and asked that they be replaced with sourced figures. [D7 §5.2; E.1 rule 36] Two of the three now have sourced bounds, and the third has the disclosed MRE Contract revenue to check it against.

WORKED EXAMPLE 8.6 — ENGINE EVENTS, MODULES AND REVENUE FROM A 300-AIRCRAFT FLEET, WITH EACH INPUT BOUNDED

Step 1, engines. Every 737NG and A320ceo has two engines: $300 \times 2 = 600$ engines (sourced: airframe configuration, Part I §2). The vehicle has "over 300 aircraft", so 600 is a floor; and the fleet was still being assembled through the first half of 2026, so it was smaller than 600 engines for the periods whose revenue is checked below.

Step 2, shop-visit events a year. The research assumed one restoration every six years (assumed: D7). The sourced intervals are: a performance restoration every 10,000–15,000 cycles for a mature CFM56 (sourced: Leeham, 8 Mar 2024) at 1,000–1,500 cycles a year (sourced: Aircraft Value News), which is roughly 6.7 to 15 years; "roughly every eight years" as a rule of thumb (sourced: Aviation Week, date not captured); and about ten years implied by 2,300 visits a year on about 24,000 engines (derived in the research from GE and fleet figures, Part III §23). Part I §6 carries the intervals and reconciliation R-008 the disagreement.

events a year = $600 \div \text{interval}$: at 6 years (D7 assumption) 100; at 8 years 75; at 10 years 60; at 6.7 years 90; at 15 years 40

Step 3, modules per event. The research assumed 1.5 modules exchanged per event (assumed: D7; FTAI replaces the time-expired modules rather than the whole engine). No disclosed ratio exists (D5 U10). The bounds are 1 (one module) and 3 (a whole-engine exchange, since FTAI's releases count three modules per engine: "450 modules (150 engines)", release of 26 Feb 2025; Part I §3 and Part VII §49 carry the convention).

modules a year at 1.5 per event: 150 (at 100 events); 112 (at 75); 90 (at 60)
modules a year at 3 per event: 300 (at 100 events); 225 (at 75)
for scale, FTAI's 2026 target is 1,050 modules (release, 25 Feb 2026) or 1,200 (call coverage, Jul 2026): 150 modules is 14% or 12.5% of it

Step 4, revenue per module. The research assumed \$1.0–2.0M (assumed: D7). Part VII §56 derives upper bounds from the segment: \$2.11M for FY2025 and \$2.34M for Q2 2026 with products revenue excluding MRE as the numerator, and \$2.96M for Q2 2026 with segment revenue including MRE (reconciliation R-046, with D5's five limits: whole engines, parts, used material and V2500 activity in the numerator; produced is not sold; core fair values; mix; no price list). For context, a conventional performance-restoration shop visit is quoted at \$1.2–1.6M or \$1.8–2.5M, and a heavy visit with life-limited parts at \$2.1–2.8M or more than \$3.5M (sourced: Safe Fly Aviation, a consultancy blog, 2026; Part I §7 and reconciliation R-013).

D7's base case: $100 \text{ events} \times 1.5 \text{ modules} \times \$1.0\text{--}2.0\text{M} = \$150\text{--}300\text{M a year}$
 at the FY2025 bound: $100 \times 1.5 \times 2.11 = \317M ; $75 \times 1.5 \times 2.11 = \237M ;
 $60 \times 1.5 \times 2.11 = \190M
 whole-engine case at the Q2 2026 bound: $100 \times 3 \times 2.34 = \702M ; $75 \times 3 \times$
 $2.34 = \$527\text{M}$

Step 5, the check against disclosed revenue. MRE Contract revenue to and from the 2025 Partnership was \$335.8M for FY2025 and \$404.0M for the first half of 2026, which annualises to about \$808M (sourced: 10-K FY2025; 10-Q Q2 2026 via the orchestrator notes).

module-equivalents implied by \$808M a year: at \$2.11M, 383; at \$2.34M,
 345; at \$2.96M, 273
 per engine on 600 engines: $383 \div 600 = 0.64 \text{ a year}$; $273 \div 600 = 0.46 \text{ a year}$
 if all were whole engines at 3 modules each: $383 \div 3 = 128 \text{ engine-equivalents}$, 21% of 600, an implied interval of about 4.7 years

The disclosed revenue sits above every case built from the sourced intervals, including the whole-engine case at 75 events. Four things in the record bear on the gap, and this Part states them without choosing among them: the 10-Q line covers "the sale and purchase of engines to and from" the Partnership, which is wider than exchanges triggered by removals; aircraft bought mid-life may have arrived with engines already due, so the first year may include catch-up events (D7 §5.2); the line may include whole-engine sales to equip the fleet rather than one-for-one exchanges; and revenue per module in the exchange sense is unknown, since the Part VII figures are bounds. The count of engines and modules behind the dollars was not disclosed (D7 U6).

Step 6, margin and elimination. At the segment's Q2 2026 Adjusted EBITDA margin of 28.5% (Part VII §53), the D7 band of \$150–300M would carry \$43–86M of segment Adjusted EBITDA a year before elimination. The disclosed first-half 2026 eliminations of \$16.6M imply gross profit on sales to the Partnership of about \$87M for the half (reading A) or \$16.6M (reading B), as worked example 8.4 shows; reading A annualises to about \$174M, above the top of the D7 band, and reading B to about \$33M, below its bottom. The gap between the two readings is the reason the inputs above are bounded rather than fixed. Limits of the example: the fleet size varied through the periods checked; every per-module figure is an upper bound; the interval sources disagree; and the modules-per-event ratio is undisclosed. [FTAI releases 26 Feb 2025, 25 Feb 2026, 17 Aug 2026; FTAI 10-K FY2025; FTAI 10-Q Q2 2026; Leeham News, 8 Mar 2024; Aircraft Value News, undated; Aviation Week, date not captured; Safe Fly Aviation, 2026; D7 §5.2; D5 §2.11; reconciliation R-008, R-013, R-046, R-047]

WHAT THE ARITHMETIC OF \$66.2 DOES AND DOES NOT SHOW

It shows that a 600-engine fleet generates somewhere between 40 and 100 engine removals a year on the sourced intervals, and that the disclosed sales to the Partnership in its first eighteen months were larger than those removals would produce at any per-module figure the research can bound. It does not show why. The candidates are stated in the example and none is confirmed. A reader who later obtains the counts of engines and modules sold to the Partnership by period, which the research could not, can close the gap in one line. [D7 U6]

66.3 RECONCILING THE SEED SALE**WORKED EXAMPLE 8.7 — THE 46 SEED ASSETS, THE QUARTERLY SALES AND THE TWO GAIN FIGURES**

Inputs: 46 aircraft at an estimated net purchase price of \$549M (sourced: release, 30 Dec 2024); sales of 4, 33 and 8 aircraft with gains of \$10.9M, \$34.6M and \$4.6M in the three quarters of 2025 (sourced: 10-Qs); FY2025 gain \$46.4M (sourced: 10-K FY2025); Q1 2025 proceeds \$58.9M (sourced: 10-Q Q1 2026, comparative).

seed price per aircraft: $549 \div 46 = \$11.9\text{M}$
 aircraft sold through Q3 2025: $4 + 33 + 8 = 45$ of 46
 gains through Q3 2025: $10.9 + 34.6 + 4.6 = \$50.1\text{M}$
 full-year gain stated: \$46.4M; difference: $50.1 - 46.4 = \$3.7\text{M}$
 Q1 2025: proceeds $58.9 \div 4 = \$14.7\text{M}$ per aircraft; net book value $(58.9 - 10.9) \div 4 = \$12.0\text{M}$ per aircraft; gain 18.5% of proceeds
 if the Q1 gain rate held for the seed portfolio: $0.185 \times 549 = \$101.6\text{M}$,
 against \$46.4M or \$50.1M stated

Three readings of the \$3.7M are possible and the record does not choose: a net loss of about \$3.7M on sales to the Partnership in the fourth quarter of 2025; a difference in what the quarterly and annual figures count; or a difference in the population, since the 10-Qs report sales to the Partnership and the seed portfolio is only part of what the Partnership bought from FTAI. The last line shows that the first four aircraft were sold at a higher gain rate than the whole-year figure implies for the seed portfolio, so the four were not representative, or the remaining 42 were sold closer to book, or the two figures count different things. The cost basis of the \$46.4M gain was not retrieved. Limits: the 45 aircraft sold through September may not all be Seed Assets; the 46th aircraft's disposition was not located; the fourth-quarter count and gain were not located. [FTAI release, 30 Dec 2024; FTAI 10-Qs Q1–Q3 2025; FTAI 10-K FY2025; D7 §5.3, T1; D6 U12; reconciliation R-054]

67. WestJet, AEI and the 2026 transactions

The year 2026 produced one transaction that shows the whole structure of this Part working at once, one collaboration that extends it into freighters, and a handful of smaller dated events. This section takes them in that order.

67.1 WESTJET: 27 737-700s, SPLIT TWO WAYS

WESTJET TRANSACTION AND AEI COLLABORATION

On 28 September 2026 FTAI announced the acquisition of 27 Boeing 737-700s from WestJet: 17 on lease to WestJet bought by the 2026 SPV in a sale-leaseback, and 10 off-lease aircraft bought by FTAI itself for their CFM56-7B engines and modules, to supply its Maintenance, Repair and Exchange customers. Price, lease terms and the holding entity for the ten were not disclosed. On 7 July 2026 FTAI and AEI announced a collaboration on 737-800 freighters pairing FTAI's lower-cycle, cargo-customised engines with AEI's conversion approvals. [FTAI release, 28 Sep 2026; FTAI and AEI release, 7 Jul 2026]

The release calls the WestJet deal "one of FTAI's largest aircraft transactions to date". [FTAI release, 28 Sep 2026] Its structure is the pattern of §60 and §61 in a single transaction. Aircraft that carry a lease go to the vehicle: the 2026 SPV bought 17 aircraft "on lease to WestJet in a sale-leaseback", which means WestJet sold aircraft it owned and leased them straight back, converting owned assets into rent obligations and releasing cash (Part II §12). Aircraft without a lease go to FTAI: the 10 off-lease 737-700s "will support FTAI's Aerospace Products business by expanding the Company's supply of CFM56-7B engines and modules available to its Maintenance, Repair and Exchange customers". [FTAI release, 28 Sep 2026] The on-lease aircraft are a rent stream for the fund; the off-lease aircraft are feedstock for the Module Factory.

The deal also sits inside the translation table. Part III §24's row on the 737 MAX production cap names WestJet, with United and Copa, among the carriers retiring 737NGs regardless of the cap. [Part III §24; D9 §5] A retiring fleet is the source of both halves of the transaction: the aircraft an airline still needs for a few more years and will lease back, and the aircraft it has already parked.

WORKED EXAMPLE 8.8 — THE TEN OFF-LEASE 737-700S AS ENGINES AND MODULES, WITH VALUE BOUNDS

Inputs: 10 aircraft, two CFM56-7B engines each (sourced: airframe configuration); FTAI's counting convention of three modules per engine (sourced: release, 26 Feb 2025); half-life market values of \$5.7M for a -7B24 and \$6.4M for a -7B27 (sourced: IBA, Sept 2025, an appraiser); green-time value \$2.8–3.4M and run-out teardown value \$0.8–1.2M per -7B (sourced: Safe Fly Aviation, 2026, a consultancy blog; the "green-time" label is internally inconsistent, reconciliation R-029).

engines: $10 \times 2 = 20$ CFM56-7B
 module-equivalents at three per engine: $20 \times 3 = 60$
 value bounds for 20 engines: at half-life market value, $20 \times 5.7 = \$114\text{M}$
 (-7B24) or $20 \times 6.4 = \$128\text{M}$ (-7B27)
 at green-time value, $20 \times 2.8\text{--}3.4 = \$56\text{--}68\text{M}$; at teardown value, $20 \times$
 $0.8\text{--}1.2 = \$16\text{--}24\text{M}$

These are bounds on what 20 engines of stated condition would be worth, not the price FTAI paid, which was not disclosed. The thrust ratings of WestJet's engines were not disclosed, and the IBA values quoted are for two specific ratings; the maintenance condition of the engines, which decides where between the teardown and half-life figures each sits (Part I §8), was not disclosed either. Sixty module-equivalents is 5.7% of the 1,050-module 2026 target, or 5% of 1,200, if every module were refurbished and sold; a run-out module becomes feedstock first (Part VII §46). Limits: configuration arithmetic only; two value sources on different bases; the airframes' own disposition (part-out, Part V §32, or conversion) was not stated. [FTAI release, 28 Sep 2026; FTAI release, 26 Feb 2025; IBA, Sept 2025; Safe Fly Aviation, 2026; reconciliation R-029]

What was not disclosed is listed in one place because two dossiers asked the same question. The purchase price for the 27 aircraft; the lease term and rent on the 17 leased back; which FTAI entity holds the 10, and whether they are booked as leasing equipment in the Aviation Leasing segment or as inventory in Aerospace Products; and whether any of the engines are destined for FTAI Power's Mod-1 generator sets (Part IX §74), which consume CFM56 cores. The Q3 2026 10-Q, not yet filed at the research date, would answer most of these. [D6 U13; D7 U14; reconciliation K-06]

67.2 AEI AND THE 737-800 FREIGHTER

On 7 July 2026 FTAI and AEI (Aeronautical Engineers, Inc.) announced a collaboration "to meet growing demand for Boeing 737-800 freighters". Part II §12 explains passenger-to-freighter conversion: a conversion house modifies a used passenger airframe under its own regulator-approved design, extending the airframe's life by a decade or more at low annual cycles. The release says FTAI "can build and maintain lower cycle engines customized for cargo, enabling FTAI and AEI to deliver aircraft at a significantly lower operating cost, while extending the CFM56 engine's lifecycle across passenger, cargo and power applications"; it describes AEI as having "625+ aircraft modified using AEI's STCs" and notes that "almost 6,000" 737-800s were delivered. [FTAI and AEI release, 7 Jul 2026]

The logic connects to Part V. A freighter flies few cycles a year, so an engine with modest green time, or a module with life-limited parts too short for a passenger operator's planned run, can serve a freighter for years (Part II §16). That is a use for exactly the part-worn modules the Module Factory takes in. The demand side is contested in the research and both sides are

recorded. Type-specialist sources count more than 260 737-800 freighters in service and a pipeline of more than 300, and an earlier IBA note reported freighter values rising while passenger narrowbody rents fell. In September 2026 IBA warned of an "oversupply of narrowbody conversions and lease rate fall". [Acumen Aviation, 18 Dec 2025; Cargo Facts; AeroIn, 2025; IBA via India Seatrade News, 6 Sep 2026; reconciliation R-034] The cost of a 737-800 conversion and the availability of conversion slots were not retrieved. [D6 U10]

67.3 THE OTHER DATED EVENTS OF 2026

EXHIBIT 8.14 — STRATEGIC CAPITAL AND LEASING EVENTS, 2026

Date	Event	What was disclosed	What was not	Source
25 Feb 2026	FY2025 results	SCI I deployment "largely completed"; SCI II fundraising launched "with anchor investor commitments"; 2026 Leasing guidance \$575M	Anchor identities and amounts	FTAI release, 25 Feb 2026
Q1 2026	10-Q	Proceeds from sales to the Partnership \$117.3M; elimination \$10.0M; carrying value \$279.4M; Russia recoveries \$44.6M; lease income \$39.9M	Aircraft count sold	FTAI 10-Q Q1 2026
22 May 2026	"FTAI Aviation Prices Inaugural Asset-Backed Securitization" (FTAI MRE 2026-1)	The 2025 Partnership's first ABS: \$612M of notes on 48 A320ceo and 737NG aircraft leased to 23 airlines; Series A expected Asf (Fitch) and A(sf) (KBRA), Series B BBB+sf (Fitch); expected close 4 Jun 2026; "significantly oversubscribed"; the vehicle "currently owns 292 aircraft"	Coupons; loan-to-value; what the proceeds repaid	FTAI release, 22 May 2026, via Finviz and AviTrader; orchestrator notes; D7 U12 (partly resolved)
30 Jun 2026	First quarterly distribution of the 2025 Partnership	That it was paid	Amount, source, FTAI's share	FTAI Q2 2026 release; D7 U13
7 Jul 2026	AEI collaboration	As \$67.2	Commercial terms; conversion cost	FTAI and AEI release, 7 Jul 2026

Date	Event	What was disclosed	What was not	Source
29 Jul 2026	Q2 2026 results	2025 SPV "fully committed"; 2026 SPV launched; Leasing guidance cut to \$475M; 2027 Leasing guidance \$450M; servicing fees \$6.99M; elimination \$6.6M; carrying value \$365.5M; MRE Contract revenue to the Partnership \$182.8M	Leasing segment Adjusted EBITDA (the \$88.2M is call coverage)	FTAI Q2 2026 release and 10-Q
14 Aug 2026	2026 SPV warehouse closed (announced 17 Aug)	\$2.0B, 13 lenders, according to \$3.0B; "\$5.5 billion of warehouse financing"; 2025 SPV "harvest phase"	Equity size; debt terms; how \$5.5B is built	FTAI release, 17 Aug 2026
15 Sep 2026	\$500M share repurchase programme	Title	Terms	FTAI release, 15 Sep 2026 (Part X §83)
28 Sep 2026	WestJet 27 × 737-700	As §67.1	Price, lease terms, holding entity	FTAI release, 28 Sep 2026

Source: FTAI Aviation Ltd. releases and filings on the dates shown; FTAI and AEI release, 7 Jul 2026. Compiled from D7 §2.7 and D6 §4.7. The repurchase programme belongs to Part X §83 and is listed here only for the calendar.

The calendar closes where the Part began. In twenty-one months FTAI sold most of its on-lease aircraft into a vehicle it manages, saw its rent fall by half, replaced it with a servicing fee of about \$7M a quarter, a 19% stake carried at \$365.5M, and \$404M of first-half related-party maintenance sales whose profit is deferred; cut its leasing guidance by \$100M and explained why in four words; and started a second vehicle whose first purchase split a retiring airline fleet into rent for the fund and engines for the factory. Every figure in that sentence is in a filing or a release cited above, and every term is undisclosed where this Part says it is.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-047	Profit elimination of \$16.6M in 1H 2026 implies ~\$87M of gross profit on MRE sales to the Partnership if proportional to the 19% stake, or \$16.6M if fully deferred; the 10-Q sentence supports either	§63.4, worked example 8.4; §66.2

Id	The two (or more) figures	Where in this Part
R-048	FTAI's stake: 19% (FY2025 10-K; 2026 10-Qs) against 20% (Q3 2025 10-Q)	§61.1, §63.5, Exhibit 8.11; worked example 8.4
R-049	2026 Aviation Leasing guidance: \$575M (25 Feb 2026) cut to \$475M (29 Jul 2026, as read by D5 and D10 and confirmed by the orchestrator) against a second reading of the July release returning \$575M (D6), established as a mis-read	§60.4, Exhibit 8.6
R-050	The \$1.4B → \$1.525B (\$1.0B Aerospace Products, \$525M Aviation Leasing) step, undated in the 10-Q quotation, dated to 27 Oct 2025 by the orchestrator notes	§60.4, Exhibit 8.6
R-051	Q2 2026 Aviation Leasing Adjusted EBITDA \$88.2M (call coverage) against no Leasing line returned from the release; implied Corporate and Other −\$46.5M unconfirmed	§58.3, Exhibit 8.3
R-052	FY2024 lease income \$234,411K against \$255,338K, two readings of the FY2025 10-K	§58.2, Exhibit 8.2; §58.3, Exhibit 8.3
R-053	FY2021 lease income \$173,864K against FY2021 Aviation Leasing segment revenue \$128,328K, both from the FY2022 10-K	§58.2, Exhibit 8.2
R-054	Gains on sale to the 2025 Partnership: \$10.9M + \$34.6M + \$4.6M = \$50.1M for 9M 2025 against \$46.4M for FY2025	§60.2, Exhibit 8.5; §63.1; worked example 8.7
R-055	"\$5.5 billion of warehouse financing" (release, 17 Aug 2026) against \$2.5B + \$2.0B = \$4.5B of disclosed commitments; matches only with the \$1.0B accordion	§62.5
R-056	Capital per aircraft: \$11.9M (seed), \$14.7M (Q1 2025), \$13.9M (first 101), \$23.6M (89 under contract), ~\$20M (\$6.0B ÷ 300)	§62.3, Exhibit 8.8; worked example 8.3
R-057	Aircraft sales to the Partnership "non-recurring in nature" (10-K, ASC 610-20) against "\$3.0+ billion of capital annually" and SCI as the primary investment route (release; 10-Q)	§60.3, §64.2
R-058	"Asset-light business model" (releases; July 2026 guidance cut) against a 19% LP interest carried at \$365.5M, ~\$380M at full commitment	§60.4, §64.1

Id	The two (or more) figures	Where in this Part
R-059	"OneIM (\$4B)" (chain map; D10 timeline) against ">\$4 billion" of deployment for the inaugural vehicle with OneIM's commitment undisclosed (release, 25 Mar 2025)	§62.2
R-060	Russia: \$120.0M write-off net of deposits (2022) against \$210.7M insured value (FY2025 10-K); recoveries \$54.3M + \$44.6M = \$98.9M	§59, Exhibit 8.4, worked example 8.2
R-043	MRE Contract revenue inside Aerospace Products segment revenue by arithmetic (B-1) against the unretrieved segment-note mapping	§63.3, §64.4
R-044	MRE Contract revenue as "stand-ready obligations to provide replacement ... engines" (Q3 2025 10-Q) against "the transaction price for sales of ... engines and related modules to the SPVs" (Q2 2026 10-Q)	§63.3
R-046	Revenue per module \$2.11M (FY2025) and \$2.34M (Q2 2026) on products revenue excluding MRE against \$2.96M (Q2 2026) on segment revenue including MRE; D7's \$1.0–2.0M assumption	§66.2, worked example 8.6
R-008	Mature CFM56 restoration interval: 10,000–15,000 cycles (Leeham) at 1,000–1,500 cycles a year, "roughly every eight years" (Aviation Week), ~10 years (derived), six years (D7 assumption)	§66.2, worked example 8.6
R-030	Lease rate factor: 0.70% a month (D3, mixed bases), 0.78%/1.0% (this Part, FY2025 on cost/net) and 0.87–1.1% (D6, 1H annualised), ~1.0% (Willis, derived), 0.9% (D7 assumption); none stated	§58.4, worked example 8.1; §66.1, worked example 8.5
R-031	Mid-life narrowbody rents: 12-year 737-800 ~\$228,000/month, –11% y/y (IBA, Jul 2025) and a "lease rate fall" warning (IBA, Sep 2026) against rising rates on new leases and extensions (Air Lease 10-K FY2025) and 10-year 737-800 rents +44% vs Jan 2022 (Cirium Ascend via Bloomberg)	§66.1, worked example 8.5
R-029	CFM56-7B value: half-life market value \$5.7M (-7B24) and \$6.4M (-7B27) (IBA, Sept 2025) against green-time \$2.8–3.4M and teardown \$0.8–1.2M (Safe Fly Aviation, consultancy blog, 2026)	§67.1, worked example 8.8
R-034	737-800 freighter demand: >260 in service and >300 pipeline (Acumen, Cargo Facts), freighter values rising (IBA, 2025) against "oversupply of narrowbody conversions" (IBA, 6 Sep 2026)	§67.2
R-013	Shop-visit cost: performance restoration (PR) \$1.2–1.6M and \$1.8–2.5M; heavy with life-limited parts (LLPs) \$2.1–2.8M and ">\$3.5M" (Safe Fly Aviation, consultancy blog); named for context only	§66.2

Id	The two (or more) figures	Where in this Part
C-21 (registry)	"Fully committed" (Jul 2026) against "fully deployed"; "currently owns 292 aircraft" (22 May 2026) against "over 300 aircraft" committed (17 Aug 2026)	\$61.1, \$62.1

Unknowns carried in this Part

From the reconciliation register (section D.1), verbatim. D6 entries that belong to Part II's mechanics (U6, ABS coupons and LTVs; U14, engine-lessor scale) are carried there.

- **U-D6-01 (D6) — U1 — Fleet by year.** Aircraft and engine counts at 31 Dec 2021, 2022, 2023, 2024 and 30 Jun 2026. Resolve: "Our Aviation Leasing segment" paragraphs in each 10-K/10-Q ([S7],[S8],[S3]).
- **U-D6-02 (D6) — U2 — Annual segment lines 2023–2025.** Lease income, maintenance revenue, asset sales revenue and cost, gain on sale of leasing equipment, equity-method pick-up, and segment Adjusted EBITDA for FY2023, FY2024, FY2025 and each 2025 quarter. Resolve: segment note and MD&A of the FY2025 10-K ([S1]) and the Q4/FY2025 release ([S6]). (The 18th search aimed at this and was refused on budget.) Partly closed by the orchestrator notes: lease income 2024 "234,411" or "255,338", 2025 "235,210"; Aviation Leasing Adjusted EBITDA 2024 "500,062", 2025 "608,912"; 2025 maintenance revenue "218,499" and asset sales revenue "106,945" (K-17).
- **U-D6-03 (D6) — U3 — Q1 and Q2 2026 segment Adjusted EBITDA.** The Q2 2026 release ([S4]) and Q2 2026 10-Q ([S3]) contain them; the \$88.2M chain-map figure needs confirmation.
- **U-D6-04 (D6) — U4 — Russia detail.** Insurance recoveries in FY2023 and FY2024; whether any recoveries occurred in 2022; the status of the 4 aircraft and 1 engine in Ukraine; whether the Russia assets remain in gross leasing-equipment cost; the gross (pre-deposit) carrying value written off in 2022; the number of claims settled versus outstanding. Resolve: "Russia" / "insurance" paragraphs in FY2023 and FY2024 10-Ks and the FY2025 10-K.
- **U-D6-05 (D6) — U5 — Lease-rate factors.** No stated LRF for FTAI or WLFC; WLFC engine count, average lease term and FY2025 totals. Resolve: WLFC FY2025 10-K ([S13]) Item 1 and Item 7. (Cluster K-14 with U-D11-01.)
- **U-D6-07 (D6) — U7 — Security deposit and maintenance deposit balances.** FTAI's balance-sheet "security deposits" and "maintenance deposits" liabilities by year, and the

reserve rates in its leases. Resolve: FY2025 10-K balance sheet and revenue-recognition note.

- **U-D6-08 (D6) — U8 — Depreciation policy.** Useful lives and residual assumptions for FTAI's aircraft and engines. Resolve: FY2025 10-K summary of significant accounting policies.
- **U-D6-09 (D6) — U9 — Half-life market values** for 12–15-year-old 737-800 and A320ceo aircraft, and full-life CFM56-7B/-5B/V2500 values, 2023–2026. Resolve: IBA/Cirium/mba published updates (D3 may hold engine values). (Cluster K-02.)
- **U-D6-10 (D6) — U10 — Conversion cost and slot availability** for 737-800 P2F (AEI, Boeing BCF, IAI). Resolve: conversion-house releases, trade press.
- **U-D6-11 (D6) — U11 — Equity-method investment** in the Leasing segment: counterparty, ownership, assets, income by year. Resolve: FY2025 10-K "Investments" note. (Cluster K-07 with U-D10-13; partly closed by the orchestrator notes, which list the equity-method investments and carrying values; which investment the segment leases "through" is still open.)
- **U-D6-12 (D6) — U12 — Seed Assets.** Number, type, sale price and transfer dates of assets sold to the 2025 Partnership; the \$46.4M gain's cost basis. Resolve: FY2025 10-K related-party / Strategic Capital note; D7.
- **U-D6-13 (D6) — U13 — WestJet transaction terms.** Purchase price, lease term and rent for the 17 leased-back aircraft, the holding entity for the 10 off-lease aircraft, and whether the 10 are booked as leasing equipment or Aerospace Products inventory. Resolve: Q3 2026 10-Q (not yet filed) and D7. (Cluster K-06 with U-D7-14.)
- **U-D6-15 (D6) — U15 — Average lease terms over time** (new-lease term at signing versus remaining term) and the mix of short- versus long-term engine leases in FTAI's book. Resolve: 10-K MD&A by year.
- **U-D7-01 (D7) — U1 — Management/servicing fee rate and base.** Only "customary, market-based compensation" [S11]. Resolve: the limited partnership agreement or servicing agreement if filed as an exhibit; the 10-K related-party note text; the Q2 2026 10-Q exhibits (ex10.8, ex10.9 were listed in the filing index but not read). (Cluster K-09 with U-D7-02.)
- **U-D7-02 (D7) — U2 — Promote / carried interest.** Only "contractual profit-sharing arrangement" [S13]. Resolve as U1.
- **U-D7-03 (D7) — U3 — Investors.** OneIM's commitment size and role; identities of other LPs beyond categories [S3, S4].
- **U-D7-04 (D7) — U4 — 2026 SPV equity.** Target size, hard cap, anchor investors, FTAI's stake, timing of final close [S6, S9].

- **U-D7-05 (D7) — U5 — Pricing of aircraft sold by FTAI to the vehicles.** Appraisal policy, third-party valuation, LP or independent-director approval. Not located.
- **U-D7-06 (D7) — U6 — MRE revenue from the vehicles.** Dollar revenue and counts of engine and module exchanges to the 2025 Partnership; expected annual volume for 300+ aircraft. Only eliminations located [S12, S13]. Partly closed by the orchestrator notes: Q2 2026 \$182.8M (vs \$69.6M), 1H 2026 \$404.0M (vs \$170.2M) (Q2 2026 10-Q); counts still open.
- **U-D7-07 (D7) — U7 — Equity-method income.** FTAI's share of Partnership earnings by period; FTAI's share of the 30 Jun 2026 distribution. (Cluster K-08 with U-D7-13.)
- **U-D7-08 (D7) — U8 — Line placement.** Which income-statement line and segment carry servicing fees and equity income (presumed Aviation Leasing / Other revenue and equity in earnings; not confirmed).
- **U-D7-09 (D7) — U9 — Guidance.** Any 2025 or 2026 guidance item specific to SCI (fees, gains, MRE volume) and the corresponding actuals. Only total and Aerospace Products figures located [S1, S6].
- **U-D7-10 (D7) — U10 — Q4 2025 and 1H 2026 aircraft sales.** Counts and gains for Q4 2025, Q1 2026 (proceeds \$117.3M known) and Q2 2026.
- **U-D7-11 (D7) — U11 — Debt terms.** Rate, advance rate, tenor and covenants of the \$2.5B facility and the \$2.0B warehouse; how \$5.5B is built; whether the 2025 facility was upsized.
- **U-D7-12 (D7) — U12 — The May 2026 ABS.** Whether "FTAI Aviation Prices Inaugural Asset-Backed Securitization" [S26] is a 2025-SPV take-out or FTAI's own balance-sheet deal; size, tranches, pricing. Partly resolved by the orchestrator notes: the 2025 Partnership's inaugural ABS (FTAI MRE 2026-1), \$612M of notes on 48 A320ceo and 737NG aircraft leased to 23 airlines, Series A expected Asf (Fitch) / A(sf) (KBRA), Series B BBB+sf (Fitch), expected close 4 Jun 2026; coupons and LTV still unknown.
- **U-D7-13 (D7) — U13 — The 30 Jun 2026 distribution.** Amount, source (rent versus sales), FTAI's share.
- **U-D7-14 (D7) — U14 — WestJet.** Disposition of the other 10 of 27 737-700s (balance sheet, part-out, FTAI Power cores).
- **U-D7-15 (D7) — U15 — Vehicle term.** Life of the 2025 Partnership, length of harvest phase, extension rights, exit (portfolio sale, ABS, piecemeal).
- **U-D7-16 (D7) — U16 — Comparables not reached.** Castlake, Apollo/Merx, AerCap fee structures; Blackbird and Thunderbolt fee rates.
- **U-D7-17 (D7) — U17 — Q1 2025 10-Q "management fees and incentive allocations".** The search summary flagged such text in S14; whether it concerns SCI or the pre-internalization Fortress management agreement is not verified.

- **U-D7-18 (D7) — U18 — VIE analysis.** Whether FTAI assessed the Partnership as a VIE and the basis for not consolidating.
- **U-D7-19 (D7) — U19 — Published SCI-specific critiques.** None located (related-party sales, transfer pricing, fund-raising dependence). Resolve: sell-side notes, short-seller updates after Jan 2025, financial press.

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PART IX

FTAI Power

This Part explains FTAI's newest business: the conversion of CFM56 engines into 25-megawatt generator sets for data centers. It covers how a jet engine becomes a stationary power plant, what FTAI has and has not said about its Mod-1 unit, the joint venture that sells it and the \$1.465 billion order it has booked, the market and the turbine shortage it sells into, the competing machines, the permitting and grid-connection rules that govern deployment, and the arithmetic that connects all of it to the CFM56 fleet described in Parts I and III. A reader who finishes it will be able to read a gas-turbine order announcement, a heat-rate claim and a dollars-per-kilowatt comparison with every term understood.

TRANSLATION-TABLE ROWS ANSWERED

Part III §24 turns each documented driver into a requirement on the aftermarket. This Part answers the rows that create a requirement for on-site electrical power rather than for engine maintenance: the row on data-center electricity demand (Part III §24, row T-10: data-center power demand) and the row on the multi-year shortage of new gas turbines (Part III §24, row T-10: gas-turbine shortage). It also bears on two rows it does not answer but draws from. The row on low CFM56 retirements (Part III §24, row T-1: low CFM56 retirements) sets the supply of retired engines from which Mod-1 units are built, and the row on the CFM56 plateau and later decline (Part III §24, row T-14: CFM56 plateau-then-decline) describes when that supply is expected to grow. FTAI Power is unusual in this primer. Every other segment turns the installed base into maintenance demand. This one removes engines from the installed base permanently, so it sits on the same side of the ledger as retirement and teardown (Part I §2.3; Part V §33).

68. Gas-turbine generator sets and the aeroderivative, functionally

Part I §1 explained what a turbofan does: it squeezes air, burns fuel in it, and uses the hot gas to spin turbines that turn a large fan, and the fan does most of the pushing. The hot middle of that engine, the core or gas generator (Part I §1), is a machine for making a stream of hot, high-pressure gas. In an aircraft the stream is spent on thrust. On the ground the same stream can be spent on turning a shaft at constant speed, and a shaft turning at constant speed is what an electrical generator needs. That is the whole idea of an aeroderivative, and this section builds it up one component at a time.

68.1 WHAT A GAS-TURBINE GENERATOR SET IS

GAS TURBINE; GAS GENERATOR

A gas turbine is a rotating engine that compresses air, burns fuel in the compressed air and expands the hot gas through turbine stages. The compressor, the combustor and the turbine that drives the compressor together form the gas generator: the assembly whose only product is hot, pressurised gas. In a jet engine that gas leaves through a nozzle and the reaction pushes the aircraft. In a power plant it is expanded through a further turbine that turns a shaft. Part I §1 boxed the core of the CFM56; the core is the gas generator of this Part. [D8 §2.1; Part I §1]

GENERATOR SET (GENSET)

A generator set is the complete unit that turns fuel into electricity: the prime mover (a gas turbine or a piston engine), the output turbine where there is one, a gearbox if the shaft speed must be changed, the electrical generator itself, and the surrounding fuel, air-intake, exhaust, lubrication, fire-protection and control systems, assembled and tested as one product. The J&F order of July 2026 is for "Mod-1 mobile gas turbine generator sets", so the thing FTAI's venture sells is the whole unit, not a bare turbine. [D8 §2.1; FTAI release, 22 Jul 2026]

UNITS OF POWER, ENERGY AND GAS USED IN THIS PART

A watt is a rate of energy flow; a megawatt (MW) is a million watts and a gigawatt (GW) is a thousand megawatts. Mod-1 is rated at 25 MW and GE Vernova's order book is measured in gigawatts. A kilowatt-hour (kWh) is one kilowatt sustained for one hour, a megawatt-hour (MWh) is a thousand of those, and a terawatt-hour (TWh) is a billion kWh; electricity consumption is measured in these. A British thermal unit (Btu) is a unit of heat; a million Btu is written MMBtu, and natural gas is priced in dollars per MMBtu. Gas volumes are given in cubic feet: a million cubic feet is MMcf, a billion is Bcf, and a daily flow is MMcf/d or Bcf/d. Pipeline gas carries roughly 1,000 Btu per cubic foot, which is the conversion the fuel-burn example below uses. [D8 §2.3; definitions from first principles]

Three families of machine compete to supply on-site power at data centers, and a fourth that is not a machine in the same sense. The first is the purpose-built stationary gas turbine, which the industry calls a frame or heavy-duty machine.

FRAME (HEAVY-DUTY) GAS TURBINE

A frame turbine is a gas turbine designed from the start to sit on a foundation and make power, typically 50 to 600 MW per unit, heavy, and built in a factory for a delivery slot years ahead. The industry labels its generations by letter: the F-, H- and J-class machines of GE Vernova, Siemens Energy and Mitsubishi Power are the large frames whose order books are described in §71.3. The size range and the classification are general industry knowledge that the research did not separately source. [D8 §2.1]

The second family is the aeroderivative, which §68.2 builds up in detail. The third and fourth are the alternatives Mod-1 competes with in §72.

RECIPROCATING ENGINE; SOLID-OXIDE FUEL CELL

A reciprocating engine is a piston engine. Large natural-gas versions of roughly 1 to 23 MW each are arranged in halls of many units; Wärtsilä describes data-center plants built from 10 to 23 MW engines reaching more than 450 MW per site, with electrical efficiency that "can exceed 50%" and "20–35% less fuel than gas turbines", both vendor claims. A solid-oxide fuel cell converts natural gas to electricity electrochemically, without combustion; Bloom Energy is the supplier the research reached, and its annual report positions the product against reciprocating engines on power density and emissions. No output or price figures for fuel cells were captured. [Wärtsilä product page, undated; Bloom Energy 10-K FY2024; D8 §2.1, U9]

Two more pairs of words recur in every turbine order announcement and need plain definitions before they are used.

SIMPLE CYCLE AND COMBINED CYCLE; BASELOAD, BACKUP AND PEAKING

A gas turbine run on its own, with its exhaust heat discarded, is in simple cycle. A combined-cycle plant adds a boiler that recovers the exhaust heat to raise steam for a second, steam-driven turbine, which raises the plant's efficiency but adds years to construction and is only built at frame scale. Mod-1 is a simple-cycle machine. Baseload duty means running continuously; backup means standing by for an outage; peaking means running only in the hours of highest demand. FTAI described Mod-1 as usable for all three. These definitions are from first principles; the research used the terms without defining them. [D8 §2.1, §2.3; registry section 2]

68.2 HOW A TURBOFAN BECOMES AN AERODERIVATIVE, FUNCTIONALLY

The CFM56 is a high-bypass turbofan with two spools (Part I §1). The low-pressure spool carries the fan, a small booster compressor and the low-pressure turbine (LPT) at the back. The high-pressure spool carries the high-pressure compressor (HPC), the combustor and the high-pressure turbine (HPT), and it is the gas generator. On the aircraft, most of the low-pressure turbine's work goes into spinning the fan, and most of the fan's air goes around the core to make thrust. For a power plant, thrust is unwanted and the fan is dead weight. The published conversion recipe, described generically in a US patent on converting turbofans, has five steps. [US patent 11,053,891, "Method for converting a turbofan engine"; D8 §2.2]

EXHIBIT 9.1 — TURNING A TURBOFAN INTO AN AERODERIVATIVE GENERATOR SET: THE FIVE FUNCTIONAL STEPS

Step	What is removed or added	Why	Source
1	Remove the fan and the bypass duct; add one or two compressor stages ("stage 00 and stage 0") in front of the existing booster	Thrust is no longer wanted; the small extra compressor replaces the fan's job of feeding the core with air at the pressure it was designed for	US patent 11,053,891
2	Replace the exhaust nozzle with a power turbine, or use the engine's own low-pressure turbine as the output turbine	The hot gas must turn a shaft instead of pushing the aircraft; a free power turbine on its own shaft lets the gas generator run at its own speed while the output shaft holds generator speed	US patent 11,053,891; the LM6000 (GE's aeroderivative built on the CF6-80C2) uses its own low-pressure spool (general industry knowledge)
3	Add a gearbox where the output shaft spins faster than the generator needs	A 60 Hz grid needs a two-pole generator at 3,600 rpm or a four-pole at 1,800 rpm; aeroderivative shafts commonly spin faster	US patent 6,895,325
4	Add the electrical generator	"Rotation of the generator's input shaft and windings produces electric power"	US patent 6,895,325
5	Add the stationary systems: natural-gas fuel nozzles and controls, air-inlet filter house, exhaust stack and silencing, emissions after-treatment where required, lubrication and cooling, fire suppression, and a control system for start-up, load-following and protection, all inside an enclosure on a trailer or skid	The aircraft engine burned kerosene in clean air at altitude; the ground unit burns pipeline gas in dirty air and must meet an air permit	General industry knowledge; FTAI has published no Mod-1 bill of materials

Source: US Patent and Trademark Office, US 11,053,891 "Method for converting a turbofan engine" and US 6,895,325 "Overspeed control system for gas turbine electric powerplant"; D8 §2.2. Which of the step-2 arrangements Mod-1 uses, and whether a booster stage replaces the fan, is undisclosed (D8 U3).

AERODERIVATIVE

An aeroderivative is an aircraft engine adapted to drive a shaft for stationary use. The fan and bypass duct are removed and extra compressor stages replace the fan; the exhaust nozzle is replaced by a power turbine, or the engine's own low-pressure turbine is used; a gearbox and a generator are fitted; and the kerosene fuel system is changed for gas. Aeroderivatives are typically 5 to 100 MW, light, modular, fast to start and fast to install, and the patent literature describes the aero gas generator as "more efficient in mid-range power than heavy-duty gas turbine". The examples in this Part are GE's LM2500 (25 to 35 MW) and LM6000 (about 48 to 50 MW), ProEnergy's PE6000 (48 MW) and FTAI's Mod-1 (25 MW). The LM2500 was built on the TF39 (a GE military turbofan) and its civil relative the CF6-6, the LM6000 on the CF6-80C2, and the PE6000 on CF6-80C2 cores taken from retired 747 engines. [US patents 11,053,891 and 5,160,080; D8 §2.1–2.2, §3]

Step 1 deserves a sentence more. The patent's words are: "In case of fan jet designs, the fan is removed and a couple of stages of compression are added in front of the existing low-pressure compressor, these additional stages usually known as stage 00 and stage 0". [US patent 11,053,891] The core was designed to receive air that the fan had already compressed a little. Take the fan away and the core would be starved. So a small compressor goes where the fan was. It does a fraction of the fan's work, because it only has to feed the core and not push a bypass stream.

Step 2 is where the two established designs differ, and where Mod-1's design is unknown.

FREE POWER TURBINE (FPT)

A free power turbine is an output turbine on its own shaft, not mechanically connected to the gas generator; it is driven only by the gas stream, which the patent calls "aerodynamically coupled". Because the two shafts are independent, the gas generator can speed up and slow down with load while the output shaft holds the fixed speed a generator needs. The alternative is to keep the engine's own low-pressure turbine and let its spool drive the generator, which is how the LM6000 works. FTAI has not said which arrangement Mod-1 uses. The acronym is this primer's, not FTAI's. [US patent 11,053,891; D8 §2.2, U3]

GEARBOX (GENSET)

A genset gearbox is the speed reducer between the turbine's output shaft and the generator. A generator synchronised to a 60 Hz grid must turn at 3,600 rpm with two poles or 1,800 rpm with four, and aeroderivative output shafts commonly spin faster, so "an external gearbox is typically required to obtain the proper input speed to a generator". It is a different device from the engine's accessory gearbox boxed in Part I §3. [US patent 6,895,325; D8 §2.2]

Step 5 is the part of the recipe that is not engine work at all, and it is the part FTAI's partner brings.

BALANCE OF PLANT; PACKAGING

The balance of plant is everything around the turbine that a stationary power unit needs: the enclosure, the trailer or skid, the generator and gearbox, the air-inlet filter house, the exhaust and silencing, the gas fuel skid, lubrication and cooling, fire suppression and the control system. Building, integrating and testing all of that around a bare engine is what the industry calls packaging. FTAI describes its venture with Jereh Group as a "packaging and distribution" joint venture and Jereh as "a global leader in gas turbine mobile packaging", so this is the work Jereh contributes (§70). [D8 §2.2, §2.7; FTAI Q1 2026 release, 29 Apr 2026; FTAI 10-Q Q2 2026]

Two terms inside that list need a gloss. Load-following means changing output to track the customer's demand minute by minute; a data center's load is not constant, and the control system must raise and lower the turbine's fuel flow to match it. Switchgear is the customer's set of breakers and busbars into which the genset's electrical output is connected. Neither term is defined in the research; both are given here from first principles.

68.3 WHAT "MOBILE" MEANS**MOBILE GENERATOR SET**

A mobile generator set is one built onto a road trailer or a skid so that it can be driven to a site, connected to a gas supply and to the customer's switchgear, commissioned in days or weeks, and later moved. The J&F order is for "mobile gas turbine generator sets", and FTAI has cited about two weeks to deploy a Mod-1 on site. Mobility matters twice: for speed, because the unit arrives finished and tested, so site work is a pad, a gas line and an electrical tie-in rather than a construction project; and for regulatory treatment, because US air rules distinguish stationary sources from "nonroad engines" (§73.2). [FTAI release, 22 Jul 2026; TheValueist summary of the Q4 2025 call, 26 Feb 2026, call coverage; D8 §2.4]

The connection to a gas supply is itself a project. A pipeline lateral is the branch line that runs from a transmission pipeline to the plant, with metering sized for the plant's flow; S&P Global reported in May 2026 that pipeline operators were striking deals "as data centers turn to colocated generation". [S&P Global Commodity Insights, 27 May 2026] The fuel-burn example below shows why the lateral matters: a single 25 MW unit at high load draws about 4.9 million cubic feet of gas a day.

68.4 HEAT RATE AND EFFICIENCY

Every power-plant comparison in this Part runs through one number, and it is worth defining with care.

HEAT RATE

The heat rate is the fuel energy a plant needs to make one kilowatt-hour of electricity, in Btu per kWh. One kWh is 3,412 Btu of energy, so a plant's efficiency is 3,412 divided by its heat rate. A lower heat rate is better. FTAI's stated figure for Mod-1 is "about 9,000", which is 37.9% efficient and sits inside the 35–40% band FTAI also stated; a 50%-efficient reciprocating engine has a heat rate of about 6,824. The heat rate converts directly into fuel cost per MWh once a gas price is chosen. [TheValueist summary of the Q4 2025 call, 26 Feb 2026, and Aviation Week's MRO Memo column (MRO means maintenance, repair and overhaul), early 2026, both call coverage; D8 §2.3]

LOWER HEATING VALUE (LHV) AND HIGHER HEATING VALUE (HHV)

A fuel's energy content can be stated on two bases. The higher heating value counts the heat recovered by condensing the water vapour in the exhaust; the lower heating value does not. Gas-turbine efficiencies are conventionally quoted on the lower basis, which gives a figure a few percentage points higher than the same machine on the higher basis. FTAI has not said which basis its 35–40% and 9,000 figures use, and they are mutually consistent only if both are on the same basis. [D8 §2.3, T8, U5]

CAPACITY FACTOR

The capacity factor is a plant's actual output over a period divided by what it would have produced running at full load for the whole period. A unit that ran flat out all year would have a factor of 100%; the fuel-burn example below uses 90% as an assumption for a baseload data-center unit. [D8 §2.3]

WORKED EXAMPLE 9.1 — WHAT A HEAT RATE OF 9,000 MEANS IN FUEL

Inputs: heat rate about 9,000 Btu/kWh and efficiency 35–40% (sourced: call coverage of the Q4 2025 call, 26 Feb 2026, basis not stated); output 25 MW (sourced: FTAI launch release, 30 Dec 2025); capacity factor 90% (assumed); 1,000 Btu per cubic foot of pipeline gas (assumed, a standard approximation); gas price \$3.50 per MMBtu (assumed, illustrative, not a forecast); reciprocating-engine efficiency "can exceed 50%" (sourced: Wärtsilä product page, a vendor claim).

$\text{efficiency} = 3,412 \div 9,000 = 37.9\%$
 $\text{fuel at full load} = 25,000 \text{ kW} \times 9,000 \text{ Btu/kWh} = 225,000,000 \text{ Btu/h} = 225 \text{ MMBtu per hour}$
 $\text{annual output} = 25 \text{ MW} \times 8,760 \text{ h} \times 0.90 = 197,100 \text{ MWh}$
 $\text{annual fuel} = 197,100 \text{ MWh} \times 9 \text{ MMBtu/MWh} = 1,773,900 \text{ MMBtu} \approx 1.77 \text{ million MMBtu}$
 $\text{annual gas} \approx 1.77 \text{ Bcf}; \text{ daily gas} = 1.77 \text{ Bcf} \div 365 \approx 4.9 \text{ MMcf/d}$
 $\text{fuel cost per MWh at } \$3.50 = 9 \times 3.50 = \$31.50$
 $\text{reciprocating engine at 50\%: heat rate} = 3,412 \div 0.50 = 6,824; \text{ fuel cost} = 6.824 \times 3.50 = \23.90 per MWh
 $\text{gap} = 31.50 - 23.90 = \$7.60 \text{ per MWh; over } 197,100 \text{ MWh} \approx \$1.5 \text{ million a year per unit}$

So one Mod-1 at 90% load burns about 1.77 Bcf of gas a year, or 4.9 MMcf a day, and 100 units would draw about 0.49 Bcf/d, a figure \$71.2 sets against the market-wide forecasts. The \$7.60 per MWh gap is the fuel penalty a turbine buyer accepts against a 50%-efficient piston plant at this gas price, in exchange for a smaller footprint, fewer units per MW and faster delivery; at a higher gas price the gap widens in proportion. Limits of the example: the capacity factor and the gas price are assumptions; the heat rate comes from call coverage, not a filing, and its heating-value basis is unstated; the 50% figure is a vendor's claim; and the \$1.5 million line is derived in this Part, not published. [D8 §2.3; Wärtsilä product page, undated; reconciliation D8 T8, T10]

68.5 EMISSIONS TREATMENT

Burning natural gas in a turbine produces nitrogen oxides, carbon monoxide and carbon dioxide. Air permits (§73.1) set limits on the nitrogen oxides in particular, because they form smog near the ground.

NITROGEN OXIDES (NOX); SELECTIVE CATALYTIC REDUCTION (SCR)

Nitrogen oxides are the pollutant that air permits limit most tightly for gas turbines. A turbine can meet a limit three ways: with a combustor designed to burn at a lower flame temperature (a low-NOx combustor), by injecting water or steam into the combustor to cool the flame, or with selective catalytic reduction, a catalyst box in the exhaust that converts the oxides to nitrogen using ammonia or urea. The permit for xAI's Memphis turbines allowed them "with certain emissions controls". FTAI has not published Mod-1's NOx level or its after-treatment. [TechCrunch, 3 Jul 2025; D8 §2.6, U5]

Whatever after-treatment a customer's permit requires sits in the balance of plant, so it falls within the packaging scope that Jereh contributes; the research recorded this as a presumption from the division of labour, not a confirmed fact. [D8 §2.6]

68.6 THE PRECEDENTS, AND THE ONE THAT WAS NOT FOUND

Converting an aircraft engine into a stationary power unit is an old practice, and the lineage matters because it tells the reader what has been done before with which cores.

EXHIBIT 9.2 — AERODERIVATIVE LINEAGE: WHICH AIRCRAFT ENGINE EACH STATIONARY UNIT CAME FROM

Stationary unit	Maker	Aircraft engine it derives from	Output class	How the derivation is sourced
LM2500	GE	TF39 (a GE military turbofan) and CF6-6 (the related civil engine)	25–35 MW; the LM2500XPRESS variant is rated 35 MW	General industry knowledge in the research; the 35 MW rating from GE Vernova's release of 22 Jul 2025
LM6000	GE	CF6-80C2; the engine's own low-pressure spool drives the generator	About 48–50 MW	General industry knowledge; output class from EEPower and Data Centre Magazine
PE6000	ProEnergy (private, non-OEM)	CF6-80C2 cores taken from retired Boeing 747 engines, "the same engine core GE Vernova uses in its LM6000"	48 MW	Data Centre Magazine; EEPower
Mod-1	FTAI Power (non-OEM)	CFM56; variant (-5B or -7B) not disclosed	25 MW	FTAI launch release, 30 Dec 2025; Aviation Week MRO Memo, early 2026
Prior production CFM56 stationary derivative	None found	The research searched for one and found none; trade press frames Mod-1 as the CFM56's first such role	—	D8 §2.2, U13; AvioRadar, 2026; AviTrader, 2 Jan 2026

Source: GE Vernova, Crusoe release, 22 Jul 2025; Data Centre Magazine, "Can Old Aeroplane Engines Powering Data Centres Take Off?", undated; EEPower, "Can repurposed jet engines solve AI data center power problems?", undated; FTAI Aviation, launch release, 30 Dec 2025; AvioRadar, 2026; AviTrader, 2 Jan 2026; D8 §2.2 and §3. Lineage rows marked "general industry knowledge" were not separately sourced in the research.

Two points in that table carry weight later. First, every established aeroderivative in the 25 to 50 MW class is built on one gas generator per unit; the research's feedstock arithmetic in §74.3 rests on the assumption that Mod-1 is too. Second, ProEnergy is the closest analogue to FTAI Power: a company outside the original equipment manufacturers (OEMs), the firms that hold the engine's design (Part I §2), that buys retired airline cores and packages them for data-center power, with 2027 delivery. [Data Centre Magazine, undated; EEPower, undated; D8 §3] The research found no production stationary derivative of the CFM56 before Mod-1 and recorded that a definitive statement from GE or CFM, or Gas Turbine World's handbook, would settle whether Mod-1 is the first. [D8 U13]

69. Mod-1 as FTAI describes it

This section gives FTAI's own description of the product, separates what came from a filing or release from what came from third-party coverage of an earnings call, and then lists what has not been said.

69.1 THE LAUNCH RELEASE, 30 DECEMBER 2025

FTAI announced FTAI Power on 30 December 2025 as a platform "focused on converting CFM56 engines to power turbines built to provide the most flexible, cost efficient and scaled solution for delivering reliable energy to data centers globally". [FTAI release, 30 Dec 2025] The release said that "the aeroderivative adapted from the CFM56 engine will provide the market with a 25-megawatt unit that offers grid operators greater flexibility and finer output control than larger units", that production was "expected to begin in 2026", and that FTAI Power "is expected to have the capacity to deliver over 100 units annually and provide service support solutions that maximize uptime by applying its modular maintenance model to power turbines". [FTAI release, 30 Dec 2025] It positioned FTAI "as one of the largest aftermarket maintenance providers and owners of the CFM56 engine", which is the Aerospace Products business of Part VII and the Leasing fleet of Part VIII. The FY2025 annual report records the launch in its business description. [FTAI 10-K FY2025]

69.2 THE CALL DESCRIPTIONS, FEBRUARY 2026

The technical figures come from the fourth-quarter 2025 earnings call of 26 February 2026, which the research reached only through a third-party summary on X and through Aviation Week's coverage, not through a transcript. [TheValueist, 26 Feb 2026, call coverage; Aviation Week MRO Memo, early 2026; D8 U15] As summarised, management described Mod-1 as 25 MW and trailer-mounted, taking about two weeks to deploy on site, usable for baseload, backup or peaking; as 35–40% efficient at a heat rate of about 9,000, "comparable to other aeroderivatives sold over the

past 30 to 40 years"; as giving the engine a second life of 10 to 20 years in power service; and as having "an unmatched turbine input cost, given access to near-fully depreciated CFM56 assets". [TheValueist, 26 Feb 2026, call coverage; Aviation Week MRO Memo, early 2026] Aviation Week noted that which CFM56 variant, the -5B or the -7B, is used was not disclosed; FTAI's aerospace business works on both. [Aviation Week MRO Memo, early 2026; FTAI 10-K FY2025]

MOD-1 (FTAI MOD-1; CFM56 AERODERIVATIVE)

Mod-1 is FTAI Power's 25 MW trailer-mounted natural-gas generator set built around a CFM56 core. FTAI launched the platform on 30 December 2025 and describes the unit as 35–40% efficient at a heat rate of about 9,000 Btu/kWh, deployable in about two weeks, usable for baseload, backup or peaking, and giving the engine a 10–20-year second life, with an input cost based on "near-fully depreciated" engines; the efficiency, heat rate, deployment time and second-life figures are from call coverage, not a filing. First delivery is targeted by the fourth quarter of 2026, production of 100 units in 2027, and capacity "over 100 units annually". If the \$1.465 billion J&F order were 100 units it would imply \$14.65 million per unit, or \$586 per kW, but the unit count is undisclosed. FTAI's 2027 guidance puts Power's Adjusted EBITDA (earnings before interest, tax, depreciation and amortisation; Part X §80), as FTAI adjusts it, at \$450 million; one call summary gives "\$450 million to \$750 million". Part III §18 glossed the term as a new non-aviation sink for the installed base. [FTAI release, 30 Dec 2025; Q4/FY2025 release, 25 Feb 2026; Q1 2026 release, 29 Apr 2026; call coverage, 26 Feb 2026 and 30 Jul 2026; D8 §1, §2.3, §2.9]

EXHIBIT 9.3 — MOD-1 AS FTAI DESCRIBES IT: EACH STATEMENT, ITS DATE, ITS SOURCE AND ITS LEVEL

Statement	Figure	Date	Source	Level
Platform launched	30 Dec 2025	30 Dec 2025	FTAI release; 10-K FY2025	Release; filing
Unit output	25 MW	30 Dec 2025	FTAI release	Release
Annual capacity	"over 100 units annually"	30 Dec 2025	FTAI release	Release
Production start	"expected to begin in 2026"	30 Dec 2025	FTAI release	Release
First delivery	"expected to be delivered by Q4 2026"	25 Feb 2026; repeated 29 Apr 2026	Q4/FY2025 release; Q1 2026 release (both 8-K exhibits)	Release filed as 8-K exhibit

Statement	Figure	Date	Source	Level
2027 production	"planned production of 100 units in 2027"	25 Feb 2026; repeated 29 Apr 2026; 10-Q Q2 2026 "planned 2027 production target of 100 Mod-1 CFM56 aeroderivative units"	Same; 10-Q Q2 2026	Release; filing
Form factor	trailer-mounted, mobile	26 Feb 2026; 22 Jul 2026	Q4 2025 call via TheValueist; J&F release	Call coverage; release
Deployment time	about two weeks on site	26 Feb 2026	Q4 2025 call via TheValueist	Call coverage
Duty	baseload, backup or peaking	26 Feb 2026	Same	Call coverage
Efficiency	35–40%	26 Feb 2026	Q4 2025 call via TheValueist; Aviation Week MRO Memo	Call coverage; trade press
Heat rate	about 9,000 (Btu/kWh implied; heating-value basis not stated)	26 Feb 2026	Same	Call coverage; trade press
Second life	10–20 years	26 Feb 2026	Q4 2025 call via TheValueist	Call coverage
Input cost	"near-fully depreciated CFM56 assets"	26 Feb 2026	Same	Call coverage
Service model	"applying its modular maintenance model to power turbines"	30 Dec 2025	FTAI release	Release
Module capacity linked to Power	3,000 modules a year "supporting its 25% market share goal and 100 Mod-1 units annually"	30 Jul 2026	Q2 2026 call via GuruFocus, Quartr	Call coverage

Statement	Figure	Date	Source	Level
2027 Power Adjusted EBITDA	\$450M within \$2.3B; "\$450 million to \$750 million" in one summary	29 Jul 2026	Q2 2026 release (text confirms "\$450 million"); Quartr, MarketBeat; GuruFocus	Release; call coverage

Source: FTAI Aviation, "FTAI Aviation Announces the Launch of FTAI Power", GlobeNewswire, 30 Dec 2025; FTAI 10-K FY2025; Q4/FY2025 earnings release, 25 Feb 2026 (8-K ex. 99.1); Q1 2026 earnings release, 29 Apr 2026 (8-K ex. 99.1); FTAI 10-Q for the quarter ended 30 Jun 2026; FTAI release of 22 Jul 2026; TheValueist (X), 26 Feb 2026; Aviation Week, "MRO Memo: FTAI Pitches CFM56 To Power AI Boom", early 2026; GuruFocus, Quartr and MarketBeat summaries of the 29–30 Jul 2026 call; orchestrator notes. Reconciliation R-062 and E.1 rule 2.

The distinction in the last column is the one the reconciliation asks every Part to keep. Where a figure came from a release or a filing it is FTAI's statement. Where it came from a summary of a call, it is a third party's rendering of FTAI's statement, and the research recorded that direct quotations on Power "should be taken from the transcripts before publication". [D8 U15; E.1 rule 2]

69.3 CAPACITY, FIRST DELIVERY AND THE 2027 TARGET

The dated sequence is short. The launch release of 30 December 2025 gave capacity for "over 100 units annually". [FTAI release, 30 Dec 2025] The fourth-quarter 2025 results release of 25 February 2026 said: "Development of FTAI Power continues on-track with first Aeroderivative product, FTAI Mod-1, expected to be delivered by Q4 2026 with planned production of 100 units in 2027", and the first-quarter 2026 release of 29 April 2026 repeated it. [FTAI Q4/FY2025 release, 25 Feb 2026; Q1 2026 release, 29 Apr 2026] The second-quarter 2026 report of 29 July 2026 added 2027 guidance with a Power line (\$74.2) and, on the call the next day, coverage reported module capacity of 3,000 a year "supporting its 25% market share goal and 100 Mod-1 units annually". [GuruFocus and Quartr, 30 Jul 2026, call coverage] Part VII §49 treats that 3,000 figure as one of four dated capacity statements with different footprints, and this Part uses it only in §74.4 with that label. [reconciliation R-037; E.1 rule 17]

69.4 SECOND LIFE, INPUT COST AND THE MODULAR SERVICE MODEL

FTAI's aerospace model, described in Part VII §46, holds inventories of serviceable modules and swaps them into a customer's engine so that the engine returns to service quickly. The launch release says FTAI will "provide service support solutions that maximize uptime by applying its modular maintenance model to power turbines". [FTAI release, 30 Dec 2025] Functionally, when a Mod-1 gas generator reaches an inspection or overhaul point, FTAI would exchange the core, or a module of it, rather than overhaul it in place. That is how aeroderivative operators already achieve high availability: an aircraft-derived core is light enough to be swapped in about a day, whereas a frame turbine is overhauled on its foundation over weeks. [D8 §2.5] Maintenance

intervals, service pricing and the contractual form of the service (a long-term agreement or time and materials) are not disclosed. [D8 U12]

MODULE (FTAI POWER USAGE)

In this Part a module is counted as FTAI's releases count it: three per engine, the fan, the core and the low-pressure turbine, which is the convention evidenced by "450 modules (150 engines)" and "1,800 modules (600 engines)" in FTAI's release of 26 February 2025. On that convention, 100 Mod-1 units built from 100 whole engines are about 300 module-equivalents a year drawn from the same production as the aftermarket. Part I §3 counts four major modules anatomically, the three above plus the accessory gearbox, and §74.4 shows what the second convention changes. [FTAI release, 26 Feb 2025, via orchestrator notes; D8 §2.10; reconciliation R-035, B-3; E.1 rule 7]

The input-cost claim can be set beside the engine values Part I §8 and Part V §32 carry, without converting it into a figure FTAI has not given. "Near-fully depreciated" describes an engine's book value on FTAI's balance sheet, not its market price. The market anchors the research holds are a run-out CFM56-7B, meaning one at the end of its usable life, at \$0.8–1.2 million (Safe Fly Aviation, 2026, a consultancy blog), and a half-life market value of \$5.7 million for a -7B24 and \$6.4 million for a -7B27 from the appraiser IBA (September 2025). [Safe Fly Aviation, 2026; IBA, Sept 2025; reconciliation C.8 item 5; E.2] FTAI's actual engine input cost per Mod-1, the engines it has earmarked for Power, and the capital the business requires are undisclosed. [D8 U6] FTAI owned 243 engines in its leasing fleet at 31 December 2025. [FTAI 10-K FY2025; E.2]

69.5 WHAT FTAI HAS NOT SAID

EXHIBIT 9.4 — MOD-1: WHAT IS DISCLOSED AND WHAT IS NOT, WITH WHAT WOULD RESOLVE EACH GAP

Question	Disclosed	Not disclosed	What would resolve it	Research id
Which CFM56 variant	FTAI works on -5B and -7B (10-K)	Whether Mod-1 uses -5B, -7B or both	A product data sheet; a call statement	D8 U3
Engines per unit	25 MW per unit	How many gas generators make one unit; one is assumed by analogy with the LM2500 class	Same	D8 U3

Question	Disclosed	Not disclosed	What would resolve it	Research id
Output arrangement	"power turbines" (release)	Free power turbine or the engine's own low-pressure turbine; whether a booster stage replaces the fan	Same; a patent filing	D8 U3
Emissions and fuel	None	NOx level; after-treatment (SCR or not); heating-value basis of the efficiency; dual-fuel capability	An air-permit application for a Mod-1 site	D8 U5
Build location	Jereh is "a global leader in gas turbine mobile packaging"	Where J&F packages units: China, the United States or elsewhere; tariff exposure; whether J&F has its own facility	FTAI or Jereh filings; a site announcement	D8 U4
Service model	"modular maintenance model" (release)	Inspection and overhaul intervals; contract form; service revenue per unit	A service agreement; a call statement	D8 U12
Capital required	None	Power capital expenditure; engines earmarked; test capacity; engine input cost per unit	The 2026 10-Qs' segment and capital notes	D8 U6
Segment reporting	FY2025 10-K has two reportable segments; 2027 guidance has three lines	Whether Power is a third reportable segment from the Q2 2026 10-Q or sits within Aerospace Products until first delivery	The segment note of the Q1 or Q2 2026 10-Q	D8 U7; D10 U-18; cluster K-10

Source: D8 §7 (unknowns U3–U7, U12); FTAI 10-K FY2025; FTAI Q1 2026 release, 29 Apr 2026; reconciliation D.1, D.2 (cluster K-10) and C.8. The orchestrator notes confirm that the FY2025 10-K has two segments and that the 2027 guidance has three lines; they do not resolve the segment question.

The build-location gap has a consequence the research stated plainly: a Chinese build location would expose the gensets to US tariff and trade-policy risk, and a US location would require Jereh to stand up or expand a facility; neither is documented in anything reached. [D8 §5]

70. J&F Power Systems and the \$1.465B order

FTAI does not sell Mod-1 to the end customer itself. It sells through a venture with a Chinese oilfield-equipment maker, and the venture is the contracting party on the one order announced so far. This section describes the partner, the venture as each parent describes it, the order's terms, and the customer as each source describes it.

70.1 JEREH GROUP

JEREH GROUP (YANTAI JEREH OILFIELD SERVICES GROUP CO., LTD.)

Jereh is a Chinese maker of oilfield equipment listed on the Shenzhen Stock Exchange (SZSE) under code 002353: drilling and oil-and-gas field engineering equipment, equipment maintenance and parts, oilfield engineering services, natural-gas compressors and gas-turbine generator sets. Its gas-turbine genset research began in 2018, and "in 2020, the first 6MW gas turbine genset in China was successfully applied in the well site". It ships trailer-mounted electric hydraulic-fracturing fleets, whose power comes from gas turbines on trailers, to US oilfield customers. FTAI calls it "a global leader in gas turbine mobile packaging". Its subsidiary GenSystems Power Solutions won a \$182 million order (February 2026) and a \$341 million order for gas-turbine generators for US data centers, deliverable by end-2027; whether those use Mod-1 or other turbines is not stated. [PitchBook profile; Jereh company history page; PRNewswire, 2024 and 2025; FTAI Q1 2026 release, 29 Apr 2026; Yicai Global, 2026; D8 §3, §4.2, U8]

The oilfield lineage is the point. Electric hydraulic fracturing runs large pumps from gas turbines mounted on trailers at a well site, moved from site to site, started and stopped often, and fuelled from field gas. That is the same packaging problem as a mobile data-center genset: an enclosure, a trailer, a generator, a fuel skid, controls and a quick connection. [D8 §2.7; PRNewswire, 2024 and 2025] The two GenSystems orders show Jereh selling gas-turbine gensets into US data centers on its own account, in parallel with the venture. The megawatt count of those orders, which would give a dollars-per-kilowatt figure, their customer and their relationship to J&F were not captured. [Yicai Global, 2026; D8 U8]

70.2 J&F: TWO PARENTS, TWO DESCRIPTIONS

J&F POWER SYSTEMS LLC (J&F)

J&F Power Systems is the entity through which Mod-1 is packaged and sold. FTAI's release of 22 July 2026 calls it "FTAI's joint venture with Jereh Group for packaging and distribution of its Mod-1 aeroderivative gas turbine", and FTAI's 10-Q for the quarter ended 30 June 2026 calls it "a strategic packaging and distribution joint venture with Jereh Group, a global leader in gas turbine mobile packaging, to support the planned 2027 production target of 100 Mod-1 CFM56 aeroderivative units". Jereh's Shenzhen exchange disclosure, as reported by FilingReader on the same day, says "its subsidiary, J&F Power Systems" signed the contract. J&F is the counterparty to a five-year master supply agreement and an initial \$1.465 billion purchase order with an unnamed cloud provider. Its ownership split, board control and which parent consolidates it are undisclosed. [FTAI release, 22 Jul 2026; FTAI 10-Q Q2 2026, via orchestrator notes; FilingReader, 22 Jul 2026; D8 §2.7, U2; reconciliation R-061]

The two descriptions are presented side by side, as the reconciliation requires, and neither is preferred. [reconciliation R-061; E.1 rule 29] On one side, FTAI's release and its quarterly filing describe J&F as FTAI's joint venture with Jereh. On the other, Jereh's exchange disclosure, reached through a secondary site rather than the Shenzhen filing itself, describes J&F as Jereh's subsidiary. The research observed that the two are compatible only if Jereh holds a controlling stake or if "subsidiary" is being used loosely. [D8 §2.7, T3] Nothing reached states a percentage.

WHY THE CONSOLIDATION QUESTION DECIDES WHAT THE \$1.465 BILLION MEANS IN FTAI'S ACCOUNTS

A purchase order is placed on J&F, not on FTAI. What appears in FTAI's revenue depends on who consolidates J&F. If FTAI controls and consolidates it, J&F's genset sales to the customer are FTAI's revenue at their full value. If Jereh consolidates it, FTAI's Power revenue is its sales of converted turbines into J&F plus its share of J&F's profit under the equity method (Part VIII §64), and the \$1.465 billion face value never appears as FTAI revenue. If neither controls it, the equity method applies to both. The division of labour FTAI describes, FTAI supplying the converted turbine and Jereh the packaging, is consistent with any of the three. The 10-Q's J&F note, which would settle it, was not returned by the research. [D8 §2.7, U2; reconciliation R-061]

The division of labour implied by FTAI's words is as follows, and it is an implication rather than a disclosed scope of work. FTAI supplies the converted turbine, the CFM56 gas generator with its power-turbine arrangement, drawing on its engine inventory, its module factories and its parts supply, which includes the multi-year materials agreement with CFM signed 22 January 2026 (Part VII §52) and the serviceable-material agreement with AAR (an aviation services company) extended to 2030. [FTAI release, 22 Jan 2026; PRNewswire, 2025; D8 §2.7] Jereh packages the turbine: the trailer or skid, enclosure, generator, gearbox, inlet, exhaust, fuel skid and controls, then tests,

distributes and commissions the complete genset. J&F contracts with the customer. [D8 §2.7] Neither the CFM nor the AAR agreement's body was read by the research. [reconciliation C.8 item 4]

70.3 THE MASTER SUPPLY AGREEMENT AND THE INITIAL ORDER

MASTER SUPPLY AGREEMENT; PURCHASE ORDER; MILESTONE PAYMENTS; PERFORMANCE ADJUSTMENT MECHANISM

A master supply agreement is a framework contract that fixes terms under which a buyer may place binding orders over its life; the orders themselves are purchase orders. Milestone payments tie the buyer's payments to events rather than to delivery alone: here an advance at signing, progress payments through production and testing, and a final payment at on-site commissioning, which FTAI's second-quarter 2026 call described as "derisking working capital". A performance adjustment mechanism is a contractual price change tied to measured performance of the delivered equipment; in this order it is "capped at 10%", and the metrics it applies to and whether it can raise as well as lower the price are undisclosed. [FTAI release, 22 Jul 2026, as reposted by Barchart and Pulse2; Quartr summary of the Q2 2026 call, call coverage; D8 §2.9, §4.1, U10, U11]

COMMISSIONING

Commissioning is the on-site testing and hand-over of installed equipment: the unit is connected to gas and to the customer's switchgear, run, and shown to meet its specification before the customer accepts it. It is the last milestone payment under the J&F order. [FTAI release, 22 Jul 2026; D8 §4.1]

EXHIBIT 9.5 — THE J&F MASTER SUPPLY AGREEMENT AND INITIAL PURCHASE ORDER, TERM BY TERM

Term	As disclosed	Source	Not disclosed
Date	22 Jul 2026	FTAI release; FilingReader on Jereh's SZSE disclosure	—
Contracting party	J&F Power Systems LLC	FTAI release	Ownership of J&F (R-061)
Counterparty	"a leading international cloud service provider" (FTAI release); "a U.S. hyperscaler" (Q2 2026 call summaries)	FTAI release; GuruFocus, Quartr	Name
Framework	Five-year master supply agreement; the customer may issue further purchase orders	FTAI release	Volume commitments beyond the initial order

Term	As disclosed	Source	Not disclosed
Initial order	\$1.465 billion purchase order for Mod-1 mobile gas turbine generator sets	FTAI release; FilingReader	Unit count; price per unit; scope (commissioning, spares, after-treatment, service) (U1)
Delivery	In batches through November 2027	FTAI release	Batch sizes and dates
Share of 2027	"a substantial number" of FTAI Power's targeted 2027 Mod-1 deliveries, the target being 100 units	FTAI release; Q4/ FY2025 release	The number
Payment	Milestone basis: advance at signing; progress payments through production, testing and on-site commissioning	FTAI release	Size of the advance; percentage at each milestone (U10)
Price adjustment	"subject to a performance adjustment mechanism capped at 10%"	Release text as reposted by Barchart and Pulse2; primary release	Metrics; whether symmetric (U11)
Working capital	Described on the Q2 2026 call as "derisking working capital"	Quartr summary, call coverage	—

Source: FTAI Aviation, "FTAI Announces \$1.465 Billion Gas Turbine Generator Set Order Through J&F Power Systems", GlobeNewswire, 22 Jul 2026; Barchart and Pulse2 reposts of the same release, Jul 2026; FilingReader, "Yantai Jereh subsidiary secures \$1.47bn turbine contract", 22 Jul 2026; GuruFocus and Quartr summaries of the 30 Jul 2026 call; FTAI Q4/FY2025 release, 25 Feb 2026; D8 §4.1 and §7. Reconciliation R-061; D8 T1.

Three mechanics in that table deserve a sentence each. The milestone structure means that cash arrives before delivery: an advance at signing funds the engines and packaging that must be bought before any unit is commissioned, which is why the call described it in working-capital terms. [Quartr, 30 Jul 2026, call coverage] The performance adjustment means that up to 10% of the order's value depends on whether delivered units meet whatever the contract measures, which could be output, heat rate, availability or schedule; the cap also bounds the exposure at 10%. [D8 §5, U11] The "substantial number" language ties the order to the 2027 target of 100 units without giving a count, and that one missing number is the input to every price and margin figure in §74.

70.4 THE CUSTOMER

HYPERSCALER

A hyperscaler is a very large cloud-computing operator that builds and runs its own data centers at a scale of gigawatts. The research's sources describe the J&F customer two ways: FTAI's release of 22 July 2026 says "a leading international cloud service provider", and summaries of the second-quarter 2026 call say "a U.S. hyperscaler". The two are not necessarily inconsistent, since a US company can operate internationally, but the primary document does not say "US", and the customer is not named in anything reached. [FTAI release, 22 Jul 2026; GuruFocus and Quartr, 30 Jul 2026, call coverage; D8 §3, T1]

Both descriptions are carried wherever the customer is mentioned in this Part, with their sources, and the primer does not choose between them. [D8 T1; E.1 rule 1]

70.5 WHAT IS NOT DISCLOSED ABOUT THE ORDER

The unit count and the price per unit are not disclosed. [D8 U1] Without them, every per-unit figure in §74 is a derivation under a stated assumption, not a fact. The research recorded that FTAI's third-quarter 2026 report or call, or Jereh's Shenzhen filing if it states quantities, would resolve it. Jereh's shares rose on both the GenSystems and the J&F announcements, which is recorded here as a reported market fact and not as evidence about the order's terms. [Yicai Global, 2026; FilingReader, 22 Jul 2026]

71. The market: data-center demand and the turbine shortage

Mod-1 sells into a market defined by two facts: electricity demand from data centers is growing faster than grids are adding supply, and the factories that make large gas turbines are sold out for years. This section gives the figures for each, with dates and sources, and then the behind-the-meter response that connects them.

71.1 DATA-CENTER ELECTRICITY DEMAND

EXHIBIT 9.6 — DATA-CENTER ELECTRICITY DEMAND: THE FIGURES THE RESEARCH REACHED, WITH DATE AND SOURCE

Measure	Figure	Date	Source and level
Global data-centre electricity consumption by 2030	More than doubles to about 945 TWh, "slightly more than" Japan's total consumption	April 2025	International Energy Agency (IEA), "Energy and AI" (intergovernmental agency report)

Measure	Figure	Date	Source and level
Electricity demand from AI-optimised data centres by 2030	More than quadruples	April 2025	IEA, same
US share	Data centres account for about half of US electricity demand growth to 2030	April 2025	IEA, same
US electricity consumption growth	+1.9% in 2026 and +2.5% in 2027; the largest four-year growth period since 2000, driven primarily by data centers	2026	US Energy Information Administration (EIA) Short-Term Energy Outlook, via Data Center Dynamics (government statistics via trade press)
Regional load growth, 2025–27	ERCOT (the Texas grid operator) about 10% a year; PJM (the mid-Atlantic grid operator) about 3.2% a year	2026	EIA via Data Center Dynamics
Price effect	Data-center demand could drive a 79% rise in ERCOT wholesale electricity prices in 2027	2026	EIA via Utility Dive

Source: IEA, "Energy and AI", executive summary and news release, April 2025; Data Center Dynamics, "EIA: US electricity use to see largest four-year growth period since 2000", 2026; Utility Dive, "Data center demand spike could drive 79% ERCOT price hike in 2027: EIA", 2026; D8 §4.3.

PJM AND ERCOT

PJM (originally Pennsylvania-New Jersey-Maryland) is the grid operator for the mid-Atlantic region of the United States, and the Electric Reliability Council of Texas (ERCOT) is the grid operator for most of Texas. Both run the wholesale electricity markets and the queues for connecting new plants in their regions. The EIA's regional growth figures above and Enverus's finding that half of behind-the-meter data-center capacity sits in these two regions are why they appear in this Part. The definitions are from first principles; the research used the names without defining them. [D8 §2.8, §4.3; registry section 2]

The IEA's 945 TWh is a global figure for all data centres, not a figure for AI alone, and the agency separately says that demand from AI-optimised data centres more than quadruples over the same period. [IEA, Apr 2025] The EIA figures are national and regional load growth, in which data centers are the primary driver rather than the whole; the Texas growth rate of about 10% a year is several times the mid-Atlantic rate. [EIA via Data Center Dynamics, 2026] None of these sources

forecasts how much of the demand will be met by on-site generation; that is Enverus's territory in §71.2.

71.2 BEHIND THE METER: "BRING YOUR OWN POWER"

BEHIND-THE-METER (BTM) GENERATION

Behind-the-meter generation is electricity generated on the customer's side of the utility meter, at the customer's site, so that it never passes through the grid. Enverus estimates that about 40% of installed data-center capacity is behind the meter, that this adds about 1.3 Bcf/d of incremental gas demand by 2030, and that half of it sits in ERCOT and PJM. A separate forecast attributed to the EIA or to S&P Global puts behind-the-meter generation at US industrial sites at 22.5 GW over 2026–2030, about 4.5 GW a year, 88% of it for data centers, with more than 80% of announced on-site generation fuelled by natural gas; the research could not confirm which of the two sources the figure belongs to. The phrase "bring your own power" describes a data-center developer that arrives with its own generation rather than waiting for a grid connection. [Enverus, "Off the grid, on the gas", 2026; EIA Today in Energy #67344 or S&P Global, 2026, attribution unverified; D8 §4.3]

Enverus attributes the behind-the-meter share to "grid interconnection congestion and long development timelines", which §73.3 takes up. [Enverus, 2026] The scale of FTAI's planned output against these figures is a matter of arithmetic.

WORKED EXAMPLE 9.2 — 100 MOD-1 UNITS A YEAR AGAINST THE MARKET-WIDE FIGURES

Inputs: 100 units a year in 2027 (sourced: FTAI Q4/FY2025 release, 25 Feb 2026); 25 MW per unit (sourced: launch release); behind-the-meter additions at US industrial sites of about 4.5 GW a year over 2026–2030, 88% for data centers (sourced: EIA or S&P Global, attribution unverified); GE Vernova backlog plus reservations 116 GW at the second quarter of 2026 (sourced: Power Engineering, Jul 2026); Enverus's 1.3 Bcf/d of incremental behind-the-meter gas demand by 2030 (sourced); 4.9 MMcf/d per unit at a 90% capacity factor (derived in worked example 9.1, with its assumptions).

$$\begin{aligned} \text{annual output capacity} &= 100 \times 25 \text{ MW} = 2,500 \text{ MW} = 2.5 \text{ GW} \\ \text{share of behind-the-meter additions} &= 2.5 \div 4.5 = 56\% \text{ of one year's} \\ &\quad \text{forecast additions} \\ \text{data-center share of those additions} &= 4.5 \times 0.88 \approx 4.0 \text{ GW a year}; 2.5 \div \\ &\quad 4.0 = 63\% \\ \text{share of GE Vernova's backlog plus reservations} &= 2.5 \div 116 = 2.2\% \\ \text{gas for 100 units} &= 100 \times 4.9 \text{ MMcf/d} = 490 \text{ MMcf/d} \approx 0.49 \text{ Bcf/d} \\ \text{share of Enverus's incremental gas} &= 0.49 \div 1.3 = 38\% \end{aligned}$$

So 100 units a year is a small fraction of what the frame-turbine makers have on order and a large fraction of one forecast of US behind-the-meter additions, and the gas those units would burn is a

large share of one forecast of incremental on-site gas demand. The two forecasts are not the same population: the 4.5 GW figure covers industrial sites in the United States, and the J&F customer is "international" in FTAI's words. Limits of the example: the 4.5 GW figure's source is unverified; the Enverus gas figure is an estimate to 2030, not an annual flow; the gas per unit rests on worked example 9.1's assumed capacity factor; and the units' location is unknown. These percentages are derived in this Part and are not published by any source. [D8 §2.10, §5; Enverus, 2026; Power Engineering, Jul 2026]

71.3 THE TURBINE SHORTAGE

The reason a buyer considers a 25 MW unit built from a used airline engine is that the machines it would otherwise buy are not available for years.

SLOT RESERVATION AGREEMENT

A slot reservation agreement is a paid reservation of a future turbine manufacturing slot, made before a firm order. GE Vernova counts reservations together with firm backlog in the figures it reports, so its 116 GW at the second quarter of 2026 and the 125 GW it expects by December 2026 are backlog plus reservations, and the company is selling slots for 2031 delivery. The term is distinct from a shop's induction slot (Part IV §27) and a parts maker's production slot (Part VI §41). [Power Engineering, Jul 2026; Industrial Info Resources, 2026; D8 §3; registry C-18]

EXHIBIT 9.7 — GAS-TURBINE ORDER BOOKS AND LEAD TIMES: A DATED SERIES, BY MANUFACTURER

Manufacturer	Measure	Figure	As of	Source and level
GE Vernova	Gas backlog	50 GW	2025	Yahoo Finance (financial press)
GE Vernova	Gas equipment backlog plus slot reservation agreements	83 GW	End-2025	Power Engineering, Jul 2026 (trade press)
GE Vernova	Same	100 GW	Q1 2026	Industrial Info Resources, 2026 (industry data vendor)
GE Vernova	Same	116 GW	Q2 2026	Power Engineering, Jul 2026, citing the 22 Jul 2026 call
GE Vernova	Same, expected	At least 125 GW	By Dec 2026	Power Engineering, Jul 2026

Manufacturer	Measure	Figure	As of	Source and level
GE Vernova	Slots	Selling 2031 delivery slots; more than half of 2031 expected contracted by end-2026	Jul 2026	Power Engineering, Jul 2026
GE Vernova	Earlier characterisation	"2026 and 2027 largely sold out"; booking 2028 and 2029	2026 (undated within year)	Oilprice (energy press)
GE Vernova	Outlook	Raised multi-year financial outlook	9 Dec 2025	GE Vernova release (company)
Siemens Energy	Gas-turbine backlog	69 GW, after booking 15 GW and shipping 6 GW in the quarter	30 Jun 2026 (fiscal Q3)	Oilprice; Energy News Beat, 2026
Siemens Energy	Lead times	"three years or more"	2026	Oilprice
Mitsubishi Power (Mitsubishi Heavy Industries)	Large-frame backlog	35 GW; quarter's orders for delivery 2028–2030; "selective in the projects we contract"	2026	Oilprice
Industry	Combined-cycle plant lead time	Five years in 2025, up from 3.5 years in 2023	2025	Oilprice
Industry	Global gas-turbine orders against manufacturing capacity	110 GW of orders at end-2025 against 60–70 GW a year of capacity	End-2025	Wood Mackenzie via Oilprice (consultancy via energy press)

Source: Yahoo Finance, "GE Vernova Backlog Hits 50 GW", 2025; Power Engineering, "Data centers drive record surge in GE Vernova power equipment orders as turbine slots tighten through 2030", Jul 2026; Industrial Info Resources, 2026; Oilprice, "The Gas Turbine Shortage Just Became AI's Biggest Constraint", 2026, and "Global Gas Turbine Orders Hit Record High as Power Demand Surges"; Energy News Beat, 2026; GE Vernova, BusinessWire release, 9 Dec 2025; D8 §3 and §4.4. Reconciliation D8 T4.

The GE Vernova rows are a time series, not a contradiction: 50 GW, then 83, 100 and 116 GW, with the later figures including slot reservations as well as firm orders. [D8 T4] The two characterisations of its sold-out horizon, "2028 and 2029" in one source and 2031 slots in another, sit at different dates in a year in which the book grew by a third, so a reader quoting either should give its date and its definition. [Oilprice, 2026; Power Engineering, Jul 2026; D8 T4] Wood

Mackenzie's comparison of 110 GW of orders against 60 to 70 GW of annual capacity is the simplest statement of the shortage: at end-2025 the industry had taken orders for roughly a year and a half to two years of its output. [Wood Mackenzie via Oilprice, 2026] A buyer that needs power in 2026 or 2027 cannot get a new frame turbine, and that is the opening for aeroderivatives, reciprocating engines and fuel cells. [D8 §5]

72. Competing supply: aeroderivatives, frame turbines, reciprocating engines, fuel cells

This section takes each alternative to Mod-1 in turn, with output, lead time and price where published, and then collects the dollars-per-kilowatt benchmarks the research reached. The frame turbines themselves were covered in §71.3; their relevance here is that they are the machines a buyer would prefer at scale and cannot get.

GE Vernova LM2500XPRESS. GE Vernova's trailer-deliverable version of the LM2500 is rated 35 MW per unit in the company's release. Crusoe, a data-center developer, ordered 10 units in December 2024 and 19 in June 2025, "nearly 1 GW" combined, for AI data centers, and the packages have been delivered. [GE Vernova release, 22 Jul 2025; Turbomachinery International, undated] Twenty-nine units at about 1 GW implies about 34.5 MW each, and a commissioning brief gives 34 MW; ratings vary with site conditions and "nearly" 1 GW is rounded, so the research carries 35 MW as GE's figure and the lower numbers as implied. [D8 T6] No price or lead time was captured.

GE Vernova LM6000. The CF6-80C2-derived LM6000 is in the 48 to 50 MW class. Trade coverage quotes its waiting list at "anywhere from three to five years". [EEPower, undated; Data Centre Magazine, undated] That the manufacturer's own aeroderivative is itself sold out for years is the context for non-manufacturer converters such as ProEnergy and FTAI. [D8 §5]

Siemens Energy SGT-A65. The SGT-A65 is Siemens Energy's aeroderivative derived from the Rolls-Royce Trent. Its output and lead time were not captured by the research, and nothing is quoted here. [D8 U9]

Solar Turbines (Caterpillar). Solar's mobile Titan-class units appear in the public record through xAI's Memphis permit, which allowed 15 "Solar SMT-130 generators" totalling up to 247 MW, about 16.5 MW each. [TechCrunch, 3 Jul 2025] Figures for the larger Titan 350 were not captured. [D8 U9]

ProEnergy PE6000. ProEnergy, a private company, builds the PE6000 at 48 MW on CF6-80C2 cores taken from retired Boeing 747 engines, "the same engine core GE Vernova uses in its LM6000". It quotes 2027 delivery and has sold 21 turbines for two data-center projects totalling

more than 1 GW, as bridging power for five to seven years until grid interconnection. [Data Centre Magazine, undated; EEPower, undated] It is the closest analogue to FTAI Power: a non-manufacturer reusing retired airline cores, limited by the supply of suitable cores and by packaging capacity. [D8 §3, §5] No price was captured.

Baker Hughes NovaLT. Baker Hughes's NovaLT line is a family of small industrial turbines, not aeroderivatives. Dynamis Power Solutions ordered 76 NovaLT16 units, about 1.3 GW, for data-center and oil-and-gas power, which implies about 17 MW each. [World Oil, 29 Jul 2026] Baker Hughes doubled its data-center order target to \$3 billion over 2025–2027 and plans to double NovaLT capacity by the first half of 2027. [Bloomberg Government, undated] No price was captured.

Wärtsilä and Caterpillar reciprocating engines. Wärtsilä builds data-center plants from 10 to 23 MW gas piston engines reaching more than 450 MW per site, and claims electrical efficiency over 50%, 20–35% less fuel than gas turbines, and 20–30% lower capital cost than turbines once reserve capacity is included; all are vendor claims from a product page. [Wärtsilä product page, undated] It announced a 507 MW US data-center order on 20 November 2025. [Wärtsilä release, 20 Nov 2025] Caterpillar's gas gensets were not captured. [D8 U9] The efficiency claim and FTAI's 35–40% figure are consistent with each other at the lower end of Wärtsilä's stated gap, but both are self-reported and the research did not test either. [D8 T10]

Bloom Energy fuel cells. Bloom's annual report positions its solid-oxide fuel cells against reciprocating engines for on-site power, including "the peak loads associated with AI data centers". [Bloom Energy 10-K FY2024] Deal sizes and dollars per kilowatt were not captured, so no comparison is drawn. [D8 U9]

EXHIBIT 9.8 — COMPARABLE UNITS AND ORDERS: OUTPUT, DELIVERY AND THE LARGEST PUBLISHED ORDER FOR EACH

Unit	Type	Output per unit	Lead time or delivery	Notable order	Source
FTAI Mod-1 (CFM56)	Aeroderivative, non-OEM, mobile	25 MW	First by Q4 2026; 100 in 2027	\$1.465B, batches to Nov 2027, unit count undisclosed	FTAI releases, 30 Dec 2025, 25 Feb 2026, 22 Jul 2026
GE Vernova LM2500XPRESS	Aeroderivative, OEM, trailer-deliverable	35 MW (GE); about 34.5 MW implied	Delivered 2025–26	Crusoe, 29 units, nearly 1 GW	GE Vernova, 22 Jul 2025; Turbomachinery International

Unit	Type	Output per unit	Lead time or delivery	Notable order	Source
GE Vernova LM6000 (CF6-80C2)	Aeroderivative, OEM	About 48–50 MW	Waiting list three to five years	—	EEPower; Data Centre Magazine
ProEnergy PE6000 (CF6-80C2)	Aeroderivative, non-OEM	48 MW	2027	21 units, more than 1 GW, two projects	EEPower; Data Centre Magazine
Siemens Energy SGT-A65 (Trent)	Aeroderivative, OEM	Not captured	Not captured	—	D8 U9
Baker Hughes NovaLT16	Small industrial turbine	About 17 MW (1.3 GW over 76 units)	Capacity doubling by 1H 2027	Dynamis, 76 units, about 1.3 GW	World Oil, 29 Jul 2026; Bloomberg Government
Solar Turbines SMT-130 (mobile)	Small industrial turbine, mobile	About 16.5 MW (247 MW over 15 units)	In service 2024–25	xAI Memphis, 15 permitted (up to 35 operated)	TechCrunch, 3 Jul 2025; Data Center Dynamics, 2025
Wärtsilä gas engines	Reciprocating	10–23 MW each; plants over 450 MW	—	507 MW US data-center order, 20 Nov 2025	Wärtsilä product page and release
Caterpillar gas gensets	Reciprocating	Not captured	Not captured	—	D8 U9
Bloom Energy fuel cells	Solid-oxide fuel cell	Not captured	Not captured	Not captured	Bloom Energy 10-K FY2024; D8 U9

Source: as listed in each row; compiled in D8 §4.5. Reconciliation D8 T6 (LM2500XPRESS rating). The per-unit figures marked "about" are divisions of a published total by a published unit count, done in the research.

72.1 WHAT A TURBINE COSTS PER KILOWATT

Dollars per kilowatt is the industry's unit price for generating equipment: the cost divided by the rated output in kilowatts. The research reached four benchmarks, and they measure four different things.

EXHIBIT 9.9 — DOLLARS-PER-KILOWATT BENCHMARKS REACHED BY THE RESEARCH, WITH SCOPE AND DATE

Benchmark	\$/kW	What it measures	Date	Source and level
Simple-cycle gas turbine, equipment only	About \$1,150 at 1 MW falling to about \$171 at about 600 MW	Bare turbine equipment; price per kW falls steeply with unit size	Undated; pre-2024 pricing	Gas Turbine World (trade publication)
100 MW aeroderivative genset, installed	\$1,175	A complete installed aeroderivative genset of 100 MW	Undated; pre-2024 pricing	Gas Turbine World
Gas-turbine prices, projected	\$600 by end-2027, "nearly tripling from 2019 levels"	Turbine equipment only	Projection to end-2027, made 2026	Wood Mackenzie via Oilprice (consultancy via energy press)
Gas capacity in regulatory filings	\$1,100–1,400 "two years ago"; \$2,000–2,500 "now"	Whole plants as approved by utility regulators, predominantly large frame and combined-cycle	2026	Power Engineering (trade press)
Headline price inflation	+195%	Gas-turbine prices, scope as the headline gives it	2026	Power Engineering
Mod-1, implied	\$586 at 100 units; \$733 at 80; \$977 at 60	A delivered mobile genset, if the \$1.465B order is the stated number of units; scope of the order unknown	Derived, 2026	D8 §2.9; worked example 9.4 below

Source: Gas Turbine World, "How much does it cost to build a Simple Cycle or Combined Cycle plant?" and "Capital Costs for Utility Scale Gas Turbine Plants", undated; Oilprice, "The Gas Turbine Shortage Just Became AI's Biggest Constraint", 2026, citing Wood Mackenzie; Power Engineering, "Gas turbine prices climb 195% as supply crunch reshapes power development", 2026; D8 §2.9 and §4.4. Reconciliation D8 T5.

The scopes do not line up, and the reconciliation asks that they never be collapsed into one figure. [D8 T5; E.1 rule 1] The Gas Turbine World figures are equipment-only for the turbine, or installed for a complete genset, at pre-2024 prices; Wood Mackenzie's is a projection for turbine equipment alone; Power Engineering's is for whole plants as regulators approve them, which includes site, civil works, interconnection and in most cases a steam cycle. A delivered mobile genset sits between a bare turbine and a plant: it includes the packaging and generator but not the site, the gas lateral or the grid connection. The Gas Turbine World size curve also matters on its

own: a small unit costs several times more per kilowatt than a large one, and a 25 MW unit sits near the expensive end of that curve. [Gas Turbine World, undated]

WORKED EXAMPLE 9.3 — WHAT EACH BENCHMARK WOULD IMPLY FOR A 25 MW UNIT

Inputs: 25 MW = 25,000 kW (sourced: launch release); the five benchmark ranges in Exhibit 9.9 (sourced as shown, scopes differ).

Gas Turbine World equipment-only, small-unit end:	$25,000 \times \$1,150 = \28.8
	million
Gas Turbine World equipment-only, large-unit end:	$25,000 \times \$171 = \4.3
	million
Gas Turbine World 100 MW aero genset, installed:	$25,000 \times \$1,175 = \29.4
	million
Wood Mackenzie turbine-only, end-2027:	$25,000 \times \$600 = \15.0
Power Engineering plant-level, current filings:	$25,000 \times \$2,000 \text{ to } \$2,500$
	$= \$50.0 \text{ to } \62.5 million

So the published benchmarks, applied to a 25 MW rating, span \$4.3 million to \$62.5 million per unit, and the spread is mostly a difference of scope and size rather than of price. The \$171 figure is for a 600 MW frame and does not describe any 25 MW machine; the \$1,150 and \$1,175 figures are for small units and complete aero gensets at older prices; the \$600 figure is the one turbine-only projection for the period in which Mod-1 units would be delivered; and the \$2,000 to \$2,500 figures include everything a mobile genset leaves out. §74.1 sets the implied Mod-1 price inside this range. Limits of the example: the benchmarks are undated or projected, the scopes differ, and none of them is for a 25 MW aeroderivative sold in 2026 or 2027, which the research listed as an unknown. [Gas Turbine World, undated; Wood Mackenzie via Oilprice, 2026; Power Engineering, 2026; D8 T5, U9]

73. Permitting and interconnection

A mobile genset can be built in weeks, but putting it to work at a data center runs through three processes that are not under the seller's control: gas supply, an air permit and, where the site also uses the grid, an interconnection agreement. The first was covered in §68.3. This section takes the other two, using the one documented US case the research reached.

73.1 THE AIR PERMIT

AIR PERMIT

An air permit is the authorisation, issued under the US Clean Air Act by the state or county agency that implements it, to construct and operate a source of air emissions. A stationary source needs a construction and operating permit before it runs, and the permit sets the emission limits and controls the source must meet. In the Memphis case the sequence ran roughly a year from first operation in mid-2024 to permit application in the winter of 2024 to a permit in July 2025. [TechCrunch, 3 Jul 2025; Tennessee Lookout, 2025; D8 §2.4, §2.8]

INTERCONNECTION AGREEMENT; INTERCONNECTION QUEUE

An interconnection agreement is the contract between a generator or a large customer and the utility or grid operator that governs its connection to the grid. The interconnection queue is the waiting list of projects seeking such agreements, which in the United States runs to years. The queue is the reason behind-the-meter generation exists: Enverus attributes the on-site share of data-center power to "grid interconnection congestion and long development timelines", and ProEnergy's customers plan on five to seven years before they expect a connection. Specific queue durations for PJM and ERCOT were not captured. [Enverus, 2026; Data Centre Magazine, undated; D8 §2.8, U14]

73.2 THE MEMPHIS PRECEDENT

The one deployment of mobile gas turbines at a data center that the research documented in detail is xAI's site in Memphis, Tennessee. It is carried here as a precedent for the process, not as a statement about FTAI, whose units have not yet been deployed.

EXHIBIT 9.10 — THE XAI MEMPHIS TIMELINE: MOBILE TURBINES, PERMIT, DISPUTE

When	What happened	Source
Mid-2024	xAI begins operating mobile gas turbines at its Memphis data center without a stationary-source air permit	TechCrunch, 3 Jul 2025; Tennessee Lookout, 2025
Winter 2024	Permit application filed with Shelby County	Tennessee Lookout, 2025
2025 (date within year not captured)	The number of turbines on site rises to as many as 35, "in violation of permit limits" per the headline, against a permit eventually written for 15	Data Center Dynamics, 2025
2025	The Southern Environmental Law Center and the NAACP (the National Association for the Advancement of Colored People) bring a Clean Air Act suit over the unpermitted turbines	Data Center Dynamics, 2025

When	What happened	Source
3 Jul 2025	Shelby County issues a permit allowing xAI "to operate 15 Solar SMT-130 generators with certain emissions controls", up to 247 MW	TechCrunch, 3 Jul 2025; E&E News, Jul 2025
Ongoing	Dispute over whether the temporary turbines were "nonroad engines" exempt from permitting, which Shelby County accepted and the plaintiffs contest	Data Center Dynamics, 2025; Tennessee Lookout, 2025; E&E News, Jul 2025

Source: TechCrunch, "xAI gets permits for 15 natural gas generators at Memphis data center", 3 Jul 2025; E&E News, "Musk's xAI gets air permit for Memphis supercomputer", Jul 2025; Data Center Dynamics, "xAI facing lawsuit over use of gas turbines at Memphis supercomputer" and "xAI doubles number of onsite gas turbines at Memphis data center in violation of permit limits", 2025; Tennessee Lookout, 2025; D8 §2.4, §2.8. Reconciliation D8 T9.

NONROAD ENGINE

A nonroad engine is a category in US air rules for engines that are mobile or temporary, generally meaning at one location for less than twelve months, and that are exempt from the permitting that applies to stationary sources. Whether trailer-mounted turbines at a data center qualify is contested. In Memphis, Shelby County accepted that xAI's temporary turbines were nonroad engines exempt from permitting. The Southern Environmental Law Center and the NAACP contest that reading as allowing an operator to "install and operate any number of new polluting turbines at any time without any written approval ... without pollution controls". The question is unresolved in litigation. [Data Center Dynamics, 2025; Tennessee Lookout, 2025; E&E News, Jul 2025; D8 §2.4, T9]

The two readings are carried without preference. [D8 T9; E.1 rule 1] On one side, a county air agency treated mobile turbines as exempt from stationary-source permitting while they were temporary. On the other, two organisations argue in court that they are stationary sources needing permits and controls. The research noted the consequence for Mod-1 in neutral terms: the outcome "could either widen or close the fast-deployment route Mod-1 is designed for". [D8 §5] A unit that must hold a stationary-source permit before it runs cannot be deployed in two weeks unless the permit was obtained in advance; a unit treated as a nonroad engine can.

73.3 INTERCONNECTION WAITS AND BRIDGING

BRIDGING POWER

Bridging power is temporary on-site generation used until a grid interconnection is available. ProEnergy's customers describe their turbine plants as "bridging power for five to seven years, which is when they expect to have grid interconnection". If queues shorten, the bridging market shrinks; if they lengthen, turbines bought as bridges become long-term plant. [Data Centre Magazine, undated; EEPower, undated; D8 §2.8, §5]

The five-to-seven-year figure is the only quantification of the interconnection wait the research reached, and it comes from one supplier's description of its customers' plans rather than from a grid operator's queue statistics. [D8 U14] It bears on Mod-1 in two ways. A 25 MW mobile unit with a stated second life of 10 to 20 years outlasts a five-to-seven-year bridge, so a buyer either keeps it as permanent plant, moves it to a new site, which is what mobility is for, or resells it. And the milestone payments and the five-year master agreement in §70.3 run on a shorter clock than the bridge, so the contract and the asset life are not the same length. These are descriptions of the time scales involved, not predictions.

74. Worked examples: implied unit price; CFM56 cores consumed

Everything quantitative that can be said about FTAI Power today is a derivation from a small set of disclosed numbers under stated assumptions, because the two numbers that would make it arithmetic rather than inference, the unit count in the order and the engines per unit, are undisclosed. This section does the derivations, labels every input, and shows each result under each base where the research holds more than one.

74.1 IMPLIED UNIT PRICE AND DOLLARS PER KILOWATT

WORKED EXAMPLE 9.4 — THE IMPLIED PRICE OF A MOD-1 UNDER THREE UNIT-COUNT ASSUMPTIONS

Inputs: order value \$1.465 billion (sourced: FTAI release, 22 Jul 2026); 2027 production target 100 units (sourced: Q4/FY2025 release, 25 Feb 2026); the order covers "a substantial number" of 2027 deliveries (sourced: FTAI release, 22 Jul 2026); unit count in the order (not disclosed; three assumptions used: 100, 80 and 60); output 25 MW = 25,000 kW (sourced: launch release).

at 100 units: $1,465,000,000 \div 100 = \14.65 million per unit; $14,650,000 \div 25,000 \text{ kW} = \$586/\text{kW}$
 at 80 units: $1,465,000,000 \div 80 = \18.3 million per unit; $\div 25,000 = \$733/\text{kW}$
 at 60 units: $1,465,000,000 \div 60 = \24.4 million per unit; $\div 25,000 = \$977/\text{kW}$

Against the benchmarks in Exhibit 9.9 and worked example 9.3: \$586 to \$977 per kW for a delivered mobile genset sits below the \$2,000 to \$2,500 per kW of current plant-level regulatory filings, near or above Wood Mackenzie's \$600 per kW turbine-only projection for end-2027, and below Gas Turbine World's \$1,150 to \$1,175 per kW for small units and installed aero gensets at older prices. The research called the comparison loose, because the scopes differ. Limits of the example: the unit count is the single undisclosed input and the result scales with it; the order's scope (commissioning, spares, after-treatment, service) is unknown, so the price may cover more than the genset; and 100

units is the whole 2027 target, whereas the release says only "a substantial number". [D8 §2.9, U1; reconciliation D8 T5]

EXHIBIT 9.11 — IMPLIED MOD-1 UNIT PRICE AND \$/KW BY ASSUMED UNIT COUNT, BESIDE THE PUBLISHED BENCHMARKS

Assumed units in the \$1.465B order	Implied price per unit	Implied \$/kW at 25 MW	Nearest published benchmarks (scope)
100 (the whole 2027 target)	\$14.65M	\$586	Wood Mackenzie \$600 turbine-only, end-2027 projection
80	\$18.3M	\$733	Between Wood Mackenzie turbine-only and Gas Turbine World small-unit equipment-only (\$1,150)
60	\$24.4M	\$977	Approaching Gas Turbine World small-unit equipment-only (\$1,150) and installed 100 MW aero genset (\$1,175), pre-2024
Any	—	—	Plant-level filings \$2,000–2,500 (whole plant, 2026) sit above every row

Source: FTAI release, 22 Jul 2026 (order value); FTAI Q4/FY2025 release, 25 Feb 2026 (100 units); Gas Turbine World, undated; Wood Mackenzie via Oilprice, 2026; Power Engineering, 2026. Unit counts are assumptions; the implied figures are derived in D8 §2.9 and this Part and are not disclosed by FTAI or J&F. Reconciliation D8 T5.

74.2 IMPLIED 2027 POWER EBITDA PER UNIT

FTAI's second-quarter 2026 report of 29 July 2026 guided 2027 business-segment Adjusted EBITDA (earnings before interest, taxes, depreciation and amortisation, as FTAI adjusts it; Part X §80) of \$2.3 billion: Aerospace Products \$1.4 billion, FTAI Power \$450 million and Aviation Leasing \$450 million. The release text confirms \$1.4 billion and "\$450 million"; the three-way split is as the dossiers and the orchestrator notes carry it. [FTAI Q2 2026 release, 29 Jul 2026; Quartr and MarketBeat, 30 Jul 2026, call coverage; orchestrator notes; D10 §4.8] One summary of the same call, by GuruFocus, gives the Power figure as "\$450 million to \$750 million". [GuruFocus, 30 Jul 2026, call coverage; reconciliation R-062] The research took the two as likely a base figure and an upside case stated on the same call, with the transcript not reached, and both are carried here as the reconciliation requires. [D8 T2; E.1 rule 30] Part X §84 carries the guidance history.

WORKED EXAMPLE 9.5 — IMPLIED 2027 POWER ADJUSTED EBITDA PER UNIT, AND WHAT IT WOULD MEAN AS A MARGIN

Inputs: 2027 Power Adjusted EBITDA \$450 million (sourced: Q2 2026 release text per the orchestrator notes; Quartr and MarketBeat call coverage) or \$450–750 million (sourced: GuruFocus call coverage); 2027 production 100 units (sourced: Q4/FY2025 release); implied revenue per unit

\$14.65 million if the order is 100 units and FTAI consolidates J&F (derived in worked example 9.4; both conditions undisclosed).

EBITDA per unit at \$450M: $450,000,000 \div 100 = \$4.5$ million
 EBITDA per unit at \$750M: $750,000,000 \div 100 = \$7.5$ million
 margin if revenue per unit were \$14.65M: $4.5 \div 14.65 = 31\%$; $7.5 \div 14.65 = 51\%$

So the guided figure is \$4.5 million of Adjusted EBITDA per unit if 100 units are delivered in 2027, or \$7.5 million at the top of the one summary's range. The margin lines hold only if FTAI's recognised revenue per unit is the full order value per unit, which requires both that the order is 100 units and that FTAI consolidates J&F. If Jerih consolidates J&F, FTAI's revenue is its turbine sale into the venture plus its share of the venture's profit, a smaller base on which the same EBITDA is a higher margin, and the figure cannot be derived from anything disclosed. If fewer than 100 units are delivered in 2027, EBITDA per unit is higher than \$4.5 million on the same guidance. Limits of the example: the unit count, the consolidation, the revenue recognition and whether Power is a reportable segment are all undisclosed; the \$450M is a guidance figure, not an outturn; and the \$450–750M range is one call summary. [D8 §2.9, U1, U2, U7; reconciliation R-061, R-062]

74.3 CFM56 CORES CONSUMED

The feedstock question connects this Part to Parts I, III and V. Each Mod-1 removes a CFM56 gas generator from the pool of engines that would otherwise fly, be torn down for parts or be rebuilt as modules. FTAI has not said how many CFM56 engines go into one unit. The simplest reading, and the one the research used, is one gas generator per 25 MW unit, by analogy with the other single-core aeroderivatives in this class: the LM2500 is one TF39 or CF6-6 core for 25 to 35 MW. [D8 §2.10, U3; GE Vernova, 22 Jul 2025] On that assumption, 100 units in 2027 is 100 CFM56 engines a year, and "over 100 units annually" at capacity is more than that. [FTAI release, 30 Dec 2025; Q4/FY2025 release, 25 Feb 2026]

The installed base and the retirement rate are the two denominators, and both are contested. Part III §18 and §19 carry the full treatment; the binding figures are repeated here. The installed base is about 23,000 in service (CFM website, undated), about 24,000 in service as of September 2025 (Aviation Business News, trade press), and more than 22,800 in Safran's civil installed base at end-2025 (Safran FY2025 results presentation, 13 February 2026), out of more than 35,000 delivered (CFM). [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; reconciliation R-001; E.2] Safe Fly Aviation, a consultancy blog, gives about 14,200 "active" engines, which one dossier read as all variants (about 6,200 -5B and 7,500 -7B) and another as the -7B fleet alone; it is a consultancy figure and never the fleet figure in this primer. [Safe Fly Aviation, 2026; reconciliation R-001, R-002; E.1 rule 14] The research's own feedstock arithmetic used the Safe Fly base, and the reconciliation asks that the OEM-level bases be shown beside it. [reconciliation C.8 item 1] The

research also described total deliveries as "well over 30,000" as general knowledge; the OEM figure of more than 35,000 is the one carried. [reconciliation R-069]

The retirement rate has its own sequence. GE's planning assumption for 2026 moved from 3–4% of the fleet, to 2–3%, to 1.5–2% (GE's chief financial officer via Aviation Week, 2026). GE reported about 1.5% retired in 2025, broadly the same as 2024, and expected about 2% in 2026, "below the 2–3% range previously anticipated" (GE fourth-quarter 2025 results via Leeham, 22 January 2026). Realised first-quarter 2026 retirements were "below 1%" (GE Aerospace investor-relations page, 2026). Safe Fly gives "near 2% per year" on its own base. [GE CFO via Aviation Week, 2026; Leeham, 22 Jan 2026; GE Aerospace investor-relations page, 2026; Safe Fly Aviation, 2026; reconciliation R-004; E.2]

WORKED EXAMPLE 9.6 — 100 CFM56 ENGINES A YEAR AGAINST EACH INSTALLED-BASE FIGURE AND EACH RETIREMENT RATE

Inputs: 100 engines a year (derived: 100 units × one gas generator per unit, the engines-per-unit figure being undisclosed); installed base about 24,000 (Aviation Business News, Sept 2025), about 23,000 (CFM website, undated), more than 22,800 (Safran, end-2025), and about 14,200 (Safe Fly, consultancy blog, 2026, basis contested); retirement rates 1.5% (GE, 2025 actual), 2% (GE, 2026 expected; Safe Fly), and "below 1%" (GE, first quarter 2026 realised, taken here as 1%).

share of the base: $100 \div 24,000 = 0.42\%$; $100 \div 23,000 = 0.43\%$; $100 \div 22,800 = 0.44\%$; $100 \div 14,200 = 0.70\%$
 retirements at 1.5%: $24,000 \times 0.015 = 360$; $23,000 \times 0.015 = 345$; $22,800 \times 0.015 = 342$
 retirements at 2%: $24,000 \times 0.02 = 480$; $23,000 \times 0.02 = 460$; $22,800 \times 0.02 = 456$; $14,200 \times 0.02 = 284$
 retirements at 1%: $24,000 \times 0.01 = 240$; $23,000 \times 0.01 = 230$
 100 engines as a share of retirements, OEM-level bases: $100 \div 480 = 21\%$ to $100 \div 342 = 29\%$ at 1.5–2%; $100 \div 240 = 42\%$ to $100 \div 228 = 44\%$ at 1%
 100 engines as a share of retirements, Safe Fly base at 2%: $100 \div 284 = 35\%$

So 100 engines a year is under half of one percent of the installed base on any OEM-level count, and between about a fifth and about three-tenths of a year's retirements at GE's 2025 actual and 2026 expected rates; it rises to about two-fifths of retirements if the first quarter of 2026's sub-1% rate were sustained for a year, and it is about 35% on the consultancy base the research itself used. The share scales inversely with the base and with the rate, which is why both are shown. Two cautions are in the mechanics. First, retirement is not the only source of cores: FTAI holds 243 engines in its leasing fleet (10-K FY2025), it buys engines and modules for its aerospace business, and the research states that what limits feedstock is the condition and price of retired cores, which Part V §33 covers. Second, if the 14,200 figure counts only -7B engines (one dossier's reading), the 35% share describes one variant and not the fleet. Limits of the example: engines per unit is undisclosed; the bases may include spare and stored engines; the retirement rates are GE's for its CFM56 fleet and Safe Fly's on

its own count; and the sub-1% figure is one quarter annualised. [D8 §2.10; reconciliation R-001, R-002, R-004, C.8 item 1; E.1 rules 13 and 14; FTAI 10-K FY2025]

EXHIBIT 9.12 — 100 MOD-1 CORES A YEAR AS A SHARE OF CFM56 RETIREMENTS, UNDER EACH BASE AND RATE

Installed-base figure (source, level)	Retirements at 1.5% (GE, 2025 actual)	Retirements at 2% (GE, 2026 expected)	Retirements at 1% (GE, 1Q 2026 "below 1%", annualised)	100 cores as share at 1.5% / 2% / 1%
~24,000 (Aviation Business News, Sept 2025, trade press)	360	480	240	28% / 21% / 42%
~23,000 (CFM website, undated, OEM page)	345	460	230	29% / 22% / 43%
>22,800 (Safran FY2025 results, 13 Feb 2026, OEM investor material)	342	456	228	29% / 22% / 44%
~14,200 (Safe Fly Aviation, 2026, consultancy blog; all variants per D8, -7B only per D3)	213	284 (Safe Fly's own "near 2%")	142	47% / 35% / 70%

Source: installed-base figures per reconciliation E.2 (CFM International website, undated; Aviation Business News, Sept 2025; Safran FY2025 Results & Investor Update, 13 Feb 2026; Safe Fly Aviation, "CFM56 Engine Market Report 2026"); retirement rates per GE Q4 2025 results via LeeHam, 22 Jan 2026, GE CFO via Aviation Week, 2026, and GE Aerospace investor-relations page, 2026. All products and shares are derived in this Part; one gas generator per unit is assumed (D8 U3). Reconciliation R-001, R-002, R-004, R-069.

Two further comparisons the research made are carried with their labels. First, GE Aerospace expected CFM56 removals in the first half of 2027 to be about 10% above 2026 (GE via Aviation Week), which bears on the supply of cores in the year Mod-1 production is planned to reach 100. [Aviation Week, "CFM56 Overhaul Demand Remains Strong", undated within 2026] Second, Part III §24's row on the CFM56 plateau and later decline describes retirements normalising later in the decade, which is the period in which the stock of retired cores would grow; the research recorded that the timing depends on a retirement rate the manufacturers revised three times in one year.

[D9 §5; reconciliation R-004]

74.4 MODULE-EQUIVALENTS: POWER AGAINST THE MODULE FACTORY

The last derivation puts Power's feedstock beside the aerospace business it shares production with. FTAI's releases count three modules per engine (\$69.4), so 100 whole engines are about 300 module-equivalents a year. [FTAI release, 26 Feb 2025, via orchestrator notes; D8 §2.10; reconciliation B-3]

WORKED EXAMPLE 9.7 — 300 MODULE-EQUIVALENTS AGAINST THE 2026 MODULE PLAN AND THE STATED CAPACITY

Inputs: 100 engines a year (derived, as above); three modules per engine (sourced: FTAI's convention, release of 26 Feb 2025) or four major modules (Part I §3's anatomical count, including the accessory gearbox); 2026 module target 1,050 (sourced: Q4/FY2025 release and call, Feb 2026) raised to 1,200 (sourced: call coverage of the Jul 2026 call, GuruFocus and Quartr; not returned from the release text); module capacity 3,000 a year (sourced: call coverage, Jul 2026, "supporting its 25% market share goal and 100 Mod-1 units annually"); FY2025 modules produced 757 (sourced: Q4 2025 call summaries).

module-equivalents, FTAI convention: $100 \times 3 = 300$ a year
 module-equivalents, anatomical count: $100 \times 4 = 400$ a year
 against the filed 2026 target: $300 \div 1,050 = 29\%$
 against the raised 2026 figure: $300 \div 1,200 = 25\%$
 against FY2025 output: $300 \div 757 = 40\%$
 against the stated 3,000 capacity: $300 \div 3,000 = 10\%$

So the engines Power would consume at 100 units a year are, in module terms, about a quarter to three-tenths of the aerospace business's 2026 plan, two-fifths of its 2025 output, and a tenth of the capacity management described in July 2026 as supporting both businesses. Part VII §49 treats those four capacity and volume figures as separate dated statements with different footprints, and this example does not compare the 3,000 figure with the 2026 targets except as management linked them on the call. One mechanical point changes the comparison and depends on an undisclosed design choice: if a Mod-1 uses only the core and discards or does not need the fan and the low-pressure turbine (§68.2), then each unit consumes one core module and releases a fan module and a low-pressure-turbine module as feedstock for the Module Factory, in which case the 300 figure overstates the drain on module production by two-thirds. If the engine's own low-pressure turbine is used as the output turbine, two of the three modules are consumed. Limits of the example: the 1,200 and 3,000 figures are call coverage; the module convention has two readings; and the design choice is unknown. [D8 §2.10, U3; reconciliation R-035, R-036, R-037; E.1 rules 2, 7 and 17; E.2]

The arithmetic of this section, taken together, is what the reconciliation calls for: every figure that bears on FTAI Power's scale is derived from a small number of disclosed inputs, each derivation is shown under each contested base, and the three undisclosed inputs that would turn the derivations into facts are the unit count in the order, the engines per unit, and who consolidates J&F.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-061	J&F is "FTAI's joint venture with Jereh Group" (FTAI release, 22 Jul 2026) and "a strategic packaging and distribution joint venture" (FTAI 10-Q Q2 2026) against "its subsidiary, J&F Power Systems" (Jereh's SZSE disclosure via FilingReader, 22 Jul 2026); ownership and consolidating party undisclosed	§70.2, Exhibit 9.5; worked example 9.5
R-062	2027 FTAI Power Adjusted EBITDA \$450M (Q2 2026 release text; Quattr, MarketBeat) against "\$450 million to \$750 million" (GuruFocus summary of the same call)	§74.2, worked example 9.5; Exhibit 9.3
R-001	CFM56 installed base ~23,000 (CFM website, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); ~14,200 and >19,000 (Safe Fly, consultancy blog)	§74.3, worked example 9.6, Exhibit 9.12
R-002	Safe Fly's ~14,200 read as all CFM56 variants (D8) or as the -7B fleet alone (D3); changes the 35% feedstock share	§74.3, Exhibit 9.12
R-004	Retirement rate: GE planning 3–4% → 2–3% → 1.5–2% (CFO via Aviation Week, 2026); 2025 actual ~1.5%, 2026 expected ~2% (GE via LeeHam, 22 Jan 2026); 1Q 2026 "below 1%" (GE investor-relations page); Safe Fly "near 2%"	§74.3, worked example 9.6, Exhibit 9.12
R-069	CFM56 deliveries >35,000 (CFM) against "well over 30,000" (D8, unsourced)	§74.3
R-035	Module count: three per engine (FTAI's convention, release 26 Feb 2025; D8) against four major modules anatomically (Part I §3)	§69.4, worked example 9.7
R-036	2026 module target 1,050 (Q4/FY2025 release, Feb 2026) against 1,200 (call coverage of the Jul 2026 call)	worked example 9.7
R-037	Module capacity 3,000 a year "supporting ... 100 Mod-1 units annually" (call coverage, Jul 2026) against the filed 1,350 and 1,800 figures (releases 2024 and 2025)	§69.3, worked example 9.7
D8 T1	Customer: "a leading international cloud service provider" (FTAI release) against "a U.S. hyperscaler" (call summaries)	§70.4, Exhibit 9.5
D8 T4	GE Vernova backlog 50 / 83 / 100 / 116 GW at successive dates, with later figures including slot reservations; "2028–29" against "2031" horizons	§71.3, Exhibit 9.7

Id	The two (or more) figures	Where in this Part
D8 T5	\$/kW: \$171–1,150 equipment-only and \$1,175 installed (Gas Turbine World, undated); \$600 turbine-only end-2027 (Wood Mackenzie); \$2,000–2,500 plant-level (Power Engineering, 2026); \$586–977 implied for Mod-1	§72.1, Exhibit 9.9, worked examples 9.3 and 9.4, Exhibit 9.11
D8 T6	LM2500XPRESS 35 MW (GE Vernova release) against ~34.5 MW implied by 29 units at nearly 1 GW and 34 MW in a commissioning brief	§72, Exhibit 9.8
D8 T8	Mod-1 35–40% efficiency and ~9,000 heat rate are consistent (37.9%) only on one heating-value basis; basis unstated	§68.4, worked example 9.1
D8 T9	Mobile turbines as "nonroad engines" exempt from permitting (Shelby County) against stationary sources needing permits and controls (Southern Environmental Law Center, NAACP); unresolved in litigation	§73.2, Exhibit 9.10
D8 T10	Reciprocating engines "20–35% less fuel" and >50% efficiency (Wärtsilä, vendor claim) against Mod-1's 35–40% (FTAI, call coverage); both self-reported	§68.4, §72

Unknowns carried in this Part

- **U-D8-01 (D8): U1. Units and price in the \$1.465B order.** Not disclosed. Would be resolved by FTAI's Q3 2026 10-Q or call, or by Jereh's SZSE filing if it states quantities.
- **U-D8-02 (D8): U2. J&F ownership, control and consolidation.** Percentages, board, which party consolidates, and how FTAI recognises Power revenue (sale of turbines into J&F, share of J&F profit, or consolidated genset sales). The Q2 2026 10-Q [10] is the document to read; its J&F note was not returned by search. The orchestrator notes add the 10-Q's description of J&F as "a strategic packaging and distribution joint venture with Jereh Group ... to support the planned 2027 production target of 100 Mod-1 CFM56 aeroderivative units"; ownership split and consolidation not returned.
- **U-D8-03 (D8): U3. Engines per unit and configuration.** How many CFM56 gas generators per 25 MW unit; -5B or -7B; whether the engine's own low-pressure turbine or a new free power turbine drives the generator; whether a booster stage replaces the fan.
- **U-D8-04 (D8): U4. Where units are built.** Jereh packaging site(s); US vs China; tariff exposure; whether J&F has its own facility.
- **U-D8-05 (D8): U5. Emissions and fuel.** NOx level, after-treatment (SCR or not), heating-value basis of the efficiency figure, dual-fuel capability.
- **U-D8-06 (D8): U6. Capital required.** Power capex, engine inventory earmarked, test capacity; FTAI's engine input cost per unit.

- **U-D8-07 (D8): U7. Segment reporting.** The 10-K FY2025 has two reportable segments [6]; call summaries speak of a "Power segment" for 2027. Whether Power is a third reportable segment from the Q2 2026 10-Q [10], or sits within Aerospace Products until first delivery, was not confirmed. (Cluster K-10 with U-D10-18, "Whether FTAI Power is a third reportable segment in the 2026 10-Qs.")
- **U-D8-08 (D8): U8. Jereh's GenSystems orders.** MW count (to derive \$/kW) and whether they use Mod-1 or other turbines; identity of the customer; relationship to J&F.
- **U-D8-09 (D8): U9. Competitor figures not captured.** Siemens Energy SGT-A65 output/lead time; Solar Turbines Titan 350; Caterpillar gas gensets; Bloom Energy deal sizes and \$/kW; published aeroderivative list prices in 2026.
- **U-D8-10 (D8): U10. Milestone schedule.** Size of the advance payment and the percentage at each milestone.
- **U-D8-11 (D8): U11. Performance adjustment.** Which metrics the 10% cap applies to (output, heat rate, availability, schedule) and whether it is symmetric (bonus as well as penalty).
- **U-D8-12 (D8): U12. Service model.** Mod-1 inspection and overhaul intervals; service contract form and pricing; FTAI's expected service revenue per unit.
- **U-D8-13 (D8): U13. Prior CFM56 stationary derivative.** None found; a definitive statement from GE/CFM or Gas Turbine World's handbook would settle whether Mod-1 is the first.
- **U-D8-14 (D8): U14. Interconnection queue durations.** PJM and ERCOT queue statistics were not captured; they quantify the "bridging" window ProEnergy's customers describe as 5–7 years [64].
- **U-D8-15 (D8): U15. Transcripts.** The Q4 2025 (26 Feb 2026) and Q2 2026 (29 Jul 2026) call transcripts were reached only through third-party summaries [15]–[19]; direct quotations on Power should be taken from the transcripts before publication.
- **U-D9-03 (D9): U3 — Definition of the ~24,000 CFM56 "in service" figure [S78]** (installed only, or installed plus spares plus stored) and its split between -5B, -7B and the older -3/-5A/-5C variants. Resolve: GE Aerospace or CFM investor materials with a fleet breakdown; the GE Bernstein presentation (27 May 2026) [S6] may contain it but its content did not surface.

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PART X

FTAI the company

This Part describes FTAI Aviation Ltd. as a corporation rather than as a set of businesses: where it came from, who ran it and on what terms, how it is owned and governed, what its consolidated accounts say, how it is financed, what it pays out, what it has promised, and what it discloses about its risks, its auditor and its litigation. A reader who finishes it will be able to read an FTAI annual report, earnings release or proxy statement and place every line in the structure described here.

TRANSLATION-TABLE ROWS ANSWERED

Parts IV to IX answer the rows of the Part III translation table directly, by describing a market or a business that a driver stresses. Part X answers them indirectly. A company's capital structure, its cash flow and its guidance are where the drivers show up as money that must be raised, spent or promised. The inventory build described in §81, and the debt that funds it in §82, are the balance-sheet form of the requirement that the low-retirement fleet creates for serviceable CFM56 material (Part III §24, row T-1: low retirements). The guidance history in §84 is management's dated statement of how much of the shop-visit plateau it expects to capture, in dollars of Adjusted EBITDA (the company's profit measure, earnings before interest, taxes, depreciation and amortization as the company adjusts it, defined in §80) and in modules (Part III §24, row T-14: shop-visit demand plateau). The margin aspirations inside that guidance, and the CFM materials agreement noted in the timeline, are the company-level response to catalogue escalation by the engine's original equipment manufacturer (OEM) (Part III §24, row T-7: OEM parts escalation). The shift of Aviation Leasing to an asset-light model, and the cut to its 2026 guidance that accompanied it, are the corporate form of the lessor-extension row (Part III §24, row T-6: lease extensions and availability). And the 2027 guidance for FTAI Power is the company's dated statement about the CFM56 cores that the durability problems of the new engines (Pratt & Whitney's GTF geared turbofan and CFM's LEAP, Part III §20) kept flying rather than retiring (Part III §24, rows T-2, T-3: GTF and LEAP durability). None of these rows is answered here with a market mechanism; each is answered with a dated corporate fact, and the Parts named above carry the mechanism.

75. History: a dated timeline

FTAI Aviation Ltd. is a Cayman Islands company listed on Nasdaq under the ticker FTAI. It is the successor to a Delaware limited liability company that Fortress Investment Group brought to the New York Stock Exchange in May 2015 under the name Fortress Transportation and Infrastructure Investors LLC. [FTAI 10-K FY2021; FTAI Infrastructure 10-K FY2024] Everything in this Part is documented in filings with the United States Securities and Exchange Commission (SEC) and in company press releases, so the forms those filings take are defined first.

SEC FILINGS: FORM 10-K, 10-Q, 8-K, DEF 14A, DEFM14A AND S-4

A Form 10-K is a US-listed company's audited annual report. A Form 10-Q is its unaudited quarterly report. A Form 8-K is a current report filed within days of a material event, and press releases are usually attached to it as Exhibit 99.1. A DEF 14A is the definitive proxy statement sent before the annual shareholder meeting, carrying director, compensation and ownership disclosure. A DEFM14A is the merger proxy for a transaction put to a shareholder vote. An S-4 is a registration statement for securities issued in a merger or an exchange offer. Foreign private issuers such as AerCap file a Form 20-F and Form 6-K instead (Part XI §87). [D10 §3, §9; D5 §8]

The research assembled the dates below from the FY2021, FY2022 and FY2025 annual reports, the quarterly reports of 2024 to 2026, the spin-off and redomiciliation documents, and the company's press releases. Where an event is known only from the orchestrator's chain map or from a filing index rather than from a read document, the table says so.

EXHIBIT 10.1 — FTAI CORPORATE TIMELINE, MAY 2015 TO 28 SEPTEMBER 2026

Date	Event	Figure or term	Source and level
15 May 2015	Fortress Transportation and Infrastructure Investors LLC lists on the NYSE as "FTAI", externally managed by Fortress's FIG LLC		FY2021 10-K; FTAI Infrastructure 10-K FY2024 (filings)
Dec 2016	\$15.0M invested in the "Advanced Engine Repair JV" (25%); a further \$13.5M in Aug 2019; whether this is the Chromalloy venture for PMA (Parts Manufacturer Approval, Part VI §37) parts is unknown	Carrying value \$22,429K at 31 Dec 2025	FY2025 10-K equity-method note, via orchestrator notes (filing)
8 Dec 2017	FTAI Aviation Ltd. formed in the Cayman Islands, the future parent		FY2022 10-K (filing)
2018	FTAI–Chromalloy "exclusive perpetual partnership" on CFM56 PMA parts, as investor blogs date it; the 10-K says only "a joint venture"	Not dated in any filing retrieved	Komodo Capital and Kairos Research blogs via D4; FY2025 10-K (investor blog; filing)
13 Dec 2021	FTAI Infrastructure formed as an LLC to hold the infrastructure businesses		FTAI Infrastructure 10-K FY2024 (filing)
2022	Russian leases terminated after the invasion of Ukraine; \$120.0M impairment net of maintenance deposits	8 aircraft and 17 engines in Russia; 4 aircraft and 1 engine in Ukraine at 31 Dec 2022	FY2022 10-K (filing)

Date	Event	Figure or term	Source and level
2022	Module Factory language first appears: Montréal "with a capacity for up to 300 shop visits per year"		FY2022 10-K language still in the FY2025 10-K (filing)
21 Jul 2022	Record date for the spin-off	One FTAI Infrastructure share per FTAI share	FTAI Infrastructure 10-K FY2024 (filing)
1 Aug 2022	FTAI Infrastructure Inc. distributed; lists on the Nasdaq Global Select Market as "FIP"		8-K, Aug 2022; FTAI Infrastructure 10-K FY2024 (filings)
10 Nov 2022	Merger completed; FTAI Aviation Ltd. becomes the parent; LLC shares exchanged automatically	DEFM14A 2022; S-4 of FTAI Finance Holdco Ltd.	8-K, Nov 2022; FY2022 10-K (filings)
4 Jan 2023	Assets of iAero Thrust (Miami) acquired with Unical Aviation as a joint venture, branded QuickTurn		GlobeNewswire 1 Dec 2023; 10-K, via orchestrator notes (release; filing)
1 Dec 2023	FTAI buys Unical's remaining interest in QuickTurn Miami	\$30.3M cash	GlobeNewswire 1 Dec 2023 (release)
28 May 2024	Internalization agreements signed; Management Agreement terminated	\$300.0M: \$150.0M cash plus 1,866,949 shares valued at \$150.0M	8-K, 28 May 2024; 10-Q Q2 2024; 10-Q Q3 2024 (filings)
6 Jun 2024	V2500 maintenance services agreement with IAE AG under EngineWise, 100-plus shop visits over five years (Part VII §51)		RTX newsroom, 6 Jun 2024 (OEM release)
17 Jun 2024	Cash leg of the internalization paid	\$150.0M	10-Q Q2 2024; 10-Q Q3 2024 (filings)
9 Sep 2024	Acquisition of Lockheed Martin Commercial Engine Solutions (LMCES), Montréal, closes	\$170.0M total consideration; 526,000 sq ft (release) or about 500,000 (10-K)	GlobeNewswire 9 Sep 2024; Aviation Week (release; trade press)
Oct 2024	Series A preferred shares redeemed in full	4,180,000 shares at \$25.00 plus \$1.6M accrued, about \$106.1M	FY2025 10-K (filing)
30 Dec 2024	Strategic Capital Initiative (SCI) launched; 2025 guidance issued	\$1.1–1.15B Adjusted EBITDA; 46 Seed Assets for about \$549M	GlobeNewswire 30 Dec 2024 (release)

Date	Event	Figure or term	Source and level
15 Jan 2025	Muddy Waters Research publishes a short report on FTAI		8-K, Jan 2025; trade press (filing; press)
18 Jan 2025	Audit Committee determines to begin a review with independent advisers		8-K, Jan 2025 (filing)
Feb 2025	Review concluded: allegations "without merit"; 10-K to be filed timely; trade coverage dates the announcement to on or about 19–20 Feb 2025		8-K Ex. 99.1, Feb 2025; StockStory via Wedbush, 20 Feb 2025 (filing; press)
26 Feb 2025	FY2024 results; QuickTurn Europe agreement announced; SCI \$2.5B asset-level debt commitment	Dividend \$0.30; 450 modules (150 engines) of added capacity	GlobeNewswire 26 Feb 2025, two releases (release)
25 Mar 2025	One Investment Management partners on the inaugural SCI vehicle	">\$4 billion" of deployment for the vehicle; OneIM's commitment undisclosed	GlobeNewswire 25 Mar 2025 (release)
5 Jun 2025	QuickTurn Europe joint venture closes, Rome Fiumicino, Italy	\$10.5M for 50%; 200,000 sq ft	GlobeNewswire 5 Jun 2025 (release)
17 Jun 2025	Audit Committee engages KPMG LLP as auditor for FY2025, after Ernst & Young LLP (2016–2025)		FY2025 10-K, via orchestrator notes (filing)
27 Oct 2025	2025 SPV equity fundraising closes at its upsized hard cap; Q3 2025 results; dividend raised	\$2.0B equity (target \$1.5B); dividend \$0.35	GlobeNewswire 27 Oct 2025, two releases (release)
30 Dec 2025	FTAI Power launched: the Mod-1 25 megawatt (MW) generator set on a CFM56 core (Part IX)		GlobeNewswire 30 Dec 2025; FY2025 10-K (release; filing)
22 Jan 2026	Multi-year materials agreement with CFM International (Part VII §52)	Terms undisclosed	GlobeNewswire 22 Jan 2026 (release)
25 Feb 2026	FY2025 results; 2026 guidance \$1.625B; dividend raised	Q4 Adjusted EBITDA \$277.2M; dividend \$0.40	GlobeNewswire and 8-K Ex. 99.1, 25 Feb 2026 (release)
27 Feb 2026	FY2025 10-K filed; auditor's report dated the same day		FY2025 10-K (filing)

Date	Event	Figure or term	Source and level
6 Mar 2026	CFO transition: Nicholas McAleese appointed Chief Financial Officer, Michael Hazan Chief Accounting Officer, succeeding Eun (Angela) Nam		GlobeNewswire 6 Mar 2026, as carried by StreetInsider, Barchart and Finviz (release)
29 Apr 2026	Q1 2026 results; dividend raised	\$0.45	GlobeNewswire 29 Apr 2026 (release)
15 May 2026	Full redemption of the 8.25% Series C preferred shares announced	Amount and redemption date not read	GlobeNewswire 15 May 2026, title (release)
22 May 2026	FTAI MRE 2026-1, the first Strategic Capital vehicle's inaugural asset-backed securitization, priced	\$612M of notes on 48 A320ceo/737NG aircraft leased to 23 airlines; Series A Asf/A(sf), Series B BBB+sf; close 4 Jun 2026; vehicle owns 292 aircraft	FTAI release 22 May 2026 via Finviz; AviTrader 26 May 2026
7 Jul 2026	Collaboration with AEI on 737-800 freighter conversions (Part VIII §67)		GlobeNewswire 7 Jul 2026 (release)
22 Jul 2026	J&F Power Systems receives an initial purchase order for Mod-1 units	\$1.465B; deliveries through Nov 2027	FTAI release 22 Jul 2026; Jereh exchange disclosure via FilingReader (release; foreign disclosure)
29 Jul 2026	Q2 2026 results; 2026 Leasing guidance cut; 2027 guidance set; dividend raised	\$2.3B for 2027; Leasing \$575M to \$475M; dividend \$0.50	GlobeNewswire and 8-K Ex. 99.1, 29 Jul 2026 (release)
14 Aug 2026	\$2.0B warehouse facility for the 2026 SPV closes (announced 17 Aug)	13 lenders; accordion to \$3.0B	GlobeNewswire 17 Aug 2026 (release)
15 Sep 2026	\$500M share repurchase program announced	Terms not read	GlobeNewswire 15 Sep 2026, title (release)
28 Sep 2026	27 Boeing 737-700 acquired from WestJet; 17 leased back through the 2026 SPV, 10 held for engines and modules	Price and lease terms not disclosed	GlobeNewswire 28 Sep 2026 (release)

Source: FTAI Aviation Ltd. Forms 10-K for FY2021, FY2022 and FY2025; Forms 10-Q for Q2 2024, Q3 2024, Q3 2025 and Q1 2026; Forms 8-K of Aug 2022, Nov 2022, 28 May 2024, Jan 2025 and Feb 2025; FTAI Infrastructure Inc. Form 10-K FY2024; GlobeNewswire releases of 1 Dec 2023, 9 Sep 2024, 30 Dec 2024, 26 Feb 2025, 25 Mar 2025, 5 Jun 2025, 27 Oct 2025, 30 Dec 2025, 22 Jan 2026, 25 Feb 2026, 6 Mar 2026, 29 Apr 2026, 15 May 2026, 22 May 2026, 7 Jul 2026, 22 Jul 2026, 29 Jul 2026, 17 Aug 2026, 15 Sep 2026 and 28 Sep 2026; RTX newsroom, 6 Jun 2024; orchestrator notes for the equity-method note, the auditor change and the CFO change. URLs in the Sources section. Accessed 2026-10-03.

75.1 THE FORTRESS YEARS, 2015 TO 2021

The company began as an externally managed investment vehicle, not as an operating business. Fortress Investment Group formed Fortress Transportation and Infrastructure Investors LLC to own transportation and infrastructure assets, and listed it on the New York Stock Exchange on 15 May 2015. [FTAI 10-K FY2021; FTAI Infrastructure 10-K FY2024] A Fortress subsidiary, FIG LLC, supplied the executives, the staff and the investment process under a management agreement, and a Fortress affiliate was entitled to a share of returns above a hurdle. Section 76 describes that structure.

In those years the vehicle owned two kinds of asset. One was aviation: aircraft and engines bought mid-life and leased out, the business that Part VIII §58 describes as Aviation Leasing. The other was infrastructure: a crude-oil and refined-products terminal at Jefferson, Texas; the Repauno site in New Jersey; the Transtar short-line railways; and the Long Ridge power and terminal site in Ohio. Those four names come from the orchestrator's brief; their individual revenues, costs and carrying values were not retrieved by the research, and the FY2021 annual report would supply them. [D10 §2.2, U-11]

The aviation side began to acquire maintenance capability before the separation. The FY2025 annual report's equity-method note records an "Advanced Engine Repair JV" in which the company invested \$15.0M in December 2016 and a further \$13.5M in August 2019 for a 25% interest, carried at \$22,429K at 31 December 2025. [FTAI 10-K FY2025, via orchestrator notes] Whether that venture is the Chromalloy PMA joint venture through which the annual report says the company "develops and manufactures PMA parts" is not stated in anything retrieved; investor blogs date an "exclusive perpetual partnership" with Chromalloy to 2018, which is neither the 2016 nor the 2019 date. [FTAI 10-K FY2025; Komodo Capital and Kairos Research via D4; reconciliation R-026] Part VI §43 carries that question. The language of the "Module Factory", a maintenance programme at the Lockheed Martin Commercial Engine Solutions facility in Montréal "with a capacity for up to 300 shop visits per year", first appears in the FY2022 annual report and is still carried in the FY2025 one. [FTAI 10-K FY2022; FTAI 10-K FY2025; reconciliation R-037]

75.2 THE SEPARATION, 2022

Three things happened in 2022 that turned the vehicle into the company this primer describes. First, the Russian leases ended. After the invasion of Ukraine the company terminated all its

leases with Russian airlines and wrote off \$120.0M, net of the maintenance deposits it already held, for the aircraft and engines it did not expect to recover. Eight aircraft and 17 engines were in Russia and four aircraft and one engine were in Ukraine at the end of the year. [FTAI 10-K FY2022] Part VIII §59 carries that episode.

Second, the infrastructure businesses left. A new company, FTAI Infrastructure Inc., was formed on 13 December 2021, and on 1 August 2022 every FTAI shareholder of record on 21 July 2022 received one share of it for each FTAI share held. [FTAI Infrastructure 10-K FY2024; FTAI 8-K, Aug 2022] Third, the company moved its legal home. On 10 November 2022 the Delaware LLC merged into a Cayman Islands company formed five years earlier, FTAI Aviation Ltd., whose shares the public received automatically. [FTAI 8-K, Nov 2022; FTAI 10-K FY2022] Section 77 takes the spin-off and the redomicile in turn.

75.3 BUILDING THE MAINTENANCE BUSINESS AND BUYING OUT THE MANAGER, 2023 TO 2024

In 2023 the company bought its first dedicated shop. On 4 January 2023 it acquired the assets of iAero Thrust in Miami together with Unical Aviation as a joint venture, branded the business QuickTurn, and on 1 December 2023 bought Unical's remaining interest for \$30.3M in cash. [GlobeNewswire, 1 Dec 2023; FTAI 10-K, via orchestrator notes] In June 2024 it signed a V2500 maintenance agreement with IAE AG under the EngineWise programme, covering more than 100 shop visits over five years; Part VII §51 describes it. [RTX newsroom, 6 Jun 2024]

On 28 May 2024 the company ended its external management. It agreed to pay its Former Manager \$300.0M, half in cash and half in 1,866,949 new shares, and the Management Agreement terminated that day. [FTAI 10-Q Q3 2024; FTAI 8-K, 28 May 2024] Three and a half months later, on 9 September 2024, it closed the acquisition of Lockheed Martin Commercial Engine Solutions in Montréal for \$170.0M, the site that houses the Module Factory. [GlobeNewswire, 9 Sep 2024; Aviation Week] The research recorded that the orchestrator's chain map and the D10 dossier had originally dated this acquisition to 2023; the release dates the closing to 9 September 2024, and this Part uses that date. [reconciliation R-039] In October 2024 it redeemed all 4,180,000 of its Series A preferred shares, and on 30 December 2024 it launched the Strategic Capital Initiative and gave 2025 guidance. [FTAI 10-K FY2025; GlobeNewswire, 30 Dec 2024]

75.4 THE SHORT REPORT, THE REVIEW AND THE CAPITAL-RAISING YEAR, 2025

On 15 January 2025 Muddy Waters Research, a short seller, published a report on FTAI's accounting. Three days later the Audit Committee determined to begin a review with independent advisers, and in February 2025 it announced that the allegations were "without merit". [FTAI 8-K, Jan 2025; FTAI 8-K Ex. 99.1, Feb 2025] Section 85 sets out the wording of each statement and the class action that followed. During the same year the Audit Committee changed

the company's auditor, engaging KPMG LLP for FY2025 with effect from 17 June 2025 after Ernst & Young LLP had audited the company from 2016. [FTAI 10-K FY2025, via orchestrator notes]

The rest of 2025 was spent raising capital for the vehicles and adding capacity. The first SCI vehicle secured a \$2.5B asset-level debt commitment on 26 February 2025, added One Investment Management as a partner on 25 March, and closed its equity at an upsized \$2.0B hard cap on 27 October. [GlobeNewswire, 26 Feb 2025, 25 Mar 2025 and 27 Oct 2025] Part VIII §62 carries the capital. The QuickTurn Europe joint venture at Rome Fiumicino closed on 5 June 2025 at \$10.5M for 50%, and on 30 December 2025 the company launched FTAI Power. [GlobeNewswire, 5 Jun 2025 and 30 Dec 2025]

75.5 GUIDANCE, REDEMPTIONS AND BUYBACKS, 2026

The first nine months of 2026 brought a materials agreement with CFM International (22 January), the FY2025 results and 2026 guidance of \$1.625B (25 February), the 10-K and auditor's report (27 February), and a new Chief Financial Officer (6 March). [GlobeNewswire, 22 Jan 2026, 25 Feb 2026 and 6 Mar 2026; FTAI 10-K FY2025] In May the company announced the full redemption of its Series C preferred shares (title located; body not read) and priced FTAI MRE 2026-1, the first Strategic Capital vehicle's \$612M inaugural asset-backed securitization on 48 aircraft (\$82.5). [GlobeNewswire, 15 May 2026, title; FTAI release, 22 May 2026, via Finviz and AviTrader] In July J&F Power Systems received a \$1.465B initial order, and the Q2 2026 results cut 2026 Aviation Leasing guidance to \$475M while setting 2027 guidance at \$2.3B. [FTAI release, 22 Jul 2026; GlobeNewswire, 29 Jul 2026] August brought the \$2.0B warehouse for the second vehicle, September a \$500M repurchase authorisation and the purchase of 27 Boeing 737-700s from WestJet. [GlobeNewswire, 17 Aug 2026, 15 Sep 2026 and 28 Sep 2026] Sections 82 to 84 take the financing, the payouts and the guidance in detail.

76. External management and the 2024 internalization

For its first nine years as a listed company FTAI had no employees of its own. That arrangement, and the price of ending it, shape how the company's expense lines read before and after mid-2024.

EXTERNAL MANAGEMENT: MANAGER, MANAGEMENT AGREEMENT, MANAGEMENT FEE, INCENTIVE ALLOCATION AND MASTER GP

External management is a structure in which a listed company has no staff of its own. A separate firm, the manager, supplies the executives, the employees and the investment process under a management agreement, and is paid a management fee, typically a percentage of the company's equity or assets. At FTAI the manager was FIG LLC, a subsidiary of Fortress Investment Group. A performance payment, the incentive allocation, went to a Fortress affiliate that the filings call the Master GP (GP, general partner) under a separate Services and Profit Sharing Agreement; it was a share of distributable returns above a hurdle. The entity name "Fortress Worldwide Transportation and Infrastructure Master GP LLC" comes from the orchestrator's chain map and was not re-verified.

[FTAI 10-Q Q3 2024; FTAI 8-K, 28 May 2024; D10 §2.1]

FORTRESS, FIG LLC AND FTAI INFRASTRUCTURE (FIP)

Fortress Investment Group is the alternative asset manager that sponsored FTAI's 2015 listing. FIG LLC, its subsidiary, was FTAI's external manager until 28 May 2024 and is called the "Former Manager" in the filings since. FTAI Infrastructure Inc., Nasdaq ticker FIP, is the company that received FTAI's infrastructure businesses on 1 August 2022; it appears in this primer only as history.

[FTAI 10-Q Q3 2024; FTAI Infrastructure 10-K FY2024]

76.1 THE TWO AGREEMENTS

Two contracts governed the relationship. The Management Agreement set the manager's fee and its duties. The Services and Profit Sharing Agreement set the incentive allocation payable to the Master GP. The Q3 2024 quarterly report names both as the agreements under which "accrued but not yet paid compensation and reimbursable expenses" were owed at termination. [FTAI 10-Q Q3 2024]

The economics of the agreements were sought and not retrieved. The management fee rate and its base, the incentive allocation formula, and the amounts paid in 2021, 2022, 2023 and the first five months of 2024 are all unknown in the research; the related-party notes of the FY2021 to FY2024 annual reports would supply them. [D10 U-5] The dossier offered one lead, a fee of 1.50% a year of total equity, and labelled it a recollection, unverified. It is repeated here only with that label and is used in no arithmetic. The practical consequence for a reader of the accounts is that the research cannot say how much of FTAI's pre-2024 operating expense was the fee and the allocation, and therefore cannot restate 2023 on a post-internalization basis.

One further open point sits in a later filing. The research flagged text about "management fees and incentive allocations" in the Q1 2025 quarterly report and could not determine whether it refers to the terminated Fortress arrangements or to the Strategic Capital vehicles' fee structure; Part VIII §63 carries the vehicle side. [D7 U17; reconciliation C.7 item 6]

76.2 THE INTERNALIZATION OF 28 MAY 2024

INTERNALIZATION

Internalization is the termination of an external management agreement, the hiring of the manager's people directly by the company, and a payment to the manager for the fee stream it gives up. FTAI's closed on 28 May 2024 for a stated \$300.0M: \$150.0M in cash paid on 17 June 2024 and 1,866,949 ordinary shares valued at \$150.0M at closing, about \$80.35 a share, plus accrued but unpaid compensation, reimbursable expenses and "certain other expenses". After that date management fees and incentive distributions ceased and were replaced by salaries, bonuses and equity awards, so operating expenses before and after mid-2024 are not like for like. [FTAI 10-Q Q3 2024; FTAI 10-Q Q2 2024; FTAI 8-K, 28 May 2024]

The mechanics, as the filings describe them, are these. On 28 May 2024 FTAI entered into definitive agreements with the Former Manager and the Master GP. The Management Agreement terminated with effect from that date, after which "the Company was no longer required to pay management fees or incentive distributions". [FTAI 10-Q Q3 2024] The consideration, in the quarterly report's words: "the Company agreed to pay \$300.0 million to its Former Manager (for itself and on behalf of the Master GP, as applicable), issued 1,866,949 ordinary shares valued at \$150.0 million at closing, with the remaining balance paid in cash on June 17, 2024". [FTAI 10-Q Q3 2024; FTAI 10-Q Q2 2024] The 8-K adds that the company also paid all accrued but unpaid compensation and reimbursable expenses under both agreements, plus certain other expenses. [FTAI 8-K, 28 May 2024]

EXHIBIT 10.2 — THE INTERNALIZATION CONSIDERATION, AS THE FILINGS FRAME IT

Component	Amount	Form	Date	Source
Equity leg	\$150.0M	1,866,949 ordinary shares valued at \$150.0M at closing	28 May 2024	10-Q Q2 2024; 10-Q Q3 2024
Cash leg	\$150.0M	Cash, "the remaining balance"	Paid 17 Jun 2024	10-Q Q2 2024; 10-Q Q3 2024
Stated total	\$300.0M			10-Q Q2 2024; 10-Q Q3 2024
Accrued but unpaid compensation and reimbursable expenses under both agreements	Not quantified in anything retrieved	Cash	At termination	8-K, 28 May 2024

Component	Amount	Form	Date	Source
"Certain other expenses"	Not quantified	Cash		8-K, 28 May 2024
Alternative framing in the same 10-Q: "agreed to pay \$150.0 million to the Former Manager"	\$150.0M	Reads as the cash leg alone		10-Q Q3 2024 (D10 T-2)
Accounting treatment (expensed as an internalization cost or capitalised)	Not retrieved			D10 U-6
Whether FIG LLC still holds the 1,866,949 shares	Not retrieved			D10 U-6

Source: FTAI Aviation Ltd., Form 10-Q for the quarter ended 30 Jun 2024 and Form 10-Q for the quarter ended 30 Sep 2024; Form 8-K and Exhibit 99.1 of 28 May 2024; Forms 8-K of June 2024 (located, not read). D10 §2.1, T-2 and U-6. Accessed 2026-10-03.

The same quarterly report frames the payment two ways. One sentence says the company "agreed to pay \$150.0 million to the Former Manager"; another describes total consideration of \$300.0M made up of the shares and the cash. [FTAI 10-Q Q3 2024; D10 T-2] The two are consistent if the first describes the cash leg alone, and that is how this Part reads them; but the total cash that left the company exceeded \$150.0M by the accrued and reimbursable amounts, which the research did not quantify.

WORKED EXAMPLE 10.1 — THE PRICE PER SHARE IMPLIED BY THE EQUITY LEG, AND THE DILUTION

Inputs: 1,866,949 shares issued (sourced: 10-Q Q2 2024 and Q3 2024); valued at \$150.0M at closing (sourced: same); shares outstanding before the issue roughly 100 million (assumed; the exact pre-deal count was not retrieved, D10 U-6; the 2026 proxy implies about 102.6 million after it, §78).

$$\begin{aligned}\text{implied value per share} &= 150,000,000 \div 1,866,949 = \$80.35 \\ \text{dilution} &= 1,866,949 \div \sim 100,000,000 \approx 1.9\%\end{aligned}$$

So the parties assigned about \$80.35 to each new share. Whether that equalled the market price on 28 May 2024 was not confirmed by the research; the 8-K exhibits would say how the share count was fixed. The issue added about 1.9% to the share count, and the count has been roughly flat since (§83). Limits of the example: the pre-deal share count is assumed; the accrued and reimbursable payments are not in the \$300.0M and are unquantified. [D10 §2.1, U-6]

76.3 WHAT THE RESEARCH DID NOT RETRIEVE ABOUT THE TRANSACTION

The task for this Part asked for the lock-up on the shares issued, the transition-services arrangements, the special committee that negotiated the deal and its advisers. None of these appears in the material the research retrieved. The 8-K of 28 May 2024 and its exhibits, and the June 2024 8-Ks that the research located but did not read, are where a lock-up or transition-services agreement would be filed and where the board's process would be described. [FTAI 8-K, 28 May 2024; FTAI 8-Ks, June 2024, located not read; D10 U-6] This Part therefore records those four items as unknown rather than describing them.

WHY THE INTERNALIZATION CHANGES HOW THE EXPENSE LINES READ

Before 28 May 2024 the cost of running the company reached the income statement as a management fee and, when earned, an incentive allocation to a Fortress affiliate. After that date the same people were FTAI's employees and their cost reached the statement as salaries, bonuses and equity awards, the last of which the proxy discloses for the named officers (\$78). A reader comparing the 2023 expense lines with those of 2025 is therefore comparing a fee contract with a payroll, and the research could not retrieve the fee amounts that would let the two be put on one basis. The one-off cost of the transaction itself, and whether it was expensed or capitalised, was also not retrieved. [FTAI 10-Q Q3 2024; D10 §2.1, U-5, U-6]

77. The 2022 spin-off and the redomicile

Two transactions in the second half of 2022 define the entity whose accounts the rest of this Part describes. The first removed the infrastructure businesses. The second changed the parent company's country and legal form.

77.1 THE SPIN-OFF OF FTAI INFRASTRUCTURE

SPIN-OFF AND RECORD DATE

A spin-off is a pro-rata distribution of a subsidiary's shares to the parent's shareholders, after which the subsidiary is a separate listed company. The record date is the day on which a shareholder must hold the parent's shares to receive the distribution. FTAI distributed one share of FTAI Infrastructure Inc. for each FTAI share held at the close of business on 21 July 2022; the distribution took place on 1 August 2022 and FTAI Infrastructure began trading on the Nasdaq Global Select Market as "FIP". [FTAI Infrastructure 10-K FY2024; FTAI 8-K, Aug 2022]

FTAI Infrastructure was formed as a limited liability company on 13 December 2021 to hold the infrastructure businesses. [FTAI Infrastructure 10-K FY2024] What went to it, per the orchestrator's brief, were the Jefferson Terminal, Repauno, Transtar and Long Ridge businesses; the research

did not retrieve their figures, and the FY2021 annual report of the parent would carry them. [D10 §2.2, U-11] What stayed was aviation: the leasing fleet and the early maintenance assets described in §75.1.

DISCONTINUED OPERATIONS

Discontinued operations is the accounting presentation under which a business that a company has disposed of, or is about to dispose of, is shown as a single net line below the results of the continuing business, with prior periods restated the same way. FTAI's FY2021 and FY2022 statements present the infrastructure businesses this way, so the net losses of those years (\$129.0M in 2021 and \$220.4M in 2022) are not comparable with the FY2023 and later results of the continuing aviation business without restatement. The D10 dossier used the term without defining it; this definition is the primer's. [FTAI 10-K FY2022, via orchestrator notes; D10 §2.2; registry section 2]

EXHIBIT 10.3 — WHAT THE FY2022 ANNUAL REPORT SHOWS FOR 2021 AND 2022, WITH THE CAUTIONS ATTACHED

Item (\$ thousands)	2021	2022	Caution	Source
Aviation Leasing segment revenues	128,328	199,040		FY2022 10-K via orchestrator notes
Aerospace Products segment revenues	"70,800"	85,113	The figure 70,800 appears three times in the search return for 2021; the notes treat the repetition as a probable extraction artefact	Same
Corporate and Other segment revenues	14,161	26,948		Same
Sum of segment revenues (derived)	213,289	311,101	Derived by addition; the 2021 sum inherits the 70,800 caution; not a reported total	Arithmetic on the rows above
Lease income (income-statement line)	173,864	179,314	A lease-income line of \$173.9M cannot sit inside Aviation Leasing segment revenue of \$128.3M; the two readings of the same filing are unreconciled (R-053)	FY2022 10-K via D6
Basic lease revenue from Russian lessees	~39,800	Leases terminated		FY2022 10-K via D6

Item (\$ thousands)	2021	2022	Caution	Source
Aviation Leasing segment Adjusted EBITDA	Not retrieved	107,556	The 2021 return of "70,800" is treated as not retrieved	FY2022 10-K via orchestrator notes
Aerospace Products segment Adjusted EBITDA	Not retrieved	27,377	Same	Same
Corporate and Other segment Adjusted EBITDA	(6,356)	(7,277)		Same
Sum of segment Adjusted EBITDA (derived)	Not computable	127,656	Derived by addition; not a reported total	Arithmetic
Net income (loss) attributable to shareholders	(128,992)	(220,374)	Includes the infrastructure businesses in discontinued operations; not comparable with 2023 onward	FY2022 10-K via orchestrator notes
Russia and Ukraine impairment, net of maintenance deposits		120,000		FY2022 10-K via D6

Source: FTAI Aviation Ltd., Form 10-K for FY2022, <https://www.sec.gov/Archives/edgar/data/1590364/000159036423000007/ftai-20221231.htm>, as read by two searches recorded in the orchestrator notes and in D6 §2.6 and §4.2. Reconciliation R-053; E.1 rule 28. Accessed 2026-10-03.

Two cautions in that exhibit are binding on the rest of the Part. The first is the artefact: the value 70,800 was returned for the 2021 Aerospace Products revenue and again for both segments' 2021 Adjusted EBITDA, and the orchestrator notes treat the repetition as a search-extraction error, so the 2021 Adjusted EBITDA by segment is carried as not retrieved. [orchestrator notes; reconciliation D. 3] The second is the contradiction: the FY2022 annual report, read by one search, gives FY2021 lease income of \$173,864K, and read by another gives Aviation Leasing segment revenues for 2021 of \$128,328K. A single income-statement line cannot exceed the revenue of the segment that contains it, so the two cannot both be right as read, and this Part presents both without choosing. [FTAI 10-K FY2022, via D6 and the orchestrator notes; reconciliation R-053] Reading the filing directly would settle both points.

77.2 THE REDOMICILIATION TO THE CAYMAN ISLANDS

CAYMAN ISLANDS EXEMPTED COMPANY AND REDOMICILIATION

A Cayman Islands exempted company is a company incorporated in the Cayman Islands that conducts its business mainly outside the islands; it is a common holding-company form for aircraft lessors. Redomiciliation is the move of a company's legal home from one jurisdiction to another. FTAI did it by merger: FTAI Aviation Ltd., formed in the Cayman Islands on 8 December 2017, became the surviving parent on completion of an Agreement and Plan of Merger dated 10 November 2022 with Fortress Transportation and Infrastructure Investors LLC, a Delaware limited liability company, whose public shares "were exchanged automatically for shares of FTAI Aviation Ltd."

[FTAI 8-K, Nov 2022; FTAI 10-K FY2022]

The merger was put to shareholders in a definitive merger proxy, a DEFM14A, in 2022. [FTAI DEFM14A 2022] In the same year an entity named FTAI Finance Holdco Ltd., SEC registrant number (CIK, the Central Index Key the SEC assigns to each filer) 1942420, filed a Form S-4, which is consistent with exchanging the old issuer's notes for notes of the new issuer. [FTAI Finance Holdco Ltd. Form S-4, 2022] The same entity filed a Schedule 13G/A, the SEC form on which a passive holder reports beneficial ownership of more than 5% of a class of shares, in 2026; its current role, whether as a note co-issuer or otherwise, was not confirmed. [D10 §3.1, U-13]

The task for this Part asked for the tax and governance consequences of the redomicile as the filings state them. The company's stated reasons are in the DEFM14A, and the research did not retrieve them; nor did it retrieve the reasons for the later move of the ordinary shares from the New York Stock Exchange to Nasdaq, where they now trade. [D10 U-12] What can be said from the material retrieved is structural. The parent is a Cayman company and the ordinary shares are its shares; the board is a classified board of seven; the proxy applies Nasdaq's independence rules (§78). The DEFM14A of 2022 and the listing-transfer 8-K are the documents that would supply the rest, and this Part records the gap rather than filling it.

WHY 2021 AND 2022 ARE NOT A BASELINE

The figures of 2021 and 2022 describe a different company. They include the infrastructure businesses in discontinued operations, they include Russian lease revenue of about \$39.8M in 2021 and its loss in 2022, they include the \$120.0M Russia write-off, and they were produced under an external manager whose fee sat in the expense lines. The FY2023 figures are the first full year of the continuing aviation company, and the FY2025 annual report's three-year segment table (§79) begins there. Any growth rate computed from 2022 to 2023 therefore compares two differently constituted companies and is not reported in this Part. [FTAI 10-K FY2022; FTAI 10-K FY2025; D10 §2.2]

78. Structure, management, board, shareholders

78.1 CORPORATE STRUCTURE

The parent is FTAI Aviation Ltd., a Cayman Islands exempted company formed on 8 December 2017 and the group's parent since the merger of 10 November 2022. Its SEC registrant number is CIK 1590364. [FTAI 10-K FY2022; FTAI 8-K, Nov 2022] Beneath it sit the operating subsidiaries listed in Exhibit 21.1 to the annual report, which the research located for the FY2024 and FY2025 filings and did not read. [FTAI 10-K FY2025, Exhibit 21.1; FTAI 10-K FY2024, Exhibit 21.1; D10 U-13] Beside the consolidated subsidiaries are four investments accounted for under the equity method, meaning that FTAI records its share of each investee's earnings as one line rather than consolidating the investee's revenue and costs; Part VIII §64 carries the accounting.

EXHIBIT 10.4 — FTAI AVIATION LTD.: PARENT, FINANCE ENTITY, SUBSIDIARIES AND EQUITY-METHOD INVESTMENTS

Entity	Relationship	Stake	Carrying value	Date	Source
FTAI Aviation Ltd.	Parent; Cayman Islands exempted company; Nasdaq: FTAI; CIK 1590364			Since 10 Nov 2022	8-K Nov 2022; 10-K FY2022
FTAI Finance Holdco Ltd. (CIK 1942420)	Filed an S-4 in 2022 (note exchange) and a Schedule 13G/A in 2026; current role not confirmed				S-4 2022; D10 U-13
Operating subsidiaries	Listed in Exhibit 21.1 to the 10-K; not read			31 Dec 2025	10-K FY2025 Ex. 21.1
2025 Partnership (first SCI vehicle)	Equity-method investment; FTAI is Servicer; \$151.6M invested in 9M 2025	19% limited-partner interest (FY2025 10-K; 2026 10-Qs); 20% (Q3 2025 10-Q)	\$281,740K	31 Dec 2025	10-K FY2025; 10-Q Q3 2025 note R13

Entity	Relationship	Stake	Carrying value	Date	Source
2025 Partnership, later dates		19%	\$279,417K; \$365,485K	31 Mar 2026; 30 Jun 2026	10-Q Q1 2026; 10-Q Q2 2026 via D7
Advanced Engine Repair JV	Equity-method investment; \$15.0M invested Dec 2016, \$13.5M Aug 2019; identity with the Chromalloy PMA venture unknown	25%	\$22,429K	31 Dec 2025	10-K FY2025 via orchestrator notes
QuickTurn Europe (Rome Fiumicino, Italy)	Equity-method joint venture; \$10.5M paid; closed 5 Jun 2025	50%	\$9,987K	31 Dec 2025	10-K FY2025; GlobeNewswire 5 Jun 2025
Prime Engine Accessories (Bristol, Connecticut; with Bauer, Inc.)	Equity-method joint venture for accessory repair	50%	Not returned		10-K FY2025 via D5
J&F Power Systems	FTAI describes it as its joint venture with Jereh Group for Mod-1 packaging and distribution; Jereh's exchange disclosure, as reported, calls it a subsidiary; ownership split and consolidating party undisclosed	Not disclosed	Not disclosed		FTAI release 22 Jul 2026; 10-Q Q2 2026; FilingReader (R-061)
FIG LLC (Former Manager); Master GP	Former external manager and incentive-allocation recipient; agreements terminated 28 May 2024				10-Q Q3 2024

Entity	Relationship	Stake	Carrying value	Date	Source
FTAI Infrastructure Inc. (Nasdaq: FIP)	Spun off 1 Aug 2022; separately listed	None			FTAI Infrastructure 10-K FY2024

Source: FTAI Aviation Ltd., Form 10-K for FY2025 and its Exhibit 21.1 (<https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai12312025exhibit211.htm>); Form 10-Q for Q3 2025, note R13; Forms 10-Q for Q1 and Q2 2026; Form 8-K of Nov 2022; Form S-4 of FTAI Finance Holdco Ltd., 2022; GlobeNewswire 5 Jun 2025 and 22 Jul 2026; orchestrator notes. Reconciliation R-026, R-048, R-061; cluster K-07. Accessed 2026-10-03.

Three points in that table carry tensions. The stake in the 2025 Partnership is 19% in the FY2025 annual report and the 2026 quarterly reports and 20% in the Q3 2025 quarterly report; the orchestrator notes suggest dilution at the vehicle's final close as a possible reason and label that a guess. [FTAI 10-K FY2025; FTAI 10-Q Q3 2025; reconciliation R-048] The Advanced Engine Repair JV may or may not be the Chromalloy venture, as §75.1 said. And J&F Power Systems is described as FTAI's joint venture by FTAI and as a subsidiary by Jerih's exchange disclosure as reported; the ownership split and the consolidating party are undisclosed, and Part IX §70 carries the consequences for who books the \$1.465B order. [FTAI release, 22 Jul 2026; FTAI 10-Q Q2 2026; FilingReader, 22 Jul 2026; reconciliation R-061] Which of the equity-method investments is the one "through" which the Aviation Leasing segment leases and sells assets, in the annual report's phrase, was not established; the 2025 Partnership is the candidate the dossiers name. [FTAI 10-K FY2025; cluster K-07]

78.2 OFFICERS

EXHIBIT 10.5 — EXECUTIVE OFFICERS AS THE SOURCES NAME THEM, WITH THE MARCH 2026 CHANGE

Name	Role	Since	Source and note
Joseph P. Adams, Jr. (68)	Chief Executive Officer and Chairman of the Board	Has led the company since the 2015 listing (dossier recollection; tenure start not re-verified)	DEF 14A 2026; director term expires 2026
David Moreno	President	Not retrieved	Q2 2026 call coverage (GuruFocus); appears in the proxy ownership table with 54,232 shares; the proxy text returned did not give his title (D10 T-5)

Name	Role	Since	Source and note
Stacy Kuperus	Chief Operating Officer	Not retrieved	DEF 14A 2026; Q2 2026 call coverage; holds 3,499 shares
Nicholas McAleese	Chief Financial Officer	Appointed 6 Mar 2026	FTAI release 6 Mar 2026, as carried by StreetInsider, Barchart and Finviz; named CFO on the Q2 2026 call (GuruFocus)
Michael Hazan	Chief Accounting Officer	Appointed 6 Mar 2026	Same release
Eun (Angela) Nam	Chief Financial Officer until 6 Mar 2026; named executive officer in the 2026 proxy, which reports 2025 pay; left for a role outside aviation per the release		DEF 14A 2026; FTAI release 6 Mar 2026 via orchestrator notes (R-067)
General Counsel and Secretary	Named in the proxy; the search rendered the name as "Yoon BoHee", unconfirmed	Not retrieved	DEF 14A 2026 (D10 U-9)

Source: FTAI Aviation Ltd., DEF 14A proxy statement 2026 (https://www.sec.gov/Archives/edgar/data/1590364/000114036126014789/ny20066354x1_def14a.htm); GlobeNewswire, "FTAI Aviation Announces CFO Transition", 6 Mar 2026, as carried by StreetInsider, Barchart and Finviz; GuruFocus transcript of the Q2 2026 call, 30 Jul 2026. Reconciliation R-067 and its resolution in the orchestrator notes. Accessed 2026-10-03.

The research recorded a tension over the Chief Financial Officer and the orchestrator later resolved it. The D10 dossier found Eun (Angela) Nam listed as a named executive officer in the 2026 proxy and could not confirm her title from the text returned; the D5 dossier found Nicholas McAleese introduced as CFO on the Q2 2026 call. [FTAI DEF 14A 2026; GuruFocus, 30 Jul 2026; reconciliation R-067] The orchestrator's later search found the company's release of 6 March 2026: McAleese was appointed Chief Financial Officer and Michael Hazan Chief Accounting Officer, succeeding Nam, who left for a role outside aviation. [GlobeNewswire, 6 Mar 2026, via StreetInsider, Barchart and Finviz] The proxy names Nam because it reports 2025 compensation, a year in which she served; the two sources describe different dates and do not conflict. Tenure start dates for the other officers, and the General Counsel's name, were not retrieved. [D10 U-9]

78.3 BOARD

CLASSIFIED BOARD

A classified board is one whose directors serve staggered terms, so that only a portion of the board stands for election in any one year. FTAI's seven directors serve three-year terms in three classes, with two or three terms expiring each year. The proxy states that a majority of the board is independent under Nasdaq rules and that the Audit, Compensation, and Nominating and Corporate Governance committees are each entirely independent. The Chief Executive Officer is also Chairman of the Board. [FTAI DEF 14A 2026]

EXHIBIT 10.6 — THE SEVEN DIRECTORS, BY CLASS

Director	Age	Term expires	Shares held (proxy)	Note
Joseph P. Adams, Jr.	68	2026	505,837	Chief Executive Officer and Chairman
Judith A. Hannaway	74	2026	3,267	
Martin Tuchman	85	2026	657,877	Largest individual insider holding
A. Andrew Levison	69	2027	16,151	
Shyam Gidumal	66	2027	1,669	
Paul R. Goodwin	83	2028	82,236	
Ray M. Robinson	78	2028	59,554	

Source: FTAI Aviation Ltd., DEF 14A proxy statement 2026. Which directors are designated independent, whether any Fortress-affiliated director remains, and whether there is a lead independent director were not retrieved (D10 U-10). Accessed 2026-10-03.

The proxy's statement that the board has an independent majority is carried as the proxy makes it. Which individual directors are the independent ones, whether any director retains a Fortress affiliation from the externally managed years, and whether a lead independent director has been designated to balance the combined chair and chief executive roles were all sought and not retrieved. [FTAI DEF 14A 2026; D10 U-10]

78.4 COMPENSATION

NAMED EXECUTIVE OFFICERS (NEOS) AND THE PROXY'S PAY DISCLOSURES

The named executive officers are the officers whose compensation a proxy statement must disclose in full, normally the chief executive, the chief financial officer and the next three highest-paid. The proxy also carries three standard disclosures that the dossier named without defining: say-on-pay, the non-binding shareholder vote on executive compensation; the CEO pay ratio, the chief executive's total pay divided by the median employee's; and pay-versus-performance, a table relating the pay actually realised by the officers to the company's financial results. Equity awards are delivered as restricted stock units (RSUs), shares delivered after a service period, and performance stock units (PSUs), shares delivered only if performance conditions are met. [FTAI DEF 14A 2026; D10 §3.4; registry section 2]

For 2025 the Chief Executive Officer's total compensation was \$5,988,406: salary \$975,000, a cash bonus under the non-equity incentive plan of \$2,340,000, stock awards of \$2,647,254, and all other compensation of \$26,152. [FTAI DEF 14A 2026] The four parts sum to the total, and the cash bonus is 2.4 times salary. The bonus metrics, the performance conditions attached to the PSUs, the totals paid to the other named officers, and whether any Fortress-era arrangement (a tax receivable agreement, under which a company pays a former owner a share of tax savings it realises, or a profit-sharing legacy) survived the internalization were not retrieved. [D10 U-10] The proxy contains a Compensation Committee Report, the compensation tables, the CEO pay ratio, the pay-versus-performance table and a say-on-pay vote; their contents beyond the CEO's total were not returned. [FTAI DEF 14A 2026]

78.5 SHAREHOLDERS AND SHARE COUNT

BENEFICIAL OWNERSHIP (THE SHAREHOLDERS TABLE)

Beneficial ownership is the proxy's measure of who holds the voting or investment power over a company's shares, whatever the registered name. The proxy lists each holder of more than 5% of the ordinary shares and each director and named officer. FTAI's 2026 table shows four institutions above 5% and nine insiders holding about 1.35% in total; dividing the largest holder's share count by its percentage implies about 102.6 million shares outstanding at the record date. [FTAI DEF 14A 2026; D10 §3.5]

EXHIBIT 10.7 — BENEFICIAL OWNERSHIP PER THE 2026 PROXY

Holder	Shares	Percent	Note
Capital International Investors	13,865,244	13.51%	

Holder	Shares	Percent	Note
Capital World Investors	12,754,798	12.43%	With Capital International, two Capital Group entities hold about 25.9%
The Vanguard Group	10,363,209	10.10%	
FMR LLC	5,390,921	5.25%	
Martin Tuchman (director)	657,877	about 0.64%	
Joseph P. Adams, Jr. (CEO)	505,837	about 0.49%	
Paul R. Goodwin	82,236		
Ray M. Robinson	59,554		
David Moreno	54,232		
A. Andrew Levison	16,151		
Stacy Kuperus	3,499		
Judith A. Hannaway	3,267		
Shyam Gidumal	1,669		
Nine insiders listed, total	1,384,322	about 1.35%	Sum of the nine rows; the proxy's "all directors and officers as a group" line was not retrieved
Implied shares outstanding	about 102.6 million		13,865,244 ÷ 13.51%, derived

Source: FTAI Aviation Ltd., DEF 14A proxy statement 2026, beneficial ownership table; percentages as the proxy states them; the insider total and the implied share count are arithmetic on the table (D10 §3.5). Whether Fortress or FIG LLC still holds the 1,866,949 internalization shares was not retrieved (D10 U-10). Accessed 2026-10-03.

Two features of the register are worth stating plainly. The four largest holders are index and active fund managers holding about 41% between them, and the two Capital Group entities alone hold about a quarter. Insider ownership is small: the nine directors and officers listed hold about 1.35%, of which the chief executive holds about 0.49%. Whether the Former Manager still holds the 1,866,949 shares it received in 2024, which would be about 1.8% of the implied count, is not visible in the table returned. [FTAI DEF 14A 2026; D10 §3.5, U-10]

CUSTOMER CONCENTRATION

Customer concentration is the share of revenue that comes from a company's largest customers, which US filings must disclose for any customer above 10%. FTAI's Q3 2025 quarterly report says two customers in Aerospace Products accounted for 16% and 11% of revenue in the quarter. Their identities, the annual figures and whether either is a related party (such as a Strategic Capital vehicle) were not retrieved. Part VII §45 and §48 carry the related-party revenue line. [FTAI 10-Q Q3 2025; D10 §4.9, U-15]

79. Consolidated financials, 2021 to 1H 2026

This section sets out what the research retrieved of FTAI's consolidated income statement and shows every cell it did not retrieve as such. The source hierarchy is simple: the FY2025 annual report for 2023 to 2025, the FY2022 annual report for 2021 and 2022, the quarterly reports and earnings releases for 2025 and 2026. The D10 dossier's searches returned the top of the FY2025 income statement and the three-year segment table; the orchestrator's later searches added the FY2025 expense lines. Annual net income attributable to shareholders, income tax and the full Adjusted EBITDA reconciliation were not returned for any year. [D10 U-1; orchestrator notes]

79.1 THE ANNUAL STATEMENT, 2021 TO 2025, AND THE HALF-YEARS

EXHIBIT 10.8 — CONSOLIDATED INCOME STATEMENT ITEMS AS RETRIEVED, 2021 TO 2025 (\$ THOUSANDS)

Line	2021	2022	2023	2024	2025	Source
Total revenues	213,289 (derived sum of segments; 70,800 caution)	311,101 (derived sum of segments)	1,170,896	1,734,901	2,507,409	10-K FY2022; 10-K FY2025
Aerospace products revenue	"70,800" (segment; caution)	85,113 (segment)	454,970	1,079,821	1,600,456	10-K FY2022; 10-K FY2025
MRE (Maintenance, Repair and Exchange) Contract revenue	not applicable	not applicable	0	0	335,788	10-K FY2025

Line	2021	2022	2023	2024	2025	Source
Lease income	173,864 (R-053)	179,314	not retrieved	234,411 or 255,338 (two readings)	235,210	10-K FY2022 via D6; 10-K FY2025
Maintenance revenue	not retrieved	not retrieved	not retrieved	not retrieved	218,499	10-K FY2025
Asset sales revenue	not retrieved	not retrieved	not retrieved	not retrieved	106,945	10-K FY2025
Other revenue (residual)					10,511 (derived; D10 T-3)	Arithmetic
Cost of sales	not retrieved	not retrieved	502,132	825,884	1,349,719	10-K FY2025
Cost of sales ÷ revenue (derived)			42.9%	47.6%	53.8%	Arithmetic
Revenue less cost of sales (derived)			668,764	909,017	1,157,690	Arithmetic
Operating expenses	not retrieved	not retrieved	not retrieved	not retrieved	152,541	10-K FY2025 via orchestrator notes
General and administrative	not retrieved	not retrieved	not retrieved	not retrieved	9,478	Same
Depreciation and amortization	not retrieved	not retrieved	not retrieved	not retrieved	225,797	Same
Interest expense	not retrieved	not retrieved	not retrieved	not retrieved	247,751	Same
Income tax; equity-method earnings; other lines	not retrieved	not retrieved	not retrieved	not retrieved	not retrieved	D10 U-1

Line	2021	2022	2023	2024	2025	Source
Net income (loss) attributable to shareholders	(128,992), incl. discontinued operations	(220,374), incl. discontinued operations	not retrieved	not retrieved	not retrieved (Q3 114,009; Q4 111,852)	10-K FY2022; releases 27 Oct 2025 and 25 Feb 2026
Adjusted EBITDA, consolidated	not retrieved	127,656 (derived sum of segments)	597,282	862,050	1,190,922	10-K FY2022; 10-K FY2025
Net cash from operating activities	not retrieved	not retrieved	128,982	(187,956)	(310,745)	10-K FY2025

Source: FTAI Aviation Ltd., Form 10-K for FY2025 (consolidated statement of operations and segment table; <https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai-20251231.htm>); Form 10-K for FY2022 (via D6 and the orchestrator notes); earnings releases of 27 Oct 2025 and 25 Feb 2026. "Not retrieved" means the research did not return the cell; it does not mean the filing lacks it. Reconciliation R-045, R-052, R-053; E.2 "Consolidated P&L 2025". Accessed 2026-10-03.

EXHIBIT 10.9 — HALF-YEAR ITEMS AS RETRIEVED, 1H 2025 AGAINST 1H 2026 (\$ THOUSANDS UNLESS STATED)

Line	1H 2025	1H 2026	Source
Total revenues; cost of sales (consolidated); net income attributable; Adjusted EBITDA (1H totals)	not retrieved	not retrieved	D10 U-1; the Q2 2026 and Q2 2025 releases carry them
Aerospace Products segment products revenue	not retrieved	1,214,800	10-Q Q2 2026 via D5
MRE Contract revenue with the 2025 Partnership	170,200	404,000	10-Q Q2 2026 via orchestrator notes
Aerospace Products segment revenue (derived)	not computable	about 1,618,800	Arithmetic in D5 §4.1
Aerospace Products segment Adjusted EBITDA (derived)	not computable	about 472,300 (Q1 222,600 + Q2 249,700)	Releases 29 Apr 2026 and 29 Jul 2026; arithmetic
Cost of sales, scope unconfirmed (segment or consolidated)	not retrieved	1,160,100 (D5 T5)	10-Q Q2 2026 via D5

Line	1H 2025	1H 2026	Source
Lease income	130,879	67,657	10-Qs via D6 (Q1 68,440 + Q2 62,439; Q1 39,892 + Q2 27,765)
Maintenance revenue	not retrieved	Q1 2026 only: 30,599	10-Q Q1 2026
Net income, quarterly	not retrieved	Q1 net income 137,899 (attribution not captured); Q2 attributable 117,585	10-Q Q1 2026; release 29 Jul 2026
Adjusted EBITDA, quarterly	not retrieved	Q1 not retrieved; Q2 291,444	Release 29 Jul 2026
Servicing fees from the 2025 Partnership	not retrieved	12,849 (Q1 5,861 derived; Q2 6,988)	10-Qs via D7
Profit eliminations on MRE sales to the Partnership	not retrieved	16,597 (Q1 10,000; Q2 6,597)	10-Qs via D7
Russia insurance recoveries	54,300 was the nine-month 2025 figure; the 1H share not retrieved	44,600 in Q1 2026	10-Q Q3 2025; 10-Q Q1 2026 via D6

Source: FTAI Aviation Ltd., Form 10-Q for Q1 2026 and Form 10-Q for Q2 2026 (via D5, D6, D7 and the orchestrator notes); earnings releases of 29 Apr 2026 and 29 Jul 2026; Forms 10-Q for Q1, Q2 and Q3 2025 (via D6 and D7). The 1H consolidated totals were not returned (D10 U-1). Accessed 2026-10-03.

The annual table reads best column by column, and the half-year table that follows it holds what was retrieved for 2025 and 2026. For 2023 to 2025 the research has total revenue, cost of sales, segment revenue and segment Adjusted EBITDA for every year, and the full expense stack for 2025 only. Revenue rose 48.2% in 2024 and 44.5% in 2025. [FTAI 10-K FY2025] Cost of sales rose faster than revenue in both years, from 42.9% of revenue in 2023 to 47.6% in 2024 and 53.8% in 2025, so the margin between revenue and cost of sales narrowed as the mix moved towards product sales. [FTAI 10-K FY2025; arithmetic] That ratio is not a gross margin in the usual sense, because FTAI's cost-of-sales line has historically also carried the carrying value of aircraft and engines sold by the Leasing segment, whose proceeds sit in asset sales revenue. [D10 §2.4; registry C-23]

For the half-years the research is thinner. Lease income fell from \$130.9M in 1H 2025 to \$67.7M in 1H 2026, a 48% drop that the Q1 2026 quarterly report attributes mainly to the sale of the Seed Assets to the 2025 Partnership. [FTAI 10-Q Q1 2026, via D6; D6 §2.6] MRE Contract revenue to

the Partnership rose from \$170.2M to \$404.0M over the same two half-years. [FTAI 10-Q Q2 2026, via orchestrator notes] The consolidated 1H totals for revenue, cost of sales, net income and Adjusted EBITDA were not returned; the Q2 2026 and Q2 2025 earnings releases carry them. [D10 U-1]

79.2 THE REVENUE LINES OF 2025 AND WHAT EACH IS

EXHIBIT 10.10 — FY2025 REVENUE BY INCOME-STATEMENT LINE, WITH THE DESCRIPTION OF EACH

Line	FY2025 (\$ thousands)	Share of total (derived)	What the line is	Source
Aerospace products revenue	1,600,456	63.8%	Sales of modules, engines and parts by the Aerospace Products segment (Part VII §47)	10-K FY2025
MRE Contract revenue	335,788	13.4%	Revenue under Maintenance, Repair and Exchange contracts with the SCI vehicles; described as "stand-ready obligations to provide replacement CFM56-5B and CFM56-7B engines to customers as they become unserviceable" in the Q3 2025 10-Q and as "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership" in the Q2 2026 10-Q (R-044; Part VII §48)	10-K FY2025; 10-Q Q3 2025; 10-Q Q2 2026
Lease income	235,210	9.4%	Rent from operating leases of aircraft and engines (Part VIII §58)	10-K FY2025
Maintenance revenue	218,499	8.7%	Maintenance reserves recognised as revenue when FTAI is no longer obliged to return them (Part II §13)	10-K FY2025
Asset sales revenue	106,945	4.3%	Proceeds from selling leased aircraft and engines, presented gross; the asset's carrying value sits in cost of sales	10-K FY2025
Sum of the five named lines	2,496,898			Arithmetic
Total revenues as reported	2,507,409	100%		10-K FY2025

Line	FY2025 (\$ thousands)	Share of total (derived)	What the line is	Source
Residual, presumed "other revenue"	10,511	0.4%	Not returned by the search; presumed (D10 T-3)	Arithmetic

Source: FTAI Aviation Ltd., Form 10-K for FY2025, consolidated statement of operations; Form 10-Q for Q3 2025; Form 10-Q for Q2 2026 (via orchestrator notes). Shares are derived. Reconciliation R-044; D10 T-3. Accessed 2026-10-03.

Two definitions of the MRE Contract revenue line sit side by side in the quarterly reports. One describes an obligation: to stand ready to supply replacement engines. The other describes a sale: the price of engines and modules sold to the vehicles' special-purpose entities. Part VII §48 and Part VIII §63 quote both and set out what each implies for revenue recognition; this Part notes only that the line is related-party revenue with vehicles FTAI itself services, and that it grew from nothing in 2024 to 13.4% of consolidated revenue in 2025. [FTAI 10-Q Q3 2025; FTAI 10-Q Q2 2026; reconciliation R-044] The five named lines fall \$10,511K short of the reported total; the residual is presumed to be an "other revenue" line that the search did not return. [D10 T-3]

79.3 THE SEGMENT SPLIT, 2023 TO 2025

The annual report reports two segments, Aerospace Products and Aviation Leasing, with a Corporate and Other column. Segment revenue is not the same thing as an income-statement line: the Aerospace Products segment's revenue of \$1,936.2M for 2025 equals the "Aerospace products revenue" line of \$1,600.5M plus the MRE Contract revenue line of \$335.8M, as the segment geographic table's total confirms. [FTAI 10-K FY2025; reconciliation B-1] The explicit segment-note sentence mapping the MRE Contract line to the Aerospace Products segment was not retrieved; the equality of the totals is the evidence for the mapping. [D10 T-4; reconciliation R-043]

EXHIBIT 10.11 — SEGMENT REVENUE AND ADJUSTED EBITDA, 2023 TO 2025 (\$ THOUSANDS)

Line	2023	2024	2025	Source
Total revenues	1,170,896	1,734,901	2,507,409	10-K FY2025 segment table
Aerospace products revenue (line)	454,970	1,079,821	1,600,456	Same
MRE Contract revenue (line)	0	0	335,788	Same

Line	2023	2024	2025	Source
Aerospace Products segment revenue (derived: the two lines above)	454,970	1,079,821	1,936,244	Arithmetic; 2025 confirmed by the segment geographic table total
Aerospace Products Adjusted EBITDA	160,009	380,636	671,252	10-K FY2025 segment table
Aerospace Products Adjusted EBITDA ÷ segment revenue (derived)	35.2%	35.2%	34.7%	Arithmetic
Lease income (Aviation Leasing)	not returned	234,411 (segment-table search) or 255,338 (earlier search)	235,210	10-K FY2025, two readings (R-052)
Aviation Leasing Adjusted EBITDA	not returned	500,062	608,912	10-K FY2025 segment table
Total Adjusted EBITDA	597,282	862,050	1,190,922	Same
Corporate and Other Adjusted EBITDA (derived residual)	not computable	(18,648)	(89,242)	Total less the two segments; see §80
Aerospace Products share of total revenue (derived)	38.9%	62.2%	77.2%	Arithmetic on segment revenue

Source: FTAI Aviation Ltd., Form 10-K for FY2025, segment table, as returned by the orchestrator's searches and recorded in the orchestrator notes; the FY2024 Aerospace Products figures agree with D5's derivation from the 10-K's "+\$520.6M" and "+76%" (reconciliation B-2). Derived rows are arithmetic. Reconciliation R-043, R-052; E.2 "Segment revenue and Adjusted EBITDA". Accessed 2026-10-03.

The segment table is the clearest statement of what changed in the company between 2023 and 2025. Aerospace Products segment revenue rose from \$455.0M to \$1,936.2M, and its share of the consolidated total from 38.9% to 77.2%. Its Adjusted EBITDA rose from \$160.0M to \$671.3M, at a margin on segment revenue that stayed within a point of 35% in all three years. [FTAI 10-K FY2025; arithmetic] Aviation Leasing's Adjusted EBITDA rose from \$500.1M in 2024 to \$608.9M in 2025 on lease income that was roughly flat, which Part VIII §58 and §63 explain through maintenance revenue, asset sales and the gains and fees from the 2025 Partnership. [FTAI 10-K FY2025; D6 §2.6; D7 §4.2] The 2024 lease income figure has two readings from the same filing, \$234,411K and \$255,338K, which the orchestrator notes attribute either to different lines (a total

lease-income line against a segment line) or to an extraction error; both are carried. [orchestrator notes; reconciliation R-052]

The FY2025 cost-of-sales figure also has two readings, and the orchestrator resolved them as two different lines. The consolidated statement of operations carries \$1,349,719K. The segment expenses table carries \$1,240,368K as the Aerospace Products segment's own cost of sales. [FTAI 10-K FY2025; orchestrator notes on R-045] The D5 dossier had warned that the smaller figure might be the consolidated line; the later search placed it in the segment table. Each is presented with its scope. [D5 T5; reconciliation R-045]

WORKED EXAMPLE 10.2 — TWO COST-OF-SALES FIGURES, TWO RATIOS, AND WHY NEITHER IS A GROSS MARGIN WITHOUT A LABEL

Inputs: consolidated cost of sales \$1,349,719K and total revenues \$2,507,409K (sourced: 10-K FY2025, statement of operations); Aerospace Products segment cost of sales \$1,240,368K and segment revenue \$1,936,244K (sourced: 10-K FY2025, segment expenses table and segment geographic table, per the orchestrator's resolution of R-045).

consolidated: $1 - 1,349,719 \div 2,507,409 = 46.2\%$ of revenue after cost of sales

segment: $(1,936,244 - 1,240,368) \div 1,936,244 = 35.9\%$ (derived; "segment gross margin")

difference between the two cost lines: $1,349,719 - 1,240,368 = 109,351$

So the consolidated line carries about \$109.4M of cost that is not the Aerospace Products segment's cost of sales. The candidate is the carrying value of aircraft and engines sold by Aviation Leasing, whose proceeds of \$106,945K sit in asset sales revenue; that identification is a reading, not a disclosure the research retrieved. The 35.9% figure is labelled derived wherever the primer uses it, and Part VII §53 carries it beside the 34.7% Adjusted EBITDA margin. Limits of the example: the composition of the \$109.4M is not confirmed; the consolidated ratio is not a product margin because it mixes rent, reserves and asset-sale proceeds with product sales. [FTAI 10-K FY2025; orchestrator notes; reconciliation R-045; E.1 rule 10]

79.4 THE QUARTERS RETRIEVED

EXHIBIT 10.12 — QUARTERLY RESULTS RETRIEVED, Q2 2025 TO Q2 2026

Quarter	Net income attributable to shareholders	Basic earnings per share (EPS)	Diluted EPS	Adjusted EBITDA, consolidated	Aerospace Products revenue; segment Adjusted EBITDA	Other lines retrieved	Source
Q2 2025	not retrieved			not retrieved	about \$491.6M; about \$165.4M (both derived from the Q2 2026 release's +78% and +51%)	Lease income \$62,439K; MRE to the Partnership \$69.6M	Release 29 Jul 2026 (derivation in D5); 10-Qs via D6 and orchestrator notes
Q3 2025	\$114,009K		\$1.10	\$297,381K	not retrieved	Two Aerospace Products customers at 16% and 11% of revenue; dividend raised to \$0.35	Release 27 Oct 2025; 10-Q Q3 2025
Q4 2025	\$111,852K	\$1.09	\$1.08	\$277,178K	about \$557M (derived from "35% margin"); \$195M at 35% (stated)	Dividend raised to \$0.40; 228 modules	Release 25 Feb 2026; call coverage via D5

Quarter	Net income attributable to shareholders	Basic earnings per share (EPS)	Diluted EPS	Adjusted EBITDA, consolidated	Aerospace Products revenue; segment Adjusted EBITDA	Other lines retrieved	Source
Q1 2026	Net income \$137,899K; attribution to shareholders not captured			not retrieved	\$743.8M (= \$522,585K products + \$221,230K MRE); \$222.6M	Lease income \$39,892K; maintenance revenue \$30,599K; dividend raised to \$0.45	10-Q Q1 2026; release 29 Apr 2026
Q2 2026	\$117,585K	\$1.15	\$1.13	\$291,444K	\$875.0M (+78%); \$249.7M (+51%)	Lease income \$27,765K; MRE to the Partnership \$182.8M; Aviation Leasing Adjusted EBITDA \$88.2M per call coverage only (R-051); dividend raised to \$0.50; 296 modules	Release 29 Jul 2026; 10-Q Q2 2026; GuruFocus and Quartr

Source: FTAI Aviation Ltd. earnings releases of 27 Oct 2025, 25 Feb 2026, 29 Apr 2026 and 29 Jul 2026 (8-K Exhibits 99.1 and GlobeNewswire); Forms 10-Q for Q3 2025, Q1 2026 and Q2 2026; D5 §4.1 for the derived Q2 2025 and Q4 2025 segment figures; call coverage (GuruFocus, Quartr) for the \$88.2M, which was not returned from the release text. Three dossiers quote the same Q2 2026 release, which is an echo and not corroboration (reconciliation B-10). Accessed 2026-10-03.

Three cautions attach to that exhibit. The Q2 2025 and Q4 2025 segment figures are derived by the D5 dossier from stated growth rates and a stated margin, not read from a release, and are rounded accordingly. The \$88.2M Aviation Leasing Adjusted EBITDA for Q2 2026 came from

call-coverage summaries; the D6 dossier's searches of the release returned the consolidated \$291.4M and the Aerospace Products \$249.7M but no Leasing line, and the arithmetic residual of \$291.4M less \$249.7M less \$88.2M, a Corporate and Other figure of about -\$46.5M, is unconfirmed. [GuruFocus and Quartr via chain map; D6 T3; reconciliation R-051] And the agreement of three dossiers on the Q2 2026 headline figures is agreement on one document, the release of 29 July 2026, not independent confirmation. [reconciliation B-10]

80. Adjusted EBITDA: the definition and the reconciliation

FTAI reports its results, sets its guidance and describes its segments in a measure it defines itself. Every Part of this primer that quotes an FTAI profit figure is quoting this measure, so its construction matters.

ADJUSTED EBITDA (CONSOLIDATED; THE COMPANY'S DEFINITION)

EBITDA is earnings before interest, taxes, depreciation and amortization. Adjusted EBITDA is FTAI's company-defined variant. It removes items management treats as non-recurring or non-cash and adds back the company's pro-rata share of Adjusted EBITDA from its unconsolidated, equity-method investments. The research expected the reconciling items to include equity-based compensation, acquisition and transaction costs, internalization costs, asset impairments, changes in the fair value of non-hedge derivatives and dividends on preferred shares, with the bridge starting from net income attributable to shareholders; the exact FY2025 definition and the line-by-line reconciliation were in the annual report and the earnings release and were not returned by the searches. FTAI reports the measure at consolidated level (\$597.3M in 2023, \$862.1M in 2024, \$1,190.9M in 2025; \$291.4M in Q2 2026) and by segment, and sets guidance in it. [FTAI 10-K FY2025; FTAI release, 29 Jul 2026; D10 §2.5, U-1]

80.1 WHAT IS KNOWN ABOUT THE CONSTRUCTION

Four things about the measure are established in the filings retrieved. First, it is reported both consolidated and by segment, and the segment figures are the ones guidance is expressed in (§84). [FTAI 10-K FY2025; FTAI release, 29 Jul 2026] Second, it includes the company's share of the Adjusted EBITDA of its equity-method investees, so the 2025 Partnership's earnings reach FTAI's Adjusted EBITDA in proportion to its 19% (or 20%) stake even though the Partnership's revenue does not reach FTAI's revenue. [D10 §2.5; D7 §2.6; reconciliation R-048] Third, in the Aviation Leasing segment the measure adds back depreciation and includes gains on the sale of leasing equipment, so a year in which the segment sells many aircraft carries the gains inside its Adjusted EBITDA; the FY2025 gain on sales to the 2025 Partnership was \$46.4M, against a nine-month sum of quarterly gains of \$50.1M, a tension Part VIII §63 carries. [FTAI 10-K FY2025; D6 §2.6; D7 §4.2; reconciliation R-054] Fourth, the measure is stated before preferred dividends, so it is

before the roughly \$6.2M a year of Series D dividends and before the Series B and Series C dividends while those were outstanding (§82). [D10 §2.5]

A fifth point sits beside those four and is recorded as the filing states it. The annual report describes the aircraft sales to the 2025 Partnership as accounted for under ASC 610-20 (the Accounting Standards Codification topic for gains on sales of non-financial assets) and as "non-recurring in nature and not considered part of the Company's ordinary activities", while the launch release describes a programme of "\$3.0+ billion of capital annually" and the Q2 2026 quarterly report says the company's primary investment activity will run through the Strategic Capital Initiative going forward. [FTAI 10-K FY2025; GlobeNewswire, 30 Dec 2024; FTAI 10-Q Q2 2026; reconciliation R-057] The strategy language and the accounting language describe the same sales differently, and Part VIII §64 sets them side by side.

80.2 SEGMENT AND TOTAL: TWO BASES

The registry records a conflict between the segment measure and the consolidated one, and between FTAI's measure and the peers'. [registry C-24] The segment measure, as the D6 dossier describes it for Aviation Leasing, "adds back depreciation and includes gains and the equity-method pick-up". The consolidated measure, as the D10 dossier describes it, is the company-defined total after the removal of items treated as non-recurring or non-cash and the add-back of the pro-rata share from unconsolidated investments and of preferred dividends. The two segments' Adjusted EBITDA do not sum to the consolidated figure; the difference is the Corporate and Other column, which carries the costs of the parent and whatever adjustments are not allocated to a segment.

EXHIBIT 10.13 — ADJUSTED EBITDA BY SEGMENT, THE TOTAL, AND THE DERIVED CORPORATE AND OTHER RESIDUAL (\$ THOUSANDS)

Period	Aerospace Products	Aviation Leasing	Sum of segments (derived)	Total Adjusted EBITDA (reported)	Corporate and Other (derived residual)	Source
FY2022	27,377	107,556	134,933	127,656 (derived from three segments)	(7,277) (reported as a segment column)	10-K FY2022 via orchestrator notes
FY2023	160,009	not returned	not computable	597,282	not computable	10-K FY2025
FY2024	380,636	500,062	880,698	862,050	(18,648)	10-K FY2025
FY2025	671,252	608,912	1,280,164	1,190,922	(89,242)	10-K FY2025

Period	Aerospace Products	Aviation Leasing	Sum of segments (derived)	Total Adjusted EBITDA (reported)	Corporate and Other (derived residual)	Source
Q4 2025	195,000 (stated, rounded)	not retrieved		277,178	not computable	Release 25 Feb 2026; call coverage
Q1 2026	222,600	not retrieved		not retrieved	not computable	Release 29 Apr 2026
Q2 2026	249,700	88,200 (call coverage only; not returned from the release)	337,900 (if the call-coverage figure is right)	291,444	(46,456) (unconfirmed; depends on the call-coverage figure)	Release 29 Jul 2026; GuruFocus, Quartr (R-051)

Source: FTAI Aviation Ltd., Form 10-K for FY2025, segment table (via orchestrator notes); Form 10-K for FY2022 (via orchestrator notes); earnings releases of 25 Feb 2026, 29 Apr 2026 and 29 Jul 2026; GuruFocus and Quartr coverage of the 30 Jul 2026 call for the \$88.2M. The residual column is arithmetic and is not a reported Corporate and Other figure for 2024 and 2025. Reconciliation R-051; E.1 rule 2. Accessed 2026-10-03.

WORKED EXAMPLE 10.3 — DERIVING THE CORPORATE AND OTHER RESIDUAL, AND WHAT THE KNOWN 2025 LINES DO AND DO NOT LET ONE RECONCILE

Inputs: total Adjusted EBITDA \$1,190,922K; Aerospace Products \$671,252K; Aviation Leasing \$608,912K (all sourced: 10-K FY2025 segment table). Depreciation and amortization \$225,797K and interest expense \$247,751K (sourced: 10-K FY2025 statement of operations via orchestrator notes).

$$\begin{aligned}
 \text{Corporate and Other residual} &= 1,190,922 - 671,252 - 608,912 = (89,242) \\
 \text{2024 residual} &= 862,050 - 380,636 - 500,062 = (18,648) \\
 \text{Adjusted EBITDA less D\&A less interest} &= 1,190,922 - 225,797 - 247,751 = 717,374
 \end{aligned}$$

The first two lines say that the costs and adjustments not allocated to a segment grew from about \$18.6M in 2024 to about \$89.2M in 2025. What is in them was not retrieved; candidates include corporate general and administrative cost and any items the company removes at consolidated level, and the segment note would say. The third line is an illustration only. It is not the company's reconciliation, because it omits income tax, the equity-method adjustments, equity-based compensation and every other reconciling item in the company's definition, none of which was retrieved; and it uses interest expense, whereas EBITDA is by construction before interest. Net income attributable to shareholders for 2025 was not retrieved, so the illustration cannot be checked against it; the two retrieved quarters give \$114.0M for Q3 and \$111.9M for Q4, about \$225.9M for the second half. Limits of the example: the residual is arithmetic, not a disclosed column; the

illustration is not a bridge. [FTAI 10-K FY2025; orchestrator notes; FTAI releases, 27 Oct 2025 and 25 Feb 2026; D10 U-1]

80.3 COMPARABILITY WITH OTHER COMPANIES

Part XI §88 compares FTAI's segment Adjusted EBITDA margin with the measures the peers report, and it begins by saying that no two of those measures are on the same basis. StandardAero reports "Segment Adjusted EBITDA", its own defined measure; HEICO reports segment operating income after depreciation; MTU and Lufthansa Technik report adjusted EBIT (earnings before interest and taxes); GE Aerospace reports segment profit; AerSale reports gross profit. [D11 §2.3, §2.5] An EBITDA margin exceeds an EBIT margin by the share of revenue that depreciation and amortization represents, which at FTAI was \$225.8M on \$2,507.4M of revenue in 2025, or 9.0 points. [FTAI 10-K FY2025; arithmetic] That is the size of the gap a reader opens by setting FTAI's 34.7% Aerospace Products margin beside an EBIT-based peer margin without adjustment, and it is why this Part states the basis every time it quotes the measure.

81. Cash flow and the inventory build

FTAI reported net income in 2024 and 2025 and consumed cash in its operations in both years. The reason is a feature of the cash flow statement that is worth setting out from first principles.

OPERATING VERSUS INVESTING CASH FLOW: INVENTORY AGAINST LEASING EQUIPMENT

Under US GAAP (generally accepted accounting principles) the cash a company spends buying assets it intends to sell is an operating outflow, because those assets are inventory and sit in working capital. The cash it spends buying assets it intends to lease is an investing outflow, because those assets are leasing equipment, capitalised and depreciated over time. A company that buys engines, modules and parts to refurbish and sell therefore shows its purchases as operating outflows, and its operating cash flow will be negative while its inventory is growing faster than its sales even if it reports a profit. FTAI's net cash from operating activities was +\$129.0M in 2023, −\$188.0M in 2024 and −\$310.7M in 2025. [FTAI 10-K FY2025; D10 §2.6]

81.1 THE FIGURES RETRIEVED

EXHIBIT 10.14 — CASH FLOW, INVENTORY AND THE LEASING-EQUIPMENT BALANCE AS RETRIEVED (\$ THOUSANDS)

Item	2023	2024	2025	Later	Source
Net cash provided by (used in) operating activities	128,982	(187,956)	(310,745)	1H 2026 not retrieved	10-K FY2025

Item	2023	2024	2025	Later	Source
Cumulative, 2023 to 2025 (derived)			(369,719)		Arithmetic
Change in inventory (working-capital line)	not retrieved	not retrieved	not retrieved		D10 U-2
Inventory, net (balance sheet)	not retrieved	not retrieved	not retrieved	1,364,300 at 31 Mar 2026	10-Q Q1 2026 via D5
Acquisition of leasing equipment (investing)	not retrieved	not retrieved	not retrieved		D10 U-2
Acquisition of property, plant and equipment (investing)	not retrieved	not retrieved	not retrieved		D10 U-2
Proceeds from asset sales (investing)	not retrieved	not retrieved	not retrieved	Proceeds from sales to the 2025 Partnership: \$58.9M in Q1 2025; \$117.3M in Q1 2026	10-Qs via D7
Leasing equipment, at cost (balance sheet)	not retrieved	2,963,452	2,057,624		10-K FY2025 via D6
Leasing equipment, net	not retrieved	2,373,730	1,545,804		10-K FY2025 via D6
Russia insurance recoveries (cash in)	not retrieved	not retrieved	54,300 in 9M 2025	44,600 in Q1 2026	10-Q Q3 2025 and 10-Q Q1 2026 via D6
Cash and cash equivalents	not retrieved	not retrieved	not retrieved	412,240 at 31 Mar 2026	10-Q Q1 2026
Whether the company defines free cash flow	not retrieved				D10 U-2

Source: FTAI Aviation Ltd., Form 10-K for FY2025 (cash flow statement and balance sheet, via D10 and D6); Form 10-Q for Q1 2026 (via D5 and D10); Forms 10-Q for Q1 2025, Q3 2025 and Q1 2026 (via D6 and D7). Cluster K-18. Accessed 2026-10-03.

The working-capital detail that would split the operating outflow into its parts, principally the change in inventory, in receivables and in deferred revenue, was not retrieved, and neither were

the investing lines. [D10 U-2; cluster K-18] So the statement that the inventory build drives the negative operating cash flow is a statement of mechanism, supported by the one inventory balance retrieved, \$1,364.3M at 31 March 2026, rather than a statement read from the cash flow statement's own lines. [FTAI 10-Q Q1 2026, via D5] The inventory balances at 31 December 2025 and 30 June 2026, and the segment's capital expenditure, were not retrieved. [D5 U12]

The investing side moved the other way. Leasing equipment at cost fell from \$2,963.5M at the end of 2024 to \$2,057.6M at the end of 2025, and net of depreciation from \$2,373.7M to \$1,545.8M, as aircraft were sold into the 2025 Partnership. [FTAI 10-K FY2025, via D6] The proceeds of those sales are investing inflows, so in 2025 the company was releasing cash from leasing equipment through investing activities while absorbing cash into inventory through operating activities. Part VIII §60 describes the asset-light shift that this pattern reflects. The releases call the shift "asset-light"; the company's limited-partner interest in the 2025 Partnership was nonetheless carried at \$281.7M at 31 December 2025 and \$365.5M at 30 June 2026, and the D7 dossier put it at about \$380M at full commitment, so the term describes owning fewer leased aircraft directly rather than holding no leasing capital. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026, via D7; reconciliation R-058]

WORKED EXAMPLE 10.4 — HOW MUCH INVENTORY THE COMPANY HOLDS RELATIVE TO ITS COST OF SALES, TWO WAYS, AND WHY THE DATES MATTER

Inputs: inventory, net \$1,364.3M at 31 Mar 2026 (sourced: 10-Q Q1 2026 via D5); consolidated cost of sales \$1,349.7M for FY2025 (sourced: 10-K FY2025); cost of sales \$1,160.1M for 1H 2026, scope (segment or consolidated) unconfirmed (sourced: 10-Q Q2 2026 via D5, with D5's T5 caveat).

months of FY2025 consolidated cost of sales: $1,364.3 \div (1,349.7 \div 12) = 12.1$ months

months of 1H 2026 cost of sales: $1,364.3 \div (1,160.1 \div 6) = 7.1$ months
cumulative operating outflow 2024–2025: $187,956 + 310,745 = \$498.7\text{M}$

So on the 2025 run-rate the company held about a year of cost of sales as inventory, and on the faster 2026 run-rate about seven months. Neither figure is an inventory-turn the company reports. The first mixes a March 2026 balance with 2025 costs; the second uses a half-year cost figure whose scope is unconfirmed; and both treat the whole consolidated cost line as the cost of inventory sold, which it is not, since it also carries the carrying value of leased assets sold. The third line shows that the two years of operating outflow together are about \$498.7M, a little over a third of the inventory balance, which is consistent with inventory being funded partly from operations' own receipts and partly from the balance sheet. Limits of the example: every ratio is derived from mixed dates and unconfirmed scopes; the inventory change itself was not retrieved. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026 and Q2 2026 via D5; D5 T5; cluster K-18]

WHY A PROFITABLE MODULE FACTORY CONSUMES CASH

The Module Factory (Part VII §46) sells serviceable modules and takes back unserviceable ones. To do that it must hold a pool of serviceable modules ahead of demand, and to build that pool it must buy engines and parts, pay for their repair, and carry them until they are sold. Every dollar spent on that pool is an operating outflow when it is spent and a cost of sales only when the module is sold. While the pool is growing, operating cash flow runs behind reported profit by the growth in the pool. The company's own statement of the mechanism, and the size of each year's inventory change, are in the cash flow statement that the research did not retrieve; the mechanism itself follows from the accounting rule in the definition above. [D10 §2.6; D5 §2.8]

82. Debt, preferred shares and leverage

FTAI finances itself with five series of senior unsecured notes, an undrawn bank revolver, and perpetual preferred shares, of which two series have been redeemed. This section sets out each instrument, the one tension in the debt figures, and three illustrations that are labelled as such because the company's own leverage definitions and its credit ratings were not retrieved.

82.1 THE SENIOR NOTES

SENIOR UNSECURED NOTES; BULLET; CALLABLE

Senior unsecured notes are bonds that rank ahead of equity and subordinated debt in a liquidation but are not backed by specific collateral. They pay a fixed coupon, normally twice a year, and repay their entire principal on the maturity date (a bullet repayment). They are typically callable, meaning the issuer may repay them early after a non-call period at a premium that declines over time. FTAI has five series, with coupons from 5.50% to 7.875% and maturities from 2028 to 2033. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; D10 §2.7]

EXHIBIT 10.15 — THE FIVE NOTE SERIES AND THE REVOLVER, AS REPORTED

Instrument	Coupon	Maturity	Balance reported (\$ thousands)	As of	Issue date; original principal	Source
Senior Notes due 2028	5.50%	1 May 2028	1,000,995	31 Dec 2025	not retrieved	10-K FY2025
Senior Notes due 2030	7.875%	1 Dec 2030	497,470	31 Dec 2025	not retrieved	10-K FY2025

Instrument	Coupon	Maturity	Balance reported (\$ thousands)	As of	Issue date; original principal	Source
Senior Notes due 2031	7.00%	1 Dec 2031	717,752	31 Mar 2026	not retrieved	10-Q Q1 2026
Senior Notes due 2032	6.875%	1 Dec 2032	823,184	31 Mar 2026	not retrieved	10-Q Q1 2026
Senior Notes due 2033	5.875%	15 Apr 2033	497,784	31 Dec 2025	not retrieved	10-K FY2025
Sum of the five balances (mixed dates)			3,537,185	mixed		Arithmetic; D10 T-1
Long-term debt, net (balance sheet)			3,451,087	31 Mar 2026		10-Q Q1 2026
Revolving Credit Facility	Base Rate + 1.75% or Adjusted Term SOFR (Secured Overnight Financing Rate) + 2.75%	22 May 2027	0 drawn	31 Dec 2025	Commitment size not retrieved	10-K FY2025
Term loans; asset-level debt consolidated from vehicles			not retrieved			D10 U-3
Total debt and total equity at 31 Dec 2025			not retrieved			D10 U-3
Credit ratings (Moody's, S&P, Fitch)			not retrieved; not searched			D10 U-7

Source: FTAI Aviation Ltd., Form 10-K for FY2025 (debt note, as returned by search); Form 10-Q for the quarter ended 31 Mar 2026 (<https://www.sec.gov/Archives/edgar/data/1590364/000162828026029335/ftai-20260331.htm>). The balances mix two dates as the searches returned them. D10 §2.7, §4.5, T-1, U-3, U-7. Accessed 2026-10-03.

The balances carry a signature. The 2028, 2030 and 2033 series sit just above or below round numbers, \$1,000M, \$500M and \$500M, which is what principal carried at a small premium or discount looks like. The 2031 and 2032 series are not round, which usually indicates partial exchanges or add-on issues. [D10 §2.7] Whether the reported figures are principal or carrying amount, when each series was issued and in what original amount, and the call schedules and covenants were not retrieved; the debt note and the note-offering 8-Ks would supply them. [D10 U-3]

The one tension in the debt figures is arithmetic. The five note balances as returned sum to \$3,537,185K, three of them at 31 December 2025 and two at 31 March 2026, while the balance sheet at 31 March 2026 reports long-term debt, net of \$3,451,087K. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; D10 T-1] The \$86.1M gap could be unamortised issuance costs and discounts netted against the principal, a change in balances between the two dates, or a series partly retired; the search did not say which, and the Q1 2026 debt note would. Both figures are carried, each with its date and its scope.

WORKED EXAMPLE 10.5 — ANNUAL COUPON ON THE NOTES, IF THE REPORTED BALANCES ARE PRINCIPAL

Inputs: the five balances and coupons in Exhibit 10.15 (sourced: 10-K FY2025 and 10-Q Q1 2026, mixed dates); the assumption that each balance is principal (assumed; D10 U-3). Reported interest expense for FY2025, \$247,751K (sourced: 10-K FY2025 via orchestrator notes).

$$\begin{aligned}
 &1,000,995 \times 5.50\% = 55,055 \\
 &497,470 \times 7.875\% = 39,176 \\
 &717,752 \times 7.00\% = 50,243 \\
 &823,184 \times 6.875\% = 56,594 \\
 &497,784 \times 5.875\% = 29,245 \\
 &\text{sum} = 230,313, \text{ about } \$230.3\text{M a year} \\
 &\text{blended} = 230,313 \div 3,537,185 = 6.5\% \\
 &\text{reported FY2025 interest expense less the illustrative coupon} = 247,751 - \\
 &\quad 230,313 = 17,438
 \end{aligned}$$

So the notes cost about \$230M a year in coupon at a blended 6.5%, on this illustration. Reported 2025 interest expense was about \$17.4M higher, which is the size of the amortisation of issuance costs, revolver commitment fees and any other interest the company carries, if the balances are principal and were unchanged through the year, neither of which is confirmed. Limits of the example: mixed dates; principal versus carrying amount unconfirmed; the 2025 balances may have differed during the year. [D10 §2.7; orchestrator notes]

82.2 THE REVOLVER

REVOLVING CREDIT FACILITY AND SOFR

A revolving credit facility, or revolver, is a bank line the company may draw and repay repeatedly up to a committed amount. It pays a floating interest rate on drawn balances and a commitment fee on the undrawn portion. FTAI's prices drawings at the banks' base rate plus 1.75% or at Adjusted Term SOFR plus 2.75%, where SOFR is the Secured Overnight Financing Rate, the US dollar benchmark that replaced LIBOR, the London Interbank Offered Rate. The facility was undrawn at 31 December 2025 and matures on 22 May 2027; its commitment size was not retrieved. [FTAI 10-K FY2025; D10 §2.7, U-3]

Two maturities fall inside the guidance horizon of §84. The revolver matures on 22 May 2027, and the 5.50% notes, about \$1.0B, on 1 May 2028. [FTAI 10-K FY2025] The terms on which either is renewed or refinanced will depend in part on the company's credit ratings, which the research did not retrieve, and on the company's leverage covenants, which it also did not retrieve. [D10 U-7] This Part records both as unknown and does not characterise them.

82.3 NET DEBT AND LEVERAGE, AS ILLUSTRATIONS

NET DEBT AND LEVERAGE RATIO

Net debt is total debt less cash. A leverage ratio divides debt, or net debt, by a measure of annual earnings, usually a form of EBITDA. The company's own leverage definitions (gross or net, whether preferred shares count as debt, which earnings measure) and the rating agencies' definitions were not retrieved, so every ratio in this section is arithmetic on two reported numbers and is labelled as such. [D10 §2.9, U-7]

WORKED EXAMPLE 10.6 — NET DEBT AT 31 MARCH 2026, AND LEVERAGE AND COVERAGE ILLUSTRATIONS THAT ARE NOT THE COMPANY'S DEFINITIONS

Inputs: long-term debt, net \$3,451,087K and cash \$412,240K at 31 Mar 2026 (sourced: 10-Q Q1 2026). Adjusted EBITDA: Q4 2025 \$277,178K (sourced: release 25 Feb 2026); Q2 2026 \$291,444K (sourced: release 29 Jul 2026); FY2025 \$1,190,922K (sourced: 10-K FY2025). FY2025 interest expense \$247,751K (sourced: 10-K FY2025 via orchestrator notes); illustrative coupon \$230,313K (derived in worked example 10.5).

$$\begin{aligned}
 \text{net debt} &= 3,451,087 - 412,240 = 3,038,847, \text{ about } \$3.04\text{B} \\
 \text{net debt} \div (\text{Q4 2025 Adjusted EBITDA} \times 4) &= 3,038,847 \div 1,108,712 = 2.7\text{x} \\
 \text{net debt} \div (\text{Q2 2026 Adjusted EBITDA} \times 4) &= 3,038,847 \div 1,165,776 = 2.6\text{x} \\
 \text{gross debt} \div \text{FY2025 Adjusted EBITDA} &= 3,451,087 \div 1,190,922 = 2.9\text{x} \text{ (mixed dates)} \\
 \text{FY2025 Adjusted EBITDA} \div \text{FY2025 interest expense} &= 1,190,922 \div 247,751 = 4.8\text{x} \\
 (\text{Q2 2026 Adjusted EBITDA} \times 4) \div \text{illustrative coupon} &= 1,165,776 \div 230,313 = 5.1\text{x}
 \end{aligned}$$

So net debt was about \$3.0B at the end of March 2026, between 2.6 and 2.7 times an annualised quarter of Adjusted EBITDA, and about 2.9 times the 2025 full-year figure on a gross basis with mismatched dates. Adjusted EBITDA covered reported interest expense about 4.8 times in 2025. None of these is a reported ratio, none uses the company's covenant definition, none treats the \$65.0M of Series D preferred shares as debt, and the Adjusted EBITDA used includes gains on asset sales and the equity-method share described in §80. A rating agency or a lender would compute something different. Limits of the example: debt at one date against earnings from another; the earnings measure is company-defined; coverage is on interest expense, not on debt service, since no principal falls due before 2027. [FTAI 10-Q Q1 2026; FTAI 10-K FY2025; FTAI releases, 25 Feb 2026 and 29 Jul 2026; D10 §2.9, U-7]

82.4 THE PREFERRED SHARES

PREFERRED SHARES: CUMULATIVE, PERPETUAL, REDEEMABLE; LIQUIDATION PREFERENCE

Preferred shares are equity that ranks ahead of the ordinary shares for dividends and in a liquidation and normally carries no vote. FTAI's are cumulative (a missed dividend accrues and must be paid before any ordinary dividend), perpetual (no maturity) and redeemable at the company's option from a first call date at the liquidation preference of \$25.00 a share plus accrued dividends. Adjusted EBITDA is stated before the preferred dividends. [FTAI 10-K FY2025; D10 §2.8]

FIXED-TO-FLOATING AND FIXED-RATE RESET PREFERRED

Two coupon designs appear. A fixed-to-floating preferred pays a fixed coupon until its first call date and then a floating rate, a reference rate plus a fixed spread, reset each quarter (Series A 8.25%, redeemed October 2024; Series B 8.00%, outstanding). A fixed-rate reset preferred pays a fixed coupon until its first reset date and then a new fixed coupon for the next five years equal to the five-year US Treasury yield plus a fixed spread (Series C 8.25%, full redemption announced 15 May 2026; Series D 9.50%, outstanding). [FTAI 10-K FY2025; GlobeNewswire, 15 May 2026, title; D10 §2.8]

EXHIBIT 10.16 — THE FOUR PREFERRED SERIES

Series	Design	Coupon	Shares; liquidation amount	Status	Quarterly dividend per share	Ticker	Source
A	Fixed- to- floating	8.25%	4,180,000; \$104.5M	Redeemed in full Oct 2024 at \$25.00 plus \$1.6M of accumulated and unpaid distributions; total outlay about \$106.1M		not retrieved	10-K FY2025
B	Fixed- to- floating	8.00%	not retrieved	Outstanding at 31 Dec 2025 per the 10-K summary returned; floating spread and reset date not retrieved (D10 T-7)	\$0.50 at the fixed rate ($8.00\% \times \$25 \div 4$); floating rate after reset not retrieved	not retrieved	10-K FY2025
C	Fixed- rate reset	8.25%	not retrieved	Outstanding at 31 Dec 2025; full redemption announced 15 May 2026; redemption date and amount not read	\$0.52 (Q1 2025), matching $8.25\% \times \$25 \div 4 = \0.515625	not retrieved	10-Q Q1 2025; GlobeNewswire 15 May 2026, title
D	Fixed- rate reset	9.50%	2,600,000; \$65.0M	Outstanding; first reset date not retrieved	\$0.59 (Q1 2025), matching $9.50\% \times \$25 \div 4 = \0.59375 ; about \$6.175M a year in total	not retrieved	10-K FY2025; 10-Q Q1 2025

Series	Design	Coupon	Shares; liquidation amount	Status	Quarterly dividend per share	Ticker	Source
Ticker note				The SEC company lookup returned "FTAIO" as a lookup key, which confirms at least one preferred ticker in the FTAI family; the mapping of FTAIO, FTAIP, FTAIN and FTAIM to series is a recollection the dossier marked unverified, and the GuruFocus transcript page for the Q2 2026 call is indexed under "FTAIN.PFD"			

Source: FTAI Aviation Ltd., Form 10-K for FY2025 (equity note, as returned); Form 10-Q for the quarter ended 31 Mar 2025 (preferred dividends declared); GlobeNewswire, "FTAI Aviation Announces Full Redemption of Outstanding 8.25% Fixed-Rate Reset Series C Cumulative Perpetual Redeemable Preferred Shares", 15 May 2026 (title only); Exhibit 4.15 to the FY2023 10-K (description of securities; located, not read). D10 §2.8, §4.6, T-7, U-4. Accessed 2026-10-03.

The redemption sequence raises a question the research could not answer. The company redeemed Series A, at 8.25%, in October 2024, and announced the redemption of Series C, also 8.25%, in May 2026, leaving Series B at 8.00% and Series D at 9.50% outstanding on the information retrieved. Nothing retrieved says why the lower-coupon Series B was kept while the two 8.25% series were retired, or what Series B's floating rate is after its reset; if its reset has passed, its current coupon is the reference rate plus its spread, not 8.00%, and that would be one explanation. [FTAI 10-K FY2025; D10 T-7, U-4] The Series D cost is small against the scale of the notes: \$6.175M a year, about \$1.54M a quarter, against about \$230M of illustrative note coupon. [D10 §2.8; worked example 10.5]

82.5 Two 2026 FINANCINGS AT THE VEHICLES

Two 2026 financings bear on the balance sheet, and both sit at the vehicles rather than at FTAI. On 22 May 2026 the company priced FTAI MRE 2026-1, which its release called the Strategic Capital vehicle's inaugural asset-backed securitization: \$612M of notes backed by 48 A320ceo and 737NG aircraft on lease to 23 airlines, in two classes, Series A expected to be rated Asf by Fitch and A(sf) by KBRA and Series B expected BBB+sf by Fitch, with closing expected on 4 June 2026 and both classes described as significantly oversubscribed. The release put the vehicle's fleet at 292 aircraft. The deal is therefore a take-out of the 2025 Partnership's asset-level debt, off FTAI's balance sheet; its coupons and loan-to-value were not in the research (D7 U12, partly resolved). [FTAI release, "FTAI Aviation Prices Inaugural Asset-Backed Securitization", 22 May 2026, via Finviz and AviTrader 26 May 2026] Part II §17 describes what such a transaction is. The second is the \$2.0B warehouse facility for the 2026 SPV closed on 14 August 2026, which the release places at the vehicle, not at FTAI, and which the same release counts within "\$5.5 billion of warehouse financing in less than two years"; the disclosed commitments of \$2.5B and \$2.0B sum to \$4.5B and reach \$5.5B only if the \$1.0B accordion is counted. [GlobeNewswire, 17 Aug 2026; reconciliation R-055] Whether any vehicle debt is consolidated onto FTAI's balance sheet was not retrieved; the 2025 Partnership is unconsolidated and carried under the equity method. [D7 §2.6; D10 U-3]

83. Dividends, share count and repurchases

ORDINARY DIVIDEND AND SHARE REPURCHASE PROGRAM

An ordinary dividend is the cash per share the board declares each quarter on the ordinary shares; it is discretionary and ranks behind the cumulative preferred dividends. A share repurchase program is a board authorisation to buy back up to a stated dollar amount of shares over time at management's discretion; it is not a commitment to buy any. FTAI's quarterly dividend was \$0.30 through the second quarter of 2025 and was raised at each of the next four results: \$0.35 (declared 27 Oct 2025), \$0.40 (25 Feb 2026), \$0.45 (29 Apr 2026) and \$0.50 (29 Jul 2026). A \$500M repurchase program was announced on 15 Sep 2026. [GlobeNewswire release titles and texts, 29 Jul 2025 to 15 Sep 2026; orchestrator notes on R-068]

83.1 DIVIDENDS PER SHARE BY QUARTER

EXHIBIT 10.17 — ORDINARY DIVIDEND PER SHARE BY QUARTER, AS RETRIEVED, 2022 TO Q2 2026

Quarter the dividend relates to	Declared	Rate per share	Source and level
2022, each quarter		not retrieved	D10 U-8
2023, each quarter		not retrieved	D10 U-8
Q1–Q3 2024		not retrieved	D10 U-8
Q4 2024	26 Feb 2025	\$0.30	Release title, 26 Feb 2025 ("Declares Dividend of \$0.30")
Q1 2025	Apr 2025	\$0.30	10-Q Q1 2025
Q2 2025	29 Jul 2025	\$0.30	Release title, 29 Jul 2025 ("Declares Dividend of \$0.30")
Q3 2025	27 Oct 2025; payable 19 Nov 2025	\$0.35	Release title and text, 27 Oct 2025 ("Increases Dividend to \$0.35")
Q4 2025	25 Feb 2026	\$0.40	Release title, 25 Feb 2026 ("Increases Dividend to \$0.40")
Q1 2026	29 Apr 2026	\$0.45	Release title, 29 Apr 2026 ("Increases Dividend to \$0.45")
Q2 2026	29 Jul 2026	\$0.50	Release title, 29 Jul 2026 ("Increases Dividend to \$0.50")
Declared for the four quarters of 2025 (derived)		\$1.35	0.30 + 0.30 + 0.35 + 0.40
Declared for the first two quarters of 2026 (derived)		\$0.95	0.45 + 0.50

Source: GlobeNewswire release titles and texts of 26 Feb 2025, 29 Jul 2025, 27 Oct 2025, 25 Feb 2026, 29 Apr 2026 and 29 Jul 2026; FTAI Aviation Ltd., Form 10-Q for the quarter ended 31 Mar 2025. The 2022 to Q3 2024 quarterly rates were not retrieved; the FY2025 10-K equity note would supply them. Reconciliation R-068 and its resolution in the orchestrator notes. Accessed 2026-10-03.

The research recorded a contradiction on the 2025 path and resolved it. The D10 dossier had inferred that the rate stayed at \$0.30 throughout 2025 because the February 2026 release

described a raise "to \$0.40", and marked the inference unconfirmed. [D10 §4.7] The release titles show otherwise: the Q3 2025 results of 27 October 2025 raised the dividend to \$0.35, so 2025 had three rates. [GlobeNewswire, 27 Oct 2025; reconciliation R-068; orchestrator notes] This Part uses the release-title path. The quarterly rates for 2022, 2023 and the first three quarters of 2024 were not retrieved. [D10 U-8]

83.2 SHARE COUNT

BASIC AND DILUTED EARNINGS PER SHARE; RSUS AND PSUS

Basic earnings per share divides net income attributable to ordinary shareholders by the weighted-average number of ordinary shares outstanding in the period. Diluted earnings per share adds the shares that would be issued if outstanding restricted stock units (RSUs, which vest with time) and performance stock units (PSUs, which vest on performance conditions) were settled. Dividing reported net income by reported EPS therefore gives the weighted share count, basic and diluted: about 102 to 104 million at FTAI in 2025 and 2026. [FTAI releases, 25 Feb 2026 and 29 Jul 2026; D10 §2.10]

EXHIBIT 10.18 — SHARE COUNT HISTORY AS RETRIEVED AND AS IMPLIED

Date or period	Shares (millions)	Basis	Source
Before 28 May 2024	about 100	Assumed from the post-deal count less the internalization issue; not retrieved	D10 §2.1, U-6
28 May 2024	+1.867	1,866,949 shares issued to the Former Manager	10-Q Q2 2024
Q3 2025	103.6 (diluted)	\$114,009K ÷ \$1.10	Release 27 Oct 2025; arithmetic
Q4 2025	102.6 (basic); 103.6 (diluted)	\$111,852K ÷ \$1.09; ÷ \$1.08	Release 25 Feb 2026; arithmetic
Record date, 2026 proxy	about 102.6	13,865,244 ÷ 13.51%	DEF 14A 2026; arithmetic
Q2 2026	102.2 (basic); 104.1 (diluted)	\$117,585K ÷ \$1.15; ÷ \$1.13	Release 29 Jul 2026; arithmetic
Weighted basic and diluted shares by year, 2021–2025	not retrieved		D10 U-8
Equity issuances other than the internalization shares	not retrieved		D10 U-8

Source: FTAI Aviation Ltd., Form 10-Q for the quarter ended 30 Jun 2024; earnings releases of 27 Oct 2025, 25 Feb 2026 and 29 Jul 2026; DEF 14A 2026. Implied counts are arithmetic on reported net income and EPS and are rounded by the EPS figures' two decimals. Accessed 2026-10-03.

The share count has been close to flat since the internalization: about 102 to 103 million basic shares in every period retrieved, with diluted shares one to two million higher for unvested RSUs and PSUs. [D10 §2.10] The company has issued no equity for cash in the period the research covers, as far as the material retrieved shows; the FY2025 equity note would confirm that and give the annual weighted counts. [D10 U-8]

WORKED EXAMPLE 10.7 — THE CASH COST OF THE ORDINARY DIVIDEND AT EACH RATE, ON THE IMPLIED SHARE COUNT

Inputs: about 102.6 million shares (derived from the 2026 proxy, Exhibit 10.18); quarterly rates of \$0.30, \$0.35, \$0.40, \$0.45 and \$0.50 (sourced: release titles, Exhibit 10.17).

at \$0.30: $0.30 \times 4 \times 102.6\text{M} = \123.1M a year
 at \$0.50: $0.50 \times 4 \times 102.6\text{M} = \205.2M a year
 declared for 2025: $1.35 \times 102.6\text{M} = \138.5M
 Series D preferred, for comparison: \$6.175M a year

So the annualised ordinary dividend rose from about \$123M at the 2024 rate to about \$205M at the mid-2026 rate, a 67% increase in the cash commitment over eighteen months, on a flat share count. Set beside the operating cash outflow of \$81 (\$310.7M in 2025) and the net debt of \$82 (\$3.04B), the dividend is one of three uses of cash the company funds from its balance sheet while inventory grows; the others are the inventory itself and, from September 2026, any shares bought under the \$500M authorisation. Limits of the example: the share count is implied, not reported; dividends paid in a calendar year differ from dividends declared for it by one quarter's timing; the repurchase program's terms, duration and funding were not read. [DEF 14A 2026; GlobeNewswire release titles; GlobeNewswire, 15 Sep 2026, title; D10 §2.10, U-8]

83.3 THE REPURCHASE PROGRAM

On 15 September 2026 the company announced a \$500M share repurchase program. [GlobeNewswire, 15 Sep 2026, title] The research read the title and not the body, so the program's duration, whether it is open-market or otherwise, how it is funded, and whether any shares had been bought by the research date are unknown. [D10 U-8] At the implied count of about 102.6 million shares, \$500M is a sum whose share-count effect depends entirely on the price paid, which this Part does not quote; the proportion of the company it could retire is therefore not computed here.

84. Guidance history

GUIDANCE

Guidance is management's published forward target for a financial measure. FTAI gives it in Adjusted EBITDA from its business segments, by segment and in total, excluding Corporate and Other, and in modules produced by Aerospace Products. It has set 2025 guidance (30 Dec 2024), raised 2026 guidance twice and then cut one segment (Oct 2025 to Jul 2026), and set 2027 guidance (29 Jul 2026). Part VII §57 carries the segment detail for Aerospace Products; this section carries the dated history of every figure. [GlobeNewswire, 30 Dec 2024, 27 Oct 2025, 25 Feb 2026 and 29 Jul 2026; D10 §4.8]

EXHIBIT 10.19 — GUIDANCE AS SET, CHANGED AND DELIVERED, DECEMBER 2024 TO JULY 2026

Date set	For	Total guided	Segment split	What happened	Source and level
30 Dec 2024	2025 Adjusted EBITDA from reportable segments	about \$1.1–1.15B	Aviation Leasing about \$500M; Aerospace Products \$600–650M; no SCI line	Delivered: Aerospace Products \$671.3M; Aviation Leasing \$608.9M; segment sum \$1,280.2M; consolidated total \$1,190.9M (which is after Corporate and Other). Both segments exceeded the guided range	Release 30 Dec 2024; 10-K FY2025 (release; filing)
2025, target	2025 modules	750		Delivered 757	Call summaries via D5 (call coverage); release 25 Feb 2026
Dec 2024 / early 2025	2026 segment Adjusted EBITDA	\$1.4B	split not retrieved	Raised	Orchestrator notes on R-049 (notes)

Date set	For	Total guided	Segment split	What happened	Source and level
27 Oct 2025 (Q3 2025 release, per the orchestrator notes; the step was undated in the Q1 2026 10-Q text D6 read)	2026 segment Adjusted EBITDA	\$1.525B	Aerospace Products about \$1.0B; Aviation Leasing \$525M	Raised	Release 27 Oct 2025; 10-Q Q1 2026 (release; filing); R-050
25 Feb 2026	2026 segment Adjusted EBITDA	\$1.625B	Aerospace Products \$1.05B; Aviation Leasing \$575M	Leasing cut in July	Release 25 Feb 2026 (release)
25 Feb 2026	2026 modules	1,050 (+39% on 757)		Raised to 1,200 on the 30 Jul 2026 call per call coverage (GuruFocus, Nasdaq.com, MarketBeat); whether the 29 Jul 2026 release text carries 1,200 is unverified	Release and call, Feb 2026; call coverage, Jul 2026 (R-036)
26 Feb 2026 call	2026 Aerospace Products margin	"approach 40%" from the mid-30s	Levers named: the PMA high-pressure turbine (HPT) blade, more used material, the CFM materials agreement	Aspiration, not a guided figure	MarketBeat call summary (call coverage)

Date set	For	Total guided	Segment split	What happened	Source and level
29 Jul 2026	2026 segment Adjusted EBITDA	\$1.525B	Aerospace Products \$1.05B reaffirmed; Aviation Leasing cut from \$575M to \$475M "reflecting its continued shift to an asset-light business model"	One earlier search reading of the same 8-K exhibit returned \$575M; the orchestrator's later check found that a mis-read and dated \$575M to February	Release 29 Jul 2026 (release); R-049
29 Jul 2026	2027 business-segment Adjusted EBITDA	\$2.3B	Aerospace Products \$1.4B; FTAI Power \$450M; Aviation Leasing \$450M	One call summary (GuruFocus) gives Power as "\$450 million to \$750 million"; the release text confirms \$450M (R-062)	Release 29 Jul 2026 (release); call coverage
Any date	2026 consolidated figure; 2026 FTAI Power figure; any SCI-specific line	not retrieved			D10 U-14; D7 U9

Source: GlobeNewswire releases of 30 Dec 2024, 27 Oct 2025, 25 Feb 2026 and 29 Jul 2026 (and the 8-K Exhibit 99.1 of 29 Jul 2026); FTAI Form 10-K FY2025 and Form 10-Q Q1 2026; call coverage (MarketBeat, GuruFocus, Quartr, Nasdaq.com) as labelled; orchestrator notes, "Resolutions after reconciliation", R-036, R-049, R-062. Reconciliation R-049, R-050, R-062; E.1 rule 21 as modified by the notes. Accessed 2026-10-03.

Three things in that table need a sentence each. First, the 2025 guidance was given for the two reportable segments and both delivered above the range: Aerospace Products at \$671.3M against \$600–650M, and Aviation Leasing at \$608.9M against about \$500M. The consolidated Adjusted EBITDA of \$1,190.9M is a different number, after the Corporate and Other residual of about \$89.2M derived in §80, and should not be set against the \$1.1–1.15B segment range as if it were the same measure. [GlobeNewswire, 30 Dec 2024; FTAI 10-K FY2025; worked example 10.3]

Second, the 2026 path went up twice and then down once. The total rose from \$1.4B to \$1.525B and then to \$1.625B, and fell back to \$1.525B when Aviation Leasing was cut by \$100M on 29 July 2026 with the stated reason of the shift to an asset-light model. [GlobeNewswire, 27 Oct 2025, 25 Feb 2026 and 29 Jul 2026; orchestrator notes on R-049] The reconciliation pass had recorded two readings of the July release, \$475M and \$575M; the orchestrator's later search found the \$575M reading to be a mis-read of the February figure, and this Part follows the notes. [reconciliation R-049; orchestrator notes] The date of the \$1.525B step was undated in the Q1 2026 quarterly-report text one dossier read and is dated to the Q3 2025 release of 27 October 2025 by the notes. [reconciliation R-050; orchestrator notes]

Third, the 2027 figure introduces a third segment line. The \$2.3B is "Business Segment Adjusted EBITDA" across Aerospace Products (\$1.4B), FTAI Power (\$450M) and Aviation Leasing (\$450M). [GlobeNewswire, 29 Jul 2026] Whether FTAI Power is a third reportable segment in the 2026 quarterly reports, or sits within Aerospace Products until its first deliveries, was not confirmed; the FY2025 annual report has two segments and the guidance has three lines. [D8 U7; D10 U-18; cluster K-10] One call summary gives the Power figure as a range of \$450M to \$750M; the release text, per the orchestrator notes, confirms \$450M, and both are carried. [GuruFocus; orchestrator notes; reconciliation R-062] The 2026 module target of 1,050, set in February, was raised to 1,200 on the July call according to call coverage; the filed figure is 1,050 and the 1,200 is labelled as coverage wherever the primer uses it. [GlobeNewswire, 25 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat, 30 Jul 2026; reconciliation R-036]

WHY THE GUIDANCE IS IN SEGMENT ADJUSTED EBITDA AND WHAT THAT LEAVES OUT

Every FTAI guidance figure is Adjusted EBITDA from the business segments, before the Corporate and Other column and before the items the company removes in its definition (§80). The 2026 figure of \$1.525B therefore has no consolidated counterpart in anything retrieved, and the 2027 figure of \$2.3B has none either. A reader converting either into a consolidated figure must subtract a Corporate and Other residual that was about \$18.6M in 2024 and about \$89.2M in 2025 on the derivation in §80, and the research cannot say what that residual will be in 2026 or 2027. The guidance also contains no line for the Strategic Capital Initiative, whose fees, gains and equity-method share reach the Aviation Leasing figure (Part VIII §63), and no figure for net income, cash flow or interest. [D10 §4.8, U-14; D7 U9; worked example 10.3]

85. Risk factors, auditor, litigation, employees

This section carries the disclosures that an annual report groups under Items 1A, 3, 9A, 1 and 2, as far as the research retrieved them, together with the dated record of the 2025 review.

85.1 RISK FACTORS

RISK FACTORS (ITEM 1A); AUDITOR; SECURITIES CLASS ACTION; LEAD PLAINTIFF

Item 1A of a Form 10-K is the company's own list of the risks it considers material, each with a heading and an explanatory paragraph. The auditor is the independent registered public accounting firm that opines on the financial statements and, for a company of FTAI's size, on internal control over financial reporting. A securities class action is a lawsuit brought on behalf of all investors who bought a company's shares in a stated class period, alleging that disclosures were false or misleading. The lead plaintiff is the investor the court appoints to direct it; the deadline for lead-plaintiff motions in FTAI's case was 18 March 2025. [D10 §5, U-15, U-16; D5 §2.9; law-firm notices via D5]

EXHIBIT 10.20 — RISK FACTORS CONFIRMED FROM THE FILINGS, AND THE LIST NOT RETRIEVED

Risk as confirmed	What the filing says, in a sentence	Source
Russia and sanctions	New or stricter export controls and sanctions against Russia, and responses to them, could "delay or prevent access to certain assets"; 8 aircraft and 17 engines remained in Russia at 31 Dec 2025 with an insured value of \$210.7M, against a \$120.0M net write-off in 2022 and \$98.9M of recoveries to Q1 2026 (Part VIII §59; R-060)	10-K FY2025; 10-K FY2022; 10-Q Q3 2025; 10-Q Q1 2026
Leverage	The 10-K lists the company's "ability to successfully utilize leverage in connection with its investments" among its risk factors; the notes and revolver of §82 are the instruments	10-K FY2025
Strategic Capital vehicles	The vehicles are related parties that buy aircraft from FTAI, pay it servicing fees and buy engines and modules from it under MRE contracts; the 10-K's own risk wording was not returned, and the D7 dossier recorded the exposure (Part VIII §63)	10-K FY2025 via D7
Customer concentration	Two Aerospace Products customers were 16% and 11% of revenue in Q3 2025	10-Q Q3 2025
MRE stand-ready obligations	The company describes "stand-ready obligations to provide replacement CFM56-5B and CFM56-7B engines to customers as they become unserviceable", which requires it to hold a pool of serviceable engines against the obligation	10-Q Q3 2025
Full Item 1A list	Headings not retrieved. The research expected, and could not confirm, risk factors on dependence on CFM, GE and other OEMs for new parts and repairs; regulatory approvals from the US Federal Aviation Administration (FAA) and the European Union Aviation Safety Agency (EASA), including PMA; key personnel; and litigation	D10 U-15

Source: FTAI Aviation Ltd., Form 10-K for FY2025 (risk-factor sentences as returned by search); Form 10-K for FY2022; Forms 10-Q for Q3 2025 and Q1 2026; D6 §2.6 for Russia; D7 for the vehicles. Item 1A in full was not retrieved (D10 U-15). Accessed 2026-10-03.

Five risks are confirmed in the company's own words or figures, and the rest of Item 1A is unknown. The research did not retrieve the headings, so it cannot say how many risk factors the company lists or how it orders them, and this Part does not supply the ones it expected to find as if they were confirmed. [D10 U-15] Reading Item 1A of the FY2025 annual report would close the gap in a single step.

85.2 THE AUDITOR

The FY2025 annual report records, as the orchestrator notes transcribe it, that Ernst & Young LLP was the company's auditor from 2016 to 2025 and that the Audit Committee engaged KPMG LLP as auditor for FY2025 with effect from 17 June 2025; the auditor's report in the FY2025 annual report is dated 27 February 2026, the filing date. [FTAI 10-K FY2025, via orchestrator notes] The 2025 proxy had asked shareholders to ratify Ernst & Young LLP as auditor for FY2024. [FTAI DEF 14A 2025, via D5] The notes' own line assigns 2025 to both firms, and the reconciliation pass recorded that which firm signed the FY2025 report was not stated. [reconciliation R-066, D.3] Read together, the sentences are consistent with Ernst & Young having audited the years to FY2024 and KPMG the year FY2025, since an engagement effective June 2025 for FY2025 would end with the report of February 2026; but the research did not read the report's signature, so this Part states the two firms, the two dates and the ambiguity, and does not resolve it. Whether the annual report disclosed any material weakness, meaning a deficiency in internal control over financial reporting that creates a reasonable possibility of a material misstatement, was not retrieved; neither was the auditor's opinion on internal control. [D10 U-16]

85.3 THE AUDIT COMMITTEE REVIEW OF 2025

The sequence is dated by the filings. Muddy Waters Research published its report on 15 January 2025. On 18 January 2025 the Audit Committee "determined to commence a review (which included the engagement of independent advisors) in response to the assertions", in the January 8-K's words as the D10 dossier read it; the D5 dossier's reading of the same 8-K is that the committee "determined to begin a review, which included engaging independent advisors". [FTAI 8-K, Jan 2025; D10 §2.11; D5 §2.9] The conclusion was announced in February 2025; trade coverage of 20 February 2025 reported the shares rising on it, which dates the announcement to on or about 19 or 20 February. [FTAI 8-K Ex. 99.1, Feb 2025; StockStory via Wedbush, 20 Feb 2025]

The wording of the conclusion differs by document, and the reconciliation pass asked that the versions be kept apart. The February 8-K exhibit, as D5 read it: "After a thorough and comprehensive review with the support of independent legal and forensic accounting advisors,

the Audit Committee determined that the allegations made against the Company are without merit. The Company expects to file its Form 10-K timely." [FTAI 8-K Ex. 99.1, Feb 2025, via D5] The same exhibit, as D10 read it: the committee "completed its review, conducted by independent legal and forensic accounting advisors ... and determined that the allegations made against the Company are without merit". [FTAI 8-K Ex. 99.1, Feb 2025, via D10] The Q3 2025 quarterly report: "the internal investigation concluded that the allegations of misconduct were all without merit". [FTAI 10-Q Q3 2025] The FY2025 annual report, as the orchestrator notes transcribe it: an Audit Committee internal investigation, with outside counsel and forensic accountants, into the January 2025 short-seller reports concluded the allegations of misconduct were "all without merit". [FTAI 10-K FY2025, via orchestrator notes] The 8-K speaks of a review and of allegations; the later filings speak of an internal investigation and of allegations of misconduct; and the annual report speaks of reports in the plural. This Part does not fuse them. [reconciliation R-063; E.1 rule 22]

The allegations themselves are carried as two secondary summaries, because the report was not read. As the amended complaint states them, per law-firm case pages: one-time engine sales reported as MRO (maintenance, repair and overhaul) revenue; whole engine sales presented as module sales; and depreciation of engines not on lease lowering reported cost of goods sold. [ktmc.com and rgrdlaw.com case pages, via orchestrator notes] As the press summarised the report in January 2025: one-time engine sales recorded as MRO revenue in Aerospace Products; the majority of the segment's Adjusted EBITDA estimated to come from gains on sales; and "misrepresentation of sales, inflation of ... margins ..., channel stuffing, and a few other accounting red flags". [Benzinga, 15 Jan 2025; Crossroads Capital letter via Hedge Fund Alpha; D5 §2.9; reconciliation R-064] Part VII §54 sets the allegations beside the segment's disclosures. No restatement was found or referenced in anything retrieved, and the FY2025 annual report presents MRE Contract revenue as a line separate from Aerospace products revenue, a change the research could not link to the review. [D5 §2.9]

85.4 THE SECURITIES CLASS ACTION

EXHIBIT 10.21 — THE REVIEW, THE LITIGATION, THE AUDITOR, THE PEOPLE AND THE PROPERTIES, IN ONE TABLE

Item	Detail	Source and level
Short report	Muddy Waters Research, 15 Jan 2025; the report itself was not read	8-K Jan 2025; trade press (filing; press)
Review start	18 Jan 2025, Audit Committee, independent advisers	8-K Jan 2025 (filing)

Item	Detail	Source and level
Review conclusion	Feb 2025, announced on or about 19–20 Feb 2025; wording per §85.3	8-K Ex. 99.1 Feb 2025; StockStory via Wedbush 20 Feb 2025 (filing; press)
Class action	Securities-fraud class action, United States District Court for the Southern District of New York, No. 25-cv-00541, Judge Jeannette A. Vargas	Law-firm case pages, ktmc.com and rgrdlaw.com, via orchestrator notes (law-firm pages; last-update date not visible)
Class period	23 Jul 2024 to 21 Jan 2025	Same
Lead-plaintiff deadline	18 Mar 2025	BFA Law notice, 12 Feb 2025 (law-firm release)
Procedural status	Defendants' motion to dismiss the amended complaint filed 20 Nov 2025; fully briefed and pending at the research date; no settlement or dismissal found	Law-firm case pages via orchestrator notes
The company's own description of the case	The 10-K text returned by search did not describe the litigation; Item 3 was not retrieved	D10 U-16; orchestrator notes
Derivative suits	Not retrieved	D10 U-16
SEC correspondence	A file dated 27 Nov 2024 exists on EDGAR, the SEC's electronic filing system; subject not retrieved; a January 2025 EDGAR document also exists	D5 U15; D10 U-16
Auditor	Ernst & Young LLP, 2016–2025; KPMG LLP engaged for FY2025 effective 17 Jun 2025; report dated 27 Feb 2026; which firm signed not confirmed	10-K FY2025 via orchestrator notes (filing); R-066
Internal control conclusion; material weakness	Not retrieved	D10 U-16
Employees	985 full-time employees and independent contractors at 31 Dec 2025, a mixed definition; 494 full-time employees in Canada, about 71% of them covered by collective bargaining agreements (about 351, derived)	10-K FY2025 via orchestrator notes (filing)
Employees by site; test-cell count	Not retrieved	Cluster K-13

Item	Detail	Source and level
Maintenance properties	100%-owned facilities in Montréal (about 500,000 sq ft per the 10-K; 526,000 per the release), Miami, Lisbon and Orange; 50% of QuickTurn Europe at Rome Fiumicino, Italy (200,000 sq ft); 50% of Prime Engine Accessories in Bristol, Connecticut; "over one million square feet" collectively	10-K FY2025 (filing); GlobeNewswire 9 Sep 2024
Headquarters; Item 2 properties list	Not retrieved; the research expected a New York headquarters and did not confirm it	D10 U-17

Source: FTAI Aviation Ltd., Forms 8-K of Jan 2025 and Feb 2025 (Ex. 99.1); Form 10-Q for Q3 2025; Form 10-K for FY2025 (via orchestrator notes); law-firm case pages (ktmc.com, rgrdlaw.com) and the BFA Law notice of 12 Feb 2025; StockStory via Wedbush, 20 Feb 2025; GlobeNewswire 9 Sep 2024. Reconciliation R-063, R-064, R-065, R-066; clusters K-11 and K-13. Accessed 2026-10-03.

The case is pending. A securities-fraud class action was filed in the Southern District of New York after the January 2025 report, docketed as No. 25-cv-00541 before Judge Jeannette A. Vargas, on behalf of investors who bought in the class period from 23 July 2024 to 21 January 2025. The defendants filed a motion to dismiss the amended complaint on 20 November 2025, and law-firm case pages describe it as fully briefed and pending; the date of those pages' last update is not visible. [ktmc.com and rgrdlaw.com via orchestrator notes] The company's own description of the proceedings, in Item 3 of the annual report, was not returned by the search, and the research did not find any derivative suit or any settlement. [D10 U-16; cluster K-11]

85.5 EMPLOYEES AND PROPERTIES

EMPLOYEES (HEADCOUNT AS THE 10-K DEFINES IT)

FTAI reports 985 "full-time employees and independent contractors" at 31 December 2025, a single figure that mixes two categories, and separately 494 full-time employees in Canada, about 71% of whom are covered by collective bargaining agreements. The Canadian figure counts employees only, so the two numbers are not on one basis. A figure of "approximately 7,800 people across over 50 facilities" that one search summary attached to FTAI is StandardAero's and is not an FTAI figure. [FTAI 10-K FY2025, via orchestrator notes; reconciliation R-065]

Two things follow from the headcount. About half the workforce is in Canada, which is where the Montréal shop and the Module Factory are, and about 351 of those 494 Canadian employees (71%, derived) are unionised. [FTAI 10-K FY2025, via orchestrator notes; arithmetic] And the company is small in people against the revenue it reports: 985 employees and contractors against \$2,507.4M of 2025 revenue, which is consistent with a business that buys, refurbishes and sells high-value

material rather than one that sells labour hours. The by-site counts and the number of test cells were not retrieved. [cluster K-13]

On properties, the annual report's business description says the company conducts engine maintenance at its 100%-owned facilities in Montréal, Miami, Lisbon and Orange, and through its 50% equity ownership in QuickTurn Europe in Rome and in Prime Engine Accessories in Bristol, and that the facilities "span over one million square feet" with test cells, tooling and engineering capability. [FTAI 10-K FY2025, via D5] The Rome site is QuickTurn Europe at Rome Fiumicino Airport, Italy; an earlier orchestrator entry placing it in New York State was an error that the notes corrected, and Part XII records the correction. [FTAI 10-K FY2025; GlobeNewswire, 5 Jun 2025; reconciliation R-040] Item 2 of the annual report, which would list the headquarters and each leased or owned property with its size, was not retrieved; Part VII §50 carries what is known of each site. [D10 U-17]

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
R-026	FTAI–Chromalloy relationship: "a joint venture" (10-K) against an "exclusive perpetual partnership" formed 2018 (investor blogs); the Advanced Engine Repair JV (25%, Dec 2016 and Aug 2019, \$22,429K) of unknown identity	§75.1, Exhibit 10.1, Exhibit 10.4
R-039	Montréal acquisition dated 2023 (chain map and D10 originally) against 9 Sep 2024, \$170.0M (release)	§75.3, Exhibit 10.1
R-040	Rome site: "Rome NY" (chain map, unsourced) against Rome Fiumicino, Italy (10-K; release 5 Jun 2025)	§85.5
R-043	Whether MRE Contract revenue sits inside the Aerospace Products segment: totals agree (\$1,936,244K = \$1,600,456K + \$335,788K) but the explicit segment-note mapping was not retrieved	§79.3, Exhibit 10.11
R-044	MRE Contract revenue defined as "stand-ready obligations" (Q3 2025 10-Q) against "the transaction price for sales of ... engines and related modules to the SPVs" (Q2 2026 10-Q)	§79.2, Exhibit 10.10
R-045	FY2025 cost of sales \$1,349,719K (consolidated line) against \$1,240,368K (Aerospace Products segment table); derived 35.9% labelled	§79.3, worked example 10.2
R-048	Stake in the 2025 Partnership 19% (FY2025 10-K; 2026 10-Qs) against 20% (Q3 2025 10-Q)	§78.1, Exhibit 10.4; §80.1
R-049	2026 Aviation Leasing guidance \$575M (25 Feb 2026) against \$475M (29 Jul 2026); an earlier \$575M reading of the July release found to be a mis-read	§84, Exhibit 10.19

Id	The two (or more) figures	Where in this Part
R-050	The \$1.4B to \$1.525B step: undated in one 10-Q reading; 27 Oct 2025 per the notes	§84, Exhibit 10.19
R-051	Q2 2026 Aviation Leasing Adjusted EBITDA \$88.2M (call coverage) against no Leasing line returned from the release; residual –\$46.5M unconfirmed	§79.4, Exhibit 10.12; §80.2, Exhibit 10.13
R-052	FY2024 lease income \$234,411K against \$255,338K, two readings of the FY2025 10-K	§79.1, Exhibit 10.8; §79.3, Exhibit 10.11
R-053	FY2021 lease income \$173,864K against FY2021 Aviation Leasing segment revenue \$128,328K, both from the FY2022 10-K	§77.1, Exhibit 10.3; Exhibit 10.8
R-054	Gains on sale to the 2025 Partnership: \$50.1M (nine-month sum of quarters) against \$46.4M (FY2025)	§80.1
R-055	"\$5.5 billion of warehouse financing" (release) against \$2.5B + \$2.0B = \$4.5B disclosed, \$5.5B only with the \$1.0B accordion	§82.5
R-057	Sales to the Partnership "non-recurring" under ASC 610-20 (10-K) against "\$3.0+ billion of capital annually" and primary investment activity through SCI (release; 10-Q)	§80.1
R-058	"Asset-light" (releases; the Jul 2026 guidance cut) against a 19% LP interest carried at \$281.7M (31 Dec 2025) and \$365.5M (30 Jun 2026), about \$380M at full commitment	§81.1
R-059	OneIM: ">\$4 billion" of deployment for the inaugural vehicle; commitment size and role undisclosed	§75.4, Exhibit 10.1
R-060	Russia: \$120.0M net write-off (2022) against \$210.7M insured value and \$98.9M recovered	§75.2; §85.1, Exhibit 10.20
R-061	J&F Power Systems: FTAI's "joint venture" (release; 10-Q) against Jerreh's "subsidiary" (exchange disclosure as reported)	§78.1, Exhibit 10.4
R-062	2027 FTAI Power Adjusted EBITDA \$450M (release text; Quartr, MarketBeat) against "\$450–750M" (GuruFocus)	§84, Exhibit 10.19
R-063	Review wording: "review ... independent advisors ... allegations ... without merit" (8-K, two readings) against "internal investigation ... allegations of misconduct ... all without merit" (10-Q, 10-K); conclusion date Feb 2025, trade coverage 19–20 Feb	§85.3, Exhibit 10.21
R-064	Allegations as the amended complaint states them (law-firm pages) against the press summaries; the report not read	§85.3

Id	The two (or more) figures	Where in this Part
R-065	Headcount 985 employees and contractors / 494 Canadian employees (10-K) against a misattributed 7,800 (StandardAero)	§85.5
R-066	Auditor: Ernst & Young LLP 2016–2025 and KPMG LLP from 17 Jun 2025 for FY2025; which firm signed the FY2025 report not stated	§85.2, Exhibit 10.21
R-067	CFO: Nicholas McAleese (Q2 2026 call) against Eun (Angela) Nam (2026 proxy, 2025 pay); resolved by the 6 Mar 2026 release	§78.2, Exhibit 10.5
R-068	2025 dividend: "\$0.30 through 2025" (D10 inference, unconfirmed) against \$0.30, \$0.30, \$0.35, \$0.40 by quarter (release titles)	§83.1, Exhibit 10.17
D10 T-1	Five note balances summing to \$3,537,185K (mixed dates) against long-term debt, net of \$3,451,087K at 31 Mar 2026	§82.1, Exhibit 10.15
D10 T-2	Internalization: "\$150.0 million to the Former Manager" against \$300.0M total, plus unquantified accrued and reimbursable amounts	§76.2, Exhibit 10.2
D10 T-3	Five revenue lines summing to \$2,496,898K against total revenues of \$2,507,409K; \$10,511K residual	§79.2, Exhibit 10.10
D10 T-5	COO: Stacy Kuperus (proxy) against David Moreno's earlier title; Moreno now President per call coverage	§78.2, Exhibit 10.5
D10 T-7	Series B 8.00% outstanding while both 8.25% series were redeemed; floating rate not retrieved	§82.4, Exhibit 10.16
R-036	2026 modules 1,050 (release, Feb 2026) against 1,200 (call coverage, Jul 2026)	§84, Exhibit 10.19

Unknowns carried in this Part

- **U-D10-01 (D10): U-1 — Full income statement 2021–2025 and 1H 2025 / 1H 2026.** Operating expense lines, depreciation and amortization, interest expense, tax, net income attributable to shareholders by year; annual Adjusted EBITDA, its definition and reconciliation items; segment revenue and Adjusted EBITDA by year; 1H totals. Resolve with the FY2025 10-K Item 8 and Item 7 segment tables [S1], the Q4/FY2025 release [S3], the Q2 2026 release [S4] and the Q2 2025 release [S25]. Also the FY2021 10-K [S15] and FY2022 annual report (EDGAR index) for 2021–2022. Partly closed by the orchestrator notes (the segment table for 2023–2025 and the FY2025 expense lines, carried in Exhibits 10.8 and 10.11).
- **U-D10-02 (D10): U-2 — Cash flow detail.** Acquisition of leasing equipment, acquisition of property, plant and equipment, proceeds from asset sales, working-capital changes

(inventory), and whether the company defines free cash flow. Resolve with the cash flow statement in [S1] and [S2].

- **U-D10-03 (D10): U-3 — Debt detail.** Issue dates and original principal of each note series; principal vs carrying amount; call schedules; covenants; revolver commitment size; any term loans or asset-level debt consolidated from SCI vehicles; total debt and equity at 31 Dec 2025. Resolve with the debt note in [S1] and [S2] and the note-offering 8-Ks.
- **U-D10-04 (D10): U-4 — Preferred detail.** Series B share count, floating spread and reset date; Series C share count, reset terms, redemption date and amount [S20]; Series D issue date and first reset; preferred tickers. Resolve with the equity note in [S1], Exhibit 4.15 to the FY2023 10-K (description of securities) [S25], and the 15 May 2026 release.
- **U-D10-05 (D10): U-5 — Management agreement economics.** The management fee rate and base, the incentive allocation formula, and fees and allocations paid in 2021, 2022, 2023 and Jan–May 2024. Recollection, unverified: a fee of 1.50% per annum of total equity. Resolve with the related-party notes in the FY2021–FY2024 10-Ks [S15][S14][S27 folder].
- **U-D10-06 (D10): U-6 — Internalization details.** FTAI's closing share price on 28 May 2024 versus the \$80.35 implied value; the amount of accrued compensation and reimbursable expenses paid; accounting treatment (expensed as "internalization" cost or capitalized); whether FIG LLC still holds the shares. Resolve with the 8-K exhibits [S10] [S11] and the Q2 2024 10-Q [S9].
- **U-D10-07 (D10): U-7 — Leverage definitions and credit ratings.** The company's leverage covenant definitions; Moody's, S&P and Fitch issuer and note ratings with dates. Not searched (budget exhausted). Resolve with the agencies' press releases and the 10-K liquidity section.
- **U-D10-08 (D10): U-8 — Dividend and share-count history.** Quarterly ordinary dividend 2022–2024 and Q1 2026; basic and diluted weighted shares by year; equity issuances other than the internalization shares; repurchase program terms [S21]. Resolve with [S1] equity note and the Sep 2026 release. Partly closed: the Q1 2026 rate is \$0.45 (release, 29 Apr 2026).
- **U-D10-09 (D10): U-9 — Officers' titles and tenure.** CFO confirmation (Eun Nam), General Counsel's name, David Moreno's current role, start dates. Resolve with the proxy's executive officers section [S6] and the 10-K Item 10. Partly closed: the 6 Mar 2026 release names Nicholas McAleese CFO and Michael Hazan Chief Accounting Officer, succeeding Nam; call coverage names Moreno President.
- **U-D10-10 (D10): U-10 — Governance detail.** Which directors are independent; any remaining Fortress affiliation; lead independent director; bonus metrics and PSU

conditions; other NEOs' pay; the "directors and officers as a group" ownership; any surviving Fortress-era arrangement. Resolve with [S6].

- **U-D10-11 (D10): U-11 — Early asset mix and the shift to aviation.** Figures for Jefferson Terminal, Repauno, Transtar and Long Ridge; dates and prices of QuickTurn, the Chromalloy JV and the Lockheed Martin Commercial Engine Solutions acquisition. Resolve with the FY2021 10-K [S15], the FY2022 annual report and the relevant 8-Ks. Partly closed by the orchestrator notes: LMCES \$170.0M, 9 Sep 2024; QuickTurn 4 Jan 2023 and 1 Dec 2023 (\$30.3M).
- **U-D10-12 (D10): U-12 — Reasons for the Cayman redomiciliation and the NYSE-to-Nasdaq move.** Resolve with the 2022 DEFM14A [S17] and the listing-transfer 8-K.
- **U-D10-13 (D10): U-13 — Subsidiaries, JV ownership percentages and equity-method carrying values; the role of FTAI Finance Holdco Ltd.** Resolve with Exhibit 21.1 [S27] and the investments note in [S1]. Partly closed by the orchestrator notes (Exhibit 10.4).
- **U-D10-14 (D10): U-14 — 2025 guidance as set on 30 Dec 2024 and the Feb 2026 2026 guidance in full (including FTAI Power and consolidated).** Resolve with [S24] and [S3]. Partly closed: the 2025 guidance is supplied by D7 (Exhibit 10.19).
- **U-D10-15 (D10): U-15 — Risk factors in full.** Supplier dependence on CFM/GE and other OEMs for new parts and repairs, regulatory approvals (FAA/EASA, PMA), SCI vehicle risks, key personnel, litigation; annual customer concentration and customer names. Only Russia, leverage, SCI, customer concentration (Q3 2025) and MRE stand-ready obligations were confirmed. Resolve with Item 1A of [S1].
- **U-D10-16 (D10): U-16 — Legal proceedings, auditor and internal control.** Status of the securities class actions filed after the Jan 2025 report and any derivative suits; the auditor's name (recollection, unverified: Ernst & Young LLP); whether any material weakness was disclosed; the content of the Jan 2025 EDGAR correspondence [S12 folder] and the later 2025 8-K [S25]. Resolve with Item 3, Item 9A and the auditor's report in [S1]. Partly closed by the orchestrator notes (Exhibit 10.21).
- **U-D10-17 (D10): U-17 — Employees and locations.** Headcount at 31 Dec 2025, by geography; properties list (New York headquarters, Montreal, Miami, Rome NY, Lisbon). Resolve with Item 1 "Human Capital" and Item 2 of [S1]. Partly closed: 985 and 494 (orchestrator notes); "Rome NY" corrected to Rome Fiumicino, Italy (R-040).
- **U-D10-18 (D10): U-18 — Whether FTAI Power is a third reportable segment in the 2026 10-Qs.** Resolve with [S2] segment note.
- **U-D5-16 (D5): U16 — Securities litigation status** (court, docket, motion to dismiss outcome, any settlement). Cluster K-11 with U-D10-16; partly closed as in Exhibit 10.21.

- **U-D5-19 (D5): U19 — FY2023 segment revenue and Adjusted EBITDA**, and the FY2022 baseline (the segment began ramping in 2022–2023 per the 10-Q language [S10 summary]). Cluster K-17 with U-D10-01; partly closed in Exhibit 10.11 and Exhibit 10.3.
- **U-D5-12 (D5): U12 — Inventory at 31 Dec 2025 and 30 Jun 2026; segment capex; the cost of building module inventory**. Cluster K-18 with U-D10-02.
- **U-D6-02 (D6): U2 — Annual segment lines 2023–2025**. Lease income, maintenance revenue, asset sales revenue and cost, gain on sale of leasing equipment, equity-method pick-up, and segment Adjusted EBITDA for FY2023, FY2024, FY2025 and each 2025 quarter. Resolve: segment note and MD&A of the FY2025 10-K ([S1]) and the Q4/FY2025 release ([S6]). (The 18th search aimed at this and was refused on budget.) Cluster K-17.
- **U-D6-11 (D6): U11 — Equity-method investment** in the Leasing segment: counterparty, ownership, assets, income by year. Resolve: FY2025 10-K "Investments" note. Cluster K-07 with U-D10-13.
- **U-D7-17 (D7): U17 — Q1 2025 10-Q "management fees and incentive allocations"**. The search summary flagged such text in S14; whether it concerns SCI or the pre-internalization Fortress management agreement is not verified.
- **U-D7-12 (D7): U12 — The May 2026 ABS**. Whether "FTAI Aviation Prices Inaugural Asset-Backed Securitization" [S26] is a 2025-SPV take-out or FTAI's own balance-sheet deal; size, tranches, pricing.
- **U-D8-07 (D8): U7. Segment reporting**. The 10-K FY2025 has two reportable segments [6]; call summaries speak of a "Power segment" for 2027. Whether Power is a third reportable segment from the Q2 2026 10-Q [10], or sits within Aerospace Products until first delivery, was not confirmed. Cluster K-10 with U-D10-18.

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2. FTAI Aviation Ltd., Exhibit 21.1 (subsidiaries) to the Form 10-K for FY2025 (located; not read). <https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai12312025exhibit211.htm>
3. FTAI Aviation Ltd., Form 10-Q for the quarter ended 31 March 2026. <https://www.sec.gov/Archives/edgar/data/1590364/000162828026029335/ftai-20260331.htm>
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11. FTAI Aviation Ltd., Forms 8-K of June 2024 (internalization closing and related; located, not read). https://www.sec.gov/Archives/edgar/data/1590364/000114036124030343/ny20031133x1_8k.htm ; https://www.sec.gov/Archives/edgar/data/1590364/000114036124028455/ny20030427x1_8k.htm ; https://www.sec.gov/Archives/edgar/data/1590364/000114036124028549/ny20030427x2_8k.htm
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13. FTAI Aviation Ltd., Form 8-K Exhibit 99.1, February 2025 (review completed; allegations without merit). https://www.sec.gov/Archives/edgar/data/1590364/000114036125005179/ef20043996_ex99-1.htm
14. FTAI Aviation Ltd., Form 10-K for the fiscal year ended 31 December 2022. <https://www.sec.gov/Archives/edgar/data/1590364/000159036423000007/ftai-20221231.htm>
15. Fortress Transportation and Infrastructure Investors LLC, Form 10-K for the fiscal year ended 31 December 2021. <https://www.sec.gov/Archives/edgar/data/1590364/000159036422000002/ftai-20211231.htm>
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17. Redomiciliation documents: FTAI Aviation Ltd., Form 8-K, November 2022, https://www.sec.gov/Archives/edgar/data/1590364/000114036122041407/brhc10044100_8k.htm ; DEFM14A 2022, <https://www.sec.gov/Archives/edgar/data/>

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PART XI

Markets and money: margin pools and peers

This Part sets FTAI Aviation beside the other companies that make money from the same engines, link by link along the chain: the manufacturer, the independent shops, the makers of approved alternative parts, the traders in used material, and the lessors. Each company's figures come from its own filing and carry their own basis. A reader who finishes it will be able to read any one of those filings against FTAI's and know, before comparing a single number, which numbers can be compared and which cannot.

TRANSLATION-TABLE ROWS ANSWERED

Part III §24 turned each demand driver into a requirement on the aftermarket. This Part answers the rows whose requirement is a question about who earns what. The row on low CFM56 retirements (Part III §24, row T-1: Low CFM56 retirements) asks who collects the shop-visit spending that a slow-retiring fleet sustains and who is starved of teardown feedstock by it; §86 and §87.5 answer with the filings of the OEM (the original equipment manufacturer), the independents and AerSale. The row on shop capacity (Part III §24, row T-8: MRO capacity shortfall and turnaround; MRO is maintenance, repair and overhaul) asks what the independents have built and what the OEM says about turnaround; §87.6 and §87.8 answer with StandardAero's and GE's filings. The row on lessor behaviour (Part III §24, row T-6: Lessor extension behaviour) asks what the lessors themselves say about demand and rates; §87.1 and §87.3 quote Willis Lease and Air Lease. The row on the CFM56 plateau, as the LEAP (CFM's successor engine, Part I §10) overtakes it (Part III §24, rows T-3, T-14: LEAP aftermarket parity / CFM56 plateau-then-decline), asks how long the pool lasts; §87.8 and §87.9 quote GE and Safran. The rows on new-engine durability (Part III §24, rows T-2, T-3: LEAP durability shortfall; row: GTF powder-metal groundings; GTF is the geared turbofan of Part III §20) appear here only as they show up in peers' margins and statements, in §86.4 and §87.9.

86. The margin pool across the chain

Part I §11 gave a first map of where money is made in keeping a CFM56 flying. This section builds the map link by link from the filings, with every margin on its stated basis, and then explains mechanically why each link's margin sits where it does. It ranks nothing. A high percentage margin and a low one can describe businesses of equal or unequal quality; the percentage is a function of what sits in the revenue line and what sits above and below the profit line, and that is what this section takes apart.

MARGIN POOL

The margin pool is this primer's term for how the profit earned in keeping an engine type flying is distributed across the links of the chain: the manufacturer's new parts and services, the shops' labour and test, the component repairers, the makers of approved alternative parts, the traders in used material and teardown, the lessors of engines and aircraft, and the servicers of third-party capital. No company reports a "margin pool"; the term names a way of reading the filings together. Its inputs are the slice-by-slice capture of a shop visit (Part IV §29) and each peer's reported margin on its own basis (this section). Part I §11 carried the preview and Part XII §92 carries the summary table.

[registry 1c; D2 §2.2; D11 §4.2]

COMPARABLE METRICS

Comparable metrics are the figures from a peer's filing that can be set beside a figure in FTAI's filings because both measure the same physical or financial quantity: owned engines, utilisation on a stated definition, maintenance-reserve revenue, gain on sale of leased equipment, a segment margin on a stated basis, approvals of alternative parts per year, gross margin on used material, and the services share of a manufacturer's revenue. The phrase is the primer's; the filings supply the quantities without naming the concept. A metric is comparable only when its definition, its period and its unit match on both sides, which §88 tests one by one. [registry 1c; D11 §2.1]

SEGMENT MARGIN BASES (FIVE)

The peers in this Part report profit on five different bases. Segment Adjusted EBITDA (StandardAero and FTAI) is segment profit before depreciation, amortisation and items each company defines as adjustments. Segment operating income (HEICO) is after depreciation and amortisation. Adjusted EBIT, where EBIT is earnings before interest and tax, is the basis used by MTU Aero Engines (MTU) and Lufthansa Technik: operating profit after depreciation and amortisation but before purchase-price-allocation effects and other defined adjustments. GE segment profit is GE's own measure, excluding certain corporate and non-operating items. Gross margin (AerSale) is sales less cost of sales, before all operating expenses. An EBITDA margin exceeds an EBIT margin for the same business by the depreciation share of revenue, so none of the five can be set beside another without an adjustment, and the adjusting figure is often not disclosed. Three fiscal calendars (December, October, May) add a second layer. [D11 §2.3, §2.5; reconciliation E.1 rule 35]

86.1 THE BILL BEFORE THE MARGIN: A SHOP VISIT, SLICE BY SLICE

Every margin in this section is earned on some slice of the money an engine owner spends. The largest single spending event is the shop visit (Part I §6 and §7), so the pool is easiest to see by starting with one visit and asking who collects each part of the bill. Air Cargo Week, a trade publication, puts material at 60–70% of a visit, labour at 20–30% and repairs and other shop work at 10–20%. [Air Cargo Week, 2025/26] Safe Fly Aviation, a consultancy blog, gives a finer cut for the CFM56: life-limited parts are 40–60% of parts cost when replaced, high-pressure turbine blades and vanes 15–25%, labour 15–20%. [Safe Fly Aviation, 2026, consultancy blog] The two breakdowns come from different secondary sources and neither names a shop-level data set; Part I §7.2 carries the tension between them. [reconciliation R-014]

EXHIBIT 11.1 — WHO COLLECTS EACH SLICE OF A CFM56 SHOP VISIT, AND THROUGH WHAT MECHANISM

Slice of the bill	Share of the visit	Who collects it	Mechanism	Peer in this Part whose filing isolates it
New OEM parts other than life-limited parts	Largest single slice within material (60–70% of the visit)	CFM International, meaning GE Aerospace and Safran, at catalogue list price	Sold through CFM's spares organisation to any shop; the shop passes the part through at list plus a handling margin, or the airline buys direct under a TrueChoice Material agreement (Part IV §28)	GE Aerospace, Commercial Engines & Services (CES); Safran (Propulsion services)
Life-limited parts	40–60% of parts cost when replaced	The OEM alone; no PMA (Parts Manufacturer Approval, Part VI §37) life-limited part exists for the CFM56 (Part VI §38)	Sold at list; the most visible escalator	GE; Safran
Used serviceable material	Substitutes for new parts	Teardown and used-material traders, and the OEM itself, which states it is "the largest used serviceable material provider" for its products	Priced as a percentage of list, 60–80% on average (Part V §34)	AerSale (AMS); AAR (Parts Supply, used-material line); AerCap Materials; Willis Aero
PMA parts	Alternative to new OEM parts	PMA makers	Sold "at lower prices than" OEM parts (Part VI §37)	HEICO (Flight Support Group)
Component repairs	10–20% of the visit	Repair vendors, OEM and independent, some under DER (Designated Engineering Representative) repairs (Part VI §40)	A repaired part costs a fraction of a new one	StandardAero (Component Repair Services); HEICO Flight Support Group (FSG), repair line

Slice of the bill	Share of the visit	Who collects it	Mechanism	Peer in this Part whose filing isolates it
Labour and test	20–30% of the visit	The engine shop: OEM, independent or airline	Shop rate times hours; labour-rate inflation 5.5–6.0% in 2025 (Oliver Wyman)	StandardAero (Engine Services); MTU; Lufthansa Technik; GE (internal shop visits)
Shop margin and risk	Embedded in every line	The shop under time-and-materials; the OEM under a flight-hour agreement, which prices the risk in	Time-and-materials versus rate per flight hour (Part IV §28)	GE (long-term service agreements)
The module exchanged instead of overhauled	Replaces part of the visit	The exchanger, who sells a refurbished module and takes back the run-out one	Part VII §46	FTAI (Aerospace Products); GE (TrueChoice Transitions)

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26; Safe Fly Aviation, 2026 (consultancy blog); GE Aerospace investor-relations page, "Recent events: your questions answered", 2026; Oliver Wyman, "The new MRO supply paradigm", Apr 2026; Aviation Week, "MRO Memo: A Seller's Market For Used Parts", date not captured. Compiled from D2 §2.2 with the peer column added in this Part. Reconciliation R-014.

Two mechanical consequences follow, and both shape every margin below. Because 60–70% of the visit is material and most material is priced off the manufacturer's catalogue, the catalogue's annual repricing flows into the cost of every visit regardless of who performs it. [D2 §2.2] And because the shop that is not the manufacturer buys those parts at list, its only levers on the bill sit on the material line: used parts instead of new, approved alternative parts instead of the manufacturer's, repair instead of replacement, and exchange instead of overhaul. Parts V, VI and VII treat those four levers; this section shows what each one's practitioner reports earning.

86.2 LINK ONE: THE MANUFACTURER'S NEW PARTS AND SERVICES

GE Aerospace reports the CFM56's manufacturer side in one segment, Commercial Engines & Services (CES). For 2025 CES had equipment revenue of \$8,304M and services revenue of \$25,010M, a total of \$33,314M, so services were 75.1% of the segment (derived in the research). Segment profit was \$8,861M, a margin of 26.6%. Revenue rose \$6.4B (24%) and profit \$1.8B (26%). [GE 10-K FY2025] GE names the drivers in two lists. Revenue grew on "increased spare parts volume, internal shop visit volume and worksopes, increased engine deliveries and pricing". Profit grew on "increased spare parts volume, internal shop visit volume and

workscores and improved pricing", partly offset by "the impact of higher install engine deliveries, inflation, higher growth investment and an unfavorable change in estimated profitability of long-term service agreements, primarily from the estimated impact from tariffs". [GE 10-K FY2025] Internal shop-visit revenue grew 24% in 2025, after 19% in 2024 and 27% in 2023. [GE 10-K FY2025, via D2 §4.3]

GE SEGMENT PROFIT

GE segment profit is GE Aerospace's own segment-level profit measure, which excludes certain corporate and non-operating items. For Commercial Engines & Services it was \$8,861M on \$33,314M in 2025, a 26.6% margin. It is a different level from GE Aerospace's total operating profit of \$9.1B, which comes after the Defense & Propulsion Technologies segment and corporate items. Whether depreciation sits above or below GE's segment profit line was not retrieved in the research, so the measure is carried as GE's own and not re-labelled as EBIT or EBITDA. [GE 10-K FY2025; D11 §2.3, §6 T4]

The two profit figures in GE's filing need to be held apart. The 10-K text returned gives CES segment profit of \$8,861M and GE Aerospace operating profit of \$9.1B, up 25%; free cash flow was \$7.7B, up 24%. [GE 10-K FY2025] The two sit at different levels, segment versus total company, and the items that reconcile them (the Defense & Propulsion Technologies segment's profit and the corporate lines) were not returned. Read one way, the CES figure is the manufacturer's commercial aftermarket profit and the \$9.1B is the whole company; read the other way, \$9.1B is the figure to set beside other companies' operating profit and \$8,861M is a sub-total that excludes items the others include. Both are recorded; the research did not retrieve the bridge, and this Part does not construct one. [D11 §6 T4] GE's total company revenue for 2025 was also not returned, so no total-company margin is computed here. [D11 U15]

Three further statements in GE's filing bear on the pool's size and life. On the fleet: "The LEAP engine, which entered into service in 2016, is expected in the coming years to overtake the mature CFM56 as the industry's largest fleet. This is expected to drive a significant increase in shop visits and need for MRO capacity as LEAP engines come due for services." On operations: about 1,450 LEAP-1A durability kits (the upgraded-parts package treated in Part III §20) had been deployed across new production and overhaul shops since certification, and turnaround times for LEAP, CFM56 and GE90 improved "by over 10% year-over-year in the fourth quarter of 2025". On deliveries: 2,386 commercial engines in 2025, 1,802 of them LEAP, up 25%. [GE 10-K FY2025] A CFM56 shop-visit count, a spare-parts price-increase percentage and any wording on a used-material programme were searched for in the filing and not returned; what the research holds on those three points comes from trade press and is carried in Part IV §26 and §28. [D11 U16; reconciliation C.11]

Safran, the other half of CFM, is not an SEC filer and is carried as context. Its Propulsion services revenue for 2025 was €10,122M, 64.6% of Propulsion revenue, against €8,430M and 61.7% in 2024, which implies Propulsion revenue of about €15.7B (derived in the research as $€10,122M \div 0.646$). Its civil installed base at end-2025 was about 29,900 engines "including more than 22,800 CFM56s", from which Safran "can expect sustained aftermarket revenue from this fleet until 2030". [Safran FY2025 results presentation, 13 Feb 2026] Civil engine spare-parts revenue in dollars rose 17.6% in 2025, "driven primarily by the CFM56". [Safran, 13 Feb 2026, via D2 §4.3] Safran's CFM56 count is one of three OEM-level fleet figures the primer carries together with the other two, about 23,000 (CFM website, undated) and about 24,000 (Aviation Business News, September 2025); Part III §18 treats the difference. [reconciliation R-001; E.1 rule 14]

The manufacturer's pricing is the one input every other link's margin is measured against, so its escalation belongs here even though Part IV §28 owns it. The trade press reports new spare-parts prices rising around 10% a year, a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023, and quotes Safran's chief executive that spares pricing has "always been above inflation by 3 to 4 points". [Aviation Week, late 2023] On life-limited parts specifically, the -7B full set compounded at about 7% a year between 2008 and 2025 on the figures Part I §4.3 carries, and one trade report put the 2023 -7B parts increase at 20–30%. [D1 example C; Magnetic via Aviation Week, 2024; reconciliation R-018] The two readings differ in scope (all spares versus life-limited parts) and in year, and Part I and Part IV carry both; none is chosen here. [E.1 rule 35; reconciliation R-018]

86.3 LINK TWO: THE INDEPENDENT SHOP

The independent shop earns the labour-and-test slice, the shop margin, and whatever it saves on material by sourcing used, alternative or repaired parts; it passes the rest of the material through. Three shops report margins the research retrieved, on two bases.

ADJUSTED EBIT

Adjusted EBIT is operating profit after depreciation and amortisation, before purchase-price-allocation effects and other adjustments each company defines. MTU Aero Engines reports its commercial maintenance business on this basis (€478M on €6.0B in 2025, 8.0%), and Lufthansa Technik reports its whole company on it (€603M on €8.049B, 7.5%). Because depreciation has already been deducted, an adjusted EBIT margin is lower than an EBITDA margin for the same business by the depreciation share of revenue, and the two cannot be compared without that share, which neither company's retrieved figures give. [MTU, 24 Feb 2026; Lufthansa Technik, FY2025 results; D11 §2.3, §8]

StandardAero, listed on the NYSE since October 2024, reports two segments. Engine Services, the shop-visit business, had revenue of \$5,354.0M in 2025, up \$709.3M (15.3%) from

\$4,644.7M, and Segment Adjusted EBITDA of \$706.9M, up \$96.0M (15.7%) from \$610.9M; the margin was 13.2% in both years (derived in the research for 2024). Component Repair Services had revenue of \$708.6M, up 19.6%, and Segment Adjusted EBITDA of \$202.7M, a 28.6% margin. [StandardAero 10-K FY2025] The filing attributes Engine Services growth to "ramping volumes from LEAP, CFM56 DFW Center of Excellence, and CF34 expansion investments" (DFW is Dallas-Fort Worth), plus business-aviation and military programmes. [StandardAero 10-K FY2025] The two segments' margins are fifteen points apart within one company, and the difference is itself a lesson in the pool: the component-repair segment sells repairs on parts, where the shop's own work is most of the price, while the engine-services segment sells whole shop visits, where new parts pass through the revenue line at list. [D11 §2.4; D2 §2.2]

MTU Aero Engines, listed in Frankfurt, reported commercial maintenance revenue of €6.0B in 2025, up 18% from €5.1B, and adjusted EBIT of €478M, up 9% from €438M, so the margin fell from 8.7% to 8.0%. MTU attributes the decline to "the higher proportion of Geared Turbofan MRO as well as the costs associated with the ramp-up of our Fort Worth site"; Geared Turbofan work was "around 40% of commercial maintenance in 2025". At group level MTU reported adjusted revenue of €8.7B and adjusted EBIT of €1.35B, a 15.5% margin (14.0% in 2024). [MTU, 24 Feb 2026] Lufthansa Technik, a segment of Deutsche Lufthansa and not separately listed, reported revenue of €8.049B, up 12% and above €8B for the first time, and adjusted EBIT of €603M, down 1%, so the margin fell from 8.5% to 7.5%; it names US tariffs, "ongoing cost increases for materials and a dollar devaluation that was unfavorable". Engine Services "generates almost half" of its revenue. [Lufthansa Technik, FY2025 results; Deutsche Lufthansa Annual Report 2025]

The three shops' margins moved differently in the same year, and the research flags the pattern as a tension rather than a trend. On one side, StandardAero's Engine Services margin held at 13.2% in both 2024 and 2025 while MTU's fell from 8.7% to 8.0% and Lufthansa Technik's from 8.5% to 7.5%, which could be read as the independent shops diverging. On the other side, the bases differ (Segment Adjusted EBITDA for StandardAero against adjusted EBIT for the two Europeans), the platform mixes differ (MTU names the Geared Turbofan at about 40% of its work; Lufthansa Technik names tariffs, materials and the dollar; StandardAero names LEAP, CFM56 and CF34 volume), and the currencies differ. A business whose depreciation share rose in the year, as a shop ramping new sites would expect, can show a flat EBITDA margin and a falling EBIT margin at once. [StandardAero 10-K FY2025; MTU, 24 Feb 2026; Lufthansa Technik, FY2025 results; D11 §6 T3] The figures cannot be read as one series, and this Part does not read them as one.

EXHIBIT 11.2 — THE INDEPENDENT-SHOP LINK: THREE SHOPS, TWO BASES, 2025 AGAINST 2024

Company and segment	Revenue 2025	Profit measure 2025	Margin 2025	Margin 2024	Basis	Stated drivers of the change	Source
StandardAero, Engine Services	\$5,354.0M (+15.3%)	Segment Adjusted EBITDA \$706.9M (+15.7%)	13.2%	13.2% (derived)	Segment Adjusted EBITDA, before D&A and defined adjustments	LEAP, CFM56 Dallas Center of Excellence, CF34 volume	StandardAero 10-K FY2025
StandardAero, Component Repair Services	\$708.6M (+19.6%)	Segment Adjusted EBITDA \$202.7M	28.6%	not returned	Segment Adjusted EBITDA	not returned	StandardAero 10-K FY2025
MTU Aero Engines, commercial maintenance	€6.0B (+18%)	Adjusted EBIT €478M (+9%)	8.0%	8.7%	Adjusted EBIT, after D&A, before purchase-price allocation (PPA) and adjustments	Geared Turbofan at about 40% of work; Fort Worth ramp-up	MTU, 24 Feb 2026
MTU Aero Engines, group	€8.7B adjusted	Adjusted EBIT €1.35B	15.5%	14.0%	Adjusted EBIT, group	not stated	MTU, 24 Feb 2026
Lufthansa Technik, whole company	€8.049B (+12%)	Adjusted EBIT €603M (−1%)	7.5%	8.5%	Adjusted EBIT, company; Engine Services "almost half" of revenue	US tariffs, material-cost increases, weaker dollar	Lufthansa Technik, FY2025 results; Deutsche Lufthansa AR 2025

Source: StandardAero, Inc., Form 10-K FY2025; MTU Aero Engines, "Figures for 2025: MTU stays on course for growth", 24 Feb 2026, and FY2025 key figures; Lufthansa Technik, "Lufthansa Technik stable on course for growth", FY2025 results; Deutsche Lufthansa AG, Annual Report 2025, MRO business segment. D11 §3.6, §3.9, §6 T3. D&A is depreciation and amortisation; PPA is purchase-price allocation, the accounting that writes up acquired assets and creates amortisation charges.

86.4 LINK THREE: THE MAKER OF APPROVED ALTERNATIVE PARTS

A PMA part (Parts Manufacturer Approval, the FAA (Federal Aviation Administration) approval that lets a company other than the manufacturer design and make a replacement part; Part VI §37) is sold under the manufacturer's list price and competes for the same slot in the shop visit as the new OEM part and the used one. HEICO's Flight Support Group (FSG) is the listed reference business. For the fiscal year ended 31 October 2025 FSG had net sales of \$3,117.3M, up 18% from \$2,639.4M, and operating income of \$750.4M, up 27% from \$593.1M. The operating margin was 24.1% (derived in the research), against 22.5% in fiscal 2024. [HEICO 10-K FY2025; HEICO release, 18 Dec 2025] HEICO describes FSG as a business that "designs, manufactures, repairs, overhauls and distributes jet engine and aircraft component replacement parts, which are approved by the FAA", using "proprietary technology to design and manufacture jet engine and aircraft component replacement parts for sale at lower prices than those manufactured by OEMs". [HEICO 10-K FY2025] Two readings of the same filing give the rate at which HEICO adds approvals: "approximately 300 to 500" a year in one dossier's search and "approximately 400 to 550 Parts Manufacturer Approvals per year" in another's. Both come from the FY2025 10-K and both are carried. [HEICO 10-K FY2025; reconciliation R-024]

Three cautions attach to the 24.1%. FSG is not a pure PMA business; it includes repair and distribution, and its "aftermarket replacement parts" line grew 15% organically in fiscal 2025. [HEICO release, 18 Dec 2025, via D4 §4.1] The operating margin is after depreciation and amortisation, so it sits on a lower base than StandardAero's or FTAI's EBITDA figures and cannot be set beside them without the depreciation share, which was not retrieved. And HEICO's stated discount to OEM price, its cumulative part count and its consolidated totals were searched for and not returned. [D11 U5; reconciliation K-05] FTAI's own PMA activity runs through a joint venture whose revenue is not separately disclosed; Part VI §43 and Part VII §52 carry what is filed and what is reported. [FTAI 10-K FY2025; reconciliation R-026]

The size of the PMA market the research reached comes from market-research vendors and is labelled so: \$11.8B for commercial-aircraft PMA in 2025 (IMARC); \$14B by 2026 (Global Industry Analysts, 2022); \$2,952.2M for engine PMA plus DER repair in 2022 rising to \$6,174.7M by 2032 (a vendor report listed on GII); and an undated FlightGlobal article describing PMA's share of engine parts as "capped at around 21%". The scope definitions differ and were not retrieved. [IMARC, 2025; Global Industry Analysts via PR Newswire, 2022; GII listing; FlightGlobal, undated; reconciliation R-023]

86.5 LINK FOUR: USED MATERIAL, TEARDOWN AND DISTRIBUTION

Used serviceable material (USM), meaning parts recovered from dismantled engines, inspected and re-released (Part V §31), is the other substitute for the new part. Its economics are a gross-

margin story: buy feedstock (whole aircraft or engines bought for teardown; Part V §33), dismantle, certify, hold inventory, sell parts at a percentage of list. AerSale's Asset Management Solutions (AMS) segment is the cleanest listed example because the filing isolates it.

GROSS MARGIN (AERSALE ASSET MANAGEMENT SOLUTIONS)

AerSale's AMS gross margin is the segment's gross profit divided by its sales: \$74.1M on \$211.6M in 2025, 35.0% (derived in the research), against \$82.5M on \$215.5M in 2024, 38.3%. The filing attributes the fall to "lower margin on USM sales resulting from changes in the product mix" and reports the aircraft line within AMS at 33.2% against 34.8%. It is the only listed margin for a business whose activity is buying whole assets and either leasing or selling them whole or dismantling them for used material. Being a gross margin, it is before every operating expense, so it is the highest of the five bases in this Part and the furthest from a profit figure. [AerSale 10-K FY2025; AerSale release, 5 Mar 2026; D11 §2.4, §3.5]

AerSale's total revenue for 2025 was \$335.3M, down 2.8% from \$345.1M; AMS sales were \$211.6M, down \$3.9M; AMS gross profit was \$74.1M, down \$8.4M (10.2%). [AerSale 10-K FY2025; AerSale release, 5 Mar 2026] Engine revenue rose on higher USM sales (+\$34.0M) and higher leasing revenue (+\$7.5M), "primarily attributable to greater activity in the PW4000 and CF6-80 product lines as the company continues to monetize its feedstock"; those are widebody engines (the PW4000 is a Pratt & Whitney engine and the CF6-80 a GE engine, both for widebody aircraft), and the CFM56 is not named in the returned text. [AerSale 10-K FY2025] The company's used-material revenue across both segments was \$137.6M in 2025, and its whole-asset sales fell to \$56.4M from 13 engines, against \$110.1M from 20 engines and one aircraft in 2024; its Adjusted EBITDA (AerSale's adjusted earnings measure, defined by the company) was \$46.1M, 13.8% of revenue, against \$33.4M and 9.7%. [AerSale release, 5 Mar 2026, via D3 §4.6] Management described "a hypercompetitive feedstock environment, marked by increased scrutiny and pressure on pricing as more companies enter the teardown sector", a statement the research holds only as relayed in search results and not from the filing text. [AerSale, as relayed; D3 §2.6; D11 U12]

The teardown margin in dollars per engine comes from a consultancy blog and is labelled so. Safe Fly Aviation gives a run-out CFM56 bought for \$0.8–1.2M yielding \$1.6–2.2M of used material, a net margin of \$0.4–0.8M per engine; the difference between yield-minus-purchase and the stated net margin implies \$0.2–0.4M of dismantling, repair, certification and selling cost, which is arithmetic on the source's endpoints rather than a figure it gives. [Safe Fly Aviation, 2026, consultancy blog; D3 §2.2] Part V §32 works the full part-out. Used-material prices sit at 60–80% of a new part's price on average, up to 85% for high-demand parts, and above list when the OEM cannot deliver, in one trade source, and at 70–75% of catalogue list for a specialist trader selling to airlines in another. [Aviation Week, date not captured; Aircraft Commerce Issue 120; reconciliation R-022] Because used material is priced off list, it inherits the manufacturer's escalation.

The market's size is contested among vendors and the figures are carried with that label: more than \$4B a year with the CFM56 the largest segment (Safe Fly, consultancy blog); \$5.87B (The Business Research Company), \$7.7B (Market.us) and \$8.55B (Fortune Business Insights, URL not isolated), all for 2025 and all from market-research vendors whose scope definitions were not retrieved. [D2 §4.6; D3 T1; reconciliation R-021]

DISTRIBUTION (NEW PARTS)

Distribution is reselling new OEM parts under a distribution agreement: the distributor holds inventory and earns the margin between the manufacturer's transfer price and its sale price. Because the manufacturer sets the cost of goods, a distribution margin is structurally thinner than a PMA or used-material margin. AAR's Parts Supply segment, about 45% of its \$3,308.0M of fiscal 2026 sales (both derived in the research), combines distribution of new parts with sales and leasing of used material; its segment operating income was not returned. Part V §35 names AAR among the players.

[AAR 10-K FY2026; D11 §2.4, §3.7]

AAR's fiscal year ends 31 May. For fiscal 2026 its commercial sales were \$2,384.1M (72.1% of the total) and its government and defense sales \$923.9M (27.9%), so consolidated sales were \$3,308.0M (derived in the research from the two). Commercial sales rose \$408.0M (20.6%), helped by ADI (American Distributors Holding Co., LLC), bought in September 2025 for \$137.1M into Parts Supply: "primarily due to strong demand and volume growth in new parts Distribution activities, including from the recent ADI acquisition, which contributed sales of \$82.2M". [AAR 10-K FY2026] Within Parts Supply, new-parts distribution grew 23.9% and used-material activities 32.1%. [AAR 10-K FY2026, via D3 §4.6] AAR also holds the exclusive agreement under which it sells CFM56 material from FTAI's engine pool through 2030, a relationship Part V §35 and Part VII §52 carry. [AAR / FTAI release, PR Newswire, 2025; reconciliation B-9] Two lessors also run used-material arms whose revenue is not disclosed: AerCap Materials, which dismantled more than 300 aircraft in 2023 and stocked more than 250,000 items for more than 850 customers at the first quarter of 2026, and Willis Aero, the parts arm of Willis Lease Finance (WLFC). [AerCap Materials factsheets, 2023 and 1Q 2026; WLFC 10-K FY2025; D11 U6]

86.6 LINK FIVE: THE LESSORS

Lessors own the engines and aircraft and collect rent, maintenance reserves and gains on sale. Their percentage margins look nothing like a shop's, for a mechanical reason set out in §86.9: a lessor's largest costs are depreciation and interest, and both sit below EBITDA.

NET BOOK VALUE (NBV)

Net book value is an asset's cost less accumulated depreciation and impairments. It is the denominator for a lessor's implied yield and the figure against which a gain on sale is measured. Air Lease's flight equipment subject to operating lease had a net book value of \$29.1B at 31 Dec 2025; FTAI's leasing equipment, net, was \$1,545.8M at the same date, down from \$2,373.7M a year earlier after the Seed Asset sales. Part VIII §58 carries FTAI's line. [Air Lease 10-K FY2025; FTAI 10-K FY2025; D11 §2.2; D6 §2.6]

Willis Lease Finance, the only US-listed company whose primary business is engine leasing, had total revenue of \$730.2M in 2025, up 28.3%, and Adjusted EBITDA of \$459.1M, up 16.6%, which is 62.9% of revenue (derived in the research). [WLFC 10-K FY2025] Its lease rent revenue was \$291.6M, up 22.4% from \$238.2M. Its "core lease rent and maintenance reserve revenues" were \$523.6M, up 15.8% from \$452.1M, so maintenance reserve revenue was \$232.0M (derived in the research by subtraction), 44% of the combined line. Gain on sale of leased equipment was \$54.0M, up 19.9%. [WLFC 10-K FY2025; D11 §2.2] Maintenance reserves (the per-hour or per-cycle payments a lessee makes so that the next shop visit is pre-funded; Part II §13) become revenue when the lessor determines they will not be paid back, typically at lease end or under a non-reimbursable lease. For a lessor of mid-life engines the reserve line is large relative to rent because the assets are close to shop visits; at Willis it was about 80% of lease rent in 2025 ($232.0 \div 291.6$, arithmetic in this Part on the filed and derived figures). [WLFC 10-K FY2025; D11 §2.2]

AerCap, the largest aircraft lessor, reports on Form 20-F. Its total revenues and other income were \$8,517M in 2025 (2024: \$7,997M). Total lease revenue was \$7,369M, of which basic lease rents were \$6,679M and "maintenance rents and other receipts" \$690M, 9.4% of lease revenue (derived in the research). Net income attributable to AerCap was \$3.8B against \$2.1B; the drivers were not returned. [AerCap 20-F FY2025; AerCap Q4 2025 release; D11 U7] AerCap's maintenance line is about 10% of its basic rents ($690 \div 6,679$, arithmetic in this Part) against about 80% at Willis; the difference is the fleet. AerCap's owned passenger fleet averaged 7.3 years (new-technology aircraft 5.4 years; current-technology 15.2 years), with 7.1 years of lease term remaining on average. [AerCap 20-F FY2025] Air Lease, whose strategy is to buy "modern, fuel-efficient new technology commercial jet aircraft", reported lease rental revenue of \$2,615.4M (2024: \$2,407.5M) on a net book value of \$29.1B, which is a 9.0% gross yield on year-end book (derived in the research); its 490 owned aircraft averaged 4.9 years of age. [Air Lease 10-K FY2025; D11 §2.2] Worked example 11.2 in §87.1 derives the yields and sets FTAI's beside them.

The task of this section asks for FTAI's Aviation Leasing EBITDA on its lease income, and the arithmetic is instructive precisely because it fails. Segment Adjusted EBITDA of \$608,912K divided by lease income of \$235,210K is 259%, a ratio above 100%, which tells the reader that the denominator is not the segment's revenue and not that the segment earns more than it

collects. [FTAI 10-K FY2025, segment table via orchestrator notes] The consolidated statement carries maintenance revenue of \$218,499K and asset sales revenue of \$106,945K alongside lease income; if all three belong to the segment (the segment-to-line mapping was not retrieved), the three sum to \$560,654K and the ratio is still 109%, because FTAI's segment Adjusted EBITDA also includes the gain on sale to the 2025 Partnership (\$46.4M in FY2025) and the equity-method pick-up, neither of which is revenue. [FTAI 10-K FY2025; D6 §2.6; D10 §2.5; reconciliation K-17] The honest statement is that no margin on lease income can be computed for FTAI's leasing segment from the retrieved lines, and that Willis's 62.9% is a whole-company ratio on a company-defined measure that the research did not decompose either. Part VIII §58 carries the segment's lines in full; Part X §80 carries FTAI's definition of Adjusted EBITDA, which the research notes was not fully retrieved. [D10 U-1]

86.7 LINK SIX: FTAI'S TWO SEGMENTS

FTAI sits on two links at once. Its Aviation Leasing segment is a lessor, treated above. Its Aerospace Products segment sells refurbished CFM56 and V2500 modules and engines through the Module Factory (Part VII §46) and so collects a slice that, in a conventional shop visit, would be split between the shop's labour, the parts pass-through and the material savings. For FY2025 the segment's revenue was \$1,936.2M, the sum of "Aerospace products revenue" of \$1,600.5M and "MRE Contract revenue" (MRE is FTAI's Maintenance, Repair and Exchange offer, Part VII §46 and §48) of \$335.8M, and its segment Adjusted EBITDA was \$671.3M, a 34.7% margin (derived in the research). [FTAI 10-K FY2025, segment table; reconciliation B-1, E.1 rule 8] On the filed segment table the margin was 35.2% in 2024 (\$380,636K on \$1,079,821K) and 35.2% in 2023 (\$160,009K on \$454,970K), both arithmetic in this Part on the 10-K's figures. [FTAI 10-K FY2025, via orchestrator notes] In 2026 the quarterly margin was 29.9% in the first quarter (\$222.6M on \$743.8M) and 28.5% in the second (\$249.7M on \$875.0M), both derived; the fourth quarter of 2025 was stated by management at "\$195 million at a 35% margin". [FTAI Q1 2026 release; FTAI Q2 2026 release, 29 Jul 2026; MarketBeat call coverage, 26 Feb 2026; D5 §4.1]

Three caveats travel with those figures and are owned by other Parts. MRE Contract revenue is revenue from the Strategic Capital vehicles, in which FTAI holds a 19% stake (20% in the Q3 2025 10-Q), and the explicit segment-note mapping of that line into Aerospace Products was not retrieved, although three documents agree on the arithmetic (Part VII §48; reconciliation R-043, B-1). The profit on those sales is eliminated through equity-method earnings by a method the filing does not disclose; \$16.6M of eliminations in the first half of 2026 implies about \$87M of gross profit if proportional to the stake or \$16.6M if fully deferred, and the 10-Q supports either reading (Part VIII §63; reconciliation R-047). And the segment's cost of sales for FY2025 appears as \$1,240,368K in one search of the 10-K and the consolidated cost-of-sales line as \$1,349,719K in another; the orchestrator's later search placed the first in the segment expenses

table and the second in the consolidated statement, which gives a segment gross margin of 35.9% on segment revenue as a derived figure carried with that scope note (Part VII §53; reconciliation R-045). [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; orchestrator notes]

EXHIBIT 11.3 — FTAI'S TWO SEGMENTS: THE FIGURES THIS PART USES, WITH BASES AND THE CAVEATS THAT TRAVEL WITH THEM

Segment and period	Revenue	Segment Adjusted EBITDA	Margin	Basis and caveat	Source
Aerospace Products, FY2023	\$454,970K (products only; no MRE line)	\$160,009K	35.2% (arithmetic in this Part)	FTAI-defined segment Adjusted EBITDA	FTAI 10-K FY2025, segment table
Aerospace Products, FY2024	\$1,079,821K (products only)	\$380,636K	35.2% (arithmetic in this Part)	As above	FTAI 10-K FY2025, segment table
Aerospace Products, FY2025	\$1,936,244K = products \$1,600,456K + MRE Contract \$335,788K	\$671,252K	34.7% (derived)	MRE is related-party revenue (R-043); segment cost of sales \$1,240,368K gives 35.9% gross, scope per R-045	FTAI 10-K FY2025
Aerospace Products, Q4 2025	about \$557M (derived from the stated margin)	\$195M	35% (stated by management)	Call coverage for the margin; release for EBITDA	MarketBeat, 26 Feb 2026; FTAI Q4/ FY2025 release
Aerospace Products, Q1 2026	\$743.8M = \$522.6M + \$221.2M	\$222.6M	29.9% (derived)	MRE 29.7% of segment revenue (derived)	FTAI Q1 2026 release; 10-Q Q1 2026
Aerospace Products, Q2 2026	\$875.0M = \$692.2M (derived) + \$182.8M	\$249.7M	28.5% (derived)	MRE to the 2025 Partnership \$182.8M (10-Q); eliminations \$6.6M in the quarter	FTAI Q2 2026 release; 10-Q Q2 2026

Segment and period	Revenue	Segment Adjusted EBITDA	Margin	Basis and caveat	Source
Aviation Leasing, FY2024	Lease income \$234,411K or \$255,338K (two readings of the 10-K)	\$500,062K	not a margin (see §86.6)	Segment Adjusted EBITDA includes gains and equity income	FTAI 10-K FY2025 (R-052)
Aviation Leasing, FY2025	Lease income \$235,210K	\$608,912K	not a margin (see §86.6)	As above; gain on sale to Partnership \$46.4M included	FTAI 10-K FY2025
Consolidated, FY2025	Total revenues \$2,507,409K	Total Adjusted EBITDA \$1,190,922K	47.5% (arithmetic in this Part)	Company-defined; D&A \$225,797K (9.0% of revenue) and interest \$247,751K sit below	FTAI 10-K FY2025

Source: FTAI Aviation Ltd., Form 10-K FY2025 (segment table and consolidated statement of operations, via the orchestrator notes and D10); Form 10-Q for the quarters ended 31 Mar 2026 and 30 Jun 2026; Q1 2026 and Q2 2026 earnings releases; Q4/FY2025 release, 25 Feb 2026; MarketBeat, "FTAI Aviation Q4 earnings call highlights", 26 Feb 2026 (call coverage). D5 §4.1; D10 §4.2. Reconciliation R-043, R-045, R-047, R-052, B-1; E.1 rules 8, 10, 12, 28.

86.8 THE POOL IN ONE TABLE

EXHIBIT 11.4 — THE MARGIN POOL ACROSS THE CHAIN: EVERY LINK, EVERY MARGIN ON ITS STATED BASIS

Link in the chain	Company and segment	Period	Revenue	Margin	Basis	Source
Manufacturer: new engines, spare parts, services	GE Aerospace, Commercial Engines & Services	CY2025	\$33,314M; services \$25,010M (75.1%)	26.6%	GE segment profit (GE's own measure)	GE 10-K FY2025

Link in the chain	Company and segment	Period	Revenue	Margin	Basis	Source
Manufacturer: total company	GE Aerospace	CY2025	not returned	Operating profit \$9.1B; no margin computed	Total company; reconciling items to CES not returned (T4)	GE 10-K FY2025
Manufacturer: services share	Safran, Propulsion	CY2025	Services €10,122M, 64.6% of about €15.7B (derived)	not reported at segment level in the retrieved text	n/a	Safran, 13 Feb 2026
Independent shop	StandardAero, Engine Services	CY2025	\$5,354.0M	13.2% (2024: 13.2%)	Segment Adjusted EBITDA	StandardAero 10-K FY2025
Component repair	StandardAero, Component Repair Services	CY2025	\$708.6M	28.6%	Segment Adjusted EBITDA	StandardAero 10-K FY2025
Independent shop (context)	MTU Aero Engines, commercial maintenance	CY2025	€6.0B	8.0% (2024: 8.7%)	Adjusted EBIT	MTU, 24 Feb 2026
Independent shop (context)	Lufthansa Technik, whole company	CY2025	€8.049B	7.5% (2024: 8.5%)	Adjusted EBIT	Lufthansa Technik, FY2025 results
PMA, with repair and distribution	HEICO, Flight Support Group	FY to 31 Oct 2025	\$3,117.3M	24.1% (FY2024: 22.5%), derived	Segment operating income	HEICO 10-K FY2025
Teardown and used material	AerSale, Asset Management Solutions	CY2025	\$211.6M	35.0% (2024: 38.3%), derived	Gross margin	AerSale 10-K FY2025

Link in the chain	Company and segment	Period	Revenue	Margin	Basis	Source
Distribution plus used material	AAR, Parts Supply	FY to 31 May 2026	about \$1,489M (derived)	not returned	n/a	AAR 10-K FY2026
Engine lessor	Willis Lease Finance, whole company	CY2025	\$730.2M	62.9% (derived)	Company-defined Adjusted EBITDA	WLFC 10-K FY2025
Aircraft lessor	AerCap	CY2025	\$8,517M; lease revenue \$7,369M	no margin basis returned; maintenance rents 9.4% of lease revenue (derived)	n/a	AerCap 20-F FY2025
Aircraft lessor, new technology	Air Lease	CY2025	Lease rental \$2,615.4M	9.0% gross yield on \$29.1B NBV (derived); no margin returned	Yield, not a margin	Air Lease 10-K FY2025
Module exchange, module and engine sales	FTAI, Aerospace Products	CY2025	\$1,936.2M (incl. MRE \$335.8M)	34.7% (derived)	FTAI-defined segment Adjusted EBITDA	FTAI 10-K FY2025
Module exchange, module and engine sales	FTAI, Aerospace Products	Q2 2026	\$875.0M (incl. MRE \$182.8M)	28.5% (derived)	As above	FTAI Q2 2026 release; 10-Q
Engine and aircraft lessor	FTAI, Aviation Leasing	CY2025	Lease income \$235.2M	Segment Adjusted EBITDA \$608.9M; no margin computable on retrieved lines (\$86.6)	FTAI-defined segment Adjusted EBITDA	FTAI 10-K FY2025

Link in the chain	Company and segment	Period	Revenue	Margin	Basis	Source
Servicer of third-party capital	FTAI, as Servicer to the 2025 Partnership	FY2025; 1H 2026	Servicing fees \$10.2M; \$12.85M	no margin disclosed	Fee income; base undisclosed (Part VIII §63)	FTAI 10-K FY2025; 10-Q Q2 2026

Source: as listed in each row; compiled from D11 §4.2 and §1, D2 §4.3, D5 §4.1, D7 §4.3 and the orchestrator notes. CY is calendar year. The five margin bases are defined at the head of §86 and are not interchangeable; the fiscal years differ (December, October, May); euro figures are not converted. Reconciliation E.1 rule 35; R-021, R-023, R-043, R-045 and D11 T3, T4 as presented in §86.2–86.7.

86.9 WHY EACH LINK'S MARGIN SITS WHERE IT DOES

The percentages in Exhibit 11.4 run from 7.5% to 62.9%, and the spread is almost entirely explained by three mechanical facts about each link: how much of the revenue line is somebody else's product passing through, how much capital the link ties up and where that capital's cost is booked, and whether the link sets a price or takes one. None of the three says anything about which link is the better business. A thin percentage on a large pass-through can be a large dollar profit; a wide percentage on a small revenue line can be a small one.

WHY A PARTS PASS-THROUGH COMPRESSES A PERCENTAGE MARGIN

Take an illustrative shop (all figures assumed) that sells \$1.0M of labour and test on a visit at a 30% margin, earning \$300,000, and buys \$2.0M of new parts at list for the same visit, passing them through at a 5% handling margin, earning \$100,000. Its revenue is \$3.0M and its profit \$400,000, a 13.3% margin. If the airline instead buys the parts directly from the manufacturer under a materials agreement, the shop's revenue is \$1.0M and its profit \$300,000, a 30% margin, on \$100,000 less profit. The shop did the same work in both cases. The percentage moved because the denominator changed, not the business. This is the arithmetic behind the fifteen-point gap between StandardAero's two segments and behind the gap between any shop's margin and a parts maker's. [mechanism from D2 §2.2; figures assumed, illustrative]

The manufacturer sets the list price and every other link prices off it. Its services revenue is three-quarters of its commercial segment, and the CFM56 part of that revenue is 85% spare parts and time-and-materials work rather than long-term contracts, so on this engine the manufacturer's capture is mostly the parts line, sold at a price it revises upward each year. [GE 10-K FY2025; Safran via Aviation Week, c. 2023/24] Its percentage margin is reduced by the new engines it sells into the fleet, which GE lists among the offsets to profit ("the impact of higher install engine deliveries"), and by changes in the estimated profitability of its flight-hour contracts, on which it bears the cost risk. [GE 10-K FY2025] The PMA maker prices under the manufacturer's list; it

carries the design and approval cost up front, at 300–550 approvals a year each requiring a separate FAA action, and then manufactures parts whose price is anchored to a list it did not set. [HEICO 10-K FY2025; D4 §2.2] The used-material trader also prices off list, at 60–80% of it, and its gross margin is what is left after the feedstock purchase; when retirements run low, feedstock gets dearer and the margin narrows, which is what AerSale's filing describes in attributing its three-point fall to mix and its management describes as a hypercompetitive feedstock market. [AerSale 10-K FY2025; Aviation Week; D3 §2.6] The distributor earns the thinnest percentage because the manufacturer sets both its cost of goods and, through list, the ceiling on its price. [D11 §2.4]

The independent shop is a price taker on material and a price setter only on its own labour and test. Its revenue line therefore carries the manufacturer's parts at list, which compresses the percentage as the callout shows, and its own levers are to substitute used, alternative or repaired parts for new ones. [D2 §2.2] Its capital is mostly people, tooling and working capital; its depreciation share of revenue is modest in a mature shop and rises when it builds new sites, which is why MTU names the Fort Worth ramp-up as a drag on an EBIT margin and why StandardAero's EBITDA margin need not move for the same reason. [MTU, 24 Feb 2026; StandardAero 10-K FY2025] The component repairer sits between the shop and the parts maker: its revenue is mostly its own work on a part it did not sell, so its percentage is wider than the shop's and narrower than a parts maker's gross margin. [StandardAero 10-K FY2025]

The lessor's capital is its product. Its revenue is rent on assets it owns, and its two largest costs, depreciation of those assets and interest on the debt that bought them, sit below EBITDA; so its EBITDA margin is high by construction and says little until depreciation and interest are deducted. Willis's 62.9% is on that basis. FTAI's consolidated statement shows the scale of the lines that sit below its own Adjusted EBITDA: depreciation and amortisation of \$225.8M and interest expense of \$247.8M in FY2025, against total Adjusted EBITDA of \$1,190.9M. [WLFC 10-K FY2025; FTAI 10-K FY2025, via orchestrator notes] A lessor's yield, not its margin, is the comparable figure, and worked example 11.2 derives it. The mid-life lessor also collects a maintenance-reserve line that a new-aircraft lessor barely has, about 80% of rent at Willis against about 10% at AerCap, because its engines are near shop visits; that line is cash the lessor will mostly spend on the next visit or pass to the next lessee, so its size is a fact about the fleet's age rather than about profit. [WLFC 10-K FY2025; AerCap Q4 2025 release; D11 §2.2]

FTAI's Aerospace Products segment sells a refurbished module or engine rather than a labour-and-material visit priced to a customer, so its revenue line is the product's price and its cost line is the feedstock, the parts it bought (new, PMA or used) and the shop work, whether its own or a partner's. Its 34.7% and 28.5% are therefore closer in construction to a parts business's margin than to a shop's, and the research warns against reading them beside StandardAero's 13.2% as the same activity measured twice. [D11 §2.3; D5 §2.2] Management named the levers it expects to move

the margin, the PMA high-pressure turbine blade, more used material and the CFM materials agreement, without a dollar figure, and the 2026 aspiration to "approach 40%" sits beside the first-half 2026 figures of 29.9% and 28.5%; Part VII §53 and §57 carry that bridge. [MarketBeat, 26 Feb 2026, call coverage; FTAI Q1 and Q2 2026 releases; reconciliation R-027]

WORKED EXAMPLE 11.1 — PUTTING AN EBITDA MARGIN AND AN EBIT MARGIN ON ONE FOOTING WITH DEPRECIATION AND AMORTISATION

The question. StandardAero's Engine Services segment reports 13.2% Segment Adjusted EBITDA (sourced: StandardAero 10-K FY2025). MTU's commercial maintenance reports 8.0% adjusted EBIT (sourced: MTU, 24 Feb 2026). Can the 5.2-point gap be read as a difference in profitability?

The identity. For any business, $\text{EBIT margin} = \text{EBITDA margin} - (\text{D\&A} \div \text{revenue})$, where D&A is depreciation and amortisation. So the two can be compared only if the D&A share of revenue is known for at least one of them, and neither company's retrieved figures give it (assumed below, in three cases).

StandardAero Engine Services (ES), moved to an EBIT basis:

D&A at 3% of revenue: $13.2 - 3.0 = 10.2\%$ EBIT

D&A at 5% of revenue: $13.2 - 5.0 = 8.2\%$ EBIT

D&A at 7% of revenue: $13.2 - 7.0 = 6.2\%$ EBIT

MTU commercial maintenance, moved to an EBITDA basis:

D&A at 3% of revenue: $8.0 + 3.0 = 11.0\%$ EBITDA

D&A at 5% of revenue: $8.0 + 5.0 = 13.0\%$ EBITDA

D&A at 7% of revenue: $8.0 + 7.0 = 15.0\%$ EBITDA

Reading. At a 5% D&A share on both sides the two businesses are within 0.2 points of each other on either basis (8.2% against 8.0% EBIT; 13.2% against 13.0% EBITDA). At 3% StandardAero sits above MTU on both bases; at 7% it sits below. The headline gap of 5.2 points is therefore inside the range that plausible D&A shares alone would produce, and the two figures cannot be ordered without the D&A line. A real data point for scale: FTAI's consolidated D&A was \$225,797K on revenue of \$2,507,409K in FY2025, 9.0% (sourced: FTAI 10-K FY2025), but most of that is depreciation of leasing equipment, so it is not a proxy for a shop's share and is shown only to illustrate the size the term can take.

Second-order limits. Even with D&A known, the two measures would still differ: StandardAero's "adjusted" removes items it defines as non-recurring, MTU's removes purchase-price-allocation effects, and the currencies and platform mixes differ (D11 T3). HEICO's 24.1% is a GAAP segment operating income, after D&A, so it sits on the EBIT side of this identity; FTAI's 34.7% and 28.5% sit on the EBITDA side. Any exhibit that lists them in one column, including Exhibit 11.4, lists five bases and not one. [StandardAero 10-K FY2025; MTU, 24 Feb 2026; HEICO 10-K FY2025; FTAI 10-K FY2025; D11 §2.3, §6 T3]

87. The peers, one by one, from their filings

Each peer below gets the same treatment: what it does in the chain, its latest fiscal year, its revenue, the segment most comparable to one of FTAI's activities with that segment's revenue and margin, its asset counts, what its filing says about CFM56 or narrowbody engine demand, pricing, capacity, module exchanges, PMA, used material or engine leasing, and the unknowns the research logged. The eight SEC filers come first; Safran, MTU and Lufthansa Technik follow as non-SEC context. The research reached each figure through searches restricted to the filer's own documents, so a figure that is absent below was searched for and not returned, and is said to be so. [D11 preamble]

87.1 WILLIS LEASE FINANCE CORPORATION (NASDAQ: WLFC)

Willis Lease owns commercial jet engines and a small number of aircraft and leases them to airlines and shops; it collects maintenance reserves, sells engines and parts out of the portfolio through Willis Aero, and manages engines for third parties. It is the only US-listed company whose primary business is engine leasing, which makes it the closest filing analogue to FTAI's Aviation Leasing segment. [WLFC 10-K FY2025; D11 §3.1]

For the year ended 31 December 2025, Willis reported total revenue of \$730.2M, up 28.3%; net income attributable to common shareholders of \$108.1M, up 3.5%; and Adjusted EBITDA of \$459.1M, up 16.6%, which is 62.9% of revenue (derived in the research). Lease rent revenue was \$291.6M, up 22.4%; core lease rent and maintenance reserve revenues together were \$523.6M, up 15.8%; gain on sale of leased equipment was \$54.0M, up 19.9%. [WLFC 10-K FY2025] At year-end the operating lease portfolio comprised \$2,801.7M of equipment held for lease, \$139.9M of notes receivable, \$30.6M of maintenance rights and \$16.6M of investments in sales-type leases, representing 363 engines, 20 aircraft, one marine vessel and other leased parts and equipment, with 69 lessees in 37 countries; 116 further engines and related equipment were managed for third parties. Average portfolio utilisation was 84.9%, against 82.9% in 2024. [WLFC 10-K FY2025]

MAINTENANCE RIGHTS

Maintenance rights is a line in Willis Lease's lease portfolio (\$30.6M at 31 Dec 2025; \$27.0M at 30 Sep 2025) that the research reports without a filing definition. From first principles, a lessor that buys an engine subject to an existing lease acquires, alongside the metal, the right to the maintenance reserves or end-of-lease compensation that the lessee has accrued but not yet paid over; that right is an asset separate from the engine and is carried at the value attributed to it. The reader should treat this as a working gloss and consult the 10-K's note for Willis's own definition. [WLFC 10-K FY2025; WLFC Q3 2025 release; registry section 2]

Two sets of Willis figures circulate in this primer and they sit at different dates. The Q3 2025 release gives utilisation of 86.0% at 30 September 2025, a lease portfolio of \$2,888.5M including \$2,700.4M of equipment, and a third quarter with \$76.6M of lease rent and \$76.1M of maintenance reserve revenue. The FY2025 10-K gives average utilisation of 84.9% for the year, equipment of \$2,801.7M at 31 December 2025 and full-year lease rent of \$291.6M. A point-in-time figure and a period average, and a quarter and a year, are different quantities; both are quoted with their dates and neither is wrong. [WLFC Q3 2025 release and 10-Q; WLFC 10-K FY2025; reconciliation R-032]

On demand, Willis's filing states: "Both Airbus and Boeing have delivery delays which have been compounded by engine manufacturer delays, shorter on-wing engine time of most new technology engines, and ongoing supply chain challenges." [WLFC 10-K FY2025, via D9 §2.8] That is a lessor's description of the drivers Part III §20 and §21 treat. The filing's CFM56 and LEAP portfolio split, its stated lease rate factor and its separately stated maintenance-services and spare-parts revenue lines for 2025 were searched for and not returned. [D11 U1, U3; reconciliation K-14, K-15] The Q2 2026 10-Q was not located; the Q1 and Q2 2026 earnings-call exhibits were located by URL only, and a 10-K/A for FY2025 was filed whose subject was not returned. [D11 U2; WLFC 10-K/A FY2025] Willis funds itself through the Willis Engine Structured Trust (WEST) series of engine securitisations (the research also carries the name Willis Engine Securitization Trust); WEST VIII in 2025 was its ninth aviation securitisation and WEST IX its tenth. Part II §17 carries the structure and Part VIII §65 the servicing fees. [KBRA (a rating agency) via D6 §2.5; D7 §3.2]

EXHIBIT 11.5 — WILLIS LEASE FINANCE, FY2025 (YEAR ENDED 31 DEC 2025)

Item	Figure	Note
Total revenue	\$730.2M (+28.3%)	
Net income attributable to common	\$108.1M (+3.5%)	
Adjusted EBITDA	\$459.1M (+16.6%)	62.9% of revenue, derived; company-defined measure
Lease rent revenue	\$291.6M (+22.4% from \$238.2M)	
Core lease rent and maintenance reserve revenues	\$523.6M (+15.8% from \$452.1M)	
Maintenance reserve revenue	\$232.0M	Derived: \$523.6M – \$291.6M; 44% of the combined line
Gain on sale of leased equipment	\$54.0M (+19.9%)	

Item	Figure	Note
Equipment held in operating lease portfolio	\$2,801.7M	At 31 Dec 2025
Notes receivable; maintenance rights; sales-type leases	\$139.9M; \$30.6M; \$16.6M	At 31 Dec 2025
Owned engines; aircraft; other	363; 20; one marine vessel and other leased parts	At 31 Dec 2025
Lessees; countries	69; 37	
Third-party managed engines	116	At 31 Dec 2025
Average utilisation (by book value)	84.9% (2024: 82.9%)	Period average; 86.0% at 30 Sep 2025 is a point-in-time figure (R-032)
Implied gross yield	10.4% on year-end book; 12.3% on the on-lease portion	Derived, illustration only (worked example 11.2)
Lease rate factor; CFM56/LEAP split; maintenance-services and parts revenue lines	not returned	D11 U1, U3

Source: Willis Lease Finance Corp, Form 10-K for FY2025; Form 10-Q for the quarter ended 30 Sep 2025 and Q3 2025 release (via D6 §4.5). D11 §3.1, §4.1. Reconciliation R-032; clusters K-14, K-15.

WORKED EXAMPLE 11.2 — DERIVING A LESSOR'S GROSS LEASE YIELD, AND ITS LEASE RATE FACTOR, FROM FILED FIGURES: WILLIS, AIR LEASE AND FTAI

Definitions. A lease rate factor (LRF), treated in Part II §14, is monthly rent divided by the asset's value. Twelve times the LRF is the gross annual yield before depreciation, interest and maintenance cost. No filing in the research states an LRF, so each is derived from a rent line and a book-value line, and each derivation carries the mismatch between a flow for the year and a stock at one date (reconciliation R-030).

Willis Lease. Inputs: 2025 lease rent revenue \$291.6M (sourced: WLFC 10-K FY2025); equipment held in operating lease portfolio \$2,801.7M at 31 Dec 2025 (sourced); average utilisation 84.9% (sourced).

$$\begin{aligned}
 \text{gross yield on year-end book} &= 291.6 \div 2,801.7 = 10.4\% \\
 \text{yield on the on-lease portion} &= 10.4\% \div 0.849 = 12.3\% \\
 \text{implied LRF} &= 12.3\% \div 12 \approx 1.0\% \text{ per month}
 \end{aligned}$$

Limits: the denominator is the year-end book, not the average over the year; it includes 20 aircraft and a vessel as well as 363 engines; and the numerator excludes the \$232.0M maintenance reserve

line, which is cash the lessor collects but mostly spends on shop visits. The result approximates the filing's own LRF, which was not returned (D11 U1).

Air Lease. Inputs: 2025 lease rental revenue \$2,615.4M (sourced: Air Lease 10-K FY2025); net book value of flight equipment subject to operating lease \$29.1B at 31 Dec 2025 (sourced).

$$\text{gross yield on year-end NBV} = 2,615.4 \div 29,100 = 9.0\%; \quad \text{implied LRF} = 9.0\% \div 12 = 0.75\% \text{ per month}$$

New-technology aircraft on long leases carry lower rate factors than mid-life engines, which is the mechanism behind the gap between 9.0% and the Willis figure; Air Lease's own stated yield was not returned (D11 U10).

FTAI Aviation Leasing. The research derived two bounded figures before the FY2025 lease income was known: 10.4% on average cost and 13.4% on average net book value using first-half 2025 lease income annualised, or 6.6% and 8.8% using first-half 2026 income against the 31 Dec 2025 base (derived in the research, D6 §2.7). The FY2025 10-K's lease income of \$235,210K (sourced, via the orchestrator notes) can now be put over D6's average 2025 bases of \$2,510.5M at cost and \$1,959.8M net (sourced: FTAI 10-K FY2025 balance sheets, averaged in D6).

$$\begin{aligned} &\text{on average cost: } 235.2 \div 2,510.5 = 9.4\%; \quad \text{on average NBV: } 235.2 \div 1,959.8 = 12.0\% \\ &\text{on the on-lease portion at 77\% utilisation (10-K definition): } 9.4\% \div 0.77 = 12.2\% \text{ on cost} \\ &\text{implied LRF on cost: } 9.4\% \div 12 = 0.78\% \text{ per month; on NBV: } 12.0\% \div 12 = 1.0\% \text{ per month} \end{aligned}$$

Limits specific to FTAI: the 77% utilisation is by equity value and days on lease, excluding airframes, not by book value as at Willis (registry C-3; worked example 11.4); the cost base may still carry the 17 engines and 8 aircraft in Russia at zero net value (D6 U4); and the base fell through 2025 as Seed Assets were sold, so an average of two year-end balances is a rough centre. Reading across: Willis about 1.0% a month on the on-lease portion; Air Lease 0.75%; FTAI 0.78% on cost or 1.0% on net, and 0.87–1.1% on the research's earlier bases. None is a stated figure; all are shown with their bases and none is chosen (E.1 rule 6). [WLFC 10-K FY2025; Air Lease 10-K FY2025; FTAI 10-K FY2025; D11 §2.2; D6 §2.7; reconciliation R-030]

87.2 AERCAP HOLDINGS N.V. (NYSE: AER)

AerCap is the largest aircraft lessor by fleet and also a large engine lessor, directly and through Shannon Engine Support, a joint venture with Safran Aircraft Engines that leases CFM engines. It owns helicopters and runs AerCap Materials, its part-out and used-material arm. It is a foreign private issuer and so files an annual report on Form 20-F and interim results on Form 6-K rather than a 10-K and 10-Qs. [AerCap 20-F FY2025; AerCap 6-K Q3 2025; D11 §3.2]

FORM 20-F / FORM 6-K

Form 20-F is the annual report a foreign private issuer files with the SEC in place of a Form 10-K, and Form 6-K is the form on which it furnishes interim results and other reports in place of 10-Qs and 8-Ks. AerCap, incorporated in the Netherlands and listed on the NYSE, files on these forms. The content differs in layout and in some disclosure requirements from a domestic filer's, which is one reason AerCap's segment-level margins were not returned in the same shape as the other peers'. [D11 §3.2, §8]

SHANNON ENGINE SUPPORT (SES)

Shannon Engine Support is AerCap's 50% engine-leasing joint venture with Safran Aircraft Engines, the Safran half of CFM, that leases CFM engines. It matters to the engine count: AerCap's "over 1,200 engines" at 30 Sep 2025 includes engines owned and managed by SES, while its "470 owned engines" at 31 Dec 2025 does not. AerCap recognised \$155M of lease rental income from SES in the first nine months of 2025. It is also a model of OEM-partnered engine leasing, with the manufacturer as co-owner of the lessor. [AerCap 6-K Q3 2025; AerCap 20-F FY2025; D11 §3.2, §6 T1]

For 2025 AerCap reported total revenues and other income of \$8,517M (2024: \$7,997M), up 7%, and net income attributable to AerCap of \$3.8B (2024: \$2.1B); the drivers of the increase were not returned. Total lease revenue was \$7,369M, of which basic lease rents were \$6,679M and maintenance rents and other receipts \$690M. [AerCap 20-F FY2025; AerCap Q4 2025 release] At 31 December 2025 the portfolio was 3,500 aircraft, engines and helicopters owned, managed or on order, and AerCap owned 470 engines, "9% of fleet". The owned passenger fleet averaged 7.3 years, split 5.4 years for new-technology and 15.2 years for current-technology aircraft, with 7.1 years of lease term remaining. By net book value the passenger and freighter fleet was 37% A320neo family, 20% 787, 8% A350, 8% 737NG and 7% A320ceo family; the five lines sum to 80% and the 737 MAX share was not among the lines returned. [AerCap 20-F FY2025; D11 U8] The filing also notes that when a lease is extended, the remaining rentals are recorded straight-line over the remaining original term plus the extension, which is how the extension behaviour Part III §22 describes reaches a lessor's revenue line. [AerCap 20-F FY2025]

The engine count carries a tension that this Part presents both ways. The Q3 2025 6-K states "over 1,200 engines (including engines owned and managed by its Shannon Engine Support JV)" at 30 September 2025. The FY2025 20-F states that AerCap "owned 470 engines (9% of fleet)" at 31 December 2025. On one reading the 1,200 figure is the scale of AerCap's engine platform, since SES engines are leased under AerCap's management; on the other the 470 figure is the only count on the same basis as Willis's 363 and FTAI's 243, which are owned engines. The bases differ (owned against owned plus managed plus joint venture) and the dates differ by a quarter. Both are recorded, neither is chosen, and any comparison with FTAI's owned engines in §89 uses 470 and says so. [AerCap 6-K Q3 2025; AerCap 20-F FY2025; D11 §6 T1; reconciliation R-033; E.1 rule 35] A

second, smaller arithmetic point: 470 engines is 13.4% of the 3,500-unit portfolio, not 9%, so the "9% of fleet" is on a basis (perhaps value, perhaps a different unit count) that the retrieved text does not state; this is arithmetic in this Part on two retrieved figures and not a finding of the research. [AerCap 20-F FY2025]

The 20-F was searched for AerCap Materials revenue, an engine-leasing revenue split, a stated lease yield or net spread, and statements on mid-life aircraft values or extension rates; none was returned. [D11 U6, U7] The one hard statement on current-technology aircraft is the age split: AerCap's current-technology owned fleet averaged 15.2 years, the age band FTAI's Strategic Capital vehicles buy (Part VIII §60). [AerCap 20-F FY2025] The Q1 2026 interim report was located by URL with no figures returned. What the research holds on AerCap Materials comes from its factsheets rather than the filing: more than 300 aircraft dismantled and more than 15,000 unique parts in 2023; more than 250,000 items stocked and more than 850 customers at the first quarter of 2026. [AerCap Materials factsheets, 2023 and 1Q 2026; D3 §3]

EXHIBIT 11.6 — AERCAP, FY2025 (YEAR ENDED 31 DEC 2025) AND THE Q3 2025 INTERIM

Item	Figure	Date and note
Total revenues and other income	\$8,517M (2024: \$7,997M; +7%)	20-F
Net income attributable	\$3.8B (2024: \$2.1B)	20-F; drivers not returned (U7)
Total lease revenue	\$7,369M	Q4 2025 release
of which basic lease rents	\$6,679M	Q4 2025 release
of which maintenance rents and other receipts	\$690M	9.4% of lease revenue (derived); about 10% of basic rents (arithmetic in this Part)
Portfolio owned, managed or on order	3,500 aircraft, engines and helicopters	31 Dec 2025
Owned engines	470, "9% of fleet"	31 Dec 2025 (R-033, side B)
Engines including SES owned and managed	"over 1,200"	30 Sep 2025 (R-033, side A)
Aircraft owned, managed or on order; owned helicopters; total assets	1,988; >300; \$72B	30 Sep 2025
SES lease rental income	\$155M	Nine months to 30 Sep 2025
NBV mix: A320neo / 787 / A350 / 737NG / A320ceo	37% / 20% / 8% / 8% / 7%	Sum 80%; 737 MAX share not returned (U8)

Item	Figure	Date and note
Average age, owned passenger fleet	7.3 years (new-technology 5.4; current-technology 15.2)	31 Dec 2025
Average remaining lease term	7.1 years	31 Dec 2025
AerCap Materials revenue; engine-leasing revenue; stated yield	not returned	U6, U7

Source: AerCap Holdings N.V., Form 20-F for FY2025; Q4 and FY2025 earnings release (6-K exhibit); Form 6-K for the quarter ended 30 Sep 2025. D11 §3.2, §4.1, §6 T1. Reconciliation R-033.

87.3 AIR LEASE CORPORATION (FORMERLY NYSE: AL; NOW SUMISHO AIR LEASE CORPORATION)

Air Lease buys new aircraft directly from Airbus and Boeing and leases them to airlines; its filing describes the strategy as purchasing "modern, fuel-efficient new technology commercial jet aircraft". It sits at the opposite end of the fleet-age spectrum from FTAI's Leasing and Strategic Capital activities, which buy mid-life 737NG and A320ceo aircraft, and it is in this Part as the contrast case and as the source of one filed statement on lease rates. [Air Lease 10-K FY2025; D11 §3.3]

For the year ended 31 December 2025 Air Lease owned 490 aircraft and managed 45, with a weighted-average fleet age of 4.9 years and a weighted-average remaining lease term of 7.2 years. The net book value of flight equipment subject to operating lease was \$29.1B, and lease rental revenue was \$2,615.4M (2024: \$2,407.5M), a 9.0% gross yield on year-end book as worked example 11.2 derives. [Air Lease 10-K FY2025] Total revenues, the pre-tax margin, a stated lease yield and any statement on engine availability were searched for and not returned. [D11 U10] On demand and rates the filing is explicit: "As global air traffic expands and aircraft production volumes remain constrained, lessors are experiencing strong demand for aircraft through new lease requests and lease extension requests. The strong demand for aircraft, combined with elevated interest rates and inflation, helped increase lease rates on new lease agreements and lease extensions during 2025. Lease rates are expected to remain strong as the supply and demand environment for commercial aircraft remains tight." [Air Lease 10-K FY2025, via D9 §2.8] That statement is one side of a tension with mid-life commentary from IBA (an appraiser), which has 12-year-old 737-800 rents down 11% year on year in July 2025; the two speak of different asset ages, and Part II §15 and Part III §22 carry both. [IBA, July 2025; reconciliation R-031]

The 10-K states that on 8 April 2026 Air Lease completed a merger under an Agreement and Plan of Merger dated 1 September 2025, with the company becoming Sumisho Air Lease Corporation. A 10-K/A for FY2025 was filed afterwards and its content was not returned; an 8-K

from 2025 and one dated 1 July 2025 were returned by URL only. The identity of the acquiring consortium beyond the surviving name was not returned and is not stated here. [Air Lease 10-K FY2025; Air Lease 10-K/A; D11 U9] The practical consequence for a peer set is that the FY2025 10-K is likely Air Lease's last annual filing as a public registrant, so a comparison for 2026 would need the acquirer's disclosures. [D11 §3.3] Air Lease also ran two third-party-capital platforms that Part VIII §65 sets beside FTAI's Strategic Capital Initiative: Blackbird Capital I and II, investment funds whose aircraft Air Lease manages and in which it holds 9.5% of the equity of each, and Thunderbolt, which sells mid-life aircraft portfolios to investors while Air Lease "continue[s] managing these aircraft for a fee", holding about 5.0% non-controlling interests in two Thunderbolt entities; the fee rates were not disclosed in the returned text. [Air Lease 10-K FY2024, via D7 §3.2; D7 U16]

EXHIBIT 11.7 — AIR LEASE, FY2025 (YEAR ENDED 31 DEC 2025), ITS LAST ANNUAL FILING BEFORE THE 8 APR 2026 TAKE-PRIVATE

Item	Figure	Note
Owned aircraft; managed aircraft	490; 45	31 Dec 2025
Weighted-average fleet age	4.9 years	31 Dec 2025
Weighted-average remaining lease term	7.2 years	31 Dec 2025
Net book value, flight equipment on operating lease	\$29.1B	31 Dec 2025
Lease rental revenue	\$2,615.4M (2024: \$2,407.5M)	
Implied gross yield on year-end NBV	9.0%	Derived (worked example 11.2)
Total revenues; pre-tax margin; stated yield; engine-availability statement	not returned	D11 U10
Statement on lease rates	"helped increase lease rates on new lease agreements and lease extensions during 2025. Lease rates are expected to remain strong"	10-K FY2025 (R-031, side B)
Merger completed	8 Apr 2026; became Sumisho Air Lease Corporation; agreement dated 1 Sep 2025	Acquirer beyond the surviving name not returned (U9)
Third-party-capital platforms	Blackbird Capital I and II (9.5% each); Thunderbolt (about 5.0% in two entities); fee rates not disclosed	10-K FY2024 (Part VIII §65)

Source: Air Lease Corporation, Form 10-K for FY2025; Form 10-K/A for FY2025; Form 8-K filings of 2025 (located by URL); Form 10-K for FY2024 (via D7 §3.2). D11 §3.3, §4.1; D9 §2.8. Reconciliation R-031; D11 U9, U10; D7 U16.

87.4 HEICO CORPORATION (NYSE: HEI)

HEICO has two groups. The Flight Support Group "designs, manufactures, repairs, overhauls and distributes jet engine and aircraft component replacement parts, which are approved by the FAA", using "proprietary technology to design and manufacture jet engine and aircraft component replacement parts for sale at lower prices than those manufactured by OEMs"; the Electronic Technologies Group is outside this chain. FSG is the listed proxy for the PMA business FTAI runs through its joint venture, and HEICO describes itself as "the largest independent supplier of non-OEM jet engine and aircraft component replacement parts". [HEICO 10-K FY2025; D11 §3.4] In 2023 HEICO acquired Wencor Group, the other large independent PMA and distribution business. [HEICO release, 2023, via D4 §3]

HEICO's fiscal year ends 31 October, so its FY2025 covers November 2024 to October 2025, ten months of calendar 2025. FSG net sales were \$3,117.3M, up 18% from \$2,639.4M; FSG operating income was \$750.4M, up 27% from \$593.1M; the derived margin was 24.1% against 22.5%. The "aftermarket replacement parts" product line grew 15% organically, by \$263.9M. [HEICO 10-K FY2025; HEICO release, 18 Dec 2025] HEICO adds approvals at a stated rate that two readings of the same 10-K give as "approximately 300 to 500" and "approximately 400 to 550 Parts Manufacturer Approvals per year"; both are carried (reconciliation R-024), and each approval is a separate FAA action (Part VI §37). Its customers are commercial airlines and air cargo carriers, repair and overhaul facilities, OEMs, and US and foreign governments. [HEICO 10-K FY2025]

The 10-K was searched for a stated discount to OEM price, a cumulative PMA count, statements on airline or lessor acceptance of PMA, and the consolidated totals; the search returned only the qualitative "lower prices" language and the approvals rate. The Q1 FY2026 release (quarter ended 31 January 2026) was located by URL with no figures; the Q3 FY2026 10-Q (quarter ended 31 July 2026) was not located. [D11 U4, U5; reconciliation K-05] Part VI §41 carries HEICO as a player in the PMA market and Part VI §42 what the research holds on acceptance; the acceptance datapoints there are airline and association statements, not HEICO's. [D4 §3]

EXHIBIT 11.8 — HEICO FLIGHT SUPPORT GROUP, FY2025 (YEAR ENDED 31 OCT 2025)

Item	Figure	Note
FSG net sales	\$3,117.3M (+18% from \$2,639.4M)	Fiscal year to 31 Oct 2025

Item	Figure	Note
FSG operating income	\$750.4M (+27% from \$593.1M)	GAAP segment operating income, after D&A
FSG operating margin	24.1% (FY2024: 22.5%)	Derived
Aftermarket replacement parts, organic growth	+15%; +\$263.9M	Product line within FSG
PMA approvals per year	"approximately 300 to 500" / "approximately 400 to 550"	Two readings of the FY2025 10-K (R-024)
Pricing statement	"lower prices than those manufactured by OEMs"; parts are "the functional equivalent" of OEM parts	10-K; no percentage discount returned (U5)
Consolidated net sales and operating income; cumulative PMA count; acceptance statements	not returned	U5
Q3 FY2026 10-Q	not located	U4

Source: HEICO Corporation, Form 10-K for FY ended 31 Oct 2025; Q4 FY2025 earnings release, 18 Dec 2025; Q1 FY2026 earnings release (located by URL). D11 §3.4, §4.1; D4 §3, §4.1. Reconciliation R-024; cluster K-05.

87.5 AERSALE CORPORATION (NASDAQ: ASLE)

AerSale has two segments. Asset Management Solutions is "comprised of activities that extract value from strategic asset acquisitions either as whole assets or by disassembling for used serviceable material"; Technical Operations (TechOps) is "MRO activities for aircraft and their components, and product sales of internally developed engineered solutions". AMS is the listed proxy for the teardown and used-material link that Part V treats. [AerSale 10-K FY2025; D11 §3.5]

For 2025 total revenue was \$335.3M, down 2.8% from \$345.1M. AMS sales were \$211.6M, down \$3.9M (1.8%); AMS gross profit was \$74.1M, down \$8.4M (10.2%), a 35.0% gross margin against 38.3%, with the aircraft line within AMS at 33.2% against 34.8%, attributed to lower margin on used-material sales from product-mix changes. Engine revenue rose on higher used-material sales (+\$34.0M) and higher leasing revenue (+\$7.5M), "primarily attributable to greater activity in the PW4000 and CF6-80 product lines as the company continues to monetize its feedstock"; the CFM56 is not named in the returned text. [AerSale 10-K FY2025; AerSale release, 5 Mar 2026] TechOps revenue was \$89.7M for the nine months to 30 September 2025 and \$123.7M for the year, a figure the release states and the research also derives as \$335.3M less \$211.6M; the two agree. [AerSale 10-Q Q3 2025; AerSale release, 5 Mar 2026; reconciliation B-8; D11 T5] Used-material

revenue across both segments was \$137.6M; whole-asset sales were \$56.4M from 13 engines against \$110.1M from 20 engines and one aircraft in 2024; revenue excluding whole-asset sales rose 18.7%; Adjusted EBITDA was \$46.1M (13.8%) against \$33.4M (9.7%). In the fourth quarter, feedstock acquisitions were \$15.4M and four engines were sold whole against six a year earlier. [AerSale release, 5 Mar 2026, via D3 §4.6 and D11 §3.5]

AerSale's filing language frames supply as "feedstock" the company "continues to monetize". A statement on retirement scarcity was searched for in the 10-K and not returned; what the research holds is management's description of "a hypercompetitive feedstock environment, marked by increased scrutiny and pressure on pricing as more companies enter the teardown sector", as relayed in search results and not read in a filing. The full-year feedstock purchase total and the 2025 net income were not returned, and the Q2 2026 10-Q was located by URL only. [D11 U11, U12; D3 §2.6; reconciliation C.11 item 5] Part V §33 sets this beside an appraiser's "lessors' market" and a trade source's "seller's market for used parts"; the three agree that asset owners have the upper hand and teardown buyers do not, and differ only in vantage point. [D3 T6]

EXHIBIT 11.9 — AERSALE, FY2025 (YEAR ENDED 31 DEC 2025)

Item	Figure	Note
Total revenue	\$335.3M (−2.8% from \$345.1M)	
Adjusted EBITDA	\$46.1M, 13.8% (2024: \$33.4M, 9.7%)	Company-defined; release of 5 Mar 2026
AMS sales	\$211.6M (−1.8%)	
AMS gross profit; gross margin	\$74.1M (−10.2%); 35.0% (2024: 38.3%)	Margin derived
AMS aircraft gross margin	33.2% (2024: 34.8%)	Stated; decline attributed to USM product mix
USM revenue, both segments	\$137.6M (\$120.1M AMS + \$17.6M TechOps)	Components sum to \$137.7M, rounding
Engine USM sales change; engine leasing revenue change	+\$34.0M; +\$7.5M	PW4000 and CF6-80 named; CFM56 not named in the returned text
Whole-asset (flight equipment) sales	\$56.4M, 13 engines (2024: \$110.1M, 20 engines and one aircraft)	
TechOps revenue	\$123.7M for 2025 (stated and derived agree, B-8); \$89.7M for 9M 2025	

Item	Figure	Note
Q4 2025 feedstock acquisitions; engines sold whole	\$15.4M; 4 (Q4 2024: 6)	Full-year feedstock total not returned (U12)
Net income 2025; Q2 2026 10-Q figures	not returned	U11
Feedstock statement	"hypercompetitive feedstock environment ... as more companies enter the teardown sector"	As relayed in search results, not from filing text

Source: AerSale Corporation, Form 10-K for FY2025; FY2025 results press release, 5 Mar 2026; Form 10-Q for the quarter ended 30 Sep 2025; Form 10-Q for the quarter ended 30 Jun 2026 (located by URL). D11 §3.5, §4.1, §6 T5; D3 §2.6, §3, §4.6. Reconciliation B-8, C.11 item 5.

87.6 STANDARDAERO, INC. (NYSE: SARO)

StandardAero is an independent engine MRO with two segments, Engine Services (shop visits across commercial, business-aviation and military platforms) and Component Repair Services (piece-part repair). It listed in October 2024. Engine Services is the listed proxy for the shop-visit link, and its CFM56 Center of Excellence in Dallas is a direct capacity competitor to FTAI's Module Factory. [StandardAero 10-K FY2025; StandardAero prospectus, Oct 2024; D11 §3.6]

CENTER OF EXCELLENCE (COE)

Center of Excellence is StandardAero's term for a shop dedicated to a single engine platform. Its CFM56-7B Center of Excellence in Dallas, Texas, opened in the second half of 2024 "to service the growing demand on that platform"; the company says it has spent more than \$100 million since 2022 to expand and to "more than double our shop visit capacity on the CFM56-7B" through the greenfield Dallas site. No shop-visit count is attached to the statement. Part IV §25 named the term; this is the full treatment. [StandardAero 10-K FY2025; D2 §3.2, §4.8; D11 §3.6, §8]

For 2025 Engine Services revenue was \$5,354.0M, up \$709.3M (15.3%) from \$4,644.7M, and Segment Adjusted EBITDA was \$706.9M, up \$96.0M (15.7%) from \$610.9M, a 13.2% margin in both years. Component Repair Services revenue was \$708.6M, up \$116.2M (19.6%), with Segment Adjusted EBITDA of \$202.7M, 28.6%. The sum of the two segments is \$6,062.5M of revenue and \$909.6M of segment Adjusted EBITDA (derived in the research; the two rounded revenue inputs sum to \$6,062.6M). [StandardAero 10-K FY2025] The consolidated figures after eliminations and corporate costs were not returned, and that gap carries a caveat the research flags twice: the segment sum is not the company's revenue, and the 1H 2026 10-Q's "approximately \$3.2B" of revenue, up 8.8%, implies about \$2.94B for 1H 2025 and therefore about \$3.12B for 2H 2025 if the full year were \$6,062.5M, which is consistent but unverified.

[StandardAero 10-Q Q2 2026; D11 U13, T6] For the six months to 30 June 2026 StandardAero reported net income of \$177.2M and Adjusted EBITDA of \$433.0M, up 12.3%, about 13.5% of the rounded revenue figure (derived). [StandardAero 10-Q Q2 2026]

The filing's capacity statements are the data points this Part sets beside FTAI's in §89. Besides Dallas, a LEAP-dedicated overhaul line at the 810,000 square-foot San Antonio, Texas facility was under industrialisation "with significant progress to date", also with more than \$100 million committed; during 2024 the company completed test-cell correlations for the LEAP-1A and LEAP-1B, obtained Chinese maintenance-organisation approval for the LEAP, developed more than 260 LEAP component repairs across its repair facilities and inducted its first LEAP engines; the LEAP and CFM56 Dallas programmes "reached profitability" in a 2025 quarter; and a 70,000 square-foot Winnipeg expansion, about 40% more footprint, came online on 3 September 2025 adding CF34 and CFM56 capacity. [StandardAero 10-K FY2025; StandardAero 10-Q Q3 2025; GlobalAir, Sept 2025; D2 §4.8] Revenue by platform and a CFM56 shop-visit count were searched for and not returned. [D11 U13] One more figure belongs to StandardAero and not to FTAI: a search summary once attributed "approximately 7,800 people across over 50 facilities" to FTAI; the research identified it as StandardAero's own profile from its IPO prospectus, and §89.6 uses it as such. [StandardAero prospectus, Oct 2024; reconciliation R-065]

EXHIBIT 11.10 — STANDARD AERO, FY2025 (YEAR ENDED 31 DEC 2025) AND 1H 2026

Item	Figure	Note
Engine Services revenue	\$5,354.0M (+15.3% from \$4,644.7M)	
Engine Services Segment Adjusted EBITDA; margin	\$706.9M (+15.7% from \$610.9M); 13.2% (2024: 13.2%, derived)	Before D&A and defined adjustments
Component Repair Services revenue; Segment Adjusted EBITDA; margin	\$708.6M (+19.6%); \$202.7M; 28.6%	
Sum of segments	\$6,062.5M revenue; \$909.6M Adjusted EBITDA	Derived; consolidated figures not returned (U13, T6)
1H 2026 revenue; Adjusted EBITDA; net income	about \$3.2B (+8.8%); \$433.0M (+12.3%); \$177.2M	Implied margin about 13.5% on a rounded figure
CFM56-7B Center of Excellence, Dallas	opened 2H 2024; >\$100M since 2022; "more than double" CFM56-7B shop-visit capacity	No visit count
LEAP line, San Antonio (810,000 sq ft)	industrialisation underway; >\$100M	

Item	Figure	Note
LEAP component repairs developed	>260	During 2024
Winnipeg expansion	+70,000 sq ft (about +40%), CF34 and CFM56; online 3 Sep 2025	GlobalAir
Headcount and sites (IPO prospectus)	about 7,800 people, over 50 facilities	StandardAero's figure, not FTAI's (R-065)
Revenue by platform; CFM56 shop-visit count	not returned	U13

Source: StandardAero, Inc., Form 10-K for FY2025; Form 10-Q for the quarters ended 30 Sep 2025 and 30 Jun 2026; prospectus (424B4), Oct 2024; GlobalAir, "StandardAero's Winnipeg expansion adds CF34 and CFM56 engine capacity", Sept 2025. D11 §3.6, §4.1, §6 T6; D2 §3.2, §4.8. Reconciliation R-065.

87.7 AAR CORP (NYSE: AIR)

AAR has four segments: Parts Supply (distribution of new OEM parts and sales and leasing of used material), Repair & Engineering (airframe maintenance, component repair, landing gear), Integrated Solutions (supply-chain and fleet programmes, largely government) and Expeditionary Services. Parts Supply is the listed proxy for the distribution-plus-used-material link, and AAR is also FTAI's partner for CFM56 material sales. [AAR 10-K FY2026; D11 §3.7]

AAR's fiscal 2026 ended 31 May 2026. Sales to commercial customers were \$2,384.1M (72.1% of the total) and to government and defense customers \$923.9M (27.9%), so consolidated sales were \$3,308.0M (derived in the research). Commercial sales rose \$408.0M (20.6%) on "strong demand and volume growth in new parts Distribution activities, including from the recent ADI acquisition, which contributed sales of \$82.2M"; ADI was acquired in September 2025 for \$137.1M into Parts Supply. Parts Supply "accounted for approximately 45% of AAR's sales in fiscal 2026", about \$1,489M (derived); within it, new-parts distribution grew 23.9% and used-material activities 32.1%. Repair & Engineering grew 8% organically excluding landing gear, with "Airframe MRO near capacity with strong production efficiency". [AAR 10-K FY2026; D3 §4.6] Segment operating income and the used-material split within Parts Supply were not returned; the Q1 FY2027 10-Q (quarter ended 31 August 2026) was located by URL only; and additional proxy material (a DEFA14A) was filed in 2026 whose subject was not returned. [D11 U14]

AAR's connections to the rest of this primer run in two directions. It holds the exclusive agreement under which it sells material from FTAI's CFM56 engine pool, extended through 2030, which is the unnamed "exclusive partnership responsible for the teardown, repair, marketing and sales of parts from the CFM56 engine pool" that FTAI's 10-K describes (Part V §35; Part VII §52). [AAR / FTAI release, PR Newswire, 2025; FTAI 10-K FY2025; reconciliation B-9] And it

distributes Chromalloy's engine parts under a supply agreement of November 2024 and an exclusive PW4000 agreement of March 2025, which places it on the PMA link as well (Part VI §41). [D4 §3]

EXHIBIT 11.11 — AAR, FY2026 (YEAR ENDED 31 MAY 2026)

Item	Figure	Note
Commercial sales	\$2,384.1M (72.1%); +\$408.0M, +20.6%	
Government and defense sales	\$923.9M (27.9%)	
Consolidated sales	\$3,308.0M	Derived from the two lines
Parts Supply share of sales; implied revenue	about 45%; about \$1,489M	Share stated; dollars derived
Parts Supply growth: new-parts distribution; USM activities	+23.9%; +32.1%	
ADI acquisition	Sept 2025, \$137.1M; \$82.2M of sales contributed	Parts Supply
Repair & Engineering organic growth ex landing gear	8%; airframe MRO "near capacity"	
Segment operating income; USM sales within Parts Supply; DEFA14A subject; Q1 FY2027 figures	not returned	U14
FTAI relationship	exclusive Serviceable Engine Products agreement for CFM56 material, through 2030	PR Newswire, 2025 (B-9)

Source: AAR Corp, Form 10-K for FY ended 31 May 2026; Form 10-Q for the quarter ended 31 Aug 2026 (located by URL); DEFA14A, 2026 (located by URL); AAR / FTAI Aviation, PR Newswire release, 2025. D11 §3.7, §4.1; D3 §4.6; D4 §3. Reconciliation B-9.

87.8 GE AEROSPACE (NYSE: GE)

GE Aerospace is the engine manufacturer. Through CFM International, its 50/50 venture with Safran, it makes the CFM56 and the LEAP; its Commercial Engines & Services segment sells new engines and spare parts and performs shop visits in its own network and under long-term service agreements (LTSAs, the per-flight-hour contracts Part IV §28 treats). It is the counterparty FTAI's PMA, used-material and module business competes with on parts and shop visits, and the supplier of the new parts the Module Factory consumes, including under the multi-year materials agreement of 22 January 2026 (Part VII §52). [GE 10-K FY2025; D11 §3.8; D4 §3]

Section 86.2 and Exhibit 11.4 carry the segment figures; this subsection adds what the filing says about demand, capacity and pricing, and what it does not.

On demand, the filing's statement that the LEAP "is expected in the coming years to overtake the mature CFM56 as the industry's largest fleet" and that this "is expected to drive a significant increase in shop visits and need for MRO capacity as LEAP engines come due for services" is the manufacturer's own horizon for the CFM56 pool, set beside Safran's "until 2030". [GE 10-K FY2025; Safran, 13 Feb 2026] On capacity, the turnaround improvement of more than 10% year on year for LEAP, CFM56 and GE90 in the fourth quarter of 2025 is a statement that makes sense only if turnaround had been a constraint, and the research sets it on one side of a tension with a consultancy survey in which two-thirds of operators cannot find enough technicians (Part IV §27; reconciliation R-019, R-020). [GE 10-K FY2025; Oliver Wyman, Apr 2026] On pricing, the filing names "pricing" and "improved pricing" among the drivers of revenue and profit but gives no percentage; the percentages the primer carries come from trade press (§86.2; Part IV §28). [GE 10-K FY2025; D11 U16] On the open market, GE's investor-relations page states that "an open network exists with multiple providers who compete and invest to win shop visits" and that GE will "continue to strengthen MRO access to OEM materials to support further CFM56 longevity", and that "GE and CFM are the largest used serviceable material provider for their products". [GE Aerospace investor-relations (IR) page, 2026, via D2 §2.3] Its TrueChoice menu (Flight Hour, Overhaul, Material, Transitions), including Ryanair's long-term materials agreement covering "their entire fleet of ~2,000 CFM56 and LEAP engines", is Part IV §28's subject. [GE 10-K FY2025; D2 §2.3]

What the filing does not give, for the record: a CFM56 shop-visit count, a spare-parts price-increase percentage, any wording on a CFM56 used-material programme by name, the CFM56 share of CES services revenue, and total company revenue for 2025. The Q1 2026 release was returned by URL with the headline that "services revenue grew 39%"; the Q2 2026 10-Q was not located. [GE Q1 2026 release; D11 U15, U16; D2 U4; reconciliation C.11 item 1] The research's substitutes for the first two gaps are trade-press figures: 2,300–2,400 CFM56 shop visits a year through 2028 (GE via Aviation Week, 2026) and the escalation percentages of §86.2. [Aviation Week, 2026; Aviation Week, late 2023]

EXHIBIT 11.12 — GE AEROSPACE, FY2025: STATEMENTS AND GAPS BEYOND THE SEGMENT FIGURES OF EXHIBIT 11.4

Topic	What the filing says	What was searched for and not returned
CFM56 pool horizon	LEAP "expected in the coming years to overtake the mature CFM56 as the industry's largest fleet"; LEAP shop visits to drive "a significant increase in shop visits and need for MRO capacity"	CFM56 shop-visit count (U16); CFM56 share of CES services (D2 U4)

Topic	What the filing says	What was searched for and not returned
Capacity and turnaround	Turnaround for LEAP, CFM56, GE90 improved "by over 10% year-over-year" in Q4 2025; about 1,450 LEAP-1A durability kits deployed	Shop-visit capacity in units
Pricing	"pricing" and "improved pricing" named as revenue and profit drivers	Any percentage (U16); trade press carries ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023 (R-018)
Contract risk	Profit reduced by "an unfavorable change in estimated profitability of long-term service agreements, primarily from the estimated impact from tariffs"	Size of the change
Used material	Investor-relations (IR) page: "GE and CFM are the largest used serviceable material provider for their products"	Programme name and scale (U16; K-16)
Open market	Investor-relations (IR) page: "an open network exists with multiple providers who compete and invest to win shop visits"	OEM share of CFM56 shop visits (D2 U3)
Deliveries	2,386 commercial engines, 1,802 LEAP (+25%)	
Company totals	Operating profit \$9.1B (+25%); free cash flow \$7.7B (+24%)	Total revenue (U15); bridge from CES segment profit \$8,861M (T4)
2026	Q1 2026 release headline: "services revenue grew 39%"	Q2 2026 10-Q not located (U15)

Source: General Electric Co. (GE Aerospace), Form 10-K for FY2025; Q1 2026 earnings release (located by URL, headline only); GE Aerospace Investor Relations, "Recent events: your questions answered", 2026; Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023, and "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026. D11 §3.8, §4.1, §6 T4, §7 U15–U16; D2 §2.3, §4.3. Reconciliation R-018, R-019, R-020, C.11.

87.9 NON-SEC FILERS, FOR CONTEXT: SAFRAN, MTU AERO ENGINES, LUFTHANSA TECHNIK

Safran (Euronext: SAF) is CFM International's other half and reports under French and European rules, so its figures are carried as context and in euros. Its 2025 results give Propulsion services revenue of €10,122M, 64.6% of Propulsion revenue (2024: €8,430M, 61.7%), implying Propulsion revenue of about €15.7B (derived); LEAP deliveries of 1,802, up 28%, matching GE's 10-K as both report the CFM total, with a backlog above 12,900 units and 2026 deliveries expected up about 15%; a civil installed base of about 29,900 engines "including more than 22,800 CFM56s", with "sustained aftermarket revenue from this fleet until 2030"; a new LEAP-1A high-pressure turbine blade certified in December 2024 that "doubles the on-wing

service life of engines operating in harsh environments"; and group revenue of €31.3B, up 14.7%, with recurring operating income of €5.2B, 16.6%. Civil spare-parts revenue in dollars rose 17.6%, "driven primarily by the CFM56". Safran tracks spare parts in dollars across CFM56, LEAP and high-thrust engines; a CFM56-only figure was not returned. [Safran FY2025 results presentation and press release, 13 Feb 2026; Safran 2025 Integrated Report; D11 §3.9, U17; D2 §4.3]

MTU Aero Engines (Xetra: MTX) is both an engine-programme partner (a shareholder in IAE, maker of the V2500) and, through MTU Maintenance, an independent shop on the CFM56 and V2500 at Hannover and Zhuhai. Its 2025 figures are in §86.3: commercial maintenance revenue €6.0B, up 18%; adjusted EBIT €478M, up 9%; margin 8.0% against 8.7%, with the Geared Turbofan at about 40% of the work and the Fort Worth ramp-up named as the drags; group adjusted revenue €8.7B, adjusted EBIT €1.35B (15.5%), adjusted net income €968M. The Geared Turbofan share is the one place in this Part where the GTF driver of Part III §20 appears in a peer's margin: more GTF work, on MTU's account, lowered the percentage. No CFM56 or V2500 share of its maintenance revenue was returned. [MTU, 24 Feb 2026; MTU FY2025 key figures; D11 §3.9, U17; D2 §3.2]

Lufthansa Technik is a segment of Deutsche Lufthansa AG (Xetra: LHA) and not separately listed. Its 2025 figures are in §86.3: revenue €8.049B, up 12%; adjusted EBIT €603M, down 1%; margin 7.5% against 8.5%, attributed to US tariffs, material-cost increases and a weaker dollar. Engine Services "generates almost half" of revenue and is "set to be a strong growth driver over the next few years, in particular through its maintenance of the new generation of engines"; Aircraft Component Services is "more than one third"; new contracts signed in 2025 were €8.8B, "for the first time evenly distributed across all three regions". No shop-visit count or CFM56 and LEAP capacity figure was returned. [Lufthansa Technik, FY2025 results; Deutsche Lufthansa AG, Annual Report 2025; D11 §3.9, U17]

EXHIBIT 11.13 — NON-SEC CONTEXT: SAFRAN, MTU AERO ENGINES AND LUFTHANSA TECHNIK, CALENDAR 2025, IN EUROS

Company	Item	Figure	Source
Safran	Propulsion services revenue; share of Propulsion	€10,122M; 64.6% (2024: €8,430M; 61.7%)	Results presentation, 13 Feb 2026
Safran	Propulsion revenue	about €15.7B (derived)	Same
Safran	Group revenue; recurring operating income	€31.3B (+14.7%); €5.2B, 16.6%	Press release, 13 Feb 2026
Safran	Civil spare parts, USD	+17.6%, "driven primarily by the CFM56"	Same

Company	Item	Figure	Source
Safran	LEAP deliveries; backlog; 2026 outlook	1,802 (+28%); >12,900; +about 15%	Results presentation
Safran	Civil installed base; CFM56	about 29,900; "more than 22,800", aftermarket "until 2030"	Results presentation
MTU	Commercial maintenance revenue; adjusted EBIT; margin	€6.0B (+18%); €478M (+9%); 8.0% (2024: 8.7%)	Release, 24 Feb 2026
MTU	Group adjusted revenue; adjusted EBIT; margin; adjusted net income	€8.7B; €1.35B; 15.5% (2024: 14.0%); €968M	Same
MTU	Geared Turbofan share of commercial maintenance	about 40%	Same
Lufthansa Technik	Revenue; adjusted EBIT; margin	€8.049B (+12%); €603M (−1%); 7.5% (2024: 8.5%)	FY2025 results release
Lufthansa Technik	Engine Services share; Component Services share	"almost half"; "more than one third"	Deutsche Lufthansa AR 2025
Lufthansa Technik	New contracts 2025	€8.8B	FY2025 results release
All three	CFM56-specific maintenance revenue, shop-visit counts, capacity	not returned	D11 U17

Source: Safran, FY2025 Results & Investor Update, 13 Feb 2026; Safran press release, 13 Feb 2026; Safran 2025 Integrated Report; MTU Aero Engines, press release and FY2025 key figures, 24 Feb 2026; Lufthansa Technik, FY2025 results release; Deutsche Lufthansa AG, Annual Report 2025, MRO business segment. D11 §3.9, §4.1; D2 §4.3.

88. The comparison table and the comparable metrics

The table below puts the eight SEC filers, the three context companies and FTAI in one frame. It is a reading aid, not a ranking. Each row carries its fiscal year, its comparable segment, that segment's revenue and margin with the basis named, and the one or two metrics that can be set beside something FTAI reports. The second exhibit lists the cautions that apply before any two cells are compared, and worked example 11.3 shows the simplest of them, period alignment, in months.

88.1 THE COMPARISON TABLE

EXHIBIT 11.14 — THE PEER SET BESIDE FTAI: FISCAL YEAR, REVENUE, COMPARABLE SEGMENT, MARGIN WITH BASIS, AND THE METRICS MOST COMPARABLE TO FTAI'S

Company	Ticker	Fiscal year	Revenue	Comparable segment	Segment revenue	Segment margin (basis)	Metrics most comparable to FTAI
FTAI (reference)	FTAI	CY2025; Q2 2026	\$2,507.4M	Aerospace Products	\$1,936.2M; \$875.0M (Q2)	34.7%; 28.5% Q2 (segment Adj. EBITDA, derived)	243 engines, 47 aircraft; 757 modules (296 in Q2); utilisation ~77% (equity basis); maintenance revenue \$218.5M
Willis Lease	WLFC	CY2025	\$730.2M	Engine leasing (whole company)	\$730.2M	62.9% (company Adj. EBITDA, derived)	363 engines; 84.9% utilisation (book basis); reserve revenue \$232.0M (derived); 116 managed
AerCap	AER	CY2025	\$8,517M	Lease revenue; engines incl. SES	\$7,369M	not returned (U7)	470 owned engines (>1,200 incl. SES, other basis); maintenance rents \$690M

Company	Ticker	Fiscal year	Revenue	Comparable segment	Segment revenue	Segment margin (basis)	Metrics most comparable to FTAI
Air Lease	AL (delisted 8 Apr 2026)	CY2025	not returned (U10)	Aircraft leasing (whole company)	\$2,615.4M rent	not returned (U10)	490 aircraft, 4.9 yrs; 9.0% implied yield (derived)
HEICO	HEI	FY to Oct 2025	not returned (U5)	Flight Support Group	\$3,117.3M	24.1% (operating income, derived)	300–500 or 400–550 PMAs/yr (R-024)
AerSale	ASLE	CY2025	\$335.3M	Asset Management Solutions	\$211.6M	35.0% (gross, derived)	USM revenue \$137.6M; 13 engines sold whole
StandardAero	SARO	CY2025	\$6,062.5M (segment sum, derived)	Engine Services	\$5,354.0M	13.2% (Segment Adj. EBITDA)	CFM56-7B capacity "more than doubled"; ~7,800 people
AAR	AIR	FY to May 2026	\$3,308.0M (derived)	Parts Supply	~\$1,489M (derived)	not returned (U14)	Distribution +23.9%; USM +32.1%
GE Aerospace	GE	CY2025	not returned (U15)	Commercial Engines & Services	\$33,314M (services \$25,010M)	26.6% (GE segment profit)	1,802 LEAP delivered; shop-visit revenue +24%
Safran (context)	SAF	CY2025	€31.3B	Propulsion services	€10,122M	16.6% group recurring operating income only	>22,800 CFM56; spares +17.6% (USD)

Company	Ticker	Fiscal year	Revenue	Comparable segment	Segment revenue	Segment margin (basis)	Metrics most comparable to FTAI
MTU (context)	MTX	CY2025	€8.7B adj.	Commercial maintenance	€6.0B	8.0% (adj. EBIT)	GTF ~40% of maintenance
Lufthansa Technik (context)	within LHA	CY2025	€8.049B	Engine Services	"almost half"	7.5% (adj. EBIT, company)	€8.8B new contracts

Source: the filings and releases listed in §87 for each company; FTAI figures from the FY2025 10-K (segment table and consolidated statement via the orchestrator notes), the Q2 2026 release and 10-Q, and D6 for the fleet. D11 §4.2, extended in this Part. CY is calendar year. Derived figures are so marked. Reconciliation R-024, R-032, R-033, R-043, R-045; D11 T1, T2, T6, T7; E.1 rule 35.

88.2 COMPARABILITY CAUTIONS

EXHIBIT 11.15 — CAUTIONS THAT APPLY BEFORE COMPARING ANY TWO CELLS OF EXHIBIT 11.14

Caution	What differs	Which cells it touches	Where it is treated
Period alignment	Three fiscal calendars: December (WLFC, AER, AL, ASLE, SARO, GE, Safran, MTU, Lufthansa Technik), October (HEICO), May (AAR); FTAI's reference quarter is Q2 2026, fifteen months after the peers' year-end	Every growth rate; HEICO and AAR against all others; FTAI Q2 2026 against every FY2025 figure	Worked example 11.3; D11 T7
Margin basis	Five bases: Segment Adjusted EBITDA (SARO, FTAI), segment operating income (HEI), adjusted EBIT (MTU, Lufthansa Technik), GE segment profit, gross margin (ASLE); company-defined "adjusted" items differ even within one base	Every margin cell	§86 head; worked example 11.1; E.1 rule 35; registry C-24
Mix: what the segment sells	A refurbished module or engine (FTAI); a labour-and-material shop visit (SARO, MTU, Lufthansa Technik); new engines plus parts plus services (GE); parts, repair and distribution (HEI); whole assets and parts (ASLE); distribution plus USM (AAR); rent and reserves (lessors)	Any margin set beside another link's	§86.9; D11 §2.3

Caution	What differs	Which cells it touches	Where it is treated
Units for "fleet"	AerCap counts owned, managed and on-order assets in one headline and owned engines separately; Willis counts owned and managed separately; FTAI counts owned assets including those in Russia	Engine and aircraft counts	§87.2; D11 T1, T2; R-033
Point in time versus period average	Willis 86.0% at 30 Sep 2025 against 84.9% average for 2025; equipment \$2,700.4M against \$2,801.7M	Utilisation and book-value cells	§87.1; R-032
Utilisation definition	Willis: share of portfolio by book value on lease over the period; FTAI: percent of days on lease weighted by equity value, excluding airframes; by count FTAI had 143 of 243 engines on lease	Utilisation cells	Worked example 11.4; registry C-3
Consolidated versus sum of segments	StandardAero's \$6,062.5M is a sum of two segments; consolidated revenue after eliminations not returned	StandardAero revenue cell	§87.6; D11 T6, U13
Derived versus stated	Every "derived" cell is a ratio of two filed numbers computed in the research or in this Part, not a figure the company reports	All derived cells	Each exhibit's source line
Related-party revenue	FTAI's segment revenue includes MRE Contract revenue from a partnership in which it holds 19% (or 20%); profit elimination method undisclosed	FTAI revenue and margin cells	§86.7; Part VII §48; Part VIII §63; R-043, R-047
Currency	Safran, MTU and Lufthansa Technik report in euros; no conversion is made here	Context rows	§87.9
Statement versus count	Capacity is stated as "more than doubled" (SARO), "over 10%" turnaround improvement (GE), or modules per year (FTAI); none is a count of shop visits	Capacity cells	§89.2; R-042
Gaps	Cells marked "not returned" were searched for in the filing and not found; they are absences in the research, not in the filings	All such cells	§87 and the unknowns list

Source: D11 §2.5 and §6; reconciliation E.1 rules 1, 35 and the ids cited in each row; registry C-3, C-24. Compiled in this Part.

WORKED EXAMPLE 11.3 — PERIOD ALIGNMENT IN MONTHS: HOW MUCH OF EACH PEER'S FISCAL YEAR OVERLAPS FTAI'S REPORTING PERIODS

Inputs (sourced: each filing's fiscal year-end as stated in D11): calendar-year filers report January to December 2025; HEICO's FY2025 runs November 2024 to October 2025; AAR's FY2026 runs June 2025 to May 2026; FTAI's reference periods are calendar 2025 and the quarter April to June 2026.

HEICO FY2025 overlap with CY2025: Jan–Oct 2025 = 10 of 12 months (83%); 2 months (Nov–Dec 2024) fall in 2024
 AAR FY2026 overlap with CY2025: Jun–Dec 2025 = 7 of 12 months (58%); 5 months (Jan–May 2026) fall in 2026
 AAR FY2026 overlap with FTAI 1H 2026: Jan–May 2026 = 5 of 6 months
 Midpoint gap, peers' CY2025 (1 Jul 2025) to FTAI Q2 2026 (15 May 2026): about 10.5 months

Reading. A growth rate quoted by HEICO is over a window that ends two months before the calendar-year filers' and AAR's over a window that ends five months after; a figure quoted for FTAI's second quarter of 2026 describes a point almost a year later than the centre of the peers' fiscal years. None of this changes a level figure (a margin, a fleet count), but it means that two growth rates in Exhibit 11.14 are never over the same twelve months, and that a comparison of FTAI's Q2 2026 margin with a peer's FY2025 margin compares two dates as well as two bases. Limits: the example assumes each company's revenue is spread evenly through its year, which is not true of any of them.

[D11 preamble, §2.5, §6 T7]

89. FTAI's numbers beside each peer's (descriptive; no ranking)

This section sets FTAI's figures beside the peers' on the metrics that are comparable, with the basis caveat attached to each pair. It describes; it does not rank. Where a pair cannot be made comparable from the retrieved figures, it says so.

89.1 ENGINES OWNED, AND UTILISATION ON TWO DEFINITIONS

At 31 December 2025 FTAI's Aviation Leasing segment owned 290 assets: 243 engines and 47 aircraft, of which 17 engines and 8 aircraft were in Russia; 143 engines and 37 aircraft were on lease. [FTAI 10-K FY2025, via D6 §4.1] Willis Lease owned 363 engines and 20 aircraft at the same date and managed 116 more engines for third parties. AerCap owned 470 engines at the same date; its "over 1,200" figure at 30 September 2025 includes engines owned and managed by Shannon Engine Support and is on a different basis (§87.2; R-033). [WLFC 10-K FY2025; AerCap 20-F FY2025; AerCap 6-K Q3 2025] On owned engines, then: FTAI 243, Willis 363, AerCap 470. The research also notes a dated trade estimate that the top five engine lessors held "over 1,000 powerplants with a book value in excess of \$5 billion" in 2019; current figures for GE Engine Leasing, SMBC Aero Engine Lease, Engine Lease Finance, Rolls-Royce & Partners Finance and MTU Maintenance Lease Services were not retrieved. [D6 §3; D6 U14]

Utilisation cannot be set side by side without naming the measure. FTAI's 10-K defines its about 77% as the percent of days on lease, weighted by the monthly average equity value of leasing equipment, excluding airframes; by count, 143 of 243 engines on lease is 59%. Willis's 84.9% is the share of the portfolio by book value on lease over the year. [FTAI 10-K FY2025; WLFC 10-K FY2025; registry C-3]

WORKED EXAMPLE 11.4 — WHY FTAI'S 77% AND 59% ARE BOTH RIGHT, AND NEITHER IS WILLIS'S 84.9%

Inputs: 243 engines owned, 143 on lease (sourced: FTAI 10-K FY2025); utilisation about 77% by equity value and days on lease, excluding airframes (sourced); Willis 84.9% by book value, period average (sourced: WLFC 10-K FY2025). The relative value of off-lease engines is not disclosed (assumed below).

by count:	$143 \div 243 = 58.8\%$
by value, if off-lease engines carry value x per engine relative to on-lease engines at 1.0:	
utilisation =	$143 \div (143 + 100x)$
$x = 1.00$ (same value):	$143 \div 243.0 = 58.8\%$
$x = 0.50$:	$143 \div 193.0 = 74.1\%$
$x = 0.33$:	$143 \div 176.3 = 81.1\%$
solving for the filed figure:	$143 \div (143 + 100x) = 0.77 \rightarrow 100x = 42.7$
	$\rightarrow x \approx 0.43$

Reading. The filed 77% is consistent with the 100 off-lease engines carrying, on average, about 43% of the equity value of an on-lease engine, which is what one would expect if the off-lease population includes run-out engines awaiting the Module Factory or teardown and the 17 engines in Russia carried at zero net book value (D6 §2.7; Part VIII §59). Willis's 84.9% weights by book value rather than equity value and includes aircraft and other equipment, so even at equal definitions of "on lease" the two percentages would not be measuring the same portfolio in the same way. The comparison the figures support is a description: Willis reported a higher on-lease share on its definition than FTAI did on its own, and FTAI's by-count share was lower still. Limits: the off-lease value share is solved for, not disclosed; the day-weighting in FTAI's definition is ignored; and Willis's engine-only utilisation was not returned. [FTAI 10-K FY2025; WLFC 10-K FY2025; D6 §2.7, §4.1; registry C-3]

89.2 MODULES BESIDE CAPACITY STATEMENTS

FTAI counts its output in modules, three to an engine by its own convention (reconciliation B-3): 757 modules in 2025 against a target of 750; 184 in the second quarter of 2025 and 296 in the second quarter of 2026, up 61%; 566 in the first half of 2026; a 2026 target of 1,050 set with the February 2026 results and raised to 1,200 on the July 2026 call per call coverage. Its stated capacity moved from 1,350 module overhauls a year (Montréal plus Miami, September 2024 release) to 1,800 modules or 600 engines (adding Rome, February 2025 release) to 3,000

modules (all sites and partners, call coverage, July 2026). [FTAI releases, 9 Sep 2024, 26 Feb 2025, 25 Feb 2026; GuruFocus and Quartr call coverage, 30 Jul 2026; reconciliation E.1 rules 2, 7, 17] At three modules an engine, 757 modules is about 252 engine-equivalents, 1,200 is 400, and 3,000 is 1,000; 400 engine-equivalents is about 17% of a 2,350-visit year, a derivation the research offers as one of three ways FTAI's share has been framed (reconciliation R-042). [D2 §3.3; D5 §2.4]

StandardAero states capacity differently: more than \$100 million spent since 2022 to "more than double our shop visit capacity on the CFM56-7B" at Dallas, a LEAP line under industrialisation at San Antonio, more than 260 LEAP component repairs developed, and a 70,000 square-foot Winnipeg expansion; no shop-visit count is given, and revenue by platform was not returned. [StandardAero 10-K FY2025; GlobalAir, Sept 2025; D11 U13] GE states a turnaround improvement of more than 10% and a 24% rise in internal shop-visit revenue, again without a count. [GE 10-K FY2025] The three companies therefore describe capacity in three units (modules, a multiple of an undisclosed base, and a revenue growth rate), and the only one of the three that can be converted to engines is FTAI's, under its own convention. Part VII §49 carries FTAI's volumes year by year and Part IV §27 the market's capacity.

89.3 SEGMENT MARGIN BESIDE HEICO'S AND STANDARDAERO'S

FTAI's Aerospace Products margin was 34.7% for FY2025 and 28.5% for the second quarter of 2026 on FTAI-defined segment Adjusted EBITDA; HEICO's Flight Support Group earned 24.1% on segment operating income after depreciation for the year to October 2025; StandardAero's Engine Services earned 13.2% on Segment Adjusted EBITDA for 2025. [FTAI 10-K FY2025; FTAI Q2 2026 release; HEICO 10-K FY2025; StandardAero 10-K FY2025] Of the three pairs only FTAI and StandardAero share a base, and even that pair is two companies' own definitions of "adjusted", applied to two different products: a refurbished module or engine sold at a price, against a shop visit whose material passes through at list (\$86.9; worked example 11.1). The HEICO pair adds the depreciation gap, in HEICO's disfavour on the percentage by an amount equal to FSG's depreciation share of sales, which was not retrieved. [D11 §2.3; D11 T3] FTAI's own gross margin, if the \$1,240,368K cost-of-sales figure is the segment's as the orchestrator's search placed it, was 35.9% on segment revenue for FY2025, carried with the scope caveat (R-045); that is the only FTAI figure on a gross basis and the only one that could be set beside AerSale's 35.0%, and the two sell different things. [FTAI 10-K FY2025; orchestrator notes; AerSale 10-K FY2025]

EXHIBIT 11.16 — FTAI'S FIGURES BESIDE EACH PEER'S, PAIR BY PAIR, WITH THE CAVEAT THAT ATTACHES TO EACH PAIR

Metric	FTAI	Peer figure	Caveat on the pair
Owned engines, 31 Dec 2025	243 (17 in Russia)	Willis 363; AerCap 470 (>1,200 incl. SES on a different basis, R-033)	Same unit; AerCap's second figure excluded
Owned aircraft, 31 Dec 2025	47 (8 in Russia)	Willis 20; Air Lease 490 (4.9 years); AerCap 1,988 owned, managed or on order (30 Sep 2025)	AerCap's figure mixes bases
Third-party assets managed	300+ aircraft committed in the 2025 Partnership (about \$6.0B, Aug 2026 release); 292 aircraft owned by the vehicle per its ABS (asset-backed securitisation) release of 22 May 2026	Willis 116 engines managed; Air Lease 45 aircraft managed	FTAI's vehicles are funds it serves as Servicer; Willis's are managed engines; Part VIII §61
Utilisation	about 77% (equity value, days on lease, ex airframes); 59% by engine count	Willis 84.9% (book value, period average); 86.0% at 30 Sep 2025	Different definitions (worked example 11.4)
Maintenance line relative to rent	Maintenance revenue \$218.5M against lease income \$235.2M (consolidated line against segment line; about 93%)	Willis \$232.0M against \$291.6M (about 80%); AerCap \$690M against \$6,679M (about 10%)	Line definitions differ; FTAI's segment mapping not retrieved
Implied gross lease yield	9.4% on average cost; 12.0% on average NBV (FY2025 lease income; this Part); 10.4%/13.4% or 6.6%/8.8% on the research's earlier bases	Willis 10.4% year-end book, 12.3% on-lease; Air Lease 9.0% year-end NBV	All derived; none stated (R-030)
Modules or shop visits	757 modules FY2025 (about 252 engine-equivalents at three per engine); 296 in Q2 2026; capacity 1,800 (release) / 3,000 (call coverage)	StandardAero: CFM56-7B capacity "more than doubled", no count; GE: internal shop-visit revenue +24%	Three units; only FTAI's converts to engines, under its own convention

Metric	FTAI	Peer figure	Caveat on the pair
PMA	Through a joint venture; revenue not separately disclosed; three or four CFM56-5B/7B approvals announced by the partner (R-025)	HEICO 300–500 or 400–550 approvals a year	FTAI's PMA figures are the partner's announcements, not FTAI's filing
Segment margin	34.7% FY2025; 28.5% Q2 2026 (segment Adjusted EBITDA)	StandardAero ES 13.2% (Segment Adjusted EBITDA); HEICO FSG 24.1% (operating income); AerSale AMS 35.0% (gross)	One shared base (StandardAero); different products throughout
Services share of a manufacturer's revenue	not applicable	GE CES 75.1%; Safran Propulsion 64.6%	FTAI is a customer of this line, not a reporter of it
Headcount	985 full-time employees and independent contractors (31 Dec 2025); 494 full-time employees in Canada	StandardAero about 7,800 people, over 50 facilities (IPO prospectus)	Mixed definition at FTAI (R-065); dates differ

Source: FTAI Aviation Ltd., Form 10-K FY2025 (via D6, D10 and the orchestrator notes), Q2 2026 release and 10-Q, releases of 9 Sep 2024 and 26 Feb 2025, and call coverage (GuruFocus, Quartr, 30 Jul 2026); Willis Lease Finance Corp, 10-K FY2025 and Q3 2025 release; AerCap, 20-F FY2025 and 6-K Q3 2025; Air Lease, 10-K FY2025; HEICO, 10-K FY2025; AerSale, 10-K FY2025; StandardAero, 10-K FY2025 and prospectus; GE, 10-K FY2025; Safran, 13 Feb 2026. D11 §2.1, §4.2; D6 §2.7, §4.1; D7 §4.3. Reconciliation R-025, R-030, R-033, R-042, R-045, R-065, B-3; E.1 rules 2, 7, 17.

89.4 THE SERVICING-FEE MODEL BESIDE WILLIS WEST AND AIR LEASE'S VEHICLES

FTAI acts as Servicer to the 2025 Partnership, in which it holds a 19% limited-partner stake (20% in the Q3 2025 10-Q), for "customary, market-based" compensation under a "contractual profit-sharing arrangement"; its servicing fees were \$10.2M in FY2025 and \$12.85M in the first half of 2026, \$6.99M of it in the second quarter. The base of the fee is not disclosed. Part VIII §61 and §63 carry the structure and the money flows, and §65 the comparables; this subsection only places the figures side by side. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; D7 §4.3, §4.4]

Willis's WEST vehicles (Willis Engine Securitization Trust, a rated engine securitisation that Willis consolidates) pay Willis 11.5% of aggregate net rents as servicer plus 2.0% as administrative agent, monthly, and 3.0% of net proceeds on engine sales, per the FY2019 and FY2020 10-Ks; the fee is internal to a vehicle Willis owns in full, so it is a transfer within the consolidated company and not third-party income. Air Lease's Blackbird Capital I and II are funds in which it held 9.5% of the equity of each and whose aircraft it managed; its Thunderbolt

platform sold mid-life portfolios while Air Lease kept managing them "for a fee" and held about 5.0% in two entities; the fee rates were not disclosed in the returned text. Carlyle Aviation Partners' securitisation vehicles pay servicing "generally in the range of 2% of rents, incentive fees up to 5% of rents in the aggregate, and 3% of sales proceeds". [WLFC 10-K FY2020 and FY2019; Air Lease 10-K FY2024; Carlyle Group 10-K FY2025; D7 §3.2, §4.4] FTAI's vehicle has since issued a securitisation of its own: on 22 May 2026 the 2025 Partnership priced \$612M of notes backed by 48 A320ceo and 737NG aircraft on lease to 23 airlines, in two classes (Series A expected Asf by Fitch and A(sf) by KBRA; Series B expected BBB+sf by Fitch), and the release stated that the vehicle "currently owns 292 aircraft"; coupons and loan-to-value were not retrieved. Part II §17 sets it beside the WEST template and Part VIII §62 carries the vehicle's debt layers. [FTAI release, 22 May 2026, via Finviz and AviTrader; orchestrator notes] Reading across, as the research does: a sponsor stake of 5–10% at Air Lease against 19% at FTAI; a servicing fee quoted as a percentage of rents at Carlyle (2%) and Willis (13.5%, inside a consolidated vehicle) against an undisclosed base at FTAI; a 3% sales fee at both Carlyle and Willis against undisclosed at FTAI. [D7 §3.2]

EXHIBIT 11.17 — FOUR SERVICING MODELS SIDE BY SIDE: FTAI'S 2025 PARTNERSHIP, WILLIS WEST, AIR LEASE'S VEHICLES, CARLYLE AVIATION PARTNERS

Platform	Structure	Sponsor stake	Servicing or management fee	Incentive	Sales fee	Disclosed fee income	Source
FTAI, 2025 Partnership	Third-party-capital partnership; FTAI is Servicer and exclusive engine-maintenance provider	19% LP (20% in Q3 2025 10-Q)	"customary, market-based"; base not disclosed	"contractual profit-sharing arrangement"; terms not disclosed	not disclosed	\$10.2M FY2025; \$12.85M 1H 2026 (\$6.99M Q2)	FTAI 10-K FY2025; 10-Q Q2 2026
Willis, WEST	Rated engine securitisation, consolidated by Willis	100% (owner)	11.5% of net rents (servicer) + 2.0% (administrative agent)	none disclosed	3.0% of net engine-sale proceeds	internal to the consolidated company	WLFC 10-K FY2019, FY2020
Air Lease, Blackbird I and II	Investment funds; Air Lease manages the aircraft	9.5% of each	rate not disclosed	not disclosed	not disclosed	not disclosed	Air Lease 10-K FY2024

Platform	Structure	Sponsor stake	Servicing or management fee	Incentive	Sales fee	Disclosed fee income	Source
Air Lease, Thunderbolt	Mid-life portfolios sold to investors; Air Lease manages "for a fee"	about 5.0% in two entities	rate not disclosed	not disclosed	not disclosed	not disclosed	Air Lease 10-K FY2024
Carlyle Aviation Partners, securitisation vehicles	Rated vehicles plus closed-end funds with carried interest	carried interest in funds	about 2% of rents	up to 5% of rents in aggregate	3% of sales proceeds	not isolated	Carlyle Group 10-K FY2025
Willis, third-party managed engines	116 engines managed at 31 Dec 2025	n/a	not returned	not returned	not returned	not returned	WLFC 10-K FY2025

Source: FTAI Aviation Ltd., Form 10-K FY2025 and Form 10-Q for the quarter ended 30 Jun 2026; Willis Lease Finance Corp, Forms 10-K for FY2019 and FY2020 and FY2025; Air Lease Corporation, Form 10-K FY2024; The Carlyle Group Inc., Form 10-K FY2025. D7 §3.2, §4.3, §4.4; D11 §3.1. Part VIII §65 carries the detail; reconciliation K-09, D7 U16.

89.5 OPERATING CASH FLOW AND INVENTORY BESIDE THE PEERS'

FTAI's net cash from operating activities was +\$128,982K in 2023, −\$187,956K in 2024 and −\$310,745K in 2025, while the company reported net income in each of the later quarters retrieved (\$111.9M attributable in Q4 2025; \$117.6M in Q2 2026). [FTAI 10-K FY2025, via D10 §4.4, §4.3] The mechanism is an accounting one: cash spent on assets held for sale, which is inventory, is an operating outflow, while cash spent on assets held for lease is an investing outflow, so a company building an inventory of engines, modules and parts to refurbish and sell shows operating outflows while it reports profit. FTAI's inventory, net, was \$1,364.3M at 31 March 2026; the 31 December 2025 and 30 June 2026 balances, the working-capital detail and the investing lines were not retrieved. [FTAI 10-Q Q1 2026; D10 §2.6; reconciliation K-18] Part X §81 carries the cash-flow statement and Part VII §53 the inventory.

The peers' comparable figures are thin because the research did not retrieve their cash-flow statements. GE Aerospace reported free cash flow of \$7.7B, up 24%, the only peer cash figure returned. AerSale's fourth-quarter feedstock acquisitions were \$15.4M, with the full-year total

not returned; AAR funded the \$137.1M ADI acquisition into Parts Supply; Willis's gain on sale of leased equipment was \$54.0M and its equipment held for lease grew to \$2,801.7M. [GE 10-K FY2025; AerSale 10-K FY2025; AAR 10-K FY2026; WLFC 10-K FY2025; D11 §5] None of these is an operating cash flow, and the research logged the lessors' cost of debt and funding terms as not searched within budget; FTAI's own notes carry coupons of 5.50% (2028), 7.875% (2030), 7.00% (2031), 6.875% (2032) and 5.875% (2033), which Part X §82 treats. [FTAI 10-K FY2025 and 10-Q Q1 2026, via D10; D11 U18; reconciliation C.11 item 2]

EXHIBIT 11.18 — CASH AND CAPITAL FIGURES: FTAI'S OPERATING CASH FLOW AND INVENTORY BESIDE THE PEER FIGURES THE RESEARCH RETRIEVED

Company	Figure	Period	What it is, and what it is not
FTAI	Net cash from operating activities +\$128,982K / -\$187,956K / – \$310,745K	2023 / 2024 / 2025	Operating cash flow; negative while net income is positive because inventory for sale is an operating outflow (D10 §2.6)
FTAI	Inventory, net \$1,364.3M	31 Mar 2026	Balance sheet; 31 Dec 2025 and 30 Jun 2026 balances not retrieved (K-18)
FTAI	Leasing equipment, net \$2,373.7M → \$1,545.8M	31 Dec 2024 → 31 Dec 2025	Investing-side asset; fell with Seed Asset sales (Part VIII §60)
FTAI	Senior notes 5.50%–7.875%, five issues; long-term debt net \$3,451,087K; cash \$412,240K	31 Dec 2025 / 31 Mar 2026	Funding cost; Part X §82
GE Aerospace	Free cash flow \$7.7B (+24%)	2025	Company-defined free cash flow; only peer cash figure returned
AerSale	Feedstock acquisitions \$15.4M	Q4 2025	A purchase of inventory, the analogue of FTAI's inventory build; full-year total not returned (U12)
AAR	ADI acquisition \$137.1M	Sept 2025	An acquisition, investing-side
Willis	Gain on sale of leased equipment \$54.0M; equipment held for lease \$2,801.7M	2025; 31 Dec 2025	Income-statement gain and balance-sheet asset; no cash-flow line returned
All lessors	Cost of debt and funding terms	n/a	Not searched within budget (U18)

Source: FTAI Aviation Ltd., Form 10-K FY2025 and Forms 10-Q for the quarters ended 31 Mar 2026 (via D10, D5, D6); GE 10-K FY2025; AerSale 10-K FY2025 and release of 5 Mar 2026; AAR 10-K FY2026; Willis Lease Finance Corp 10-K FY2025. D10 §2.6, §4.4, §4.5; D11 §5, U18. Reconciliation K-18, C.11 item 2.

89.6 SCALE IN PEOPLE

FTAI reported 985 full-time employees and independent contractors at 31 December 2025, of whom 494 were full-time employees in Canada, about 71% of them covered by collective bargaining agreements; the figure mixes employees and contractors. StandardAero's prospectus profile is about 7,800 people across more than 50 facilities, a figure a search summary once misattributed to FTAI. [FTAI 10-K FY2025, via orchestrator notes; StandardAero prospectus, Oct 2024; reconciliation R-065] The two numbers describe different businesses at different dates: a module factory with partner shops and a 50%-owned site in Rome, against an engine MRO performing whole shop visits across commercial, business-aviation and military platforms in its own facilities. Headcount for the other peers was not retrieved. The pair is given because the research flagged the misattribution, and so that the reader has both figures with their owners.

Tensions carried in this Part

Id	The two (or more) figures	Where in this Part
D11 T4	GE: CES segment profit \$8,861M (26.6%) against GE Aerospace operating profit \$9.1B; reconciling items not returned	§86.2, Exhibits 11.4, 11.12
D11 T3	Independent-shop margins 2024 → 2025: StandardAero ES 13.2% → 13.2% (Segment Adj. EBITDA); MTU 8.7% → 8.0% (adj. EBIT); Lufthansa Technik 8.5% → 7.5% (adj. EBIT); bases and mixes differ	§86.3, Exhibit 11.2; worked example 11.1
R-033 / D11 T1	AerCap engines: "over 1,200" incl. SES (6-K, 30 Sep 2025) against 470 owned (20-F, 31 Dec 2025)	§87.2, Exhibit 11.6; §89.1; Exhibit 11.16
D11 T2	Engine-leasing comparable: Willis (363 owned, whole company) against AerCap's engine book (470 owned, 9% of AerCap)	§87.2; Exhibit 11.15
R-032	Willis: 86.0% utilisation and \$2,700.4M equipment at 30 Sep 2025 against 84.9% average and \$2,801.7M at 31 Dec 2025	§87.1, Exhibit 11.5; Exhibit 11.15
R-030	Lease rate factor: 0.70% (D3 mixed bases); 0.87–1.1% (FTAI, D6); about 1.0% (Willis, derived); 0.9% assumed (D7); 0.75% (Air Lease, this Part); 0.78%/1.0% (FTAI FY2025, this Part)	Worked example 11.2; Exhibit 11.16
R-024	HEICO PMAs a year: "300–500" (D4's reading) against "400–550" (D11's reading), one filing	§86.4, §87.4, Exhibits 11.8, 11.14

Id	The two (or more) figures	Where in this Part
R-023	PMA market size: \$11.8B (IMARC, 2025); \$14B by 2026 (GIA); \$2.95B → \$6.17B engine PMA plus DER (GII listing); share "capped at ~21%" (FlightGlobal)	\$86.4
R-021	USM market size: >\$4B/yr (Safe Fly); \$5.87B (TBRC); \$7.7B (Market.us); \$8.55B (Fortune Business Insights), 2025	\$86.5
R-022	USM price: 60–80% of new, up to 85%, above list when OEM constrained (Aviation Week) against 70–75% of list, trader to airline (Aircraft Commerce)	\$86.5
R-018	OEM escalation: spares ~10%/yr, double-digit Nov 2022, high-single-digit Aug 2023, inflation +3–4 points (Aviation Week) against LLP set ~7%/yr compounded and -7B parts +20–30% in 2023 (D1; Magnetic)	\$86.2
R-014	Cost shares: material 60–70% / labour 20–30% / repairs 10–20% (Air Cargo Week) against LLPs 40–60% of parts / HPT (high-pressure turbine) blades and vanes 15–25% / labour 15–20% (Safe Fly)	\$86.1, Exhibit 11.1
R-001	CFM56 installed base: ~23,000 (CFM); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025)	\$86.2, Exhibit 11.4
R-043 / B-1	FTAI segment revenue \$1,936.2M = \$1,600.5M + \$335.8M; explicit segment-note mapping of MRE not retrieved	\$86.7, Exhibit 11.3; Exhibit 11.15
R-045	FTAI cost of sales FY2025: \$1,240,368K (segment expenses table, per the notes) against \$1,349,719K (consolidated line); 35.9% segment gross margin derived with scope caveat	\$86.7, Exhibit 11.3; §89.3
R-047	Profit elimination on MRE sales: \$16.6M in 1H 2026 implies about \$87M of gross profit if proportional to 19%, or \$16.6M if fully deferred	\$86.7
R-052	FTAI FY2024 lease income: \$234,411K or \$255,338K (two readings of the FY2025 10-K)	Exhibit 11.3
R-042	FTAI share: ~5% of a \$22B market (release, 2024); 25% of CFM56 shop visits (call goal); ~17% of a 2,350-visit year at 1,200 modules (derived)	\$89.2
R-036 / R-037	2026 module target 1,050 (release, Feb 2026) against 1,200 (call coverage, Jul 2026); capacity 1,800 (release) against 3,000 (call coverage)	\$89.2; Exhibit 11.16
R-031	Lease rates: Air Lease 10-K "increase ... expected to remain strong" against IBA's 12-year 737-800 rents –11% y/y (Jul 2025)	\$87.3, Exhibit 11.7
R-065	Headcount: 985 (FTAI, mixed definition) against "approximately 7,800" (StandardAero's figure, once misattributed)	\$87.6, §89.6, Exhibit 11.16

Id	The two (or more) figures	Where in this Part
D11 T5 / B-8	AerSale TechOps FY2025: \$123.7M stated (release) and derived (\$335.3M – \$211.6M); agree	§87.5, Exhibit 11.9
D11 T6	StandardAero: sum of segments \$6,062.5M against consolidated revenue not returned; 1H 2026 "\$3.2B" consistent but unverified	§87.6, Exhibit 11.10; Exhibit 11.15
D11 T7	Fiscal years: December (most), October (HEICO), May (AAR)	§88, Exhibit 11.15; worked example 11.3
R-019 / R-020	GE turnaround +10% and material input +40% (10-K) against trade-press 90–120 days and Oliver Wyman's labour survey	§87.8, Exhibit 11.12

Unknowns carried in this Part

- **U-D11-01 (D11) — U1 — Willis Lease lease rate factor and CFM56/LEAP portfolio split.** Searched the FY2025 10-K [S1]. Not returned. Resolve: read the 10-K's "Lease Portfolio" and "Market" sections directly.
- **U-D11-02 (D11) — U2 — Willis Lease Q2 2026 10-Q figures.** The Q2 2026 earnings-call 8-K exhibit was located by URL [S3b] but no figures came back; the 10-Q itself was not located. Resolve: search "wlfc-20260630".
- **U-D11-03 (D11) — U3 — Willis Lease maintenance services and spare-parts revenue lines for FY2025.** The 10-K's revenue table was not returned beyond lease rent, the combined core line and gain on sale. Resolve: read the consolidated statement of income in [S1].
- **U-D11-04 (D11) — U4 — HEICO Q3 FY2026 10-Q (quarter ended 31 Jul 2026).** Not located; only the Q1 FY2026 release URL [S7b]. Resolve: search "hei-20260731".
- **U-D11-05 (D11) — U5 — HEICO consolidated FY2025 net sales and operating income; PMA cumulative count; stated discount to OEM; acceptance statements.** Searched [S7, S7a, S8]. Only FSG figures and the 400–550 PMA rate were returned. Resolve: read Item 1 and Item 7 of [S7].
- **U-D11-06 (D11) — U6 — AerCap Materials revenue and AerCap engine-leasing revenue.** Searched the 20-F [S4]. Not returned. Resolve: read the 20-F's business description and the segment or revenue note.
- **U-D11-07 (D11) — U7 — AerCap FY2025 net income drivers (\$3.8B versus \$2.1B) and any stated lease yield or net spread.** Not returned. Resolve: read the Q4 2025 earnings release [S4a] in full.

- **U-D11-08 (D11) — U8 — AerCap 737 MAX share of NBV and total narrowbody share.** The five NBV lines returned sum to 80%; the remainder was not itemised. Resolve: read the fleet table in [S4].
- **U-D11-09 (D11) — U9 — Air Lease acquiring consortium.** The 10-K text returned names only the surviving entity, Sumisho Air Lease Corporation, and the agreement date [S6]. Resolve: read the 1 Sep 2025 merger 8-K.
- **U-D11-10 (D11) — U10 — Air Lease FY2025 total revenues, pre-tax margin, stated lease yield, engine-availability statements.** Not returned. Resolve: read Item 7 of [S6].
- **U-D11-11 (D11) — U11 — AerSale FY2025 net income; Q2 2026 10-Q figures.** The Q2 2026 10-Q was located by URL [S12b]. Resolve: read [S12a] and [S12b].
- **U-D11-12 (D11) — U12 — AerSale full-year feedstock purchases and any statement on retirement scarcity.** Only the Q4 figure (\$15.4M) was returned. Resolve: read the AMS discussion and cash-flow statement in [S11].
- **U-D11-13 (D11) — U13 — StandardAero FY2025 consolidated revenue, net income and Adjusted EBITDA; revenue by platform; CFM56 shop-visit count.** Not returned. Resolve: read Item 7 and the segment note of [S10].
- **U-D11-14 (D11) — U14 — AAR segment operating income; USM sales within Parts Supply; subject of the 2026 DEFA14A [S13b]; Q1 FY2027 10-Q figures [S13a].** Not returned. Resolve: read the segment note in [S13] and the DEFA14A.
- **U-D11-15 (D11) — U15 — GE Aerospace FY2025 total revenue; Q2 2026 10-Q.** Not returned or located. Resolve: read the consolidated statement in [S14]; search "ge-20260630".
- **U-D11-16 (D11) — U16 — GE CFM56 shop-visit count, spare-parts price increases, and any "material solutions" or used-serviceable-material programme wording.** Not returned from [S14]. Resolve: read the CES segment discussion in [S14] and the Q4 2025 and Q2 2026 earnings materials.
- **U-D11-17 (D11) — U17 — Safran USD spare-parts growth; MTU CFM56 and V2500 share of MRO; Lufthansa Technik shop-visit count and CFM56/LEAP capacity.** Not returned from [S16]–[S19a]. Resolve: read the Safran URD Propulsion section, MTU's annual report segment note, and LHT's annual report.
- **U-D11-18 (D11) — U18 — Cost of debt and funding terms for each lessor.** Not searched for within the budget. Resolve: D6 and D10 cover FTAI's; peers' are in their debt notes.
- **U-D6-05 (D6; cluster K-14 with U-D11-01) — U5 — Lease-rate factors.** No stated LRF for FTAI or WLFC; WLFC engine count, average lease term and FY2025 totals. Resolve: WLFC FY2025 10-K ([S13]) Item 1 and Item 7.

- **U-D3-12** (D3; cluster K-15 with U-D11-03) — **U12** — Willis Lease portfolio value, number of engines, and Willis Aero spare-parts revenue for 2025.
- **U-D4-02** (D4; cluster K-05 with U-D11-05) — **U2** — **PMA price discount for the Chromalloy HPT blade and vane** (and HEICO's typical discount). Resolve with Chromalloy/FTAI pricing disclosure or MRO price catalogues (the SR Technics 2026 CFM56-7B catalogue may list PMA options [71]).
- **U-D4-09** (D4; cluster K-05 with U-D11-05) — **U9** — **HEICO portfolio split** by part category (consumables vs rotating vs hot-section) and HEICO's PMA discount.
- **U-D6-14** (D6) — **U14** — **Current engine-lessor scale**. Portfolio size and value for GE Engine Leasing, SMBC Aero Engine Lease, ELF, AerCap Engines, RRPf, MTU Maintenance Lease Services. Resolve: company websites and parent filings (D11 for the listed ones).
- **U-D7-16** (D7) — **U16** — **Comparables not reached**. Castlake, Apollo/Merx, AerCap fee structures; Blackbird and Thunderbolt fee rates.
- **U-D2-04** (D2) — **U4. CFM56 share of GE CES services revenue**. Not disclosed in the 10-K. GE has historically described the CFM56 as the largest services contributor; a number would need an investor presentation or call transcript (the 22 Jan 2026 webcast transcript URL was returned, not read).
- **U-D10-02** (D10) — **U-2** — **Cash flow detail**. Acquisition of leasing equipment, acquisition of property, plant and equipment, proceeds from asset sales, working-capital changes (inventory), and whether the company defines free cash flow. Resolve with the cash flow statement in [S1] and [S2].

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PART XII

Reference

This Part is the primer's back matter: a glossary of every term the registry assigns, the chain map with the Parts that carry each box, the margin pool in one table, an index of every company and institution named, the full register of what the research could not find, and the consolidated list of sources. A reader who has a term, a company or a figure in mind can start here and be sent to the section that treats it.

HOW THIS PART IS BUILT

Part XII contains almost no new prose. Its six sections are compiled from files that already exist: the term registry (research/term-registry.md), the chain map drawn for the orchestrator, the margin exhibits of Part XI §86, the eleven Part files themselves, and section D of the reconciliation (research/reconciliation.md). Small scripts did the compiling, so that each entry can be traced to the file it came from and the compilation can be re-run when a Part changes. Where this Part states a figure, it is a figure another Part already states, with that Part named; where it states a definition, it is the definition the owner Part boxed. Nothing here is ranked and nothing is a view. Cross-references use the outline's section numbers, so "Part VII §46" means section 46 of the primer, which sits in Part VII.

90. Glossary

The glossary lists every term in section 1 of the term registry, 302 entries in all, in alphabetical order. The registry sorted terms under the Part that owns them; this section sorts them by name and gives the owner at the end of each entry. Each entry has the term, its acronym or alias where the registry records one, a definition of one or two sentences, and an owner line of the form "Owner: Part N §M", which names the section where the owner Part boxes the term. Where the owner Part did not box the term but another Part did (because that Part used it first), the owner line names both: "Owner: Part VIII; boxed in Part II §17". Where no Part boxed the term, the owner Part alone is given.

The definitions are pared, not rewritten. For the terms a Part boxed, the definition is the first one or two sentences of that Part's box, with citations and cross-reference sentences removed. For the dozen terms no Part boxed (company and timeline entries, aliases, and a few labels), the definition is the registry's own, tightened to two sentences. Figures that appear in a definition are the owner Part's figures and carry that Part's sources; the owner section should be read for the basis and date of any number quoted here. Twenty-four terms are defined differently by two or more dossiers; the registry's section 3 quotes both definitions and names the form the primer uses, and the owner Parts flag the difference where it matters (for example "core", which means the hot module in Part I and the returned unserviceable unit in Part VII §47). The Spine's own vocabulary is the registry's section 4, forty grouped entries, and every one of them appears below under its owner.

2025 Partnership (2025 SPV; FTAI SCI I; inaugural Strategic Capital vehicle)

The first SCI vehicle, launched 30 December 2024: a closed-end partnership whose asset-owning special-purpose vehicles buy on-lease mid-life 737NG and A320ceo aircraft. Owner: Part VIII §61.

2026 SPV (SCI II; second investment vehicle)

The second SCI vehicle. Its \$2.0B warehouse facility from 13 lenders, with a \$1.0B accordion to \$3.0B, closed on 14 August 2026; it buys on-lease mid-life 737NG and A320ceo aircraft from August 2026, and its first disclosed deal is 17 737-700s bought on a sale-leaseback with WestJet (§67). Owner: Part VIII §62.

25% share goal

FTAI's stated goal of 25% of CFM56 shop visits, which the 3,000-module capacity is said to support alongside 100 Mod-1 units a year. The denominator (the industry's annual shop-visit count, or a module equivalent of it) and the target year were not disclosed in anything the research retrieved. Owner: Part VII §49.

737NG / A320ceo (737 Next Generation; A320 current engine option)

The 737NG (Next Generation) is the Boeing 737-600, -700, -800 and -900ER, built from about 1997 to 2019 and powered only by the CFM56-7B. The A320ceo (current engine option) is the Airbus A318, A319, A320 and A321 family built from about 1988 to 2020, powered by either the CFM56-5B or the V2500 (the alternative engine from the IAE consortium, §10). Owner: Part I §2.

Accessory gearbox (AGB; accessory drive)

The accessory gearbox is a train of gears driven off the high-pressure spool by a radial shaft running out through the fan frame. It turns the fuel pump, the oil pumps, the hydraulic pump and the aircraft's electrical generator. Owner: Part I §3.

Accordion

A pre-agreed option in a credit agreement to enlarge the facility, exercisable if lenders agree to fund the increase; the borrower is not entitled to the extra amount until they do. The 2026 warehouse is \$2.0B of committed capacity with a \$1.0B accordion, so up to \$3.0B. Owner: Part VIII §62.

Adjusted EBIT

Adjusted EBIT is operating profit after depreciation and amortisation, before purchase-price-allocation effects and other adjustments each company defines. MTU Aero Engines reports its commercial maintenance business on this basis (€478M on €6.0B in 2025, 8.0%), and Lufthansa Technik reports its whole company on it (€603M on €8.049B, 7.5%). Owner: Part XI §86.

Adjusted EBITDA (consolidated; the definition) (EBITDA)

EBITDA is earnings before interest, taxes, depreciation and amortization. Adjusted EBITDA is FTAI's company-defined variant. Owner: Part X §80.

Advance rate / borrowing base

The advance rate is the percentage of each asset's value that the banks will lend against it under a warehouse facility. The borrowing base is the total credit available at any moment, recomputed as assets are added, sold or revalued: the sum over the assets of value times advance rate. Owner: Part II §17.

Aeroderivative

An aeroderivative is an aircraft engine adapted to drive a shaft for stationary use. The fan and bypass duct are removed and extra compressor stages replace the fan; the exhaust nozzle is replaced by a power turbine, or the engine's own low-pressure turbine is used; a gearbox and a generator are fitted; and the kerosene fuel system is changed for gas. Owner: Part IX §68.

Aerospace products revenue (income-statement line)

The consolidated income-statement line for the segment's sales of modules, engines and parts to customers other than the Strategic Capital vehicles: \$454,970 thousand in 2023, \$1,079,821 thousand in 2024 and \$1,600,456 thousand in 2025. Owner: Part VII §47.

Aerospace Products segment (AP)

FTAI's reportable segment that develops, repairs and sells CFM56 and V2500 engines, modules and parts through the Module Factory; FY2025 segment revenue \$1,936.2M and segment Adjusted EBITDA (FTAI's adjusted earnings measure) \$671.3M; Adjusted EBITDA is earnings before interest, taxes, depreciation and amortisation as the company adjusts them, defined in Part X §80. Owner: Part VII; boxed in Part I §11.

Aftermarket demand

Aftermarket demand for an engine type is the sum of the work and assets needed to keep that type flying: shop visits, modules, spare engines, used serviceable material and leases of engines and aircraft. It is generated by the installed base multiplied by how hard that base flies, net of the engines that leave. Owner: Part III §18.

Air permit

An air permit is the authorisation, issued under the US Clean Air Act by the state or county agency that implements it, to construct and operate a source of air emissions. A stationary source needs a construction and operating permit before it runs, and the permit sets the emission limits and controls the source must meet. Owner: Part IX §73.

Aircraft on ground (AOG)

An aircraft on ground is one that cannot fly for want of a serviceable part or engine. Short-term engine leases exist to cover AOG events and planned shop visits; a lessor that can deliver an engine within days is selling the airline the difference between a grounded aircraft and a flying one. Owner: Part II §14.

Airworthiness Directive / Service Bulletin (AD / SB)

A Service Bulletin is a manufacturer's instruction to modify or inspect; an Airworthiness Directive is a regulator's order making a fix or inspection mandatory. Compliance records for both must travel with life-limited and hard-time parts. Owner: Part V; boxed in Part I §3.

Airworthiness Limitations Section / Instructions for Continued Airworthiness (ALS / ICA)

The Instructions for Continued Airworthiness are the manufacturer's approved maintenance instructions for a type-certificated product, attached to the type certificate: the Engine Shop Manual and the component maintenance manuals (Part I §9) are part of them. Owner: Part VI §38.

Anchor investor

An early, large limited partner whose commitment seeds a fund-raise and signals to later investors that the terms have been negotiated by someone with scale. The February 2026 results release said fundraising for the second vehicle had launched "with anchor investor commitments"; the anchors' identities and amounts were not disclosed. Owner: Part VIII §62.

Anticipated repayment date / legal final maturity (ARD)

The anticipated repayment date is the date after which a securitization sweeps all excess cash to pay down principal faster and steps up the coupon on the notes, so that the deal is designed to be refinanced or called at that date. The legal final maturity is the much later last date by which the notes must be repaid. Owner: Part II §17.

Appraiser (IBA; mba Aviation; Ascend by Cirium; AVITAS)

An appraiser is an independent valuer that publishes base and market values and lease rates for aircraft and engines, at half-life unless stated; the four this primer cites are IBA, mba Aviation, Ascend by Cirium and AVITAS (a US appraisal firm). The market trades on their figures, and securitisation rating agencies test deals against them. Owner: Part II; boxed in Part I §8.

ASC 606 / noncash consideration

ASC 606, Topic 606 of the US Accounting Standards Codification (ASC), is the US revenue-recognition standard for contracts with customers. Under it, noncash consideration received from a customer is measured at fair value and included in the transaction price. Owner: Part VII §47.

ASC 610-20

Topic 610-20 of the US Accounting Standards Codification (ASC), the standard for gains and losses on the derecognition of nonfinancial assets that are not part of a company's ordinary revenue-generating activities. FTAI books aircraft sales to the 2025 Partnership under it, describing them as "non-recurring in nature and not considered part of the Company's ordinary activities". Owner: Part VIII §64.

Asset-backed securitization (ABS)

An asset-backed securitization is a transaction in which a lessor sells a portfolio of leased aircraft or engines to a bankruptcy-remote SPV, which issues notes to investors secured on the

assets and on their lease cash flows. The notes are divided into tranches paid in order of seniority. Owner: Part II §17.

Asset-level debt (Non-recourse debt)

Borrowing by the asset-owning SPVs, secured on the aircraft and their leases, with no recourse to the sponsor. If the aircraft and their rent do not cover the loan, the lenders' claim stops at the SPV; FTAI's own balance sheet is not exposed. Owner: Part VIII §61.

Asset-light model

Management's term for owning fewer leased assets directly and earning instead servicing fees, maintenance work and equity income from vehicles that own them. The Strategic Capital releases say the initiative lets FTAI "maintain an asset-light business model", and the July 2026 cut to Aviation Leasing guidance was explained as "reflecting a continued shift to an asset-light business model". Owner: Part VIII §60.

Audit Committee / Audit Committee review

The board committee of independent directors that oversees financial reporting and the external auditor. On 18 January 2025 FTAI's Audit Committee "determined to begin a review, which included engaging independent advisors, in response to the assertions alleged in the January 15, 2025 report issued by Muddy Waters Research". Owner: Part VII §54.

Auxiliary power unit (APU)

The auxiliary power unit is the small gas turbine in the aircraft's tail that supplies electrical power and compressed air on the ground and in some flight phases. It is one of the highest-value airframe rotables in a part-out and one of the components for which lessors collect maintenance reserves (Part II §13). Owner: Part V §32.

Average age

The average age of a fleet is the mean number of years since delivery of its aircraft. Maintenance need rises with age in steps rather than smoothly: the first shop visit, the first life-limited-part replacement, the first heavy airframe check. Owner: Part III §18.

Aviation Leasing segment (AL segment)

FTAI's reportable segment that owns, leases and sells aircraft and engines, directly and through an equity-method investment; 290 assets at 31 Dec 2025 and segment Adjusted EBITDA \$608.9M for FY2025. Owner: Part VIII; boxed in Part I §11.

Backlog

Backlog is the count of firm orders not yet delivered. Airbus's stood at a record 8,754 aircraft at the end of December 2025, more than ten years of production at 2026 delivery rates; the 737 MAX backlog exceeds 4,800 aircraft. Owner: Part III §21.

Back-to-birth traceability (BTB)

Back-to-birth traceability is a continuous record, for each life-limited part, of every cycle it has flown on every engine since manufacture, with no gaps. A gap means the part's remaining life cannot be proven and it is treated as unusable. Owner: Part I §9.

Balance of plant / packaging

The balance of plant is everything around the turbine that a stationary power unit needs: the enclosure, the trailer or skid, the generator and gearbox, the air-inlet filter house, the exhaust and silencing, the gas fuel skid, lubrication and cooling, fire suppression and the control system. Owner: Part IX §68.

Base value

Base value is an appraiser's opinion of an asset's underlying economic value in an open, unrestricted, stable market with balanced supply and demand. Market value is the appraiser's opinion of the most likely trading price in the actual current market between willing, informed parties; it sits above base value in a tight market and below it in a soft one. Owner: Part II; boxed in Part I §8.

Basic / diluted EPS; RSU / PSU

Basic earnings per share divides net income attributable to ordinary shareholders by the weighted-average number of ordinary shares outstanding in the period. Diluted earnings per share adds the shares that would be issued if outstanding restricted stock units (RSUs, which vest with time) and performance stock units (PSUs, which vest on performance conditions) were settled. Owner: Part X §83.

Behind-the-meter (BTM)

Behind-the-meter generation is electricity generated on the customer's side of the utility meter, at the customer's site, so that it never passes through the grid. Enverus estimates that about 40% of installed data-center capacity is behind the meter, that this adds about 1.3 Bcf/d of incremental gas demand by 2030, and that half of it sits in ERCOT and PJM. Owner: Part IX §71.

Beyond economic repair (BER)

A removed part is beyond economic repair when it is outside limits and not worth repairing: the repair would cost more than the part would fetch, or no approved repair exists. It is scrapped. Owner: Part V §31.

Blackbird / Thunderbolt (Air Lease)

Air Lease Corporation's two routes to third-party capital. Blackbird Capital I and II are investment funds whose aircraft Air Lease manages and in which it holds non-controlling interests of 9.5% of the equity of each. Owner: Part VIII §65.

Booster (Low-pressure compressor (LPC))

The booster is a small three-stage compressor mounted directly behind the fan on the low-pressure shaft. It gives the air bound for the core a first squeeze before the main compressor takes over. Owner: Part I §3.

Borescope / boroblend

A borescope inspection looks inside the engine with a camera passed through ports in the casing, on wing or in a shop. A boroblend repair grinds out small blade damage found that way, through the same ports, without disassembly. Owner: Part I §6.

Bridging power

Bridging power is temporary on-site generation used until a grid interconnection is available. ProEnergy's customers describe their turbine plants as "bridging power for five to seven years, which is when they expect to have grid interconnection". Owner: Part IX §73.

Build standard (Configuration)

The build standard is the set of service-bulletin states a module has been built to. The manufacturer issues service bulletins that change hardware over the years, and two modules at different standards may not mate, or may not be approved to run together. Owner: Part I §3.

Bypass / turbofan / high-bypass turbofan

A turbofan is a jet engine in which a large front fan pushes most of the incoming air around the hot core rather than through it. The air that goes around is the bypass stream. Owner: Part I §1.

Capacity factor

The capacity factor is a plant's actual output over a period divided by what it would have produced running at full load for the whole period. A unit that ran flat out all year would have a factor of 100%; the fuel-burn example below uses 90% as an assumption for a baseload data-center unit. Owner: Part IX §68.

Capacity (shop) (Inductions per year)

A shop's capacity is the number of engine inductions a year it can take. Owner: Part IV §25.

Carlyle Aviation Partners (fee comparable)

The aviation arm of The Carlyle Group, which runs closed-end funds that earn carried interest and also sponsors aircraft securitizations that it services. Carlyle's FY2025 10-K states that servicing fees on its aviation securitizations are "generally in the range of 2% of rents, incentive fees up to 5% of rents in the aggregate, and 3% of sales proceeds". Owner: Part VIII §65.

Catalogue list price (CLP; list price; catalog price)

The catalogue list price is the manufacturer's published price for every new spare part, repriced at least annually. A shop passes it through at list plus a handling margin, and used and alternative parts are priced as discounts to it. Owner: Part IV; boxed in Part I §4.

Cayman Islands exempted company / redomiciliation (Redomicile)

A Cayman Islands exempted company is a company incorporated in the Cayman Islands that conducts its business mainly outside the islands; it is a common holding-company form for aircraft lessors. Redomiciliation is the move of a company's legal home from one jurisdiction to another. Owner: Part X §77.

Center of Excellence (CoE)

Center of Excellence is StandardAero's term for a shop dedicated to a single engine platform. Owner: Part XI §87.

CFM International (CFM)

CFM International is the joint venture of GE Aerospace and Safran Aircraft Engines (formerly Snecma) that is the OEM and type-certificate holder of the CFM56 and of its successor, the LEAP (CFM's new-generation engine, treated in §10). Owner: Part I §2.

CFM materials agreement

A multi-year agreement between FTAI and CFM International announced on 22 January 2026 under which, in the release's words, FTAI "secures OEM replacement part supply, thrust performance upgrades and component repair" for CFM56 engines. Owner: Part VII §52.

CFM56 family (-3, -5A, -5B, -5C, -7B) (CFM56-7B; CFM56-5B)

The CFM56 is a two-shaft, high-bypass turbofan of which more than 35,000 have been delivered since entry into service in 1982, with roughly 23,000 to 24,000 in service depending on the source (§2.2). The -7B is the sole engine of the 737NG, with more than 15,000 built and the last engine for a commercial 737NG delivered in 2019. Owner: Part I §2.

Channel stuffing

Recognising sales to intermediaries or related parties ahead of end demand, so that reported revenue runs ahead of what final customers have bought. It appears among the allegations as third parties summarised the report; the counterparty the report named was not retrieved.

Owner: Part VII §54.

Chromalloy

Chromalloy is a Palm Beach Gardens, Florida, company that repairs and coats turbine-engine parts, makes investment castings (a casting method in which a ceramic mould is formed around a wax pattern of the part), and designs and manufactures PMA parts. Owner: Part VI §41.

Classified board

A classified board is one whose directors serve staggered terms, so that only a portion of the board stands for election in any one year. FTAI's seven directors serve three-year terms in three classes, with two or three terms expiring each year. Owner: Part X §78.

Closed-end vehicle

A fund that raises a fixed amount of equity commitments up to a hard cap, invests them during a deployment period, then stops buying and enters a harvest phase in which it collects rent, distributes cash to its investors and sells assets until it winds up. Owner: Part VIII §61.

Combustor (Combustion chamber)

The combustor is the chamber between the high-pressure compressor and the high-pressure turbine in which fuel is sprayed into the compressed air and burned. It is part of the core module. Owner: Part I §3.

Commercial Engines & Services (CES)

Commercial Engines & Services is the GE Aerospace segment that sells new commercial engines and spare parts and performs shop visits, in GE's own shops and under long-term service agreements. In 2025 it reported revenue of \$33,314 million, of which \$25,010 million (75.1%) was services, and segment profit of \$8,861 million, a 26.6% margin. Owner: Part IV §25.

Commissioning

Commissioning is the on-site testing and hand-over of installed equipment: the unit is connected to gas and to the customer's switchgear, run, and shown to meet its specification before the customer accepts it. It is the last milestone payment under the J&F order. Owner: Part IX §70.

Comparable metrics

Comparable metrics are the figures from a peer's filing that can be set beside a figure in FTAI's filings because both measure the same physical or financial quantity: owned engines, utilisation on a stated definition, maintenance-reserve revenue, gain on sale of leased equipment, a segment margin on a stated basis, approvals of alternative parts per year, gross margin on used material, and the services share of a manufacturer's revenue. Owner: Part XI §86.

Component Maintenance Manual (CMM)

The Component Maintenance Manual is the manufacturer's document that sets inspection and repair limits for an individual component; the Engine Shop Manual is the engine-level equivalent. Owner: Part V; boxed in Part I §9.

Conduct Policies (CFM–IATA, 2018)

The Conduct Policies are the commitments CFM International published in 2018 to settle a complaint the International Air Transport Association (IATA) had brought to the European Commission about CFM's aftermarket conduct. Owner: Part VI §42.

Consignment

Consignment is teardown and sale of parts on the owner's behalf for a fee or a commission, with title to the parts retained by the owner, who may be an airline, a lessor, a bank or an insurer. It is the alternative to buying feedstock outright and bearing the inventory risk. Owner: Part V §32.

Core (Gas generator)

The core is the hot, fast, expensive middle of the engine: the high-pressure compressor, the combustor and the high-pressure turbine, treated as one module. The industry also calls it the gas generator. Owner: Part I §1.

Cost build (shop visit)

The cost of a shop visit built up as the sum of labour, new OEM parts, component repairs, life-limited-part replacement (new or used) and the test-cell run, plus freight, consumables and the shop's margin. Material is about 60 to 70% of the total; Part I §7 works the example.

Owner: Part I.

Cost of sales (segment versus consolidated)

Two cost-of-sales figures for 2025 exist in the FY2025 10-K. One is \$1,240,368 thousand, which the orchestrator's later reading identified as the Aerospace Products segment's cost of sales in the segment expenses table; one dossier had returned it against a segment query but could not confirm its scope. Owner: Part VII §53.

Critical component / critical part

Under the FAA–EASA bilateral's Technical Implementation Procedures, a critical component is "a part identified as critical by the design approval holder during the product certification process or otherwise by the Authority for the State of Design. Owner: Part VI §38.

Customer concentration

Customer concentration is the share of revenue that comes from a company's largest customers, which US filings must disclose for any customer above 10%. FTAI's Q3 2025 quarterly report says two customers in Aerospace Products accounted for 16% and 11% of revenue in the quarter. Owner: Part X §78.

Cycles remaining (CR)

Cycles remaining is a life-limited part's certified limit minus the cycles it has consumed; used-part packages are advertised on it ("6235 CR"). Cycles since new is the number of cycles an engine or part has flown since manufacture. Owner: Part I §4.

Cycles since new (CSN)

Cycles remaining is a life-limited part's certified limit minus the cycles it has consumed; used-part packages are advertised on it ("6235 CR"). Cycles since new is the number of cycles an engine or part has flown since manufacture. Owner: Part I §4.

Deployment / fully committed / harvest phase

Deployment is the buying phase, during which commitments are called and aircraft acquired. "Fully committed" means every dollar of equity has been allocated to signed acquisitions, even if some closings are still pending; it is a different state from "fully deployed", which

means closed and funded, and the releases use both words at different dates (§62.1). Owner: Part VIII §61.

DER repair

A DER repair is a major repair whose data a Designated Engineering Representative approved on FAA Form 8110-3, rather than being taken from the manufacturer's shop manual. The approved data may be one-time, covering a specific serial-numbered part, or multiple-use, repeatable on any eligible part; a multiple-use approval is what turns a repair into a product a shop can sell. Owner: Part VI §40.

Designated Engineering Representative (DER (company DER / consultant DER))

A Designated Engineering Representative is an individual engineer appointed by the FAA under 14 CFR Part 183 to examine engineering data and make findings of compliance on the FAA's behalf within a delegated technical area, such as structures, powerplant, engines or systems. Owner: Part VI §40.

Discontinued operations

Discontinued operations is the accounting presentation under which a business that a company has disposed of, or is about to dispose of, is shown as a single net line below the results of the continuing business, with prior periods restated the same way. Owner: Part X §77.

Disk / spool

A disk is the heavy wheel to which a row of blades attaches. A spool is several disks machined as a single piece. Owner: Part I §4.

Distribution (new parts)

Distribution is reselling new OEM parts under a distribution agreement: the distributor holds inventory and earns the margin between the manufacturer's transfer price and its sale price. Because the manufacturer sets the cost of goods, a distribution margin is structurally thinner than a PMA or used-material margin. Owner: Part XI §86.

Dividend (ordinary) / share repurchase program

An ordinary dividend is the cash per share the board declares each quarter on the ordinary shares; it is discretionary and ranks behind the cumulative preferred dividends. A share repurchase program is a board authorisation to buy back up to a stated dollar amount of shares over time at management's discretion; it is not a commitment to buy any. Owner: Part X §83.

Dual release

A dual release is one tag signed under both FAA and EASA authority, so that a part can move between US- and EU-registered aircraft without re-certification. The research reached industry descriptions only. Owner: Part I §9.

Durability Improvement Package / LEAP durability kit (DIP; HPT durability kit)

The durability kit is CFM's package of hardware changes to lengthen the LEAP's time on wing: a new high-pressure turbine stage-1 blade and nozzle and a forward inner nozzle support, "designed to more than double time on wing, especially in hot and harsh environments". The LEAP-1A kit is certified and "now incorporated into all deliveries and shop visits". Owner: Part III; boxed in Part I §10.

EASA Form 1 (Form 1)

EASA Form 1 is the European agency's authorised release certificate, the equivalent of Form 8130-3, issued by an EASA-approved maintenance or production organisation. New parts made under a European production organisation approval (POA) "are typically ineligible for an 8130-3 tag and are more properly released on an EASA Form 1". Owner: Part I §9.

EGT margin (EGTM)

EGT margin is "the difference between the maximum temperature limit on the engine and the actual temperature read at the time of take-off", measured at full take-off thrust on a hot day so that engines can be compared. It shrinks as the engine wears, fastest in the first 1,000 to 2,000 cycles after a shop visit. Owner: Part I §5.

Employees

FTAI reports 985 "full-time employees and independent contractors" at 31 December 2025, a single figure that mixes two categories, and separately 494 full-time employees in Canada, about 71% of whom are covered by collective bargaining agreements. The Canadian figure counts employees only, so the two numbers are not on one basis. Owner: Part X §85.

End-of-lease compensation (EOL compensation)

End-of-lease compensation is the alternative regime, used for lessees whose credit is strong enough that the lessor does not demand monthly reserves. Owner: Part II §12.

Engine flight cycle (EFC; FC; cycle)

An engine flight cycle (EFC), or simply a flight cycle (FC) or "cycle", is one take-off and landing. An engine flight hour (EFH), or flight hour (FH), is one hour of engine operation. Owner: Part I §1.

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Engine lease: short-term / long-term

A short-term engine lease runs from a few months to about three years and covers a shop visit or an aircraft grounded for want of an engine. A long-term engine lease runs for years and provides a permanent spare. Owner: Part II; boxed in Part I §6.

Engine removal forecast

The engine removal forecast is an airline's rolling schedule of each installed engine's projected date of removal for a shop visit. Each engine's date is set by whichever of its limits is reached first: remaining exhaust-gas-temperature margin (Part I §5), remaining life-limited-part cycles (Part I §4), or a hard inspection. Owner: Part III §22.

Engine sale / outright module sale / exchange

The segment's three product transactions. An engine sale books the whole-engine price as revenue. Owner: Part VII §47.

Engine shop

An engine shop is a facility holding a regulatory repair-station approval (§9) and, in practice, the manufacturer's technical data and tooling. It disassembles engines on stands, cleans and inspects every piece, replaces or repairs what is out of limits, reassembles the engine and runs it in a test cell before release. Owner: Part IV; boxed in Part I §6.

Engine Shop Manual (ESM)

The Engine Shop Manual is the manufacturer's manual defining how the engine is disassembled, inspected, repaired and rebuilt, including the limits a part must meet to stay in service. Regulators treat it as the reference: an Australian airworthiness directive notes that the "CFM56-7B ESM contains instructions for calculating remaining life of each engine stationary part within certain time frames and thresholds". Owner: Part I §9.

Engine stand

An engine stand is the wheeled fixture that holds a roughly 2.5-tonne engine, or a module, while it is worked on, stored or transported. Owner: Part IV §27.

EngineWise

EngineWise is Pratt & Whitney's aftermarket services brand, under which IAE's network performs V2500 shop visits. Owner: Part IV; boxed in Part I §10.

Equity commitment / hard cap

An equity commitment is an investor's promise to fund capital when the vehicle calls it; capital is drawn as assets are bought, not on signing. A hard cap is the maximum amount of commitments the sponsor will accept. Owner: Part VIII §61.

Equity method (Equity-method investment; equity pick-up; equity in earnings of unconsolidated entities)

An accounting method for an investment over which the investor has significant influence but not control. Owner: Part VIII §64.

Equity-method investment in Aviation Leasing

The unnamed investment through which the Aviation Leasing segment "leases and sells" assets, in the FY2025 10-K's segment description. Its counterparty, FTAI's ownership

percentage, the assets inside it and the income it has contributed by year were not retrieved.

Owner: Part VIII §64.

Escalation (spare parts and LLPs) (LLP escalation; catalogue escalation)

Escalation is the manufacturer's annual increase in catalogue list prices. Trade press put new spare-parts price increases at around 10% a year in 2022–23 (a double-digit increase on 1 November 2022 and a high-single-digit increase in August 2023), and Safran's chief executive described a rule of pricing "above inflation by 3 to 4 points". Owner: Part IV; boxed in Part I §4.

Exchange fee / core (exchange sense)

The exchange fee is the cash price a customer pays to receive a serviceable module in return for an unserviceable one. The returned core is the unserviceable module the customer hands over, which has value to the exchanger as feedstock. Owner: Part VII §47.

Exhaust-gas temperature (EGT)

Exhaust-gas temperature is the temperature of the gas leaving the turbine, measured by probes behind the low-pressure turbine. Every engine has a certified EGT red line that it may not exceed. Owner: Part I §5.

External management / manager / management agreement / management fee / incentive allocation / Master GP

External management is a structure in which a listed company has no staff of its own. A separate firm, the manager, supplies the executives, the employees and the investment process under a management agreement, and is paid a management fee, typically a percentage of the company's equity or assets. Owner: Part X §76.

FAA Form 8110-3 (8110-3; Statement of Compliance with Airworthiness Standards)

Form 8110-3 is the document on which a Designated Engineering Representative records an approval, or a recommendation for approval, of engineering data, including the data for a repair or alteration. The designee keeps the original and sends copies to the FAA's managing office and to the owner, operator or repair station that requested the approval. Owner: Part VI §40.

FAA Form 8130-3 (8130-3; Airworthiness Approval Tag; Authorized Release Certificate)

Form 8130-3 is the US tag. For a new part it states that the part, produced under an FAA production approval, conforms to its design and is in a condition for safe operation; for a used part it is the maintenance release by an FAA-approved repair station. Owner: Part I §9.

Fan frame (Fan frame sub-module)

The fan frame is the structural front assembly of the engine, built from four main pieces: the containment case that would catch a released fan blade, the outlet guide vanes that straighten the fan's airflow, the frame itself, and the housing for the radial shaft that drives the gearbox.

Owner: Part I §3.

Feedstock

Feedstock is the trade's word for the aircraft and engines bought, or taken on consignment, to be torn down for parts. Later Parts extend it to the unserviceable module a customer returns in an exchange and to engine inventory held for module production, and say so when they do.

Owner: Part V; boxed in Part I §3.

Finance lease / sales-type lease

A finance lease transfers substantially all the economic risks and rewards of owning the asset to the lessee, so that in substance it is a loan secured on the asset rather than a rental. Owner:

Part II §12.

First run / mature run

The first run is the interval from new to the first shop visit. Mature runs are the later intervals, which are shorter because repaired hardware does not match new hardware and because operators plan them around the life-limit clock. Owner: Part I §6.

Fixed-to-floating / fixed-rate reset preferred (Reset preferred)

Two coupon designs appear. A fixed-to-floating preferred pays a fixed coupon until its first call date and then a floating rate, a reference rate plus a fixed spread, reset each quarter (Series A 8.25%, redeemed October 2024; Series B 8.00%, outstanding). Owner: Part X §82.

Fleet maturity

Fleet maturity is the distribution of an installed base by the number of shop visits each engine has had: none, one, or more. GE reported in June 2019 that 57% of CFM56 engines had not had a first shop visit and 21% had had one; Safran reported in 2024 that about 70% had seen zero or one. Owner: Part IV §26.

Fleet states: in service / parked (stored) / retired (withdrawn) (Fleet in service)

An aircraft in service is assigned to an operator and flying, or in short-term maintenance. A parked or stored aircraft is not flying but has not been formally withdrawn; an aircraft parked for more than about 90 days usually counts as stored, and a stored aircraft can return to service or go on to retirement. Owner: Part III §18.

FLIGHT DECK

FLIGHT DECK is the name GE Aerospace gives its proprietary operating system for running its factories and shops. Its FY2025 annual report credits it with increasing material input from priority suppliers by more than 40% year on year, enabling a 26% rise in services revenue, and with the turnaround improvement quoted above. Owner: Part IV §27.

Forensic accounting

Investigative accounting that tests transactions and records for misstatement, as distinct from the recurring audit of financial statements. The Audit Committee's advisors were described as "independent legal and forensic accounting advisors" in the February 2025 announcement and as "outside counsel and forensic accountants" in the FY2025 10-K. Owner: Part VII §54.

Form 10-K / 10-Q / 8-K / DEF 14A / DEFM14A / S-4 (SEC forms)

A Form 10-K is a US-listed company's audited annual report. A Form 10-Q is its unaudited quarterly report. Owner: Part X §75.

Form 20-F / Form 6-K

Form 20-F is the annual report a foreign private issuer files with the SEC in place of a Form 10-K, and Form 6-K is the form on which it furnishes interim results and other reports in place of 10-Qs and 8-Ks. AerCap, incorporated in the Netherlands and listed on the NYSE, files on these forms. Owner: Part XI §87.

Fortress / FIG LLC / FTAI Infrastructure (Former Manager; FIP)

Fortress Investment Group is the alternative asset manager that sponsored FTAI's 2015 listing. FIG LLC, its subsidiary, was FTAI's external manager until 28 May 2024 and is called the "Former Manager" in the filings since. Owner: Part X §76.

Frame (heavy-duty) turbine (F-, H-, J-class)

A frame turbine is a gas turbine designed from the start to sit on a foundation and make power, typically 50 to 600 MW per unit, heavy, and built in a factory for a delivery slot years ahead. Owner: Part IX §68.

Free power turbine (FPT)

A free power turbine is an output turbine on its own shaft, not mechanically connected to the gas generator; it is driven only by the gas stream, which the patent calls "aerodynamically coupled". Because the two shafts are independent, the gas generator can speed up and slow down with load while the output shaft holds the fixed speed a generator needs. Owner: Part IX §68.

Freighter conversion (P2F (passenger-to-freighter); conversion STC)

Freighter conversion is the structural modification of a used passenger airframe into a cargo aircraft: a main-deck cargo door, a strengthened floor, a rigid barrier between the cockpit and the hold, and a cargo loading system. It extends the airframe's revenue life by 10 to 15 years at low cycles a year. Owner: Part II §15.

FTAI–Chromalloy joint venture (Chromalloy JV; PMA JV)

FTAI's FY2025 annual report says only that the company "develops and manufactures Parts Manufacturer Approval ('PMA') parts through a joint venture", as part of a "proprietary portfolio of products that enables it to provide cost savings and flexibility to airline, lessor, and maintenance, repair, and operations customers"; the partner, the ownership split, the accounting treatment, the revenue and the savings are not in the filing text retrieved. Owner: Part VI §43.

Full-life value (Full life)

Full-life value is the value with every maintenance item fresh and every life-limited part at zero cycles: "full-life engines are brand new or in recently restored condition". It equals the half-life value plus half the cost of a full restoration and a full set of life-limited parts. Owner:

Part I §8.

Gain on sale of leased equipment / to the 2025 Partnership

The sale price of an asset sold from the leasing portfolio minus its net book value. FTAI recorded \$46.4M of such gains on sales to the 2025 Partnership in FY2025; the three quarterly gains disclosed for 2025 sum to \$50.1M for nine months, which is a tension (\$60.2).

Owner: Part VIII §60.

Gas turbine / gas generator

A gas turbine is a rotating engine that compresses air, burns fuel in the compressed air and expands the hot gas through turbine stages. The compressor, the combustor and the turbine that drives the compressor together form the gas generator: the assembly whose only product is hot, pressurised gas. Owner: Part IX §68.

GE segment profit

GE segment profit is GE Aerospace's own segment-level profit measure, which excludes certain corporate and non-operating items. For Commercial Engines & Services it was \$8,861M on \$33,314M in 2025, a 26.6% margin. Owner: Part XI §86.

Gearbox (genset)

A genset gearbox is the speed reducer between the turbine's output shaft and the generator. A generator synchronised to a 60 Hz grid must turn at 3,600 rpm with two poles or 1,800 rpm with four, and aeroderivative output shafts commonly spin faster, so "an external gearbox is typically required to obtain the proper input speed to a generator". Owner: Part IX §68.

Geared turbofan (GTF; PW1100G (A320neo); PW1500G, PW1900G)

The geared turbofan is Pratt & Whitney's new-generation engine. A reduction gearbox between the fan and the low-pressure turbine lets each turn at its own best speed, which the CFM56's direct-drive design does not allow. Owner: Part III §20.

Generator set (Genset)

A generator set is the complete unit that turns fuel into electricity: the prime mover (a gas turbine or a piston engine), the output turbine where there is one, a gearbox if the shaft speed must be changed, the electrical generator itself, and the surrounding fuel, air-intake, exhaust, lubrication, fire-protection and control systems, assembled and tested as one product. Owner:

Part IX §68.

Glider

A glider is a completed airframe parked at the factory awaiting engines. Airbus had about 60 at the end of the first half of 2025 and 32 at the end of the third quarter, and its chief executive said the number would reach zero by year-end. Owner: Part III §21.

Global Replacement Ratio

The Global Replacement Ratio is the number of deliveries needed to replace retirements, expressed as a share of total deliveries over a period. Cirium put it at 46.1% for 2025 on 2015–2024 data: fewer than half of new deliveries were replacing a retired aircraft, and the rest were growth. Owner: Part III §19.

GMF AeroAsia / EgyptAir partnerships (Capacity partnerships (Indonesia, Egypt))

Capacity partnerships referenced on the Q2 2026 call with Garuda Indonesia's MRO arm and with EgyptAir Maintenance & Engineering, adding module capacity in Indonesia and Egypt. Their terms, capacity and economics were not retrieved. Owner: Part VII.

Green time / green-time engine / green-time lease

Green time is the remaining usable life on an engine or module before its next mandatory shop visit, set by whichever runs out first: the life-limited part with the fewest cycles remaining, the EGT margin, or a hard inspection or directive. Owner: Part II; boxed in Part I §8.

Gross leasing yield

Gross leasing yield is lease income for a period divided by the average leasing equipment at cost over the period, or at net book value (cost less accumulated depreciation). Owner: Part II §14.

Gross margin (AerSale Asset Management Solutions) (AMS gross margin)

AerSale's AMS gross margin is the segment's gross profit divided by its sales: \$74.1M on \$211.6M in 2025, 35.0% (derived in the research), against \$82.5M on \$215.5M in 2024, 38.3%. The filing attributes the fall to "lower margin on USM sales resulting from changes in the product mix" and reports the aircraft line within AMS at 33.2% against 34.8%. Owner: Part XI §86.

GTF Advantage

The GTF Advantage is Pratt & Whitney's upgraded PW1100G, certified by EASA in 2026 for entry into service later in 2026. It is the fix for the fleet going forward; the discs already in service are dealt with by the inspection programme. Owner: Part III §20.

GTF powder-metal issue (Powder metal)

Certain turbine discs in the GTF are made by a powder-metallurgy route, in which metal powder is consolidated into a disc. Powder produced over a period of years was found to be contaminated. Owner: Part III §20.

Guidance

Guidance is management's published forward target for a financial measure. FTAI gives it in Adjusted EBITDA from its business segments, by segment and in total, excluding Corporate and Other, and in modules produced by Aerospace Products. Owner: Part X §84.

Half-life (base) value (Half life; HLBV)

Half-life is an appraisal convention, not a physical midpoint. In ISTAT's words it is "a standard appraisal industry term to indicate that no value adjustment has been made for the actual maintenance status ... the assumption being that the airframe, engines (modules & LLPs), landing gear, and other major maintenance events are in half-life status. Owner: Part I §8.

Half-life trade

The half-life trade is this primer's term, not the sources', for the trade in engines, modules and life-limited parts that are part-way through their lives and priced on the life they have left: a part-worn engine leased for its remaining months or torn down now; a module valued on the remaining cycles of the life-limited parts inside it; a used disk priced at its life fraction times list price less a discount (§34). Owner: Part V §32.

Hard-time part (HT part)

A hard-time part is one with a fixed overhaul or replacement interval, as distinct from an on-condition part that is replaced when inspection finds it worn. It needs complete operating-history records and evidence of compliance with every applicable Airworthiness Directive and Service Bulletin. Owner: Part V §31.

Heat rate (Btu/kWh)

The heat rate is the fuel energy a plant needs to make one kilowatt-hour of electricity, in Btu per kWh. One kWh is 3,412 Btu of energy, so a plant's efficiency is 3,412 divided by its heat rate. Owner: Part IX §68.

HEICO Flight Support Group (HEI; FSG)

The Flight Support Group is the larger of HEICO Corporation's two operating groups and the largest independent PMA business. Owner: Part VI §41.

High-pressure compressor (HPC)

The high-pressure compressor is the nine-stage compressor on the high-pressure spool, driven by the high-pressure turbine behind it. It raises the core airflow to the pressure at which fuel can be burned efficiently. Owner: Part I §3.

High-pressure turbine (HPT)

The high-pressure turbine is the single-stage turbine immediately behind the combustor. It takes power from the hottest gas in the engine and uses it to drive the high-pressure compressor. Owner: Part I §3.

History timeline

The dated sequence of corporate events that Part X assembles: the 2015 IPO of Fortress Transportation and Infrastructure Investors, the 2022 spin-off and redomiciliation, the 2023 QuickTurn and 2024 Montréal acquisitions, the 2024 internalization, the launch of the Strategic Capital Initiative and of FTAI Power, the 2025 short report, the 2026 CFM materials agreement, the J&F order, the 2026 SPV warehouse, the \$500M buyback authorisation and the WestJet purchase of 28 September 2026. Part X §75 carries the table with each date and source. Owner: Part X.

Hospital (quick-turn) visit (Quick-turn; QTSV)

A hospital or quick-turn visit fixes one thing and returns the engine: a borescope finding, a damaged fan blade, a module exchange, an accessory change. Trade press reports that such visits "are being used as an alternative to avoid substantial financial and time investments into full-performance shop visits", at about 45 days' turnaround. Owner: Part VII; boxed in Part I §6.

Hot section

The hot section is the combustor, the high-pressure turbine blades and vanes (the vanes are also called nozzles; the stage-1 vane is the HPT nozzle), and the adjacent parts that combustion gas touches directly. It is where margin is lost and where most of it is recovered.

Owner: Part I §5.

Hurdle / preferred return

The return limited partners receive on their capital before the sponsor shares in profit beyond its own pro-rata interest; an 8% annual preferred return is a common private-fund convention, used here only as an illustration in §66. FTAI's filings do not disclose whether the 2025 Partnership has one. Owner: Part VIII §63.

Hydro-mechanical unit (HMU)

The hydro-mechanical unit is the engine's fuel-metering unit, the accessory that sets fuel flow under the engine control's command. It is a high-value engine accessory and, with CFM56 fan disks and structural engine parts, one of the most searched-for teardown items in one 2026 consultancy report. Owner: Part V §32.

Hyperscaler

A hyperscaler is a very large cloud-computing operator that builds and runs its own data centers at a scale of gigawatts. The research's sources describe the J&F customer two ways: FTAI's release of 22 July 2026 says "a leading international cloud service provider", and summaries of the second-quarter 2026 call say "a U.S. hyperscaler". Owner: Part IX §70.

Identity (with / without a licensing agreement)

Identity is the PMA basis in which the applicant shows that its design is identical to the type-certificate holder's in every respect that matters. Without a licence, the applicant usually

holds the manufacturer's drawings legitimately, for example as a former supplier; FAA Notice 8110.55 of 1996 set the policy. Owner: Part VI §37.

Independent MRO / MRO (Maintenance, Repair and Overhaul)

MRO is the industry's generic term for maintenance, repair and overhaul work and for the companies that do it. Owner: Part IV; boxed in Part I §6.

Induction / induction slot (Slot)

Induction is the acceptance of an engine into work at a shop, the point from which turnaround time is counted. An induction slot is the date a shop commits to start a given engine. Owner: Part IV §27.

Installed base

The installed base is the population of engines fitted to aircraft in the operating fleet. Published figures for the CFM56 "in service" range from about 23,000 to about 24,000 and may include spare and stored engines as well as installed ones, since none of the sources states its basis. Owner: Part III; boxed in Part I §2.

Interconnection agreement / interconnection queue

An interconnection agreement is the contract between a generator or a large customer and the utility or grid operator that governs its connection to the grid. The interconnection queue is the waiting list of projects seeking such agreements, which in the United States runs to years. Owner: Part IX §73.

Internalization

Internalization is the termination of an external management agreement, the hiring of the manager's people directly by the company, and a payment to the manager for the fee stream it gives up. Owner: Part X §76.

Inventory (Module Factory working capital)

Serviceable modules and parts awaiting exchange or sale, plus unserviceable engines, modules and parts in work: \$1,364.3M net at 31 March 2026. Owner: Part VII §53.

ISTAT (International Society of Transport Aircraft Trading)

ISTAT, the International Society of Transport Aircraft Trading, is the trade body whose definitions standardise appraisal terms such as half-life and full-life. Owner: Part I §8.

Jereh Group (Yantai Jereh Oilfield Services Group (SZSE 002353))

Jereh is a Chinese maker of oilfield equipment listed on the Shenzhen Stock Exchange (SZSE) under code 002353: drilling and oil-and-gas field engineering equipment, equipment maintenance and parts, oilfield engineering services, natural-gas compressors and gas-turbine generator sets. Its gas-turbine genset research began in 2018, and "in 2020, the first 6MW gas turbine genset in China was successfully applied in the well site". Owner: Part IX §70.

J&F Power Systems LLC (J&F)

J&F Power Systems is the entity through which Mod-1 is packaged and sold. Owner: Part IX §70.

LEAP (LEAP-1A / LEAP-1B)

The LEAP is CFM's new-generation engine: the LEAP-1A is one of two options on the Airbus A320neo family, and the LEAP-1B is the sole engine of the Boeing 737 MAX (the re-engined 737). About 5,500 were in service at the date of one trade-press piece, with about 50% under long-term service agreements against about 15% of the CFM56 fleet. Owner: Part III; boxed in Part I §10.

Lease extension / extension share

A lease extension renews an expiring operating lease with the incumbent airline instead of taking the aircraft back for transition, sale or part-out. The extension share is the proportion of scheduled lease expiries that are extended rather than returned. Owner: Part III §22.

Lease income / maintenance revenue / asset sales revenue

The Aviation Leasing segment's three income-statement lines. Lease income is rent from operating leases of aircraft and engines (\$235.2M for 2025). Owner: Part VIII §58.

Lease rate (Rent)

The lease rate is the periodic rent, quoted per month. Owner: Part II §12.

Lease rate factor (LRF)

The lease rate factor is the monthly rent divided by the asset's current market value, expressed as a percentage. An LRF of 1.0% means annual rent is 12% of value. Owner: Part II §12.

Lease transition

A lease transition is the move of an aircraft from one lessee to the next. Between the two leases the lessor pays for reconfiguration of the cabin, for the maintenance needed to meet the next lessee's delivery condition, and for the downtime while the aircraft earns no rent. Owner: Part III §19.

Lease-attached trading

Lease-attached trading is buying an aircraft from another lessor with the existing lease transferred to the buyer, so that the buyer acquires a rent stream and a return-condition obligation rather than an empty aircraft. The lease is what makes the asset financeable: lenders and securitization investors underwrite contracted rent from a known airline. Owner: Part II §12.

Lessor / lessee

The lessor is the owner that rents out an aircraft or engine; the lessee is the operator that rents it. Plain words that need no box; Part II §12 uses them from its first paragraph. Owner: Part II.

Letter of intent (LOI)

A signed agreement to transact that is non-binding, or binding only on conditions, ahead of a definitive purchase contract. At the hard-cap date the 2025 Partnership had "190 aircraft closed or under LOI", of which 101 had closed for \$1.4B and the rest were under contract for \$2.1B. Owner: Part VIII §62.

LHV / HHV (Lower / higher heating value)

A fuel's energy content can be stated on two bases. The higher heating value counts the heat recovered by condensing the water vapour in the exhaust; the lower heating value does not.

Owner: Part IX §68.

Licensed / joint-venture shop

A licensed shop is an engine shop that performs CFM56 work under a licence from the manufacturer; a joint-venture shop is one in which the manufacturer holds equity, often alongside an airline. Both operate with guaranteed access to the manufacturer's parts and technical data, and both are counted as part of the OEM network rather than as independents.

Owner: Part IV §25.

Life-expired

A life-limited part is life-expired when it has reached the cycle limit the manufacturer certified for it. It has no used-material value, however good its condition, and it must be mutilated so that it cannot re-enter the supply chain. Owner: Part V §31.

Life-limited part (LLP)

A life-limited part is a rotating or pressure-bearing part (a disk, a spool, a shaft, certain seals) whose failure the engine case cannot be designed to contain. It is certified for a fixed number of flight cycles and must be removed and scrapped at that limit regardless of its condition.

Owner: Part I §4.

Limited partner / general partner (LP / GP)

A limited partner is an investor who commits capital to a partnership, has limited liability and takes no part in managing it. A general partner manages the partnership and bears liability.

Owner: Part VIII §61.

Line-replaceable unit / quick-engine-change kit (LRU / QEC)

A line-replaceable unit is an accessory that can be changed on the flight line without opening the engine. The quick-engine-change kit is the set of mounts, plumbing and accessories that dresses a bare engine for installation on a particular aircraft. Owner: Part I §6.

LLP stack (Full LLP set; shipset (2008 usage))

Another name for the full set of life-limited parts in an engine, which Part I calls the full LLP set; its remaining life largely sets a used engine's value. A 2008 price table calls the same thing a shipset. Owner: Part I.

LLP stagger

The mismatch of remaining lives across an engine's life-limited parts: replacing only the limiting part brings the next LLP-forced removal sooner. It is the same mechanism Part I §4 treats as stub life. Owner: Part I.

Loan-to-cost

A vehicle's debt divided by its total capital (equity plus debt). It is distinct from loan-to-value, which divides debt by the appraised value of the assets (Part II §17). Owner: Part VIII §61.

Loan-to-value (LTV)

Loan-to-value is the note balance divided by the appraised value of the assets in the pool. The senior notes might be sized to 60–70% of appraised value and the senior and mezzanine notes together to 75–80%, leaving equity of 20–25%; those ranges are illustrative, because the two deals the research retrieved did not return their loan-to-value ratios (D6 U6). Owner: Part II §17.

Lockheed Martin Commercial Engine Solutions (LMCES; Montréal)

Lockheed Martin Commercial Engine Solutions (LMCES) was the Montréal engine shop that FTAI acquired from Lockheed Martin Canada, closing on 9 September 2024 for total consideration of \$170.0 million; the shop is about 500,000 square feet in the annual report and 526,000 in the closing release. Owner: Part VII; boxed in Part IV §25.

Long-term service agreement (LTSA; LTA (D9); rate-per-flight-hour (RPFH); flight-hour agreement)

A long-term service agreement is a multi-year contract under which the operator pays a fixed amount per engine flight hour and the manufacturer, or a shop, commits to perform or pay for the engine's shop visits. The provider takes the volume and cost risk and in return controls the workscope, the parts sourcing and the choice of shop. Owner: Part IV §28.

Low-pressure turbine (LPT)

The low-pressure turbine is the four-stage turbine at the rear of the engine. It takes the remaining energy from the gas after the high-pressure turbine and drives the fan and booster through the long low-pressure shaft. Owner: Part I §3.

LP spool / HP spool; two-shaft (two-spool) design (LP; HP)

A spool is a rotating assembly of compressor and turbine stages on one shaft. The CFM56 has two, one inside the other. Owner: Part I §1.

Maintenance cliff

The maintenance cliff is the point at which the next heavy maintenance event, an engine shop visit, a life-limited-part replacement or a heavy airframe check, costs more than the aircraft is worth kept. An owner at the cliff retires, converts or sells the aircraft rather than pay for the event. Owner: Part III §19.

Maintenance, Repair and Exchange (MRE)

MRE is FTAI's name for its offer to engine owners, chosen as a contrast with the industry's MRO. Owner: Part VII §45.

Maintenance reserves (Supplemental rent; maintenance rents (AerCap))

Maintenance reserves are periodic payments the lessee makes to the lessor in addition to basic rent, "calculated with reference to the utilization of airframes, engines and other major life-limited components during the lease". Owner: Part II §12.

Maintenance rights

Maintenance rights is a line in Willis Lease's lease portfolio (\$30.6M at 31 Dec 2025; \$27.0M at 30 Sep 2025) that the research reports without a filing definition. Owner: Part XI §87.

Maintenance-adjusted value (Maintenance adjustment)

Maintenance-adjusted value is "the fair market value of an Aircraft or Engine based on half-life values as adjusted by the relevant Aircraft Appraiser for the actual maintenance conditions and specifications". For each item the adjustment is (actual remaining fraction – 0.5) multiplied by the cost to restore that item, summed module by module. Owner: Part I §8.

Major module / sub-module

The manufacturer's shop manual arranges the engine as a hierarchy. The major modules (the fan and booster, the core, the low-pressure turbine, and the accessory gearbox) are each built from numbered sub-modules. Owner: Part I §3.

Major repair

A major repair is a repair the manufacturer's manual does not cover, or a repair of a part the manual says must be scrapped, that does not qualify as minor. Major repairs "must be accomplished in accordance with technical data approved by the Administrator", meaning the FAA or someone acting under its delegation. Owner: Part VI §40.

Management fee / incentive fee / sales fee

Three fee types in third-party-capital vehicles: a management or servicing fee on capital committed, assets managed or rents collected; an incentive fee for performance; and a sales fee on disposal proceeds. Owner: Part VIII §63.

Margin pool

The margin pool is this primer's term for how the profit earned in keeping an engine type flying is distributed across the links of the chain: the manufacturer's new parts and services, the shops' labour and test, the component repairers, the makers of approved alternative parts, the traders in used material and teardown, the lessors of engines and aircraft, and the servicers of third-party capital. Owner: Part XI §86.

Market value

Base value is an appraiser's opinion of an asset's underlying economic value in an open, unrestricted, stable market with balanced supply and demand. Market value is the appraiser's

opinion of the most likely trading price in the actual current market between willing, informed parties; it sits above base value in a tight market and below it in a soft one. Owner: Part II; boxed in Part I §8.

Master supply agreement / purchase order / milestone payments / performance adjustment mechanism

A master supply agreement is a framework contract that fixes terms under which a buyer may place binding orders over its life; the orders themselves are purchase orders. Owner: Part IX §70.

Material solutions

Material solutions is the generic trade phrase for a shop's or a manufacturer's package of used serviceable material, repaired parts and approved alternative parts assembled to reduce the new-parts bill of a shop visit. The whole commercial logic of a shop that is not the manufacturer sits on this line, since material is 60–70% of a visit. Owner: Part IV §28.

Mid-life narrowbody (Mid-life)

A mid-life narrowbody is a 737NG or A320ceo roughly 10 to 18 years old, or more loosely one in the middle of a 25-to-30-year economic life, when an increasing share of the aircraft's value sits in its engines and those engines are approaching or past a shop visit. Owner: Part II; boxed in Part I §4.

Mini-pack (Records mini-pack)

A mini-pack is an engine records summary that lists each life-limited part with its certified limit and the cycles it has consumed. Traders publish them so that a buyer can see the engine's remaining life part by part before asking for the full records. Owner: Part I §4.

Mobile genset

A mobile generator set is one built onto a road trailer or a skid so that it can be driven to a site, connected to a gas supply and to the customer's switchgear, commissioned in days or weeks, and later moved. Owner: Part IX §68.

Mod-1 (FTAI Mod-1; CFM56 aeroderivative)

Mod-1 is FTAI Power's 25 MW trailer-mounted natural-gas generator set built around a CFM56 core. Owner: Part IX §69.

Module

A module is a section of the engine designed to be removed and replaced as a unit. It is bolted to its neighbours at flanges, and it carries its own part number, its own serial number and its own maintenance record. Owner: Part I §3.

Module as unit of trade

A serviceable module from a dismantled engine is valued like a whole engine in miniature: its serviceable parts at a discount to list, plus the remaining life of the life-limited parts inside it, less the repair needed to make it serviceable. Owner: Part V; boxed in Part I §8.

Module exchange / engine exchange

A module exchange separates the engine at the major-module flanges, removes a time-expired or damaged module and installs a serviceable one from a pool, so the aircraft returns to service in days to weeks instead of months. The incoming unserviceable module becomes the provider's feedstock. Owner: Part VII; boxed in Part I §3.

Module Factory

The Module Factory is FTAI's name for the CFM56-7B and -5B modular repair and refurbishment programme quoted above from its FY2025 annual report, based at its Montréal shop. Functionally it is a production-line approach that builds and holds an inventory of serviceable modules for exchange. Owner: Part VII; boxed in Part I §3.

Module (FTAI Power usage)

In this Part a module is counted as FTAI's releases count it: three per engine, the fan, the core and the low-pressure turbine, which is the convention evidenced by "450 modules (150 engines)" and "1,800 modules (600 engines)" in FTAI's release of 26 February 2025. Owner: Part IX §69.

Modules produced / refurbished (count)

FTAI's quarterly count of CFM56 modules its Module Factory has refurbished and made serviceable: 184 in the second quarter of 2025, 228 in the fourth, 757 for 2025 against a target of 750, 296 in the second quarter of 2026 and 566 for the first half. Owner: Part VII §49.

MRE Contract revenue

The income-statement line, new in FY2025, for FTAI's sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the special-purpose vehicles of the 2025 Partnership under the MRE Contract: \$335.8M for 2025, \$221.2M in the first quarter of 2026, \$182.8M in the second (against \$69.6M a year earlier) and \$404.0M for the first half of 2026 (against \$170.2M). Owner: Part VII §48.

MRO market / engine MRO

MRO is maintenance, repair and overhaul, the generic term for aircraft and engine upkeep (Part IV §25). The MRO market is the dollar value of that spending in a year, and engine MRO is the engine share of it. Owner: Part III §23.

Muddy Waters report / short report / short seller

A short seller is an investor positioned to profit if a share price falls; a short report is research such an investor publishes after taking that position. Muddy Waters Research published a report on FTAI on 15 January 2025. Owner: Part VII §54.

Named executive officers and compensation (NEO; say-on-pay; CEO pay ratio; pay-versus-performance)

The named executive officers are the officers whose compensation a proxy statement must disclose in full, normally the chief executive, the chief financial officer and the next three highest-paid. Owner: Part X §78.

Net book value (NBV)

Net book value is an asset's cost less accumulated depreciation and impairments. It is the denominator for a lessor's implied yield and the figure against which a gain on sale is measured. Owner: Part XI §86.

Net debt / leverage ratio

Net debt is total debt less cash. A leverage ratio divides debt, or net debt, by a measure of annual earnings, usually a form of EBITDA. Owner: Part X §82.

Non-incident statement

A non-incident statement is the seller's declaration that a part was not installed on an aircraft involved in an accident, a fire or an immersion. It is a standard element of a used-material documentation package, beside the removal tag, the teardown report and the certificate of conformance. Owner: Part V §31.

Nonroad engine

A nonroad engine is a category in US air rules for engines that are mobile or temporary, generally meaning at one location for less than twelve months, and that are exempt from the permitting that applies to stationary sources. Whether trailer-mounted turbines at a data center qualify is contested. Owner: Part IX §73.

NOx / selective catalytic reduction (SCR)

Nitrogen oxides are the pollutant that air permits limit most tightly for gas turbines. Owner: Part IX §68.

OEM (Original equipment manufacturer)

The OEM is the company that holds the type certificate for the engine, meaning the regulator's approval of its design, and that built it. Owner: Part IV; boxed in Part I §2.

OEM aftermarket model

The OEM aftermarket model is the way the engine's manufacturer earns from its installed base after delivery: spare parts sold at an escalating catalogue list price to every shop, plus branded services (shop visits, parts and repair packages, used material, repair management) and long-term agreements priced per flight hour. Owner: Part IV §28.

OEM-only clause (No-PMA / no-DER clause; TCH-configuration clause)

An OEM-only clause is a lease return condition requiring that at redelivery the engine contain only parts produced by the type-certificate holder and only repairs from the type-certificate holder's manuals. A lessee that used PMA parts or DER repairs during the term must then

either remove them at the last heavy check before return or pay compensation. Owner: Part VI §42.

OneIM (One Investment Management)

An investment manager announced on 25 March 2025 as FTAI's partner "on the inaugural vehicle", in a release that described more than \$4 billion of deployment for the vehicle. OneIM's commitment size, and whether its role is that of an anchor limited partner or a co-general partner, are not disclosed. Owner: Part VIII §62.

On-lease

Already under an operating lease to an airline. A buyer of an on-lease aircraft acquires a contracted rent stream and a return-condition obligation rather than an empty aircraft, and lenders can underwrite the contracted rent. Owner: Part VIII §61.

Operating lease

An operating lease is a contract under which the owner (the lessor) rents an aircraft or engine to an operator (the lessee) for a fixed term in exchange for periodic rent, while the lessor keeps title to the asset and keeps the risk that it is worth more or less than expected when it comes back. Owner: Part II §12.

Operating versus investing cash flow (inventory versus leasing equipment)

Under US GAAP (generally accepted accounting principles) the cash a company spends buying assets it intends to sell is an operating outflow, because those assets are inventory and sit in working capital. The cash it spends buying assets it intends to lease is an investing outflow, because those assets are leasing equipment, capitalised and depreciated over time.

Owner: Part X §81.

Orange / Lisbon

Two 100%-owned maintenance sites listed in the FY2025 10-K. Orange (California) has no retrieved description or date; Lisbon is described in the chain map as 113,000 sq ft and 300-plus modules a year, unverified against a primary source, and was still ramping in July 2026.

Owner: Part VII.

Overhaul (full disassembly; heavy or full shop visit) (Zero-time (D2))

An overhaul disassembles every module to piece parts, inspects, repairs and rebuilds, and replaces the life-limited parts that would fall due within the next run. It is done when the life limits are due, at roughly 20,000 to 25,000 cycles. Owner: Part I §6.

Part 145 repair station (Part-145)

A Part 145 repair station is a maintenance organisation certificated by the FAA, or its European equivalent under the European Union Aviation Safety Agency (EASA), to perform maintenance and to issue release tags. It is the approval an engine shop or a component repair shop holds. Owner: Part IV; boxed in Part I §9.

Part 21 Subpart K (14 CFR Part 21 Subpart K)

Part 21 Subpart K is the section of the FAA's certification regulations under which Parts Manufacturer Approvals are issued. Owner: Part VI §37.

Part-out / teardown

Part-out is the owner's decision to dismantle a retired aircraft or engine and sell its components individually; teardown is the physical disassembly. A run-out CFM56 bought for \$0.8–1.2M is reported to yield \$1.6–2.2M of used material, net \$0.4–0.8M before repair cost (consultancy blog, 2026). Owner: Part V; boxed in Part I §8.

Parts Manufacturer Approval (PMA)

A Parts Manufacturer Approval is the FAA's combined design-and-production approval, issued under 14 CFR Part 21 Subpart K, that lets a company other than the type-certificate holder legally make and sell a replacement or modification part for a type-certificated aircraft, engine or propeller. Owner: Part VI §37.

Peer set (WLFC, AER, AL, HEI, ASLE, SARO, AIR, GE (SEC filers); Safran, MTU, Lufthansa Technik (context))

The companies Part XI sets beside FTAI from their own filings: Willis Lease Finance (engine lessor), AerCap and Air Lease (aircraft lessors), HEICO (PMA), AerSale (teardown and used material), StandardAero (independent engine MRO), AAR (distribution and used material) and GE Aerospace (OEM), with Safran, MTU Maintenance and Lufthansa Technik as non-SEC context. They were chosen so that each FTAI activity has a filing that isolates it. Owner: Part XI.

Performance restoration (PR; core PR)

A performance restoration is a shop visit in which the core is opened and the high-pressure compressor, combustor and high-pressure turbine are restored to recover EGT margin; the fan and low-pressure turbine may be inspected but are not fully disassembled. A core PR is this scope confined to the core module. Owner: Part I §6.

Preferred shares (cumulative, perpetual, redeemable; liquidation preference) (Series A, B, C, D)

Preferred shares are equity that ranks ahead of the ordinary shares for dividends and in a liquidation and normally carries no vote. FTAI's are cumulative (a missed dividend accrues and must be paid before any ordinary dividend), perpetual (no maturity) and redeemable at the company's option from a first call date at the liquidation preference of \$25.00 a share plus accrued dividends. Owner: Part X §82.

Prime Engine Accessories (Bristol JV)

A 50/50 joint venture with Bauer, Inc. in Bristol, Connecticut, described by FTAI as an MRE repair facility for accessory parts expected to deliver up to \$75,000 of average savings per

shop visit. It is accounted for under the equity method and was expected to be operational at the end of 2025. Owner: Part VII.

Production certificate / production approval (PC)

A production certificate is the FAA's approval of a manufacturer's quality system, under which that manufacturer may produce parts to an approved design and release them as airworthy. Owner: Part VI §37.

Production rate cap (737 MAX)

The production rate cap is the monthly ceiling on 737 output imposed by the FAA after the Alaska Airlines door-plug event of January 2024: 38 aircraft a month, raised to 42 in October 2025, with the review to move to 47 completed in May 2026. Owner: Part III §21.

Profit elimination (Intra-entity profit elimination)

When a company sells to an entity it accounts for under the equity method, the profit on the sale is not all earned yet from the group's point of view, because the company still owns a share of the buyer and so a share of the asset. Owner: Part VIII §63.

Promote / carried interest (Carry)

The sponsor's share of a fund's profit above its pro-rata interest, paid once the hurdle is cleared through the distribution waterfall. Carlyle describes its carried interest as "a special residual allocation of profit on third-party capital", generally calculated on realised gains for closed-end funds. Owner: Part VIII §63.

Quarantine

Quarantine is the segregation of a part whose documentation is suspect, pending investigation. The triggers named in the trade press are mismatched part or serial numbers, unusual formatting on a release tag, and an issuing organisation the buyer does not recognise. Owner: Part V §31.

QuickTurn (Miami) / QuickTurn Europe (Rome)

QuickTurn is the brand of FTAI's Miami engine business, formed from the assets of iAero Thrust acquired on 4 January 2023 with Unical Aviation as partner; FTAI bought Unical's remaining interest on 1 December 2023 for \$30.3 million in cash. Owner: Part VII; boxed in Part IV §25.

Reciprocating engine / solid-oxide fuel cell

A reciprocating engine is a piston engine. Large natural-gas versions of roughly 1 to 23 MW each are arranged in halls of many units; Wärtsilä describes data-center plants built from 10 to 23 MW engines reaching more than 450 MW per site, with electrical efficiency that "can exceed 50%" and "20–35% less fuel than gas turbines", both vendor claims. Owner: Part IX §68.

Related-party transaction

A transaction between a company and an entity it controls, manages or significantly influences, which accounting rules require to be disclosed because it is not necessarily on arm's-length terms. Every SCI stream in §63 is one: FTAI sells aircraft to, services, sells engines and modules to, and holds equity in the same partnership. Owner: Part VIII §64.

Release tag / serviceable tag (8130-3 or Form 1 tag)

The release tag is the signed airworthiness release, on FAA Form 8130-3 or on the European Union Aviation Safety Agency (EASA) Form 1, by which an approved organisation makes a part legally serviceable. It is necessary and not sufficient: the buyer also needs the part's documented history, and because the forms "can be easily generated" buyers vet the supplier too. Owner: Part V; boxed in Part I §9.

Repair management

Repair management is a service the manufacturer sells to a shop that is not its own: it manages the routing of the shop's piece-part repairs through the manufacturer's repair network and data, so the independent shop buys repair capability rather than building it. GE sold such a contract to Aero Norway, a CFM56 independent. Owner: Part IV §28.

Repair vendors / component repairs

Component repairs are the restoration of parts that can be repaired rather than scrapped, such as turbine airfoils, cases and seals, by welding, re-coating, machining or re-plating. Owner: Part IV §29.

Residual value / residual-value risk

Residual value is the lessor's forecast of what the asset will be worth, or fetch in a part-out, at the end of its depreciation life. Residual-value risk is the risk that the realised figure differs.

Owner: Part II §12.

Retirement rate

The retirement rate is the share of a fleet withdrawn from service in a year. GE's planning assumption for CFM56 retirements in 2026 moved from 3–4% of the installed base, to 2–3%, to 1.5–2%; GE reported about 1.5% retired in 2025 and expected about 2% in 2026, and first-quarter 2026 retirements ran "below 1%". Owner: Part III; boxed in Part I §2.

Return conditions (Redelivery conditions)

Return conditions are the contractual minimum state of a leased engine at redelivery: minimum hours and cycles to the next restoration, minimum cycles remaining on each life-limited part, a borescope and a test-cell run, and records in order. They may exclude parts and repairs that are not the manufacturer's. Owner: Part II; boxed in Part I §9.

Revenue and EBITDA per module (derived)

Aerospace products revenue divided by modules produced (\$2.11M for 2025; \$1.94M for Q1 2026; \$2.34M for Q2 2026) and segment Adjusted EBITDA divided by modules (\$0.89M;

\$0.82M; \$0.84M). A second version divides segment revenue including MRE Contract revenue by modules (\$2.96M for Q2 2026). Owner: Part VII §56.

Reverse bleed system (RBS)

The reverse bleed system is a LEAP cooling arrangement that reduces the need for on-wing fuel-nozzle replacements, which CFM describes as "bringing on-wing maintenance needs in line with CFM56 standards". It received initial engine-level certification for the LEAP-1B alongside the HPT durability kit on 18 July 2026, and was already being deployed on the LEAP-1A. Owner: Part III §20.

Revolving credit facility / SOFR (Revolver)

A revolving credit facility, or revolver, is a bank line the company may draw and repay repeatedly up to a committed amount. It pays a floating interest rate on drawn balances and a commitment fee on the undrawn portion. Owner: Part X §82.

Risk factors / auditor / securities class action / lead plaintiff (Class action)

Item 1A of a Form 10-K is the company's own list of the risks it considers material, each with a heading and an explanatory paragraph. The auditor is the independent registered public accounting firm that opines on the financial statements and, for a company of FTAI's size, on internal control over financial reporting. Owner: Part X §85.

Rotable

A rotable is a part that is repaired and reused rather than consumed: landing gear, the auxiliary power unit, thrust reversers, avionics boxes, flight-control actuators, wheels and brakes. Rotables are the airframe items a teardown sells; consumables and expendables are not worth the paperwork. Owner: Part V §32.

Run-out / exhausted engine

A run-out or exhausted engine is one with no usable time left before a shop visit: a life-limited part is at its limit, or the temperature margin is gone, or a hard inspection is due. It is the only kind of engine that becomes teardown feedstock, because an engine with life is worth several times its teardown yield as a whole (§32.5). Owner: Part V §32.

Russia assets, impairment and insurance recoveries (Contingent cover)

In 2022 FTAI "terminated all lease agreements with Russian airlines" and recorded a \$120.0M impairment, "net of maintenance deposits, to write-off the entire carrying value of leasing equipment assets that the company did not expect to recover from Ukraine and Russia". At 31 December 2025, 8 aircraft and 17 engines were "still located in Russia" with an insured value of \$210.7M. Owner: Part VIII §59.

Sale-leaseback

A sale-leaseback is a transaction in which an airline sells aircraft it owns to a lessor and leases them straight back, converting owned assets into rent obligations and releasing cash to the airline. Owner: Part II §17.

Security deposit

A security deposit is cash, or a bank letter of credit, that the lessee posts at signing. The lessor holds it against default and returns it at redelivery if the lessee has performed. Owner: Part II §12.

Seed Assets

FTAI's term for the on-lease aircraft sold from its own balance sheet to start the 2025 Partnership: 46 aircraft "at an estimated net purchase price of \$549 million" in the launch release, about \$11.9M each. The 10-Qs attribute the fall in aircraft lease revenue in 2026 to "the sale of Seed Assets to the 2025 Partnership". Owner: Part VIII §60.

Segment Adjusted EBITDA

FTAI's non-GAAP profit measure reported for each segment: Aerospace Products \$160.0M for 2023, \$380.6M for 2024 and \$671.3M for 2025, and \$249.7M in the second quarter of 2026. It includes the profit on MRE Contract revenue (before the equity-method elimination, which runs through a different line) and FTAI's share of the Rome and Bristol ventures. Owner: Part VII §53.

Segment margin bases (five)

The peers in this Part report profit on five different bases. Segment Adjusted EBITDA (StandardAero and FTAI) is segment profit before depreciation, amortisation and items each company defines as adjustments. Owner: Part XI §86.

Senior unsecured notes / bullet / callable (Senior notes)

Senior unsecured notes are bonds that rank ahead of equity and subordinated debt in a liquidation but are not backed by specific collateral. They pay a fixed coupon, normally twice a year, and repay their entire principal on the maturity date (a bullet repayment). Owner: Part X §82.

Serviceable / unserviceable

A part or module is serviceable when an approved organisation has inspected and released it and it is eligible for installation. It is unserviceable when it has been removed pending repair, or scrapped. Owner: Part I §9.

Serviceable as removed

A removed part is serviceable as removed when it has been inspected to the applicable manual and found within limits. It is tagged and sold without repair. Owner: Part V §31.

Servicer

The party contracted to manage a vehicle's assets: leasing and remarketing the aircraft, collecting rent and reserves, and overseeing their technical condition, for a fee. In a

securitization the lessor that sold the assets is usually the servicer (Willis Lease for its WEST trusts, Part II §17). Owner: Part VIII §61.

Servicing fee

FTAI's compensation as Servicer of the 2025 Partnership: \$10.2M for FY2025, \$6.988M for the second quarter of 2026 and \$12.849M for the first half of 2026 (so \$5.861M for the first quarter, derived). The 10-K describes it only as "customary, market-based compensation"; its rate, its base (committed equity, invested cost, or rents) and the income-statement line it sits on are not disclosed. Owner: Part VIII §63.

Shannon Engine Support (SES)

Shannon Engine Support is AerCap's 50% engine-leasing joint venture with Safran Aircraft Engines, the Safran half of CFM, that leases CFM engines. It matters to the engine count: AerCap's "over 1,200 engines" at 30 Sep 2025 includes engines owned and managed by SES, while its "470 owned engines" at 31 Dec 2025 does not. Owner: Part XI §87.

Shareholders

FTAI's largest reported holders per the 2026 proxy: Capital International Investors 13.51%, Capital World Investors 12.43%, The Vanguard Group 10.10% and FMR LLC 5.25%, with nine insiders at about 1.35%. About 102.6M shares are implied. Owner: Part X.

Shop visit (SV)

A shop visit is any removal of the engine from the aircraft for work in an engine shop, from a visit to fix one finding to a complete rebuild. Part VIII uses one dossier's phrase "teardown and restoration" for the same event; this primer reserves "teardown" for dismantling an engine to sell its parts (Part V §32). Owner: Part I §6.

Shop-visit demand (CFM56 shop visits per year)

The market's size in visits a year: just above 2,000 CFM56 visits in 2019, and about 2,300 to 2,400 projected for each of 2026 to 2028 with the apex around 2027 to 2028 (GE and Safran). It is driven by the installed base, utilisation and the retirement assumption, which was over-forecast by 1.5 to 2.5 points in 2025 and 2026; Part III §23 carries the forecast curve and Part IV §26 the market reading. Owner: Part IV.

Shop-visit forecast / first shop visit (Shop-visit curve)

A shop-visit forecast counts the expected engine inductions per year for a type. A first shop visit is the first removal of an engine for shop maintenance since new; first runs are usually the longest, so a fleet's first-visit peak comes years after its delivery peak. Owner: Part III §23.

Simple cycle / combined cycle; baseload / backup / peaking; PJM / ERCOT

A gas turbine run on its own, with its exhaust heat discarded, is in simple cycle. A combined-cycle plant adds a boiler that recovers the exhaust heat to raise steam for a second, steam-

driven turbine, which raises the plant's efficiency but adds years to construction and is only built at frame scale. Owner: Part IX §68.

Slot reservation agreement

A slot reservation agreement is a paid reservation of a future turbine manufacturing slot, made before a firm order. Owner: Part IX §71.

Spare engine / spare ratio (Spare-engine ratio)

A spare engine is one held or leased to replace an installed engine while it is in the shop. Owner: Part II; boxed in Part I §6.

Special-purpose vehicle (SPV)

A special-purpose vehicle is a bankruptcy-remote legal entity that exists only to own a defined pool of assets and the debt secured on them, insulated from the sponsor's other liabilities. Bankruptcy-remote means that if the sponsor fails, the SPV's assets are not caught up in the sponsor's insolvency, so a lender to the SPV looks only to those assets and their leases. Owner: Part II §17.

Spin-off / record date

A spin-off is a pro-rata distribution of a subsidiary's shares to the parent's shareholders, after which the subsidiary is a separate listed company. The record date is the day on which a shareholder must hold the parent's shares to receive the distribution. Owner: Part X §77.

Stand-ready obligation

A contractual duty to supply on demand. Under the MRE Contracts FTAI must provide replacement CFM56-5B and -7B engines to the customer as the customer's engines become unserviceable, so it must hold a pool of serviceable engines against the obligation whether or not any particular engine fails in a given period. Owner: Part VII §48.

Strategic Capital Initiative (SCI)

FTAI's asset-management arm, launched 30 December 2024. It raises third-party capital into closed-end vehicles that buy on-lease, mid-life 737NG and A320ceo aircraft; FTAI manages each vehicle as Servicer, co-invests as a limited partner, and supplies every engine shop visit for the fleet through module and engine exchanges with its Maintenance, Repair and Exchange business (Part VII §46). Owner: Part VIII §61.

Stub life

Stub life is "the shortest life remaining of all LLPs installed in an engine". In its pejorative sense it is the "several thousand EFCs of remaining LLP life that will not be used and the parts will be scrapped" when life-limited parts are replaced early so that they do not cut short the next run. Owner: Part I §4.

Sum of modules / sum of parts

Sum of modules is the engine's value computed module by module, or as parts. Owner: Part I §8.

Supplemental Type Certificate (STC)

A Supplemental Type Certificate is the FAA's approval of a change to a type-certificated design. As a PMA basis, the part is the subject of its own design approval, the STC, and the PMA becomes the production approval for that STC's parts. Owner: Part VI §37.

Syndicated / structuring agent / lead arranger

A syndicated loan is one shared across many lenders under a single agreement, so no bank carries the whole exposure; the 2026 warehouse is syndicated among 13 banks. The structuring agent designs the deal; the lead arranger assembles the lender group. Owner: Part VIII §61.

Technical Implementation Procedures (TIP (Revision 7))

The Technical Implementation Procedures are the document by which the FAA and the European Union Aviation Safety Agency (EASA) implement the US–EU Bilateral Aviation Safety Agreement for airworthiness and environmental certification; the current edition is Revision 7 with Amendment 1. Among much else, they set the rules under which EASA accepts FAA Parts Manufacturer Approvals for installation on EASA-certificated or EASA-validated products. Owner: Part VI §39.

Test and computation (T&C)

Test and computation is the PMA basis an applicant uses when it has none of the manufacturer's design data. It reverse-engineers the part: it measures examples of the manufacturer's part, determines the material and any coatings, and writes its own drawings and process specifications. Owner: Part VI §37.

Test cell / test-cell run

A test cell is an instrumented enclosure in which a rebuilt engine is run at power to confirm its thrust, EGT margin and vibration before release. Test-cell hours are a physical capacity limit for a shop: FTAI's releases quote capacity in engine tests a year alongside modules (Part VII §49). Owner: Part I §6.

Thrust / thrust rating (lbf; -7B24, -7B26, -7B27)

Thrust is the engine's forward push, in pounds-force (lbf). A thrust rating is the thrust level at which a given set of hardware is sold; the engine's electronic control limits how hard it may run, and the model suffix gives the rating in thousands of lbf (a -7B24 is rated at 24,000 lbf, a -7B27 at 27,300). Owner: Part I §1.

Time on wing

Time on wing is the number of cycles or hours between an engine's installation on an aircraft and its removal. It is the smaller of the life-limit clock (§4) and the EGT-margin clock (§5), or, less often, a hard inspection falling due. Owner: Part I §5.

Time-and-materials (T&M)

Under a time-and-materials arrangement the operator pays for each shop visit as it occurs: parts at the catalogue list price and labour by the hour, with the workscope agreed visit by visit. Owner: Part IV §28.

Top-case repair

In a top-case repair the high-pressure compressor case is split horizontally and the upper half lifted off, giving access to the compressor blades and vanes for repair or replacement without removing the rotor. StandardAero lists top-case and bottom-case repairs as separate services.

Owner: Part I §6.

Tranche: senior / mezzanine / equity (Class or Series A, B, C notes; E-certificates)

A tranche is one layer of the claims on a securitization, paid in order of seniority. The senior notes (Class or Series A) are paid first, carry the lowest coupon and the highest rating. Owner:

Part II §17.

Translation table

The translation table is Part III's organising device: one row per documented demand driver, stating the aftermarket requirement the driver creates and what it does not stress. "Stress" means that the driver raises the quantity or the price of that requirement; the last column records what the driver leaves untouched or pushes the other way. Owner: Part III §24.

TrueChoice (Flight Hour, Overhaul, Material, Transitions)

TrueChoice is GE Aerospace's branded menu of CFM56 and LEAP services: Flight Hour (a rate-per-flight-hour agreement), Overhaul (time-and-materials shop visits in GE's network), Material (new and used OEM parts and repairs sold to third-party shops and airlines) and Transitions (green-time leases, exchanges, material buy-back and shorter builds using more used material). Owner: Part IV; boxed in Part I §9.

TRUEngine

TRUEngine is CFM's designation for an engine whose configuration and material content conform to CFM's own definition: OEM parts and OEM-approved repairs. CFM states that the designation "is being embraced by industry's leading asset valuation providers", so appraisers can treat a non-conforming configuration as a discount (§8). Owner: Part I §3.

Turnaround time (TAT)

Turnaround time is the calendar time from an engine's induction into a shop to its release. For a full CFM56 overhaul it was about 60 days before the pandemic and 90 to 120 days in 2025; light work takes about 45 days. Owner: Part IV; boxed in Part I §6.

Type certificate / type-certificate holder (TC / TCH)

A type certificate is the Federal Aviation Administration's (FAA) approval, issued under Title 14 of the Code of Federal Regulations (CFR), of the design of an aircraft, an engine or a

propeller, issued after the design has been shown to meet the applicable airworthiness standards. Owner: Part VI §37.

Unapproved parts episode (AOG Technics, 2023)

In 2023 parts with forged documentation from the supplier AOG Technics were found on CFM56 engines at Delta, United, Southwest and American. They are a different category from PMA parts, which are FAA-approved. Owner: Part VI.

Unconditional MRO demand

Aviation Week's 2026 forecast gives "unconditional MRO demand of \$853bn over ten years". The forecast's own definition was not retrieved. Owner: Part III §23.

Up-cycling

APOC Aviation's own term for buying an engine in order to tear it down for material. Owner: Part V.

Used LLPs (Part-life LLPs; LLP package)

Used LLPs are life-limited parts removed from another engine with full records and remaining life, sold as packages with a stated number of cycles remaining and chosen to match the buyer's planned next run. Dealers advertise them as such: "CFM56-7B27E/B1F Engine LLP Package ... Owner: Part I §4.

Used serviceable material (USM)

Used serviceable material is any aircraft or engine part that has been in service, has been removed, and has been inspected (and repaired if needed) to the manufacturer's manual and released on an airworthiness tag with its full history, so that it is legally fit to be installed again. Owner: Part V §31.

USM pricing versus new (USM discount to CLP)

Used serviceable material is priced as a percentage of the OEM's catalogue list price (CLP) for the same part: on average 60–80% of a new part's price; 70–75% of list when a specialist engine trader sells repaired material to an airline; up to 85% of list for high-demand parts early in a platform's life; and above list when the manufacturer cannot deliver. Owner: Part V §34.

Utilization (FTAI definition) (Utilisation)

FTAI's utilization is "the percent of days on-lease in the quarter weighted by the monthly average equity value of aviation leasing equipment, excluding airframes". Each engine counts in proportion to its book equity value, not one-for-one, and airframes are left out. Owner: Part VIII §58.

V2500 / IAE (IAE V2500-A5; IAE International Aero Engines)

The IAE V2500-A5 is the A320ceo family's alternative engine, a two-shaft modular turbofan built by the IAE consortium, IAE International Aero Engines. Owner: Part I §10.

V2500 program (IAE EngineWise agreement)

On 6 June 2024 IAE and FTAI signed a five-year EngineWise agreement covering more than 100 full performance-restoration shop visits on V2500 engines in IAE's network, described by RTX as "one of IAE's largest engine maintenance agreements by number of engines with a non-airline customer". Owner: Part VII; boxed in Part I §10.

Variable interest entity / primary beneficiary (VIE)

Under US GAAP, an entity whose equity holders lack the power to direct its activities, or whose equity is too thin to fund it without support, is a variable interest entity, and the party with the power to direct its most significant activities and exposure to its economics (the primary beneficiary) must consolidate it even without a majority of the votes. Owner: Part VIII §64.

Warehouse facility (Warehouse line)

A warehouse facility is a revolving bank loan to an SPV, secured on whatever assets the SPV buys during an availability period, with an advance rate and a borrowing base, and with a term after which the balance must be repaid or refinanced, commonly by issuing a securitization. Owner: Part II §17.

Waterfall (ABS)

The payment waterfall is the ordered allocation of the pool's collections on each payment date: fees and expenses first, then interest on the A notes, then interest on the B notes, then scheduled principal on the A and B notes, then any C notes, with any surplus to the equity. Owner: Part II §17.

WestJet transaction / AEI collaboration

On 28 September 2026 FTAI announced the acquisition of 27 Boeing 737-700s from WestJet: 17 on lease to WestJet bought by the 2026 SPV in a sale-leaseback, and 10 off-lease aircraft bought by FTAI itself for their CFM56-7B engines and modules, to supply its Maintenance, Repair and Exchange customers. Owner: Part VIII §67.

Whole-asset sale (Flight-equipment sale (AerSale))

A whole-asset sale is the sale of a complete aircraft or engine from a trader's inventory rather than its disassembly into parts; AerSale's filings call the line "flight equipment sales". AerSale sold 13 engines this way in 2025 against 20 engines and one aircraft in 2024. Owner: Part V §35.

Willis Engine Structured Trust (WEST)

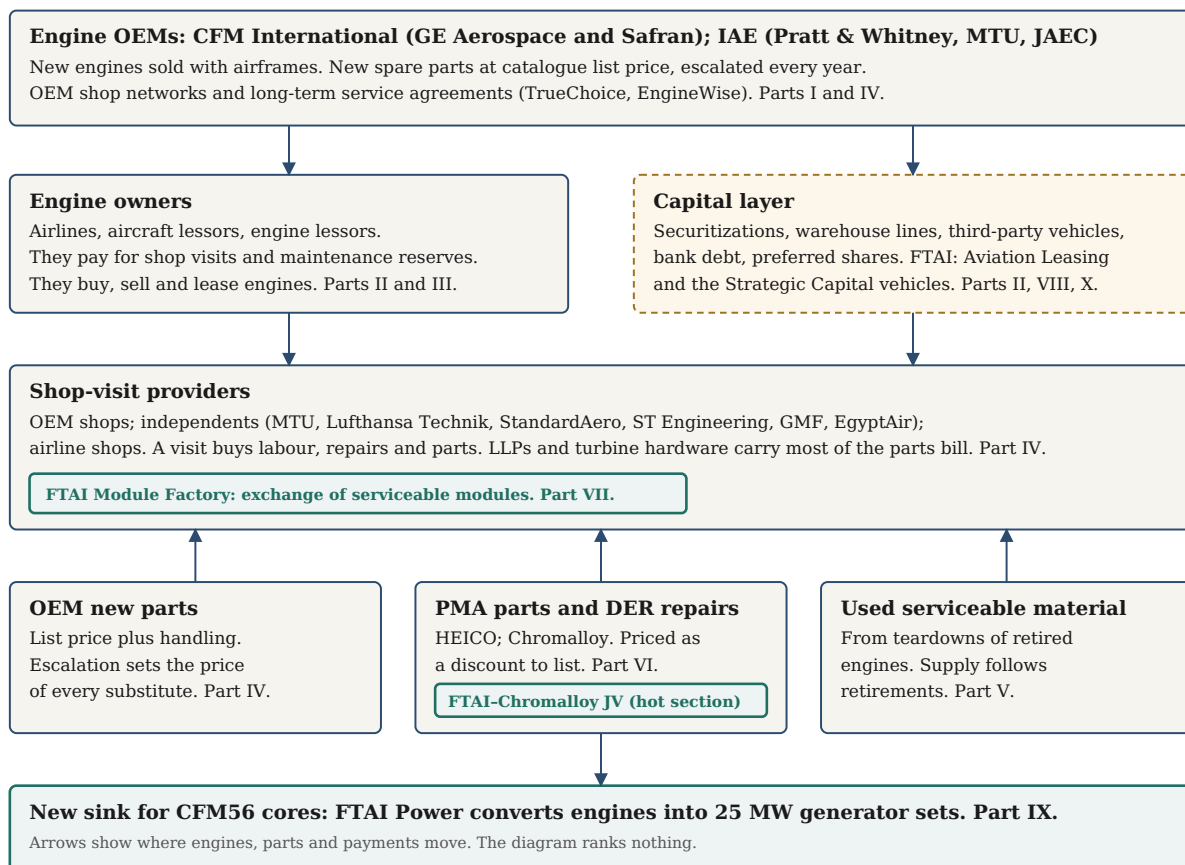
The Willis Engine Structured Trust deals are Willis Lease's series of rated engine securitizations: WEST VIII, rated in June 2025, raised \$596M against 64 assets; WEST IX, rated in December 2025, raised \$392.9M against 49 assets. Owner: Part II §17.

Workscope

The workscope is the specification of what a shop visit will do: which modules are opened, how deep the work goes in each, and which parts are replaced new, repaired, or swapped for used material. It runs from a light quick-turn to a full overhaul. Owner: Part I §6.

91. Chain map

The chain map is the one-page picture the orchestrator drew before the dossiers were commissioned, redrawn here from the research file `research/chain-map.md`. It shows where engines, parts and payments move between the links of the CFM56 aftermarket, and it fixed the segment list the eleven dossiers were written against. The paragraphs after the figure take each box in turn and say which Parts carry it. They say nothing about which boxes matter more; the diagram ranks nothing, and neither does this section.



The CFM56 aftermarket chain as the orchestrator mapped it: engine OEMs at the top, engine owners and the capital layer beneath them, shop-visit providers in the middle, the three sources of material below, and FTAI Power as a new sink for cores at the bottom. Green boxes mark FTAI's positions. Arrows show where engines, parts and payments move. Source: `research/chain-map.md`, drawn 2026-10-03 and corrected for the Rome site (§94).

Engine OEMs. The top box is the two manufacturers: CFM International, the joint venture of GE Aerospace and Safran that holds the type certificate for the CFM56 and the LEAP, and IAE, the consortium of Pratt & Whitney, MTU and the Japanese Aero Engines Corporation (JAEC) that holds it for the V2500. New engines are sold with airframes; new spare parts are sold at catalogue list price, revised upward each year; and the OEMs run shop networks and branded service agreements, TrueChoice and EngineWise. Part I §2 introduces CFM and the CFM56 family and Part I §10 the V2500 and the LEAP. Part III §20 carries the new engines' durability record. Part IV §25 carries the OEM shop network and Part IV §28 the OEM aftermarket model, including list prices, escalation and the service brands. Part VI §42 carries the OEM's responses to approved alternative parts, and Part XI §86.2 and §87 carry GE's Commercial Engines & Services segment and Safran's Propulsion figures from their filings.

Engine owners. The left-hand box in the second row is the airlines, aircraft lessors and engine lessors that own the engines. They pay for shop visits and maintenance reserves, and they buy, sell and lease engines. Part II §12 to §16 carries the contracts between them: the operating lease, maintenance reserves and return conditions, engine leasing and spare ratios, mid-life economics and green time. Part III §19 carries how an owner decides to retire an aircraft and Part III §22 what lessors and airlines did about extensions, availability and lease rates. Part VIII §58 and §59 carry FTAI's own position as an owner, including Russia, and Part XI §87 carries the three listed lessors' filings (Willis Lease, AerCap and Air Lease).

Capital layer. The right-hand box in the second row is the money behind the owners: securitizations, warehouse lines, third-party vehicles, bank debt and preferred shares. Part II §17 carries the instruments from first principles, with the WEST engine securitizations as the sourced template and a worked tranching example. Part VIII §60 to §66 carries FTAI's Strategic Capital Initiative, its two vehicles, their two layers of debt, the Servicer role, the money flows between FTAI and the vehicles and the comparable platforms. Part X §82 carries FTAI's own senior notes, revolver and preferred shares, and Part X §81 the inventory build that the balance sheet funds. The FTAI MRE 2026-1 securitization, priced in May 2026, is carried in Part II §17 and Part VIII §61 to §62.

Shop-visit providers. The wide box in the middle is where the engine is opened: the OEM's own shops, the independents (MTU, Lufthansa Technik, StandardAero, ST Engineering, GMF AeroAsia, EgyptAir and others) and the airlines' shops. A visit buys labour, repairs and parts, and life-limited parts and turbine hardware carry most of the parts bill. Part I §6 and §7 carry what a shop visit is and what it costs, with the cost build and a worked example. Part IV §25 to §30 carries the market: who performs visits, how many there are, what limits capacity, where the money in a visit goes and the four options for an engine at its life limit. Part XI §86.3 and §87 carry the independents' margins and statements from their filings.

FTAI Module Factory. The green tag inside the shop-visit box is FTAI's position in that link: the exchange of serviceable modules in place of a conventional visit. Part I §3 first boxes the module and the module exchange, and Part II §14 glosses FTAI's Maintenance, Repair and Exchange offer as a variant of shop-visit cover. Part VII §45 to §57 is the full treatment: what the segment sells, how an exchange works, pricing and revenue recognition, the MRE Contract revenue with the Strategic Capital vehicles, volumes and capacity, facilities and partners, the V2500 programme, the segment financials, the short report and the litigation, two worked examples and the guidance. Part X §79 to §81 carries the segment in the consolidated statements, and Part XI §86.7 sets its margin beside the other links on its stated basis.

OEM new parts. The left-hand box in the fourth row is the manufacturer's new parts at list price plus handling. Escalation of that list sets the price of every substitute, because used and approved alternative parts are priced as discounts to it. Part I §4 carries the life-limited-part prices by year and derives the escalation rate. Part IV §28 and §29 carry the catalogue, the escalation steps and the slice of a visit's bill they land on. Part III §24 turns escalation into a demand driver (row T-7), and Part XI §86.2 shows what the manufacturer reports earning on the parts and services line.

PMA parts and DER repairs. The middle box in the fourth row is the two legal routes to a part or a repair the manufacturer did not sell: a part made under a Parts Manufacturer Approval, by HEICO, Chromalloy and others, and a repair approved by a Designated Engineering Representative. Both are priced as a discount to list. Part I §5 boxes PMA briefly and Part I §9 boxes the DER repair. Part VI §37 to §44 is the full treatment: the four approval bases, critical parts and the hot section, EASA and the bilateral, DER repairs, the PMA market, acceptance by airlines and lessors and the OEM's responses, and a worked parts bill with and without PMA. Part XI §86.4 carries HEICO's Flight Support Group margin. The green tag inside the box, the FTAI–Chromalloy joint venture for hot-section parts, is carried in Part VI §43 (what is filed and what is reported), Part VII §52 (its place in the segment) and Part IV §29 (a first gloss).

Used serviceable material. The right-hand box in the fourth row is material recovered from torn-down engines, whose supply follows retirements. Part II §16 carries green time and the trade in part-worn engines. Part V §31 to §36 is the full treatment: what makes a used part serviceable, teardown and part-out economics with a worked 737-800 example, supply and the 2023 to 2026 shortage, pricing against new parts, the players and the module as a unit of trade. Part III §19 carries the retirements that did not happen and Part III §24 the feedstock-scarcity row (T-9). Part XI §86.5 carries AerSale's and AAR's filings, and Part VII §52 carries used material within FTAI's segment, including the exclusive serviceable-parts agreement with AAR.

New sink for CFM56 cores. The bottom box is FTAI Power, which converts CFM56 engines into 25 MW generator sets and so removes cores from the aviation pool permanently. Part IX §68 to §74 is the full treatment: the generator set and the aeroderivative from first principles, Mod-1

as FTAI describes it, J&F Power Systems and the \$1.465B order, the data-center market and the turbine shortage, competing supply, permitting and interconnection, and two worked examples (implied unit price; cores consumed). Part III §18 glosses Mod-1 as a demand driver and Part III §24 carries it as row T-10. Part VII §49 notes that Mod-1 draws on the same module production as the aftermarket, and Part X §84 carries the 2027 guidance for the Power segment.

The translation table and the Parts that answer it. Part III §24 (Exhibit 3.14) turned each documented demand driver into a requirement on the aftermarket, in fourteen numbered rows. Parts IV to XI each open with a paragraph naming the rows they answer. Exhibit 12.1 was compiled by searching those opening paragraphs for the row identifiers; it says which Parts name each row, not which Part treats the underlying requirement, which Part III's Exhibit 3.15 gives.

EXHIBIT 12.1 — TRANSLATION-TABLE ROWS T-1 TO T-14 AND THE PARTS THAT OPEN BY NAMING THEM

Row	Driver (Part III §24, Exhibit 3.14)	Parts that open by naming the row
T-1	Low CFM56 retirements	Part IV, Part V, Part VI, Part VII, Part VIII, Part IX, Part X, Part XI
T-2	GTF powder-metal groundings	Part IV, Part V, Part VII, Part VIII, Part X, Part XI
T-3	LEAP durability shortfall and the July 2026 kit	Part IV, Part V, Part VII, Part X, Part XI
T-4	737 MAX production rate cap	Part V, Part VII, Part VIII
T-5	Airbus delivery shortfall and the rate-75 deferral	Part V, Part VII, Part VIII
T-6	Lessor extension behaviour	Part V, Part VI, Part VII, Part VIII, Part X, Part XI
T-7	OEM parts-price escalation	Part IV, Part V, Part VI, Part X
T-8	MRO capacity shortfall and turnaround	Part IV, Part VII, Part VIII, Part XI
T-9	Used-material and feedstock scarcity	Part VI
T-10	Data-center power demand and the gas-turbine shortage	Part IX
T-11	Freighter conversion	Part V, Part VII, Part VIII
T-12	Higher utilisation of mature narrowbodies	Part VII
T-13	GTF groundings easing	Part VII
T-14	CFM56 plateau-then-decline and LEAP aftermarket parity	Part VII, Part IX, Part X, Part XI

Source: the "Translation-table rows answered" paragraph at the head of each of Parts IV to XI, searched for the identifiers "row T-n" and "rows T-n, T-m"; row labels from Part III §24, Exhibit 3.14. Compiled for this Part on 2026-10-03.

92. Margin pool table

Part XI §86 built the margin pool link by link from the filings and ended with one table (Exhibit 11.4). This section consolidates every margin figure that section carries into one exhibit, with the company, the segment, the fiscal period, the margin, the basis and the source, and then lists FTAI's own margins as Parts VII and X state them. The caveats travel with the figures. The peers report profit on five bases (segment Adjusted EBITDA, segment operating income, adjusted EBIT, GE segment profit and gross margin), which Part XI §86 defines at its head; an EBITDA margin exceeds an EBIT margin for the same business by the depreciation share of revenue, so no two bases can be set beside each other without that share, which is mostly not disclosed. Three fiscal calendars are in use (December, October and May), euro figures are not converted, and "derived" marks a ratio computed in the research from two reported numbers. The rows below follow the order of the chain as Part XI walks it, from the manufacturer to the servicer; they are not ordered by margin, and the exhibit ranks nothing. The shares of a shop visit's bill in Part XI's Exhibit 11.1 (material 60 to 70%, labour 20 to 30%, repairs 10 to 20%) are shares of spending, not margins, and are not repeated here.

EXHIBIT 12.2 — EVERY MARGIN FIGURE IN PART XI §86, BY LINK IN THE CHAIN, ON ITS STATED BASIS

Company	Segment or scope	Fiscal period	Margin (and the figures behind it)	Basis	Source (Part XI section)
GE Aerospace	Commercial Engines & Services	CY2025	26.6% (\$8,861M segment profit on \$33,314M; services \$25,010M, 75.1% of the segment, derived)	GE segment profit, GE's own measure; whether depreciation sits above or below it was not retrieved	GE 10-K FY2025 (§86.2)
GE Aerospace	Total company	CY2025	No margin computed: operating profit \$9.1B stated; total revenue not returned; items reconciling CES to the total not returned	Total-company operating profit	GE 10-K FY2025 (§86.2; D11 T4)

Company	Segment or scope	Fiscal period	Margin (and the figures behind it)	Basis	Source (Part XI section)
Safran	Propulsion	CY2025	No margin reported in the retrieved text; services €10,122M were 64.6% of Propulsion revenue (about €15.7B, derived), against 61.7% in 2024	Services share of revenue, not a margin	Safran FY2025 results, 13 Feb 2026 (§86.2)
StandardAero	Engine Services	CY2025	13.2% (\$706.9M on \$5,354.0M)	Segment Adjusted EBITDA, before depreciation, amortisation and defined adjustments	StandardAero 10-K FY2025 (§86.3)
StandardAero	Engine Services	CY2024	13.2% (\$610.9M on \$4,644.7M, derived)	Segment Adjusted EBITDA	StandardAero 10-K FY2025 (§86.3)
StandardAero	Component Repair Services	CY2025	28.6% (\$202.7M on \$708.6M)	Segment Adjusted EBITDA	StandardAero 10-K FY2025 (§86.3)
MTU Aero Engines	Commercial maintenance	CY2025	8.0% (€478M on €6.0B)	Adjusted EBIT, after depreciation and amortisation, before purchase-price-allocation effects	MTU, 24 Feb 2026 (§86.3)
MTU Aero Engines	Commercial maintenance	CY2024	8.7% (€438M on €5.1B)	Adjusted EBIT	MTU, 24 Feb 2026 (§86.3)
MTU Aero Engines	Group	CY2025	15.5% (€1.35B on €8.7B adjusted revenue); 14.0% in 2024	Adjusted EBIT, group	MTU, 24 Feb 2026 (§86.3)

Company	Segment or scope	Fiscal period	Margin (and the figures behind it)	Basis	Source (Part XI section)
Lufthansa Technik	Whole company	CY2025	7.5% (€603M on €8.049B); 8.5% in 2024	Adjusted EBIT, company; engine services "almost half" of revenue	Lufthansa Technik FY2025 results; Deutsche Lufthansa Annual Report 2025 (§86.3)
HEICO	Flight Support Group	FY to 31 Oct 2025	24.1% (\$750.4M on \$3,117.3M, derived); 22.5% in FY2024	Segment operating income, after depreciation and amortisation; the segment includes repair and distribution as well as PMA	HEICO 10-K FY2025; HEICO release, 18 Dec 2025 (§86.4)
AerSale	Asset Management Solutions	CY2025	35.0% (\$74.1M on \$211.6M, derived); 38.3% in 2024 (\$82.5M on \$215.5M)	Gross margin, before all operating expenses	AerSale 10-K FY2025; release 5 Mar 2026 (§86.5)
AerSale	AMS, aircraft line	CY2025	33.2%; 34.8% in 2024	Gross margin	AerSale 10-K FY2025 (§86.5)
AerSale	Whole company	CY2025	13.8% (\$46.1M on \$335.3M); 9.7% in 2024 (\$33.4M)	AerSale-defined Adjusted EBITDA	AerSale release, 5 Mar 2026, via D3 (§86.5)
Safe Fly Aviation (consultancy blog)	A run-out CFM56 teardown	2026	Not a percentage: an engine bought for \$0.8–1.2M yields \$1.6–2.2M of used material, a net margin of \$0.4–0.8M per engine	Dollars per engine from a consultancy blog, labelled so	Safe Fly Aviation, 2026 (§86.5)
AAR	Parts Supply	FY to 31 May 2026	Not returned; the segment is about 45% of \$3,308.0M of sales (both derived)	n/a	AAR 10-K FY2026 (§86.5)

Company	Segment or scope	Fiscal period	Margin (and the figures behind it)	Basis	Source (Part XI section)
Willis Lease Finance	Whole company	CY2025	62.9% (\$459.1M on \$730.2M, derived)	Company-defined Adjusted EBITDA; depreciation and interest, a lessor's largest costs, sit below the line	WLFC 10-K FY2025 (\$86.6)
AerCap	Whole company	CY2025	No margin basis returned; maintenance rents \$690M were 9.4% of lease revenue of \$7,369M (derived); net income \$3.8B against \$2.1B, drivers not returned	n/a	AerCap 20-F FY2025; Q4 2025 release (\$86.6)
Air Lease	Whole company	CY2025	No margin returned; lease rental revenue \$2,615.4M on net book value of \$29.1B is a 9.0% gross yield (derived)	Yield on book, not a margin	Air Lease 10-K FY2025 (\$86.6)
FTAI	Aerospace Products	CY2025	34.7% (\$671,252K on \$1,936,244K, derived; revenue includes MRE Contract revenue of \$335,788K from the 2025 Partnership)	FTAI-defined segment Adjusted EBITDA	FTAI 10-K FY2025 (\$86.7); Exhibit 12.3
FTAI	Aerospace Products	Q2 2026	28.5% (\$249.7M on \$875.0M, derived; MRE \$182.8M)	FTAI-defined segment Adjusted EBITDA	FTAI Q2 2026 release; 10-Q (\$86.7); Exhibit 12.3

Company	Segment or scope	Fiscal period	Margin (and the figures behind it)	Basis	Source (Part XI section)
FTAI	Aviation Leasing	CY2025	No margin computable on the retrieved lines: segment Adjusted EBITDA \$608.9M against lease income \$235.2M is 259%, because the measure includes gains on sale and equity income and the segment's full revenue mapping was not retrieved	FTAI-defined segment Adjusted EBITDA	FTAI 10-K FY2025 (§86.6)
FTAI	Consolidated	CY2025	47.5% (\$1,190,922K on \$2,507,409K, arithmetic in Part XI); depreciation and amortisation \$225,797K (9.0% of revenue) and interest \$247,751K sit below	Company-defined Adjusted EBITDA	FTAI 10-K FY2025 (§86.7, Exhibit 11.3)
FTAI	Servicer to the 2025 Partnership	FY2025; 1H 2026	No margin disclosed; servicing fees \$10.2M and \$12.85M; rate and base undisclosed	Fee income	FTAI 10-K FY2025; 10-Q Q2 2026 (§86.8; Part VIII §63)

Source: Part XI §86.2 to §86.8 and Exhibits 11.2, 11.3 and 11.4, which cite GE Aerospace Form 10-K FY2025; Safran FY2025 results presentation, 13 Feb 2026; StandardAero, Inc. Form 10-K FY2025; MTU Aero Engines, 24 Feb 2026; Lufthansa Technik FY2025 results and Deutsche Lufthansa AG Annual Report 2025; HEICO Corporation Form 10-K FY2025 and release of 18 Dec 2025; AerSale Corporation Form 10-K FY2025 and release of 5 Mar 2026; AAR Corp Form 10-K FY2026; Willis Lease Finance Corporation Form 10-K FY2025; AerCap Holdings N.V. Form 20-F FY2025; Air Lease Corporation Form 10-K FY2025; FTAI Aviation Ltd. Form 10-K FY2025 and Form 10-Q Q2 2026; Safe Fly Aviation, 2026 (consultancy blog). The five bases are defined at the head of Part XI §86 and are not interchangeable. Reconciliation E.1 rule 35; R-021, R-023, R-043, R-045; D11 T3, T4.

Exhibit 12.3 collects FTAI's own margins, which Part VII §53 (Exhibits 7.11 to 7.13) and Part X §79 and §80 (Exhibits 10.8 to 10.13) state. The main series is Aerospace Products segment Adjusted EBITDA divided by segment revenue, where segment revenue is the "Aerospace products revenue" line plus, from 2025, the "MRE Contract revenue" line, a sum the FY2025 10-

K's geographic table confirms for 2025. Three caveats travel with the series. MRE Contract revenue is related-party revenue from vehicles in which FTAI holds a 19% stake (20% in the Q3 2025 10-Q), and the profit on it is eliminated through equity-method earnings by a method the filing does not disclose (Part VII §48; Part VIII §63). The segment's cost of sales for FY2025 appears as \$1,240,368K in the segment expenses table and the consolidated cost-of-sales line as \$1,349,719K, and the gross margins below carry each figure's scope (Part VII §53; reconciliation R-045). And the quarterly figures for Q2 2025 and Q4 2025 are derived from growth rates or a stated margin rather than read from a filing, as their rows say. Consolidated Adjusted EBITDA is the company's own measure, whose exact FY2025 definition and reconciliation the research did not retrieve (Part X §80).

EXHIBIT 12.3 — FTAI'S OWN MARGINS AS PARTS VII AND X STATE THEM: SEGMENT ADJUSTED EBITDA ON SEGMENT REVENUE BY YEAR AND QUARTER, GROSS MARGINS BY SCOPE, AND THE CONSOLIDATED FIGURES

Measure	Period	Revenue base (\$ thousands unless stated)	Profit measure	Margin	Basis and caveat	Stated in
Aerospace Products segment Adjusted EBITDA ÷ segment revenue	FY2022	85,113 (products line; no MRE line)	27,377	32.2% (derived)	Segment figures from the FY2022 10-K; the segment was ramping	Part VII Exhibit 7.11
Same	FY2023	454,970	160,009	35.2% (derived)	FY2025 10-K segment table	Part VII Exhibit 7.11; Part X Exhibit 10.11
Same	FY2024	1,079,821	380,636	35.2% (derived)	FY2025 10-K segment table	Part VII Exhibit 7.11; Part X Exhibit 10.11

Measure	Period	Revenue base (\$ thousands unless stated)	Profit measure	Margin	Basis and caveat	Stated in
Same	FY2025	1,936,244 = products 1,600,456 + MRE Contract 335,788	671,252	34.7% (derived; the release gives +76% growth in segment Adjusted EBITDA)	MRE Contract revenue is related-party revenue (R-043); FTAI- defined measure	Part VII Exhibits 7.11 and 7.12; Part X Exhibit 10.11; Part XI Exhibit 11.3
Same	Q2 2025	≈491,600 (derived: 875.0 ÷ 1.78)	≈165,400 (derived: 249.7 ÷ 1.51)	≈33.6% (derived)	Both figures back-derived from the Q2 2026 release's year-on-year growth rates	Part VII Exhibit 7.12; Part X Exhibit 10.12
Same	Q4 2025	≈557,000 (derived: 195 ÷ 0.35)	195,000 (stated, rounded)	35% (stated by management)	Margin from call coverage (MarketBeat, 26 Feb 2026); EBITDA from the release	Part VII Exhibit 7.12; Part X Exhibits 10.12 and 10.13
Same	Q1 2026	743,800 = 522,585 products + 221,230 MRE	222,600	29.9% (derived)	MRE was 29.7% of segment revenue (derived)	Part VII Exhibit 7.12; Part X Exhibit 10.12

Measure	Period	Revenue base (\$ thousands unless stated)	Profit measure	Margin	Basis and caveat	Stated in
Same	Q2 2026	875,000 = ≈692,200 products (derived) + 182,800 MRE	249,700	28.5% (derived)	MRE to the 2025 Partnership \$182.8M (10- Q); eliminations \$6.6M in the quarter	Part VII Exhibit 7.12; Part X Exhibit 10.12; Part XI Exhibit 11.3
Same	1H 2026	≈1,618,800 (derived: 1,214,800 products + 404,000 MRE)	≈472,300 (derived: 222,600 + 249,700)	≈29.2% (derived)	Sum of two quarters; no half-year segment figure was retrieved	Part VII Exhibit 7.12; Part X Exhibit 10.9
Management's stated aspiration for the segment margin	2026	n/a	n/a	"approach 40%"	Call coverage (MarketBeat, 26 Feb 2026); an aspiration, not a result; levers named were the PMA HPT blade, more used material and the CFM materials agreement	Part VII §53 and §57; Part XI §86.9
Aerospace Products gross margin (segment scope)	FY2025	1,936,244	Segment cost of sales 1,240,368	35.9% (derived)	Scope per the orchestrator's reading of the segment expenses table (R-045)	Part VII Exhibit 7.13

Measure	Period	Revenue base (\$ thousands unless stated)	Profit measure	Margin	Basis and caveat	Stated in
Gross margin, scope unconfirmed	1H 2026	≈1,618,800	Cost of sales 1,160,100 returned against the same segment query	28.3% (derived)	Segment or consolidated scope not confirmed	Part VII Exhibit 7.13; Part X Exhibit 10.9
Consolidated revenue less cost of sales ÷ revenue	FY2023	1,170,896	Cost of sales 502,132	57.1% (derived; cost of sales 42.9% of revenue)	Consolidated cost of sales includes the carrying value of leased assets sold	Part VII Exhibit 7.13; Part X Exhibit 10.8
Same	FY2024	1,734,901	Cost of sales 825,884	52.4% (derived; 47.6%)	As above	Part VII Exhibit 7.13; Part X Exhibit 10.8
Same	FY2025	2,507,409	Cost of sales 1,349,719	46.2% (derived; 53.8%)	As above; the difference from the segment figure, 109,351, is close to asset sales revenue of 106,945, presented gross	Part VII Exhibit 7.13; Part X Exhibit 10.8
Consolidated Adjusted EBITDA ÷ total revenues	FY2023	1,170,896	597,282	51.0% (arithmetic in this Part on Part X's figures)	Company- defined measure; definition and reconciliation not fully retrieved (D10 U-1)	Part X Exhibits 10.8 and 10.11 (figures)

Measure	Period	Revenue base (\$ thousands unless stated)	Profit measure	Margin	Basis and caveat	Stated in
Same	FY2024	1,734,901	862,050	49.7% (arithmetic in this Part on Part X's figures)	As above	Part X Exhibits 10.8 and 10.11 (figures)
Same	FY2025	2,507,409	1,190,922	47.5% (arithmetic in Part XI)	As above; D&A 225,797 and interest 247,751 sit below the measure	Part XI Exhibit 11.3; Part X Exhibit 10.8
Aviation Leasing segment Adjusted EBITDA against lease income	FY2024	Lease income 234,411 or 255,338 (two readings of the 10-K, R-052)	500,062	Not a margin	The measure includes gains on sale and equity income; the segment's other revenue lines for 2024 were not retrieved	Part X Exhibit 10.11; Part XI Exhibit 11.3
Same	FY2025	Lease income 235,210 (maintenance revenue 218,499 and asset sales revenue 106,945 sit on the consolidated statement; their segment mapping was not retrieved)	608,912	Not a margin (259% of lease income; 109% of the three lines summed)	Includes the \$46.4M gain on sale to the 2025 Partnership and the equity- method pick- up	Part XI §86.6 and Exhibit 11.3; Part VIII §58

Measure	Period	Revenue base (\$ thousands unless stated)	Profit measure	Margin	Basis and caveat	Stated in
Aviation Leasing segment Adjusted EBITDA	Q2 2026	not retrieved	88,200 (call coverage only; not returned from the release, R-051)	Not computable	Call coverage (GuruFocus, Quartr)	Part X Exhibits 10.12 and 10.13
Segment share of consolidated Adjusted EBITDA (Aerospace Products)	FY2023; FY2024; FY2025	Consolidated Adjusted EBITDA	Segment Adjusted EBITDA	26.8%; 44.2%; 56.4% (derived)	A share of profit, not a margin; shown because Part VII states it	Part VII Exhibit 7.11

Source: Part VII §53, Exhibits 7.11, 7.12 and 7.13; Part X §79 and §80, Exhibits 10.8, 10.9, 10.11, 10.12 and 10.13; Part XI §86.6, §86.7 and Exhibit 11.3. Underlying filings: FTAI Aviation Ltd. Form 10-K FY2025 (segment table, segment expenses table, consolidated statement of operations), Form 10-K FY2022, Forms 10-Q for Q1 and Q2 2026, Q4/FY2025 release of 25 Feb 2026, Q1 2026 release, Q2 2026 release of 29 Jul 2026; MarketBeat call coverage, 26 Feb 2026; GuruFocus and Quartr call coverage, 30 Jul 2026. Reconciliation R-043, R-045, R-047, R-051, R-052, B-1; E.1 rules 8, 10, 12, 28.

93. Company index

The index lists every company, joint venture, investment vehicle and institution named in the eleven Parts: FTAI's own entities and vehicles, the engine and airframe manufacturers, the shops, the parts makers and traders, the lessors and the capital providers, the airlines, the appraisers and data providers, the consultancies and market-research vendors, the regulators and courts, the law firms, auditors and rating agencies, and the trade publications and news services the Parts cite. The index has 224 entries. It was built with a script that searched a curated list of names, each as a pattern allowing for the variants the Parts use, across the eleven Part files, and recorded the numbered section in which each match fell. The entries are alphabetical. Each gives the ticker where the company is listed (tickers are identifiers added for the reader's convenience and are not drawn from the research files), one line on the role the company plays in the primer, and the Parts and sections in which it appears. Where a company appears in more than seven sections of one Part, the entry gives the range and the count. "Unknowns list only" means the name appears in that Part only in its closing list of unknowns. The role lines are descriptive and carry no view of any company.

EXHIBIT 12.4 — COMPANY INDEX: EVERY COMPANY, VEHICLE AND INSTITUTION NAMED IN PARTS I TO XI, WITH ROLE AND LOCATION

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
2025 Partnership (FTAI SCI I)	The first Strategic Capital vehicle: \$2.0B equity hard cap, \$2.5B asset-level debt, about \$6B across 300-plus aircraft; FTAI is Servicer and 19% LP.	Part II §17; Part III §22; Part VI §43; Part VII §45, §48, §50, §51, §53; Part VIII §58–§67 (9 sections); Part X §75, §78, §79, §80, §81, §82; Part XI §86, §89
2026 SPV (SCI II)	The second Strategic Capital vehicle, with a \$2.0B warehouse facility from 13 lenders closed 14 August 2026.	Part II §17; Part VIII §62, §65, §67; Part X §75, §82
AAR Corp (NYSE: AIR)	Distribution, USM and MRO company; exclusive Serviceable Engine Products partner for FTAI's CFM56 pool through 2030.	Part II §16; Part V §32, §35; Part VI §41; Part VII §45, §52; Part IX §70; Part XI §86, §87, §88, §89
Acumen Aviation	Consultancy cited on retirements and freighter conversion.	Part II §14, §15; Part III §18, §19, §22, §23, §24; Part IV §26; Part V §32; Part VIII §67
Advanced Engine Repair JV	A 25% equity-method investment (\$22.4M carrying value at 31 Dec 2025); whether it is the Chromalloy PMA venture is unresolved.	Part IV §29; Part VI §43; Part VII §52; Part VIII §64; Part X §75, §78
AEI (Aeronautical Engineers, Inc.)	Freighter-conversion house (625-plus conversions); July 2026 collaboration with FTAI.	Part II §15; Part III §22; Part VIII §67; Part X §75
AerCap Holdings N.V. (NYSE: AER)	Largest aircraft lessor; 20-F filer; peer in Part XI.	Part II §12, §13, §14; Part V §32, §35; Part VI §42; Part VII (unknowns list only); Part VIII §65; Part X §75; Part XI §86, §87, §88, §89
AerCap Materials	AerCap's used-material arm; dismantled more than 300 aircraft in 2023.	Part V §35; Part XI §86, §87
Aero Norway	Independent CFM56 shop; buyer of GE repair management.	Part III §23; Part IV §25, §27, §28

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Aerodynamic Advisory	Consultancy cited at the IATA Maintenance Cost Conference on retirements and part-outs.	Part III §19; Part V §33
AerSale Corporation (NASDAQ: ASLE)	Listed teardown and USM company; Asset Management Solutions gross margin.	Part I §11; Part II (unknowns list only); Part III §19, §24; Part IV §29; Part V §32, §33, §35; Part X §80; Part XI §86, §87, §88, §89
AFI KLM E&M	Air France Industries KLM Engineering & Maintenance, named among airline MROs.	Part IV §25; Part VII (unknowns list only)
Air Cargo Week	Trade publication; shop-visit cost breakdown.	Part I §4, §7, §11; Part III §22, §23; Part IV §27, §28, §29; Part VI §37, §40, §44; Part VII §52; Part XI §86
Air Data News	Trade publication cited on Airbus deliveries.	Part III §21, §24; Part V §33
Air Lease Corporation (formerly NYSE: AL)	New-aircraft lessor taken private as Sumisho Air Lease Corporation on 8 April 2026; peer in Part XI.	Part II §13, §14, §15; Part III §21, §22, §24; Part VI §42; Part VII (unknowns list only); Part VIII §58, §65, §66; Part XI §86, §87, §88, §89
Airbus SE (Euronext Paris: AIR)	Airframer of the A320ceo and A320neo families; delivery shortfall and gliders treated in Part III.	Part I §2, §10; Part II §12, §15; Part III §18, §20, §21, §22, §24; Part V §33, §35; Part XI §87
Aircraft Commerce	Trade publication; maintenance analyses and owner's guides for the CFM56.	Part I §3, §4, §5, §6, §7, §10; Part III §22; Part IV §28, §29; Part V §32, §34, §36; Part VI §42; Part VII §51, §52; Part XI §86
Aircraft Value News	Trade publication on values and lease rates.	Part I §1, §4, §5, §7, §8; Part II §13, §14, §16; Part III §20, §22, §23; Part IV §30; Part V §32, §33, §34, §36; Part VI §42; Part VIII §66
AJW Group	Parts and engine services company whose records mini-packs are cited.	Part I §3, §4; Part II §14; Part V §31, §35; Part VI §38
American Airlines (NASDAQ: AAL)	CFM56 operator affected by the AOG Technics episode.	Part III §22

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
ANAC (Brazil)	Regulator whose PMA acceptance is an open question.	Part VI §39
AOG Technics	Supplier of parts with forged documentation found on CFM56s in 2023.	Part VI §42
APOC Aviation	Engine lessor and up-cycler named in Part V.	Part II §16; Part V §32, §35
Apollo Global Management / Merx Aviation (NYSE: APO)	Apollo's structured-products business ATLAS SP led the 2025 Partnership's debt; Merx is its aviation platform.	Part VIII §61, §65; Part XI (unknowns list only)
Argus Media	Price-reporting agency cited in Part III.	Part III §21
Asset Securitization Report	Structured-finance publication cited in Part II.	Part II §17
ATLAS SP Partners	Structuring agent and lead arranger of the \$2.5B asset-level debt commitment (February 2025).	Part VIII §61, §62
Aviation Business News	Trade publication.	Part I §2, §6, §9; Part II §14, §16; Part III §18, §19, §22, §23; Part IV §26, §27, §30; Part V §32, §33, §35; Part VII §46, §55; Part IX §74; Part XI §86
Aviation Fleet Support	Parts dealer whose core LLP package listing is cited in Parts I and V.	Part I §3, §4; Part VII §46
Aviation Suppliers Association	Trade association cited on parts documentation.	Part I §9; Part VI §37, §42
Aviation Week	Trade publication; the most-cited press source in the primer.	Part I §2, §4, §5, §6, §10, §11; Part II §14; Part III §18, §19, §20, §23, §24; Part IV §25, §26, §27, §28, §29; Part V §31, §33, §34, §35, §36; Part VI §37, §40, §41, §42, §44; Part VII §46, §49, §50, §51, §52; Part VIII §66; Part IX §68, §69, §74; Part X §75; Part XI §86, §87

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
AVITAS	Appraiser named among value sources.	Part I §8; Part II §12
AviTrader	Trade publication.	Part I §4, §8, §9; Part II §12, §14, §15, §16, §17; Part III §19, §22, §24; Part IV §26, §30; Part V §31, §32, §33, §35; Part VI §41, §43; Part VII §51, §52; Part VIII §61, §62, §67; Part IX §68; Part X §75, §82; Part XI §89
AVM Magazine	Trade publication cited on used-material supply.	Part III §19, §23, §24; Part IV §25, §27; Part V §33; Part VII §52
Avolon	Aircraft lessor named in Part VI among those from which no current return-condition statement was retrieved.	Part VI §42
Bain & Company	Consultancy; slot-wait and capacity-shortfall figures.	Part I §6; Part II §14; Part III §19, §22, §23, §24; Part IV §27, §30; Part V §33; Part VII §46, §55
Baker Hughes (NASDAQ: BKR)	Maker of the NovaLT small industrial turbine line, noted as not aeroderivative.	Part IX §72
Barchart	Financial news site carrying FTAI releases.	Part IX §70; Part X §75, §78
Bauer, Inc.	FTAI's partner in Prime Engine Accessories.	Part VI §43; Part VII §50; Part X §78
BELAC	Chromalloy's early hot-section PMA venture (CFM56-3 HPT blade).	Part VI §41, §42
Benzinga	Financial news site cited on the short report.	Part VII §54; Part X §85
Bernstein	Broker whose Strategic Decisions Conference hosted GE Aerospace's 27 May 2026 presentation.	Part I §6; Part III §23; Part IV §27; Part VII §55; Part IX (unknowns list only)
Blackbird Capital I and II	Air Lease's third-party-capital funds.	Part VIII §65; Part XI §87, §89

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Bleichmar Fonti & Auld (BFA Law)	Plaintiffs' law firm cited on the class action.	Part VII §54; Part X §85
Bloom Energy (NYSE: BE)	Solid-oxide fuel-cell maker named among competing supply.	Part IX §68, §72
Bloomberg	Financial news service.	Part II §12, §15; Part III §19, §22; Part V §33; Part VI §41; Part VIII §66; Part IX §72
Boeing (NYSE: BA)	Airframer of the 737NG and 737 MAX; subject of the FAA production rate cap; Boeing Converted Freighter programme.	Part I §2, §10; Part II §12, §14, §15, §17; Part III §18, §21, §22; Part V §33; Part VII §45; Part VIII §62, §67; Part IX §68, §72; Part X §75; Part XI §87
The Business Research Company	Market-research vendor; USM market size.	Part V §35; Part XI §86
CAAC (China)	Regulator whose PMA acceptance is an open question.	Part VI §39
CAPA (Centre for Aviation)	Trade publication and data provider.	Part III §20, §21, §24; Part V §33
Capital Group (Capital International Investors; Capital World Investors)	FTAI's two largest reported shareholders (13.51% and 12.43%).	Part X §78
Cargo Facts	Trade publication on freighter conversion.	Part I §4, §8; Part II §15; Part III §18, §19, §22, §24; Part V §32, §36; Part VIII §67
Carlyle Aviation Partners (The Carlyle Group) (NASDAQ: CG)	Aviation investment manager; fee comparable for the Strategic Capital vehicles.	Part VI §41; Part VIII §63, §65, §66; Part XI §89
Castlelake	Aviation investment manager named as a comparable not reached.	Part VIII §65; Part XI (unknowns list only)
Caterpillar Inc. (NYSE: CAT)	Gas-genset and reciprocating-engine maker named among competing supply.	Part IX §72

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
CFM International	50/50 joint venture of GE Aerospace and Safran; OEM and type-certificate holder of the CFM56 and LEAP.	Part I §1–§11 (8 sections); Part II §14; Part III §18, §19, §20, §21, §23, §24; Part IV §25, §26, §27, §28, §29; Part V §33, §34, §35; Part VI §37, §39, §40, §41, §42, §43; Part VII §46, §52, §55, §57; Part IX §68, §70, §74; Part X §75, §84, §85; Part XI §86, §87
Chromalloy	Turbine-part repair, casting and PMA company; holder of the only FAA PMA HPT blade for the CFM56-5B/-7B; FTAI's PMA partner.	Part I §2, §5, §7; Part III §18; Part IV §29; Part V (unknowns list only); Part VI §37–§44 (8 sections); Part VII §45, §46, §51, §52; Part X §75, §78; Part XI §87
Cirium (Ascend by Cirium)	Fleet-data and appraisal provider; retirement counts, extension shares and the Global Replacement Ratio.	Part I §8; Part II §12, §15; Part III §18, §19, §20, §21, §22, §24; Part IV §25, §26; Part V §33; Part VIII §66
Condor	German airline; lessee of an APOC CFM56-5A engine.	Part II §16; Part V §32, §35
Copa Airlines (NYSE: CPA)	737NG operator retiring aircraft, named as a counter-case.	Part III §19, §24; Part VIII §67
Crossroads Capital	Investor whose letter on FTAI is cited in Parts VII and X.	Part VII §54; Part X §85
Data Center Dynamics	Data-center trade publication.	Part IX §71, §72, §73
Delta Air Lines (NYSE: DAL)	CFM56 operator affected by the AOG Technics unapproved-parts episode.	Part V §31; Part VI §42
Delta TechOps	Delta Air Lines' MRO arm, named among airline shops.	Part IV §25; Part V §31; Part VII (unknowns list only)
Deutsche Bank AG (NYSE: DB)	Co-lender on the 2025 Partnership's asset-level debt.	Part VIII §61, §62
Deutsche Lufthansa AG (Xetra: LHA)	Parent of Lufthansa Technik.	Part III §19; Part XI §86, §87
DGCA India	Regulator whose PMA acceptance is an open question.	Part VI §39

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
E&E News	Energy and environment news service.	Part IX §73
EgyptAir Maintenance & Engineering	Airline MRO; capacity partner of FTAI (Egypt); buyer of GE overhaul consulting.	Part IV §25, §28; Part VII §50
Embraer (NYSE: ERJ)	Airframer of the E2 family, powered by the GTF.	Part III §20
Engine Lease Finance Corporation (ELFC)	Engine lessor whose 2016 statement on PMA in return conditions is cited.	Part II §14; Part VI §42; Part XI §89
ePlaneAI	Trade and market commentary site.	Part I §1, §2, §8, §10; Part II §16; Part III §18, §20, §23, §24; Part IV §26, §27; Part V §32, §33, §35
ERCOT	Texas grid operator; interconnection queue.	Part IX §71, §73
Ernst & Young LLP	FTAI's auditor 2016 to 2025.	Part VII §54; Part X §75, §85
European Commission	Recipient of IATA's 2018 complaint settled by CFM's Conduct Policies.	Part VI §42
European Union Aviation Safety Agency (EASA)	EU regulator; Form 1; PMA acceptance under the bilateral.	Part I §9, §10; Part III §20; Part V §31; Part VI §37, §38, §39, §40; Part X §85
Federal Aviation Administration (FAA)	US regulator: type and production certificates, PMA, DER, Part 145, the 737 MAX rate cap.	Part I §5, §9, §10; Part III §20, §21, §24; Part IV §29; Part V §31, §33; Part VI §37, §38, §39, §40, §41, §42, §43; Part VII §45, §51, §52; Part X §85; Part XI §86, §87
Finviz	Financial news site carrying FTAI releases.	Part II §17; Part VIII §61, §62, §67; Part X §75, §78, §82; Part XI §89
Fitch Ratings	Rating agency on FTAI MRE 2026-1.	Part II §17; Part VIII §61, §67; Part X §82; Part XI §89
FL Technics	Independent MRO named in Part IV.	Part IV §25
FlightGlobal	Trade publication.	Part I §5, §6; Part II §14; Part III §22; Part IV §26; Part VI §40, §41; Part XI §86

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
FMR LLC	FTAI shareholder (5.25%).	Part X §78
Forecast International	Forecasting firm cited on shop-visit demand.	Part III §23; Part IV §25, §28
Fortress Investment Group / FIG LLC	FTAI's external manager until the internalization of 28 May 2024.	Part VIII (unknowns list only); Part X §75, §76, §77, §78
Fortress Transportation and Infrastructure Investors LLC	FTAI's predecessor, listed on the NYSE in May 2015 and externally managed by Fortress.	Part X §75, §77
Fortune Business Insights	Market-research vendor; USM market size.	Part V §35; Part XI §86
Frontier Airlines (NASDAQ: ULCC)	Airline whose 10-K fleet plan is cited as a source not read.	Part III (unknowns list only)
FTAI Aviation Ltd. (NASDAQ: FTAI)	The subject of the primer: Cayman-incorporated owner of the Aerospace Products and Aviation Leasing segments, the Strategic Capital Initiative and FTAI Power.	Every Part; the company itself is Part X, Aerospace Products Part VII, Aviation Leasing and Strategic Capital Part VIII, FTAI Power Part IX
FTAI Finance Holdco Ltd.	FTAI finance subsidiary and note co-issuer named in SEC filings; role not fully retrieved.	Part X §75, §77, §78
FTAI Infrastructure Inc. (NASDAQ: FIP)	The infrastructure businesses spun off to shareholders on 1 August 2022; appears only as history.	Part X §75, §76, §77, §78
FTAI MRE 2026-1	The 2025 Partnership's inaugural asset-backed securitization, priced 22 May 2026 (\$612M of notes on 48 aircraft).	Part II §17; Part VIII §61, §67; Part X §75, §82
FTAI Power	FTAI's power platform launched 30 December 2025; builds the Mod-1 generator set around a CFM56 core.	Part III §19, §24; Part VII §49, §57; Part VIII §58, §60, §67; Part IX §68, §69, §70, §72, §74; Part X §75, §84

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
FTAI–Chromalloy joint venture (PMA JV)	The joint venture through which FTAI develops and manufactures PMA parts; terms not disclosed in retrieved filings.	Part IV §29; Part VI §38, §43; Part VII §52; Part X §75
GA Telesis	USM, MRO and leasing group named among the players.	Part IV §25, §29; Part V §32, §33, §35; Part VII (unknowns list only)
Garuda Indonesia (IDX: GIAA)	Parent of GMF AeroAsia.	Part VII §50
Gas Turbine World	Energy trade publication and handbook.	Part IX §68, §72, §74
GE Aerospace (NYSE: GE)	CFM partner; builds the CFM56 core; Commercial Engines & Services segment; TrueChoice services. Formerly GE Aviation.	Part I §2, §3, §6, §9, §10, §11; Part II §14, §16; Part III §18, §19, §20, §23, §24; Part IV §25, §26, §27, §28, §29, §30; Part V §33, §34, §35, §36; Part VI §37, §41, §42; Part VII §46, §47, §49, §50, §52, §53, §55; Part VIII §66; Part IX §68, §72, §74; Part X §80, §85; Part XI §86, §87, §88, §89
GE Engine Leasing	GE's engine-leasing arm named among engine lessors.	Part II §14; Part XI §89
GE Vernova (NYSE: GEV)	Frame and aeroderivative gas-turbine maker; 116 GW backlog plus reservations at Q2 2026.	Part III §24; Part IX §68, §71, §72, §74
GECAS	GE's former aircraft-leasing arm; the 2019 GE Aviation and GECAS investor day is a source for fleet figures.	Part III §18; Part IV §26; Part V §35
GenSystems	Jereh subsidiary with \$182M and \$341M US data-center turbine orders.	Part IX §70
Global Industry Analysts	Market-research vendor; PMA market size (2022).	Part VI §41; Part XI §86
Global Information Inc. (GII)	Report reseller listing an engine PMA and DER forecast.	Part V §35; Part VI §41; Part XI §86

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Global Material Solutions	Historical Pratt & Whitney CFM56-3 PMA venture, found only via headlines.	Part IV §28; Part V (unknowns list only); Part VI §41
GlobeNewswire	Press-release distributor for FTAI's releases.	Part I §3; Part V §35; Part VI §42, §43; Part VII §45–§57 (9 sections); Part VIII §62; Part IX §69, §70; Part X §75–§85 (8 sections)
GMF AeroAsia (IDX: GMFI)	Garuda Indonesia's MRO arm; capacity partner of FTAI (Indonesia).	Part III (unknowns list only); Part IV §25; Part VII §50
GuruFocus	Call-coverage aggregator; source of several call figures labelled as such.	Part IV §25; Part VII §49, §50, §57; Part VIII §58; Part IX §69, §70, §74; Part X §78, §79, §80, §82, §84; Part XI §89
Hedge Fund Alpha	Investment commentary site cited on the short report.	Part VII §54; Part X §85
HEICO Corporation (Flight Support Group) (NYSE: HEI)	Largest independent PMA business; FSG operating margin carried in Part XI.	Part I §11; Part IV §29; Part VI §38, §41; Part VII (unknowns list only); Part X §80; Part XI §86, §87, §88, §89
Howmet Aerospace (NYSE: HWM)	Casting and forging supplier whose commentary could resolve lead-time unknowns.	Part IV (unknowns list only)
IAE International Aero Engines	Consortium (Pratt & Whitney, P&W Aero Engines International, JAEC, MTU) that is the OEM of the V2500.	Part I §2, §10; Part III §18, §19, §23; Part IV §28; Part VII §45, §49, §51; Part X §75; Part XI §87
iAero Thrust	The Miami engine business whose assets FTAI and Unical acquired on 4 January 2023.	Part IV §25; Part V §35; Part VII §50; Part X §75
IAG Engine Center (International Airlines Group) (LSE: IAG)	Former operator of the Rome Fiumicino shop that became QuickTurn Europe.	Part IV §25; Part VI §43; Part VII §50
IATA	International Air Transport Association; complainant in the 2018 CFM Conduct Policies settlement; Maintenance Cost Conference.	Part III §18, §19, §23; Part V §33; Part VI §40, §42, §43

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
IBA	Appraiser; engine values, lease rates and conversion-oversupply warning cited throughout.	Part I §1, §6, §8; Part II §12, §13, §14, §15, §16, §17; Part III §18, §19, §22, §23, §24; Part IV §26, §27, §30; Part V §32, §33, §36; Part VI §42; Part VII §55; Part VIII §66, §67; Part IX §69; Part XI §87
IMARC	Market-research vendor; PMA market size.	Part VI §41; Part XI §86
In Practise	Interview publisher; FTAI module-swap interviews located but not read.	Part I §7; Part IV §29; Part V (unknowns list only); Part VI §37, §40, §43, §44; Part VII §47, §52
Industrial Info Resources	Industrial data provider cited in Part IX.	Part IX §71
Informa Markets	Owner of Aviation Week and the MRO conference series.	Part III §19, §21; Part IV §26; Part V §33; Part IX §71
Insider Monkey	Investment commentary site; call transcripts.	Part VI §43, §44; Part VII §45, §47, §50, §52
International Energy Agency (IEA)	Source of the data-center electricity forecast (about 945 TWh by 2030).	Part III §24; Part IX §71
Israel Aerospace Industries (IAI)	Freighter-conversion house named with AEI and Boeing.	Part II (unknowns list only); Part VIII (unknowns list only)
ISTAT	International Society of Transport Aircraft Trading; standardises appraisal terms such as half-life.	Part I §2, §8; Part II (unknowns list only); Part III §18; Part V §32; Part VI §42; Part VII (unknowns list only)
Japanese Aero Engines Corporation (JAEC)	IAE consortium member.	Part I §10; Part IV §28
Jefferson Terminal	Infrastructure business spun off with FTAI Infrastructure in 2022; history only.	Part VI (unknowns list only); Part VII (unknowns list only); Part X §77
Jereh Group (SZSE: 002353)	Chinese oilfield-equipment maker and mobile gas-turbine packager; FTAI's partner in J&F Power Systems.	Part IX §68, §69, §70, §74; Part X §75, §78

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
JetBlue Airways (NASDAQ: JBLU)	Airline whose 10-K fleet plan is cited as a source not read.	Part III (unknowns list only)
J&F Power Systems LLC	FTAI's joint venture with Jerreh Group for packaging and distribution of Mod-1; counterparty to the \$1.465B order of 22 July 2026.	Part IX §68, §69, §70, §71, §74; Part X §75, §78
Kairos Research	Investor blog cited on the Chromalloy PMA release.	Part VI §38, §41, §43, §44; Part VII §52; Part X §75
KBRA (Kroll Bond Rating Agency)	Rating agency on the WEST deals and FTAI MRE 2026-1.	Part II §17; Part VIII §61, §67; Part X §82; Part XI §87, §89
Kessler Topaz Meltzer & Check	Plaintiffs' law firm whose case page is the source for the class-action status.	Part VII §54; Part X §85
Komodo Capital	Investor blog cited on the Chromalloy PMA release.	Part VI §38, §41, §43, §44; Part VII §52; Part X §75
KPMG LLP	Engaged as FTAI's auditor for FY2025, effective 17 June 2025.	Part VII §54; Part X §75, §85
Leeham News	Trade publication covering OEM results and engine technology.	Part I §2, §4, §5, §6, §7, §8, §10; Part II §14; Part III §19, §20, §21, §22, §23, §24; Part IV §25, §26, §28, §30; Part V §33; Part VIII §66; Part IX §74
Lockheed Martin Commercial Engine Solutions (LMCES)	The 526,000 sq ft Montréal engine shop FTAI bought from Lockheed Martin Canada on 9 September 2024 for \$170.0M.	Part IV §25; Part VI (unknowns list only); Part VII §46, §49, §50; Part X §75
Lockheed Martin Corporation (NYSE: LMT)	Seller of the Montréal shop through Lockheed Martin Canada.	Part IV §25
Long Ridge	Power and terminal business spun off in 2022; history only.	Part VI (unknowns list only); Part VII (unknowns list only); Part X §75, §77
Lufthansa Technik	MRO segment of Deutsche Lufthansa; engine services almost half of revenue; adjusted EBIT basis.	Part I §11; Part IV §25; Part VI §41; Part VII §53; Part X §80; Part XI §86, §87, §88

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Magnetic Group (Magnetic MRO)	Independent MRO and trader quoted on CFM56 market conditions.	Part I §4; Part IV §25, §28; Part V §31, §34; Part VI §37; Part XI §86
MarketBeat	Call-coverage aggregator; source of several call figures labelled as such.	Part IV §25, §28; Part VI §42, §43; Part VII §49, §52, §53, §57; Part IX §69, §74; Part X §84; Part XI §86
Market.us	Market-research vendor; USM market size.	Part V §35; Part XI §86
Master GP	Fortress affiliate that received the incentive allocation under the pre-2024 management structure.	Part X §76, §78
mba Aviation	Appraiser named among value sources.	Part I §8; Part II §12, §13; Part IV (unknowns list only); Part V (unknowns list only); Part VIII (unknowns list only)
McKinsey & Company	Consultancy; retirement-rate normalisation from 2028.	Part III §19, §24; Part V §33
Mitsubishi Power	Frame gas-turbine maker; 35 GW backlog cited in Part IX.	Part II §14; Part IX §68, §71
Module Factory (Montréal)	FTAI's module repair and refurbishment programme and site; the operating heart of Aerospace Products.	Part I §3, §10, §11; Part III §24; Part IV §25, §28; Part V §35; Part VII §45, §46, §49, §50, §51, §53; Part VIII §58, §61, §67; Part IX §74; Part X §75, §81, §85; Part XI §86, §87, §89
Moody's (NYSE: MCO)	Rating agency whose FTAI ratings were not retrieved.	Part X §82
Motley Fool	Investment commentary site; call transcripts.	Part VI §43; Part VII §46, §49, §52, §57
MTU Aero Engines AG (Xetra: MTX)	IAE partner and independent MRO (MTU Maintenance); reports commercial maintenance on an adjusted EBIT basis.	Part I §6, §10, §11; Part II §14, §16; Part III §23; Part IV §25, §28, §30; Part VII §51, §53; Part X §80; Part XI §86, §87, §88
MTU Maintenance	MTU's engine MRO arm, independent on the CFM56; also MTU Maintenance Lease Services.	Part I §6, §10; Part II §14; Part IV §25; Part VII §47, §51; Part XI §87, §89

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Muddy Waters Research	Short seller that published the 15 January 2025 report on FTAI.	Part VII §47, §54; Part VIII §64; Part X §75, §85
NAACP	Challenger of the nonroad-engine reading in Memphis.	Part IX §73
Nasdaq	FTAI's listing exchange since the 2022 redomiciliation.	Part IV §25; Part V §35; Part VII §49, §57; Part X §75, §76, §77, §78, §84; Part XI §87
New York Stock Exchange (NYSE)	FTAI's predecessor's listing exchange (2015); several peers' exchange.	Part V §35; Part VI §41; Part X §75, §77; Part XI §86, §87
Oilprice.com	Energy news site.	Part III §24; Part IX §71, §72, §74
Oliver Wyman	Consultancy; MRO forecasts and labour-rate inflation figures.	Part I §7; Part III §23, §24; Part IV §25, §27, §29; Part V §33; Part XI §86, §87
OneIM (One Investment Management)	Partner on the inaugural Strategic Capital vehicle, announced 25 March 2025.	Part VIII §62; Part X §75
Persistence Market Research	Market-research vendor cited in Part VI.	Part VI §41
PJM Interconnection	Mid-Atlantic grid operator; interconnection queue.	Part IX §71, §73
Power Engineering	Energy trade publication.	Part III §24; Part IX §71, §72, §74
PR Newswire	Press-release distributor (AAR, Chromalloy releases).	Part II §16; Part V §35; Part VI §41, §42; Part VII §45; Part IX §70; Part XI §86, §87
Pratt & Whitney	RTX engine unit: GTF and V2500 maker; EngineWise aftermarket brand.	Part I §10; Part III §18, §20, §21, §23, §24; Part IV §28; Part V §33, §35; Part VI §41; Part VII §45, §51; Part XI §86
Precision Castparts (PCC)	Casting and forging supplier named with Howmet.	Part IV (unknowns list only)
Prime Engine Accessories	50/50 accessory-repair joint venture with Bauer, Inc. in Bristol, Connecticut.	Part VI §43, §44; Part VII §45, §46, §47, §50, §52, §53, §56; Part VIII §64; Part X §78, §85

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
ProEnergy	Aeroderivative packager (PE6000 from CF6-80C2 cores) whose customers plan five to seven years of bridging power.	Part IX §68, §72, §73
Quartr	Call and capital-markets-day summary service.	Part III §23; Part IV §25, §26; Part VII §49, §50; Part VIII §58; Part IX §69, §70, §74; Part X §79, §80, §84; Part XI §89
QuickTurn Europe	50%-owned, equity-method CFM56 shop at Rome Fiumicino (former IAG Engine Center), closed 5 June 2025.	Part I §3; Part IV §25; Part VI §43; Part VII §46, §49, §50; Part VIII §64; Part X §75, §78, §85
QuickTurn (Miami)	FTAI's Miami engine maintenance business, formed from iAero Thrust assets in 2023.	Part I §6; Part IV §25; Part V §35; Part VI (unknowns list only); Part VII §46, §50; Part X §75
Repauno	Infrastructure business spun off in 2022; history only.	Part VI (unknowns list only); Part VII (unknowns list only); Part X §75, §77
Robbins Geller Rudman & Dowd	Plaintiffs' law firm whose case page is the source for the class-action status.	Part VII §54; Part X §85
Rolls-Royce (LSE: RR)	Former IAE shareholder (32.5% per Aircraft Commerce, 2010) whose exit was not retrieved; parent of the engine lessor RRPf.	Part I §10; Part II §14; Part IX §72; Part XI §89
Rolls-Royce & Partners Finance (RRPF)	Engine lessor named among engine lessors.	Part II §14; Part XI §89
RTX Corporation (NYSE: RTX)	Parent of Pratt & Whitney; disclosed the GTF powder-metal issue in 2023.	Part I §10; Part III §18, §20, §23, §24; Part IV §28; Part VII §45, §51; Part X §75
Ryanair (NASDAQ: RYAAY)	CFM56-7B operator with a TrueChoice Material agreement covering about 2,000 engines.	Part III §19, §22; Part IV §25, §28; Part V §33; Part VI §42; Part VII §52; Part XI §87

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Safe Fly Aviation	Consultancy blog; CFM56 market figures carried with that label throughout.	Part I §2–§11 (8 sections); Part II §12, §13, §14, §15, §16; Part III §18, §19, §22, §23, §24; Part IV §26, §27, §28, §29, §30; Part V §32, §33, §34, §35, §36; Part VI §38, §40, §44; Part VII §46, §47, §52, §55; Part VIII §66, §67; Part IX §69, §74; Part XI §86
Safran SA (Euronext Paris: SAF)	CFM partner through Safran Aircraft Engines; builds the fan, booster, LPT and gearbox; Propulsion services revenue carried in Part XI.	Part I §2, §4, §10, §11; Part II §14; Part III §18, §19, §20, §23, §24; Part IV §25, §26, §28, §29; Part V §33, §34, §35; Part VI §37, §42, §43; Part VII §49; Part VIII §65; Part IX §74; Part XI §86, §87, §88, §89
Seeking Alpha	Investment commentary site.	Part VII §47, §54
Sequa Corporation	Chromalloy's parent holding company.	Part IV §29; Part VI §41
Setna iO	USM trader named in Part V.	Part V §35
Shannon Engine Support (SES)	AerCap's 50% engine-leasing joint venture with Safran Aircraft Engines.	Part II §14; Part V §35; Part VIII §65; Part XI §87, §88, §89
Shelby County Health Department	Tennessee permitting authority that accepted the nonroad-engine reading for xAI's Memphis turbines.	Part IX §73
Siemens Energy (Xetra: ENR)	Gas-turbine maker; lead times of three-plus years cited in Part IX.	Part III §24; Part IX §68, §71, §72
SMBC Aero Engine Lease	Engine lessor named among the engine lessors whose scale was not retrieved.	Part II §14; Part XI §89
Solar Turbines (Caterpillar)	Industrial gas-turbine maker (Titan 350) named among competing supply.	Part IX §72
Southern Environmental Law Center	Challenger of the nonroad-engine reading in Memphis.	Part IX §73
Southwest Airlines (NYSE: LUV)	CFM56-7B operator; TrueChoice Flight Hour reference customer.	Part III (unknowns list only); Part IV §28; Part VI §40, §41, §42, §44

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
S&P Global (NYSE: SPGI)	Rating agency whose FTAI ratings were not retrieved.	Part IX §68, §71; Part X §82
Spirit Airlines	A320ceo operator whose 10-K language on maintenance reserves is quoted in Part II; its fleet plan is cited as a source not read.	Part II §12, §13; Part III §22
SR Technics	Independent MRO; its 2026 CFM56-7B price catalogue is cited as a possible source.	Part IV §25; Part VI §44; Part VII (unknowns list only); Part XI (unknowns list only)
ST Engineering (SGX: S63)	Singapore engineering group with an engine MRO arm named among independents.	Part IV §25, §27; Part VII (unknowns list only)
StandardAero, Inc. (NYSE: SARO)	Independent engine MRO; CFM56-7B Center of Excellence in Dallas; Engine Services and Component Repair Services segments.	Part I §3, §4, §6, §7, §11; Part IV §25, §27; Part VI §38; Part VII §46, §50, §53; Part X §80, §85; Part XI §86, §87, §88, §89
Strategic Capital Initiative (SCI)	FTAI's asset-management arm, launched 30 December 2024, raising third-party vehicles that buy mid-life 737NG and A320ceo aircraft.	Part II §17; Part III §22; Part V §35; Part VII §45, §48, §49, §54; Part VIII §60–§67 (8 sections); Part X §75, §78, §79, §80, §84; Part XI §87
StreetInsider	Financial news site carrying FTAI's CFO release.	Part X §75, §78
Sumisho Air Lease Corporation	Surviving entity after Air Lease's take-private.	Part VIII §65; Part XI §87
Tarmac Aerosave	Storage and dismantling facility at Toulouse-Francazal.	Part V §32, §35
Tennessee Lookout	News site covering the Memphis turbine permitting dispute.	Part IX §73
Thunderbolt	Air Lease's mid-life portfolio-sale platform.	Part VIII §65; Part XI §87, §89

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
TipRanks / The Fly	Financial news service cited on the short report and sell-side reaction.	Part VII §54; Part VIII §64
Transport Canada (TCCA)	Regulator whose PMA acceptance is an open question.	Part VI §39
Transtar	Short-line railroad business spun off in 2022; history only.	Part VI (unknowns list only); Part VII (unknowns list only); Part X §75, §77
Turkish Airlines (BIST: THYAO)	Parent of Turkish Technic; named for grounding more than 40 aircraft in the GTF episode.	Part III §20
Turkish Technic	Turkish Airlines' MRO arm, named among airline MROs.	Part IV §25; Part VII (unknowns list only)
UK Civil Aviation Authority	Regulator whose PMA acceptance is an open question.	Part VI (unknowns list only)
Unical Aviation	USM trader; FTAI's former joint-venture partner in QuickTurn Miami (interest bought out 1 December 2023).	Part IV §25; Part V §35; Part VII §50, §53; Part X §75
United Airlines (NASDAQ: UAL)	737NG operator retiring aircraft; affected by the AOG Technics episode.	Part III §19, §24; Part VI §41, §42; Part VIII §67
US District Court, Southern District of New York	Court hearing the securities class action No. 25-cv-00541 (Judge Jeannette A. Vargas).	Part VII §54; Part X §85
US Patent and Trademark Office	Cited in Part IX.	Part IX §68
US Securities and Exchange Commission (SEC)	Regulator receiving FTAI's filings; EDGAR is the primer's filing source.	Part I §11; Part III (unknowns list only); Part IV §25; Part VII §45, §54; Part X §75, §77, §78, §82, §85; Part XI §86, §87, §88
Utility Dive	Energy trade publication.	Part IX §71
The Vanguard Group	FTAI shareholder (10.10%).	Part X §78
VAS Aero Services	USM trader named among the players.	Part V §35

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
Veritas Capital	Private-equity owner of Chromalloy since December 2022.	Part IV §29; Part VI §41
Visual Approach Analytics	Analytics firm forecasting the 2027 peak in CFM56 and V2500 first shop visits.	Part I §10; Part III §20, §23, §24; Part IV §26
Wencor Group	PMA and distribution company acquired by HEICO in 2023.	Part VI §41; Part XI §87
WestJet	Canadian airline; sold 27 737-700s to FTAI and the 2026 SPV on 28 September 2026.	Part II §13, §15, §17; Part III §19, §24; Part V §35; Part VII §46; Part VIII §62, §67; Part X §75
Willis Aero	The parts arm of Willis Lease Finance.	Part V §32, §35; Part XI §86, §87
Willis Engine Structured Trust (WEST)	Willis Lease's series of rated engine ABS vehicles (WEST VIII, WEST IX).	Part II §14, §17; Part VIII §61, §63, §65, §66; Part XI §87, §89
Willis Lease Finance Corporation (NASDAQ: WLFC)	Pure-play engine lessor; WEST securitizations; peer in Part XI.	Part II §12, §13, §14, §15, §17; Part III §20, §22, §24; Part V §32, §35; Part VI §42; Part VII (unknowns list only); Part VIII §58, §60, §61, §63, §65, §66; Part XI §86, §87, §88, §89
Wizz Air (LSE: WIZZ)	GTF operator named for its A321ceo phase-out plan and its compensation agreement with Pratt & Whitney.	Part III §19, §20
Wolfe Research	Sell-side firm quoted after the January 2025 short report.	Part VII §54
Wood Mackenzie	Energy consultancy; gas-turbine order and capacity comparison.	Part IX §71, §72, §74
Wärtsilä (Nasdaq Helsinki: WRT1V)	Maker of large reciprocating gas engines competing for data-center power.	Part IX §68, §72

Company, vehicle or institution (ticker where listed)	Role in the primer	Where it appears
xAI	Operator of the Memphis data center whose trailer-mounted turbines tested the nonroad-engine permitting question.	Part IX §68, §72, §73
Yahoo Finance	Financial news aggregator.	Part VI §43; Part IX §71

Source: the eleven Part files in parts/, searched on 2026-10-03 for a curated list of names and their variants; section numbers are those of the outline. Roles are summarised from the Parts named in each row. Tickers are listing identifiers and are not research findings; Air Lease's ticker is given as former because the company was taken private on 8 Apr 2026 (Part XI §87).

94. Unknowns register

The reconciliation's section D is the consolidated register of what the eleven dossiers could not find: 176 entries, each copied verbatim from its dossier with an id of the form U-D5-07 (dossier D5, its seventh unknown). The reconciliation grouped the entries that ask the same question into 19 clusters covering 44 entries and left the other 132 as singletons. This section reproduces the register in that order: the clusters first (§94.1), then the singletons grouped by the Part that owns them (§94.2), then a table of every entry the orchestrator notes or the Parts resolved in whole or in part during the build, with the resolving fact and its source (§94.3). Every question is given in the words its dossier used, including the dossier's own numbering (U1, U-1) and its internal source references in square brackets, which point to that dossier's source list. Each entry also says which Parts carry it in their closing unknowns lists. An entry that §94.3 lists as resolved is left unchanged in §94.1 or §94.2, so that the register remains the record of what was asked; the resolution sits only in the table. The register is long, and that is its purpose: the primer never fills an absence, so the absences must be listed.

94.1 CLUSTERS: ENTRIES THAT ASK THE SAME QUESTION

The reconciliation found 19 clusters covering 44 of the 176 entries. Each cluster below gives the verbatim question of every entry in it, the dossiers that raised it, what the research already holds, and what would resolve it. The cluster titles and the resolution notes are the reconciliation's; nothing in the quoted entries has been altered.

K-01 — Module-exchange pricing

Raised by D2, D3, D5. Rationale, as the reconciliation states it: all three ask what a CFM56 module exchange costs the customer above the LLP life transferred; D5's entry also asks the turnaround, which no other dossier asks.

- **U-D2-07 (D2) — U7. Module-exchange pricing.** No source gives the exchange fee or a worked price for a CFM56 HPT or core exchange; Safe Fly's "\$150,000–\$400,000 removal cost per module" is ambiguous. D5 may find it in FTAI's investor materials or the In Practise interview ("FTAI Aviation: CFM56 repair and module swap", URL returned, not read). *Carried in Parts IV, V, VII.*
- **U-D3-20 (D3) — U20 — Pricing of a module exchange (exchange fee, LLP-life true-up)** from any provider, including FTAI. D5 may hold FTAI's own disclosures. *Carried in Parts IV, V, VII.*
- **U-D5-03 (D5) — U3 — Exchange turnaround time and exchange pricing.** FTAI's quoted turnaround and any published exchange-fee ranges. Trade press (e.g., the In Practise interview [S38]) likely has estimates; not read. *Carried in Parts IV, V, VII.*

What would resolve it. FTAI investor materials or the In Practise interviews (located by D2 and D5, not read), or an MRO price catalogue listing exchange options (the SR Technics 2026 catalogue, located, not read). Related, not merged: U-D1-14 (whether an exchange needs a test-cell run).

K-02 — Appraiser values for CFM56 engines

Raised by D1, D2, D3, D6. Rationale, as the reconciliation states it: one appraiser engine-value table (IBA, mba or Cirium, with half-life, full-life and by rating) answers every one of these; D6's entry additionally asks for aircraft values, which no engine table gives.

- **U-D1-09 (D1) — U9 Current half-life base values for -7B and -5B by rating (IBA, mba, Cirium).** Resolve with an appraiser publication; this session's searches returned definitions, not values. *Carried in Parts I, II, V.*
- **U-D2-08 (D2) — U8. Value of a freshly overhauled full-stack CFM56-7B** versus a green-time unit. Needed to close the worked example. Resolve via IBA/Ascend/mba engine value tables. *Carried in Parts IV, V.*
- **U-D3-04 (D3) — U4 — Any mba Aviation (or Ascend by Cirium) CFM56-5B/-7B value or lease-rate figure.** None returned. *Carried in Parts II, V.*
- **U-D3-05 (D3) — U5 — A current appraiser half-life value for the CFM56-5B (only trend commentary and a trade-press lease-rate range were captured).** *Carried in Parts I, II, V.*

- **U-D6-09 (D6) — U9 — Half-life market values** for 12–15-year-old 737-800 and A320ceo aircraft, and full-life CFM56-7B/-5B/V2500 values, 2023–2026. Resolve: IBA/Cirium/mba published updates (D3 may hold engine values). *Carried in Parts II, V, VIII.*

What the research holds. Partly closed: D3 holds IBA's half-life market values for the -7B24 (\$5.7M) and -7B27 (\$6.4M), Sept 2025; nothing holds a -5B value, a base value, a full-life value or an aircraft value. **What would resolve it.** the IBA Engine Values release in full, mba's CFM56-5B Market Insight (located by D6, not read), or a Cirium Ascend values extract.

K-03 — CFM catalogue escalation by year

Raised by D1, D2. Rationale, as the reconciliation states it: both ask for the dated percentage of each CFM spare-parts price revision; D1 asks for LLPs specifically and D2 for the catalogue generally, but the same annual revision sets both.

- **U-D1-10 (D1) — U10** Years and amounts of each CFM LLP escalation step 2019–2025 (only "Aug 2023, +20–30% on -7B parts" and "~12% LLP escalation" without a year were reached) [33][34]. *Carried in Parts I, IV.*
- **U-D2-01 (D2) — U1. Catalog escalation 2024, 2025, 2026.** Only the Nov 2022 (double-digit) and Aug 2023 (high-single-digit) increases and Safran's "inflation + 3–4 points" rule were found. Resolve with Aviation Week/Aircraft Commerce coverage of each year's CFM price revision, or GE/Safran earnings call Q&A on "price". *Carried in Parts I, IV.*

What would resolve it. Aviation Week or Aircraft Commerce coverage of each year's revision, or GE/Safran call Q&A on price. Related, not merged: U-D11-16 (whose price-increase clause is the same question but which bundles a shop-visit count and a programme-wording question).

K-04 — FTAI's disclosure of the Chromalloy joint venture

Raised by D4, D5. Rationale, as the reconciliation states it: both are answered by the same 10-K footnote on the JV (its terms and its equity-method line); the identity question raised by the orchestrator notes (whether the "Advanced Engine Repair JV" is this venture) sits in the same footnote.

- **U-D4-03 (D4) — U3 — FTAI–Chromalloy JV terms:** formation date, ownership split, accounting (consolidated vs equity method), exclusivity, "at cost" pricing, revenue recognized. Resolve by reading FTAI's FY2018–FY2025 10-Ks for the JV footnote and any 8-K. *Carried in Parts VI, VII.*
- **U-D5-11 (D5) — U11 — Chromalloy JV contribution in FTAI's numbers:** equity-method earnings line, PMA share of parts, FTAI-stated savings per module. *Carried in Parts VI, VII.*

What would resolve it. the investments note in the FY2018–FY2025 10-Ks and any 8-K on the venture. Related, not merged: U-D10-13 (all JV ownership percentages), U-D4-12 (Chromalloy's production capacity).

K-05 — A stated PMA discount to OEM list price

Raised by D4, D11. Rationale, as the reconciliation states it: each asks for a disclosed percentage discount of a PMA part to the OEM part; the HEICO portfolio and consolidated-figure questions ride on the same filing read.

- **U-D4-02 (D4) — U2 — PMA price discount for the Chromalloy HPT blade and vane** (and HEICO's typical discount). Resolve with Chromalloy/FTAI pricing disclosure or MRO price catalogues (the SR Technics 2026 CFM56-7B catalogue may list PMA options [71]). *Carried in Parts VI, VII, XI.*
- **U-D4-09 (D4) — U9 — HEICO portfolio split** by part category (consumables vs rotating vs hot-section) and HEICO's PMA discount. *Carried in Parts VI, XI.*
- **U-D11-05 (D11) — U5 — HEICO consolidated FY2025 net sales and operating income; PMA cumulative count; stated discount to OEM; acceptance statements.** Searched [S7, S7a, S8]. Only FSG figures and the 400–550 PMA rate were returned. Resolve: read Item 1 and Item 7 of [S7]. *Carried in Parts VI, XI.*

What would resolve it. a Chromalloy or FTAI price statement; HEICO's 10-K Items 1 and 7; an MRO catalogue listing PMA alternatives.

K-06 — The WestJet transaction

Raised by D6, D7. Rationale, as the reconciliation states it: both ask what happened to the 27 737-700s (price, lease terms, and where the ten off-lease aircraft sit).

- **U-D6-13 (D6) — U13 — WestJet transaction terms.** Purchase price, lease term and rent for the 17 leased-back aircraft, the holding entity for the 10 off-lease aircraft, and whether the 10 are booked as leasing equipment or Aerospace Products inventory. Resolve: Q3 2026 10-Q (not yet filed) and D7. *Carried in Part VIII.*
- **U-D7-14 (D7) — U14 — WestJet.** Disposition of the other 10 of 27 737-700s (balance sheet, part-out, FTAI Power cores). *Carried in Part VIII.*

What would resolve it. the Q3 2026 10-Q (not yet filed at the research date).

K-07 — The investments note: which equity-method stakes, at what percentage and carrying value

Raised by D6, D10. Rationale, as the reconciliation states it: one note in the FY2025 10-K answers both; D10 also asks for the subsidiary list and the finance co-issuer, which the same exhibit set covers.

- **U-D6-11 (D6) — U11 — Equity-method investment** in the Leasing segment: counterparty, ownership, assets, income by year. Resolve: FY2025 10-K "Investments" note. *Carried in Parts VIII, X.*
- **U-D10-13 (D10) — U-13 — Subsidiaries, JV ownership percentages and equity-method carrying values; the role of FTAI Finance Holdco Ltd.** Resolve with Exhibit 21.1 [S27] and the investments note in [S1]. *Carried in Parts VIII, X.*

What the research holds. Partly closed by the orchestrator notes: 2025 Partnership 19% (20% in the Q3 2025 10-Q), \$281,740K; Advanced Engine Repair JV 25%, \$22,429K; QuickTurn Europe 50%, \$9,987K; Prime Engine Accessories 50%, carrying value not returned; Exhibit 21.1 URL located. Still open: which investment the Aviation Leasing segment leases "through"; FTAI Finance Holdco's role.

K-08 — FTAI's share of the 2025 Partnership's earnings and its first distribution

Raised by D7. Rationale, as the reconciliation states it: both ask for FTAI's share of the Partnership's cash and earnings in 1H 2026; the distribution's amount and FTAI's share appear in both entries.

- **U-D7-07 (D7) — U7 — Equity-method income.** FTAI's share of Partnership earnings by period; FTAI's share of the 30 Jun 2026 distribution. *Carried in Part VIII.*
- **U-D7-13 (D7) — U13 — The 30 Jun 2026 distribution.** Amount, source (rent versus sales), FTAI's share. *Carried in Part VIII.*

What would resolve it. the Q2 2026 10-Q's equity-method and related-party notes.

K-09 — Economic terms of FTAI's servicing arrangement with the 2025 Partnership

Raised by D7. Rationale, as the reconciliation states it: both are terms of the same limited-partnership and servicing agreements; D7 itself says "Resolve as U1".

- **U-D7-01 (D7) — U1 — Management/servicing fee rate and base.** Only "customary, market-based compensation" [S11]. Resolve: the limited partnership agreement or servicing agreement if filed as an exhibit; the 10-K related-party note text; the Q2 2026 10-Q exhibits (ex10.8, ex10.9 were listed in the filing index but not read). *Carried in Part VIII.*

- **U-D7-02 (D7) — U2 — Promote / carried interest.** Only "contractual profit-sharing arrangement" [S13]. Resolve as U1. *Carried in Part VIII.*

What would resolve it. the agreements if filed as exhibits (Q2 2026 10-Q ex10.8 and ex10.9 were listed in the filing index and not read; the orchestrator's gap-fill item 5 was not executed). Related, not merged: U-D7-05 (how aircraft sold into the vehicles are priced).

K-10 — Whether FTAI Power is a reportable segment

Raised by D8, D10. Rationale, as the reconciliation states it: identical question.

- **U-D8-07 (D8) — U7. Segment reporting.** The 10-K FY2025 has two reportable segments [6]; call summaries speak of a "Power segment" for 2027. Whether Power is a third reportable segment from the Q2 2026 10-Q [10], or sits within Aerospace Products until first delivery, was not confirmed. *Carried in Parts IX, X.*
- **U-D10-18 (D10) — U-18 — Whether FTAI Power is a third reportable segment in the 2026 10-Qs.** Resolve with [S2] segment note. *Carried in Parts IX, X.*

What would resolve it. the segment note of the Q1 or Q2 2026 10-Q.

K-11 — Status of the securities class action

Raised by D5, D10. Rationale, as the reconciliation states it: the litigation-status question is identical; D10's entry also asks the auditor's name and the internal-control conclusion, which are different questions and remain partly open.

- **U-D5-16 (D5) — U16 — Securities litigation status** (court, docket, motion to dismiss outcome, any settlement). *Carried in Parts VII, X.*
- **U-D10-16 (D10) — U-16 — Legal proceedings, auditor and internal control.** Status of the securities class actions filed after the Jan 2025 report and any derivative suits; the auditor's name (recollection, unverified: Ernst & Young LLP); whether any material weakness was disclosed; the content of the Jan 2025 EDGAR correspondence [S12 folder] and the later 2025 8-K [S25]. Resolve with Item 3, Item 9A and the auditor's report in [S1]. *Carried in Parts VII, X.*

What the research holds. Partly closed by the orchestrator notes: S.D.N.Y. No. 25-cv-00541, Judge Vargas, class period 23 Jul 2024 to 21 Jan 2025, motion to dismiss filed 20 Nov 2025 and pending (law-firm pages); auditor line with the ambiguity recorded in R-066; internal-control conclusion not returned. **What would resolve it.** Item 3, Item 9A and the auditor's report in the FY2025 10-K.

K-12 — Deal terms for FTAI's maintenance facilities

Raised by D5, D10. Rationale, as the reconciliation states it: the dates and prices of LMCES, QuickTurn and the Chromalloy JV are asked in both; D10's entry also asks for the infrastructure businesses' figures and D5's for Orange and Lisbon, which the other does not.

- **U-D5-07 (D5) — U7 — Facility deal terms.** LMCES purchase price; QuickTurn Miami date and price; what and where "Orange" is and when acquired; Lisbon's size, capacity and opening date (chain map: 113,000 sq ft, 300+ modules/yr); verification of Rome at \$10.5M for 50%. *Carried in Part VII.*
- **U-D10-11 (D10) — U-11 — Early asset mix and the shift to aviation.** Figures for Jefferson Terminal, Repauno, Transtar and Long Ridge; dates and prices of QuickTurn, the Chromalloy JV and the Lockheed Martin Commercial Engine Solutions acquisition. Resolve with the FY2021 10-K [S15], the FY2022 annual report and the relevant 8-Ks. *Carried in Parts VI, VII, X.*

What the research holds. Partly closed by the orchestrator notes: LMCES \$170.0M, 9 Sep 2024, 526,000 sq ft; QuickTurn 4 Jan 2023 (with Unical) and 1 Dec 2023 (\$30.3M for Unical's interest); Rome \$10.5M for 50%, 200,000 sq ft, closed 5 Jun 2025. Still open: Orange (location unsourced, size and date not returned), Lisbon (unverified), the Chromalloy JV's deal terms, and the infrastructure figures. **What would resolve it.** the relevant 8-Ks, the FY2021 10-K and the FY2022 annual report.

K-13 — Headcount

Raised by D5, D10. Rationale, as the reconciliation states it: both ask the company's headcount; D5 also asks test cells and by-site counts, D10 also asks the properties list.

- **U-D5-09 (D5) — U9 — Test-cell count; employees by site and in total.** *Carried in Part VII.*
- **U-D10-17 (D10) — U-17 — Employees and locations.** Headcount at 31 Dec 2025, by geography; properties list (New York headquarters, Montreal, Miami, Rome NY, Lisbon). Resolve with Item 1 "Human Capital" and Item 2 of [S1]. *Carried in Parts VII, X.*

What the research holds. Partly closed by the orchestrator notes: 985 full-time employees and independent contractors at 31 Dec 2025; 494 full-time employees in Canada, about 71% under collective bargaining agreements; test capacity "more than 500" (Sep 2024) and "more than 600" (Feb 2025) engine tests a year. Still open: employees by site; test-cell count; the properties list. **What would resolve it.** Items 1 ("Human Capital") and 2 of the FY2025 10-K.

K-14 — Willis Lease's stated lease rate factor

Raised by D6, D11. Rationale, as the reconciliation states it: both ask whether Willis discloses a lease rate factor (D6 also asks FTAI's, which no filing states; D11 also asks the CFM56/LEAP split).

- **U-D6-05 (D6) — U5 — Lease-rate factors.** No stated LRF for FTAI or WLFC; WLFC engine count, average lease term and FY2025 totals. Resolve: WLFC FY2025 10-K ([S13]) Item 1 and Item 7. *Carried in Parts II, VIII, XI.*
- **U-D11-01 (D11) — U1 — Willis Lease lease rate factor and CFM56/LEAP portfolio split.** Searched the FY2025 10-K [S1]. Not returned. Resolve: read the 10-K's "Lease Portfolio" and "Market" sections directly. *Carried in Parts II, VIII, XI.*

What the research holds. Partly closed: D11 derives about 1.0% a month for Willis from the 10-K; D6 derives 0.87–1.1% for FTAI. No stated figure exists. **What would resolve it.** WLFC 10-K FY2025 Items 1 and 7.

K-15 — Willis Aero and Willis Lease's parts revenue

Raised by D3, D11. Rationale, as the reconciliation states it: both ask for Willis Lease's spare-parts revenue line for 2025; D3's portfolio value and engine count are supplied by D11 (363 engines, \$2,801.7M equipment).

- **U-D3-12 (D3) — U12 — Willis Lease portfolio value, number of engines, and Willis Aero spare-parts revenue for 2025.** *Carried in Parts V, XI.*
- **U-D11-03 (D11) — U3 — Willis Lease maintenance services and spare-parts revenue lines for FY2025.** The 10-K's revenue table was not returned beyond lease rent, the combined core line and gain on sale. Resolve: read the consolidated statement of income in [S1]. *Carried in Parts II, V, XI.*

What would resolve it. the consolidated statement of income in the WLFC 10-K FY2025.

K-16 — GE's own used-material and "material solutions" programme

Raised by D3, D4. Rationale, as the reconciliation states it: both ask what GE's CFM56 used-material or material-solutions offering is and how large; D4's entry also asks for a CFM statement on the Chromalloy blade, which no dossier found.

- **U-D3-15 (D3) — U15 — GE Aerospace's used-material arm: name, scale and CFM56 role not captured** (the only hit was a TrueEngine presentation URL whose content was not read). *Carried in Parts IV, V.*
- **U-D4-10 (D4) — U10 — Any CFM/GE statement specifically on the Chromalloy HPT blade PMA, and any GE "material" programme** (e.g. TrueChoice Material) positioning

against PMA. The brief's phrase "material solutions" was not matched to a current CFM/GE programme in this session; the historical P&W "Global Material Solutions" CFM56-3 PMA venture was found only via headlines [51][52]. *Carried in Parts IV, V, VI.*

What the research holds. Partly closed: D2 quotes GE (2026) that "GE and CFM are the largest used serviceable material provider for their products" and describes TrueChoice Material and TrueChoice Transitions. Name and scale remain open. **What would resolve it.** GE's CES segment discussion and the TrueEngine/TrueChoice materials (located by D3 and D4, not read). Related, not merged: U-D11-16 (programme wording plus shop-visit count plus pricing).

K-17 — Segment revenue and Adjusted EBITDA by year

Raised by D5, D6, D10. Rationale, as the reconciliation states it: all three are answered by the FY2025 10-K's segment tables (Item 7 and the segment note); D6 also asks for each Leasing line and the 2025 quarters, D10 also asks the full statement of operations and the Adjusted EBITDA reconciliation.

- **U-D5-19 (D5) — U19 — FY2023 segment revenue and Adjusted EBITDA**, and the FY2022 baseline (the segment began ramping in 2022–2023 per the 10-Q language [S10 summary]). *Carried in Parts VII, X.*
- **U-D6-02 (D6) — U2 — Annual segment lines 2023–2025.** Lease income, maintenance revenue, asset sales revenue and cost, gain on sale of leasing equipment, equity-method pick-up, and segment Adjusted EBITDA for FY2023, FY2024, FY2025 and each 2025 quarter. Resolve: segment note and MD&A of the FY2025 10-K ([S1]) and the Q4/FY2025 release ([S6]). (The 18th search aimed at this and was refused on budget.) *Carried in Parts VIII, X.*
- **U-D10-01 (D10) — U-1 — Full income statement 2021–2025 and 1H 2025 / 1H 2026.** Operating expense lines, depreciation and amortization, interest expense, tax, net income attributable to shareholders by year; annual Adjusted EBITDA, its definition and reconciliation items; segment revenue and Adjusted EBITDA by year; 1H totals. Resolve with the FY2025 10-K Item 8 and Item 7 segment tables [S1], the Q4/FY2025 release [S3], the Q2 2026 release [S4] and the Q2 2025 release [S25]. Also the FY2021 10-K [S15] and FY2022 annual report (EDGAR index) for 2021–2022. *Carried in Parts VII, X.*

What the research holds. Partly closed by the orchestrator notes: the segment table for 2023–2025 (revenues, total Adjusted EBITDA, Aerospace products revenue, MRE Contract revenue, Aerospace Products and Aviation Leasing Adjusted EBITDA, lease income with two readings for 2024) and FY2025 expense lines (operating expenses, G&A, D&A, interest). Still open: 2023 Leasing lines; maintenance and asset-sales lines for 2023–2024; the Adjusted EBITDA reconciliation; net income attributable by year; 1H totals; 2021–2022 with the "70,800" artefact

and the R-053 contradiction unresolved. **What would resolve it.** the FY2025 10-K Items 7 and 8 read directly, and the Q4/FY2025 release.

K-18 — Capital tied up in module inventory

Raised by D5, D10. Rationale, as the reconciliation states it: both ask how much cash the inventory build consumes (balance-sheet balances in D5; the working-capital and investing lines in D10); D10's entry also asks whether the company defines free cash flow.

- **U-D5-12 (D5) — U12 — Inventory at 31 Dec 2025 and 30 Jun 2026; segment capex; the cost of building module inventory.** *Carried in Parts VII, X.*
- **U-D10-02 (D10) — U-2 — Cash flow detail.** Acquisition of leasing equipment, acquisition of property, plant and equipment, proceeds from asset sales, working-capital changes (inventory), and whether the company defines free cash flow. Resolve with the cash flow statement in [S1] and [S2]. *Carried in Parts VII, X, XI.*

What the research holds. Known: inventory \$1,364.3M at 31 Mar 2026 (D5); operating cash flow −\$310.7M (2025), −\$188.0M (2024), +\$129.0M (2023) (D10). **What would resolve it.** the cash-flow statement and inventory note in the FY2025 10-K and the Q2 2026 10-Q.

K-19 — Terms of the January 2026 CFM materials agreement

Raised by D4, D5. Rationale, as the reconciliation states it: identical question about one agreement.

- **U-D4-14 (D4) — U14 — Terms of the FTAI–CFM Jan 2026 materials agreement** (duration, volumes, pricing, whether it restricts FTAI's PMA use). *Carried in Parts IV, VI, VII.*
- **U-D5-18 (D5) — U18 — Terms of the CFM materials agreement** mentioned on the Q4 2025 call. *Carried in Parts IV, VII.*

What the research holds. Known: the 22 Jan 2026 release's description (OEM replacement parts, thrust performance upgrades, component repair; multi-year). **What would resolve it.** the release body (located, not read) and any 10-Q description.

94.2 SINGLETONS, BY THE PART THAT OWNS THEM

The remaining 132 entries stand alone. They are grouped by the Part whose dossier raised them (D6 entries on leasing mechanics sit under Part II and D6 entries on FTAI's own fleet under Part VIII, following the Parts' own unknowns lists). The note after each entry says which Parts carry it in their closing unknowns lists.

Part I, first principles: the CFM56 as a machine (dossier D1): 12 entries

- **U-D1-01** (D1) — **U1** Full numbered sub-module list for the -7B and -5B (the ESM breakdown). Resolve with the CFM56-7B/-5B Engine Shop Manual or a CFM training manual. *Carried in Part I.*
- **U-D1-02** (D1) — **U2** Exact list of part numbers common to the -5B and -7B core (basis for cross-variant module and LLP interchange). Resolve with the Illustrated Parts Catalogues or a CFM commonality bulletin. *Carried in Part I.*
- **U-D1-03** (D1) — **U3** Per-part LLP table for the -5B (only the 18-set, 20,000–30,000 range was reached). Resolve with a -5B records mini-pack (AJW, StandardAero publish them). *Carried in Part I.*
- **U-D1-04** (D1) — **U4** Prices of used LLP packages with stated cycles remaining. Resolve with dealer quotes or D3's sources. *Carried in Part I.*
- **U-D1-05** (D1) — **U5** As-new EGT margin by -7B/-5B thrust rating and typical loss per 1,000 cycles. Resolve with Aircraft Commerce Issue 58 (-7B) and Issue 50 (-5B) maintenance analyses in full. *Carried in Part I.*
- **U-D1-06** (D1) — **U6** First-run and mature-run intervals by thrust rating and region. Same sources as U5. *Carried in Part I.*
- **U-D1-07** (D1) — **U7** Labour hours per workscope and shop labour rates (US\$/hour) for CFM56 visits. Resolve with an MRO price catalogue or Aircraft Commerce's maintenance budgets. *Carried in Parts I, IV.*
- **U-D1-08** (D1) — **U8** Core-only LLP group price as a share of the full set. Resolve with a CFM catalogue extract. *Carried in Part I.*
- **U-D1-11** (D1) — **U11** Formal regulatory definition of back-to-birth traceability and of dual release; this session reached industry descriptions only [14][40][41]. *Carried in Part I.*
- **U-D1-12** (D1) — **U12** Current LEAP delivery total and the dates/years of -5A and -5C production end. *Carried in Parts I, III.*
- **U-D1-13** (D1) — **U13** V2500 module list and current per-visit cost (2010 figures only) [38]. *Carried in Parts I, VII.*
- **U-D1-14** (D1) — **U14** Whether a module exchange requires a test-cell run under the CFM56 ESM. *Carried in Parts I, IV, VII.*

Part II, first principles: how money moves around an engine (D6 mechanics): 2 entries

- **U-D6-06** (D6) — **U6** — **ABS coupons and LTVs.** WEST VIII and IX note coupons, initial LTVs, and E-certificate sizes. Resolve: KBRA pre-sale reports ([S16],[S17]) and WLFC 8-Ks. *Carried in Part II.*

- **U-D6-14 (D6) — U14 — Current engine-lessor scale.** Portfolio size and value for GE Engine Leasing, SMBC Aero Engine Lease, ELF, AerCap Engines, RRPf, MTU Maintenance Lease Services. Resolve: company websites and parent filings (D11 for the listed ones). *Carried in Parts II, XI.*

Part III, the context and the translation table (D9): 11 entries

- **U-D9-01 (D9) — U1 — 2026 in-service counts and average ages for the A320ceo family and the 737NG, separately, with parked share.** Only a combined A320 family in-service figure (11,308, July 2026 [S13]) and a 2023 stored-A320ceo figure (~570, avg 17.1 years [S14]) surfaced. Resolve: Cirium Fleets Analyzer or IBA Insight extracts, or the fleet tables in Aviation Week's 2026 Fleet & MRO Forecast. *Carried in Part III.*
- **U-D9-02 (D9) — U2 — Date of the "47% of 737NGs / 45% of A320ceos aged ≤10 years" statistic [S77/S78].** Given production ended in 2019/2022, the proportion in 2026 must be lower; the figure is probably from 2021-22. Resolve: read the ISTAT article. *Carried in Part III.*
- **U-D9-03 (D9) — U3 — Definition of the ~24,000 CFM56 "in service" figure [S78]** (installed only, or installed plus spares plus stored) and its split between -5B, -7B and the older -3/-5A/-5C variants. Resolve: GE Aerospace or CFM investor materials with a fleet breakdown; the GE Bernstein presentation (27 May 2026) [S6] may contain it but its content did not surface. *Carried in Parts I, III, IX.*
- **U-D9-04 (D9) — U4 — V2500 engines in service, 2025-26, from IAE/RTX or Cirium.** Only a low-grade figure surfaced (T8). Resolve: IAE/Pratt & Whitney V2500 programme page or RTX investor day fleet slide. *Carried in Part III.*
- **U-D9-05 (D9) — U5 — A year-by-year series of GTF AOG counts (average and peak) for 2024, 2025 and 2026 from RTX's own disclosures.** RTX's statements surfaced as percentages (-20%, -25%) rather than levels [S31]. Resolve: RTX Q4 2024, Q4 2025 and Q2 2026 earnings slides/transcripts [S32]. *Carried in Part III.*
- **U-D9-06 (D9) — U6 — Historical retirement rates for the A320ceo and 737NG specifically (annual counts 2015-2025).** Only aggregate all-type retirements and the CFM56 percentage surfaced. The sentence "retirements in relation to the share of the active fleet as of 2023 was 31% for the 737 and 26% for the A320" appeared in a result [S59/S60 context] but its meaning (share of total retirements by type?) is not clear. Resolve: Cirium or IBA type-level retirement tables. *Carried in Part III.*
- **U-D9-07 (D9) — U7 — The 2026 Aviation Week Fleet & MRO Forecast's type-level shop-visit curve for CFM56 through 2030, and Oliver Wyman's narrowbody-engine subtotal by year.** Only headline totals surfaced [S47, S51]. Resolve: the forecast documents themselves. *Carried in Part III.*

- **U-D9-08 (D9) — U8 — Labour figures (engine MRO technician headcount, shortfall, wage inflation).** Oliver Wyman cites skilled-worker retirements qualitatively [S47]. D2 may hold these. *Carried in Parts III, IV.*
- **U-D9-09 (D9) — U9 — Airline-level fleet plans with CFM56 retirement dates.** Spirit's 2023 fleet plan exhibit surfaced as a URL [SEC fleet plan, Aug 2023] but was not read; the FY2025 10-Ks of Southwest, JetBlue and Frontier surfaced but their fleet-plan tables were not extracted [S73-S75]. Resolve: read the fleet-plan sections. *Carried in Part III.*
- **U-D9-10 (D9) — U10 — Boeing/Airbus 20-year market outlook replacement-demand figures for single-aisles (the share of deliveries that replace retirements).** Cirium's 2024 fleet forecast (44,500 deliveries, \$2.7tn, 20 years) surfaced [S83] but without the replacement share. Resolve: Boeing CMO and Airbus GMF 2025/2026 editions. *Carried in Part III.*
- **U-D9-11 (D9) — U11 — Average AOG duration (wing-to-wing days) for a GTF powder-metal shop visit, by year.** RTX gave a 23% reduction in turnaround [S31] without the base. Resolve: RTX transcripts. *Carried in Part III.*

Part IV, the shop-visit market and the OEM aftermarket (D2): 7 entries

- **U-D2-02 (D2) — U2. Shop-visit counts 2020–2023.** Only the 2019 level (~2,000+) and the 2024 forecast ("topping 2,000") were found. Resolve from Safran capital-markets-day decks (the safran-group.com media downloads 397180 / 448184 returned in search but not read) or the GE Aerospace Bernstein presentation of 27 May 2026. *Carried in Parts III, IV.*
- **U-D2-03 (D2) — U3. OEM versus independent share of CFM56 shop visits.** No estimate found. Resolve with Aviation Week/Cirium shop-visit-by-provider data or the Oliver Wyman fleet forecast's provider split. *Carried in Part IV.*
- **U-D2-04 (D2) — U4. CFM56 share of GE CES services revenue.** Not disclosed in the 10-K. GE has historically described the CFM56 as the largest services contributor; a number would need an investor presentation or call transcript (the 22 Jan 2026 webcast transcript URL was returned, not read). *Carried in Parts IV, XI.*
- **U-D2-05 (D2) — U5. Capacity of the named airline MROs.** AFI KLM E&M, Delta TechOps, Turkish Technic, GMF AeroAsia, EgyptAir M&E, GA Telesis, Lufthansa Technik and ST Engineering: no CFM56 inductions- per-year or expansion figures returned. Resolve via each company's annual report or trade-press shop profiles. *Carried in Parts IV, VII.*
- **U-D2-06 (D2) — U6. Engine-stand shortages and HPT blade / casting / forging lead times.** No numbers found. Resolve via Aviation Week supply-chain coverage, PCC/Howmet commentary, or GE call Q&A. *Carried in Part IV.*

- **U-D2-09 (D2) — U9. IAE "FleetCare"**. Not found as a current product; only EngineWise. Resolve on the P&W services site. *Carried in Part IV.*
- **U-D2-10 (D2) — U10. Dates**. Several Aviation Week articles were returned without publication dates (noted "c. 2023/24" or "2025"). The writer should confirm before quoting. *Carried in Parts III, IV.*

Part V, used serviceable material and teardown (D3): 15 entries

- **U-D3-01 (D3) — U1** — Sourced split of a 737-800 part-out value between engines and airframe. Would be resolved by an appraiser part-out report or an AerSale/AAR investor presentation with a teardown waterfall. *Carried in Part V.*
- **U-D3-02 (D3) — U2** — Total sell-through period for teardown inventory (months to recover, say, 80% of proceeds). AerSale's 10-K inventory-turnover disclosure or a trade-press teardown case study would answer it. *Carried in Parts II, V.*
- **U-D3-03 (D3) — U3** — Whether Calver's \$9,878,733 / \$4,939,366 CFM56-5B LLP figure is per engine or per aircraft (two engines). The original Cargo Facts presentation would settle it. *Carried in Parts I, V.*
- **U-D3-06 (D3) — U6** — Current CLP of representative CFM56-7B LLPs (e.g. HPT disk, fan disk) and of a full -7B LLP set. Needed for a precise module-value example. *Carried in Parts I, V.*
- **U-D3-07 (D3) — U7** — The identity and terms of FTAI's "exclusive partnership" for teardown, repair, marketing and sale of CFM56 parts. The 10-K excerpt does not name it. The FTAI 10-K text or an FTAI press release would answer it. *Carried in Part V.*
- **U-D3-08 (D3) — U8** — Purchase price of a 737-800 bought for part-out in 2025–2026 (and of the airframe alone). *Carried in Part V.*
- **U-D3-09 (D3) — U9** — Teardown labour and facility cost per aircraft and per engine. *Carried in Part V.*
- **U-D3-10 (D3) — U10** — Values of a 737-800 APU, avionics suite, thrust reversers and other rotables. *Carried in Part V.*
- **U-D3-11 (D3) — U11** — AAR FY2026 total revenue and Parts Supply segment revenue in dollars (only the share and growth were captured). *Carried in Part V.*
- **U-D3-13 (D3) — U13** — AJW Group scale. *Carried in Part V.*
- **U-D3-14 (D3) — U14** — Unical Aviation: nothing returned. *Carried in Parts V, VII.*
- **U-D3-16 (D3) — U16** — Actual retirement counts for 2023, 2024 and 2025 from Cirium (only 2019–2022 counts and the 2024–26 shortfall percentage were captured). *Carried in Parts III, V.*
- **U-D3-17 (D3) — U17** — Number of CFM56-5B and -7B engines torn down per year, 2023–2025. *Carried in Part V.*

- **U-D3-18 (D3) — U18** — Scope definitions behind the three USM market-size estimates (what counts as USM; whether engines, airframes and components are all included). *Carried in Part V.*
- **U-D3-19 (D3) — U19** — Oliver Wyman's 2020 forecast figures and any updated Oliver Wyman USM forecast (title only captured). *Carried in Part V.*

Part VI, PMA parts and DER repairs (D4): 12 entries

- **U-D4-01 (D4) — U1** — **Current lessor return-condition practice (2024–2026).** All lessor statements found are 2009–2016. Resolve with recent lessor 10-K/20-F lease descriptions (AerCap, Air Lease, Willis Lease), ISTAT or IBA surveys, or lessor conference remarks. *Carried in Parts II, VI, VII.*
- **U-D4-04 (D4) — U4** — **Time from PMA application to approval** for hot-section parts. Resolve via FAA PMA database (DRS) entries with dates, or company statements. *Carried in Part VI.*
- **U-D4-05 (D4) — U5** — **Whether the CFM56 HPT blade/vane PMAs are "critical components"** under the TIP and therefore which EASA acceptance clause applies; whether EASA acceptance has been obtained. *Carried in Part VI.*
- **U-D4-06 (D4) — U6** — **Acceptance by other regulators** (UK CAA, TCCA, ANAC, CAAC, DGCA India) of these PMAs. *Carried in Part VI.*
- **U-D4-07 (D4) — U7** — **Any PMA for CFM56 life-limited rotating parts.** None found. *Carried in Parts VI, VII.*
- **U-D4-08 (D4) — U8** — **How DER-approved repair data is accepted on EU-registered engines** under the TIP. *Carried in Part VI.*
- **U-D4-11 (D4) — U11** — **PMA policies of Delta, American, United, Ryanair, and non-US CFM56 operators** for hot-section parts. *Carried in Part VI.*
- **U-D4-12 (D4) — U12** — **Chromalloy production capacity** (blade sets per year) and FTAI's share of it. *Carried in Part VI.*
- **U-D4-13 (D4) — U13** — **Quantified residual-value effect** of PMA/DER content on engine or aircraft appraisals. *Carried in Part VI.*
- **U-D4-15 (D4) — U15** — **Customer uptake figures for the Chromalloy blade** (named airlines/MROs, sets delivered). Only "sold out for 2025" was found. *Carried in Part VI.*
- **U-D4-16 (D4) — U16** — **CFM56-7B HPT blade count per set** (80 is a recollection, unverified here) and the OEM list price for a full HPT blade and vane set. *Carried in Parts I, IV, VI.*
- **U-D4-17 (D4) — U17** — **Dates for the LPT vane PMA and the V2500 HPT blade PMA.** *Carried in Part VI.*

Part VII, FTAI Aerospace Products (D5): 11 entries

- **U-D5-01 (D5) — U1 — Revenue-recognition text for exchanges.** The 10-K's exact policy for exchange transactions (how the core is valued, whether exchange fees are separately quantified). Resolves with the 10-K revenue note [S1]. *Carried in Part VII.*
- **U-D5-02 (D5) — U2 — Disaggregation of "Aerospace products revenue."** Engines vs modules vs parts vs repairs for 2023, 2024, 2025 and each 2025–2026 quarter; margin by line; whether disclosure changed after the February 2025 review. Resolves with the 10-K and 10-Q revenue notes [S1–S3] or with the Q2 2026 release [S5]. *Carried in Part VII.*
- **U-D5-04 (D5) — U4 — Module counts for 2022, 2023, 2024 and Q1/Q3 2025.** Only Q4 2024 (≈ 136) is derivable. Resolves with the Q4 2024 release [S43], the Q1 2025 release [S42], the Q3 2025 release [S11] and the FY2023 release. *Carried in Part VII.*
- **U-D5-05 (D5) — U5 — Denominator of the 25% share goal** and the year by which it is targeted. *Carried in Parts IV, VII.*
- **U-D5-06 (D5) — U6 — Engine sales counts; V2500 shop visits completed; current V2500 fleet vs 200 target; whether any V2500 module work is done in-house.** *Carried in Part VII.*
- **U-D5-08 (D5) — U8 — GMF AeroAsia and EgyptAir terms** (capacity committed, economics, duration). *Carried in Parts IV, VII.*
- **U-D5-10 (D5) — U10 — Whether Rome's modules are counted in FTAI's module totals,** given equity-method accounting. *Carried in Part VII.*
- **U-D5-13 (D5) — U13 — Muddy Waters' own wording** on expense classification, channel stuffing and the "other red flags". *Carried in Part VII.*
- **U-D5-14 (D5) — U14 — Exact FY2024 10-K filing date** (accession 0001590364-25-000006; company says timely). *Carried in Part VII.*
- **U-D5-15 (D5) — U15 — Subject of the 27 Nov 2024 SEC correspondence** [S20]. *Carried in Part VII.*
- **U-D5-17 (D5) — U17 — Named customers, long-term agreements, customer concentration** (any customer $\geq 10\%$ of revenue), and whether SCI-vehicle revenue is disclosed as related-party. *Carried in Part VII.*

Part VIII, FTAI Aviation Leasing and the Strategic Capital Initiative (D6 FTAI sections, D7): 22 entries

- **U-D6-01 (D6) — U1 — Fleet by year.** Aircraft and engine counts at 31 Dec 2021, 2022, 2023, 2024 and 30 Jun 2026. Resolve: "Our Aviation Leasing segment" paragraphs in each 10-K/10-Q ([S7],[S8],[S3]). *Carried in Part VIII.*

- **U-D6-03 (D6) — U3 — Q1 and Q2 2026 segment Adjusted EBITDA.** The Q2 2026 release ([S4]) and Q2 2026 10-Q ([S3]) contain them; the \$88.2M chain-map figure needs confirmation. *Carried in Part VIII.*
- **U-D6-04 (D6) — U4 — Russia detail.** Insurance recoveries in FY2023 and FY2024; whether any recoveries occurred in 2022; the status of the 4 aircraft and 1 engine in Ukraine; whether the Russia assets remain in gross leasing-equipment cost; the gross (pre-deposit) carrying value written off in 2022; the number of claims settled versus outstanding. Resolve: "Russia" / "insurance" paragraphs in FY2023 and FY2024 10-Ks and the FY2025 10-K. *Carried in Part VIII.*
- **U-D6-07 (D6) — U7 — Security deposit and maintenance deposit balances.** FTAI's balance-sheet "security deposits" and "maintenance deposits" liabilities by year, and the reserve rates in its leases. Resolve: FY2025 10-K balance sheet and revenue-recognition note. *Carried in Parts II, IV, VIII.*
- **U-D6-08 (D6) — U8 — Depreciation policy.** Useful lives and residual assumptions for FTAI's aircraft and engines. Resolve: FY2025 10-K summary of significant accounting policies. *Carried in Parts II, VIII.*
- **U-D6-10 (D6) — U10 — Conversion cost and slot availability** for 737-800 P2F (AEI, Boeing BCF, IAI). Resolve: conversion-house releases, trade press. *Carried in Parts II, VIII.*
- **U-D6-12 (D6) — U12 — Seed Assets.** Number, type, sale price and transfer dates of assets sold to the 2025 Partnership; the \$46.4M gain's cost basis. Resolve: FY2025 10-K related-party / Strategic Capital note; D7. *Carried in Part VIII.*
- **U-D6-15 (D6) — U15 — Average lease terms over time** (new-lease term at signing versus remaining term) and the mix of short- versus long-term engine leases in FTAI's book. Resolve: 10-K MD&A by year. *Carried in Parts II, VIII.*
- **U-D7-03 (D7) — U3 — Investors.** OneIM's commitment size and role; identities of other LPs beyond categories [S3, S4]. *Carried in Part VIII.*
- **U-D7-04 (D7) — U4 — 2026 SPV equity.** Target size, hard cap, anchor investors, FTAI's stake, timing of final close [S6, S9]. *Carried in Part VIII.*
- **U-D7-05 (D7) — U5 — Pricing of aircraft sold by FTAI to the vehicles.** Appraisal policy, third-party valuation, LP or independent-director approval. Not located. *Carried in Part VIII.*
- **U-D7-06 (D7) — U6 — MRE revenue from the vehicles.** Dollar revenue and counts of engine and module exchanges to the 2025 Partnership; expected annual volume for 300+ aircraft. Only eliminations located [S12, S13]. *Carried in Parts VII, VIII.*

- **U-D7-08 (D7) — U8 — Line placement.** Which income-statement line and segment carry servicing fees and equity income (presumed Aviation Leasing / Other revenue and equity in earnings; not confirmed). *Carried in Part VIII.*
- **U-D7-09 (D7) — U9 — Guidance.** Any 2025 or 2026 guidance item specific to SCI (fees, gains, MRE volume) and the corresponding actuals. Only total and Aerospace Products figures located [S1, S6]. *Carried in Parts VII, VIII.*
- **U-D7-10 (D7) — U10 — Q4 2025 and 1H 2026 aircraft sales.** Counts and gains for Q4 2025, Q1 2026 (proceeds \$117.3M known) and Q2 2026. *Carried in Part VIII.*
- **U-D7-11 (D7) — U11 — Debt terms.** Rate, advance rate, tenor and covenants of the \$2.5B facility and the \$2.0B warehouse; how \$5.5B is built; whether the 2025 facility was upsized. *Carried in Parts II, VIII.*
- **U-D7-12 (D7) — U12 — The May 2026 ABS.** Whether "FTAI Aviation Prices Inaugural Asset-Backed Securitization" [S26] is a 2025-SPV take-out or FTAI's own balance-sheet deal; size, tranches, pricing. *Carried in Parts II, VIII, X.*
- **U-D7-15 (D7) — U15 — Vehicle term.** Life of the 2025 Partnership, length of harvest phase, extension rights, exit (portfolio sale, ABS, piecemeal). *Carried in Part VIII.*
- **U-D7-16 (D7) — U16 — Comparables not reached.** Castlake, Apollo/Merx, AerCap fee structures; Blackbird and Thunderbolt fee rates. *Carried in Parts VIII, XI.*
- **U-D7-17 (D7) — U17 — Q1 2025 10-Q "management fees and incentive allocations".** The search summary flagged such text in S14; whether it concerns SCI or the pre-internalization Fortress management agreement is not verified. *Carried in Parts VIII, X.*
- **U-D7-18 (D7) — U18 — VIE analysis.** Whether FTAI assessed the Partnership as a VIE and the basis for not consolidating. *Carried in Part VIII.*
- **U-D7-19 (D7) — U19 — Published SCI-specific critiques.** None located (related-party sales, transfer pricing, fund-raising dependence). Resolve: sell-side notes, short-seller updates after Jan 2025, financial press. *Carried in Part VIII.*

Part IX, FTAI Power (D8): 14 entries

- **U-D8-01 (D8) — U1. Units and price in the \$1.465B order.** Not disclosed. Would be resolved by FTAI's Q3 2026 10-Q or call, or by Jereh's SZSE filing if it states quantities. *Carried in Part IX.*
- **U-D8-02 (D8) — U2. J&F ownership, control and consolidation.** Percentages, board, which party consolidates, and how FTAI recognises Power revenue (sale of turbines into J&F, share of J&F profit, or consolidated genset sales). The Q2 2026 10-Q [10] is the document to read; its J&F note was not returned by search. *Carried in Part IX.*
- **U-D8-03 (D8) — U3. Engines per unit and configuration.** How many CFM56 gas generators per 25 MW unit; -5B or -7B; whether the engine's own low-pressure turbine or

a new free power turbine drives the generator; whether a booster stage replaces the fan. *Carried in Part IX.*

- **U-D8-04 (D8) — U4. Where units are built.** Jerreh packaging site(s); US vs China; tariff exposure; whether J&F has its own facility. *Carried in Part IX.*
- **U-D8-05 (D8) — U5. Emissions and fuel.** NOx level, after-treatment (SCR or not), heating-value basis of the efficiency figure, dual-fuel capability. *Carried in Part IX.*
- **U-D8-06 (D8) — U6. Capital required.** Power capex, engine inventory earmarked, test capacity; FTAI's engine input cost per unit. *Carried in Part IX.*
- **U-D8-08 (D8) — U8. Jerreh's GenSystems orders.** MW count (to derive \$/kW) and whether they use Mod-1 or other turbines; identity of the customer; relationship to J&F. *Carried in Part IX.*
- **U-D8-09 (D8) — U9. Competitor figures not captured.** Siemens Energy SGT-A65 output/lead time; Solar Turbines Titan 350; Caterpillar gas gensets; Bloom Energy deal sizes and \$/kW; published aeroderivative list prices in 2026. *Carried in Part IX.*
- **U-D8-10 (D8) — U10. Milestone schedule.** Size of the advance payment and the percentage at each milestone. *Carried in Part IX.*
- **U-D8-11 (D8) — U11. Performance adjustment.** Which metrics the 10% cap applies to (output, heat rate, availability, schedule) and whether it is symmetric (bonus as well as penalty). *Carried in Part IX.*
- **U-D8-12 (D8) — U12. Service model.** Mod-1 inspection and overhaul intervals; service contract form and pricing; FTAI's expected service revenue per unit. *Carried in Part IX.*
- **U-D8-13 (D8) — U13. Prior CFM56 stationary derivative.** None found; a definitive statement from GE/CFM or Gas Turbine World's handbook would settle whether Mod-1 is the first. *Carried in Part IX.*
- **U-D8-14 (D8) — U14. Interconnection queue durations.** PJM and ERCOT queue statistics were not captured; they quantify the "bridging" window ProEnergy's customers describe as 5–7 years [64]. *Carried in Part IX.*
- **U-D8-15 (D8) — U15. Transcripts.** The Q4 2025 (26 Feb 2026) and Q2 2026 (29 Jul 2026) call transcripts were reached only through third-party summaries [15]–[19]; direct quotations on Power should be taken from the transcripts before publication. *Carried in Part IX.*

Part X, FTAI the company (D10): 11 entries

- **U-D10-03 (D10) — U-3 — Debt detail.** Issue dates and original principal of each note series; principal vs carrying amount; call schedules; covenants; revolver commitment size; any term loans or asset-level debt consolidated from SCI vehicles; total debt and equity at

31 Dec 2025. Resolve with the debt note in [S1] and [S2] and the note-offering 8-Ks. *Carried in Part X.*

- **U-D10-04 (D10) — U-4 — Preferred detail.** Series B share count, floating spread and reset date; Series C share count, reset terms, redemption date and amount [S20]; Series D issue date and first reset; preferred tickers. Resolve with the equity note in [S1], Exhibit 4.15 to the FY2023 10-K (description of securities) [S25], and the 15 May 2026 release. *Carried in Part X.*
- **U-D10-05 (D10) — U-5 — Management agreement economics.** The management fee rate and base, the incentive allocation formula, and fees and allocations paid in 2021, 2022, 2023 and Jan–May 2024. Recollection, unverified: a fee of 1.50% per annum of total equity. Resolve with the related-party notes in the FY2021–FY2024 10-Ks [S15][S14][S27 folder]. *Carried in Part X.*
- **U-D10-06 (D10) — U-6 — Internalization details.** FTAI's closing share price on 28 May 2024 versus the \$80.35 implied value; the amount of accrued compensation and reimbursable expenses paid; accounting treatment (expensed as "internalization" cost or capitalized); whether FIG LLC still holds the shares. Resolve with the 8-K exhibits [S10] [S11] and the Q2 2024 10-Q [S9]. *Carried in Part X.*
- **U-D10-07 (D10) — U-7 — Leverage definitions and credit ratings.** The company's leverage covenant definitions; Moody's, S&P and Fitch issuer and note ratings with dates. Not searched (budget exhausted). Resolve with the agencies' press releases and the 10-K liquidity section. *Carried in Part X.*
- **U-D10-08 (D10) — U-8 — Dividend and share-count history.** Quarterly ordinary dividend 2022–2024 and Q1 2026; basic and diluted weighted shares by year; equity issuances other than the internalization shares; repurchase program terms [S21]. Resolve with [S1] equity note and the Sep 2026 release. *Carried in Part X.*
- **U-D10-09 (D10) — U-9 — Officers' titles and tenure.** CFO confirmation (Eun Nam), General Counsel's name, David Moreno's current role, start dates. Resolve with the proxy's executive officers section [S6] and the 10-K Item 10. *Carried in Part X.*
- **U-D10-10 (D10) — U-10 — Governance detail.** Which directors are independent; any remaining Fortress affiliation; lead independent director; bonus metrics and PSU conditions; other NEOs' pay; the "directors and officers as a group" ownership; any surviving Fortress-era arrangement. Resolve with [S6]. *Carried in Part X.*
- **U-D10-12 (D10) — U-12 — Reasons for the Cayman redomiciliation and the NYSE-to-Nasdaq move.** Resolve with the 2022 DEFM14A [S17] and the listing-transfer 8-K. *Carried in Part X.*

- **U-D10-14 (D10) — U-14 — 2025 guidance as set on 30 Dec 2024 and the Feb 2026 2026 guidance in full (including FTAI Power and consolidated).** Resolve with [S24] and [S3]. *Carried in Part X.*
- **U-D10-15 (D10) — U-15 — Risk factors in full.** Supplier dependence on CFM/GE and other OEMs for new parts and repairs, regulatory approvals (FAA/EASA, PMA), SCI vehicle risks, key personnel, litigation; annual customer concentration and customer names. Only Russia, leverage, SCI, customer concentration (Q3 2025) and MRE stand-ready obligations were confirmed. Resolve with Item 1A of [S1]. *Carried in Part X.*

Part XI, markets and money: margin pools and peers (D11): 15 entries

- **U-D11-02 (D11) — U2 — Willis Lease Q2 2026 10-Q figures.** The Q2 2026 earnings-call 8-K exhibit was located by URL [S3b] but no figures came back; the 10-Q itself was not located. Resolve: search "wlfc-20260630". *Carried in Part XI.*
- **U-D11-04 (D11) — U4 — HEICO Q3 FY2026 10-Q (quarter ended 31 Jul 2026).** Not located; only the Q1 FY2026 release URL [S7b]. Resolve: search "hei-20260731". *Carried in Part XI.*
- **U-D11-06 (D11) — U6 — AerCap Materials revenue and AerCap engine-leasing revenue.** Searched the 20-F [S4]. Not returned. Resolve: read the 20-F's business description and the segment or revenue note. *Carried in Part XI.*
- **U-D11-07 (D11) — U7 — AerCap FY2025 net income drivers (\$3.8B versus \$2.1B) and any stated lease yield or net spread.** Not returned. Resolve: read the Q4 2025 earnings release [S4a] in full. *Carried in Part XI.*
- **U-D11-08 (D11) — U8 — AerCap 737 MAX share of NBV and total narrowbody share.** The five NBV lines returned sum to 80%; the remainder was not itemised. Resolve: read the fleet table in [S4]. *Carried in Part XI.*
- **U-D11-09 (D11) — U9 — Air Lease acquiring consortium.** The 10-K text returned names only the surviving entity, Sumisho Air Lease Corporation, and the agreement date [S6]. Resolve: read the 1 Sep 2025 merger 8-K. *Carried in Part XI.*
- **U-D11-10 (D11) — U10 — Air Lease FY2025 total revenues, pre-tax margin, stated lease yield, engine-availability statements.** Not returned. Resolve: read Item 7 of [S6]. *Carried in Part XI.*
- **U-D11-11 (D11) — U11 — AerSale FY2025 net income; Q2 2026 10-Q figures.** The Q2 2026 10-Q was located by URL [S12b]. Resolve: read [S12a] and [S12b]. *Carried in Part XI.*
- **U-D11-12 (D11) — U12 — AerSale full-year feedstock purchases and any statement on retirement scarcity.** Only the Q4 figure (\$15.4M) was returned. Resolve: read the AMS discussion and cash-flow statement in [S11]. *Carried in Part XI.*

- **U-D11-13 (D11) — U13 — StandardAero FY2025 consolidated revenue, net income and Adjusted EBITDA; revenue by platform; CFM56 shop-visit count.** Not returned. Resolve: read Item 7 and the segment note of [S10]. *Carried in Part XI.*
- **U-D11-14 (D11) — U14 — AAR segment operating income; USM sales within Parts Supply; subject of the 2026 DEFA14A [S13b]; Q1 FY2027 10-Q figures [S13a].** Not returned. Resolve: read the segment note in [S13] and the DEFA14A. *Carried in Part XI.*
- **U-D11-15 (D11) — U15 — GE Aerospace FY2025 total revenue; Q2 2026 10-Q.** Not returned or located. Resolve: read the consolidated statement in [S14]; search "ge-20260630". *Carried in Part XI.*
- **U-D11-16 (D11) — U16 — GE CFM56 shop-visit count, spare-parts price increases, and any "material solutions" or used-serviceable-material programme wording.** Not returned from [S14]. Resolve: read the CES segment discussion in [S14] and the Q4 2025 and Q2 2026 earnings materials. *Carried in Parts III, IV, XI.*
- **U-D11-17 (D11) — U17 — Safran USD spare-parts growth; MTU CFM56 and V2500 share of MRO; Lufthansa Technik shop-visit count and CFM56/LEAP capacity.** Not returned from [S16]–[S19a]. Resolve: read the Safran URD Propulsion section, MTU's annual report segment note, and LHT's annual report. *Carried in Part XI.*
- **U-D11-18 (D11) — U18 — Cost of debt and funding terms for each lessor.** Not searched for within the budget. Resolve: D6 and D10 cover FTAI's; peers' are in their debt notes. *Carried in Part XI.*

94.3 RESOLVED DURING THE BUILD

The orchestrator ran a small number of its own searches after the dossiers were written, recorded in research/orchestrator-notes.md, and the Parts' authors found a few further facts while writing. Exhibit 12.5 lists every register entry those efforts resolved, in whole or in part, with the resolving fact and its source. "Part" in the extent column means the entry still has open components, which the row names. The entries themselves stand unaltered above; a reader checking a question should read the entry there and the resolution here.

EXHIBIT 12.5 — REGISTER ENTRIES RESOLVED IN WHOLE OR IN PART DURING THE BUILD, WITH THE RESOLVING FACT AND SOURCE

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D5-07, U-D10-11 (K-12)	Deal terms for FTAI's maintenance facilities	LMCES, Montréal: acquired 9 Sep 2024 from Lockheed Martin Canada, total consideration \$170.0M, 526,000 sq ft; at closing Montréal plus Miami gave capacity for 1,350 CFM56 module overhauls and more than 500 engine tests a year. QuickTurn, Miami: iAero Thrust assets acquired 4 Jan 2023 with Unical Aviation; Unical's remaining interest bought 1 Dec 2023 for \$30.3M cash. QuickTurn Europe, Rome Fiumicino: \$10.5M for 50%, closed 5 Jun 2025, 200,000 sq ft, adding capacity for 450 modules (150 engines) a year	GlobeNewswire, 9 Sep 2024; Aviation Week, "FTAI Aviation Snaps Up LMCES For \$170 Million"; GlobeNewswire, 1 Dec 2023; GlobeNewswire, 26 Feb 2025 and 5 Jun 2025; FTAI 10-K FY2025	Part: Orange (nature, size, date) and Lisbon (size, capacity, opening) remain open, as do the Chromalloy JV's deal terms and the infrastructure businesses' figures
U-D5-09, U-D10-17 (K-13)	Headcount, test cells, properties	985 full-time employees and independent contractors at 31 Dec 2025; 494 full-time employees in Canada, about 71% of them under collective bargaining agreements. Engine-test capacity "more than 500" a year (Sep 2024) and "more than 600" (Feb 2025)	FTAI 10-K FY2025; GlobeNewswire, 9 Sep 2024 and 26 Feb 2025	Part: employees by site, test-cell count and the properties list remain open
U-D10-17, U-D5-07	The Rome site's location ("Rome NY" in the dossier's properties list and the chain map)	FTAI's Rome site is QuickTurn Europe at Rome Fiumicino Airport, Italy, the former IAG Engine Center. "Rome NY" was an orchestrator error in the chain map, copied by two dossiers, and was corrected in the research files; no Part repeats it as fact	FTAI 10-K FY2025; GlobeNewswire, 5 Jun 2025; reconciliation R-040; orchestrator notes	Whole, for the location

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D5-16, U-D10-16 (K-11)	Status of the securities class action; auditor; internal control	Class action: S.D.N.Y. No. 25-cv-00541, Judge Jeannette A. Vargas; class period 23 Jul 2024 to 21 Jan 2025; defendants' motion to dismiss the amended complaint filed 20 Nov 2025, fully briefed and pending. Auditor: Ernst & Young LLP 2016 to 2025; the Audit Committee engaged KPMG LLP for FY2025 effective 17 Jun 2025; the auditor's report is dated 27 Feb 2026. The Audit Committee's internal investigation concluded the short report's allegations were "all without merit"	Law-firm case pages (ktmc.com, rgrdlaw.com; last-update date not visible); FTAI 10-K FY2025	Part: the internal-control conclusion, any derivative suits and the 10-K's own description of the case remain open (R-066 records an ambiguity in the auditor line)

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D5-19, U-D6-02, U-D10-01 (K-17)	Segment revenue and Adjusted EBITDA by year; the full income statement	FY2025 10-K segment table (\$ thousands, 2023 / 2024 / 2025): total revenues 1,170,896 / 1,734,901 / 2,507,409; total Adjusted EBITDA 597,282 / 862,050 / 1,190,922; Aerospace products revenue 454,970 / 1,079,821 / 1,600,456; MRE Contract revenue 0 / 0 / 335,788; Aerospace Products Adjusted EBITDA 160,009 / 380,636 / 671,252; lease income not returned / 234,411 (or 255,338, two readings) / 235,210; Aviation Leasing Adjusted EBITDA not returned / 500,062 / 608,912. FY2025 consolidated lines: cost of sales 1,349,719; operating expenses 152,541; G&A 9,478; D&A 225,797; interest expense 247,751. FY2022 10-K: segment revenues 2022 / 2021 Aviation Leasing 199,040 / 128,328, Aerospace Products 85,113 / 70,800 (2021 figure flagged as a probable extraction artefact), Corporate and Other 26,948 / 14,161; 2022 segment Adjusted EBITDA 107,556 / 27,377 / (7,277); net income attributable 2022 (220,374), 2021 (128,992), not comparable with 2023 onward because of discontinued operations	FTAI 10-K FY2025; FTAI 10-K FY2022	Part: 2023 Leasing lines, 2023 to 2024 maintenance and asset-sales lines, the Adjusted EBITDA reconciliation, net income attributable by year, 1H totals and the 2021 segment figures remain open

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D10-01, D5 T5 (R-045)	Which cost-of-sales figure is which	Both FY2025 figures are in the 10-K: \$1,240,368K is the Aerospace Products segment's cost of sales in the segment expenses table; \$1,349,719K is the consolidated cost-of-sales line. The segment gross margin of 35.9% on segment revenue is a derived figure	FTAI 10-K FY2025; orchestrator notes	Whole, for the scopes; the 10-K's mapping of the \$109,351K difference was not retrieved
U-D7-06	MRE revenue from the vehicles	MRE Contract revenue with the 2025 Partnership: Q2 2026 \$182.8M (Q2 2025 \$69.6M); 1H 2026 \$404.0M (1H 2025 \$170.2M). The 10-Q defines the line as "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership"	FTAI 10-Q Q2 2026	Part: counts of engine and module exchanges remain open
U-D5-17	Named customers, concentration, related-party disclosure	The related-party revenue is identified and quantified as above. Two Aerospace Products customers accounted for 16% and 11% of revenue in Q3 2025	FTAI 10-Q Q2 2026; FTAI 10-Q Q3 2025	Part: customer names and annual concentration remain open
U-D6-11, U-D10-13 (K-07)	Equity-method stakes, percentages and carrying values	At 31 Dec 2025: Advanced Engine Repair JV 25%, \$22,429K (\$15.0M invested Dec 2016, \$13.5M Aug 2019); 2025 Partnership 19% (20% in the Q3 2025 10-Q), \$281,740K, \$151.6M invested in 9M 2025, FTAI as Servicer; QuickTurn Europe 50%, \$9,987K; Prime Engine Accessories 50%, carrying value not returned. Exhibit 21.1 (subsidiaries) located	FTAI 10-K FY2025; FTAI 10-Q Q3 2025, note R13	Part: which investment the Aviation Leasing segment leases "through", and FTAI Finance Holdco's role, remain open

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D4-03, U-D5-11 (K-04)	FTAI–Chromalloy joint venture terms and contribution	The Advanced Engine Repair JV figures above apply only if that venture is the Chromalloy PMA joint venture; the 10-K describes the PMA venture only as "a joint venture", and the identity is unknown	FTAI 10-K FY2025; orchestrator notes	Part, conditionally: terms, accounting and contribution remain open
U-D3-07	FTAI's exclusive partner for teardown, repair, marketing and sale of CFM56 parts	AAR Corp, under a Serviceable Engine Products agreement that the two companies extended through 2030, covering teardown, repair, marketing and sale of material from FTAI's pool of over 450 CFM56 engines	AAR release, 27 Mar 2025 (PR Newswire; body as carried by ePlaneAI); reconciliation B-9	Part: the agreement's economic terms remain open
U-D3-14	Unical Aviation	Unical Aviation was FTAI's joint-venture partner in QuickTurn Miami until 1 Dec 2023	GlobeNewswire, 1 Dec 2023; FTAI 10-K FY2025	Part: Unical's own scale remains open
U-D5-13	Muddy Waters' own wording	The complaint's allegations as summarised on law-firm pages: one-time engine sales reported as MRO revenue; whole-engine sales presented as module sales; depreciation of engines not on lease lowering reported cost of goods sold	Law-firm case pages (ktmc.com, rgrdlaw.com)	Part: the report's own wording remains open
U-D8-02	J&F ownership, control and consolidation	Description added: the Q2 2026 10-Q calls J&F "a strategic packaging and distribution joint venture with Jereh Group, a global leader in gas turbine mobile packaging, to support the planned 2027 production target of 100 Mod-1 CFM56 aeroderivative units"	FTAI 10-Q Q2 2026	Description only; ownership split and consolidation remain open

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D10-14, U-D7-09	Guidance in full, including the 2026 path and any Leasing-specific item	2027 Business Segment Adjusted EBITDA guidance \$2.3B: Aerospace Products \$1.4B, FTAI Power \$450M, Aviation Leasing \$450M. The 2026 path: \$1.4B (Dec 2024 and early 2025); \$1.525B = \$1.0B Aerospace Products + \$525M Leasing (27 Oct 2025); \$1.625B = \$1.05B + \$575M (25 Feb 2026); \$1.525B = \$1.05B + \$475M (29 Jul 2026), the Leasing cut "reflecting its continued shift to an asset-light business model". The 2026 module target of 1,050 (25 Feb 2026 release) was raised to 1,200 on the 30 Jul 2026 call, a figure that comes from call coverage and was not confirmed in the release text	FTAI Q2 2026 release, 29 Jul 2026 (text as carried by Barchart); Q3 2025 release, 27 Oct 2025; Q4/FY2025 release, 25 Feb 2026; GuruFocus, Nasdaq.com, MarketBeat call coverage, 30 Jul 2026; reconciliation R-036, R-049	Part: the 2025 guidance as set on 30 Dec 2024 is supplied by dossier D7; whether the 29 Jul 2026 release itself carries 1,200 modules is unverified
U-D10-08	Dividend and share-count history	Ordinary dividend per share: \$0.30 for Q1 and Q2 2025; \$0.35 for Q3 2025 (declared 27 Oct 2025, payable 19 Nov 2025); \$0.40 for Q4 2025 (declared 25 Feb 2026); \$0.45 for Q1 2026 (declared Apr 2026); \$0.50 for Q2 2026 (declared 29 Jul 2026)	GlobeNewswire release titles and texts; reconciliation R-068	Part: 2022 to 2024 dividends, weighted share counts by year and the repurchase program's terms remain open
U-D10-09	Officers' titles and tenure (CFO confirmation)	Nicholas McAleese was appointed Chief Financial Officer and Michael Hazan Chief Accounting Officer on 6 Mar 2026, succeeding Eun (Angela) Nam, who left for a role outside aviation; the 2025 proxy named Nam because it predates the change	FTAI release, 6 Mar 2026 (carried by StreetInsider, Barchart, Finviz); reconciliation R-067	Part: the General Counsel's name, David Moreno's current role and start dates remain open

Entry (cluster)	The question, in brief	Resolving fact	Source	Extent
U-D6-03	Q1 and Q2 2026 Leasing segment Adjusted EBITDA	\$88.2M for Q2 2026, from call-coverage summaries only; the figure was not returned from the release text	GuruFocus; Quartr; reconciliation R-051	Part, and only as call coverage; Q1 2026 remains open
U-D7-12	The May 2026 ABS: whose deal, and its terms	FTAI MRE 2026-1, priced 22 May 2026, is the 2025 Partnership's inaugural asset-backed securitization: \$612M of notes backed by 48 A320ceo and 737NG aircraft on lease to 23 airlines; Series A expected Asf (Fitch) and A(sf) (KBRA), Series B expected BBB+sf (Fitch); expected close 4 Jun 2026; both classes "significantly oversubscribed"; the vehicle "currently owns 292 aircraft"	FTAI release, 22 May 2026 (investor-relations node; carried by Finviz and by AviTrader, 26 May 2026)	Part: coupons and loan-to-value remain open
Gap-fill queue	Nine entries the orchestrator queued for its own searches: U-D5-04, U-D5-02, U-D1-09, U-D7-01, U-D7-02, U-D10-15, U-D5-14, U-D5-15 and U-D11-16	No result recorded; the queue was not completed because the shared search budget was exhausted	Orchestrator notes, "Search budget" and "Gap-fill queue"	Not resolved; listed so the reader knows they were attempted

Source: reconciliation.md section D.3; orchestrator-notes.md, "Gap-fill results", "Resolutions after reconciliation" and "Late addition" (all 2026-10-03); Part V §35 and Part VII §52 (AAR agreement); Part VIII §61 to §62 (FTAI MRE 2026-1); Part VII §50 and Part X §85 (Rome Fiumicino); Part X §78 (CFO). Each resolving fact is cited to the underlying filing, release or page named in the source column; the entries themselves are unchanged in §94.1 and §94.2.

95. Consolidated sources

Each Part ends with a numbered list of the sources it used. This section merges the eleven lists into one. The merge was done by a script: it read every numbered entry in each Part's "Sources" section, keyed entries by their URL (or by title where an entry has no URL, after normalising case and punctuation), kept one entry per key with the shortest of the descriptions the Parts gave it, sorted the result by publisher and then by date, numbered it, and recorded which Parts cite each entry. The eleven lists held 849 citations of external sources; after the merge there are 442 distinct sources. Entries that point to the primer's own research files rather than to an external document are listed separately in §95.2, so that the numbered list contains only external sources. The numbering here is this section's own and does not match the numbers inside the Parts, which each Part's inline citations use. Where an entry gives two URLs, the Part cited both forms of the same document (for example an SEC exhibit and the GlobeNewswire release). Publisher names were normalised only where the same publisher appeared under two spellings; the descriptions are otherwise as the Parts wrote them, including notes such as "located, not read" and "date not visible".

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95.2 RESEARCH FILES CITED BY THE PARTS

The Parts cite the primer's own research files only for derived figures and definitions the research assembled, as the writer brief required. Those citations are listed here so the merge above is complete; the files themselves are published alongside the primer under research/.

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- Part I of this primer, "First principles: the CFM56 as a machine", §4.3 (LLP set price series and derived escalation), §5, §7.2 (Exhibit 1.10) and §9; Part IV §28–29; Part V §31, §34; Part VII §52; reconciliation.md sections A.2, A.4, C.4, D and E; term-registry.md Part VI rows and sections 1b, 1c, 2 and 3. Cited in Part VI.
- Project research files (derivations and assembled definitions cited as such): D3 (/home/user/SPOT/ftai-primer/research/dossiers/D3-usm-teardown.md), D6 (D6-leasing.md), D7 (D7-strategic-capital.md), D9 (D9-demand-context.md), D11 (D11-peers.md); reconciliation.md; term-registry.md. Cited in Part II.
- Project research files cited for derived figures and definitions only: dossiers D5, D6, D7, D9, D10 and D11 in /home/user/SPOT/ftai-primer/research/dossiers/; reconciliation.md; term-registry.md; orchestrator-notes.md (including the FY2025 10-K segment-table search and the late addition on the 22 May 2026 securitization). Cited in Part VIII.
- Research dossiers and notes for this primer: D8 (FTAI Power), D9 §2.2 and §5 (installed base; translation table), D5 §2.4 (module capacity), D10 §4.8 (guidance history), D1 §2.1 (the machine); orchestrator notes (10-Q Q2 2026 wording on J&F; 2027 guidance split);

reconciliation sections A.1, A.4, A.6, C.8, D and E. Cited only where the dossier itself produced a derived figure or a definition. Cited in Part IX.

- Research dossiers cited for derived figures and definitions they assembled: D1 (the CFM56 as a machine), D2 (the shop-visit market), D3 (used serviceable material and teardown), D4 (PMA and DER), D5 (Aerospace Products), D7 (Strategic Capital), D9 (demand context), D10 (the company), D11 (peers), /home/user/SPOT/ftai-primer/research/dossiers/. Cited in Part VII.

95.3 SOURCES AND BASIS

The research for this primer is as of 3 October 2026. Every figure carries its source and date where the Parts state them, and every disagreement between sources is presented with both figures; nothing is averaged and nothing is chosen. The sources fall into a small number of kinds, and the Parts label each. Filings with the US Securities and Exchange Commission (Forms 10-K, 10-Q, 8-K, 20-F, 6-K, DEF 14A and their exhibits) were reached through search restricted to sec.gov, because direct access to EDGAR was blocked in the build environment; each filing figure is cited to the EDGAR URL the search returned, and where a search returned two readings of one line (for example FTAI's 2024 lease income) both are carried. Company releases were reached on GlobeNewswire, PR Newswire or the company's investor-relations pages. Earnings-call content was reached through third-party coverage (GuruFocus, MarketBeat, Quartr, Nasdaq.com, Barchart and transcript services) where the Parts say "call coverage", and the figures that rest on it alone, such as the 1,200-module 2026 target and the \$88.2M Q2 2026 Leasing Adjusted EBITDA, are labelled so wherever they appear. Trade press (Aviation Week, Leeham News, Aircraft Commerce, Cargo Facts, AviTrader and others), regulators' documents (FAA, EASA, the eCFR) and appraisers' publications (IBA, mba Aviation, Cirium) are cited by name and date. Consultancy blogs, above all Safe Fly Aviation, are labelled "consultancy blog" at every use, and market-research vendors' market sizes are labelled as vendor figures whose scope definitions were not retrieved. A few entries are marked "located, not read" or "title located; body not read": the document's existence and title were established but its text was not retrieved, and the Parts use only the title. The eleven dossiers (D1 to D11), the reconciliation and the term registry from which the Parts were written are published alongside under research/, with the orchestrator notes and the chain map.

Build notes

The primer was built from eleven research dossiers (D1 to D11 in research/dossiers/), one per segment of the chain map, each written by a separate research agent against a common brief; from a reconciliation pass (research/reconciliation.md) that cross-checked the dossiers and

produced the tensions, corroborations, information-needs and unknowns registers and the binding instructions to writers; from a term registry (research/term-registry.md) that assigned every term to an owner Part; and from an outline (outline.md) that fixed the section numbers used throughout. Facts the orchestrator gathered after the dossiers were written sit in research/orchestrator-notes.md. The Spine was written last and placed first.

The render pipeline is in build/. Each Part is one markdown file in parts/ with a small block of metadata at the top (title, running header, date line, in the YAML key-and-value format), a part-divider block and the components the writer brief defines (definition boxes, worked examples, callouts and exhibits as blocks of HTML, the web page markup language, with markdown inside). build/render.py converts a Part with Python-Markdown (tables, definition lists, footnotes, attribute lists, a table of contents and markdown-in-HTML) and paginates it with WeasyPrint under build/assets/primer.css, an EB Garamond stylesheet with running headers, a page-numbered contents list and tables that flow across pages. build/validate.py checks each rendered PDF for near-blank pages, exhibits without tables and unbalanced components, and reports page and word counts. build/compile_master.sh renders the cover, the Spine and the twelve Parts into one PDF. This Part's compiled sections were generated by a script that reads the registry, the Part files and the reconciliation, and fills a template; the script is reproducible and was run on 2026-10-03.

Three limits are known and should be read with the primer. First, there is no staged corpus: the research environment could not download filings or documents into a local store, so every figure was read through search results and is cited to the URL the search returned, and some documents were located by title but never read. Second, the research agents shared a pool of 200 web searches that was exhausted during the first wave; dossier D5 (Aerospace Products, the deepest Part) and dossier D10 (the company) were affected most, and the orchestrator's gap-fill queue for them was only partly executed (§94.3 lists what was and was not closed). Third, several figures rest on third-party coverage of earnings calls rather than on filings or releases; the Parts label every such figure "call coverage", and §95.3 names the two that matter most. Within those limits the primer is descriptive and complete as to what was found, and §94 is complete as to what was not.